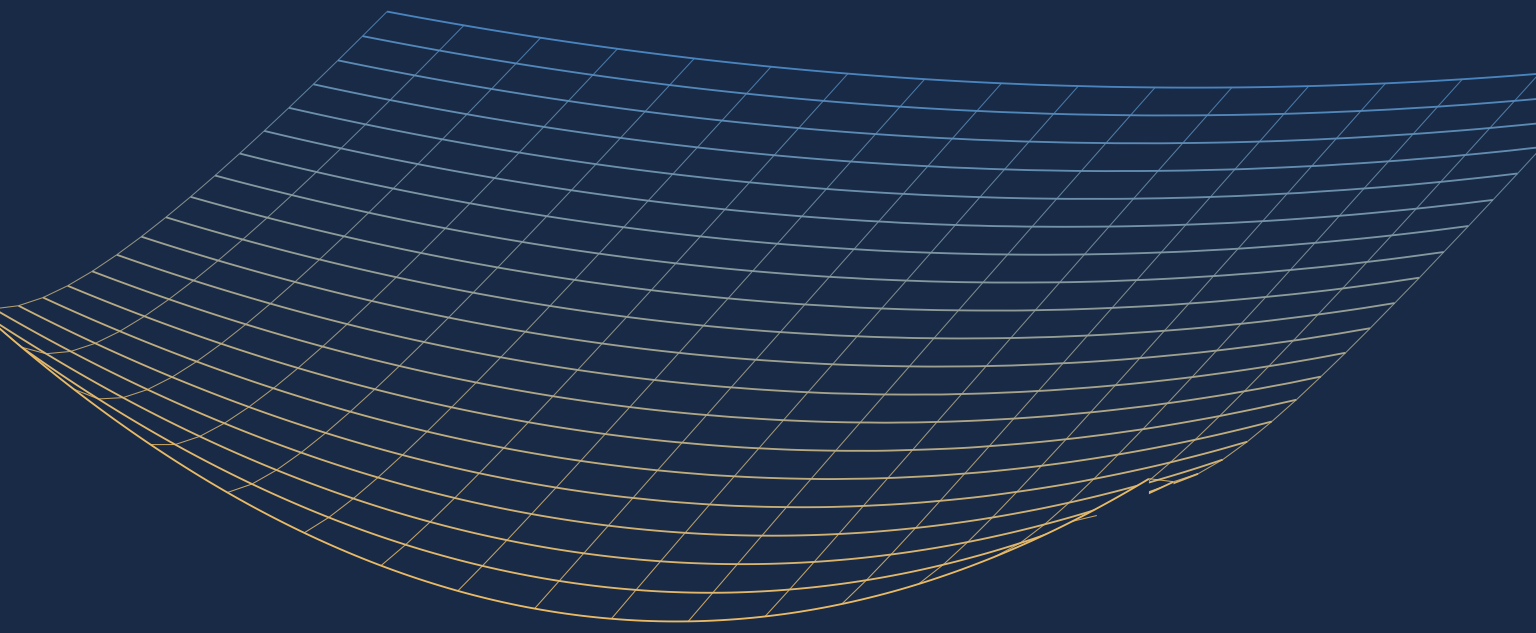
DRAFT EDITION

DERIVATIVES

CONTRACTS, MODELS, AND MARKETS

WULIN SUO



About the Cover

The cover is an equity implied volatility surface, the object of Chapter 19 and the one picture that says what this book is about.

The curves running across the sheet are smiles, one to a maturity: implied volatility against moneyness, out-of-the-money puts to the left, calls to the right. The lines running the other way join them along expiration, so the surface reads as a mesh rather than as a set of separate curves. Maturity runs back to front, two years at the back and six weeks at the front, and read in that direction you watch the skew steepen as expiration approaches — the most durable empirical regularity in equity option markets, and the one thing a single Black–Scholes volatility cannot accommodate, since it would draw this picture as a flat plane. The colors run with maturity, the amber of the near smiles cooling to blue at the long end; they are the two colors the interior uses for its worked examples and its case studies.

The shape is not artistic license. It is drawn from

$$\sigma(k,T) \;=\; 0.18 \;+\; \frac{0.11\,k^2}{T+0.30} \;-\; \frac{0.075\,k}{T+0.50},$$

with k the log-moneyness and T the maturity in years: a convexity term and a skew term, each decaying with maturity at its own rate. Curvature and tilt therefore fade at different speeds, which is what makes the surface twist as it flattens rather than simply subside.

That it is computed rather than drawn is the point, and it is the rule the whole book follows. Every figure here is generated in TEX from the formula the text derives, so no curve can drift out of step with the equation beside it; the cover is held to the same standard as the figures inside it.

The surface is also the natural emblem for the title. It is a market observable, quoted and traded in its own right. It is the output of a model, and the place where models are most visibly asked to do more than they can. And its shape is a fact about market structure: who buys crash protection, and who is left short it. Contracts, models, and markets meet here, which is why this is the picture on the front.

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Derivatives: Contracts, Models, and Markets

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Draft — October 29, 2026

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Derivatives: Contracts, Models, and Markets
Wulin Suo

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Preface

This book teaches derivatives the way the subject is actually organized: instruments and the infrastructure that clears them, then options and their valuation, then volatility, credit, rates, and the places where models are asked to do more than they can. That sequence is not mine. It was established by John Hull in *Options, Futures, and Other Derivatives*, refined across eleven editions, and it is the order in which the subject is taught almost everywhere for the good reason that it works. A student who understands forward pricing before futures, futures before swaps, and binomial trees before Black–Scholes is a student who has never had to accept a result without proof. *I have followed that sequence deliberately and I want to say so at the outset*, because a reader who knows Hull's book will recognize the architecture and should not have to wonder whether the resemblance is accidental.

What follows is not a revision of that book, and it is not authorized by or affiliated with it. It is a new one, written because the subject moved. Professor Hull passed away on January 31, 2026 and the eleventh edition, published in 2023, is the last. The years around it were not quiet ones. LIBOR was replaced by overnight rates compounded in arrears, and with it the single discount curve that a generation of textbooks assumed. Collateral became the thing that determines what a derivative is worth, so that valuation adjustments moved from a specialist annex into the price itself. Volatility modeling absorbed a genuinely new idea in rough volatility, the first structural change to the subject since stochastic volatility. Clearing houses became systemically central, and their margin models became a market risk in their own right. Commodities and crypto each acquired enough substance to deserve a chapter rather than a remark, and machine learning enough to deserve the online technical note that accompanies the book. *A book that stopped in 2023 would now mislead a practitioner about the mechanics of the instruments they trade every day*, and that is the gap this one is written to fill.

I owe Professor Hull a great deal more than a sequence, and I came to the subject through him. My first doctorate was in applied mathematics (stochastic control theory), and I spent the year after it as a post-doctoral fellow in the operations management group at the Faculty of Management, University of Toronto, now the Joseph L. Rotman School of Management. By the end of that year I had decided against a career in mathematics, without much idea of what else the training was for. Someone there told me that a professor named John Hull was well known in financial derivatives, and that a mathematical background was useful there. I bought his book, then in its second edition, and the subject turned out to be a great deal more interesting than a way of salvaging a training. He took me on as a doctoral student in finance at the University of Toronto. That copy was the second edition; this book is written after the eleventh.

Much of what I understand about how to explain a model — and about the difference between a model that fits and a model that is right — I learned from him. Readers who want the original statements of results this book uses will find his research papers in Appendix D, cited where the results are used, alongside the rest of the literature.

There is something further that no citation captures. *More than any other book, his brought people into this subject.* A great many of the practitioners and academics now working in derivatives first understood what a derivative was from its pages, and would say so without being asked; I am one of them. That is a rarer achievement than any particular result, and it is

the standard this book is written against rather than one it expects to meet.

Where this book departs from its predecessor, it does so on purpose, and the departures are the reason it exists.

It argues. A textbook can present a model and leave the reader to infer what it is worth. This one states what each model captures, where it fails, and how a practitioner would tell, often with the failure quantified. Chapter 21 measures how badly Heston misses the short-dated skew; Chapter 26 shows that a senior tranche's expected loss moves by a factor of twenty-eight for a tripling of the default rate. The claims are meant to be checkable, and where I make one I have tried to give the number that supports it.

Every figure is computed, not drawn. The plots are generated from the same formulas the text derives, so a curve cannot drift out of step with the equation beside it. Where a worked example runs long, the script that produced it is kept, and a reader can rerun it.

It is bilingual, and the Chinese edition is a superset rather than a translation. It covers everything here, on the same terms, and adds the domestic market where it differs (China's benchmark architecture, its bond futures conventions, its structured products), because a Chinese reader needs both the international convention and the local one, and needs to understand the reason for the gap.

And it takes market structure seriously as subject matter. Clearing, margin, collateral, auctions, and the definitions inside contracts are not preliminaries to the mathematics. They are where a large share of actual losses come from, and several chapters are built around episodes (2008, the 2020 dash for cash, Archegos, the LME nickel squeeze, Credit Suisse's additional tier 1) in which the mechanism mattered more than the model.

The mathematical level is deliberately unchanged from its predecessor's: accessible, and worked through with intuition and numbers rather than asserted. *Five practical notes on how the book is put together.*

Elective material. A dagger on a chapter or section title (Chapters 14, 21, 28, 30 and 32, and one section of Chapter 20) marks material aimed at readers with stronger quantitative preparation. The mark appears in the table of contents and in the running head as well as on the heading, so a path through the book can be planned from the contents alone. Elective material can be skipped without breaking continuity: an MBA reader can pass from Chapter 13 to Chapter 15 without Chapter 14, for instance. A financial-engineering reader should not skip it, and nothing in it is optional in the sense of being less settled or less used—it is harder, not softer. Section 1.13 maps the dependencies between chapters and is the place to start if you are assembling a path rather than reading straight through.

Technical notes. Derivations that would interrupt the main argument are collected in a companion set of technical notes, published separately and referenced in the text as [TN15], numbered to match the chapter that defers to them. The main text states results and explains why they hold; the notes prove them. *The two the argument leans on hardest are reproduced in Appendix A*, so a reader working from the book alone is never stopped by one.

Examples and case studies. Every formula whose magnitude matters is carried through actual numbers, because a reader who has not seen a convexity adjustment expressed in basis points does not yet know whether it is worth computing. Several worked examples deliberately verify a result two ways, and those cross-checks are part of the method being taught rather than decoration. The case studies carry the episodes: Metallgesellschaft, Amaranth, Archegos, the LME nickel squeeze, and the rest.

Companion notebooks. Twenty-three chapters (3, 9, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 31, 32, 33, 34, 35, 36 and 38) come with a Python notebook that reproduces their numbers; the online note on machine learning, TN39, has one as well. The code is written out rather than delegated: a calculation usually left to a library is set down step by step, because

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seeing it is the point of reading it. *No chapter depends on its notebook*, since each is complete as it stands, but a reader who wants to change a parameter and watch what moves will get further with one open. The two editions run identical code; only the narrative and the printed labels are translated.

Boxed equations. A box around an equation marks a result that later chapters build on and that a reader should be able to reproduce from memory. A chapter has one or two at most, and many have none; intermediate steps, approximations and special cases are never boxed, however often they are used.

Errors that remain are mine. I would be grateful to hear of them.

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† indicates an elective chapter or section; the preface explains what that means.

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Notation

The literatures this book spans grew up separately and their symbols collide; these are the conventions used here, followed by the symbols that carry more than one meaning.

PRICES, RATES, AND TIME

t	the present, or a general date
T	maturity, expiration, or settlement date
τ	accrual fraction between two dates, in years
S, S_0, S_T	spot price of the underlying: generally, now, at maturity
F, F_0	forward or futures price
K	strike price, or the delivery price of a forward
r	continuously compounded risk-free rate
q	dividend yield; the foreign risk-free rate for a currency
y	convenience yield on a commodity
L, N	notional principal

OPTIONS AND VOLATILITY

c, p	prices of a European call and put (Chapters 11, 15, 16)
C, P	prices of an American call and put (Chapter 11); P standing alone is never a bond price
σ	volatility; implied volatility where the context is a quote
$N(x)$	standard normal cumulative distribution function
$N'(x)$	standard normal density, $e^{-x^2/2}/\sqrt{2\pi}$; written ϕ or φ in Chapters 24, 25, 29 and 36
d_1, d_2	the arguments of the Black–Scholes and Black formulas
Δ, Γ, Θ, $\mathcal{V}$	delta, gamma, theta, vega
H	barrier level
κ	speed of mean reversion of variance in a stochastic volatility model
H (Chs. 14, 21)	Hurst parameter of fractional Brownian motion, and of a rough volatility model

TERM STRUCTURE

$P(t,T)$	price at t of a zero-coupon bond maturing at T
$A(t,T)$, $B(t,T)$	the coefficients of an affine bond price, $P = A\,e^{-Br}$
$B(t)$	value at t of one unit invested from time 0 at the overnight rate used for discounting; the accumulation account
A	annuity, or PV01 leg, of a swap
a	speed of mean reversion of the short rate
Q	the risk-neutral measure
Q^T, Q^N	the T-forward measure and the measure induced by numeraire N
$\mathbb{E}$	expectation, with the measure carried as a superscript
Λ	a Radon–Nikodym derivative: dQ/dP in Chapter 28, and the ratio of two forward measures in Chapter 30; Λ_t is its restriction to $\mathcal{F}_t$, the density process

CREDIT

λ	hazard rate, or default intensity
τ	time of default
$Q(t)$	survival probability to t (Chapters 24–26); not the measure
R	recovery rate
A	risky annuity: the present value of one basis point paid until default
ρ	default correlation

RISK

α	confidence level of a risk measure
μ	drift, or mean
ρ	correlation
β	sensitivity to a common factor

SYMBOLS WITH MORE THAN ONE MEANING

- ρ: a *correlation* nearly everywhere (including price against variance in a stochastic volatility model), and the Greek *rho* in Chapter 18.
- H: a *CSA threshold* in Chapter 9, a *barrier level* in Chapter 34, the *Hurst parameter* in Chapters 14 and 21, a *futures price* in Chapter 28; no chapter uses two at once.
- B: the *accumulation account* $B(t)$ (Chapters 28 and 30) and an *affine coefficient* $B(t,T)$ (Chapter 31); the arguments separate them.
- τ: an *accrual fraction* in the rates chapters, the *time of default* in Chapters 9, 24 and 25.
- A: an *annuity* in the swap chapters, a *risky annuity* in the credit chapters; the same construction, stopping or not at default.

The normal distribution function is $N(\cdot)$, not $\Phi(\cdot)$, except in one formula of Chapter 28, where N is a notional. $P(t,T)$ is always a bond price; P alone is the American put of Chapter 11, or in Chapters 19, 21 and 22 a European put as a function of strike, $P(K)$.

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Part I

Markets, Instruments, and Infrastructure

The first nine chapters build the market before they build any model. They introduce the four instrument families (forwards, futures, swaps, and options) and then the infrastructure that decides what a derivative is actually worth: the risk-free-rate benchmarks that replaced LIBOR, the collateral agreement, and the valuation adjustments that follow from it. Nothing here needs stochastic calculus. A reader who takes only this part should still be able to read a term sheet, say who is on the other side of it, and know which discount curve applies.

1 **Introduction: Derivatives and Their Markets**
what the instruments are, who trades them, and how large the market is

2 **Futures Markets, Clearing, and Central Counterparties**
the exchange, the margin cycle, and the default waterfall

3 **Hedging Strategies Using Futures**
basis risk, the minimum-variance ratio, and when a hedge fails

4 **Interest Rates in the Risk-Free-Rate Era**
SOFR and its peers, and what replaced a term rate formed from quotes

5 **Determination of Forward and Futures Prices**
cost of carry, and why forwards and futures are not the same

6 **Interest Rate Futures**
the Treasury contract, the conversion factor, and cheapest-to-deliver

7 **Swaps and the Multi-Curve Framework**
the instrument, and why one curve became several

8 **Collateral, Margin, and Funding**
the CSA as a pricing input rather than paperwork

9 **XVA: Valuation Adjustments in Practice**
CVA, DVA, FVA, and what each one charges for

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Chapter 1
Introduction: Derivatives and Their Markets

A derivative is a contract whose value depends on something else. That is nearly the whole definition, and its emptiness is the point: the something else can be a commodity price, an interest rate, an exchange rate, a share price, a credit event, an index of volatility, a quantity of rainfall, or the price of another derivative. What unites the instruments in this book is not what they reference but what they do—they separate exposure to a risk from ownership of the thing that carries it.

That separation is the entire economic content of the subject. A farmer who wants the price of wheat fixed in September need not sell the wheat in March. A pension fund that wants thirty-year interest rate exposure need not buy thirty-year bonds. A bank that has made a loan it wants to keep, to a borrower whose credit it no longer wants, need not sell the loan. In each case a contract does what a transaction in the underlying asset would otherwise have done, usually with less capital, less disruption, and more precision—and, as the rest of this book documents at length, with risks of its own that are easy to underestimate precisely because so little cash changes hands at the outset.

This chapter sets out what the instruments are, how the markets that trade them are organized, how large those markets actually are once the headline numbers are read correctly, who is on each side of a trade and why, and what has repeatedly gone wrong. It is the map; every claim in it is developed properly later.

Sections 1.1 to 1.3 set out the four instruments, where they came from, and why they exist. Sections 1.4 to 1.7 describe the markets: exchange-traded against over-the-counter, the legal architecture of master agreements and netting, how large the market is once gross notional is read correctly, and what can serve as an underlying. Sections 1.8 to 1.10 are the participants: who uses derivatives and for what, why the labels hedger, speculator, and arbitrageur blur, and the leverage that makes every position larger than its cash. Sections 1.11 and 1.12 record what has gone wrong and how regulation responded after 2008, and Section 1.13 explains how the chapters fit together.

1.1 THE FOUR BUILDING BLOCKS

Almost every derivative in existence is a combination of four contracts. Learning the four well is more useful than memorizing the hundreds of named structures assembled from them.

A forward obliges one party to buy, and the other to sell, a specified asset on a specified future date at a price agreed now, the **forward price**. No money changes hands at inception; the forward price is set so that the contract is worth zero to both sides on day one. It is the simplest derivative and the subject of Chapter 5.

A futures contract is a forward that has been standardized, listed on an exchange, and settled daily in cash; the price at which it trades is the **futures price**. Daily settlement is more than a technicality: it converts a single payment at maturity into a stream of payments along the way, so that whichever side the price moves against must find cash now rather than at maturity, and Chapters 2 and 8 are largely about the consequences.

A swap exchanges two streams of payments (most commonly a fixed interest rate for a floating one) over a series of dates. A swap is economically a portfolio of forwards, one per payment date, bundled into a single contract with a single price. Chapter 7 develops them.

An option gives its holder the *right*, not the obligation, to buy (a call) or sell (a put) at a set price, the strike price. Because the holder will exercise only when it pays to do so, the option can never be worth less than zero to them, and they must therefore pay for it at the outset with a **premium**. Part II is devoted to options.

The essential distinction is between the first three and the last. Forwards, futures, and swaps are *symmetric*: both parties are obliged, gains and losses run in both directions, and the **payoff**, what the contract delivers at maturity, is a straight line. Options are *asymmetric*: one party has a choice, the payoff is kinked, and the premium is the price of the kink. Figure 1.1 shows the difference in terms of **profit**, which is the payoff net of any premium paid or received.

Three features of Figure 1.1 are worth fixing now. The forward's line passes through zero at the forward price, so the contract starts life worth nothing and can end worth a great deal in either direction. The long call's worst case is bounded at the premium, however far the market falls. And the short call, in the fourth panel, has the mirror-image profile: a gain capped at the premium and an unbounded loss. That asymmetry is why option writing is a business for firms that can hedge and post collateral, and why it recurs in almost every chapter about things going wrong.

1.2 WHERE THESE MARKETS CAME FROM

Derivatives are often presented as a modern invention, and the impression is wrong in a way that obscures how the market works. Contracts for future delivery are ancient; what is modern is the standardization, the clearing, and the range of risks that can be referenced. More useful than a chronology is the pattern the chronology reveals: each major instrument appeared shortly after a risk became intolerable and a way of measuring it became available.

The Dojima rice market in Osaka, formally sanctioned in 1730, traded standardized claims on rice for future delivery, with a clearing house and margin: the essential apparatus of Chapter 2, two centuries early. The Chicago Board of Trade, founded in 1848, standardized grain contracts in the 1860s for the same reason: buyers and sellers separated by a season and a railroad needed a price agreed in advance.

The modern market dates from the breakdown of the Bretton Woods system. When fixed exchange rates ended in the early 1970s, currency risk became a problem every multinational suddenly had, and in May 1972 the Chicago Mercantile Exchange listed the first financial futures, on foreign currencies. Inflation and interest rate volatility followed, and with them the first interest rate futures in 1975–77 and the Eurodollar contract in 1981, on three-month US dollar deposits held at banks outside the United States—the first futures contract settled purely in cash because its underlying could not be delivered at all.

Options followed the same logic on a different track. Options had traded over the counter for centuries, illiquidly and with unreliable pricing. The Chicago Board Options Exchange opened in April 1973 with listed calls on sixteen stocks; the Black–Scholes–Merton papers appeared the same year. It is tempting to say the model created the market. It is more accurate

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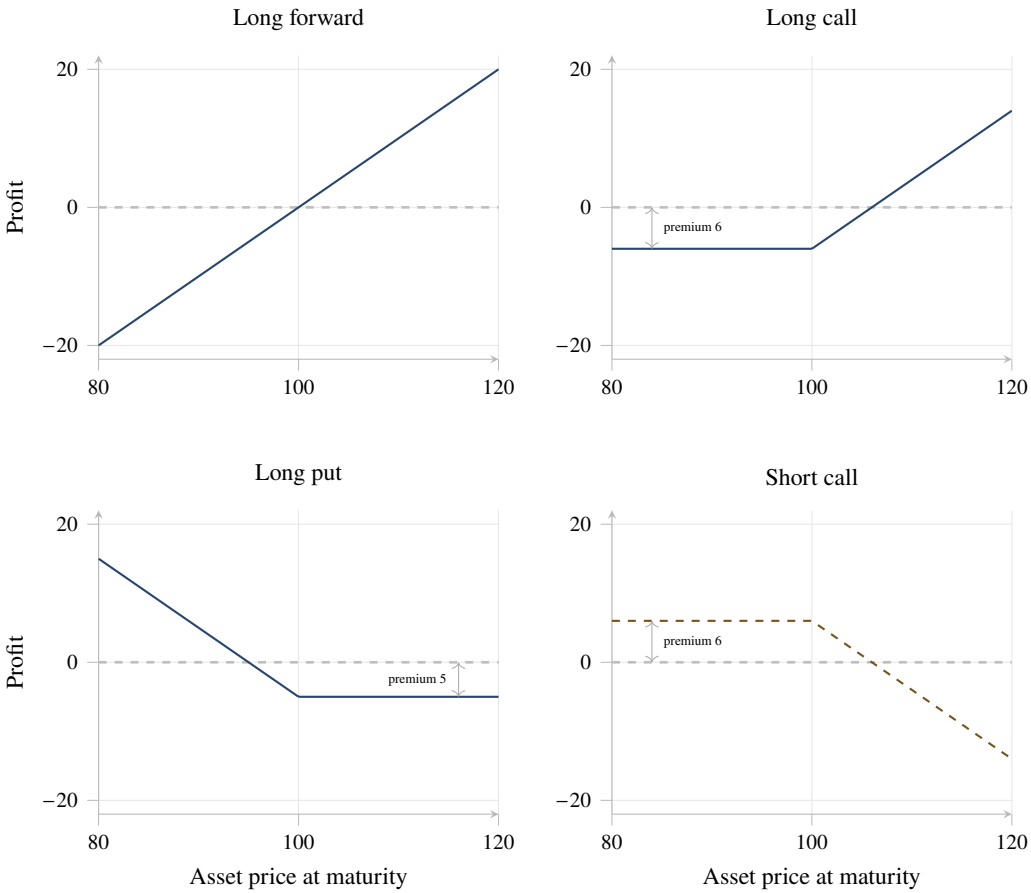


Figure 1.1: Profit at maturity, premium included. Strike and forward price 100; call premium 6, put premium 5.

to say that standardization and a workable valuation method arrived together, and that neither would have sufficed alone, a point Chapter 15 returns to.

Swaps arrived from a different direction: not an exchange, but a single negotiated trade. The 1981 currency swap between IBM and the World Bank, arranged by Salomon Brothers, is conventionally cited as the first. Within a decade the swap market was larger than the exchange-traded market, entirely bilateral and documented under agreements that had to be invented as it grew (Section 1.5). Credit default swaps, which pay the buyer if a specified borrower defaults, emerged from the same bilateral tradition in the mid-1990s, developed by banks that wanted to keep lending to clients whose credit they had already taken too much of.

The last two decades reverse the direction of travel. Rather than new instruments, the innovations have been structural: central clearing extended to over-the-counter (OTC) contracts after 2009, mandatory margining completed in 2022, and the replacement of the London Interbank Offered Rate (LIBOR) by overnight risk-free rates between 2021 and 2023—a change of foundations rather than of instruments. Newer underlyings continue to appear, and Section 1.7 surveys them, but the striking feature of the modern era is that the four building blocks of Section 1.1 have not changed. What changed is who stands between the two parties, and what they must post.

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1.3 WHY DERIVATIVES EXIST

Why anyone bothers is a fair question, because the alternatives are real and sometimes better.

A firm exposed to a price it dislikes has several options that do not involve derivatives at all. It can hold physical inventory, which costs storage and capital. It can buy insurance, which works for events that are verifiable and idiosyncratic but not for market prices, since an insurer cannot diversify away a move that hits every policyholder at once. It can restructure operations (an airline can hedge fuel by owning a refinery, as Delta did in 2012), which works but ties up capital and management attention in a business the firm did not choose to be in. It can hold more equity, absorbing the risk rather than transferring it. Or it can simply accept the exposure, which for a diversified shareholder may be the correct answer and is discussed in Section 3.2.

Derivatives compete with these on three grounds:

Precision. A contract can be written on exactly the risk in question (one currency pair, one maturity, one delivery month) rather than on a bundle that happens to contain it.

Capital efficiency. Transferring $100 million of exposure requires posting margin measured in single-digit millions rather than committing $100 million. This is a genuine advantage and simultaneously the source of most of the disasters in Section 1.11; Section 1.10 treats it as one topic rather than two.

Reversibility. A hedge can be lifted when the exposure changes. Selling the underlying business cannot.

Against these sits the standing objection: because so little cash is required at inception, the size of a derivatives position is not disciplined by the size of the balance sheet behind it. Every episode in Section 1.11 is a variation on that sentence.

1.4 EXCHANGE-TRADED AND OVER-THE-COUNTER MARKETS

Derivatives trade in two structurally different places, and the difference used to be sharp. It is now more subtle, because regulation has pushed the two toward each other.

Exchange-traded contracts are standardized: the exchange fixes the size, the underlying, the delivery months, and the settlement terms, and participants choose only price and quantity. Every trade is novated to a central counterparty (CCP), which replaces it with two contracts, one between each party and the CCP, so the CCP becomes buyer to every seller and seller to every buyer, and no participant bears the credit risk of another. Standardization concentrates liquidity (any two participants wanting the same contract meet in the same order book) at the cost of fit: the contract may not match the exposure a hedger actually has, which is the basis risk of Chapter 3.

Over-the-counter contracts are negotiated bilaterally between two institutions and can specify anything the parties agree on: a non-standard maturity, an unusual underlying, a bespoke payoff. The fit is exact. Historically the price of that fit was credit risk: each party depended on the other's solvency for the life of the contract, which for a thirty-year swap is a long time.

Figure 1.2 shows the two, and the hybrid that now dominates.

The middle panel is where most of the market now sits. Following the G20's 2009 commitments (Section 1.12), standardized OTC contracts, above all interest rate swaps, must be cleared through a central counterparty. Such a trade is negotiated bilaterally, like an OTC contract, and then novated to a CCP, like a futures trade. It inherits the flexibility of one structure and the credit mechanics of the other, and that is why the old question "exchange-traded or OTC?" now has three answers rather than two. Chapter 2 works through what a CCP actually does with the risk it absorbs, which is not the same as making the risk disappear.

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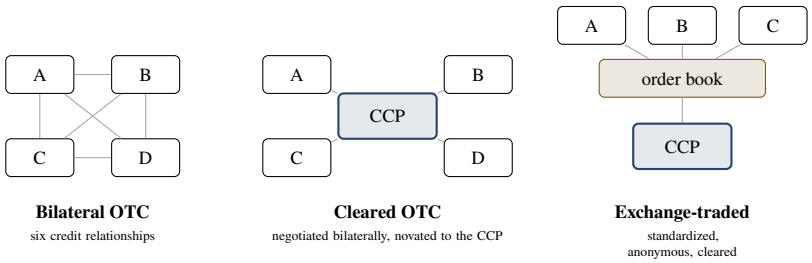


Figure 1.2: How a trade reaches its counterparty. Four participants shown.

1.5 THE LEGAL ARCHITECTURE: MASTER AGREEMENTS AND NETTING

A bilateral derivative is a promise extending years into the future, and what makes such a promise financeable is not the pricing model but the document. This is not a legal aside. The reduction from $23 trillion to $3.4 trillion in the next section is entirely an artifact of the arrangement described here, and Chapters 8 and 9 assume it throughout.

Two institutions did not exist when the swap market began and had to be invented as it grew. The first is the contract published by the International Swaps and Derivatives Association (ISDA), the **ISDA Master Agreement**, now the standard for essentially all bilateral derivatives. Its central provision is deceptively simple: all transactions between the two parties, however many and however different, form a **single agreement**. A party cannot default on one swap and perform on another.

That provision exists to enable **close-out netting**. If one party defaults, every outstanding transaction between them terminates at once, each is valued, and the values are summed to a single net amount owed one way or the other. Without it, an insolvency administrator could do what the market calls *cherry-picking*: enforce the contracts that favor the estate and disclaim the rest. Example 1.1 shows what that difference is worth.

Example 1.1 Why the single agreement clause exists

Bank A has five outstanding trades with counterparty B. Marked to market (valued at current market prices) from A's perspective:

$$+40, \quad -35, \quad +20, \quad -15, \quad -8 \qquad (\$ \text{ million})$$

The positives are amounts B owes A; the negatives are amounts A owes B. A's gross positive exposure is $60 million; the net is $2 million in A's favor.

With close-out netting, B's default terminates all five. They sum to $2 million, which A claims from the estate. A's loss is at most $2 million, and if collateral has been posted under the arrangement described below, close to nothing.

Without close-out netting, the administrator enforces the three contracts on which A owes money and disclaims the two on which A is owed. A must pay 35 + 15 + 8 = $58 million in full, and joins the queue of unsecured creditors for its $60 million of claims. At a 20% recovery rate (the share of its claims an unsecured creditor eventually collects), A collects $12 million:

$$\text{A's net payment} = 58 - 12 = \$46 \text{ million, in place of a } \$2 \text{ million claim}$$

The trades did not change. The document did. This is why the enforceability of close-out netting in a given jurisdiction is a precondition for a derivatives market existing there at all, and why regulators permit netted exposures to be used in capital calculations only where that enforceability has been established.

The second institution is the Credit Support Annex (CSA), an optional schedule to the

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Master Agreement under which the parties post collateral against the net amount as it moves. The CSA converts a credit exposure into a daily cash flow, which largely removes default risk and puts liquidity risk, the risk of having to find cash at short notice, in its place. That substitution is the subject of Chapter 8, because the second risk turned out to be the one that broke firms in 2022.

Together the two documents explain the structure of the numbers in the next section: gross market value is what the contracts are worth, netting under the Master Agreement collapses it to a fraction, and collateral under the CSA covers most of what remains.

1.6 HOW BIG IS THE MARKET?

Derivatives statistics are quoted in numbers so large that they are routinely misused. The Bank for International Settlements (BIS) puts the notional amount outstanding of over-the-counter derivatives at roughly $850 trillion in recent years ($845 trillion at the end of 2025), something like seven times world gross domestic product (GDP). The figure is correct and almost entirely uninformative.

Notional is the reference amount used to *calculate* payments. It is not an amount anyone owes, expects to receive, or can lose. On a $100 million ten-year interest rate swap, the $100 million is never paid by anyone: it is multiplied by a rate difference to determine a payment of a few hundred thousand dollars. Two more meaningful measures follow.

Gross market value is what the outstanding contracts are actually worth: the sum of what every position with a positive value would be paid if all contracts were settled today. Across the market this runs at roughly $23 trillion, under 3% of notional.

Gross credit exposure nets each pair of counterparties' offsetting positions against one another, as their master agreements permit them to do. That reduces the figure again, to roughly $3.4 trillion, under half a percent of the headline. Most of that remainder is collateralized, by the mechanisms of Chapter 8.

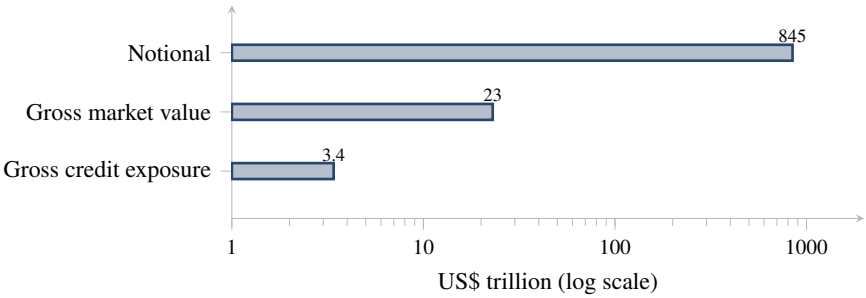


Figure 1.3: OTC derivatives at the end of 2025 (BIS). Log scale.

Figure 1.3 needs a log scale to show all three bars at once, which is itself the lesson: each step down is roughly an order of magnitude. A commentator who reports the first number and implies the third has overstated the risk by a factor of some two hundred and fifty. Example 1.2 traces the same three measures through a single swap.

Example 1.2 What $100 million of notional actually risks

A bank enters a five-year interest rate swap on $100 million notional, paying 4.00% fixed annually and receiving the floating rate.

The notional is never exchanged. In a year when the average floating rate sets at 4.60%, the bank receives

$$100{,}000{,}000 \times (0.0460 - 0.0400) = \$600{,}000$$

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and in a year when floating sets at 3.50% it pays $500,000. The annual cash flows are measured in hundreds of thousands on a $100 million "position."

Now suppose rates move sharply and the swap's market value rises to $4 million in the bank's favor. That $4 million—not $100 million—is what the bank would lose if the counterparty failed and nothing had been collateralized. If the same counterparty also owes the bank $1.5 million on one swap and is owed $3 million on another, netting under the master agreement reduces the exposure again, to $2.5 million.

So the same trade supports three "sizes": $100 million of notional, $4 million of market value, $2.5 million of net exposure, and after collateral very likely a few hundred thousand. Each is the right answer to a different question, and confusing them is the single most common error in commentary about this market.

Case Study 1.1 Why the headline number will not go away

Notional outstanding is a poor measure of risk, and everyone who works in the market knows it. It nevertheless remains the number that gets quoted, for two reasons worth understanding.

The first is that it is the only figure that can be reported consistently. Market value depends on the valuation date and on models; notional is written into the contract. The second is that notional is not *meaningless*: it scales operational load, and it is the base on which several regulatory measures are calculated. Two portfolios with the same market value but very different notionals are not equally easy to manage, close out, or transfer.

A useful discipline: whenever a derivatives figure is quoted, ask which of the three it is. If the source does not say, it is almost certainly notional, and almost certainly being used to suggest something the number does not support.

What the market is made of

The composition is as lopsided as the size is large. Figure 1.4 gives OTC notional by asset class at the end of 2025.

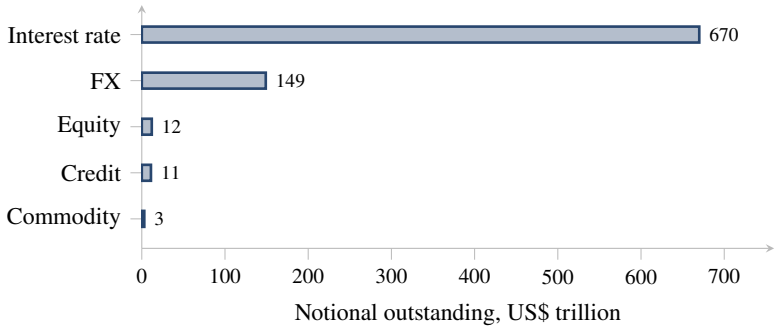


Figure 1.4: OTC notional by asset class, end of 2025 (BIS).

Interest rate contracts are roughly 80% of the total and foreign exchange most of the rest. Credit, equity, and commodity derivatives together are a rounding error on notional—which is precisely why notional misleads, since credit derivatives caused a financial crisis while representing about 1% of it. This lopsidedness explains the shape of this book: Chapters 4 through 9 and 28 through 33 concern rates, because that is where the market is.

Exchange-traded volumes tell a different and more recent story. In 2024 the Futures Industry Association counted a record 206.5 billion futures and options contracts traded worldwide, and the largest single venue was India's National Stock Exchange. In 2025 the count fell 42% to

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119.3 billion, most of the fall in equity contracts, though Asia–Pacific exchanges still accounted for well over half of the total. Contract counts are not comparable to notional (an Indian index option and a short-term interest rate futures contract on $1 million differ in size by orders of magnitude), and a swing of that size in a single year is a warning against reading a trend off them. Measured by *activity*, the center of gravity in listed derivatives has nonetheless moved toward Asia, and toward small-size, short-dated, retail-accessible contracts. Chapter 10 returns to this, including the growth of options expiring the same day they are traded, which now account for a large share of US index option volume.

1.7 WHAT CAN BE AN UNDERLYING

The five asset classes of Figure 1.4 account for almost all the outstanding notional, but they do not exhaust what a derivative can reference, and the exceptions are instructive about why derivative markets succeed or fail.

Weather. The Chicago Mercantile Exchange (CME) lists futures and options on heating and cooling **degree days** for major cities, settling against temperature data published by a national meteorological service. The index is the number of degrees by which each day's average temperature falls below (heating) or rises above (cooling) 65°F, summed over a month or season. An energy utility's revenue falls in a mild winter; a weather contract pays when the winter is mild. The market is small but it works, and it works for a specific reason: the settlement index is public, verifiable, and impossible for either party to influence.

Freight. Forward freight agreements, which fix the cost of carrying a cargo on a given route, settle against published shipping route indices. Their users, ship owners with capacity to fill and charterers with cargo to move, have genuinely opposite exposures, which is the second condition for a market to function.

Emissions. Futures on European Union allowances (permits to emit one metric ton of carbon dioxide under the EU's emissions trading system) are now among the more liquid commodity contracts in Europe. Here the underlying exists only because policy created it, which makes the contract's value dependent on the durability of the policy—a risk with no analog in a wheat contract.

Volatility. A derivative can reference the market's expectation of future variability rather than any price. The Cboe Volatility Index (VIX) measures the volatility that S&P 500 option prices imply for the next 30 days; VIX futures, listed in 2004, and options on them made volatility an asset class in its own right; Part IV is devoted to it.

Cryptoassets. CME listed bitcoin futures in December 2017 and ether futures subsequently. After US spot bitcoin exchange-traded funds (ETFs) were approved in January 2024, listed options on those funds followed later that year, adding options on a spot product to the options on bitcoin futures that CME had listed since 2020. The instruments are entirely conventional; what is novel is the underlying and the extreme volatility that pricing must accommodate.

Other derivatives. An option on a futures contract, or a swaption (an option to enter a swap), references a derivative rather than an asset. These are not curiosities: swaptions are among the largest option markets in the world, and Chapter 29 prices them.

What makes a derivative market possible. The list suggests three conditions, and they explain the failures as well as the successes. There must be an *observable, verifiable settlement reference* that neither party can manipulate. There must be participants with *genuinely opposite exposures*, so that trade is risk transfer rather than a wager between two people who both merely have opinions. And the exposure must be *material enough to be worth managing*, since a contract no one needs will not attract the liquidity that makes it usable.

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Proposals for derivatives on macroeconomic aggregates (GDP growth, inflation-adjusted national income, house price indices for individual homeowners) have repeatedly satisfied the first condition and failed the second. Almost everyone is exposed to a recession in the same direction, so there is no natural party to take the other side, and the contracts have not sustained liquidity despite decades of advocacy. When a proposed contract seems obviously useful and does not exist, the missing party is usually the explanation.

1.8 WHO USES DERIVATIVES, AND WHAT FOR

Textbooks traditionally divide participants into hedgers, speculators, and arbitrageurs. The taxonomy is useful and Section 1.9 takes it seriously, but on its own it tells a reader almost nothing about who is actually on the other side of a trade. Table 1.1 is more concrete: real institution types, the exposure each has, and the instrument each reaches for.

Table 1.1: Who trades what, and against which exposure.

Participant	Exposure	Typical instrument
Airlines, shippers	Fuel cost	Crude and refined product swaps, options
Exporters, importers	Receivables and payables in foreign currency	FX forwards, currency options
Corporate borrowers	Floating-rate debt service	Pay-fixed swaps, interest rate caps
Bond issuers	Locking a coupon before issue	Forward-starting swaps, swaptions
Pension funds, insurers	Long-dated liabilities discounted at market rates	Receive-fixed swaps, gilt and Treasury futures
Asset managers	Portfolio duration and equity beta	Bond and index futures
Banks (loan books)	Concentrated credit exposure	Credit default swaps, index tranches
Dealers	Inventory acquired from clients	Whatever offsets it, plus the futures strip
Hedge funds	Deliberately acquired relative-value positions	Everything in this book

Two observations about the table matter more than its contents.

First, most of these participants are *removing* an exposure they acquired by doing something else: flying aircraft, writing insurance, lending money. Derivatives markets exist mainly to serve them, and the profits earned by the other side are, in the aggregate, the price those participants pay for the service. Example 1.3 follows one of them through a quarter.

Second, the dealer row is different in kind. A dealer does not begin with an exposure; it acquires one by accommodating clients, and then manages it. That is a business of intermediation, and understanding it explains a great deal that is otherwise puzzling about market prices: why a heavily demanded contract trades rich, why hedging costs rise when everyone hedges at once. This book returns to it repeatedly, and Chapters 8 and 9 are about what that intermediation costs.

Example 1.3 An airline hedging fuel

An airline expects to burn 12 million gallons of jet fuel next quarter. Fuel is roughly 25% of its operating cost, and a 30-cent move in the price per gallon changes quarterly fuel spend by $3.6 million, enough to swing a thin operating margin to a loss.

There is no liquid jet fuel futures contract, so the airline hedges with heating oil futures, whose underlying is refined similarly. Each contract covers 42,000 gallons. Jet fuel has historically moved about 0.64 of a cent for each cent of heating oil (Example 3.3 estimates that ratio), so the airline

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buys

$$\frac{12{,}000{,}000}{42{,}000} \times 0.64 = 182.9 \rightarrow 183 \text{ contracts}$$

Suppose jet fuel then rises 30 cents a gallon and heating oil futures rise 34 cents. The airline pays $3.6 million more for fuel and gains

$$183 \times 42{,}000 \times 0.34 = \$2{,}613{,}240$$

on the hedge, leaving about $0.99 million of the increase uncovered. Jet fuel rose 30 cents for every 34 of heating oil, a ratio of 0.88 against the 0.64 the hedge was sized on, so in this quarter the hedge was too small. The ratio is set on the average relationship, and in any one quarter the two prices can move more closely together than that.

Three things in this example run through the whole book. The hedge is *approximate*, because the traded instrument is not the exposure (the cross-hedging problem of Section 3.5). The residual is not small relative to the margin being protected. And the gain on the futures leg arrives as cash, daily, whether or not the fuel has yet been bought—which is a benefit here and, in Chapter 3's failure cases, a catastrophe.

1.9 HEDGING, SPECULATION, AND ARBITRAGE—AND WHY THE LABELS BLUR

With the participants in view, the traditional triad can be stated properly.

A **hedger** holds an existing exposure and enters a derivative that offsets it. The test is not the trade but the portfolio: the same short futures position is a hedge for a producer and a directional bet for someone with no production.

A **speculator** takes a position to profit from a view. Speculators are frequently described apologetically, as though tolerated. They should not be. Hedgers as a group are net sellers of risk, and someone must be the net buyer; a market of hedgers alone would not clear. What speculation supplies is the willingness to hold an exposure no one else wants, and the price of that service is a real cost to hedgers, as Section 5.14 shows when it works out who pays whom.

An **arbitrageur** exploits a price inconsistency between related instruments, in principle without taking risk. This is the category that requires the most caution, and Section 5.15 is devoted to why: the textbook arbitrage assumes unlimited borrowing at a single risk-free rate, and the real trade requires a balance sheet, financing that may be withdrawn, and the ability to survive the position moving against you before it converges. Covered interest parity, the rule that a forward exchange rate equals the spot rate adjusted for the two currencies' interest rates, is violated persistently in the deepest currency markets in the world, and the violations exist because that assumption is false. Example 1.4 applies all three labels to a single trade.

Example 1.4 One trade, three labels

A trader buys 100 Treasury bond futures contracts. Which of the three is it?

If the trader runs a pension fund whose liabilities are longer-dated than its bonds, so that a fall in yields would raise the liabilities by more than the assets, it is a hedge: the futures offset an exposure the fund already carries.

If the trader believes long yields will fall and holds nothing else, it is speculation.

If the trader simultaneously sells the cheapest-to-deliver cash bond short (the bond a futures seller would choose to deliver), borrowing it through a reverse repo (lending cash against the bond as collateral; Section 4.1 describes the repo market), having computed that the futures are cheap relative to the bond after carry (the bond's coupon income less the cost of financing it), it is arbitrage: the basis trade of Section 6.5, run in reverse.

The three are the same instrument, the same direction, and the same quantity. Only the rest of

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the balance sheet distinguishes them, which is why regulators, risk managers, and accountants all find these categories so difficult to apply, and why a firm's own description of a position is weak evidence about what it is. Chapter 3 describes a firm whose "hedge" was economically sound and failed anyway, and Section 5.14 describes a fund whose "spread trade" carried more risk than an outright position would have.

1.10 LEVERAGE

The feature that makes derivatives efficient is the same one that makes them dangerous, and it is worth isolating.

To gain $10 million of exposure to the S&P 500 by buying shares requires $10 million. To gain the same exposure through index futures requires posting initial margin, a good-faith deposit sized to cover a few days of adverse moves, and returned when the position closes. If that margin is 5% of notional, the same exposure costs $500,000 to establish. Example 1.5 sets this route beside three others.

Table 1.2: $10 million of index exposure, four ways.

Route	Cash required	Loss on 10% fall	% of cash
Buy the shares	$10,000,000	$1,000,000	10%
Buy on 50% margin	$5,000,000	$1,000,000	20%
Index futures (5% IM)	$500,000	$1,000,000	200%
At-the-money call options	$600,000	$600,000	100%

Example 1.5 The same exposure, four ways

A fund wants $10 million of exposure to an equity index. Table 1.2 shows what each route requires and what a 10% market fall does to the capital committed.

The futures row is the one to study. A 10% fall produces a loss of twice the cash committed. The position is not merely wiped out—the fund owes $500,000 more, and will be asked for it in cash the following morning as variation margin, the daily settlement of gains and losses that Section 1.1 describes. Initial margin is not the maximum loss; it is a deposit, and Chapter 2 explains what happens when it is exhausted.

The option row (calls whose strike equals the current index level, which is what *at the money* means) differs in kind rather than degree. The premium is the maximum loss in every state of the world, which is why Section 12.10 argues that buying options is sometimes the right answer even when it looks expensive. The cost of that protection is the $600,000 premium, which is lost entirely if the index is unchanged.

Table 1.2 also explains a structural fact about the market: derivatives let a position be built that no lender would finance directly. A firm that wanted $10 billion of equity exposure would find it difficult to borrow $10 billion; it can obtain the same exposure through total return swaps with several dealers, none of whom sees the whole picture. Whether the resulting position is prudent is a separate question from whether anyone can observe it, and Archegos Capital Management showed in March 2021 what happens when the answer to both is no. It built concentrated positions in a few US and Chinese stocks almost entirely through total return swaps with several prime brokers, the banks that finance and execute trades for hedge funds and similar clients. The banks owned the shares, so Archegos filed no ownership disclosures, and each broker saw only its own exposure, with no way to know that competitors had written substantially the same trade against the same client. When the stocks fell, margin calls arrived at every broker at once, Archegos could not meet them, and the banks' simultaneous liquidations

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drove prices further down; the banks lost more than $10 billion between them. Section 27.14 works the numbers and Section 40.6 tells the episode.

1.11 WHAT GOES WRONG

Derivatives have a reputation for producing spectacular losses, and it is deserved. What is less well understood is that the losses follow a small number of repeated patterns. Table 1.3 lists the episodes this book returns to, and Chapter 40 takes them up together.

Table 1.3: Episodes and the mechanism each illustrates.

Episode	Year	Mechanism
Metallgesellschaft	1993	A long-dated hedge sound in value but funded with short-dated futures; rolls and margin calls exhausted liquidity first (Chs. 3, 40)
Barings	1995	Unsupervised trader, unreconciled accounts; the loss was in plain futures and options (Ch. 40)
Long-Term Capital Management	1998	Convergence trades that were right eventually, financed in a way that could not survive being wrong first (Ch. 40)
Amaranth	2006	Calendar spreads with low margin and no exit; size itself was the risk (Ch. 5)
Global financial crisis	2007–09	Credit derivatives on correlated collateral, valued with models calibrated on a benign sample (Chs. 24–26)
Negative WTI settlement	2020	Retail products whose design assumed prices could not go below zero (Ch. 5)
Archegos	2021	Concentrated leverage invisible across dealers (Chapters 27, 40)
LME nickel	2022	A short-covering spike that broke the clearing house's own rules (Chs. 2, 40)
European energy	2022	Solvent hedgers unable to fund variation margin (Ch. 8)
UK pension LDI	2022	Leveraged rate hedges, held in liability-driven investment mandates, liquidated into the market they were hedging against (Chs. 8, 40)

Read down the mechanism column and a pattern emerges that is not the expected one. Very few of these were valuation failures. The models mostly worked. What failed was *funding*: a position that was economically correct required cash at a moment when cash was unavailable, and had to be closed at the worst possible time.

This is the book's most persistent theme, and it recurs in almost every chapter. A hedge that is right about price can still fail through liquidity. That sentence would have been an odd thing to build a derivatives textbook around in 2005. After 2008, 2020, and 2022, it is the central practical lesson of the subject, and it is why Chapter 8 (collateral, margin, and funding) exists as a chapter in its own right rather than as an appendix to the swaps chapter.

The second pattern is that scale and invisibility travel together. Barings, Amaranth, and Archegos each involved a position that had grown too large to exit, and in each case those who could have stopped it never saw its full size: senior management at Barings, the New York Mercantile Exchange, which saw only the part of Amaranth's position held there, and the dealers facing Archegos, each of which saw only its own slice.

1.12 REGULATION SINCE 2008

The market described in this book is substantially a creation of policy decisions taken after 2008, and a reader who learned the pre-crisis structure would find much of current practice

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inexplicable.

At the Pittsburgh summit in September 2009, the G20 agreed four commitments for OTC derivatives:

- standardized contracts should be centrally cleared;
- they should trade on exchanges or electronic platforms where appropriate;
- all contracts should be reported to trade repositories;
- non-centrally-cleared contracts should attract higher capital.

Implementation took over a decade and reshaped the market:

Clearing mandates moved the great majority of interest rate and index credit derivatives to CCPs, which is why Chapter 2 treats clearing houses as systemically important institutions rather than as infrastructure.

Trade reporting created the data underlying the statistics of Section 1.6, and, less happily, a compliance burden that consolidated dealing into fewer firms.

Uncleared margin rules, phased in from 2016 and completed in September 2022, require two-way initial margin on bilateral trades between large participants. This changed pricing, not merely operations: posting initial margin has a funding cost, which is the margin valuation adjustment (MVA) of Chapter 9.

Capital rules (the credit valuation adjustment charge, the standardized approach for counterparty credit risk, and the leverage ratio) made dealer balance sheets an explicit input into derivative prices. Section 5.15's argument that arbitrage requires a balance sheet is in large part an argument about these rules.

Who writes the capital rules is worth stating once, because every later chapter assumes it. The **Basel Committee on Banking Supervision**, a body of central banks and supervisors hosted at the Bank for International Settlements, agrees standards (Basel I in 1988, Basel II in 2004, and the post-crisis Basel III), which have no legal force until each jurisdiction writes them into its own rules, as the European Union, the United States, the United Kingdom, Canada, and China separately do. A bank must then hold capital, meaning common equity first and other loss-absorbing instruments after it, of at least 8% of its risk-weighted assets (each exposure multiplied by a weight meant to reflect its riskiness), plus buffers that take the requirement above 10%, and separately at least a few percent of its *unweighted* exposure under the leverage ratio. Everything Chapters 9, 26, 27, and 39 say about capital is an elaboration of those two ratios, Section 27.11 assembles them for a whole bank, and the divergences they report (Europe exempting a charge the United States does not) come from the national step rather than from the accord.

Two developments are more recent and still unfolding. Benchmark reform retired LIBOR in favor of overnight risk-free rates, an event that changed the foundations of interest rate derivative pricing rather than its details; Chapter 4 is written from the new foundation rather than adapting the old one. And clearing continues to expand: the mandate from the Securities and Exchange Commission (SEC) to centrally clear US Treasury cash and repo transactions, adopted in December 2023 and phasing in over the following years, extends the post-crisis model to the largest government bond market in the world. Section 2.11 takes it up, with its compliance dates, and Section 6.5 its effect on the basis trade.

1.13 HOW THE CHAPTERS FIT TOGETHER

The book serves three audiences with different needs, and the shape it takes is the same for all three. Part I (Chapters 1–9) covers markets, forwards, futures, swaps, and the collateral and

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valuation machinery that modern practice requires. Part II develops options mechanics and strategies, and Part III their valuation, from the binomial tree through Black–Scholes–Merton, the Greeks (the sensitivities of an option's value to its inputs), and the volatility surface. Part IV treats volatility as an asset class. Parts V–IX cover credit, interest rate derivatives in depth, exotics and structured products, commodity and digital markets, and risk and regulation.

Figure 1.5 is the map. Within a part the chapters run in order and each assumes the one before it; the arrows are the principal cross-part dependencies, and they are what a reader assembling a path actually needs.

Three chapters are shaded because the rest of the book stands on them. Chapter 4's risk-free-rate conventions are assumed by every rates chapter that follows. Chapter 13's binomial tree is assumed by everything in valuation, exotics, and computation. And Chapter 15 is the gate between those and the rest: later chapters open by naming it as a prerequisite almost as often as they name Chapter 13.

A hub in this sense is a chapter you must pass through, not one you keep turning back to, and the distinction is worth drawing because the two rankings disagree. By the number of later chapters that cite it at all, the busiest chapter in the book is Chapter 19, whose volatility surface is referred to from most of the chapters that follow it, but almost none of them need it as a prerequisite. *Chapter 15 is the reverse*: cited less often, and skippable by no one.

SUMMARY

A derivative is a contract whose value depends on something else, and its economic function is to separate exposure to a risk from ownership of the thing that carries it.

Four building blocks account for nearly everything: forwards, futures, swaps, and options. The first three are symmetric (both parties are obliged and the payoff is linear), while options are asymmetric, giving one party a choice and requiring a premium in exchange. The distinction organizes the whole subject.

Markets are exchange-traded, bilateral OTC, or—increasingly—bilaterally negotiated and centrally cleared. The third category is a post-2009 creation and now holds most of the outstanding interest rate risk in the world.

The market's size is routinely misstated. Notional outstanding is the figure usually quoted, but gross market value is a small fraction of it and exposure after netting a smaller fraction still, most of it collateralized. Interest rate contracts make up the large majority of the notional total, which is why this book weights them heavily.

Participants are better described by what they do than by the hedger–speculator–arbitrageur triad, though the triad remains useful once it is understood that the same trade falls into different categories depending on the rest of the balance sheet. Dealers are a distinct case: they do not begin with an exposure but acquire one by serving clients, and the cost of that intermediation shows up in prices.

Leverage is the source of both the efficiency and the danger. Initial margin is a deposit, not a maximum loss; a futures position can lose several times the cash committed, while a purchased option cannot lose more than its premium.

Most derivative disasters were not valuation failures. They were funding failures: positions that were economically correct but required cash when none was available. That theme recurs throughout the book and justifies the prominence given to collateral and margin.

FURTHER READING

The BIS publishes semiannual OTC derivatives statistics and quarterly exchange-traded data; they are the authoritative source for Section 1.6 and are worth consulting directly, since the definitions matter as much as the numbers. The Futures Industry Association's annual volume

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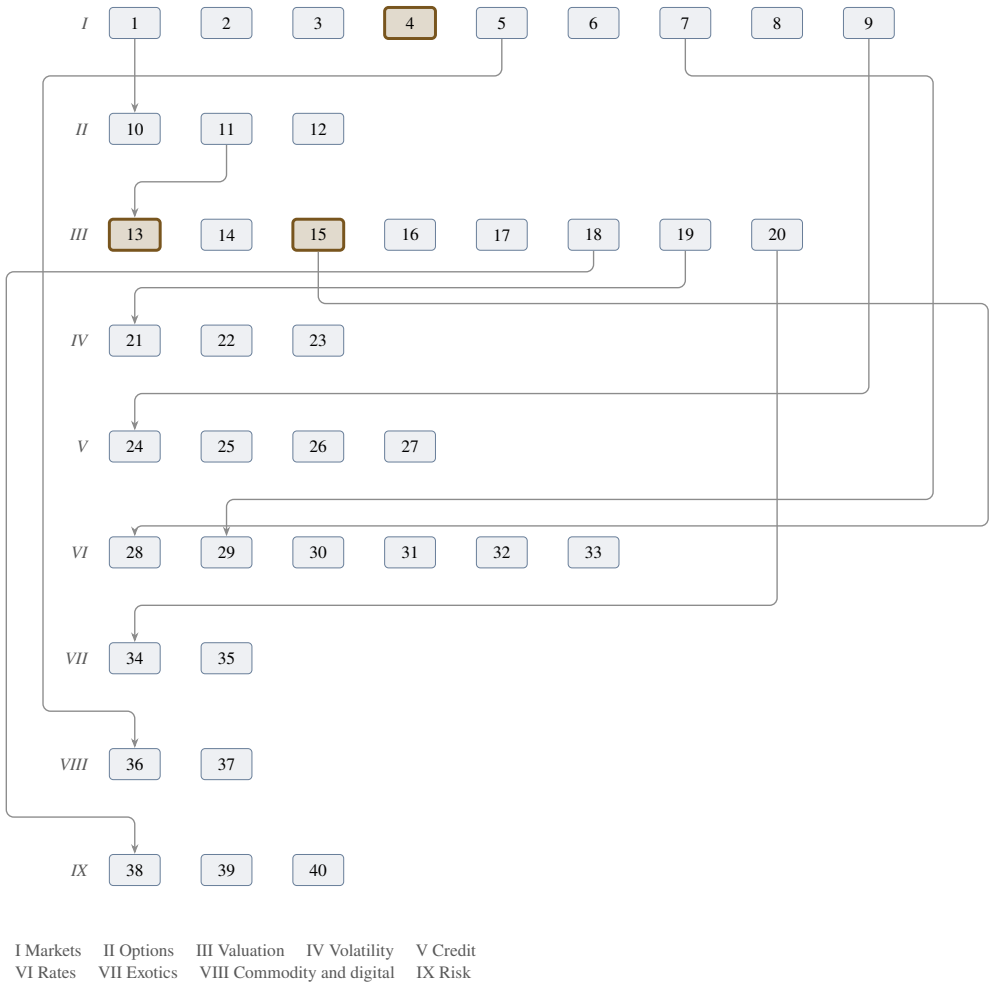


Figure 1.5: Chapters run in order within a part; arrows are the principal cross-part dependencies. Shaded chapters are the three hubs.

survey covers listed markets. For the post-crisis reform program, the Financial Stability Board's annual progress reports on OTC derivatives reform track what was promised against what was delivered. On Archegos, the Credit Suisse board's external investigation report sets out how the exposure went unmeasured.

PRACTICE QUESTIONS

1.1 Explain why a forward contract has zero value at inception but an option does not, and what this implies about when money changes hands in each.

1.2 A company enters a $50 million notional five-year swap paying 3.8% fixed and receiving floating. In the first year floating averages 5.1%. Compute the net payment and state who pays whom. What fraction of notional is it?

1.3 Distinguish notional amount, gross market value, and gross credit exposure. A newspaper reports that a bank "has $3 trillion of derivatives." What further question should a reader ask?

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1.4 An investor obtains $4 million of index exposure through futures requiring 6% initial margin. The index falls 12%. Compute the loss, express it as a multiple of the initial margin, and state what the investor owes.

1.5 Explain why a market consisting only of hedgers would not function, and identify what speculators are being paid for.

1.6 Classify each as hedging, speculation, or arbitrage, and state what additional information you would need in each case: (a) a copper producer selling copper futures; (b) a fund buying gold futures and selling gold forward at a higher price; (c) an importer buying dollars forward; (d) a bank buying protection on a borrower to whom it has lent.

1.7 Using Figure 1.1, explain why the writer of a call is in a different business from the buyer of one, and what the writer must be able to do to be in it safely.

1.8 Explain what changes when the same economic trade is executed on an exchange rather than bilaterally. Identify which of those changes transfers risk to someone else and which merely changes when cash moves, and state one exposure that the exchange does not remove.

1.9 Using Example 1.1, explain what the single agreement clause prevents an administrator from doing in a bankruptcy, and use it to account for the fall from $23 trillion of gross market value to $3.4 trillion of gross credit exposure.

1.10 Archegos, as this chapter describes it, held a position no counterparty could see in full. Identify the instrument that made it invisible, state what each prime broker could observe and what it could not, and explain what would have had to change (in the instrument, the reporting, or both) for the aggregate position to be visible to any one of them.

FURTHER QUESTIONS

1.11 Table 1.3 attributes most of the listed episodes to funding rather than valuation failure. Choose two episodes and explain precisely what cash was required, when, and why it was not available. Then state what a risk report would have had to measure to give warning.

1.12 Central clearing was intended to reduce systemic risk by replacing a web of bilateral exposures with a hub. Construct an argument that it instead concentrates risk, and identify what would have to be true for the concentration to be worth accepting. Chapter 2 gives the full treatment; attempt it first.

1.13 In the Archegos episode, each dealer's individual exposure was defensible. Explain how a set of individually prudent positions produced a collectively imprudent one, and evaluate whether a disclosure rule attaching to economic exposure rather than share ownership would have prevented it.

1.14 An airline hedges fuel and the fuel price then falls, so the hedge loses money and the airline's unhedged competitor reports lower costs. Construct the argument that the hedging program should be continued, and the argument that it should not. Which turns on facts about the airline rather than about fuel?

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Chapter 2
Futures Markets, Clearing, and Central Counterparties

A forward contract is a promise. Its value depends entirely on the promisor being willing and able to perform, years after the promise was made, at a price that may by then look catastrophic. Futures markets exist to remove that dependence, and the machinery they use to do so—standardization, daily settlement, and a central counterparty standing between every buyer and seller—is the subject of this chapter.

That machinery works well enough that it is easy to take for granted. This chapter deliberately does not. The events of 2022 turned questions that had been footnotes in earlier textbooks into first-order concerns: how much margin a clearing house should demand, what happens when it demands too much too fast, and who absorbs the loss if a large member fails. Sections 2.8 through 2.10 take them up directly.

Sections 2.1 and 2.2 are the contract: what a futures specification fixes and why the futures price converges to spot. Sections 2.3 to 2.6 are the machinery: margin and daily settlement, the central counterparty, clearing members and their clients, and delivery. Section 2.7 covers regulation and manipulation; Sections 2.8 to 2.10 are the three questions above; Section 2.11 is the OTC markets, cleared and bilateral; Section 2.12 is who trades and how orders reach the market; and Section 2.13 closes by setting forwards against futures.

2.1 FUTURES CONTRACTS AND THEIR SPECIFICATION

A **futures contract** is an agreement to buy or sell an asset at a specified future time for a price agreed today. What distinguishes it from a forward is that it is standardized and exchange-traded, which permits a liquid secondary market: a position can be closed by taking the opposite position rather than by negotiating with the original counterparty.

Standardization means the exchange, not the parties, specifies everything except the price:

- **The asset.** For commodities, a grade and a set of permitted substitutions with price adjustments. This is not a detail, since a contract's deliverable grade determines what it actually tracks.
- **The contract size.** Too small and commissions dominate; too large and smaller hedgers are excluded. Exchanges sometimes list "mini" contracts for this reason.
- **Delivery arrangements.** Where delivery may be made, and for financial contracts whether delivery occurs at all or the contract is cash-settled against a published index.
- **Delivery months and the last trading day.**
- **Price quotation, tick size (the smallest permitted price change), and on some contracts daily price limits and position limits.**

Table 2.1 shows the specification of one such contract.

Table 2.1: The light sweet crude oil futures contract. Terms as of press date.

Exchange	New York Mercantile Exchange
Ticker	CL
Contract size	1,000 barrels
Deliverable grade	West Texas Intermediate light sweet crude, with listed substitutable grades
Delivery location	Cushing, Oklahoma
Delivery months	Every month, several years forward
Price quotation	US dollars and cents per barrel
Tick size	$0.01 per barrel, or $10 per contract
Last trading day	Third business day before the 25th of the preceding month
Position limits	Set by the exchange, tightening near delivery

Two entries repay attention. The *deliverable grade* determines what the contract tracks: a producer whose output differs from the deliverable grade is exposed to the difference, which is the basis risk of Chapter 3. And the *delivery location* matters as much as the grade: a contract deliverable at an inland hub prices differently from one deliverable at a port, because the two are connected only by pipeline capacity that can bind.

Table 2.2 shows that crude oil contract once it is trading.

Table 2.2: Quotes by delivery month. Illustrative, not market data.

Delivery	Settle ($/bbl)	Change	Volume	Open interest
December	78.29	+0.24	412,880	128,450
January	78.59	+0.21	198,340	265,910
February	78.88	+0.18	76,220	142,300
March	79.18	+0.16	41,150	98,660
June	80.07	+0.11	12,480	54,120
December (next)	81.90	+0.07	6,310	71,940

Five things can be read off it, and most are taken up again later:

- *The settle column is a forward curve*: the price of the same asset for delivery at each of six dates, generated here from Chapter 5's cost-of-carry relation at a spot price of $78.00 a barrel and a net carry rate of 4.5% a year, continuously compounded, with the six contracts delivering 1, 2, 3, 4, 7 and 13 months out. It is the object Section 2.12 says these markets exist to produce. It slopes upward here because storage and financing costs exceed any convenience yield (the benefit of holding the physical commodity rather than a claim on it), which is the storable-commodity case of Section 2.2.
- *Volume and open interest are different quantities and peak in different months*: December trades most because positions are being rolled out of it (closed in the expiring month and reopened in the next), while January holds the most open interest because that is where they are going.
- *Liquidity thins with maturity*, which is why the stack-and-roll hedge of Chapter 3 exists at all: a hedger wanting ten-year protection cannot simply buy it in the ten-year contract, so it holds the whole exposure in the liquid near contract and rolls it forward as each one expires.
- The December contract a year out carries more open interest than the nearer June, because annual hedging programs settle on December dates.
- The tick from Table 2.1 converts the change column into money: December's $0.24 is $240 a contract.

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Price limits and position limits. A **daily price limit** halts trading if the price moves more than a specified amount from the previous settlement. The intention is to prevent large moves driven by disorderly trading. The effect is contested: a limit does not remove the information causing the move, and a market that is limit-locked is one in which no one can trade at all, which converts a price problem into a liquidity problem. **Position limits** cap the contracts a speculator may hold, to reduce the scope for cornering the market, that is, holding a position large enough that the shorts must buy back from the party that also controls the deliverable supply, which Section 2.7 takes up. They typically tighten as delivery approaches, when the deliverable supply is most easily squeezed.

Volume and open interest

Two quantities are reported for every contract and are constantly confused. **Volume** is the number of contracts traded during the day. **Open interest** is the number of contracts outstanding at its end, counted on one side only, since every long is matched by a short.

They differ because a trade may open positions, close them, or transfer one from one holder to another. Only the first and second change open interest.

Example 2.1 Volume and open interest

Trace a newly listed contract over four days.

Day	Trade	Volume	Change	Open interest
1	A opens long 10; B opens short 10	10	+10	10
2	C opens long 5; D opens short 5	5	+5	15
3	A closes 4; C opens 4	4	0	15
4	A closes 6; B closes 6	6	−6	9

On day 3 one participant left and another arrived, so the number of outstanding contracts was unchanged: the position was transferred, not created or extinguished. On day 4 both sides closed, so six contracts ceased to exist.

The distinction carries information. Rising prices on rising open interest suggest new money entering on the long side; rising prices on *falling* open interest suggest shorts closing out, which is a different phenomenon and typically less persistent.

2.2 CONVERGENCE TO THE SPOT PRICE

As delivery approaches, the futures price converges to the spot price. The argument is a simple arbitrage.

Suppose during the delivery period the futures price is above the spot price. A trader can sell the futures, buy the asset, and deliver immediately, earning the difference. This is riskless and will be done until the gap closes. If instead the futures price is below the spot price, those who wish to acquire the asset will buy futures and wait for delivery, pushing the futures price up.

Neither argument requires any view about where the price is going. It follows that at expiration the two prices must be essentially equal, and the futures contract can be treated as a claim on the spot asset at that date.

The futures price may approach the spot price from above (the normal pattern for a storable commodity, where the futures price includes storage and financing costs that shrink as delivery nears) or from below, which happens when holding the physical asset conveys a benefit that the futures contract does not. Chapter 5 develops what determines the gap *before* expiration; the point here is only that it must close.

What converges is the gap, not the prices. Spot and futures prices both move throughout,

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and neither approaches delivery along any predictable line. The gap between them (the spot price minus the futures price, called the basis) *narrows to zero anyway*, because the arbitrage that forces it does not care about the route.

The convergence is what makes a futures contract useful as a hedge at all. A hedger who holds the physical asset and shorts futures relies on the two prices ending up together; if they did not, the hedge would leave a residual whose size was unknown in advance. Chapter 3 examines what happens when the hedge is lifted *before* delivery, when convergence is incomplete and the residual is exactly the basis risk. Figure 3.1 draws the convergence with that residual marked.

2.3 MARGIN AND DAILY SETTLEMENT

The exchange's central problem is that a futures contract can accumulate large losses for one side over its life, and the losing party may prefer to walk away. The solution is not to check creditworthiness but to make the question irrelevant: **losses are collected daily, in cash, before they can grow**.

The margin account. On opening a position, a trader deposits **initial margin** with the broker. At the end of each trading day the account is adjusted for that day's gain or loss, called **marking to market**, or **daily settlement**. Gains are credited to the account, and any balance above the initial margin may be withdrawn; losses are debited from it.

The balance need not stay at the initial margin. It may fall as far as a lower threshold, the **maintenance margin**, before anything happens; only if it falls below that does the trader receive a **margin call**, and must then restore the balance to the *initial* margin level. If the call is not paid, the broker closes the position. The daily gains and losses themselves, settled in cash, are the **variation margin**; older texts give that name to the deposit that answers a call, and Example 2.2 shows why the two are better kept apart.

What margin is posted in. Initial margin need not be cash. Clearing houses also accept Treasury bills and other government securities, valued after a haircut (a deduction from their market value; Section 4.1), and many hedgers post them for exactly that reason: a bill keeps earning its yield while it sits in the account. Cash may earn interest too, but at a rate the broker or clearing house sets, usually below the overnight benchmark, so on a large position the forgone interest is a real cost of hedging. Variation margin is different. It is settled in cash every day, and the interest earned on gains or paid to finance losses is what can make a futures price differ from a forward price (Section 5.12).

Example 2.2 Operation of a margin account

An investor buys two gold futures contracts, each for 100 ounces, at a futures price of \$3,200 per ounce, a position of 200 ounces and \$640,000 of notional. The clearing house sets maintenance margin at \$15,000 per contract and initial margin ten percent above it at \$16,500, so the investor posts \$33,000 and must keep the account above \$30,000. The initial requirement is about 5% of notional, the order of magnitude a liquid futures contract attracts, and it exceeds the maintenance level by a set proportion.

Day	Futures price	Daily gain	Cumulative gain	Balance	Margin call
0	3,200.00	—	—	33,000	—
1	3,188.00	−2,400	−2,400	30,600	—
2	3,176.00	−2,400	−4,800	28,200	4,800
3	3,168.00	−1,600	−6,400	31,400	—
4	3,190.00	+4,400	−2,000	35,800	—

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On day 2 the balance of \$28,200 falls below the \$30,000 maintenance level, so the investor must deposit \$4,800 to restore the \$33,000 initial level. Day 3 therefore starts from \$33,000 and closes at \$31,400, still above maintenance, so no further call; on day 4 the price recovers to leave \$35,800.

Check the arithmetic overall: the price fell \$10 from 3,200 to 3,190, a cumulative loss of $10 \times 200 = \$2{,}000$. The investor deposited $33{,}000 + 4{,}800 = \$37{,}800$, and $37{,}800 - 2{,}000 = \$35{,}800$.

The daily gain column is the variation margin. Each entry is the day's mark-to-market, debited from the account when the position loses and credited when it gains, and it is paid every day whether or not any call is made. The \$4,800 on day 2 is not variation margin but a top-up of *initial* margin. The two are indistinguishable in this table because a futures account runs them through one balance; Section 8.2 separates them. The separation matters most to a hedger: variation margin on the futures leg must be paid in cash every day, even while the position it hedges gains as much on paper, so a hedge that is working economically can still run its owner out of cash.

Two further features deserve attention. The investor was called for cash on day 2 even though the position was still open and might recover, as it did. And the \$2,800 standing above the initial requirement on day 4 could be withdrawn immediately. Futures positions consume and release cash continuously, which is a different property from owning the underlying asset.

What the daily settlements add up to. The example's cross-check is an instance of a general fact. A position opened at F_1 and closed at F_2, settled every day in between, collects

$$F_2 - F_1 \text{ per unit if long}, \qquad F_1 - F_2 \text{ if short}, \tag{2.1}$$

whatever route the price took in between. Each intermediate price enters two consecutive settlements with opposite signs and cancels, so the sum telescopes to the difference between the first and last. Equation (2.1) is what allows a futures position to be treated as a single payoff at the date it is closed, which is how Chapter 3 reads a hedge and how Chapter 5 prices one. What does *not* cancel is the timing: the same total arrives as a stream rather than a lump, which is the subject of Section 2.9 and of Chapter 8, and the reason a futures price and a forward price are not quite the same number.

Why this matters beyond the mechanics. Example 2.2 looks like bookkeeping, and for a two-contract position it is. Scaled up, it is the mechanism behind several of the episodes in this book. A hedger whose futures leg loses money must find cash *today*, while the offsetting gain on the hedged item may not be realizable for years. The hedge is economically sound and the firm can still fail. Chapter 3 describes this happening to Metallgesellschaft; Section 2.9 describes it happening across an entire market in 2022.

And what it does to a speculator. Read the other way, the same table is the risk in a futures position taken for a view. Example 2.2 commits \$33,000 against \$640,000 of exposure, so gold need fall only 5.2% to consume the deposit, and past it the trader owes the rest of the move in cash the next morning; the option of Example 1.5, whose premium bounds the loss, differs in kind. The consequence is not that leverage is large but that it is *continuous*: a view expressed in futures must be solvent every day until it is vindicated, because a call unmet is a position closed by the broker at whatever price the market offers that morning. Section 40.4 is this arithmetic at institutional scale, and Chapter 38 explains why a risk measure built on the distribution of outcomes does not see it: the loss that matters is on the path, not at the horizon.

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How initial margin is set

Example 2.2 took the margin requirement as given. How it is arrived at determines a great deal of what follows in this chapter.

Initial margin is intended to cover the loss a position could suffer over the time it would take to close it out if the holder defaulted (one or two days for a liquid futures contract, five or more for a cleared swap portfolio) at a high confidence level, typically 99% to 99.7%. Clearing houses estimate the distribution of moves from historical data, often by simulating the portfolio's value under several years of observed daily moves and reading off the appropriate quantile. Strictly, this sets what a clearing member posts to the CCP. Exchange rules turn that figure into the customer's maintenance margin, with initial margin set above it (by ten percent in Example 2.2), and a broker may require more of its own clients.

These models have names, and a practitioner meets them before meeting any of the theory. The **Standard Portfolio Analysis of Risk** (SPAN) has been the long-standing framework for listed futures and options: it revalues the portfolio under a fixed grid of price and volatility scenarios and charges the worst outcome, which is why a margin requirement can be reproduced exactly from a published parameter file. Cleared swaps are margined instead by simulation across several years of historical moves, and the listed markets are migrating the same way. *The simulation-based models are the historical simulation of Section 38.5 under another name*, and the Standard Initial Margin Model (SIMM) of Chapter 8 is the bilateral cousin: the same problem, solved by an industry committee rather than by a clearing house.

The feature with the largest practical consequence is that margin is computed on the *portfolio*, not contract by contract. Positions that offset receive credit for it, as Example 2.3 shows.

Example 2.3 Portfolio margining and spread positions

A trader holds 100 long contracts in the March delivery month and 100 short in June. Taken separately, each leg might require \$2,000 per contract:

$$200 \text{ contracts} \times \$2{,}000 = \$400{,}000$$

But the two legs nearly offset: a parallel move in the underlying gains on one and loses on the other, and what remains is exposure to the *spread* between the two months, which is far less volatile. The clearing house might set the requirement for the paired position at \$200 per contract pair:

$$100 \text{ pairs} \times \$200 = \$20{,}000$$

a twentieth of the gross figure.

This is why calendar spreads, basis trades, and relative-value strategies generally can be run at leverage that would be impossible outright. Chapter 6 makes the point about the cash–futures basis trade, where a hedged position financed by repo (Section 4.1), with the lender withholding only a small haircut of the bond's value, runs at close to fifty times leverage.

It also identifies the danger. The offset is granted on the assumption that the two legs move together. When they stop doing so (when the basis itself dislocates, as in March 2020), the margin credit is withdrawn at the same moment the position starts losing money. This is a more violent version of the procyclicality of Section 2.9.

2.4 CENTRAL COUNTERPARTIES

Daily settlement limits how large a loss can grow. It does not answer who bears the loss if a party fails to pay. That is the role of the **central counterparty**, or clearing house.

When a trade is cleared, the original contract between buyer and seller is replaced, or **novated**, by two contracts, each facing the CCP. The CCP becomes the buyer to every seller and the seller to every buyer. Neither original party has any further exposure to the other; both

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are exposed to the CCP.

Two things the rest of this chapter takes for granted follow from that substitution, and neither would exist without it. The first is **offset**. Section 2.6 observes that most positions are closed before delivery, and Example 2.1 watches open interest fall as traders do it, but closing out is only possible because the contract is with the CCP rather than with whoever sold it. A trader who is long sells to *anyone*, the CCP nets the two positions against each other, and the position is gone. Without novation the trader would hold two live contracts with two different parties, remain exposed to both, and be able to exit only by persuading the original counterparty to tear the trade up.

The second is that the order book can be anonymous. Section 2.1 credits standardization with a liquid secondary market, and the reason is that any two participants wanting the same contract meet in the same place. That works only if neither has to price the other's credit before trading, and novation is what makes the contracts fungible: a March contract is one instrument whatever the identity of the seller. *The CCP is therefore not an adjunct to the futures market but a precondition for it*: liquidity, offset, and the daily settlement of Section 2.3 all rest on it.

The change in structure, from the web in the left panel of Figure 1.2 to the hub in the middle one, is not cosmetic. With n participants trading bilaterally there are up to $n(n-1)/2$ exposures (6 for the four participants drawn there, 15 for six, and tens of thousands for a real market). Clearing replaces them with n. Three benefits follow.

Multilateral netting. A member long with one counterparty and short with another has no net position, and after novation no net exposure. Bilaterally it would have had two.

Uniform, transparent margin. The CCP sets margin by rule rather than by bilateral negotiation, which removes the tendency for the strongest firms to demand collateral from the weakest while extending credit to each other.

Orderly default management. A CCP has a rehearsed procedure for closing out a failed member's positions, typically by auctioning the portfolio to survivors. In a bilateral market each of a failed firm's hundreds of counterparties scrambles simultaneously to replace its own hedge, which is what made September 2008 as disorderly as it was.

The netting benefit is the largest of the three and is worth computing rather than asserting, because its size is what justifies the concentration of risk that comes with it, and because the same arithmetic shows how easily it is given away.

Example 2.4 What multilateral netting is worth, and what fragmentation costs

Five dealers, with every pair's obligations already netted bilaterally, so these ten figures are what a bilateral market leaves outstanding:

Owes		Owes	
A → B	120	C → A	80
A → D	45	C → D	90
B → C	60	D → B	110
B → E	25	D → E	70
E → A	30	E → C	55

Gross bilateral exposure is the sum of these, **685**, because each is owed to a different party and none offsets another.

Novate all ten to one CCP and each dealer holds a single net position:

	A	B	C	D	E
Net position	−55	+145	−55	−45	+10

The five sum to zero, as they must. Total exposure is now the sum of the positive positions,

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155, 22.6% of the bilateral figure, a 77.4% reduction, and no netting agreement, collateral, or credit assessment produced it. It came from replacing ten counterparty relationships with five.

Now split the market. Suppose A, B, and C clear at one CCP, D and E at another, and the six trades that cross the boundary stay bilateral. Exposure rises to **485**, 70.8% of the bilateral figure, giving back 330 of the 530 that netting had saved. Fragmenting a cleared market across two CCPs destroys nearly two-thirds of the benefit of clearing it, and nothing either CCP does (margin, default fund, or waterfall) can win it back, because the lost offsets run between the two.

That arithmetic is the argument in every dispute about where a CCP should sit and about interoperability (links between CCPs that let members of one clear trades against members of another), and it explains why dealers resist being required to clear the same product in two places. The case for one CCP is quantitatively strong, and it is the same calculation that makes that CCP systemically important.

What clearing does not do. It does not eliminate risk; it relocates and concentrates it. Every participant now depends on one institution. That institution's resilience is therefore a matter of public interest, which is what the rest of this chapter is about.

2.5 CLEARING MEMBERS, CLIENTS, AND PORTING

Figure 1.2 implies that everyone faces the CCP directly. Almost no one does. Access is two-tier: a limited number of well-capitalized firms are **clearing members** of the CCP, and everyone else (asset managers, pension funds, corporates, hedge funds) clears *through* a member as a **client**. A client's positions reach the CCP only through its member, so if the member fails, what the client recovers turns entirely on how those positions and their collateral were held.

Segregation. Client collateral must be kept separate from the member's own. Under **individual segregation** each client's collateral sits in its own account; under **omnibus segregation** clients are pooled, which is cheaper and offers less protection, since a shortfall caused by one client in the pool can reach the others. Many clients do not know which they have.

What nets against what. A member holds more than one position at the CCP. Its own trades sit in a **house account**, where Example 2.4's netting applies in full; its clients' sit in separate accounts, and the two never net, which is what segregation means. Whether clients net against one another depends on how the account is margined: margined **gross**, the CCP computes initial margin client by client and adds the results; in a **net** omnibus account the pool offsets first, which is why it is cheaper. *Variation margin nets in every regime*: it settles a loss already suffered, so offsetting it gives nothing away. The choice governs initial margin alone.

Porting. If a clearing member fails, the CCP will attempt to **port** its clients (transfer their positions and collateral to a surviving member) rather than close them out. Porting preserves the client's hedge; closing out destroys it and leaves the client to rebuild in a market that is, by construction, disorderly. *It is also what gross margining buys*: collateral attributable to a client is collateral that can move with them, whereas a netted pool has a single margin figure out of which no one client can be lifted.

But porting depends on a surviving member being willing to accept the client, and that willingness is scarcest exactly when it is needed. A member contemplating a failed competitor's client book during a crisis is asked to take on unfamiliar positions and consume its own capital at the worst moment. Clients therefore hold a protection that is real but conditional—and its condition correlates with the event it protects against.

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The two-tier structure also concentrates the market further than Figure 1.2 suggests: the number of firms willing to offer client clearing has fallen steadily, since the business consumes balance sheet under the leverage ratio (Section 1.12), while earning thin fees. A system designed to reduce concentration risk rests on a shrinking number of intermediaries.

2.6 DELIVERY

Most futures positions are closed before delivery (for financial contracts, the overwhelming majority). But the possibility of delivery is what ties the futures price to the spot price, so the mechanics matter even to those who never use them.

For a physically delivered contract the process runs from the short side. During the delivery month the short may issue a **notice of intention to deliver** to the exchange. The exchange assigns the notice to a long (typically the one with the oldest outstanding position), who must accept delivery and pay the invoice amount. The short chooses *when* within the delivery period to issue notice, and for some contracts *what* to deliver from a permitted set. Where that set spans grades or delivery points, the invoice amount is not the futures price alone: published **price adjustments** add a premium or discount for what was actually delivered, measured against the par grade and location, the ones the quoted futures price refers to. Without them the short would simply deliver the cheapest admissible item and keep the difference. Chapter 6 shows what those choices are worth in the Treasury market, where the adjustment is the conversion factor.

For a physical commodity, everything it costs to hold the asset falls on the short until delivery. Storing it, insuring it, and moving it to an approved delivery point are the short's expense, which is why those costs appear in the forward price of Section 5.8 rather than anywhere in the contract's terms. What happens at delivery then depends on whether the exchange licenses somewhere to put the asset. For metals and grains it does, and delivery is the transfer of a **warehouse receipt** (title to a specified lot already sitting in a licensed warehouse) rather than the movement of anything: storage comes with the title, and the long inherits the charges from that day. Getting the metal or grain physically out is a separate operation with a queue of its own, and Section 36.12 shows that the length of that queue moves the price.

Where no such warehouse stands behind the contract, the obligation is harder than it looks, and it belongs to the long. Crude oil is delivered into tankage at Cushing, Oklahoma, and a long that has not arranged capacity before expiration must find it in a market where everyone else needs it at the same moment. Section 5.9 describes the April 2020 expiration at which there was none to be had and the price settled below zero. *This is the reason most participants close out rather than deliver*, and the reason a retail product that mechanically holds the front contract is not the safe instrument it appears to be.

Financial contracts are usually **cash-settled** instead, against a published settlement index. There is no deliverable asset for a stock index or an overnight interest rate, and cash settlement avoids the artificiality of constructing one. It also removes the squeeze risk that delivery obligations create, at the cost of making the contract dependent on the integrity of the settlement index, which is its own vulnerability, as Chapter 4 shows for LIBOR.

2.7 REGULATION AND MARKET MANIPULATION

Futures markets are regulated to ensure prices are set by genuine supply and demand rather than by participants able to distort them. The characteristic abuse is the **corner** or **squeeze**: a trader accumulates a dominant long futures position *and* control of the deliverable supply, so that shorts approaching delivery must buy back from the same party that controls the physical market. The price rises not because the commodity has become scarce in any economically meaningful sense but because the shorts cannot obtain it.

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The defenses are position limits, tightening into the delivery period; surveillance of large positions; and the exchange's power to intervene by raising margin, imposing liquidation-only trading (positions may be closed but not opened), or in extremis altering delivery terms.

The nickel episode of Section 2.9 is instructive precisely because it was not a manipulation. The large short was a genuine producer hedge, and the price spike was driven by shorts scrambling to cover. Yet the market effect was much the same: prices dislocated from any plausible physical value, participants unable to transact. Regulation aimed at intent does not necessarily protect against outcomes that arise without it.

2.8 THE DEFAULT WATERFALL

A CCP that guarantees performance must be able to absorb the failure of a member. The resources it can call on are arranged in a strict sequence, the **default waterfall**, shown in Figure 2.1. Beyond the defaulter's own margin, the main resource is the **default fund**: a pool to which every clearing member contributes, held to cover losses larger than a defaulter's margin can absorb.

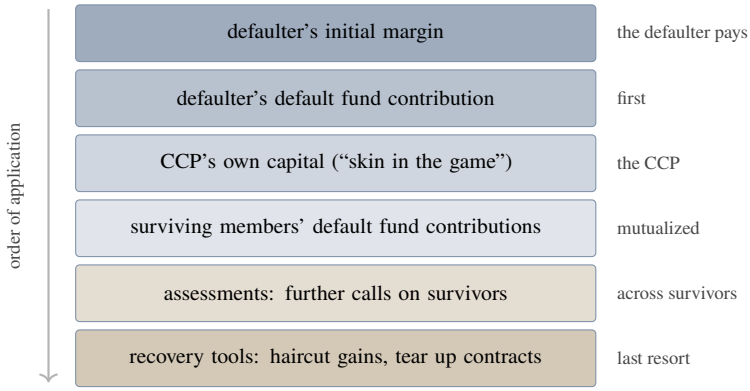


Figure 2.1: The sequence in which a CCP applies resources after a member default.

The ordering embodies a principle: *the defaulter pays first*. Only when the defaulter's own resources are exhausted does anyone else contribute, and the CCP's own capital (its *skin in the game*) sits ahead of surviving members' mutualized funds, so that a CCP which sets margin too low bears the first consequence itself. That layer is small relative to the fund below it, and how small it should be is one of the live disputes in the field. Example 2.5 takes one default through the layers.

Example 2.5 Applying the waterfall

A clearing member defaults. Closing out its portfolio in the market realizes a loss of $900 million. The CCP's resources are:

Layer	Amount ($m)
Defaulter's initial margin	500
Defaulter's default fund contribution	80
CCP skin in the game	50
Surviving members' default fund	2,000

Applying the waterfall in order: the defaulter's margin absorbs $500m, leaving $400m. Its default fund contribution absorbs $80m, leaving $320m. The CCP's own capital absorbs $50m,

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leaving \$270m. That remainder is mutualized across the surviving members, consuming

$$\frac{270}{2{,}000} = 13.5\%$$

of their pooled contributions.

The survivors did nothing wrong, had no exposure to the defaulter bilaterally, and have lost 13.5% of the funds they posted, at a moment when a major member has just failed and markets are presumably under stress. Clearing did not make the loss disappear; it distributed it. That is what "mutualization" means, and members' willingness to accept it is what the system rests on.

2.9 MARGIN PROCYCLICALITY

Initial margin is calibrated to cover a large adverse move over the time it would take to close out a defaulted portfolio, conventionally a high confidence level over a few days. Since estimates of how far prices move are based on recent volatility, **margin requirements rise when volatility rises**. Because volatility is highest in a falling, stressed market, margin moves with the market cycle (lowest in calm, highest in a crisis), and a requirement that moves with the cycle reinforces it. That property is **procyclicality**.

This is prudent for the CCP taken alone and destabilizing for the system taken together. Consider what it means. Volatility spikes; every member's margin requirement increases at once; all of them must find cash simultaneously; and the natural way to raise cash is to sell assets into a market that is already falling. Margin models designed to protect the clearing house can amplify the very move they are protecting against. Example 2.6 puts a size on the jump.

Example 2.6 How fast margin can move

A fund holds a portfolio whose initial margin is set to cover a three-standard-deviation adverse move over three days, about 99.9% one-sided. Under normal conditions, with annualized volatility of 15%, the three-day standard deviation on a \$1 billion position is

$$1{,}000 \times 0.15 \times \sqrt{3/252} = \$16.4\text{m}$$

and at three standard deviations the requirement is about \$49m.

Volatility then triples to 45%, not extreme by the standards of 2022 in energy or rates. The same calculation gives a three-day standard deviation of \$49.1m and a requirement of about \$147m.

The fund must find an additional \$98 million in cash, on a position it has not changed, within a day. If it holds the cash, nothing happens. If it does not, it must sell—and so must everyone else in the same position at the same moment.

Case Study 2.1 LME nickel, March 2022

On March 7 and 8, 2022 the price of nickel on the London Metal Exchange (LME) more than tripled: from about \$29,000 a metric ton at the previous Friday's close it rose some 66% on the first day and then, in a few hours of the second morning, spiked above \$100,000.

Behind the move was a very large short position held by a producer that had hedged its output. As prices rose, the margin calls on that position grew to a size that threatened both the holder and the brokers standing between it and the exchange. Nickel is not an unusually large market, and the losses were plausibly beyond what the default waterfall could have absorbed without severe consequences for members.

The LME's response was extraordinary: it suspended trading in nickel and *canceled*

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several billion dollars of trades that had already executed that morning, resetting the market to the previous evening's close. Trading did not resume normally for over a week.

The cancellation protected the clearing system at the direct expense of those who had bought during the spike, who had transacted in good faith on a functioning exchange and had their profits annulled. Litigation followed. Whatever the legal merits, the episode established something uncomfortable that the textbook description of futures markets omits: when the alternative is the failure of the clearing house, the finality of a trade is not absolute.

Two further episodes the same year made the same point without any cancellation of trades. European energy utilities that had sold futures against future output faced margin calls in the tens of billions of euros when gas and power prices rose several-fold, and UK pension funds running liability-driven investment met collateral calls by selling gilts, which drove yields higher and triggered further calls. In each case a hedge correct in every economic sense was defeated by the timing of the cash it demanded; Sections 8.9 and 8.10 work both episodes (Example 8.9 for energy, the liability-driven investment case study and Example 8.11 for the pension funds), and Sections 40.8 and 40.12 tell them.

What can be done. Regulators have pressed CCPs toward margin models that are less reactive: floors on the volatility estimate, buffers built in calm periods, and caps on how fast requirements may increase. Each reduces procyclicality at the cost of holding more margin permanently, which is a real cost borne in normal times to avoid a rare one. Where the balance should sit is unresolved, and reasonable people place it differently.

2.10 WHO ABSORBS A CCP'S LOSSES?

The waterfall of Section 2.8 is finite. If losses exhaust every layer, the CCP faces its own failure, and since clearing is now mandatory for large parts of the derivatives market, the failure of a major CCP would be a systemic event of the first order.

The tools available at the bottom of the waterfall are uncomfortable. **Assessment rights** allow the CCP to demand further contributions from surviving members, transmitting stress to firms that may already be struggling. **Variation margin gains haircutting** withholds part of the gains owed to members whose positions have profited, taking money from the winners to pay for the losers. **Tear-up** cancels contracts outright, leaving members who thought they were hedged suddenly unhedged.

Each of these is a way of allocating a loss that has already occurred. None creates resources. And each would be deployed in precisely the conditions in which it does most damage.

This has produced an unresolved policy debate with three broad positions. Some argue CCPs should hold much more of their own capital, so that they bear more of the consequence of setting margin too low. CCPs respond that their capital is small relative to the risks they manage and that heavier requirements would make clearing uneconomic. Others argue for an explicit public backstop, on the grounds that a systemically essential institution will be rescued in practice and that pretending otherwise mainly obscures who bears the risk. The objection is that an explicit backstop invites exactly the underpricing of risk it would then have to cover.

The question matters for this book because so much of what follows assumes cleared markets function. It is worth holding in mind that the assumption is a judgment about institutional resilience, not a mathematical fact.

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2.11 OTC MARKETS: CLEARED AND BILATERAL

Not everything trades on an exchange. The over-the-counter market, in which participants negotiate directly, remains larger than the exchange-traded market by notional and is where most of the instruments in this book are transacted.

Post-crisis reform pushed it toward the futures model along two tracks.

Standardized OTC derivatives must be centrally cleared. Interest rate swaps in the major currencies and index credit default swaps are cleared through CCPs that operate essentially as described above. This is why Chapter 7 can treat cleared swaps as the standard case. Where they are cleared has become a policy question as well: since June 2025 the European Union's revised European Market Infrastructure Regulation (EMIR) has required EU firms above the clearing threshold in euro- and zloty-denominated interest rate derivatives and euro short-term interest rate derivatives to hold an active account at a CCP in the EU. Under technical standards in force since February 2026, they must also clear at least five trades on average through it in each of their most-used contract types, a rule aimed at reducing the EU's reliance on clearing houses in London.

Non-cleared derivatives face mandatory margin. Trades that cannot be cleared (because they are bespoke) are subject to uncleared margin rules requiring both variation margin and, above certain thresholds, initial margin posted to a segregated third party. The intent is to remove the incentive to avoid clearing simply to avoid posting collateral.

The result is that the practical distinction between exchange-traded and OTC has narrowed considerably. What remains is the contract's flexibility: an OTC trade can be tailored to a hedging need exactly, while a futures contract cannot. That flexibility is why the OTC market persists, and the basis risk it eliminates is real, as Chapter 3 discusses.

And clearing has now been extended beyond derivatives altogether. In December 2023 the US Securities and Exchange Commission adopted rules requiring that eligible secondary-market transactions in US Treasury securities be cleared through a registered clearing agency: both outright purchases and sales, and the repurchase agreements that finance them. The requirement is being phased in, with the cash leg first and repo six months later: after a one-year extension in 2025, cash transactions must be cleared from December 31, 2026 and repo from June 30, 2027.

The reasoning belongs to this book rather than to securities-market infrastructure. A large share of Treasury repo was bilateral and uncleared, which meant leverage extended on privately negotiated haircuts, to positions no regulator could total. The cash–futures basis trade of Section 6.5 is financed that way, and March 2020 is what happens when the financing withdraws from a crowded one. The rule is a market-structure answer to a derivatives problem, which is why it appears in this chapter and not in a chapter on bonds.

The mechanism is the one already described. It is novation to a central counterparty, margin set by a published model rather than agreed across a telephone, and multilateral netting across members, applied to a market where the incumbent clearing agency, the Fixed Income Clearing Corporation, has long cleared a substantial share of interdealer activity while much client activity sat outside it. Access models matter as much as the mandate: a client can be sponsored into clearing by a member, and rules permitting execution with one dealer and clearing through another determine how much competition the change actually produces.

And the questions this chapter has already raised follow it across. Section 2.8's default waterfall and Section 2.9's procyclicality now apply to the financing market for the asset that prices every other dollar asset. Concentrating that risk in a clearing house does not remove it; it makes it measurable, mutualizes it among members, and raises the stakes on the margin model. Whether that is a good trade is the same question Section 2.10 asks, on a larger underlying.

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2.12 TRADERS, ORDERS, AND THE SHAPE OF THE MARKET

The three motives for trading, hedging, speculation, and arbitrage, are those of Section 1.9, which also makes the case that hedgers and speculators need each other. In a futures market the categories are less clean than they look (a dealer hedging inventory is simultaneously speculating on the basis), and what this section adds is how traders place their orders and what their trading produces.

Orders. A **market order** executes immediately at the best available price. A **limit order** executes only at a specified price or better, and may not execute at all; the trade-off between certainty of execution and certainty of price is the fundamental one, and no order type escapes it. A **stop order** becomes a market order once the price reaches a trigger, and is commonly used to cap losses, though in a fast market it executes at whatever price is available *after* the trigger, which may be far worse. A **stop-limit order** adds a limit to that price (a floor, for a stop to sell) at the risk of not executing at all, which in a falling market is precisely the outcome the stop was meant to prevent.

The nickel episode is a reminder that every order type assumes a functioning market, and inherits the fragility of that assumption.

Price discovery

Hedging is not the only function these markets serve, and arguably not the most important one. A liquid futures market with contracts extending years forward produces something no individual participant could: a continuously updated, publicly visible **forward curve** for the underlying: the settlement prices of the successive delivery months plotted against those months, so that a single line shows what the market will pay for the asset at every date it trades. The process by which trading assembles many participants' scattered information into such a price is **price discovery**. Chapter 5 works out what determines its shape, and Figure 5.2 draws the characteristic ones.

That curve is used by firms that never trade a single contract. An oil refiner deciding how much crude to process next quarter, a farmer choosing what to plant, and a utility deciding whether to contract for supply all read prices off a curve established by others' trading. The information is a public good produced as a by-product of private speculation, which is the strongest argument for tolerating speculation that serves no hedging purpose of its own.

Two cautions attach. A futures price is not a forecast of the future spot price; it is a price that clears today's market for future delivery, and Chapter 5 shows it equals the expected spot price only when the asset carries no systematic risk, differing from it otherwise by a risk premium. And the signal degrades exactly when it matters most: a limit-locked or suspended market produces no price at all, and the LME's nickel cancellation produced prices that were subsequently declared not to have happened.

2.13 FORWARDS VERSUS FUTURES

Table 2.3 summarizes the differences.

The economically significant difference is daily settlement: because gains and losses are reinvested or financed at the short rate prevailing on the day they arrive, a futures price can differ from the forward price on the same terms, as Section 5.12 shows and Chapter 30 quantifies through the convexity adjustment.

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Table 2.3: Forward and futures contracts compared.

Forward	Futures
Private contract between two parties	Traded on an exchange
Not standardized	Standardized
Usually one delivery date	Range of delivery dates
Settled at the end of the contract	Settled daily
Delivery or final cash settlement usual	Position usually closed before delivery
Some credit risk	Virtually no credit risk

SUMMARY

A futures contract standardizes everything except price, which permits a liquid secondary market in which positions are closed by offset rather than negotiation, and arbitrage forces convergence to spot at delivery. Volume counts contracts traded and open interest counts contracts outstanding, so the two series read together carry information a price series alone does not.

The exchange manages credit risk not by assessing it but by making it irrelevant: positions are marked to market daily and losses collected in cash before they accumulate, with a fall below the maintenance level triggering restoration to the initial level. Initial margin covers a high-confidence adverse move over the assumed close-out period and is computed on the portfolio, so a calendar spread can require a small fraction of what its two legs would need outright. That credit is what makes relative-value strategies viable at high leverage and what makes them fragile, since it is withdrawn precisely when the two legs stop moving together.

A central counterparty novates each trade into two, collapsing up to $n(n-1)/2$ bilateral exposures into n, permitting multilateral netting and an orderly default procedure. It concentrates risk rather than eliminating it. Access is two-tier, so most participants' protection depends on how their collateral is segregated and on whether they can be ported to a surviving member, which requires someone willing to take the book at the worst possible moment. Losses are absorbed in a fixed sequence on the principle that the defaulter pays first, and in the worked example the default reaches into the surviving members' pooled contributions: clearing distributed the loss rather than removing it. Beyond the waterfall, assessments, gain haircutting, and tear-ups allocate losses rather than creating resources.

Initial margin rises roughly in proportion to volatility, which is prudent individually and destabilizing collectively. The three episodes of 2022 (LME nickel, European energy, and UK liability-driven investment) shared that mechanism despite different causes: economically sound hedges were defeated by the timing of the cash they demanded. The nickel cancellation established that trade finality yields when the alternative is the clearing house failing. OTC markets have converged on this model, and what survives the convergence is flexibility: an OTC contract can match a hedging need exactly and a futures contract cannot.

FURTHER READING

The Committee on Payments and Market Infrastructures and the International Organization of Securities Commissions (CPMI–IOSCO) *Principles for Financial Market Infrastructures* (2012), together with the later guidance on CCP resilience and recovery, is the primary source for the clearing machinery of this chapter, and it is written for supervisors rather than for lawyers. Pirrong's ISDA discussion paper on the economics of central clearing is the skeptical counterweight to official-sector enthusiasm and is worth reading for exactly that reason. On the margin spiral of 2022, the BIS bulletin on margin dynamics in centrally cleared commodities markets and the Financial Stability Board's *Liquidity Preparedness for Margin and Collateral*

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Calls (2024) are the official-sector accounts, and Duffie's Jackson Hole paper on Treasury market resilience shows the same mechanism at work in the largest market there is.

PRACTICE QUESTIONS

2.1 Explain, using an arbitrage argument in both directions, why the futures price converges to the spot price at delivery.

2.2 An investor sells three silver futures contracts, each for 5,000 ounces, at \$28.00 per ounce. Initial margin is \$9,000 per contract and maintenance margin \$7,000 per contract. Construct the margin account over five days if the settlement prices are \$28.20, \$28.55, \$28.30, \$28.95, and \$28.70. On which days is there a margin call, for how much, and why is the account restored to the initial rather than the maintenance level?

2.3 Six firms trade with one another bilaterally. How many bilateral exposures exist, how many remain after novation to a CCP, and what is the general result for n firms?

2.4 A clearing member defaults, generating a close-out loss of \$1.4 billion. The defaulter's margin is \$600m, its default fund contribution \$120m, the CCP's skin in the game \$75m, and the surviving members' default fund \$2.5 billion. Work through the waterfall in order, state what percentage of the surviving members' fund is consumed, and explain why the CCP's own capital sits ahead of surviving members' contributions.

2.5 A \$500 million position has an initial margin requirement set at three standard deviations of a three-day move. Calculate the requirement at annualized volatilities of 12% and 30%, state how much additional cash must be found, and explain why a practice that is prudent for one clearing house may be destabilizing for the system.

2.6 Distinguish volume from open interest over three days: (i) two new participants open a 20-contract position between them; (ii) one of them closes 8 to a new entrant; (iii) both original parties close 12. Give both figures for each day.

2.7 A trader is long 200 September contracts and short 200 December, with outright margin of \$1,500 per contract and a spread rate of \$250 per pair. Compare the portfolio and contract-by-contract requirements, and explain what happens to the position if the spread relationship breaks down.

2.8 Explain the difference between individual and omnibus segregation of client collateral, why a client might not know which it has, and why the chance of being ported is lowest exactly when it is most needed.

2.9 Explain what a corner or squeeze is and how position limits are meant to prevent it. Why did the nickel episode produce similar market effects without being a manipulation, and what does the cancellation of executed trades imply about finality?

2.10 Futures markets are defended on the grounds that they produce price discovery. Explain what this means, who benefits, why it is an argument for tolerating speculation that serves no hedging purpose, and under what conditions the argument fails.

FURTHER QUESTIONS

2.11 The 2022 episodes in energy, nickel, and UK pensions had quite different causes. Identify the mechanism common to all three, explain why being correctly hedged did not protect the firms involved, and state what a risk manager should have measured that a mark-to-market framework does not capture.

2.12 Set out the case that CCPs should hold substantially more of their own capital in the waterfall, then the case against. What evidence would move you?

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2.13 Variation margin gains haircutting takes money owed to members whose positions have profited. Explain when it would be used, why it is preferred to simply failing, and what it does to the incentive to clear.

2.14 Central clearing concentrates exposures in a single institution. Construct the argument that this makes the system *more* fragile rather than less, and identify what would have to be true for the argument to be right.

2.15 A commodity producer hedges three years of output by selling futures. Prices triple. Show that the firm is economically better off than if it had not hedged, yet may fail. What financing arrangements would have prevented this, and what do they cost in normal times?

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Chapter 3
Hedging Strategies Using Futures

Most participants in futures markets are there to reduce a risk they already have, not to acquire one. This chapter is about how they do it and, more usefully, about the ways in which it does not quite work.

Futures are only one way to do it, and it is worth being explicit about why a firm chooses them. They are *liquid*, so a large hedge can be put on and taken off the same day. They are *cheap*, with spreads a fraction of the equivalent bank quote. They carry *no bilateral exposure to the other side of the trade*, because the clearing house of Chapter 2 stands between the two sides (exposure to the clearing house and to the clearing broker remains, as Chapter 2 explains). And they cost *nothing at inception*, which matters more to a treasurer defending a budget than it should. The alternatives each give one of these up: a forward matches the exposure exactly but is illiquid and carries the dealer's credit, a swap does the same across many dates, and an option keeps the upside but has to be paid for.

A hedge that worked perfectly would be uninteresting. The producer would lock in a price, the consumer would lock in a cost, and there would be nothing further to say. In practice four things intervene: the futures contract may not match the exposure, the hedge may be lifted before the contract expires, the exposure may last longer than any liquid contract, and (the theme Chapter 2 introduced and Chapter 8 develops) the hedge generates cash flows on a schedule the hedged item does not. *Each is the shadow of one of the advantages just listed.* Standardization is what makes futures liquid and what makes them fit imperfectly; a small number of listed maturities is what concentrates the liquidity and what forces the roll; daily settlement is what removes the bilateral credit risk and what demands the cash. A hedger does not get to take the advantages without their costs, and choosing futures is choosing that bundle.

Each of these is a source of residual risk, and each has cost real firms large sums. Sections 3.3 through 3.9 take them in turn, and Section 3.10 examines what happens when they combine.

3.1 SHORT AND LONG HEDGES

A **short hedge** is appropriate when a firm already owns an asset, or expects to own one, and wants to protect against a fall in its price. The firm sells futures. If the price falls, the loss on the physical position is offset by a gain on the futures.

A **long hedge** is appropriate when a firm knows it will need to buy an asset in future and wants to protect against a rise in price. The firm buys futures.

Example 3.1 An oil producer's short hedge

On May 15 a producer contracts to sell 1 million barrels of crude on August 15 at whatever the spot price is on that day. Spot is currently \$80 and the September futures price is \$80.50. Each contract covers 1,000 barrels, so the producer sells 1,000 contracts.

On August 15 the spot price has fallen to \$72 and the September futures price is \$72.10.

Revenue from selling the oil	\$72,000,000
Gain on the futures position, $(80.50 - 72.10) \times 1{,}000{,}000$	\$8,400,000
Total	\$80,400,000

The effective price received is \$80.40 per barrel: close to, but not exactly, the \$80.50 futures price at which the hedge was placed.

Note the discrepancy of 10 cents. It is not an error, and it is the subject of Section 3.3.

3.2 ARGUMENTS FOR AND AGAINST HEDGING

Before the mechanics comes a question that matters more than it is usually given credit for: should a firm hedge at all?

The case against. Shareholders can diversify. If an investor holds an oil producer alongside an airline, the oil price risk partly cancels within the portfolio, and the investor did not need either company to hedge. This is the Modigliani–Miller argument applied to risk: in frictionless markets, whatever a firm does to its risk its shareholders could do or undo on their own account, so hedging cannot add value. Hedging costs money and management attention, and a firm that hedges away the exposure its shareholders bought it for may be destroying the thing they wanted. Gold mining is the clearest case. Producers sold their output forward through the 1990s, and when the gold price rose through the decade that followed, shareholders who had bought the miners precisely to own gold discovered the exposure had been sold out from under them. The industry then spent years, and a great deal of money, buying those hedge books back.

There is also a competitive argument. If an entire industry faces the same input cost and passes it through, hedging does not reduce a firm's risk relative to its competitors—it *creates* it, because the hedged firm's costs now diverge from everyone else's. Airlines are the standing example: when fuel is expensive every carrier raises fares together, so the hedged carrier gains only until prices fall, at which point it is the one carrier that cannot cut.

The case for. The strongest arguments are about frictions the diversification story ignores. *Financial distress is costly*: a firm that runs out of cash loses customers, suppliers, and staff in ways that a shareholder's diversified portfolio does not compensate for, so reducing the probability of that state has value even if the expected outcome is unchanged. Lenders act on this directly: a reserve-based loan to an oil and gas producer, whose size (the **borrowing base**) is set by the value of the producer's reserves, commonly *requires* that a share of production be hedged as a condition of that borrowing base, because what the lender is buying is the reduced probability of distress and not a view on the oil price. Froot, Scharfstein, and Stein (1993) sharpen the point: when outside finance costs more than internal cash, a shortfall in cash flow forces a firm to pass up good investments, so a hedge that keeps internal funds available in bad states protects investment, not merely earnings. Taxes argue the same way, as Smith and Stulz (1985) show: where the tax schedule is convex, as progressive rates and limits on the use of losses make it, a smoother taxable income lowers the expected tax bill. *Managers know more than shareholders* about the firm's exposures, and can hedge more cheaply than shareholders can replicate. And *planning is easier* when input costs are known: a firm that can commit to a price can sign contracts it otherwise could not. An airline sells tickets months before it buys the fuel to fly them, and a utility files tariffs it cannot revisit for a year; neither can quote a price today without knowing a cost it will not pay until later.

Hedging is worth doing when it reduces the probability of genuinely bad states, and is closer to a wash when it merely smooths outcomes within a range the firm can absorb.

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Case Study 3.1 The hedger who is punished for hedging

A hedge that works generates a loss on one leg. That is what a hedge is.

The difficulty is that the two legs are usually not reported the same way. The futures loss is realized, visible, and appears in a specific line of the accounts with someone's name against it. The offsetting gain sits in an operating result, unlabeled, and looks like ordinary business performance. When prices move against the futures leg, the person who put the hedge on has a demonstrable loss to explain, and the person who did nothing has a good year.

Airlines have provided the clearest examples in both directions. Carriers that hedged fuel were criticized for the cost when prices fell, and praised when prices rose; carriers that did not hedge faced the reverse. The underlying decision, whether to accept a known cost or an unknown one, was the same in every year, but it was judged each time by an outcome that the decision was explicitly designed not to depend on.

The practical consequence is that hedging programs need to be justified and measured on the combined position, before the fact, with the mandate agreed in advance. A program evaluated leg by leg after the fact will be abandoned at the worst possible moment. That is what Section 3.10 describes happening to a real firm.

3.3 BASIS RISK

Example 3.1 did not lock in \$80.50. The reason is that the hedge involved two prices (the spot price the producer received and the futures price at which the hedge was closed), and they need not move together.

The **basis** is defined as

$$\text{basis} = \text{spot price of the asset to be hedged} - \text{futures price of the contract used} \tag{3.1}$$

Chapter 2 showed the basis converges to zero at delivery. Before delivery it does not, and it varies.

Write S_1, F_1 for the spot and futures prices when the hedge is placed, S_2, F_2 for their values when it is lifted, and $b_1 = S_1 - F_1$, $b_2 = S_2 - F_2$ for the corresponding bases. For a short hedge the effective price received is

$$S_2 + (F_1 - F_2) = F_1 + b_2 \tag{3.2}$$

and the identical expression holds for a long hedge, as the effective price paid.

Equation (3.2) is the central result of this chapter. The hedger locks in F_1, which is known today, plus the basis at the time the hedge is closed, which is not. All the residual uncertainty is in b_2.

Figure 3.1 shows the two prices converging. The gap narrows to zero at delivery, but a hedger who lifts the hedge before then is exposed to wherever the gap happens to be at that moment.

The figure is also equation (3.2) drawn. The hedger sells the physical at S_2 and closes the futures for $F_1 - F_2$, so S_2 plus the futures gain $F_1 - F_2$ reaches the dashed horizontal line above the two marked points, and that line sits above F_1 by exactly the gap b_2 between them. Both readings give the same height, which is what the identity says. Notice what has dropped out: the sinuous common movement of the two paths appears in S_2 and in F_2 and cancels between them, so the effective price does not depend on where prices went, only on the gap that remained.

Who gains from which move. An increase in the basis is called a **strengthening**; a decrease is a **weakening**. From (3.2), a short hedger receives $F_1 + b_2$ and therefore **benefits from**

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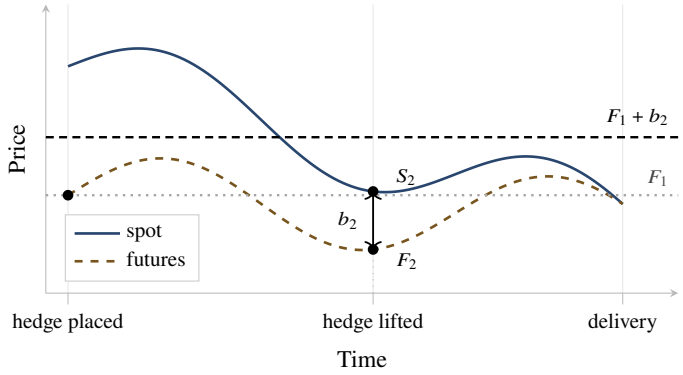


Figure 3.1: Spot and futures prices converging at delivery; b_2 is the basis at the lift date. Schematic.

an unexpected strengthening. A long hedger pays the same quantity and benefits from a weakening. Example 3.2 puts numbers on a weakening.

Example 3.2 Basis risk in numbers

A hedge is placed when the spot price is \$62 and the futures price is \$60, so $b_1 = +\$2$. It is lifted when the spot price is \$58 and the futures price is \$57.50, so $b_2 = +\$0.50$.

For a short hedger the effective price is

$$F_1 + b_2 = 60 + 0.50 = \$60.50$$

Had the basis held at \$2, the effective price would have been \$62. The basis weakened by \$1.50, and the short hedger lost exactly that amount per unit relative to expectation—despite the hedge having done its job in every other respect.

Compare the unhedged outcome. The spot price fell \$4 from the \$62 at which the hedge was placed. The hedge absorbed \$2.50 of the fall and left \$1.50, and that residual is basis risk.

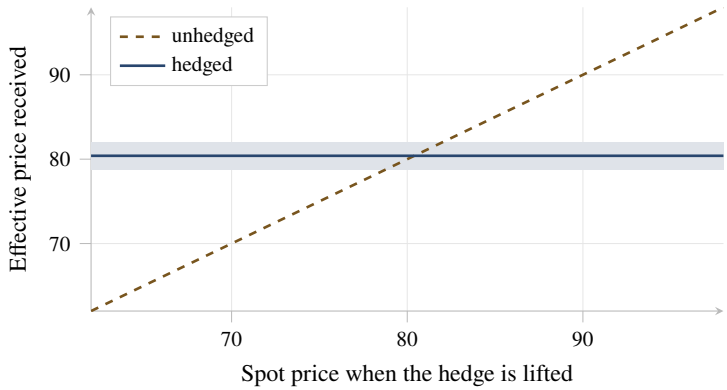


Figure 3.2: Effective price received against the spot price at the time the hedge is lifted. The band is basis risk.

Figure 3.2 makes the point visually. Unhedged, the effective price is whatever the spot price turns out to be (the diagonal). Hedged, it is $F_1 + b_2$, which is flat in the spot price: the hedge has removed the exposure entirely. What remains is the vertical thickness of the band, which is the variability of b_2.

The figure also shows what the hedger has given up. Everything to the right of the crossing

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point is upside the hedger will not receive, and every hedging program has to survive a year in which that region is where the market went. This is the organizational problem of Case Study 3.1, drawn.

Where basis risk comes from. Basis risk has three sources, each corresponding to a mismatch:

- **Asset mismatch.** The asset hedged differs from the deliverable grade: jet fuel hedged with heating oil (since 2013 the New York Mercantile Exchange (NYMEX) contract of that name has specified ultra-low-sulfur diesel delivered in New York Harbor), a corporate bond portfolio hedged with Treasury futures. Chapter 2's remark that the deliverable grade determines what a contract tracks is the same point.
- **Location mismatch.** The same commodity at a different delivery point, connected only by transport capacity that can bind (the reason Chapter 2 flagged delivery location as economically significant).
- **Timing mismatch.** The hedge is lifted before the contract expires, so convergence is incomplete.

Choosing the contract. Two decisions reduce basis risk. Choose the asset whose price is most closely correlated with the exposure, and choose a delivery month *after* the hedge will be lifted, but not much after, since liquidity concentrates in near contracts. Hedging a June exposure with a July contract is usually better than with a June contract, since holding into the delivery month exposes the hedger to delivery mechanics and to erratic prices.

3.4 BASIS RISK BEYOND COMMODITIES

Basis risk is usually introduced with commodities, where the mismatches are physical and easy to picture. It is at least as important in financial hedges, where it is less visible and therefore more dangerous.

Credit. A portfolio of corporate bonds hedged with Treasury futures is protected against the level of interest rates and not at all against credit spreads. In calm markets the two move together enough that the hedge looks effective; in a credit sell-off, government yields *fall* while corporate yields rise, so both legs lose simultaneously. The hedge's measured R^2 over a quiet sample (the share of the exposure's variance the hedge removed, which Section 3.5 defines) says nothing about this, because the sample contains no such episode.

This is the general form of the problem, and it shapes how the next section should be read: a hedge ratio estimated over a period in which the basis behaved will fail in the period when the basis is what moves.

Currency. A firm with an exposure in a currency that has no liquid futures contract hedges with a correlated one: a Norwegian krone receivable hedged with euro futures, say. The hedge works while the two currencies track and fails when they do not, and they most reliably fail to track when something specific is happening to one of them, which is when the exposure matters.

Rates. Chapter 7's swap spread is a basis. A dealer hedging a swap book with Treasury futures, as Chapter 6 describes, is exposed to the spread between the swap curve and the government curve, and Chapter 7 shows that spread moving to a place no one expected it to go.

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The common structure. In each case a hedger has substituted a liquid instrument for the exposure it actually holds and retained the difference. The difference is typically small, mean-reverting, and well behaved—until it is none of those things. A hedger should be able to state, before placing a hedge, what the basis is, what would make it move, and how much it could move; a hedger who can only state the historical correlation has not answered the question.

3.5 CROSS HEDGING AND THE MINIMUM VARIANCE HEDGE RATIO

When the hedged asset differs from the futures asset (a **cross hedge**), a one-for-one position is generally not optimal. If the futures price is more volatile than the spot price, a smaller futures position gives a better hedge.

The **hedge ratio** h is the size of the futures position relative to the exposure. The value minimizing the variance of the hedged position, the **minimum-variance hedge ratio**, is

$$\boxed{h^* = \rho \frac{\sigma_S}{\sigma_F}} \tag{3.3}$$

where σ_S and σ_F are the standard deviations of changes in the spot and futures prices over the hedge's life, and ρ their correlation. [TN3] derives it, extends it to several futures contracts at once, and shows what the ratio costs when its three inputs are estimated rather than known. That cost decides whether a second hedging contract is worth using at all.

If the two prices move identically, $\rho = 1$ and $\sigma_S = \sigma_F$, giving $h^* = 1$: hedge one-for-one. Any imperfection moves the optimal ratio away from one. A correlation below one always moves it downward, relative to a perfectly correlated hedge with the same volatility ratio, because a futures position that does not reliably offset should be smaller. A volatility mismatch can move it either way: above one when the spot price is the more volatile, below one when the futures price is.

The connection to regression. Equation (3.3) is the slope coefficient from regressing changes in the spot price on changes in the futures price. This is not a coincidence: minimizing squared deviations is what both problems do. It is also how h^* is estimated in practice. The regression's R^2 then measures the proportion of variance the hedge removes.

The optimal number of contracts. With Q_A units of exposure and Q_F units per contract,

$$N^* = \frac{h^* Q_A}{Q_F} \tag{3.4}$$

Example 3.3 applies both formulas.

Example 3.3 Hedging jet fuel with heating oil

An airline expects to buy 2 million gallons of jet fuel in one month. There is no liquid jet fuel futures contract, so it uses heating oil futures, each covering 42,000 gallons, as a cross hedge.

From historical data on monthly changes: the standard deviation of the jet fuel price change is \$0.032 per gallon, that of the heating oil futures price change is \$0.040, and their correlation is 0.8. From (3.3),

$$h^* = 0.8 \times \frac{0.032}{0.040} = 0.64$$

and from (3.4),

$$N^* = \frac{0.64 \times 2{,}000{,}000}{42{,}000} \approx 30.5$$

so the airline buys 30 contracts.

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Two observations follow. The hedge ratio is well below one (the airline's 30 contracts cover 1.26 million gallons against a 2 million gallon exposure) because heating oil futures are more volatile than jet fuel and imperfectly correlated with it. And the hedge is far from perfect: with $\rho = 0.8$, the regression R^2 is 0.64, so about 36% of the variance of the jet fuel cost remains after hedging. That the R^2 equals the hedge ratio here is a coincidence of the inputs: $R^2 = \rho^2$ while $h^* = \rho\,\sigma_S/\sigma_F$, and the two agree only because $\sigma_S/\sigma_F = 0.8$ happens to equal ρ.

Estimating the hedge ratio in practice

Equation (3.3) requires three numbers that must be estimated, and the estimation choices matter more than the formula.

Use changes, not levels. Regressing the spot price on the futures price rather than their *changes* typically produces an R^2 near one, and a hedge ratio near one whenever the two levels share a trend of similar slope. Both series are nonstationary and trend together; the regression is measuring that shared trend, not the co-movement a hedge relies on. (Cointegration, a long-run tie between the two levels, is a separate question and does not change the point.) This is among the most common errors in applied hedging, and it flatters the result badly.

Match the data frequency to the hedge horizon. A hedge held for a month should be estimated on monthly changes. Daily data offers more observations and a tighter standard error, but it measures a different thing: short-horizon co-movement is often weaker than long-horizon co-movement, because temporary dislocations between spot and futures wash out over time. Example 3.4 shows how large the difference can be.

Example 3.4 The hedge ratio depends on the horizon

The same spot and futures series are used to estimate (3.3) at two frequencies.

Data frequency	σ_S	σ_F	ρ	h^*
Daily changes	0.0070	0.0094	0.62	0.46
Monthly changes	0.0320	0.0400	0.80	0.64

The two estimates differ by nearly 40%. The daily figure is not wrong; it is the right hedge ratio for a hedge held one day. Applying it to a one-month hedge underhedges substantially, because it has been calibrated on a horizon over which spot and futures diverge more.

Most of the rise comes from the correlation, which climbs from 0.62 to 0.80; the rest comes from the volatility ratio σ_S/σ_F, which edges up from 0.745 to 0.800. The general rule is that correlation between related assets usually rises with the measurement horizon, so the hedge ratio does too. A hedger who has a monthly exposure and a daily dataset should aggregate the data, not the reasoning.

The sample period is a judgment, not a technicality. A long sample gives precision; a short one gives relevance. The minimum-variance ratio is not a constant of nature: it shifts when market structure changes, when a new delivery point opens, when refining capacity changes the relationship between two grades. A ratio estimated over ten years spanning a structural break is precisely estimated and wrong.

Overlapping observations. Using monthly changes measured weekly, a common device for increasing sample size, creates overlapping windows and correlated residuals. The point estimate is usually acceptable; the standard errors are not, and a hedger who concludes the ratio is precisely determined on this basis is mistaken.

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Tailing the hedge. Equation (3.4) works in units, and it slightly overhedges. Because futures are settled daily, a gain realized today earns interest until the hedge is lifted, so a smaller position delivers the same offset at the horizon. Discounting over the remaining hedge life $T - t$ at the interest rate r gives

$$N^* = \frac{h^* Q_A}{Q_F} e^{-r(T-t)} \tag{3.5}$$

a correction known as **tailing the hedge**; [TN3] derives it and tabulates the factor. The correction in (3.5) is small for short horizons and grows with maturity and with the level of interest rates: at $r = 4\%$ it removes 0.3% of the position over one month but 7.7% over two years, and the position is scaled back up as the horizon shortens. It comes from daily settlement, as the forward–futures price difference of Section 5.12 does, though unlike that difference it does not require interest rates to be uncertain.

Ratios estimated on returns. Futures hedge ratios are more often estimated on *proportional* changes than on absolute ones (returns on the spot against returns on the futures), which is the natural scale when the position is resized as prices move. Write $\hat{h}$ for a ratio estimated that way, and let S and F be the spot and futures prices when the hedge is placed (the S_1 and F_1 of Section 3.3, with the subscript dropped because only one date is in play here). Then $\hat{h}$ is not the h^* of (3.3). A proportional change becomes an absolute one by multiplying by the price, so $\sigma_{\Delta S/S} = \sigma_S/S$ and $\sigma_{\Delta F/F} = \sigma_F/F$, while ρ is unchanged because correlation does not depend on scale. Hence

$$\hat{h} = \rho \frac{\sigma_S/S}{\sigma_F/F} = h^* \frac{F}{S}.$$

A ratio estimated on returns is therefore applied to *values* rather than quantities,

$$N^* = \frac{\hat{h} V_A}{V_F} \tag{3.6}$$

where $V_A = SQ_A$ is the value of the position being hedged and $V_F = FQ_F$ the value of one contract. Substituting takes (3.6) straight back to (3.4):

$$\frac{\hat{h} V_A}{V_F} = h^* \frac{F}{S} \cdot \frac{SQ_A}{FQ_F} = \frac{h^* Q_A}{Q_F}.$$

The two are the same hedge written two ways, and either is correct on its own terms. What is not correct is to mix them: an h^* estimated on price changes and then applied to values gives

$$\frac{h^* V_A}{V_F} = \frac{h^* Q_A}{Q_F} \cdot \frac{S}{F},$$

rescaling the position by S/F, shrinking the hedge in contango, when futures prices stand above spot and rise with maturity, and enlarging it in backwardation, when they stand below spot and fall with maturity, for no reason at all.

3.6 HEDGING AN EQUITY PORTFOLIO

Stock index futures let a manager change a portfolio's market exposure without trading its constituents. This is the use Chapter 6 identifies for Treasury futures and duration.

The relevant hedge ratio is the portfolio's **beta**, its sensitivity to the index. To hedge fully,

$$N^* = \beta \frac{V_A}{V_F} \tag{3.7}$$

where V_A is the portfolio's value and V_F the value of one futures contract.

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Example 3.5 Hedging a portfolio

A manager holds a \$5,050,000 portfolio with a beta of 1.5 against an index standing at 4,200. Futures contracts are worth \$50 per index point, so each contract is worth $4{,}200 \times 50 = \$210{,}000$. From (3.7),

$$N^* = 1.5 \times \frac{5{,}050{,}000}{210{,}000} = 36.07$$

so the manager shorts 36 contracts.

Suppose the index then falls 10% to 3,780. The gain on the futures is

$$36 \times (4{,}200 - 3{,}780) \times 50 = \$756{,}000$$

while the portfolio, with a beta of 1.5, falls by roughly 15%, or \$757,500. The two offset to within \$1,500.

Removing the market also removes the market's return. Setting aside dividends and the gap between futures and spot prices, a fully hedged portfolio earns roughly the risk-free rate plus whatever the stocks earn beyond what their beta predicts (their alpha).

Changing beta rather than eliminating it. A manager who wants to reduce exposure without removing it shorts

$$N^* = (\beta - \beta^*)\,\frac{V_A}{V_F} \tag{3.8}$$

contracts, where β^* is the target. In Example 3.5, where $V_A/V_F = 24.05$, equation (3.8) moves beta from 1.5 to 0.75 by shorting $0.75 \times 24.05 = 18.04$, so 18 contracts; *raising* beta to 2.0 gives a negative count, and the manager goes long $0.5 \times 24.05 = 12.02$, so 12.

Why hedge rather than sell? If the manager wanted no market exposure, selling the portfolio would achieve it. Futures are used instead for three reasons: the manager may believe the specific stocks will outperform and want to keep that bet while removing the market, transaction costs on index futures are far lower than on the underlying holdings, and the position may be intended as temporary. The first of these is the most important: hedging with index futures is how a manager separates stock selection from market timing.

Hedging is not a one-off decision

Equations (3.4) and (3.7) give a position to put on today. Both quantities they depend on then move.

The portfolio's value changes, so the hedge sized against V_A becomes wrong; the beta changes as holdings change and as the market's factor structure shifts; and for a commodity hedger the exposure itself changes as production forecasts are revised. A hedge placed once and left is correct only on the day it was placed.

How fast does it drift? For the equity hedge the answer is available in closed form. If the index moves by a proportion m, the portfolio moves by βm and one contract by m, so the position required by (3.7) changes by the factor

$$\frac{1 + \beta m}{1 + m}.$$

Two things follow. At $\beta = 1$ the factor is exactly one: market moves alone never call for rebalancing, because the exposure and the contract are revalued in step. What a manager is chasing is the amount by which β differs from one. And even away from $\beta = 1$ the drift is slow.

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In Example 3.5 the 10% fall leaves a portfolio of $4,292,500 and a contract worth $189,000, so the position now required is

$$1.5 \times \frac{4{,}292{,}500}{189{,}000} = 34.07$$

against the 36 contracts held, a two-contract adjustment after a 10% move in the market.

Rebalancing has a cost (each adjustment crosses a spread), so in practice firms use a schedule and a trigger together: monthly for an equity overlay (a futures position laid over a portfolio that is itself left untouched), on the planning cycle for a commodity hedger whose forecast exposure is the thing that moves, with a band of a few percent of the required position to catch large moves in between and a forced adjustment at every roll. The ratio itself is re-estimated far less often than the position is resized, quarterly at most, because re-estimating on each new observation feeds sampling noise straight into the position, and [TN3] puts a number on that cost. The choice of band is the same trade-off that recurs throughout this book: continuous adjustment is accurate and expensive, infrequent adjustment is cheap and approximate.

This is also the first appearance of an idea that dominates Part III. Chapter 18 shows that hedging an option requires exactly this kind of continuous rebalancing, because an option's sensitivity to the underlying changes as the underlying moves. The difference is one of degree: here the hedge ratio drifts slowly, and there it moves whenever the market does.

3.7 HOW MUCH TO HEDGE

Equation (3.3) answers a narrow question: what ratio minimizes the variance of a single exposure over a single horizon? Corporate hedging programs rarely answer only that question, and it is worth seeing what else enters.

Partial hedging is normal and often deliberate. Many firms hedge a fixed fraction of expected exposure (half the next year's production, a quarter of the year after) rather than the variance-minimizing amount. Three reasons follow, all defensible:

The exposure itself is uncertain. A producer hedging 100% of *expected* output is overhedged if output disappoints, and then holds a naked short futures position that loses if prices rise, which an industry-wide shortfall tends to make them do. Hedging a fraction leaves room for the quantity to come in low.

Shareholders may want some exposure. An investor buying a copper miner is buying copper exposure. A fully hedged miner is a fabrication business with a mine attached, which is not what was bought.

Cash constraints bind. Section 3.10 is about this, and it is the reason a variance-minimizing ratio can be the wrong answer.

Layered hedging. A common structure hedges a declining fraction across a rolling horizon (perhaps 75% of the next six months, 50% of the following six, 25% of the year after), adding a new layer each quarter. This averages the entry price across many dates, which removes the risk of having placed the entire program on a single bad day, and it degrades gracefully if forecasts change.

The point is that the entry price is itself a risk. A firm that hedges its full three-year exposure in one transaction has replaced price risk with timing risk, and Case Study 3.1's organizational problem then attaches to a single decision made on a single day.

Or hedge with options instead. Futures are not the only instrument. A firm that buys an **option** (a producer a put, a consumer a call) removes the unfavorable tail and keeps the favorable

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one, in exchange for a premium. A purchased option whose premium is paid up front then owes no variation margin at all. That difference bears directly on the failures of Section 3.10, but options are not developed until Chapter 10, so the comparison is made in Section 12.10.

3.8 ACCOUNTING FOR HEDGES

Case Study 3.1 described the organizational difficulty of hedging. There is a formal version of it in the accounts, and it shapes which hedges firms actually choose.

Derivatives are carried at fair value with changes recognized in profit or loss. The hedged item usually is not: inventory sits at cost, future production is not on the balance sheet at all, and a forecast purchase does not exist as an accounting item. So a perfectly effective economic hedge produces *earnings volatility from the derivative alone*, with nothing offsetting it until the hedged transaction occurs.

Hedge accounting exists to fix this, in one of two ways depending on what is hedged. In a **fair value hedge**, of an asset or liability already on the balance sheet, the hedged item is remeasured for the hedged risk and the change goes to profit or loss beside the derivative's, so nothing is deferred; inventory hedged with futures, the case above, is of this kind. In a **cash flow hedge**, of a forecast transaction, there is no item to remeasure, so the effective portion of the derivative's gain or loss is held in other comprehensive income, a component of equity outside profit, and moved into profit when the hedged transaction affects earnings. Either way the two legs meet in the same period, which is what the economics always said.

The requirements are not trivial. Documentation must exist at inception, and the designation constrains what the firm may subsequently do with the position. The effectiveness test differs between the two main standards. International Financial Reporting Standard (IFRS) 9 requires an economic relationship between the hedged item and the hedging instrument, assessed prospectively, with any ineffectiveness recognized in profit or loss as it arises; it dropped the retrospective 80–125% test of its predecessor. US generally accepted accounting principles (US GAAP), in Accounting Standards Codification (ASC) Topic 815, still require the hedge to be highly effective, but after inception allow that to be assessed qualitatively.

The consequence for hedging choices. Because qualifying is onerous and failing is visible, firms sometimes prefer a hedge that qualifies for hedge accounting over one that is economically better. IFRS 9 eases one version of the problem by allowing a risk component to be designated as the hedged item, such as the crude oil component of a jet fuel purchase, so a crude hedge of jet fuel can qualify for that component. But a cross hedge whose link to the hedged item is real yet hard to evidence may still fail to qualify, or qualify with large recorded ineffectiveness, where a worse-matched but simpler instrument passes. A dynamic strategy that adjusts as exposures change may be harder to document than a static one that hedges less well.

This is worth stating plainly: the accounting treatment is a real input into hedging decisions, and it does not always point the same way as the economics. A reader who finds a firm's hedging program puzzling should check whether the puzzle dissolves once the accounting constraint is added.

And the tax rule is a third answer

Tax authorities also have a view, and it is not the accounting one. The specifics are jurisdictional and change, so what follows is the shape of the problem rather than any country's current rules; a reader acting on it should check the rules in force.

Three features recur across jurisdictions and matter to a hedger. *Timing*: gains on an exchange-traded futures position are commonly taxed as they accrue (marked to market at the year end whether or not the position was closed), while the hedged item is taxed when it is

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sold. *Character*: the derivative's gain may be a capital gain while the hedged item produces ordinary income, and capital losses often cannot be set against ordinary income. *Qualification*: some systems, notably the United States, provide a hedging exception that aligns both, taxing the derivative on the same timing and character as the item it hedges, but it must be claimed by identifying the hedge on the day it is entered into; others largely follow the accounting treatment.

The failure mode is the one this chapter keeps meeting. A hedge that is economically exact can be taxed asymmetrically: the loss leg deductible in one year at one rate, the gain leg taxable in another at another. The firm is then paying tax on a profit it did not make, in a year in which its economic position did not change—and it is paying cash, on the schedule of Section 3.10's margin calls rather than on the schedule of its production. Metallgesellschaft, whose collapse Section 3.10 describes, met the same asymmetry in its accounts rather than its tax bill: German accounting rules, which recognize unrealized losses but not unrealized gains, booked the losses on its futures while ignoring the offsetting gains on its customer contracts.

So a hedging decision faces three tests and they do not agree. The economics ask whether the exposure is offset, hedge accounting asks whether the offset can be documented and demonstrated, and the tax rule asks whether the hedge was identified in time and in the right form. *A hedge can pass any two and fail the third*, and the firm discovers which one it failed only afterward.

3.9 ROLLING THE HEDGE FORWARD

Section 3.3 assumed a contract exists spanning the exposure. Often none does. Liquidity in futures concentrates in the nearest few contracts, while a firm's exposure may extend years.

The response is to **stack and roll**: place the entire exposure in a near contract, and as it approaches expiration, close it and open the equivalent position in the next. A five-year exposure hedged with quarterly contracts requires nineteen rolls.

Each roll introduces basis risk of its own: the hedger closes at one contract's price and opens at another's, and the spread between them is not known in advance. Nineteen rolls means nineteen draws from that distribution. Example 3.6 works one sequence of rolls through, and Example 3.7 shows what the draws can cost when they run against the hedger.

Example 3.6 A rolling hedge, roll by roll

A producer will sell 1 million barrels in twelve months, but only three-month contracts are liquid. It shorts the near contract and rolls quarterly. The market is in mild contango throughout.

Month	Action	Price	Gain on leg
0	Short contract A	80.00	
3	Close A	78.00	+2.00
	Short contract B	78.60	
6	Close B	76.50	+2.10
	Short contract C	77.00	
9	Close C	75.00	+2.00
	Short contract D	75.40	
12	Close D	74.00	+1.40
	Sell the oil at spot	73.90	
	Total futures gain		**+7.50**

The effective price is 73.90 + 7.50 = \$81.40 per barrel—\$1.40 *better* than the \$80.00 at which the hedge was placed.

The improvement decomposes cleanly. Each roll closed a contract and opened a more

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expensive one: +0.60, +0.50, +0.40, totaling \$1.50 of roll gains. The final basis contributed 73.90 − 74.00 = −\$0.10. Together, 80.00 + 1.50 − 0.10 = \$81.40.

The producer gained because it was *short* in a market in contango. Reverse the direction (a consumer hedging with long futures in the same market) and the identical sequence of prices produces \$1.50 of roll *losses*.

Example 3.7 The cost of rolling

A producer must hedge 1 million barrels a month for three years, but only the front three contracts are liquid. It stacks the entire remaining exposure in the front contract and rolls monthly: 35 rolls, since the last month's barrels are delivered from the front contract without a further roll.

Suppose each roll costs 15 cents a barrel, because the market is in backwardation, which works against a short hedger. The position rolled shrinks from 35 million barrels to 1 million, an average of 18 million,

$$35 \text{ rolls} \times \$0.15 \times 18{,}000{,}000 = \$94.5 \text{ million}$$

of cumulative roll cost, against a hedge that was placed to protect a price.

Example 3.6 showed which direction the cost runs: a short hedger *gains* from rolling in contango and loses in backwardation, and a long hedger the reverse (Section 5.11 works the long investor's case). The producer above is short in backwardation and so pays. The mirror case, a firm hedging the other way with long futures, faces exactly this bill when the market moves into contango.

That is not a hypothetical. It is what happened to Metallgesellschaft.

3.10 WHEN HEDGING FAILS

The mechanisms above (basis risk, roll risk, and the cash flow timing of Chapters 2 and 8) combine badly. The clearest illustration remains a case from the early 1990s, told in Case Study 3.2.

Case Study 3.2 Metallgesellschaft

The US subsidiary of the German firm Metallgesellschaft sold long-term contracts to deliver heating oil and gasoline at fixed prices, some running as long as ten years. It had committed to sell roughly 160 million barrels at prices fixed in advance.

That is a short physical position, and it was hedged by going *long* futures. Since no liquid ten-year contract existed, the firm stacked the exposure in near-dated futures, about a third of the position, and short-dated over-the-counter swaps, and rolled them.

Three things then happened at once. Oil prices fell sharply through 1993, so the long futures position lost money—and because futures are settled daily, those losses had to be paid in cash immediately. The market shifted from backwardation into contango, so each roll now cost money rather than earning it. And the offsetting gains, sitting in customer contracts running out to a decade, produced no cash at all.

The margin calls ran to roughly \$900 million by the end of 1993, and the losses eventually came to about \$1.3 billion. The parent, MG AG, was itself in difficulty and was rescued by its banks in early 1994. Facing a cash demand of that size against gains it could not realize, it chose not to fund its subsidiary's margin: it liquidated the futures positions and unwound many of the customer contracts, crystallizing the losses while abandoning the position that would have offset them.

On the reading of Culp and Miller (1995) the hedge largely held in value, the futures losses being matched by gains on the supply contracts, and what it lacked was the cash

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> to carry that offset through time. Mello and Parsons (1995) and Edwards and Canter (1995) disagree: a stack of short-dated contracts sized one for one against commitments a decade long was not a sound hedge in the first place. Whether the parent was right to liquidate rather than fund the position remains debated. What is not in dispute is the mechanism, and it is the one this book keeps meeting: a position hedged against price risk, defeated by the timing of the cash it demanded, and abandoned at the point of maximum loss because the combined position was judged one leg at a time.

The 2022 episodes of Chapters 2 and 8 are the same story with different commodities and a different resolution. European utilities that had sold output forward faced margin calls in the tens of billions against physical gains years away. The difference in outcome was not that those firms hedged better; it was that governments extended credit lines to them, while Metallgesellschaft's parent chose not to fund its subsidiary's margin.

What to take from this. Three practical conclusions follow, and they are the reason this chapter exists in the form it does.

Size the hedge to the cash you can produce, not only to the exposure. A hedge ratio derived from (3.3) minimizes price variance and says nothing about liquidity. A firm that cannot fund the margin on its variance-minimizing hedge should hedge less—and should know that in advance rather than discovering it. Often the cash problem is solved by a different instrument rather than a smaller hedge: the reserve-based borrowers of Section 3.2 usually hedge with swaps from their lending banks, secured by the same collateral as the loan and carrying no variation margin.

Stack and roll converts a long-dated exposure into a sequence of short-dated ones, and the sequence has its own risk. It is not a substitute for a matched-maturity hedge; it is a different strategy with different failure modes.

Judge the combined position. Every failure in this section involved someone looking at the futures leg alone.

SUMMARY

A short hedge protects a firm that owns or will own an asset; a long hedge protects one that must buy. Whether to hedge at all is a real question, since shareholders can diversify; the stronger arguments concern frictions, above all the cost of financial distress. Hedges are also hard to sustain, because the loss on the futures leg is visible while the offsetting gain is buried in operating results, so a program must be mandated in advance and judged on the combined position.

A hedge lifted before delivery locks in $F_1 + b_2$, so the risk that remains is the basis at the close, which arises from mismatches of asset, location, and timing. Basis risk is at least as important in financial hedges and less visible: a corporate bond portfolio hedged with Treasury futures keeps its credit spread risk, and a hedge in a correlated proxy currency fails exactly when the two currencies part company.

The variance-minimizing ratio $h^* = \rho\,\sigma_S/\sigma_F$ is a regression slope, and its R^2 says how much of the risk the hedge removes. It must be estimated on changes rather than levels and at the frequency of the hedge horizon. An equity portfolio is hedged with index futures in proportion to its beta, which separates stock selection from market exposure.

Firms rarely hedge the full variance-minimizing amount, and purchased options whose premium is paid up front, which then owe no variation margin, are the alternative that avoids the funding failures. Accounting and tax can make an economically sound hedge look bad: the derivative is marked to market while the hedged item often is not, hedge accounting realigns

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the two only for hedges that qualify, and a tax rule may treat the two legs differently. Exposures longer than the liquid contracts are stacked and rolled, which adds a fresh basis at every roll. Metallgesellschaft combined every failure mode, and the 2022 energy episodes repeated the mechanism; what separated the outcomes was whether someone would fund the margin, not better hedging.

FURTHER READING

Ederington (1979) is the origin of the minimum-variance hedge ratio in the form this chapter uses, and it is short. Froot, Scharfstein, and Stein (1993) make the modern case for why a firm hedges at all—the question behind every hedge ratio and the one a formula cannot answer. On Metallgesellschaft the literature is a genuine disagreement and should be read from both sides: Culp and Miller (1995) argue the strategy was sound and the liquidation was the error, while Edwards and Canter (1995) and Mello and Parsons (1995) take the opposing view. Smith and Stulz (1985) set out the tax and distress-cost arguments for hedging that Froot, Scharfstein, and Stein build on.

PRACTICE QUESTIONS

3.1 Set out the argument that a firm should not hedge because its shareholders can diversify, and give two reasons the argument may fail.

3.2 Define the basis, show that a short hedger who lifts a hedge before delivery receives $F_1 + b_2$, and explain what this implies about where the residual risk lies.

3.3 A hedge is placed when the spot price is \$44 and the futures price is \$41.50, and lifted when the spot price is \$47 and the futures price is \$45.20. Compute the effective price for a short hedger, state whether the basis strengthened or weakened, and explain why a June exposure is usually better hedged with a July contract than a June one.

3.4 A company will buy 800,000 units of a commodity in three months. The standard deviation of the change in its price over three months is \$0.65, that of the futures price \$0.81, and the correlation 0.72. Each contract covers 50,000 units. Compute the minimum-variance hedge ratio and the number of contracts, state what proportion of the variance remains, and explain why that ratio is the slope of a regression.

3.5 A \$12 million portfolio has a beta of 0.9. Index futures are worth \$50 per point and the index stands at 4,000. How many contracts hedge it fully, and how many reduce beta to 0.4?

3.6 An analyst estimates a hedge ratio of 0.46 from daily data and 0.64 from monthly data for a hedge held one month. State which should be used, explain why the two differ in this direction, and explain what goes wrong if the regression is run on levels rather than changes.

3.7 Give three reasons a firm might deliberately hedge less than the variance-minimizing amount, and explain what layered hedging achieves that a single transaction does not.

3.8 A manager hedges an equity portfolio with index futures using the beta-adjusted contract count. Show that if the index moves by a proportion m, the position required changes by the factor $(1 + \beta m)/(1 + m)$. Explain why a portfolio with a beta of one never needs rebalancing on account of a market move alone, state the condition on β and m under which the hedge must be *increased*, and evaluate the factor for a portfolio with a beta of 0.8 after the index rises 12%.

3.9 A bank hedges a portfolio of corporate bonds with Treasury futures. Explain what basis

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risk remains, why a historical regression over a calm period would understate it, and what happens to both legs in a credit sell-off.

3.10 Using the roll table of Example 3.6, recompute the effective price if the market had been in backwardation, with each roll closing at a price 0.40 *above* the next contract, and state who gains in each case.

FURTHER QUESTIONS

3.11 A fund manager wishes to keep a portfolio's stock selection but remove its market exposure for three months. Explain why index futures are preferred to selling the portfolio, and identify what risk remains after the hedge is in place.

3.12 Explain what stack and roll is, why a firm would use it, and the two distinct risks it introduces that a matched-maturity hedge would not. Relate the direction of the roll cost to the shape of the forward curve.

3.13 Set out the Metallgesellschaft episode as a sequence of events, identifying at each stage which of this chapter's risks was operating. Was the strategy unsound, or was the liquidation the error? Defend a position.

3.14 A producer's variance-minimizing hedge would require it to post margin that could reach 40% of its annual revenue in a severe price move. Explain why the hedge ratio of equation (3.3) does not capture this, what the firm should do, and how you would set the hedge ratio instead.

3.15 A firm chooses a hedging instrument that matches its exposure less well than an available alternative, on the grounds that the alternative would not qualify for hedge accounting. Set out the argument for and against that decision, and say what you would need to know about the firm to judge it.

3.16 European utilities in 2022 and Metallgesellschaft in 1993 faced the same mechanism with different outcomes. Identify the mechanism precisely, explain what differed, and state what a treasurer should arrange in advance so that the difference does not matter.

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Chapter 4
Interest Rates in the Risk-Free-Rate Era

Interest rates are the raw material of derivatives pricing. Every valuation in this book discounts something, and most of what we discount is itself a function of rates. It is worth spending a chapter on how rates are defined, measured, and assembled into a curve before we start using them.

A reader who learned interest rates from an older textbook will find this chapter organized differently from the way the subject was taught for the preceding thirty years. That is deliberate, and it reflects a genuine change in the market rather than a change in taste.

Until recently, the reference rate at the center of the derivatives market was LIBOR: a rate with a term structure built into it, quoted for one month, three months, six months, and a year, and fixed at the *start* of each interest period. Everything about how we taught interest rate derivatives assumed a rate of that shape. Forward rate agreements, floating-rate notes, caps, floors, and swaps were all defined against a rate you knew in advance.

LIBOR is gone. The rates that replaced it are **overnight** rates. They have no term structure of their own, they are published one day at a time, and a three-month interest amount computed from them is not known until the three months have elapsed. Every convention in the market had to be rebuilt around that fact, and a great deal of the machinery in Chapters 6, 7, 29, and 30 rests on getting these mechanics right.

This chapter therefore builds from the overnight rate upward. Sections 4.1 through 4.6 cover the rates themselves: what they measure, why the old ones were replaced, and how an overnight rate is compounded into the term rate a contract actually needs. Sections 4.7 through 4.12 cover the traditional apparatus (compounding conventions, zero curves, forward rates, duration, and convexity), which the transition left essentially intact. Readers who have met that apparatus elsewhere can move through the later sections quickly, but should not skip Sections 4.4 and 4.5, which have no counterpart in older treatments.

4.1 TYPES OF INTEREST RATES

An interest rate in any situation defines the amount of money a borrower promises to pay a lender. The higher the credit risk of the borrower, the higher the rate. Rates are quoted with a compounding frequency and a day count convention attached; ignoring either is a common and expensive error, and we return to both in Section 4.7. A rate also has a **tenor** (the length of the period it applies to, such as overnight, one month, or three months), which is not the same thing as the maturity of the instrument that pays it. A ten-year swap whose floating rate resets every three months has a maturity of ten years and a tenor of three months. The distinction runs through the whole book: the tenor is what the rate measures, the maturity is how long the exposure lasts.

Treasury rates. These are the rates at which a government borrows in its own currency. Because a government can in principle raise taxes or print money to repay such debt, these are conventionally treated as having negligible credit risk. Treasury rates are nevertheless *not* the rates derivatives markets discount at, for reasons that are partly technical and partly practical. Treasury yields differ from funding rates in either direction, depending on the supply of government debt, the balance-sheet cost of holding it, and the demand for it as collateral; Section 7.9 shows long dollar swap spreads negative since 2008, which puts Treasury yields above the swap curve. And a derivatives dealer cannot fund itself at the Treasury rate.

Repo rates. A repurchase agreement is a secured loan: one party sells securities and agrees to buy them back at a slightly higher price, the difference being the interest. The collateral need not be government securities. It can be almost any security the two sides agree on (agency mortgage-backed securities, corporate bonds, equities), and the rate depends on which it is. Better collateral commands a lower rate and a smaller **haircut** (the margin by which the collateral's market value exceeds the cash lent against it) because the lender of cash is less exposed if the borrower fails; repo against government bonds trades at the lowest rate, and repo against equities pays materially more. Credit risk is small but not zero, and it is a joint condition: the borrower must default *and* the collateral must have fallen in value before the lender loses anything.

Two further distinctions matter for what follows. A **general collateral** (GC) repo is one in which the lender of cash will take any security from an agreed class, and is therefore lending cash; a security in particular demand trades **special**, at a repo rate below general collateral and occasionally near zero, because the lender of cash is giving up interest for the privilege of borrowing *that* security. And repo trades at maturities from overnight out to several months. The dollar's reference rate is built from one narrow corner of all this (overnight, general collateral, US Treasury), and Section 4.2 explains why that corner and not another.

Overnight risk-free rates (RFRs). These are the successors to LIBOR and the subject of most of this chapter. Each is an average of actual overnight borrowing transactions in its currency, published once per business day by a central bank or exchange. The abbreviation **RFR** is used throughout this book, and in the market, for a rate of this kind.

Reference rates in derivatives. A **reference rate** is a published benchmark that contracts point at rather than negotiate individually. Its usefulness depends entirely on being both widely accepted and hard to manipulate. The story of the past decade is the market's discovery that its principal reference rate satisfied the first condition and not the second.

4.2 THE OVERNIGHT RISK-FREE RATES

Each major currency now has a designated overnight rate, selected by a working group of market participants and regulators: the Secured Overnight Financing Rate (SOFR) in dollars, the Sterling Overnight Index Average (SONIA), the Euro Short-Term Rate (€STR), the Tokyo Overnight Average Rate (TONA), and the Swiss Average Rate Overnight (SARON). Table 4.1 sets out how each is built.

Three features are common to all of them, and each is a direct response to a defect in LIBOR.

They are transaction-based. Each rate is computed from actual trades, not from estimates. SOFR, SONIA, and €STR are published the next morning from the previous day's transactions; SARON is computed in real time and fixed at the close of the same day from trades and binding

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Table 4.1: The principal overnight risk-free rates.

Currency	Rate	Basis	Administrator
USD	SOFR	Repo, US Treasuries	Federal Reserve Bank of New York
GBP	SONIA	Unsecured	Bank of England
EUR	€STR	Unsecured	European Central Bank
JPY	TONA	Unsecured	Bank of Japan
CHF	SARON	Repo, SNB GC basket	SIX Swiss Exchange

quotes; TONA has a same-day preliminary figure. SOFR is a volume-weighted median of overnight Treasury repurchase transactions and rests on well over $2 trillion of daily activity. LIBOR, at the end of its life, was in some tenors supported by a small number of transactions per day or none at all.

They are overnight. There is no one-month or three-month SOFR fixing in the way there was a one-month or three-month LIBOR. This is the feature with the largest practical consequences, and Section 4.4 is devoted to it.

They are close to risk-free. An overnight loan to a creditworthy counterparty carries almost no credit risk, and a secured overnight loan carries almost none even if the counterparty is weak. LIBOR, by contrast, was the rate at which *banks* could borrow unsecured for a term, and therefore embedded bank credit risk. That embedded credit component behaved unhelpfully: it widened exactly when the financial system was under stress, which is exactly when the contracts referencing it mattered most.

The policy rate is not the benchmark

A reader who knows the US money market will object that the table omits an obvious overnight dollar rate: the **effective federal funds rate** (EFFR), at which banks lend reserves to one another, published by the same New York Fed that publishes SOFR. It is not the benchmark, for two reasons worth separating.

Volume. Fed funds trading collapsed after 2008. Once reserves became abundant banks no longer needed to borrow them, and most of what remains is Federal Home Loan Banks (which cannot earn interest on reserve balances) lending to institutions that can. Daily volume runs on the order of $100 billion against SOFR's more than $2 trillion. A benchmark carrying hundreds of trillions of notional should not rest on the narrower market, which is the defect that ended LIBOR in a different form.

A policy rate makes a poor contract rate. EFFR is the rate the Federal Open Market Committee sets a target range for and steers toward, and it moves when policy decides it should. A benchmark ought to describe the market rather than be administered by an authority pursuing objectives of its own.

That second property is what keeps fed funds useful elsewhere. Being the policy rate makes it the rate in which expectations of policy are expressed (fed funds futures are how the market prices the next meeting), and the **fed funds–SOFR basis**, the spread between the two rates, trades actively through swaps that exchange one rate for the other, because a bank funds itself in one market and runs its derivative book in the other. The two rates can diverge with neither market under credit stress, for instance when Treasury collateral is scarce and repo rates fall relative to unsecured ones. Section 7.4 describes the October 2020 switch of dollar discounting from EFFR to SOFR, and the compensation it required.

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Secured versus unsecured

Table 4.1 shows that the rates are not all built the same way, and the distinction matters. SOFR and SARON are *secured*: they measure the cost of borrowing against collateral. SONIA, €STR, and TONA are *unsecured*: they measure the cost of borrowing overnight without collateral, and so retain a very small credit component.

The two secured rates are not secured against the same thing. SOFR is Treasury repo and nothing else. SARON is computed from the Swiss repo market against the general collateral basket of the Swiss National Bank (SNB), which admits cantonal and foreign government bonds, bonds of supranational institutions (for example, the European Investment Bank and the World Bank), and covered bonds alongside Confederation debt. "Secured" describes the structure of the loan, not the identity of the collateral, and a benchmark's designers choose both.

Which repo rate? A repo market is not one rate, and turning it into a benchmark takes choices. SOFR is drawn from three segments of the overnight Treasury market (tri-party repo, the interdealer General Collateral Finance market, and cleared bilateral transactions) and is a volume-weighted *median* rather than a mean, so that a small number of extreme trades cannot move it. The cheapest quarter of the bilateral volume is discarded before the median is taken, and that is how specials are kept out: a trade struck at a low rate because the lender wanted one particular bond says something about that bond, not about the price of cash. The same data supports two further published rates, the Tri-party General Collateral Rate and the Broad General Collateral Rate, which are repo rates but are not benchmarks. Not every repo rate is a reference rate, and what separates them is these choices rather than anything in the underlying trades.

It follows that SOFR is a proxy for the risk-free rate of an arbitrage argument rather than the thing itself. What a given firm can actually finance a given security at depends on that security and on that firm, which is where Section 5.10 picks the question up.

Being secured is not the same as being immune to stress. A secured rate reflects conditions in the collateral market, and those conditions can move violently for reasons having nothing to do with credit.

Case Study 4.1 September 2019: when the risk-free rate spiked

On September 17, 2019, SOFR fixed at 5.25%, up from 2.43% the day before. Some overnight repo trades executed at rates approaching 10% intraday. Nothing had happened to the creditworthiness of any borrower; the collateral in question was US Treasury securities.

What had happened was a collision of cash flows. A large corporate tax payment date coincided with the settlement of a heavy Treasury auction, draining reserves from the banking system at the moment dealers needed to finance a large inventory of new securities. Demand for overnight cash overwhelmed supply, and the price of cash, the repo rate, went to whatever level cleared the market. The Federal Reserve intervened with repo operations within a day.

The episode carries two lessons for anyone using SOFR. First, a rate can be free of *credit* risk and still be exposed to *funding* and *liquidity* conditions; "risk-free" is a statement about default, not about volatility. Second, the effect was concentrated in a single day. A contract referencing an overnight rate compounded over three months would have absorbed the spike into an average and barely noticed it. That is a real, if incidental, argument for the compounded-in-arrears conventions of Section 4.4. Predictable calendar effects of the same kind, milder but recurring, appear at each quarter-end and year-end.

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Negative rates

A feature of these rates that would have seemed impossible when LIBOR was designed is that they can be, and have been, negative. €STR was negative from its launch until 2022, following the European Central Bank's deposit facility rate below zero from 2014. TONA was negative from 2016 until the Bank of Japan ended its negative rate policy in March 2024, and SARON was negative from 2015 until 2022 and has fixed slightly below zero again since the SNB cut its policy rate to zero in June 2025.

This is not a curiosity. A lognormal model of rates cannot represent a negative rate, so when European rates went below zero the market for rate options moved to **normal** (or Bachelier) and **shifted lognormal** volatilities, a change Chapter 29 takes up.

A second consequence is contractual. Many floating-rate loans carry a zero floor, so the borrower never receives interest. A floor is an option, and a loan with an embedded floor is a loan plus a short position in that option, a position many borrowers did not realize they had sold until it moved into the money. It showed in hedges. A borrower that had swapped a zero-floored loan to fixed with an unfloored swap found that, once the floating rate went negative, the loan's floating payment stopped at zero while the swap's floating leg kept falling, so the borrower paid the fixed rate plus the size of the negative rate. The hedge leaked, which is why such hedgers bought floors to close the gap.

4.3 WHY LIBOR ENDED

LIBOR was, for nearly four decades, the most important number in finance. At its peak it was referenced by contracts with a notional value in the hundreds of trillions of dollars: swaps, futures, floating-rate notes, syndicated loans, student loans, and adjustable-rate mortgages.

Its construction was simple and, as it turned out, fatally so. Each business day a panel of banks was asked a question: at what rate could you borrow funds, were you to do so by asking for and then accepting interbank offers in a reasonable market size? The highest and lowest quarter of the submissions were discarded and the rest averaged, and the result published as that day's fixing for each currency and tenor.

The question is hypothetical. It asks what a bank *could* borrow at, not what it *did* borrow at. During normal conditions the distinction was immaterial. Two developments made it fatal.

The submissions were manipulable, and were manipulated. Because a bank's own derivatives book was sensitive to the fixing, a submitter had a direct financial interest in the answer. Beginning in 2012, investigations across several jurisdictions established that traders at multiple panel banks had requested and obtained submissions favorable to their positions, and that some banks had understated their submissions during the financial crisis to avoid signaling funding difficulty. The resulting fines exceeded $9 billion and several individuals were prosecuted, though several of the convictions were later overturned on appeal (by the US Court of Appeals for the Second Circuit in 2022 and the UK Supreme Court in 2025).

The underlying market disappeared. More fundamentally, banks largely stopped funding themselves through term unsecured interbank borrowing after the crisis. By the late 2010s, some LIBOR tenors were supported by only a small number of transactions in a day. A benchmark that hundreds of trillions of dollars of contracts depend upon cannot rest on the expert judgment of panels of 11 to 16 banks per currency answering a hypothetical question about a market that no longer trades.

In July 2017 the UK Financial Conduct Authority (FCA) announced that it would not compel panel banks to make submissions beyond the end of 2021, which amounted to setting a date for LIBOR's death. Most settings ceased at the end of 2021. The remaining and most

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important ones (overnight, one-, three-, six-, and twelve-month US dollar LIBOR) ceased on June 30, 2023. A temporary "synthetic" version (no longer a bank submission but term SOFR plus the fixed spread adjustment described below) was published for the one-, three-, and six-month US dollar settings only, for use in legacy contracts only, until September 30, 2024; the sterling and yen synthetic settings had ended earlier.

Not every interbank offered rate was retired. The Euro Interbank Offered Rate (EURIBOR), reformed in 2019 to a hybrid methodology that anchors submissions in transactions where they exist, remains the principal floating index for euro loans and swaps, and the Tokyo Interbank Offered Rate (TIBOR), Australia's bank bill swap rate (BBSW), the Hong Kong Interbank Offered Rate (HIBOR), and the Shanghai Interbank Offered Rate (Shibor) all continue. The Canadian Dollar Offered Rate (CDOR) ceased at the end of June 2024 in favor of the Canadian Overnight Repo Rate Average (CORRA). Where such a rate survives, it sits beside an RFR rather than in place of one, and the two-curve arithmetic of Section 4.10 applies to it directly.

Fallbacks: how existing contracts moved

Announcing that a rate will stop is straightforward. Moving the existing stock of contracts is not. A thirty-year swap written in 2015 against three-month LIBOR needs an answer to the question of what it references in 2030.

The answer for derivatives was the **ISDA 2020 IBOR Fallbacks Supplement and Protocol** (IBOR being the generic name for LIBOR-type interbank offered rates), which took effect in January 2021 and which the great majority of the market adhered to. It specifies that on cessation, a contract referencing LIBOR instead references

$$\text{compounded RFR in arrears} \;+\; \text{a fixed spread adjustment}$$

The spread adjustment is necessary because LIBOR contained bank credit risk and the RFR does not; simply substituting one for the other would transfer value from one side of every contract to the other. The adjustment was set as the five-year historical median of the spread between LIBOR and the compounded RFR, and, crucially, was fixed permanently on March 5, 2021, the date the FCA announced the cessation timetable. Table 4.2 gives the US dollar values.

Table 4.2: ISDA fallback spread adjustments for US dollar LIBOR, fixed March 5, 2021.

LIBOR tenor	Spread adjustment (bp)
1 month	11.448
3 months	26.161
6 months	42.826
12 months	71.513

Fixing the spread on a single date was the decisive design choice. It made every legacy contract's future economics knowable in advance, which is what allowed the transition to happen without a wave of disputes. It also means those numbers are now permanent features of the legacy book: a three-month dollar LIBOR swap that fell back in 2023 will reference SOFR plus 26.161 basis points for the rest of its life, regardless of what bank credit spreads do. Example 4.1 follows one such swap through a single payment.

Example 4.1 A legacy swap after fallback

A \$50 million interest rate swap written in 2015 pays three-month USD LIBOR quarterly. Following cessation, its floating leg references compounded SOFR in arrears plus the fixed adjustment of 26.161 basis points.

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For an accrual period of 92 days over which compounded SOFR works out to 4.28%, the floating rate is

$$4.28\% + 0.26161\% = 4.54161\%$$

and, on an actual/360 basis, the interest amount is

$$\$50{,}000{,}000 \times 0.0454161 \times \frac{92}{360} = \$580{,}317$$

Had LIBOR survived and fixed at 4.55% for the same period, the payment would have been \$581,389, a difference of about \$1,072, roughly 0.2% of the coupon.

That difference is deliberate. The spread adjustment was set to match the historical median gap between the two rates over five years; it was never intended to reproduce LIBOR period by period. What fixing the adjustment secures is that its level was known in advance and beyond either party's influence: whatever value it transferred between the two sides was fixed in advance and therefore priceable, which was the condition needed for the transition to proceed without renegotiating millions of contracts.

The protocol covered derivatives, whose counterparties could adhere to it. Cash products (loans, floating-rate notes, securitizations) carry their own fallback language, and many older US contracts had none that worked once LIBOR stopped. The Adjustable Interest Rate (LIBOR) Act of March 2022 and the Federal Reserve's implementing Regulation ZZ moved those contracts by statute, replacing US dollar LIBOR with a SOFR-based rate plus the same ISDA spread adjustment; for most cash contracts the replacement rate is CME term SOFR, the forward-looking rate of Section 4.6.

4.4 COMPOUNDING AN OVERNIGHT RATE IN ARREARS

We now come to the mechanical heart of the chapter. A contract needs an interest amount for a period (a quarter, say). The reference rate supplies one number per business day. How do we get from one to the other?

The answer adopted almost universally is to **compound the daily rates over the interest period**. The economic logic is that of rolling an overnight deposit: lend for one day at today's rate, reinvest principal and interest tomorrow at tomorrow's rate, and continue to the end of the period. The compounded rate is whatever single rate would have produced the same terminal value.

Formally, for an interest period running from T_1 to T_2 and containing d_b business days and d_c calendar days, with day-count base B:

$$R = \left[\prod_{i=1}^{d_b} \left(1 + \frac{r_i \, n_i}{B} \right) - 1 \right] \times \frac{B}{d_c} \tag{4.1}$$

where r_i is the overnight rate published for business day i and n_i is the number of calendar days that rate applies for. On most days $n_i = 1$; on a Friday $n_i = 3$, because Friday's rate accrues across the weekend; around holidays it can be larger. The base depends on the currency: SOFR and €STR use actual/360, so $B = 360$, while SONIA and TONA use actual/365, so $B = 365$.

Two things about (4.1) deserve emphasis. First, the product is taken over *business* days, each rate weighted by the calendar days it covers, but the final scaling is by *calendar* days; this is what makes weekends accrue correctly. Second, and more consequentially:

> *R is not known until the last business day of the interest period.* A payment based on it cannot be calculated at the start of the period, only at the end. This is what "in arrears" means, and it is the single largest operational difference between the RFR world and the LIBOR world.

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Figure 4.1 contrasts the two arrangements. Under LIBOR a single fixing at T_1 determined the entire coupon, so both parties knew the amount for the whole period. Under an RFR the coupon accumulates one day at a time and is complete only at T_2—which is why every convention in Section 4.5 exists.

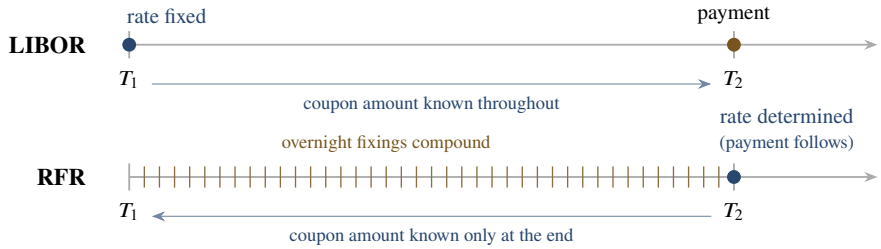


Figure 4.1: A rate set in advance against a rate determined in arrears.

Example 4.2 Compounding SOFR over one week

Consider a one-week interest period running from Monday the 6th to Monday the 13th: seven calendar days containing five business days, with Friday's rate applying for three days. SOFR accrues on actual/360, so $B = 360$.

Business day i	Date	r_i	n_i
1	Monday 6	4.31%	1
2	Tuesday 7	4.30%	1
3	Wednesday 8	4.32%	1
4	Thursday 9	4.33%	1
5	Friday 10	4.30%	3

The daily accrual factors are $1 + r_i n_i/360$:

$$1.000119722, \quad 1.000119444, \quad 1.000120000, \quad 1.000120278, \quad 1.000358333$$

Their product is 1.000838036, so from (4.1)

$$R = (1.000838036 - 1) \times \frac{360}{7} = 0.000838036 \times 51.428571 = 4.3099\%$$

For comparison, the simple day-weighted average of the same fixings is

$$\frac{4.31 + 4.30 + 4.32 + 4.33 + 3 \times 4.30}{7} = \frac{30.16}{7} = 4.3086\%$$

The compounded rate exceeds the simple average by 0.13 basis points over one week. The gap widens with the period: at a 4% rate level, compounding daily over a full year adds about 8 basis points relative to simple accrual. Over a quarter it is about 2 basis points: small, but not small enough to ignore in a market that quotes to a tenth of one. The gap grows roughly with the square of the rate level and in proportion to the term, so it is negligible when rates are near zero and becomes material quickly as they rise.

Published averages and indices

Because the arithmetic in Example 4.2 is tedious and error-prone, administrators publish it. The Federal Reserve Bank of New York publishes 30-, 90-, and 180-day compounded SOFR averages, and a **SOFR Index** that accumulates the daily compounding from a fixed start date. The index makes the compounded rate between any two dates a two-lookup calculation:

$$R = \left[\frac{\text{Index}(T_2)}{\text{Index}(T_1)} - 1\right] \times \frac{B}{d_c} \tag{4.2}$$

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where, as in (4.1), T_1 and T_2 are the start and end of the interest period, d_c is the number of calendar days between them, and $B = 360$ for SOFR. This is worth knowing not as a shortcut but because it eliminates a class of dispute: two parties reading the same two index values cannot disagree about the interest amount, whereas two parties each compounding sixty daily fixings can and do.

4.5 CONVENTIONS THAT MAKE COMPOUNDING WORKABLE

The problem with a rate known only on the final day of a period is that payment systems need notice. A borrower with a coupon due on the last day of a quarter cannot wait until the close of business that day to learn the amount; by then it is too late to arrange the payment.

The market's response was a family of conventions, each trading a small amount of economic precision for a usable notice period. A practitioner needs to recognize all of them, because a contract's value depends on which one it uses.

Payment delay. The simplest fix: compound over the full interest period, then pay a few business days after it ends. It is economically the cleanest, since the observation window matches the accrual window exactly. Its cost is a small timing adjustment, treated in Chapter 30.

Lookback. The rate for each day of the accrual period is taken from a fixed number of business days earlier, typically five. The final observation therefore occurs five days before period end, giving the notice required. In a **lookback without observation shift**, each observed rate is weighted by the calendar days of the accrual date it is applied to; in a **lookback with observation shift**, it is weighted by the calendar days of the observation date. The distinction sounds pedantic and is not. Without the shift, each rate is weighted by days that belong to a different date, a weighting mismatch: the compounded rate can no longer be reproduced from discount factors or from the SOFR Index, so it is not exactly the rate the curve prices. With the shift the observation period is a genuine compounding period, which is why the shifted convention is generally preferred.

Lockout (rate cutoff). The rate is observed normally until a cutoff a few days before period end, after which the last observed rate is frozen and repeated for the remaining days. It is simple to operate, at the cost of ignoring genuine rate movements in the final days, an awkward property if the cutoff spans a central bank meeting.

Last reset. This abandons in-arrears determination entirely, using the *previous* period's compounded rate as the current period's coupon. The borrower knows the coupon at the start of the period, as under LIBOR. The cost is that the rate is paid one period after it is determined, a timing adjustment priced with the machinery of Section 30.2 (Chapter 30 discusses this case at the end of Section 30.4).

Table 4.3 sets them side by side, writing k for the lookback, lockout, or delay length in business days, typically two to five. The column that matters operationally is the third: how much notice the payer gets. The column that matters for valuation is the fourth.

Figure 4.2 shows the same information as windows on a timeline, which makes the family resemblance clearer than the table can: each convention is the accrual period with its observation window slid, truncated, or displaced. Payment delay observes exactly the accrual period and defers only the payment. A lookback slides the whole window earlier. A lockout truncates it and repeats the last observed rate across the pale segment. Last reset abandons the current period altogether and takes the coupon from the previous one.

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Table 4.3: Conventions for compounding an overnight rate in arrears.

Convention	Observation window	Notice before payment	Principal cost
Payment delay	Matches the accrual period exactly	k days	Timing adjustment (Section 30.2)
Lookback, no observation shift	Shifted back k days; weights from accrual dates	k days	Weighting mismatch; rate not reproducible from discount factors or the index
Lookback with observation shift	Shifted back k days; weights from observation dates	k days	Minimal; observation period length may differ slightly from accrual
Lockout	Accrual period, but the final rate is frozen for the last k days	k days	Ignores rate moves late in the period, awkward across a policy meeting
Last reset	The *previous* accrual period	Full period	Coupon reflects a stale rate; timing adjustment (Section 30.2)

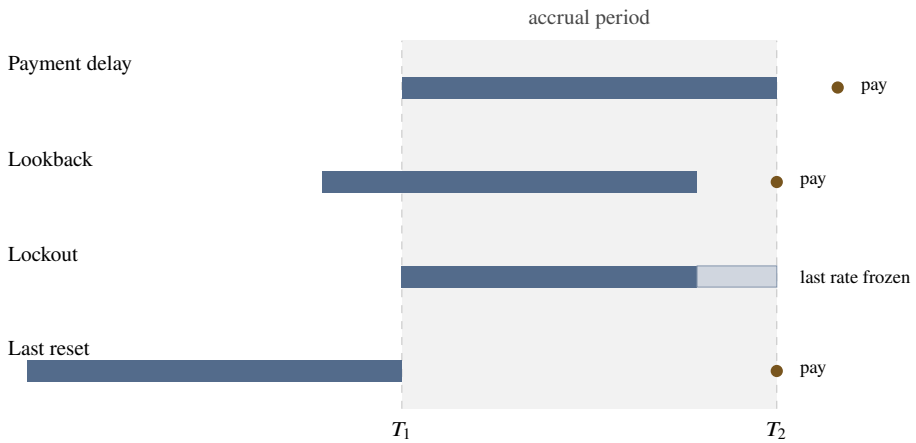


Figure 4.2: Observation windows (solid bars) against the accrual period (shaded).

The choice among these is not merely operational. Each convention implies a slightly different exposure, and a portfolio hedged on the assumption of one convention while transacted under another carries a small unhedged basis.

Cleared SOFR swaps use payment delay, usually two business days, and the interdealer market is uniform in this. Floating-rate notes use a lookback: five banking days with observation shift is the sterling standard for SONIA notes, while dollar notes have used both the shifted and unshifted forms, and the first agency SOFR notes of 2018 and 2019 paid a simple average of daily SOFR with a lockout before the market settled on compounding. Loans and securitizations, constrained by what a servicing system can compute, use lookbacks or fall back on last reset, or escape the question altogether by referencing the term rate of Section 4.6. The convention the Alternative Reference Rates Committee (ARRC), the market group that steered the dollar transition, recommended for SOFR loans is **daily simple SOFR**, which accrues each day's rate without compounding and so lets a borrower repay or draw down on any day without recomputing the period's interest.

Futures draw a distinction of their own. Three-month SOFR futures settle against daily SOFR *compounded* over the reference quarter, while one-month contracts settle against a

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simple arithmetic average over the calendar month. The underlying rate is the same and the accumulation is not, which is part of what the basis between the two contracts prices.

The division between derivatives and loans is not accidental. The derivatives market could rebuild its infrastructure once and centrally, through its clearing houses, and so adopted the convention that is cleanest economically. The loan market took whatever its existing servicing systems could support, fragmented across lookbacks and last reset, and in the end reached for a forward-looking term rate instead—which is where the next section begins.

4.6 TERM RATES AND CREDIT-SENSITIVE RATES

Two categories of rate sit alongside the compounded overnight rates, and both exist because some users genuinely wanted what LIBOR provided.

Term SOFR

A **term rate** is a forward-looking rate for a period, known at the start of that period (LIBOR's defining feature). Term SOFR is constructed not from submissions but from the SOFR derivatives market: CME publishes one-, three-, six-, and twelve-month term rates derived from SOFR futures prices, which embed the market's expectation of the average overnight rate over each period.

This is a sounder construction than LIBOR's, because it rests on traded prices. But it carries a structural vulnerability, which is circularity: term SOFR is built from derivatives that reference overnight SOFR, so it is only as robust as the volume of those derivatives. If exposures migrated from overnight SOFR to the term rate, with hedges following them, the market the term rate is computed from would thin while the exposures depending on it grew, which is the failure mode that ended LIBOR.

For that reason the ARRC recommended term SOFR for a deliberately restricted set of uses: business loans, and derivatives entered into to hedge those loans. It explicitly did *not* recommend it for the general derivatives market. The LIBOR Act added a third use by statute, as the replacement rate for legacy cash contracts (Section 4.3). The result is a two-tier structure, with loans referencing a forward-looking term rate and the derivatives market referencing compounded overnight rates, a basis that must itself be managed.

Credit-sensitive rates

A bank that funds itself unsecured and lends at a rate tied to SOFR has a genuine problem: in a crisis its funding costs rise while SOFR, being risk-free, does not. LIBOR's embedded credit component hedged that exposure automatically.

Several credit-sensitive benchmarks were launched to fill the gap, of which the most prominent was Bloomberg's Short-Term Bank Yield Index (BSBY). Regulators, including the SEC chair, questioned publicly whether it rested on sufficient transaction volume, and raised the concern that it reproduced LIBOR's central defect: a thin underlying market supporting enormous referencing volume, with the added feature that the thinness would be worst precisely during the stress the index was supposed to measure. Bloomberg announced its discontinuation in 2023 and ceased publication in November 2024. Smaller credit-sensitive indices such as Ameribor continue on a modest scale.

The episode is worth remembering when evaluating any proposed benchmark. The question is never only whether the index measures the right economic quantity. It is whether the market underneath it is deep enough to survive being referenced at scale, and whether it stays deep when conditions deteriorate.

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4.7 MEASURING INTEREST RATES

We now turn to the traditional apparatus, beginning with a point that causes more confusion than any other: a quoted rate is meaningless without its compounding frequency.

Suppose \$100 is invested for one year at 10%. With annual compounding the terminal value is \$110. With semiannual compounding it is $100 \times 1.05^2 = \$110.25$; with quarterly compounding $100 \times 1.025^4 = \$110.38$; with monthly, \$110.47. The same quoted number describes four different investments.

In general, an amount A invested for n years at rate R with compounding m times per year grows to

$$A\left(1 + \frac{R}{m}\right)^{mn} \tag{4.3}$$

Letting $m \to \infty$ gives **continuous compounding**, under which the terminal value is

$$A\, e^{Rn} \tag{4.4}$$

Continuous compounding is used throughout the option-pricing chapters of this book, because it makes the algebra dramatically simpler: discount factors multiply as $e^{-R_1 t_1} e^{-R_2 t_2} = e^{-(R_1 t_1 + R_2 t_2)}$, with no adjustment terms. It is a mathematical convenience, not a market convention—no instrument actually pays interest continuously.

Converting between conventions is essential and straightforward. Equating (4.3) and (4.4), a rate R_m with compounding frequency m corresponds to a continuously compounded rate

$$R_c = m \ln\left(1 + \frac{R_m}{m}\right) \tag{4.5}$$

and conversely

$$R_m = m\left(e^{R_c/m} - 1\right) \tag{4.6}$$

Example 4.3 applies each in turn.

Example 4.3 Converting compounding frequencies

Consider a rate of 4% per annum with semiannual compounding. From (4.5), the equivalent continuously compounded rate is

$$R_c = 2\ln(1.02) = 2 \times 0.019803 = 3.9605\%$$

Conversely, a rate of 4% with continuous compounding corresponds, from (4.6), to

$$R_2 = 2\left(e^{0.02} - 1\right) = 2 \times 0.020201 = 4.0403\%$$

with semiannual compounding. The continuously compounded rate is always the lower of the two, since more frequent compounding achieves a given terminal value with a smaller quoted rate.

Day count conventions. A rate quotation is also incomplete without a day count, which specifies how a partial year is measured. The interest for a period is the notional times the rate times a year fraction, and the day count sets that fraction:

$$\text{interest} = N \times R \times \frac{\text{days in the period}}{\text{days in the year}},$$

with both counts defined by the convention. Four are in common use:

- **Actual/360**: actual days over 360. US dollar and euro money-market instruments and floating legs, the overnight rates SOFR and €STR, the annual fixed legs of swaps on SOFR and €STR, and China's Shibor;

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- **Actual/365 (fixed)**: actual days over 365. Sterling (SONIA), the overnight rates TONA and CORRA, Japanese government bonds, and Chinese repo, swaps on the seven-day repo fixing FR007, and renminbi fixed legs;
- **Actual/actual**: actual days over the actual length of the year or coupon period. US Treasury coupons and most other government bonds;
- **30/360**: every month counts as 30 days and the year as 360. US corporate bonds, and the fixed legs of euro swaps and of the older US dollar LIBOR swaps.

The difference is not trivial: an actual/360 rate of 4% earns interest on 365 days of a normal year, giving an actual/365-equivalent of $4\% \times 365/360 = 4.0556\%$. Mismatched day counts across the legs of a trade are a persistent source of small, systematic pricing errors.

4.8 ZERO RATES, BOND PRICING, AND BOOTSTRAPPING

The n-year **zero rate** (or spot rate) is the rate earned on an investment made today that pays nothing until year n. It is the term structure's fundamental object: every other rate is built from it.

A coupon-bearing bond's price is the sum of its cash flows discounted at the zero rate appropriate to each:

$$B = \sum_{i=1}^{n} c_i \, e^{-R_i t_i} \tag{4.7}$$

where c_i is the cash flow at time t_i and R_i the continuously compounded zero rate for maturity t_i. The bond's **yield** is the single rate y that, used for all maturities in (4.7), reproduces the observed price. The **par yield** is the coupon rate that makes the bond price equal to par.

Bootstrapping the OIS curve

Zero rates are not directly observable beyond the shortest maturities; they must be extracted from traded instruments. In the current market the natural instruments are **overnight indexed swaps** (OIS), swaps exchanging a fixed rate for the compounded overnight rate, because these are the most liquid and are what collateralized derivatives discount at.

The extraction procedure, **bootstrapping**, works outward from the shortest maturity, using each newly determined discount factor to solve for the next.

Example 4.4 Bootstrapping a zero curve from OIS rates

Suppose annual-paying OIS rates are quoted as 4.00%, 4.20%, and 4.30% for one, two, and three years. Because a swap has zero value at inception, and its floating leg plus a final notional payment is worth par (Section 7.3 shows why), its fixed rate (the swap's **par rate**) is by construction a par yield: each quote says that a bond paying that coupon prices at par.

One year. A single payment of 1.04 at $t = 1$ must be worth 1, so

$$\mathrm{DF}_1 = \frac{1}{1.04} = 0.961538, \qquad z_1 = 4.0000\%$$

Two years. The two-year par bond pays 0.042 at $t = 1$ and 1.042 at $t = 2$:

$$0.042 \times 0.961538 + 1.042\,\mathrm{DF}_2 = 1 \implies \mathrm{DF}_2 = \frac{1 - 0.040385}{1.042} = 0.920936$$

so $(1 + z_2)^2 = 1/0.920936 = 1.085852$ and $z_2 = 4.2042\%$.

Three years.

$$0.043\,(0.961538 + 0.920936) + 1.043\,\mathrm{DF}_3 = 1 \implies \mathrm{DF}_3 = \frac{1 - 0.080946}{1.043} = 0.881164$$

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so $(1+z_3)^3 = 1.134863$ and $z_3 = 4.3072\%$.

At one year the zero and par rates coincide, since there is only one payment; beyond it each zero rate sits slightly above the corresponding par rate, which is the general pattern for an upward-sloping curve.

In practice the procedure is complicated by instruments that do not mature on convenient dates, by the need to interpolate between them, and by the requirement that the resulting curve be smooth enough that forward rates implied from it are not absurd. Curve construction is a specialty in its own right, and the choice of interpolation method is a genuine source of valuation differences between dealers.

4.9 FORWARD RATES AND FORWARD RATE AGREEMENTS

Forward rates are the future rates implied by today's zero curve. If investing for T_2 years must give the same result as investing for T_1 years and then reinvesting the proceeds at the forward rate until T_2, then the forward rate for the period $[T_1, T_2]$ satisfies

$$(1+z_1)^{T_1}(1+F_a)^{T_2-T_1} = (1+z_2)^{T_2}, \qquad 1+F\tau = \frac{\mathrm{DF}_1}{\mathrm{DF}_2} \tag{4.8}$$

where z_1 and z_2 are the annually compounded zero rates to T_1 and T_2, DF_1 and DF_2 the corresponding discount factors, and $\tau = T_2 - T_1$. The two forms state the same forward with different compounding: F_a is compounded annually, and F is a simple rate, the convention forward rate agreements use and the one meant below by "the forward rate." They coincide only for a one-year period, as in the applications below; otherwise $1+F_a = (1+F\tau)^{1/\tau}$. In continuously compounded terms the relationship (4.9) is cleaner still:

$$e^{R_1 T_1} \times e^{F(T_2-T_1)} = e^{R_2 T_2}, \qquad \text{so} \qquad F = \frac{R_2 T_2 - R_1 T_1}{T_2 - T_1} \tag{4.9}$$

where R_1 and R_2 are the continuously compounded zero rates and F here denotes the continuously compounded forward.

Applying (4.8) to Example 4.4 gives the one-year forward rate starting in one year, and the one-year rate starting in two years:

$$\frac{0.961538}{0.920936} - 1 = 4.4088\%, \qquad \frac{0.920936}{0.881164} - 1 = 4.5136\%.$$

Note how much more steeply the forward curve rises than the zero curve, as Figure 4.3 shows: zero rates are averages of forwards, so a modestly rising zero curve implies a considerably steeper forward curve. This amplification is a recurring feature and a frequent source of surprise.

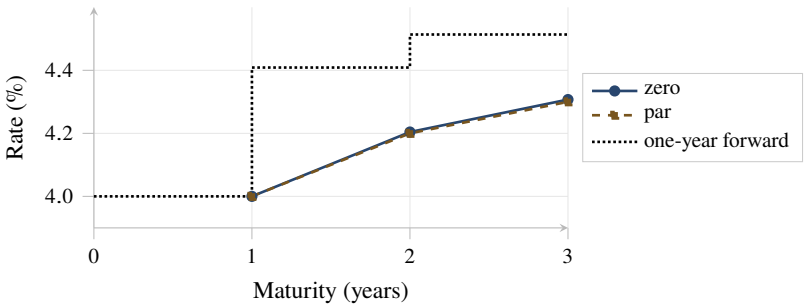


Figure 4.3: Par, zero and one-year forward rates from Example 4.4.

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Forward rate agreements. A forward rate agreement (FRA) is an agreement to exchange a fixed rate for a floating rate on a notional principal for a future period. Its value follows from a single observation: an FRA is worth zero when the fixed rate equals the forward rate. More generally, an FRA in which the fixed rate R_K is received in exchange for the realized rate, with notional L over accrual τ, is worth

$$V = L\,\tau\,(R_K - F)\,\mathrm{DF}_2 \tag{4.10}$$

to the party receiving fixed, where F is the current forward rate. Equation (4.10) is the simplest instance of the general principle developed in Chapter 30: we value the contract by substituting the forward rate for the unknown future rate, an operation whose validity is not automatic and whose exceptions occupy that chapter.

An FRA traditionally settles at the start of the period, T_1, as soon as the rate has fixed, rather than at T_2 when the interest would be due. The payment is therefore the period's interest difference discounted back over the period at the realized rate R: the party paying fixed receives $L\,\tau\,(R - R_K)/(1 + R\tau)$ at T_1, a negative amount meaning it pays.

FRAs were a major product in the LIBOR era, when a single fixing determined the settlement, and they still trade where a term rate survives: banks and corporate borrowers use EURIBOR FRAs to hedge the next few resets on euro loans. In the RFR currencies their role has narrowed considerably, since a short OIS achieves the same exposure with conventions that match the rest of the market. The short OIS market is where dealers, banks, and funds position for central-bank decisions, often through meeting-dated OIS whose accrual periods run from one policy meeting to the next.

4.10 THE MULTI-CURVE REALITY

One assumption made implicitly above needs flagging, because Chapter 7 dismantles it.

We have spoken of "the" zero curve, as though a single term structure describes a currency. Before 2008 that was approximately true. It is not true now, for a simple reason: discounting a collateralized derivative correctly requires the rate paid on the collateral, which under a standard credit support annex is the overnight rate. Meanwhile the rate a contract's cash flows are *indexed* to may be something else: EURIBOR (the largest surviving interbank rate), term SOFR, a foreign currency's RFR, or a legacy fallback rate.

The consequence is that valuation requires at least two curves: a discount curve built from OIS, and one or more forecast curves for the indices the contract references. Their difference is not noise. Cross-currency basis spreads (the additional spread required to swap one currency's RFR against another's) have at times exceeded 50 basis points and are themselves traded.

For the present chapter the single-curve treatment is adequate and keeps the mechanics visible. Chapter 7 develops the multi-curve framework properly, and treats it as the standard case rather than as a correction to a single-curve baseline.

4.11 DURATION AND CONVEXITY

We close with the two standard measures of a bond's interest rate sensitivity, both of which recur throughout the book. Convexity in particular is the subject of Chapter 30.

Duration measures the sensitivity of a bond's price to a parallel shift in the zero curve. With continuous compounding, if the bond has price B and cash flows c_i at times t_i, its duration is the cash-flow-weighted average time to payment:

$$D = \frac{\sum_{i=1}^{n} t_i\, c_i e^{-y t_i}}{B} \tag{4.11}$$

from which

$$\Delta B \approx -B D\,\Delta y \tag{4.12}$$

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When rates are quoted with compounding frequency m rather than continuously, the corresponding sensitivity uses **modified duration**, $D^* = D/(1 + y/m)$.

Equation (4.12) is a first-order approximation and inherits the usual limitation: it is a straight line fitted to a curve. The curvature is **convexity**,

$$C = \frac{1}{B}\frac{\partial^2 B}{\partial y^2} \tag{4.13}$$

giving the improved approximation

$$\Delta B \approx -BD\,\Delta y + \frac{1}{2}B\,C\,(\Delta y)^2 \tag{4.14}$$

Since $C > 0$ for an ordinary bond, the convexity term is positive for a rate move in either direction: a bondholder gains more when rates fall than they lose when rates rise by the same amount. Convexity is therefore desirable, and, as Chapter 30 shows at length, it is priced.

Figure 4.4 shows why. The duration approximation is the tangent to the price–yield curve at the current yield, and because that curve is convex the tangent lies *below* it on both sides. Duration alone therefore understates the price whichever way rates move; the shaded region between the two lines is what the convexity term in (4.14) recovers.

Two features of the figure repay attention. The bond is the one of Example 4.5. Plotting the percentage change rather than the price level makes the tangent's slope exactly minus the modified duration (here −2.7429 per percentage point of yield), independent of the bond's price, so the picture illustrates the definition rather than one particular bond. And the yield range shown is deliberately far wider than anything a three-year bond would realistically see: over a plausible range the two lines are almost indistinguishable, which is precisely why duration alone suffices for small moves and fails for large ones.

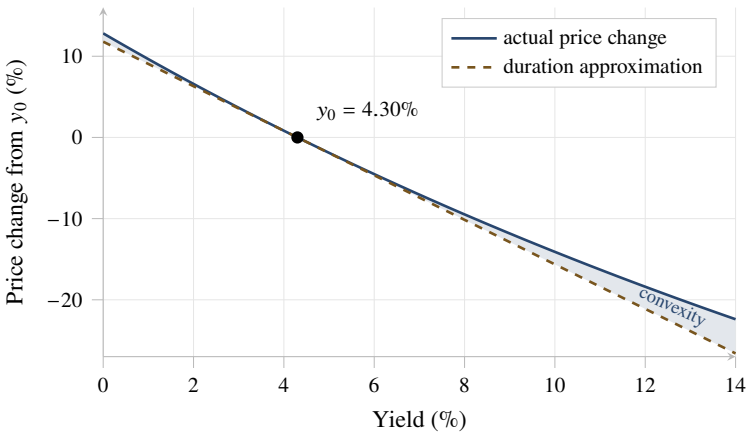


Figure 4.4: Percentage price change against yield for the three-year 5% bond of Example 4.5.

Example 4.5 Duration and convexity of a three-year bond

A three-year bond pays a 5% annual coupon and yields 4.30% with annual compounding. Discounting the cash flows of 5, 5, and 105 gives

$$B = 4.79386 + 4.59623 + 92.54147 = 101.93156$$

The duration from the discrete analog of (4.11), with each cash flow discounted by $(1 + y)^{t_i}$ in

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place of $e^{y t_i}$, is

$$D = \frac{1 \times 4.79386 + 2 \times 4.59623 + 3 \times 92.54147}{101.93156} = \frac{291.61071}{101.93156} = 2.8608 \text{ years}$$

and the modified duration is $D^* = 2.8608/1.043 = 2.7429$. The convexity from (4.13) works out to $C = 10.3499$.

Now suppose the yield rises 50 basis points to 4.80%. The duration-only estimate is

$$\Delta B \approx -101.93156 \times 2.7429 \times 0.005 = -1.39794 \implies B \approx 100.5336$$

Adding the convexity term from (4.14),

$$\frac{1}{2} \times 101.93156 \times 10.3499 \times 0.005^2 = +0.0132 \implies B \approx 100.5468$$

Repricing the bond exactly at 4.80% gives 100.5467. The duration estimate is off by 1.3 cents per 100 of face; adding convexity reduces the error to a hundredth of a cent. For a 50 basis point move on a three-year bond the refinement is marginal—but the convexity term grows with the square of the rate move and with the square of maturity, and on a thirty-year bond in a volatile market it is indispensable.

4.12 THEORIES OF THE TERM STRUCTURE

Why is the zero curve shaped the way it is? Three explanations are traditionally offered.

Expectations theory. Forward rates equal expected future spot rates. An upward-sloping curve then means the market expects rates to rise. This is clean, and Chapter 30 explains precisely why it cannot be exactly right: forward rates are expectations only under a particular probability measure, and the adjustment between that measure and the real world is not zero.

Market segmentation theory. Short, medium, and long maturities are distinct markets with distinct participants (money funds at the short end, pension funds and insurers at the long end), and rates in each are set by local supply and demand. The theory explains persistent curve anomalies that expectations alone cannot, and it received strong support from the 2022 UK gilt episode, in which forced selling by liability-driven investors moved long gilt yields far more than the change in rate expectations.

Liquidity preference theory. Borrowers prefer to lock in long-term funding while lenders prefer to keep money short and available. Clearing that mismatch requires paying lenders a premium to extend, so forward rates exceed expected future spot rates and the curve slopes upward more often than not. This is the explanation most consistent with the historical record: yield curves have been upward-sloping most of the time, which the pure expectations theory would require to mean the market has persistently expected rising rates.

None of the three is complete, and modern practice treats the curve as an observable to be fitted rather than a prediction to be explained. But the intuitions matter when curves behave strangely, and knowing which force is operating is often the difference between recognizing an opportunity and taking a position the market is about to move against.

SUMMARY

The reference rate at the center of the derivatives market has changed, and with it the conventions that everything else is built on. LIBOR was a forward-looking term rate, fixed at the start of each interest period, derived from banks' estimates of their own unsecured funding costs, and therefore carrying an embedded bank credit component. It was manipulated, and the market

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underneath it had largely ceased to exist. It was replaced by overnight rates (SOFR, SONIA, €STR, TONA, SARON), each computed from actual transactions and each close to free of credit risk.

The defining consequence is that a term rate must now be built by compounding daily fixings across an interest period, which means the rate is not known until the period ends. The compounding formula weights each business day's rate by the calendar days it covers and scales the result to the full period, on a day-count base that differs by currency: actual/360 for SOFR and €STR, actual/365 for SONIA and TONA. Because in-arrears determination leaves no time to settle payments, the market developed conventions (payment delay, lookbacks with and without observation shift, lockouts, last-reset structures) that each trade a little economic precision for operational notice, and that each imply slightly different exposures.

Legacy contracts moved to the new rates through the ISDA fallback protocol, which substitutes the compounded RFR plus a spread adjustment fixed permanently in March 2021. Forward-looking term rates exist, built from SOFR futures, but their recommended use is deliberately restricted to loans and their hedges, to avoid recreating LIBOR's vulnerability. Credit-sensitive benchmarks intended to restore banks' funding hedge have largely failed for the same reason, BSBY's discontinuation being the clearest case.

The traditional apparatus survives the transition intact. Rates must always be quoted with a compounding frequency and day count; continuous compounding simplifies the algebra of later chapters. Zero curves are bootstrapped from traded instruments, now principally OIS, and forward rates are implied from them, rising more steeply than the zero curve itself. Duration measures first-order rate sensitivity and convexity the curvature, the latter being both a desirable property of a bond position and, as Chapter 30 develops, a quantity the market prices explicitly.

One simplification made here is corrected in Chapter 7: there is no longer a single curve per currency. Discounting a collateralized trade uses the rate paid on its collateral, while its cash flows may be indexed to something else, so valuation requires a discount curve and one or more forecast curves.

FURTHER READING

Schrimpf and Sushko's "Beyond LIBOR: A Primer on the New Reference Rates" in the March 2019 *BIS Quarterly Review* is a short account of why the benchmarks changed and what replaced them. The Financial Stability Board's progress reports on *Reforming Major Interest Rate Benchmarks* (2014–2023) record the transition as it happened rather than in retrospect. For the mechanics, the Alternative Reference Rates Committee's fallback-language recommendations and its guidance on the scope of use for term SOFR are the operative documents, and Henrard's 2019 note treats the fallback spread methodology in the detail a practitioner needs.

PRACTICE QUESTIONS

4.1 Explain the two distinct defects in LIBOR's construction that led to its replacement, and state which of the two would have justified replacement even if no manipulation had ever occurred.

4.2 An investment pays 12% per annum with quarterly compounding. Give the equivalent continuously compounded and annually compounded rates, and give the semiannual equivalent of a rate quoted as 5% continuously compounded.

4.3 Explain why SOFR can be described as risk-free even though it rose from 2.43% to 5.25% in a single day in September 2019, and state what kind of risk that episode illustrates.

4.4 An interest period runs Monday to Monday, seven calendar days. The overnight rates on Monday through Friday are 3.95%, 3.96%, 3.94%, 3.97%, and 3.96%. Using (4.1),

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calculate the compounded rate on an actual/360 basis and compare it with the day-weighted simple average.

4.5 Explain the difference between a lookback with an observation shift and one without, and why the market generally prefers the shifted convention.

4.6 Explain why it mattered that the ISDA fallback spread adjustments were fixed permanently on a single date rather than recalculated periodically.

4.7 Annual-paying OIS rates are 3.50%, 3.70%, and 3.85% for one, two, and three years. Bootstrap the zero rates and discount factors, calculate the one-year forward rates beginning in one and two years, and explain why an upward-sloping zero curve implies a forward curve that rises more steeply.

4.8 A five-year bond pays a 6% annual coupon and yields 5%. Calculate its price, duration, and modified duration, estimate the price change for a 25 basis point increase in yield, and compare with the exact repricing.

4.9 A forward rate agreement fixes a rate on a notional for a future period. Explain how the forward rate is extracted from two zero rates, why the contract settles at the start of the period rather than at its end, and what that settlement convention requires of the discounting.

4.10 Explain why a single discount curve was adequate before 2008 and is not now. State which curve discounts a collateralized swap and which forecasts its floating payments, and explain why the two being different is a statement about collateral rather than about the credit of the counterparties.

4.11 State the theories of the term structure the chapter sets out, and give what each predicts about the shape of the curve when short rates are expected to stay flat. Explain which prediction the historical record supports, and what that implies for reading a forward rate as a forecast.

FURTHER QUESTIONS

4.12 A bank makes floating-rate loans indexed to term SOFR and funds itself with unsecured deposits. Describe the exposure this creates, explain why LIBOR did not create it, and discuss why credit-sensitive benchmarks intended to solve the problem have largely failed.

4.13 Using the SOFR Index relationship (4.2), show that the compounded rate between any two dates can be obtained from two index values. Explain why this eliminates a class of settlement disputes, and identify what could still be disputed.

4.14 Term SOFR is derived from SOFR futures rather than from submissions. Explain why this makes it more robust than LIBOR, and why regulators nonetheless restricted its recommended scope of use. What would have to be true of the futures market for the restriction to be unnecessary?

4.15 A thirty-year swap written in 2015 referenced three-month USD LIBOR. Describe precisely what it references today, and explain how the value of the contract was preserved through the transition. Under what circumstances would a party have been economically disadvantaged by the fallback?

4.16 Consider a floating-rate note paying compounded SOFR in arrears with a five-day payment delay, and an otherwise identical note using a five-day lookback without observation shift. Explain qualitatively how their values differ, which adjustment from Chapter 30 applies to the first, and why the second cannot be valued exactly from the discount curve.

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Chapter 5
Determination of Forward and Futures Prices

Chapter 2 established that the futures price converges to the spot price at delivery. This chapter answers the question that leaves open: what determines it *before* delivery?

The answer, for a large class of assets, is remarkably tight. The forward price is not a forecast, not a consensus view, and not the result of supply and demand for the contract. It is pinned by arbitrage to a specific relationship with the spot price, and a trader who quotes anything else can be taken apart by someone with a bank account and no opinion whatsoever about the future direction of the underlying price.

That is the good news. The rest of the chapter is about where the argument weakens: assets that cannot be sold short, assets held for use rather than investment, and the fact that the arbitrage requires a balance sheet, which is not free, a theme that recurs through Chapters 6, 7, and 8.

What arbitrage means. An **arbitrage opportunity** is a trading strategy that costs nothing to set up, can never lose money, and makes money in at least one state of the world; equivalently, it pays money today and can never lose thereafter. The argument of this chapter, and the pricing in every chapter after it, rests on the **no-arbitrage assumption**: such opportunities cannot persist, because anyone who found one would trade it in unlimited size and prices would move until it was gone. It is an assumption about prices, not about people. It does not say that arbitrages never appear, only that the market does not leave them lying about, and Section 5.15 examines where it fails in practice.

Section 5.1 draws the distinction the chapter turns on, between assets held for investment and assets held for use, and sets out what short selling requires. Sections 5.2 to 5.5 are the arbitrage argument, its extensions to known income and known yield, and the value of a forward already on the books. Sections 5.6 to 5.8 apply it to the three markets where it is tested daily (index futures, currencies, and commodities), and Sections 5.9 to 5.11 collect the pieces into the cost of carry, the implied repo rate, and the roll yield of a position held across contracts. Sections 5.12 and 5.13 settle two questions that are often confused: whether forward and futures prices differ, and whether either forecasts the spot price. Section 5.14 shows how the shape of the curve is traded, and Section 5.15 is where the arbitrage argument stops.

5.1 INVESTMENT ASSETS, CONSUMPTION ASSETS, AND SHORT SELLING

The arbitrage arguments that follow depend on a distinction.

An **investment asset** is held by a significant number of investors purely for investment (gold, silver, stocks, bonds, currencies). It need not be held *only* for investment; gold is also used industrially. What matters is that enough holders are indifferent between holding the physical asset and holding a claim on it, so that arbitrage can proceed by borrowing the asset from them.

A **consumption asset** is held primarily to be used: oil, copper, wheat, live cattle. No one holds crude oil as a store of value. Its holders are precisely the negation of the condition above: they are *not* indifferent between the physical asset and a claim on it, because what they want from it is the use. A refinery needs barrels in the tank to keep the plant running, and a claim on barrels next month will not run it.

Two consequences follow, and the second matters more than the first. Such a holder will not lend the asset, so the short sale that the arbitrage of Section 5.2 depends on is simply unavailable: you cannot readily borrow oil from a refinery that intends to process it next week. But even a holder who owns the asset outright, and needs to borrow nothing from anyone, will not *sell it now and buy it back later*, however high today's price, because the whole point of holding it is to have it in the meantime. The trade that would keep a forward price from falling too far below what it costs to hold the asset until delivery is therefore not merely hard to finance: no one who has the asset wants to do it. That is what turns this chapter's equalities into one-sided inequalities for commodities, and it is the economic content of the convenience yield in Section 5.8.

Short selling. To sell short, an investor borrows the asset from a holder, sells it, and later buys it back to return. The mechanics matter for pricing. The short seller must pass through any income the asset generates (dividends, coupons) to the lender, and pays a **borrowing fee** for the loan. For liquid stocks that fee is a few basis points; for hard-to-borrow stocks it can be tens of percent, and the position may be recalled at any time.

Every result in this chapter assumes short selling is costless where it is possible. Where it is not possible at all, the arbitrage works in only one direction, and the results become inequalities rather than equalities, as Section 5.8 shows.

Notation and assumptions. Throughout, S_0 is today's spot price, S_T the spot price at delivery, F_0 the forward or futures price, T the time to delivery, and r the continuously compounded risk-free rate for maturity T (Section 4.7). The standing assumptions are that participants face no transaction costs, are subject to the same tax rate, can borrow and lend at r, and take arbitrage opportunities when they arise.

The third of these deserves skepticism from the outset: no one borrows at the risk-free rate. We keep the assumption because it makes the structure clear, and return to what it costs in Section 5.15.

5.2 THE BASIC ARBITRAGE

Consider an investment asset providing no income and costing nothing to store (a non-dividend-paying stock is the standard example). Then

$$\boxed{F_0 = S_0 e^{rT}} \tag{5.1}$$

The argument is worth setting out in both directions, because the pair of trades is the template for everything that follows. Each direction is an arbitrage opportunity in the sense defined at the start of the chapter: it costs nothing, cannot lose, and makes money if the forward price is out of line.

If $F_0 > S_0 e^{rT}$, borrow S_0 at rate r, buy the asset, and short a forward contract. At T, deliver the asset for F_0 and repay $S_0 e^{rT}$. The profit is $F_0 - S_0 e^{rT} > 0$, with no investment and no risk. This is the **cash-and-carry** trade: borrow cash, buy the asset, carry it to delivery. It is the trade the rest of the book refers to by that name.

If $F_0 < S_0 e^{rT}$, short the asset for S_0, invest the proceeds at r, and take a long forward. At T, receive $S_0 e^{rT}$ from the investment, pay F_0 for the asset under the forward, and return it to

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the lender. The profit is $S_0 e^{rT} - F_0 > 0$. This is the **reverse cash-and-carry**, and Section 5.1's question of who will part with the asset is the question of who can do this one.

Since both cannot be available, equality must hold.

Example 5.1 A forward on a non-dividend-paying stock

A stock trades at \$50, the six-month risk-free rate is 4%, and the stock pays no dividend. From (5.1) the six-month forward price is

$$F_0 = 50\, e^{0.04\times 0.5} = 50 \times 1.020201 = \$51.01$$

Suppose a dealer quotes \$52. Borrow \$50, buy the stock, short the forward. Six months later, deliver the stock for \$52 and repay \$51.01, keeping \$0.99 per share, with no capital committed and no exposure to where the stock went.

Suppose instead the quote is \$50. Short the stock, invest the \$50 to receive \$51.01, and buy the stock back at \$50 under the forward. Profit \$1.01.

No view on the stock is required, and no probability distribution enters. The forward price is determined by the spot price and the interest rate alone, which is why Chapter 2's remark that a futures price is not a forecast is more than a caution.

The constant rate is a convenience, not an assumption of the argument. Financing the purchase by selling zero-coupon bonds maturing at T, rather than by borrowing at r, the same two trades show that

$$F_0 = \frac{S_0}{P(0,T)} \tag{5.2}$$

where $P(0,T)$ is today's price of a zero-coupon bond paying \$1 at T. This holds whatever rates do in between; only the bond's traded price enters, not a model for it. A constant rate makes $P(0,T) = e^{-rT}$, and (5.2) is (5.1) again. Section 28.7 starts from (5.2), in an economy where rates are stochastic by assumption.

When short selling is costly

Example 5.1 used both arbitrages, and the second borrows the stock. It is natural to expect that an expensive borrow degrades the equality into a band, and eventually it does, but not for the reason it first appears, and not as readily.

The lower bound does not need a short sale at all. If $F_0 < S_0 e^{rT}$, an investor who *already owns* the asset can sell it, invest the proceeds at r, and take a long forward. At T that investor has the asset back, having received $S_0 e^{rT}$ and paid F_0, and is better off than by holding throughout. Nothing has been borrowed and no fee has been paid. This is exactly why an investment asset was defined in Section 5.1 by holders who are indifferent between the asset and a claim on it: their willingness to exchange one for the other is what defends the lower bound, and the cost of borrowing is nothing to them.

A borrowing fee therefore does not by itself widen the no-arbitrage range. What widens it is holders who will not or cannot make that exchange: an index fund mandated to hold the constituents, an insider under lockup, a taxable holder who would realize a gain by selling, or a holder already lending the shares and collecting the fee, for whom selling means surrendering that income. Where the sale route is closed and only borrowed stock is available, write b for the continuously compounded borrowing fee. The cash-and-carry trade is unaffected, so the upper bound stands; the reverse trade surrenders b to the lender, and

$$S_0\, e^{(r-b)T} \;\le\; F_0 \;\le\; S_0\, e^{rT} \tag{5.3}$$

It is worth being clear about which case is which. If the fee accrues to the *marginal* holder (the one whose readiness to sell would close the gap), the asset is an investment asset paying

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income at rate b, and Section 5.4 gives the equality $F_0 = S_0 e^{(r-b)T}$, not a band. Equation (5.3) is the intermediate case: some holders earn the fee, but the holder who would otherwise arbitrage cannot act. Example 5.2 puts numbers on the band.

Example 5.2 A hard-to-borrow stock

A stock trades at \$50 with a six-month rate of 4%. For an easy-to-borrow stock, Example 5.1 pins the forward at \$51.01.

Now suppose the stock is heavily shorted and the borrowing fee is 3% a year. From (5.3) the bounds are

$$50\, e^{(0.04-0.03)\times 0.5} = \$50.25 \qquad \text{and} \qquad 50\, e^{0.04\times 0.5} = \$51.01$$

Any forward price between \$50.25 and \$51.01 is free of arbitrage: a band 76 cents wide where previously there was a point. A forward quoted at \$50.40 is not mispriced to anyone who must borrow the stock to act on it.

The band is reached only if no holder will sell: a shareholder who owns the stock outright still closes the gap at \$50.40 by selling at \$50, investing to receive \$51.01, and buying the forward. Heavily shorted stocks tend to be markets where few holders will, which is why the two conditions arrive together: the borrow is expensive *because* the free float is already lent, and what remains is held by people who will not sell. So on a stock where fees exceed 20% a year, a persistently "cheap" forward is offering a profit only to whoever already owns the stock and will part with it, and the price is telling you how few of those there are.

5.3 KNOWN INCOME

If the asset provides a known cash income (coupons on a bond, a declared dividend), the holder of the physical asset receives it and the holder of a forward does not. The forward price must be lower by the present value of that income.

Writing I for the present value of the income received during the contract's life,

$$F_0 = (S_0 - I)\, e^{rT} \tag{5.4}$$

The arbitrage is the same as before, with one adjustment: the trader who buys the asset and shorts the forward receives the income along the way and can invest it. Example 5.3 applies it to a coupon-bearing bond.

Example 5.3 A forward on a coupon-bearing bond

A bond trades at \$900. It pays a \$40 coupon in four months, and a nine-month forward contract is being priced. The risk-free rate is 4% for all maturities.

The present value of the coupon is

$$I = 40\, e^{-0.04\times 4/12} = 40 \times 0.986755 = \$39.47$$

so from (5.4)

$$F_0 = (900 - 39.47)\, e^{0.04\times 0.75} = 860.53 \times 1.030455 = \$886.74$$

The forward price is below the spot price, which looks odd until one recalls that the forward buyer forgoes a \$40 coupon to save nine months of financing on \$900—and here the coupon is worth more.

5.4 KNOWN YIELD

Sometimes income is better expressed as a proportion of the asset's price than as a fixed amount: a stock index paying a dividend yield, or a foreign currency earning interest. Writing q for the

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continuously compounded yield,

$$F_0 = S_0\, e^{(r-q)T} \tag{5.5}$$

The intuition is direct: holding the asset earns q and costs r to finance, so the forward price grows at the net rate $r - q$. If the yield exceeds the financing rate, the forward price is *below* spot.

Example 5.4 A stock index futures price

An equity index stands at 4,200. The dividend yield on the index is 1.8% and the three-month risk-free rate is 4.3%. From (5.5),

$$F_0 = 4200\, e^{(0.043-0.018)\times 0.25} = 4200 \times 1.006270 = 4{,}226.3$$

Index arbitrage. When the futures price departs from (5.5), the trade is to buy the index basket and sell futures, or the reverse. It is executed by computer, since it requires simultaneous transactions in hundreds of stocks, and the profit per unit is small. It is the staple of index market makers and of bank *delta-one* desks (the desks that run exposures moving one-for-one with the underlying, as against the option books of Part II), for whom a few basis points is a business only because the volume is enormous. Example 5.5 shows how little of a quoted gap survives the costs.

This is an institutional trade, and only an institutional one. Simultaneity across hundreds of stocks requires automated basket execution and direct market access; the short direction requires standing stock-borrow relationships; and the financing has to be arranged at institutional rates. Equation (5.5) is enforced by the algorithmic desks of dealers, index funds, and quantitative managers. Read it as a description of how the price comes to be disciplined, not as an opportunity available on a retail platform.

The position is then held to settlement rather than traded out of. Index futures settle in cash against a published index level, so the basket must be liquidated *at that level*, which is why expiration days carry a concentration of orders to trade at the closing price and why volume in the constituents spikes at the moment the settlement level is set. A basket sold a few minutes early has stopped being a hedge and become a directional position in five hundred stocks.

The two directions are not equally available. Buying the basket against short futures needs cash and a broker. Selling the basket against long futures needs a borrow in every constituent: the constraint of Section 5.1 reappearing in an equity market, as borrowing arrangements for hundreds of stocks, at fees that differ from stock to stock, on loans that can be recalled without notice. The no-arbitrage band is therefore wider below theoretical value than above it: a futures price that is too cheap can persist, where one equally rich is arbitraged away more readily. When index futures nonetheless sit slightly rich (above theoretical value), the cause is not this asymmetry but the demand for exposure in futures form and the cost of the balance sheet needed to meet it, which Section 5.6 takes up.

Example 5.5 Index arbitrage with costs

The futures in Example 5.4 are quoted at 4,240, against a theoretical 4,226.3, a gap of 13.7 index points. The trade is to buy the basket and short the futures.

Round-trip transaction costs (commissions, bid–ask spreads, and market impact across the basket) come to perhaps 8 index points. The net is 5.7 points, and on a contract sized at \$50 per index point,

$$5.7 \times \$50 = \$285 \text{ per contract}$$

Two lessons follow. The arbitrage does not enforce equality but a *band*: no-arbitrage requires

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only that the futures price stay within transaction costs of the theoretical value. And the width of that band depends on execution capability, so the same mispricing is an opportunity for one participant and not for another.

When index arbitrage becomes impossible, as in a market where the underlying stocks cannot be traded or where trading halts break the simultaneity the trade requires, the band widens without limit, and futures and cash can decouple entirely. This happened in October 1987, and it is the reason the relationship is a mechanism rather than a law.

5.5 VALUING AN EXISTING FORWARD CONTRACT

A forward contract is worth zero when entered. Later it is not, because the forward price has moved while the contract's delivery price K has not.

For a long forward position with delivery price K and current forward price F_0 for the same maturity, the value today, f, is

$$\boxed{f = (F_0 - K)\, e^{-rT}} \tag{5.6}$$

Derivation. Compare two portfolios. Portfolio A is the existing long forward at delivery price K; portfolio B is a newly entered long forward at the current forward price F_0, which by definition is worth zero.

At T both portfolios deliver the same asset, worth S_T. Portfolio A pays K for it and portfolio B pays F_0, so A's terminal value exceeds B's by exactly $F_0 - K$, whatever S_T turns out to be. Since this difference is certain, it can be discounted at the risk-free rate:

$$f_A - f_B = (F_0 - K)e^{-rT}$$

and since $f_B = 0$, equation (5.6) follows.

The step to notice is that the asset price cancels. The value of an existing forward does not depend on any model of the underlying, only on how far today's forward price has moved from the contract's own delivery price. This is why forward and swap books can be revalued from a curve alone. Example 5.6 values one such contract.

Example 5.6 Valuing a forward after the market moves

An investor is long a forward on a non-dividend-paying stock with a delivery price of \$48 and three months remaining. The stock now trades at \$52 and the risk-free rate is 4%.

The current three-month forward price is

$$F_0 = 52\, e^{0.04\times 0.25} = \$52.52$$

so from (5.6)

$$f = (52.52 - 48)\, e^{-0.04\times 0.25} = 4.52 \times 0.990050 = \$4.48$$

Equation (5.6) is the same structure as the swap valuation of Chapter 7: an off-market contract is worth the difference between its own rate and the current market rate, discounted. Chapter 7's $L(S - K)A$ is the annuitized version of the same idea.

5.6 STOCK INDEX FUTURES IN PRACTICE

Equation (5.5) treats the dividend yield as a known constant. For an index over a few months this is a reasonable approximation, since the constituent companies' dividends are largely announced in advance, but three practical points are worth making.

Dividends are lumpy, not continuous. The yield q is an approximation to a sequence of discrete payments concentrated in particular weeks. For contracts spanning a heavy dividend

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period the approximation matters, and desks price index futures off the actual projected dividend stream rather than a smoothed yield, returning in effect to equation (5.4).

Dividends are forecasts. Unlike a bond coupon, a dividend can be cut. In 2020 many companies suspended dividends within weeks, and index futures prices moved for that reason alone. The dividend assumption embedded in a futures price is a traded quantity in its own right: dividend futures and swaps exist precisely so it can be hedged.

Index futures are the cheapest way to hold the index. Because the contract requires only margin, a fund can obtain index exposure without buying the constituents. This is the same observation Chapter 6 makes about Treasury futures and duration, and it has the same consequence: steady demand for exposure in futures form tends to leave futures slightly rich to fair value. That richness is not an unclaimed riskless profit; it is what an arbitrageur is paid for committing the cash and the balance sheet to close the gap, which is why it survives (Section 5.15). The demand, not the borrow asymmetry of Section 5.4, decides which side the gap appears on: the asymmetry on its own would let a cheap futures price last longer than a rich one.

5.7 CURRENCY FORWARDS AND COVERED INTEREST PARITY

A foreign currency is an investment asset paying a known yield, the foreign risk-free rate r_f. Applying (5.5) with $q = r_f$ and S_0 quoted as units of domestic currency per unit of foreign currency,

$$F_0 = S_0\, e^{(r - r_f)T} \tag{5.7}$$

This is **covered interest parity**, and it says something economically strong: the forward exchange rate is determined entirely by the two interest rates, with no reference to anyone's view about where the currency is heading. A currency with a high interest rate trades at a forward *discount*, by exactly enough to offset the interest advantage.

How they are quoted. Equation (5.7) is stated in terms of "domestic" and "foreign," and the market does not use those words. A spot rate is quoted as units of the **quote** currency per one unit of the **base** currency: EUR/USD at 1.0850 is 1.0850 dollars per euro, and USD/JPY at 150.25 is 150.25 yen per dollar. Which currency is the base is pure convention: the euro, sterling, and the Australian and New Zealand dollars are quoted as base against the dollar, while the dollar is the base against most other currencies. Mapping this onto (5.7), *domestic is the quote currency and foreign is the base*, so for USD/JPY the domestic rate is the yen rate and the foreign rate is the dollar rate.

Forwards are not quoted outright. The interbank market quotes **forward points**: the difference between the forward and the spot rate, expressed in the last decimal place of the spot quote, which the client adds to spot to get the outright rate. Points are quoted rather than outright rates because they move far more slowly. Spot ticks continuously, while the interest differential driving the points does not, so a dealer can hold a points quote long after a spot quote has gone stale, and a foreign exchange (FX) swap, which is a spot leg against an offsetting forward leg, trades the points alone with no exposure to spot at all. The sign of the points carries the differential directly: negative points mean the base currency earns the higher interest rate.

Futures are quoted differently again, and for some pairs the other way up. Exchange-traded currency futures are quoted *outright*, in US dollars per unit of the other currency. For the euro that agrees with the interbank convention; for the yen it inverts it, so CME yen futures trade near 0.006650 dollars per yen where the interbank market quotes 150 yen per dollar. The two numbers are reciprocals of each other, and taking one for the other is a standard and expensive error.

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Example 5.7 A currency forward

The spot rate is 150 yen per dollar. The one-year dollar rate is 4.3% and the one-year yen rate is 0.5%. The quote currency is the yen and the base currency is the dollar, so the yen is domestic and the dollar foreign, and

$$F_0 = 150\, e^{(0.005-0.043)\times 1} = 150 \times 0.962713 = 144.41 \text{ yen per dollar}$$

A dealer would not quote 144.41. It would quote forward points of −5.59 yen against a spot of 150.00, or −559 points in a market where a **pip** (one unit in the last decimal place of the quote) is 0.01 yen, leaving the client to add them. The negative sign says at once that the base currency, the dollar, carries the higher interest rate.

The dollar is worth fewer yen forward than spot: a forward discount of 3.7%, or 3.8% continuously compounded, which exactly offsets its 3.8% interest advantage. An investor who borrows yen, converts to dollars, invests at 4.3%, and sells the dollars forward ends up exactly where borrowing yen at 0.5% would have left them. Borrowing the low-rate currency to invest in the high-rate one is the **carry trade**, and there is no riskless profit in it once the currency risk is hedged, and equation (5.7) is the statement of that.

An unhedged carry trade is a different matter entirely: it earns the interest differential and bears the currency risk, and whether that is attractive is an empirical question, not an arbitrage one.

Who uses them. Companies are the most familiar users. An American exporter invoicing in euros sells the euros forward to fix the dollar value of its receivables, and an importer buys its foreign currency forward to fix the cost of its payables; either way the forward removes the exchange rate risk between invoice and payment. The largest volume comes from asset managers, pension funds and insurers that hold foreign bonds and equities and hedge the currency back into their own, usually by rolling one- to three-month forwards or FX swaps, which removes the currency risk of the holding and leaves its local return. Banks use FX swaps to fund themselves: a bank whose deposits are in euros and whose loans are in dollars swaps the one into the other for a fixed term rather than borrowing dollars unsecured. When the demand to obtain dollars through FX swaps outruns the balance sheet dealers will commit to supplying them, the forward stops obeying equation (5.7), and that gap is the cross-currency basis that Section 7.8 prices.

And yet it fails. Chapter 7 shows that covered interest parity has been persistently violated since 2008, by tens of basis points, in the most liquid currency pairs in existence. That is a striking thing to say about an arbitrage relationship derived in three lines. Section 5.15 explains why, and the explanation is the most important qualification in this chapter.

Which foreign rate? Equation (5.7) contains r_f, and the choice is less obvious than it looks. It should be the rate at which the arbitrageur can actually invest the foreign currency for the contract's life, which for a bank executing the trade is not a government bond yield but a money market or repo rate, and may differ from the rate available to a different institution.

This is the same observation as Section 5.10: the relationship is enforced by a specific trade done by a specific participant, and the rate that belongs in the formula is the one that participant faces. Much of the apparent breakdown in covered interest parity dissolves, and the remainder becomes interpretable, once one stops asking what "the" risk-free rate is and starts asking what the marginal arbitrageur can actually borrow and lend at.

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5.8 COMMODITIES, STORAGE, AND CONVENIENCE YIELD

Commodities cost money to store, which is negative income. Writing u for the storage cost expressed as a continuous proportional rate,

$$F_0 = S_0\, e^{(r+u)T} \tag{5.8}$$

for an investment commodity such as gold.

Storage is often charged as an amount rather than a proportion of value: so many dollars an ounce a year for a vault, so many cents a bushel a month at a grain elevator. It is then handled exactly as known income was in Section 5.3, with the sign reversed. Writing U for the present value of the storage costs incurred during the contract's life,

$$F_0 = (S_0 + U)\, e^{rT} \tag{5.9}$$

The arbitrage is the basic one of Section 5.2: a trader who buys the commodity and sells it forward must also pay to store it, so the forward price has to repay both the purchase and the storage, with interest. Example 5.8 prices the same contract both ways.

Example 5.8 A gold forward

Gold trades at \$2,000 per ounce, the one-year rate is 4.3%, and storage and insurance cost 0.5% of value per year. From (5.8),

$$F_0 = 2000\, e^{(0.043+0.005)\times 1} = 2000 \times 1.049171 = \$2{,}098.34$$

Suppose instead the vault charges a flat \$10 an ounce, payable at the end of the year: the same 0.5% of today's price, stated as an amount. Its present value is

$$U = 10\, e^{-0.043\times 1} = 10 \times 0.957911 = \$9.58$$

so from (5.9)

$$F_0 = (2000 + 9.58)\, e^{0.043\times 1} = 2009.58 \times 1.043938 = \$2{,}097.88$$

Since the fee is paid at delivery, Ue^{rT} is the \$10 itself, and the forward price is simply the financed spot price, \$2,087.88, plus the fee. It sits slightly below the proportional figure because a proportional rate charges storage on a value that grows over the year, while a flat fee does not grow.

Consumption commodities and the convenience yield

For a consumption commodity the argument works in only one direction, and this is where the chapter's tidy equalities break down.

If $F_0 > S_0 e^{(r+u)T}$, the usual arbitrage applies: buy the commodity, store it, sell forward. Anyone can do this, so this inequality cannot persist.

If $F_0 < S_0 e^{(r+u)T}$, the reverse trade requires *selling the commodity short* (borrowing oil from someone who holds it). But holders of consumption commodities hold them because they intend to use them. A refinery will not lend its crude, at any price, if lending it means shutting down. The arbitrage is therefore unavailable, and we can conclude only that

$$F_0 \le S_0\, e^{(r+u)T} \tag{5.10}$$

or, with storage stated as a present value, $F_0 \le (S_0 + U)e^{rT}$.

The gap that (5.10) leaves open is filled by defining the **convenience yield** y, the benefit of holding the physical commodity rather than a claim on it, so that equality is restored:

$$F_0 = S_0\, e^{(r+u-y)T} \tag{5.11}$$

With storage as a present value, the same definition reads $F_0\, e^{yT} = (S_0 + U)e^{rT}$.

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It should be said plainly that y is not observed and not modeled: it is *defined* as whatever makes the equation hold. That does not make it useless, because what it measures is real: the value of having the physical commodity available when you need it, which rises when supply is tight.

Example 5.9 Backing out a convenience yield

Crude oil trades at \$80 and the one-year futures price is \$76. The risk-free rate is 4.3% and storage costs 3% a year. From (5.11),

$$76 = 80\, e^{(0.043+0.03-y)} \implies \ln(0.95) = 0.073 - y$$

so

$$y = 0.073 + 0.051293 = 12.43\%$$

A convenience yield of over 12% says the market places a large premium on having oil now rather than in a year. That is a statement about physical tightness (low inventories, constrained logistics), and traders read the term structure of convenience yields as a supply indicator rather than as a pricing artifact.

5.9 THE COST OF CARRY

All the results above are one formula. Define the **cost of carry** c as the cost of holding the asset: financing cost, plus storage, minus income. Then for an investment asset

$$F_0 = S_0\, e^{cT} \tag{5.12}$$

and for a consumption asset

$$F_0 = S_0\, e^{(c-y)T} \tag{5.13}$$

For a non-dividend stock $c = r$; for a stock index $c = r - q$; for a currency $c = r - r_f$; for a storable commodity $c = r + u$.

The cost of carry is a term structure, not a number. Writing c with no argument hides the fact that every ingredient varies with maturity: r is the zero rate to T, storage is contracted at different rates for different horizons, and the convenience yield moves most of all. What (5.12) needs at maturity T is $c(T)$, the *average* carry between now and T. It stands to the carry over each intervening month exactly as Chapter 4's zero rate stands to the forward rates underneath it (Figure 4.3), and inherits the same amplification: a mildly rising $c(T)$ implies a considerably steeper profile of carry between adjacent delivery months. This is why a real forward curve is not the single exponential of Figure 5.1, and why it can slope one way at the front and the other further out, which is what Figure 5.2 shows for four markets.

Figure 5.1 shows the two shapes this produces. When carry exceeds convenience yield the curve slopes upward (**contango**) and holding the physical asset is expensive relative to holding the futures. When convenience yield dominates the curve slopes downward (**backwardation**) and the market is paying a premium for immediate availability. Backwardation is the normal state for commodities in tight supply and, for a financial asset, where there is no convenience in holding the physical, arises only when the asset's income yield exceeds the financing rate, as for the dollar against the yen in Example 5.7.

What the curve looks like, market by market

Figure 5.1 draws the two shapes as though a market chooses between them. Real markets do not: each commodity has a characteristic curve, set by how cheaply it can be stored and how badly it is needed now. Table 5.1 summarizes, and Figure 5.2 draws four representative shapes.

Read that figure twice. The first reading is the shape. The second is the gap between the markers (the contracts that actually trade, at listed delivery months) and the dashed line, which

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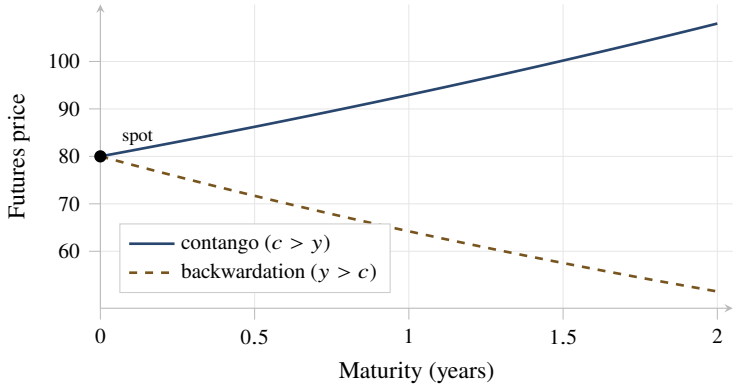


Figure 5.1: Forward curves when the cost of carry exceeds the convenience yield and when it does not. Constant carry, with the curvature exaggerated.

is (5.13) with a single convenience yield applied to every maturity. *A constant y fits none of them*: gold's markers sit a shade below the curve and drift further as its small convenience yield accumulates, crude's pull away as the front-month scarcity decays, gas swings with the season, and corn runs below full carry because commercial storage caps the slope. The convenience yield is a term structure, not a number, and the only honest way to read y off a market is one maturity at a time, which is what Example 5.10 does at a single point, and what Section 5.15 says cannot be pushed too far.

Table 5.1: Characteristic forward curve shapes.

Market	Typical shape	What sets it
Gold, precious metals	Contango, almost always	Above-ground stock is decades of mine supply, so $y \approx 0$ and the curve is the financing curve
Base metals	Mild contango; sharp backwardation in squeezes	Storable at moderate cost; shape tracks exchange inventory
Crude oil	Alternates; backwardated more often than not	y swings with inventory, and storage capacity is finite
Natural gas	Seasonal; neither label applies	Winter above summer; storage is costly and capacity-constrained
Grains, oilseeds	Rises within a crop year, breaks at the new crop	Harvest cycle; commercial storage cost caps the slope
Livestock	No carry relation	Not storable; animals must be fed and change in value
Electricity	No carry relation	Not storable at all; the forward is expectation plus risk premium

Gold: the pure carry case. Gold is the cleanest illustration of equation (5.12) in the whole commodity complex, because it barely behaves like a commodity at all. Roughly sixty years of mine production sits above ground in vaults and jewelry, so no plausible shortage can develop; the convenience yield of holding physical gold instead of a claim on it is close to zero. Storage and insurance run a fraction of a percent a year. What remains is financing cost, and the gold forward curve is therefore essentially the interest rate curve drawn in dollars per ounce.

This is why gold is almost never backwardated. When it briefly is, the cause is not a shortage of gold but a dislocation in the market for *immediately deliverable* gold in a particular location (refineries closed, flights grounded, bars of the wrong specification), and it resolves in

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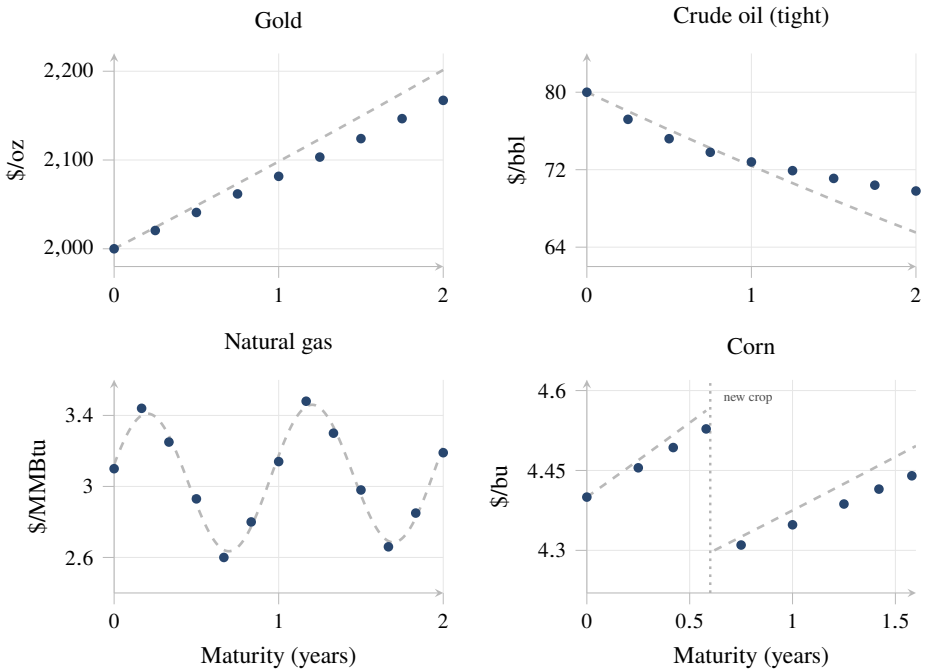


Figure 5.2: Four representative shapes. Markers are listed delivery months; the dashed line is what a constant convenience yield would give. Schematic.

weeks.

Example 5.10 Backing out the gold lease rate

Example 5.8 computed the full-carry forward price of gold, assuming no convenience yield: at a spot of \$2,000, financing of 4.3%, and storage and insurance of 0.5%, it came to \$2,098.34.

Suppose the one-year forward actually trades at \$2,081.60. The gap is small but it is not noise. The market reads it as a lease rate, defined as the financing rate less the forward premium on metal:

$$l = 0.043 - \ln\left(\frac{2081.60}{2000}\right) = 0.043000 - 0.039990 = 0.003010 = 0.30\%$$

Those 30 basis points are the **gold lease rate**: what a holder earns by lending metal to a bullion bank, and equivalently what a borrower pays to obtain metal now rather than in a year. It is a quoted market rate, and this residual calculation is how it is quoted. Storage does not enter it: the convention compares the forward premium with the financing rate alone. Solving (5.11) for the convenience yield adds storage back, $y = l + u = 0.30\% + 0.50\% = 0.80\%$.

The magnitude carries the lesson. A lease rate of 30 basis points, and a convenience yield under one percent, is what "essentially zero convenience yield" looks like in a real market, and it should be set beside the 12.43% of Example 5.9. The same equation governs both, and one parameter separates them by a factor of fifteen.

Crude oil: the regime-switching case. Oil spends most of its time backwardated. The reason is structural: refiners cannot run without crude, inventories are small relative to consumption, and a genuine shortfall is expensive in a way no financial hedge repairs. That sustains a positive convenience yield most of the time. But when demand falls or production overshoots, inventories build, the convenience yield collapses, and the curve flips into contango, and then the arithmetic of storage takes over.

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Suppose the front contract is \$42 and the twelve-month is \$54, a \$12 contango. A trader can buy physical crude, charter a tanker, and sell the deferred contract. A very large crude carrier holds about two million barrels; at \$60,000 a day the charter costs \$21.9 million a year, or about \$10.95 a barrel, and financing \$42 for a year at 2% adds \$0.84. Total carry is close to \$12, and the trade is marginal—which is the point. *Contango is capped near full carry*, because as soon as it exceeds the cost of storage, storage gets bought until it does not. This is not a hypothetical: floating storage on chartered tankers was a large and visible trade in 2009 and again in 2020.

The cap has a failure mode, and it is the one the next Case Study describes. The equilibrating mechanism works by *buying* storage. When there is no more storage to buy, nothing caps the contango, and prices do things that pricing formulas assuming a finite u cannot represent.

Natural gas: seasonal, so neither word applies. Gas demand is concentrated in winter heating and summer cooling, while production is roughly flat. Storage bridges the gap: gas is injected into underground facilities from April to October and withdrawn from November to March. The forward curve is therefore not monotone at all: it oscillates, with winter contracts above the summer contracts on either side, and calling it "in contango" or "in backwardation" is a category error.

The interesting point on such a curve is the March/April spread, which straddles the end of the withdrawal season. March is the last winter contract and April the first injection contract, so the spread is a direct bet on whether the winter ends with storage tight or comfortable. It is volatile enough to have earned the name *widow-maker*, and Section 5.14 returns to it: the largest hedge fund collapse of its era was a bet on this one spread.

Agriculture: carry within a crop year, a break across it. Grain arrives all at once at harvest and is consumed all year, so it must be stored, and within a crop year the curve normally rises at close to the cost of storing it. Grain markets have a name for the ceiling, **full carry** (interest plus commercial storage), and quote the spread as a percentage of it, as Example 5.11 does for corn.

What makes agricultural curves distinctive is that the carry relation *resets* at the new crop. A December corn contract and the following December contract are claims on different harvests, and no amount of storage links them, because grain stored from one harvest competes with grain that has not yet been planted. This is the discontinuity in the fourth panel of Figure 5.2, and it is why old-crop and new-crop contracts are quoted and traded as different markets.

Example 5.11 Reading a grain spread as a percentage of full carry

December corn trades at \$4.40 a bushel and March corn at \$4.55, both in the same crop year. Financing costs 5.0% and commercial elevator storage is 5 cents a bushel a month.

Over the three months from December to March, full carry is

$$\underbrace{4.40 \times 0.050 \times 0.25}_{\text{interest} = \$0.055} + \underbrace{3 \times 0.05}_{\text{storage} = \$0.150} = \$0.205$$

The market spread is $4.55 - 4.40 = \$0.15$, or

$$\frac{0.150}{0.205} = 73\% \text{ of full carry}$$

Read this as a signal about stocks. Near 100% of full carry, the market is paying anyone who stores grain their full cost, which it only needs to do when supplies are ample and someone must be induced to hold them. At 73% the market is paying most but not all of the cost: the 15-cent spread must first cover 5.5 cents of interest, so an elevator whose own storage cost for the three months is below the remaining 9.5 cents still profits by storing, one at full commercial rates does

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not, and grain moves out of the highest-cost storage first.

An *inverse* (March below December) reverses the message entirely. The market is then paying a premium to pull grain out of storage immediately, which is the agricultural version of backwardation and signals old-crop tightness. Storage cannot arbitrage it away, because the one thing carry cannot do is move grain backward in time.

Livestock and electricity: where the relation breaks. Two categories in Table 5.1 have no cost-of-carry relation at all, and they are worth naming because they mark the boundary of everything in this chapter.

Live cattle cannot be stored. An animal held for three months is not the same animal: it has been fed, at a cost, and has gained weight, changing what is delivered. Electricity is more extreme: outside limited battery and pumped-hydro capacity it cannot be stored for even an hour, so there is no trade that links today's price to next July's. In both markets the forward price is not spot compounded at anything. It is the market's expectation of the future spot price, plus a risk premium: the relationship of Section 5.13, standing alone because the arbitrage argument that usually dominates it is unavailable. This is why power forward curves show shapes no storable commodity ever displays, including negative prices in hours when the grid must dispose of generation.

Case Study 5.1 When storage runs out

On April 20, 2020, the expiring May contract for West Texas Intermediate (WTI) crude settled at −$37.63 a barrel. Buyers were paid to take oil.

Equation (5.8) assumes storage is available at a cost u. In April 2020, with demand collapsed by the pandemic and production still running, the tanks at the delivery point at Cushing, Oklahoma were effectively full. Longs approaching expiration faced physical delivery of oil they had nowhere to put, and could not roll into the next month fast enough. Storage was not expensive; it did not exist. The carry relation itself cannot produce a negative futures price, since it makes F a positive multiple of the spot price. What went negative was the spot value of a barrel at Cushing: a barrel that had to be taken with nowhere to put it had become a disposal cost, and the expiring contract, converging to that spot value, settled below zero. Section 36.1 returns to the episode.

Two lessons follow. First, the parameters in these formulas are not always finite, and a model that treats u as a small constant will be wrong in precisely the situations that matter. Second, this is what a *consumption* asset looks like when the consumption stops: the convenience yield collapsed to something deeply negative, and holding the physical was a liability, not a benefit.

The episode also broke a great deal of software. Option pricing models assuming lognormal prices cannot represent a negative underlying, exactly the problem Section 4.2 describes for interest rates (and Chapter 29 prices), and exchanges had to switch commodity options to normal models within days.

It broke retail products too, and more painfully. The largest losses fell on holders of a Bank of China product called Crude Oil Treasure, which gave retail clients exposure to WTI futures. Its client trading window closed hours before the exchange struck the settlement price, so holders could neither see the collapse nor exit it. And because the product was unleveraged (clients funded it in full), it had no margin close-out mechanism at all, on the reasoning that an unleveraged position cannot lose more than its principal. That reasoning is sound only while prices stay positive. Roughly sixty thousand investors lost their principal and finished owing the bank money, and Chinese

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regulators subsequently barred banks from offering such products to individuals.

The general lesson is worth more than the episode. Every product and every model contains variables that someone has quietly assumed to be nonnegative. It is worth asking which ones, and what happens if they are not.

5.10 THE IMPLIED REPO RATE

Practitioners rarely compute a theoretical forward price and compare it with the market. They invert the question: given the two prices, what return would the cash-and-carry trade actually earn? That return is the **implied repo rate**.

Buy the asset today at S_0, finance it, and deliver it at T for the invoice amount F_0, the sum the futures buyer pays on delivery. Ignoring intermediate income, the annualized return is

$$r_{\text{implied}} = \left(\frac{F_0}{S_0} - 1 \right) \times \frac{\text{days per year}}{\text{days to delivery}} \tag{5.14}$$

Comparing this with the rate at which the position can genuinely be financed, the *actual* repo rate, answers the question directly. If the implied repo rate exceeds the actual repo rate, the cash-and-carry trade earns more than it costs and should be done. If it is lower, the trade loses, and the reverse is indicated for anyone able to borrow the asset. Example 5.12 makes the comparison for a deliverable bond.

Example 5.12 Implied against actual repo

A bond has a cash price of 98.50. It can be delivered into a futures contract in 90 days for an invoice amount of 99.05. Financing in the repo market costs 4.30%.

From (5.14), using a 365-day year,

$$r_{\text{implied}} = \left(\frac{99.05}{98.50} - 1 \right) \times \frac{365}{90} = 0.0055838 \times 4.0556 = 2.26\%$$

The trade returns 2.26% and costs 4.30% to finance: a loss of 2.04% annualized. Buying the bond and selling the futures is not profitable here: the futures are *cheap* relative to the cash bond, and the indicated trade is the reverse, subject to being able to borrow the bond.

The implied repo rate is the practitioner's version of everything in this chapter. It restates the cost-of-carry relationship as a comparison between two interest rates, which is how the trade is actually assessed. Chapter 6 uses the same quantity under a different name: the net basis of Section 6.4 is the gross basis less carry, and it is negative precisely when the implied repo rate exceeds the actual one.

Which financing rate? Equation (5.14) compares against the *repo* rate, not the risk-free rate, and the distinction is the practical content of Section 5.15. The relevant rate is what this particular asset can be financed at by this particular participant, which depends on the asset's quality as collateral, on whether it is trading "special" in the repo market, and on the borrower's own standing. Two firms looking at the same two prices can correctly reach opposite conclusions.

5.11 ROLLING A POSITION AND ROLL YIELD

Futures contracts expire; an investor wanting continuous exposure must roll, selling the expiring contract and buying the next. The shape of the curve in Figure 5.1 then has a direct and often underappreciated effect on returns, which Example 5.13 isolates by holding the spot price fixed.

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Example 5.13 The cost of rolling in contango

An investor wants continuous exposure to a commodity trading at \$80 spot. The one-month futures contract trades at \$80.50, a market in mild contango.

The investor buys the one-month contract at \$80.50. Suppose the spot price does not move at all over the month. By Chapter 2's convergence argument, the contract must approach \$80 as it expires. The investor therefore loses \$0.50, sells at \$80, and buys the new one-month contract at \$80.50 to continue.

Repeating twelve times, the position loses

$$12 \times \$0.50 = \$6.00$$

or 7.5% of the initial \$80—on a commodity whose price never changed.

In backwardation the arithmetic reverses. With the one-month contract at \$79.50, the same investor gains \$0.50 each month and earns 7.5% a year from a flat spot price.

This is the **roll yield**, and for long-horizon commodity exposure it frequently dominates the spot price move entirely. It is the reason exchange-traded products tracking commodity futures can fall over years in which the underlying commodity rose, and why the offering documents of such products describe their roll methodology at length.

The roll yield is not a market inefficiency and there is no arbitrage in it. It is the cost of carry, collected month by month by whoever holds the physical and paid month by month by whoever wants exposure without holding it. What flows is the *net* carry $c - y$, so the direction can reverse. In Example 5.9 the convenience yield exceeds financing and storage: the holder of physical oil takes that yield in kind, and the futures price, set below spot, hands the futures investor a positive roll yield instead.

5.12 FORWARD VERSUS FUTURES PRICES

We have used "forward price" and "futures price" interchangeably. Chapter 2 explained why they are not the same instrument: futures are settled daily, forwards at maturity.

The consequence is precise. If interest rates were certain, the two prices would be identical: daily settlement would move cash around but the amounts would be known. When rates are stochastic the timing matters, and the direction depends on the correlation between the asset price and interest rates.

Consider an asset *positively* correlated with rates. A trader long futures makes money when the asset rises, and the gain arrives as variation margin at a moment when rates are high, so it is reinvested well. Losses arrive when rates are low and are financed cheaply. Being long futures is therefore better than being long a forward, and the futures price must be higher to compensate.

For an asset negatively correlated with rates the argument reverses and the futures price is lower.

The magnitude is negligible for short-dated contracts on assets weakly correlated with rates, which is why the distinction is usually ignored. It is emphatically not negligible for interest rate contracts, where the asset *is* the rate and the correlation is one. Chapter 30 develops this into the convexity adjustment, which reaches tens of basis points, and Chapter 6 shows the consequences for curve construction. Section 28.7 states the same result in one line: daily settlement makes the money market account the futures contract's numeraire, so the futures price is a martingale under the risk-neutral measure while the forward price is one under the T-forward measure.

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5.13 FUTURES PRICES AND EXPECTED SPOT PRICES

We come finally to the question every newcomer asks: is the futures price the market's forecast of the future spot price?

The answer is no, and the reason is instructive.

Consider a speculator who takes a long futures position, holds it to delivery, and sets aside the present value of the futures price, $F_0 e^{-rT}$, in a risk-free investment. At T the investment pays F_0, which buys the asset under the contract, so the speculator ends with an asset worth S_T. The outlay today is certain and is discounted at r; the payoff is risky and is discounted at k, the rate appropriate to the asset's risk. Under standard asset pricing the present value of the whole position must be zero:

$$-F_0\, e^{-rT} + E[S_T]\, e^{-kT} = 0$$

so that

$$F_0 = E[S_T]\, e^{-(k-r)T} \tag{5.15}$$

If the asset has no **systematic risk** (risk that moves with the market as a whole and so cannot be diversified away), then $k = r$ and (5.15) gives $F_0 = E[S_T]$. If it has positive systematic risk (most assets), then $k > r$ and $F_0 < E[S_T]$: the futures price sits below the expected spot price, and the speculator earns the difference as compensation for bearing risk.

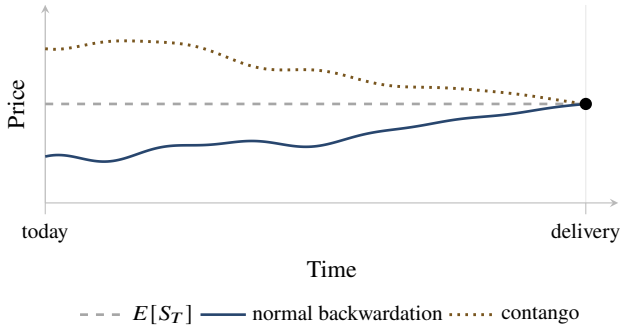


Figure 5.3: The futures price drifting toward the expected spot price as delivery approaches. Illustrative paths.

Figure 5.3 shows what this implies over time. If $F_0 < E[S_T]$, the futures price must drift *upward* on average as delivery approaches, since it must converge to S_T. Keynes called this **normal backwardation**, and explained it by hedging pressure: if producers are net short (as they typically are, hedging future output), speculators must be net long, and will require compensation. Where hedgers are net long instead, the argument reverses and $F_0 > E[S_T]$, which Keynes called **contango**.

Two warnings on terminology. The words are used in two different senses and the ambiguity causes real confusion. Section 5.9 used contango and backwardation to describe the *shape of the forward curve* today, which is observable. Keynes used them to describe the relationship between the futures price and the *expected* spot price, which is not observable, since no one sees $E[S_T]$. Most market usage means the first. Statements that a market "in contango" implies traders expect prices to rise confuse the two, and are wrong: the shape of the curve is determined by carry and convenience, not by expectations.

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5.14 TRADING THE SHAPE OF THE CURVE

Two results in this chapter describe money changing hands. Section 5.11 showed that the shape of the curve determines a roll yield, and Section 5.13 showed that a risk premium is compensation somebody is paying. Both are therefore things a speculator can try to earn, and three expressions are common.

Calendar spreads. Buying one delivery month and selling another is a direct bet on the shape of the curve, with no outright exposure to the level of the price. If the front contract is expected to richen relative to the second, because inventories are tightening, the trade is long the front and short the deferred. Section 2.3 showed why this is efficient: the two legs offset, so the margin requirement is a fraction of the outright, and Example 2.3's calendar spread required a twentieth of the gross amount.

Case Study 5.2 Amaranth and the widow-maker

In September 2006 Amaranth Advisors, a multi-strategy hedge fund managing about $9.2 billion, lost roughly $6.6 billion in a matter of weeks and closed. Almost all of it came from natural gas calendar spreads: principally long March against short April, the spread of Section 5.9 that straddles the end of the withdrawal season.

The position was not obviously foolish. A cold winter drains storage, March scarcity bids the last winter contract above the first injection contract, and the spread widens. It had done so before. But the 2005–06 winter had been mild, storage ended the season full, and the 2006 hurricane season passed without disrupting Gulf production. The spread collapsed instead of widening.

What turned a bad trade into a fatal one was size. Amaranth's positions were large enough that the Senate Permanent Subcommittee on Investigations later found it had at times held on the order of 40% of the open interest in NYMEX winter contracts. A position that large cannot be exited: every attempt to sell moved the spread further against the remaining position, because there was no one on the other side but the firm itself. The fund's positions were eventually transferred in bulk to JPMorgan and Citadel at a steep discount.

Three lessons follow, each pointing at a different chapter. **Low margin is not low risk.** Spread positions receive the margin offsets of Section 2.3 precisely because the legs hedge each other against parallel moves, which is exactly what lets a trader accumulate a notional position that a directional book could never finance. **Liquidity is part of the position.** A trade you cannot exit is a different trade from the one you put on. **Margin arrives before the thesis resolves.** The variation margin mechanism of Section 2.3 does not wait to find out whether you were right about the winter.

The episode also had a regulatory afterlife. Amaranth had shifted positions from NYMEX to the then-unregulated Intercontinental Exchange (ICE) to escape NYMEX's position accountability levels (the position sizes above which the exchange can demand an explanation and order a position cut), and closing that gap was a direct motivation for extending Commodity Futures Trading Commission (CFTC) oversight to economically equivalent contracts traded off-exchange.

Carry harvesting. A systematic strategy holds long positions in backwardated markets and short positions in contangoed ones, collecting the roll yield of Section 5.11 across many commodities. This is the commodity version of a carry trade, and like all carry trades it earns a steady return punctuated by sharp losses when the curve inverts.

Taking the other side of hedging pressure. If Keynes was right that producers' hedging demand pushes F_0 below $E[S_T]$, a speculator who simply holds long positions in commodities

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where producers dominate should earn the difference.

Does the premium exist? The empirical answer is that it is real but neither large nor reliable. Studies find returns consistent with a risk premium in some commodities and periods and not others, and the premium appears to move with the balance of hedging pressure, which is itself unstable, since whether producers or consumers dominate a market changes with industry structure.

This is a genuine risk premium and not an arbitrage. It is compensation for bearing a risk that someone else wanted to shed, it is earned on average and lost in specific episodes, and it disappears when hedging demand switches direction.

Why a hedger should care. This is not a digression from hedging; it is the same fact seen from the other side. The speculator's expected return is the hedger's expected cost. A producer hedging in a persistently backwardated market where producers dominate is systematically paying a premium to transfer risk, and that premium is the price of the insurance, not evidence that hedging is a mistake.

It also disposes of a common confusion. Contango is not "bad" and backwardation is not "good." Example 3.6 showed a short hedger gaining from rolling in contango, and Section 5.11 showed a long investor losing from it in the same market. Any statement about the shape of the curve that does not specify which side of the market one is on is empty.

Chapter 36 develops these strategies properly for energy and commodity markets, where the term structure carries the most information.

5.15 THE LIMITS OF ARBITRAGE

Every result in this chapter rests on the assumption stated in Section 5.1: that participants can borrow and lend freely at the risk-free rate and will take arbitrage opportunities when they see them. It is time to examine what that assumption costs.

The arbitrage in Example 5.1 requires borrowing \$50 and holding a stock for six months. That is a balance sheet position. Post-crisis capital rules (the leverage ratio of Section 1.12 in particular) make holding it expensive, and the expense does not shrink because the position is hedged.

This is not an abstraction, and the next two chapters meet it three times:

- **Covered interest parity** (Section 5.7) has been violated continuously since 2008. Chapter 7 shows the deviation is the price of the balance sheet required to close it.
- **Swap spreads** (Chapter 7) turned negative, offering an apparently riskless \$811,100 on a \$100 million ten-year position, which sat available for years because the trade requires financing a bond.
- **The Treasury cash–futures basis** of Chapter 6 is the arbitrage of Example 5.1 applied to government bonds. It persists, and is harvested by funds willing to run fifty times leverage—which is what the balance sheet constraint looks like when someone is prepared to bear it.

Case Study 5.3 Arbitrage requires a balance sheet

The three dislocations above are usually discussed separately (one in FX, one in rates, one in Treasuries) and are usually presented as puzzles. They are the same phenomenon.

In each case the textbook arbitrage exists and is riskless in the sense this chapter means. In each case executing it requires holding an asset and financing it. And in each

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case post-crisis regulation made that holding costly in a way that does not depend on the position's risk, so the cost cannot be hedged away.

The consequence is a genuine change in what "no-arbitrage" means. Before 2008 it was reasonable to treat these relationships as identities, because the intermediaries who enforced them faced negligible costs for doing so. They are now better understood as relationships that hold up to a spread, where the spread is the price of intermediation. That price is highest when dealers have the least balance-sheet capacity to spare, at reporting dates and in periods of stress, and those are the times when the dislocations are largest.

For a reader of this chapter, the practical implication is not to distrust the formulas. They remain the right starting point and the right way to think. It is to be precise about what enforces them, and to expect the enforcement to weaken under stress rather than to strengthen.

One consequence deserves naming here, because this chapter has used r throughout without saying whose rate it is. For a *collateralized* derivative the question has a definite answer, and it is not a matter of judgment: the discount rate is the rate paid on the collateral, fixed by the collateral agreement rather than chosen by the valuer. Section 7.4 derives it. Where a position is collateralized in its own currency, this chapter's r is that rate; where it is not (an uncollateralized trade, or collateral posted in another currency), the difference becomes an explicit valuation adjustment in Chapter 9.

SUMMARY

For an investment asset, arbitrage in both directions forces $F_0 = S_0 e^{rT}$, and no view about the asset is needed. Income lowers the forward price, whether it is known in amount, $F_0 = (S_0 - I)e^{rT}$, or as a yield, $F_0 = S_0 e^{(r-q)T}$; storage raises it; a currency is an asset yielding the foreign rate, so a hedged carry trade earns nothing. An existing forward is worth $(F_0 - K)e^{-rT}$, with no model of the asset required.

The relation is an equality only where holders are willing to sell. A borrowing fee opens a band when they are not, and for a consumption commodity they never are, so only an upper bound survives and the convenience yield closes the gap. Everything reduces to the cost of carry, itself a term structure: carry above the convenience yield gives contango, below it backwardation. April 2020 showed what happens when storage is unavailable rather than merely costly.

Practitioners run the relation in reverse: the implied repo rate says whether a cash-and-carry pays, and the cost of rolling in contango is the cost of carry, paid by whoever wants exposure without the physical. Neither the forward nor the futures price is a forecast; the futures price sits below the expected spot price when the asset carries systematic risk, so contango does not mean prices are expected to rise. Finally, arbitrage needs a balance sheet, which post-crisis regulation made expensive, so the formulas hold only up to a spread, and the spread is widest when dealers can least afford to close it.

FURTHER READING

Working (1949) is the origin of the theory of storage and states why a forward curve slopes the way it does; Keynes's *A Treatise on Money* (1930, vol. 2) is where normal backwardation comes from, and the distinction between the two ideas is worth getting straight. Cornell and Reinganum (1981) is the standard early test of forward against futures prices. On why the arbitrage this chapter relies on can fail, Shleifer and Vishny (1997) is the general argument

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and Du, Tepper, and Verdelhan (2018) the specific one, documenting deviations from covered interest parity that ought not to exist.

PRACTICE QUESTIONS

5.1 A non-dividend-paying stock trades at \$72 and the four-month risk-free rate is 3.6%. Compute the four-month forward price, and describe the arbitrage if the forward is quoted at \$74, giving the profit per share.

5.2 A stock will pay a \$1.50 dividend in two months and \$1.50 in five months. It trades at \$64 and the six-month rate is 4%, flat. Compute the six-month forward price.

5.3 An index stands at 3,800 with a dividend yield of 2.1% and a six-month rate of 4.0%. Compute the six-month futures price, state what it would be if the dividend yield exceeded the interest rate and why, and explain why that yield is a forecast rather than a known quantity.

5.4 An investor is short a forward on a non-dividend-paying stock with a delivery price of \$95 and five months remaining. The stock trades at \$88 and the rate is 3.5%. Value the position.

5.5 The spot rate is 1.08 dollars per euro, the one-year dollar rate 4.2% and the euro rate 2.6%. Compute the one-year forward rate, state which currency is at a forward discount and why, and explain why a hedged carry trade earns nothing while an unhedged one is a different proposition entirely.

5.6 Copper trades at \$4.20 per pound with a one-year futures price of \$4.05, a rate of 4.3%, and storage costs of 1.5%. Compute the implied convenience yield and interpret it. Repeat the carry calculation for silver at \$29 per ounce with storage of 0.4% and a rate of 4.3%, and explain why the two cases require different arguments.

5.7 Explain the difference between the two uses of the word "contango" in this chapter, and why it is wrong to say that a market in contango expects prices to rise.

5.8 A stock trades at \$120 with a three-month rate of 4% and a borrowing fee of 5% a year. Compute the no-arbitrage band for the three-month forward price, and explain why a quote inside the band is not a mispricing.

5.9 A bond has a cash price of 101.20 and can be delivered into a futures contract in 60 days for an invoice amount of 101.90. Compute the implied repo rate on a 365-day basis. If the bond can be financed at 4.10%, should the cash-and-carry trade be done?

5.10 Derive the value of a long forward contract by comparing two portfolios. Explain why the underlying asset price does not appear in the result, and what this permits operationally.

5.11 A commodity trades at \$60 spot with the one-month futures price at \$60.75, and an investor holds continuous long exposure by rolling monthly. If the spot price is unchanged after a year, what is the return? Repeat for a one-month futures price of \$59.40, and identify who receives what the futures investor pays.

5.12 Silver trades at \$29 per ounce, and storage and insurance cost \$0.12 per ounce per quarter, payable at the start of each quarter. The interest rate is 4.3% for all maturities. Compute the nine-month futures price. If the nine-month futures contract trades at \$30.40, describe the arbitrage and give the profit per ounce.

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FURTHER QUESTIONS

5.13 Explain why the arbitrage argument for a consumption commodity works in only one direction, and derive the resulting inequality. Is the convenience yield a model, a measurement, or a definition? Defend your answer.

5.14 In April 2020 the expiring WTI contract settled below zero. Explain which assumption in equation (5.8) failed, what it implies about the convenience yield in that month, and what pricing models had to be changed as a result.

5.15 Explain why forward and futures prices differ when interest rates are stochastic, deriving the direction of the difference from the correlation between the asset and rates. Why is the effect negligible for equity index futures and material for interest rate futures?

5.16 A trader observes that index futures are 20 points above their theoretical value. Set out the trade, identify every cost and risk it involves, and explain what would have to be true for the trade to be genuinely riskless. Under what market conditions would you decline to do it?

5.17 A stock is impossible to borrow at any price. Explain why this alone does not put the forward price below $S_0 e^{rT}$, state the condition on the shareholder register under which it does, and say why that condition and an expensive borrow tend to arrive together.

5.18 Explain why the no-arbitrage band around an index futures price is wider below theoretical value than above it. State what a participant needs to close the gap in each direction, and explain why, although index futures are more often rich than cheap, the asymmetry is not the reason.

5.19 A dealer quotes USD/JPY spot at 148.50 and one-year forward points of -480. Give the outright one-year forward rate, identify the base and quote currencies, say which currency carries the higher interest rate and how you know, and state roughly what a CME yen futures contract for the same date would be quoted at.

5.20 Equation (5.12) writes the cost of carry as a single number c. Identify which of its components vary with maturity, explain what $c(T)$ must therefore mean, and state the analogy with the zero and forward rates of Chapter 4. What does the analogy predict about the carry implied between two adjacent delivery months when $c(T)$ is rising gently?

5.21 Covered interest parity, negative swap spreads, and the Treasury cash–futures basis are usually discussed separately. Argue that they are one phenomenon. What does your answer imply about how the no-arbitrage results of this chapter should be taught and used?

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Chapter 6
Interest Rate Futures

Interest rate futures are the most heavily traded derivatives in existence, and they are where most of the hedging that the rest of this book describes actually gets done. A dealer who has warehoused the risk of a swap, a structured note, or a mortgage portfolio does not usually offload it into another bespoke trade; it hedges in the futures market, where the liquidity is.

Two families dominate. **Short-term interest rate futures** settle against an overnight rate averaged over a reference period, and are used to trade and hedge the front of the curve. **Treasury futures** require physical delivery of a government bond, and are used further out. The two are constructed on completely different principles, and each has a wrinkle that repays attention: the short-term contracts carry the convexity adjustment of Chapter 30, and the Treasury contracts embed a delivery option that decides which bond the contract tracks.

The chapter closes with a trade that arises directly from that second wrinkle: the cash–futures basis trade, which hedge funds have built to the high hundreds of billions of dollars, and which in March 2020 took roughly $1 trillion of Federal Reserve purchases to arrest. It is a recurring subject of financial stability reports for that reason.

Section 6.1 sets out the quotation conventions without which none of the prices can be read. Section 6.2 covers the short-term contract, and Sections 6.3 and 6.4 the Treasury contract and the delivery option that decides which bond it tracks. Section 6.5 turns to the basis trade built on that option, and Sections 6.6 to 6.8 to the uses: hedging a swap book with the strip, duration-based hedging, and building the front of the curve from futures prices.

6.1 DAY COUNTS AND QUOTATION CONVENTIONS

Chapter 4 noted that a rate is meaningless without its day count. Futures markets add quotation conventions on top, and the combination is the subject's main source of confusion.

Day counts. US Treasury bonds accrue on actual/actual; corporate and municipal bonds on 30/360; money market instruments including SOFR on actual/360. The 30/360 convention treats every month as thirty days and gives no month a 31st, which produces the small anomaly that a bond accrues interest on February 29 and 30 and none on the 31st of a long month.

Price quotation for bonds. US Treasury bonds and notes are quoted in points and thirty-seconds of a point, per $100 of face value. A quote of `110-16` means $110 + 16/32 = 110.50$. Finer increments appear as a third digit: `110-165` means $110 + 16.5/32 = 110.5156$, and the shorter note futures are quoted in quarters of a thirty-second.

Clean and dirty price. A coupon is paid in full to whoever holds the bond on the payment date, although a seller may have held it for most of the period. The seller is compensated through **accrued interest**, the part of the next coupon earned since the last coupon date on the

bond's day count:

$$\text{accrued interest} = \text{coupon} \times \frac{\text{days since the last coupon}}{\text{days in the coupon period}}$$

The buyer pays it to the seller on top of the price and receives the whole coupon back on the payment date. The quoted price is the **clean price**, which excludes accrued interest. The amount actually paid is the **dirty price**, or cash price:

$$\text{cash price} = \text{quoted price} + \text{accrued interest since the last coupon} \tag{6.1}$$

A 4% Treasury paying coupons on March 1 and September 1, bought at `101-08` for June 15 settlement, has accrued 106 of its 184-day coupon period on actual/actual, so per \$100 of face the buyer pays

$$101.25 + 2.00 \times \frac{106}{184} = 101.25 + 1.1522 = 102.4022$$

The convention exists so that a bond's quoted price does not sawtooth upward between coupons and drop on each payment date, which would obscure genuine price movement.

Price quotation for short-term rate futures. Short-term interest rate futures are quoted as

$$\text{quoted price} = 100 - R \tag{6.2}$$

where R is the rate in percent. A quote of 95.72 corresponds to a rate of 4.28%. The convention inverts the usual direction: a trader who expects rates to *rise sells* the futures contract. It makes the contract behave like a price, rising when the holder profits, as price-based margining systems require.

6.2 SOFR FUTURES

The Eurodollar futures contract, which settled against three-month LIBOR, was for decades the most actively traded futures contract in the world. Its replacement settles against SOFR, and the difference in construction has consequences.

Case Study 6.1 Retiring the largest futures contract ever traded

The Eurodollar contract launched in 1981 and became the benchmark for the front end of the dollar curve, with open interest at times above ten million contracts, over \$10 trillion of notional.

It could not simply be delisted: millions of open positions, some running for years, referenced a rate that would cease to exist. CME's solution, executed in April 2023, was to convert open Eurodollar positions into the corresponding SOFR futures, applying the same ISDA fallback spread adjustment that Chapter 4 described for swaps (26.161 basis points for the three-month tenor), so that the economics of each position carried over.

The conversion was, by the standards of such things, uneventful. That is the point worth noting: the transition succeeded because the compensation mechanism was fixed in advance and applied uniformly, so that no participant had to negotiate. Where fallback language was ambiguous, disputes followed instead.

Contract construction

Two contracts cover the front end.

Three-month SOFR futures have a notional of \$1,000,000 and settle against the *compounded* daily SOFR over a three-month reference period running between consecutive

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International Money Market (IMM) dates (the third Wednesdays of March, June, September, and December). Since a basis point on \$1,000,000 for three months is

$$\$1{,}000{,}000 \times 0.0001 \times 0.25 = \$25$$

each basis point is worth \$25 per contract; Example 6.1 follows a position to its profit.

One-month SOFR futures have a notional of \$5,000,000 and settle against the *arithmetic* average of daily SOFR over a calendar month. A basis point is worth

$$\$5{,}000{,}000 \times 0.0001 \times \frac{1}{12} = \$41.67$$

Example 6.1 Profit on a SOFR futures position

A trader expects the central bank to tighten more than the market has priced, and sells 200 three-month SOFR futures at 95.72, an implied rate of 4.28%.

Rates subsequently rise 12 basis points and the contract settles at 95.60, an implied rate of 4.40%. Because the trader is short the price, the gain is

$$200 \text{ contracts} \times 12 \text{ bp} \times \$25 = \$60{,}000$$

The trader profited from rates rising while being *short* the contract. The initial margin required would have been a small fraction of the \$200 million notional involved, which is what makes futures the instrument of choice for expressing rate views.

The averaging feature and delta decay

These contracts have a property with no counterpart in the Eurodollar world, one that Sections 29.4 and 30.3 return to, and it is worth seeing directly.

A Eurodollar contract settled against a rate fixed on a single date. Until that date its sensitivity to rates was full; after it, the contract was over. A SOFR contract settles against an average accumulating across its reference period. Each day that passes locks in one more daily fixing, so the contract's remaining sensitivity falls, approximately linearly, from full at the start of the reference period to zero at its end. Rates desks call a first-order sensitivity of this kind a **delta**, which is why the effect is known as *delta decay*; the same word names an option's sensitivity to its underlying in Chapter 18, and the two are the same idea applied to different variables. Figure 6.1 contrasts the two contracts.

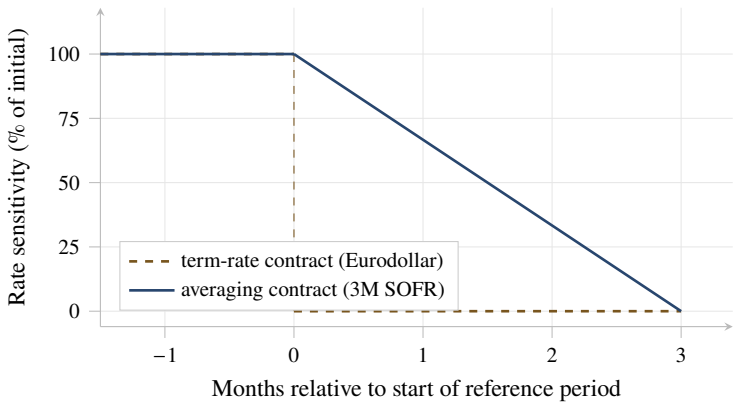


Figure 6.1: Rate sensitivity through the reference period.

The practical consequence is that a hedge built from SOFR futures must be rebalanced *through* the reference period rather than rolled at a single date. A position that is correctly

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hedged on the first day of the quarter is under-hedged by the last, because the futures leg has lost sensitivity that the hedged item has not.

Who uses them, and for what

The contracts are rarely traded one at a time. A **strip** is a set of contracts of the same type spanning a run of consecutive dates (or, in the option markets of Chapter 22, a run of strikes) bought or sold together and priced as a single position. The SOFR strip is the sequence of quarterly contracts stretching out along the curve, and most of what follows is about the strip rather than about any one contract in it.

Four groups trade the SOFR strip, and their separate motives explain the contract's liquidity.

Borrowers and lenders hedging near-term funding. A corporate treasurer expecting to roll commercial paper (short-term unsecured company borrowing) in six months, or a bank funding a loan book at short rates, locks in the cost by selling futures for the relevant quarter. This is the textbook hedge and the contract's original purpose.

Dealers hedging swap books. This group is by far the largest source of volume. A dealer that has received fixed on a five-year swap holds front-end delta it does not want, and offloads it by selling a strip of SOFR futures covering the swap's early accrual periods. Chapter 7 describes the residual risks this leaves; the futures market is where the first-order risk goes.

Traders positioning on monetary policy. The strip of consecutive contracts is a direct reading of the market's expected path for the policy rate, so a view that the central bank will move faster or slower than priced is expressed by buying one contract and selling another. Trades of this kind (the **steepener**, which profits if later rates rise relative to earlier ones, and the **flattener**, which profits if they fall) are among the most common macro trades.

Curve builders. The futures strip supplies the market's most liquid information about the forward rates of Section 4.9 between roughly three months and five years, and is a primary input to the curve construction of Chapter 7, after the convexity correction described next.

Futures rates are not forward rates

The final property, developed at length in Section 30.3, must be flagged here because it governs how these contracts are used to build a curve. Daily settlement favors the short side of the contract, which gains when rates rise: its gains arrive when rates are high and can be reinvested at those rates, and its losses arrive when rates are low and are cheap to finance. A forward rate agreement, settled once, offers neither side that advantage, so the futures rate must sit *above* the forward rate to compensate the long. The gap is the **convexity adjustment**. How large it is depends on the model used for the short rate; the simplest form, and the one this chapter quotes, comes from the Ho–Lee model of Chapter 31, in which the short rate moves by normally distributed amounts with constant volatility:

$$\text{Forward rate} = \text{Futures rate} - \frac{1}{2}\sigma^2 T_1 T_2 \tag{6.3}$$

where T_1 is the time until the contract's rate is determined, T_2 the time until the end of the period that rate covers, and σ the standard deviation of the absolute change in the short rate over a year (a normal volatility, which is the market convention for this parameter). Strictly, (6.3) adjusts the continuously compounded rate; Section 30.3 adds the small term for the simply compounded rate the contract settles on. At $\sigma = 1.00\%$ the adjustment is under a basis point on the one-year contract and about 51 basis points on the ten-year, growing with the *square* of maturity—though Ho–Lee has no mean reversion, no pull of the rate back toward a long-run level, and Section 30.3 shows that this overstates the long end once mean reversion is allowed for. Curve construction that treats quoted futures rates as forward rates will be visibly wrong beyond the first two years or so.

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6.3 TREASURY FUTURES

Treasury futures rest on a different principle: physical delivery of a government bond.

Two features of that delivery shape what follows, and both are the short's choice. *Delivery is not a date but a month.* Any business day of the contract's delivery month will serve, and it is the **short** who declares which one, by giving notice; the long takes delivery when told. *And the short chooses the bond as well*, from a basket set by the exchange. Neither choice is a technicality; together they are worth a material amount, priced in Section 6.4.

Physical delivery raises a further difficulty: no single bond stays suitable. A contract referencing one specific ten-year note would become a contract on a nine-year note within a year, and would be vulnerable to a squeeze if someone accumulated the issue. The exchanges' solution is to define a **deliverable basket**: the short may deliver any bond satisfying a maturity range on the delivery date. Table 6.1 lists the principal US contracts.

Table 6.1: Principal US Treasury futures contracts.

Contract	Approximate deliverable maturity	Face value
2-year note	1¾–2 years	$200,000
5-year note	4¼–5¼ years	$100,000
10-year note	6½–10 years	$100,000
Treasury bond	15–25 years	$100,000
Ultra bond	25 years and over	$100,000

Conversion factors

Allowing any bond in the basket to be delivered creates an obvious problem: a high-coupon bond is worth more than a low-coupon one, so the short would always deliver the lowest-coupon bond, and the contract would track that one bond alone.

The remedy is the **conversion factor**. Each deliverable bond is assigned a factor equal to its price per $1 of face value if it were priced to yield a notional standard rate (6% in the US contracts), so that a bond paying more than 6% is scaled up and one paying less is scaled down. The amount the long pays on delivery, writing CF for the conversion factor, is

$$\text{invoice} = (\text{settlement price} \times \text{CF} + \text{accrued interest}) \times \frac{\text{face}}{100} \tag{6.4}$$

Example 6.2 computes it for one delivery.

Example 6.2 The invoice amount on delivery

A ten-year note futures contract settles at `110-16`, that is 110.50. The short delivers a bond with conversion factor 0.8456 on which accrued interest is $1.20 per $100 of face; the contract covers $100,000.

From (6.4),

$$(110.50 \times 0.8456 + 1.20) \times 1{,}000 = (93.4388 + 1.20) \times 1{,}000 = \$94{,}638.80$$

The factor scales the payment down because the delivered bond's coupon is below the 6% standard.

Who uses them, and for what

Treasury futures are the principal tool for moving duration around, and the identity of the users matters because it explains a persistent feature of the price.

Asset managers adjusting duration. A bond fund that wants to shorten duration ahead of an expected sell-off can sell futures rather than sell bonds: faster, cheaper in transaction costs,

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and without disturbing a carefully constructed credit portfolio. A fund that wants *more* duration than its cash allows buys futures, obtaining the exposure for margin rather than the full purchase price. This second use is important: it means a large and fairly price-insensitive stream of buying interest sits in the futures market, which tends to leave futures *rich* (expensive) relative to the cash bond. That richness is precisely what the basis trade of Section 6.5 harvests.

Dealers hedging inventory. A dealer holding bonds bought from customers hedges the price risk with futures until the position is distributed. Because the hedge is against a basket rather than the specific bond, what remains is basis risk, the subject of Section 6.4.

Mortgage portfolios. Holders of mortgage-backed securities face duration that *shortens* when rates fall, as borrowers prepay, and lengthens when rates rise, the property called **negative convexity**, since the price gains less in a rally than it loses in a sell-off. Keeping such a portfolio hedged requires constant rebalancing in the same direction as the market is already moving, and Treasury futures are the usual instrument. This is a well-known amplifier of rate moves.

Curve and spread traders. Buying one contract and selling another expresses a view on the shape of the curve (long two-year against short ten-year futures is a steepener) without an outright duration position. Trading futures against the bonds expresses a view on the basis itself.

Liability-driven investors. Pension funds and insurers with long-dated liabilities use long-maturity contracts to lengthen asset duration toward liability duration. Section 8.9 works through what happened in 2022 when this exposure was held with leverage.

How conversion factors are calculated

The definition is operational, and worth carrying out, because the two conventions buried in it are what make conversion factors imperfect.

Step 1: round the maturity. Measure the bond's remaining life from the first day of the delivery month and round it *down* to the nearest three months. A bond with 8 years and 5 months remaining is treated as having 8 years and 3 months; one with 8 years and 2 months is treated as having exactly 8 years. This rounding is what allows a single factor to be published for the whole delivery month rather than recomputed daily.

Step 2: price the bond at 6%. Discount the bond's coupons and principal at exactly 6% per annum, semiannually compounded, using the rounded maturity and \$1 of face; the result is the conversion factor.

For a bond whose rounded maturity is a whole number of half-years, this is an ordinary bond price calculation.

Example 6.3 Conversion factor for a bond with 8 years remaining

A bond pays a 5% annual coupon semiannually and has exactly 8 years remaining after rounding: $n = 16$ half-years, a coupon of \$2.50 per \$100 of face, and a discount rate of 3% per half-year.

The annuity factor is

$$a = \frac{1 - 1.03^{-16}}{0.03} = \frac{1 - 0.623167}{0.03} = 12.5611$$

so the price per \$100 of face is

$$2.50 \times 12.5611 + \frac{100}{1.03^{16}} = 31.4027 + 62.3167 = 93.7194$$

The conversion factor is this divided by 100:

$$\text{CF} = 0.9372$$

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It is below 1 because the bond's 5% coupon is below the 6% standard.

Now take an odd quarter. Suppose the same bond has 8 years and 3 months remaining after rounding. Three months from now it will have exactly 8 years to run and will just have paid a coupon, so at that date it is worth the \$93.7194 above plus the \$2.50 coupon:

$$93.7194 + 2.50 = \$96.2194$$

Discount that back one quarter at 3% per half-year. Today the bond is already three months into a six-month coupon period, and the interest accrued since the last (notional) coupon date is paid separately by the buyer, as accrued interest, so it must not be charged for twice; deduct it:

$$\frac{96.2194}{1.03^{1/2}} - 2.50 \times \frac{3}{6} = 94.8078 - 1.25 = 93.5578 \qquad \Longrightarrow \qquad \text{CF} = 0.9356$$

The factor is slightly below the 0.9372 of the whole-half-year case, because the extra quarter carries a below-coupon bond a little further from par. The general formula below is built for this case.

The general formula. When the rounded maturity ends on an odd quarter, the bond is valued at the next coupon date, three months away, discounted back, and reduced by the interest accrued since. The exchange publishes a single formula covering both cases. Writing c for the annual coupon rate, n for the number of whole years, and z for the remaining months ($z \in \{0, 3, 6, 9\}$), set $v = z$ if $z < 7$ and $v = z - 6$ otherwise. Then

$$\text{CF} = \frac{1}{1.03^{\,v/6}} \left[\frac{c}{2} + \kappa + \frac{c}{0.06}\,(1 - \kappa) \right] - \frac{c}{2} \cdot \frac{6 - v}{6} \tag{6.5}$$

where $\kappa = 1.03^{-2n}$ if $z < 7$ and $\kappa = 1.03^{-(2n+1)}$ otherwise. The final term deducts the interest accrued over the $6 - v$ months since the last coupon; the leading factor discounts back over the odd quarter. Example 6.4 checks the formula against Example 6.3.

Example 6.4 Applying the formula

Check against Example 6.3. Here $c = 0.05$, $n = 8$, $z = 0$, so $v = 0$ and $\kappa = 1.03^{-16} = 0.623167$:

$$\text{CF} = 1 \times \left[0.025 + 0.623167 + \frac{0.05}{0.06} \times 0.376833 \right] - 0.025 = 0.962194 - 0.025 = 0.9372$$

which reproduces the direct calculation exactly.

Now raise the coupon. With $c = 0.07$ and the same maturity,

$$\text{CF} = \left[0.035 + 0.623167 + \frac{0.07}{0.06} \times 0.376833 \right] - 0.035 = 1.0628$$

Above 1, as it must be for a coupon above 6%.

Finally, take an odd quarter. With $c = 0.05$, $n = 8$, $z = 3$, so $v = 3$ and κ is unchanged at 0.623167:

$$\text{CF} = \frac{1}{1.03^{0.5}}\,[0.962194] - 0.025 \times \frac{3}{6} = 0.985329 \times 0.962194 - 0.0125 = 0.9356$$

reproducing the direct construction at the end of Example 6.3. The bracket is that example's \$96.2194 per \$100, the leading factor is its discount back over the odd quarter, and the final term is its accrued deduction. The formula is that construction written out once.

Coupon	Rounded maturity	Conversion factor
5%	8 years	0.9372
7%	8 years	1.0628
5%	8 years 3 months	0.9356

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Why the factors do not equalize perfectly. Two approximations are now visible. The factors are computed at a *fixed* 6% yield, so they correctly equalize the basket only when the curve is flat at 6%; at any other level the scaling is systematically off, and increasingly so the further the curve sits from 6%. And the maturity rounding discards up to three months of a bond's life, which distorts factors slightly and in a direction that varies across the basket. Both errors are small but not zero, and the residue decides which bond is cheapest to deliver.

6.4 THE CHEAPEST-TO-DELIVER BOND

Because conversion factors equalize the basket only approximately, some bonds end up cheaper to deliver than others. The short, who chooses, will deliver whichever is cheapest—and since the contract is then priced off that bond, identifying it comes first.

For each candidate the short compares the cost of buying the bond against what it receives on delivery. Ignoring carry, the relevant quantity is the **gross basis**:

$$\text{gross basis} = \text{quoted bond price} - \text{futures price} \times \text{conversion factor} \tag{6.6}$$

and the **cheapest-to-deliver** (CTD) bond is the one that minimizes it.

Example 6.5 Identifying the cheapest-to-deliver bond

The futures price is 110.50 and three bonds are deliverable:

Bond	Quoted price	Conversion factor	Gross basis
A	98.50	0.8912	0.02240
B	104.25	0.9430	0.04850
C	112.75	1.0195	0.09525

For bond A, $98.50 - 110.50 \times 0.8912 = 98.50 - 98.4776 = 0.0224$; the others follow the same way. Bond A is cheapest to deliver, at a cost 2.24 cents per \$100 of face above the futures invoice.

The differences look trivial (seven hundredths of a point separates A from C), but on the \$100 million position of Example 6.8 the gap between choosing correctly and incorrectly is \$72,850.

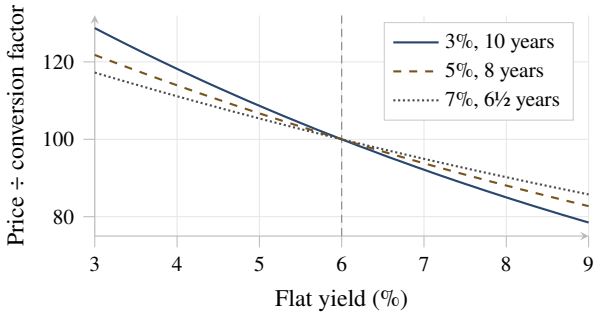


Figure 6.2: Price divided by conversion factor for three deliverable bonds of the ten-year basket, at a flat yield.

Rules of thumb. Two regularities recur. When yields are above the 6% notional coupon, low-coupon long-maturity bonds tend to be cheapest to deliver; when yields are below 6%, high-coupon short-maturity bonds tend to be. The reason is duration. A factor fixes each bond's value as if it yielded 6%; when yields are higher, every bond is worth less than that, and the long-duration bonds fall furthest below their factor, while when yields are lower every bond is worth more, and the short-duration bonds rise least above it. Since yields have been below

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6% for most of the contract's recent history, the CTD has usually sat at the short end of the basket. Figure 6.2 draws this for three bonds of the ten-year basket, the middle one being the 5% bond of Example 6.3. Divided by its conversion factor, each bond's price is exactly 100 at a 6% yield, so the three lines cross there, and the cheapest to deliver, the lowest line, switches from the short high-coupon bond to the long low-coupon one as yields rise through 6%. Near 6% the lines sit close together and a small move can change which is lowest, which is why the quality option described below is worth most when yields are near the notional coupon.

Carry and the net basis

The gross basis of (6.6) ignores what happens between now and delivery. Meanwhile the bond's holder earns its coupon and pays repo to finance it; the difference is **carry**:

$$\text{carry} = \text{coupon income over the holding period} - \text{repo financing cost} \tag{6.7}$$

and the quantity that actually measures the cost of the delivery option is the **net basis**:

$$\text{net basis} = \text{gross basis} - \text{carry} \tag{6.8}$$

Example 6.6 computes both for one deliverable bond.

Example 6.6 Carry and net basis

A deliverable bond carries a 4.5% coupon, is quoted at 101.20, and can be financed at a repo rate of 4.30% for the 90 days to delivery. Its gross basis is 0.0500 per \$100 of face.

Coupon income accrues on an actual/actual basis over 90 days:

$$4.50 \times \frac{90}{365} = 1.1096 \text{ per \$100 face}$$

Repo interest is charged at actual/360, here on the quoted price for simplicity (strictly it is charged on the cash price, accrued interest included):

$$101.20 \times 0.0430 \times \frac{90}{360} = 1.0879$$

so carry is $1.1096 - 1.0879 = +0.0217$, and from (6.8)

$$\text{net basis} = 0.0500 - 0.0217 = 0.0283$$

That 0.0283 is not a residual to be explained away: it is what the short's delivery options cost. A trader who buys this bond and sells futures against it acquires those options and pays 0.0283 per \$100 of face for them (0.028% of the cash price, or about 11 basis points annualized over the 90 days). The gross basis of 0.0500 confuses that price with the carry; the net basis separates them, and is the number a basis trader watches.

The sign of carry depends on whether the coupon exceeds the financing rate. In the low-rate decade carry was reliably positive and the distinction was a refinement; once short rates rose above the coupons on older bonds, carry turned negative for much of the deliverable basket, and net basis exceeded gross basis. Any statement about basis-trade profitability that ignores carry is incomplete, including the deliberately simplified Example 6.8 below.

The same ingredients give a theoretical futures price. Set the delivery options aside and treat the contract as a forward on the cheapest-to-deliver bond, which the short could buy today, finance in repo, and deliver. The bond's forward price is its cash price (quoted price plus accrued interest) carried to delivery at the repo rate, less the value of any coupon paid in between, as for an asset with known income (Section 5.3). Deducting the accrued interest the long will pay separately at delivery gives a quoted forward price, and dividing by the conversion factor turns that into a futures price, since the invoice of (6.4) pays the futures price times the factor. Example 6.7 carries it out.

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Example 6.7 The futures price from the cheapest-to-deliver bond

Take the bond of Example 6.6 as the cheapest to deliver, with a conversion factor of 0.9180. Its last coupon was 60 days ago, none falls in the 90 days to delivery, and accrued interest is counted as there.

Today's cash price is the quoted price plus 60 days of accrued interest:

$$101.20 + 4.50 \times \frac{60}{365} = 101.20 + 0.7397 = 101.9397$$

Carried to delivery at the 4.30% repo rate (actual/360), it becomes the bond's forward cash price:

$$101.9397 \times \left(1 + 0.0430 \times \frac{90}{360}\right) = 103.0356$$

At delivery the long pays 150 days of accrued interest separately, so the quoted forward price is

$$103.0356 - 4.50 \times \frac{150}{365} = 103.0356 - 1.8493 = 101.1863$$

and the theoretical futures price is

$$\frac{101.1863}{0.9180} = 110.2247$$

The gross basis of 0.0500 in Example 6.6 implies a market futures price of $(101.20 - 0.0500)/0.9180 = 110.1852$, which is 0.0395 lower. Multiplied by the conversion factor, the gap is 0.0363 per \$100 of face: the net basis of 0.0283, plus 0.0080 of repo interest on the accrued interest, which that example left out for simplicity. The futures contract trades below its forward-based value by what the short's delivery options are worth.

The delivery options

The short's right to choose which bond to deliver is a genuine option, and it is not the only one. Four are conventionally distinguished, all belonging to the short and all therefore depressing the futures price below what a no-arbitrage calculation on today's CTD alone would give.

The quality option is the one just described: the right to deliver any bond in the basket. Its value comes from the possibility that the CTD *changes* before delivery. Since the identity of the CTD depends on the level and shape of the curve, a large enough move switches it, and the short benefits from switching to whichever bond has become cheapest. The option is therefore worth more when the basket is wide and yields sit near the 6% notional coupon, where small moves can flip the ranking, and worth less when one bond is cheapest by a wide margin.

The timing option is the right to choose which business day of the delivery month to deliver. When carry is positive the short prefers to wait, earning more coupon than financing costs; when carry is negative it prefers to deliver early. Since the sign of carry is known, this option is worth little in practice—but it is not zero, because carry can change.

The wild card option arises from a mismatch in trading hours. The futures settlement price is struck at 2:00 p.m. Chicago time, but the short's notice of intention to deliver is not due until 6:00 p.m., while the cash market's New York session runs to 5:00 p.m. New York time (4:00 p.m. in Chicago). So the invoice amount is fixed for four hours, of which the first two have a liquid cash market to act in. If bond prices fall in that window, the short buys the deliverable cheaply and delivers it against the higher invoice already fixed at 2:00. It recurs every day of the delivery month, so a small daily edge adds up.

The end-of-month option extends the same idea: futures trading ends some days before the last delivery date, so for those days the invoice price is fixed while the cash market moves.

None of these is large individually. Collectively they are the reason a Treasury futures contract cannot be priced as a simple forward on the current cheapest-to-deliver bond, and the reason that basis traders speak of being short an option rather than holding an arbitrage.

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6.5 THE CASH–FUTURES BASIS TRADE

The relationship between a Treasury futures contract and its cheapest-to-deliver bond gives rise to a trade that has grown from an arbitrage niche into a staple of financial stability reports: buy the cash bond, sell futures against it, finance the bond in repo, and hold to delivery.

The profit is worth stating carefully, because the sign is easy to get wrong. The trade pays away the gross basis (you bought at the cash price and will receive the futures invoice) and earns the carry along the way. Its profit and loss (P&L) to delivery is therefore

$$\text{profit} = \text{carry} - \text{gross basis} = -\,\text{net basis} \tag{6.9}$$

using the net basis of Section 6.4. The trade is profitable precisely when the *net basis is negative*, which happens when the futures are rich relative to the cash bond, the condition that the asset-manager duration demand of Section 6.3 tends to create. Equivalently, and as Section 5.10 puts it, it is profitable when the implied repo rate exceeds the actual repo rate.

A *positive* net basis means the reverse: the cash-and-carry loses, and what it loses is the value of the short's delivery options from Section 6.4. That is the textbook state of affairs, and it is why the trade is not simply a riskless profit waiting in the market.

Two forces therefore set the sign, and they push opposite ways. The delivery options belong to the short and *depress* the futures price, driving the net basis positive. Demand for duration in futures form (from asset managers, and from liability-driven investors who want exposure without balance sheet) makes futures *rich*, driving it negative. Which dominates is an empirical matter, and since the 2010s the demand side has usually won in the liquid front contracts. That is what turned this from an occasional arbitrage into a standing business large enough to be a financial-stability question, and it is also why the trade is not riskless: the sign it depends on is set by a balance of flows, not by an identity.

Figure 6.3 sets out the three legs: the bond bought in the cash market, the futures sold against it, and the repo that finances the position until delivery closes all three.

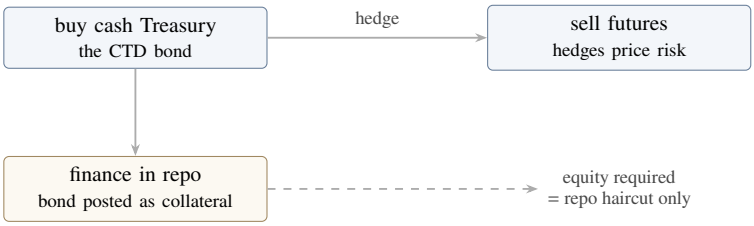


Figure 6.3: Structure of the cash–futures basis trade.

Example 6.8 Leverage in the basis trade

A fund buys \$100 million face of a cheapest-to-deliver bond at 98.50 and sells the corresponding futures. Asset-manager demand has left the futures rich, so the net basis is −0.0180 per \$100 of face. From (6.9) the position earns, held to delivery,

$$\$100{,}000{,}000 \times \frac{0.0180}{100} = \$18{,}000$$

On its own this is unimpressive: \$18,000 on a \$98.5 million position is 1.8 basis points. What transforms it is financing: the repo lender advances the bond's value less a haircut (Section 4.1), which the borrower funds itself. Treasury repo haircuts are small (take 2%), so the fund posts

$$0.02 \times \$98{,}500{,}000 = \$1{,}970{,}000$$

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of its own money and borrows the rest. The return on equity is

$$\frac{18{,}000}{1{,}970{,}000} = 0.91\%$$

over the life of the trade. If that is three months, the annualized return on equity is roughly 3.7%—from a position whose market risk is hedged.

The leverage is close to fifty times. At a 1% haircut it would be a hundred times.

Why it matters beyond the fund

Example 6.8 explains both why the trade is attractive and why regulators watch it. The economics work only at scale and with high leverage, and hedge funds' aggregate position has been estimated in the high hundreds of billions of dollars. Three features make that uncomfortable:

- **The leverage is procyclical.** Repo haircuts widen under stress. A move from 2% to 4% halves the position a given equity can support, forcing sales when markets are illiquid.
- **Margin on the futures leg moves the other way.** A sharp rally raises variation margin calls on the short futures position, and that margin must be met in cash even though the offsetting gain on the bond is unrealized. This is the mechanism of Chapter 8, appearing here.
- **Everyone is in the same trade.** The positions are concentrated in a small number of funds running similar strategies, so unwinding is correlated.

Case Study 6.2 March 2020

In the second week of March 2020, the US Treasury market (the world's deepest securities market, and the benchmark risk-free asset) stopped functioning properly. Bid–ask spreads widened by an order of magnitude, and on some days prices for on-the-run (most recently issued) and off-the-run (older) issues diverged without precedent.

The causes were several: a global dash for cash, foreign official selling, and dealer balance sheets already at their limits. A widely cited contributor was the forced unwinding of leveraged basis positions. As volatility spiked, haircuts rose and futures margin calls landed, and funds sold cash Treasuries into a market that could not absorb them, which pushed cash prices further from futures, widened the basis, and triggered more selling.

The Federal Reserve ended the episode by buying Treasuries on a scale not previously contemplated, roughly $1 trillion within weeks. The lesson is that a trade genuinely hedged against *price* risk can still fail catastrophically through *funding* risk, and leverage that looks prudent at normal haircuts does not survive stressed ones.

The clearing response

The principal regulatory response has been to push Treasury cash and repo activity into central clearing, under the SEC rules that Section 2.11 sets out with their compliance dates. Their intended effect is the multilateral netting of Example 2.4, together with uniform margin standards in place of bilaterally negotiated haircuts.

For this trade the rules bite through the financing leg. Once the repo that funds the position is cleared, its haircut comes from the clearing house's published margin model rather than from a negotiation with one dealer, and that puts a ceiling on the leverage in Example 6.8: a haircut that a dealer might set at 1% for a good client becomes a number the clearing house sets for everyone, and the example's fifty-to-a-hundred-times leverage needs a small haircut. The offsetting benefit is netting (a fund whose futures and repo positions are margined together

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posts against the net exposure, not the gross), and the practical question, unresolved as this is written, is which of the two effects dominates for a trade whose economics are a few basis points.

6.6 HEDGING A SWAP BOOK WITH THE STRIP

Section 6.2 named dealers hedging swap books as the largest source of SOFR futures volume. Chapter 7 treats swaps; only their outline is needed here. An interest rate swap exchanges a fixed rate for a floating one on a notional principal, period by period, and only the difference between the two payments changes hands. In each period the party that receives fixed is paid the fixed rate and pays the floating rate set for that period, here SOFR compounded over the quarter, so it gains when rates fall and loses when they rise. The party that pays fixed holds the opposite position.

For a dealer that has received fixed, the exposure is not a single number: each accrual period of the swap carries its own forward rate risk, and the futures strip lets those be hedged period by period. Example 6.9 hedges the first year of a two-year swap this way.

Example 6.9 Hedging the front of a swap

A dealer has received fixed on a \$500 million two-year swap with quarterly floating periods. Consider the first four accrual periods, each with $\tau = 0.25$.

A one basis point rise in any one period's forward rate costs the dealer about

$$\$500{,}000{,}000 \times 0.0001 \times 0.25 = \$12{,}500$$

before discounting. At \$25 per basis point per contract, the dealer sells

$$\frac{\$12{,}500}{\$25} = 500 \text{ contracts}$$

against each accrual period, 2,000 across the first year.

That match needs the dates to line up, and usually they do not. It assumes the swap's accrual periods coincide with the futures' reference periods, which holds only if the swap starts on one of the quarterly IMM dates (the third Wednesdays of March, June, September, and December, on which those reference periods begin and end). Suppose instead the first period runs from May 10 to August 10: 92 days, of which 39 fall before the third Wednesday of June and 53 after. The March contract covers the first stretch and the June contract the second, so the period's \$12,500 of sensitivity is split between them in proportion to days,

$$\frac{39}{92} \times \$12{,}500 = \$5{,}299 \to 212 \text{ March}, \qquad \frac{53}{92} \times \$12{,}500 = \$7{,}201 \to 288 \text{ June}$$

still 500 contracts, but spread across two delivery months. The residual is real: the two contracts do not move identically, so the hedge retains a small basis between the two reference periods, and a dealer whose book is mostly off-IMM swaps carries it on every one.

Three qualifications follow. Discounting reduces the required position slightly for later periods, since the \$12,500 arrives at the end of the accrual period rather than today. The futures hedge the futures rate, which by Section 30.3 exceeds the forward rate, so the hedge inherits convexity risk that must be managed separately. And as Figure 6.1 showed, each contract's sensitivity decays through its reference quarter while the swap's does not, so the ratio must be revisited within the period rather than only at the roll.

Beyond two years or so the strip thins out and dealers switch to hedging with other swaps or with Treasury futures, accepting the basis risk that comes with hedging a swap exposure using a government bond instrument. That basis, the swap spread of Section 7.9, is itself a traded risk.

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6.7 DURATION-BASED HEDGING

The most common use of Treasury futures is to adjust the interest rate exposure of a bond portfolio without trading the bonds themselves.

Chapter 4 gave the first-order relationship $\Delta B \approx -BD\Delta y$. Applying it to both the portfolio and the futures contract, and requiring the changes to offset, gives the number of contracts to short:

$$N^* = \frac{P\,D_P}{F\,D_F} \tag{6.10}$$

where P is the value of the portfolio, D_P its duration at the hedge horizon, F the contract value of one futures contract, and D_F the duration of the cheapest-to-deliver bond at contract maturity.

Example 6.10 Hedging a bond portfolio

A fund holds \$10 million of bonds with a duration of 6.8 years and wishes to hedge with ten-year note futures. The futures price is `110-16`, so with \$100,000 face per contract,

$$F = \frac{110.50}{100} \times \$100{,}000 = \$110{,}500$$

The cheapest-to-deliver bond has a duration of 7.6 years at contract maturity. From (6.10),

$$N^* = \frac{10{,}000{,}000 \times 6.8}{110{,}500 \times 7.6} = \frac{68{,}000{,}000}{839{,}800} = 80.97$$

so the fund shorts 81 contracts.

Check: a one basis point rise in yields costs the portfolio about $10{,}000{,}000 \times 6.8 \times 0.0001 = \$6{,}800$, while 81 contracts gain about $81 \times 110{,}500 \times 7.6 \times 0.0001 = \$6{,}802$.

What this hedge does not do. Equation (6.10) immunizes against a *parallel* shift only. A portfolio hedged this way remains exposed to changes in the shape of the curve (the bucketed exposures of Section 29.10) and to the possibility that the CTD bond changes, which alters D_F discontinuously. It also inherits the limitation of duration itself: it is a first-order approximation, and Chapter 4's convexity term reappears for large moves.

Hedging to a target duration

Managers more often want to *change* duration than to eliminate it. If the target duration is D_T rather than zero, the required position becomes

$$N^* = \frac{P\,(D_P - D_T)}{F\,D_F} \tag{6.11}$$

which is positive when duration must be reduced (short futures) and negative when it must be raised (long futures); Example 6.11 works the second case.

Example 6.11 Raising portfolio duration

The fund of Example 6.10 decides its \$10 million portfolio should carry a duration of 9.0 years rather than 6.8, because it expects yields to fall. From (6.11),

$$N^* = \frac{10{,}000{,}000\,(6.8 - 9.0)}{110{,}500 \times 7.6} = \frac{-22{,}000{,}000}{839{,}800} = -26.2$$

The negative sign means buying: the fund goes long 26 contracts.

The alternative, selling short-dated bonds and buying long ones, achieves the same duration but requires \$10 million of transactions, several days of execution in less liquid issues, and realized gains or losses on every position touched. The futures route requires margin on 26 contracts and

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takes seconds.

Futures contracts expire while the exposures they hedge often do not, so a five-year liability hedged with three-month contracts is stacked and rolled, as Section 3.9 describes: each roll crosses a bid–ask spread and is made at whatever spread between the two contracts prevails on the day, which is not known in advance. Section 3.10 tells the Metallgesellschaft episode, in which such a hedge failed through roll costs and margin calls while the hedged position performed as designed, the same funding and roll risk that recurs in Section 6.5.

6.8 BUILDING THE FRONT OF THE CURVE

Section 4.8 bootstrapped a zero curve from OIS quotes. In practice the region from roughly three months to five years is pinned down instead by the futures strip, because that is where the liquidity concentrates. The procedure has one wrinkle, and it is the subject of Chapter 30.

Each contract gives a futures rate for a specific three-month period. Converting the strip into forward rates requires subtracting the convexity adjustment of (6.3); only then can the forwards be chained into discount factors. Example 6.12 does so for three contracts.

Example 6.12 From futures quotes to forward rates

Three consecutive contracts settle at 95.72, 95.55, and 95.41, implying futures rates of 4.28%, 4.45%, and 4.59% for periods beginning in 1.00, 1.25, and 1.50 years. Take a short-rate volatility of 1.00% per year and apply the Ho–Lee adjustment $\frac{1}{2}\sigma^2 T_1 T_2$ from Section 30.3:

T_1	T_2	Futures rate	Adjustment	Forward rate
1.00	1.25	4.280%	0.63 bp	4.2738%
1.25	1.50	4.450%	0.94 bp	4.4406%
1.50	1.75	4.590%	1.31 bp	4.5769%

For the first contract, (6.3) gives $\frac{1}{2} \times 0.0001 \times 1.00 \times 1.25 = 0.0000625$, or 0.63 basis points; the others follow.

At this point on the curve the corrections are under a basis point and a curve builder might be tempted to skip them. The temptation should be resisted, because the adjustment grows with the square of maturity: by the eight-year contract it reaches 33 basis points.

Beyond the liquid futures region, the curve is completed with swap quotes, and the two must be made consistent; that is the simultaneous calibration described in Section 7.6.

SUMMARY

Interest rate futures come in two families: short-term contracts settle in cash against an overnight rate averaged over a reference period, and Treasury futures require delivery of a bond from a defined basket. Short-term contracts are quoted as $100 - R$, so a trader expecting higher rates sells; because the settlement rate accrues through the reference period, a contract's sensitivity decays toward zero across it, and hedges must be rebalanced within the quarter rather than rolled on one date. Bonds are quoted clean, with accrued interest added to reach the cash price.

Each contract's users explain much of its behavior. The SOFR strip serves borrowers hedging funding, dealers hedging the front of their swap books, and macro traders positioning on the policy path. Treasury futures serve asset managers and liability-driven investors moving duration, dealers hedging inventory, and mortgage holders rebalancing negative convexity, and the steady demand for duration in futures form tends to leave futures rich to the cash bond.

Conversion factors set at a 6% notional coupon make the basket only approximately equivalent, so one bond is cheapest to deliver and the contract tracks it, and which bond that is

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switches as yields cross the notional coupon. The futures price is that bond's forward price divided by its conversion factor, less the value of the short's options to choose the bond and the delivery day; the net basis, the gross basis less carry, is what those options cost.

The basis trade buys the cheapest bond, sells futures against it and finances it in repo. It earns a convergence of a few basis points, and thin repo haircuts make the leverage high, so it runs at scale and is fragile under stress. March 2020 showed that a trade hedged against price can still fail through funding, and central clearing of Treasury cash and repo is the main regulatory response.

Duration-based hedging sells $N^* = PD_P/(FD_F)$ contracts against a parallel shift and leaves curve reshaping, cheapest-to-deliver switches and convexity unhedged; a dealer instead matches the strip to a swap book period by period. Hedges that outlive their contracts must be rolled, which exposes the hedger to the roll spread every time. Finally, the strip anchors the front of the curve, once the convexity adjustment, negligible for near contracts and material for distant ones, is subtracted from the futures rates.

FURTHER READING

Burghardt and Belton's *The Treasury Bond Basis* is the standard practitioner treatment of conversion factors, the cheapest-to-deliver, and the delivery options, and it is the book to own if this material is your job. The Committee on the Global Financial System's *Repo Market Functioning* (2017) covers the financing leg that the basis trade depends on. On what happens when that financing withdraws, Duffie's Brookings paper on redesigning the Treasury market after March 2020 and the Financial Stability Board's thematic reviews of leverage in non-bank financial intermediation (2023–2024) are the two accounts to read together.

PRACTICE QUESTIONS

6.1 A Treasury bond is quoted at 104-20 with accrued interest of \$0.85 per \$100 of face. Express the quoted price as a decimal and give the cash price.

6.2 A trader buys 150 three-month SOFR futures at 96.10 and the contract settles at 96.34. Calculate the profit, state what happened to the implied rate, and explain why the contract is quoted as $100 - R$ in the first place.

6.3 Explain why the rate sensitivity of a three-month SOFR futures contract declines through its reference period, why this did not happen with Eurodollar futures, and what it implies for a hedger.

6.4 A bond pays a 6% coupon semiannually and has exactly 10 years to maturity after rounding. Calculate its conversion factor at the 6% notional coupon rate and explain the result. Then use equation (6.5) to calculate the conversion factor of a 4% coupon bond with 7 years and 3 months remaining after rounding.

6.5 The futures price is 108.75 and three bonds are deliverable, quoted at 96.20, 101.85, and 118.40, with conversion factors 0.8825, 0.9350, and 1.0880. Identify the cheapest-to-deliver bond.

6.6 Distinguish gross basis from net basis. A deliverable bond has a 3.0% coupon, is quoted at 94.60, can be financed at 4.40% for the 60 days to delivery, and has a gross basis of 0.0410. Calculate carry and the net basis, and comment on the sign.

6.7 A portfolio worth \$25 million has a duration of 5.4 years, ten-year note futures are priced at 109-00, and the CTD has a duration of 8.1 years at contract maturity. How many contracts should be shorted to immunize it? What position instead moves a \$40 million portfolio of duration 4.2 years to a target of 7.5 years, with futures at 111-00 and a CTD

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duration of 8.0?

6.8 Describe the cash–futures basis trade, and state where the profit comes from and where the leverage comes from.

6.9 A ten-year note futures contract settles at `112-08`. The delivered bond has a conversion factor of 0.9125 and accrued interest of \$0.60 per \$100. Calculate the invoice amount for one \$100,000 contract.

6.10 Explain why conversion factors do not make all deliverable bonds exactly equivalent, and state which bonds tend to be cheapest to deliver when yields are below 6% and why.

6.11 Explain why steady demand from asset managers for duration in futures form tends to leave Treasury futures rich relative to the cash bond, and connect this to Section 6.5.

6.12 A dealer has received fixed on a \$300 million swap with quarterly floating periods. How many three-month SOFR futures should it sell against each accrual period, ignoring discounting, and give two reasons the true hedge ratio differs.

6.13 Drawing on Sections 3.9 and 3.10, explain what a stack-and-roll hedge is and the two costs it incurs, and why a hedge of this kind can fail even when the underlying exposure behaves exactly as expected.

FURTHER QUESTIONS

6.14 A fund runs a \$500 million basis position at a net basis of -0.031 per \$100 with a repo haircut of 1.5%. Calculate the profit on full convergence, the equity required, the return on equity, and the leverage ratio. If the haircut widens to 4%, how much additional equity must the fund find, and what are its options if it cannot?

6.15 The basis trade is hedged against the price of Treasuries, yet it failed in March 2020. Explain precisely which risk was unhedged, identify the two distinct channels through which it materialized, and explain why both worsened simultaneously.

6.16 Explain why the short's delivery options depress the Treasury futures price. Would you expect the effect to be larger when the yield curve is flat or steep, and when yields are near 6% or far from it?

6.17 A dealer hedges a swaps book with SOFR futures. Explain three distinct reasons the hedge will not be exact, drawing on this chapter, Chapter 7, and Chapter 30.

6.18 Central clearing of Treasury cash and repo is intended to reduce systemic risk. Set out the mechanisms by which it should do so, and then argue the opposite case as strongly as you can. What would you want to know about the clearing house before deciding?

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Chapter 7
Swaps and the Multi-Curve Framework

The interest rate swap is the largest derivatives market in the world by notional outstanding, and it is the instrument through which almost every other rate derivative is hedged. It is also, for our purposes, the place where a simplification made in Chapter 4 has to be paid for.

That chapter spoke of "the" zero curve for a currency, as though a single term structure described it. The assumption was flagged as provisional, and this chapter retires it. A modern swap valuation requires *at least two* curves: one that says what cash flows are worth, and one that says how large the floating cash flows will be. They are built from different instruments, they move independently, and the gap between them is traded in large volume.

It would be easy to present this as a technical correction added to a simpler theory—which is how it entered the market, in the years after 2008, and how most textbooks still frame it. That framing is misleading. The single-curve world was not a simpler truth later refined; it was a coincidence. Two things happened to line up: the rate at which banks lent to each other for three months, and the rate at which a derivative's collateral earned interest. Once those separated, they never rejoined, and there is no reason they should have. This chapter therefore develops the two-curve structure as the standard case and treats the single-curve world as the special case it always was.

Sections 7.1 to 7.3 are the contract, why it exists, and how it is valued. Sections 7.4 to 7.6 are the two-curve structure: why the collateral agreement determines the discount curve, why one curve is not enough, and how the curves are built. Sections 7.7 to 7.9 are the instruments and prices that structure produces (basis swaps, cross-currency swaps and their basis, and swap spreads as the price of balance sheet), and Sections 7.10 to 7.12 close with credit, netting and clearing, what the transition did to the swap book, and the other swaps the market trades.

7.1 MECHANICS OF AN INTEREST RATE SWAP

A **swap** is an agreement to exchange cash flows on a schedule. In the standard interest rate swap, the **plain vanilla** swap, two parties agree to exchange, on a notional principal L, a stream of fixed interest payments for a stream of floating payments. The notional is never exchanged; it exists only to scale the payments, which is why an enormous notional outstanding corresponds to a much smaller amount of money at risk.

Since the transition described in Chapter 4, the floating leg of a standard swap in dollars, sterling, Swiss francs, yen and Canadian dollars references a compounded risk-free rate in arrears. Such a contract is the overnight indexed swap of Section 4.8: a **SOFR OIS** in dollars, a SONIA OIS in sterling, and so on for each currency's designated rate. In those currencies it is now the market standard rather than a specialist product. The euro is the large exception: EURIBOR survived the transition (Section 4.3), and most euro swaps still pay an annual fixed rate against six-month EURIBOR, with €STR OIS trading beside them; Australian, Hong Kong and Chinese swaps likewise still reference term rates. In dollars the name OIS once

meant a different contract. Before October 2020 the OIS curve was built from federal funds swaps, not SOFR, and Section 7.4 describes the switch; where this chapter writes OIS without qualification, the currency's current rate is meant. On each payment date the floating payer owes

$$L\,\tau_i\,R_i \tag{7.1}$$

where τ_i is the length of accrual period i as a fraction of a year, R_i is the compounded RFR over that period computed as in Section 4.4, and the fixed payer owes $L\,\tau_i\,K$ for the agreed fixed rate K. Only the difference changes hands. Figure 7.1 shows the arrangement.



Figure 7.1: Cash flows in a plain vanilla interest rate swap. The notional L is not exchanged.

Example 7.1 A swap's cash flows over its life

Company B enters a three-year annual swap on a notional of \$100 million, paying 4.20% fixed and receiving the compounded RFR. Suppose the compounded rate turns out to be 4.00% over the first year, 4.35% over the second, and 4.60% over the third. All figures below are in millions of dollars.

Payment date	Compounded RFR	Floating received	Fixed paid	Net to B
End of year 1	4.00%	+4.00	−4.20	−0.20
End of year 2	4.35%	+4.35	−4.20	+0.15
End of year 3	4.60%	+4.60	−4.20	+0.40
Total				**+0.35**

Three features of the table repay attention.

Only the net column changes hands: the gross \$4 million legs are never exchanged, and neither is the \$100 million notional.

The rate in the second column is not known at the start of the period it applies to. Under the conventions of Chapter 4 it accumulates from daily fixings across the year and is complete only at the end of the period, a few days at most before it is paid. Under the LIBOR conventions this table would have replaced (where each rate was fixed twelve months in advance), B would have known each payment a year ahead. This is the single largest operational change the transition brought.

B did well here only because rates rose. Had they fallen to 3.5%, B would have paid \$700,000 in that year. That is not a loss in any meaningful sense: B entered the swap to *fix* its cost, and Example 7.2 shows the combined position doing exactly that.

Conventions. A swap confirmation specifies far more than the fixed rate: the payment frequency on each leg (usually annual on both legs for an OIS, while swaps on the surviving term rates mix frequencies: annual fixed against semiannual EURIBOR in euros, quarterly against quarterly BBSW in Australian dollars), the day count on each leg, the business day convention, the calendar, and, critically after Chapter 4, the exact compounding convention and payment lag on the floating leg. Two swaps quoted at the same rate under different conventions are not the same trade, and the difference is not always small. Table 7.1 sets out a confirmation for a five-year dollar swap, and later sections return to it: Section 7.3 values it, and Section 7.4 shows that its last line, the collateral, is what fixes the curve used to do so.

The fixed rate is 4.31% rather than the 4.295% midpoint of the dealer's quote because the customer is paying fixed and therefore deals at the higher of the dealer's two rates, as the next

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Table 7.1: Terms of a five-year dollar swap. Illustrative, not a market quote.

Notional	$100,000,000
Effective date	March 18, 2026
Maturity date	March 18, 2031
Fixed rate	4.31%, paid by the customer
Fixed leg	Annual, actual/360
Floating index	SOFR, compounded daily in arrears
Floating leg	Annual, actual/360
Payment lag	Two business days
Business day convention	Modified following
Calendar	New York
Clearing	Cleared at a CCP
Collateral	Variation margin in USD cash, interest at SOFR

subsection explains. Every other line is a convention, and every one of them changes the cash flows.

How swaps are quoted

A dealer quotes a swap by quoting the *fixed* rate; the floating leg is understood to be the index flat, with no spread. A quote of "4.28 / 4.31" on the five-year means the dealer will receive fixed at 4.31% and pay fixed at 4.28%, and the three-basis-point difference is the bid–ask spread. The customer pays a slightly higher fixed rate than the mid when paying fixed and receives a slightly lower one when receiving. This is the usual arrangement, with the notable feature that the spread is quoted in rate rather than in price. Three basis points sounds negligible and is not: on the notional of Table 7.1 it is worth about $134,000 over the life of the trade, by the calculation Section 7.3 sets out.

Swaps are often traded on the IMM dates of Section 6.2, the standardized quarterly dates used by futures markets, so that a swap position can be hedged against futures without a date mismatch.

A particular class of standardized contracts, **market agreed coupon** swaps, fixes the coupon at a round number in increments of 25 basis points and settles the difference from the par rate as an upfront payment. These are one product among many rather than the general case: the swap of Table 7.1 is an ordinary par swap, struck at whatever rate makes it worth zero, and most of the market trades that way. What the round coupon buys is *fungibility*. Two par swaps struck a day apart carry different fixed rates and net against each other only in value; two market agreed coupon swaps of the same coupon and dates are the same instrument, so they compress and clear more cheaply. That matters most to dealers and to hedge funds and other active traders, who open and close positions often and want them to net; a company hedging one particular loan has no use for fungibility and trades a par swap matched to the loan. What it costs is cash on day one. Written that way, Table 7.1's swap would carry a 4.25% coupon (the nearest quarter-point to its 4.295% par rate), and the 4.5 basis point difference, multiplied by the swap's annuity of 4.48 (the present value of the year fraction accrued in each of its five years, about 1.014 under actual/360, discounted at the par rate; Section 7.3), gives

$$\$100{,}000{,}000 \times 0.00045 \times 4.48 = \$201{,}600$$

half as much again as the entire bid–ask. Because the coupon is the nearest quarter-point and not the one below, the payment runs in either direction: here the fixed payer makes it, having been given a coupon below the market rate, but at a par rate of 4.205% the same coupon would sit above par and the fixed payer would receive it. Half an increment also bounds it at inception (about $560,000 on this trade), though a standing coupon the market has since moved away

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from can exceed that.

The dealer running such a book is not taking a view. Having received fixed from one customer, it hedges by paying fixed elsewhere: to another customer, in the interdealer market, or with futures. What remains after hedging is a residual exposure to the shape of the curve, to the basis spreads of Section 7.5, and to the counterparty credit of Section 7.10. Managing those residuals, rather than predicting rates, is the business. The residual is measured bucket by bucket, as the dollar value of one basis point (DV01) of each quoted maturity separately: the book's change in value for a one-basis-point move in that maturity alone. This bucketed or key-rate DV01 shows exposure to the shape of the curve that a single parallel-shift number nets away. The short buckets are typically hedged with the futures of Chapter 6, the longer ones with swaps.

7.2 WHY SWAPS EXIST

A swap changes the interest rate character of an exposure without disturbing the contract that created it. That single sentence covers most of the market, but it is worth working through the four standard uses concretely, because the cash flow accounting is what makes the idea concrete.

Transforming a liability

This is the most common use. A company has borrowed at a floating rate and wants certainty about its interest cost. Figure 7.2 shows the three flows and where the floating rate cancels.

Example 7.2 Converting floating debt to fixed

Company B has a three-year loan of $50 million on which it pays the RFR plus 80 basis points. It would rather know its cost. It enters a swap on the same notional, paying 4.20% fixed and receiving the RFR.

Its three cash flows are now:

Pays on the loan	−(RFR + 0.80%)
Pays on the swap	−4.20%
Receives on the swap	+RFR
Net cost	−5.00% fixed

The RFR appears twice with opposite signs and cancels. B has converted a floating liability into a fixed one at 5.00%, or $2.5 million a year, without renegotiating the loan or informing the lender.



Figure 7.2: Company B pays its lender on the left and swaps on the right.

Note what has *not* changed. B still owes the loan, still bears the 80 basis point credit spread, and, as Section 7.10 discusses, has added a counterparty to its balance sheet. What it has removed is one specific risk: the level of the reference rate.

Transforming an asset

The same operation applies to the other side of the balance sheet, and its diagram is the mirror image of Figure 7.2, with every arrow reversed. An investor holding a floating-rate asset who wants a fixed return receives fixed and pays floating.

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Example 7.3 Converting a floating-rate asset to fixed

Company A holds \$50 million of floating-rate notes yielding the RFR plus 25 basis points, and wants a fixed return to match a fixed obligation. It enters a swap receiving 4.20% fixed and paying the RFR:

Receives on the notes	+(RFR + 0.25%)
Receives on the swap	+4.20%
Pays on the swap	−RFR
Net return	+4.45% fixed

A has synthesized a 4.45% fixed-rate asset out of a floating one.

These two examples are mirror images, and in principle A and B could face each other directly. In practice they do not, for the obvious reason that they do not know each other and would not want each other's credit.

The role of the financial intermediary

A dealer stands between them, as in Figure 7.3 and Example 7.4, entering offsetting swaps with each and quoting a bid–ask spread on the fixed rate, say 4.18 / 4.22, read as in Section 7.1: the dealer pays fixed at 4.18% and receives fixed at 4.22%.

Example 7.4 The dealer in the middle

The dealer pays A 4.18% fixed and receives the RFR; it receives 4.22% fixed from B and pays the RFR. The floating legs offset exactly, so the dealer is left with

$$4.22\% - 4.18\% = 4 \text{ basis points}$$

on \$50 million, or \$20,000 a year, for as long as both swaps run.

The all-in outcomes shift slightly. A now earns 4.18% + 0.25% = 4.43% rather than 4.45%, and B pays 4.22% + 0.80% = 5.02% rather than 5.00%. Between them they have paid the dealer four basis points for finding the other side and standing between them.

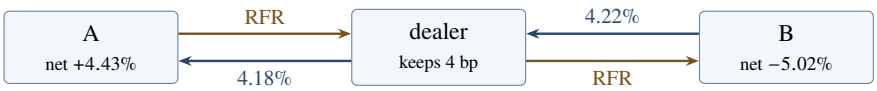


Figure 7.3: The dealer faces both customers. Its two floating legs offset, leaving the four basis point spread.

The dealer is not merely a matchmaker. It takes B's credit risk and A's credit risk, and it need not have found A and B simultaneously: in practice it warehouses one side, hedges the rate risk in the futures market of Chapter 6, and waits for offsetting business. What it earns is compensation for that warehousing and for the credit it has taken, not a broker's fee.

The comparative advantage argument, and what is wrong with it

The fourth traditional motivation is comparative advantage, and it deserves both a fair statement and a critique.

Suppose Company A and Company B can each borrow for five years in either market, on these terms:

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	Fixed	Floating
Company A (stronger credit)	5.0%	RFR + 0.3%
Company B (weaker credit)	6.4%	RFR + 1.0%
Difference	1.4%	0.7%

A is cheaper in both markets, but its advantage is larger in the fixed market (1.4%) than in the floating market (0.7%). Suppose further that A *wants* floating and B wants fixed, which is the inconvenient case, since each wants the market in which it is relatively worse off.

The apparent total gain is $1.4\% - 0.7\% = 0.7\%$. Splitting it evenly, each should end up 35 basis points better off than by borrowing directly. Example 7.5 shows how.

Example 7.5 Splitting the comparative advantage gain

A borrows fixed at 5.0%, its stronger market; B borrows floating at RFR + 1.0%. They then swap, with A receiving 5.0% fixed and paying RFR − 0.05%.

Company A:

$$\underbrace{-5.0\%}_{\text{loan}} + \underbrace{5.0\%}_{\text{swap received}} - \underbrace{(\text{RFR} - 0.05\%)}_{\text{swap paid}} = -(\text{RFR} - 0.05\%)$$

A borrows floating at RFR − 0.05%, against RFR + 0.30% directly, better by 35 basis points.

Company B:

$$\underbrace{-(\text{RFR} + 1.0\%)}_{\text{loan}} - \underbrace{5.0\%}_{\text{swap paid}} + \underbrace{(\text{RFR} - 0.05\%)}_{\text{swap received}} = -6.05\%$$

B borrows fixed at 6.05%, against 6.4% directly, also better by 35 basis points.

The two gains sum to the 0.7% predicted, and the RFR cancels for both parties.

The arithmetic is correct. The economics is not, and it is worth seeing why, because the flaw illustrates something general about apparent riskless profits.

The two rate pairs are not comparable commitments. A five-year fixed loan is a five-year commitment by the lender. A floating-rate loan is typically rolled at intervals, and the spread over the index is reset (or the facility withdrawn) if the borrower's credit deteriorates. The floating lender therefore has an option the fixed lender does not, and the weak borrower B pays a smaller *quoted* spread in the floating market precisely because the lender retains the right to walk away.

When B enters a swap to convert its floating borrowing into synthetic fixed, it does not thereby acquire a five-year fixed rate. It acquires a five-year swap plus a rolling floating loan that may be repriced against it. The "gain" is compensation for retaining exactly the risk that made the floating quote look cheap. If B's credit deteriorates, the swap keeps paying but the loan spread widens, and the synthetic fixed rate rises.

Comparative advantage is therefore a poor explanation of why the swap market is the size it is. The better explanation is the mundane one contained in Examples 7.2 and 7.3: swaps let institutions separate the interest rate character of an exposure from the funding decision that created it, and they do so with far less balance sheet than transacting in the underlying bonds would require.

Other users

Beyond the four classical uses, four groups account for much of the market's size.

Banks managing the gap. A bank whose deposits reprice overnight while its loan book or securities portfolio is fixed is structurally exposed to rates rising, and closes that mismatch

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with swaps rather than by changing its lending. This is one of the largest sources of end-user demand. Silicon Valley Bank in March 2023 is the case of a bank that left the mismatch largely open: it held long fixed-rate securities against deposits that could leave overnight, the 2022 rise in rates cut the securities' value well below cost, and the deposits left when depositors saw it.

Mortgage hedgers. A mortgage servicer or a holder of mortgage-backed securities loses duration when rates fall, because borrowers refinance, and gains it when rates rise. To stay hedged it must pay fixed in swaps as rates rise and receive as they fall, which is buying high and selling low on a large scale; in a sharp move that flow pushes swap rates further in the direction they were already going.

Pension funds and insurers matching liabilities. Long-dated liabilities discounted at market rates create enormous duration exposure. Receiving fixed in long-dated swaps lengthens asset duration toward liability duration without buying bonds, which is why, as Section 7.9 notes, this demand compresses swap spreads, and why the 2022 crisis in UK pension funds' liability-driven investment (LDI) strategies, which Chapter 8 examines, was driven by collateral calls on swap and repo positions rather than by losses on bonds held outright.

Issuers arbitraging their own funding. A company that can issue a fixed-rate bond cheaply but prefers floating exposure issues the bond and swaps it. This is what remains of the comparative advantage idea: the gain is real, but it comes from the issuer's access to a particular investor base, not from a mispricing between two markets.

What makes all of this practical for a nonfinancial company is hedge accounting. A swap is a derivative and would otherwise be marked to market through earnings each quarter while the debt it hedges sits at cost, so the hedged position would show volatility it does not have. Under IFRS 9 and its US counterpart, ASC 815 (Section 3.8), a swap documented as a hedge of specific debt has its gains and losses recognized alongside the hedged item, and reported earnings show the fixed or floating cost the company chose.

7.3 VALUING A SWAP

There are two equivalent ways to value a swap, and it is worth having both.

As an exchange of bonds. A swap is a long position in one bond and a short position in another. The fixed payer is short a bond paying coupon K and long a floating-rate note.

As a portfolio of forward contracts. Each exchange of payments is an FRA of the kind valued in Section 4.9. The swap is their sum. This is the more useful view, because it extends directly to the multi-curve case.

Take the second route, and write DF_i for the discount factor to payment date t_i and F_i for the forward rate for accrual period i. The ith exchange alone is an FRA: the fixed payer owes $L\tau_i K$ and is owed $L\tau_i R_i$, and valuing it means substituting the forward rate F_i for the unknown R_i and discounting, so it is worth

$$f_i = L\,\tau_i\,(F_i - K)\,\mathrm{DF}_i \tag{7.2}$$

which is the FRA value of Section 4.9 seen from the other side, there written for the party receiving fixed, here for the party paying it. Summing (7.2) over the payment dates gives the swap:

$$V_{\text{payer}} = L\sum_{i=1}^{n}\tau_i\,F_i\,\mathrm{DF}_i \;-\; L\,K\sum_{i=1}^{n}\tau_i\,\mathrm{DF}_i \tag{7.3}$$

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The second sum is the **annuity**

$$A = \sum_{i=1}^{n} \tau_i \, \mathrm{DF}_i \tag{7.4}$$

the present value of one unit of fixed rate received on the swap's schedule; multiplied by $0.0001L$ it is the present value of one basis point of fixed rate on the notional, which is why it is also called the PV01 leg. It will be used repeatedly: it is what converts a difference between rates into an amount of money, it is the swap's sensitivity to its own fixed rate, and Chapter 29 will take it as a numeraire (the unit in which other prices are measured) in its own right. A five-year annual swap discounted at 4%, with each year fraction taken as one, has $A \approx 4.45$, so a one-basis-point move in the par rate changes the value of $100 million of notional by about $44,500.

Setting $V = 0$ and solving gives the **par swap rate**, the fixed rate at which a new swap has zero value:

$$S = \frac{\sum_{i=1}^{n} \tau_i \, F_i \, \mathrm{DF}_i}{\sum_{i=1}^{n} \tau_i \, \mathrm{DF}_i} \tag{7.5}$$

so the par rate is a weighted average of the forward rates, with the discount factors as weights. From (7.3) and (7.5) the value of an existing swap has the compact form

$$V_{\text{payer}} = L \, (S - K) \, A \tag{7.6}$$

which says that a swap is worth the difference between the current par rate and its own fixed rate, annuitized. Most swap risk management reduces to this expression.

A telescoping identity—and a warning

Something convenient happens if the floating index and the discount curve come from the *same* curve. In that case the forward rate satisfies $\tau_i F_i = \mathrm{DF}_{i-1}/\mathrm{DF}_i - 1$ by definition, with $\mathrm{DF}_0 = 1$ for today, so

$$\sum_{i=1}^{n} \tau_i F_i \, \mathrm{DF}_i = \sum_{i=1}^{n} (\mathrm{DF}_{i-1} - \mathrm{DF}_i) = 1 - \mathrm{DF}_n \tag{7.7}$$

The whole floating leg collapses to $L(1 - \mathrm{DF}_n)$: the forward rates cancel, and only the first and last discount factors survive. This is the classic result that a floating-rate note prices at par on its reset dates.

Remember this identity, and remember equally that it is *false in general*. It holds only when the rate being forecast is the same rate being used to discount. That was approximately true before 2008 and is not true now. Section 7.5 shows what replaces it.

Example 7.6 Valuing an existing swap

A three-year annual swap on a notional of $100 million pays fixed at 4.00% and receives the compounded RFR. Using the OIS curve bootstrapped in Chapter 4, the discount factors are 0.961538, 0.920936, and 0.881164, and the corresponding forward rates are 4.0000%, 4.4088%, and 4.5136%.

Floating leg. Discounting each projected payment,

$$100\big[0.040000 \times 0.961538 + 0.044088 \times 0.920936 + 0.045136 \times 0.881164\big] = \$11{,}883{,}596$$

The telescoping identity (7.7) gives the same answer in one step, to the rounding in the forward rates, since index and discount curve agree here:

$$100\,(1 - 0.881164) = \$11{,}883{,}600$$

Fixed leg. The annuity is $A = 0.961538 + 0.920936 + 0.881164 = 2.763638$, so the fixed leg

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is worth

$$100 \times 0.04 \times 2.763638 = \$11{,}054{,}552$$

Value. To the fixed payer the swap is worth 11,883,600 − 11,054,552 = \$829,048.

Checking with (7.6): the par rate is $(1 - 0.881164)/2.763638 = 4.30\%$, so the swap is worth $100 \times (0.0430 - 0.0400) \times 2.763638 = \$829{,}091$, agreeing to rounding. The payer is receiving a floating stream now worth 4.30% while paying only 4.00%.

7.4 THE COLLATERAL AGREEMENT DETERMINES THE DISCOUNT CURVE

We now come to the question the previous section left open: discount at *what*?

The answer is not a matter of preference. Consider a swap that is fully collateralized under a CSA, the standard arrangement described in Chapter 8. Each day, the party for whom the swap has negative value posts collateral equal to that value, and receives interest on the posted collateral at a rate specified in the CSA, almost always the overnight rate of the collateral currency.

Now think about what a dealer holding such a position actually owns. If the swap's value rises by \$1, the dealer receives \$1 of collateral, on which it owes the poster the overnight rate until the position reverses. The dealer may use that cash, so the gain on the swap is financed by a liability that accrues at the collateral rate: the position is, in cash flow terms, self-financing at the overnight rate. Replicating it therefore costs the overnight rate, and by the arbitrage arguments of Chapter 5, the correct discount rate for a fully collateralized derivative is **the rate paid on its collateral**.

This has a consequence that surprises people the first time they meet it: the same contract, with the same cash flows, has a different value under different CSAs. A swap collateralized in dollars earning SOFR and an otherwise identical swap collateralized in euros earning €STR are not worth the same. The CSA is part of the instrument.

A cleared swap, such as the one in Table 7.1, works the same way with the clearing house's rulebook in place of a CSA: variation margin is paid in cash and earns price alignment interest at the overnight rate, so cleared dollar swaps are discounted on the SOFR curve. Bilateral CSAs change more slowly, and some older dollar CSAs still pay the effective federal funds rate; trades under them are discounted on a federal funds curve, a live case of the same contract carrying two values.

Case Study 7.1 The day discounting changed

For years after the crisis, cleared swap markets discounted at the effective federal funds rate in dollars and the euro overnight index average (EONIA) in euros, because those were the rates paid on cleared collateral. When the market moved to SOFR and €STR, the collateral rate changed—and therefore so did the value of every cleared swap.

The clearing houses did not simply re-mark and move on. LCH (originally the London Clearing House) and CME executed the dollar switch over a weekend in October 2020, changing the discounting rate on their entire cleared portfolios at once. Because the change transferred value between members (some portfolios gained, others lost), each house paid cash compensation to make members whole, and additionally distributed basis swaps to offset the change in risk profile that the discounting switch produced. The euro switch, in July 2020, needed only the cash: EONIA had been redefined in 2019 as €STR plus a fixed 8.5 basis points, so the spread between the old and new discount rates was known for every future date and left no basis risk to hand back.

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> Two lessons follow. First, the choice of discount curve is not an accounting convention; it is worth enough money that changing it required a coordinated, compensated, market-wide operation. Second, discounting risk is a real risk: a portfolio can be flat to rates and still gain or lose from a change in the rate its collateral earns.

Uncollateralized trades. If a swap is not collateralized, as many corporate trades are not, the replication argument above does not apply, because the dealer must fund the position from its own resources. The discount rate then involves the dealer's funding cost, which introduces the funding valuation adjustment of Chapter 9. It also means the trade's value depends on *who is holding it*, which is uncomfortable but true.

7.5 WHY ONE CURVE IS NOT ENOUGH

We now have two distinct roles for a curve, and no reason for one curve to fill both.

- The **discount curve** answers: what is a cash flow at time t worth today? Section 7.4 settled this: it is determined by the collateral rate in the CSA, and is built from OIS.
- A **forecast curve** answers: how large will the floating payment be? This is determined by whatever index the contract references, which may be the same overnight rate, or term SOFR, or a foreign currency's RFR, or a legacy fallback rate.

Figure 7.4 sets out the division of labor. When the index and the collateral rate coincide, the two curves coincide and everything collapses to the single-curve formulas, including the telescoping identity (7.7). When they do not, the curves must be built and used separately, and Example 7.7 measures the error of letting one curve do both jobs.

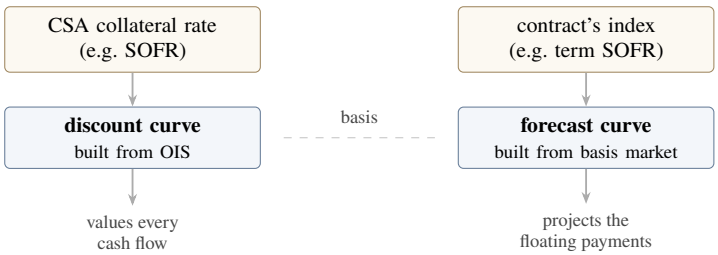


Figure 7.4: The two roles a curve plays. Before 2008 the same curve filled both.

The basis is not noise

The gap between the two curves is the **curve basis**, and it is a traded quantity with its own market. Historically the largest example was the spread between LIBOR and OIS, which measured the credit and liquidity premium in interbank lending. Before 2007 it sat at a small number of basis points and was widely ignored; in late 2008 it exceeded 350 basis points. A dealer valuing a swap book on the assumption that the two were the same would have been wrong by an amount that dwarfed every other modeling choice in the portfolio.

That particular basis has gone with LIBOR. Others remain and are structural rather than crisis-driven:

- **Term versus compounded.** Term SOFR is a forward-looking rate derived from futures; compounded SOFR is backward-looking. They are not equal, and the difference is traded,

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though narrowly: under the ARRC's scope-of-use rules, term SOFR derivatives are confined to end users hedging loans and other cash products that reference it, and the dealers who face them lay the exposure off in term SOFR against SOFR basis swaps.

- **Tenor basis.** Where a currency retains term indices of different lengths, a swap referencing the shorter index against one referencing the longer trades at a spread. The euro is the main case: three-month against six-month EURIBOR is quoted as a basis swap (the "3s6s" basis), and euro swap systems carry a separate forecast curve for each tenor. The economics is the same as the old LIBOR tenor basis: rolling a short index carries different risk than fixing a long one.
- **Cross-currency basis.** The subject of Section 7.8.

Example 7.7 Where the single-curve identity breaks

Take the swap of Example 7.6, but suppose the floating leg references an index whose forward rates sit 10 basis points above the OIS forwards (a term rate, say). The forecast forwards become 4.1000%, 4.5088%, and 4.6136%, while the discount factors, being set by the CSA, are unchanged.

The floating leg is now

$$100\big[0.041000 \times 0.961538 + 0.045088 \times 0.920936 + 0.046136 \times 0.881164\big] = \$12{,}159{,}960$$

against \$11,883,596 from the same calculation in Example 7.6. The difference of \$276,364 is exactly the basis times the annuity:

$$100 \times 0.0010 \times 2.763638 = \$276{,}364$$

The telescoping identity (7.7) would have given \$11,883,600 and been wrong by more than a quarter of a million dollars—because the rate being forecast is no longer the rate being used to discount. The par swap rate rises from 4.30% to 4.40%: the 10 basis point basis passes through one-for-one into the par rate.

7.6 BUILDING THE CURVES

Curve construction in a multi-curve world proceeds in a definite order, because the pieces depend on one another.

Step 1: the discount curve. Bootstrap from OIS quotes, exactly as in Section 4.8. These instruments have a single index, and that index is the collateral rate, so the construction is self-contained.

Step 2: the forecast curves. For each index the book references, build a curve that reprices the instruments referencing that index, while discounting them on the curve from Step 1. This is the step that makes the procedure genuinely multi-curve: the forecast curve is defined by what it must reprice, given a discounting choice already fixed.

Step 3: check consistency. Basis swaps quoted between two indices must be repriced by the pair of forecast curves. In practice this is not a sequential bootstrap but a simultaneous calibration: the curves are solved jointly so that every quoted instrument (OIS, index swaps, basis swaps, futures with their convexity corrections from Chapter 30) is repriced at once. Example 7.8 carries the three steps through by hand.

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Example 7.8 Two curves from two sets of quotes

Payments are annual and $\tau_i = 1$ throughout, so the construction can be followed by hand.

Step 1: the discount curve. OIS quotes are 4.00%, 4.20%, and 4.30% for one, two, and three years. An OIS forecasts the same rate it discounts at, so its floating leg telescopes and the par condition is simply $S \sum_i \mathrm{DF}_i = 1 - \mathrm{DF}_n$. Solving in order,

$$\mathrm{DF}_1 = \frac{1}{1.04} = 0.961538, \qquad \mathrm{DF}_2 = \frac{1 - 0.042 \times 0.961538}{1.042} = 0.920936$$

$$\mathrm{DF}_3 = \frac{1 - 0.043 \times (0.961538 + 0.920936)}{1.043} = 0.881164$$

which are the discount factors of Example 7.6. The forward rates this curve implies are 4.0000%, 4.4088%, and 4.5136%.

Step 2: the forecast curve. The book also holds swaps referencing a twelve-month term index, quoted at 4.10% for one year and 4.32% for two. These are not OIS: the index is not the collateral rate, so the floating leg does not telescope and the forwards have to be solved for, while discounting on the Step 1 curve, which is already fixed. The one-year swap gives $F_1 = 4.1000\%$ directly. The two-year swap then gives F_2 from

$$0.0410 \times 0.961538 + F_2 \times 0.920936 = 0.0432 \times 1.882475$$

where $1.882475 = \mathrm{DF}_1 + \mathrm{DF}_2$ is the two-year annuity factor, so $F_2 = 4.5497\%$. The term index is forecast 10.00 basis points above the overnight rate in the first year and 14.09 in the second. The basis is not flat, and no single curve can carry both sets of forwards.

Step 3: the consistency check. A two-year basis swap pays the term index and receives the overnight rate plus a spread b. Setting its value to zero,

$$b = \frac{(0.0410 - 0.0400) \times 0.961538 + (0.045497 - 0.044088) \times 0.920936}{1.882475}$$

which comes to 0.001200, or 12.00 basis points. That is what the market quotes, so the pair of curves is consistent.

Notice what that check did *not* establish. With one schedule and one discount curve, b is forced to the difference between the two par rates, 4.32% − 4.20% = 12 basis points, and the basis swap has told the calibration nothing the outright quotes had not. Basis swaps are needed where that equality breaks (legs on different payment frequencies, or maturities and indices at which no outright swap trades), which is the usual reason a curve is pinned by basis quotes at the long end and by outright quotes at the short end.

Two practical warnings carry over from Chapter 4 and become sharper here. Interpolation choices matter more, because errors in one curve propagate into every other through the basis instruments. And the resulting object is not unique: two desks calibrating to identical quotes with different interpolation and different instrument selection will produce curves that agree at the quoted points and disagree between them. That disagreement is a genuine source of valuation difference, not a rounding artifact.

7.7 BASIS SWAPS

A **basis swap** exchanges two floating legs. Neither leg is fixed; the contract trades the spread between two indices, quoted as a number of basis points added to one leg to make the swap worth zero at inception.

Basis swaps serve two purposes. Commercially, they let an institution move exposure from one index to another without taking a directional rate position: a bank whose assets reference one index and whose liabilities reference another closes the mismatch with a basis swap, not with an ordinary swap. A euro borrower whose loan resets on three-month EURIBOR while its existing hedge pays six-month uses one to align the two, and a dealer that has written term SOFR

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swaps to loan borrowers lays off the term-against-overnight exposure the same way. Technically, they pin down the forecast curves of Steps 2 and 3 above wherever no outright swap on an index trades, and there the relationship between the curves would otherwise be unobservable. Within a single currency the euro 3s6s basis is now the main example; Section 33.9 returns to basis swaps as a product and to what their spreads price.

The spread is not arbitrary. It reflects the genuine economic difference between the two indices: differences in credit content, in the term over which the rate is fixed, and in the supply and demand for exposure to each. When the two indices are close substitutes the spread is small and stable; when they differ in credit character it widens under stress, exactly when it matters.

7.8 CROSS-CURRENCY SWAPS AND THE CROSS-CURRENCY BASIS

A **cross-currency swap** exchanges payments in two different currencies. Unlike a single-currency swap, the notional principals *are* exchanged, at the start and again at maturity, because the two legs are denominated in different money and there is nothing to net. Figure 7.5 shows the structure.



Figure 7.5: A EUR/USD cross-currency basis swap, from the perspective of a party raising dollars against euros. Notional is exchanged at both ends.

The economic content is a collateralized loan of one currency against another. A European bank needing dollars can borrow euros at home and swap them for dollars for five years, returning the dollars and reclaiming the euros at maturity. It is, in substance, secured dollar borrowing.

Three kinds of user account for most of the market. Banks outside the United States use it, as here, to fund dollar lending and dollar securities. Companies and supranational issuers borrow in whichever market is cheapest for them, often by issuing dollar or euro bonds, and swap the proceeds and the coupons back into the currency their revenues are in. And investors holding foreign bonds, such as a Japanese insurer holding US corporate debt, swap the foreign coupons and principal into their own currency so that the bond's return is not swamped by the exchange rate.

The cash flows are easiest to see in the fixed-for-fixed version, where neither leg floats, as in Example 7.9.

Example 7.9 Cash flows on a fixed-for-fixed currency swap

A European company raises dollars for five years. At a spot rate of 1.10 dollars per euro it pays €100 million and receives $110 million at the start, pays 4.2% on the dollars and receives 2.5% on the euros annually, and reverses the principal exchange at maturity. Amounts are in millions.

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Year	USD cash flow	EUR cash flow
0	+110.00	−100.00
1	−4.62	+2.50
2	−4.62	+2.50
3	−4.62	+2.50
4	−4.62	+2.50
5	−114.62	+102.50

The interest amounts are $110 \times 4.2\% = 4.62$ and $100 \times 2.5\% = 2.50$; year 5 adds the principal back.

Compare this with Example 7.1. Three differences stand out.

The principal is exchanged, twice. In a single-currency swap the two notionals are identical and cancel, so exchanging them would be pointless. Here they are in different money and cannot be netted, so they must actually change hands—and because they do, the contract carries exchange rate risk that an interest rate swap does not.

Nothing nets. Both legs are paid gross, in their own currency, on every date.

It is a bond issued and a bond bought. Read the dollar column alone and it is exactly the cash flow of issuing a $110 million 4.2% bond; the euro column is exactly that of buying a €100 million 2.5% bond. That is the whole instrument: borrow in one currency, lend in the other.

The exchange rate risk sits overwhelmingly in the final row. The annual interest amounts are small and partly offsetting, but the $114.62 million repayment against €102.50 million received is where a move in spot is felt, which is what Example 7.11 quantifies.

What actually trades in volume is the floating-for-floating version of Figure 7.5, where each leg pays its own currency's overnight rate and the whole price of the transaction is compressed into a single spread: the basis.

Two distinctions matter in practice. Between dealers the standard contract is the **mark-to-market** (resettable) cross-currency swap, in which the dollar notional is reset to the current spot equivalent at each payment date and the difference is paid in cash, so the exchange rate exposure that Example 7.11 measures is settled period by period instead of accumulating to maturity; Section 33.9 prices the difference. The constant-notional swap of the examples here is what a company hedging a fixed-size bond issue or foreign investment uses. And a cross-currency swap is not an **FX swap**, which is a spot exchange of currencies reversed by a forward exchange, with no interest payments in between: the interest differential is built into the forward points of Section 5.7. FX swaps are by far the larger market, concentrated at maturities under a year, and are how banks and investors fund and hedge foreign-currency positions from day to day; cross-currency swaps carry the same funding trade out to many years.

The cross-currency basis

Textbook theory says the two floating legs should exchange flat: covered interest parity holds, and any spread would be arbitrageable by borrowing in one currency, swapping, and lending in the other.

Since 2008 this has been persistently false. A spread, the **cross-currency basis**, is quoted on one leg, and for the euro, yen, Swiss franc and sterling against the dollar it has been negative for most of the period since, meaning the party *providing dollars* is paid a premium. (The Australian and New Zealand dollars, whose banks and companies borrow abroad and swap the proceeds home, have often run a positive basis: there the scarce side is the local currency.) The basis widened dramatically during the 2008 crisis and again during the 2011–12 European sovereign crisis, and although it narrowed afterward it did not return to zero. It also spikes predictably at quarter-ends and year-ends.

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Case Study 7.2 The end of covered interest parity

Covered interest parity is not a theory about expectations or risk preferences. It is an arbitrage relationship: borrow dollars, convert to euros at spot, invest in euros, and sell the euro proceeds forward for dollars. If the resulting dollar return differs from the dollar interest rate, someone has been handed a riskless profit.

For decades the relationship held to within a basis point or two. Since 2008 it has failed continuously—not by a basis point but by tens of basis points, for years at a time, in the most liquid currency pairs in the world.

The explanation is not that traders stopped noticing, and it has two phases. In 2008 and again in 2011–12 the basis measured stress: banks outside the United States needed dollars, their credit was in doubt, and dollar lenders would supply them only at a premium. Dollar swap lines between the Federal Reserve and other central banks, opened in 2008 and widened again in March 2020, are what capped those spikes. The persistence afterward has a different cause: the arbitrage requires balance sheet. Post-crisis regulation (the leverage ratio of Section 1.12 in particular, phased in from 2015) made it expensive for banks to expand their balance sheets to capture small spreads. Against that limited supply sits steady demand: Japanese and European banks, insurers and pension funds holding dollar assets fund or hedge them through FX and cross-currency swaps, and they pay the premium. The trade is still there; no one wants to hold it. This is also why the basis widens at reporting dates, when balance sheet is scarcest.

The episode is an instance of the limits of arbitrage set out in Section 5.15.

Example 7.10 The cost of a negative basis

A European company swaps €100 million into dollars for five years at a cross-currency basis of −20 basis points. It has lent euros and borrowed dollars, so it pays SOFR flat on the dollar leg and *receives* €STR less 20 basis points on the euro leg. Receiving 20 basis points less than the euro overnight rate is precisely how it pays for the dollars.

With euro discount factors of 0.975610, 0.951814, 0.928599, 0.905951, and 0.883854, the euro annuity is

$$A_{\text{EUR}} = 4.645828$$

The present value of the 20 basis point spread is

$$100{,}000{,}000 \times 0.0020 \times 4.645828 = €929{,}166$$

So the negative basis is worth just under a million euros on this trade, paid by the party that wanted dollars, earned by the party that supplied them. It is precisely the quantity that a single-curve valuation would set to zero.

Valuation

A cross-currency basis swap is most easily valued as a pair of floating-rate notes: long a note in the currency lent, short a note in the currency borrowed, with the two converted at the current spot rate. At a reset date each note prices at par *provided* its spread equals the current market basis; when the market basis has moved, the off-market spread is worth its difference times the annuity, exactly as in Section 7.3.

Example 7.11 Valuing a cross-currency swap after the market moves

The company of Example 7.10 entered its five-year swap three years ago: it lent €100 million and borrowed $110 million at a spot rate of 1.10 dollars per euro, receiving €STR less 20 basis points.

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Two years remain. The euro discount factors for those two years are 0.975610 and 0.951814, so the remaining annuity is $A_{\text{EUR}} = 1.927424$.

Case 1: only the exchange rate has moved. Spot is now 1.05, and the market basis is still −20 basis points. Because the contract's spread equals the market spread, the euro note prices at par and so does the dollar note:

$$V = €100{,}000{,}000 - \frac{\$110{,}000{,}000}{1.05} = 100{,}000{,}000 - 104{,}761{,}905 = -€4{,}761{,}905$$

The company borrowed dollars and the dollar has strengthened, so the liability has grown in euro terms.

Case 2: the basis has also widened. Suppose in addition the market basis has moved to −45 basis points. The contract pays €STR less 20, where the market now requires €STR less 45, so the euro note pays 25 basis points *more* than the market demands and is worth more than par. The company receives that leg, so the move is in its favor:

$$100{,}000{,}000\,(1 + 0.0025 \times 1.927424) = €100{,}481{,}856$$

and the swap is worth

$$100{,}481{,}856 - 104{,}761{,}905 = -€4{,}280{,}049$$

The widening basis has offset €481,856 of the exchange rate loss, because the company locked in dollar funding when dollars were cheaper than they are now.

Two lessons follow. The exchange rate move is an order of magnitude larger than the basis move (€4.76 million against €0.48 million), which is why cross-currency swap books are managed first as FX exposures and only then as basis exposures. And the basis is a genuine mark-to-market risk in its own right: a position that is fully hedged in both currencies' rates and in spot FX still moves when the basis does.

Valuation also requires a decision about the collateral currency, because Section 7.4 applies here too. A trade collateralized in dollars and a trade collateralized in euros have different values even with identical cash flows. Many CSAs permit the poster to choose among several eligible currencies, which makes the collateral a rolling option to deliver whichever currency is cheapest, an option that must itself be valued, and one that Section 8.5 takes up where it belongs, alongside the rest of collateral practice.

7.9 SWAP SPREADS AND THE PRICE OF BALANCE SHEET

The **swap spread** is the difference between the swap rate and the yield on a government bond of the same maturity. For most of the swap market's history it was positive, and the explanation was obvious: the swap curve carried bank credit risk through its LIBOR index, while the government curve did not, so the swap rate stood above the government yield by roughly the market's assessment of that credit difference.

Dollar swap spreads have now been negative for years: at thirty years since late 2008, at ten years since the middle of the 2010s, and, since the switch to SOFR, across most of the curve. The switch itself lowered the series, because a SOFR swap carries no bank credit at all, so spreads measured before and after 2021 are not directly comparable. Read literally, this says the market accepts a lower fixed rate from a bank than from the sovereign (that banks are better credits than the government), which would be an extraordinary claim, and is not what it means.

The resolution is the argument of the previous section, transplanted. Example 7.12 sets out the trade a negative swap spread appears to offer.

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Example 7.12 The arbitrage no one wants

The ten-year swap rate is 4.00% and the ten-year government bond yields 4.10%, so the swap spread is −10 basis points. The apparent trade is:

1. Buy $100 million of the ten-year bond, yielding 4.10%, funding it in the repo market at approximately the overnight rate.
2. Pay fixed at 4.00% in a ten-year swap, receiving the compounded overnight rate.

The floating leg received on the swap approximately offsets the repo cost of funding the bond. What remains is a fixed 4.10% received against a fixed 4.00% paid: 10 basis points, apparently for no market risk at all. With a ten-year annuity of 8.111 at 4%, the present value is

$$100{,}000{,}000 \times 0.0010 \times 8.111 = \$811{,}100$$

The trade is real, and it sat available for years. What it requires is a balance sheet: the bond must be held and financed. Under the leverage ratio (Section 1.12), holding $100 million of government bonds consumes capital whether the position is hedged or not. A dealer weighing $811,100 against the capital that position ties up for a decade may reasonably decline.

So a negative swap spread is not a statement about relative creditworthiness. It is the price of the balance sheet needed to close it—the same phenomenon, measured a different way, as the cross-currency basis of Section 7.8. Heavy government bond supply pushes yields up; regulation makes the offsetting arbitrage expensive; and demand from pension funds and insurers to receive fixed in swaps, which requires almost no balance sheet, pushes swap rates down. The spread settles wherever those forces balance, and there is no reason that point should be positive.

Two recent regulatory changes bear directly on it. In November 2025 the US banking agencies recalibrated the enhanced supplementary leverage ratio for the largest banks, in force since April 1, 2026, with the stated aim of making it a backstop rather than a binding constraint; the more balance sheet that releases for holding Treasuries, the less negative swap spreads need be. And the SEC's Treasury clearing mandate of Section 2.11, which applies to cash trades from December 31, 2026 and to repo from June 30, 2027, lets dealers net cleared Treasury and repo positions against one another, which also reduces the balance sheet the trade consumes. Relative-value funds that do put the trade on, financed in repo, carry the two risks a dealer avoids: that the spread moves further against them before it converges, and that their repo funding is withdrawn. In March 2020 both happened at once.

Swap spreads, like the cross-currency basis, are the limits of arbitrage of Section 5.15 at work, and they can stay dislocated for years without anyone behaving irrationally.

7.10 CREDIT RISK, NETTING, AND CLEARING

A swap creates credit exposure, but of an unusual shape. Unlike a loan, where one party is always the creditor, a swap can have positive value to either side depending on where rates go. Exposure is therefore *two-sided* and changes sign over the contract's life.

It is also far smaller than the notional suggests. The notional is never exchanged; what is at risk is the swap's replacement cost (what it would cost to replace the contract on current terms if the counterparty defaulted), which by (7.6) is $L(S - K)A$. A $100 million ten-year swap that has moved 50 basis points in your favor exposes you to roughly $4 million, not $100 million. This is why aggregate notional statistics (hundreds of trillions) are a poor guide to systemic exposure.

Three mechanisms reduce what remains.

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Netting. The ISDA Master Agreement provides for close-out netting, which Section 1.5 and Example 1.1 set out, cherry-picking included. Netting enforceability is a legal question decided jurisdiction by jurisdiction, which is why China's Futures and Derivatives Law, in force since August 2022, mattered so much to the dealers trading with Chinese counterparties: it was the first Chinese statute to recognize close-out netting. Until a jurisdiction settles the question, the netted figure a dealer reports is not necessarily the figure that would survive its counterparty's bankruptcy court.

Collateral. A credit support annex requires the out-of-the-money party to post collateral against current exposure, reducing it toward zero between margin calls. Section 7.4 showed the valuation consequence; Chapter 8 examines what posting collateral does to liquidity, which is a different and sometimes larger problem.

Central clearing. Post-crisis reforms require standardized swaps to be cleared, which novates each one to a central counterparty by the mechanism of Section 2.4. Bilateral credit risk is replaced by exposure to the CCP, and Chapter 2 examines what that concentration achieves and what it costs.

The same reforms changed how swaps are traded and recorded. In the United States, standard swaps that the regulator has declared "made available to trade" must be executed on a swap execution facility (SEF), an electronic venue where requests for quotes go to several dealers at once, and the European Union imposes a matching trading obligation. Every swap, cleared or not, is reported to a trade repository, which is what makes the market's size measurable at all. Dealers periodically take part in portfolio compression, tearing up offsetting trades among many participants to cut gross notional without changing anyone's net risk (Section 33.12). And swaps that are not cleared carry mandatory initial margin under the uncleared margin rules of Section 8.4, which is deliberately more expensive than clearing and is one of the main reasons standard trades clear.

What survives all three is the residual counterparty risk priced as a credit valuation adjustment (CVA) in Chapter 9 and capitalized under the framework of Chapter 27. It is smaller than it once was, but it is not zero, and the 2021 episode discussed in Chapter 27 shows how quickly it returns when a portfolio is concentrated and margining is lax.

7.11 WHAT THE TRANSITION DID TO THE SWAP BOOK

Chapter 4 described the ISDA fallback mechanism in the abstract. It is worth seeing what it meant for an actual portfolio, because the swap book was where the transition would succeed or fail.

A dealer entering 2021 held millions of swaps referencing LIBOR, with maturities running decades out. Three things had to happen. New trades had to move to RFR references, which market convention accomplished by coordinated "big bang" dates on which interdealer trading switched. Cleared legacy trades were converted by the clearing houses before each LIBOR setting ended (sterling, yen and Swiss franc in December 2021, dollars in April and May 2023): each LIBOR swap was replaced by a standard OIS, with the fallback spread adjustment folded into the fixed rate and cash paid for the remaining differences in convention. Bilateral legacy trades fell back under the protocol.

The two routes left different things behind. A converted cleared swap is an ordinary OIS, indistinguishable from one traded new. A bilateral LIBOR swap that fell back references compounded SOFR plus the fixed spread of Table 4.2 for the rest of its life, under the protocol's own conventions (Example 4.1), so those spreads remain a structural feature of the legacy book. Because the spread is a constant over compounded SOFR, such a trade needs no curve of its

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own; what it needs is a booking and confirmation system that records it correctly, and the stock of these trades shrinks only as they mature or are restructured.

The broader observation is the one from Chapter 4, now visible at portfolio scale: the transition did not eliminate the multi-curve structure. It changed which curves the structure contains.

7.12 OTHER SWAPS

The vanilla structure generalizes in many directions, of which a few deserve naming here; Chapter 33 treats them in full.

Amortizing and accreting swaps vary the notional over time, typically to match an underlying loan (Section 33.1). **Forward-starting swaps** begin at a future date (Section 33.2); a company planning to issue a bond in six months pays fixed on one today to lock in the rate, and they are the underlying for the swaptions of Chapter 29. **Constant maturity swaps** reference a long-tenor swap rate on each reset; Section 33.5 describes them and Section 30.5 derives the convexity adjustment they need. **Inflation swaps** exchange a fixed rate for the realized change in a consumer price index and are how pension funds hedge inflation-linked liabilities (Section 33.10).

Equity swaps exchange the total return on an equity or index for a floating rate. **Total return swaps** extend the same exchange to any reference asset (a bond, a loan portfolio, a single stock), and they transfer the full economics of an asset without transferring the asset. Hedge funds use them to hold exposure they would otherwise have to finance and disclose, and banks provide them as a form of secured lending, which is why they are the instrument through which leverage accumulates quietly. Section 33.11 treats both, and Chapter 27 examines what total return swaps made possible in 2021.

Commodity swaps exchange a fixed price for a floating commodity index (Section 33.8), and are how airlines fix their fuel cost and producers their selling price for years ahead; **variance swaps** exchange a fixed variance for realized variance and are treated in Chapter 22. **Credit default swaps** (CDS), despite the name, are structurally quite different (one leg is a periodic premium, the other pays only if a specified issuer defaults) and belong to Chapter 25.

What unites them is the mechanism of this chapter: identify the cash flows, project the floating ones on the appropriate forecast curve, and discount all of them on the curve the collateral agreement dictates.

SUMMARY

An interest rate swap exchanges fixed for floating payments on a notional that is never exchanged; in most major currencies the standard contract is now an overnight indexed swap referencing a compounded risk-free rate in arrears, while euro swaps still reference EURIBOR. Valued as a portfolio of forward rate agreements, it is worth $L(S - K)A$, the gap between the current par rate and its own fixed rate, annuitized. The comparative advantage argument traditionally used to motivate swaps does not survive inspection, because the fixed and floating borrowing rates it compares are not comparable commitments: the floating lender retains the right to reprice or withdraw, and the apparent gain is compensation for that. Swaps are better explained by their ability to separate the interest rate character of an exposure from the funding decision that created it.

The discount curve is not a modeling choice. For a collateralized derivative the correct discount rate is the rate paid on its collateral, because that is what replicating the position costs; the same cash flows are therefore worth different amounts under different credit support annexes, which is why the move to SOFR discounting in October 2020 required coordinated compensation across cleared markets. Valuation needs at least two curves: a discount curve

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determined by the CSA and one or more forecast curves determined by the indices the contracts reference. Where the two coincide the floating leg telescopes to $L(1 - \mathrm{DF}_n)$ and the familiar single-curve results follow; where they do not, that identity fails and the error equals the basis times the annuity. Curves are built discount first, and in practice by simultaneous calibration.

Basis swaps exchange two floating legs, hedge the mismatch between indices, and are the instruments that pin down the forecast curves. Cross-currency swaps exchange payments and notionals in two currencies and are economically secured borrowing across currencies; their basis has been persistently nonzero since 2008, first as a measure of dollar funding stress and then because the arbitrage that would close it consumes balance sheet that post-crisis regulation made expensive, which is also why it widens predictably at reporting dates. Negative long-dated dollar swap spreads are the same phenomenon measured a second way: closing that gap means financing a bond, and the leverage ratio charges capital for it regardless of the position's risk.

Swaps create two-sided credit exposure far smaller than the notional, being the replacement cost $L(S - K)A$ rather than the principal. Close-out netting under the ISDA Master Agreement collapses a defaulted counterparty's trades to a single net amount and prevents cherry-picking; collateral under a CSA reduces the residual toward zero between margin calls; and central clearing replaces bilateral exposure with exposure to a CCP that nets multilaterally. What remains is priced as CVA (Chapter 9) and capitalized under the framework of Chapter 27.

FURTHER READING

Bianchetti's "Two Curves, One Price" (*Risk*, August 2010) is an early statement of the multi-curve framework, and Piterbarg's "Funding Beyond Discounting" (*Risk*, February 2010) derives the collateral-rate discounting result that the framework rests on. Henrard's *Interest Rate Modelling in the Multi-Curve Framework* is the book-length treatment. Du, Tepper, and Verdelhan (2018) is the standard reference on the cross-currency basis, and it is the empirical counterpart to this chapter's claim that the basis is a price rather than an error.

PRACTICE QUESTIONS

7.1 State the comparative advantage argument for swaps, explain the flaw in it, and give the better explanation for the size of the swap market.

7.2 A two-year annual swap on a notional of \$50 million pays fixed at 3.50% and receives floating. Discount factors are 0.9709 and 0.9426; forward rates are 3.00% and 3.30%. Value the swap to the fixed payer and find the par swap rate.

7.3 Show that when the floating index and the discount curve are the same, the floating leg is worth $L(1 - \mathrm{DF}_n)$, and state precisely the condition under which this fails.

7.4 Explain why the correct discount rate for a fully collateralized derivative is the rate paid on its collateral, and state what follows about two otherwise identical swaps collateralized in different currencies.

7.5 Repeat Question 7.2 assuming the floating index has forward rates 15 basis points above those given, with discount factors unchanged, and verify that the change in value equals the basis times the annuity.

7.6 Describe the order in which curves must be built in a multi-curve framework, explain why the discount curve must come first, and explain why a single currency needs more than one forecast curve.

7.7 Explain what a basis swap is, give two distinct reasons the market needs them, and explain why notional principals are exchanged in a cross-currency swap but not in a single-currency one.

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7.8 The eight-year swap rate is 3.60% and the eight-year government bond yields 3.72%. Compute the swap spread, describe the trade it appears to offer, calculate its present value on a $250 million position using an eight-year annuity of 6.85, and explain both what prevents the trade from being done and why a negative spread does *not* imply that banks are better credits than the government.

7.9 Explain why a swap creates two-sided credit exposure and why the exposure is far smaller than the notional. For the swap of Question 7.2, state the exposure of each party.

7.10 Explain what close-out netting achieves, what would happen without it, and why its legal enforceability is a jurisdiction-by-jurisdiction question.

FURTHER QUESTIONS

7.11 A five-year cross-currency swap on a notional of €200 million carries a basis of −35 basis points. Using the euro discount factors of Example 7.10, find the present value of the basis. Who pays it, and what are they paying for?

7.12 Covered interest parity is an arbitrage relationship, not a behavioral theory. Explain carefully why it has failed persistently since 2008, and why the failure is largest at quarter-ends and year-ends.

7.13 When clearing houses switched dollar discounting from the federal funds rate to SOFR in October 2020, they paid cash compensation to members and also distributed basis swaps. Explain what each of the two was compensating for, and why cash alone would not have been sufficient.

7.14 A dealer's swap book is flat to a parallel shift in rates but holds large positions in different tenor buckets and in two currencies. Identify at least three distinct risks the book still carries that Chapter 4's single-curve framework could not have represented.

7.15 A CSA permits the collateral poster to deliver any of three currencies. Explain why this makes the collateral an option, why that option has value, and how you would begin to quantify it. Which chapter's machinery would you need?

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Chapter 8
Collateral, Margin, and Funding

Several chapters have now deferred a question to this one. Chapter 2 showed margin calls transmitting stress across a market; Chapter 6 showed them threatening a hedged basis position; Chapter 7 showed the collateral agreement determining a derivative's discount curve. Each treated collateral as a means to some other end. This chapter treats it as the subject. The reason is a proposition that would have seemed eccentric twenty years ago and is now the organizing fact of derivatives risk management:

> A position can be perfectly hedged against price risk and still destroy the firm holding it, because the hedge demands cash on a schedule the hedged item does not supply.

This is not a hypothetical. In 2022 it happened three times in one year (in European energy, in LME nickel, and in UK pensions), to firms in unrelated businesses, none of which had misjudged the direction of any market. Sections 40.7, 40.8, and 40.12 tell all three episodes. The mechanism is always the same, and it lives entirely in the collateral arrangements that earlier chapters passed over.

Sections 8.1 to 8.3 cover the collateral agreement itself (the CSA, variation and initial margin, haircuts), Sections 8.4 and 8.5 the rules for uncleared trades and the optionality a CSA can carry, and Sections 8.6 to 8.8 what happens to collateral once posted (transformation, rehypothecation, segregation, disputes). Sections 8.9 and 8.10 work the proposition above through the 2022 energy and pension episodes and show how the resulting liquidity risk is measured and managed. Section 8.11 asks whether there is enough collateral in the system, and Section 8.12 says where all this leaves valuation, the starting point of Chapter 9.

8.1 THE CREDIT SUPPORT ANNEX

Derivatives between two parties are documented under an ISDA Master Agreement, which Section 1.5 described as providing close-out netting. Collateral is governed by the CSA, a separate document attached to it.

The CSA is where the economics of a trade are quietly decided. Its principal terms are:

- **Eligible collateral.** Which assets may be posted (cash in which currencies, which government bonds, occasionally corporate bonds or equities), and the haircut applied to each, the percentage by which its market value is marked down when it is counted (Section 8.3).
- *The interest rate paid on collateral.* Usually the overnight rate of the currency posted. Section 7.4 showed that this single line determines the discount curve for the whole trade.
- **Threshold.** An amount of exposure below which no collateral is required, in effect an unsecured credit line. The uncleared margin rules of Section 8.4 require a zero threshold on variation margin between the firms they cover, so a threshold now survives only on initial margin (the allowance of Section 8.4) and on legacy or uncovered CSAs.

- **Independent amount.** Collateral one party posts on top of whatever the mark-to-market requires, fixed in advance or set by formula. Before the uncleared margin rules it was how dealers took initial margin from hedge funds.
- **Rating triggers.** Credit-dependent terms: a threshold that falls, or an independent amount that rises, when a party is downgraded, so that a downgrade becomes a collateral call. Such clauses turned the 2008 downgrade of American International Group (AIG) into calls it could not meet (Section 40.9).
- **Minimum transfer amount.** A floor on the size of any single call, to avoid transferring trivial sums daily.
- *Frequency and timing.* How often the portfolio is valued and calls are made, how quickly a call must be met, and how disputes are handled.
- *Direction.* Whether both parties post (a two-way CSA) or only one (a one-way CSA).

These terms combine into a single number each day. The **credit support amount** is the collateral the receiver is entitled to hold:

$$\text{credit support amount} = \text{exposure} + \text{independent amount} - \text{threshold},$$

floored at zero, and the call is the difference between it and the collateral already held, valued after haircuts. Example 8.1 works one through.

Example 8.1 Computing a margin call

A dealer faces a corporate client under a CSA outside the uncleared margin rules. The client posts an independent amount of \$5 million and has a threshold of \$10 million; the minimum transfer amount is \$500,000, and calls are rounded up to the next \$100,000. The dealer holds collateral worth \$12.0 million after haircuts.

At today's valuation the portfolio is worth \$23.47 million to the dealer. The credit support amount is

$$23.47 + 5 - 10 = \$18.47\text{m},$$

so the shortfall is $18.47 - 12.0 = \$6.47$ million. That exceeds the minimum transfer amount, and rounding makes the call \$6.5 million. The dealer issues it on the valuation date and the client delivers within a business day or two, the settlement period the CSA sets. The dealer then holds \$18.5 million.

The next day the portfolio is worth \$23.2 million. The credit support amount falls to \$18.2 million and the dealer holds \$0.3 million more than it is entitled to, but that is below the minimum transfer amount, so nothing moves.

Even when every call is met, the dealer is collateralized only to within the threshold net of the independent amount: on the first day it held \$18.5 million against \$23.47 million of exposure. That gap of nearly \$5 million is the unsecured credit line Chapter 9 prices.

Two-way and one-way. In a one-way CSA the party that posts is traditionally the dealer, when the client is a sovereign, supranational, or large corporate with the bargaining power to insist. Since the post-crisis CVA charge and funding costs made such arrangements expensive for dealers, a number of sovereign debt offices have moved to two-way CSAs to obtain better pricing, but one-way arrangements remain common for sovereigns and supranationals, which the uncleared margin rules exempt.

One-way CSAs look like a commercial concession and are in fact a funding commitment. The dealer hedges the client trade with an offsetting trade that is itself collateralized. When the client trade moves in the dealer's favor, the hedge moves against it, so the dealer posts collateral

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on the hedge and receives nothing from the client; it therefore funds that collateral for the life of the trade, out of its own resources, at its own cost of funds. This is the origin of the funding valuation adjustment of Chapter 9, and it is why a trade that appears to be priced at the mid-market level can be substantially loss-making once the CSA is read.

Title transfer and security interest. Two legal forms exist. Under **title transfer**, collateral becomes the recipient's property outright, to be returned in equivalent form later; the recipient may use it freely, including by passing it on to a third party. Under a **security interest**, the poster retains ownership and the recipient holds a charge, a legal claim on the assets that it can enforce only if the poster defaults. The English-law CSA is a title transfer; the New York-law CSA creates a security interest, but its Paragraph 6(c) gives the recipient a right to use the collateral, including to rehypothecate it (Section 8.7), unless the parties disapply that right. The distinction is invisible until a default, at which point it determines whether the poster is an owner reclaiming property or an unsecured creditor in a queue.

8.2 VARIATION MARGIN AND INITIAL MARGIN

Two quite different things are both called margin, and conflating them is the most common error in this area.

Variation margin (VM) covers exposure that has *already arisen*. Each day the portfolio is revalued and the party for whom it has moved unfavorably transfers that amount. VM is a transfer of value that has already been lost; it does not make anyone worse off in economic terms, since the payer has suffered the loss whether or not any collateral changes hands.

Initial margin (IM) covers exposure that *might arise in the future*: specifically, the further move that could occur between a counterparty's last VM payment and the completion of close-out. It is posted at inception, recalculated daily by a model, and held, in a changing amount, until the trade matures. The distinction matters enormously for funding.

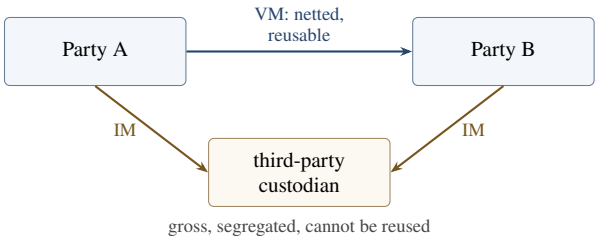


Figure 8.1: Variation margin passes between the parties; initial margin is posted gross to a custodian.

As Figure 8.1 shows, VM flows between the parties and is netted: A pays B or B pays A, never both. The recipient may use it, so it is not withdrawn from the system. IM is different: *both* parties post, *gross*, to a segregated account with a third-party custodian, and neither may touch it. The figure shows the bilateral arrangement under the uncleared margin rules; at a clearing house only the member posts initial margin. IM is therefore genuinely encumbered for the life of the trade, and it must be funded.

Example 8.2 Variation and initial margin on the same portfolio

A fund and a dealer hold a swap portfolio. At inception it is worth zero, and each posts initial margin of $25 million to a custodian. Neither can use the other's.

Rates then move and the portfolio is worth −$40 million to the fund. The fund pays $40 million

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of variation margin to the dealer, who may use it.

The two obligations are not comparable:

	Variation margin	Initial margin
Covers	loss already incurred	possible future move
Direction	one way, netted	both ways, gross
Amount	\$40m from fund	\$25m each
Reusable by recipient	yes	no
Economic cost to poster	none (value already lost)	funding spread, life of the trade

Posted IM is not idle in every sense: cash earns interest and bonds keep their yield. What it costs is the poster's funding spread over that return. At the 60 basis point spread Chapter 9 uses for this cost, the fund's IM costs it

$$\$25\text{m} \times 0.0060 = \$150{,}000 \text{ a year},$$

and the dealer bears the same on its own posting: a real cost of doing the trade that appears nowhere in the swap's valuation.

8.3 HAIRCUTS

Collateral is not counted at its full market value. A haircut (Section 4.1) discounts it, so that the collateral remains sufficient even if its own price falls before it can be liquidated.

If the haircut is h, then covering an exposure E requires collateral with market value

$$M = \frac{E}{1 - h} \tag{8.1}$$

Haircuts reflect the collateral's price volatility, its liquidity, and the time it would take to sell. Cash in the CSA's own currency carries none. Short-dated government bonds carry half a percent to two percent; long-dated government bonds more, because their prices move more; corporate bonds and equities considerably more. For firms that do not model their own, the uncleared margin rules of Section 8.4 prescribe a standardized schedule:

- cash, none;
- high-quality government debt, 0.5% up to one year to maturity, 2% from one to five years, and 4% beyond five;
- either of these in a currency other than the one the trade settles in, a further 8% currency add-on.

The currency add-on bears on the currency choice of Section 8.5. It is waived for cash variation margin in a major currency, where most of that option lives, but it applies to bonds and to initial margin posted in another currency. Example 8.3 shows what happens when a haircut widens.

Example 8.3 Haircuts and how they behave under stress

A fund must cover a £100 million exposure with gilts (UK government bonds) subject to a 4% haircut. From (8.1) it must deliver

$$\frac{100}{0.96} = £104.2 \text{ million of gilts}$$

Markets then become volatile and the haircut is raised to 10%. Covering the same £100 million now requires

$$\frac{100}{0.90} = £111.1 \text{ million}$$

an additional £6.9 million of gilts—on an exposure that has not changed at all.

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The example holds the exposure fixed; a real stress does not. When markets turn volatile enough to raise the haircut, three things happen together, and they compound. The exposure itself has probably grown, because the volatility that widened the haircut also moved the portfolio. The gilts have probably fallen in price, so the fund holds fewer pounds of collateral than it did. And the haircut has widened. The fund must find more assets, from a smaller pool, to cover a larger exposure. This is the same procyclicality Section 2.9 identified in initial margin models, operating through a second channel. A firm that stress-tests its margin requirements using *current* haircuts has tested the easy case.

8.4 UNCLEARED MARGIN RULES AND SIMM

Before the financial crisis, initial margin on bilateral derivatives was rare between dealers. Dealers did collect it from hedge funds, as independent amounts, but held it commingled with their own assets, which the failure of Lehman Brothers exposed. The post-crisis reforms of Section 2.11 pushed standardized products into clearing; for what remained bilateral, regulators imposed the uncleared margin rules (UMR), phased in over several years by size of counterparty and now applying broadly.

UMR requires variation margin on uncleared trades between covered firms, and initial margin as well once both groups hold more than €8 billion of uncleared derivatives notional ($8 billion in the United States), averaged over a prescribed window. Even then the parties may agree on a threshold of up to €50 million per counterparty group and post only the excess, to a segregated third-party custodian. The intent is that staying bilateral should not be a way to avoid posting collateral. Initial margin reached the largest dealers first and, in the final phase in September 2022, firms on the investing side such as asset managers and pension funds above the notional threshold. Some funds deliberately keep their uncleared notional below it so as never to post initial margin.

Calculating IM. Firms may use a model (which some regulators must approve) or a standardized schedule. Almost all use the model, because the schedule is punitive, and almost all use the same one: **ISDA SIMM**, an industry-standard methodology. The industry converged on one model deliberately, for a reason Section 8.8 explains. Adopting it is not automatic: in the European Union, the European Market Infrastructure Regulation as revised in 2024 brings in supervisory authorization of initial margin models, starting with the largest groups, and gives the European Banking Authority a central role in validating industry-wide models such as SIMM.

SIMM works from sensitivities rather than simulation. Each risk factor sensitivity is multiplied by a prescribed risk weight, and the weighted sensitivities are aggregated using prescribed correlations.

Example 8.4 A simplified SIMM calculation

A portfolio has interest rate sensitivity in two tenor buckets, the maturity ranges into which SIMM groups rate sensitivities:

Bucket	Sensitivity per bp	Risk weight (bp)	Weighted sensitivity
2-year	$80,000	60	$4.8m
10-year	$120,000	50	$6.0m

With a prescribed correlation of $\rho = 0.7$ between the two buckets, the margin is

$$\text{IM} = \sqrt{4.8^2 + 6.0^2 + 2 \times 0.7 \times 4.8 \times 6.0} = \sqrt{99.36} = \$9.97\text{m}$$

Compare this with the two figures that bracket it. With zero correlation the two charges would

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combine as $\sqrt{4.8^2 + 6.0^2}$ = \$7.68m. Simple addition of the weighted sensitivities gives \$10.8m, what would be required with no diversification recognized.

The diversification benefit here is modest, only 8% below the undiversified sum, because rate buckets are highly correlated. The benefit is larger where correlations are lower, as between interest rates and FX, which SIMM treats as separate risk classes within one product class. Across its four product classes (rates and FX, credit, equity, commodity) it recognizes none: their margins are simply added. Netting, too, stops at the edge of a relationship, since each counterparty margins only its own portfolio. This is why firms concentrate trades with fewer counterparties: splitting an offsetting book across two dealers means posting IM twice, on two portfolios each less diversified than the whole.

Cleared against bilateral

Because IM is model-driven, the same trade can require quite different amounts depending on where it is booked. Two features drive the gap.

The **margin period of risk** (the assumed time to close out a defaulted portfolio) is conventionally five days for cleared trades and ten for bilateral ones, on the reasoning that a CCP has a rehearsed auction process and a bilateral counterparty does not. Since margin scales roughly with the square root of the holding period, that alone makes bilateral IM about $\sqrt{10/5} = 1.41$ times larger.

Netting then compounds it. At a CCP a member's trades net across all its counterparties; bilaterally, each relationship nets only within itself. Example 8.5 puts numbers on both features.

Example 8.5 The same trade, cleared and bilateral

A fund executes a swap whose five-day 99% risk (the loss over five days that is exceeded with only 1% probability) is \$12 million.

Cleared. IM is \$12 million, posted once to the CCP. The dealer on the other side faces the CCP as well, but its leg offsets against the rest of its large cleared book and adds little to its own IM, so the trade's own cost is the fund's \$12 million.

Bilateral. The ten-day margin period raises it to roughly $12\sqrt{2}$ = \$17 million. Under UMR both parties post, gross and segregated. Other trades in the relationship have already used up its €50 million threshold, so each posts the full \$17 million and \$34 million is immobilized against the same economic risk.

Immobilized collateral is therefore nearly three times larger in the bilateral case. Without that offset (a dealer whose cleared book did not absorb the leg), the comparison would be \$24 million against \$34 million, and the gap would be the margin period's factor of 1.41 alone. Add the diversification point of Example 8.4 (the fund's other trades net at the CCP but not across separate bilateral relationships) and the difference widens further.

The rules were designed to make clearing cheaper than not clearing. The corollary is that genuinely bespoke hedges, which cannot be cleared, now carry a collateral cost that may exceed the value of the tailoring—which is why some hedges that would reduce a firm's risk are no longer worth doing.

Some firms were spared the clearing mandate altogether. European pension schemes were exempt from it, because clearing would have obliged them to hold cash for variation margin. The EU exemption expired in June 2023; the United Kingdom made its own permanent in 2025, so UK schemes may still keep their swaps bilateral.

The wider consequence of UMR is that initial margin has become a large, permanent, system-wide demand for high-quality assets, locked away with custodians where neither party may use it (Figure 8.1). Section 8.11 takes up that demand.

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8.5 CSA OPTIONALITY

Section 7.8 noted in passing that a CSA permitting several collateral currencies gives the poster an option. It deserves explicit treatment as a real position that must be valued and hedged.

Suppose a CSA permits cash in dollars, euros, or sterling, each earning its own overnight rate. The poster chooses which to deliver, and may switch. It will deliver whichever is cheapest to fund. Since the currencies' overnight rates and the cross-currency basis of Section 7.8 both move, the cheapest currency changes over time.

Because the collateral rate determines the discount curve, this is an option over the discount curve itself. The tools for valuing it come from two places: it is an option on the best of several rates, of the kind Section 34.10 treats, and comparing a foreign overnight rate with a domestic one requires the changes of measure of Sections 30.1 and 30.6. Neither alone is enough, which is part of why the position is so often left unpriced. Example 8.6 gives a first estimate of its size.

Example 8.6 The value of collateral choice

A dealer owes $1 billion on a portfolio under a CSA permitting dollars, euros, or sterling, so the dealer is the poster. Dollar collateral earns SOFR. To post euros instead, the dealer swaps dollars into euros: it lends dollars and receives SOFR, borrows euros and pays €STR plus the basis b (the quoting convention of Example 7.10), and earns €STR on the euros it posts. Its net return, in dollar terms, is

$$\text{SOFR} - (€STR + b) + €STR = \text{SOFR} - b$$

The euro overnight rate cancels, so what a currency's collateral earns relative to dollar collateral is its basis against the dollar with the sign reversed. With the basis at -15 basis points for euros and -10 for sterling, euro collateral earns the equivalent of SOFR plus 15 basis points, sterling SOFR plus 10, and dollars SOFR flat. The dealer posts euros, the cheapest currency to fund, and discounting at the euro collateral rate converted into dollars, rather than at SOFR, raises the effective discount rate by 15 basis points.

Take a portfolio annuity of 8, the annuity of Section 7.3: the present value of one unit a year over the portfolio's life. Posting euros rather than dollars then lowers the dealer's liability by roughly

$$\$1{,}000\text{m} \times 0.0015 \times 8 = \$12\text{m}$$

Using the annuity assumes the collateral balance stays at $1 billion over the portfolio's life, which overstates the value as a swap book amortizes.

This is only a first approximation: it prices the *current* cheapest currency, not the option to switch as relative rates move, which is worth more. Valuing it properly requires modeling the joint dynamics of the overnight rates and the cross-currency basis, a genuine multi-currency term structure problem. A dealer that ignores this and discounts on the domestic curve alone is systematically mispricing every trade under a multi-currency CSA. Two firms quoting the same trade under the same CSA can differ by millions purely on this treatment.

Similar optionality arises wherever the CSA offers a choice: eligible bonds as well as cash, a range of maturities, or substitution rights. In every case the poster will deliver the cheapest, and the receiver should price on that assumption. How much of this optionality survives depends on the agreement's vintage. The ISDA Standard CSA of 2013 was designed to remove it, and the variation margin CSAs signed since the uncleared margin rules are mostly cash-only, often in few currencies, so the option now lives mainly in legacy CSAs.

8.6 COLLATERAL TRANSFORMATION

The margin requirements of Sections 8.2 and 8.4 together create a problem for a particular kind of institution. Variation margin on cleared trades, and increasingly on bilateral ones, must be

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posted in *cash*. But the institutions with the largest derivatives hedges (pension funds, insurers, asset managers) hold almost no cash. They hold bonds, because holding cash would defeat their purpose. When margin is called they have the assets but not the form.

Collateral transformation is the business of converting bonds into cash: the fund repos its bonds, receiving cash to post as margin, and pays the repo rate while continuing to earn the bond's yield. Figure 8.2 shows the transaction, which leaves the fund with a financing leg and a rollover it did not have before. Insurers also use securities lending for the same purpose, lending corporate bonds or equities that a clearing house would not accept against government bonds or cash in a collateral upgrade trade.

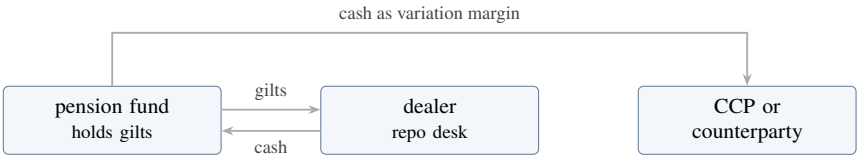


Figure 8.2: A fund repos bonds for cash to meet a margin call.

Example 8.7 The cost of transformation

A pension fund must post £500 million of cash variation margin. It holds gilts and repos them with a 2% haircut. Take illustrative rates: the repo costs 4.25%, and the cash, once posted, earns SONIA at 4.00% (the CSA rate, or price alignment interest at a clearing house).

To raise £500 million it must deliver

$$\frac{500}{0.98} = \text{£510.2 million of gilts}$$

The gilts keep earning their yield whether or not they are repoed, so the yield drops out of the cost. The fund pays the repo rate on the cash and receives SONIA on the same cash as posted margin, a drag of 25 basis points, or

$$\text{£500m} \times 0.0025 = \text{£1.25 million a year}$$

The haircut adds no running cost, since the fund owns the extra £10.2 million of gilts outright, but those gilts are encumbered and cannot raise cash anywhere else.

Two features make this fragile rather than merely expensive. The haircut means the fund needs *more* assets than the cash it raises, and haircuts widen under stress exactly as Example 8.3 showed: if the haircut is raised to 8% when the repo is next rolled, the same £500 million needs $500/0.92 = \text{£543.5}$ million of gilts, £33.3 million more. And the transformation depends on a dealer being willing to roll the repo, a service that consumes balance sheet under the leverage ratio of Section 1.12, and that dealers withdraw when balance sheet is scarce, which is precisely when the fund needs it.

Collateral transformation therefore converts a market risk the fund has hedged into a dependence on the continued functioning of the repo market. That is a different risk, not a smaller one.

8.7 REHYPOTHECATION, SEGREGATION, AND WHAT HAPPENS ON DEFAULT

Section 8.1 distinguished title transfer from security interest. The distinction becomes concrete when the recipient of collateral fails. **Rehypothecation** is the reuse of collateral received. A dealer that receives bonds as variation margin may post them onward to meet its own obligations, and the same security can support several transactions in a chain. Both legal forms allow it: a title transfer outright, and the New York-law security interest through the right of use that Section 8.1 described. This is efficient: it means the system needs less collateral in total to

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support a given volume of activity. It is also fragile, because a chain of claims on one security unwinds messily when a link fails, and because the original poster may find that its property has traveled somewhere it cannot follow.

The protections differ sharply by margin type, and the arrangement in Figure 8.1 is what produces the difference:

- *Variation margin* is transferred outright, or under a New York-law CSA held with a right of use, and may be reused. Either way the poster has parted with control of it: if the recipient fails, the poster is an unsecured creditor for the return of equivalent assets, ranking with everyone else.
- *Initial margin* under UMR must be held by a third-party custodian and may *not* be rehypothecated. The poster remains the owner; if the recipient fails, the IM is returned rather than joining the estate.

That asymmetry is deliberate and was learned expensively. Client assets held by failed intermediaries, Lehman Brothers in 2008 and MF Global in 2011, proved far harder to recover than clients had assumed, and in several cases had been reused in ways the clients did not know about and could not have discovered. The segregation requirement for initial margin exists because the alternative was tested and failed.

Wrong-way risk in collateral. A separate hazard is taking collateral whose value falls precisely when the counterparty defaults. The extreme case, accepting a counterparty's own securities, is obvious and prohibited, but weaker versions are easy to miss: taking the bonds of a bank's sovereign as collateral against exposure to that bank, or taking energy-sector securities against exposure to an energy trading firm. In each case the collateral and the exposure share a driver, so the protection evaporates in the state where it is needed. Chapter 27 develops this as wrong-way risk in general.

8.8 DISPUTES AND RECONCILIATION

A margin call is a demand for money based on one party's valuation of a portfolio. The other party has its own valuation, and the two will differ: over the marks on illiquid trades, over which trades are in the portfolio at all, over the discount curve of Section 7.4, and over the terms of the CSA itself.

Disputes are therefore routine rather than exceptional, and the procedure for handling them is part of the arrangement. Small differences never become calls at all, because no collateral moves below the CSA's minimum transfer amount (Section 8.1). The uncleared margin rules cap it at €500,000 across initial and variation margin combined ($500,000 in the United States), which is small enough that it removes only the noise. Above it, the undisputed amount is transferred while the difference is investigated, and the CSA specifies a resolution procedure: recalculation, and failing that, polling reference dealers for quotations. Firms also reconcile portfolios on a schedule set by regulation (daily for the largest books) to catch discrepancies before they become material, and a dispute large enough and old enough must be reported to the supervisor. Three observations make this more than an operational footnote.

First, a dispute is a *funding* event. If a firm calls for $80 million and receives $50 million pending resolution, it is short $30 million while the disagreement lasts, and must fund that gap.

Second, disputes cluster exactly when they are most dangerous. Illiquid positions are hardest to mark when markets are disorderly, and both parties have an incentive to mark in their own favor. A firm relying on incoming collateral to meet its own outgoing calls has built a chain whose weakest link is a valuation disagreement, at the moment when everyone is disagreeing.

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Third, and least obviously, a dispute is expensive long after it is settled. A **netting set** (the trades with one counterparty that a single enforceable netting agreement covers) that produces repeated disputes lasting longer than the margin period of risk has that period *doubled* for the following two quarters in the dealer's counterparty capital calculation. The add-on for potential future exposure measured there (a high quantile of how large the exposure could grow, which Section 9.1 defines) scales with the square root of the period, through the maturity factor of Section 27.3, as margin does in Section 8.4. Disputing is therefore not free even for the party whose valuation is right; Example 8.8 prices the funding cost and compares the two.

Example 8.8 What a dispute costs, and when

A dealer calls a counterparty for $80 million of variation margin. The counterparty accepts $50 million and pays that, disputing the rest. The $30 million gap is outstanding for twelve days before the two sides reconcile, and the dealer funds it at 5.0%:

$$\$30\text{m} \times 0.050 \times \frac{12}{360} = \$50{,}000$$

That is the visible cost, and it is trivial. The cost that can matter more is in capital: if this is the third dispute on the netting set in two quarters to outlast the margin period of risk, the dealer's counterparty capital calculation doubles that period for the next two quarters, raising the capital held against the netting set whether or not the dealer's valuation was the correct one (Section 27.3 works through the rule).

The lesson is not that firms should concede disputes. It is that the cost of a valuation disagreement is mostly not in the disputed amount, so the investment that pays is in not disagreeing: reconciled portfolios, agreed curves, and a common margin model.

The cost of disputes also explains why the industry converged on SIMM, a choice that would otherwise look like bureaucratic uniformity. Initial margin is model-driven, and two counterparties running their own proprietary models would disagree about the number on every single call: not occasionally, but always, and by amounts far larger than the marks. A common model does not make either party's risk estimate better. It makes the two estimates the *same*, which is what converts initial margin from a standing dispute into an operational transfer. SIMM is best understood as dispute-avoidance infrastructure that happens to also measure risk.

8.9 MARGIN CALLS AS LIQUIDITY RISK

The sections above describe how collateral works. This section states the chapter's thesis, which is why any of it matters.

A hedge and the item it hedges generate cash on different schedules. The hedge is marked daily and settles daily. The hedged item (physical output to be delivered over years, a pension liability payable over decades, a bond held to maturity) does not. When the hedge loses money, the loss is a cash outflow *today*. When it gains, the offsetting loss on the hedged item is generally not a cash outflow at all. This asymmetry is invisible in a mark-to-market framework, which nets the two and reports approximately zero. It is the entire problem.

Example 8.9 The liquidity cost of a correct hedge

A power producer expects to generate 10 TWh over the next three years and hedges by selling forward at €80 per MWh. Its revenue is locked in at

$$10{,}000{,}000\ \text{MWh} \times €80 = €800\ \text{million}$$

Prices then rise to €250 per MWh. The producer's short forward position has lost

$$10{,}000{,}000 \times (€250 - €80) = €1.7\ \text{billion}$$

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and, because the position is margined, it must pay that €1.7 billion in cash, now.

Against it stands a gain of exactly €1.7 billion on the electricity it will generate. But that electricity does not exist yet. It will be produced and sold over three years, and the cash will arrive then.

The firm is better off than if it had not hedged: it has locked in €800 million of revenue in a market that might equally have fallen to €30. It is also, quite possibly, unable to pay what it owes on the day: illiquid, though not insolvent.

Set the two sides of Example 8.9 against time and they sum to approximately zero: €1.7 billion out today, about €567 million a year in over the next three years. Figure 8.3 plots the cumulative cash: the net position opens €1.7 billion short and climbs back to zero only as the last of the power is sold. The difficulty is entirely in the timing. The hedge remains economically sound right up until the firm fails. Solvency and liquidity are different questions, and derivatives collateral is the mechanism that most reliably separates them.

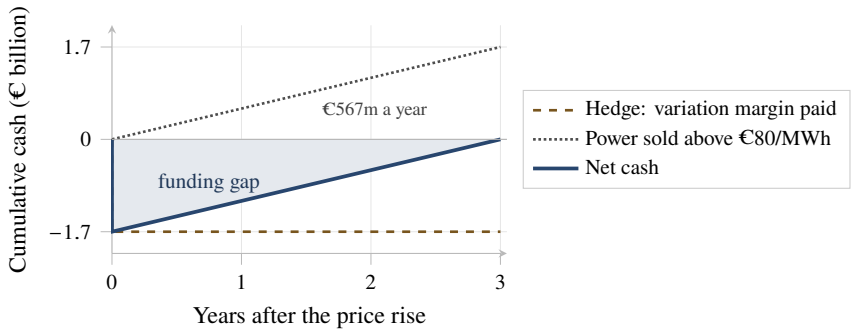


Figure 8.3: Cumulative cash in Example 8.9 (schematic: even sales, prices held at €250, no discounting).

This is why several European governments extended emergency credit lines to energy companies in 2022: the utilities in difficulty were the ones that had sold their output forward and faced Example 8.9's arithmetic at national scale, while those that had not hedged faced no margin calls and took windfall profits. The European Union's regulatory answer, in late 2022, was to widen what its clearing houses could accept: as a temporary measure, bank guarantees not backed by collateral became eligible from non-financial firms that clear directly, and public guarantees from financial and non-financial clearing members alike. The 2024 revision of the European Market Infrastructure Regulation made the relief permanent and extended it to non-financial firms that clear through a member. Section 40.12 tells the episode, and what it says about a system that prices risk correctly and liquidity not at all.

Case Study 8.1 Liability-driven investment, 2022

UK pension funds face a structural problem: liabilities extending fifty years, and not enough long-dated gilts in existence to match them. The standard solution, liability-driven investment, uses leverage (gilt repo and interest rate swaps) to obtain the required duration from a smaller pool of assets.

The leverage was collateralized, and the collateral was gilts. When the September 2022 fiscal announcement pushed gilt yields sharply higher, the funds' swap and repo positions moved against them and collateral was called. To meet the calls they sold gilts. Selling gilts pushed yields higher. Higher yields generated further calls.

The loop was not a panic: the mechanism was structural. The assets pledged as

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collateral were the same assets whose price the forced selling was driving down, so the more collateral was called, the less the remaining collateral was worth.

The Bank of England intervened by buying gilts on a temporary basis, explicitly to break the loop rather than to set a yield. The subsequent regulatory response set collateral buffers: the Bank's Financial Policy Committee (March 2023) and the Pensions Regulator (April 2023) required funds to withstand a rise of about 250 basis points in gilt yields, which caps leverage indirectly by fixing the collateral held against it. The target was liquidity rather than solvency. In January 2025 the Bank opened a contingent repo facility, activated at its discretion, through which it can lend cash against gilts to eligible insurers, defined benefit pension schemes, and liability-driven investment funds.

8.10 MEASURING AND MANAGING MARGIN LIQUIDITY

The obvious response to the previous section is to hold enough cash. The difficulty is knowing how much, since the requirement depends on a market move that has not happened. Firms therefore run **margin liquidity stress tests**, which differ from ordinary market risk measurement in three ways.

They measure gross outflow, not net loss. A value-at-risk (VaR) framework reports the net change in portfolio value. The relevant quantity here is the *cash that must leave the firm*, and the offsetting gain does not count, because it produces no cash.

They model the initial margin response, not only variation margin. A large move generates a VM call equal to the loss, *and* raises the IM requirement because volatility has risen (the procyclicality of Section 2.9), at a clearing house within days and bilaterally at the next recalibration. At a clearing house both land at once.

They assume the transformation channel narrows. A stress test that assumes normal repo access (Example 8.7) has assumed away the problem.

Example 8.10 Sizing a liquidity buffer

A fund holds a swap portfolio of \$2 billion notional with a DV01 of \$1.6 million, so a one basis point move in rates changes its value by \$1.6 million.

It stresses a 100 basis point adverse move over one week, which is severe but well within the range observed in 2022:

$$\text{VM outflow} = 100 \times \$1.6\text{m} = \$160 \text{ million}$$

Volatility rises with the move, and the initial margin model responds. The portfolio is cleared, and a clearing house's margin model reacts to a volatile week within days; SIMM, by contrast, is recalibrated only periodically. If the clearing house's initial margin on the book rises 50%, from \$60 million to \$90 million, that is a further \$30 million call.

The total requirement is \$190 million of cash within days, against a portfolio whose *value* may be unchanged once the hedged item is included. If the fund holds \$100 million of cash and expects to repo bonds for the rest, it should ask what haircut it will face at that moment, and whether dealers will still lend against its bonds at all.

The tools. Firms manage this with cash buffers, committed repo or credit lines, CSAs that permit non-cash collateral where possible, and leverage limits that keep the plausible call within reach. Each has a cost, paid continuously, to avoid an event that may not arrive. That trade-off

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is the same one Section 2.9 identified for CCP margin models, and it is resolved no more cleanly here. Sizing them requires anticipating the call, which is why the Basel Committee, CPMI, and IOSCO's final reports of January 2025 ask clearing houses to give members margin simulators that show requirements under stress scenarios, and to disclose a standardized measure of how strongly their initial margin responds to market volatility.

Why the problem is worse than each firm's own numbers suggest

Three effects make the aggregate picture worse than the sum of single-firm calculations like Example 8.10.

Calls are correlated. A large market move calls margin from everyone positioned the same way at the same moment. Each firm's stress test may be individually adequate while the system's total demand for cash exceeds what can be raised, which is precisely what turns a price move into a liquidity event.

The supply of cash contracts as demand for it rises. Repo capacity depends on dealer balance sheet (Example 8.7), so demand and supply move in opposite directions at once.

Selling to raise cash moves the price that generated the call. Where the asset sold is the asset driving the exposure (gilts sold by gilt-collateralized funds), the loop closes on itself, as the 2022 pension episode of Case Study 8.1 showed.

A single firm can estimate the second, and its own contribution to the third, as Example 8.11 does; the first, and the rest of the third, depend on what everyone else is doing.

Example 8.11 The loop, quantified

A pension fund holds £1.2 billion of gilts with a duration of 12 and closes its duration gap with £3 billion notional of twenty-year receive-fixed swaps. The twenty-year annuity at 4% is 13.590, so the swap DV01 is

$$£3\text{bn} \times 13.590 \times 0.0001 = £4.077 \text{ million per basis point}$$

Rates rise 100 basis points. On a funding-ratio basis (assets divided by liabilities, the measure a pension fund is managed to) the fund is fine: its liabilities fall by more than its assets, which is what the hedge was for. On a cash basis, assuming the swaps are cleared so that their initial margin rises with volatility (swaps margined bilaterally for variation only would call the first line alone):

Variation margin on the swaps	£407.7m
Initial margin increase as volatility doubles	£120.0m
Cash required within days	**£527.7m**

At the same instant the gilt pool has fallen 12% to £1,056m, and the repo haircut on it has widened from 2% to 8%:

	Before	After
Gilt pool	£1,200m	£1,056m
Cash raisable against it	£1,176m	£972m

Capacity fell 17.4% in the same move that raised the requirement from nothing to £527.7m. That requirement is now 54.3% of what remains. Neither number appears in a net-loss stress test, which reports approximately zero.

And the loop has not started yet. Raising £527.7m at an 8% haircut would tie up £573.6m of gilts, and on a day like this the repo may not be there. Suppose the fund sells £527.7m of gilts instead, and that selling on that scale moves twenty-year yields a further 15 basis points (modest against what was actually observed). That is another £61.2m of variation margin, met by selling another £61.2m of gilts, which moves yields 1.74 basis points more, and so on; the further fall in the gilt pool from these later rounds is ignored:

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Round	Yield move	Extra VM
1	15.00 bp	£61.2m
2	1.74 bp	£ 7.1m
3	0.20 bp	£ 0.8m

The series converges to £596.9m, a 13.1% amplification of the fund's own requirement, and it converges only because one fund's selling has a bounded price impact. The series diverges when enough funds sell at once, which is the precise sense in which this is a system-level property rather than a risk management failure: the same calculation performed by every fund individually gives the right answer for each of them and the wrong answer for all of them.

No firm-level stress test, however conservative, captures this, which is why the supervisory response, including the Financial Stability Board's work on margin preparedness, has come through financial stability channels rather than individual firm requirements.

No one behaved badly. In the energy and pension episodes of 2022 the utilities hedged as textbooks instruct, the pension funds matched their liabilities as their mandates required, and the clearing houses raised margin as their models and their regulators demanded. Every participant did what it was supposed to do, and the aggregate outcome was a liquidity crisis. Frameworks that locate such failures in imprudence will not anticipate the next one.

8.11 IS THERE ENOUGH COLLATERAL?

Clearing mandates, uncleared margin rules, and higher haircuts have each increased the high-quality assets immobilized against derivatives exposures. Is the supply adequate?

The pessimistic case is straightforward. Initial margin is segregated and cannot be reused (Section 8.2), so every dollar of IM is a dollar of government bonds or cash withdrawn from productive use for the life of a trade. Bank regulation has simultaneously increased demand for the same assets as liquidity buffers. And collateral transformation, which lets bond-holders meet cash calls, depends on dealer balance sheet that regulation has made expensive.

The optimistic case is equally straightforward. Government debt outstanding has grown enormously over the same period, so eligible collateral has expanded alongside demand. Netting and compression (tearing up offsetting trades among many participants at once) have reduced gross exposures substantially. And the pre-crisis alternative, in which large exposures went uncollateralized, was demonstrably worse.

The aggregate quantity has not been the binding constraint; the *distribution* and *timing* have. The 2022 episodes were caused not by a shortage of eligible assets but by particular firms needing cash on particular days and unable to convert what they held quickly or cheaply enough. A system can hold ample collateral in aggregate and still fail at the moment of transformation, which is why Section 8.10's stress tests focus on the channel rather than the aggregate.

8.12 WHERE THIS LEAVES VALUATION

Collateral changes what a derivative is worth, in three distinct ways that later chapters develop.

The *discount curve* is set by the CSA's collateral rate, as Section 7.4 established, and by the cheapest-to-deliver option of Section 8.5 where the CSA permits choice.

Funding enters when collateral does not fully cover the exposure. A one-way CSA, an uncollateralized trade, or a CSA with a large threshold all leave the dealer funding an exposure from its own resources. Chapter 9 prices this as a funding valuation adjustment (FVA).

Initial margin must itself be funded for the life of the trade, at a cost that no discounting convention captures. Chapter 9 prices this as MVA; Example 8.2 gave its size.

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The phrase "the value of the swap" is therefore incomplete: a swap's value depends on who holds it, under what CSA, funded how, and Chapter 9 makes that dependence explicit.

What these three do *not* capture is the proposition this chapter opened with. All of them price the *cost* of collateral: what it is worth to discount on one curve rather than another, to fund an uncollateralized exposure, to leave initial margin encumbered for ten years. None of them prices the risk of being unable to deliver collateral on the day it is called. The producer of Example 8.9 owed €1.7 billion in cash against a hedge that was working exactly as designed; no valuation adjustment would have shown a problem, because in valuation terms there was not one. That exposure is a liquidity risk, measured by the stress tests of Section 8.10 and not by any adjustment to a price. Collateral changed what derivatives are worth; it also changed what can bring a firm down, and the two are not the same list.

SUMMARY

The credit support annex sets what may be posted, at what haircut, above what threshold, and at what rate. These terms are not administrative: the collateral rate sets the discount curve, a one-way CSA is a funding commitment, and a choice of collateral currency gives the poster an option over the discount curve itself.

Variation margin settles a loss already incurred. It is netted, may be reused, and costs the poster nothing beyond the loss. Initial margin covers a possible future move. It is posted gross by both parties, segregated and barred from reuse so that it survives the failure of whoever holds it, and funded for the life of the trade, so it is a running cost. A bilateral trade immobilizes several times the collateral of a cleared one by design, which makes genuinely bespoke hedges expensive to hold.

Haircuts widen under stress, so a firm must find more assets, from a pool that has itself shrunk, to cover an exposure that has grown: the procyclicality of margin models, through a second channel. Disputes are funding events: the party that made the call funds the disputed amount until it is resolved, and disputes multiply when markets are disorderly. A firm holding bonds rather than cash depends on the repo market to transform them, and that market narrows in the same stress.

The chapter's central proposition is that a hedge and the hedged item generate cash on different schedules. A producer that sells its output forward owes margin in cash today against revenue that arrives later, so it can be better off for having hedged and still unable to pay. That is why European governments lent to energy firms in 2022, and why the UK pension episode was a collateral loop rather than a solvency problem.

Managing this requires stress tests of gross cash outflow that assume the repo channel narrows, and even those understate the problem, because calls are correlated across firms and forced sales move the prices that caused them. No participant in the energy and pension episodes of 2022 behaved imprudently, and the result was still a liquidity crisis. The system holds enough collateral in aggregate; what fails is converting it into cash on the day it is called. The valuation adjustments of Chapter 9 price the cost of collateral, not the risk of being unable to deliver it.

FURTHER READING

Gregory's *The xVA Challenge* is the standard practitioner reference for collateral, margin, and the funding that follows from them. The ISDA SIMM methodology document is the operative specification for initial margin and changes often enough that the current version is the one to use. On margin procyclicality, the Basel Committee on Banking Supervision and CPMI–IOSCO's *Review of Margining Practices* (2022) is the official record. Two final reports of January 2025 set out the policy that followed from it: the same bodies' *Transparency and*

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Responsiveness of Initial Margin in Centrally Cleared Markets: Review and Policy Proposals for clearing houses, and the Basel Committee and IOSCO's *Streamlining Variation Margin Processes and Initial Margin Responsiveness of Margin Models in Non-centrally Cleared Markets* for bilateral trades. The Financial Stability Board's *Liquidity Preparedness for Margin and Collateral Calls* (2024) is the supervisory response. The Bank of England's December 2022 *Financial Stability Report* is the place to start on liability-driven investment and gilt market functioning in the 2022 episode.

PRACTICE QUESTIONS

8.1 Explain the difference between a one-way and a two-way CSA, why a one-way CSA makes an apparently mid-market trade loss-making for the dealer, and why the rate paid on collateral is the most economically important term in the document.

8.2 A fund posts $40 million of variation margin and $25 million of initial margin, and its funding costs 60 basis points more than posted margin earns. Give the annual economic cost of each, and explain why they differ along the dimensions of what each covers, direction, and reusability.

8.3 Explain why initial margin is posted gross by both parties to a third-party custodian rather than netted between them.

8.4 A portfolio has sensitivities of $50,000 per basis point in a bucket with a risk weight of 55 bp and $90,000 per basis point in a bucket with a risk weight of 45 bp, with a prescribed correlation of 0.6. Calculate the initial margin, compare it with the undiversified sum, and explain why the result leads firms to concentrate their trades with one counterparty.

8.5 Explain why a CSA permitting several collateral currencies gives the poster an option, and why that option is an option over the discount curve.

8.6 A pension fund must post £300 million of cash margin, which earns SONIA at 4.0% once posted. It holds bonds yielding 3.8% and can repo them at 4.3% with a 3% haircut. How many bonds must it deliver, and what is the annual cost of the transformation?

8.7 An airline expects to buy 500 million gallons of jet fuel evenly over the next two years and hedges by paying a fixed $2.80 a gallon in swaps that are margined daily. Jet fuel then falls to $2.10 a gallon. Calculate the variation margin the airline must pay and the offsetting saving on its fuel bill, and state when each becomes cash. Explain why the airline can be perfectly hedged against price risk and still fail.

8.8 A fund must cover a $250 million exposure with bonds subject to a 5% haircut. How much collateral must it deliver? Recalculate at a 12% haircut, and explain why the two other things happening at the same moment make the position worse than the comparison suggests. Now suppose the $250 million is initial margin under the uncleared margin rules, posted in euro-denominated government bonds with more than five years to maturity against trades that settle in dollars. Using the standardized schedule (4% on the bonds plus the 8% currency add-on), how much must the fund deliver, and how much of that is due to the currency?

8.9 A swap has a five-day 99% risk of $20 million. Compare the collateral immobilized if it is cleared with the amount immobilized bilaterally under UMR, stating your assumptions. Why were the rules calibrated this way, and what is the cost of that design?

8.10 Explain what rehypothecation is, why it is economically efficient, and why initial margin may not be rehypothecated while variation margin may.

8.11 Give two examples of wrong-way risk in the choice of collateral, and explain the common

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feature that makes them dangerous.

8.12 A dealer calls a counterparty for $60 million of variation margin. The counterparty pays $35 million and disputes the rest, and the gap is outstanding for 20 days, which the dealer funds at 4.8%. Calculate the funding cost of the dispute. The dispute is the third on the netting set in two quarters to outlast the margin period of risk, so the period doubles from ten days to twenty for the next two quarters. If the netting set's potential future exposure add-on is $150 million and capital costs 40 basis points a year on the exposure, what does the doubling cost? Explain why a margin dispute is a funding event that clusters at the worst possible time.

8.13 A CSA gives a counterparty a threshold of $5 million and requires it to post an independent amount of $2 million. The minimum transfer amount is $500,000, and calls are rounded up to the next $100,000. The dealer's exposure is $14.23 million and it holds collateral worth $8.0 million after haircuts. Calculate the margin call. The next day the exposure is $14.6 million: is there a further call? How much of the exposure is then uncollateralized, and why?

8.14 A dealer posts $600 million of cash collateral under a CSA permitting dollars, euros, or sterling. Dollar collateral earns SOFR, the cross-currency basis against the dollar is -20 basis points for euros and -8 for sterling, and the portfolio annuity is 6. Which currency does the dealer post, and what is the approximate value of that choice relative to posting dollars? In what respect does this estimate understate the value of the CSA's optionality, and in what respect does it overstate it?

8.15 A dealer's portfolio with one counterparty holds an interest rate swap whose SIMM margin on its own is $8 million and an equity swap whose margin on its own is $5 million, and the two positions have tended to move in opposite directions. What initial margin does SIMM require on the two together? Replace the equity swap with an FX forward whose margin on its own is $5 million. SIMM prescribes a correlation of about 0.3 between the interest rate and FX margins, whatever the positions' signs: what is the margin now? Explain the difference.

FURTHER QUESTIONS

8.16 A gas producer hedges 25 TWh of output at €60 per MWh, and prices then rise to €200. What financing arrangement would have allowed the firm to meet its margin calls and survive, and what would that arrangement have cost in a year when prices did not move?

8.17 Explain the liability-driven investment loop of 2022 step by step. Identify the feature that made it self-reinforcing rather than merely costly, and explain why the regulatory response limited leverage through a minimum collateral buffer rather than by capping it directly.

8.18 A margin liquidity stress test differs from value at risk in three ways. Set them out, and explain why a firm passing a VaR-based test could still fail the liquidity one.

8.19 A $5 billion portfolio has a DV01 of $3.2 million. Stress a 150 basis point move over a week and assume the clearing house's initial margin rises 60%, from a base of $120 million. Calculate the total cash requirement, and discuss what the firm should assume about repo haircuts and dealer willingness at that moment.

8.20 Collateral was introduced to reduce credit risk, and it does. Argue that it has substituted a liquidity risk that is harder to measure and more correlated across firms than the credit risk it replaced. Then argue the other side. Which view do the 2022 episodes support?

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Chapter 9
XVA:
Valuation Adjustments in Practice

Chapters 7 and 8 arrived, by different routes, at the same uncomfortable conclusion. A swap's value depends on the curve used to discount it, and the curve depends on the collateral agreement. Where collateral does not fully cover the exposure (under a one-way CSA, a threshold, or no CSA at all), the dealer funds the position itself; where initial margin is required, that too must be funded for the life of the trade. Behind all of it stands the possibility that the counterparty defaults. The phrase "the value of the swap" therefore does not identify a number: the same cash flows are worth different amounts to different holders, under different agreements, funded differently. The adjustments that convert a textbook valuation into a price someone will actually trade at are collectively called **XVA**, and this chapter is about what they are, who charges them, and where they overlap.

Default needs a precise meaning, because every adjustment prices it. A counterparty **defaults** when it fails to meet its obligations on any of its contracts, or becomes bankrupt or insolvent. Default belongs to the counterparty, not to a trade, and it reaches all of its liabilities at once: cross-default and acceleration clauses put every obligation into default when one is missed, and bankruptcy stays them all together. Creditors differ only in how much each recovers, which depends on seniority and collateral. A swap that is performing perfectly is therefore caught when the counterparty defaults on a bond, a loan or another dealer. For derivatives, the ISDA Master Agreement sets the consequences. An **event of default** (among other things, a failure to pay or deliver under the agreement that is not cured within a short grace period, bankruptcy or insolvency, or, where the parties have agreed it, a default on other borrowing above a threshold amount) entitles the other party to terminate every trade under the agreement and replace them with a single net close-out amount (Section 1.5). That amount, less the collateral held, is what the counterparty may never pay. Default is a legal event, not a market price: a downgrade or a wider credit spread is not a default, although a rating trigger in the CSA can turn one into a collateral call (Section 8.1). It also differs from two neighboring definitions. Bank capital rules treat an obligor as defaulted when it is more than 90 days past due on a material obligation or is judged unlikely to pay, and a credit default swap pays on its own list of credit events (Chapter 25).

Section 9.1 defines exposure, the quantity every adjustment is built on. Sections 9.2 to 9.7 take the adjustments one at a time (credit, debit, funding, margin, collateral, and capital), and Section 9.8 adds them up on one swap. Section 9.9 treats wrong-way risk and Section 9.10 the desk that manages all of it. Section 9.11 confronts the fact that the same trade now carries three defensible values, and Section 9.12 asks what that does to the idea of a price, where the framework falls short, and where it came from.

Two warnings come first. First, XVA is not a refinement: on a long-dated uncollateralized trade the adjustments can exceed the bid–ask spread by more than an order of magnitude and dominate the economics of the trade. Second, this is the least settled material in the book: the

components are not agreed, some may double-count, and one of them, the subject of Section 9.4, has been argued to have no business existing at all. The disputes are set out rather than resolved.

9.1 EXPOSURE

An **exposure** is the amount a party would stand to lose if its counterparty failed at a given moment, before any recovery. It is not the notional, which in a swap is never at risk, nor the portfolio's value, which can be negative. It varies with the date, and collateral held against it reduces it; the definition below leaves collateral out until equation (9.2) subtracts it. If a counterparty defaults at time t, the loss depends on what the portfolio is worth at that moment. If the portfolio has *negative* value (the dealer owes money), there is no loss; the obligation survives the counterparty's default and must still be paid. If the portfolio has *positive* value, the dealer joins the queue of unsecured creditors and recovers a fraction.

The exposure is therefore

$$E(t) = \max\left(V(t),\, 0\right) \tag{9.1}$$

where $V(t)$ is the value at time t of the trades held with that counterparty, taken together and signed from the dealer's point of view, so that $V(t) > 0$ means the counterparty owes the dealer. Which trades may be taken together is a legal question, taken up below.

Equation (9.1) is the payoff of a call option on that value struck at zero: *counterparty risk is an option*, and everything that makes options hard (volatility, correlation, the need to model a distribution rather than a point) applies to it.

Three summaries of the exposure distribution are used:

- **Expected exposure** (EE), the average over scenarios at each future date: $\mathrm{EE}(t)$ is the average of $\max(V(t), 0)$.
- **Expected positive exposure** (EPE), the average of $\mathrm{EE}(t)$ over the life of the trade, a single number summarizing the profile. Banks using the internal model method capitalize a variant, effective EPE (Section 27.5).
- **Potential future exposure** (PFE), a high quantile of the exposure distribution, used for credit limits rather than valuation.

How exposure is computed

Obtaining $\mathrm{EE}(t)$ means obtaining the distribution of the portfolio's value at every future date, by a Monte Carlo simulation in three stages.

1. *Evolve the risk factors.* Simulate paths of the underlying risk factors (interest rate curves, FX rates, credit spreads, commodity prices) from today to T on a chosen time grid.
2. *Revalue the portfolio.* On every path, at every grid date, revalue every trade in the netting set. This is the expensive step: N paths $\times$ m dates $\times$ K trades valuations.
3. *Aggregate.* On each path and date, net across trades, subtract collateral, take the positive part, and average across paths.

Formally, for path j and date t_i,

$$E_j(t_i) = \max\left(\sum_{k=1}^{K} V_k^{(j)}(t_i) - C^{(j)}(t_i),\; 0\right), \qquad \mathrm{EE}(t_i) = \frac{1}{N}\sum_{j=1}^{N} E_j(t_i) \tag{9.2}$$

where $C^{(j)}(t_i)$ is the collateral held on that path and date.

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The maximum in (9.2) is taken *after* netting across trades and subtracting collateral. Taking it trade by trade and summing overstates exposure, by convexity (the netting inequality (9.4) below), in some portfolios by a large multiple.

Computing cost is what limits the calculation, which is among the largest a dealer runs: 5,000 paths and 100 dates on a netting set of 2,000 trades is a billion trade valuations for one counterparty, repeated across thousands of counterparties nightly. Two techniques make it feasible. **American Monte Carlo** (the least-squares regression of Section 20.11, borrowed for its regression step rather than for the early-exercise decision) replaces exact revaluation with a fitted function of the state variables. **Adjoint algorithmic differentiation** (AAD, Section 20.10) produces every sensitivity in a small multiple of one valuation's cost, and the desk of Section 9.10 hedges the charge's sensitivity to every market input, not only its level.

The shape of an exposure profile

For an interest rate swap the exposure profile has a characteristic hump.

Two forces work against each other. *Diffusion*: the longer rates have had to move, the wider the range of possible values, which pushes exposure up roughly with $\sqrt{t}$. *Amortization*: as payment dates pass, the swap's sensitivity to rates falls, which pulls exposure down roughly linearly. The product peaks at about one third of the way through the trade.

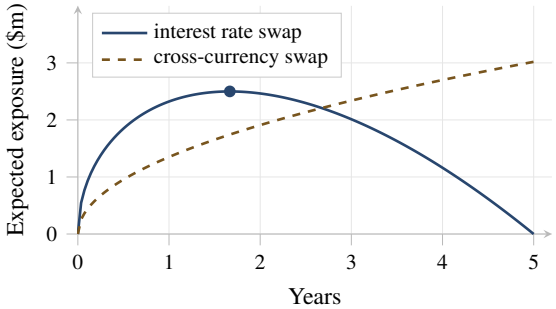


Figure 9.1: Expected exposure for a five-year interest rate swap and for a cross-currency swap (the latter schematic).

Figure 9.1 contrasts this with a cross-currency swap. There the notional is exchanged at maturity (Chapter 7), so the largest cash flow is the last one and nothing amortizes: diffusion operates unopposed and exposure rises to maturity. A cross-currency swap therefore carries more counterparty risk than an interest rate swap of the same notional and tenor, and carries it later, when the counterparty has had longest to default. The curve drawn is schematic and understates the difference (on Example 9.2's inputs the two curves give CVAs only about 9% apart), because the final exchange of notional is exposed to the whole move in the exchange rate, which at ordinary currency volatilities puts a real profile well above the one drawn. Between dealers the effect is usually contained by a mark-to-market resetting cross-currency swap, which adjusts one notional to the current exchange rate at each reset and so returns the currency move to the counterparty period by period (Section 7.8; Chapter 33 prices it); the corporate swaps that carry most cross-currency CVA rarely reset.

Other products have other shapes. A purchased option has exposure equal to its full value throughout, since that value can never be negative, while a sold option has none once the premium is received, so two trades that look symmetric carry radically asymmetric counterparty risk. An FX forward rises with $\sqrt{t}$ to maturity like the cross-currency swap, because nothing amortizes. A repo or fully collateralized financing is exposed only to the extent that the collateral can lose more than the haircut over the margin period of risk, which is why Chapter 8's haircuts

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belong to credit as well as to liquidity. In general, exposure grows with the uncertainty remaining and shrinks with the obligations discharged.

Expected exposure and potential future exposure

Expected exposure and potential future exposure serve different purposes, and confusing them produces numbers a factor of four apart (Example 9.1).

Suppose that at the peak of the profile the portfolio value is normally distributed about zero with standard deviation σ. Then the expected exposure has a closed form,

$$\mathrm{EE} = \int_0^\infty v \, \frac{1}{\sigma\sqrt{2\pi}} \, e^{-\frac{v^2}{2\sigma^2}} \, dv = \frac{\sigma}{\sqrt{2\pi}} = 0.3989\,\sigma \tag{9.3}$$

while the potential future exposure at the 95% level is the corresponding quantile of the positive part, $1.645\,\sigma$.

Example 9.1 Expected against potential future exposure

The swap of Figure 9.1 peaks at an expected exposure of \$2.5 million. From (9.3) the implied standard deviation of the portfolio's value at that date is

$$\sigma = \frac{2.5}{0.3989} = \$6.27 \text{ million}$$

so the 95% potential future exposure is

$$\mathrm{PFE}_{95} = 1.645 \times 6.27 = \$10.3 \text{ million}$$

roughly four times the expected exposure.

Both numbers describe the same trade, and both are used, for different things. *Pricing* uses expected exposure, because the hedge of Section 9.2 is sized to the average and its cost is what the client is charged. *Credit limits* use potential future exposure, because a limit is meant to cap a bad case rather than an average one. In practice the two are usually simulated differently as well: exposure for pricing from risk factors calibrated to market prices such as implied volatilities, exposure for limits from dynamics estimated on history.

A credit officer refusing a trade on a \$10.3 million PFE and a trader pricing it off a \$2.5 million EE are not disagreeing; they are answering different questions.

Netting matters, and it matters at the right level. Exposure is computed on the netting set (the group of trades covered by one master agreement), not trade by trade:

$$E(t) = \max\left(\sum_i V_i(t),\, 0\right) \;\le\; \sum_i \max\left(V_i(t),\, 0\right) \tag{9.4}$$

The inequality is strict whenever some trades have positive values and others negative ones, so a new trade's contribution to exposure can be *negative*: a trade that offsets the existing portfolio reduces the netting set's exposure and with it the CVA charge. In the limiting case, a client that already pays fixed on a swap and now asks to receive fixed on identical terms closes the netting set's exposure to zero, and the new trade's incremental CVA is minus the whole CVA of the existing one. A dealer that quotes a better price for a trade that hedges its existing exposure to a client than for one that adds to it is not giving a discount; it is pricing correctly.

9.2 CVA

The **credit valuation adjustment** is the market value of the counterparty's option to default: the amount by which a trade is worth less than the same cash flows promised by a counterparty that cannot fail.

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Everything in this book is priced the same way: the value of a position is what it costs to hedge it, a principle Chapter 13 makes exact. Here it applies directly, because the hedge for this option already trades. A credit default swap on the counterparty (introduced in Chapter 1, named in Chapter 7, and analyzed in Chapter 25) pays its buyer the loss on a stated amount if that counterparty defaults, in exchange for a running premium, so the premium rate, the **CDS spread**, is the market's price for insuring one dollar owed by that counterparty for one year. CVA is the cost of insuring the exposure. No probability is estimated along the way; the spread is a price, and it is used as one.

The formula is worth building rather than quoting, because everything questionable about CVA enters as an assumption along the way.

What a default costs. Suppose the counterparty defaults at some time τ before the final maturity T. Write $V(t)$ for the portfolio's value assuming no default, and R for the **recovery rate**, the fraction of its claim that a creditor eventually collects from the estate, so that a default costs it $1 - R$, the fraction the credit chapters call the loss given default. Recovery is treated here as a known constant, which it is not: it varies with seniority and with the state of the economy, and Chapter 25 describes how the market fixes it by auction. As Section 9.1 showed, the loss is one-sided, which is what makes it an option:

$$L = (1 - R)\max\left(V(\tau),\, 0\right) \quad \text{if } \tau \le T, \qquad \text{and zero otherwise.} \tag{9.5}$$

What insuring it costs. A CDS with notional N pays $(1 - R)N$ on default, the same $1 - R$, because the protection seller makes good the loss on a bond of that face amount. A dealer that knew the exposure path in advance could buy protection on an amount equal to $E(t)$ for each future date, and the payout would match the loss in (9.5) exactly. The premium is the spread s times the amount covered, per year, paid only while the counterparty is still alive. The cost of the hedge is therefore a stream of premiums, $s \times E(t)$ a year, running until maturity or default, whichever comes first, and CVA is the present value of that stream.

Two assumptions turn this into something a desk can compute.

Assume the amount to insure is the average exposure. The exposure is a distribution at each date, not a known path, and the dealer insures its average, the expected exposure $\mathrm{EE}(t)$. That is the right amount only if the size of the exposure when the counterparty fails is unrelated to the failure itself; where the exposure is largest in the states in which the counterparty is likeliest to fail, the average understates what needs cover (Section 9.9).

Assume discounting separates from exposure. This is exact under fixed rates and otherwise a good approximation, the link between the discount factor and the exposure being second order.

Discretizing on a grid, $0 = t_0 < t_1 < \cdots < t_n = T$, and writing $c(t_{i-1}, t_i)$ for the cost of covering one dollar of exposure over the interval (the premium for that interval, scaled by the fraction of scenarios in which the counterparty is still alive to be insured), and $\mathrm{DF}(t_i)$ for the discount factor to t_i:

$$\boxed{\mathrm{CVA} = \sum_{i=1}^{n} \mathrm{EE}(t_i)\; c(t_{i-1}, t_i)\; \mathrm{DF}(t_i)} \tag{9.6}$$

The spread is the *counterparty's*: CVA prices the risk that it fails, and Section 9.3 turns the same calculation on the dealer's own. It comes from market spreads (the counterparty's own CDS or bonds where they trade, otherwise a proxy curve for its rating, sector, and region) rather than from historical default rates.

Chapter 24 arrives at the same cost from the other side. A spread of s with a loss given default of $1 - R$ means the market is charging as though the counterparty fails at a rate of about

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$\lambda = s/(1-R)$ a year, a relation Chapter 24 calls the credit triangle; with that rate, the cost of cover for an interval is the loss on default times the chance of a default falling in it,

$$c(t_{i-1}, t_i) = (1-R)\, q(t_{i-1}, t_i), \qquad q(t_{i-1}, t_i) = e^{-\lambda t_{i-1}} - e^{-\lambda t_i}.$$

The two readings must agree, because a CDS is priced so that its premiums and its payouts balance, and Example 9.2 checks that they do.

Example 9.2 Computing CVA on a swap

A dealer has an uncollateralized five-year swap with a corporate client on a notional of \$100 million. The expected exposure profile is the interest rate swap curve of Figure 9.1, and equation (9.3) produces it. By date t the swap's value has moved by roughly the change in the swap rate times the notional L times the annuity of the payments still to come, about $T - t$ years of them; the rate change has a standard deviation of $\sigma_r \sqrt{t}$, so the value is roughly normal about zero with standard deviation $L\, \sigma_r \sqrt{t}\, (T - t)$, and

$$\mathrm{EE}(t) = 0.3989 \times L\, \sigma_r \sqrt{t}\, (T - t)$$

With L = \$100 million, T = 5 years and a rate volatility σ_r of 1.456% a year, the profile rises with $\sqrt{t}$ as the rate diffuses and falls as the swap amortizes, and at mid-year points it gives the expected exposure column of the table below. Section 20.6 confirms it by simulation, at the same volatility. The client's CDS trades at 120 basis points a year; recovery is assumed to be 40%, so the loss given default is 60%, and the credit triangle puts the market's default rate for this issuer at $0.012/0.60 = 2\%$ a year. Discounting is at 4% continuously compounded.

Equation (9.6) is evaluated on an annual grid, using mid-year exposures and discount factors. The cost of cover in each year is the 120 basis point premium scaled by the fraction of scenarios in which the client is still alive to be insured, $e^{-0.02 \times 0.5} = 0.990$ in the first year and falling by 2% a year after that:

Year	EE (\$m)	Cost of cover	Discount factor	Contribution (\$m)
1	1.848	1.188%	0.9802	0.02152
2	2.490	1.165%	0.9418	0.02731
3	2.296	1.141%	0.9048	0.02371
4	1.630	1.119%	0.8694	0.01585
5	0.616	1.097%	0.8353	0.00564
			CVA	0.09404

$$\mathrm{CVA} = \$94{,}000$$

That is 9.4 basis points of notional, charged up front. It is not a reserve the dealer sets aside from prudence; it is the market price of an option the dealer has written (the client's option to walk away), and if the dealer does not charge it, the trade is mispriced from inception.

The cost-of-cover column is also what Chapter 24's reading gives. In the first year the chance of a default is $1 - e^{-0.02} = 1.980\%$, and $0.60 \times 1.980\% = 1.188\%$, the figure in the table; the same holds in every row to the third decimal. Insuring the exposure and bearing the expected loss on it cost the same, as they must when the insurance is fairly priced.

Years 2 and 3 dominate, because that is where the exposure hump sits. A trade whose exposure peaked later, like the cross-currency swap of Figure 9.1, would carry a larger CVA for the same notional and the same counterparty: about \$102,000 with that curve as drawn, some 9% more, and more again at realistic currency volatility.

Why the spread, and not the record. The cost of cover in the table is about 1.2% a year. Borrowers that trade at that kind of spread have, historically, cost their creditors much less, by a factor of several for an issuer of this quality and far more for the safest credits (Section 24.9).

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Concluding that the CDS market overprices, and charging the client the historical figure, would be a mistake. The spread is not a forecast of how often the counterparty will fail; it is what a seller of protection demands for a risk that pays off in exactly the bad times, when the seller least wants to pay, and so it includes compensation for that timing, for the seller's own funding and capital, and for the possibility that this counterparty turns out worse than its history. CVA uses the spread because CVA is what the hedge costs, not what the loss is expected to be. A dealer that charged historical CVA would undercharge by that same several-fold factor, and would find every hedge it bought costing more than the revenue meant to pay for it. Chapter 28 shows that the same is true of every price in this book: derivatives are priced by what they cost to replicate, and the probabilities a hedging cost implies are not the probabilities a statistician would estimate.

9.3 DVA

If the counterparty's option to default has value, so does the dealer's. The **debit valuation adjustment** (DVA) is the mirror image of CVA: the value to the dealer of its own option to default, computed on the *negative* exposure (the states in which the dealer owes money and would not pay it) at the dealer's own CDS spread, exactly as the counterparty would compute CVA on the dealer.

DVA has an impeccable derivation and an absurd implication. It is required for symmetry: if the client charges the dealer CVA for the dealer's credit risk, and the dealer charges the client CVA for the client's, then each party's CVA is the other's DVA, and the two agree on the trade's value. The implication is that a deterioration in the dealer's own credit is a gain. As the dealer's spread widens, its own option to default becomes more valuable, DVA rises, and the dealer books a profit.

Case Study 9.1 Profiting from your own decline

In 2011, as bank credit spreads widened sharply in the euro area crisis, several large banks on both sides of the Atlantic reported substantial DVA gains. The accounting was correct and the logic unimpeachable: their liabilities, including their derivative liabilities, had become less likely to be paid in full and were therefore worth less, and a fall in the value of a liability is a gain.

The presentational problem was obvious (institutions reporting profits derived from the market's declining confidence in them), and the prudential problem was worse. Capital exists to absorb losses, and capital that increases as the firm's creditworthiness falls does the opposite: a firm approaching failure would report its largest DVA gains just as its capital was most needed.

Regulators resolved this by filtering DVA out of regulatory capital: the gain may be recognized in accounting profit but is deducted from common equity. The effect is a permanent gap between the accounting value of a derivatives book and its regulatory value.

Much of the 2011 gain was not on derivatives at all but on the banks' own bonds carried at fair value, and the accounting standards have since moved that part out of profit and into other comprehensive income (IFRS 9 and, in the United States, Accounting Standards Update (ASU) 2016-01, both required from 2018). The DVA on derivative liabilities still runs through profit.

Most dealers treat DVA as real for valuation and hard to use for management. It cannot be hedged directly, since that would mean selling protection on oneself, which no one will buy in the size required; banks have hedged it approximately by selling protection on peer banks

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whose spreads move with their own, taking on those banks' credit. And it is realized, if at all, mostly by defaulting, or in part through the funding benefit of Section 9.4, where the two adjustments overlap. It therefore appears in the accounts and carries little weight in pricing.

Two refinements matter for a precise bilateral calculation. Each party's charge should be conditioned on the other surviving (the first-to-default effect), which makes bilateral CVA and DVA each slightly smaller than their one-sided versions. And the amount claimed on default depends on the close-out convention: if the surviving party may claim the cost of replacing the trade with a new counterparty, that replacement value includes the surviving party's own DVA, and the loss on default is smaller than a risk-free close-out implies. The 2002 ISDA Master Agreement frames its close-out amount as a replacement cost, which points toward the replacement reading, though how far in practice is not settled.

9.4 FVA

The **funding valuation adjustment** is the cost of funding an uncollateralized derivative position, and it is the most contested item in this chapter. Section 8.1 gave the mechanism: the dealer hedges the uncollateralized client trade with a margined one, so when the client trade moves in the dealer's favor the hedge calls variation margin in cash that nothing from the client offsets, and the difference has to be borrowed. If the trade is against the dealer, the dealer keeps cash that a CSA would have required it to hand over, while the hedge returns margin. Either way the trade leaves a funding position (a cost in the first case, a benefit in the second), carried at the dealer's own cost of funds, which is above the risk-free rate by its funding spread. Under the one-way CSA of Section 8.1, which sovereigns, supranationals and central banks typically insist on, the dealer must pass to the client the margin its hedge returns, so the second case disappears and FVA is all cost.

Example 9.3 Computing FVA

Take the swap of Example 9.2 and suppose the dealer's funding spread over the risk-free rate is 60 basis points. The discounted expected exposure uses the exposure and discount columns of Example 9.2 with the cost of cover left out, each year weighted by its length $\Delta t_i = 1$:

$$\sum_i \mathrm{EE}(t_i)\,\mathrm{DF}(t_i)\,\Delta t_i = 1.848 \times 0.9802 + 2.490 \times 0.9418 + 2.296 \times 0.9048 + 1.630 \times 0.8694 + 0.616 \times 0.8353 = \$8.17 \text{ million-years}$$

The funding cost is

$$\mathrm{FVA} = 0.0060 \times \$8.17\text{m} = \$49{,}000$$

So on this trade FVA is about half of CVA. This is the cost side alone, the funding of positive exposure; the benefit on negative exposure is taken up below. The ratio is not general. It is roughly the funding spread against the counterparty's credit spread, here 60 basis points against 120, and it comes out a little above one-half because the cost of cover decays with survival while the funding cost, as computed here, does not. But it makes the point that funding is the same order of magnitude as credit, not a rounding item.

Case Study 9.2 The FVA debate

In 2012 Hull and White argued, in a widely read article, that FVA should not exist.

Their argument applies a principle this book has used throughout: the value of an asset is determined by its risk, not by how the buyer chose to finance it. A firm with a high funding cost does not get to charge more for the same cash flows; that the cheapest-funded firm would win every trade is a statement about competition, not about value. Charging

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FVA, on this view, confuses the dealer's funding problem with the trade's worth, and amounts to passing the dealer's own credit risk to the client twice, since the client is already compensated through DVA.

The practitioner response is that this is correct in a frictionless world and irrelevant in this one. A dealer that ignores FVA and prices at theoretical value will win exactly the trades that are expensive for it to fund, and will discover the cost in its funding line rather than in its trading book. Whatever the theory says, the cash outflow is real.

The debate was settled, in the way such debates usually are, by accounting. Firms began recognizing FVA in their published results (most visibly when JPMorgan took a $1.5 billion FVA charge in the fourth quarter of 2013), and once one major dealer had recognized it, the others followed rather than explain the discrepancy.

The theoretical question was answered later, and the answer gives each side part of what it claimed. Andersen, Duffie, and Song (2019) showed that funding an uncollateralized position transfers value from shareholders to the bank's creditors, so FVA is a real cost to the shareholders who decide whether to trade, and belongs in the price the desk quotes; but it is not a property of the cash flows, and the trade's value to all of the bank's stakeholders together does not contain it. It is worth understanding both positions, because the same argument recurs whenever a cost that is real to a firm is not a property of the instrument it trades.

FVA and DVA overlap. A genuine technical problem underlies the dispute. The dealer's funding spread reflects its own credit risk, which DVA also prices, so charging both counts the same thing twice, at least in part. Practice varies: some firms compute FVA using only the liquidity component of their funding spread; others restrict FVA to a funding cost adjustment on positive exposure and set the corresponding funding benefit adjustment on negative exposure against DVA. No treatment is universally agreed, and two dealers can quote materially different prices on identical trades for this reason alone.

9.5 MVA

The **margin valuation adjustment** is the cost of funding initial margin over the life of a trade. It exists because of the rules Chapter 8 described: initial margin is posted gross, segregated, and cannot be reused, so it must be funded from inception to maturity.

Example 9.4 Computing MVA

Suppose the swap of Example 9.2 is instead cleared. The clearing house sizes initial margin to cover the swap's value change over a five-day close-out at a high confidence level, 99% or more; at the rate volatility behind Figure 9.1 the five-day standard deviation of the value is about $1 million, so the margin starts at about $2.5 million and declines roughly linearly to zero as the trade amortizes, an average of $1.25 million. MVA is charged on the margin the trade adds to the dealer's cleared portfolio, and this dealer's cleared book is assumed not to offset the trade, so the increment is the whole stand-alone amount; Example 8.5's dealer, whose cleared leg nets against its existing book, would pay far less. The dealer funds the margin at the same 60 basis point spread.

A charge accruing at a constant rate over the trade's life is worth that rate times the trade's discounted life, which at the 4% used throughout this chapter is

$$\int_0^5 e^{-0.04t}\,dt = \frac{1 - e^{-0.2}}{0.04} = 4.53 \text{ years}$$

Using that discount-weighted duration,

$$\text{MVA} = 0.0060 \times \$1.25\text{m} \times 4.53 = \$34{,}000$$

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Compare this with the figures already computed: CVA of \$94,000 and FVA of \$49,000 (Examples 9.2 and 9.3). On this trade the cost of funding initial margin is about a third of the credit charge it replaces.

The proportion is not fixed. MVA scales with the margin model's output and with the funding spread, and both rise in exactly the conditions that raise everything else: a stressed look-back period in the margin model, a ten-day close-out for an uncleared trade instead of five, a longer maturity, or a dealer whose own funding has become expensive. Practice Question 8 doubles the margin and raises the spread by half, and MVA overtakes CVA. Clearing and the uncleared margin rules replaced part of the credit risk they removed with a funding requirement, and a firm that dropped its CVA desk after mandatory clearing, on the grounds that counterparty risk had gone, would have been charging nothing for a cost that had changed form rather than disappeared.

MVA is also awkward to compute, because it requires forecasting a *margin model's output* on future dates and scenarios: a clearing house's own initial margin model for a cleared trade, SIMM for an uncleared one under the rules of Section 8.4. Under the uncleared rules the cost is also marginal in a way the cleared one is not: two firms exchange no initial margin until the requirement between them passes the €50 million threshold, so a trade that fits under it carries no MVA, and once the threshold is used up every further trade carries its full margin.

On the same grid as (9.6),

$$\mathrm{MVA} = \sum_{i=1}^{n} s_F(t_i)\ \overline{\mathrm{IM}}(t_i)\ \mathrm{DF}(t_i)\ \Delta t_i \tag{9.7}$$

where $s_F(t)$ is the dealer's funding spread over the rate paid on posted margin (60 basis points in Example 9.4), $\overline{\mathrm{IM}}(t_i)$ is the initial margin outstanding at t_i, averaged across scenarios, and $\Delta t_i = t_i - t_{i-1}$. Like the other adjustments, this is written as a positive cost; Section 9.8 carries the signs.

The difficulty in (9.7) is entirely in $\overline{\mathrm{IM}}(t_i)$: computing it exactly means running the margin model *inside* the Monte Carlo, evaluating sensitivities on every path and date and aggregating them through the margin formulas, which multiplies an already expensive calculation by the cost of a full sensitivity run. Three approximations are used.

A deterministic profile scales today's initial margin by the remaining risk (for a swap, roughly by remaining duration). That is what Example 9.4 does with its linear decline, and it is adequate while the portfolio's composition is stable.

Regression fits IM as a function of the simulated state variables on a subset of paths and evaluates the fitted function everywhere: the regression Section 9.1 borrowed from American Monte Carlo for exposure, turned on the margin model rather than on value.

Sensitivity scaling applies the model's risk weights to the sensitivities simulated on each path, but replaces the step that combines them through the model's correlation formulas, which depends on the whole portfolio on that path, with a simpler fixed rule.

All three approximate a quantity that is itself an approximation, a standardized margin model being no more than a proxy for a risk measure.

9.6 COLVA AND THE TERMS OF THE CSA

Chapter 8 showed that a CSA is rarely the idealized arrangement in which collateral exactly tracks the mark-to-market in cash, daily, in one currency. Real CSAs have thresholds, minimum transfer amounts, non-cash eligible collateral, and currency choice, and each departure has a value. Strictly, the **collateral valuation adjustment** (ColVA) is the value of the terms on which collateral is remunerated and chosen: a collateral rate (or, for non-cash collateral, a return) other than the overnight rate the curve is built on, a floor on that rate, and the right to choose what to post, such as the cheapest of several currencies, an option over the discount curve that

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Example 8.6 values. Thresholds, minimum transfer amounts, and the margin period of risk (the gap between the last collateral movement and close-out) act differently. They leave exposure uncovered, a $5 million threshold being a $5 million unsecured credit line, so their cost is CVA and FVA computed on the exposure that remains; adding it again under ColVA would count it twice. Desks discuss the whole set together, as the terms of the CSA, and so does this section.

Since the end of LIBOR the CSA's cash rate has been an overnight rate such as SOFR or €STR, and since 2016 the uncleared margin rules have required daily variation margin between the firms they cover, so the standard CSA has drifted toward the idealized one and the residual terms matter most on legacy agreements.

Thresholds, minimum transfer amounts, and the margin period of risk enter the simulation through one term, the $C^{(j)}(t_i)$ of (9.2), and the modeling has a single essential feature: collateral reflects the portfolio's value at an earlier date. Writing δ for the margin period of risk, H for the threshold and M for the minimum transfer amount, and setting to zero the independent amount that Chapter 8's credit support amount also includes,

$$C(t) = \max\left(V(t-\delta) - H,\ 0\right) \quad \text{subject to changes of at least } M \tag{9.8}$$

Equation (9.8) describes collateral the dealer *holds*. Under a two-way CSA the dealer also posts when V is negative, symmetrically, and posted collateral matters in two ways the equation omits: it is an exposure of its own if the counterparty does not segregate it, and it must be funded, which is where FVA enters a collateralized trade.

The consequence is that a perfect CSA does not give zero exposure. Even at $H = M = 0$, equation (9.8) leaves $C(t) = \max(V(t-\delta), 0)$: the collateral held reflects the value one margin period ago, so the collateralized exposure is at most the portfolio's move over that window, and exactly that whenever the dealer was holding collateral,

$$E(t) = \max\left(V(t) - V(t-\delta),\ 0\right)$$

At Chapter 8's ten days bilateral and five days cleared, this residual is why a fully collateralized trade still carries a small CVA rather than none. Initial margin is sized to cover exactly this move, which is what it is for, and why it costs what Section 9.5 says it does.

Example 9.5 The cost of a threshold

The swap of Example 9.2 is with a corporate client outside the uncleared margin rules, which would otherwise require a zero threshold (Chapter 8). The client insists on a $5 million threshold in its CSA, below which no collateral moves. The dealer agrees, believing the CSA to be otherwise standard.

For exposures up to $5 million the trade is effectively uncollateralized. In most scenarios the exposure stays below that level, so on the normal model of Section 9.1, averaged over the mid-year dates, the threshold leaves about $1.4 million of uncollateralized exposure over the life of the trade. Insuring that amount at the client's 120 basis point spread for the trade's discounted life carries

$$\text{CVA on the threshold} \approx 0.012 \times \$1.4\text{m} \times 4.53 = \$76{,}000$$

plus funding on the same amount at 60 basis points,

$$0.0060 \times \$1.4\text{m} \times 4.53 = \$38{,}000$$

A concession that sounded operational ("we would rather not exchange small amounts") costs about $114,000 on a single trade, most of the way to the uncollateralized CVA and FVA. (Weighting the premiums by survival, as Example 9.2 does, would take a few thousand dollars off; at this precision it does not matter.) Thresholds are priced, and clients are frequently surprised by how much.

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9.7 KVA

The **capital valuation adjustment** (KVA) is the cost of the regulatory capital a trade requires over its life: the equity a bank must hold against the position because the rules say so, rather than because it chose to. How much capital a trade requires is Chapter 27's calculation (counterparty exposure under the standardized approach for counterparty credit risk (SA-CCR) in Section 27.3, the charge against CVA's own volatility in Section 27.6), and Example 9.6 takes the result as given.

Capital is not free. Shareholders require a return above the risk-free rate (a hurdle of perhaps 8 to 12% over it), and a trade that consumes capital for ten years consumes that return for ten years. If the trade does not earn it, the shareholders are worse off even though the trading book shows a profit.

Example 9.6 Computing KVA, and its size

Example 27.3 computes the counterparty credit capital for the uncollateralized swap of Example 9.2 at \$0.33 million at inception, against a corporate at a 100% risk weight. Counterparty capital rises in the first years as replacement cost builds, then falls as the swap runs off. The example takes the case in which the client, a nonfinancial corporate, is exempt from the CVA capital charge, as it is under the EU rules (Section 27.7), so the capital is counterparty credit capital alone; suppose it averages \$0.4 million over the swap's life, and that the firm's hurdle rate over the risk-free rate is 10%. Then

$$\text{KVA} = 0.10 \times \$0.4\text{m} \times 4.53 = \$181{,}000$$

This is about twice the CVA and nearly four times the FVA. Where the CVA capital charge applies, as it does to the same trade in the United States and Canada (Section 27.7), it adds roughly as much capital again and KVA is materially higher. On long-dated uncollateralized trades KVA is often the largest adjustment of all, and it is the reason such trades became expensive and less common after the post-crisis capital rules took effect. Practitioners compute it on a projected capital profile, trade by trade within the portfolio, and include the leverage ratio where it binds, since a bank must hold capital against whichever rule is the stricter.

CVA the price and CVA the capital charge

The same three letters denote two different things, a persistent source of confusion.

CVA as used in Section 9.2 is a *price*: the market value of the counterparty's default option, deducted from the value of the trade.

The **CVA capital charge** is a *regulatory requirement*: capital held against the risk that CVA itself *moves*. Its rationale is empirical: in the financial crisis, most losses on counterparty exposures came not from counterparties actually defaulting but from the mark-to-market deterioration of CVA as credit spreads widened. Firms held capital against default and were hit by volatility.

The charge is computed under the Basel framework of Section 1.12, either by the basic approach (BA-CVA), a formula on exposures and supervisory risk weights, or, with supervisory approval, by the standardized approach (SA-CVA), which works from CVA's sensitivities to credit spreads and to the market factors driving the exposure. The Basel standard took effect in 2023, in the European Union in 2025 and in the United Kingdom in 2027; the US agencies re-proposed theirs in March 2026, and they were not final at the time of writing. Section 27.6 sets out the approaches and Section 27.7 the exemptions. The charge feeds into KVA, since it is part of the capital the trade consumes.

The distinction matters because the two respond to different actions. Single-name CDS protection on a counterparty hedges the price and reduces the charge (under BA-CVA by at most three-quarters), and novating the trade to a clearing house eliminates both. But for most

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counterparties no single-name CDS trades (Section 9.10), and the index or proxy protection a desk can actually buy hedges the *price* while earning only partial relief from the *capital charge*. Tightening a CSA lowers both by different amounts: the price by what the firm's own exposure model says, the charge by what the standardized formulas allow.

KVA is the most conceptually troubled of the adjustments. Capital requirements are set by regulators and change; the hurdle rate is a management choice, not a market price; and including a firm's cost of equity in the valuation of an instrument sits uneasily with the idea that value is a property of the instrument. Some firms treat KVA as a pricing hurdle (a floor below which they will not trade) rather than a valuation adjustment carried in the accounts, which is arguably the more honest treatment.

9.8 PUTTING THEM TOGETHER

Table 9.1 collects the adjustments for the trade used throughout this chapter, under two collateral arrangements. Two entries rest on inputs the examples did not state. DVA is computed at the dealer's own spread, taken as 49 basis points, on the mirror image of the client's exposure profile: it assumes the dealer's negative exposure has the same profile as its positive one, which holds for an at-market swap under the symmetric normal model of Example 9.2. And the cleared trade's capital is taken as averaging $20,000, set by the leverage ratio rather than by the risk-based rule: Example 27.3's trade exposure to the clearing house needs only about $1,400 at the 2% risk weight, but at a 3% leverage requirement it needs about $20,000 (Section 27.10). The totals are net of DVA.

Table 9.1: XVA on a five-year $100m swap under two collateral regimes, net of DVA.

Adjustment	Uncollateralized	Cleared	Driver
CVA	−94,000	≈ 0	counterparty default
DVA	+39,000	≈ 0	own default
FVA	−49,000	≈ 0	funding the exposure
MVA	—	−34,000	funding initial margin
KVA	−181,000	−9,000	regulatory capital
Total	**−285,000**	**−43,000**	

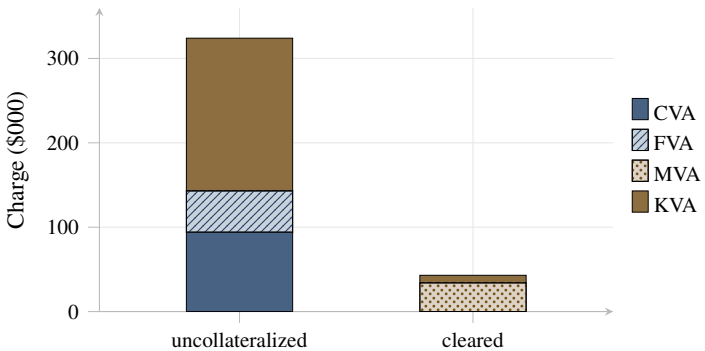


Figure 9.2: Composition of the charge in Table 9.1, gross of DVA.

Figure 9.2 plots the composition, and shows what the totals conceal: the two columns are not the same charge in different sizes but largely different charges.

Three observations follow.

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The adjustments are large. On the uncollateralized trade they total 28.5 basis points of notional net of DVA. DVA carries little weight in pricing (Section 9.3), so the charge a client is actually quoted is nearer the $324,000 before it: 32.4 basis points of notional, or about 7 basis points a year on the swap rate, against an interdealer bid–ask spread on a five-year swap of a fraction of a basis point. The XVA is not a correction to the price; on this trade it substantially *is* the price.

Clearing changes the composition of the charge rather than eliminating it. Credit and funding charges collapse toward zero, and are replaced by the cost of funding initial margin. The total falls (clearing is genuinely cheaper, as Chapter 8 explained it was designed to be), but not to zero. For this client the cleared column is a comparison rather than a choice: a nonfinancial corporate hedging its own business is usually exempt from mandatory clearing and from the uncleared margin rules (the end-user exception in the United States, the clearing threshold in the European Union), which is why uncollateralized corporate swaps still exist and why the left-hand column is the one a corporate treasurer is quoted.

Capital dominates. On the uncollateralized trade KVA is larger than everything else combined. That explains market structure changes that otherwise look puzzling: why dealers pushed clients into clearing, why long-dated uncollateralized corporate trades became expensive, and why balance-sheet-light business models proliferated.

9.9 WRONG-WAY RISK

Wrong-way risk arises when exposure increases in the same states in which the counterparty is more likely to default. The textbook example is buying credit protection on a sovereign from a bank in that country: the protection is worth most exactly when the sovereign is in trouble, which is exactly when the bank is least able to pay. Chapter 25 shows this was not hypothetical.

Less dramatic cases are common. A dealer that has bought protection against falling oil prices from an oil producer has exposure that rises when oil prices fall, which is when the producer's credit is worst. A dealer receiving fixed from a highly levered borrower has exposure that rises as rates fall, which may correlate with the conditions that stress the borrower.

Right-way risk is the reverse and reduces CVA: a dealer that has bought protection against *rising* oil prices from the producer has exposure rising when oil rises, which is when the producer is strongest.

The insurance argument shows exactly where the first assumption of Section 9.2 did its work. What the dealer needs cover for is the exposure *at the moment the counterparty fails*, the expected exposure *given that* the default falls in each interval, and the standard calculation replaces it with the unconditional average. Under wrong-way risk the exposure in the scenarios where the counterparty actually defaults is systematically larger than the average across all scenarios, so the amount insured is too small, and (9.6) is a lower bound rather than an estimate.

Two modeling approaches are standard, and regulation avoids the question:

- *Intensity models* let the counterparty's default rate rise and fall with the driver of the exposure.
- *Copula approaches* join the default time and the exposure while leaving each one's own distribution as it was, the device Section 26.6 turns on portfolio credit.
- *Regulation* sidesteps both, covering general wrong-way risk with the multiplier of 1.4 (Section 27.5) and specific wrong-way risk with a rule that treats a legally connected reference entity as already defaulted (Section 27.9).

Example 27.4 puts a size on the effect: a dependence strong enough to double the exposure expected at default doubles the CVA. Where the dependence is definitional rather than statistical,

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no correlation parameter is adequate. Protection bought on a sovereign from a bank domiciled in it is not an exposure with a correlation adjustment; the exposure and the default are the *same event*, and the trade should be assessed as a jointly contingent claim.

9.10 THE XVA DESK

The organizational response to all this is a central **XVA desk**.

When a trader books a client trade, the XVA desk charges the relevant adjustments and takes the corresponding risk onto its own book. The originating desk is left with a clean, hedgeable market risk position; the XVA desk holds the credit, funding, and capital exposures across the whole firm.

The centralization exists because these risks are *portfolio* properties, not trade properties: exposure is a netting-set quantity (9.4), so a new trade's CVA depends on everything already facing that counterparty, and only a desk that sees the whole book can price the increment.

Hedging. The XVA desk hedges CVA by doing what Section 9.2 priced: buying credit protection on its counterparties, and hedging the sensitivity of the exposure itself to rates and FX. Because the exposure is an option, CVA has a delta and a vega: sensitivities to the underlying market variable and to its volatility. Hedging CVA is harder than it sounds: liquid CDS exists for a few hundred issuers, while a dealer faces thousands of counterparties, most with no traded credit, so much CVA is hedged by proxy, with index CDS and correlation assumptions, leaving basis risk whenever a specific issuer diverges from its proxy.

Other instruments fill parts of the gap. A **contingent CDS** pays the loss on a notional that is itself defined by a referenced derivative's value at default, so it insures the exposure exactly rather than its average, and is bought where one large exposure justifies the bespoke cost. Banks also transfer CVA risk in portfolio form, through credit-linked notes and guarantees sold to investors such as insurers and credit funds that want the spread and can carry the risk. And what a desk hedges follows the capital charge as well as the price: SA-CVA recognizes hedges of the exposure's rate and FX sensitivities and BA-CVA does not (Section 27.6), so a desk under SA-CVA hedges the market side of CVA as well as the credit side.

FVA and MVA are hedged, to the extent they are, by managing the firm's own funding profile; in most banks the treasury charges them to the desks through funds-transfer pricing, at an internal rate for the term and liquidity of the funding each trade consumes. KVA is generally not hedged at all; it is managed by not doing the trades.

Why the XVA desk loses money in a crisis

An XVA desk that has hedged carefully can still report large losses, for three recurring reasons.

Proxy basis. When a specific counterparty's spread widens while the index used to hedge it does not, the CVA rises and the hedge does not respond. In a broad credit sell-off the proxy works; in an idiosyncratic default it does not, and idiosyncratic defaults are exactly when CVA matters.

Cross-gamma. CVA depends on exposure *and* on credit, and the exposure depends on rates and FX, so a rate move changes CVA even with credit spreads unchanged, and a spread move changes how much a rate move matters. That interaction is CVA's **cross-gamma**. Hedging the credit sensitivity alone leaves it exposed, and the cross-terms are largest when both move together, which is the wrong-way risk of Section 9.9 as the hedging desk sees it.

Hedging costs. Buying CDS protection consumes capital and funding of its own, so a desk that hedges its CVA fully has converted a credit exposure into a capital and funding exposure, which is an improvement but not a nullity.

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The deeper issue is that CVA hedging is an attempt to hedge an option on a portfolio whose composition changes daily, written on a counterparty whose credit may not trade, in a market where correlations move when it matters. Doing it imperfectly is not a failure of execution; it is the nature of the problem.

9.11 THREE DIFFERENT NUMBERS

A recurring source of confusion inside banks is that the same trade carries three XVA figures that are all correct and none of which agree.

The accounting number is what appears in published financial statements. It must satisfy fair value accounting standards (IFRS 13 internationally and ASC 820 in the United States), which require an exit price, what a market participant would pay, including the risk that either party does not perform, and allow a portfolio of trades with one counterparty to be valued net. That admits CVA, DVA, and now FVA, which any market participant would price, and generally excludes KVA, because a firm's own hurdle rate is not a market participant's.

The regulatory number is what drives capital. It includes the CVA capital charge of Section 9.7, applies prudential filters (removing DVA, as Section 9.3 described), and follows rules that may differ from the firm's own models.

The pricing number is what the desk charges the client. It includes everything the trade actually costs the firm, KVA included, because a trade that does not clear its capital hurdle destroys value regardless of how it is booked.

The client has an accounting number of its own, seen from the other side: its fair value carries CVA on the dealer and DVA on itself. Where it has designated the swap as a hedge of a loan, those adjustments have no counterpart in the hedged item, so they show up in its profit as hedge ineffectiveness.

The three diverge systematically, and the divergence is not an error to be reconciled away. A trade can be profitable on the accounting number, capital-destroying on the pricing number, and punitive on the regulatory one. Firms that manage to only one of the three make predictable mistakes: managing to accounting alone produces trades that consume capital without earning it, to regulatory alone an optimization of rules rather than of economics, and to pricing alone published results that surprise the market.

The practical resolution is that pricing decisions use the pricing number, published accounts use the accounting number, and the difference is explained rather than eliminated.

9.12 XVA AND THE MEANING OF VALUE

The chapter began with the claim that "the value of the swap" does not identify a number. What survives can be stated precisely. The cash flows on their own still have a well-defined value: the forward rates of Chapter 7, discounted on the collateral curve, a number two dealers will agree on to within a rounding error. What is not objective is the price, which depends on who the counterparty is, what the master agreement and CSA say, how the dealer funds itself, what capital the trade attracts, and what is already on the books between the two parties, so two dealers quoting the same client on the same trade may legitimately differ by tens of basis points, and neither is wrong. This is a genuine change in what a derivative *is*. Before the crisis, a swap was a bundle of cash flows and its value was a property of those cash flows. It is now more accurate to say that a swap is a bundle of cash flows plus a relationship, and that a substantial part of its price is the cost of maintaining the relationship.

The framework that prices the relationship remains contested, for three reasons.

The components overlap and the overlaps are not agreed. Besides the FVA–DVA problem of Section 9.4, capital is held against counterparty risk that CVA has already priced, and MVA

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and FVA both price funding. Every firm decides for itself what to include, so the totals are not comparable across firms.

Some components are not market prices. CVA is disciplined by hedging: a dealer that charges too little can be arbitraged. KVA is not, for the reasons Section 9.7 gave, and calling it a valuation adjustment gives it a status it has not earned.

The precision is illusory. Simulations over years, credit spreads taken from thin markets, and unobservable correlations cannot support a total reported to the nearest thousand dollars, as Table 9.1 reports it.

None of this argues for ignoring the adjustments. The costs they represent are real, and a firm that does not charge them will discover them elsewhere. It argues for treating XVA as a disciplined framework for allocating known costs, rather than as a valuation in the sense that Chapters 13 and 15 use the word.

Counterparty credit adjustments predate the crisis: Sorensen and Bollier (1994) and Duffie and Huang (1996) already treated a swap's counterparty exposure as an option that is not symmetric between the two parties, and by the mid-2000s large dealers ran CVA desks. What came after 2008 was everything else, collateral-rate discounting, funding, margin and capital, and the works that introduced each are set out in the Further Reading.

SUMMARY

Exposure is $\max(V(t), 0)$: counterparty risk is an option on the portfolio's value struck at zero, so it needs a distribution rather than a point estimate, and it is measured across the netting set, where an offsetting trade attracts a lower charge. An interest rate swap's exposure humps early in its life, while a cross-currency swap's rises to maturity because the notional is exchanged then. Expected exposure serves pricing and potential future exposure serves credit limits, and much internal disagreement comes from not saying which is meant.

CVA is the market price of the counterparty's option to default, priced the way everything in this book is priced, by what the hedge costs: CDS protection on the counterparty, sized to the expected exposure. It therefore uses market-implied rather than historical default probabilities, because it is the cost of a hedge and not a forecast of a loss. DVA is its mirror image, required if two parties are to agree on a value, but it turns a deterioration in a firm's own credit into a reported gain, so regulators filtered it out of capital and it survives in accounts rather than in pricing.

Three further adjustments price funding and capital. FVA charges an uncollateralized position at the dealer's own funding spread; Hull and White argued that funding cost is not a property of the instrument, and practice settled the question through accounting recognition. MVA funds initial margin, so clearing and margin rules converted part of the charge for counterparty risk into a funding charge rather than removing it. KVA prices capital at the firm's hurdle rate and is the most strained of the set, since it embeds a management choice in an instrument's value.

Together the adjustments can come to many times the interdealer spread: the XVA is not a correction to the price but substantially is the price. Clearing reduces the total without removing it. ColVA prices how collateral is remunerated and chosen; thresholds and the margin period of risk appear as CVA and FVA on the exposure they leave.

CVA the price and the CVA capital charge are different objects, the second held against the volatility of the first, so reducing one need not reduce the other. Wrong-way risk makes the standard calculation a lower bound. And the same trade carries three defensible numbers, for accounting, for regulatory capital and for pricing, and they differ by design rather than by error. XVA is managed centrally because the quantities are portfolio properties, and hedged imperfectly because most counterparties have no traded credit. The framework remains

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contested, but the costs are real, and a firm that does not charge them meets them somewhere else.

FURTHER READING

Gregory's *The xVA Challenge* is the standard practitioner reference and Green's *XVA* treats the whole set of adjustments in one volume; Brigo, Morini, and Pallavicini's *Counterparty Credit Risk, Collateral and Funding* gives the derivations this chapter states, in full. Burgard and Kjaer (2011) is the source of the partial-differential-equation representation with bilateral credit and funding costs. The FVA argument is worth reading in the original: Hull and White's "The FVA Debate" and their later "Collateral and Credit Issues in Derivatives Pricing" set out one side, Laughton and Vaisbrot's 2012 reply in *Risk* the other, and Andersen, Duffie, and Song's "Funding Value Adjustments" the resolution in terms of who bears the cost. Piterbarg's "Funding Beyond Discounting" is the source of collateral-rate discounting, and Green, Kenyon, and Dennis's "KVA: Capital Valuation Adjustment by Replication" introduced the capital charge. Capriotti (2011) explains the adjoint algorithmic differentiation that makes a nightly exposure run affordable at all.

PRACTICE QUESTIONS

9.1 Explain why counterparty exposure is $\max(V(t), 0)$ rather than $|V(t)|$, and why this makes counterparty risk an option.

9.2 Explain why an interest rate swap's expected exposure peaks partway through its life while a cross-currency swap's rises to maturity, and state which carries more counterparty risk and why.

9.3 Explain why exposure is computed on the netting set rather than trade by trade, and why a dealer might quote a better price for a trade that offsets its existing position with a client.

9.4 A three-year uncollateralized swap has expected exposures of \$1.2m, \$1.5m, and \$0.9m at mid-year points. The counterparty's CDS trades at 180 basis points, recovery is 40%, and discount factors are 0.97, 0.93, and 0.89. Compute the CVA, first as the cost of insuring the exposure and then as loss given default times the default rate the spread implies, and confirm the two agree.

9.5 Explain why CVA is computed from the counterparty's CDS spread rather than from its historical default rate, and what would happen to a dealer that used the historical rate.

9.6 State the argument for DVA and the objection to it, explain why regulators removed DVA from capital, and state what that leaves banks managing.

9.7 Set out the Hull–White argument against FVA and the practitioner response. In what sense was the debate settled and in what sense was it not? Identify where FVA and DVA arguably double-count, and state what a firm would have to believe to include both.

9.8 Using the trade of Example 9.4, recompute MVA at a funding spread of 90 basis points and average initial margin of \$2.5 million, compare it with the CVA of Example 9.2, and explain why KVA is nonetheless often the largest adjustment on a long-dated uncollateralized trade.

9.9 A portfolio's value at the peak of its exposure profile is normally distributed about zero with a standard deviation of \$9 million. Compute the expected exposure and the 95% potential future exposure, and explain which is used for pricing, which for credit limits, and why.

9.10 A client requests a \$3 million threshold in its CSA. Using a CDS spread of 150 basis

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points, a funding spread of 70 basis points, an average uncollateralized exposure of $1 million, and a discount-weighted duration of 4.0, estimate the cost of the concession. How would the funding cost change if the client were instead a sovereign that insisted on a one-way CSA, under which only the dealer posts collateral?

9.11 Distinguish CVA the valuation adjustment from the CVA capital charge, explain why the latter exists, and give one action that reduces both and one that hedges the price while leaving most of the charge in place.

9.12 An XVA desk reports a large loss in a quarter during which it was fully hedged on credit spreads. Give three distinct explanations consistent with the hedging having been done competently.

FURTHER QUESTIONS

9.13 A dealer prices a ten-year uncollateralized swap for a corporate client at theoretical value plus CVA only, ignoring FVA, MVA, and KVA. Describe what happens to the firm over the following decade, identifying which cost appears where in its accounts.

9.14 Give three examples of wrong-way risk, one specific and two general, and explain in each case why insuring the average exposure fails. How would you adjust the calculation in each case?

9.15 Argue that KVA does not belong in a valuation at all, and then argue that omitting it guarantees that a dealer's derivatives business destroys shareholder value. Which treatment would you adopt, and how would you present it to a trading desk?

9.16 Post-crisis reform moved derivatives into clearing to reduce counterparty risk. Using Table 9.1, assess how far it succeeded, and explain precisely what was reduced and what was substituted for it.

9.17 Most of a dealer's counterparties have no traded CDS. Explain how CVA is hedged in practice, what basis risk this leaves, and when that basis risk would be most damaging. Relate your answer to the default correlation of Chapter 24.

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Part II

Options: Mechanics and Strategies

Three chapters that take an option apart without pricing it. The mechanics come first (how contracts are listed, adjusted, and exercised), then the arbitrage bounds that hold whatever model one believes, and finally the standard strategies. Chapter 11 deserves more attention than its lack of a model suggests: those bounds are the constraints every later model must respect, and they follow from no-arbitrage alone.

10 **Mechanics of Options Markets**
listing, adjustment, exercise, and the life cycle of a contract

11 **Properties of Options and Arbitrage Bounds**
put–call parity, and the limits every model must respect

12 **Trading Strategies Involving Options**
spreads, combinations, and the profit diagrams that describe them

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Chapter 10
Mechanics of Options Markets

Apart from the delivery options embedded in Treasury futures (Section 6.4), everything in Part I was symmetric. A forward, a futures contract, and a swap all oblige both parties, and both can gain or lose without limit. This chapter begins the study of instruments that are not symmetric, and the asymmetry changes almost everything: how they are priced, how they are margined, who can safely write them, and what happens at expiration.

The economics were sketched in Section 1.1. An option confers a right rather than an obligation, so its holder will exercise only when it pays; the payoff can therefore never be negative to the holder, and they must pay for it at the outset. What Part III must supply is the price of that right. First, though, comes the machinery: what exactly the contract says, how it is quoted, who guarantees it, what the writer must post, and what happens at expiration. Getting this wrong is a common source of loss, and unlike model error it is avoidable.

Sections 10.1 to 10.4 are the contract and how it is read: the four positions, the specifications, the underlyings beyond single stocks, and the option chain. Sections 10.5 and 10.6 are how options trade and are quoted, including the volatility quotation Chapter 15 explains. Sections 10.7 to 10.9 are the mechanics of short selling, margin, and exercise and assignment; Section 10.10 looks beyond the standard contract and Section 10.11 at who trades options and why.

This chapter also covers a development with no counterpart in earlier textbooks: options markets have been transformed since 2020 by retail participation and by contracts that expire the same day they are traded. Section 10.12 treats it as a market-structure topic rather than a curiosity, because it now shapes intraday behavior in the largest equity index in the world.

10.1 CALLS, PUTS, AND THE FOUR POSITIONS

A **call** gives its holder the right to buy the underlying asset at the **strike price** K; a **put** gives the right to sell at K. The right lasts until the **expiration date**. A **European** option can be exercised only at expiration; an **American** option can be exercised at any time up to it. The names, coined by Paul Samuelson in the mid-1960s, have nothing to do with geography: both types trade on both continents.

Every option contract has two sides. The holder is said to be **long** and has **bought** the option; the writer is **short** and has **sold** it. There are four elementary positions, shown in Figure 10.1. Like Figure 1.1, the figure plots profit, the payoff net of the premium, because that is what a trader experiences.

The two right-hand panels are the ones to dwell on. A writer's best case is the premium, achieved when the option expires worthless, and their worst case is unbounded for a call and bounded only by the stock reaching zero for a put. This is why the two sides of an option trade are not mirror images in any practical sense: they face different margin requirements, different capital treatment, and different credit risk. The buyer of an option has no obligations after paying the premium and can be a retail client with a small account. The writer has an open-ended obligation and must be collateralized, which is the subject of Section 10.8.

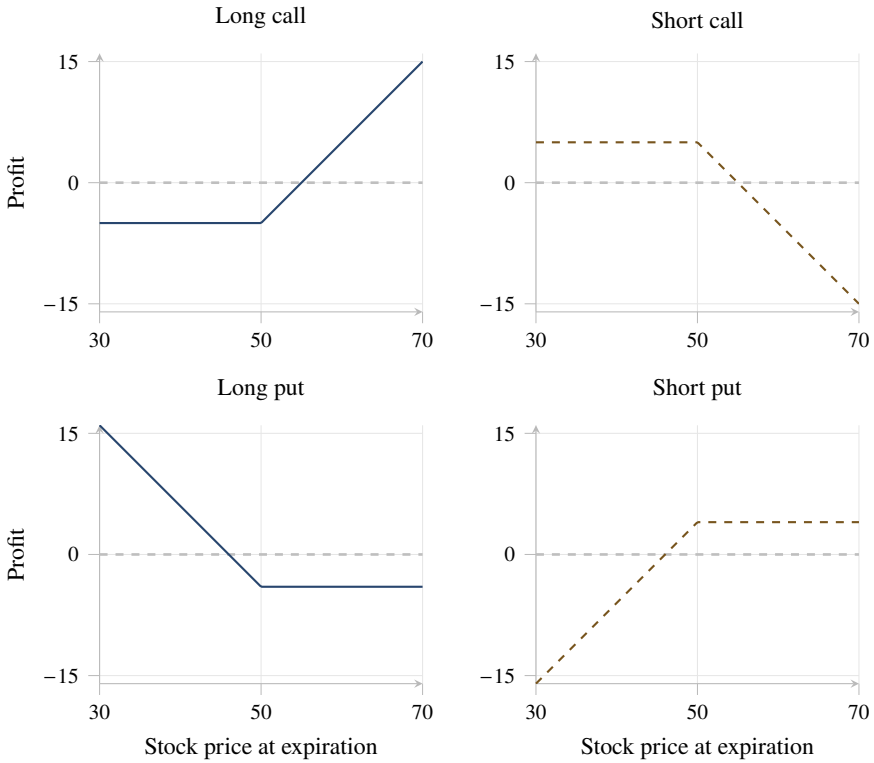


Figure 10.1: Profit at expiration. Strike 50; call premium 5, put premium 4.

Moneyness. An option's **moneyness** is where the stock price S stands relative to the strike K. An option is **in the money** if exercising it immediately would produce a positive cash flow, **at the money** if the stock price equals the strike, and **out of the money** otherwise. For a call these correspond to $S > K$, $S = K$, and $S < K$; for a put the inequalities reverse. The terms describe the option, not the trade: an out-of-the-money option can be an excellent purchase and an in-the-money one a poor purchase, depending entirely on price.

Intrinsic and time value. The **intrinsic value** of a call is $\max(S - K, 0)$, and of a put $\max(K - S, 0)$: what the option would be worth if it expired now. An American option must trade for at least its intrinsic value, since otherwise one could buy it and exercise immediately for a riskless profit. The difference between the market price and the intrinsic value is the **time value**, and it exists because the remaining life gives the stock further opportunity to move in the holder's favor while the downside stays capped.

Figure 10.2 shows the relationship that Part III spends several chapters making precise. Suppose the stock is at 50 and a call struck at 50 trades at 5. Its intrinsic value is zero, so all 5 is time value. With the stock at 60 the curve puts the call at about 12: 10 of intrinsic value and about 2 of time value. With the stock at 40 it is worth about 2, again all time value. Two features generalize. Time value is greatest near the money, because that is where uncertainty about whether the option finishes in or out of the money is greatest; deep in-the-money and out-of-the-money options have little of it. And time value decays to zero at expiration, when the curve collapses onto the kink, the phenomenon Chapter 18 quantifies as theta.

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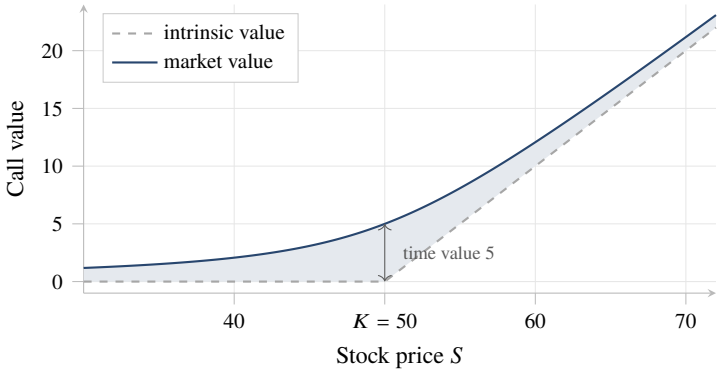


Figure 10.2: A call before expiration, assumed to trade at 5 with the stock at the strike of 50. Shaded region is time value. Schematic.

10.2 CONTRACT SPECIFICATIONS

An exchange-traded option is a standardized contract, and the standard terms matter because a trader chooses among them rather than negotiating them.

Contract size. A standard US equity option covers 100 shares. A quoted premium of $3.20 therefore costs $320 per contract, and a novice who confuses the two has misjudged their position by two orders of magnitude.

Strike prices. The exchange lists strikes at fixed intervals around the current price and adds new ones as the stock moves. The base rule for US equity options sets the interval by price level: $2.50 for strikes up to $25, $5 from $25 to $200, and $10 above $200. Listing programs have narrowed that considerably. The $2.50 Strike Price Program extends the $2.50 interval on selected classes to $50, and to $100 within $10 of the stock price; most actively traded stocks carry $1 intervals below $50, and the lowest-priced stocks $0.50; the largest ETFs and indices are denser still: SPY, the largest S&P 500 ETF, is listed at $1 intervals whatever its price, and S&P 500 index options at 5-point intervals near the money, widening to 25 and 50 points farther out. An actively traded $48 stock therefore shows strikes at 45, 46, 47, and so on up to 50, then 52.50, 55, 57.50, and 60. New strikes are added as the stock nears either end of the ladder, so the price stays bracketed, and old ones stay listed until they expire. This is why a stock that has doubled carries a long tail of near-worthless low-strike puts beneath it.

Expiration dates. US equity options historically expired monthly, on the third Friday. That is still the anchor for the largest contracts, but the calendar has filled in dramatically. Weekly expirations were introduced in 2005 and extended across the most liquid stocks, and since the week of January 26, 2026, after SEC approval that month, stocks with a market value above $700 billion and the largest bitcoin ETF have also carried Monday and Wednesday expirations. For the S&P 500 index, expirations now occur *every trading day*, following the addition of Tuesday and Thursday expirations in 2022. Figure 10.3 shows what this looks like over a typical month, and Section 10.12 explains why it matters.

Exercise style. Options on individual US stocks are American; options on broad indices are typically European. The distinction is not cosmetic. Chapter 11 shows that an American call on a non-dividend-paying stock should never be exercised early, that an American put may well be, and that the difference in value between the two styles can be substantial for puts.

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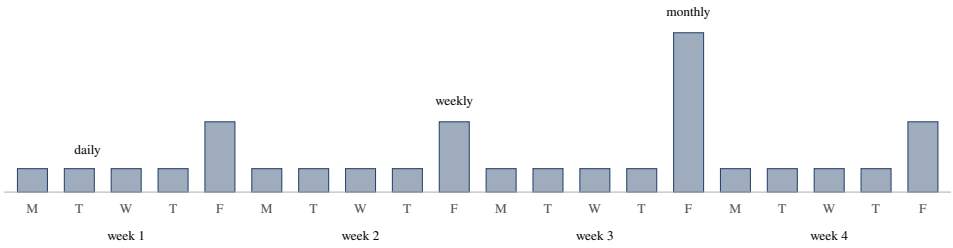


Figure 10.3: S&P 500 index option expirations in a month. Bar heights schematic.

Settlement. Equity options deliver shares. Index options settle in cash against a value computed from opening or closing prices, because delivering a 500-stock basket is impractical.

Corporate actions and contract adjustment

An option is a claim on a specified quantity of a specified security. When a corporate action changes what a share *is*, the contract must be adjusted, or the option would be arbitrarily enriched or destroyed by an event unrelated to the company's value.

The governing principle is to preserve the aggregate value of the position. For a stock split or stock dividend, the contract multiplier and strike are adjusted so that the total strike cost is unchanged. For a merger, the deliverable becomes whatever a holder of 100 shares receives. Ordinary cash dividends, paid under a regular policy, do *not* trigger an adjustment, however large; a special dividend of \$0.125 a share or more does, normally by reducing the strike. This is important and often misunderstood, and Chapter 11 shows that it is precisely why dividends can make early exercise of American calls rational.

Example 10.1 A three-for-two split

An investor holds 10 call contracts on a stock, strike \$60, each covering 100 shares. The company declares a 3-for-2 stock split; each old share becomes 1.5 new shares, and the stock price falls from about \$66 to about \$44.

Without adjustment the options would be destroyed: a \$60 strike on a \$44 stock is deep out of the money, though nothing of economic substance has happened. The adjustment multiplies the deliverable by $3/2$ and divides the strike by the same factor:

$$\text{new contract size} = 100 \times \frac{3}{2} = 150 \text{ shares}, \qquad \text{new strike} = \frac{60}{3/2} = \$40$$

Check that the position is unchanged. Before, exercising cost $10 \times 100 \times 60 = \$60{,}000$ for 1,000 shares worth \$66,000. After, exercising costs $10 \times 150 \times 40 = \$60{,}000$ for 1,500 shares worth $1{,}500 \times 44 = \$66{,}000$. The strike cost, the share value, and the moneyness are all preserved.

The number of contracts stayed at 10 and the contract size changed. For splits producing whole-number ratios (2-for-1, 3-for-1) the convention runs the other way: the contract count is multiplied and the size stays at 100. Either way the aggregate is preserved, but a trader reading a position report must know which convention was applied.

How an option is named

Every listed option needs an identifier, and unlike a stock ticker it must encode four things at once: the underlying, the expiration, the type, and the strike. Since 2010 the US market has used the format established by the Options Symbology Initiative, shown in Figure 10.4.

The symbol in the figure reads: Apple, expiring January 19, 2024, call, strike \$190.000. Nothing has to be looked up: the identifier is the contract description.

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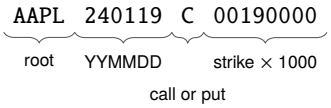


Figure 10.4: US option symbol format.

The strike is carried in eight digits with three implied decimals, so `00190000` is \$190.000 and `00002500` is \$2.50. The three decimal places exist precisely because adjustments of the kind in Example 10.1 can produce strikes that are not round numbers: dividing \$60 by $3/2$ gave a clean \$40, but dividing by $7/5$ would not have.

This scheme replaced a much cruder one. Before 2010, an option was identified by a root plus two letters, one encoding the expiration month and type and one encoding the strike. With only twenty-six letters available for strikes, the system could not survive the proliferation of strike prices and expirations described earlier in this section, and by the late 2000s it was producing genuine ambiguity about which contract a symbol referred to.

The adjusted-contract trap. One convention deserves emphasis because it causes real losses. When a contract is adjusted for a corporate action, it does not keep the ordinary root but receives a *numbered* one: `AAPL1`, then `AAPL2` for a second adjustment, and so on. The digit is appended whatever the length of the root; the root field of the symbol is six characters wide, so even a five-letter root such as `GOOGL` has room for it. The two roots then trade side by side, and they are different contracts. `AAPL` delivers 100 shares; `AAPL1` may deliver 150 shares, or shares plus cash, or shares in an acquiring company.

A trader who sees an unusually cheap quote on what appears to be a familiar option is very often looking at an adjusted contract with a different deliverable, and a position report that lists both roots under the same underlying will misstate the exposure. The rule is simple: a numbered root means read the adjustment memorandum before trading.

10.3 OPTIONS ON INDICES, ETFS, CURRENCIES, AND FUTURES

Options on individual shares are the natural teaching case but not the largest market. Table 10.1 sets out how the specifications differ across underlyings, because the differences change what a position actually is.

Table 10.1: Specifications across underlyings, US market.

Underlying	Size	Exercise	Settlement	Note
Single stock	100 shares	American	Deliver shares	Dividends drive early exercise
Stock index	Multiplier × index	European	Cash	Settlement value may be untradeable
Index ETF	100 shares	American	Deliver shares	Index economics, stock mechanics
Currency	Fixed currency amount	European	Cash or delivery	Foreign rate acts like a dividend yield
Futures	One futures contract	American	Futures position	Exercise yields a margined position

Index options. A stock index option has a multiplier (\$100 for the main US index contracts), so a contract on an index at 5,250 controls $5{,}250 \times 100 = \$525{,}000$ of notional. These are

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the largest option contracts most participants will encounter. Example 10.2 works through a position.

Cash settlement introduces a subtlety. Monthly index options settle against a value computed from the *opening* prices of the constituent stocks, which do not all open at once. The resulting settlement value can differ from any index level published during the session, and a holder cannot trade at it. A position carried into settlement is therefore exposed to a price that is calculated rather than observed.

Example 10.2 An index option position

An index stands at 5,250. A fund buys 10 call contracts struck at 5,300, expiring in one month, quoted at 42.00. The multiplier is \$100.

The position costs $10 \times 100 \times 42.00 = \$42{,}000$ and controls $10 \times 100 \times 5{,}250 = \$5{,}250{,}000$ of index.

The fund has obtained exposure to \$5.25 million of index for \$42,000 (eight-tenths of one percent) and, unlike the futures position of Example 1.5, it can never be asked for another cent. That is the main argument for using options when funding is uncertain.

If the index settles at 5,380, the payoff is $(5{,}380 - 5{,}300) \times 100 \times 10 = \$80{,}000$ and the profit is \$38,000. If it settles anywhere at or below 5,300 the options expire worthless and \$42,000 is lost. The index must rise about 1% merely to reach the strike, and about 1.8% to break even: buying out-of-the-money options requires being right about direction, size, and timing simultaneously, which is why Chapter 12 builds structures that relax one of those requirements.

ETF options. Options on an index-tracking ETF give near-identical economic exposure but with the mechanics of a stock option: American exercise, delivery of ETF shares, and dividends paid by the fund. Since November 2024 they have included the spot bitcoin ETFs, the largest now with a position limit of 1,000,000 contracts . A trader who is short ETF calls can be assigned early (made to deliver because a holder has exercised, the mechanism of Section 10.9) around an **ex-dividend date**, the first day the shares trade without the coming dividend, because a holder can exercise the day before to collect it; a trader short European index calls such as SPX cannot be assigned at all before expiration. (The S&P 100 trades both ways: OEX, the first index option, listed by the CBOE in 1983, is American, and XEO is its European twin.) The two positions look interchangeable on a payoff diagram and are not interchangeable in a risk system.

There is also a tax distinction, and in the US it is large enough to drive the choice between the two. Broad-based index options are "Section 1256" contracts: every gain is split, with 60% taxed at the long-term capital gains rate and 40% at the short-term rate, however briefly the position was held, and open positions are marked to market at year end. Options on an index ETF are ordinary equity options and receive no such treatment: a position held for a month is taxed entirely at the short-term rate. For a taxable US investor trading short-dated index exposure, this is often a larger consideration than the bid–ask spread.

Currency options. A currency option is written on a fixed amount of foreign currency, and its valuation treats the foreign interest rate exactly as a dividend yield, the reasoning of Section 5.7 carried into option pricing. Chapter 16 develops it.

Options on futures. Exercising an option on a futures contract does not deliver an asset; it delivers a *position* in the futures contract, marked to market from the start. Exercising a futures call therefore produces a long futures position at the most recent settlement price plus a cash payment of that settlement price less K; a holder who closes the position at once collects

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the intrinsic value at the futures price then prevailing. Because the resulting position is itself margined, an option on futures never requires the holder to fund the underlying, which is why these contracts dominate exchange-traded options on commodities and interest rates, where producers and processors use them to set a floor or a ceiling on what they sell or buy, and banks and asset managers to hedge rate moves. Chapter 16 prices them with Black's model, and Chapter 6's SOFR and Treasury contracts are the underlyings that matter most.

10.4 READING AN OPTION CHAIN

Options are quoted in a grid (strikes down, expirations across) known as the **option chain**. Table 10.2 shows one expiration for a stock at \$48.30. The chain is shown at \$2.50 intervals for compactness; an active stock at this price would also carry the \$1 strikes between them.

Table 10.2: Calls and puts on a stock at \$48.30, 45 days to expiration. Vol is volume and OI open interest.

Calls				Strike	Puts			
Bid	Ask	Vol	OI		Bid	Ask	Vol	OI
6.55	6.75	120	3,104	42.50	1.55	1.62	890	9,220
4.35	4.50	410	5,870	45.00	1.81	1.89	2,140	14,505
2.62	2.72	3,880	21,440	47.50	2.52	2.63	3,410	18,760
1.44	1.51	5,260	28,115	50.00	3.80	3.95	1,505	11,230
0.68	0.74	2,940	19,680	52.50	5.55	5.75	240	4,010
0.28	0.33	1,110	12,450	55.00	7.60	7.90	60	1,845

Example 10.3 What the chain is telling you

Read Table 10.2 carefully and several facts emerge that no formula supplied.

Where the trading is. Volume and open interest peak at the 50 and 47.50 strikes, just above and below the \$48.30 spot. Liquidity concentrates near the money, and a trader wanting the 55-strike call must accept a bid–ask spread that is wider in proportion to the premium.

The cost of the spread. The 50-strike call is 1.44 bid, 1.51 ask. A trader who buys and immediately sells loses \$0.07 per share, or \$7 per contract, or 4.7% of the position's value. On the 55-strike call the spread is \$0.05 on a \$0.30 option, which is over 16%. Bid–ask spreads on cheap options are enormous in percentage terms, and a strategy that trades far out-of-the-money options frequently can be defeated by transaction costs alone, regardless of whether the view is right.

Volume versus open interest. Volume is contracts traded today; open interest is contracts outstanding. At the 42.50 put, volume of 890 against open interest of 9,220 says most holders are sitting still. High volume relative to open interest (the 47.50 call, 3,880 against 21,440) indicates active repositioning.

An internal check. Put–call parity, derived in Chapter 11, requires that a call and put at the same strike be linked through the stock price and the discounted strike. At the 47.50 strike, the call's midpoint, 2.67, and the put's, 2.575, differ by \$0.095. With no dividends and a 5% rate over 45 days, the difference would have to be

$$S - Ke^{-rT} = 48.30 - 47.50\,e^{-0.05\times 45/365} = 48.30 - 47.208 = \$1.092$$

The shortfall of about \$1.00 is mostly the market's expectation of dividends payable before expiration, together with any cost of borrowing the stock, which the quotes alone cannot separate from them. Every strike in the chain implies about the same \$1.00, so the chain is internally consistent, and Section 11.6 shows how to read the number off the quotes rather than guessing at it.

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10.5 TRADING AND MARKET MAKING

US listed options trade on more than a dozen exchanges, all clearing to the same clearing house, with orders routed among them by price and by economics that the end investor does not see. The SEC regulates them and FINRA the brokers that sell them; options on futures are futures products, regulated by the CFTC. Two features of this structure have direct consequences.

Market makers quote a bid and an ask continuously and are the counterparty to most retail trades. They do not take directional views; they earn the spread and hedge the resulting exposure, dynamically, using the underlying and other options. Chapter 18 is about how that hedging works. What matters here is the consequence: a market maker who has sold calls is short the option and must buy stock as the stock rises, which means dealer hedging *transmits* option activity into the underlying market. Section 10.12 shows what happens when the option activity becomes large enough relative to the stock.

Payment for order flow. Retail brokers in the US route customer orders to wholesalers (large market makers that fill retail orders), which pay for the privilege, and since 2019 this has funded zero-commission trading. The trade is not free; the cost has moved from an explicit commission into the execution price, and it is the harder cost to see. Whether retail investors are better off on net is contested: execution quality improved for stocks, but the picture is murkier for options, where spreads are wider and **price improvement** (the amount by which a fill beats the best displayed quote) is measured against a quote so wide that beating it says little.

Position limits, order types, and legging risk

Three further pieces of trading mechanics have practical consequences.

Position and exercise limits. Exchanges cap the number of contracts any one participant may hold on the same side of the market (calls held plus puts written, or puts held plus calls written), with limits ranging from tens of thousands of contracts on less actively traded stocks to 250,000 on the most liquid stocks, and millions on the largest ETFs. Exercise limits cap how many may be exercised within any five consecutive business days, generally at the same number as the position limit. The purpose is to prevent the option market being used to corner the underlying: a call position large relative to a stock's free float (the shares actually available for trading), exercised at once, can force short sellers to buy back shares at inflated prices. Bona fide hedgers can seek exemptions, as from the futures limits of Section 2.1.

Order types. Because option bid–ask spreads are wide relative to the premium (Example 10.3 showed 4.7% and 16%), a market order is genuinely dangerous. It executes at the ask when buying and the bid when selling, on a quote that may itself have widened in the instant before it arrives. Professional practice is to place limit orders at or near the midpoint and accept non-execution; a retail trader who submits market orders in options pays the full spread on every round trip.

Legging risk. A strategy with two or more option legs can be executed leg by leg or as a package. Executing separately exposes the trader to the market moving between fills, which on a two-option position worth \$0.40, built from options worth \$2.60 and \$2.20, can easily exceed the value of the position itself. Exchanges therefore maintain complex order books that match multi-leg orders as a single package at a net price. Chapter 12 builds strategies from several legs throughout, and every one of them should be understood as a package rather than as a sequence of trades.

10.6 QUOTING IN VOLATILITY

A feature of professional option markets surprises newcomers: dealers do not negotiate the price. They negotiate the implied volatility, which Section 15.7 defines as the volatility that, put into the pricing formula, returns the option's price. The price follows once the trade is agreed.

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The reason is practical. An option's price depends on the underlying price, the strike, the time remaining, interest rates, and volatility. Four of those five move continuously or are fixed by the contract; only volatility expresses a view. If a dealer quotes "$1.47" and the stock moves a penny before the client responds, the quote is stale. If the dealer quotes "21.5 volatility," the quote survives the stock moving, and the price is struck at the end from the agreed volatility and an agreed reference spot.

Two consequences follow, and both matter for Part III.

First, large OTC option trades are commonly done with a **delta exchange**: the option trades together with an agreed quantity of the underlying, sized to the option's directional exposure (its delta, in the language of Chapter 18), so that neither side takes on a position in the stock they did not want. The trade is a pure volatility transaction, which is what both sides intended.

Second, and more fundamentally: implied volatility is a price, not a model output. It is tempting to regard it as something extracted from an option price by inverting a formula. In the professional market the causation runs the other way: volatility is quoted, negotiated, and traded, and the dollar price is derived from it. Chapter 19 studies the resulting surface of quoted volatilities across strikes and maturities as a market object in its own right, and this is why.

10.7 SHORT SELLING, BORROW COSTS, AND PUTS

One more piece of mechanics connects the option market to Part I, and it explains a pattern that otherwise looks like mispricing.

Buying a put and writing a call at the same strike and expiration creates a position that gains and loses exactly as a short stock position does, a **synthetic short**. Chapter 11 derives this from put–call parity. It means an investor who wants short exposure has two routes: borrow the stock and sell it, or build the synthetic in the option market.

The two routes must cost about the same, which has a direct consequence. Section 5.2 showed that when a stock is hard to borrow and its holders will not or cannot sell it, the no-arbitrage relation degrades from an equality into a band whose width is set by the borrow fee. That band appears in the option market too: on a hard-to-borrow stock, puts trade above the price that put–call parity with the calls implies, by roughly the present value of the borrow cost. Example 10.4 puts a number on it.

Example 10.4 What a borrow fee does to the option chain

A stock trades at $50 and can be borrowed only at a fee of 3% a year. Consider six-month options struck at $50.

An arbitrageur who saw the put priced above what put–call parity implies would sell it, buy the call, and short the stock (the reversal of Chapter 11), but shorting requires borrowing at 3%, which over six months costs

$$50\left(1 - e^{-0.03\times 0.5}\right) = 50 \times 0.014888 = \$0.74$$

per share. The trade is therefore unprofitable unless the put is more than 74 cents rich, and the option market can sit anywhere inside that band without offering a profit to anyone who must borrow the stock. (A holder willing to sell shares already owned avoids the fee, which is why the band appears only where such holders are scarce, as they usually are on a heavily shorted stock.)

So a chain showing puts 60 cents rich to naive put–call parity is not mispriced and does not indicate bearish sentiment. It is quoting the borrow fee. A trader who reads it as a signal has read a financing cost as an opinion. Chapter 19 finds something similar in the volatility skew, the pattern in which low-strike options trade at higher implied volatilities than high-strike ones: part of it is a price set by one-sided demand for protection rather than a forecast.

The magnitude scales with the fee, and for genuinely hard-to-borrow stocks the fee can reach tens of percent, at which point the put becomes the only practical way to be short and its price says

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almost nothing about the expected direction of the stock.

10.8 MARGIN REQUIREMENTS

The asymmetry of Figure 10.1 produces an asymmetry in collateral, and it is the cleanest illustration in the book of margin following risk rather than notional.

The buyer posts nothing. A purchased option is paid for in full at inception (US rules let a listed option with more than nine months to run be bought with 75% paid and the rest borrowed) and creates no further obligation. The maximum loss is the premium, already paid. This is what makes options valuable to a hedger with limited access to liquidity (Section 12.10). (The exception is an option margined futures-style, Section 16.5, whose value change is settled daily.)

The writer posts margin. A **naked** call writer, one who has sold calls without owning the shares, has unlimited exposure and must collateralize it. The standard US requirement is the greater of two formulas:

$$\text{(a)}\quad 20\%\text{ of underlying value} - \text{out-of-the-money amount} + \text{option value} \tag{10.1}$$

$$\text{(b)}\quad 10\%\text{ of underlying value} + \text{option value} \tag{10.2}$$

The option value is the premium on the day of the trade and the market price thereafter, so the requirement is recomputed daily as the option moves. The second formula is a floor that binds for far out-of-the-money options, where the first would permit a very large position on very little collateral. A naked put writer posts the greater of the same two formulas, with the out-of-the-money amount measured as how far the stock stands above the strike and the floor in (b) taken as 10% of the strike price rather than of the underlying value. Example 10.5 applies both to a call.

Example 10.5 Margin on a naked call

A trader writes 10 naked call contracts, strike \$55, on a stock at \$48.30, collecting \$0.30, about the midpoint of the 0.28–0.33 quote in Table 10.2. The premium received is $10 \times 100 \times 0.30 = \$300$.

Underlying value is $10 \times 100 \times 48.30 = \$48{,}300$, and the options are \$6.70 out of the money, or $10 \times 100 \times 6.70 = \$6{,}700$ in aggregate. From (10.1),

$$0.20 \times 48{,}300 - 6{,}700 + 300 = 9{,}660 - 6{,}700 + 300 = \$3{,}260$$

and from (10.2),

$$0.10 \times 48{,}300 + 300 = \$5{,}130$$

The requirement is the greater, \$5,130.

The trader has collected \$300 and must post \$5,130, seventeen times the premium, to support a position whose maximum gain is that \$300 and whose maximum loss is unbounded. If the stock rallies to \$70, the calls are worth \$15 and the position has lost $10 \times 100 \times 15 - 300 = \$14{,}700$, nearly three times the margin originally posted. As in Example 1.5, *margin is a deposit and not a maximum loss*, and the writer will be called for the difference.

This is the arithmetic behind a standing piece of market wisdom: selling far out-of-the-money options produces a long run of small gains punctuated by losses that erase them. The strategy requires capital, discipline, and the ability to survive the tail. It is not a yield-enhancement technique, whatever it is marketed as.

Covered positions require less. A call writer who owns the underlying shares has no unbounded exposure (the shares are the deliverable) and posts no additional margin beyond the stock itself. A cash-secured put writer who holds K per share in cash is similarly covered. These structures dominate retail option writing for exactly this reason.

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Portfolio margining evaluates the whole position rather than each leg, stressing the portfolio across a grid of underlying prices and volatilities and requiring collateral for the worst outcome. The methodology is the option-market cousin of the portfolio margining of Section 2.3 and of the initial margin models of Chapter 8. It substantially reduces requirements for genuinely hedged positions and, as Section 5.14 illustrated with Amaranth, thereby permits positions whose size becomes the risk.

10.9 THE CLEARING HOUSE, EXERCISE, AND ASSIGNMENT

Every US listed option is cleared by the Options Clearing Corporation (OCC), which was established in 1973 alongside the Chicago Board Options Exchange (CBOE) and remains the sole clearing house for the market. The OCC becomes buyer to every seller and seller to every buyer, so an option holder never depends on the solvency of the person who wrote it. The mechanism and the risk it concentrates are those of Section 2.4.

Exercise. Most positions are never exercised but closed by an offsetting trade, the holder selling the option and the writer buying it back, which also captures any remaining time value. A holder exercising an American option notifies their broker, who notifies the OCC. At expiration, options that are in the money by one cent or more are exercised *automatically* ("exercise by exception") unless the holder instructs otherwise. This default prevents value being abandoned through inattention, but it does not apply to an option that finishes exactly at the strike, the trap of Example 10.6.

Assignment. The OCC does not know which particular writer should be assigned, and it does not try to. It allocates exercise notices *randomly* among clearing members with short positions, who in turn allocate randomly or first-in-first-out among their clients. A writer therefore cannot know in advance whether they will be assigned, only estimate the probability. A writer of an in-the-money American option can have it exercised against them at any moment.

Example 10.6 Pin risk at expiration

A market maker is short 200 put contracts struck at \$50 and has hedged by shorting 10,000 shares, roughly the hedge Chapter 18 will call the delta hedge: with the stock near \$50 the puts behave like about half of their 20,000 shares. As expiration approaches, though, the puts are almost certain to be exercised if the stock finishes below \$50 and certain to expire worthless if it finishes above.

The stock closes at exactly \$50.00 on expiration Friday.

If every holder exercises, the writer is assigned 20,000 shares at \$50, which covers the 10,000-share short and leaves 10,000 shares to sell. If no one exercises, the writer is left short 10,000 shares they did not want. In practice some holders exercise and some do not, for reasons the writer cannot observe, and the writer learns the outcome only after the market has closed: on Saturday morning, with the next opportunity to trade on Monday.

The exposure is therefore an unhedged overnight-plus-weekend position of unknown size and unknown direction, on up to 10,000 shares. This is **pin risk**, and it is why traders close positions at strikes the stock is likely to finish near, rather than carrying them into expiration for the last few cents of premium.

A related effect runs the other way. A dealer who is *long* options struck near the spot (long gamma, in Chapter 18's terms) hedges by buying stock when it falls below the strike and selling when it rises above, which mechanically damps movement around that price. Stocks with large open interest at a round strike tend to gravitate toward it as expiration approaches. This "pinning" effect is a hedging artifact rather than a market view. The writer in this example, being short, would trade the other way, which is why pinning depends on who holds the options at the strike.

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10.10 BEYOND THE STANDARD CONTRACT

Three families of option-like instruments sit outside the listed market and are worth distinguishing, because their differences are exactly the features a reader should be alert to.

Warrants are options issued by a company on its own stock, usually attached to a bond or share issue so that it raises capital more cheaply, while holders gain long-dated upside. Exercise creates new shares, so the payoff is diluted, which changes the valuation (Chapter 17). A listed call, by contrast, is written by another participant and exercise transfers existing shares.

Convertible bonds are debt with an embedded option to convert into equity. The holder owns a bond plus a call; the issuer has sold that call in exchange for a lower coupon. They form a large market in embedded options, and Chapter 25 treats them together with credit.

Employee stock options are non-transferable, forfeited if unvested when the employee leaves, and long-dated. Because holders cannot sell them, they exercise early for liquidity and diversification reasons that no arbitrage-based model captures, which is why the valuation of employee options is a distinct problem, the subject of Chapter 17.

Alongside them sits the over-the-counter market in options proper. These are options in the full sense of Section 10.1, negotiated bilaterally with a dealer rather than listed. The listed market is the visible one; by notional the OTC market is larger, and it is where most currency options, most interest rate options (the caps, floors, and swaptions of Chapter 29), and every exotic structure of Chapter 34 trade. The reason is customization. An exchange lists a grid of strikes and dates on a hundred shares of one stock; a dealer will write any strike and any date, on a basket, with a barrier, on an average price, or paid in another currency, and will do so in an amount that would move a listed market visibly. The terms are documented under the ISDA Master Agreement of Section 1.5, the price is negotiated as a volatility in the manner of Section 10.6, and the buyer is exposed to the dealer's default for the life of the trade unless the agreement is collateralized under a CSA of the kind Chapter 8 describes or the contract is standardized enough to clear. The credit valuation adjustment of Chapter 9 is the price of that difference. The users are the ones the listed market cannot serve: a treasurer hedging a receivable whose date and amount match no listed contract, an asset manager buying a swaption against a pension fund's interest rate exposure, an issuer of the structured notes of Chapter 35 hedging the option it has sold to retail investors. What the buyer gives up is the exchange's transparency: there is no public last trade, the dealer's quote is the only price, and a position is unwound with the same dealer or not at all.

Flexible exchange (FLEX) options, introduced in 1993, are the listed market's answer: exchange-listed, but with terms the parties choose, namely any strike, any expiration, either exercise style. They occupy the middle ground the cleared-OTC market occupies in Section 1.4, offering customization with a clearing house behind it. **Long-term equity anticipation securities (LEAPS)** are simply long-dated listed options, extending up to about three years (five on the S&P 500).

10.11 WHO TRADES OPTIONS, AND WHY

Options serve purposes the linear instruments of Part I cannot, and it is worth naming who is on each side before Part III begins pricing them.

Protective put buyers own an asset and want a floor under it. The put is insurance: a premium is paid, the downside is capped, the upside is retained. Pension funds, insurers, and founders and executives with concentrated stock positions are the natural buyers, and Section 12.10 explains why a futures hedge is not a substitute.

Covered call writers own an asset and sell part of the upside for income. Asset managers run this as a systematic strategy, and it is among the most widely used retail option strategies. It converts an uncertain equity return into a more certain but capped one, and it performs poorly

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in the strong markets that investors hold equities for, as Example 10.7 shows.

Example 10.7 Writing a covered call

An investor owns 500 shares of the stock in Table 10.2, trading at \$48.30, and writes 5 call contracts struck at \$52.50 at the \$0.68 bid, 45 days out. Premium received is $5 \times 100 \times 0.68 = \$340$.

Three outcomes at expiration follow, on the 500-share position:

Stock at \$46. The calls expire worthless. The shares have lost $500 \times 2.30 = \$1{,}150$, offset by the \$340 premium, for a net loss of \$810 against \$1,150 unhedged.

Stock at \$52.50. The calls expire just at the money and worthless. The shares have gained $500 \times 4.20 = \$2{,}100$, plus \$340, for \$2,440. This is the best possible outcome, and every price at or above the strike earns it.

Stock at \$58. The calls are assigned at \$52.50. The result is the same \$2,440, while the unhedged position would have gained $500 \times 9.70 = \$4{,}850$. The investor gave up \$2,410.

The premium is \$0.68 on a \$48.30 stock: 1.4% of downside protection, and the breakeven falls only to \$47.62. The strike is 8.7% above spot, so all upside beyond 8.7% has been sold for that 1.4%.

The maximum return is \$2,440 on \$24,150 of stock, or 10.1% in 45 days, which annualizes to a number that appears prominently in marketing material. Annualizing it is misleading: the maximum cannot be earned repeatedly, because it requires the stock to gain 8.7% or more every 45 days.

The strategy is not unsound: the writer is selling insurance and collecting a premium for it, and Chapter 19 shows that the risk premium is positive on average. But "income" misdescribes the strategy: it is the sale of upside.

Corporate issuers embed options in securities. A callable bond is a bond plus a call sold to the issuer, who pays for the right to refinance with a higher coupon; a convertible is the reverse, a call sold to the holder in exchange for a lower coupon.

Volatility traders take positions in the option without wanting exposure to the underlying at all, hedging the direction away and trading the volatility that remains. Part IV is about this.

Dealers intermediate all of the above and hedge what accumulates. Their aggregate position determines whether dealer hedging damps or amplifies moves in the underlying, the mechanism of Example 10.6 operating at market scale.

Retail investors have become, since 2020, a large enough share of activity to change market structure, which is the subject of the next section.

10.12 THE RETAIL TRANSFORMATION AND SAME-DAY OPTIONS

The US options market more than doubled in volume between 2019 and 2023, and the composition of that growth is as important as its size. Three changes drove it.

The first was the elimination of retail commissions in late 2019, which removed the fixed cost that had made small option trades uneconomic. The second was the arrival, during 2020, of a large cohort of new retail participants trading through mobile applications. The immediate effect was a surge in single-stock call buying, concentrated in short maturities and small orders.

The second-order effect is the interesting one. Retail buying of short-dated calls leaves dealers short those calls and therefore short gamma, in the language Chapter 18 introduces: they are obliged to buy the stock as it rises and sell as it falls, the opposite of the damping behavior in Example 10.6. Chapter 23 develops the mechanism in full. In heavily traded stocks during 2021, this hedging demand was large relative to the free float of the stock, and the feedback from option market to stock market became visible rather than theoretical.

A third change came from the exchange side. Once S&P 500 options expired every trading day, every session contained contracts with hours rather than weeks to run. These are

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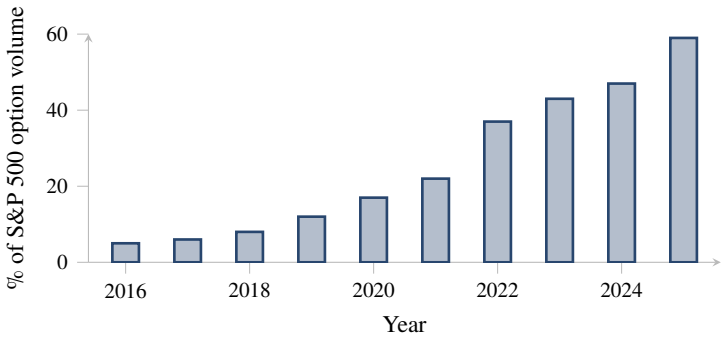


Figure 10.5: Share of S&P 500 index option volume expiring the same day. Approximate.

zero-days-to-expiration (0DTE) options, and Figure 10.5 shows their share of index option volume rising from about 5% to more than half.

Why 0DTE options behave differently. An option with hours to live has almost no time value and enormous gamma, so a dealer hedging a book of them must trade the underlying frequently and in large amounts, carrying option trading into the index during the day. Customers both buy and sell them and the positions close within the day, so dealers carry no large net position overnight; that is why studies so far have found less effect on the index than early commentary feared. Chapter 23 develops the mechanism and weighs that evidence.

Case Study 10.1 What retail bought, and what it cost

The retail options boom is usually discussed in terms of market stability. The more immediate question is what it did to the participants themselves.

Retail option trading is overwhelmingly *buying*, and heavily concentrated in short-dated, out-of-the-money calls: the cheapest contracts on the chain and, as Example 10.3 showed, those with the widest proportional spreads. A \$0.30 option with a \$0.05 spread costs over 16% of its value to enter and exit. A trader turning such a position over weekly faces a transaction cost burden that no directional skill plausibly overcomes.

Layered on top is a structural point from Part III. Buying options is buying volatility, and Chapter 19 documents that implied volatility exceeds subsequent realized volatility on average across most markets and periods. Persistent option buyers therefore pay a risk premium on top of the spread, and should know they are paying it.

None of this argues against retail participation. The cost is simply incurred where no statement shows it: in the spread and in the volatility premium.

SUMMARY

A call confers the right to buy at the strike and a put the right to sell; European options are exercisable only at expiration, American options at any time. A buyer's loss is capped at the premium and the buyer posts no margin; a writer's gain is capped at the premium and the writer's open-ended loss must be collateralized. This asymmetry, not the payoff formula, distinguishes the two sides in practice. An option's value is intrinsic value (what it would be worth exercised now) plus time value, which is largest near the money and decays to zero at expiration.

A standard US equity option covers 100 shares. Contracts are adjusted for splits, mergers, and special dividends to preserve aggregate value, but not for ordinary cash dividends, and an

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adjusted contract trades under a numbered root. The same right differs by underlying: index options settle in cash against a computed value, ETF options can be assigned early around dividends and are taxed differently, and an option on a futures contract delivers a margined position rather than an asset.

Reading an option chain reveals what no model supplies: where liquidity is, how large the proportional bid–ask spread is on cheap options, and whether the quotes are consistent with put–call parity. Professional markets quote options in implied volatility, so a volatility is a traded number and not a model output. On a hard-to-borrow stock the borrow fee appears as puts priced above parity, a financing cost rather than a view.

Naked writers post the greater of two formulas, recomputed daily and far exceeding the premium collected; covered and cash-secured writers post much less, and portfolio margining reduces requirements for hedged positions and thereby enables larger ones.

The OCC clears all US listed options and allocates exercise notices randomly among writers. Options in the money by one cent are exercised automatically at expiration, and positions struck near the closing price create pin risk: an unhedged position of unknown size discovered after the market has closed. The larger over-the-counter market holds most currency and interest rate options and all exotic options, and its buyer is exposed to the dealer's default unless the trade is collateralized or cleared.

Since 2020 zero-commission retail participation and index options expiring every trading day have reshaped the market, and same-day options now account for more than half of S&P 500 option volume. Their hedging carries option trading into the index during the day, and the retail cost lies in bid–ask spreads and the volatility risk premium rather than in commissions.

FURTHER READING

The OCC's *Characteristics and Risks of Standardized Options*, the disclosure document every US options customer receives, states the contract, exercise, and margin rules; the OCC's adjustment memoranda show the principle of Example 10.1 applied case by case, and exchange rulebooks give the current margin formulas. Bryzgalova, Pavlova, and Sikorskaya (2023) is the empirical study of retail option buying and payment for order flow behind Section 10.12's case study. Cboe's research note "The Rise of SPX 0DTE Options" documents the growth that Figure 10.5 plots, and Brogaard, Han, and Won's "Does 0DTE Options Trading Increase Volatility?" is the place to start on the open question Chapter 23 weighs. On payment for order flow, the brokers' Rule 606 routing reports, which disclose what each wholesaler pays for options orders, are the primary data.

PRACTICE QUESTIONS

10.1 An investor buys 5 call contracts at a quoted premium of \$2.85. What is the total cost, and what is the maximum loss?

10.2 Using Table 10.2, compute the proportional bid–ask spread for the 45-strike call and the 55-strike call. Explain why they differ and what it implies for strategy selection.

10.3 A stock trades at \$74 and pays no dividend. A trader writes 8 naked call contracts struck at \$85 for a premium of \$1.10. Compute the margin requirement from both formulas and state which binds.

10.4 A company declares a 2-for-1 stock split. State how the contract terms of an outstanding option are adjusted under each of the two conventions described in Section 10.2, and verify that the aggregate strike cost is preserved in both.

10.5 Explain why an option buyer posts no margin while a writer must, and relate the answer to Figure 10.1.

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10.6 A trader is short 50 call contracts struck at $30 and the stock closes at $30.00 on expiration Friday. Describe the trader's position on Saturday morning and explain what could have been done on Friday to avoid it.

10.7 An option expiring in three hours and one expiring in three months are both at the money. Compare their intrinsic value, time value, and sensitivity to a $1 move in the underlying.

10.8 Distinguish a warrant from a listed call option on the same stock at the same strike. Which is worth more, and why?

10.9 An index option, a currency option, and an option on a futures contract differ in what changes hands on exercise. State what each delivers, and explain why exercising an option on a futures contract leaves the holder with a position to manage rather than an asset to keep.

10.10 Dealers quote many options in volatility rather than in price. Explain what the convention buys them, state what else must be agreed before a volatility quote determines a cash amount, and explain why such a quote stays meaningful while the underlying moves and a price quote does not.

10.11 An investor wants six months of downside protection on a $5 million S&P 500 portfolio, with the index at 5,250 and SPY at $525. Compare buying index puts (SPX) with buying SPY puts: the number of contracts, settlement and exercise, what each leaves the investor holding at expiration, and US tax treatment.

10.12 Name three distinct institution types that buy options and state, for each, the exposure it is buying or removing. For one of them, explain how its demand shows up in the prices the rest of the market sees.

FURTHER QUESTIONS

10.13 Using the 50-strike quotes in Table 10.2, repeat the put–call parity check of Example 10.3 at that strike. Does the chain remain consistent? Identify what you must assume about dividends for the two strikes to agree, and comment on whether that assumption is reasonable.

10.14 Section 10.12 argues that dealer hedging of short-dated retail call buying amplifies moves in the underlying, while Example 10.6 argues that dealer hedging near expiration damps them. Reconcile these. Precisely what determines the sign of the effect, and what would you need to observe to know which regime prevails today?

10.15 Two weeks before the covered call in Example 10.7 expires, the stock is at $55, the calls are quoted $2.50–$2.60, and tomorrow the stock goes ex-dividend with the $1.00 dividend of Example 10.3, after which the calls would be worth about $2.35. Explain why the investor may be assigned tonight, compute what an assignment costs against holding through the ex-dividend date, and state what the investor could have done today to avoid it.

10.16 Contract adjustments preserve aggregate value for splits and mergers but not for ordinary cash dividends. Explain the rationale for treating the two differently, and construct a case where the distinction transfers substantial value from one side of a contract to the other.

10.17 Same-day options do not accumulate an overnight position, which Section 10.12 offers as a reassuring feature. Construct the scenario in which this feature fails to protect the market, and identify what a regulator would need to monitor to see it developing.

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Chapter 11
Properties of Options and Arbitrage Bounds

This chapter contains no model. Nothing in it assumes a distribution for the stock price, a volatility, or a stochastic process, and none of its results will be revised when Chapter 15 supplies a formula. Everything here follows from the absence of arbitrage, and the results are therefore stronger than anything a model can deliver: a model can be wrong, whereas these relationships hold for any model that is not internally inconsistent.

That is worth stating plainly because the temptation is to skip ahead. A reader who knows the Black–Scholes–Merton formula but not put–call parity will price options adequately and misunderstand the market: they will read a price relationship as sentiment, exercise an American call that should have been sold, or construct a "riskless" arbitrage that carries an assignment risk they did not know existed.

Chapter 10 deferred four questions to this chapter: why the option chain in Example 10.3 had to be internally consistent, why a put and a written call together behave like short stock, why dividends make early exercise of an American call rational, and why an American put is worth more than a European one. All four are answered by arbitrage arguments alone.

Section 11.1 lists what determines an option price. Sections 11.2 and 11.3 derive the bounds on a single option and across strikes; Sections 11.4 to 11.7 are put–call parity and what follows from it: synthetic positions, the implied forward, and the box spread as a financing trade. Sections 11.8 to 11.11 are early exercise: why an American call on a stock paying no dividend is never exercised early, why a put may be, what dividends change, and the parity bounds that survive for American options. Section 11.12 asks what an apparent violation actually means, and Section 11.13 who trades these relationships.

11.1 WHAT DETERMINES AN OPTION PRICE

Six quantities determine the value of a stock option: the current stock price S_0, the strike K, the time to expiration T, the volatility σ, the risk-free rate r, and the dividends expected during the option's life. Table 11.1 gives the direction of each effect; the reasoning matters more.

Table 11.1: Effect of increasing one variable, holding the others fixed.

Variable increases	European call	European put	American call	American put
Stock price S_0	+	−	+	−
Strike price K	−	+	−	+
Time to expiration T	?	?	+	+
Volatility σ	+	+	+	+
Risk-free rate r	+	−	+	−
Dividends	−	+	−	+

Figure 11.1 shows the shapes. Two are worth noting beyond their direction, though their

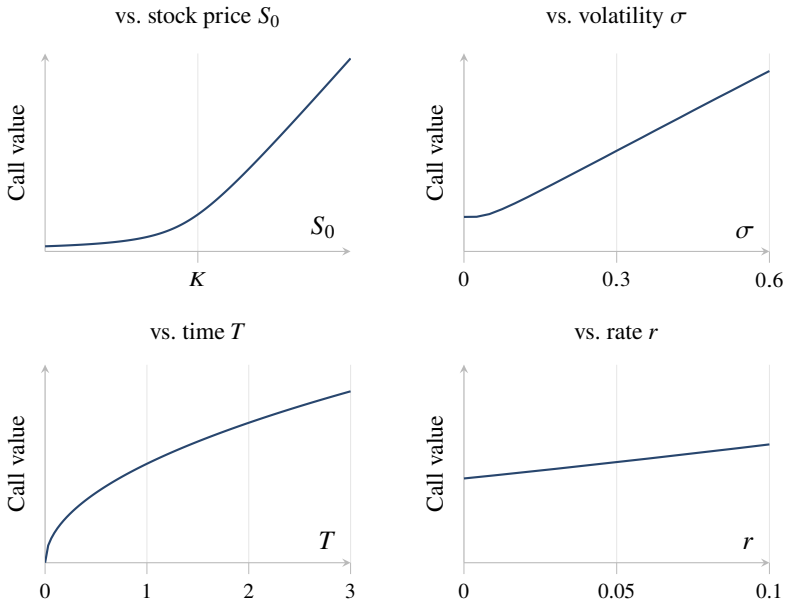


Figure 11.1: A European call, varying one variable at a time. Schematic.

shapes come from the model of Chapter 15 rather than from arbitrage: an at-the-money call's value is flat in volatility at first and then close to linear (the slope is what Chapter 18 names vega, and Chapter 19 relies on the near-linearity when it treats implied volatility as a quoted price), while the dependence on time grows roughly with $\sqrt{T}$, so doubling the maturity does not double the time value.

Stock and strike prices. A call pays $\max(S_T - K, 0)$, which rises with S_T and falls with K.

Volatility. This entry separates options from every instrument in Part I and deserves care. For a forward contract, volatility is irrelevant to the price: the payoff is linear, so increased dispersion raises the good outcomes and lowers the bad ones symmetrically, and they cancel. An option's payoff is not linear. Its downside is truncated at zero, so greater dispersion raises the expected payoff without a matching penalty. *Both* calls and puts increase in value with volatility, which surprises readers who think of a call and a put as opposite bets.

Risk-free rate. Two effects run in the same direction for a call. A higher rate reduces the present value of the strike the holder will eventually pay, and it raises the forward price, $S_0 e^{rT}$ for a stock paying no dividend, on which the call is effectively written. Both increase call values and decrease put values. This is one of the few places where intuition from other markets misleads: rising rates *raise* call prices, holding the stock price fixed.

Dividends. A dividend reduces the stock price on the ex-dividend date without compensating an option holder, who receives no dividends. Calls fall in value, puts rise. Section 11.10 shows this is also what makes early exercise of American calls rational.

Time to expiration, and why two entries are question marks. An American option with more time cannot be worth less, because its holder can do everything the shorter-dated holder can do and more: the longer option's exercise opportunities are a superset.

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European options do not have this property, and the reason is instructive. Consider two European calls on a stock paying a large dividend in one month, one expiring in two weeks and one in six weeks. The shorter option expires before the stock drops on the ex-dividend date; the longer one does not. The shorter option can be worth more. A European put can lose value with more time for a different reason when rates are high and the put is deep in the money: the extra time defers the receipt of K. Superset reasoning requires the freedom to act, and European options do not have it.

11.2 UPPER AND LOWER BOUNDS

Before any question of what an option is worth, there is the question of what it cannot be worth.

Notation. Throughout this chapter c and p are the prices of a European call and put, and C and P the prices of the American call and put on the same stock with the same strike K and expiration T; S_0 is the current stock price and r the risk-free rate. A capital P standing alone is always the American put; the bond price of later chapters carries its two arguments, $P(t, T)$.

The assumption behind everything that follows. The chapter assumes no arbitrage, as defined at the opening of Chapter 5, and frictionless trading (borrowing and lending at r, shorting without cost, no bid–ask spread), which Section 11.12 relaxes. Each bound below is proved the same way: suppose it fails, write down the trade that then costs nothing and cannot lose, and conclude the prices cannot stand.

Upper bounds. A call gives the right to buy the stock, so it can never be worth more than the stock: $c \le S_0$ and $C \le S_0$. If it were, selling the call and buying the stock would collect $c - S_0 > 0$ today, and the share covers any exercise. A put gives the right to sell at K, so it can never be worth more than K; and since a European put pays at most K at time T, its value is capped by the present value: $p \le Ke^{-rT}$, while $P \le K$. An American option can be exercised at once, so it is never worth less than its intrinsic value: $C \ge \max(S_0 - K, 0)$ and $P \ge \max(K - S_0, 0)$.

Lower bound for a European call. Four portfolios, defined once here, carry every proof in the chapter:

Portfolio A	one European call, plus cash of Ke^{-rT}
Portfolio B	one share
Portfolio C	one European put, plus one share
Portfolio D	cash of Ke^{-rT}

Compare A with B. The cash in A grows to exactly K by time T. If $S_T > K$ the call is exercised, using the K to buy the share, and A is worth S_T. If $S_T \le K$ the call lapses and A is worth K. So A is worth $\max(S_T, K)$, which is never less than B's S_T. A portfolio that is never worth less at T must not cost less today, so $c + Ke^{-rT} \ge S_0$, and since $c \ge 0$,

$$c \ge \max\left(S_0 - Ke^{-rT},\ 0\right) \tag{11.1}$$

Comparing Portfolio C, worth $\max(S_T, K)$, with Portfolio D, worth K, gives

$$p \ge \max\left(Ke^{-rT} - S_0,\ 0\right) \tag{11.2}$$

If dividends with present value D are paid during the option's life, the share collects them and the option does not, so S_0 becomes $S_0 - D$ in both bounds (Table 11.2).

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Figure 11.2 draws the region a call's bounds permit without dividends, and Example 11.1 trades a call priced below it.

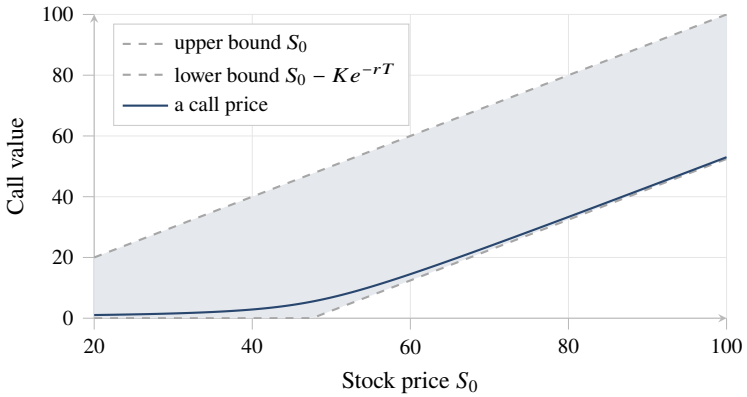


Figure 11.2: Permitted region. $K = 50$, $r = 5\%$, $T = 1$.

Example 11.1 Arbitrage when the lower bound fails

A non-dividend-paying stock trades at \$20. A one-year European call struck at \$18 trades at \$3.00. The rate is 10%.

The lower bound (11.1) is

$$20 - 18e^{-0.10} = 20 - 18 \times 0.904837 = 20 - 16.287 = \$3.713$$

so \$3.00 is too cheap. Buy the call and short the stock:

$$\text{cash received} = 20 - 3 = \$17, \qquad \text{grown for one year} = 17e^{0.10} = \$18.788$$

At expiration there are two cases. If $S_T > 18$, exercise the call, pay \$18 for the share, return it, and keep $18.788 - 18 = \$0.788$. If $S_T \le 18$, let the call lapse, buy the share in the market for $S_T \le 18$, return it, and keep at least \$0.788.

The profit is at least \$0.788 in every state, from an initial investment of nothing. The mispricing is $3.713 - 3.00 = \$0.713$, and $0.713 \times e^{0.10} = \$0.788$. The arbitrage profit is exactly the mispricing carried forward at the risk-free rate, a useful check on any such calculation.

11.3 BOUNDS ACROSS STRIKES

The bounds so far constrain one option. Three further relations constrain options at *different* strikes on the same underlying and maturity, and they matter disproportionately: they are the conditions a whole option chain must satisfy, and Chapter 19 will use them as the test of whether a volatility surface is arbitrage-free.

Write $c(K)$ for the price of a call struck at K, with $K_1 < K_2 < K_3$.

Monotonicity. $c(K_1) \ge c(K_2)$. A call with a lower strike confers the right to buy more cheaply, so it cannot be worth less. If it were, buy the low-strike call, sell the high-strike one, collect the difference, and hold a spread whose payoff is never negative.

The spread bound. $c(K_1) - c(K_2) \le (K_2 - K_1)e^{-rT}$. The payoff difference between the two calls never exceeds $K_2 - K_1$, so their price difference cannot exceed its present value.

Convexity. For equally spaced strikes, $K_2 = (K_1 + K_3)/2$,

$$c(K_1) + c(K_3) \ge 2\,c(K_2) \tag{11.3}$$

Buying the outer two calls and selling two of the middle one is a **butterfly**, and its payoff at expiration is zero outside $[K_1, K_3]$ and positive inside, peaking at K_2. A position whose

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payoff is never negative cannot be established for a credit, which is exactly what (11.3) says. With monotonicity, this makes $c(K)$ a convex, decreasing function of K, the shape drawn in Figure 11.3. Both results hold for American calls too, since the long calls can be exercised whenever a short one is assigned; the spread bound loses its discount, $C(K_1) - C(K_2) \le K_2 - K_1$, because assignment can demand $K_2 - K_1$ at once.

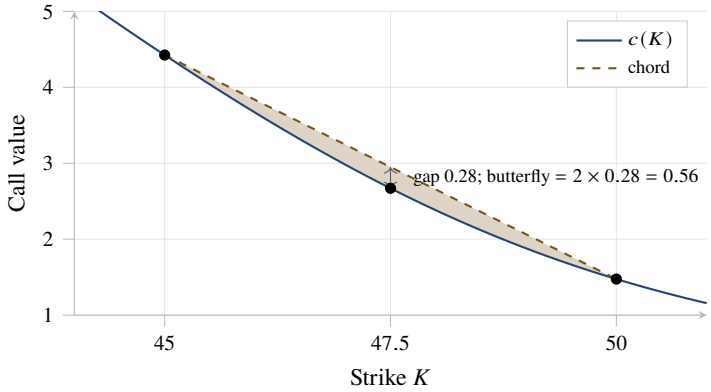


Figure 11.3: Convexity in the strike, at the three strikes of Example 11.2. Shaded: the gap between chord and curve.

Geometrically, convexity says the curve lies below any chord joining two of its points. The vertical gap between the chord and the curve at K_2 is half the value of the butterfly, since the butterfly sells the middle call twice, and it must be nonnegative.

Example 11.2 Testing a chain for convexity

Take the calls from Table 10.2, using mid prices. Options on individual US stocks are American, and monotonicity and convexity hold for both styles, so write $C(45) = 4.425$, $C(47.50) = 2.67$, $C(50) = 1.475$. The strikes are equally spaced, so (11.3) requires

$$4.425 + 1.475 = 5.90 \;\ge\; 2 \times 2.67 = 5.34$$

The chain passes, with \$0.56 to spare, which is the butterfly's value.

Check the other two conditions as well. Monotonicity holds: 4.425 > 2.67 > 1.475. Because the options are American, the spread bound takes its undiscounted form:

$$C(45) - C(50) = 2.95 \;\le\; 50 - 45 = \$5$$

with about \$2 to spare.

Now suppose the 47.50 call were instead quoted 3.00 bid / 3.10 ask. Convexity would fail: $2 \times 3.05 = 6.10$ against 5.90. Trade it at the quotes: buy the 45 call at its \$4.50 ask, buy the 50 call at its \$1.51 ask, sell two 47.50 calls at the \$3.00 bid:

$$\text{net cash} = 2 \times 3.00 - 4.50 - 1.51 = 6.00 - 6.01 = -\$0.01$$

That is a penny of cost, not a credit, so the violation at mid prices does *not* survive the bid–ask spread, and there is no trade. Had the 47.50 bid been \$3.05, the position would have been established for a \$0.09 credit with a guaranteed payoff between \$0 and \$2.50: an arbitrage.

11.4 PUT–CALL PARITY

The chapter's central result relates the call and the put at the same strike and maturity, and it compares Portfolios A and C of Section 11.2. At expiration, A is worth $\max(S_T, K)$ as shown there. Portfolio C is worth S_T if $S_T > K$ (the put lapses) and K if $S_T \le K$ (the put is exercised),

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also $\max(S_T, K)$. Two portfolios worth the same at T in every state must cost the same today:

$$\boxed{c + Ke^{-rT} = p + S_0} \tag{11.4}$$

This is **put–call parity**. It holds for European options on a non-dividend-paying stock, and it requires no assumption whatever about how the stock behaves. If dividends with present value D are expected during the option's life, the share in Portfolio C also delivers those, and the relation becomes

$$c + D + Ke^{-rT} = p + S_0 \tag{11.5}$$

Because $c - p$ is linear in K in either form, a put butterfly is worth what the call butterfly at the same strikes is worth, so puts are convex in the strike too, and the call's monotonicity and spread bound become $0 \le p(K_2) - p(K_1) \le (K_2 - K_1)e^{-rT}$. Example 11.3 shows what a violation of (11.4) is worth and how it is traded.

Example 11.3 Trading a parity violation

A stock trades at \$31 and pays no dividends. Three-month European options struck at \$30 are quoted: the call at \$3.00, the put at \$2.25. The rate is 10%.

Parity (11.4) says the put should be

$$p = c + Ke^{-rT} - S_0 = 3.00 + 30e^{-0.025} - 31 = \$1.2593$$

The put is quoted \$0.99 too high. Sell it, and hedge by buying the replicating portfolio: buy the call and short the stock, investing the proceeds.

$$\text{cash in} = 2.25 - 3.00 + 31 = \$30.25, \qquad 30.25\, e^{0.025} = \$31.0158$$

At expiration, if $S_T > 30$, exercise the call. If $S_T \le 30$, the put is exercised against you and you buy at \$30. Either way the short is closed for \$30, leaving

$$31.0158 - 30 = \$1.0158$$

in every state. Again this is the mispricing carried forward: $0.9907 \times e^{0.025} = \$1.0158$.

This trade has a name. Long call, short put, short stock is a **reversal**; the opposite is a **conversion**.

Parity for other underlyings

Chapter 10 cataloged options on indices, currencies, and futures. Parity extends to each of them by replacing the share in Portfolio C with whatever position replicates the underlying at time T, and the results are worth having explicitly:

$$\text{Index, dividend yield } q: \quad c + Ke^{-rT} = p + S_0 e^{-qT} \tag{11.6}$$

$$\text{Currency, foreign rate } r_f: \quad c + Ke^{-rT} = p + S_0 e^{-r_f T} \tag{11.7}$$

$$\text{Futures, price } F_0: \quad c + Ke^{-rT} = p + F_0 e^{-rT} \tag{11.8}$$

Equations (11.6) and (11.7) are the same statement: an index yielding q and a currency yielding r_f both require holding less than one unit today to have one unit at T, which is the reasoning of Sections 5.4 and 5.7 carried into option pricing.

The futures version (11.8) rearranges to something that holds for every underlying once F_0 is read as its forward price to T, and is used constantly on trading desks:

$$c - p = (F_0 - K)e^{-rT} \tag{11.9}$$

Set $K = F_0$ and the right-hand side vanishes: a call and a put struck at the forward price have equal value. This is why the strike equal to the forward price, called **at-the-money forward**, is the usual reference point for quoting options, and why a straddle at that strike (Chapter 12) is

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the standard way to trade volatility without taking a directional position: the call and the put cost the same, and their deltas (Chapter 18) nearly offset. Chapter 19 measures moneyness and log-moneyness from this strike.

11.5 SYNTHETIC POSITIONS

Equation (11.4) has four terms, so it can be rearranged four ways, and each rearrangement is a recipe for manufacturing one instrument from the others:

$$S_0 = c - p + Ke^{-rT} \qquad \text{synthetic long stock} \tag{11.10}$$

$$c = p + S_0 - Ke^{-rT} \qquad \text{synthetic call}$$

$$p = c - S_0 + Ke^{-rT} \qquad \text{synthetic put}$$

$$-S_0 = p - c - Ke^{-rT} \qquad \text{synthetic short stock} \tag{11.11}$$

Figure 11.4 shows the most useful of these graphically: a long call combined with a short put at the same strike has exactly the payoff of a long forward.

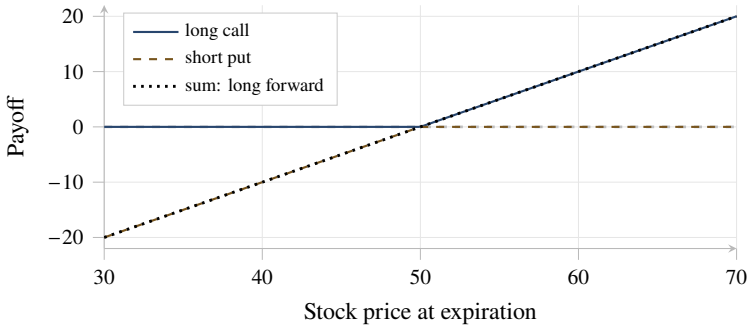


Figure 11.4: Long call plus short put at the same strike. $K = 50$.

Equation (11.11) is the synthetic short Section 10.7 promised, whose put trades above parity by the borrow fee (Example 10.4).

11.6 READING THE IMPLIED FORWARD

Parity is usually taught as a test. In practice it is a *measurement device*: rearranged, it extracts from observable option prices a quantity no one quotes directly and everyone needs.

Rearranging (11.9) gives the option market's own forward price,

$$F_0 = K + (c - p)\, e^{rT} \tag{11.12}$$

It requires no view on dividends, no borrow-cost assumption, and no model.

Example 11.4 What the chain says about dividends

Take the 47.50 strike from Table 10.2: call mid 2.67, put mid 2.575, 45 days to expiration, $r = 5\%$. From (11.12),

$$F_0 = 47.50 + (2.67 - 2.575)\, e^{0.05\times 45/365} = 47.50 + 0.095 \times 1.006183 = \$47.596$$

The spot price is \$48.30, so the option market's 45-day forward is 70 cents below spot. That is striking, because interest alone would put the forward *above* spot, by $48.30(e^{0.0061644} - 1) = \$0.30$. Something is pulling it down by a dollar.

That something is mostly dividends, the roughly \$1.00 that Example 10.3 found from the same quotes (1.092 − 0.095). So the chain says the market expects about \$1.00 of dividends before expiration, roughly 2% of the share price in 45 days, which suggests a high-yielding stock with an

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ex-dividend date inside the option's life. The option prices contain the market's forecast, and where that forecast differs from the announced dividend, the gap is itself information: about a dividend the market expects to change, or, as the second caveat explains, about the cost of borrowing the stock.

Two caveats keep this honest. First, the options of Table 10.2 are American, and Section 11.11 shows that arbitrage alone confines $C - P$ here only to a band about \$1.29 wide. The reading is usable because at 47.50, nearest the money, both options are liquid and neither carries much early-exercise premium, so $C - P$ stays close to the European $c - p$; the \$1.00 is good to ten cents or so, not to the cent. Second, parity gives *one* equation, and the forward embeds dividends, borrow costs, and financing together. Having computed \$1.00, one cannot separate "dividends of \$1.00" from "dividends of \$0.60 plus a 40-cent borrow cost" without outside information. Practitioners therefore quote the implied forward and stop there: decomposing it further is a modeling choice, not a measurement.

11.7 BOX SPREADS: PARITY AS A FINANCING TRADE

Combining a synthetic long at one strike with a synthetic short at another produces something with no equity exposure at all. Buy the call and sell the put at K_1; sell the call and buy the put at K_2, with $K_1 < K_2$. From (11.10) and (11.11) the stock cancels, and the payoff at expiration is $K_2 - K_1$ regardless of where the stock finishes. This is a **box spread**, and it is a zero-coupon bond assembled from four options; Example 11.5 reads off the interest rate its price implies.

Example 11.5 Borrowing and lending through the option market

An investor constructs a box on a stock index using one-year European options at strikes 4,900 and 5,000. Whatever the index does, the position pays 100 index points at expiration, \$10,000 per contract at the \$100 multiplier.

If the box can be bought for 95.12 index points (\$9,512 per contract), the implied lending rate is

$$r = \ln\left(\frac{100}{95.12}\right) = \ln(1.051304) = 5.00\%$$

If it can be bought for 93.00 points, the rate is $\ln(100/93.00) = 7.26\%$, and the investor lends at 7.26% on a fully collateralized, centrally cleared position. *Selling* the box borrows at that rate.

Institutions use box spreads as a financing instrument, competing with repo, because the position is cleared, requires no counterparty credit assessment, and can be executed in large amounts on an exchange. The implied rate is a directly observable financing rate, worth comparing against the rates of Chapter 4, since discrepancies between them are the option market's version of the basis trades of Chapter 6.

Case Study 11.1 The box that was not riskless

A box spread is riskless *if the options are European*. In 2019 a retail trader on a US brokerage constructed box spreads using options on an ETF (which, as Section 10.3 noted, are American) and treated the position as a guaranteed loan. The account was reported to have lost more than ten times its \$5,000 of capital.

The mechanism is exactly the assignment risk of Section 10.9. Two of the four legs are short options. If the ETF moves so that one of the short options is deep in the money, its holder may exercise it early. The trader is then assigned, must deliver or receive shares immediately, and is left holding a position that no longer has the offsetting structure the box depended on—at a moment chosen by someone else, not by them.

Two lessons follow, and both generalize past this episode. First, every result so far must be re-examined before it is applied to American options. Second, a position that is

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riskless at expiration is not necessarily riskless *before* expiration. Chapter 3 made this point about funding and Chapter 6 about the cash–futures basis trade. Here it appears as a counterparty's right to exercise early, when it suits them rather than you.

11.8 EARLY EXERCISE: AMERICAN CALLS WITHOUT DIVIDENDS

An American call on a non-dividend-paying stock should never be exercised early. The result is stronger than it first appears, and it has two independent proofs worth knowing because they generalize differently.

The bounds argument. From (11.1), an American call is worth at least $S_0 - Ke^{-rT}$, since it is worth at least as much as the European call. Exercising yields only $S_0 - K$. Because $Ke^{-rT} < K$ whenever $r > 0$,

$$\underbrace{S_0 - Ke^{-rT}}_{\text{value if sold}} > \underbrace{S_0 - K}_{\text{value if exercised}}$$

The holder always does better selling the option than exercising it. Note what this proves: at every date before expiration the call is worth more alive than exercised, so the right to exercise early is never used and adds nothing, and $C = c$ for a non-dividend-paying stock.

The economic argument. Exercising early does two things, both bad. It pays K now rather than at T, giving up the interest, $K - Ke^{-rT} = K\left(1 - e^{-rT}\right)$, which is the difference between the two sides of the display above. And it discards insurance: a holder who exercises owns the stock outright and suffers fully if it falls, whereas a holder who waits keeps a floor at zero. The second point is why the result holds even at $r = 0$, where the interest argument vanishes. Example 11.6 puts numbers on both.

Example 11.6 Why not to exercise

A stock trades at \$50 and pays no dividend. An investor holds an American call struck at \$40 with one month to run. The rate is 6%. Exercising immediately yields $50 - 40 = \$10$.

But the call must be worth at least

$$50 - 40e^{-0.06\times 1/12} = 50 - 40 \times 0.995012 = 50 - 39.800 = \$10.20$$

so selling it yields at least \$10.20: twenty cents more, guaranteed, with no view on the stock.

Suppose instead the investor wants to own the stock and holds the call for that reason. Even then, exercising is wrong: they should sell the call for at least \$10.20 and buy the stock at \$50, which costs \$39.80 net against \$40 for exercising. And if what they want is to keep the exposure for the remaining month rather than own the stock, holding the call is better still, because exercising would also throw away the protection against the stock falling below \$40 during that month.

In a frictionless market no preference makes exercising right, which is unusual: most results in finance depend on what the investor wants.

11.9 EARLY EXERCISE: AMERICAN PUTS

For puts the conclusion reverses, and seeing why shows what the call case relied on.

Take the extreme case. A stock has fallen to \$0.05 and an American put struck at \$50 has six months to run. Exercising now yields \$49.95 immediately. Since the stock cannot fall below zero, waiting can add at most the last five cents, and it costs interest on the strike for as long as the holder waits: held to expiration, the \$50 is worth $50e^{-rT}$ today, less than \$49.95 at

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any rate above about 0.2%. Waiting a shorter time pays only if the stock reaches zero almost at once, which no realistic model allows. Exercise now.

The general principle is that exercising a put brings forward the receipt of K, which earns interest, whereas exercising a call brings forward the payment of K, which costs interest. The insurance argument still cuts the other way (exercising discards the chance of the stock falling further), but for a sufficiently deep in-the-money put the interest gain dominates.

There is therefore a **critical stock price** $S^*(t)$ below which immediate exercise is optimal, and it rises toward K as expiration approaches. Figure 11.5 shows the consequence for value.

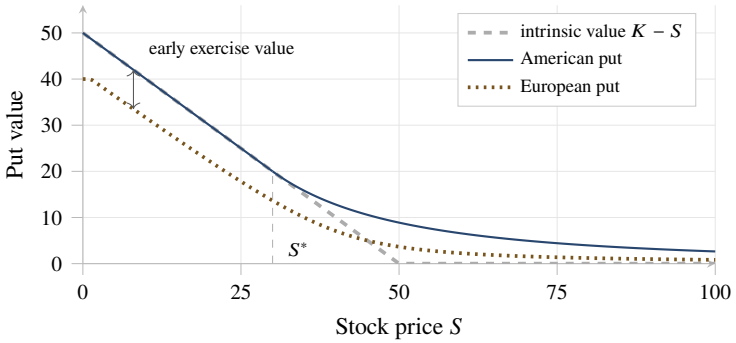


Figure 11.5: Put values against the stock price. $K = 50$; schematic, with the gap between the two exaggerated.

Three features of Figure 11.5 carry the argument. The American put coincides with intrinsic value below S^*, where it is exercised. The European put falls *below* intrinsic value for low stock prices (it is worth less than $K - S$ because the K arrives only at T), which is possible precisely because it cannot be exercised. And the European put is capped at Ke^{-rT} while the American put is capped at K. The gap between the two curves is the value of the right to exercise early, and unlike the call case it is genuinely positive: $P > p$.

11.10 DIVIDENDS AND EARLY EXERCISE

Now consider the third question Chapter 10 deferred. Ordinary cash dividends do not adjust the contract (Section 10.2), so an option holder simply suffers the price drop on the ex-dividend date. In a frictionless market that, and only that, can make early exercise of an American call rational. Frictions supply a second reason: a stock that is costly to borrow earns its owner a lending fee that a call holder does not receive, so it behaves like a stock paying a dividend, and a deep in-the-money call on it can be worth exercising early.

The trade-off is sharp. Exercising just before the stock goes ex-dividend captures the dividend D, because the exerciser owns the stock across the ex-date. The costs are the interest forgone on K over the remaining life $T - t$, where t is just before the ex-dividend date (only up to the next ex-dividend date, if one falls before expiration), and the downside protection given up. Comparing the dividend with the interest alone gives a clean condition: early exercise cannot be optimal unless

$$D > K\left(1 - e^{-r(T-t)}\right) \tag{11.13}$$

Example 11.7 applies the condition to a holder facing an ex-dividend date.

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Example 11.7 Should the call be exercised for the dividend?

A stock trades at \$60 and goes ex-dividend tomorrow, paying \$1.50. An investor holds American calls struck at \$55 with two months remaining. The rate is 5%.

The interest forgone by paying the strike two months early is

$$K\left(1-e^{-r(T-t)}\right) = 55\left(1-e^{-0.05\times 2/12}\right) = 55\,(1-0.991701)$$
$$= 55\times 0.008299 = \$0.456$$

The dividend is \$1.50, comfortably larger. Condition (11.13) is satisfied and exercise may well be optimal: capturing \$1.50 outweighs giving up \$0.46 of interest.

Now change one number. If the dividend were \$0.30 instead, the condition fails (\$0.30 against \$0.456) and exercise is *definitely* wrong, whatever else is true.

Condition (11.13) is necessary, not sufficient. When it holds, exercise *may* be optimal, but the comparison above ignored the insurance the holder abandons, which is worth something whenever the stock might fall below \$55 in the remaining two months. A holder deep in the money, where that protection is nearly worthless, should exercise; a holder near the money should think harder.

The practical rule is that early exercise of American calls, when it happens at all, happens immediately before an ex-dividend date and essentially never otherwise, which is why option desks maintain dividend calendars.

For American puts, dividends work the opposite way. A dividend lowers the stock price, which raises the put's value and makes waiting more attractive. Dividends therefore *discourage* early exercise of puts, and the critical price S^* falls as a dividend approaches.

11.11 PARITY BOUNDS FOR AMERICAN OPTIONS

Put–call parity is an equality for European options and fails for American ones, because either side may be exercised at a time the arbitrage argument does not control. What survives is a pair of inequalities. For a non-dividend-paying stock,

$$S_0 - K \;\le\; C - P \;\le\; S_0 - Ke^{-rT} \tag{11.14}$$

For the right-hand inequality, hold Portfolio C and be short Portfolio A, both in American form: hold the put and one share, write the call, and borrow Ke^{-rT}. If the call is assigned early, the share is delivered for K, which more than repays the loan, and the put is kept; otherwise everything nets to zero at T, and any dividends collected along the way are extra. So $P + S_0 - C - Ke^{-rT} \ge 0$, with or without dividends. The left-hand inequality follows from a portfolio argument of the same kind. With dividends of present value D, only the left edge moves, down by D:

$$S_0 - D - K \;\le\; C - P \;\le\; S_0 - Ke^{-rT} \tag{11.15}$$

The practical content is that $C - P$ sits in a band of width $K - Ke^{-rT}$, the interest on the strike over the option's life, plus D when the stock pays a dividend. Without dividends a short-dated band is narrow and parity is very nearly an equality. For the Table 10.2 stock at the 47.50 strike, with D = \$1.00, the band runs from $48.30 - 1.00 - 47.50 = -\$0.20$ to $48.30 - 47.208 = \$1.09$, \$1.29 wide; the European value $S_0 - D - Ke^{-rT}$ is \$0.092, and the quoted $C - P$ of \$0.095 sits almost exactly on it. Table 11.2 collects every bound in the chapter: the European difference is a point, and the American one a band around it.

11.12 WHAT VIOLATIONS ACTUALLY MEAN

Everything above assumed frictionless trading. Real chains appear to violate these relations routinely, and reading those violations correctly is the practical skill this chapter is for.

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Table 11.2: Bounds from arbitrage alone; D is the present value of dividends (0 if none).

	Lower bound	**Upper bound**
European call c	$\max(S_0 - D - Ke^{-rT},\ 0)$	S_0
American call C	$\max(S_0 - D - Ke^{-rT},\ S_0 - K,\ 0)$	S_0
European put p	$\max(D + Ke^{-rT} - S_0,\ 0)$	Ke^{-rT}
American put P	$\max(D + Ke^{-rT} - S_0,\ K - S_0,\ 0)$	K
$c - p$	$S_0 - D - Ke^{-rT}$	$S_0 - D - Ke^{-rT}$
$C - P$	$S_0 - D - K$	$S_0 - Ke^{-rT}$

Borrow costs. A borrow fee opens a band above parity (Example 10.4). Inside it there is no arbitrage, and puts priced above parity are quoting the borrow fee, not a bearish view.

Transaction costs. A conversion involves four legs (call, put, stock, financing), each crossing a spread. Table 10.2's spreads at the 50 strike, \$0.07 on the call and \$0.15 on the put, cost about \$0.11 against mid prices on the option legs alone, and twice that if the position is unwound early: more than many apparent violations.

The financing rate is not r. Every relation in this chapter used the risk-free rate. Section 5.15 argued at length that no participant borrows at it, and the relevant rate is the one *this* participant can actually finance at. Two firms observing the same chain can correctly disagree about whether an arbitrage exists.

Dividends are estimates. Parity with dividends, (11.5), requires knowing D. For an option spanning an uncertain dividend, the apparent violation reflects a different dividend forecast rather than a mispricing.

The lesson is the one that recurs from Chapter 5: *these relations define a band, not a point*, and the width of the band is a real economic quantity. A trader who assumes it is zero and trades every apparent violation will discover the width empirically, which is more expensive.

11.13 WHO TRADES THESE RELATIONSHIPS

Several kinds of participant trade these relationships directly, and who they are explains why real chains are as consistent as they are.

Option market makers are the largest users. A market maker who has accumulated an unwanted position uses conversions and reversals (Section 11.4) to transform it (turning a short put into a long stock position plus a short call) at a cost they can compute in advance. This is not speculation; it is inventory management, and it is why quotes across a chain move together rather than independently.

Dispersion and volatility desks (Chapter 22) use parity to strip the directional component out of a volatility position.

Financing desks trade box spreads (Section 11.7); when box rates diverge from repo rates, the reason is usually balance-sheet capacity rather than mispricing, as Section 5.15 argued.

Structured-product desks use synthetics (11.10)–(11.11) to manufacture exposures they cannot trade directly. Where shorting is restricted, prohibited, or prohibitively expensive, the synthetic short is the only route.

Index and dividend traders use the implied forward of Section 11.6 as a traded quantity in its own right. Dividend expectations extracted from option chains are traded directly in dividend futures and dividend swaps, which pay the dividends actually distributed over a stated period; there the number computed in Example 11.4 is the price being traded.

The common thread is that none of these participants is forecasting the stock. They are trading the relationships among instruments written on it, and the fact that they compete to do so is the mechanism that makes the relationships hold.

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SUMMARY

Every result in this chapter follows from the absence of arbitrage (a strategy that costs nothing, cannot lose, and can gain), without any model of how the stock behaves. The book assumes throughout that no such opportunity exists, as a statement about prices rather than about people.

Six factors determine an option's value. Calls and puts both rise with volatility, because the payoff is truncated at zero and greater dispersion carries no symmetric penalty; higher rates raise calls and lower puts; and more time helps American options unambiguously but not necessarily European ones, which cannot act on the extra time. Arbitrage alone then bounds each price above and below, and across strikes it imposes monotonicity, the spread bound and convexity. Convexity says a butterfly cannot have negative value, a test that must be run at prices one can trade rather than at mid prices, and Chapter 19 uses it to test a volatility surface for arbitrage.

Put–call parity, $c + Ke^{-rT} = p + S_0$, links the call and put at the same strike and maturity. Rearranged, it manufactures synthetic stock and synthetic options, and the synthetic short explains why puts trade above parity on hard-to-borrow stocks; two synthetics at different strikes make a box spread, a zero-coupon bond built from options, riskless only when the options are European, and used as a financing instrument. Parity is also a measurement: the implied forward read off an option chain is the chain's most useful single number, and its shortfall from the carried-forward spot reflects mostly expected dividends, mixed with borrow costs.

An American call on a non-dividend-paying stock should never be exercised early, because selling always beats exercising: the option is worth more than its exercise value, and exercising discards downside protection. An American put may be exercised early, because deep in the money the interest earned on receiving the strike now outweighs the option value given up, so below a critical stock price exercise is optimal. In a frictionless market, dividends are the only reason to exercise an American call early, and when they make it optimal, it happens immediately before an ex-dividend date.

For American options parity becomes a pair of inequalities rather than an identity. In practice all these relations define bands rather than points, widened by borrow costs, transaction costs, financing that is not at the risk-free rate, and uncertain dividends. The width is an economic quantity worth measuring, not an error term to ignore.

FURTHER READING

Merton's 1973 paper on rational option pricing established most of the results in this chapter and states them in full. Stoll's 1969 paper introduced put–call parity to the modern literature. Natenberg's *Option Volatility and Pricing* describes conversions, reversals, and boxes as a market maker trades them. Ofek, Richardson and Whitelaw (2004) is the evidence for Section 11.12: apparent parity violations track borrow costs and short-sale constraints. Jensen and Pedersen (2016) is the place to start on when frictions make early exercise of a call rational, and van Binsbergen, Diamond and Grotteria (2022) read risk-free rates off index-option prices through the parity relation a box spread locks in.

PRACTICE QUESTIONS

11.1 Explain why both a call and a put increase in value with volatility, and why a forward contract's value does not depend on volatility at all.

11.2 A non-dividend-paying stock trades at \$45. A six-month European call struck at \$42 trades at \$3.10 and the rate is 6%. Is the lower bound violated? If so, construct the arbitrage and compute the profit.

11.3 A stock trades at \$28 with no dividends. Four-month European options struck at \$27 are

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quoted: call \$2.60, put \$1.10, rate 5%. Test parity and, if it fails, state the trade and the profit.

11.4 Using put–call parity, express a synthetic long put in terms of a call, the stock, and cash. Verify the payoff in both states at expiration.

11.5 Explain why an American call on a non-dividend-paying stock should never be exercised early, giving both the bounds argument and the economic argument. Which of the two still works when $r = 0$?

11.6 A stock has fallen to \$0.25 and an American put struck at \$40 has three months remaining. Should it be exercised? Give the numerical comparison with $r = 4\%$.

11.7 A stock at \$80 goes ex-dividend tomorrow, paying \$0.90. An investor holds American calls struck at \$75 with three months remaining; the rate is 4%. Apply condition (11.13) and state what conclusion is and is not justified.

11.8 Construct a box spread from options struck at \$40 and \$50 expiring in six months, list the four legs, and state the payoff. If the box costs \$9.70, what rate is implied?

11.9 State the three relations that constrain call prices at different strikes on the same underlying and maturity, and for each identify the position that profits if it is violated.

11.10 Using the mid prices of Table 10.2 at every strike, with $r = 5\%$, compute the implied forward (11.12) and the dividend it implies, $S_0 - F_0 e^{-rT}$. At which strikes does the reading drift, in which direction, and why? Which strike would you quote?

11.11 State the parity bounds for American options, explain why the European equality becomes a band, identify which side of the band binds when the put is deep in the money and why, and state how the band changes when the stock pays dividends of present value D.

11.12 A chain shows puts trading 40 cents above parity on a \$50 stock with six months to expiration. Using Example 10.4, state the borrow fee that would account for it, and explain why observing a violation of this size is not the same as being able to profit from it.

FURTHER QUESTIONS

11.13 Section 11.1 states that more time to expiration can *reduce* the value of a European option. Construct two explicit examples, one for a call and one for a put, in which this occurs, and identify precisely which feature of European exercise makes it possible.

11.14 The case study in Section 11.7 describes a box spread that was not riskless because it used American options. Work out exactly which of the four legs can be assigned early, under what stock price conditions, and what position the trader is left holding. Then state a rule for when a box may safely be treated as a financing trade.

11.15 Condition (11.13) is necessary but not sufficient. Construct a case in which the condition holds and early exercise is nevertheless wrong, and explain what quantity the condition omits. How would you decide the question in practice?

11.16 An option chain on a hard-to-borrow stock shows puts persistently above parity by an amount implying a 15% borrow fee. A colleague proposes selling puts and buying calls to "harvest the mispricing," delta-hedging with stock in the manner of Chapter 18. Explain what exposure this position actually carries, and what would have to happen for it to lose money.

11.17 Derive the left-hand inequality in (11.14) by constructing two portfolios and comparing their values. State which step uses the put's right to early exercise, and show that for European options the inequality follows from the equality (11.4).

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Chapter 12
Trading Strategies Involving Options

Options can be combined into an unlimited number of payoff shapes, and textbooks that catalog them tend to leave readers able to draw a great many diagrams and unable to say why anyone would want one. This chapter is organized around the opposite question. Every position is a bet on something, and a well-chosen strategy is one whose bet matches the view its holder actually has.

There are only four things to bet on. **Direction**—where the underlying goes. **Magnitude**—how far it moves, regardless of which way. **Time**—whether the move happens before the option expires. And **skew**, the relative price of downside and upside protection. Most strategies are a bet on two or three of these simultaneously, which is why traders lose money on positions whose directional view turns out to be right.

Every one of these bets is levered. A spread that costs $2.04 on a $48.30 stock earns up to 1.45 times its cost if the stock rises 8.7%, and loses all of it if the stock falls 1.7%. Leverage is not a fifth thing to bet on; it is the multiplier on the other four, the same multiplier Example 1.5 measured for futures, and it is why a position whose diagram looks tame can be anything but tame in a portfolio. The examples below therefore report what each position costs alongside what it can pay, because the ratio of the two is where the leverage sits.

Chapter 11 has already supplied the tools. Put–call parity means many strategies that appear distinct are the same position in a different form, and Section 12.1 begins there because it eliminates a great deal of apparent complexity before any of it accumulates.

Sections 12.2 to 12.6 take the strategies by the view they express, from spreads for direction to risk reversals for skew; the rest of the chapter prices one view four ways, builds arbitrary payoffs, turns to the institutional uses (the collar and the portfolio hedge), and closes with structured products, position management and costs.

In the worked examples, the numbers come from the option chain of Table 10.2 (a stock at $48.30, 45 days to expiration), and they are computed at the *bid and ask*, not the mid. Section 12.14 shows why that decision changes the conclusions.

12.1 STRATEGIES THAT ARE EQUIVALENT

Two of the most widely used retail strategies are not new positions at all.

A **covered call** is long stock plus a written call. Rearranging parity (11.4), which holds exactly for European options,

$$S_0 - c = Ke^{-rT} - p$$

so the covered call has the payoff of a *written put* plus cash. A **protective put** is long stock plus a purchased put, and the same rearrangement gives

$$S_0 + p = c + Ke^{-rT}$$

which is a *purchased call* plus cash. Figure 12.1 shows both.

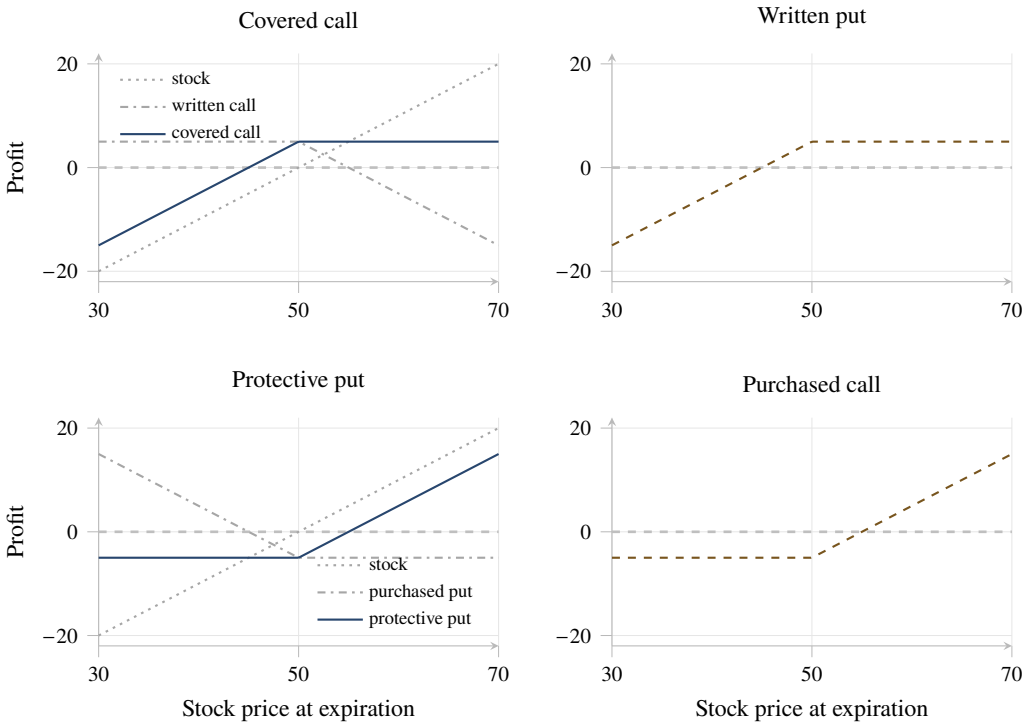


Figure 12.1: Profit at expiration of each strategy, with its components in gray, and of its parity equivalent. $K = 50$.

The panels in each row are identical, and this is not a curiosity: it has three practical consequences that recur through the chapter.

It disciplines the marketing. A covered call is often sold as a conservative income strategy and a written put as a speculative one. Held to expiration, the covered call is a written put plus cash, a **cash-secured put**, and brokers, who set options-approval levels, commonly approve the two at the same lowest level; the higher level is for the *naked* put, the same exposure without the cash. The *market risk* is identical.

It tells you which to trade. If two routes to the same exposure exist, take the cheaper. Writing the put is one transaction rather than two, so it pays the bid–ask spread once instead of twice.

It identifies the real bet. A covered call writer is short a put: they are selling insurance, and they profit when the market is calm and lose when it falls. Describing this as "generating income from a stock you already own" is accurate about the cash flow and silent about the risk.

12.2 SPREADS: A DIRECTIONAL VIEW WITH BOTH ENDS CAPPED

A **spread** combines two or more options of the same type, each of them a **leg** of the spread. The simplest is a directional bet financed by selling away the part of the move you do not expect.

A **bull call spread** buys a call and sells a higher-strike call. It costs less than the call alone, because the sale pays for part of it, and in exchange the profit stops at the higher strike. A **bear put spread** is the mirror image (Example 12.2). Each can also be built from the other type of option: a **bull put spread** sells a put and buys a lower-strike put, and a **bear call spread** sells a call and buys a higher-strike call. These bring in a net premium, so for each view one version is a **debit** spread, which costs a net premium to establish, and the other a **credit** spread.

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By put–call parity the two versions of the same view have the same profit diagram apart from interest on the strike difference, so the choice between them is about cash up front and margin, not about the view; put credit spreads are a staple of investors who sell options for income.

Example 12.1 A bull call spread

Using Table 10.2, buy the 47.50 call at its \$2.72 ask and sell the 52.50 call at its \$0.68 bid:

$$\text{net debit} = 2.72 - 0.68 = \$2.04 \text{ per share, or } \$204 \text{ per spread}$$

The outcomes are:

Stock at expiration	Payoff	Profit
≤ 47.50	0.00	−2.04
49.54	2.04	0.00
≥ 52.50	5.00	+2.96

Maximum profit \$2.96, maximum loss \$2.04, breakeven \$49.54. The stock must rise 2.6% in 45 days merely to break even, and 8.7% to earn the maximum. The **reward-to-risk ratio** (maximum profit divided by maximum loss, a trader's shorthand that says nothing about how likely either is) is 1.45 to 1.

Compare with buying the 47.50 call outright at \$2.72. That costs 33% more and has unlimited upside. The spread does better at every price below \$53.18, rise or fall, and the outright call above it. The spread is not a cheaper way to be bullish; it is a bet on a specific range.

Example 12.2 A bear put spread

The mirror image, for a trader who expects a fall. Using Table 10.2, buy the 50 put at its \$3.95 ask and sell the 45 put at its \$1.81 bid:

$$\text{net debit} = 3.95 - 1.81 = \$2.14 \text{ per share, or } \$214 \text{ per spread}$$

The outcomes are:

Stock at expiration	Payoff	Profit
≥ 50	0.00	−2.14
47.86	2.14	0.00
≤ 45	5.00	+2.86

Maximum profit \$2.86, maximum loss \$2.14, breakeven \$47.86. The stock must fall 0.9% in 45 days to break even and 6.8% to earn the maximum, and the reward-to-risk ratio is 1.34 to 1, against 1.45 to 1 for the bull spread of Example 12.1. Part of the gap is moneyness (the bull spread buys a strike \$0.80 below spot, the bear spread one \$1.70 above) and part is the chain's dividend; Example 12.5 separates the forward from the skew.

12.3 BUTTERFLIES AND CONDORS: A VIEW THAT THE PRICE STAYS PUT

Section 11.3 introduced the butterfly as the constraint that makes the call price convex in the strike. As a trade, it profits if the underlying stays near the middle strike until expiration.

Buy one call at K_1, sell two at K_2, buy one at K_3, with equal spacing. The payoff is zero outside $[K_1, K_3]$ and peaks at $K_2 - K_1$ in the middle. A **condor** spreads the two middle strikes apart, producing a plateau instead of a peak: a wider target for a lower maximum. As traded, most condors are **iron condors**: an out-of-the-money put spread and an out-of-the-money call spread, both sold, entered for a credit and, by parity, with the call condor's profit profile. This credit form is what premium-selling funds run, and Chapter 23 finds it a large and growing share of same-day index options volume.

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Figure 12.2 collects the four profit profiles, and reading them together shows what every spread buys and sells: the strike that was sold caps the profit, and the premium it brings in is exactly what makes the structure cheaper than the option bought alone.

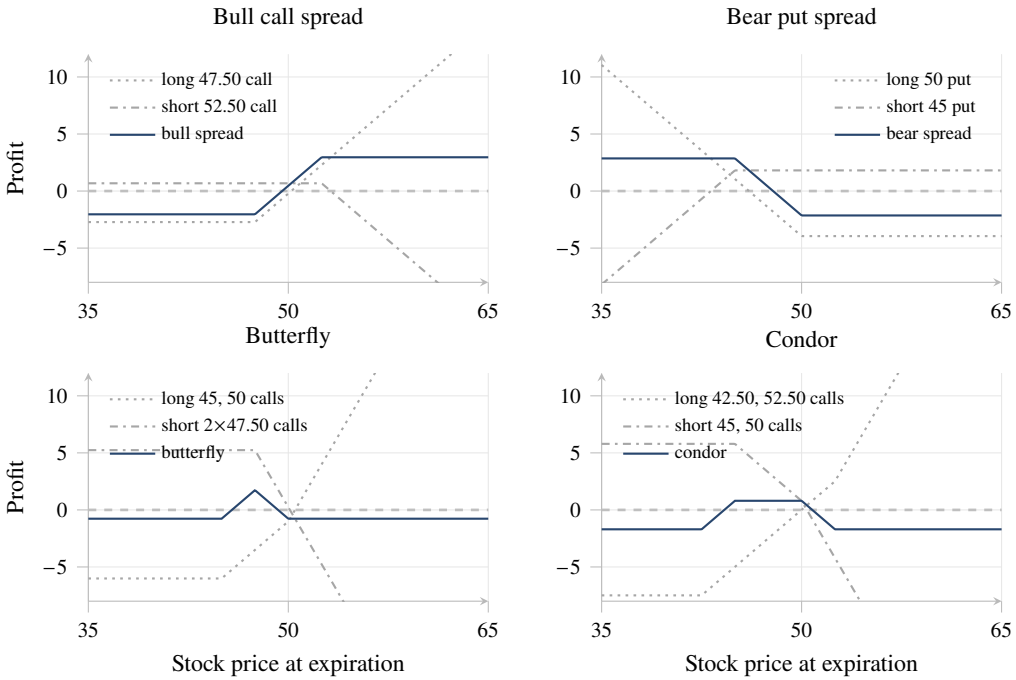


Figure 12.2: Profit at expiration for four spread structures at the quotes of Table 10.2, with the legs in gray.

Example 12.3 A butterfly, priced at the quotes

From Table 10.2, buy the 45 call at \$4.50, sell two 47.50 calls at \$2.62, buy the 50 call at \$1.51:

$$\text{cost} = 4.50 - 2 \times 2.62 + 1.51 = 6.01 - 5.24 = \$0.77$$

The maximum payoff is \$2.50, achieved if the stock finishes at exactly \$47.50, so the maximum profit is \$1.73 against a maximum loss of \$0.77, a reward-to-risk ratio of 2.25 to 1. The breakevens are \$45.77 and \$49.23.

That ratio looks attractive until the window is examined. The stock is at \$48.30, so it must finish between 5.2% below and 1.9% above its current level, in 45 days, for the position to make anything at all. Butterflies pay well precisely because that is unlikely.

Now comes the point Section 11.3 set up. At mid prices Example 11.2 found the same butterfly worth \$0.56, so the trader paid \$0.77 for something worth \$0.56: execution consumed \$0.21 (27% of the price paid, and 38% of the position's theoretical value) before the market moved at all. Section 12.14 shows that this cost recurs in every multi-leg strategy.

12.4 STRADDLES AND STRANGLES: A VIEW ON THE SIZE OF THE MOVE

To bet that something will happen without a view on which way, buy both a call and a put. A **straddle** uses the same strike; a **strangle** uses a lower put strike and a higher call strike, which is cheaper and needs a move at least as large to pay.

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Example 12.4 How far the stock must move

Buy the 47.50 straddle at the asks: call \$2.72 plus put \$2.63 equals \$5.35.

The breakevens are 47.50 ± 5.35, that is \$42.15 and \$52.85. From the current \$48.30 the stock must fall 12.7% or rise 9.4% within 45 days for the position to make a cent.

A strangle using the 45 put (\$1.89 ask) and the 50 call (\$1.51 ask) costs \$3.40, 36% less, but its breakevens are \$41.60 and \$53.40, outside the straddle's on both sides. The strangle costs less and risks less, but it needs a larger move, because the market has priced both structures off the same volatility surface.

This is the general result. An option position's cost and the move it requires are two views of one number, the implied volatility. Buying a straddle is not a bet that the stock will move; the market already expects it to. It is a bet that the stock will move *more than the implied volatility says*—and Chapter 19 documents that implied volatility exceeds subsequent realized volatility on average. The straddle buyer is systematically paying a premium for that bet.

Figure 12.3 draws the two positions of Example 12.4 side by side.

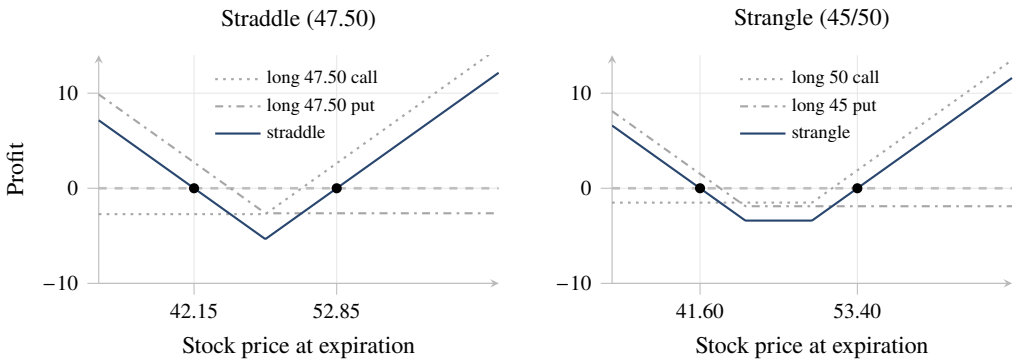


Figure 12.3: Profit at expiration for the straddle and strangle of Example 12.4, at the quotes of Table 10.2, with the legs in gray.

The raw straddle is not a pure volatility trade. An at-the-money straddle has almost no exposure to the direction of a small move, because the call gains about as much as the put loses. That balance holds only while the stock stays near the strike. If the stock rises, the call moves into the money and comes to behave more like the stock, while the put moves out of the money and matters less, so the position as a whole now gains when the stock rises and loses when it falls: it has become net long. Chapter 18 measures that sensitivity as the option's delta and shows why it changes with the stock price. The consequence needs no measurement to see: a straddle held through a large move ends up as a directional position in whichever direction the stock happened to go.

A trader who genuinely wants only volatility exposure must **delta-hedge**: trade the underlying continuously to keep the net directional exposure at zero, so that what remains is a bet on realized against implied volatility alone. Chapters 18 and 22 develop the trade: Chapter 18 measures the gamma (how fast the delta changes as the stock moves) and the theta (the value lost each day to time decay) that the hedge trades against each other, and Section 22.1 shows that the hedging profit accumulated along the way is the difference between realized and implied variance, weighted by gamma. The unhedged straddle of Example 12.4 is a coarser instrument: it contains the volatility bet, plus a directional bet the trader did not intend and is not being paid for.

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12.5 CALENDAR SPREADS AND THE TERM STRUCTURE

A **calendar spread** sells a near-dated option and buys a longer-dated one at the same strike. It is not a directional trade and not straightforwardly a volatility trade; it is a bet on two things at once.

The first is **time decay**. A near-dated option loses time value faster than a long-dated one: Section 11.1 noted that value grows roughly with $\sqrt{T}$, so the decay per day is steepest at the end. Selling the fast-decaying option against the slow one collects that difference, provided the stock stays near the strike. Figure 12.4 shows both halves of the argument: the left panel is the spread's profit at the near expiration, a hump that peaks at the strike and falls away on either side, and the right panel is time value against remaining life, with the near-dated option sitting on the steep part of the curve.

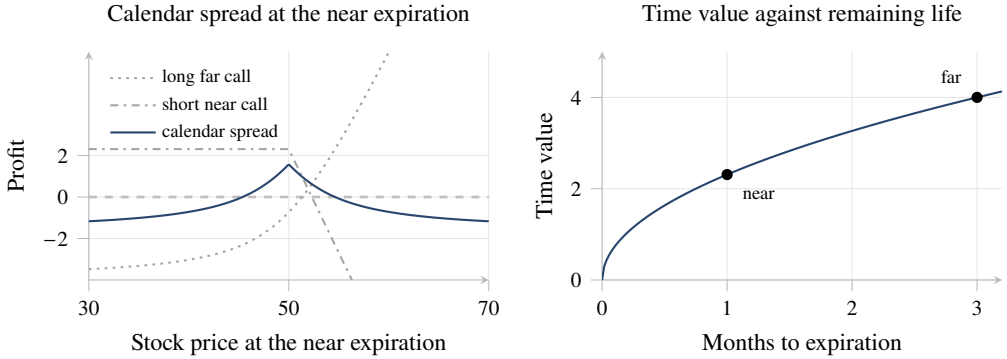


Figure 12.4: Left: profit on a calendar spread at the near option's expiration, $K = 50$, schematic. Right: time value against remaining life, proportional to $\sqrt{T}$.

The second is the **term structure of implied volatility**. The two legs are usually priced at different implied volatilities, and the calendar spread gains if the far option's implied volatility rises and loses if the near one's does. When near-dated implied volatility is elevated (before an earnings announcement, say), selling the near option at that level against a far-dated option priced at a calmer one is the trade the structure is designed for. Chapter 19 treats that term structure properly.

A **diagonal spread** changes the strike as well as the maturity, combining the calendar bet with a directional one. It is popular and frequently misunderstood, because its holder often has a clear view on only one of the two bets they have taken.

12.6 RISK REVERSALS AND RATIO SPREADS: A VIEW ON SKEW

A **risk reversal** buys an out-of-the-money call and sells an out-of-the-money put. Section 11.5 showed that at the same strike this is a synthetic long; at different strikes it is a long position with a gap in the middle, and it can often be established for zero cost.

What makes it interesting is what determines that cost. Buying the call and selling the put trades one implied volatility against another, so the price of a risk reversal *is* the skew: the amount by which downside protection is more expensive than upside participation. In equity markets the lower-strike put almost always carries the higher implied volatility, because puts are persistently in demand, so the trader who buys the call and sells the put is *paid* the skew in volatility terms: the option sold is the dear one and the option bought the cheap one. In dollars the one in Section 12.7 is entered for a small credit, and since its two strikes sit about equally far from the forward, that credit is mostly the skew. Chapter 19 explains why it is there.

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Risk reversals are therefore the standard way to trade a view on skew rather than on the level of volatility. They are also the standard corporate hedge: a firm wanting downside protection at zero premium sells away the upside to pay for it, which is a collar (a put bought and a call sold against a holding; Section 12.9) and is the same structure viewed from the other side. That side pays the skew; Examples 12.5 and 12.6 measure what it costs.

A **ratio spread** buys one option and sells more than one further out. It is cheap to establish, sometimes for a credit, and profits over a range, but the extra short options leave the position exposed beyond them. The shape looks safe on a profit diagram and is not, because the loss is unbounded (for calls; very large for puts) in exactly the tail the trader expects not to occur. Figure 12.5 shows both: the risk reversal is a long position with a flat stretch between the strikes, and the ratio spread's profit turns down without limit beyond the strike it sold twice.

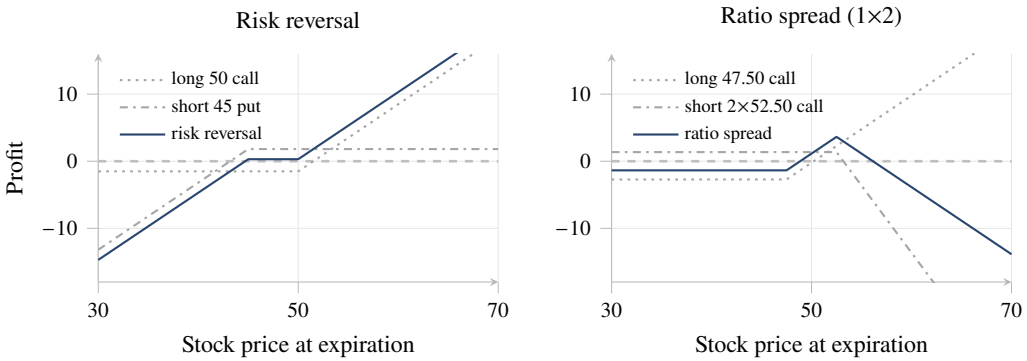


Figure 12.5: Profit at expiration for a risk reversal and a 1×2 ratio spread at the quotes of Table 10.2, with the legs in gray.

12.7 EXPRESSING ONE VIEW FOUR WAYS

The abstract discussion becomes concrete when a single view is priced through several instruments. Suppose a trader believes the stock, at \$48.30, will rise to around \$52 within 45 days. Table 12.1 shows what happens under four ways of acting on that: buying the stock, buying the 47.50 call outright, the 47.50/52.50 bull spread of Section 12.2, and a risk reversal. The outright call is the spread's own long leg, so the second and third columns differ only by the call the spread sold.

The risk reversal here buys the 50 call at \$1.51 and sells the 45 put at \$1.81, for a net credit of \$0.30. Its capital line is not that credit. The written put must be margined, and formula (a) of Section 10.8 applied to it (20% of \$48.30, less the \$3.30 by which the put is out of the money, plus the \$1.81 premium) requires \$8.17 a share, so the route commits $817 - 30 = \$787$. Four things in the table are worth extracting.

The structure tailored to the view wins when the view is exactly right. At \$52, the trader's actual target, the bull spread earns \$246, more than either other option route, and returns 121% on its capital, more than any route; only the stock earns more in dollars, \$370, with twenty-four times the capital. This is what it means for a strategy to match a view.

It loses when the view is right about direction and wrong about size. At \$56 the spread earns \$296 while the outright call earns \$578. The spread sold that outcome away, which is precisely how it was made cheap.

Zero cost is not zero risk. The risk reversal was entered for a \$30 credit and still loses \$270 at \$42, almost as much as the outright call, which cost \$272. Its cheapness comes from the written put, and the written put is the whole downside. A structure that costs nothing has not

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Table 12.1: Profit per 100 shares of exposure, at the quotes of Table 10.2, and the return on the capital each route commits.

Stock at expiration	Buy stock	47.50 call	47.50/52.50 spread	Risk reversal
Profit, $				
$42.00	−630	−272	−204	−270
$45.00	−330	−272	−204	+30
$48.30	0	−192	−124	+30
$52.00	+370	+178	+246	+230
$56.00	+770	+578	+296	+630
Capital committed	$4,830	$272	$204	$787
Return on capital committed				
$42.00	−13.0%	−100%	−100%	−34.3%
$45.00	−6.8%	−100%	−100%	+3.8%
$48.30	0	−70.6%	−60.8%	+3.8%
$52.00	+7.7%	+65.4%	+120.6%	+29.2%
$56.00	+15.9%	+212.5%	+145.1%	+80.1%

eliminated risk; it has financed one exposure by selling another.

Capital efficiency and risk run in opposite directions. Buying the stock ties up $4,830 and loses $630 in the bad case; the spread ties up $204 and loses $204. The return block makes the leverage explicit: at the target the spread returns 121% on its capital and the stock 7.7%, and at $42 the same spread returns −100% while the stock loses 13%. A return on capital is meaningless unless the loss column is read at the same time, which is the point Example 1.5 made about leverage, appearing here as a choice among strategies.

12.8 BUILDING AN ARBITRARY PAYOFF

The structures above are conventional, but nothing restricts a trader to them. A general result underlies the whole subject and is worth stating explicitly, because it explains where exotic payoffs come from and is the bridge to Chapters 19, 22 and 34.

Any piecewise-linear payoff can be replicated exactly by a bond, a forward, and a portfolio of European calls. The recipe is mechanical: place a call at each kink, with a quantity equal to the *change in slope* at that kink.

Take the bull call spread. Its payoff is flat at zero below $47.50, rises with slope 1 to $52.50, and is flat above. The slope changes by +1 at 47.50 and by −1 at 52.50, so the replicating portfolio is one long call at 47.50 and one short call at 52.50—which is what a bull spread is. The butterfly's slopes are 0, +1, −1, 0, giving changes of +1, −2, +1: long one, short two, long one.

The same recipe builds anything. A note paying 70% of an index's rise, capped once the index has risen 30% (a maximum return of 21%), is a payoff with slope changes of +0.7 at the initial level and −0.7 at 130% of it: buy 0.7 calls at the first strike, sell 0.7 at the second. Figure 12.6 shows the construction.

Digitals as the limit of a spread. A **digital** (or binary) option pays a fixed amount if the underlying finishes above a strike and nothing otherwise. Its payoff has a jump rather than a kink, so no finite portfolio of calls reproduces it, but a tight call spread approaches it. Buying a call at $K - \varepsilon$ and selling one at $K + \varepsilon$, scaled by $1/(2\varepsilon)$, gives a payoff that rises from 0 to 1 across a narrow band. As $\varepsilon \to 0$ the position becomes a digital, and its price becomes

$$\text{digital}(K) = -\frac{\partial c}{\partial K}$$

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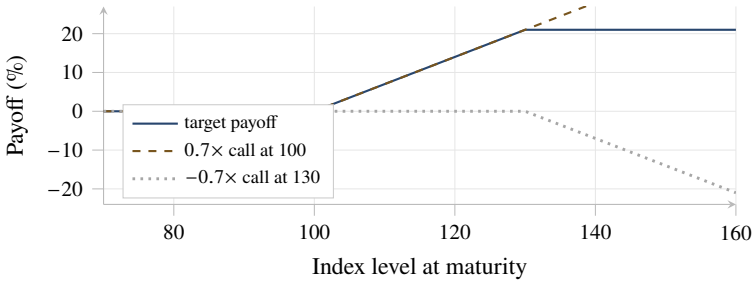


Figure 12.6: A capped participation payoff as two call positions.

This is not merely a pricing device. It is how digitals are hedged: a dealer short a digital holds a tight call spread, and the tighter the spread the larger the position required, which is why digitals are risky to hedge near the strike at expiration: the same pin risk of Example 10.6 in a more acute form. The autocallable coupons of Section 12.12 are digitals, which is why those products are difficult to hedge precisely when the index sits near a barrier. Chapter 19 pushes the same construction one derivative further, to a butterfly of vanishing width, and reads the market's probability distribution for the underlying off the option chain.

12.9 COLLARS AND THE CONCENTRATED SHAREHOLDER

The most common institutional use of an option combination is not speculative. A founder, executive, or family office holding a large position in a single stock wants to reduce the risk without selling, because selling would trigger tax, breach a lockup (an agreement not to sell for a set period, typical after a public offering), or signal something unwanted.

The **collar** buys a put and sells a call to pay for it. Section 12.6 noted it is a risk reversal seen from the other side; here it is a hedge rather than a view.

Example 12.5 A collar, and why its strikes are not symmetric

A shareholder holds 200,000 shares at \$48.30, worth \$9.66 million. Using Table 10.2, they buy the 45 put at its \$1.89 ask and sell the 50 call at its \$1.44 bid:

$$\text{net} = 1.44 - 1.89 = -\$0.45 \text{ per share, a debit of } \$90{,}000$$

The call pays for three-quarters of the put. Leaving that \$90,000 aside, the position is floored at \$45 and capped at \$50:

$$\text{worst case} = 200{,}000 \times 45 = \$9.0\text{m}, \qquad \text{best case} = 200{,}000 \times 50 = \$10.0\text{m}$$

The put is 6.8% below spot; the call is only 3.5% above it. The holder gave up 3.5% of upside to protect against everything below 6.8% down, and still paid.

Two things impose that asymmetry. The first is the forward: the dividend Example 10.3 found in the chain puts the 45-day forward near \$47.60, from which the strikes are nearly symmetric, 5.5% below and 5.1% above, so on this chain the forward sets most of the gap in the strikes. The second is skew: measured from the forward the two options are nearly equidistant, yet the put costs \$0.38 more at mid (\$0.45 at the quotes), because the chain prices it at a 46% implied volatility against the call's 37% (Section 15.11). A zero-cost collar must therefore set its put further from the forward than its call, as Example 12.6 shows on index quotes. Chapter 19 explains why, and the reason connects directly to who is on each side: shareholders like this one persistently buy downside protection, and someone must be paid to supply it.

A practical corollary follows. Because skew is part of what sets the terms, a collar is cheapest to establish when skew is flat.

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12.10 HEDGING A PORTFOLIO WITH INDEX OPTIONS

Section 3.6 hedged an equity portfolio with index futures. Options offer an alternative with a different shape, and the choice between them is the clearest institutional application of everything in this chapter.

The contract count is the same beta-adjusted calculation as for futures:

$$N = \beta \frac{V_P}{S \times \text{multiplier}} \tag{12.1}$$

where V_P is the value of the portfolio, β its beta against the index, S the index level, and $S \times$ multiplier the value of the index underlying one option contract (the role V_F played for a futures contract in Section 3.6). What differs is that a put costs money but can lose no more than that, while a futures hedge is free to establish and removes the upside along with the downside.

Example 12.6 What protecting a portfolio actually costs

A fund holds \$50 million of equities with a beta of 1.1 against an index at 5,250. The index option multiplier is \$100. From (12.1),

$$N = 1.1 \times \frac{50{,}000{,}000}{5{,}250 \times 100} = 1.1 \times 95.24 = 104.8 \rightarrow 105 \text{ contracts}$$

Buying three-month puts 5% out of the money, struck at 4,987.5 and quoted at 60.00 points:

$$\text{cost} = 60 \times 100 \times 105 = \$630{,}000 = 1.26\% \text{ of the portfolio, per quarter}$$

The floor is set on the index, not on the portfolio. With a beta of 1.1 a 5% fall in the index is roughly a 5.5% fall in the portfolio, and the beta-adjusted contract count is what makes the puts pay at that rate. The beta scaling is an approximation that ignores interest and dividends; Section 16.2 derives the strike that corrects for them. If the index falls 10%, to 4,725, the portfolio loses about \$5.5 million while the 105 puts pay $262.5 \times 100 \times 105 = \$2.76$ million, so the loss net of the put payoff is \$2.74 million, or 5.5% of the portfolio, 6.7% once the \$630,000 premium is counted. The floor stays near 5.5% however far the index falls beyond the strike, drifting only with the rounding of 104.8 contracts to 105.

Renewed every quarter, that is roughly 5% a year. Set that against a long-run expected equity return in the high single digits and the conclusion is uncomfortable but unavoidable: continuous protection at a strike this close costs most of the expected return. This single number explains why most institutional hedging structures look the way they do.

The put-spread collar is the usual response. Buy the 4,987.5 put at 60, sell a 4,462.5 put (15% down) at 22, and sell a 5,512.5 call (5% up) at 48:

$$\text{net} = -60 + 22 + 48 = +10 \text{ points, a credit of } \$105{,}000$$

Look first at the two 5% strikes. The put costs 60 points and the call 48. Over three months the index forward sits only about 1% above spot (a 5% interest rate less a dividend yield near 1.5%), which on its own would make the call the dearer of the two, so the gap comes from skew and if anything understates it. This is the same skew Example 12.5 found on a single stock, here with the forward slightly above spot rather than pulled below it by a dividend. A zero-cost collar on this index has to set its call nearer to spot than its put.

The fund is now protected between −5% and −15% on the index (about −5.5% to −16.5% on the portfolio) and capped at +5% on the index, about +5.5% on the portfolio; and it has been paid to enter the structure. What it has given up is protection *below* −15%—precisely the scenario the hedge was ostensibly for. Selling the deep put is part of what pays for the structure, so a budgeted program keeps protection against ordinary declines and gives up protection against catastrophes.

With the bought put moved to the money, wrapped in an exchange-traded fund and reset once a year, this structure is the **buffer ETF**, or defined-outcome fund: over each period the fund absorbs the first 10 or 15% of the index's loss and caps its gain, and such funds held about \$78 billion at the end of 2025. Financial advisers buy them for clients near retirement who want to stay in

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equities with the first losses absorbed, and who accept the cap, and the losses beyond the buffer, in exchange.

Tail hedging inverts the choice: buy puts 30% out of the money on the index (a fall of about 33% in the portfolio) at, say, 6 points, costing $63,000 a quarter, or about 0.5% a year. The premium is small and the payoff enormous when it comes. The difficulty is that Chapter 19 shows deep out-of-the-money puts carry the *highest* implied volatilities of any options on the surface, so it is the dearest insurance relative to its expected payoff, although its premium is the smallest. A strategy costing 0.5% a year that pays off only in a rare crash must pay a great deal on those occasions to justify itself, and whether it does is genuinely contested.

As Section 3.2's case study warned, an option premium sits in a named budget line every year it pays nothing, which is part of why collars, which can be struck at no cost, are popular.

Options instead of futures: the margin asymmetry

A commercial hedger faces the same choice, and there the decisive difference is not the premium at all. A futures hedge fixes the price in *both* directions and demands cash on every day the market moves against it. A purchased option (a producer's put, a consumer's call) whose premium is paid up front removes only the unfavorable tail, and owes no variation margin (unless the option itself is margined futures-style, Section 16.5), because the worst outcome for a purchased option is that it expires worthless.

That asymmetry is what Section 3.10's hedging failures are made of. Every one of them runs through variation margin on a futures leg, and a hedger holding purchased options instead—puts for a producer, calls for a seller of fixed-price supply like Metallgesellschaft—would have faced no margin call in any of them. The premium therefore buys two distinct things: the retained upside, and the elimination of funding risk. Whether the second is worth paying for depends on the firm's access to liquidity, which is exactly the judgment Section 3.10 says firms get wrong.

Writing options is a different matter entirely. A firm that sells options to reduce its hedging cost (selling calls to fund puts, the collar above) takes the margin obligation back on the written leg, along with an exposure on that leg that is not capped. The premium saved is not free, and such a structure's cash flows can come to resemble the futures hedge it was meant to improve on.

12.11 WHAT EACH STRATEGY IS ACTUALLY A VIEW ON

Table 12.2 is the chapter in one place. It records what each structure bets on, what it costs, and what it can lose.

Three observations organize the table.

The time column is almost the volatility column reversed. Positions that are short volatility are helped by the passage of time, and positions that are long volatility are hurt by it. This is not a coincidence: Chapter 18 shows that gamma and theta are tied together: a position cannot hold gamma without paying theta, or sell it without collecting theta. The practical meaning is immediate: you are paid to wait or you pay to wait, and which one is determined by whether you are net long or short options.

Almost nothing is a pure directional bet. Only the outright option, the **vertical spreads** of Section 12.2 (spreads whose legs share an expiration and differ only in strike), and the risk reversal have direction as their dominant exposure. Everything else requires the trader to be right about magnitude or timing as well, and a strategy chosen for its payoff shape frequently embeds a volatility view its holder never formed.

The neutral strategies are all short a sharp move. Butterflies, condors, and calendars all profit from stillness, which means they all lose in a sharp move, even the calendar, which Table 12.2 lists as "term" because its far-dated option gains when implied volatility rises. A

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Table 12.2: Exposures of the main structures, with the net premium and the maximum loss per share at the quotes of Table 10.2 (the covered call writes the 50 call; the protective put buys the 45 put).

Strategy	Direction	Volatility	Time	Net premium	Maximum loss
Covered call	mildly up	short	+	credit 1.44	46.86
Protective put	up	long	−	debit 1.89	5.19
Bull call spread	up	mixed	mixed	debit 2.04	2.04
Bear put spread	down	mixed	mixed	debit 2.14	2.14
Butterfly	neutral	short	+	debit 0.77	0.77
Condor	neutral	short	+	debit 1.70	1.70
Straddle	none	long	−	debit 5.35	5.35
Strangle	none	long	−	debit 3.40	3.40
Calendar spread	neutral	term	+	debit	the debit
Risk reversal	up	skew	mixed	credit 0.30	44.70
Ratio spread	mildly up	short	+	debit 1.36	unlimited

portfolio of "diversified" neutral strategies is a concentrated short-volatility position, and it will be discovered to be one on the day it matters.

The table also says whose book each structure ends up on. Vertical spreads belong to directional traders who want a view at a known maximum loss, which is also what the position is margined at (Section 12.14). Straddles and strangles are bought around earnings and central-bank meetings by event traders, and delta-hedged by volatility funds; market makers are usually on the other side. Butterflies and condors, which profit from stillness, suit traders who expect a quiet market, and premium-selling funds run condors in the iron form of Section 12.3. Calendars and diagonals are used by term-structure traders and by dealers managing the maturity profile of their volatility exposure, selling options in the nearest expiration at elevated implied volatility against later ones. Ratio spreads are mostly professional: a dealer or hedge fund uses one to reduce the cost of a directional view, and a put ratio spread sells the low-strike puts whose implied volatility the skew has raised.

12.12 STRUCTURED PRODUCTS ARE PACKAGED STRATEGIES

Most people who own an option strategy did not choose it. They bought a product, and the product is a strategy with a wrapper, a fee, and a name.

A principal-protected note promises the return of capital plus participation in an index. It is a zero-coupon bond plus a call.

A reverse convertible pays a high coupon and repays principal in shares if the stock finishes below its initial level (in the common barrier form, only if it has also traded through a lower barrier). It is a bond plus a written put: the high coupon is the put premium, as Example 12.8 shows.

An autocallable pays a coupon and redeems early if the index is at or above a level on an observation date, and can return less than principal if it breaches a lower one. It is a written **down-and-in put** (a put that comes into existence only if the index falls through the lower level, the barrier) plus a series of digital coupons. The Chinese "snowball" products of Section 35.7 are exactly this structure.

Example 12.7 Taking apart a principal-protected note

A five-year note returns 100% of principal at maturity plus α times any rise in an index. What can α be?

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The issuer must set aside enough to repay principal. At a five-year rate of 4.5%,

$$\text{zero-coupon bond} = 100\,e^{-0.045\times 5} = 100\times 0.798516 = \$79.85$$

leaving \$20.15 per \$100 to buy call options. If a five-year at-the-money index call costs 25% of notional,

$$\alpha = \frac{20.15}{25} = 80.6\%$$

Subtract a 2% distribution fee and the budget falls to \$18.15, giving $\alpha = 72.6\%$. The 2% fee cost the investor 8 percentage points of participation, because it is taken out of the \$20.15 option budget, a tenth of it, rather than out of the whole note.

Now vary the rate, holding the call's cost at 25%. At a 1% rate the bond costs $100e^{-0.05} =$ \$95.12, leaving \$4.88, and after the fee \$2.88, for participation of 11.5%. At a 5% rate the bond costs \$77.88, leaving \$22.12, and after the fee \$20.12, for participation of 80.5%.

This explains an entire product cycle. Principal-protected notes were nearly impossible to build attractively through the years of near-zero interest rates, and higher interest rates make them attractive to build again. No change in investor preferences is needed to explain it. The option budget is what is left after the bond, and when the bond is expensive there is nothing left.

Example 12.8 Where a 12% coupon comes from

A one-year reverse convertible on a stock at \$100 pays a 12% coupon. If the stock finishes at or above \$100 the investor receives \$100; if below, they receive the shares instead. The one-year rate is 4.5% and a one-year at-the-money put on the stock is worth \$9.00.

Decompose what the investor owns. They hold a bond, they receive the coupon, and they have written a put struck at \$100:

$$\begin{aligned}\text{bond} &= 100\,e^{-0.045} = \$95.60\\ \text{coupon} &= 12\,e^{-0.045} = \$11.47\\ \text{written put} &= -\$9.00\\ \text{total} &= \$98.07\end{aligned}$$

The investor pays \$100 for a package worth \$98.07. The \$1.93 difference is the fee, and it is not disclosed as one.

More important is where the 12% comes from. It is not compensation for credit risk and has nothing to do with the issuer's creditworthiness beyond the bond leg. Of the coupon, roughly 4.5 points are the risk-free rate and the remainder is the premium on a put the investor has sold, less the fee just computed. The high coupon is insurance income, and the risk being insured is exactly the risk the investor is taking.

The product is therefore attractive in the same conditions as the covered call of Example 10.7, and unattractive in the same ones. It performs worst when the stock falls sharply, which is when the investor most wanted the \$100 back and instead receives shares worth considerably less. A useful test for any high-coupon structured product: work out what option the buyer has written, and ask whether they would have written it deliberately.

Case Study 12.1 The covered-call fund boom

Funds that systematically write calls against an equity portfolio (sold as "derivative income" or "buy-write" strategies) grew from a niche into one of the fastest-gathering categories in the ETF market during 2022–2024, with assets approaching \$100 billion by the end of 2024.

The appeal is a headline distribution yield far above anything available in equities or, for much of the period, in bonds. The mechanics of Section 12.1 explain where it comes

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from: the fund is short puts, and the yield is insurance premium. Chapter 19 shows the premium is on average positive, so this is a real source of return and not an illusion.

Two things are nevertheless misdescribed. First, the distribution is not income in the sense investors usually mean. It is partly the option premium and partly a return of the investor's own capital, and a fund can keep paying a high distribution while its net asset value falls, because whatever the premium and dividends do not cover is paid out of the fund's own assets. Second, and more important, the strategy gives up the strong markets (Example 10.7). A fund that caps the upside every month will, over a long horizon containing a few large rallies, badly trail the index it writes against—while having delivered exactly the yield it advertised throughout.

The fund has no maturity and promises nothing about what an investor ends up with: it writes new calls every month indefinitely, and its net asset value floats with the index, less the upside it sells and the capital it pays out. These products convert an uncertain total return into a higher, steadier distribution and a lower expected terminal value. For an investor who needs cash flow and can accept that trade, they do something useful. For an investor who reads the distribution yield as income without cost, they are a slow disappointment, and the disappointment arrives only after several years.

12.13 MANAGING A POSITION AFTER ENTRY

Every profit diagram in this chapter describes a position held to expiration. Most are not, and the decisions taken in between determine the outcome at least as much as the structure chosen at entry.

Rolling. A covered call writer whose stock has risen through the strike faces assignment and the loss of the shares. The standard response is to **roll**: buy back the written call and sell another at a higher strike or a later date. Usually this costs money, since the call being repurchased is now in the money.

Rolling once is ordinary position management. Rolling repeatedly is something else. Each roll realizes a loss on the option leg and re-establishes a short call further out, so a writer who rolls to avoid assignment has converted a defined, capped strategy into an open-ended bet that the stock will stop rising—the one thing they have so far been wrong about. The strategy that was chosen for its conservatism becomes, through a sequence of individually reasonable adjustments, a persistent short call on a rising stock.

Repairing. A trader holding a long call on which the stock has fallen can sell a higher-strike call against it, converting the position into a spread and recovering part of the premium. The arithmetic is unkind: the stock has fallen, so the higher-strike call is now cheap, and the repair raises very little.

Pre-committing to exits. A stock position can be held indefinitely; an option position cannot. Time decay means "wait and see" is a decision with a price, and Table 12.2 gives its sign for each structure. The discipline that follows is to decide before entry what will cause the position to be closed (a target, a stop, or a date), because after entry the decay makes every delay look like a reason to keep waiting for the move that has not yet come.

Adjustments accumulate costs. Each adjustment adds legs, and Section 12.14 shows what legs cost. A butterfly adjusted twice has crossed eight more bid–ask spreads on top of the four at entry, and on the numbers of Example 12.3 the twelve together exceed the position's entire

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theoretical value. Frequent adjustment is not risk management; past a point it is a transfer to the market maker.

12.14 THE COSTS THAT DECIDE WHETHER A STRATEGY WORKS

Every profit diagram in this chapter is drawn as though the position could be established at a single price. It cannot, and the gap between the diagram and the outcome is usually decisive.

Each leg pays the bid–ask spread. Example 12.3's butterfly lost 38% of its value at entry, and a four-legged condor crosses four spreads. The number of legs multiplies the transaction cost, and the structures with the most attractive reward-to-risk ratios usually have the most legs.

Exit costs the same again. A butterfly held to expiration avoids the exit spread but accepts whatever the stock does. Closing early pays the spread a second time, which on the numbers above would consume three-quarters of the theoretical value.

Legging risk is real. The risk of Section 10.5, that the market moves between fills, is larger here: on a \$0.56 butterfly built from options worth \$4.425, \$2.67, and \$1.475, a move of about 50 cents in the underlying between fills (1% of the stock price) can exceed the entire position value.

Margin treatment varies. A spread whose loss is capped requires margin equal to that maximum loss; a ratio spread with uncovered short options requires the naked-writing margin of Section 10.8. Two structures with similar diagrams can have very different capital requirements, and the more attractive-looking one is often the more expensive to hold.

Short American legs can be assigned. The case study in Section 11.7 showed a supposedly riskless box destroyed by early assignment. A short call is most likely to be exercised just before an ex-dividend date, when the dividend exceeds its remaining time value (Example 11.7); a short put, once it is deep in the money (Section 11.9). Either way one leg disappears at a moment chosen by someone else: a bull call spread whose short call is assigned becomes a long call against short stock, a synthetic long put.

Put together, these determine which strategies are practical for whom. A market maker with negligible transaction costs, the portfolio margining of Section 10.8 (margin set on the net risk of the whole account rather than leg by leg), and the ability to trade packages can hold structures that are not practical for a retail trader crossing full spreads on every leg. The correct strategy depends on who is trading it—a question textbooks usually leave implied.

SUMMARY

Every option position is a bet on some combination of direction, magnitude, time, and skew. Choosing a strategy means matching those exposures to a view actually held.

Put–call parity collapses much of the apparent variety. A covered call is a written put plus cash; a protective put is a purchased call plus cash. Where two routes to one exposure exist, the one with fewer option legs is usually cheaper to trade.

Spreads cap the payoff in exchange for a lower cost and are bets on a range rather than on a direction. Butterflies and condors bet on stillness. Straddles and strangles bet on magnitude—but because both the cost and the required move come from the same implied volatility, a straddle is a bet that realized volatility will exceed implied, not that the stock will move.

Calendar spreads bet on time decay differentials and on the term structure of volatility. Risk reversals bet on skew, and are the same structure as the corporate collar.

Strategies that are neutral on direction are almost all short volatility, so a portfolio of them is concentrated rather than diversified.

Any piecewise-linear payoff can be replicated by a bond, a forward, and calls, with weights equal to the change in slope at each kink. A digital is the limit of a tight spread, priced at $-\partial c/\partial K$, and that is how dealers hedge it.

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Structured products are packaged strategies: a principal-protected note is a bond plus a call, a reverse convertible is a bond minus a put, an autocallable is a written barrier put plus digital coupons. Because the option budget is whatever remains after buying the bond, participation rates depend sharply on the level of interest rates, which is why principal-protected notes became scarce in the years of near-zero interest rates.

Transaction costs decide which strategies are practical. Paying the bid–ask spread on every leg can consume a large share of a butterfly's mid-price value, and structures with the best reward-to-risk ratios generally have the most legs and therefore the highest costs. Margin treatment, legging risk, and early assignment on American legs complete the picture, and together they mean the right strategy depends on who is trading it.

FURTHER READING

The OCC's *Characteristics and Risks of Standardized Options* and McMillan's *Options as a Strategic Investment* set out the standard structures and their margin treatment, which changes more often than textbooks do. On covered calls, Whaley (2002) built the buy-write index and reports its first long-horizon record, and Israelov and Nielsen (2015) decompose the strategy into the equity, short-volatility and timing exposures this chapter identifies. Henderson and Pearson (2011) take a retail structured note apart into a bond and options, as Example 12.8 does, and find it priced about 8% above its value; it is the model for doing the exercise on any term sheet. Carr and Madan (2001) state the replication result of Section 12.8 in general, and Breeden and Litzenberger (1978) take it to the butterfly of vanishing width and the risk-neutral density.

PRACTICE QUESTIONS

12.1 Show, using put–call parity, that a covered call has the same profit profile as a written put. State one respect in which the two positions nevertheless differ in practice.

12.2 Using Table 10.2, construct a bear put spread from the 47.50 and 45.00 puts at the quoted bid and ask. Compute the debit, the maximum profit, the maximum loss, and the breakeven.

12.3 Using Table 10.2, price the 47.50/50/52.50 call butterfly at the quoted bid and ask and at mid prices. Compute what execution costs as a percentage of the mid value, both breakevens at the quotes, and where the profitable range lies relative to the \$48.30 stock price.

12.4 Explain why a strangle that costs 36% less than a straddle can require a larger move to break even. What does this imply about "cheap" option strategies generally?

12.5 A five-year principal-protected note is offered when the five-year rate is 3% and a five-year at-the-money call costs 22% of notional. Compute the participation rate with no fee and with a 1.5% fee.

12.6 Identify what each of the following is a bet on, using the four things to bet on named in the chapter's introduction: (a) a calendar spread; (b) a risk reversal; (c) a condor; (d) a covered call.

12.7 Explain why a portfolio containing butterflies, condors, and calendar spreads is not diversified.

12.8 A trader establishes a four-legged condor by executing each leg separately. Describe two distinct ways this can cost more than expected, and state how each is avoided.

12.9 A calendar spread sells the near-dated option and buys the longer-dated one at the same

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strike. Explain why this is neither a directional trade nor straightforwardly a volatility trade, identify the two effects that determine its profit, and state the move in the underlying that hurts it most.

12.10 A risk reversal buys an out-of-the-money call and sells an out-of-the-money put. Explain why its cost is a direct reading of the skew rather than of the level of volatility, and state what a risk reversal that costs nothing to enter tells you about the two implied volatilities.

12.11 A founder holds a large single-stock position and wants to reduce the risk without selling. Describe the collar that achieves this, explain in whose favor the skew works when the collar is on an equity, and state what the shareholder remains exposed to that a sale would have removed.

12.12 A manager hedges an equity portfolio with index puts rather than index futures. State the beta-adjusted contract count, explain what shape the option hedge buys that the futures hedge does not, and explain why the floor it sets on the portfolio is only approximate even when the beta is correct.

12.13 A producer can hedge by selling futures or by buying puts. Compare the two on cost at inception, retained upside, and cash demands over the life of the hedge, and identify which consideration the textbook comparison usually omits.

FURTHER QUESTIONS

12.14 Example 12.3 found that execution consumed 38% of a butterfly's theoretical value. Derive a general expression for the fraction of a multi-leg strategy's value consumed by spreads, in terms of the number of legs and the proportional spread on each. Then use it to explain why market makers can trade structures retail traders cannot.

12.15 A fund writes one-month at-the-money calls against an equity index every month, collecting a premium of 1.5% of the index each time. Construct the sequence of monthly outcomes over a period containing eleven months of +1% and one month of +15%, and compare the fund's total return with the index's. Then explain why a backtest beginning in a calm period will overstate the strategy's appeal.

12.16 Section 12.12 decomposes an autocallable into a written barrier put plus digital coupons. Using that decomposition, explain what happens to the issuer's hedge as the index approaches the knock-in barrier, and connect your answer to the 2024 episode described in Section 35.7.

12.17 A corporate treasurer is offered a zero-cost collar on a large shareholding. Identify what the treasurer is selling, what determines whether the collar can genuinely be struck at zero cost, and under what market conditions the terms will be most attractive.

12.18 Table 12.2 lists the calendar spread as "term" rather than long or short volatility. Explain precisely what exposure this means, construct a case in which implied volatility rises across all maturities and the calendar spread nevertheless loses money, and state what the position is really long.

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Part III

Option Valuation

The valuation engine, built twice. Chapter 13 builds it discretely, where the argument is visible; Chapters 14 and 15 build it in continuous time and arrive at the same answer. What follows extends the result to other underlyings, to the hedging problem, to the smile the model cannot produce, and to the numerical methods used when no formula exists. Chapter 14 carries the dagger: a reader who wants the Black–Scholes–Merton formula and its use rather than its derivation can pass from Chapter 13 straight to Chapter 15.

13 Binomial Trees
replication and risk-neutral valuation, in the case where you can see them

14 Wiener Processes and Itô's Lemma
the continuous-time machinery behind the formula

15 The Black–Scholes–Merton Model
the equation, the formula, and the assumptions that carry them

16 Options on Indices, Currencies, and Futures
one formula, adapted to three underlyings

17 Employee Stock Options and Equity Compensation
grant-date fair value, vesting, and dilution

18 The Greek Letters
the sensitivities, and what a hedged book costs to run

19 Volatility Smiles and Surfaces
the evidence that one volatility is not enough

20 Numerical Procedures
trees, finite differences, and Monte Carlo, including early exercise

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Chapter 13
Binomial Trees

Chapters 10 to 12 described what options are, what relationships they must satisfy, and how they are combined. None of them priced one. Chapter 11 established bounds, and bounds are not a price.

This chapter supplies the missing element, and it does so with almost no mathematics.

The tree here is a teaching device before it is a computational method. It is used because it is the simplest possible market in which the entire argument of option pricing can be seen at once (one period, two states, two traded instruments), and because in that setting every step can be checked by hand. How to build trees efficiently, how many steps they need, and how to make them converge quickly are questions about computation, and they belong to Chapter 20.

What the device makes visible is the point of it: **replication**, **no arbitrage**, and **risk-neutral valuation**: the argument by which a derivative acquires a price at all. That argument is the foundation of everything in Part III, and it is genuinely difficult to motivate once the machinery of continuous time is in the way. Here it fits on a page.

Sections 13.1 to 13.4 are the argument in its smallest setting: one step, risk-neutral valuation, state prices, and why two states suffice. Sections 13.5 to 13.7 extend it to many steps, to the general form, and to American exercise. Sections 13.8 to 13.10 are the choices that connect the tree to a real underlying (the up and down moves, convergence, and dividends), and Sections 13.11 and 13.12 close with the identity of pricing and hedging and with what the tree assumes.

Later chapters depend on this one. Chapters 15 and 16 take risk-neutral valuation as given; Chapter 18 builds on the identity of pricing and hedging established in Section 13.11; Chapter 19 works in the incomplete markets that Section 13.4 introduces.

13.1 A ONE-STEP BINOMIAL MODEL

Suppose a stock trades at \$20 and that in three months it will be worth either \$22 or \$18. Nothing else can happen. We want the value f of a European call struck at \$21, which pays \$1 in the up state and nothing in the down state.

Figure 13.1 carries both on one tree (the stock's two possible prices above, the call's two possible values below), and the rest of this chapter shows how the stock's two prices force the call's value today.

The method is to construct a portfolio whose value is the same in both states. Hold Δ shares and write one call. In the up state it is worth $22\Delta - 1$; in the down state, 18Δ. These are equal when

$$22\Delta - 1 = 18\Delta \quad \Longrightarrow \quad \Delta = 0.25$$

At $\Delta = 0.25$ the portfolio is worth $18 \times 0.25 = \$4.50$ in *either* state. It is riskless. A riskless portfolio must earn the risk-free rate, or there is an arbitrage, so its value today is $4.50\, e^{-rT}$. With $r = 4\%$ and $T = 0.25$,

$$4.50\, e^{-0.04\times 0.25} = 4.50\, e^{-0.01} = \$4.455$$

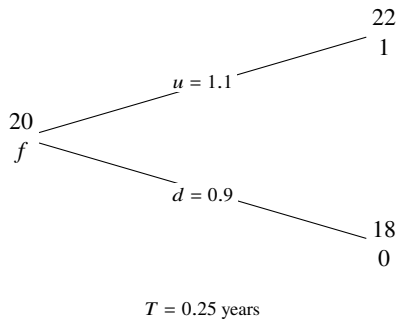


Figure 13.1: Stock price above, call value below. $K = 21$.

The portfolio today costs $20\Delta - f = 5 - f$, so

$$5 - f = 4.455 \quad \Longrightarrow \quad f = \$0.545$$

Read the same construction the other way and it is a recipe. Holding 0.25 shares and owing \$4.455 is worth $0.25 \times 22 - 4.50 = \$1$ in the up state and $0.25 \times 18 - 4.50 = \$0$ in the down state: it *replicates* the call. Its cost is $5 - 4.455 = \$0.545$, and that is the price: the call is worth what it costs to manufacture. Example 13.1 shows what happens if it trades at any other price.

Example 13.1 Checking the arbitrage

Suppose the call traded at \$0.60 rather than \$0.545. Write the call, buy 0.25 shares:

$$\text{cash out} = 0.25 \times 20 - 0.60 = \$4.40$$

Borrow the \$4.40 at 4%. In three months repay $4.40\, e^{0.01} = \$4.444$, and the portfolio is worth \$4.50 whichever way the stock moves. The profit is \$0.056 in every state, from nothing invested—and \$0.056 is $(0.60 - 0.545)e^{0.01}$, the mispricing carried forward, as in Example 11.1.

If instead the call traded at \$0.50, run the trade in reverse: buy the call, short 0.25 shares, invest the proceeds.

Notice what never entered the calculation: no one was asked how likely the up state is, or what return investors require for bearing risk. The argument works for a trader who thinks the stock will certainly rise and for one who thinks it will certainly fall. That is the feature to hold on to, because Section 13.2 shows it is not an accident of this example.

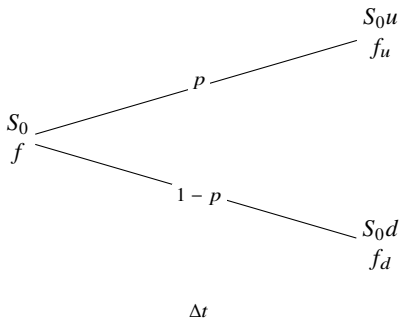


Figure 13.2: The general one-step tree. Stock price above, derivative value below; p and $1 - p$ are the risk-neutral probabilities of (13.2).

The general one-step formula. Write the stock as moving from S_0 to S_0u or S_0d over a step of length Δt, with the derivative paying f_u or f_d, as in Figure 13.2. The same construction

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gives the number of shares held per option, the hedge ratio known as the option's delta,

$$\Delta = \frac{f_u - f_d}{S_0 u - S_0 d} \tag{13.1}$$

and, after substituting and rearranging,

$$\boxed{f = e^{-r\,\Delta t}\,[p f_u + (1-p) f_d]} \qquad \text{where} \qquad p = \frac{e^{r\,\Delta t} - d}{u - d} \tag{13.2}$$

For no arbitrage we need $d < e^{r\Delta t} < u$; otherwise the stock always beats or always loses to cash. That condition also guarantees $0 < p < 1$.

Check the example:

$$p = \frac{e^{0.04\times 0.25} - 0.9}{1.1 - 0.9} = 0.5503,$$
$$f = e^{-0.04\times 0.25}\,[0.5503 \times 1 + 0.4497 \times 0] = \$0.545,$$

as before. Here and throughout, intermediate values are carried unrounded and printed to four places, so the last digit of a product can differ from the product of the printed factors.

13.2 RISK-NEUTRAL VALUATION

Equation (13.2) has the form of a discounted expected payoff, with p playing the role of a probability. What p is needs care, because most of the confusion in derivative pricing starts here.

What p is. Compute the expected stock price using p:

$$\mathbb{E}[S_T] = pS_0 u + (1-p)S_0 d = S_0\,[pu + (1-p)d] = S_0 e^{r\Delta t}$$

(the last step is algebra: substitute p and simplify). So p is the probability under which the stock is expected to earn the risk-free rate. It is called the **risk-neutral probability**, and the corresponding expectation is taken in the **risk-neutral world**.

What p is not. It is not the real-world probability of the stock rising, and it is not anyone's forecast. In the example $p = 0.5503$; the true probability might be 0.7 or 0.3, and the option price would be \$0.545 either way. This is exact, not an approximation: the true probability is irrelevant.

Why this works. The reason is not that anyone is risk-neutral. It is that the option can be *replicated*: Section 13.1 built a portfolio of stock and cash that reproduces the option's payoff exactly. Two things with identical payoffs must have identical prices, whatever anyone's attitude to risk. Attitudes to risk do set the interest rate and the price of the underlying asset, but once those two prices are given they have no further say in the option's. Preferences and real-world probabilities drop out of the problem, and the only assignment of probabilities that survives (the only *measure*, in the language of Part III) is the one under which no risk premium is required, which is p.

The result generalizes into the principle underlying all of Part III:

> **Risk-neutral valuation.** The value of a derivative is the expected value of its payoff in a world where all assets earn the risk-free rate, discounted at the risk-free rate.

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The risk-neutral probability is a tool for calculating derivative prices, not a description of the world. We do not believe investors are risk-neutral. We use the risk-neutral measure because it gives the right answer to a question about relative prices, and it gives it far more easily than the alternative. Chapter 15 shows the same principle at work in continuous time; Chapter 28 shows what happens when one changes to a different measure entirely. Example 13.2 shows what the alternative involves.

Example 13.2 The real-world calculation, for comparison

Suppose the true probability of the up state is 0.7. Then the expected stock price is $0.7\times22+0.3\times18 = \$20.80$, an expected return of

$$\frac{20.80}{20} - 1 = 4\% \text{ over three months}$$

The expected call payoff is $0.7 \times 1 + 0.3 \times 0 = \$0.70$. To discount it we need the return investors require *on the option*, which is not 4%: the option is a leveraged position in the stock and is far riskier.

That discount rate cannot be looked up. It depends on how leveraged the option is, and the leverage depends on the option's price, which is what we were trying to find. The real-world approach is circular unless a preference model is supplied from outside.

Run backward from the known answer, and the circle closes. Discounting \$0.70 back to the known price \$0.545 implies a required return of $0.70/0.5448 - 1 = 28.5\%$ per quarter, an enormous risk premium, correctly reflecting how leveraged the option is. Risk-neutral valuation sidesteps this entirely. The required return on the option, which no one can observe, never has to be found: the replication argument makes the risk premium cancel out, and what remains is the risk-free rate, which everyone can observe.

13.3 STATE PRICES AND THE ABSENCE OF ARBITRAGE

Equation (13.2) can be read a second way, and the second reading connects this chapter to Chapters 11 and 12 and states the principle in its most general form.

Group the discount factor with the probabilities:

$$f = \underbrace{e^{-r\Delta t}p}_{\psi_u} f_u + \underbrace{e^{-r\Delta t}(1-p)}_{\psi_d} f_d \tag{13.3}$$

The two quantities ψ_u and ψ_d are **state prices**. Each is the price, today, of a security that pays \$1 if a particular state occurs and nothing otherwise. Such a security is called an **Arrow–Debreu security**, and once the price of each is known, *every* derivative on this underlying can be priced by multiplying its payoffs by the corresponding state prices and adding.

In the running example,

$$\psi_u = e^{-0.04\times0.25} \times 0.5503 = 0.5448,$$
$$\psi_d = e^{-0.04\times0.25} \times 0.4497 = 0.4453,$$

and Example 13.3 confirms both by replication.

Example 13.3 Building and using state prices

Verify that $\psi_u = 0.5448$ really is the price of \$1 in the up state. Replicate that payoff with Δ shares and B in cash:

$$22\Delta + Be^{0.04\times0.25} = 1, \qquad 18\Delta + Be^{0.04\times0.25} = 0$$

Subtracting gives $\Delta = (1-0)/(22-18) = 0.25$, which is (13.1) with the payoffs 1 and 0, and then

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$Be^{0.04\times0.25} = -4.50$, so $B = -\$4.4552$. The cost today is

$$0.25 \times 20 - 4.4552 = \$0.5448$$

which is ψ_u, as claimed. In this example the security paying \$1 up and \$0 down *is* the call struck at 21, which is why $f = \psi_u$. Similarly, replicating \$1 in the down state takes $\Delta = -0.25$ and $B = \$5.4453$, so $\psi_d = -0.25 \times 20 + 5.4453 = \$0.4453$.

Now price something else with no further work. A claim paying \$3 in the up state and \$7 in the down state is worth

$$3 \times 0.5448 + 7 \times 0.4453 = 1.6343 + 3.1169 = \$4.7512$$

Checking by replication, (13.1) gives $\Delta = (3-7)/(22-18) = -1$, then $B = 25e^{-0.04\times0.25} = \$24.7512$, and the cost is $-20 + 24.7512 = \$4.7512$.

Two state prices value every derivative on this stock: the replication is done once, and every payoff afterward is a weighted sum.

Note also that $\psi_u + \psi_d = e^{-r\Delta t} = 0.99005$. A claim paying \$1 in *both* states is a riskless bond, so the state prices must sum to the discount factor, which is where the discounting in (13.2) comes from, and why p and $1-p$ behave like probabilities: they are state prices *normalized* by the bond price.

The arbitrage-free condition. State prices make precise what "no arbitrage" requires. A security paying \$1 in one state and nothing otherwise cannot have a negative price (the buyer would be paid to take a payoff that is never negative), and it cannot be free, since it sometimes pays. So both state prices must be strictly positive, which given $\psi_u + \psi_d = e^{-r\Delta t}$ is exactly the condition

$$0 < p < 1 \quad \Longleftrightarrow \quad d < e^{r\Delta t} < u$$

met earlier in Section 13.1.

This is a special case of a general result:

> **The market admits no arbitrage if and only if a set of strictly positive state prices exists**; equivalently, if and only if there is a risk-neutral measure, a probability measure that gives every state a positive probability and under which every asset's expected return is the risk-free rate.

Read in that direction, risk-neutral valuation stops looking like a trick. The risk-neutral measure is not an assumption added to the model; it is another name for the absence of arbitrage. A market with no arbitrage always has such a measure, and a market with an arbitrage never does.

The connection to Chapter 12. Section 12.8 built a digital option as the limit of a tight call spread. Push that construction one derivative further: a butterfly of vanishing width centered at K pays only if the underlying finishes at K, and its price, divided by the square of its width, tends to $\partial^2 c/\partial K^2$. A butterfly of that kind *is* an Arrow–Debreu security, and its price *is* a state price. The risk-neutral density of the terminal price, which Chapter 19 reads off a continuum of traded option prices, and the state prices of this section are the same object seen from two directions, one observed in the market, the other constructed from a replication argument. Chapter 19 uses the first; most of the rest of Part III uses the second.

13.4 WHY TWO STATES? COMPLETENESS

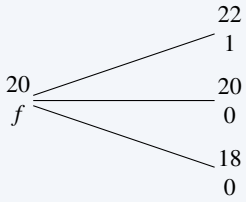
An obvious objection has been left standing since Section 13.1: a stock does not have two possible prices in three months. Section 13.9 answers part of that by letting the steps shrink, but the deeper answer is not about realism at all, and it is the most important idea in this chapter.

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The two-state assumption is what makes the market **complete**: every payoff can be replicated by trading the instruments available. There are two traded instruments (the stock and the bond) and two states, so the replicating equations are two equations in two unknowns and have a unique solution. *Every* payoff can therefore be manufactured, so every payoff has a price. Add a third state and that stops being true, as Example 13.4 shows.

Example 13.4 Three states, and the price is no longer unique

Suppose the stock at \$20 can move to \$22, \$20, or \$18. Everything else is as before: $r = 4\%$, $\Delta t = 0.25$, and a call struck at \$21 paying \$1, \$0, \$0.



Try to replicate it with Δ shares and B in cash:

$$22\Delta + Be^{0.01} = 1, \qquad 20\Delta + Be^{0.01} = 0, \qquad 18\Delta + Be^{0.01} = 0$$

The last two give $2\Delta = 0$, so $\Delta = 0$ and then $B = 0$, which contradicts the first. The call cannot be replicated, and the argument that produced a price collapses.

What survives? The state prices must still be positive and must still price the two traded instruments correctly:

$$22\psi_1 + 20\psi_2 + 18\psi_3 = 20, \qquad \psi_1 + \psi_2 + \psi_3 = e^{-0.01} = 0.99005$$

Two equations in three unknowns have a whole family of solutions rather than one. Eliminating ψ_2 gives $\psi_1 = \psi_3 + 0.0995$, and requiring $\psi_2 > 0$ and $\psi_3 > 0$ confines ψ_3 to the open interval $(0,\ 0.445274)$.

The call's value is ψ_1, so it can be anything in

$$\$0.0995 \ < \ f \ < \ \$0.5448$$

Every price in that interval is consistent with no arbitrage. The theory no longer delivers a price—it delivers bounds, and the upper one is more than five times the lower.

Note what sits at the two ends. As ψ_3 approaches 0.445274, ψ_2 goes to zero, the middle state is priced at nothing, and the value approaches \$0.5448, the binomial answer of Section 13.1. As ψ_3 approaches zero instead, the down state is priced at nothing and the value approaches \$0.0995, the binomial answer for a tree whose two moves are to \$22 and \$20. Each bound is a two-state model, the limiting case of the three-state one.

This example is worth dwelling on because it identifies precisely what the binomial assumption is doing. It is not a claim about how stocks behave. It is the assumption that makes the number of states equal the number of traded instruments, and that equality, not realism, is what converts Chapter 11's bounds into a price.

The general statement is the second half of the result quoted above:

> **The market admits no arbitrage and is complete if and only if there is exactly one risk-neutral measure**; equivalently, if and only if there is exactly one set of strictly positive state prices. Where it is unique, every derivative has one arbitrage-free price. Where it is not, derivatives have a range of arbitrage-free prices, and selecting one requires something beyond arbitrage: a model, or a view about preferences.

This is not a technicality confined to contrived examples. Real markets are incomplete for exactly this reason: prices jump, volatility moves unpredictably, and the number of things that can happen exceeds the number of instruments available to hedge them. The consequence

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appears throughout the rest of the book. It is why Chapter 19's volatility smile (options at different strikes implying different volatilities for the same underlying) cannot be argued away, why Chapter 34's exotic payoffs carry model risk that vanilla options do not, and why two desks using different models can quote different prices for the same structure without either committing an arbitrage.

Completeness is the assumption that lets this chapter say "the price." Everywhere it fails, the accurate phrase is "a price, given a model."

13.5 TWO-STEP TREES AND BACKWARD INDUCTION

With more than one period, apply (13.2) repeatedly, working backward from expiration one step at a time, the procedure called **backward induction**. Figure 13.3 extends the example to two three-month steps.

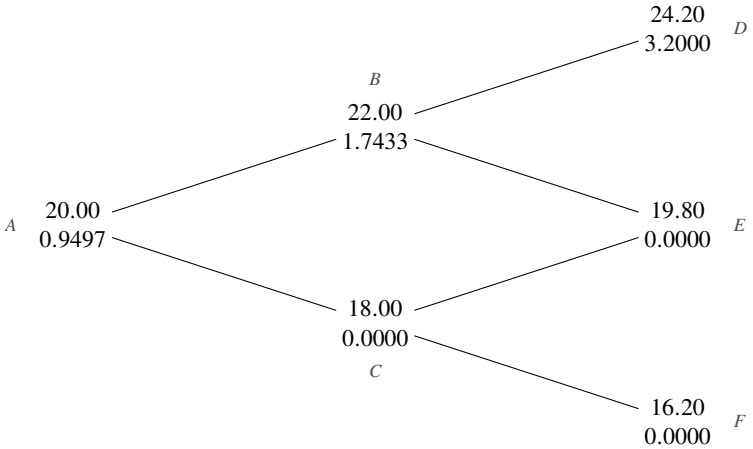


Figure 13.3: Stock price above, European call value below. $K = 21$, $r = 4\%$.

At the terminal nodes D, E, and F the call is worth $\max(S_T - 21, 0)$: \$3.20, \$0, \$0. Because u, d, and r are the same at every step, $p = 0.5503$ is the same at every node too. Working back with it and $e^{-r\Delta t} = 0.99005$:

$$\begin{aligned}
\text{node B}: &\quad e^{-0.01}\,[0.5503 \times 3.20 + 0.4497 \times 0] = \$1.7433 \\
\text{node C}: &\quad e^{-0.01}\,[0.5503 \times 0 + 0.4497 \times 0] = \$0 \\
\text{node A}: &\quad e^{-0.01}\,[0.5503 \times 1.7433 + 0.4497 \times 0] = \$0.9497
\end{aligned}$$

Two features of the procedure matter more than the arithmetic.

The tree recombines. An up-then-down move and a down-then-up move both give $S_0ud = 19.80$, so the number of nodes grows linearly rather than exponentially: $n + 1$ nodes after n steps rather than 2^n. This is what makes trees computationally practical, and Section 13.10 shows what happens when it fails.

The replication is dynamic. The riskless portfolio of Section 13.1 must be *rebalanced* at each node, because Δ from (13.1) differs at each one. At node B, $\Delta = (3.20 - 0)/(24.20 - 19.80) = 0.7273$; at node C, $\Delta = 0$. Pricing the option assumes the hedger can and does trade at every step. Section 13.12 returns to what that assumption costs.

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Example 13.5 Following the hedge through the tree

Take the tree of Figure 13.3 and follow a trader who writes the call and hedges it.

From (13.1) the deltas are $\Delta_A = (1.7433 - 0)/(22 - 18) = 0.4358$ at node A, $\Delta_B = (3.20 - 0)/(24.20 - 19.80) = 0.7273$ at node B, and $\Delta_C = 0$ at node C. The arithmetic below carries the deltas unrounded.

Today (node A). Sell the call for \$0.9497. Buy 0.4358 shares at \$20, costing \$8.7164. Borrow the difference:

$$8.7164 - 0.9497 = \$7.7667$$

Three months on. The debt has grown to $7.7667\, e^{0.01} = \$7.8448$. If the stock is at 22 (node B), the shares are worth $0.4358 \times 22 = \$9.5881$, so the position is worth

$$9.5881 - 7.8448 = \$1.7433$$

which is exactly the call's value at node B. If instead the stock is at 18 (node C), the shares are worth \$7.8448 and the position is worth precisely nothing, again the call's value there.

Rebalancing at node B. Delta must rise from 0.4358 to 0.7273, so buy 0.2915 more shares at \$22, costing \$6.4119, financed by borrowing. Total debt becomes $7.8448 + 6.4119 = \$14.2567$.

At expiration. The debt grows to $14.2567\, e^{0.01} = \$14.4000$. If the stock reaches 24.20, the shares are worth $0.7273 \times 24.20 = \$17.6000$ and the position is worth

$$17.6000 - 14.4000 = \$3.2000$$

which is exactly the call's payoff. If the stock falls to 19.80, the shares are worth \$14.4000, the position is worth zero, and the call expires worthless.

The hedge closes to the penny in every state, having been financed entirely by the \$0.9497 received at the outset. That is what it means for \$0.9497 to be the price—it is the cost of manufacturing the option's payoff.

Two things in the walkthrough deserve emphasis. The trader bought more shares at 22 and would have sold shares at 18: the hedge *buys high and sells low* by construction, and the accumulated cost of doing so is precisely the option premium. And nothing was ever at risk (no step required a view), which is the replication argument made operational rather than asserted.

13.6 THE MODEL IN GENERAL FORM

The examples have been worked one node at a time. Set out formally, the model is compact, and its structure becomes visible in a way node-by-node arithmetic hides.

The lattice. Divide the derivative's life, T years, into n steps of length $\Delta t = T/n$. The underlying starts at S_0 and at each step is multiplied by u or by d. After i steps it has taken j up moves and $i - j$ down moves, so it stands at

$$S_{i,j} = S_0\, u^j d^{\,i-j}, \qquad j = 0, 1, \ldots, i \tag{13.4}$$

Because u and d multiply, only the count j matters, not the order of the moves. That is the recombining property, and it is why step i has $i + 1$ nodes rather than 2^i.

The measure. Under the risk-neutral measure Q, each step is an up move with probability p from (13.2), independently of the others. The number of up moves in n steps is therefore *binomially distributed*, which is the model's name:

$$Q(j \text{ up moves}) = \binom{n}{j} p^j (1 - p)^{n-j}$$

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The pricing formula. For a European claim paying $h(S_T)$ at maturity, backward induction through the whole tree collapses to a single sum:

$$f_0 = e^{-rT} \sum_{j=0}^{n} \binom{n}{j} p^j (1-p)^{n-j}\, h\big(S_0 u^j d^{n-j}\big) \tag{13.5}$$

and for a European call, $h(S) = \max(S - K, 0)$, so the terms with $S_0 u^j d^{n-j} \le K$ vanish and the sum runs only over the nodes finishing in the money.

Equation (13.5) is the two-step calculation of Section 13.5 written out in one line. Applying it there with $n = 2$ reproduces \$0.9497 directly, without the intermediate node values, though those values are exactly what an American option needs, which is why Section 13.7 cannot use this shortcut.

The martingale property. There is a more revealing way to state what Q does. Under it, the *discounted* price of any traded asset is a **martingale**: its expected future value, discounted, equals its value today:

$$e^{-r\Delta t}\, \mathbb{E}^Q[S_{i+1} \mid S_i] = S_i, \qquad e^{-r\Delta t}\, \mathbb{E}^Q[f_{i+1} \mid S_i] = f_i \tag{13.6}$$

The first equality is the calculation of Section 13.2 restated; the second is equation (13.2) restated. That both hold under the same measure is the entire content of risk-neutral valuation, and it is the form in which the principle carries over to continuous time in Chapter 15 and to other numeraires (other choices of the asset in which prices are measured) in Chapter 28.

The continuous limit. With the parameter choice (13.8), Section 13.9 shows that as $n \to \infty$ the terminal distribution becomes lognormal (the logarithm of the price is normally distributed) and the sum (13.5) for a European call converges to the Black–Scholes–Merton formula, which Chapter 15 derives directly and which is the continuous-time counterpart of everything in this chapter.

13.7 AMERICAN OPTIONS

Trees remain in use because they handle early exercise, which Chapter 11 showed matters for puts and for calls on dividend-paying stocks, and which, outside a few special cases, no closed-form formula can address.

The modification is small. At each node, compute the value of continuing (equation (13.2) applied to the two successor nodes) and compare it with the value of exercising immediately. The node's value is the greater:

$$f = \max\Big(\underbrace{e^{-r\Delta t}\,[p f_u + (1-p) f_d]}_{\text{continue}},\ \underbrace{\text{intrinsic value}}_{\text{exercise}}\Big) \tag{13.7}$$

Example 13.6 applies the comparison to a two-step put. The same comparison handles any decision embedded in a security, including an issuer's right to call (redeem early) a convertible bond: each right is a max or a min at the nodes where it can be exercised, and because the rights interact node by node, the bond and its embedded options cannot be valued separately and added.

Example 13.6 An American put, and what early exercise is worth

A stock trades at \$50, with $u = 1.2$ and $d = 0.8$ over each of two one-year steps. Consider an American put struck at \$52, with $r = 5\%$.

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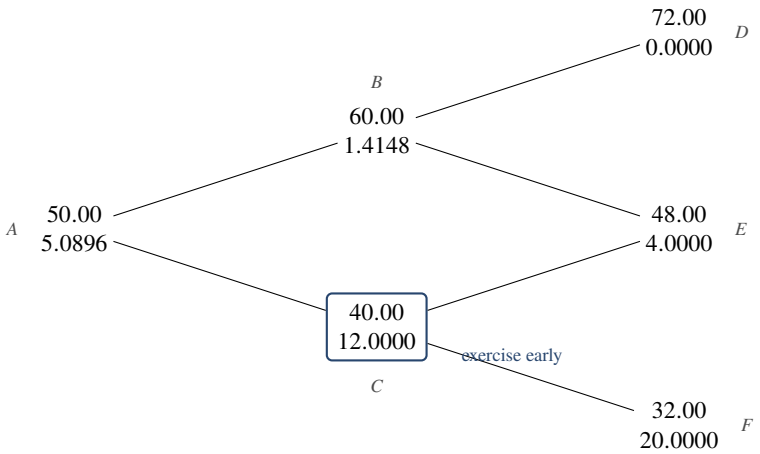


Figure 13.4: Stock price above, American put value below. $K = 52$, $r = 5\%$.

First, $e^{0.05\times 1} = 1.051271$ and

$$p = \frac{1.051271 - 0.8}{1.2 - 0.8} = \frac{0.251271}{0.4} = 0.6282$$

Figure 13.4 sets out the tree. At the terminal nodes D, E, and F the put is worth \$0, \$4, and \$20.

Node B ($S = 60$). Continuation value is

$$e^{-0.05}\,[0.6282 \times 0 + 0.3718 \times 4] = 0.951229 \times 1.4873 = \$1.4148$$

Exercising yields $\max(52 - 60, 0) = 0$. Hold. Value \$1.4148.

Node C ($S = 40$). Continuation value is

$$e^{-0.05}\,[0.6282 \times 4 + 0.3718 \times 20] = 0.951229 \times 9.9492 = \$9.4639$$

Exercising yields $52 - 40 = \$12$. **Exercise.** Value \$12.

Node A (today). Continuation value is

$$e^{-0.05}\,[0.6282 \times 1.4148 + 0.3718 \times 12] = 0.951229 \times 5.3506 = \$5.0896$$

Exercising yields $52 - 50 = \$2$. Hold. Value \$5.0896.

Now price the European put for comparison. The only difference is at node C, where the European put is worth its continuation value of \$9.4639 rather than \$12:

$$p_{\text{Eur}} = e^{-0.05}\,[0.6282 \times 1.4148 + 0.3718 \times 9.4639] = \$4.1927$$

The early-exercise premium is $5.0896 - 4.1927 = \$0.897$, more than 21% of the European value. Section 11.9 argued that an American put is worth more than its European counterpart; here is the gap, computed.

The tree also *locates* the exercise decision, which is its practical advantage. Exercise is optimal at node C and not at nodes A or B, so the critical price S^* of Figure 11.5 lies between 40 and 60 at that date, and below 50 today. Figure 13.5 traces it date by date on a 1,000-step tree with $\sigma = \ln(1.2/0.8)/2 = 20.3\%$, the volatility of the two-step tree; the curve $S^*(t)$ it draws is the put's **early-exercise boundary**. The boundary starts just above 40, rises to about 42 by the one-year date the two-step tree bracketed, and climbs the rest of the way to the strike only in the final weeks.

13.8 CHOOSING u AND d

So far u and d have been given. In practice they are chosen so that the tree reproduces the volatility of the underlying, which is the only parameter of the model that must be estimated.

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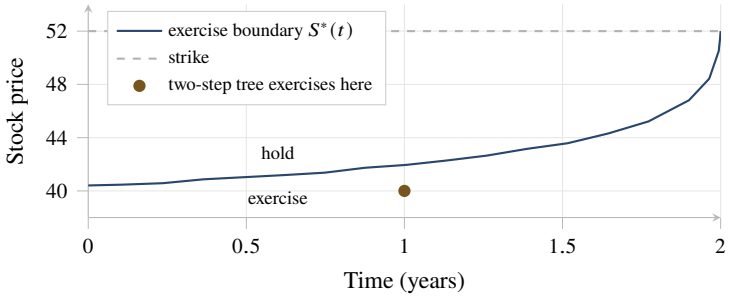


Figure 13.5: Early-exercise boundary of the Example 13.6 put on a 1,000-step tree, $\sigma = 20.3\%$.

The standard choice, due to Cox, Ross, and Rubinstein, is

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = \frac{1}{u} = e^{-\sigma\sqrt{\Delta t}} \tag{13.8}$$

where σ is the stock's **volatility**, the annualized standard deviation of its continuously compounded return, and Δt is the length of a step in years.

These match the variance of the stock's return over each step to $\sigma^2\Delta t$, to first order in Δt, which is the same variance the continuous-time model of Chapter 15 assumes. Setting $d = 1/u$ also centers the tree, so that an up move followed by a down move returns exactly to the starting price.

This leaves σ itself, the one input that cannot be read off a screen. It is the same number in the real world and in the risk-neutral one: moving from the true probability to p changes the tree's drift but, to first order in Δt, not its variance, so a volatility measured from historical prices can go into the tree unchanged. Estimating it, from history or by inverting a traded option's price, is taken up once, in Chapter 15; here σ is given, and Example 13.7 builds a tree from it.

Example 13.7 Building a tree from a volatility

A stock has volatility 30% a year. Build a tree with monthly steps, $\Delta t = 1/12$:

$$u = e^{0.30\sqrt{1/12}} = e^{0.086603} = 1.09046, \qquad d = 1/1.09046 = 0.91704$$

With $r = 4\%$,

$$p = \frac{e^{0.04/12} - 0.91704}{1.09046 - 0.91704} = \frac{1.003339 - 0.91704}{0.17342} = 0.49762$$

One sanity check is worth performing every time: $d < e^{r\Delta t} < u$, equivalently $0 < p < 1$. Here $0.917 < 1.0033 < 1.090$. It fails when the step is too long for the volatility and rate, and the symptom is a negative probability.

Note also how little p moved from one-half. Expanding $e^{r\Delta t}$, u, and d to order Δt gives

$$p \approx \frac{1}{2} + \frac{1}{2}\left(\frac{r - \sigma^2/2}{\sigma}\right)\sqrt{\Delta t} = \frac{1}{2} + \frac{1}{2} \times \frac{0.04 - 0.3^2/2}{0.30} \times 0.2887 = 0.4976,$$

the exact value to four places. So the risk-neutral probability is close to a coin flip and the drift enters only as a small correction. The volatility sets u and d, and with them how far the tree spreads at each step.

13.9 CONVERGENCE

A two-step tree is a crude description of a stock. The model is justified because refinement removes the crudeness: as the number of steps rises, the tree value settles down to a limit, the

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Black–Scholes–Merton value that Chapter 15 derives. Here the point is only that the limit exists, so that the arbitrary-looking choice of two branches does not determine the answer.

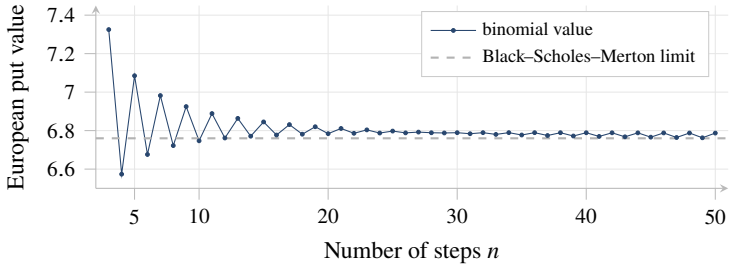


Figure 13.6: European put value against number of steps, $n \geq 3$. $S_0 = 50$, $K = 52$, $\sigma = 30\%$, $r = 5\%$, $T = 2$.

Figure 13.6 shows the characteristic behavior, and two features of it matter in practice.

Convergence is slow. The error decays roughly as $1/n$, so halving it requires doubling the number of steps, and since a tree of n steps has about $n^2/2$ nodes, that quadruples the work.

Convergence oscillates. Successive values zigzag, because where the strike falls between adjacent terminal nodes changes with the parity of n; in the figure the even-n values sit near 6.79 while the odd-n values drift down toward the limit, and both are still above it at $n = 50$. Reading convergence off a single value of n is therefore unreliable, and at small n adding one step can move the price by more than a one-dollar move in the stock would.

Both are numerical rather than conceptual difficulties, and Chapter 20 addresses them. What matters here is what convergence establishes about the *argument*: the price does not depend on the artificial two-branch assumption, because refining that assumption away leaves the answer essentially unchanged.

Why use a tree at all, then? Not for European values, which have a formula. The tree's worth is that it puts the pricing principle on display (replication, no arbitrage, risk-neutral valuation) and that the principle survives when a decision has to be made at an intermediate date, which is Section 13.7's situation and what a closed-form formula cannot express.

Why the limit is lognormal

It is worth seeing, without any stochastic calculus, why the tree converges to the model of Chapter 15 rather than to something else.

Over n steps the stock price is $S_0 u^j d^{n-j}$, where j is the number of up moves. Taking logarithms,

$$\ln S_T = \ln S_0 + j \ln u + (n - j) \ln d$$

so $\ln S_T$ is a *linear function of* j, and j is a sum of n independent binary draws. The central limit theorem then does the rest: as n grows, j becomes normally distributed, so $\ln S_T$ becomes normal, so S_T becomes **lognormal**.

Two things follow. The lognormal distribution that Chapter 15 assumes is not an extra assumption imposed from outside; it is what any model built from small independent multiplicative shocks produces. And the parameter choice (13.8) was designed to make the variance come out at $\sigma^2 T$ in the limit, so the limiting distribution has exactly the volatility that was put in.

The same reasoning identifies what would have to change to get a different limit. Shocks that are not independent, whose size depends on the price level or varies at random, or that are occasionally very large break the conditions under which the limit is lognormal; the last three are precisely the modifications that Chapters 19 and 21 make to reproduce the volatility smile.

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13.10 DIVIDENDS AND OTHER UNDERLYINGS

A known dividend yield. If the stock pays a continuous yield q, the stock itself grows at $r - q$ in the risk-neutral world, so only p changes:

$$p = \frac{e^{(r-q)\Delta t} - d}{u - d} \tag{13.9}$$

Everything else is untouched. The same substitution handles a stock index (with q the dividend yield), a currency (with $q = r_f$, the foreign rate, following Section 5.7), and a futures contract.

Entering a futures contract costs nothing, so in the risk-neutral world a futures price must have zero expected growth. Setting $q = r$ in (13.9) gives

$$p = \frac{1 - d}{u - d} \tag{13.10}$$

Table 13.1 collects the cases, and Example 13.8 applies the last of them.

Table 13.1: The probability p for different underlyings.

Underlying	Risk-neutral probability p
Non-dividend-paying stock	$\left(e^{r\Delta t} - d\right)/(u - d)$
Stock or index with yield q	$\left(e^{(r-q)\Delta t} - d\right)/(u - d)$
Currency, foreign rate r_f	$\left(e^{(r-r_f)\Delta t} - d\right)/(u - d)$
Futures contract	$(1 - d)/(u - d)$

Example 13.8 An American call on futures, exercised early

A futures price is 100, with $\sigma = 25\%$, $r = 4\%$, and two three-month steps. Value an American call struck at 100.

From (13.8), $u = e^{0.25\sqrt{0.25}} = 1.133148$ and $d = 0.882497$. From (13.10),

$$p = \frac{1 - 0.882497}{1.133148 - 0.882497} = 0.4688$$

Terminal futures prices are 128.40, 100.00, and 77.88, giving payoffs 28.4025, 0, and 0.

Upper node ($F = 113.3148$). Continuation is $e^{-0.01}\,[0.4688 \times 28.4025] = 13.1824$. Exercising yields $113.3148 - 100 = 13.3148$. **Exercise.**

Lower node. Both successors pay nothing; value 0.

Today. Continuation is $e^{-0.01}\,[0.4688 \times 13.3148] = 6.1798$, against an exercise value of 0. Hold.

The American call is worth 6.1798; the European is worth 6.1183.

Section 11.8 proved that an American call on a non-dividend-paying stock should never be exercised early—and here is one exercised early, with no dividend anywhere in the problem.

There is no contradiction, only a limit to that result's scope. The proof relied on the underlying being an asset that must be paid for, so that deferring payment of the strike earns interest. Any positive yield weakens the argument, because waiting now costs the income the underlying pays. A futures contract costs nothing to enter, which is exactly what (13.10) encodes by setting $q = r$: the yield equals the interest rate, and deferring the strike earns nothing net. Early exercise of a deep-in-the-money American call is therefore a live decision on futures options, on currencies with a high foreign interest rate, and on high-yielding indices.

A known cash dividend. A fixed dollar dividend is harder, and the difficulty is worth noting because it is the first place the method's convenience breaks. If the stock drops by D on an ex-dividend date, the tree *stops recombining*: an up-then-down path and a down-then-up path

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no longer meet, because the dividend is subtracted at different price levels. The node count no longer grows linearly; with a dividend at every step it reaches 2^n. Nothing about the pricing principle changes (the replication argument at each node is untouched), but the computation becomes impractical, and Chapter 20 deals with it.

13.11 PRICING AND HEDGING ARE ONE CALCULATION

One consequence of the argument deserves to be stated on its own, because it is easy to pass over and it governs everything in Part III.

The hedge ratio Δ of (13.1) is not a by-product of the pricing calculation. It *is* the pricing calculation: the price was obtained by finding the Δ that makes the portfolio riskless, and the option's value fell out of that construction. There is no separate step in which one first prices the option and then asks how to hedge it.

This is why the sensitivities of an option are not, in this theory, sensitivity analysis. They are the recipe for manufacturing it, and Example 13.5 carried that recipe out node by node. A trader who cannot hedge a position cannot claim to know what it is worth, because the price was defined as the cost of the hedge. Chapter 18 develops the full set of sensitivities on that understanding.

13.12 WHAT THE TREE ASSUMES

The argument of Section 13.1 is valid given its assumptions. They are the assumptions of the continuous-time model the tree converges to, and Section 15.1 lists them and says where each fails: volatility is neither known nor constant, prices jump rather than moving by u or d, and borrowing is not at r. One consequence belongs to the tree. Rebalancing at every node is what makes the hedge riskless, so with transaction costs the imperfection grows with the number of rebalances, and the limit that makes the model converge is also the limit that makes hedging infinitely expensive. Chapter 18 confronts this directly.

None of this makes the model useless. What survives is the *principle* that a derivative replicable from traded assets has a price independent of anyone's beliefs, and that principle is what the rest of Part III builds on.

SUMMARY

A binomial tree lets the underlying move to one of two values over a short interval. Holding $\Delta = (f_u - f_d)/(S_0u - S_0d)$ shares against a written derivative is riskless, so the portfolio must earn the risk-free rate, and that fixes the price. Pricing and delta hedging are one calculation, not two.

The same price is a discounted expectation, $f = e^{-r\Delta t}[p f_u + (1-p) f_d]$ with $p = (e^{r\Delta t} - d)/(u-d)$. The real-world probability of an up move never appears: p is the probability under which the stock earns the risk-free rate, and the device works because the derivative can be replicated, so preferences cancel out. With the discount factor absorbed, the weights are state prices, which are positive exactly when there is no arbitrage; the risk-neutral measure is another name for the absence of arbitrage.

Two states and two traded assets make the market complete, and that is what turns Chapter 11's bounds into a single price. With a third state the option cannot be replicated and the theory delivers only a range; wherever completeness fails, the accurate phrase is "a price, given a model."

Multi-period trees are solved backward. They recombine, so nodes grow linearly, and an American option takes the greater of continuation and exercise value at each node, which prices the early-exercise premium and locates the boundary. With $u = e^{\sigma\sqrt{\Delta t}}$ and $d = 1/u$ only volatility must be estimated, and as the steps shrink the terminal distribution becomes lognormal

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and the value converges, oscillating, to the Black–Scholes–Merton formula. Dividend yields, currencies, and futures change only p; for futures $p = (1 - d)/(u - d)$, which is why an American call on futures can be worth exercising early with no dividend anywhere.

The tree assumes costless rebalancing, constant known volatility, no jumps, and borrowing at the risk-free rate, and all four fail in practice. What survives is the principle: a derivative that can be replicated from traded assets has a price independent of anyone's beliefs.

FURTHER READING

The original papers are the place to start: Cox, Ross, and Rubinstein (1979) for the model itself, and Rendleman and Bartter (1979), published independently in the same year. Cox and Ross (1976) is where risk-neutral valuation first appears as a general principle. For convergence behavior, Broadie and Detemple (1996), a comparison of American option valuation methods, documents the odd–even oscillation and sets out which refinements are worth their cost.

PRACTICE QUESTIONS

13.1 A stock at \$40 will move to \$44 or \$36 in six months. The rate is 5%. Value a European call struck at \$40 by constructing the riskless portfolio, then verify the answer using (13.2).

13.2 In the previous question, compute the risk-neutral probability and verify that the expected stock price under it equals $S_0 e^{rT}$.

13.3 Explain why the real-world probability of an up move does not appear in the option price. Would the answer change if it were 0.9 rather than 0.5?

13.4 A stock at \$100 has $u = 1.15$, $d = 0.87$, and two six-month steps, with $r = 6\%$. Value a European put struck at \$100.

13.5 Repeat the previous question for an American put and compute the early-exercise premium. At which nodes is exercise optimal?

13.6 A stock has volatility 25% a year. Compute u, d, and p for a tree with three-month steps when $r = 3\%$. Verify that $d < e^{r\Delta t} < u$.

13.7 Explain why the binomial value oscillates as n increases, and why the existence of a limit matters for the argument of Section 13.1.

13.8 Show that for a futures contract $p = (1 - d)/(u - d)$, and explain in words why the interest rate disappears.

13.9 A stock at \$50 can move to \$56, \$50, or \$44 over one period, with $r = 4\%$ and $\Delta t = 0.5$. Show that a call struck at \$52 cannot be replicated, and find the range of arbitrage-free prices by imposing positive state prices.

13.10 Explain why a one-period model with two states can be completed by the stock and the bond, and what fails when a third state is added. State what completeness has to do with the option having a single arbitrage-free price rather than a range of them.

13.11 Explain why pricing an option and hedging it are one calculation in a binomial tree rather than two. State where the hedge ratio appears in the pricing arithmetic, and what that implies about a model that prices well but hedges badly.

13.12 State four assumptions the binomial tree makes, and for each name a market condition that violates it. Identify which of the four the convergence result of Section 13.9 is most sensitive to, and why.

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FURTHER QUESTIONS

13.13 Example 13.2 computed a 28.5% quarterly required return on the call by working backward from its known price. Repeat this for the put in the same one-step model, and explain why the two required returns differ so sharply in sign and magnitude. What does this tell you about using risk-adjusted discount rates for options?

13.14 A known cash dividend breaks the recombining property. Demonstrate this explicitly with a two-step tree, a \$50 stock, $u = 1.2$, $d = 0.8$, and a \$2 dividend paid between the steps. Then explain why the pricing principle is unaffected even though the computation is not.

13.15 Section 13.12 notes that the limit making the tree converge is also the limit making hedging infinitely expensive under transaction costs. Formalize this: with a proportional cost k per trade, estimate how total hedging cost scales with n, and identify the trade-off between model error and trading cost. (Chapter 18's gamma sets the size of each rebalancing trade.)

13.16 Using a three-step tree, trace the early-exercise boundary for an American put and compare it with the critical price described in Section 11.9. How does the boundary move as you approach expiration, and why must it converge to the strike?

13.17 An analyst prices a convertible bond by valuing the bond and the conversion option separately and adding them. Explain why this is wrong when the bond is callable, and identify which feature of backward induction the additive approach discards.

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Chapter 14
Wiener Processes and Itô's Lemma †

Chapter 13 priced options without any continuous-time mathematics. It could do so because the binomial world has finitely many states, and finitely many states can be handled with algebra. Chapter 15 does the same job in continuous time, where the states form a continuum and algebra is no longer enough. This chapter supplies the language for that. A reader who wants the Black–Scholes–Merton formula and its use rather than its derivation can pass from Chapter 13 straight to Chapter 15 without loss of continuity. Chapters 15, 16, 19, and 22 do cite four results here worth taking even on a skip: the lognormal property (Section 14.9: the median grows at $\mu - \sigma^2/2$, more slowly than the mean, a gap behind common errors in performance reporting), the $\sigma/16$ rule (Section 14.6), the volatility of a ratio (Section 14.11) and the model's failures (Section 14.12). Chapters 20, 21, 28, and 30 assume the whole chapter.

Sections 14.1 to 14.4 build the process for a stock price, from the Markov property through Wiener processes to geometric Brownian motion; Sections 14.5 and 14.6 say what its two parameters mean. Sections 14.7 and 14.8 are Itô's lemma and why it delivers a hedge; Sections 14.9 and 14.10 are the lognormal property and the risk-neutral process. Sections 14.11 and 14.12 extend the model to two sources of uncertainty and state what it assumes, and Section 14.13 introduces fractional Brownian motion, which Chapter 21's rough volatility will need.

14.1 THE MARKOV PROPERTY

A **Markov process** is one in which only the present value of the variable is relevant for predicting the future. The path taken to get there carries no information.

Stock prices are usually assumed to be Markov. The motivation is **weak-form market efficiency**, the hypothesis that past prices cannot be used to predict future ones: if the history of prices contained information about future prices beyond what today's price contains, that information could be traded on, and trading on it would remove it. The assumption is not exactly true: under volatility clustering, which Chapter 38 models when it forecasts volatility, the size of tomorrow's move is partly predictable even though its direction is not. The Markov assumption is nevertheless close enough to be the standard starting point.

The model adds one assumption this chapter relies on throughout: changes over non-overlapping periods are independent, so their variances add. The Markov property alone does not deliver this: the mean-reverting interest rates of Chapter 31 are Markov, yet their changes are negatively correlated, and their variance levels off instead of growing linearly.

Example 14.1 Why variances, not standard deviations, add

Suppose a stock at \$40 has a price change with standard deviation \$10 over one year. What is the uncertainty over two years?

The change over two years is the change in year one plus the change in year two. These are assumed independent, so their *variances* add:

$$\text{Var}(2\text{ years}) = 100 + 100 = 200, \qquad \text{standard deviation} = \sqrt{200} = \$14.14$$

Uncertainty, measured by standard deviation, grows with $\sqrt{T}$, not with T; it is the variance that grows with T. Over six months the standard deviation is $10\sqrt{0.5} = \$7.07$; over three months, $10\sqrt{0.25} = \$5.00$.

This fact is why Section 11.1 found option value growing roughly as $\sqrt{T}$. It is why volatility is quoted per annum and rescaled by $\sqrt{\Delta t}$. And it is why the binomial parameters of Section 13.8 were $u = e^{\sigma\sqrt{\Delta t}}$ rather than $e^{\sigma \Delta t}$. The square root was not a modeling choice but a consequence of independence.

14.2 WIENER PROCESSES

The building block is a process W with two properties. Over a short interval Δt,

$$\Delta W = \varepsilon\sqrt{\Delta t}, \qquad \varepsilon \sim N(0,1) \tag{14.1}$$

and the values of ΔW for any two non-overlapping intervals are independent. A process with these properties is a **Wiener process**, also called **Brownian motion**.

Two consequences follow immediately. The mean of ΔW is zero and its variance is Δt, so the variance per unit time is exactly 1, which is what makes W a natural unit of randomness. And over a long interval T, made of $T/\Delta t$ independent steps,

$$W(T) - W(0) \sim N(0, T)$$

by the same variance-addition argument as Example 14.1.

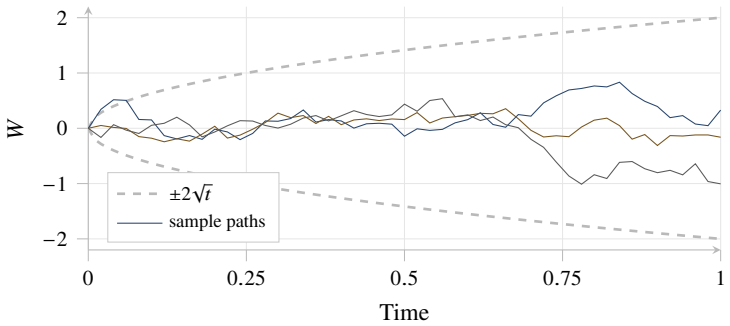


Figure 14.1: Three illustrative simulated Wiener paths and the $\pm 2\sqrt{t}$ envelope.

Two features of the paths in Figure 14.1 recur throughout Part III.

The paths are continuous but extremely rough. As Δt shrinks, the standard deviation of ΔW shrinks only as $\sqrt{\Delta t}$, so that of $\Delta W/\Delta t$ is $1/\sqrt{\Delta t}$ and grows without bound. A Wiener path has, with probability one, no derivative anywhere. This is why the calculus of Section 14.7 has to be different from the ordinary kind: one cannot simply write dW/dt.

And the envelope widens as $\sqrt{t}$, not linearly. The typical distance from zero grows as time passes, but at a decreasing rate.

14.3 GENERALIZED WIENER PROCESSES

A Wiener process has zero **drift**, the expected change per unit time, and a unit **variance rate**, the variance per unit time. Both can be generalized. A variable x follows a **generalized Wiener process** if

$$dx = a\,dt + b\,dW \tag{14.2}$$

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with a and b constants. The $a\,dt$ term is a deterministic trend (x drifts at rate a per unit time), and the $b\,dW$ term, the **diffusion**, adds noise with variance rate b^2.

Over an interval T,

$$x(T) - x(0) \sim N\left(aT,\ b^2T\right)$$

The mean grows linearly in T and the standard deviation as $\sqrt{T}$, so the trend eventually dominates the noise, but only eventually, and the crossover can be a long way off, as Example 14.2 shows.

Example 14.2 A cash position

A company's cash balance, in millions, follows a generalized Wiener process with drift 20 per year and variance rate 900 per year. It currently holds \$50 million.

After one year the balance is distributed $N(50 + 20,\ 900)$, that is, mean \$70 million with standard deviation \$30 million. After six months it is $N(60,\ 450)$: mean \$60 million, standard deviation $\sqrt{450}$ = \$21.2 million.

Writing $N(x)$ for the standard normal cumulative distribution function, the probability that the balance is negative at the end of one year is

$$\Pr(x < 0) = N\left(\frac{0 - 70}{30}\right) = N(-2.333) = 0.98\%$$

and at the end of six months, $N(-60/21.2) = N(-2.83) = 0.23\%$. The balance is expected to *grow*, yet the probability is *higher* at one year than at six months, because uncertainty accumulates faster than the drift does over short horizons.

Running out of cash, though, means touching zero at any time, not being below it on a given date. That probability is larger, 2.7% within the year (by the reflection argument Chapter 34 uses for barrier options), and it never falls: over an unlimited horizon it is $e^{-2\times20\times50/900} = 10.8\%$. This is the same structure as the funding failures of Chapter 8: a position with a positive expected trend can still fail on the way, and the distribution on any one date understates how often.

An **Itô process** generalizes (14.2) once more by letting the coefficients depend on the current value and on time:

$$dx = a(x,t)\,dt + b(x,t)\,dW \tag{14.3}$$

This is the form most of the rest of the book uses. It is shorthand for an integral equation,

$$x(T) - x(0) = \int_0^T a(x,t)\,dt + \int_0^T b(x,t)\,dW$$

and the second integral is not an ordinary one, because W has no derivative to integrate against. It is an **Itô integral**, the limit of the sums

$$\sum_{i=0}^{n-1} b\big(x(t_i), t_i\big)\,\Delta W_i, \quad \Delta W_i = W(t_{i+1}) - W(t_i), \quad 0 = t_0 < \cdots < t_n = T,$$

as the longest interval shrinks to zero, in which b is evaluated at the *start* of each interval, before the increment ΔW_i is known. In finance this is not a convention but a fact about trading: a hedge ratio is chosen before the price move it is exposed to, so a trading gain is a sum of exactly this kind. The choice makes the integral a fair game (a martingale, the term Section 13.6 defined), with expected value zero whatever b does, and it is the source of everything that separates Itô calculus from the ordinary kind, beginning with the extra term in Section 14.7. Evaluating b at the midpoint of each interval instead gives the Stratonovich integral, which obeys the ordinary chain rule but is not a fair game and does not describe what a trader can earn. The derivation in [TN14] works through the **quadratic variation** of W, the limit of the sum of its squared increments, which over $[0, T]$ is exactly T; Section 14.7 shows why.

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14.4 THE PROCESS FOR A STOCK PRICE

What process should a stock price follow? A generalized Wiener process is the obvious candidate, and it is wrong for two reasons that determine the model that replaces it.

Constant absolute drift is not how returns work. A \$10 stock and a \$1,000 stock do not have the same expected dollar gain. Investors require a *percentage* return, so the drift should be proportional to the price.

Constant absolute volatility is not how risk works either. A \$1 move means something very different at \$10 than at \$1,000. And a constant b permits the price to go negative, which for a limited-liability share it cannot.

Both objections are met by making the coefficients proportional to S:

$$\boxed{dS = \mu S\,dt + \sigma S\,dW} \tag{14.4}$$

This is **geometric Brownian motion**, and it is the model underlying everything from Chapter 15 onward. Dividing through, the *proportional* return over a short Δt is close to normal:

$$\frac{dS}{S} = \mu\,dt + \sigma\,dW, \qquad \frac{\Delta S}{S} \sim N\left(\mu\,\Delta t,\ \sigma^2\Delta t\right).$$

Here μ is the expected return per year and σ is the volatility, the same σ that Section 13.8 fed into the binomial tree.

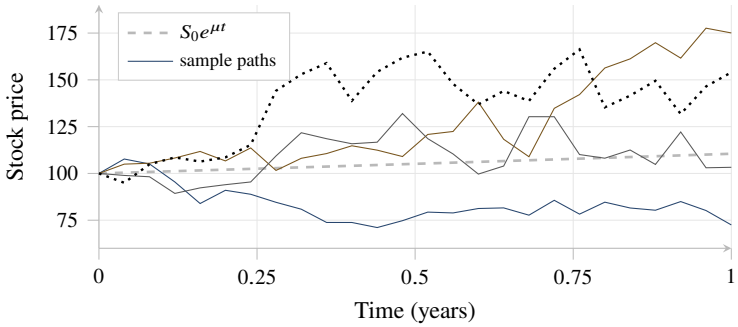


Figure 14.2: Four illustrative simulated paths. $S_0 = 100$, $\mu = 10\%$, $\sigma = 40\%$.

Figure 14.2 shows four paths from this process. Note that none of them tracks the dashed line $S_0e^{\mu t}$ at all closely, and that half of them finish below it despite μ being positive. Section 14.9 explains why that is exactly what should be expected.

Example 14.3 Reading the process over a short interval

A stock trades at \$50 with $\mu = 16\%$ and $\sigma = 30\%$. Over one week, $\Delta t = 1/52 = 0.01923$:

$$\frac{\Delta S}{S} \sim N\left(0.16 \times 0.01923,\ 0.30^2 \times 0.01923\right) = N(0.003077,\ 0.001731)$$

so the weekly return has mean 0.31% and standard deviation $\sqrt{0.001731} = 4.16\%$. In dollar terms,

$$\Delta S \sim N(0.154,\ 2.08^2)$$

The noise term is about thirteen and a half times the drift term over one week, and since the drift scales with Δt while the noise scales with $\sqrt{\Delta t}$, the ratio gets worse as the interval shortens: over one day it is about thirty to one.

This is why estimating μ from historical returns is hopeless in a way that estimating σ is not: to measure a drift buried under thirty times its own size in noise, one needs decades of data, over which the drift will not have stayed constant.

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The discrete-time version

Equation (14.4) is a statement about infinitesimals. To simulate it, or to connect it to Chapter 13's tree, it must be discretized, and there are two ways of doing so that are not equivalent.

The direct reading of (14.4) gives

$$\Delta S \approx \mu S\,\Delta t + \sigma S\,\varepsilon\sqrt{\Delta t} \tag{14.5}$$

which is accurate only for small Δt and can produce a negative price if ε is sufficiently negative. The alternative uses the solution of the equation rather than the equation itself, which Section 14.9 will derive:

$$S_{t+\Delta t} = S_t \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)\Delta t + \sigma\varepsilon\sqrt{\Delta t}\right] \tag{14.6}$$

This is *exact* for any Δt, and it cannot go negative. Figure 14.2 was generated with (14.6).

The comparison is worth keeping in mind because the same choice arises whenever a process is discretized, and Chapter 20 returns to it. Note also that (14.6) is recognizably Chapter 13's tree with the two-point distribution of ε replaced by a normal one: $u = e^{\sigma\sqrt{\Delta t}}$ was the same expression with $\varepsilon = \pm 1$ and the drift moved out of the step and into p.

Where the model came from. Louis Bachelier's thesis of 1900 used the arithmetic process of Section 14.3, which this section has just rejected for stocks. Norbert Wiener gave the process its rigorous construction in 1923. Kiyoshi Itô defined the integral of Section 14.3 in 1944 and set out the calculus built on it, including the lemma of Section 14.7, in 1951. Section 15.12 traces the economics, from Bachelier to Black, Scholes, and Merton.

14.5 EXPECTED RETURN

The parameter μ was introduced as "the expected return," and that phrase turns out to name two different numbers. The distinction causes more confusion than any other point in this chapter, and it is not confined to theory.

From (14.4), the expected *proportional* return over a short interval satisfies

$$\mathbb{E}\left[\frac{\Delta S}{S}\right] \approx \mu\,\Delta t$$

so μ is the expected return in the sense of an arithmetic average of short-period returns. But from (14.10), which Section 14.9 derives, the expected *continuously compounded* return over $[0, T]$ is

$$\mathbb{E}\left[\frac{1}{T}\ln\frac{S_T}{S_0}\right] = \mu - \frac{\sigma^2}{2}$$

which is the expected rate earned by an investor who holds the asset throughout. Example 14.4 shows the gap in five years of returns.

Example 14.4 Arithmetic and geometric averages

An asset returns +30%, −10%, +20%, −5%, and +15% over five years. The arithmetic average is

$$\frac{30 - 10 + 20 - 5 + 15}{5} = 10.0\%$$

But an investor who held throughout finished with

$$1.30 \times 0.90 \times 1.20 \times 0.95 \times 1.15 = 1.53387$$

times their money, an annual compounded return of $1.53387^{1/5} - 1 = 8.93\%$.

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In log returns the five years average 8.56% ($e^{0.0856} - 1 = 8.93\%$), 1.44 percentage points below the arithmetic average, and half the sample variance of the five simple returns is $0.0288/2 = 1.44$ percentage points, a match to the second decimal that in general holds only approximately. This is the $\sigma^2/2$ that separates (14.13) from (14.14), seen in five numbers.

The arithmetic average is not wrong; it answers a different question. It is the right input for an expected one-period return, and therefore the right thing to compare with μ. The compounded figure is the right answer to "what did an investor make," and it corresponds to $\mu - \sigma^2/2$.

The practical consequence is a reporting convention that repays skepticism. A track record summarized by the arithmetic average of its annual returns, common in strategy marketing and backtests, exceeds what an investor who held throughout received, and the excess grows with volatility. The standardized figure a US mutual fund must publish is not of this kind: the SEC's Form N-1A defines its average annual total return as the compounded rate.

14.6 VOLATILITY

As Section 13.8 defined it, σ is the annualized standard deviation of the continuously compounded return. Over a period Δt it is $\sigma\sqrt{\Delta t}$, by the variance-addition property of Section 14.1.

One rule of thumb is used constantly on trading desks. With about 252 trading days in a year and $\sqrt{252} \approx 15.9$, the daily standard deviation is approximately

$$\sigma_{\text{daily}} \approx \frac{\sigma}{16}$$

so a stock with 32% annual volatility has a daily standard deviation of about 2%, and a 6% move is a three-standard-deviation event. It converts an abstract parameter into something a trader can check against the screen.

How σ is estimated (from historical returns, or by inverting an option price) is taken up in Chapter 15, where it is needed for the first time.

14.7 ITÔ'S LEMMA

Assume that a derivative's value depends only on the current stock price and time, so that it can be written $f(S, t)$. That is an assumption, not a general truth: it holds for standard calls and puts, and fails for contracts whose payoff depends on the path, such as Chapter 34's Asian and lookback options, whose value depends on where the stock has been and not only on where it is now. If S follows (14.4), what process does $f(S, t)$ follow? Ordinary calculus gives the wrong answer, and the reason it does is the single technical fact this chapter exists to deliver.

Consider a Taylor expansion of f:

$$\Delta f = \frac{\partial f}{\partial S}\Delta S + \frac{\partial f}{\partial t}\Delta t + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\Delta S^2 + \cdots$$

In ordinary calculus the ΔS^2 term is discarded, because ΔS is proportional to Δt and Δt^2 is negligible. Here it is not. From (14.4), $\Delta S \approx \mu S\,\Delta t + \sigma S\varepsilon\sqrt{\Delta t}$, so

$$\Delta S^2 = \sigma^2 S^2 \varepsilon^2\,\Delta t + \text{terms of order } \Delta t^{3/2} \text{ and smaller}$$

whose leading term is of order Δt, *not* Δt^2. It survives. A single $\varepsilon^2\Delta t$ is still random: its mean is Δt and its standard deviation $\sqrt{2}\,\Delta t$, of the same size. But over an interval T, made of $T/\Delta t$ steps, the squared increments add to a total with mean T and variance $2T\Delta t$, which vanishes as the steps shrink. The sum, the quadratic variation of Section 14.3, becomes non-random and equal to T, as Figure 14.3 shows on one path. In the notation that results,

$$dW^2 = dt \tag{14.7}$$

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a rule about sums over many steps, not about any single increment. [TN14], reproduced in Appendix A, makes this precise.

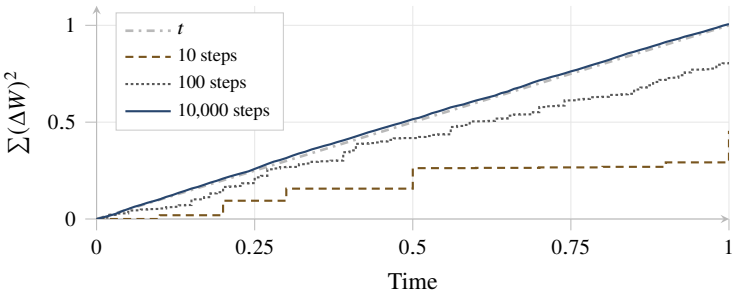


Figure 14.3: Running sum of squared increments along one simulated Wiener path, sampled on grids of 10, 100 and 10,000 steps.

Substituting dS and $dS^2 = \sigma^2 S^2\,dt$ into the expansion gives **Itô's lemma**:

$$\boxed{df = \left(\frac{\partial f}{\partial S}\mu S + \frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2 S^2\right)dt + \frac{\partial f}{\partial S}\sigma S\,dW} \tag{14.8}$$

Everything except the third term in the drift is what ordinary calculus would give. That third term is the entire difference, and it is the reason a derivative's expected return is not simply the chain rule applied to the stock's. It is proportional to the second derivative, the convexity, which is why it will reappear in Chapter 18 as gamma and in Chapter 30 as the source of convexity adjustments.

Example 14.5 Itô's lemma applied to a forward price

The forward price of a stock that pays no dividend, for delivery at T is $F = Se^{r(T-t)}$, from Section 5.2. What process does F follow?

The partial derivatives are

$$\frac{\partial F}{\partial S} = e^{r(T-t)}, \qquad \frac{\partial F}{\partial t} = -rSe^{r(T-t)}, \qquad \frac{\partial^2 F}{\partial S^2} = 0$$

Substituting into (14.8), and noting that the second-derivative term vanishes:

$$dF = \left(e^{r(T-t)}\mu S - rSe^{r(T-t)}\right)dt + e^{r(T-t)}\sigma S\,dW = (\mu - r)F\,dt + \sigma F\,dW$$

So the forward price also follows geometric Brownian motion, with *the same volatility* σ as the spot but a drift of $\mu - r$ rather than μ.

The equality of volatilities is why a volatility quoted on the spot can be used to price an option on the forward, which Chapter 16 relies on throughout. And the drift being the *excess* return $\mu - r$ is the continuous-time statement of Section 13.10's observation that a futures price has zero growth in the risk-neutral world: setting $\mu = r$, as risk-neutral valuation does, makes the forward's drift exactly zero.

Common applications

Table 14.1 collects the cases used elsewhere in the book. Each is (14.8) with the relevant derivatives substituted, and each is worth being able to reproduce.

14.8 WHY THE LEMMA DELIVERS A HEDGE

The last row of Table 14.1 is the bridge between Chapter 13 and Chapter 15.

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Table 14.1: Processes followed by functions of a stock price obeying (14.4).

Function	Process	Where used
$\ln S$	$\left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma\, dW$	The lognormal property, Section 14.9
S^n	$S^n\left[\left(n\mu + \frac{n(n-1)}{2}\sigma^2\right) dt + n\sigma\, dW\right]$	Power payoffs; $n = 2$ in Practice Question 5
$Se^{r(T-t)}$	$(\mu - r)F\, dt + \sigma F\, dW$	Forward prices, Example 14.5
general $f(S,t)$	drift as in (14.8), diffusion $\dfrac{\partial f}{\partial S}\sigma S\, dW$	Every derivative, Section 14.8

For *any* derivative $f(S,t)$, Itô's lemma gives

$$df = (\cdots)\, dt + \frac{\partial f}{\partial S}\, \sigma S\, dW \tag{14.9}$$

Whatever the drift term turns out to be, the *random* part of the derivative's movement is the stock's own shock dW, scaled by $\sigma S\, \partial f/\partial S$. And the random part of the stock's movement, from (14.4), is $\sigma S\, dW$: the same dW.

Consider a portfolio that is short one derivative and long $\partial f/\partial S$ shares. Its change over an infinitesimal time interval is

$$-df + \frac{\partial f}{\partial S}\, dS = -(\cdots)\, dt - \frac{\partial f}{\partial S}\sigma S\, dW + \frac{\partial f}{\partial S}\, (\mu S\, dt + \sigma S\, dW)$$

and the two dW terms cancel exactly. The portfolio has no random component at all. It is riskless, so it must earn r, and imposing that condition is precisely what Chapter 15 does to obtain the Black–Scholes–Merton equation.

Three observations follow, and they connect this chapter back to the last one.

It is Chapter 13's argument in continuous time. Section 13.1 found the Δ that made a portfolio worth the same in both states. Here $\partial f/\partial S$ makes a portfolio's change contain no dW. The discrete hedge ratio

$$\Delta = \frac{f_u - f_d}{S_0 u - S_0 d}$$

is the difference quotient whose limit is $\partial f/\partial S$.

μ cancels too. The derivative's drift, the $(\cdots)$ above, contains the term $\mu S\, \partial f/\partial S$; the short derivative subtracts it and the shares add it back, so μ drops out of the portfolio's change along with dW, leaving only the time and convexity terms for Chapter 15 to set equal to the riskless return. This is the continuous-time version of Chapter 13's observation that the real-world probability never entered, and it is why an option can be priced without forecasting the stock.

The hedge is instantaneous, not permanent. The cancellation holds over a single dt. Since $\partial f/\partial S$ changes as S and t change, the position must be adjusted continuously: the dynamic replication of Section 13.5, now requiring infinitely frequent trading. That requirement is harmless in the model and expensive in reality, which is the subject of Chapter 18 and the reason the assumptions of Section 14.12 matter.

14.9 THE LOGNORMAL PROPERTY

The most useful application of Itô's lemma is to $G = \ln S$. Here

$$\frac{\partial G}{\partial S} = \frac{1}{S}, \qquad \frac{\partial^2 G}{\partial S^2} = -\frac{1}{S^2}, \qquad \frac{\partial G}{\partial t} = 0$$

and (14.8) gives

$$d\ln S = \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma\, dW \tag{14.10}$$

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The right-hand side has constant coefficients, so $\ln S$ follows a *generalized* Wiener process, and Section 14.3 already told us the answer for those:

$$\ln S_T \sim N\left(\ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right)T,\ \ \sigma^2 T\right) \tag{14.11}$$

Because its logarithm is normal, S_T is lognormal. This is the result Chapter 13 anticipated by a different route: Section 13.9 showed that a product of many small independent multiplicative shocks becomes lognormal by the central limit theorem. Equation (14.11) is that conclusion in continuous time, and the agreement is not a coincidence: the binomial model was built to converge to this one.

Moments of the lognormal

Two results are used constantly from Chapter 15 onward. If $\ln X \sim N(m, s^2)$, then

$$\mathbb{E}[X] = e^{m+s^2/2}, \qquad \mathrm{Var}(X) = e^{2m+s^2}\left(e^{s^2} - 1\right) \tag{14.12}$$

Applying the first to (14.11), with $m = \ln S_0 + (\mu - \sigma^2/2)T$ and $s^2 = \sigma^2 T$:

$$\mathbb{E}[S_T] = \exp\left(\ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right)T + \frac{\sigma^2 T}{2}\right) = S_0 e^{\mu T}$$

The two rates are consistent: μ is the expected growth rate of the price, and Itô's lemma makes the logarithm, and with it the median, grow $\sigma^2/2$ more slowly.

Where the $\sigma^2/2$ comes from, and why it matters

The term $-\sigma^2/2$ in (14.10) is the Itô correction term of (14.8), and it is the most consequential piece of notation in this chapter.

From (14.11), two different quantities both have a claim to be called "the" growth rate:

$$\mathbb{E}[S_T] = S_0 e^{\mu T} \tag{14.13}$$

$$\mathrm{median}(S_T) = S_0 e^{(\mu - \sigma^2/2)T} \tag{14.14}$$

The expected value grows at μ. The median (and the continuously compounded return actually realized on a typical path) grows only at $\mu - \sigma^2/2$. The gap is entirely due to volatility, and it is sometimes called **volatility drag**.

Example 14.6 Expected value against typical outcome

A stock trades at \$100 with $\mu = 10\%$ and $\sigma = 40\%$. After one year:

$$\mathbb{E}[S_T] = 100e^{0.10} = \$110.52, \qquad \text{median} = 100e^{0.10-0.08} = 100e^{0.02} = \$102.02$$

The expected price is \$110.52 and yet half of all outcomes fall below \$102.02. Worse, ask how often the expected value is reached at all. Since $\ln S_T \sim N(\ln 100 + 0.02,\ 0.40^2)$,

$$\Pr(S_T > 110.52) = 1 - N\left(\frac{\ln 1.1052 - 0.02}{0.40}\right) = 1 - N(0.20) = 42.1\%$$

The stock finishes below its own expected value 58% of the time. Nothing is wrong with the arithmetic: the mean is dragged up by a long right tail that occurs rarely.

A 95% interval for the terminal price makes the dispersion concrete. With probability 0.95,

$$\ln 100 + 0.02 - 1.96 \times 0.40 \le \ln S_T \le \ln 100 + 0.02 + 1.96 \times 0.40,$$

that is, $\$46.58 \le S_T \le \$223.45$ at 40% volatility over a single year.

Two common errors follow directly from ignoring the distinction, besides the fund-reporting one of Example 14.4. Quoting an "expected return" of 10% invites investors to expect \$110.52,

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when the typical outcome is \$102.02. And a structured product promising participation in an index's *average* growth is offering something systematically better than participation in its *realized* growth, a difference of $\sigma^2/2$ per year, which at 40% volatility is eight percentage points.

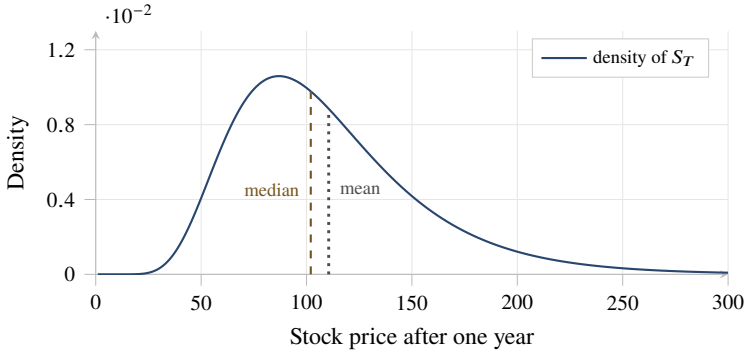


Figure 14.4: Density of S_T. $S_0 = 100$, $\mu = 10\%$, $\sigma = 40\%$, $T = 1$.

Figure 14.4 shows why. The distribution is bounded below by zero and has a long right tail; the mean sits to the right of the median, dragged there by outcomes that are large but rare. The mode is lower still, at $S_0 e^{(\mu - 3\sigma^2/2)T} = \$86.94$.

Which rate does an option price use? Neither: Section 14.10 replaces μ by r, and the $-\sigma^2/2$, a property of the lognormal distribution rather than of the drift, survives.

Case Study 14.1 Leveraged exchange-traded funds

A widely sold class of funds promises a fixed multiple (usually two or three times) of the *daily* return of an index. Investors often assume this delivers the same multiple over longer horizons. It does not, and Itô's lemma says exactly by how much it fails.

If the index follows (14.4), a fund whose value V delivers k times its instantaneous return has $dV/V = k\,dS/S = k\mu\,dt + k\sigma\,dW$, a geometric Brownian motion with drift $k\mu$ and volatility $k\sigma$, before the financing cost of the borrowed $k-1$ units of exposure, which lowers the drift by a further $(k-1)r$. From (14.14), its median grows at

$$k\mu - \frac{(k\sigma)^2}{2} = k\mu - \frac{k^2\sigma^2}{2}$$

while k times the index's own median growth would be $k\left(\mu - \sigma^2/2\right)$. The shortfall is

$$\frac{k(k-1)}{2}\,\sigma^2 \text{ per year}$$

which for $k = 2$ is σ^2 and for $k = 3$ is $3\sigma^2$.

Numbers make the size clear. Take an index with $\mu = 8\%$ and $\sigma = 25\%$. Over one year the index's median return is $e^{0.08 - 0.03125} - 1 = 5.0\%$, so an investor might expect roughly 10% from a two-times fund. The fund's median return is

$$e^{0.16 - 0.125} - 1 = 3.6\%$$

At $\sigma = 40\%$ the arithmetic turns brutal: the index's median return is $e^{0.08 - 0.08} - 1 = 0\%$, and the two-times fund's median is $e^{0.16 - 0.32} - 1 = -14.8\%$. In the median case the index goes nowhere and the leveraged fund loses fifteen percent, before financing, fees, and the cost of daily rebalancing.

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Nothing here is a defect in the product, which does precisely what its prospectus says: it delivers k times the return *each day*. The gap is a mathematical consequence of compounding a scaled-up volatility, and it is the same $\sigma^2/2$ that separates the mean from the median in Example 14.6. These funds are sound instruments for a position held for days and poor ones for a position held for years, and the boundary between the two is set by σ.

14.10 THE RISK-NEUTRAL PROCESS

Chapter 13 established that a derivative may be valued as though every asset earned the risk-free rate. In the language of this chapter that statement becomes a single substitution: under the risk-neutral measure Q (Section 13.6), with W^Q a Wiener process under Q,

$$dS = rS\,dt + \sigma S\,dW^Q \tag{14.15}$$

and therefore, repeating Section 14.9 with μ replaced by r,

$$\ln S_T \sim N\left(\ln S_0 + \left(r - \frac{\sigma^2}{2}\right)T,\ \ \sigma^2 T\right), \qquad \mathbb{E}^Q[S_T] = S_0 e^{rT} \tag{14.16}$$

Equation (14.16) is the entire input to Chapter 15. Everything the Black–Scholes–Merton formula knows about the stock is contained in it.

Only the drift changes. The volatility σ is identical under both measures, as Section 13.8 found for the tree. The formal statement is Girsanov's theorem (a change to an equivalent measure adds a drift to a Wiener process and changes nothing else), which [TN14] sketches; the intuition is that changing the measure re-weights which paths are likely but cannot change how rough they are.

The correction term survives. Substituting r for μ does not remove the $-\sigma^2/2$, because that term came from Itô's lemma rather than from the drift. This is why the Black–Scholes–Merton formula contains $r - \sigma^2/2$; the correction is the Itô term of the logarithm, not a property of the expected return.

The connection back to Chapter 13 is exact. There, p made the discounted stock price a martingale (Section 13.6), the condition that the stock's expected growth be r. Here, $\mathbb{E}^Q[S_T] = S_0 e^{rT}$ is the same condition, written for a continuum of states.

14.11 TWO SOURCES OF UNCERTAINTY

Many instruments depend on more than one uncertain quantity: a quanto (a foreign asset's payoff converted at an exchange rate fixed in advance, Chapter 16) on both an asset and an exchange rate, a spread option (an option on the difference between two prices) on two prices. The apparatus needs one addition.

Let W_1 and W_2 be Wiener processes whose increments have correlation ρ, so that, by the argument behind (14.7),

$$dW_1\,dW_2 = \rho\,dt \tag{14.17}$$

alongside $dW_1^2 = dW_2^2 = dt$ from (14.7). For a function $f(x, y, t)$ of two Itô processes $dx = a_1 dt + b_1 dW_1$ and $dy = a_2 dt + b_2 dW_2$, Itô's lemma becomes

$$\begin{aligned} df = &\left(\frac{\partial f}{\partial x}a_1 + \frac{\partial f}{\partial y}a_2 + \frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}b_1^2 + \frac{1}{2}\frac{\partial^2 f}{\partial y^2}b_2^2 + \frac{\partial^2 f}{\partial x\,\partial y}b_1 b_2 \rho\right)dt \\ &+ \frac{\partial f}{\partial x}b_1\,dW_1 + \frac{\partial f}{\partial y}b_2\,dW_2 \end{aligned} \tag{14.18}$$

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The only new term, the last in the drift, exists for the same reason as the two second-derivative terms: the cross-product $\Delta x\,\Delta y$ is of order Δt rather than Δt^2. It is also the source of the correlation term in Chapter 30's quanto adjustment. Example 14.7 applies it to a price ratio.

Example 14.7 The volatility of a ratio

Two assets follow geometric Brownian motions with expected returns μ_1 and μ_2, volatilities σ_1 and σ_2, and correlation ρ. What process does the ratio $G = S_1/S_2$ follow?

Applying (14.18) and collecting terms gives

$$\frac{dG}{G} = \left(\mu_1 - \mu_2 + \sigma_2^2 - \rho\sigma_1\sigma_2\right) dt + \sigma_1\, dW_1 - \sigma_2\, dW_2$$

so the ratio is itself a geometric Brownian motion, with volatility

$$\sigma_G = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2} \tag{14.19}$$

Put numbers on it. With $\sigma_1 = 20\%$, $\sigma_2 = 12\%$, and $\rho = 0.4$:

$$\sigma_G = \sqrt{0.04 + 0.0144 - 2 \times 0.4 \times 0.20 \times 0.12} = \sqrt{0.0352} = 18.8\%$$

The ratio is *less* volatile than S_1 alone, because the two assets partly move together and the common movement cancels in the ratio. Reverse the correlation to $\rho = -0.4$ and $\sigma_G = \sqrt{0.0544 + 0.0192} = \sqrt{0.0736} = 27.1\%$, more volatile than either input.

Equation (14.19) is used more often than almost any other result in this chapter:

- It gives the volatility of a cross exchange rate from two rates against a common currency.
- It gives the volatility of a stock measured in a foreign currency, which Chapter 16 needs for composite options (options on a foreign stock whose payoff is converted at the expiration exchange rate). A product S_1S_2 is the ratio of S_1 to $1/S_2$, whose correlation with S_1 is $-\rho$, so the minus sign in (14.19) becomes a plus.

It is the formula for the standard deviation of a difference, which is what a ratio becomes after taking logarithms. That is the quickest way to remember it.

14.12 WHAT THE MODEL ASSUMES

Geometric Brownian motion is a modeling choice, not a fact. Section 15.1 collects the assumptions of the pricing model built on it and says where each fails; three ways real prices depart from it are worth having here. Volatility is not constant but clusters (calm follows calm and violent follows violent), with consequences Chapter 19 documents in the volatility surface. Prices are not continuous: the negative oil settlement of Section 5.9, the LME nickel suspension of Chapter 2 and any earnings announcement produce moves no continuous path can accommodate, and a model with jumps assigns much more probability to extreme outcomes than (14.11) does. And returns are independent across time in direction but not in magnitude: clustering violates the Markov property of Section 14.1 for the price alone, though not for price and volatility jointly, and Section 14.13 introduces the standard process whose increments are not independent. Of the two parameters, σ must be estimated (Chapter 15 takes up how), while Example 14.3 showed that μ effectively cannot be, which does not matter for option pricing and matters a great deal for everything else.

14.13 FRACTIONAL BROWNIAN MOTION

Rough volatility, the finding that the path of an asset's volatility is far more jagged than a Wiener path, together with the models built on that finding, appears in Sections 21.8 and 23.6 and in [TN21], so the process behind it is introduced here. It is also the standard process for which Section 14.1's variance rule fails.

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Definition. **Fractional Brownian motion** B_t^H is a continuous process, jointly normal at any set of dates, with $B_0^H = 0$, mean zero, and covariance

$$\operatorname{Cov}\left[B_s^H, B_t^H\right] = \frac{1}{2}\left(s^{2H} + t^{2H} - |t-s|^{2H}\right), \qquad 0 < H < 1 \tag{14.20}$$

Its increments are **stationary**, as a Wiener process's are: the variance of a change depends only on the length of the interval,

$$\operatorname{Var}\left[B_t^H - B_s^H\right] = |t-s|^{2H} \tag{14.21}$$

The exponent H is the **Hurst parameter**, after the hydrologist who first measured it in the flood records of the Nile. At $H = 1/2$ the covariance (14.20) is $\min(s,t)$, which is the Wiener process of Section 14.2: the variance over T is T and increments over disjoint intervals are independent.

The increments are correlated. Take two consecutive years. The change over the second year, $B_2^H - B_1^H$, has by stationarity the same variance as the change over the first, and the two-year variance is the sum of the two plus twice their covariance. With (14.21) on both sides, $2^{2H} = 1 + 1 + 2\rho$, so the correlation between successive equal increments is

$$\rho = 2^{2H-1} - 1 \tag{14.22}$$

which is -0.43 at $H = 0.1$, zero at $1/2$, and $+0.74$ at $H = 0.9$. Below one-half the process is **anti-persistent**: each move partly undoes the last. Above one-half it is **persistent**, with a long memory: a rise makes a further rise more likely. Section 14.1's argument that variances add needed exactly this cross term to vanish, and it vanishes only at one-half.

Scaling and roughness. Equation (14.21) says the process is **self-similar**: stretching time by a factor c scales the path's distribution by c^H, so an increment over a short interval Δt has standard deviation Δt^H. For the Wiener process that is $\sqrt{\Delta t}$, the scaling Section 14.2 built on. For $H < 1/2$, Δt^H is larger than $\sqrt{\Delta t}$ for small Δt, so the path moves more over short intervals and is *rougher*; for $H > 1/2$ it is smoother. Figure 14.5 shows the three cases built from the same random draws: the rough path reverses constantly, the persistent one trends, and the Wiener path sits between them.

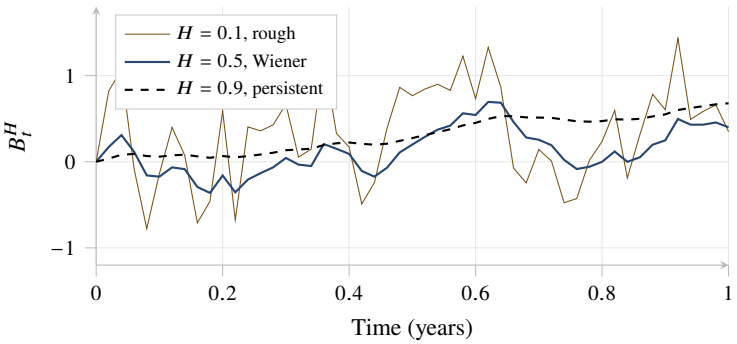


Figure 14.5: Fractional Brownian motion at $H = 0.1$, 0.5 and 0.9, built from the same random draws; each has unit variance at one year.

The roughness matters for the calculus of this chapter. Section 14.7 rested on $dW^2 = dt$: the statement that the squared increments of a Wiener path over $[0,T]$ add up to T. For fractional Brownian motion the n squared increments over a grid of step T/n add up to about

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$n\,(T/n)^{2H} = T^{2H}\,n^{1-2H}$, which grows without bound for $H < 1/2$ and shrinks to zero for $H > 1/2$. There is no $dB^2 = dt$, and Itô's lemma in the form of Section 14.7 does not apply. Nor is the process Markov: by (14.20) the next increment is correlated with every past increment, so the current value does not summarize the path and forecasting needs the whole history.

Example 14.8 Variances that do not add

Take $H = 0.1$, the value Chapter 21 estimates for log-volatility, and unit variance over one year.

Two years. By (14.21) the two-year variance is $2^{0.2} = 1.15$, against 2 for a Wiener process, and the two-year standard deviation is $2^{0.1} = 1.07$ against $\sqrt{2} = 1.41$. Doubling the horizon adds 7% to the uncertainty, not 41%, because the second year's move is correlated at −0.43 with the first and partly undoes it.

Annualizing daily data. Section 14.6 turned a daily standard deviation into an annual one by multiplying by $\sqrt{252}$. For a process with Hurst parameter H the factor is 252^H instead: $252^{0.1} = 1.74$ at $H = 0.1$, against $252^{0.5} = 15.9$. Applying the square-root rule to such a series would overstate the one-year standard deviation by $252^{0.4} = 9.1$ times. The same arithmetic at $H = 0.9$ understates it by the same factor. The square-root-of-time rule is the $H = 1/2$ case of T^H, and it is only as good as the independence it assumes.

Built from a Wiener process. One standard construction writes the process as a weighted sum of past Wiener increments,

$$B_t^H = \sqrt{2H} \int_0^t (t-s)^{H-1/2}\, dW_s \tag{14.23}$$

The kernel $(t-s)^{H-1/2}$ weights each past shock by its age. At $H = 1/2$ it is one, and the integral is W_t. Below one-half the exponent is negative and the weight on the most recent shocks rises without bound as $s \to t$, which is where the roughness comes from, while the weight on old shocks decays as a power of their age rather than dying off exponentially, which is where the memory comes from. The $\sqrt{2H}$ makes the variance at each date exactly t^{2H}, but this version, started at zero, has stationary increments only approximately; it is the Riemann–Liouville form, a close relative. Mandelbrot and Van Ness's two-sided integral has the exact covariance (14.20) and stationary increments, and nothing in this book turns on the difference. Section 21.8 uses this kernel, with the integral inside an exponential and a volatility-of-volatility in front.

What it can and cannot model. As a model for a traded price it fails, and for a reason the correlation (14.22) makes plain: a process whose next move is predictable from its last can be traded on. With $H > 1/2$ a rise predicts a rise, with $H < 1/2$ a rise predicts a fall, and in either case a strategy that trades on the prediction at ever shorter intervals earns a profit with a risk that shrinks toward zero. Trading gains cannot be defined as Itô integrals against it (in the technical term, it is not a semimartingale), no risk-neutral measure exists, and the apparatus of Chapter 15 onward has nothing to attach to. It is therefore not a candidate to replace (14.4). Mandelbrot proposed it for prices in the 1960s, with $H > 1/2$, and the proposal did not survive. The arbitrage is fragile in practice: Cheridito (2003) shows that a minimal delay between trades removes it, and Guasoni (2006) that proportional transaction costs of any size do, so the process is not absurd as a description of prices; it is merely useless as a pricing model. A "fractional Black–Scholes" formula exists and is sometimes quoted. It is not a price, its implied volatilities look nothing like a market surface, and no strategy replicates the option at that price; Section 8 of [TN21] explains all three.

Volatility is different. Spot volatility is not itself a traded asset, so no arbitrage argument rules out a rough path for it, and nothing forbids modeling it with a process that would be

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arbitrageable as a price. Gatheral, Jaisson, and Rosenbaum (2018) measured the roughness of realized volatility across many assets and found $H \approx 0.1$, far rougher than a Wiener path, and Section 21.8 shows what follows for option prices: the one number reproduces a feature of the volatility surface that two decades of classical models could not, and Example 14.8's arithmetic is the reason.

Simulating it. A Wiener path is simulated one independent increment at a time, and a fractional path cannot be, because the increments are not independent. The exact method writes down the covariance matrix (14.20) on the time grid, takes its Cholesky factor (the lower-triangular matrix L with $LL^\top$ equal to the covariance matrix), and multiplies a vector of standard normal draws by it, which is how Figure 14.5 was made, and why its three paths are the same randomness seen through three kernels. Each path costs the square of the number of steps and the factorization its cube, and Chapter 21's case study on rough volatility explains what replaced the method in production.

SUMMARY

A Markov process is one in which only the current value matters for the future. With independent changes over separate periods, variances add, so uncertainty grows with $\sqrt{T}$ rather than T. That is why volatility is scaled by $\sqrt{\Delta t}$ and why Chapter 13's tree used $u = e^{\sigma\sqrt{\Delta t}}$.

A Wiener process has increments $\Delta W = \varepsilon\sqrt{\Delta t}$ with ε standard normal and non-overlapping increments independent, so $W(T) - W(0) \sim N(0, T)$. Its paths are continuous everywhere and differentiable nowhere, which is why a different calculus is required.

A generalized Wiener process adds a constant drift and variance rate, $dx = a\,dt + b\,dW$; an Itô process lets both depend on the state and time.

Stock prices are modeled by geometric Brownian motion, $dS = \mu S\,dt + \sigma S\,dW$, because returns rather than dollar changes are what scale with the price level, and because the proportional form keeps the price positive. Over short intervals the noise term dominates the drift by an order of magnitude, which is why μ is hard to estimate; the hedge of Section 14.8 removes it from pricing.

Itô's lemma gives the process followed by a function $f(S, t)$. It differs from ordinary calculus by one term, $\frac{1}{2}\sigma^2 S^2\,\partial^2 f/\partial S^2\,dt$, which survives because $dW^2 = dt$. That term is proportional to convexity and reappears as gamma in Chapter 18 and as the convexity adjustment in Chapter 30.

Applying the lemma to $\ln S$ gives $d\ln S = (\mu - \sigma^2/2)dt + \sigma\,dW$, so $\ln S_T$ is normal and S_T is lognormal. The expected price grows at μ while the median grows only at $\mu - \sigma^2/2$, so the gap widens with volatility and a volatile stock finishes below its own expected value more often than not. Confusing the two rates produces real and common errors in return reporting and product design.

Because a derivative and its stock are driven by the same dW, a position long $\partial f/\partial S$ shares and short the derivative carries no randomness, and μ cancels with it: Chapter 13's hedge in continuous time. Under the risk-neutral measure only the drift changes, from μ to r; the volatility carries over and the $-\sigma^2/2$ survives.

The model assumes constant volatility, continuous paths, independent returns, and known parameters. All four fail; knowing which one a given phenomenon violates identifies where in Parts III and IV its repair is found. Fractional Brownian motion is the standard process without independent increments: its variance over T is T^{2H}, so variances no longer add, and below $H = 1/2$ its paths are rougher than a Wiener path. It cannot model a traded price without admitting arbitrage, but it can model volatility, which Chapter 21 does with H far below $1/2$.

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FURTHER READING

The derivation of Itô's lemma is in [TN14], reproduced in Appendix A, which also makes the $dW^2 = dt$ argument precise. Beyond that, Neftci's introduction to the mathematics of financial derivatives bridges this level and a rigorous treatment, and Baxter and Rennie's account is a short exposition of why the change of measure works. Karatzas and Shreve's *Brownian Motion and Stochastic Calculus* is the standard graduate text on stochastic calculus, for a reader who wants the construction of the Itô integral itself rather than the rule that follows from it. For the empirical failures of geometric Brownian motion, Chapter 19's references are the place to start, since they organize the evidence around what the option market prices. Fractional Brownian motion entered finance through Mandelbrot and Van Ness (1968), who introduced the process and the Hurst parameter; Rogers (1997) is the short proof that it admits arbitrage as a price process, and Cheridito (2003) and Guasoni (2006) show the arbitrage vanishing under a minimal trading delay and under transaction costs respectively; the literature on the fractional Black–Scholes formula is cited in [TN21]. Gatheral, Jaisson, and Rosenbaum (2018) is the paper that found the process, with $H \approx 0.1$, in volatility.

PRACTICE QUESTIONS

14.1 A variable's change over one year has standard deviation 5. What is the standard deviation of its change over three years, and over one month? State the property you used.

14.2 A variable follows a generalized Wiener process with drift 3 per year and variance rate 16 per year, starting at 25. Give its distribution after two years and compute the probability that it is negative.

14.3 Explain why $dS = \mu\,dt + \sigma\,dW$ is unsuitable as a model for a stock price, giving two distinct reasons.

14.4 A stock at \$80 has $\mu = 12\%$ and $\sigma = 25\%$. Give the distribution of the proportional change over two weeks, and compare the sizes of the drift and noise terms.

14.5 Using Itô's lemma, find the process followed by $G = S^2$ when S follows (14.4).

14.6 A stock at \$60 has $\mu = 14\%$ and $\sigma = 35\%$. Compute the expected price and the median price after two years, and explain why they differ.

14.7 For the stock in the previous question, find the range within which the price after two years lies with 95% probability.

14.8 Explain in words why the term $\frac{1}{2}\sigma^2 S^2\,\partial^2 f/\partial S^2$ appears in Itô's lemma but not in the ordinary chain rule.

14.9 State the Markov property and explain why it is an assumption about information rather than about volatility. Explain what weak-form market efficiency has to do with it, and give one observed feature of returns that sits uneasily with it.

14.10 A stock has expected return μ. Explain why the expected continuously compounded return per year is $\mu - \sigma^2/2$ rather than μ, and state which of the two a manager quoting the arithmetic average of past annual returns is reporting.

14.11 Explain why Itô's lemma delivers a hedge and not merely a change-of-variable formula. State which term in the lemma the hedge removes, which term survives it, and why the surviving term is what the option is paid for.

14.12 State the risk-neutral process for a stock price. Explain what changes and what does not when moving to it from the real-world process, and why the volatility is untouched while the drift is not.

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FURTHER QUESTIONS

14.13 Example 14.6 found that a stock finishes below its expected value 58% of the time. Derive a general expression for $\Pr(S_T > \mathbb{E}[S_T])$ in terms of σ and T alone, show that it does not depend on μ, and evaluate it for $\sigma\sqrt{T} = 0.2$, 0.6, and 1.0. What happens as volatility rises?

14.14 A fund reports the arithmetic average of its annual returns over ten years as 12%. Explain what an investor's compounded return would have been if the volatility was 30%, state which of the two figures is the relevant one for an investor, and identify the circumstances in which the gap is largest.

14.15 Apply Itô's lemma to the forward price when the underlying pays a continuous dividend yield q, and show that under risk-neutral valuation the forward's drift is again zero. Explain the connection to Section 13.10.

14.16 Section 14.12 notes that volatility clustering violates the Markov property. Construct an argument that this does *not* contradict weak-form market efficiency, and identify precisely what a trader could and could not profit from if returns were unpredictable but their magnitudes were not.

14.17 Section 14.13 states that a change in fractional Brownian motion over any interval of length T has variance T^{2H}. Extend the argument that gives the correlation $2^{2H-1} - 1$ between consecutive yearly increments to the first year's increment and the increment over year $k+1$, and show that the correlation is $\tfrac{1}{2}\left[(k+1)^{2H} - 2k^{2H} + (k-1)^{2H}\right]$. Evaluate it for $k = 1, 5, 20$ at $H = 0.9$, and explain why a decay of order k^{2H-2}, slower than exponential, is the long memory the section names.

14.18 The Itô correction term is proportional to $\partial^2 f / \partial S^2$. Using this, explain without calculation whether the correction raises or lowers the drift of a long call position, and connect your answer to the statement in Chapter 12 that long-volatility positions are hurt by the passage of time.

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Chapter 15
The Black–Scholes–Merton Model

Chapter 13 established the principle: a derivative that can be replicated has a price independent of anyone's beliefs. Chapter 14 supplied the mathematics for carrying that principle into continuous time, and ended one step short of the result. This chapter takes the step.

What emerges is a closed-form formula for European options—the most widely used equation in finance, and the one that made options a mass market rather than a specialist's trade. Its importance is not that it is right. Chapter 19 documents at length that it is not, in a specific and measurable way. Its importance is that it turned option pricing into a calculation with five inputs, of which four are read off a screen and the fifth became a traded quantity in its own right.

Sections 15.1 to 15.3 are the assumptions, the differential equation, and the formula. Sections 15.4 and 15.5 say what its terms mean and how it behaves. Sections 15.6 and 15.7 are the fifth input, volatility, estimated from history and implied from prices; Sections 15.8 to 15.10 are the formula in practice, with dividends, and for American options. Section 15.11 is what the model gets wrong and Section 15.12 where it came from.

Section 15.2 can be read for its conclusions alone; the formula in Section 15.3 is self-contained.

15.1 THE ASSUMPTIONS

The model rests on assumptions that Chapter 14 has already named. They are collected here because knowing them is how one knows when the answer will be wrong.

1. The stock follows geometric Brownian motion with constant μ and σ.
2. Short selling is permitted, with full use of the proceeds.
3. There are no transaction costs or taxes, and securities are perfectly divisible.
4. No dividends are paid over the option's life (relaxed in Section 15.9).
5. There are no arbitrage opportunities.
6. Trading is continuous.
7. The risk-free rate r is constant and the same for borrowing and lending.

Every one of these fails in reality, and the book has already said where. Assumption 1 fails through volatility clustering and jumps (Section 14.12; Chapter 19 measures what they do to option prices). Assumptions 2 and 7 fail in the way Section 5.15 documented at length: borrowing is not at r, and shorting is neither free nor always possible. Assumption 3 is what Chapter 18 is about. Assumption 6 fails whenever a market closes or a stock is suspended.

The list is not a disclaimer; it is a diagnostic tool. When a market price disagrees with the formula, the useful question is which of the seven has broken, because that identifies the repair.

15.2 THE BLACK–SCHOLES–MERTON DIFFERENTIAL EQUATION

Section 14.8 supplies the derivation. Take a stock following

$$dS = \mu S\,dt + \sigma S\,dW$$

and a derivative on it whose price at time t can be expressed as $f(S,t)$, a function of the current stock price and time alone. Itô's lemma gives

$$df = \left(\frac{\partial f}{\partial S}\mu S + \frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2 S^2\right)dt + \frac{\partial f}{\partial S}\sigma S\,dW$$

and a portfolio short one derivative and long $\partial f/\partial S$ shares has no dW term at all. Writing Π for its value,

$$\Pi = -f + \frac{\partial f}{\partial S}S, \qquad d\Pi = \left(-\frac{\partial f}{\partial t} - \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2 S^2\right)dt$$

The portfolio is riskless over dt, so it must earn r, or Section 13.1's arbitrage is available. Setting $d\Pi = r\Pi\,dt$ and rearranging gives the **Black–Scholes–Merton differential equation**:

$$\boxed{\frac{\partial f}{\partial t} + rS\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 f}{\partial S^2} = rf} \tag{15.1}$$

Three observations matter more than the derivation.

μ is absent. The stock's expected return entered through Itô's lemma and canceled when the hedge was imposed. This is Chapter 13's central result (the real-world probability is irrelevant) appearing in continuous time. Two investors who disagree completely about where the stock is going must still agree on the option's value.

The equation is the same for every derivative whose value depends only on S and t. Nothing in the derivation used the payoff, so forwards, calls, puts, digitals, and the barrier options of Chapter 34 all satisfy (15.1). What distinguishes them is the **boundary condition**, the value at $t = T$:

European call: $f = \max(S-K,0)$, forward: $f = S-K$, digital: $f = \mathbf{1}\{S > K\}$

The instrument is the boundary condition, not the equation. The qualification "only on S and t" matters. A payoff that depends on the path (an average, a running maximum) needs an extra state variable and the equation gains a term; an American option satisfies it only in the region where holding beats exercising, which the free boundary of Section 15.10 encloses; and a derivative whose value also moves with volatility or interest rates needs those as variables too. Each of those is a different equation, but the argument that produces it is this one.

The hedge is instantaneous. The portfolio is riskless over dt only, because $\partial f/\partial S$ changes as S and t change. Section 14.8 flagged this and Chapter 18 confronts its cost.

15.3 THE FORMULA

Solving (15.1) subject to the European call's boundary condition is a piece of applied mathematics carried out in [TN15], reproduced in Appendix A. The route taken there is the one Chapter 13 makes natural: rather than solving the equation directly, take the risk-neutral expectation of the payoff.

Under the risk-neutral measure, Section 14.10 gives

$$\ln S_T \sim N\left(\ln S_0 + \left(r - \frac{\sigma^2}{2}\right)T,\ \sigma^2 T\right)$$

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and taking the expected payoff $\max(S_T - K, 0)$ under that distribution and discounting it at r produces the **Black–Scholes–Merton formulas**, which the market calls simply Black–Scholes:

$$\boxed{c = S_0 N(d_1) - K e^{-rT} N(d_2)} \tag{15.2}$$

$$p = K e^{-rT} N(-d_2) - S_0 N(-d_1) \tag{15.3}$$

where $N(\cdot)$ is the standard normal cumulative distribution function and

$$d_1 = \frac{\ln(S_0/K) + \left(r + \sigma^2/2\right) T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T} \tag{15.4}$$

The formula takes five inputs: S_0, K, T, r, and σ. Four are observable; the fifth is not, and Sections 15.6 and 15.7 are about it.

Two routes to the same answer

There are two ways to get from (15.1) to (15.2), and both are worth recognizing because different parts of the book use different ones.

The **differential-equation route** solves (15.1) directly, subject to $f = \max(S - K, 0)$ at T. A change of variables turns it into the heat equation of physics, whose solution has been known since the nineteenth century. This is Black and Scholes's original route, and it is the one that generalizes to the numerical methods of Chapter 20.

The **risk-neutral route** takes the discounted expectation instead:

$$c = e^{-rT} \int_K^\infty (S_T - K)\, g(S_T)\, dS_T$$

where g is the lognormal density of (14.16). Splitting the integrand,

$$c = \underbrace{e^{-rT} \int_K^\infty S_T\, g\, dS_T}_{S_0 N(d_1)} - \underbrace{K e^{-rT} \int_K^\infty g\, dS_T}_{K e^{-rT} N(d_2)}$$

The second integral is $Q(S_T > K)$, the source of $N(d_2)$; the first is an expectation of S_T restricted to the exercise region, the source of $N(d_1)$. [TN15] evaluates both.

This route is the one to remember, because it makes the formula's two terms mean something (Section 15.4 unpacks them), and because it is the route that extends to every payoff in Chapter 34 without modification.

A check on the equation. It is worth confirming that (15.1) really does govern instruments other than options. Take a forward contract with delivery price K, worth $f = S - K e^{-r(T-t)}$ by Section 5.5. Then

$$\frac{\partial f}{\partial t} = -rK e^{-r(T-t)}, \qquad \frac{\partial f}{\partial S} = 1, \qquad \frac{\partial^2 f}{\partial S^2} = 0$$

and substituting into the left-hand side of (15.1),

$$-rK e^{-r(T-t)} + rS \times 1 + 0 = r\left(S - K e^{-r(T-t)}\right) = rf$$

The equation is satisfied identically. The forward contract's value from Chapter 5, derived there from a cash-and-carry argument with no model at all, is a solution of the Black–Scholes–Merton equation, which it must be, since both rest on the same no-arbitrage principle.

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Example 15.1 Pricing a call and a put

A non-dividend stock trades at \$58. Value six-month options struck at \$55 with $r = 4\%$ and $\sigma = 24\%$.

First $\sigma\sqrt{T} = 0.24\sqrt{0.5} = 0.16971$, and

$$d_1 = \frac{\ln(58/55) + (0.04 + 0.0288) \times 0.5}{0.16971} = \frac{0.053110 + 0.034400}{0.16971} = 0.5157$$

$$d_2 = 0.5157 - 0.1697 = 0.3460$$

From normal tables, $N(0.5157) = 0.69697$ and $N(0.3460) = 0.6353$. Then

$$c = 58 \times 0.69697 - 55e^{-0.02} \times 0.6353 = 40.42 - 34.25 = \$6.17$$

and for the put, $N(-d_1) = 0.30303$, $N(-d_2) = 0.3647$:

$$p = 55e^{-0.02} \times 0.3647 - 58 \times 0.30303 = 19.66 - 17.58 = \$2.08$$

Always check parity. From Section 11.4, $c - p$ must equal $S_0 - Ke^{-rT}$:

$$6.1731 - 2.0840 = 4.0891, \qquad 58 - 55e^{-0.02} = 58 - 53.9109 = 4.0891$$

They agree exactly, and they must: the formulas satisfy parity identically, because $N(d)+N(-d) = 1$ for every d. A discrepancy in a spreadsheet is an arithmetic error, not a modeling question, and this is the fastest test of any implementation.

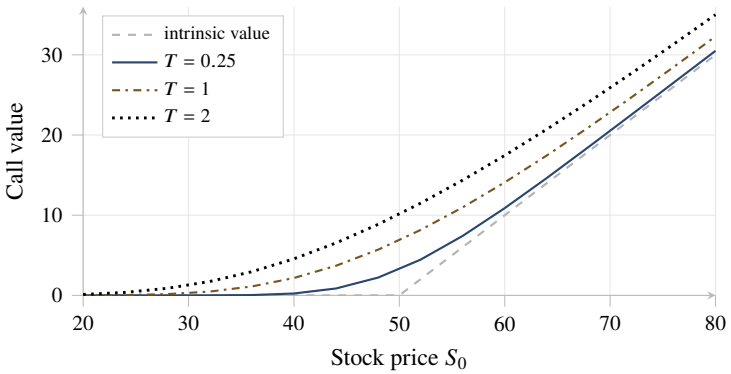


Figure 15.1: Call value against the stock price. $K = 50$, $r = 4\%$, $\sigma = 30\%$.

Figure 15.1 plots (15.2) against S_0 for three maturities. This is Figure 10.2 with a formula behind it: the curves lie above intrinsic value by the time value, the gap is widest near the money, and it shrinks toward zero as $T \to 0$. The model has not produced a new shape—it has produced a number for a shape Chapter 10 already described.

15.4 WHAT $N(d_1)$ AND $N(d_2)$ MEAN

The formula is more memorable, and more useful, once its terms are read rather than computed.

Rewrite (15.2) as

$$c = e^{-rT}\Big[\underbrace{S_0 e^{rT} N(d_1)}_{\text{expected stock received}} - \underbrace{K\,N(d_2)}_{\text{expected strike paid}}\Big]$$

Under the risk-neutral measure, the terms read as follows:

$N(d_2)$ **is the probability that the option is exercised.** It is $Q(S_T > K)$, the risk-neutral probability of finishing in the money. In Example 15.1, $N(d_2) = 0.6353$, so the market prices a 64% chance of exercise. This is not the real-world probability, for the reason Chapter 13 gave:

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Q weights states so that the stock earns r, not μ. A stock with a high expected return has a higher true probability of exercise than $N(d_2)$ suggests.

$N(d_1)$ **is the option's delta**, the hedge ratio $\partial c/\partial S$ that Sections 13.1 and 14.8 built the entire theory around. In the example it is 0.6970, so replicating the call requires holding 0.6970 shares per option written. Chapter 18 takes this as its starting point.

$S_0N(d_1)$ **is the present value of receiving the stock in the states where the option is exercised.** It is not S_0 times the exercise probability. The option is exercised only in the high-price states, those with $S_T > K$, so the stock's average value across those states exceeds its average over all states, the forward price S_0e^{rT}; $N(d_1)$ is $N(d_2)$ scaled up by that ratio, which is why $N(d_1) > N(d_2)$.

That last point resolves a persistent confusion. Only $N(d_2)$ is the exercise probability under the risk-neutral measure; $N(d_1)$ is the exercise probability under the measure that uses the stock as numeraire ([TN15]; Chapter 28).

15.5 PROPERTIES OF THE FORMULA

Any formula should be tested at the edges. Equation (15.2) passes every test, and each reproduces a result Chapter 11 obtained *without* a model.

Deep in the money. As $S_0 \to \infty$, both d_1 and $d_2 \to \infty$, so $N(d_1), N(d_2) \to 1$ and

$$c \to S_0 - Ke^{-rT}$$

The call becomes a forward: exercise is certain, so the holder is committed to buy at K.

Deep out of the money. As $S_0 \to 0$, both arguments $\to -\infty$ and $c \to 0$.

Zero volatility. As $\sigma \to 0$, the option's outcome is certain. Then $d_1, d_2 \to \pm\infty$ according to whether S_0 exceeds Ke^{-rT}, and

$$c \to \max\left(S_0 - Ke^{-rT},\, 0\right)$$

which is exactly the lower bound (11.1) of Section 11.2.

Infinite volatility. As $\sigma \to \infty$, $d_1 \to +\infty$ while $d_2 \to -\infty$, so $N(d_1) \to 1$, $N(d_2) \to 0$, and

$$c \to S_0$$

the upper bound of Section 11.2. The formula cannot be made to violate the bounds however the inputs are abused.

At expiration. As $T \to 0$, $c \to \max(S_0 - K, 0)$.

The put runs the same checks in mirror image. As $S_0 \to \infty$, both $N(-d_1)$ and $N(-d_2)$ vanish and $p \to 0$; as $S_0 \to 0$ they tend to one and $p \to Ke^{-rT} - S_0 \to Ke^{-rT}$, the put's upper bound of Section 11.2, which is also its value as $\sigma \to \infty$. As $\sigma \to 0$, $p \to \max(Ke^{-rT} - S_0, 0)$, the put's lower bound (11.2), and as $T \to 0$, $p \to \max(K - S_0, 0)$. None of these needs a separate calculation: since (15.3) follows from (15.2) by parity, each put limit is the corresponding call limit plus $Ke^{-rT} - S_0$.

The agreement with Chapter 11 is not a coincidence. Those bounds followed from no-arbitrage alone; this formula followed from no-arbitrage *plus* a distributional assumption. A model that added an assumption and thereby violated a conclusion reachable without it would be internally inconsistent. Extra assumptions may narrow a range of admissible prices; they must never move it.

Computing the normal distribution function

Equation (15.2) requires $N(x)$, which has no elementary closed form. Pricing libraries compute it as $\operatorname{erfc}(-x/\sqrt{2})/2$ from the double-precision complementary error function, to machine

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precision. By hand or in a spreadsheet a polynomial serves; a standard one, accurate to 7.5×10^{-8}, sets $\gamma = 0.2316419$ and $k = 1/(1 + \gamma x)$ for $x \geq 0$, then

$$N(x) \approx 1 - \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \left(a_1 k + a_2 k^2 + a_3 k^3 + a_4 k^4 + a_5 k^5 \right)$$

with $a_1 = 0.319381530$, $a_2 = -0.356563782$, $a_3 = 1.781477937$, $a_4 = -1.821255978$, $a_5 = 1.330274429$. For negative arguments, $N(-x) = 1 - N(x)$.

That accuracy is far beyond what pricing near the money requires (a bid–ask spread is typically five orders of magnitude wider), but not in the far wings, where $N(-d)$ is around 10^{-6}, nor when implied volatility is computed by iteration, where an error in N is divided by the option's sensitivity to volatility, which is small far from the money.

15.6 ESTIMATING VOLATILITY FROM HISTORY

Volatility must be estimated, and the two ways to do it answer different questions.

The historical estimator. Take $n + 1$ successive closing prices $S_0, S_1, \ldots, S_n$ and form the log returns $u_i = \ln(S_i/S_{i-1})$. The sample standard deviation

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (u_i - \bar{u})^2} \tag{15.5}$$

where $\bar{u}$ is the mean of the u_i, estimates volatility per observation interval, and is annualized by multiplying by $\sqrt{252}$ for daily data, $\sqrt{52}$ for weekly, $\sqrt{12}$ for monthly.

Table 15.1: Ten daily returns.

Day	Close	u_i	Day	Close	u_i
0	50.00	—	6	51.10	+0.013793
1	50.75	+0.014889	7	51.85	+0.014570
2	50.20	−0.010897	8	51.60	−0.004833
3	51.30	+0.021676	9	52.30	+0.013475
4	51.05	−0.004885	10	52.00	−0.005753
5	50.40	−0.012814			

Example 15.2 Computing a historical volatility

From Table 15.1, $\sum u_i = 0.039221$ and $\sum u_i^2 = 0.00163889$. Then

$$s^2 = \frac{1}{9} \left[0.00163889 - \frac{(0.039221)^2}{10} \right] = 0.000165007$$

so $s = 0.012846$ per day, and

$$\sigma = 0.012846\sqrt{252} = 0.012846 \times 15.8745 = 20.4\%$$

Three cautions apply.

Ten observations is far too few. The standard error of a volatility estimate is approximately $\sigma/\sqrt{2n}$, here $20.4/\sqrt{20} = 4.6$ percentage points. A two-standard-error band runs from about 11% to 30%, which is not a usable input. Ninety to 180 observations is the usual compromise: long enough to reduce the error, short enough that volatility has not yet changed.

Trading days, not calendar days. Volatility accumulates while markets are open, not while time passes: French and Roll (1986) found the variance over a weekend far below three days' worth. So the annualization factor is $\sqrt{252}$; using 365 would overstate volatility by about 20%.

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The mean is usually dropped. Over short intervals the drift is negligible next to the noise (Example 14.3 showed it is outweighed about thirteen and a half to one over a week), so setting $\bar{u} = 0$ in (15.5), and dividing by n rather than $n - 1$ since no mean is now estimated, typically improves the estimate by removing a noisily estimated parameter. Here it gives $s = 0.012802$, changing the annualized figure by 0.1 percentage points.

What an error in volatility costs

The standard error above is easy to accept and easy to underrate; Table 15.2 prices it.

Table 15.2: Value of the call of Example 15.1 at different volatilities.

σ	Call value	Difference	Percent
14%	4.8292	−1.3440	−21.8%
19%	5.4736	−0.6995	−11.3%
24%	6.1731	—	—
29%	6.9003	+0.7272	+11.8%
34%	7.6428	+1.4697	+23.8%

The outer rows are roughly two standard errors of a ten-observation estimate on either side of 24% (the same $\pm 2\sigma/\sqrt{2n}$ band Example 15.2 computed, here about ±10 points): a 14% to 34% range for volatility corresponds to a \$4.83 to \$7.64 range for the option. The historical estimate from ten observations does not determine the price to better than about ±20%.

Two conclusions follow. Historical volatility needs far more data than a casual calculation suggests. And the relationship is close to linear over this range, which is the sensitivity Chapter 18 names vega and which Section 15.7 exploits to invert the formula.

The deeper objection to historical volatility is not statistical. It is that the estimator answers the wrong question: it measures what the stock *did*, and an option's value depends on what it *will do*.

15.7 IMPLIED VOLATILITY

The alternative inverts the problem. Take the option's market price as given and ask what volatility makes (15.2) reproduce it. That number is the **implied volatility**.

The inversion is well posed because a call's value is increasing in σ, as Section 11.1 argued without a model, and strictly so in the formula, whose vega is positive for every $\sigma > 0$; Figure 15.2 shows it for the call of Example 15.3. A strictly increasing function takes each value at most once, so an implied volatility is unique, and it exists whenever the price lies strictly between the bounds of Section 11.2. No closed form exists, so it is found numerically, which on modern hardware is instantaneous.

Figure 15.2 also shows a feature that matters in practice. Below about 5% volatility the curve is indistinguishable from the lower bound $S_0 - Ke^{-rT} = 1.149$: the call is in the money, and with so little volatility it almost never finishes out of the money, so extra volatility adds almost nothing. The function is still strictly increasing, but it is nearly flat there, so a small error in the price moves the implied volatility a long way. The put at the same strike has the same vega, so switching to it does not steepen the curve; what it changes is the price being read. Most of the in-the-money call's price is the lower bound, and only the remainder carries information about volatility, whereas the out-of-the-money put's price is all time value and usually trades on a narrower spread. That is why implied volatility at each strike is read from the out-of-the-money option, here the put, which by parity implies the same volatility as the call.

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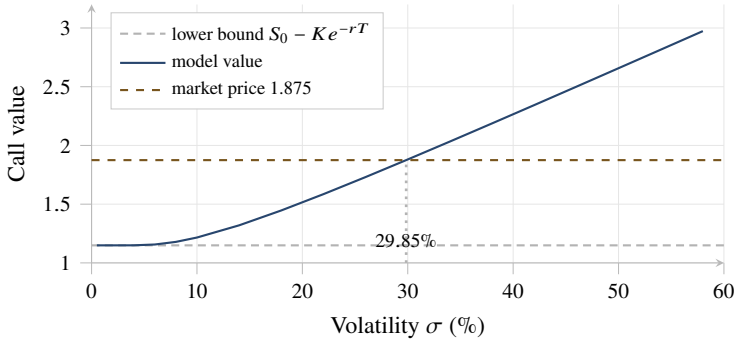


Figure 15.2: Call value against volatility, and the implied volatility. $S_0 = 21$, $K = 20$, $r = 3\%$, $T = 0.25$.

Example 15.3 Backing out an implied volatility

A three-month call on a \$21 stock, struck at \$20 with $r = 3\%$, trades at \$1.875.

Try $\sigma = 20\%$: the formula gives \$1.5148, too low. Try 30%: \$1.8807, slightly too high. Try 29%: \$1.8429. Narrowing further gives

$$\sigma_{\text{implied}} = 29.85\%$$

The market price is consistent with 29.85% volatility over the next three months. Had this stock's own history given a figure near 20%, as it did for the stock of Example 15.2, two readings would be possible: the market anticipates something the past does not contain (an earnings announcement, a pending decision), or option buyers are paying the **volatility risk premium**, the amount by which implied volatility tends to exceed the volatility later realized, which Chapter 19 documents. Usually both are true.

No one used (15.2) to compute a price here; the price was given, and the formula converted it into a volatility. That is what it is mostly used for in practice, and Section 10.6 already reported the consequence: dealers quote in volatility because volatility is the coordinate in which prices are stable.

The formula as a language. This reverses the usual account of what the formula is for. As a device for computing prices from a volatility forecast, it is unreliable, because volatility is neither constant nor reliably forecastable over an option's life. As a device for converting between prices and volatilities (a common language in which options of different strikes and maturities can be compared), it is indispensable, and its failures are informative rather than fatal.

Chapter 19 shows that implied volatility varies systematically with strike and maturity, so the model is wrong; the market uses it anyway, because a wrong but agreed transformation is still a usable quoting convention. The VIX of Chapters 19 and 22 carries the same idea to an index: one volatility number read from a strip of option prices.

15.8 USING THE FORMULA IN PRACTICE

Three conventions decide whether two people pricing the same option agree.

Which interest rate? The model says "the" risk-free rate, and Section 5.15 explained why no single such rate exists: which one a participant faces depends on who they are and what they can pledge. In practice, the rate used is the one at which the hedge can be financed over the option's life, which for a dealer's OTC trade, cleared or collateralized under a CSA, is the collateral rate: the OIS rate of Chapter 4. For listed options, desks commonly take r and q

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(with any borrow cost) jointly from the implied forward of Section 11.6 and the box-spread rate of Section 11.7. Using a government yield instead is a small error for short-dated equity options and a large one for long-dated rate derivatives, which is why Chapters 7 and 8 built the discounting apparatus they did.

Which maturity? Section 15.6 annualized volatility using *trading* days, and the two clocks must be kept apart: variance accrues on trading days, interest on calendar days. An option expiring in 30 calendar days has standard deviation $\sigma\sqrt{21/252}$ over its life but discount factor $e^{-30r/365}$, so d_1 carries rT_{cal} beside $\sigma^2 T_{\text{trd}}/2$. Mixing the conventions is a common implementation error. On average the two agree, since $252/365 \approx 21/30$, but for an option with a week or two to run the placement of weekends and holidays inside its life moves the variance by several percent. Some desks go further and weight days unequally (Chapter 23's business time: a day containing an earnings announcement carries more variance than a quiet Tuesday), which is a refinement of the same idea.

Which volatility? This is the substantive one. The formula assumes a single σ governs the stock, so any option on it should use the same number. The market disagrees: implied volatilities differ across strikes and maturities, as Section 15.11 notes and Chapter 19 documents.

The working resolution is to abandon the pretense of one volatility and use, for each option, the implied volatility of that option, interpolated from the traded surface if the exact strike and maturity are not quoted. The formula is then a quoting convention, strike by strike, rather than a model of the stock. A desk that prices a book with a single volatility is not using Black–Scholes conservatively; it is using a model the market has already rejected.

15.9 DIVIDENDS

Assumption 4 can be relaxed in two ways, according to how dividends are described.

A continuous dividend yield. If the asset pays a known yield q, its holder receives q and the option holder does not. Section 5.4 showed the effect: the asset behaves like one worth $S_0 e^{-qT}$ today. Replacing S_0 by $S_0 e^{-qT}$ throughout gives

$$c = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2) \tag{15.6}$$

$$d_1 = \frac{\ln(S_0/K) + \left(r - q + \sigma^2/2\right) T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}$$

This single substitution covers stock indices (q = dividend yield), currencies ($q = r_f$, the foreign rate), and futures ($q = r$), and Chapter 16 develops each; Example 15.4 applies it to an index.

The differential equation changes as well, and the change explains the substitution. The shares in the hedge portfolio now pay dividends of

$$qS\frac{\partial f}{\partial S}\,dt$$

over each interval, which the riskless return must include, so the argument of Section 15.2 gives

$$\frac{\partial f}{\partial t} + (r - q)S\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} = rf \tag{15.7}$$

The asset's risk-neutral drift is $r - q$ while discounting is still at r: the option holder forgoes the yield, so the price the option sees grows more slowly than the riskless rate. Solving (15.7)

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with the call's boundary condition gives (15.6), and setting $q = r$ removes the first-derivative term altogether, which is the form Chapter 16 uses for options on futures.

Example 15.4 An index option

An index stands at 1,500 with a 2.5% dividend yield. Value a six-month call struck at 1,550 with $r = 4\%$ and $\sigma = 18\%$.

$$d_1 = \frac{\ln(1500/1550) + (0.04 - 0.025 + 0.0162) \times 0.5}{0.18\sqrt{0.5}} = -0.135056, \qquad d_2 = -0.262335$$

$$c = 1500e^{-0.0125} \times 0.446284 - 1550e^{-0.02} \times 0.396532 = 661.11 - 602.45 = 58.66$$

At the \$100 multiplier of Section 10.3, one contract costs \$5,866.

Here $N(d_2) = 0.3965$: the market prices a 40% risk-neutral chance that the index gains more than 3.3% in six months.

Known cash dividends. For an individual stock the dividend is a known amount on a known date, not a yield. The standard treatment subtracts the present value of the dividends payable during the option's life and applies (15.2) to what remains (Example 15.5).

Example 15.5 A call on a dividend-paying stock

A stock trades at \$50 and will pay \$0.80 in two months and \$0.80 in five months. Value a six-month call struck at \$50 with $r = 4\%$ and $\sigma = 30\%$.

The present value of the dividends is

$$0.80e^{-0.04(2/12)} + 0.80e^{-0.04(5/12)} = 0.7947 + 0.7868 = \$1.5815$$

so the adjusted stock price is $S^* = 50 - 1.5815 = \$48.4185$. Applying (15.2) to S^* gives

$$c = \$3.83$$

against \$4.70 for an otherwise identical option on a stock paying nothing, so the dividends have cost the call 18.5% of its value.

The volatility in this calculation is that of S^*, not of S, and the two differ because S^* is a smaller number carrying the same absolute risk. For a modest yield the difference is negligible; for a high yield over a long horizon it is not. [TN15] quantifies it.

15.10 AMERICAN OPTIONS

Equations (15.2) and (15.3) price *European* options. Chapter 11 established when that matters.

American calls on non-dividend stocks need nothing: Section 11.8 proved early exercise is never optimal, so $C = c$ and (15.2) is exact.

American puts have no closed-form solution, and this is not a gap awaiting a cleverer derivation—it is a consequence of the problem's structure. The holder's exercise decision must be made at every date before expiration, so the boundary condition of Section 15.2 is not fixed at T but applies along a **free boundary** (the critical stock price at each date below which immediate exercise is optimal), whose location is itself part of the answer. Equation (15.1) still holds; what is missing is a closed form for where it stops holding.

Three responses are used, in ascending order of cost.

Use the European value as a lower bound. Since $P \geq p$ always (Section 11.9), the formula gives a floor. How slack the floor is depends on how deep in the money the option is and on the level of rates. The two-year put of Section 13.9's convergence figure is worth 6.7601 as a European, the limit the tree settles to, and 7.4719 as an American on a fine tree, an early-exercise premium of 10.5%, far too large to ignore. For a one-month out-of-the-money put on the same

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stock it is under a percent.

Use an analytic approximation. Several exist. The best known, the Barone-Adesi–Whaley quadratic approximation, drops a small term from the early-exercise premium's equation so that it solves in closed form, then finds the boundary by iteration. Such approximations are fast and typically accurate to a few tenths of a percent, inside the bid–ask spread for most listed options.

Use a lattice. Chapter 13's backward induction handles the free boundary exactly by testing exercise at every node, and Chapter 20 makes it efficient. This, or the finite-difference grid of Chapter 20, is what most dealers' pricing systems use, and a large part of why the formula has not retired the tree.

American calls on dividend-paying stocks may, in a frictionless market, be exercised early, but only immediately before an ex-dividend date (Section 11.10). That restriction permits a serviceable approximation, worked through in Example 15.6.

Example 15.6 Black's approximation

A stock trades at \$52 and pays a \$1.50 dividend in five months. Value an American call struck at \$50 expiring in six months, with $r = 4\%$ and $\sigma = 30\%$.

First check whether early exercise is even possible, using condition (11.13):

$$D = 1.50 \quad \text{against} \quad K\left(1 - e^{-r(T-t_1)}\right) = 50\left(1 - e^{-0.04/12}\right) = \$0.17$$

where $t_1 = 5/12$ is the ex-dividend date. The condition is comfortably satisfied, so exercise just before the ex-dividend date may be optimal.

Black's approximation values the American call as the greater of two European calls: one expiring at T on the dividend-adjusted price, and one expiring just before the ex-dividend date at t_1 on the unadjusted price.

European to T. Subtracting the dividend's present value, $\text{PV}(D) = 1.50e^{-0.04(5/12)} = \$1.4752$, gives $S^* = \$50.5248$, and (15.2) with $T = 0.5$ gives \$5.00.

European to $t_1 = 5/12$. Here the option is assumed exercised before the dividend, so no adjustment applies. With $S_0 = 52$ and $T = 5/12$, (15.2) gives \$5.48.

Black's approximation takes the larger:

$$C \approx \max(5.00,\ 5.48) = \$5.48$$

so, of the two policies, exercising just before the ex-dividend date is worth more than holding to expiration.

The approximation is crude, and it always understates: it chooses between the two policies today, while the holder chooses at the dividend date after seeing the price; that option to decide later is what it omits. Its merit is that it requires two evaluations of a formula rather than a lattice, and with one dividend it is usually within a few cents of the tree value.

15.11 WHAT THE MODEL GETS WRONG

The formula was published in 1973, the options market adopted it within two or three years, and it has been known to be wrong since at least 1987. Both facts are worth taking seriously.

Volatility is not constant across strikes. If the model were right, every option on a stock would imply the same σ. They do not, and the failure can be measured on the book's own data.

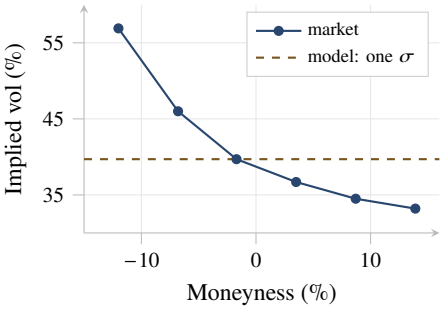
Table 15.3 inverts (15.2) across the option chain of Table 10.2, using the \$1.00 dividend that Section 11.6 extracted from the same quotes.

The implied volatility falls monotonically from 56.9% to 33.2% as the strike rises. This is a single stock, on a single day, at a single maturity, and the model requires all six numbers to be identical. They differ by a factor of 1.7. The calls are American, so part of each price, most at the lowest strikes, is early-exercise premium; inverting a European formula folds it into the implied volatility, which a first reading accepts and Chapter 19 refines.

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Table 15.3: Call implied volatilities from the chain of Table 10.2. $S_0 = 48.30$, $r = 5\%$, discounting over 45 calendar days, variance over 31 trading days; moneyness is $K/S_0 - 1$.

Strike	Moneyness	Call mid	Implied vol
42.50	−12.0%	6.650	56.9%
45.00	−6.8%	4.425	46.0%
47.50	−1.7%	2.670	39.7%
50.00	+3.5%	1.475	36.7%
52.50	+8.7%	0.710	34.5%
55.00	+13.9%	0.305	33.2%



The shape (high implied volatility at low strikes, falling as the strike rises) is the equity skew. It has been a stable feature of index and single-stock options since the crash of October 1987, and it says something specific: the market assigns more probability to large downward moves than the lognormal distribution of (14.11) allows. Chapter 19 is devoted to reading it.

Put–call parity (Section 11.4) holds identically for the Black–Scholes–Merton formulas, so a call and a put at the same strike and maturity must imply the same volatility. If European options do not, the cause is a data problem (a stale quote, a wrong dividend or borrow-fee assumption, a mismatched timestamp) and not a market phenomenon; on American-style listed options a residual gap is the early-exercise premium, which Section 11.6 already met in this chain. Checking it is the first test on any option dataset.

Volatility is not constant across time either. Implied volatility varies with maturity, and that term structure (the pattern of implied volatility across maturities) itself moves over time.

Returns are not lognormal. Real returns have fatter tails than the lognormal distribution behind (15.2), and jumps that no continuous path allows. The skew of Table 15.3 is that disagreement showing up in prices: low strikes are priced for a fatter lower tail than the model allows.

The frictionless assumptions (2, 3, and 7) fail as Section 15.1 noted. The model remains the market's language for the reason Section 15.7 gave: it is used not as a price generator but as an invertible map between prices and a single number, and for that purpose being wrong in a stable, well-understood way is tolerable. The volatility surface of Chapter 19 is not a correction to the model—it is a catalog of exactly how the model is wrong, quoted in the model's own units.

Case Study 15.1 The formula that arrived with the market

The Chicago Board Options Exchange opened on April 26, 1973. Black and Scholes's paper appeared in the *Journal of Political Economy* that May, having been rejected twice before publication; Merton's paper, developed alongside theirs, appeared in the *Bell Journal* the same spring. A market and its pricing model were born within weeks of each other.

Adoption was extraordinarily fast for an academic result. Traders on the CBOE floor were using the formula within two or three years, from about 1975 first on Black's printed sheets of model values and then on a handheld calculator sold pre-programmed with it. By the end of the decade it was the market's common language, and Scholes and Merton received the Nobel Prize in 1997. Black had died in 1995, and the prize is not awarded posthumously.

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What makes the episode instructive is that the model's assumptions were violated on the day it was published, and everyone involved knew it. Stocks pay dividends, volatility moves, markets close overnight, borrowing is not at the risk-free rate. Black said so himself, at length, in a 1988 essay titled "The Holes in Black–Scholes"—a list of the model's defects written by one of its authors, which remains accurate today.

The formula spread anyway, as a quoting convention rather than a pricing model, and no better convention has displaced it in fifty years.

15.12 WHERE THIS CAME FROM

Bachelier (1900) is the beginning, and it is remarkable twice over: he modeled the price itself, not its logarithm, as Brownian motion (the arithmetic Wiener process $dS = \sigma\, dW$ of Section 14.3) five years before Einstein described the physics, and he derived option values from it. The normal model's flaw (prices can go negative) kept it out of use for most of the twentieth century, and it returned in the 2010s when negative rates made it the natural choice for rate options, and again in April 2020, when WTI futures settled below zero and CME moved its crude-oil options to it for four months.

The intervening attempts show what was missing. Samuelson (1965), adopting the geometric Brownian motion Osborne had proposed, got the distribution right, but priced the option by discounting its expected payoff at a rate that had to be assumed. No one could say what that rate should be, and that single gap is what Black and Scholes (1973) closed: a continuously rebalanced hedge removes the risk, so no risk premium is needed and no view about the stock is required. Merton (1973) reached the same formula by a different route in the same year and extended it to dividends and to stochastic rates.

Cox and Ross (1976) then extracted the principle from the formula. If the hedge argument works, the option can be valued as though every asset drifted at the risk-free rate—not because investors are risk-neutral but because the answer does not depend on whether they are. Cox, Ross and Rubinstein (1979), with Rendleman and Bartter (1979) independently in the same year, turned that principle into Chapter 13's tree, and Harrison and Kreps (1979) with Harrison and Pliska (1981) gave the general statement Chapter 28 uses: no arbitrage holds if and only if an equivalent martingale measure exists, a risk-neutral measure of Chapter 13's kind under which every discounted asset price has zero drift.

SUMMARY

The Black–Scholes–Merton equation follows from Chapter 14's observation that a portfolio short one derivative and long $\partial f/\partial S$ shares has no random component. Setting its return to r gives $\partial f/\partial t + rS\,\partial f/\partial S + \frac{1}{2}\sigma^2 S^2\,\partial^2 f/\partial S^2 = rf$. The stock's expected return μ cancels, which is Chapter 13's principle in continuous time, and the equation governs every derivative whose value depends only on S and t: the instrument is the boundary condition, not the equation. A forward contract satisfies it identically, which confirms that Chapter 5 and this chapter rest on the same principle.

Solving it for a European call gives $c = S_0 N(d_1) - Ke^{-rT}N(d_2)$, with the put following from parity. The formulas satisfy put–call parity identically, which is the fastest test of any implementation.

Of the two normal terms, $N(d_2)$ is the risk-neutral probability of exercise, not the real-world one. $N(d_1)$ is the delta, and $S_0 N(d_1)$ is the present value of receiving the stock in the states where the option is exercised.

Volatility is the one input that must be estimated. The historical estimator uses the sample standard deviation of log returns, annualized by $\sqrt{252}$; over a short sample its standard error

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is large, and it answers the wrong question, measuring what the stock did rather than what it will do. Implied volatility inverts the formula against a market price. The inversion is unique because option value is strictly increasing in volatility.

The formula's real role is as a language. It converts prices into a single comparable number, which is why dealers quote in volatility and why a model known to be wrong remains universal.

A continuous dividend yield is handled by replacing S_0 with $S_0 e^{-qT}$ (equivalently, the equation's drift term becomes $(r - q)S\,\partial f/\partial S$), which also covers indices, currencies, and futures. Known cash dividends are subtracted at present value.

American calls on non-dividend stocks are priced exactly by the European formula. American puts need a lattice or an analytic approximation. American calls on dividend-paying stocks can be valued by Black's approximation, the greater of two European values.

The model is wrong in documented ways: volatility varies with strike and maturity, returns have fat tails and jumps, and the frictionless-market assumptions fail. Measured on the book's own option chain, implied volatility differs widely across strikes. Chapter 19 catalogs the first of these in the model's own units.

FURTHER READING

[TN15] derives (15.2) from the risk-neutral expectation and shows the equivalence with solving the differential equation. The original papers, Black and Scholes (1973) and Merton (1973), are worth reading for how differently the two derive the same result: Black and Scholes through a hedged portfolio and the heat equation; Merton through a self-financing replicating portfolio (one that needs no cash added or withdrawn after it is set up) under weaker assumptions; the risk-neutral route this chapter prefers is Cox and Ross (1976). Black's 1988 essay "The Holes in Black–Scholes" is a short account of the model's failures by one of its authors, and its list still applies nearly four decades on.

PRACTICE QUESTIONS

15.1 A non-dividend stock trades at \$30. Value a three-month European call struck at \$29 with $r = 5\%$ and $\sigma = 25\%$. Then value the put and verify put–call parity.

15.2 Explain why the stock's expected return does not appear in (15.1), and state which chapter established the same result in a discrete setting.

15.3 In Example 15.1, state the risk-neutral probability that the option is exercised and explain why this is not the probability an investor should use when forecasting.

15.4 Compute the historical volatility from the last six observations of Table 15.1 and compare it with the estimate from all ten. What does the difference tell you about the reliability of short samples?

15.5 A call on a \$60 stock struck at \$60 with six months to run and $r = 4\%$ trades at \$5.20. Estimate the implied volatility to the nearest half percent by trial and error.

15.6 An index at 2,000 with a 2% dividend yield has three-month calls struck at 2,050 trading at 42. With $r = 3\%$, find the implied volatility.

15.7 A stock at \$40 pays a \$0.60 dividend in three months. Value a five-month European call struck at \$40 with $r = 5\%$ and $\sigma = 28\%$.

15.8 Explain why the European formula prices an American call on a non-dividend stock exactly, but not an American put.

15.9 Verify the limiting behavior of (15.2) as $\sigma \to \infty$ and confirm that it reproduces the upper bound of Section 11.2. Do the same as $\sigma \to 0$.

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15.10 Show that a forward contract satisfies (15.1), and explain what this demonstrates about the relationship between Chapter 5 and this chapter.

15.11 Using Table 15.3, state by what factor the implied volatilities differ across strikes and explain in one sentence what the model requires them to be.

FURTHER QUESTIONS

15.12 Show algebraically that (15.2) and (15.3) satisfy put–call parity for all parameter values, using $N(d) + N(-d) = 1$. Then explain why this makes parity useless as a test of whether the *model* is correct, though useful as a test of whether an *implementation* is.

15.13 Section 15.4 states that $N(d_1) > N(d_2)$ always. Prove this from (15.4), and give the economic reason in one sentence.

15.14 Example 15.3 found an implied volatility of 29.85%; suppose the same stock's own history gave 20.4%, as Example 15.2's did. Construct two distinct explanations consistent with both numbers, describe an observation that would distinguish them, and state which is more likely if the gap persists across all strikes and maturities.

15.15 Black's approximation always understates the American call value. Explain why, identify the choice available to the holder that it omits, and construct a case in which the understatement would be largest.

15.16 Example 15.5 notes that the volatility in the dividend adjustment refers to S^* rather than S. Quantify this: for a stock with a 4% annual yield and a five-year option, estimate the proportional difference between σ_{S^*} and σ_S, and state whether ignoring it overstates or understates the option value.

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Chapter 16
Options on Indices, Currencies, and Futures

Chapter 15 derived the Black–Scholes–Merton formula for options on a non-dividend-paying stock, a small corner of the option market by notional. The largest option markets in the world are written on interest rates, currencies, and futures contracts, and almost none of them involve a share that pays nothing. Currency options are also the first option market in this book that trades mainly over the counter, dealer to client, rather than on an exchange; Section 10.10 set out what that changes.

The extension needed is smaller than it looks. Section 15.9 already stated it: replacing S_0 with $S_0 e^{-qT}$ covers every case, with q interpreted differently in each. This chapter works through what q means in each market, what conventions surround the resulting prices, and one genuinely new structure, the quanto, where an option's payoff is fixed in a currency other than the one its underlying trades in.

It also introduces Black's model, the form the Black–Scholes–Merton formula takes when written on a forward or futures price. That form, not the original, is the market standard almost everywhere outside single-stock equity options, and Chapter 29's caps and swaptions are written in it or in its normal-model variant (Section 29.8).

Section 16.1 states the one substitution. Sections 16.2 to 16.4 apply it to index options and to currency options, including how the FX market actually quotes. Sections 16.5 and 16.6 are Black's model, on futures and beyond them, and Section 16.7 is the quanto. Sections 16.8 and 16.9 close with the two practical questions: where the forward comes from, and which convention each market uses.

16.1 ONE SUBSTITUTION, THREE MARKETS

The observation that unifies the chapter is that a *yield* on the underlying and a *dividend* on a stock enter the pricing problem identically. Both mean that holding the asset produces income the option holder does not receive, so the asset behaves as though it were worth $S_0 e^{-qT}$ today.

With that substitution, equations (15.2)–(15.4) become

$$c = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2) \tag{16.1}$$

$$p = K e^{-rT} N(-d_2) - S_0 e^{-qT} N(-d_1) \tag{16.2}$$

$$d_1 = \frac{\ln(S_0/K) + \left(r - q + \sigma^2/2\right) T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}$$

Table 16.1 gives the interpretation of q in each market. Nothing else changes.

The last row gets a section of its own: Section 16.5 shows that setting $q = r$ gives Black's model.

Table 16.1: What q means in equation (16.1).

Underlying	q	Reason
Non-dividend stock	0	Holding it produces no income
Dividend-paying stock or index	Dividend yield	Holder receives dividends; option holder does not
Currency	Foreign rate r_f	Foreign currency earns interest abroad
Futures contract	r	Costs nothing to hold, so it grows at zero under Q

16.2 OPTIONS ON STOCK INDICES

For an index, q is the dividend yield of the constituent stocks over the option's life: an average, weighted by index weight, and known reasonably well in advance because dividend policies change slowly.

Section 15.9 priced a straightforward index call. The more instructive application is the one Section 12.10 introduced without a model: using index puts to insure a portfolio.

Example 16.1 Insuring a portfolio, and why the strike is not obvious

A fund holds \$30 million tracking an index at 1,500, with a dividend yield of 2%. The manager wants a floor at \$28.5 million (a 5% loss) over six months, with $r = 4\%$ and $\sigma = 18\%$.

If the portfolio mirrors the index, the natural instrument is a put struck 5% below the current level, at 1,425. The contract multiplier is \$100, so

$$\text{contracts} = \frac{30{,}000{,}000}{1{,}500 \times 100} = 200$$

and from (16.2) with $K = 1{,}425$, the put is worth 37.27 index points, costing

$$200 \times 100 \times 37.27 = \$745{,}400$$

or 2.48% of the portfolio for six months.

A floor at 1,425 does not produce a floor at \$28.5 million. The portfolio also collects dividends over the six months (about 1% at a 2% annual yield), so at index 1,425 the fund holds roughly $30\text{m} \times 0.95 + 0.30\text{m} = \$28.8$ million before the put pays anything, and about \$28.05 million net of the premium, below the floor that was wanted. Separately, the put's payoff arrives at expiration while the insurance was wanted throughout.

A manager who wants a genuine floor on *terminal wealth* must solve for the strike that delivers it, accounting for dividends received and the premium paid. If the portfolio is not an index tracker its beta enters as well; the next paragraph works that out. The mechanical calculation is easy and the specification of what is being insured is not, which is where most portfolio-insurance mandates go wrong.

When the portfolio is not the index. A portfolio with beta β against the index does not fall 5% when the index does, so the strike that protects it has to be found through the beta. Write V for the portfolio's value (Section 3.6's V_A), and keep S for the index the put is written on. The capital asset pricing model (CAPM) relates the two total returns over the option's life,

$$\frac{V_T}{V_0} + q_V T - e^{rT} = \beta \left(\frac{S_T}{S_0} + qT - e^{rT} \right)$$

and solving for the index level gives

$$\frac{S_T}{S_0} = \frac{1}{\beta} \left(\frac{V_T}{V_0} + q_V T \right) - qT + \left(1 - \frac{1}{\beta} \right) e^{rT} \tag{16.3}$$

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where q_V and q are the dividend yields of the portfolio and the index, and S_T, the index level at which the portfolio reaches its floor, is the put strike. Take the fund of Example 16.1 with $\beta = 1.5$, dividend yields of 2% on both portfolio and index, and $r = 4\%$. A \$28.5 million floor on terminal wealth, dividends included but before the premium, is $V_T/V_0 + q_V T = 0.95$, so

$$\frac{S_T}{S_0} = \frac{0.95}{1.5} - 0.01 + \left(1 - \frac{1}{1.5}\right) e^{0.02} = 0.9634$$

and the puts are struck at $0.9634 \times 1{,}500 = 1{,}445$, not 1,425: a portfolio with beta 1.5 has lost 5%, dividends included, by the time the index level is down only 3.7%. The number of contracts scales with beta as well, $1.5 \times 200 = 300$, exactly as Section 3.6's hedge ratio did. At $\beta = 1$ equation (16.3) reduces to $S_T/S_0 = V_T/V_0 + (q_V - q)T$, the dividend point made in the example.

Example 16.1 bought a put, but through the mid-1980s a large body of institutional money instead manufactured one, a technique marketed as **portfolio insurance**: selling index futures as the market fell, in the proportion a put's delta prescribes (Section 18.11). Section 40.2 tells how that played out on October 19, 1987. The equity skew dates from that day, and Chapter 19's case study "The day the skew appeared" asks why.

Who trades index options

Index options are the largest equity option market by notional, and the participants divide cleanly.

Protection buyers are the users of Example 16.1: pension funds, insurers, and asset managers buying puts or put spreads against a portfolio. Section 12.10 established the arithmetic that governs them: continuous protection struck 5% out of the money costs roughly 5% a year, so almost everyone buys a spread or a collar instead.

Overwriters systematically sell calls against index holdings. Section 12.12's case study on derivative-income funds described the largest such program, and Example 10.7 gave the shape: an uncertain equity return converted into a capped one.

Volatility traders take positions in index volatility itself, delta-hedged as Section 12.4 described. They are the natural counterparty to both groups above, and Part IV treats them.

Dispersion traders exploit the relationship between index volatility and the volatilities of its constituents. An index is less volatile than the average of its members because they are imperfectly correlated (Section 14.11's two-asset volatility in miniature), so selling index volatility and buying single-stock volatility is a position in correlation. Chapter 22 develops it.

The structure of that list explains a persistent feature of the market. Protection buyers and overwriters both create one-directional demand, for puts and against calls respectively, and dealers absorb the imbalance. The equity skew is partly the price of that imbalance, which is a market-structure explanation to set alongside the crash-risk explanation that dates the skew to 1987. Chapter 19 weighs them.

16.3 CURRENCY OPTIONS

A foreign currency is an asset that pays interest at the foreign rate, so $q = r_f$. Substituting into (16.1) gives what is often called the Garman–Kohlhagen formula, though it is the same equation:

$$c = S_0 e^{-r_f T} N(d_1) - K e^{-rT} N(d_2) \tag{16.4}$$

with S_0 the spot exchange rate expressed as units of domestic currency per unit of foreign.

That last phrase carries more difficulty than the formula. Every quantity in an FX option must be labeled with which currency it is measured in, and errors of this kind are a common source of loss in currency options among people who understand the mathematics perfectly.

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Example 16.2 A euro call

A US firm expects to pay €1 million in six months and wants protection against a rising euro. Spot is 1.0850 dollars per euro, US rates are 4%, euro rates 2.5%, and volatility is 8%. Value a call on the euro struck at 1.1000.

Here the domestic currency is the dollar, so $r = 4\%$, $q = r_f = 2.5\%$, $S_0 = 1.0850$, $K = 1.1000$.

$$d_1 = \frac{\ln(1.0850/1.1000) + \left(0.04 - 0.025 + 0.08^2/2\right) \times 0.5}{0.08\sqrt{0.5}} = -0.0819, \qquad d_2 = -0.1384$$

$$c = 1.0850e^{-0.0125} \times 0.467383 - 1.1000e^{-0.02} \times 0.444954 = 0.50081 - 0.47976 = 0.02105$$

On a €1 million notional the premium is $21,053.

Check the forward, since it is the reference every FX desk uses:

$$F = 1.0850\, e^{(0.04-0.025)\times 0.5} = 1.09317$$

The strike of 1.1000 is above the forward, so the call is out of the money *forward*, and indeed $d_2 < 0$, a risk-neutral exercise probability of 44.5%.

The premium is in dollars per euro of notional; Table 16.2 sets out the three other ways the same trade is quoted.

Who buys currency options, and what they buy

Currency options are bought mostly by people with a currency exposure they did not choose, and the structures reflect that.

Importers and exporters face the exposure of Example 16.2: a known foreign-currency payment or receipt at a known date. A plain option caps the cost, but the premium is a visible expense that must be defended, and Section 12.10 explained why that matters more than it should. The usual response is a **range forward**, also called a zero-cost collar or cylinder, which Example 16.3 constructs.

Example 16.3 A zero-cost range forward

The importer of Example 16.2 owes €1 million in six months and does not want to pay a premium. They buy a euro call struck at 1.1300 and sell a euro put struck so the premiums offset.

The call at 1.1300 is worth 0.01063 dollars per euro. Solving for the put strike with the same value gives $K_p = 1.0587$, and the structure costs nothing at inception.

The result is a band:

Euro at expiration	What happens	Cost of €1m
Below 1.0587	Put exercised against them	$1,058,748
1.0587 to 1.1300	Both expire; buy at spot	$1,058,748–$1,130,000
Above 1.1300	They exercise the call	$1,130,000

Compare the outright forward, which locks in $1,093,168 with certainty. The collar gives up the chance of paying less than $1,058,748 in exchange for capping the cost at $1,130,000, and costs nothing today.

The trade-off is the one Section 12.10 identified for equity index collars, with one difference. In equities the skew forces an asymmetric band: the put bought is further from spot than the call sold. Here, at a flat 8% volatility, the band is nearly symmetric around the forward (3.1% below it, 3.4% above); with a smile like Figure 16.1's, where euro puts carry the higher volatility, the put the importer sells is worth more and can be struck further out. The shape of the collar a client can be offered is a direct read on the shape of the smile, which is why Section 16.4's quoting conventions matter commercially and not just descriptively.

Cross-border investors hedge the currency exposure that comes with a foreign asset.

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Options are used rather than forwards when the size of the foreign holding is uncertain: an equity portfolio's value moves, so the required hedge moves with it, and an option's payoff scales in a way a fixed forward does not.

Carry traders buy protection on the funding currency, the low-yield currency borrowed to finance the position. A position long a high-yield currency against a low-yield one earns the differential and loses if the funding currency rallies, which tends to happen abruptly, when many such positions are unwound at once. An out-of-the-money option on the funding currency converts an open-ended tail into a premium, and the demand for it is visible in the risk reversal.

Structured product issuers embed currency options in notes sold to retail investors. The simplest is the dual-currency deposit, in which the investor sells a currency option in exchange for an enhanced deposit rate. Section 12.12's test applies: work out what option the buyer has written, and ask whether they would have written it deliberately.

Macro funds trade currency volatility and the risk reversal outright, as views on an exchange rate rather than hedges of it.

16.4 HOW THE FX MARKET ACTUALLY QUOTES

Currency options are quoted differently from every other option market, and the conventions are worth knowing because Chapter 19 uses them and they are more sensible than they look.

Which currency is the premium in? Table 16.2 sets out the four correct ways to quote the premium of Example 16.2.

Table 16.2: One premium, four quotations. €1m call struck at 1.1000; spot 1.0850.

Convention	Computed as	Value
Domestic per unit foreign	the formula's direct output	0.02105
% of foreign notional	premium / S_0	1.940%
% of domestic notional	premium / K	1.914%
Domestic cash	premium × foreign notional	\$21,053

The first and second differ by whether the premium is paid in dollars or euros, the second and third by which leg's notional is the base, and the fourth is the first multiplied by the notional. Every one is correct and they are not equal, which is why FX confirmations state the premium currency, amount, and settlement date rather than a percentage.

A related trap concerns delta. A euro call is also a dollar put, so a position has two deltas depending on which currency is treated as the asset, and the two are related through the **premium-adjusted delta**: the ordinary delta less the premium expressed in units of the foreign currency, $\Delta - c/S_0$, which is the delta seen by a holder who pays the premium in that currency. Desks additionally distinguish *spot* delta from *forward* delta, and whether the premium has been deducted. The number of ways to be right is large, and the number of ways to be consistent with a counterparty is one.

Volatility, not price. As Section 10.6 described generally, FX options trade on implied volatility. But the FX market goes further: it identifies options by *delta* rather than by strike. A "25-delta put" is the put whose delta is −0.25, whatever strike that turns out to be today.

The reason is that a strike ages badly. What matters for an option's price is how far its strike sits from the forward in standard deviations,

$$\frac{\ln(K/F_0)}{\sigma\sqrt{T}}$$

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and for a fixed strike that distance changes every day: the forward moves with spot, and even if it did not, the shrinking $\sigma\sqrt{T}$ pushes the same strike further from the money in standard-deviation terms as expiration approaches. A strike set at today's forward is at the money *forward* today (the sense Example 16.2 used, which coincides with the spot sense only when $r = r_f$) and is unlikely to be at the money in either sense next month. A 25-delta option is defined by that distance itself, so it is always the same distance out of the money. Delta is a moneyness measure that is automatically rescaled by volatility and time, and quoting in it keeps a surface comparable across days. Figure 19.4 draws exactly this, marking one strike on the smile before and after a move in spot.

Three quotes per maturity. Rather than a volatility for each strike, the FX market quotes three numbers at each maturity, the at-the-money (ATM) volatility, the risk reversal (RR), and the butterfly (BF), from which the whole smile is reconstructed, as Example 16.4 does:

$$\text{ATM} = \text{volatility of the delta-neutral straddle}$$

$$\text{RR}_{25} = \sigma_{\text{25-delta call}} - \sigma_{\text{25-delta put}} \tag{16.5}$$

$$\text{BF}_{25} = \frac{1}{2}\left(\sigma_{\text{25-delta call}} + \sigma_{\text{25-delta put}}\right) - \text{ATM} \tag{16.6}$$

The delta-neutral straddle is struck close to the forward but not at it; long-dated and some emerging-market pairs use the at-the-money-forward straddle instead. Strictly, the quoted butterfly fixes the price of a strangle valued at the single volatility ATM + BF, and (16.5) and (16.6) are the first-order reading desks use for a quick reconstruction.

The risk reversal quote measures the smile's *skew* (whether calls or puts are more expensive in volatility terms), and Section 12.6 showed it is the instrument that trades exactly that. The butterfly quote measures the smile's *curvature*, how much the wings exceed the middle, and the instrument that isolates it is a strangle bought against a straddle in vega-neutral proportions.

This is not a coincidence: the market quotes what it trades, and it trades level, skew, and curvature because those are the three things a smile can do.

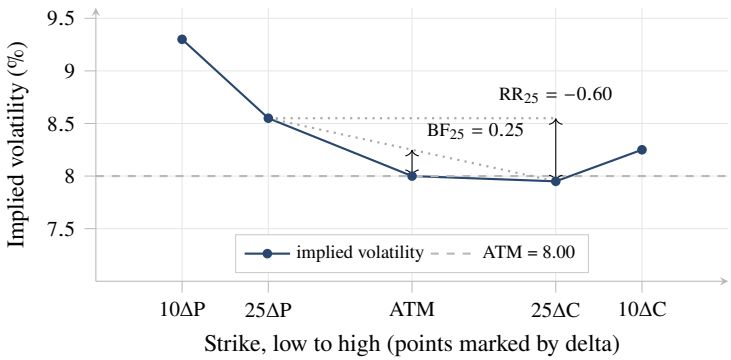


Figure 16.1: Smile built from ATM 8.00, $\text{RR}_{25} = -0.60$, $\text{BF}_{25} = 0.25$; the 10-delta points are extrapolated. Dotted: the 25-delta put level, and the chord joining the two 25-delta points. Illustrative.

Example 16.4 Reconstructing a smile from three numbers

A dealer quotes six-month EUR/USD as ATM 8.00, 25-delta risk reversal −0.60, 25-delta butterfly 0.25.

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Solving (16.5) and (16.6) for the two wing volatilities:

$$\sigma_{25C} = \text{ATM} + \text{BF} + \frac{1}{2}\text{RR} = 8.00 + 0.25 - 0.30 = 7.95\%$$

$$\sigma_{25P} = \text{ATM} + \text{BF} - \frac{1}{2}\text{RR} = 8.00 + 0.25 + 0.30 = 8.55\%$$

The three quotes fix three points on the smile, the at-the-money and the two 25-delta points of Figure 16.1; the figure's 10-delta points are extrapolated. Adding 10-delta quotes fixes five points, which is what most desks use.

Read the risk reversal. It is negative, meaning euro puts are more expensive than euro calls in volatility terms: the market is paying more for protection against a falling euro. In equity markets the skew has this sign essentially always (Section 15.11); in FX it changes sign as sentiment moves, which makes it a genuinely informative price rather than a structural constant.

16.5 OPTIONS ON FUTURES AND BLACK'S MODEL

The last row of Table 16.1 produces the most widely used version of the formula.

A futures contract costs nothing to enter, so in the risk-neutral world its price has zero expected growth; Section 13.10 derived this from the tree, and Section 14.7 from Itô's lemma. Setting $q = r$ in (16.1) makes the $S_0 e^{-qT}$ term equal $F_0 e^{-rT}$, and the formula collapses to

$$c = e^{-rT}\,[F_0 N(d_1) - K N(d_2)] \tag{16.7}$$

$$p = e^{-rT}\,[K N(-d_2) - F_0 N(-d_1)] \tag{16.8}$$

$$d_1 = \frac{\ln(F_0/K) + \sigma^2 T/2}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}$$

This is **Black's model**, and Example 16.5 applies it. Three features make it the market's preferred form.

The interest rate appears only as a discount factor. It does not enter d_1 or d_2 at all. Everything about the underlying is contained in F_0, which is directly observable, so the model requires no view on carry, dividends, convenience yields, or foreign rates, all of which are already embedded in the futures price.

The forward is the natural reference. Setting $K = F_0$ makes $c = p$, which Section 11.4 derived from parity alone. Black's model reproduces it immediately, and this is why the at-the-money-forward straddle is the usual volatility reference point outside FX (Section 16.4).

It generalizes. Anything with an observable forward price can be priced this way, whether or not a futures contract exists; Section 16.6 shows how far that reaches.

Example 16.5 An option on a futures contract

A futures price is 100, with $r = 4\%$, $\sigma = 25\%$, and six months to expiration. Value calls and puts struck at 100.

$$d_1 = \frac{0 + 0.0625 \times 0.5/2}{0.25\sqrt{0.5}} = 0.0884, \qquad d_2 = -0.0884$$

$$c = e^{-0.02}\,[100 \times 0.535216 - 100 \times 0.464784] = 0.9802 \times 7.043 = 6.904$$

and the put comes to the same 6.904, as it must at $K = F_0$.

As a consistency check, note that for a non-dividend stock $F_0 = S_0 e^{rT}$, so Black's model on the futures price and the Black–Scholes–Merton formula on the spot must agree. Taking Example 15.1's option ($S_0 = 58$, $K = 55$, $r = 4\%$, $\sigma = 24\%$, $T = 0.5$) gives $F_0 = 58e^{0.02} = 59.1717$, and (16.7) returns \$6.1731, identical to four decimals. The two are the same formula written in different variables.

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How exercise settles. Exercise does not deliver the commodity, and it does not happen at an intraday price. At CME exercise notices are processed after the close: the holder of a call is given a long futures position opened at the strike, and that position is marked to the exercise day's settlement price like any other. The first mark-to-market therefore pays the holder that settlement price less the strike, and leaves a long futures position at the settlement price. The writer, on assignment, pays the cash and takes the short position.

Together the cash and the position are a futures contract bought at K: the cash is the settlement price less K, and the position, worth nothing once marked, earns only the moves after the close, so it is a new futures position rather than part of the payoff. The book therefore takes the payoff as $\max(F - K, 0)$ with F the settlement price on the day of exercise. The price at which the holder decided to exercise does not enter: the move from it to the settlement is added to a call's payoff and subtracted from a put's, which is one reason many contracts now settle straight to cash. Example 16.6 works through a call and a put.

Example 16.6 Exercising a futures call, and a futures put

A trader holds one call on the December crude oil futures contract struck at \$70.00, on 1,000 barrels. With the futures at \$73.40 during the day, when the call's intrinsic value is

$$1{,}000 \times (73.40 - 70.00) = \$3{,}400,$$

the trader submits an exercise notice. The contract settles that evening at \$73.85, and the notice is processed after the close: a long futures position opened at \$70.00 and marked to \$73.85. The move after the decision adds

$$1{,}000 \times (73.85 - 73.40) = \$450$$

for a total of \$3,850, which is $1{,}000 \times (73.85 - 70.00)$: the exercise day's settlement price less the strike, exactly as (16.7) assumes.

The put is the mirror image. A put on the same contract struck at \$76.00, exercised at the same moment, is worth

$$1{,}000 \times (76.00 - 73.40) = \$2{,}600$$

at \$73.40, but it opens a *short* futures position at \$76.00, marked to \$73.85. The same \$450 the call gained is lost, leaving \$2,150, which is $1{,}000 \times (76.00 - 73.85)$. The move from the intraday price to the settlement price helps one and hurts the other by the same amount, and both holders end with the intrinsic value at the settlement price.

Keeping the futures position leaves the cash and an outright position at \$73.85, long for the call holder and short for the put holder. That is a bet on crude, not a larger payoff, and it is the choice an assigned writer faces in reverse.

One settlement feature, not universal, does change the value. Where the exchange margins the option itself (**futures-style margining**, used at Eurex and ICE Futures Europe), the premium is not paid up front but posted and marked daily, so there is nothing to discount and Black's model loses its discount factor:

$$c = F_0 N(d_1) - K N(d_2)$$

At the rates of the 1980s the missing discount factor was a large difference; at recent rates it is small but not zero. Exercising early would also release nothing, so an American futures-style option is worth its European value.

Futures options versus spot options. Where both exist, they are not interchangeable, for four reasons, the first of which Section 10.3 gave.

Exercise delivers a margined position, not an asset to fund. That changes early exercise, so the agreement checked in Example 16.5 holds only for European options. Example 13.8

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showed why an American call on a futures contract can be exercised early with no dividend anywhere in the problem: a futures contract costs nothing to hold, which is exactly what $q = r$ encodes, so the premise of Section 11.8's argument, that deferring the strike payment earns interest, fails, and exercise releases the intrinsic value as margin that can earn interest. Two practical consequences follow. Listed commodity and rate options with the premium paid up front are usually American, so Black's model gives a lower bound rather than the value, and desks price them on lattices as Section 15.10 described. And a writer of American futures options should expect assignment whenever the option is deep in the money, not merely near expiration.

Where the futures contract expires with the option, a European futures option and a European spot option are worth the same. American ones are not: when the futures price stands above spot (contango), the American call on the futures is worth more than the American call on the spot and the American put less, and in backwardation the ranking reverses.

The underlying futures contract usually expires shortly after the option, so the option's underlying is a specific contract month rather than the spot. On a commodity with a steep curve, that difference is material.

And the futures price already contains the carry. Where the carry is uncertain (an unknown convenience yield, an unpredictable dividend), pricing off the futures removes an estimation problem that the spot-based formula would have to solve.

Who uses them. Section 10.3 named the hedgers; a refiner buying crude calls or a grower buying corn puts gets a cap or a floor where a futures hedge would be a lock. Asset managers and mortgage servicers use options on Treasury and rate futures to hedge duration and the convexity of Chapter 26's prepayment risk. By notional, the largest listed rate options are those on SOFR futures, which absorbed the Eurodollar options in 2023; since the futures price is 100 less the rate (Section 6.1), a call on the price is a put on the rate, which is why a SOFR call hedges a fall in rates. Exchanges also list weekly and daily expirations on equity index, Treasury, crude oil and gold futures, the futures-market counterpart of Chapter 23's short-dated options. Commodity trading advisors and volatility funds trade them for the volatility itself, and they are the natural counterparty to the hedgers.

Where commodity and energy options live

Black's model is not a special case in commodities—it is the whole market. Almost every listed option on oil, gas, metals, or grain is written on a *futures contract* rather than on the physical, and there are good reasons.

The spot price may not exist in a usable form. Section 2.1's contract specification showed that a commodity's price is a location, a grade, and a delivery window, not a single number, and Section 36.9 develops the point. A futures contract fixes all three, so it is the only unambiguous price to write an option on.

Delivery is avoided. Exercising a futures option produces a futures position, which can be closed. Exercising a physical option produces an obligation to take delivery of crude oil, which most option users are not equipped to do.

Wherever the underlying is awkward to hold, the market writes the option on the forward and prices it with Black's model; Section 16.6 carries this to interest rates.

16.6 BLACK'S MODEL BEYOND FUTURES

Nothing in (16.7) requires a futures contract. All it needs is a forward price that is observable and that has zero expected growth under the measure used for discounting. That condition holds far beyond the futures market, and the world's largest option market runs on it.

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A caplet is a call on a forward interest rate: Black's model applies with F_0 the forward rate and K the cap rate, scaled by the accrual fraction and discounted to the payment date at the end of the period. **A swaption** is an option on a forward swap rate, and the same form applies with the discount factor replaced by the *annuity*, the present value of one unit paid on each fixed-leg date (Chapter 7's PV01 leg). Chapter 29 derives both formulas and the normal-model form in which the market now mostly quotes them (Section 29.8), and covers who trades them: borrowers capping floating funding, issuers of callable debt, and pension funds hedging a fall in rates.

The condition that must be checked. Black's model requires the forward to have zero drift under the discounting measure, and this is where the apparatus can be misapplied. A forward rate is a martingale under the measure associated with *its own* payment date. A forward swap rate is a martingale under the measure associated with *the annuity*. Use one where the other belongs (pay a rate at the wrong date, observe it in the wrong currency, or average it over a period) and the forward acquires a drift.

That drift is exactly the **convexity, timing, and quanto adjustments** of Chapter 30, and Section 16.7 shows the simplest case next. Black's model is not wrong in those situations; it is being fed the wrong forward, and the adjustment corrects the forward rather than the formula.

16.7 QUANTOS AND COMPOSITE OPTIONS

Only one structure in this chapter needs new machinery: an option on a foreign asset whose payoff is settled in the domestic currency.

Two versions exist, and confusing them is expensive, as Example 16.8 shows.

A **composite** (or "compo") option pays the foreign-currency payoff converted at the *prevailing* exchange rate at expiration. The holder is exposed to both the asset and the currency, and the option is effectively written on the asset measured in domestic currency. Write V for the asset's foreign-currency price and S for the exchange rate in domestic units per foreign unit, as in Section 16.3. The option's volatility is that of the product VS, which by Section 14.11 is

$$\sigma_{\text{compo}} = \sqrt{\sigma_V^2 + \sigma_S^2 + 2\rho\sigma_V\sigma_S}$$

where σ_V and σ_S are the volatilities of the asset and of the exchange rate, and ρ is their correlation.

A **quanto** pays the foreign-currency payoff converted at a *fixed* rate agreed in advance, typically 1.0, so that a foreign index moving 100 points pays 100 domestic currency units. The holder has no currency exposure at all, which is what makes quantos popular: an investor gets foreign equity exposure without an FX position. CME's dollar-denominated Nikkei 225 futures contract is a listed quanto; over the counter, structured-note issuers are major buyers of quanto calls, and the dealers who sell them carry the correlation risk described below.

That convenience has a price, and the price is not obvious.

The quanto adjustment. The issuer of a quanto must hedge by holding the foreign asset, and the size of that foreign holding, measured in domestic currency, changes as the exchange rate moves. If the asset and the currency are correlated, the hedge must be adjusted in a way that is systematically profitable or costly, and the option's price must reflect it.

The result, derived in Chapter 30, is that the foreign asset's forward price must be adjusted:

$$F_0^{\text{quanto}} = V_0\, e^{\left(r_f - q_f - \rho\,\sigma_V\sigma_S\right)T} \tag{16.9}$$

where V_0 is the asset's foreign-currency price today, r_f the foreign rate, and q_f its dividend yield. The option is then priced with Black's model (16.7), discounting at the domestic rate.

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The correction term $-\rho\sigma_V\sigma_S$ is the **quanto adjustment**. Its sign follows the correlation: when the foreign asset and the foreign currency move together, the adjustment reduces the forward and cheapens a call.

Example 16.7 Pricing a quanto call

A foreign index stands at 3,000 with a 2% dividend yield. Foreign rates are 1%, domestic rates 4%. Index volatility is 20%, exchange-rate volatility 10%, and their correlation is +0.3. Value a one-year quanto call struck at 3,100, paying one domestic unit per index point.

From (16.9),

$$F_0^{\text{quanto}} = 3000\, e^{(0.01-0.02-0.3\times0.20\times0.10)\times1} = 3000\, e^{-0.016} = 2{,}952.38$$

Without the correction the forward would be $3000e^{-0.01} = 2{,}970.15$, so the quanto adjustment lowers the forward by 17.77 points, or 0.60%.

Applying Black's model with $\sigma = 20\%$ and discounting at the domestic 4%:

$$d_1 = -0.1439, \qquad d_2 = -0.3439, \qquad c = 167.52$$

against 175.18 if the adjustment were omitted: an error of 4.6% in the option's value from a 0.6% error in the forward. Options magnify errors in the forward, so the adjustment cannot be waved away as second order.

Two practical points follow. The price depends on ρ, a correlation that is not directly traded and must be estimated, so quantos carry a risk that vanilla options do not. That risk is why quanto books carry **correlation reserves**: part of the book's model value held back against the chance that the correlation has been misestimated. And the sign matters: had the correlation been −0.3, the adjustment would have *raised* the forward and the call.

Example 16.8 The same view, two structures

Take the market of Example 16.7 and price the composite alongside the quanto. Assume the exchange rate starts at 1.00 domestic unit per foreign unit, so both structures start from an index level of 3,000 in domestic terms, and both are struck at 3,100 for one year.

Composite. The underlying is the index measured in domestic currency, whose volatility is

$$\sigma_{\text{compo}} = \sqrt{0.20^2 + 0.10^2 + 2\times0.3\times0.20\times0.10} = 24.90\%$$

Its forward is $3000e^{(r-q_f)T} = 3000e^{0.02} = 3{,}060.60$, and Black's model gives

$$c_{\text{compo}} = 274.69$$

Quanto. From Example 16.7, $c_{\text{quanto}} = 167.52$.

The composite costs 64% more, for two reasons: it carries a higher volatility (24.9% against 20%), and its forward is higher, 3,060.60 against 2,952.38. Most of that gap is the foreign currency's own forward premium, since the domestic rate exceeds the foreign rate by three percentage points; the remaining 0.6% is the quanto adjustment the composite escapes.

The payoffs explain the price gap better than the formulas do:

Index	FX rate	Quanto pays	Composite pays
3,400	0.90	300	0
3,400	1.00	300	300
3,400	1.10	300	640
3,100	1.10	0	310
2,800	0.90–1.10	0	0

The quanto pays on the index alone. The composite pays on the index *and* the currency, so it can pay when the index ends at the strike and the currency has risen—and pay nothing when the index has risen and the currency has fallen.

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These are different views, not different prices for the same view. An investor who wants foreign equity exposure and no currency exposure buys the quanto and pays for the correlation adjustment. An investor who is bullish on both buys the composite and pays for the extra volatility. Selling one to a client who wanted the other is an unsuitable sale that the payoff table makes obvious and the price comparison does not.

16.8 GETTING THE FORWARD RIGHT

Every formula in this chapter takes a forward as its real input. Black's model says so explicitly; equation (16.1) says so in disguise, since $S_0 e^{-qT}$ is the discounted forward $F_0 e^{-rT}$. The practical question in each market is not which formula to use but where the forward comes from.

Table 16.3 gives the answer, and one rule governs all of it.

Table 16.3: Where F_0 comes from in each market.

Market	Computed as	But in practice
Equity index	$S_0 e^{(r-q)T}$	Read from index futures
Single stock	$S_0 e^{rT}$ − FV(divs)	Implied from the option chain
Currency	$S_0 e^{(r-r_f)T}$	Read from the FX forward market
Commodity	Not computable	The futures price
Interest rate	From the curve	From the curve (Ch. 4)

The rule is to use the traded forward wherever one exists, never the computed one. Earlier chapters gave the reasons, and they are not small.

For currencies, Section 5.7 showed that covered interest parity has been persistently violated since 2008, by tens of basis points in the deepest markets in existence. A forward computed from $S_0 e^{(r-r_f)T}$ will therefore differ from the forward one can actually trade at, and the difference is the cross-currency basis of Chapter 7. Pricing an FX option off the computed forward embeds an error that counterparties can trade against.

For commodities the forward is not computable even in principle (Section 5.8); the futures price is the primary datum.

For single stocks, Section 11.6 showed the implied forward can be read directly from the option chain; Example 16.9 does the same for an index and shows what that forward contains.

Example 16.9 Extracting the dividend yield from index options

An index stands at 1,500. Six-month options struck at 1,500 trade at 68.50 (call) and 61.20 (put), with $r = 4\%$. What dividend yield is the market assuming?

Put–call parity with a yield, from Section 11.4, requires

$$c - p = S_0 e^{-qT} - K e^{-rT}$$

Substituting,

$$7.30 = 1500\, e^{-0.5q} - 1500\, e^{-0.02} = 1500\, e^{-0.5q} - 1470.30$$

so $1500 e^{-0.5q} = 1477.60$, giving

$$e^{-0.5q} = 0.985067 \quad \Longrightarrow \quad q = 3.01\%$$

Three points follow, and the third is the one that matters.

The calculation needs no volatility and no model: it is parity, which Chapter 11 established without either.

It gives the market's dividend expectation, which for an index is often better than a published forecast, because index dividends are actively traded.

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> And the number is not purely a dividend yield. It is whatever makes the forward come out right, which bundles expected dividends with the cost of financing or borrowing the index basket. What has been measured is the implied forward, expressed as a yield: the same warning Section 11.6 gave for single stocks. A desk that treats 3.01% as a dividend forecast and separately adds a financing spread has counted the same thing twice.

16.9 WHICH CONVENTION FOR WHICH MARKET

Table 16.4 collects what a reader needs to move between markets without importing the wrong assumption.

Table 16.4: Pricing form and quoting convention by market.

Market	Form used	Quoted in	Moneyness by
Single-stock equity	Black–Scholes–Merton, discrete dividends	Price	Strike
Equity index	Black–Scholes–Merton with q	Price or volatility	Strike
Currency	Black–Scholes–Merton with r_f	Volatility	Delta
Commodity, energy	Black's model on futures	Volatility	Strike or delta
Interest rate (caps, swaptions)	Black or normal model on forward rates	Volatility	Strike

Two habits follow from Table 16.4.

Ask what the forward is before asking what the option is worth. In every row, the model takes a forward as its input, and errors in the forward are magnified in the option; Example 16.7 put the magnification at roughly eight to one. Most disputes about an option's value turn out to be disputes about the forward.

Ask which currency each number is in. Table 16.2 showed one premium admitting four correct expressions. The same discipline applies to notionals, strikes, and deltas in any cross-currency structure.

Why volatilities differ so much between markets

A reader moving between the markets of Table 16.4 meets volatility numbers that differ by an order of magnitude, and the differences are structural rather than accidental.

Table 16.5: Indicative implied volatility ranges.

Market	Typical range
Major currency pairs	5–12%
Emerging-market currencies	8–25%
Equity indices	12–25% calm, 30–80% in crisis
Single stocks	20–60%
Commodities	20–60%, energy higher

The most striking entry in Table 16.5 is the first, and Section 14.11 already explains it. An exchange rate is a *ratio* of two currencies, and by equation (14.19) the volatility of a ratio is

$$\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$$

Two large economies share most of their shocks (global growth, commodity prices, risk appetite), so ρ is high and the common component largely cancels. What remains in an exchange rate is the difference between two economies, not the level of either, and differences are smaller than levels.

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Equities sit at the other end for the opposite reason. A share is a residual claim on a leveraged balance sheet, so the volatility of the assets is amplified by the leverage before it reaches the equity. Single stocks exceed indices because diversification removes the idiosyncratic part but not the leverage.

The practical use of Table 16.5 is as a sanity check. A quoted volatility outside its market's range is far more often a units or convention error than a discovery, and Section 14.6's rule of thumb (daily move $\approx \sigma/16$) converts any of these into something checkable against a screen in seconds.

SUMMARY

Replacing S_0 with $S_0 e^{-qT}$ covers indices, currencies, and futures, with q the dividend yield, the foreign rate, and the domestic rate. Nothing else in the formula changes.

Portfolio insurance is mechanically simple and conceptually not: a floor on the index level is not a floor on terminal wealth, because dividends are received, premium is paid, and a portfolio with beta other than one moves further than the index.

In currencies the difficulty is not the formula but the labeling: a premium can be correctly quoted four ways, and every quantity must carry its currency. The market quotes volatility rather than price and identifies options by delta rather than strike, because a strike ages and a delta does not. Each maturity is three numbers (ATM, 25-delta risk reversal, 25-delta butterfly) measuring the smile's level, skew, and curvature, which are what the corresponding instruments trade. Unlike equity skew, the FX risk reversal changes sign.

Setting $q = r$ gives Black's model, $c = e^{-rT}[F_0 N(d_1) - K N(d_2)]$: the rate is only a discount factor, and everything about the underlying sits in the observable forward. At $K = F_0$ the call and put are equal. It prices anything with an observable forward, caplets and swaptions included, provided that forward has no drift under the discounting measure; where it does, Chapter 30's adjustments correct the forward, not the formula. Every formula here takes a forward as its real input, and the traded forward is to be used wherever one exists: covered interest parity fails, commodity carry is unobservable, and an implied dividend yield bundles financing with dividends.

Exercising a futures option opens a futures position at the strike, marked to the exercise day's settlement price, so the holder collects the intrinsic value at that settlement price and is left holding the futures. Early exercise is possible without dividends, and the underlying is a specific contract month.

A composite option converts the foreign payoff at the prevailing rate and carries volatility $\sqrt{\sigma_V^2 + \sigma_S^2 + 2\rho\sigma_V\sigma_S}$; a quanto converts at a fixed rate and removes the currency exposure, at the cost of a $-\rho\sigma_V\sigma_S$ correction to the foreign forward's drift. Even a small correction to the drift moves the price materially, and the dependence on an untraded correlation is why quanto books carry correlation reserves.

FURTHER READING

Merton (1973) gives the continuous-yield extension (16.1), Garman and Kohlhagen (1983) is the standard citation for the currency-option form (16.4), and Black (1976) introduces the futures form. On FX quoting conventions (premium currency, spot or forward delta, premium adjustment, and which straddle is at the money), Reiswich and Wystup (2010) set them out in one paper, and Clark's *Foreign Exchange Option Pricing* (2011) works them through as a desk uses them. Chapter 30 derives the quanto adjustment.

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PRACTICE QUESTIONS

16.1 An index stands at 800 with a dividend yield of 3%. Value a four-month European call struck at 820 with $r = 4\%$ and $\sigma = 22\%$.

16.2 Explain why $q = r$ for a futures contract, and state which chapters established this in discrete and continuous time.

16.3 Spot GBP/USD is 1.2600, US rates are 4%, UK rates 4.5%, and six-month volatility is 9%. Value a call on sterling struck at 1.2800, and express the premium both in dollars per pound and as a percentage of the sterling notional.

16.4 A dealer quotes three-month USD/JPY as ATM 9.50, $\mathrm{RR}_{25} = +1.20$, $\mathrm{BF}_{25} = 0.30$. Compute the two 25-delta wing volatilities and state which currency the market is paying more to protect against.

16.5 Using Black's model, value a six-month call on a futures price of 60 struck at 58, with $r = 3\%$ and $\sigma = 30\%$.

16.6 Show that at $K = F_0$ Black's model gives $c = p$, and identify the result from Chapter 11 that this reproduces.

16.7 Explain the difference between a quanto and a composite option, and state which one an investor wanting foreign equity exposure without currency risk should buy.

16.8 A foreign index at 2,500 has $q_f = 1.5\%$, $r_f = 2\%$, $\sigma_V = 22\%$, $\sigma_S = 9\%$, and $\rho = -0.25$. Compute the one-year quanto forward and state whether the adjustment raises or lowers it.

16.9 Every formula in this chapter takes a forward as its real input. Using Table 16.3, state the rule that governs how the forward is obtained for an equity index, a currency, and a commodity, and explain why a desk that computes the forward from spot and a carry rate can be wrong even when its rates are right.

16.10 Using Table 16.4, state which convention applies to an equity index option, a currency option, and an option on a futures contract. Explain what goes wrong when a convention is carried across from the market it belongs to, and give one case where two conventions would both be defensible.

FURTHER QUESTIONS

16.11 Example 16.1 notes that a floor on the index level is not a floor on terminal wealth. Set up the equation that determines the strike delivering a genuine \$28.5 million floor, accounting for dividends received and premium paid, and explain why the answer depends on whether the premium is funded from the portfolio or from outside it.

16.12 The FX market quotes by delta rather than strike. Construct the argument for why this makes a volatility surface more stable across days, then identify a circumstance in which it makes the surface *less* stable and state what desks do about it.

16.13 Example 16.7 found that a 0.60% adjustment to the forward changed the option value by 4.6%. Derive the general relationship between a proportional error in the forward and the resulting proportional error in an at-the-money option's value, and explain why the magnification is largest for short-dated options.

16.14 A trader observes that Black's model on a stock index futures price and the Black–Scholes–Merton formula on the index spot give different answers. List the possible causes in order of likelihood, and state what data you would examine to distinguish them.

16.15 Quanto pricing depends on a correlation that is not directly traded. Explain what this

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implies for how a quanto book should be risk-managed, what a correlation reserve is intended to cover, and why the problem is worse for long-dated quantos than for short-dated ones.

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Chapter 17
Employee Stock Options and Equity Compensation

Every option in this book so far has been a traded contract. Someone wrote it, someone bought it, and either could close the position at any time by trading it back. Chapters 13 through 16 rest on that: a price is the cost of a replicating strategy, and the replicating strategy requires that positions can be established and unwound.

None of this holds for an employee stock option. It cannot be sold. It is forfeited on departure. Its holder is unhedged, undiversified, and forbidden from shorting the stock. It runs ten years when listed stock options run at most about three. And it is issued by the company on its own shares, so exercise creates stock rather than transferring it.

The result is an option whose value depends on human behavior, and the models in this chapter say so explicitly. That is not a retreat from Chapters 13–16. It is those chapters applied consistently: when replication is impossible, the price is no longer pinned by arbitrage, and what fills the gap must be stated as an assumption rather than left implicit.

The chapter also covers the instruments that have largely displaced options in listed-company pay, restricted stock units and performance shares, and the accounting rules that drove the change. Section 10.10 promised this treatment for both warrants and employee options, and the two are the same problem: an option written by the issuer on its own equity.

Sections 17.1 and 17.2 are the contract and why it is not a traded call. Sections 17.3 to 17.5 value it: the models, the dilution that exercise causes, and the exercise decision itself. Section 17.6 is the accounting, Section 17.7 the restricted stock units and performance shares that displaced options, and Section 17.8 private-company equity. Sections 17.9 and 17.10 are warrants and the backdating episode, and Section 17.11 who uses equity pay and why.

17.1 THE CONTRACT

A grant of employee stock options specifies more terms than a listed contract does.

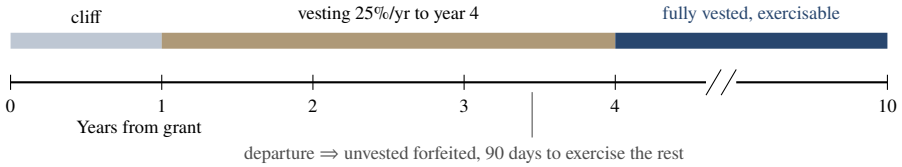


Figure 17.1: A typical grant: one-year cliff, then vesting to year four, ten-year term. The axis is broken between years 4 and 10.

Vesting. The option cannot be exercised until it vests. The common US pattern is a one-year **cliff** (nothing vests until the first anniversary, then 25% at once) followed by monthly or quarterly vesting to year four. Figure 17.1 shows the shape. Unvested options are forfeited outright on departure, which makes vesting the retention mechanism the grant exists for.

Term. Ten years from grant is standard, though four-year and seven-year terms appear. Compare this with the roughly three-year outer limit on single-stock LEAPS from Section 10.10: an employee option outlasts any standard listed option on a stock.

The post-termination exercise window. On leaving, an employee typically has 90 days to exercise vested options or lose them. This clause does most of the damage to the value, because it turns a ten-year option into one that expires 90 days after an event neither party controls.

Non-transferability. The option cannot be sold, pledged, or given away. There is no secondary market and no way to monetize it except by exercising and selling.

Strike. The strike is almost always the market price on the grant date. Sections 17.6 and 17.10 explain why: an at-the-money grant used to carry no accounting charge at all, and still carries the cleanest tax treatment.

Dilution. Exercise creates new shares. The company receives the strike price in cash and issues stock, so the share count rises. Section 17.4 works out what this does to the valuation.

17.2 WHY AN EMPLOYEE OPTION IS NOT A TRADED CALL

Four of the restrictions above break the arbitrage argument, all in the same direction.

The holder cannot hedge. Chapter 13's replication argument requires that a position be constructed and maintained. An employee cannot short the company's stock (most equity plans forbid it and insider-trading policy forbids the rest), so the option's value to the holder is not the cost of replicating it. It is whatever an undiversified, risk-averse person with a concentrated position thinks it is worth, which is *less*.

The holder is undiversified. An employee's human capital is already invested in the firm. Adding options on the same firm concentrates rather than diversifies, so the marginal option is worth less to the employee than to a diversified investor. The size of the gap depends on how risk-averse the holder is and on how much of their wealth is already tied to the firm, and the utility-based estimates run wide: Hall and Murphy (2002) put an executive's **certainty-equivalent** value (the sure sum of cash the executive would accept in place of the option) at roughly half to two-thirds of the Black–Scholes–Merton cost for a typical at-the-money grant, and Meulbroek (2001) found discounts of the same order for concentrated holdings. Whatever the number, it is a real cost of paying in options, and it appears nowhere in the accounts.

Forfeiture is a random early expiration. Departure forfeits unvested options and leaves 90 days to exercise vested ones. Modeling this requires an assumption about employee turnover, and the assumption matters: Section 17.3 shows a value moving from 16.10 to 12.59 as the assumed exit rate goes from zero to 6% a year.

Early exercise is the norm, not the exception. Section 11.8 proved that an American call on a non-dividend-paying stock should never be exercised early, and employees exercise early constantly. Bettis, Bizjak and Lemmon (2005) find ten-year options exercised a median of 4.25 years before expiration, under six years into the term; Carpenter's (1998) sample of top executives gives a mean of 5.83 years after grant, at a stock price 2.75 times the strike. Five to six years at two to three times the strike is the working summary, and it is where the exercise multiple of Section 17.3 comes from.

This is not employees being irrational. The proof in Section 11.8 assumes the holder can sell the option instead. An employee cannot. For a holder choosing between exercising and holding an unsellable, unhedgeable, forfeitable claim on a single company, exercising to diversify is the sensible choice. The theorem does not fail; its hypothesis does.

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That said, exercise-and-hold is common and usually a mistake: it gives up the remaining time value and, in the US, prepays ordinary income tax on the spread to acquire exposure the option already provided (Example 17.4).

17.3 VALUING AN EMPLOYEE OPTION

Three approaches are in use. They disagree, and by how much is the main finding.

1. Black–Scholes with the contractual term. Put the ten-year term into the formula of Chapter 15 and read off the answer. This overstates the value, because it ignores forfeiture and early exercise entirely.

2. Black–Scholes with an expected term. Replace the contractual term with the expected time to exercise or forfeiture, estimated from the company's own history. This is the workhorse: it is what most disclosures use, it is auditable, and it is defensible precisely because it makes no claim to be a model of behavior. A company whose own history gives no reasonable basis for the estimate may use the simplified method of the SEC's Staff Accounting Bulletins (SABs) 107 and 110, which sets the expected term of a plain at-the-money grant midway between vesting and the contractual term: 6.5 years for Example 17.1's grant.

Its weakness is that expected life is not a parameter of the option but an *outcome*, and it depends on the stock price path. A company whose stock triples will see exercises much earlier than one whose stock stagnates, so the expected term estimated from history is conditioned on a history that will not repeat.

3. A lattice with an exercise rule. Build the tree of Chapter 13 and replace optimal exercise with a behavioral rule: the employee exercises voluntarily once the stock reaches a multiple M of the strike, and is forced to exercise (or forfeit, if unvested) on departure, which arrives at rate λ. This is the Hull–White approach, and it is the one that spells out what the other two conceal.

Example 17.1 Three answers for one grant

A company grants at-the-money options: $S_0 = K = 40$, ten-year term, three-year cliff vesting, $\sigma = 30\%$, $r = 4\%$, dividend yield $q = 1\%$.

Method	Value	% of grant price
Black–Scholes, 10-year contractual term	16.704	41.8%
Black–Scholes, 6.5-year simplified term	13.808	34.5%
Black–Scholes, 6-year expected term	13.295	33.2%
Black–Scholes, 4.5-year expected term	11.552	28.9%
Lattice, $M = 2.5$, exit rate 3%/yr	14.22	35.5%

The spread is 16.704 to 11.552, a factor of 1.45. ASC 718 rules out the first row, which ignores early exercise, but the methods it allows still span 15.56 (the lattice's value conditional on vesting, the standards' basis; Example 17.5) to 11.552, and on one million options that vest that is a difference of $4.0 million in reported compensation expense, produced entirely by choices about the model rather than about the contract.

The lattice makes the sensitivity explicit:

	exit 0%	exit 3%	exit 6%
$M = 2.0$	—	13.54	—
$M = 2.5$	16.10	14.22	12.59
$M = 3.0$	—	14.52	—

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Turnover matters more than the exercise multiple: moving the exit rate from 0% to 6% costs 3.51 of value, while moving the exercise multiple from 2.0 to 3.0 gains only 0.98, less of it from 2.5 to 3.0 than from 2.0 to 2.5, because a higher threshold matters only on the paths that reach it.

The practical implication is that the most important input is a human-resources (HR) statistic. The company's own attrition rate drives the number more than volatility does over the plausible range, and a firm with high turnover is genuinely granting less, option for option.

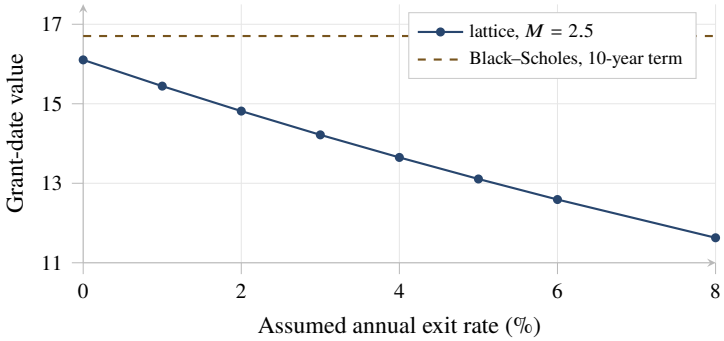


Figure 17.2: Lattice value against the assumed annual exit rate. $S_0 = K = 40$, $\sigma = 30\%$, ten-year term.

Figure 17.2 makes the point visually: the reported expense is close to linear in an assumption that no market quotes and no auditor can verify beyond checking that the company applied its own history consistently.

This is the position of the whole chapter. An employee option cannot be traded, so its reported value is a model output whose most sensitive input is behavioral.

17.4 DILUTION

Exercise creates shares. Existing holders own a smaller fraction of a company that has received the strike price in cash, and **dilution** is the size of that transfer: the fraction of the company passing to the option holders. Three measures of it appear in this section and they are not equal. The *share-count* measure is the new shares as a fraction of the enlarged total, and it is what governs a warrant's value. The *economic* measure is the grant's value as a fraction of market capitalization. The *reported* measure counts only the incremental shares the treasury stock method leaves after assumed buybacks, and Example 17.3 finds it running at about half the economic figure.

Take a company with N shares and no debt, whose assets, excluding anything received from the option holders, will be worth V_T at date T. It has issued M warrants with strike K (in this section M counts warrants; it is no longer the exercise multiple of Section 17.3). If they are exercised, the company receives MK and issues M shares, so the post-exercise share price is

$$\frac{V_T + MK}{N + M}$$

and each warrant is worth that less K. A line of algebra gives

$$\text{warrant payoff} = \frac{N}{N+M}\,\max\!\left(\frac{V_T}{N} - K,\ 0\right) \tag{17.1}$$

So a warrant is $N/(N+M)$ of a call option on V/N, the per-share value of the company *excluding* the warrant proceeds. Both parts of that sentence matter, and the second is where practitioners go wrong. Example 17.2 works through the first.

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Example 17.2 What dilution costs, and what it does not

A company has 10 million shares. It issues 2 million five-year warrants struck at \$40, with $\sigma = 30\%$, $r = 4\%$, $q = 1\%$, and a per-share value of \$40 before the warrants exist.

A five-year call on \$40 struck at \$40 is worth **12.171**. The dilution factor is $10/12 = 0.8333$, so each warrant is worth

$$\frac{10}{12} \times 12.171 = \mathbf{10.143}$$

16.7% less than the equivalent call, which is exactly the fraction of the company the new shares represent.

The total is $2\text{m} \times 10.143 = \$20.3$ million. If the warrants are handed out for nothing, the share price falls by $20.3/10 = \$2.03$ on announcement, from \$40.00 to \$37.97. The value did not vanish; it moved from the existing shareholders to the warrant holders, and the price move is the transfer.

The mistake worth avoiding. Once the warrants or options are outstanding and known, the observed share price already reflects them. Applying (17.1) to the market price then double-counts: the market price is V/N net of the warrant liability, not gross of it.

This has a direct consequence for employee options, and it is the opposite of what intuition suggests. An outstanding employee option grant requires no dilution adjustment at all. The market knows the options exist (they are disclosed), so the share price already embeds them, and the grant-date fair values of Section 17.3 are computed on the observed price with no $N/(N + M)$ factor. The adjustment applies only when valuing an instrument the market has not yet been told about, which for a public company is essentially never.

The dilution that matters for employee options is therefore a governance question about the size of the grant, not a valuation adjustment: how much of the company is being transferred, over what period, and in exchange for what.

Diluted earnings per share

Dilution is quantified and published in only one place, and it ignores the valuation above.

Diluted earnings per share (EPS) is computed by the **treasury stock method**: assume the options are exercised, assume the proceeds are used to buy back shares at the average market price, and count the difference as incremental shares.

Example 17.3 Reported dilution against economic dilution

A company has 10 million shares, net income of \$50 million, and 1 million options struck at \$30. The average share price for the period is \$50. The options were granted at the money with the stock at \$30, worth about \$10 each on the parameters below, and \$6 million of that \$10 million grant-date cost, which is expensed over the vesting period (Section 17.6), remains unrecognized on average over the period.

Basic EPS is $50/10 = \mathbf{\$5.00}$.

Assumed proceeds are $1\text{m} \times 30 = \$30$ million, plus the \$6 million of unrecognized cost (the value of the service employees still owe, which ASC Topic 260 also treats as proceeds): \$36 million, which buys back $36/50 = 0.72$ million shares. Incremental shares are $1.00 - 0.72 = 0.28$ million, so

$$\text{diluted EPS} = \frac{50}{10.28} = \mathbf{\$4.864}, \qquad \text{reported dilution } \frac{0.28}{10} = 2.8\%$$

The reported figure is the increase in the share count. Earnings per share itself falls by 2.7%: a share count 2.8% larger divides into the same earnings.

Valued with the stock at \$50, six years, $\sigma = 30\%$, $r = 4\%$ and $q = 1\%$, each option is worth \$25.68, so the grant is worth \$25.7 million against a market capitalization of \$500 million, or **5.1%**

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of the company.

The reported dilution is roughly half the economic dilution, and the reason is structural: the treasury stock method counts the option at its intrinsic value less unrecognized compensation cost, which ignores time value entirely. It is an accounting convention for allocating this period's earnings, not a measure of what has been given away.

The gap widens exactly where it matters most. For a volatile company whose options are near the money, intrinsic value is small and time value is nearly the whole of the option, so the treasury stock method reports almost no dilution on grants that transfer a substantial fraction of the equity.

An analyst who wants the economic number must compute it, and the inputs are disclosed: option counts, strikes, and grant-date fair values all appear in the equity compensation footnote. Section 17.11 returns to why this matters for reading buyback announcements.

17.5 THE EXERCISE DECISION

Section 17.2 said employees exercise early and that this is not irrational. Example 17.4 works through what the choice actually looks like, because the arithmetic is not obvious and the common advice is wrong in both directions.

Example 17.4 Three things to do with a vested option

An employee holds 10,000 vested non-qualified options (NSOs) struck at \$20: the ordinary kind, taxed as income on the spread at exercise (Section 17.8 covers the incentive kind). The stock is \$50, four years remain, $\sigma = 35\%$, $r = 4\%$, no dividend. Their marginal tax rate is 40%.

The corresponding traded call is worth **33.483** against an intrinsic value of **30.00**, so the time value is **3.483**, or 11.6% of the intrinsic value, and \$34,826 across the position.

Action	Cash today	Position afterward
Hold	\$0	options worth \$334,826 before tax
Exercise and sell	+\$180,000	nothing
Exercise and hold	−\$320,000	\$500,000 of stock

Read the third row carefully. Exercising costs \$200,000 in strike and creates \$300,000 of ordinary income, so \$120,000 of tax comes due whether or not any stock is sold. Holding would defer that tax, not avoid it, so exercise-and-hold prepays it: the employee spends \$320,000 in cash to acquire exposure the option already gave them, and gives up the \$34,826 of time value.

What the prepayment buys is that later appreciation is taxed as a capital gain rather than as ordinary income. That is the real content of "I want to start the capital gains clock," an offset worth weighing on a stock expected to rise far, though the time value, the deferral and the concentration usually outweigh it. "I believe in the company" is no reason at all: belief is an argument for keeping the exposure, and the option *is* the exposure, with leverage and without the cash outlay.

The case for exercising is the case for diversifying, and it is the first two rows that are in genuine tension. They must be compared after tax on both sides, since the options also owe the tax when eventually exercised: at a constant 40% rate the \$334,826 is worth $0.6 \times \$334{,}826 = \$200{,}896$ after tax, against \$180,000 of cash from exercising and selling. Diversifying therefore costs about \$20,900, the after-tax time value. An employee whose net worth is already concentrated in the firm may rationally pay that, and Section 17.2's undiversified-holder discount is exactly the statement that the model value is not what the position is worth to them.

Two things sharpen the decision. *Time value does not disappear as the stock rises*: at \$120 the same option's time value would still be 2.98, barely changed from the 3.483 here. It never falls to zero: for a non-dividend-paying stock it converges to the interest on the deferred strike, $K(1 - e^{-rT})$, which is 2.96 for this option and is Section 11.8's result seen from the other side. What does fall is the time value measured against what is at stake: 3.0% of the intrinsic value rather than 11.6%, so an employee exercising to diversify surrenders proportionally less at the higher

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price. *A near departure forces it*: the 90-day window converts the choice into a deadline, which is why so many exercises cluster around resignations rather than around stock prices.

17.6 ACCOUNTING AND EXPENSING

Until 2006 a US company with a calendar fiscal year could grant at-the-money options and record no expense whatsoever. Accounting Principles Board Opinion 25 (APB 25, 1972) measured the charge as the option's *intrinsic* value at grant, which for an at-the-money option is zero, and Financial Accounting Standard 123 (FAS 123, 1995), though it encouraged expensing fair value, let companies keep that method with footnote disclosure; nearly all did. Companies that would never have paid $14 million in cash bonuses granted one million options worth $14 million and reported nothing.

FAS 123R (issued in December 2004), now ASC 718, and IFRS 2, in force from 2005, ended that. In outline, the current rules are these:

Measurement at grant-date fair value. The charge is the model value on the grant date, computed as in Section 17.3 but conditional on vesting: departures before vesting reduce the number of awards expensed, not the value of each. It is never remeasured for subsequent stock price moves. An option valued at $15 at grant that expires worthless was still a $15 expense, and so was one that ends $200 in the money.

Recognition over the vesting period. The total is spread across the service period, but the pattern depends on whether the grant is treated as one award or several. **Straight-line recognition** treats it as one award and expenses equal amounts each year; **graded recognition** treats each vesting tranche as a separate award expensed over its own vesting period, so the early years carry a share of every tranche. Example 17.5 sets the two side by side.

Equity or liability classification. Everything above assumes settlement in shares. An award settled in cash is a liability. It may be a **stock appreciation right (SAR)**, which pays the spread without the holder paying the strike, or phantom stock, which pays a share's value. A liability is remeasured to fair value every reporting date until it settles, so a cash-settled SAR, unlike the equity-settled option above, rises and falls with the share price for years and ends at whatever the award pays. Companies accept that volatility to pay the award without issuing shares, though the cost to existing shareholders is the same: the cash is theirs too, so settling in it keeps the share count fixed but transfers the same value. SARs are common where local law restricts issuing shares to employees and among private companies with no market to settle into.

Forfeiture. Expense on awards forfeited for failure to vest is reversed. Since ASU 2016-09, ASC 718 lets a company either estimate forfeitures or account for them as they occur; IFRS 2 requires an estimate. A company that estimates uses the exit rate *twice*, for different purposes: once inside the model, after vesting, to shorten the option's life, and again outside it to reduce the number of awards expected to vest.

Pay versus performance. For fiscal years ending on or after December 16, 2022, US proxy statements must also report "compensation actually paid" (Item 402(v) of Regulation S-K, adopted in August 2022), which remeasures equity awards at fair value at each year end and at vesting. It is the remeasurement the expense never makes, published beside it.

Example 17.5 Straight-line against graded recognition

A grant of 100,000 options vests 25% a year over four years. Each annual tranche is a separate option with its own vesting date, so each has its own value (Example 17.1's lattice, with the cliff set at the tranche's vesting date). The values are the expected cost per option granted, including the chance of leaving before vesting:

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Tranche	Options	Value	Cost
1-year cliff	25,000	14.52	$363,000
2-year cliff	25,000	14.37	$359,250
3-year cliff	25,000	14.22	$355,500
4-year cliff	25,000	14.05	$351,250
Total	100,000		**$1,429,000**

Per option granted, longer vesting costs *less*, not more, because vesting later means more exposure to forfeiture before the option becomes exercisable. The effect is small (3.2% from the first tranche to the fourth), but it runs the opposite way from the usual "more time is worth more" intuition: the extra time is time the holder cannot use. The standards measure fair value conditional on vesting instead, and put pre-vesting departures into the number of awards. On that basis each value is divided by the probability of staying until the cliff, $e^{-0.03c}$ for a cliff of c years, and the tranches are worth 14.96, 15.26, 15.56 and 15.85. These rise with the cliff: once pre-vesting departures are set aside, what costs value is early exercise after vesting, voluntary at the multiple or forced by departure, and a later cliff postpones both. The expected totals are the same on either basis.

Now recognize that $1,429,000 over the vesting period:

	Year 1	Year 2	Year 3	Year 4
Straight-line	$357,250	$357,250	$357,250	$357,250
Graded	$748,938	$385,938	$206,312	$87,812

Graded recognition puts 52% of the total cost in the first year against straight-line's 25%. Both are permitted under ASC 718 for awards with only a service condition (vesting on continued employment alone); IFRS 2 requires the graded pattern. The total is identical and the timing is not, so two companies with identical pay can report materially different first-year expense.

The difference washes out in steady state for a company that grants every year, but *not* for one that has just started granting or made a large one-off award, precisely the cases where the number is being looked at.

Repricing and modification. When the stock falls far enough that outstanding options are hopelessly underwater, they stop retaining anyone. Companies respond by repricing (lowering the strike) or by exchanging old options for fewer new ones or for restricted stock.

Accounting treats this as a modification: the incremental charge is the fair value *added* by the change, measured immediately before and after, on top of any unrecognized original cost. So a repricing is not free, and shareholders usually must approve it. The economics are worse than the accounting suggests: an employee who learns that underwater options get reset has learned that the downside of the option was never real, which removes most of the incentive the grant was supposed to create.

Repricing and exchanging are usually compared on cost; Example 17.6 shows that this framing gets the incentive question backward.

Example 17.6 Repricing against a value-for-value exchange

A company granted 500,000 ten-year options struck at $80. The stock is now $30 with six years remaining; volatility is 45%, the rate 4%, and the stock pays no dividend. The existing options are worth $6.5944 each and a new six-year at-the-money option would be worth $14.6527.

Repricing all 500,000 to a $30 strike adds $14.6527 − $6.5944 = $8.0583 of fair value per option. The alternative is a value-for-value exchange: cancel the underwater options and grant fewer new ones at $30, in a ratio chosen so the fair value handed over matches the fair value taken

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back: \$6.5944/\$14.6527 = 0.45005 new per old, or 225,025 in all. The two compare as follows:

	Reprice	Exchange
Options held afterward	500,000	225,025
Fair value before	\$3,297,207	\$3,297,207
Fair value after	\$7,326,327	\$3,297,207
Incremental expense	**\$4,029,120**	**\$0**

The exchange delivers exactly what it took away, so nothing is recognized. That is why boards prefer it, and why proxy advisers (firms advising institutional shareholders how to vote) accept it.

Now look at what each does to the employee's exposure rather than to the reported number. Incentive intensity is the delta of the grant, how many shares of exposure the employee actually holds:

	Delta	Shares of exposure
Underwater \$80 options	0.452	225,931
Repriced \$30 options	0.779	389,507
Exchanged \$30 options	0.779	175,297

The exchange that costs nothing to report leaves the employee with 22% *less* exposure to the share price than the underwater options they surrendered. A deeply out-of-the-money option has a low value and a surprisingly high delta (nearly half a share), so trading it for a smaller number of at-the-money options is a bad trade in delta terms even when it is a fair trade in value terms.

So the two alternatives fail in opposite directions. The repricing multiplies the value of the employee's grant by 2.2 and their share exposure by 1.7, and charges shareholders \$4.0 million for doing it, having taught everyone that strikes follow the stock down. The exchange reports nothing, costs nothing, and quietly reduces the incentive it was adopted to restore: the one that is easy to disclose does less of what the grant was for.

17.7 RESTRICTED STOCK UNITS AND PERFORMANCE SHARES

Options have been displaced. Most large listed companies now grant the majority of equity compensation as restricted stock units (RSUs), and the reason is visible in one calculation.

An **RSU** is a promise to deliver one share on vesting. Having no strike, it is not an option but a prepaid forward on the stock (equivalently, a zero-strike call) with delivery contingent on service. Its grant-date fair value is the share price, less the present value of dividends forgone during vesting if the units carry no **dividend equivalents**: cash or extra units matching the dividends paid on the shares while the units are unvested.

Example 17.7 Why RSUs displaced options

Take the same company and grant terms: $S_0 = 40$, $\sigma = 30\%$, $q = 1\%$, Example 17.1's three-year cliff and 3% annual exit rate.

The two must be valued on one basis. The option's 14.22 is an expected cost per option granted and carries the chance of leaving before the cliff (Example 17.5), so the RSU must carry it too: a unit without dividend equivalents is a prepaid forward on the share that is forfeited on departure before vesting, worth $40e^{-(0.01+0.03)\times 3} = \mathbf{35.48}$. So one RSU costs the same as **2.50 options** (a ratio unchanged conditional on vesting, where both values rise by $e^{0.09}$), which is why a company moving from options to RSUs grants one unit in place of two to three options.

Now suppose the stock falls immediately after the grant, with the same strike and the same vesting still ahead:

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Stock price	RSU value	change	Option value	change
40	35.48	—	14.22	—
30	26.61	−25%	8.41	−41%
20	17.74	−50%	3.68	−74%

A halving of the stock destroys 74% of the option grant's value and exactly 50% of the RSU grant's. Figure 17.3 shows why: the RSU is a straight line (a prepaid forward is linear in the share price), while the option curve bends away from it and flattens toward zero, so the same fall in the stock takes proportionally more of the option. After 2008 and again after 2022, boards that had granted options found their entire retention apparatus nearly worthless at exactly the moment they most needed to retain people, while boards that had granted RSUs found theirs merely diminished.

That is the whole argument, and it is an argument about the *purpose* of the grant. An option pays only for outperformance; an RSU is deferred stock that pays for staying. If the objective is retention, as it is for most employees below the executive level, the RSU does the job with less accounting expense per unit of retention delivered.

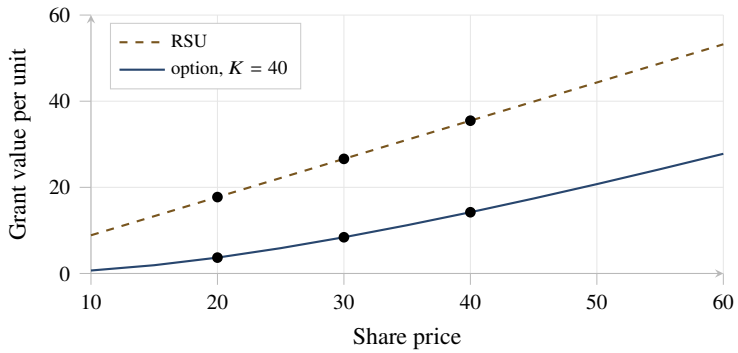


Figure 17.3: Grant value against the share price, on Example 17.7's terms. Marks show the table's three cases.

Performance share units. At executive level, grants increasingly vest on a performance condition rather than on service alone, and the accounting treats two kinds differently.

A **non-market condition** (revenue, earnings per share, a regulatory approval) is not built into the fair value. The grant-date value is computed as if the condition will be met, and the number of awards expected to vest is estimated and trued up each period. If the target is missed, the expense is reversed entirely.

A **market condition**, such as total shareholder return (TSR: price appreciation plus dividends) measured against an index or a peer group, *is* built into the fair value and is never trued up. The award is valued at grant with a Monte Carlo simulation of the company's stock against the benchmark, and that value is expensed whether or not the target is hit, provided the employee completes the service period.

The asymmetry is easily missed. A relative TSR award that pays nothing still generates expense for its full grant-date value, because the possibility of paying was what was purchased. An EPS-target award that pays nothing generates no expense at all. Two grants that look identical in the proxy statement can behave completely differently in the income statement, and Chapter 20's simulation machinery is what values the first of them.

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17.8 PRIVATE COMPANIES AND STARTUP EQUITY

Everything above assumes an observable share price. A private company has none, with consequences beyond valuation.

The 409A valuation. Section 409A of the US Internal Revenue Code requires that options be struck at or above fair market value, and for a private company that value is in practice established by an independent appraisal (a **409A valuation**, which the tax authorities presume reasonable) refreshed at least annually or after any material event. If the strike turns out to be below fair market value, the option fails Section 409A, and the statute puts the tax on the employee, not the company: the in-the-money amount is taxed as the option vests, plus a 20% additional tax and interest, on a gain the employee cannot sell. The employee cannot check the valuation, which is why companies buy the appraisal (the tax authorities can overturn it only by showing it grossly unreasonable) and why some reimburse employees when one fails.

The appraisal is a genuine exercise in valuation under uncertainty: comparable-company multiples, discounted cash flow, and an allocation of the resulting equity value across the preferred and common share classes, since the common stock employees receive sits behind the preferred holders' **liquidation preferences**, their right to be paid back first when the company is sold or wound up. That subordination is why the 409A price routinely sits far below the price paid in the latest preferred round (often under half at an early-stage company, a gap that narrows only as an exit approaches), and why an employee's strike price can be far below the company's headline valuation.

Illiquidity has no offsetting benefit. The employee holds an option on an asset with no market, so the diversification argument of Section 17.2 bites hardest here. There is not even the escape of exercising and selling; Example 17.8 shows the cost to a leaver.

Example 17.8 The exercise-and-hold trap

An engineer holds 100,000 vested incentive stock options (ISOs) struck at \$0.50. The latest 409A valuation is \$8.00. They leave the company, and the 90-day window starts.

Exercising costs $100{,}000 \times 0.50 = \$50{,}000$ in cash. That is the visible cost. The invisible one is the tax. Exercising an ISO creates no regular taxable income, but the alternative minimum tax (AMT), a parallel US tax computation, counts the spread as income:

$$(8.00 - 0.50) \times 100{,}000 = \$750{,}000$$

which at the 28% AMT rate produces a bill on the order of **\$210,000** the following April, on an unrealized gain in shares that cannot be sold.

The total cash required is roughly \$260,000, against a position that may be worth nothing. The alternative is to let the options lapse and walk away from a nominal \$750,000.

This is not a rare edge case; it is the standard situation for anyone leaving a private company with appreciated options, and it is why several structural responses exist: extended post-termination windows (some companies now allow seven or ten years, though an option exercised more than three months after departure loses ISO status), company-run tender offers that create periodic liquidity, and third-party financing against the shares. Each of these is a response to the same defect: an instrument that generates a tax liability before it generates cash.

Incentive against non-qualified options. Incentive stock options avoid ordinary income tax at exercise but trigger AMT as above, and keep the favorable treatment only if the shares are held two years from grant and one from exercise; at most \$100,000 of stock, valued at grant, may first become exercisable as ISOs in a year. Non-qualified options are taxed as ordinary income on the spread at exercise, withheld through payroll, with no AMT complication, and the company deducts that income; an ISO gives it no deduction unless the shares are sold early. The choice turns on whether the employee will hold long enough to qualify.

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Why startups grant options at all. Cash-constrained companies pay in equity because they have equity and not cash, and the option's leverage means a given accounting expense buys a larger notional bet on the company's success, which is precisely the alignment a startup wants and a mature company does not. The instrument matches the situation. It is when a mature, cash-generating company keeps using it out of habit that the mismatch of Section 17.7 appears.

17.9 WARRANTS BEYOND EMPLOYEE OPTIONS

Section 10.10 introduced warrants alongside employee options. Three families are worth distinguishing, because only two of them dilute.

Corporate warrants are issued by the company, usually attached to a bond or preferred issue as a sweetener, and exercise creates shares. Equation (17.1) applies. The attachment is the point: a warrant lets an issuer sell debt at a lower coupon by handing the buyer equity upside, the trade of Chapter 25's convertible bond with the two parts separable.

Special purpose acquisition company (SPAC) warrants were the highest-volume example of the 2020–21 boom. A SPAC is a listed shell company that raises cash from public investors to merge with a private operating company, which thereby becomes listed. It sells units (typically one share plus a fraction of a warrant), with the warrant struck around 15% above the unit price and running five years from the merger. They dilute, they are usually redeemable by the issuer once the stock trades well above the strike, and that redemption right, a call the issuer holds on the warrant, caps the holder's upside. Redemption and, before the merger, the risk that no deal closes and the warrants expire worthless are why SPAC warrants trade well below the plain Black–Scholes–Merton value of their terms, and a valuation that ignores them is badly wrong.

Covered warrants are issued by a bank, not by the company, on someone else's stock. They are the dominant retail option instrument in Hong Kong, Germany, and several Asian markets, and they are simply calls and puts in a retail wrapper: exercise settles in cash or in existing shares, so there is no dilution and (17.1) does not apply. What they carry instead is issuer credit risk (the uncollateralized counterparty exposure that Chapter 9's CVA prices) and a bid–ask spread set by a single market maker who is also the issuer.

The test is always the same question: does exercise create a share or move one? If it creates one, the dilution factor applies and the instrument is a claim on the company. If it moves one, the instrument is an ordinary option and the issuer is just a counterparty.

17.10 BACKDATING

Between roughly 1996 and 2005, a substantial number of US companies granted options and then chose the grant date afterward, selecting a day on which the stock had been low, and reported the grant as at-the-money. It was not.

The mechanism was pure accounting arbitrage. Under APB 25 an at-the-money grant carried zero expense and an in-the-money grant carried an expense equal to its intrinsic value. Backdating converted the second into the first in the records while delivering the second in substance. Example 17.9 puts numbers on both halves of that statement.

Example 17.9 The arithmetic of choosing the lowest price

Take a stock with 45% annual volatility, $r = 4\%$ and $q = 1\%$. Over a 21-trading-day month, the expected minimum is **8.3%** below the expected price on a randomly chosen day.

Grant 500,000 ten-year options honestly at the month's typical price of \$40 and the strike is \$40. Backdate to the low and the strike is \$36.67.

The option value rises from 21.478 to 22.190, only 3.3%, or \$356,000 across the grant, because a ten-year option at 45% volatility is already worth 54% of the stock price and an 8% strike

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reduction moves it little.

The intrinsic value is the real number. The backdated option is immediately $3.33 in the money, so under APB 25 it carried a required expense of

$$3.33 \times 500{,}000 = \$1.67 \text{ million}$$

which was reported as zero.

That gap is the point: little value to the executive, a great deal of concealed expense to the company. It was the misreporting, not the transfer, that made it a securities-law violation rather than a compensation decision.

Case Study 17.1 How backdating was found, and what ended it

The practice was uncovered statistically rather than by a whistleblower. Erik Lie, an academic studying option grant dates, noticed that stock returns were systematically *negative* before grant dates and *positive* after, across hundreds of companies, over years.

No plausible innocent story fits that pattern. Executives cannot reliably forecast a bottom, and boards do not schedule compensation meetings for market troughs. A pattern of buying at the low across a large sample is neither skill nor luck. It is hindsight.

What ended it was not enforcement but a change in reporting infrastructure. The Sarbanes–Oxley Act of 2002 required insiders to report option grants within **two business days** rather than, as previously, up to 45 days after the end of the fiscal year. Backdating requires a reporting lag, and the lag was removed. Heron and Lie (2007) show the anomaly shrinking sharply after 2002 and surviving mainly among grants reported late.

The episode matters for a reason beyond the fraud: it is a clean case of a disclosure rule making a behavior mechanically difficult rather than merely illegal, as Section 2.7's position limits and Section 2.11's clearing mandate did.

By March 2007 the research firm Glass Lewis counted 257 companies with internal reviews or SEC or Justice Department investigations, the SEC staff put its own count at about 140, dozens restated earnings, and a small number of executives were prosecuted. The much larger effect was on practice: grants are now made on pre-scheduled dates by formula, a lasting improvement bought with a scandal. A restatement now carries a further cost: under the SEC's Rule 10D-1, adopted in 2022 and in force through the exchanges' listing standards since October 2, 2023, a listed company must recover, regardless of fault, the incentive pay its executive officers received in the three fiscal years before a required restatement in excess of what the restated figures would have paid.

Grant timing today. The successor to backdating is **spring-loading**, granting just before good news is released. SAB 120 (November 2021) says such an award's fair value should reflect the pending news, and Item 402(x), adopted in December 2022, requires a company to state its grant-timing policy and tabulate executive option grants made from four business days before to one business day after a 10-Q, a 10-K or an 8-K disclosing material nonpublic information. The same release imposed a 90- to 120-day cooling-off period on officers' and directors' Rule 10b5-1 trading plans, which carry much of the selling of Section 17.11.

17.11 WHO USES EQUITY COMPENSATION, AND WHAT FOR

The instruments in this chapter exist to solve specific problems, and each solves a different one.

Retention is the dominant purpose below executive level, and RSUs do it best. Unvested units are forfeited on departure, so the employee carries a rolling balance of value they lose by leaving. An option grant achieves the same thing only while it is above water, which is why the

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drawdown behavior shown in Example 17.7 drove the shift to RSUs.

Alignment is the argument for options and for performance shares. An option pays only if the share price rises above the grant-date level, so the holder's payoff resembles a shareholder's, levered. Whether this genuinely changes behavior is contested; what is not contested is that it changes the *distribution* of the executive's payoff, and a convex payoff rewards volatility. An executive paid in options has a personal interest in the firm taking more risk, aligned with shareholders only when they want more risk, and never with creditors (Chapter 24).

Cash conservation is the startup's reason (Section 17.8).

Tax and accounting arbitrage was a reason and is much less of one. From 1994, Section 162(m) of the Internal Revenue Code capped a public company's deduction for each top executive's pay at $1 million but exempted performance-based pay, at-the-money options included, a push toward options that the 2017 tax act removed. The zero-expense era ended in 2006 and the reporting lag closed in 2002. What remains is genuine but modest: the corporate deduction on option exercise can exceed the recorded expense when the stock has risen, producing a windfall tax benefit.

Where this shows up in market prices. Two channels connect the chapter to the rest of the book. Large option and RSU programs generate predictable, calendar-driven *selling* (vesting dates and same-day-sale exercises, in which the shares are sold at once to cover the strike and the tax), which single-stock market makers track explicitly. And companies that grant heavily buy back stock to offset dilution, making buyback programs partly a mechanical consequence of the compensation plan rather than a signal about valuation. Reading a buyback announcement as a statement about intrinsic value, when it is offsetting a vesting schedule, is the same category of error as reading a borrow cost as a directional view.

SUMMARY

An employee stock option is non-transferable, forfeited on departure, exercisable only after vesting, and long-dated. Those restrictions break the replication argument on which Chapters 13–16 rest, so its value is not enforced by arbitrage and must be modeled with a behavioral assumption stated explicitly. Employees exercise early (typically about halfway through the term, once the stock has risen far above the strike), which does not contradict Section 11.8's theorem: its hypothesis, that the holder can sell the option instead, does not hold. Defensible methods give materially different values for the same grant, and the most sensitive input is turnover: an HR statistic, not a market observable.

Dilution is measured three ways and they disagree. An outstanding, disclosed grant needs no valuation adjustment at all, because the share price already reflects it, and applying the $N/(N+M)$ factor to the market price double-counts. The reported measure, the treasury stock method behind diluted EPS, understates economic dilution, and the gap is widest exactly where time value dominates: near-the-money options on volatile stocks.

For the holder, exercising and holding is usually the mistake: it spends the strike, prepays the tax and gives up the time value to acquire exposure the option already provided, in exchange for capital-gains treatment of later appreciation. The case for exercising is diversification, not belief in the company, and its price is the after-tax time value; belief in the company is an argument for keeping the option.

Expensing at grant-date fair value replaced intrinsic-value accounting in 2005–06, and the value is never remeasured, so an option that expires worthless was still an expense. Graded recognition moves the cost well forward of where straight-line puts it. Performance awards divide on the same principle: a market condition is priced into the grant-date value and never trued up, while a non-market condition is trued up and reversed if missed. RSUs have displaced

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options at most large listed companies, and the reason is drawdown behavior: a fall in the stock destroys a larger fraction of an option grant's value than of an RSU's, which loses value exactly in proportion to the price. Repricing underwater options is expensed as the value it adds; a value-for-value exchange reports nothing but leaves the employee with less exposure to the share price than the options surrendered, because a deep out-of-the-money option carries a high delta for its value.

Private companies strike at a 409A value below the preferred round, because common stock is subordinate, and leaving with appreciated ISOs creates an AMT liability on a gain that cannot be sold to pay it. Warrants divide by whether exercise creates a share or moves one: corporate and SPAC warrants dilute, while a covered warrant issued by a bank on someone else's stock is an ordinary option carrying issuer credit risk. Backdating exploited the old intrinsic-value rule to conceal a substantial expense. It was detected statistically and ended by Sarbanes–Oxley's two-day reporting rule, a disclosure change that made the practice mechanically difficult rather than merely illegal.

FURTHER READING

Hull and White's (2004) paper on employee stock option valuation sets out the exercise-multiple lattice of Section 17.3 and is the direct source for the approach, with Huddart and Lang (1996), Carpenter (1998) and Bettis, Bizjak and Lemmon (2005) supplying the exercise evidence it is calibrated to; it is also explicit about what the model does and does not claim. Hall and Murphy (2002) and Meulbroek (2001) are where the undiversified-holder discounts quoted in Section 17.2 come from; both make plain how much the answer depends on the risk aversion assumed. Lie (2005) and Heron and Lie (2007) are the empirical backdating papers and repay reading as an exercise in inference from a distribution rather than from a document. On the accounting, ASC 718 and IFRS 2 differ in ways that matter for the graded-recognition question of Example 17.5, and any comparison of US and European compensation disclosures should start there. The practitioner literature on 409A valuation is largely produced by appraisal firms and should be read with that in mind.

PRACTICE QUESTIONS

17.1 Explain why Section 11.8's theorem on early exercise does not apply to an employee option, being precise about which assumption fails.

17.2 A grant has a ten-year term but an expected life of five years. Explain why using the expected life in Black–Scholes is an approximation rather than a correction, and whether its error has a definite sign.

17.3 Using Example 17.1, state which input the reported expense is most sensitive to, and explain why that is awkward for an auditor.

17.4 A company with 20 million shares issues 5 million warrants, and the equivalent call is worth \$8.00. What is each warrant worth? Explain why an outstanding, disclosed employee grant instead requires no dilution adjustment, and state which of a covered and a corporate warrant needs the factor in (17.1).

17.5 A grant of 200,000 options has a grant-date fair value of \$12 each and vests 25% a year over four years. Compute the first-year expense under straight-line recognition, and compare it with graded recognition.

17.6 Explain why one RSU is worth two to three options, why the ratio would be smaller for a more volatile company, and what happens to the expense of a performance award with a market and with a non-market condition if the target is missed.

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17.7 An employee leaves a private company holding 50,000 ISOs struck at $1.00 with a 409A value of $6.00. Compute the cash needed to exercise and the AMT income created.

17.8 Explain how backdating was detected, and why the Sarbanes–Oxley two-day reporting rule stopped it where prohibition had not.

17.9 A vested option is worth $18.40 with the stock at $60 and a strike of $45. State the time value, explain what an employee gives up by exercising, and use Example 17.4 to explain why "I believe in the company" is an argument against exercising rather than for it.

17.10 A company has 40 million shares, 4 million options struck at $25, and an average share price of $40. Compute the incremental shares under the treasury stock method ignoring unrecognized cost, explain why the method understates economic dilution, and identify the type of company for which the understatement is largest.

17.11 Using Example 17.6, state the incremental expense of the repricing and of the value-for-value exchange, and explain why the exchange is cheaper to report and weaker as an incentive.

17.12 A company grants 100,000 cash-settled SARs, fully vested at grant, worth $8 each; a year later they are worth $13. Explain how the expense differs from an equity-settled option grant of the same value, and what the pay-versus-performance table of Item 402(v) would show for the year.

FURTHER QUESTIONS

17.13 Example 17.1 shows the lattice value falling from 16.10 to 12.59 as the assumed exit rate rises from 0% to 6%. Explain the mechanism by which turnover destroys option value, distinguishing its effect on vested from unvested options, and discuss whether a company with high turnover should regard its option program as cheap or as ineffective.

17.14 Section 17.2 states that an employee values the option below its model value because they are undiversified, and that the gap appears nowhere in the accounts. Construct the argument that the accounting is nonetheless correct, then the argument that it is not, and state what each side takes the expense to be measuring.

17.15 An executive is paid entirely in at-the-money options. Show that their payoff is convex in the share price, and explain what this implies about their preferred level of firm risk. Extend the argument to creditors, and explain why the same grant that aligns the executive with shareholders sets them against bondholders.

17.16 Example 17.7 shows an option grant losing 74% of its value on a halving of the stock while an RSU grant loses 50%. A board argues that this makes options the better instrument because they punish failure. Evaluate the argument, distinguishing the retention objective from the incentive objective, and state what the board's conclusion should be for a senior executive and for a mid-level engineer.

17.17 Example 17.9 finds that backdating raised the option's value by only 3.3% while concealing $1.67 million of expense. Explain why the accounting effect so greatly exceeded the value transferred, and use this to argue that the practice is better understood as a financial-reporting offense than as a compensation abuse. Then explain why the two are hard to separate in a company where executive pay is benchmarked against reported earnings.

17.18 Section 17.11 claims that buyback announcements from heavy option grantors are partly mechanical rather than informational. Design a test that would separate the two motives using publicly available data, stating what you would need and what each hypothesis predicts.

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Chapter 18
The Greek Letters

Every chapter since Chapter 13 has rested on one idea: an option's price is the cost of manufacturing its payoff. This chapter verifies it by simulation: a delta-hedging program run 20,000 times costs, on average, exactly the formula's value.

The same simulation produces a second number, and it is this chapter's subject: at five rebalances the standard deviation of the hedging cost is $1.21 on an option worth $3.19. The model assumes continuous, costless hedging; the market permits neither.

The Greeks are the vocabulary for managing that gap. They are not sensitivity analysis added afterward to a pricing model; Section 13.11 established that delta *is* the pricing calculation. This chapter takes the remaining derivatives of the same formula and shows that each answers a specific question a dealer must answer before the end of the day.

Section 18.1 states the dealer's problem. Sections 18.2 to 18.6 take the Greeks one at a time: delta, gamma, theta, vega, rho, and the rest. Sections 18.7 to 18.9 are the book: managing it, what desks actually hedge, and how often to rebalance. Sections 18.10 to 18.13 go past the Greeks to synthetic options, the feedback when hedging moves the market, and static hedging.

18.1 THE DEALER'S PROBLEM

Start where a dealer starts. A client buys 100,000 call options on a stock at $63, struck at $64, with eighteen weeks to expiration: $S_0 = 63$, $K = 64$, $T = 18/52$, $\sigma = 22\%$, $r = 4\%$, and no dividend. These are the inputs behind every number in this chapter. Chapter 15's formula values each option at $3.19438, so the dealer receives

$$100{,}000 \times 3.19438 = \$319{,}438$$

The dealer now has a liability that will be worth $\max(S_T - 64, 0) \times 100{,}000$ in eighteen weeks. What should they do?

Do nothing: a naked position. If the stock finishes at $54 the options expire worthless and the dealer keeps $319,438. If it finishes at $74 they owe $1 million, for a loss of $680,562.

Buy the stock: a covered position. Buying 100,000 shares at $63 costs $6.3 million. If the stock finishes at $74 the shares gain $1.1 million and the options cost $1 million, leaving a profit of $419,438. But if it finishes at $54, the shares have lost $900,000 and the dealer is down $580,562.

Stock at expiration	Naked	Covered
$54	+$319,438	−$580,562
$74	−$680,562	+$419,438

Figure 18.1 plots the two. *Neither is a hedge.* Each converts the dealer's position into a directional bet, merely in opposite directions, and each can lose roughly twice the premium collected. A dealer who intermediates thousands of such trades cannot run either.

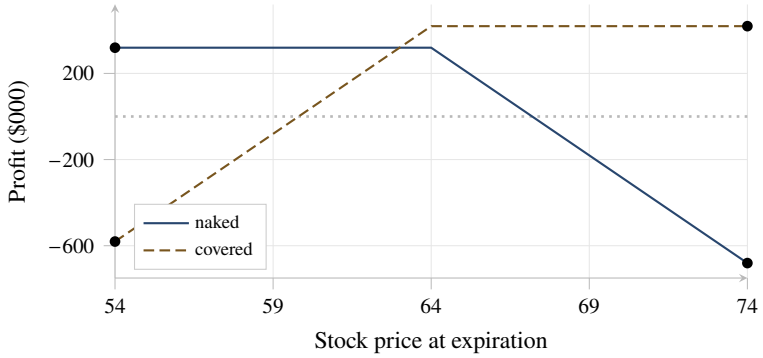


Figure 18.1: Profit at expiration on 100,000 written calls, $K = 64$, against the two unhedged responses. Marks are the table's two cases.

Switch between them: a stop-loss. The obvious repair is to run naked below \$64, buy the 100,000 shares when the stock rises through the strike, and sell them when it falls back. If every trade were done at exactly \$64, the dealer would hold the shares precisely when the option finishes in the money, the hedge would cost nothing, and the premium would be riskless profit. It cannot work. Purchases are made just above \$64 and sales just below, a stock near the strike crosses it again and again, and watching more closely makes each crossing cheaper but the crossings more numerous, so the hedge's error does not shrink. It is the naive hedge that delta hedging improves on.

What is needed is a position that holds up in *both* rows of the table, whichever way the stock moves, and Chapter 13 already found it: hold a quantity of stock that makes the portfolio insensitive to the stock's movement. That quantity is delta.

Delta is the first of several quantities the dealer has to control. Over a short interval Δt the value Π of the position changes because the stock moves by ΔS, because time passes, and because implied volatility moves by $\Delta\sigma$. Expanding that change in the things that move gives

$$\Delta\Pi \approx \Delta\,\Delta S + \frac{1}{2}\Gamma\,(\Delta S)^2 + \Theta\,\Delta t + \mathcal{V}\,\Delta\sigma \tag{18.1}$$

where delta Δ, gamma Γ, theta Θ (per year, with Δt in years) and vega $\mathcal{V}$ are the position's partial derivatives with respect to S, S twice, t and σ. A change Δr in the interest rate adds a term $\rho\,\Delta r$ in the same way, and for equity options it is usually small because the rate barely moves over the interval. Each Greek is the coefficient of one term, and hedging is the choice of offsetting positions that cancel terms of this expansion. Sections 18.2 to 18.6 take the terms in turn.

18.2 DELTA

Delta, the coefficient of the first term of (18.1), is the rate of change of an option's value f with respect to the price S of the underlying:

$$\Delta = \frac{\partial f}{\partial S} \tag{18.2}$$

For a European call on a non-dividend stock, Section 15.4 identified it already:

$$\Delta_{\text{call}} = N(d_1), \qquad \Delta_{\text{put}} = N(d_1) - 1 \tag{18.3}$$

so a call's delta lies in $(0, 1)$ and a put's in $(-1, 0)$. With a dividend yield, both are multiplied by e^{-qT}.

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In the dealer's problem, $N(d_1) = 0.5199$. Holding 51,995 shares against the 100,000 written calls makes the portfolio delta-neutral: a small move in the stock changes the value of the shares and the options by equal and opposite amounts. Example 18.1 reads delta as a multiplier and converts it into the option's leverage.

Every Greek in this chapter is evaluated at the option's own implied volatility, the convention Section 15.8 set for pricing, so a book's delta adds Black–Scholes deltas computed at different volatilities. That is exact only if each option's implied volatility stays put as the stock moves; Section 19.11 shows what changes when the smile moves with the spot, and equation (19.4) corrects the delta for it.

Example 18.1 Delta as a multiplier, and an option's leverage

Delta as a multiplier. Delta converts a move in the stock into a move in the option. If the stock rises by \$1 to \$64, the call should gain about $0.5199 \times 1 = \$0.52$; repricing at \$64 gives 3.7385, a rise of 0.5442. The gap, 0.0242, is curvature: the gamma term of (18.1), $0.04886/2 = 0.0244$ with the gamma of Section 18.3. For a \$3 fall delta predicts $-3 \times 0.5199 = -1.5597$, but repricing at \$60 gives 1.8568, a fall of only 1.3376: the value curve stays above delta's tangent.

Leverage. Dividing by the option's value turns the multiplier into percentages:

$$\frac{\Delta f}{f} \approx \Omega \frac{\Delta S}{S}, \qquad \Omega = \frac{S\,\Delta}{f}$$

where Ω, the option's **elasticity**, is its leverage: the percentage change in its value for a 1% change in the stock. Here $\Omega = 0.5199 \times 63/3.19438 = 10.25$: a 1% rise to \$63.63 raises the call by 10.56% exactly, and the call's volatility is about $\Omega\sigma \approx 226\%$. At the same stock price Ω is 5.71 at a strike of 54 and 15.98 at 74. Out-of-the-money calls carry the most leverage, which is why they are the speculator's instrument; Ω is also the $N(d_1)V/E$ that Section 24.6 computes for equity as a call on the firm.

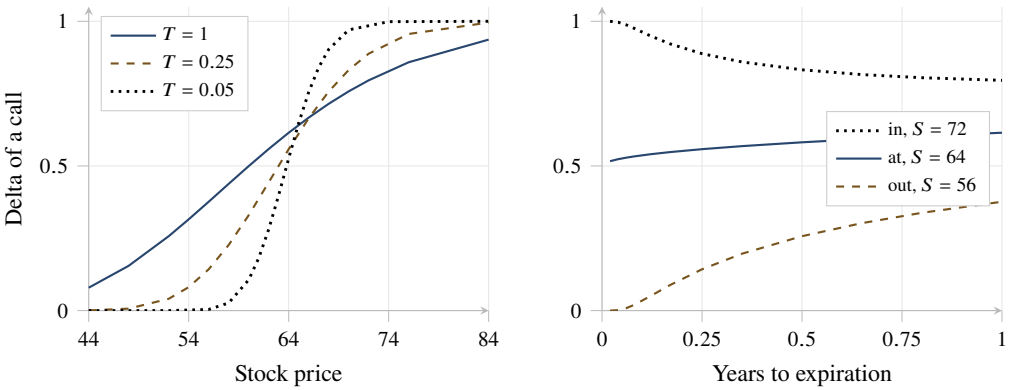


Figure 18.2: Call delta against the stock price and against remaining life. $K = 64$, $r = 4\%$, $\sigma = 22\%$.

Figure 18.2 shows the feature that makes hedging difficult. As expiration approaches, delta stops being a smooth function and approaches a step: an option that will finish in the money has delta 1, one that will finish out has delta 0, and near the strike the transition becomes almost vertical. A hedge tracking a near-vertical function must buy or sell almost the entire position on a small move in the stock. The right panel says it along the other axis: hold the stock still and delta walks to 1, to 0, or to neither as expiration nears. This is the pin risk of Example 10.6 and the 0DTE mechanism of Section 10.12, both now visible as statements about $N(d_1)$.

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A hedging program, week by week

The dealer's position is best understood by running it. Tables 18.1 and 18.2 follow the same written position (100,000 calls, $K = 64$, eighteen weeks) along two simulated price paths, rebalancing weekly to restore delta neutrality. The two paths differ in one respect only: the first finishes in the money and the second finishes out.

The bookkeeping is the same in both. The dealer borrows to buy $\Delta \times 100{,}000$ shares at week 0, and each week borrows or repays to bring the holding back to the new delta. The cumulative cost column compounds the previous week's balance at 4% and then adds the week's trade, so it is the dealer's outstanding debt.

Table 18.1: Weekly rebalancing against 100,000 written calls, path finishing in the money. $S_0 = 63$, $K = 64$, $r = 4\%$, $\sigma = 22\%$, 18 weeks. Costs in \$000.

Week	Stock price	Delta	Shares purchased	Cost of shares	Cumulative cost	Interest
0	63.00	0.520	51,995	3,275.7	3,275.7	2.5
1	61.12	0.421	−9,889	−604.4	2,673.8	2.1
2	62.51	0.488	6,651	415.8	3,091.6	2.4
3	60.86	0.394	−9,362	−569.7	2,524.2	1.9
4	58.46	0.261	−13,343	−780.0	1,746.1	1.3
5	61.57	0.418	15,788	972.1	2,719.5	2.1
6	62.02	0.438	1,915	118.8	2,840.4	2.2
7	62.03	0.431	−692	−43.0	2,799.6	2.2
8	61.55	0.391	−3,959	−243.7	2,558.1	2.0
9	63.48	0.513	12,183	773.4	3,333.5	2.6
10	61.94	0.396	−11,724	−726.2	2,609.8	2.0
11	64.41	0.574	17,822	1,147.9	3,759.8	2.9
12	64.71	0.597	2,360	152.7	3,915.4	3.0
13	63.86	0.523	−7,417	−473.7	3,444.7	2.6
14	63.25	0.455	−6,797	−429.9	3,017.5	2.3
15	66.43	0.781	32,561	2,163.0	5,182.9	4.0
16	70.36	0.988	20,695	1,456.1	6,642.9	5.1
17	69.67	0.998	975	67.9	6,716.0	5.2
18	66.04	1.000	238	15.7	6,736.8	
Shares delivered on exercise, proceeds $64 \times 100{,}000$					−6,400.0	
Cost of hedging the option					**336.8**	

Example 18.2 Reading the two hedging programs

The premium collected in Section 18.1 was \$319,438. Invested at 4% for eighteen weeks it would be worth \$323,900 at expiration, the number to compare the hedging cost against.

The hedge cost \$336,800 on the first path and \$309,300 on the second. One is 4.0% above the premium and the other 4.5% below. The dealer lost money on one path and made money on the other, having done nothing different in either case.

That is the whole content of the simulation below, seen transaction by transaction: discrete hedging does not reproduce the option payoff; it reproduces it *on average*, and the residual is the standard deviation Table 18.3 measures. Weekly rebalancing over eighteen weeks is eighteen adjustments, and eighteen is not many.

The trades themselves. In weeks 4 and 5 of Table 18.1 the stock fell to 58.46 and the dealer *sold* 13,343 shares, then rose to 61.57 and the dealer *bought* 15,788 back, \$3.11 higher, a loss of

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Table 18.2: The same position along a path finishing out of the money. Costs in $000.

Week	Stock price	Delta	Shares purchased	Cost of shares	Cumulative cost	Interest
0	63.00	0.520	51,995	3,275.7	3,275.7	2.5
1	61.60	0.446	−7,443	−458.5	2,819.7	2.2
2	63.45	0.536	9,080	576.1	3,398.0	2.6
3	61.36	0.421	−11,549	−708.6	2,692.0	2.1
4	61.57	0.426	474	29.2	2,723.2	2.1
5	61.08	0.390	−3,530	−215.6	2,509.7	1.9
6	59.75	0.305	−8,524	−509.3	2,002.3	1.5
7	58.01	0.201	−10,372	−601.7	1,402.2	1.1
8	58.38	0.205	346	20.2	1,423.5	1.1
9	61.22	0.358	15,323	938.1	2,362.6	1.8
10	58.88	0.197	−16,084	−947.1	1,417.4	1.1
11	62.68	0.440	24,278	1,521.8	2,940.3	2.3
12	64.54	0.584	14,383	928.3	3,870.8	3.0
13	64.24	0.558	−2,599	−167.0	3,706.9	2.9
14	66.55	0.765	20,684	1,376.5	5,086.2	3.9
15	64.57	0.594	−17,061	−1,101.6	3,988.5	3.1
16	63.73	0.484	−11,027	−702.8	3,288.8	2.5
17	62.86	0.292	−19,212	−1,207.7	2,083.7	1.6
18	60.90	0.000	−29,162	−1,776.0	309.3	
Option expires worthless; no shares held					0.0	
Cost of hedging the option					**309.3**	

about $41,500 on the shares just let go. Nothing was mismanaged: as Example 13.5 showed, delta hedging sells low and buys high by construction, and the accumulated cost of doing so is the option premium.

Why the paths end where they do. In Table 18.1 delta walks up to 1.000 and stays there, so the dealer ends holding the full 100,000 shares and delivers them on exercise. In Table 18.2 delta walks down to zero and the last trade liquidates the remaining position. The program converges to the terminal payoff automatically (no decision is made at expiration) because $N(d_1)$ becomes the step function of Figure 18.2.

Where the cost came from. On the out-of-the-money path the option expired worthless and the hedge still cost $309,300. In weeks 13 to 15 the stock went 64.24, 66.55, 64.57, ending within a third of a dollar of where it started, while the dealer bought 20,684 shares at 66.55 and sold 17,061 of them back at 64.57: about $33,800, a tenth of the premium, lost on one whipsaw. That is the gamma cost of Section 18.3, arriving before the concept is defined: a delta hedge pays for movement, not direction, and most for movement near the strike, where gamma peaks.

Portfolio delta. Delta is additive. A book of options on the same underlying has

$$\Delta_{\text{portfolio}} = \sum_i w_i \Delta_i$$

where w_i is the number of options of type i held, negative for those written. A desk holding thousands of positions therefore computes one number and trades one quantity of stock.

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The price is the cost of the hedge

The two paths above showed the same program landing once above and once below the premium. Over many paths the claim behind every chapter since Chapter 13, verified on a two-step tree in Example 13.5, can be tested against the formula itself. Take the dealer's option of Section 18.1, worth \$3.1944, and run the program of Table 18.1 over 20,000 simulated paths, rebalancing n times and paying out $\max(S_T - K, 0)$ at expiration. The paths carry a *real-world* drift of $\mu = 10\%$, deliberately different from $r = 4\%$. Table 18.3 reports the cost in time-0 dollars.

Table 18.3: Cost of hedging 1 written call with no trading costs, in time-0 dollars. 20,000 paths; $\mu = 10\%$; formula value 3.1944.

Rebalances n	Mean cost	Std. dev.	Mean − formula
5	3.2079	1.2050	+0.0135
10	3.1946	0.8560	+0.0002
20	3.1950	0.6265	+0.0006
50	3.1972	0.4011	+0.0028
100	3.1942	0.2873	−0.0002
250	3.1949	0.1830	+0.0005
Formula (15.2)	3.1944	—	—

The mean cost is the formula's value. Once rebalancing is frequent enough the two agree to within a few tenths of a cent, which is the simulation's own sampling error. The dealer who charges \$3.19 and hedges diligently breaks even on average; that is what the number means, and why the formula is a price rather than a forecast. And the drift did not matter: the paths were simulated at 10% while the formula uses 4%, which is Chapter 13's principle verified numerically.

Discrete hedging leaves risk, and it falls as $1/\sqrt{n}$. At five rebalances the standard deviation is \$1.21, more than a third of the option's value; at 250 it is \$0.18. The hedge is exact only in the continuous limit, and the residual is what Chapter 15's assumption of continuous trading conceals. Kamal and Derman (1999) give its standard deviation as approximately $\sqrt{\pi/4}\,\sigma\mathcal{V}/\sqrt{n}$, with $\mathcal{V}$ the option's vega (14.77 here; Section 18.5). That is 0.407 at $n = 50$ and 0.288 at $n = 100$, within 2% of the table from 50 rebalances on; at five, where the approximation is coarsest, it overstates by 7%. That residual is where Section 18.9 begins: the model says hedge continuously, the market charges for every trade, and Table 18.7 adds the charge to exactly this simulation.

18.3 GAMMA

Delta hedging is exact only for infinitesimal moves, because delta itself changes. **Gamma** measures how fast:

$$\Gamma = \frac{\partial^2 f}{\partial S^2} = \frac{N'(d_1)}{S_0\sigma\sqrt{T}} \tag{18.4}$$

where N' is the standard normal density, the slope of N:

$$N'(x) = \frac{1}{\sqrt{2\pi}}\, e^{-x^2/2}.$$

Gamma is positive for both purchased calls and purchased puts, and negative for written ones.

The practical meaning is direct. A delta-neutral portfolio has no first-order exposure, so its profit and loss $\Delta\Pi$ over a move ΔS, to second order, is what remains of (18.1) once the delta

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term is hedged away and time and volatility are held still:

$$\Delta\Pi \approx \frac{1}{2}\Gamma\,(\Delta S)^2 \tag{18.5}$$

Figure 18.3 is the same statement geometrically. The delta approximation is the tangent at the hedge point; the option's value curves away from it on both sides, so a written option hedged with stock loses whichever way the stock goes.

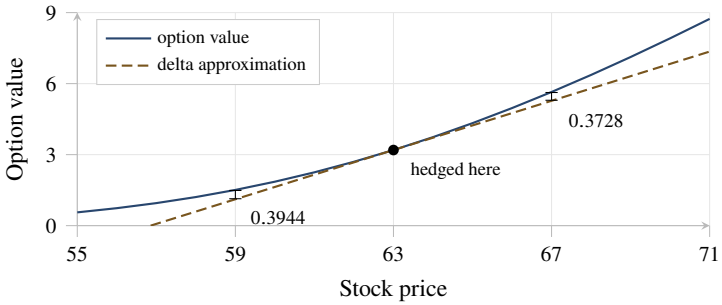


Figure 18.3: The dealer's option against its delta approximation at \$63. The marked gaps are Example 18.3's losses.

Example 18.3 What gamma costs a short option position

The dealer of Section 18.1 is delta-neutral at $S = 63$, with $\Gamma = 0.04886$ per option, so the book's gamma is $-0.04886 \times 100{,}000$, negative because the options are written.

Suppose the stock jumps before the hedge can be adjusted:

Stock	Actual P&L	$-\frac{1}{2}\Gamma(\Delta S)^2$
59	−0.3944	−0.3909
61	−0.0987	−0.0977
63	0	0
65	−0.0959	−0.0977
67	−0.3728	−0.3909

per option. The approximation is accurate to within two cents, and on 100,000 options a move to \$59 costs \$39,443 and a move to \$67 costs \$37,283.

Both columns are negative. A delta-neutral short-option position loses money whichever way the stock moves, and loses in proportion to the *square* of the move. Being short gamma means being short large moves regardless of direction, the same statement as Section 12.4's observation that a straddle buyer is betting on magnitude, seen from the other side of the trade.

What gamma is worth. Equation (18.5) puts a value on movement, and it depends on two things. The *sign* decides whether movement pays or costs: a book long gamma gains whichever way the stock goes, a book short gamma loses either way. The *size* decides how much, and because the relation is quadratic, doubling the move quadruples the amount. Section 18.4 sets that against what holding the position costs.

Figure 18.4 shows where the trouble concentrates. Gamma peaks at the money and rises sharply as expiration nears: at one year the peak is 0.031, at three months 0.058, at under three weeks 0.127, four times the one-year peak. At the money the contrast is sharper: with a week to run gamma is 0.20 against 0.027 for a one-year option (Figure 18.5), more than seven times. Gamma is the rate at which delta moves, so on the dealer's 100,000 options that means

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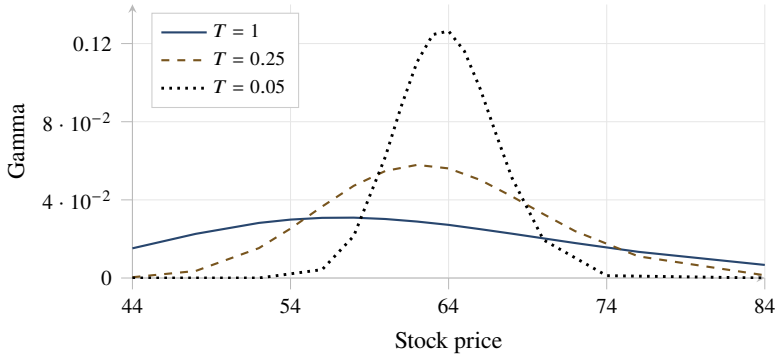


Figure 18.4: Gamma. $K = 64$, $r = 4\%$, $\sigma = 22\%$.

rehedging some 20,000 shares for every dollar the stock travels, against 2,700 for the one-year option, which explains most of what Section 10.12 said about same-day options.

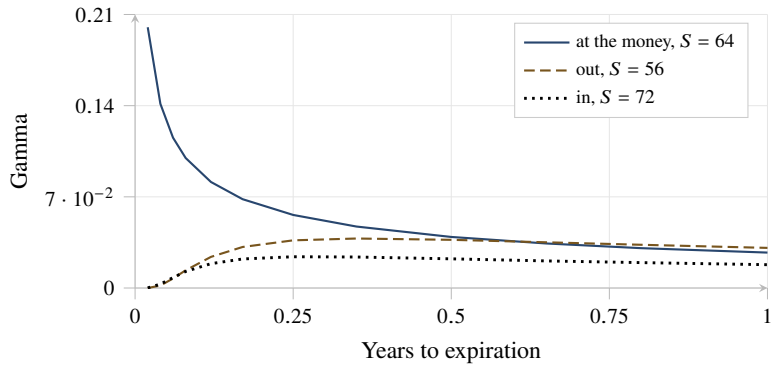


Figure 18.5: Gamma as expiration approaches. $K = 64$, $r = 4\%$, $\sigma = 22\%$.

Figure 18.5 shows the other dimension. At the money gamma rises without limit as expiration approaches, while away from the money it collapses toward zero. An option that will finish decisively in or out of the money has nothing left to hedge; one that will finish near the strike has all of it left, and no time in which to do it.

Neutralizing gamma requires options. Delta can be hedged with the stock, because the stock's delta is 1. Gamma cannot: the stock's gamma is zero, so no quantity of it changes the book's gamma. Only another option can offset gamma, which is why dealers trade with each other and not only with clients, and why a desk short gamma across its whole book cannot reduce that exposure cheaply.

18.4 THETA, AND WHY IT IS GAMMA UNDER ANOTHER NAME

Theta, the coefficient of Δt in (18.1), is the rate at which an option loses value as time passes:

$$\Theta = \frac{\partial f}{\partial t} \tag{18.6}$$

For a European option on a non-dividend stock,

$$\Theta_{\text{call}} = -\frac{S_0 N'(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}N(d_2), \qquad \Theta_{\text{put}} = -\frac{S_0 N'(d_1)\sigma}{2\sqrt{T}} + rKe^{-rT}N(-d_2) \tag{18.7}$$

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Both share the first term, the decay of time value, and differ in the second, the interest on the strike, negative for the call and positive for the put. Theta is usually quoted per calendar day, and for the dealer's option $\Theta = -5.8757$ per year, or about $-\$0.016$ a day.

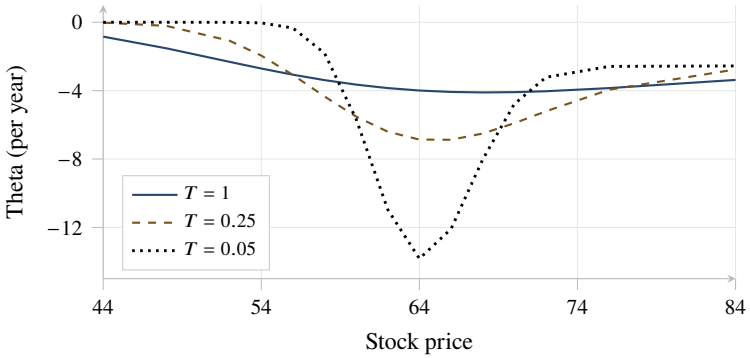


Figure 18.6: Call theta, per year. $K = 64$, $r = 4\%$, $\sigma = 22\%$.

Figure 18.6 shows the shape. Set beside Figure 18.4 it is close to a mirror image (both concentrate near the money, and both sharpen as expiration nears), and the rest of this section shows that the resemblance is an identity rather than a coincidence.

Theta is not a risk in the sense delta and gamma are, because the passage of time is not uncertain; what matters is its tie to gamma.

Recall the Black–Scholes–Merton equation (15.1):

$$\Theta + rS\Delta + \frac{1}{2}\sigma^2 S^2 \Gamma = rf$$

This holds for every derivative the model prices. For a *delta-neutral* portfolio, $\Delta = 0$ and the equation reduces to

$$\boxed{\Theta + \frac{1}{2}\sigma^2 S^2 \Gamma = r\Pi} \tag{18.8}$$

Theta and gamma are two sides of one quantity. A delta-neutral book long gamma normally pays theta and one short gamma normally earns it, with the trade-off fixed by the model rather than chosen by the trader. The word is *normally* because, where gamma is nearly zero, the interest term in (18.8) is all that remains, which is how a deep in-the-money put carries positive gamma and positive theta at once. Example 18.4 checks the identity on the dealer's option.

Example 18.4 Verifying the identity

For the dealer's option: $\Theta = -5.8757$, $\Delta = 0.51995$, $\Gamma = 0.048862$, $S = 63$, $\sigma = 0.22$, $r = 0.04$, $f = 3.1944$.

$$\Theta + rS\Delta + \frac{1}{2}\sigma^2 S^2 \Gamma = -5.8757 + 0.04 \times 63 \times 0.51995 + \frac{1}{2} \times 0.0484 \times 63^2 \times 0.048862$$
$$= -5.8757 + 1.3103 + 4.6932 = 0.1278$$

and $rf = 0.04 \times 3.1944 = 0.1278$, exactly.

A trader long gamma is paid whenever the stock moves, by (18.5), and pays theta every day it does not. A trader short gamma collects theta daily and pays whenever the stock moves. Neither is a riskless profit; the model prices the exchange, and (18.8) is the price.

This also explains the standing warning of Section 12.11. A portfolio of butterflies, condors, and calendar spreads is not diversified: all are short gamma, so all collect theta and all lose together on a large move. Equation (18.8) says they are the same position.

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The break-even move. That exchange has a break-even. A delta-hedged position also carries its financing, the interest on the borrowing in Table 18.1's cumulative cost column, and that is the $r\Pi$ of (18.8). Net of it, the position's carry is $\Theta - r\Pi = -\frac{1}{2}\sigma^2 S^2 \Gamma$. Setting the gamma gain of (18.5) against that carry over one trading day, $\Delta t = 1/252$ on Section 15.8's clock, gives the move at which a delta-hedged position comes out level:

$$\frac{1}{2}\Gamma(\Delta S)^2 = \frac{1}{2}\sigma^2 S^2 \Gamma \, \Delta t \qquad \Longrightarrow \qquad |\Delta S| = \sigma S \sqrt{\Delta t} \tag{18.9}$$

For the dealer's option that is $63 \times 0.22/\sqrt{252} = \$0.87$, or 1.39% of the \$63 spot. On a day the stock moves less than that the written position is ahead; on a day it moves more it is behind, and the whipsaw of Example 18.2 was a run of the second kind. Setting the gain against the option's theta alone, $|\Theta|/365$, would give \$0.81, which leaves out the financing and counts weekends as days on which the stock moves.

Gamma cancels, so the break-even is one standard deviation of the stock's daily move at the implied volatility, whatever the option. That is what it means for the option to have been fairly priced, and why a desk that has sold gamma is short realized volatility against the implied volatility it sold.

Which volatility to hedge at. Hedged at its implied volatility σ_i, a long option earns the gap between realized and implied variance day by day, weighted by gamma (Further Question 18.15), so its total depends on the path as well as on how much the stock moves. If the forecast is right, hedging at the volatility σ_r the trader expects to be realized locks in the total instead, at $f(\sigma_r) - f(\sigma_i)$, the option's value at that volatility less its value at the implied one; the position's marked value still swings until expiration (Ahmad and Wilmott, 2005). Most desks hedge at implied, the volatility the book is marked at.

18.5 VEGA

Vega measures sensitivity to volatility and is the coefficient of $\Delta\sigma$ in (18.1):

$$\mathcal{V} = \frac{\partial f}{\partial \sigma} = S_0 \sqrt{T}\, N'(d_1) \tag{18.10}$$

It is positive for all purchased options, calls and puts alike, as Section 11.1 established without a model.

For the dealer's option, $\mathcal{V} = 14.769$, meaning a one-percentage-point rise in volatility raises each option by \$0.148, or \$14,769 across 100,000 options. Set that beside the \$319,438 premium: a two-point move in implied volatility changes the position's value by a tenth of what the dealer was paid to take it on.

Two features matter in practice.

Vega is largest for long-dated options, scaling roughly with $\sqrt{T}$, which is the opposite of gamma. A book can therefore be short gamma and long vega at once, and desks routinely are.

Vega should not exist. The Black–Scholes–Merton model assumes σ is a known constant, so a sensitivity to changes in it is a sensitivity to something the model says cannot happen. That the market computes and hedges it anyway is the clearest admission that the model is being used as a quoting convention rather than as a description, which is Section 15.7's point, now visible in the risk report. Chapter 19 confronts the consequences.

Like gamma, vega peaks near the money (Figure 18.7); unlike gamma, it does not spike as expiration approaches but shrinks. Comparing with Figure 18.4 makes the point that governs how books are constructed: the short end carries the gamma and the long end carries the vega, so a desk's two largest risks live at opposite ends of the maturity curve and must be hedged with different instruments.

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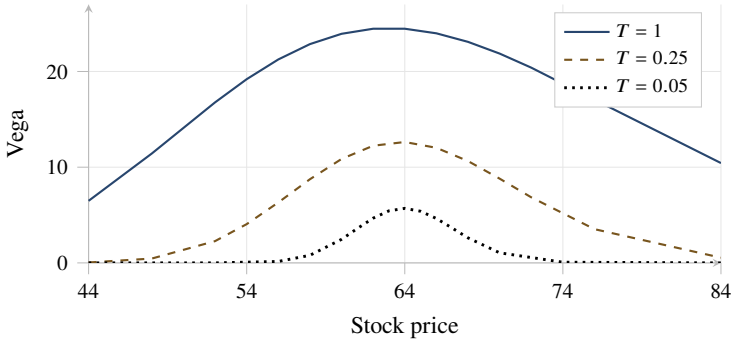


Figure 18.7: Vega. $K = 64$, $r = 4\%$, $\sigma = 22\%$.

18.6 RHO AND THE REST

Rho measures sensitivity to the interest rate and supplies the $\rho\,\Delta r$ term that (18.1) sets aside:

$$\rho_{\text{call}} = KTe^{-rT}N(d_2), \qquad \rho_{\text{put}} = -KTe^{-rT}N(-d_2) \tag{18.11}$$

Only the discounted strike in Chapter 15's formula carries r, so rho is the time to expiration multiplied by the present value of what the holder expects to pay. For the dealer's option $\rho = 10.233$, so a one-point rise in rates is worth \$0.102 an option, or \$10,233 across the 100,000.

Set that beside vega: a one-point move in implied volatility is worth \$14,769 on the same position, the same order of magnitude. Rho is ignored on short-dated equity books not because the sensitivity is small but because the input barely moves: implied volatility travels several points in a week, policy rates in 25-basis-point steps a few times a year.

Maturity reverses that. Rho grows with T while an option's value grows roughly with $\sqrt{T}$, so the exposure per dollar of premium climbs with the term:

At-the-money call	Value	Rho	Per 100bp, % of value
18 weeks	3.739	11.29	3.0%
1 year	6.841	32.51	4.8%
5 years	18.008	147.71	8.2%
10 years	27.301	253.04	9.3%

So rho is watched wherever the term is long or the rate is the point: the ten-year employee options and warrants of Chapter 17, the convertible bonds of Chapter 25, the structured notes of Chapter 35; and the conversions, reversals and boxes of Chapter 11, which are option structures assembled so that the financing rate is the only exposure left, and which market makers use to lend and borrow. In Chapter 29's rate derivatives rho stops being a side-Greek altogether, because there the curve is the underlying.

In currency options rho is first-order. An option on an exchange rate has two rhos, one per currency, and Chapter 16's substitution gives both:

$$\rho = KTe^{-rT}N(d_2), \qquad \rho_f = -STe^{-r_fT}N(d_1)$$

where r is the domestic interest rate, r_f the foreign one, and S the exchange rate, as in Chapter 16. They almost cancel. Adding them gives exactly $-Tc$, so a parallel move in both rates does no more than discount the option, and what survives is the exposure to the *differential*, which is what sets the forward. Take Chapter 16's euro–dollar inputs: spot 1.0850, dollar rates 4%, euro rates 2.5%, volatility 8%. A one-year euro call struck at spot gains 14.4% of its premium when dollar rates rise 100 basis points and euro rates do not; the equity option of Section 18.1 gains 3.2% on the same shift, and even a three-month currency option gains 7.5%. That is why a

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central bank meeting is a scheduled volatility event in the currency market in a way it is not for a single stock.

Beyond the five, desks compute higher-order and cross sensitivities as the book requires: **vanna** ($\partial^2 f/\partial S\,\partial\sigma$, how delta moves with volatility) and **volga** ($\partial^2 f/\partial\sigma^2$, how vega moves with volatility) matter wherever the smile matters, which is Chapter 19's territory and essential for the barrier options of Chapter 34. **Charm** ($\partial^2 f/\partial S\,\partial t$, how delta moves with the passage of time) is a rounding error for a monthly option and the dominant hedging flow for the same-day options of Chapter 23, where a dealer's delta can fall by two-thirds over a session in which the index does not move.

Table 18.4 collects the five.

Table 18.4: Definitions, signs for a purchased option, and what each answers.

Greek	Derivative	Sign	Question it answers
Delta Δ	$\partial f/\partial S$	+ call, − put	How much stock to hold
Gamma Γ	$\partial^2 f/\partial S^2$	+	How often to rebalance, and what a jump costs
Theta Θ	$\partial f/\partial t$	−	What waiting costs per day
Vega $\mathcal{V}$	$\partial f/\partial\sigma$	+	What a change in implied volatility costs
Rho ρ	$\partial f/\partial r$	+ call, − put	What a rate move costs

18.7 MANAGING A BOOK

A desk does not hedge options one at a time. Every Greek is a derivative, so all are additive across positions on the same underlying, and a book of thousands of trades is five numbers.

Additivity means the book has a single expansion (18.1), whose coefficients are the summed Greeks, and hedging the book is canceling its terms in order. The order is set by which instrument reaches which term. Stock carries only the first term, and only options carry the second, so gamma is cleared with options before delta is cleared with stock; Example 18.6 does it in that order. Example 18.5 first checks the expansion against a full revaluation of the book in Table 18.5.

Table 18.5: Three positions on one stock. $S = 63$, $r = 4\%$, $\sigma = 22\%$.

Position	Quantity	Delta	Gamma	Vega
Calls K=64, 18 weeks	−100,000	−51,995	−4,886	−1,476,869
Puts K=58, 18 weeks	+50,000	−10,441	+1,761	+532,343
Calls K=70, 40 weeks	−40,000	−15,435	−1,259	−845,404
Book		**−77,870**	**−4,384**	**−1,789,929**

Desks report gamma in dollars rather than shares: the change in the position's dollar delta for a 1% move in the stock, $\Gamma S^2/100$. For this book that is $-4{,}384 \times 63^2/100 = -\$174{,}000$, the stock it must buy after every 1% rise. Vega is quoted per volatility point, one hundredth of the figures in the table, and theta per calendar day.

Example 18.5 The book's expansion against a full revaluation

One day passes ($\Delta t = 1/365$), the stock falls from \$63 to \$61 ($\Delta S = -2$), and implied volatility rises from 22% to 23% ($\Delta\sigma = 0.01$), with r unchanged at 4%. Table 18.5 gives the book's delta, gamma and vega. Its theta, from (18.7), is +601,812 per year, positive because the book is net short options. The terms of (18.1) are

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Term	Book Greek	Change	P&L ($)
$\Delta\,\Delta S$	−77,870	$\Delta S = -2$	+155,740
$\frac{1}{2}\Gamma(\Delta S)^2$	−4,384	$(\Delta S)^2 = 4$	−8,768
$\Theta\,\Delta t$	+601,812	$\Delta t = 1/365$	+1,649
$\mathcal{V}\,\Delta\sigma$	−1,789,929	$\Delta\sigma = 0.01$	−17,899
Sum of the four terms			+130,722
Full revaluation			+133,001
Residual			+2,279

The full revaluation prices each of the three positions by the Black–Scholes–Merton formula at $S = 61$, $\sigma = 23\%$ and one day less to expiration. The expansion accounts for all but $2,279 of the $133,001, or 1.7%. The delta term carries nearly everything. The day's theta of $1,649 is, by (18.8), largely the book's payment for being short 4,384 gamma, and by (18.9) its gamma and theta break even on a move of one daily standard deviation, $0.87; the $2 move costs more than five times the day's theta.

The residual is what the expansion leaves out: terms of third order in ΔS, and cross terms. About $1,850 of it is the cross term $(\partial^2\Pi/\partial S\,\partial\sigma)\,\Delta S\,\Delta\sigma$, the change in the book's vega as the stock moves, which the separate delta and vega terms cannot see. The expansion is local, and at this size what it misses is small; Section 18.10 takes up the moves where it is not.

Example 18.6 Neutralizing a book, in the right order

Table 18.5 gives a book that is short 77,870 delta, short 4,384 gamma, and short 1.79 million vega. Hedge it with one traded option, the 40-week call struck at 64:

	Delta	Gamma	Vega
The book	−77,870	−4,384	−1,789,929
40-week call, K=64, each	0.5692	0.03232	21.711

Gamma first, because it cannot be hedged with stock. Zeroing the book's gamma requires buying

$$\frac{4{,}384}{0.03232} = 135{,}644 \text{ options}$$

Then delta, with stock. Those 135,644 options carry $135{,}644 \times 0.5692 = 77{,}209$ of delta, which takes the book from −77,870 to −661. Buying 661 shares restores neutrality.

Vega is what remains. The hedging option carried $135{,}644 \times 21.711 = 2{,}944{,}967$ of vega, so the book moves from −1,789,929 to +1,155,038. Neutralizing gamma has flipped the sign of the vega exposure and left it large.

That is the general problem. There are three exposures and, on a single underlying, only two hedging instruments: the stock and one option. Neutralizing all three requires a *second* option of a different maturity, because gamma and vega have different term structures (Figures 18.4 and 18.7). Short-dated options are the gamma instrument and long-dated options are the vega instrument, and a desk needs both. Example 18.7 builds that hedge.

Example 18.7 Clearing gamma and vega together

Example 18.6 ended gamma-neutral but $1.16 million long vega. Two options of *different* maturities clear both at once. Keep the 40-week call and add an 8-week call at the same strike:

Instrument	Delta	Gamma	Vega	Gamma per unit vega
A: K=64, 8 weeks	0.4729	0.07321	9.835	0.00744
B: K=64, 40 weeks	0.5692	0.03232	21.711	0.00149

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The last column is the whole reason two maturities are enough. The short-dated option delivers five times as much gamma per unit of vega as the long-dated one, so the two are not redundant and any combination of gamma and vega can be reached.

Writing n_A and n_B for the quantities bought, neutrality is a 2×2 system:

$$0.07321\, n_A + 0.03232\, n_B = 4{,}384$$
$$9.835\, n_A + 21.711\, n_B = 1{,}789{,}929$$

with solution $n_A = 29{,}357$ and $n_B = 69{,}145$. Both are positive: the book is short both exposures, so both options are bought.

Then delta, with stock. The two positions carry $29{,}357 \times 0.4729 + 69{,}145 \times 0.5692 = 53{,}240$ of delta, taking the book from −77,870 to −24,630. Buying 24,630 shares closes it, and all three exposures are now zero.

Gamma and vega must be solved *together*, because every option moves both; delta is solved last, because the stock moves delta alone and so cannot disturb what has just been set. A hedge built in the other order has to be redone.

It is not free. The two option positions cost $421,218 to establish and carry a combined theta of about −$1,475 per day. That is Section 18.4 arriving as a cash flow: the desk has bought gamma, and buying gamma means paying theta every day the stock fails to move. Neutrality is a position with a running cost, not a state of rest.

Books across many underlyings

Table 18.5 aggregated positions on *one* stock. A single-stock desk may hold options on hundreds of stocks, and hedging each separately would multiply Table 18.7's transaction costs by the number of underlyings.

The standard response reuses Section 3.6's device. Convert each position's delta into an equivalent index exposure using the stock's beta,

$$\Delta_{\text{index-equivalent}} = \sum_i \beta_i\, \Delta_i\, S_i \tag{18.12}$$

where Δ_i is the book's delta in stock i, S_i is its price and β_i its beta against the index. Each term is therefore a dollar exposure rather than a share count, and the sum takes the place of βV_A in Section 3.6's hedge ratio, which divides it by the value of one futures contract to give the number to sell. One trade replaces hundreds.

The hedge removes the *market* component of the book's delta and leaves the idiosyncratic component, the part of each stock's movement unexplained by the index. That residual is small on any one day and is not diversified away, because the book's positions are not random: a desk accumulates whatever clients want to sell.

The same decomposition defines a trade. A book long single-stock options and short index options is long idiosyncratic volatility and short market volatility, which is the dispersion position Section 16.2 named. A desk that hedges by (18.12) without noticing has taken that position by accident, and it pays off when correlations fall and loses when they rise, which is when everything else is losing too.

Greeks for other underlyings

Chapter 16 extended the pricing formula to indices, currencies, and futures by the substitution $S_0 \to S_0 e^{-qT}$. The Greeks follow the same route, and Table 18.6 gives the results. The simplest case is a forward contract: its delta is e^{-qT}, or 1 without a yield, and its gamma is zero, so its hedge never has to chase the stock.

Two consequences follow. Delta is no longer bounded by 1 but by e^{-qT}: for an index option with $q = 3\%$ and $T = 2$, the factor $e^{-qT} = 0.942$ means a deep in-the-money call requires

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Table 18.6: Call Greeks for an underlying with yield q.

Greek	Formula
Delta	$e^{-qT}N(d_1)$
Gamma	$\dfrac{e^{-qT}N'(d_1)}{S_0\sigma\sqrt{T}}$
Theta	$-\dfrac{S_0N'(d_1)\sigma e^{-qT}}{2\sqrt{T}} + qS_0N(d_1)e^{-qT} - rKe^{-rT}N(d_2)$
Vega	$S_0\sqrt{T}\,e^{-qT}N'(d_1)$

only 0.94 units of the index, not 1. Currency desks often quote the premium-adjusted delta of Section 16.4 instead, so a quoted delta must be read with its convention. And for a *futures* option, $q = r$ makes every Greek a derivative with respect to the futures price F: the delta is $e^{-rT}N(d_1)$, the discounting of a payoff settled on F, and the hedge is held in the futures contract itself. The factor in Section 18.9's futures hedge (18.13) is a different adjustment, needed only when a spot-based option is hedged with futures.

18.8 WHAT DESKS ACTUALLY HEDGE

A textbook can neutralize every Greek. A desk cannot, and the choices it makes are informative.

Delta is hedged, continuously and almost automatically. It is cheap, since the underlying is the most liquid instrument available, and an unhedged delta swamps every other exposure.

Gamma is managed, not neutralized. Neutralizing it requires buying options, which costs theta by (18.8). Desks therefore run a gamma position deliberately, tightening it before events and letting it drift otherwise, and set limits rather than targets.

Vega is hedged by bucket. A single vega number is nearly useless because volatility does not move in parallel: the front of the curve moves more than the back, as Chapter 19 documents. Desks compute vega by maturity bucket and hedge each, which is the same bucketing logic Section 29.10 applies to interest rate deltas.

Rho is usually ignored in equities. It dominates in rates, where Chapter 29 takes it up.

18.9 HOW OFTEN TO REBALANCE

Table 18.3 showed hedging error falling as $1/\sqrt{n}$ with no transaction costs. Introducing costs changes the problem qualitatively, because they accumulate with every trade. Table 18.7 repeats that simulation, charging 0.10% of the value of each rebalancing trade, though not of the final settlement at expiration.

Table 18.7: Cost of hedging 1 written call, with a 0.10% proportional trading cost. 20,000 paths; formula value 3.1944.

Rebalances n	Mean cost	Std. dev.	Mean + s.d.
5	3.2628	1.1924	4.4552
10	3.2926	0.8634	4.1560
20	3.3158	0.6212	3.9370
50	3.3685	0.4075	3.7760
100	3.4266	0.2976	**3.7242**
250	3.5457	0.2232	3.7689

The two columns move in opposite directions (Figure 18.8). Trading more often reduces the *uncertainty* of the outcome (the standard deviation falls from 1.19 to 0.22) while raising

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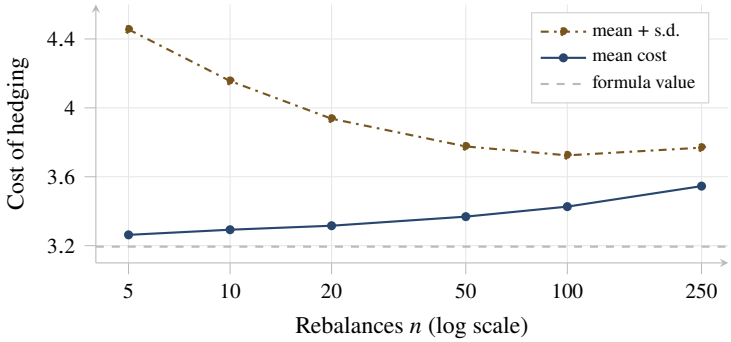


Figure 18.8: Hedging cost against rebalancing frequency, with a 0.10% trading cost.

its *expected* cost, from 3.26 to 3.55, as commissions accumulate. The total is minimized near $n = 100$.

Three conclusions follow.

There is a finite optimum. Once costs exist, continuous rebalancing is not merely impractical but wrong: the limit is strictly worse than a finite frequency.

Hedging is not free even at the optimum. At $n = 100$ the mean cost is 3.4266 against a formula value of 3.1944, so the dealer must charge about 7.3% above the model price to break even on trading costs. That margin is one component of the bid–ask spreads of Example 10.3, and it is why illiquid underlyings have wider option markets.

A clock is the wrong trigger. Table 18.7 rebalances at fixed intervals, which trades when nothing has happened and fails to trade when a great deal has. Desks instead rebalance when delta drifts outside a band, which saves trades in quiet markets and trades more often in volatile ones. Figure 18.4 shows what the band means for the stock: where gamma is high, each dollar the stock travels moves delta further, so the stock can travel less far before the band is crossed.

Hedging with futures instead

A desk hedging an index option uses futures rather than the basket, with Section 3.6's adjustment.

Because a futures price moves $e^{(r-q)(T^*-t)}$ times as far as the spot for a maturity T^*, the futures position required, in units of the underlying (dividing by the contract size gives contracts), is

$$N_{\text{futures}} = e^{-(r-q)(T^*-t)}\,\Delta_{\text{portfolio}} \tag{18.13}$$

Futures are more liquid and cheaper to trade than a basket, and they require margin rather than the full purchase price. The cost is basis risk (Section 3.3): the hedge tracks the futures price, and the option settles against the index. Where index futures trade persistently away from fair value, as China's index futures did after the 2015 trading restrictions, that basis is a running cost of a short futures hedge, charged at every roll.

18.10 BEYOND THE GREEKS

The Greeks are derivatives, so they describe behavior *near* the current state, where they are most reliable and least needed.

They fail for large moves. Equation (18.5) is a second-order approximation. Example 18.3 showed it accurate to within two cents for \$4 moves on a \$63 stock. For a 20% move it predicts a loss of \$3.88 an option either way, where repricing gives \$3.48 on a fall and \$3.07 on a rise, because gamma itself changes over so large a move. A risk report built entirely from Greeks

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describes a world in which nothing much happens, which is why every desk supplements them with **scenario analysis**: revalue the whole book at a grid of underlying prices and volatilities, and read the worst cell.

A typical grid moves the underlying by ±2%, ±5%, ±10%, ±20% and volatility by ±2, ±5, ±10 points, and reports the full revaluation rather than an approximation. The largest losses often sit in the corners, a big move *and* a volatility spike, where the cross term in $\Delta S\,\Delta\sigma$, which (18.1) omits, grows with the product of the two.

They assume you can trade. Every number in this chapter assumes the hedge can be adjusted at a quoted price. A gap denies it: a stock that closes at 63 and opens at 50 never traded at 56, so the delta is corrected after the loss rather than during it. Rebalancing more often does not help: Table 18.3's error falls as $1/\sqrt{n}$ only because the simulated path is continuous. On the dealer's position a 20% gap down costs $347,821 against the $319,438 collected, and a 30% one costs $663,836, the "roughly twice the premium" for which Section 18.1 rejected the naked position. The hedge is also financed and consumes liquidity: Table 18.1's cumulative cost column is borrowing, and Sections 18.11 and 18.12 describe what happens when every desk needs the same trade at the same moment.

They assume the inputs move independently. They do not: equity volatility rises when prices fall, which is the skew of Section 15.11 restated as a dynamic. The cross-derivative vanna of Section 18.6 captures this, and a book that is delta- and vega-neutral but vanna-exposed will discover the exposure on the day both move. Hull and White (2017), estimating the correction (19.4) from index and stock option data, find a minimum-variance delta below the Black–Scholes delta for calls and puts alike: fewer shares against a call, more against a put.

They assume the model. Every Greek in Table 18.4 is a derivative *of the Black–Scholes–Merton formula*. If the formula is the wrong map from prices to values, as Section 15.11 measured, its derivatives inherit the error. Where no closed form exists, the Greeks are themselves estimated, by bumping and revaluing, by pathwise or likelihood-ratio estimators, or by algorithmic differentiation, and Sections 20.10 and 32.13 show what each costs.

The Greeks remain indispensable, but they are a local linearization of an inexact model.

18.11 CREATING OPTIONS SYNTHETICALLY

Section 18.2 described a dealer hedging a written option. Running the same machinery in reverse manufactures an option that no one sold. That is what portfolio insurance did in 1987 (Section 40.2).

To create a synthetic long put on an index portfolio with yield q, hold the portfolio and maintain a short index position worth

$$e^{-qT}\,[1 - N(d_1)] \times \text{portfolio value}$$

taken in futures through (18.13) and adjusted continuously. As the market falls, $N(d_1)$ falls and the short grows; as it rises, the short shrinks. At expiration the combination has the payoff of portfolio-plus-put.

The attraction is that no premium is paid up front; Section 12.10 explained why that matters more to a committee than it should. The cost appears instead as the accumulated trading losses of Example 18.2. Two differences from buying a put are decisive.

The cost is uncertain rather than fixed. Buying a put fixes the price at inception. Replicating one pays the realized cost, which by Table 18.7 has a standard deviation that does not vanish. If

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realized volatility exceeds the volatility the program assumed, replication costs more than the put would have, exactly when protection is needed most.

The replication requires a liquid market, and it consumes liquidity. Selling futures into a decline is the mechanism, so the strategy needs a market able to absorb the sales at the moment everyone running the same program is selling. Buying a put transfers this problem to the writer; replicating one keeps it.

18.12 WHEN HEDGING MOVES THE MARKET

The Greeks describe how a dealer responds to the market. In aggregate those responses *are* part of the market, and the mechanism earlier chapters met without naming is gamma.

Consider the sign. A dealer who is **short gamma**, having sold options, must buy the underlying as it rises and sell as it falls, by (18.3). That is trading *with* the move, which amplifies it; a **long-gamma** dealer does the opposite, which damps it.

Five episodes in this book are now one phenomenon.

Pinning (Example 10.6). Near expiration, dealers long gamma at a strike buy below it and sell above it, holding the stock toward the strike. This is long-gamma damping, concentrated by the enormous gamma of Figure 18.4's shortest-dated curve.

Retail call buying (Section 10.12). Retail investors buy calls, leaving dealers short gamma, and dealer hedging amplifies moves: the opposite sign, and no contradiction, since which one prevails depends on which side of the options dealers hold.

Snowball knock-ins (Sections 12.12 and 35.7). Here the sign changes with the index. Issuers hold the barrier put the note's buyer has written, so above the barrier they are long gamma and buy as the index falls, damping the decline exactly as pinning does. At the barrier that reverses discontinuously: delta jumps, part of the hedge is sold, and, larger still, the bid that had been supporting the market stops.

Portfolio insurance (Section 40.2). This is the 1987 case, and the largest. Manufacturing a long put by trading means trading with the move, selling into declines: a short-gamma dealer's flow, executed by the option's would-be buyer rather than its writer.

Inverse volatility products (Section 22.8's case study). To keep a constant inverse exposure to VIX futures, an exchange-traded product must buy those futures when VIX rises, a short-gamma hedger's flow written into a prospectus; on February 5, 2018, that buying pushed the futures higher still, and the largest such product was terminated. Here the hedge was the product itself, so no dealer's risk report showed it.

Case Study 18.1 Reading dealer gamma

Because the sign of dealer gamma determines whether hedging damps or amplifies, estimates of aggregate dealer gamma have become a widely followed market indicator.

The estimates require assumptions that deserve skepticism. Open interest is observable, but who is long and who is short is not, and the usual convention (that dealers are long the calls, bought from investors writing calls against stock they hold, and short the puts, sold to institutions buying protection) is a guess that can be badly wrong. Retail call buying since 2020 has left dealers short many of the calls the convention counts as long; Chapter 23 shows how much of the answer the assumption decides.

The mechanism is nevertheless real: dealer hedging transmits option-market flow into the underlying, in a direction that depends on a position no one observes. That is enough to explain why volatility around expirations behaves differently and why large structured-product programs leave visible marks, but not to forecast a market.

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18.13 STATIC HEDGING

Everything so far has been **dynamic** hedging: hold a position, watch it drift, trade to restore it. Chapter 12 supplied the alternative, for when dynamic hedging breaks down.

Static hedging replicates a payoff with a portfolio of options bought once and held. Section 12.8 gave the recipe: any piecewise-linear payoff is a bond, a forward, and calls weighted by the change in slope at each kink. Such a hedge needs no rebalancing, incurs no gamma cost, and is indifferent to tomorrow's liquidity.

Static hedges use options, which are expensive and available only at traded strikes and maturities; dynamic hedges use the underlying, which is cheap and always available. One pays up front for certainty, the other as it goes for flexibility.

The choice is forced at discontinuities. A digital payoff is a step, so its gamma diverges near the strike at expiration. There, dynamic hedging becomes impractical, as it does near a barrier (Chapter 34); the usual answer is a static hedge, a tight call spread (Section 12.8), accepting a small mismatch for never having to trade. That is what went wrong in the snowball episodes of Sections 12.12 and 35.7: the issuers hedged dynamically where gamma diverged.

SUMMARY

A dealer who writes options and does nothing, or who buys the underlying outright, holds a directional bet in either case. Neither is a hedge. The Greeks are the coefficients that say how a book's value responds to each of the things that move it, and hedging is the business of controlling them.

Delta, $\partial f/\partial S$, is $N(d_1)$ for a call, and holding delta shares per option written makes a book insensitive to small moves. Scaled by S/f, delta becomes the option's elasticity, its leverage, which is highest out of the money. A delta hedge buys high and sells low by construction, and the accumulated cost of doing so is the option premium: that is what the premium pays for. Against a gap, though, there is no moment at which to trade.

Gamma measures how fast delta changes. A delta-neutral book gains or loses in proportion to the square of the move, so a short-gamma position loses on moves in either direction. Gamma peaks at the money near expiration and can be neutralized only with other options. Theta is gamma under another name: for a delta-neutral portfolio the Black–Scholes–Merton equation reduces to $\Theta + \frac{1}{2}\sigma^2 S^2\Gamma = r\Pi$, so a book long gamma pays for waiting and is paid when the market moves, and a short-gamma book is the reverse. With its financing counted, a delta hedge comes out level when the stock moves by exactly the amount its implied volatility assigns.

Vega is largest for long-dated options, the opposite of gamma, so books are commonly short gamma and long vega at once, and that vega exists at all admits that the model is a quoting convention rather than a description of the world. Rho is ignored on short-dated equity books because rates barely move, but it is first-order in currency options, where the forward is a rate differential.

Desks hedge delta continuously, manage gamma within limits rather than neutralizing it, and hedge vega bucket by bucket because volatility does not move in parallel. Hedging error falls with the number of rebalances while transaction costs rise with it, so optimal rebalancing is finite and is usually triggered by a delta band rather than a clock. A book's gamma and vega are cleared together with options of two maturities, and delta last with stock. The Greeks are local: they understate large moves and say nothing about gaps, so desks revalue the book on a grid of scenarios.

Dealer hedging feeds back into the underlying, with the sign set by gamma. Pinning, retail-driven amplification, snowball knock-ins, 1987's portfolio insurance, and the inverse volatility products of February 2018 are one mechanism seen five times, differing only in who holds which side. Manufacturing a put by trading replaces a known premium with an uncertain

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cost; where gamma diverges, at a digital's strike or a barrier, a static hedge in options is the answer.

FURTHER READING

Taleb's *Dynamic Hedging* is the standard practitioner account and covers what textbooks smooth over, particularly the behavior of the Greeks near barriers and expirations. On transaction costs and optimal rebalancing, the Leland (1985) adjustment is the place to start and shows explicitly how the effective volatility at which a writer must price rises with trading costs. For the feedback effects of Section 18.12, the Brady Commission's 1988 report is the primary source on the mechanism operating at scale, and Grossman (1988) analyzed what dynamic hedging does to the market it trades in, which Section 18.12's argument follows.

PRACTICE QUESTIONS

18.1 A dealer writes 50,000 calls with delta 0.62. How many shares must be held to be delta-neutral, and what happens to that number as the stock rises?

18.2 Explain why neither a naked nor a covered position is a hedge, using the two outcomes in Section 18.1.

18.3 A delta-neutral portfolio has gamma $-4{,}000$. Estimate the profit or loss if the underlying moves by \$3, and state whether the direction of the move matters.

18.4 Using (18.8), find the theta of a delta-neutral portfolio worth \$500,000 with gamma $-2{,}000$, when $S = 80$, $\sigma = 25\%$, and $r = 4\%$.

18.5 Explain why gamma cannot be hedged with the underlying asset but delta can.

18.6 An option has vega 18. By how much does its value change if implied volatility rises from 22% to 24%? On a position of 20,000 options, what is the total?

18.7 Explain why a book can be short gamma and long vega simultaneously, and which maturities contribute each.

18.8 A trader holds a portfolio of butterflies and calendar spreads and describes it as diversified. Using (18.8), explain why it is not.

18.9 Using Table 18.7, state which rebalancing frequency minimizes the mean cost plus one standard deviation, and explain why the mean cost rises with the number of rebalances while the standard deviation falls. Explain also why the formula value of 3.1944 is not attainable at any frequency.

18.10 A pension fund holds an index portfolio and wants the payoff of portfolio-plus-put without paying a premium. State the futures position it must run and how that position changes as the market falls, and give two respects in which the result differs from buying the put outright.

18.11 A dealer who is short gamma must buy the underlying as it rises and sell as it falls. State whether that damps or amplifies the move, and explain why pinning near expiration (Example 10.6) and retail call buying (Section 10.12) push in opposite directions even though both are dealer delta-hedging.

18.12 An exchange-traded product offering inverse exposure to VIX futures must buy those futures when VIX rises. Explain why its rebalancing rule has that sign, why that rule trades the way a short-gamma hedger does, and what distinguishes this case from a dealer whose hedging incidentally moves the market.

18.13 For a European put with the same inputs as the dealer's call ($S_0 = 63$, $K = 64$, $T = 18/52$,

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$r = 4\%$, $\sigma = 22\%$), find its value by put–call parity, then its delta and the elasticity Ω of Example 18.1. Interpret the sign of each, and compare the put's leverage with the call's.

FURTHER QUESTIONS

18.14 Section 18.9 shows hedging error falling as $1/\sqrt{n}$ while expected transaction costs rise, roughly as $\sqrt{n}$ for a proportional cost. Write down the total expected cost as a function of n with a proportional trading cost k, find the n that minimizes it, and describe how the optimum moves with volatility and with k.

18.15 Example 18.4 verifies the Black–Scholes–Merton equation from which (18.8) follows. Show that a trader who is long gamma and delta-hedges continuously earns the gamma-weighted difference between realized and implied variance, $\int \frac{1}{2}\Gamma S^2(\sigma^2_{\text{realized}} - \sigma^2_{\text{implied}})\,dt$ over the option's life, and explain why the weighting matters to someone who buys options because they expect a specific event.

18.16 Section 18.12 argues that dealer hedging amplifies moves when dealers are short gamma and damps them when long. Construct the conditions under which the aggregate position flips sign during a single trading day, and explain why this makes gamma-based indicators unreliable precisely when they are most consulted.

18.17 A desk is short a large quantity of short-dated at-the-money options into an earnings announcement. Using Figures 18.2 and 18.4, describe what happens to the book's delta and gamma across the announcement, and state what the desk should have done beforehand and why it might not have.

18.18 Vega measures sensitivity to a parameter the model assumes constant. Construct the argument that computing vega is therefore incoherent, then the argument that it is nevertheless the right thing to do, and state which position a risk manager should take.

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Chapter 19
Volatility Smiles and Surfaces

Five chapters have deferred to this one. Section 11.3 derived conditions a whole option chain must satisfy and said they would be used here. Section 12.8 built a digital as the limit of a call spread and left the next derivative, the butterfly, which Section 13.3 met as a state price, to be read off the market here. Section 15.11 measured implied volatilities ranging from 56.9% to 33.2% across six strikes on a single stock on a single day, and observed that the model requires all six to be identical. Section 16.4 gave the FX market's quoting conventions, and Section 18.10 warned that the Greeks inherit the model's error.

The subject of this chapter is that error: the model requires one volatility, and the market's prices imply a different one at every strike. It is large, stable, well documented, and *informative*. The pattern of implied volatilities across strikes and maturities is not noise around a correct model. It is the market's statement about the distribution of the underlying, written in the only language the market has agreed on.

Sections 19.1 and 19.2 describe the two shapes the smile takes and what they say about the distribution. Section 19.3 introduces volatility indices and variance swaps, Section 19.4 the models that fit a smile, and Section 19.5 the constraints a surface must obey. Sections 19.6 to 19.10 cover the term structure, index against single-stock smiles, trading, the smile's causes and the volatility risk premium; Sections 19.11 and 19.12 cover how the smile moves and how the surface is used.

19.1 TWO SHAPES

An implied volatility is not observed. It is Section 15.7's construction: the σ that makes Chapter 15's formula reproduce a price the market actually traded at, so each one is a single traded option inverted. Plot those numbers against strike for a fixed maturity and one of two characteristic shapes appears. That curve is a **smile**, and the same option chain has one for every listed maturity. Taken together they define a function of two variables, one implied volatility for each strike K and maturity T, and that function is the **volatility surface**. It is what a desk quotes, marks and hedges from.

Put–call parity makes a call and a put at the same strike and maturity imply the same volatility (Section 15.11), so a surface carries one number per strike, read from the out-of-the-money side, where the bid–ask spread is tighter.

Equities: a skew. Figure 19.1 shows the shape that has characterized index and single-stock options since the crash of October 1987. Implied volatility falls as the strike rises, so that, measured in implied volatility, low-strike puts are expensive and high-strike calls cheap relative to at-the-money options; only at short maturities does the far call wing (the curve's high-strike end) turn up again. The pattern is called the **skew** or "smirk" rather than a smile because it is not symmetric.

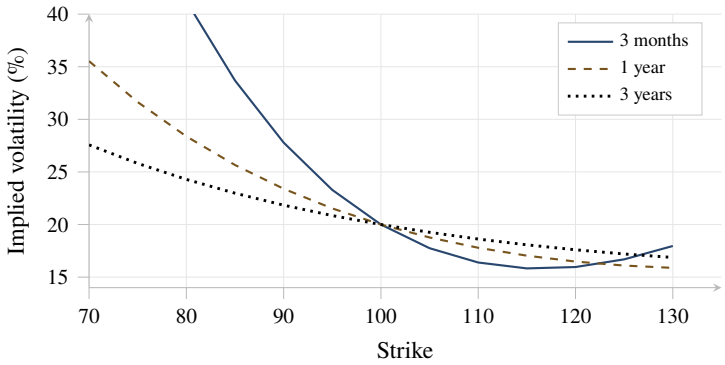


Figure 19.1: Equity implied volatility against strike, at three maturities. $S_0 = 100$; illustrative.

Currencies: a smile. Section 16.4 gave the FX shape: implied volatility is lowest near the money and rises in *both* wings, which is the symmetric curve the word "smile" describes. It is tilted (the risk reversal quoted there measures by how much), but the tilt changes sign as sentiment moves, whereas equity skew does not; only in pairs with a known crash direction, an emerging-market currency or a carry trade's funding currency, does it keep one sign for years.

The difference is structural, and Section 19.9 explains it. An exchange rate is a ratio of two currencies, and a crash in one is a spike in the other, so the distribution has no preferred direction. An equity is a levered claim on a firm: it can be wiped out in a crash, but it has no matching way to multiply in value just as suddenly.

The term structure. Figure 19.1 also shows that the skew flattens as maturity lengthens. Measure the skew by its slope, the fall in implied volatility, in volatility points, for each 10 points of strike between strikes of 90 and 110: the slope falls from 5.7 at three months to 2.8 at one year to 1.6 at three years.

Most of that is the unit of measurement: the same 20 points of strike are about two standard deviations at three months and little more than half of one at three years, as the next subsection measures. The central limit theorem of Section 13.9 would flatten it faster than the change of unit alone, since independent shocks (jumps above all) average out over a long horizon, which is why jump models generate skew mainly at the short end (Sections 21.5 and 21.7). Volatility shocks are persistent and do not average out so quickly, and they keep the long-dated skew alive.

Measuring moneyness

The axis of Figure 19.1 is the strike, which is what the contract names and the worst of the available coordinates for comparing one option with another. Six are in use, set out in Table 19.1, and they are not interchangeable.

They are one axis in different units, and three things follow.

The coordinate decides what "the same option" means from one day to the next. Section 19.11's two rules for how the smile moves with the spot, sticky strike and sticky delta, are a disagreement about which coordinate it stands still in, not about today's prices, and (19.4) is its cost to a hedger.

A fixed strike is a different option at every maturity. At 20% volatility the 90 strike of Figure 19.1 is 1.05 standard deviations below the money at three months, 0.53 at one year and 0.30 at three, because $\sigma\sqrt{T}$ shrinks as expiration approaches. Measured per unit of standardized moneyness rather than per point of strike, the figure's three curves have slopes of 5.68, 5.59 and 5.56 volatility points per standard deviation: they are one curve in that coordinate. Real

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Table 19.1: The coordinates a surface is quoted in.

Measure	Definition	What it is for
Strike	K	The contract term, and how an exchange lists the option
Spot moneyness	K/S	Section 19.11's sticky-delta rule; the columns of Table 19.3
Forward moneyness	K/F	Removes carry, so at the money means $K = F$
Log-moneyness	$\ln(K/F)$	Scale-free: an underlying at 40 and one at 4,000 report the same skew for the same shape, which is why Chapter 21 measures in it
Standardized moneyness	$\ln(K/F)/(\sigma\sqrt{T})$	Distance from the money in standard deviations; comparable across maturities, as delta is
Delta	$N(d_1)$	The FX quoting convention of Section 16.4 (spot or forward delta, premium-adjusted in some pairs), already rescaled by volatility and time

surfaces flatten more slowly than that: Section 21.7 measures the decay as $T^{-0.4}$ against the $T^{-1/2}$ a curve stationary in standardized moneyness would give, so in that coordinate the skew slowly *steepens* with maturity.

Delta is the only one a trader can act on directly, since it is the hedge ratio, already scaled by volatility and time. That is why FX quotes in it, and it is also the circular one: delta depends on the volatility being quoted, so a 25-delta option's strike and its implied volatility depend on each other and have to be solved for together.

Shapes across asset classes

Table 19.2 sets out the surface's shape in each market; the differences are systematic.

Table 19.2: Characteristic surface shapes and quoting conventions.

Market	Shape	Why
Equity index	Steep skew	Leverage, crash risk, one-sided protection demand
Single stock	Milder skew	Same forces, plus takeover possibility lifting the right wing
Currency	Smile, tilt varies	Ratio of two economies; no preferred direction
Commodity	Often *reverse* skew	Supply shocks push prices up, not down
Interest rates	Skew, quoted in normal volatility	Rates can be negative, so lognormal fails outright

Commodities frequently skew the other way. A grain, natural-gas or power price is cushioned on the way down by production cuts and unbounded above when supply fails, so the fat tail is on the *upside*; crude oil and base metals, driven more by demand, often carry a put skew instead. Section 5.9's negative oil settlement does not undercut the cushioning argument: it required storage to run out.

Rates are quoted in normal volatility: the standard deviation of the rate's change measured in basis points, rather than of its percentage change. Chapter 29 explains why: once policy rates went negative, a lognormal model assigning zero probability to $r < 0$ became untenable, and the market moved to quoting basis-point volatility.

19.2 WHAT THE SMILE SAYS ABOUT THE DISTRIBUTION

Section 12.8 built a digital as the limit of a tight call spread, priced at $-\partial c/\partial K$. Section 13.3 pushed the same idea one derivative further, to a state price. An infinitesimal butterfly centered at K (the call spread with strikes $K - \varepsilon$ and K bought against the one with strikes K and $K + \varepsilon$

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sold, scaled by $1/\varepsilon^2$) pays only if the underlying finishes at K, so its price is the state price for that outcome, and

$$\boxed{g(K) = e^{rT}\,\frac{\partial^2 c}{\partial K^2}} \tag{19.1}$$

where g is the risk-neutral density of S_T and c is the market call price as a function of strike, so the derivatives run along the smile. This is the Breeden–Litzenberger result, and it means a complete set of option prices contains the market's entire probability distribution for the underlying. [TN19] derives it in two differentiations, shows why nonnegativity of the density is the convexity condition of Section 11.3, and sets out what changes when the chain is finite.

Applied to a smile, not a single volatility, it makes the chapter's central point visible.

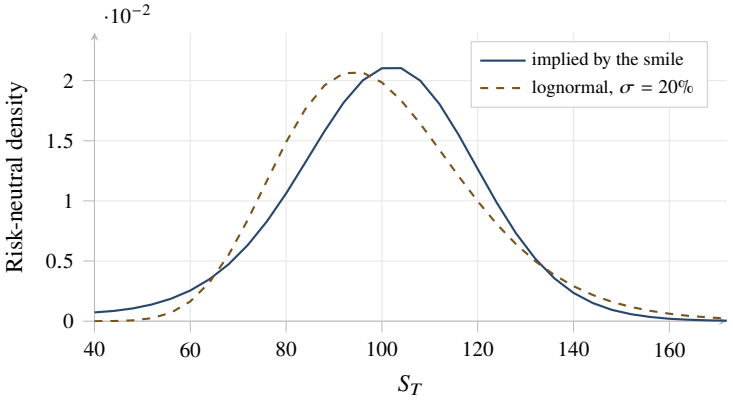


Figure 19.2: Density implied by a one-year equity skew, against the lognormal it would be if the model held. $S_0 = 100$, $r = 0$.

Example 19.1 Reading the tails

Figure 19.2 plots the density obtained from (19.1) applied to the one-year skew

$$\sigma(K) = 0.20 - 0.15\ln(K/100) + 0.15\ln^2(K/100)$$

the shape of Figure 19.1 with a milder left wing, against the lognormal density that a constant 20% volatility would produce.

Each probability is read off as a digital, $\Pr(S_T < K) = 1 + e^{rT}\,dc/dK$ with $r = 0$, where dc/dK is the call price's slope along the smile; this is the density integrated from zero:

	Implied by smile	Lognormal
$\Pr(S_T < 60)$	3.0%	0.7%
$\Pr(S_T < 70)$	6.9%	4.6%
$\Pr(S_T < 80)$	14.8%	15.5%
$\Pr(S_T > 130)$	6.0%	7.9%

The market assigns more than four times as much probability as the lognormal does to a 40% decline and 1.5 times as much to a 30% decline, and less probability to a 30% rise. Between the tails the implied density runs below the lognormal from about 65 to 97 and above it near the center, which is why the two columns nearly agree at 80: the distribution is peaked, with mass drawn from the left shoulder into the center and the far tail.

Expensive low-strike puts and cheap high-strike calls are not a pricing anomaly: they are what a fat left tail and a thin right tail look like when translated into Black–Scholes volatilities.

One caveat matters for practice. A tail probability at a quoted strike is the digital $1 + e^{rT}\,dc/dK$, so it depends only on the smile's level and slope there. Below the lowest traded strike the smile

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must be extrapolated, but the probability of finishing below that strike is still fixed: it is a digital put, which the put spread between the two lowest traded strikes approximates. What the extension decides is how that probability is spread over lower prices, and so anything integrated over the tail, and it must be free of arbitrage to do even that. Extended as a quadratic, this smile is not: the density turns negative below a strike of about 6, and integrated upward from that point it totals 1.06 rather than 1.00, a 6% error created entirely outside the observable range.

The same calculation on real quotes

Figure 19.2 used a smooth parameterized smile. The point of (19.1) is that it needs no such thing: six quoted prices and arithmetic are enough, as Example 19.2 shows.

Example 19.2 Extracting a distribution from six quotes

Return to the option chain of Table 10.2: a stock at \$48.30, 45 days to expiration, six strikes \$2.50 apart. Section 15.11 solved each call's midquote (the bid–ask midpoint) for implied volatility:

Strike	Call mid	Implied vol	$C_{i-1} - 2C_i + C_{i+1}$	Density $g(K)$
42.50	6.650	56.9%	—	—
45.00	4.425	46.0%	0.4700	0.0757
47.50	2.670	39.7%	0.5600	0.0902
50.00	1.475	36.7%	0.4300	0.0692
52.50	0.710	34.5%	0.3600	0.0580
55.00	0.305	33.2%	—	—

The fourth column is the price of a butterfly with wings \$2.50 wide. Every entry is positive, so the chain passes Section 11.3's convexity test and admits no butterfly arbitrage. The fifth column divides by the wing width squared and compounds forward, which is (19.1) in difference form: $g(K) = e^{rT}(C_{i-1} - 2C_i + C_{i+1})/h^2$ with $h = 2.50$, $r = 5\%$ and $T = 45/365$.

Multiplying each density by \$2.50 gives, approximately, the probability the market assigns to each bucket, the \$2.50 band around a strike: 18.9%, 22.5%, 17.3%, 14.5%. The four together cover \$43.75 to \$53.75 and carry 73.3% of the probability.

The tails come from the outer call spreads. A call spread's price, compounded forward and divided by its width, approximates the digital at the midpoint of its strikes, so $\Pr(S_T > 43.75) \approx e^{rT}(6.650 - 4.425)/2.50 = 89.6\%$ and $\Pr(S_T > 53.75) \approx e^{rT}(0.710 - 0.305)/2.50 = 16.3\%$. The left tail is the complement of the first: $\Pr(S_T < 43.75) = 10.4\%$.

Why the total is exactly 100%. 10.4% + 73.3% + 16.3% = 100.0%, but the total is an *identity*, not a test: the middle differences telescope, leaving only the two outer call spreads, which are precisely the tail digitals, so *any* six numbers would sum to 100%.

A lognormal at the at-the-money implied volatility of 39.7%, centered on Example 11.4's forward of \$47.60, puts only 53.0% of its mass in that same central band. The lognormal that *does* put 73.3% there has $\sigma = 25.6\%$.

So the market is pricing the middle of the distribution as if volatility were about 26%, while the strike 12% below spot is quoted near 57%: the distribution is peaked in the center, and the little mass below \$43.75 must lie far enough out to justify that price. More mass in the center and in the far tails than a normal distribution carries is called **leptokurtosis**, and this is what it looks like when you measure it rather than assume it.

Two limits apply. The calls are American and the stock pays a dividend before expiration, so the deep-in-the-money \$42.50 quote, the one the left tail is read from, carries early-exercise premium a European call would not. Every \$0.10 of it lowers the measured $\Pr(S_T < 43.75)$ by about 4 points, so the 10.4% understates the European tail and the 57% overstates the wing. And the whole left tail rests on that one call spread: it fixes the 10.4%, but six strikes spanning −12% to +14% of spot cannot say how that 10.4% is distributed below them, as Example 19.1 warned. A chain tells you a tail's mass, and through the lowest put its mean, but never its shape.

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19.3 VOLATILITY INDICES AND VARIANCE SWAPS

Chapter 1 introduced VIX and Section 18.12 used it; this section defines it. **VIX** is Cboe's volatility index for the S&P 500: the annualized volatility, in percentage points, that the index's listed options imply over the next 30 days. It is a construction from the surface, and the construction is Chapter 12's replication result applied once more.

The static portfolio that pays variance. Suppose one wants a payoff equal to the *realized variance* of the underlying over $[0, T]$ (a traded contract defines it as a discrete sum of squared daily log returns, Section 22.2; the continuous-time definition used here is for pricing and hedging). Section 12.8 showed that any payoff can be built from options; the specific answer here is that realized variance is replicated by a static portfolio of out-of-the-money options weighted by $1/K^2$, together with a dynamic position in the underlying. The fair value of that portfolio gives the **fair variance swap rate**:

$$\sigma_{\text{var}}^2 = \frac{2e^{rT}}{T}\left[\int_0^F \frac{P(K)}{K^2}\,dK + \int_F^\infty \frac{C(K)}{K^2}\,dK\right] \tag{19.2}$$

where F is the forward price of the underlying to expiration, and P and C are put and call prices at strike K. VIX is 100 times the square root of (19.2) at 30 days, interpolated in total variance $\sigma^2 T$ between the two listed expirations that bracket 30 days. Cboe computes the same index at other horizons, from nine days (VIX9D) to three months (VIX3M) and, since April 2023, one day (VIX1D), the reference for Chapter 23's same-day options.

The $1/K^2$ weighting is not a modeling choice; it is what makes the portfolio's payoff proportional to variance rather than to something else. Section 22.3 derives this strip of options, and [TN22] takes up the correction the exchange applies to it in practice. The weighting has a consequence that decides how the index behaves: low strikes receive far more weight than high ones. Because the skew makes those low strikes expensive, a volatility index built this way systematically exceeds the at-the-money implied volatility, by more than four volatility points on Figure 19.1's one-year skew, as Section 22.6 measures. And because real quotes do not run to zero and infinity, Cboe truncates the strip by a mechanical rule, so the published index depends on how far out options happen to be quoted that day.

The instrument behind the formula is the **variance swap**, which pays the difference between realized variance and the fixed rate σ_{var}^2. It became the standard volatility instrument because its payoff is the realized variance whatever price level the moves occurred at, and because its replicating options are held *statically*. The direction of the logic matters here: VIX is not an input to the surface; it is an output of it.

19.4 MODELS THAT FIT THE SMILE

The Black–Scholes–Merton model assumes one volatility, and the market quotes many. Three families of model resolve the contradiction, and each buys something at a price.

Local volatility. Let volatility be a deterministic function of the price and time, $\sigma(S, t)$. Dupire showed that for any arbitrage-free surface, given at every strike and maturity, there is exactly one such function reproducing it. Local volatility therefore fits the smile *perfectly*, by construction, and it preserves completeness: the market has one source of risk, so Section 13.4's replication argument survives and every option is hedgeable.

The cost is its dynamics. A local-volatility model that matches today's surface generally predicts that tomorrow's smile will flatten and shift in ways the market does not display, which matters for anything whose value depends on how the surface moves, such as Section 19.11's delta and the exotic options of Chapter 34.

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Stochastic volatility. Let volatility be a second random process, correlated with the price. Heston's model is the standard example, and the negative correlation between price and volatility generates the equity skew directly, the leverage explanation of Section 19.9 as a model.

The cost is completeness. There are now two sources of risk and one hedging instrument, so by Section 13.4 the market is incomplete, no unique price exists, and options must be used to hedge options. That is not a defect of the model; it is the model reporting a fact about the world that the Black–Scholes–Merton model concealed.

Jumps. Allow the price to move discontinuously. Jumps produce fat tails at short maturities without requiring an implausibly high volatility for the continuous moves, which is exactly where Figure 19.1 shows the steepest skew. They also break completeness, and for a sharper reason: a jump of unknown size cannot be hedged by any position in the underlying taken before it.

SABR. In interest rate markets the standard is SABR (stochastic alpha, beta, rho), a stochastic-volatility model with a closed-form implied-volatility approximation, which lets a desk fit a smile with four intuitive parameters rather than a numerical procedure. Chapter 29 uses it for swaptions, and Section 29.9 fits one to market data.

No model dominates: each is used where its virtue is the requirement, and exotics desks mostly run a hybrid of the first two, the local-stochastic volatility of Section 21.9, and compare it against the others. Section 19.12 explains why the disagreement between them is itself the number to look at.

19.5 CONSTRAINTS ON A SURFACE

Because the implied density must be a density, a volatility surface cannot take just any shape. Section 11.3 derived the constraints without a model; they now become tests for a surface.

Nonnegative density. From (19.1), $\partial^2 c/\partial K^2 \geq 0$, which is exactly the convexity condition (11.3). A surface that makes call prices nonconvex in the strike implies negative probabilities and admits the butterfly arbitrage of Example 11.2.

Monotonicity. $c(K)$ must be decreasing in K, though by no more than the present value of the strike gap (Section 11.3).

Calendar consistency. For a fixed strike (strictly, a fixed forward moneyness K/F, which is the same thing when there is no carry), total variance $\sigma^2(K,T)\,T$ must be nondecreasing in T. If it were not, a calendar spread would have a negative price for a nonnegative payoff.

These are the conditions any interpolation must respect, and respecting them is harder than it sounds. Fitting a smooth curve through quoted volatilities is not the same as fitting an arbitrage-free surface, and a naive spline through market data routinely produces negative densities somewhere between the quotes. Dealers' pricing systems parameterize the surface instead, most often with Gatheral's stochastic-volatility-inspired (SVI) parameterization, which fits each expiration's total variance with five parameters and is linear in log-strike in the wings; its surface extension, surface SVI (SSVI; Gatheral and Jacquier 2014), comes with explicit conditions under which the whole surface passes the tests above. The linear wings are required: Lee (2004) showed that without arbitrage total implied variance can grow at most linearly in log-strike, $\sigma^2 T$ at most about $2\,|\ln(K/F)|$ far out in either wing. A quadratic smile like Example 19.1's grows faster and must eventually imply an arbitrage; that one does so near a strike of 6, where its density turns negative.

In a rising call wing the monotonicity condition is the one to watch, and Example 19.3 turns it into a number a trader can carry.

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Example 19.3 How fast a wing may steepen

Take spot at 100, six months to expiration and a 3% rate. The strike enters the price twice, directly and through the implied volatility the smile assigns to it, so differentiating along the smile is a total derivative:

$$\frac{dc}{dK} = \frac{\partial c}{\partial K} + \frac{\partial c}{\partial \sigma}\frac{d\sigma}{dK} = -e^{-rT}N(d_2) + \mathcal{V}\frac{d\sigma}{dK}$$

The first partial is minus the digital a flat volatility would give (Section 12.8), negative because a higher strike is worth less; the second is vega, positive. A rising wing pushes against the first term, and the call price stops falling in the strike (the call spread turns negative) once

$$\frac{d\sigma}{dK} > \frac{e^{-rT}N(d_2)}{\mathcal{V}}$$

The right-hand side is a hard ceiling on the wing's slope, and it is not constant:

Strike	σ	Vega	Maximum steepening per 5 of strike
110	22.5%	25.77	5.34 volatility points
115	24.0%	22.83	4.45 volatility points
120	26.0%	20.19	3.86 volatility points
125	29.0%	18.60	3.49 volatility points

The ceiling tightens as the strike moves out, because $N(d_2)$ falls faster than vega does, the opposite of an extrapolation that steepens the wing as it goes.

Now breach it. Take a chain quoting 24.0%, 23.2%, and 22.8% at 100, 105, and 110, then an aggressive wing at 25.0% and 31.0% for 115 and 120:

Strike	σ	Call	Call spread
100	24.0%	7.4824	
105	23.2%	5.0715	+2.4109
110	22.8%	3.3237	+1.7478
115	25.0%	2.6563	+0.6673
120	31.0%	3.0442	**−0.3878**

The 115/120 call spread has a payoff that is never negative and a price of −\$0.3878: you are paid \$0.39 to receive something worth between \$0 and \$5.

The chain tolerates $\sigma(120)$ up to 29.27% before the spread turns, so the quoted 31.0% was 1.7 volatility points past the boundary, far too small a discrepancy to see by eye on a plotted smile, and an arbitrage nonetheless. That is why the check is arithmetic rather than visual, and why it is run on the quotes before any curve is fitted through them.

19.6 THE TERM STRUCTURE AND ITS DYNAMICS

The surface has a second dimension, and it moves differently from the first.

Figure 19.3 shows the characteristic pattern. In calm markets the term structure slopes *upward*: short-dated volatility is low and long-dated volatility is higher. In stressed markets it *inverts*, sometimes violently, with one-month volatility far above two-year.

The mechanism is mean reversion. Volatility is not a random walk. It clusters, as Section 14.12 noted and Section 38.10 models, but it also returns toward a long-run level. A short-dated option prices volatility over a period in which today's level dominates; a long-dated option prices an average over a period in which mean reversion has had time to act. The slope simply reports which side of the long-run level the market is currently on.

Two consequences follow for the desk.

Vega buckets are not interchangeable. Section 18.8 said desks compute vega by maturity bucket rather than as one number, and Figure 19.3 is the reason: short maturities move several times as much as long ones. Hedging total vega hedges a parallel shift that does not occur.

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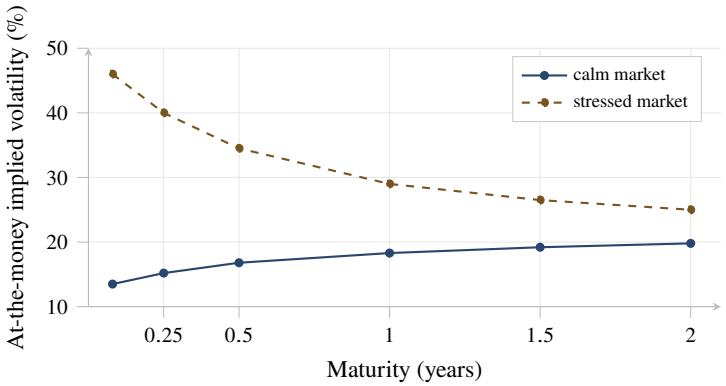


Figure 19.3: At-the-money implied volatility against maturity, in two regimes. Illustrative.

Calendar spreads are a term-structure trade. Section 12.5 described the calendar spread as a bet on time decay and on the term structure. Now the second half is precise: selling near-dated and buying far-dated volatility is a bet that short-dated volatility stays low or, when it is high, falls back toward its long-run level, and it loses in exactly the stress episodes that invert the curve.

A scheduled event bends both dimensions. When a single large move is expected on a known date (an earnings release, a court ruling), the distribution of the price just after it can be bimodal; near-the-money options are then worth more relative to the wings than a lognormal implies, and the smile becomes a frown. Only expirations after the date contain the event. If the one just before trades at 25% and one fifteen trading days out at 35%, and the fourteen ordinary days carry 25%, subtracting their variance from the later expiration's total leaves the event day's: $(0.35^2 \times 15 - 0.25^2 \times 14)/252 = 0.0038$. Its square root, 6.2%, is the standard deviation the market prices for the announcement-day move, against $0.25/\sqrt{252} = 1.6\%$ on an ordinary day.

19.7 THE INDEX SMILE AND THE SINGLE-STOCK SMILE

Table 19.2 recorded that an index skews more steeply than its own constituents. That is not an observation about two unrelated markets: the index *is* the constituents, so the two surfaces are linked, and what links them is correlation.

For a portfolio with weights w_i and component volatilities σ_i,

$$\sigma_I^2 = \sum_i w_i^2 \sigma_i^2 + \sum_{i \neq j} w_i w_j \, \rho_{ij} \, \sigma_i \sigma_j \tag{19.3}$$

Everything on the right except the correlations is observable from single-stock option prices, and the left-hand side is observable from index option prices. If all pairs are assumed to share one correlation, that one unknown can be solved for, and the answer is called the **implied correlation**, which Example 19.4 computes.

Example 19.4 Implied correlation and what dispersion trades

Take an index of 50 equally weighted stocks, each with a one-year implied volatility of 30%, and a one-year index implied volatility of 18%. With a common correlation ρ, (19.3) collapses to

$$\sigma_I^2 = \sigma^2 \left[\frac{1}{n} + \rho \left(1 - \frac{1}{n} \right) \right]$$

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so $(0.18/0.30)^2 = 0.36 = 0.02 + 0.98\rho$ and $\rho = 34.7\%$.

Where the trade is. Suppose realized correlation over the past year has been 25%. At that correlation the index volatility would be $0.30\sqrt{0.02 + 0.25 \times 0.98} = 15.4\%$, against the 18% the index options actually cost. Selling index volatility and buying the components, the **dispersion trade** of Sections 16.2 and 18.7, collects that gap of 2.6 volatility points, a difference in volatility rather than a 2.6% return, if correlation stays at 25% and the components realize their 30%.

Why the gap persists. It is Section 19.10 again, in a specific form. Index puts are the instrument institutions buy for portfolio protection, and protection is a demand for correlated downside, so the premium concentrates in the index rather than in the constituents. Implied correlation exceeding realized correlation is the price of that demand.

And why the trade is not free. In a crash component volatilities and correlation rise together, so index volatility rises proportionally more than single-stock volatility, and the long single-stock leg covers only part of the loss. The dispersion seller is short exactly the thing that moves most, which is why the strategy earns steadily and loses in large, infrequent amounts (Example 22.8).

The same mechanism explains the shape difference the table recorded. Correlation is not a constant: it rises when markets fall and falls when they rise. On the downside the index gets hit twice, by higher component volatility and by higher correlation together; on the upside dispersion increases and damps the index move. So the index inherits its constituents' skew and then adds a correlation skew on top of it, and the sum is steeper than any single stock's.

19.8 TRADING THE SMILE

A single surface has three dimensions of shape (level, skew, and curvature), and Section 16.4 already observed that the FX market quotes precisely those three because they are what it trades. The instruments are Chapter 12's, and correlation, which spans two surfaces, is traded through the dispersion trade of Section 19.7.

Level is traded with straddles or, more cleanly, with the variance swaps of Section 19.3.

Skew is traded with the risk reversal of Section 12.6: long an out-of-the-money call, short an out-of-the-money put, or the reverse. A trader who thinks puts are too expensive relative to calls buys the risk reversal, selling the expensive put and buying the cheap call, and must include Section 19.11's delta correction in the hedge.

Curvature is traded with the vega-weighted butterfly, the position the foreign-exchange market's butterfly quote of Section 16.4 prices: a strangle bought against an at-the-money straddle, sized so the two vegas cancel. Buying the wings against the at-the-money straddle is a bet that the market will come to price more extreme outcomes in both directions.

Who is on each side. Most of the participants are the ones Section 16.2 named, and their positions in surface terms are now legible. Protection buyers are structurally long the put wing. Covered-call writers are structurally short the call wing. Structured-product issuers hold the other side of what their products embed: long the barrier puts that buyers of Section 12.12's reverse convertibles and autocallables write, which sit low in the strike dimension, and short the calls inside principal-protected notes. Each of these leaves a persistent mark on a specific region of the surface, and dealers hold whatever the end users leave unmatched.

A region expensive relative to its neighbors usually reflects who has been buying it rather than a superior forecast, and Section 19.10 shows that, on average, the buyers overpay.

19.9 WHY THE SMILE EXISTS

Four explanations are offered; all four contribute, and their relative weights are contested.

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The distribution really is not lognormal. Section 14.12 gave the reasons: prices jump, and volatility is neither constant nor independent of the price level. A model with jumps or with stochastic volatility produces fatter tails than the lognormal, and fitting the Black–Scholes–Merton model to options priced under such a model generates a smile as an artifact. This explanation is certainly part of the answer, and it explains the FX smile well.

Leverage. A share is a residual claim on a levered balance sheet. When the equity value falls, leverage rises, so the equity becomes more volatile: volatility and price are negatively correlated by construction. This produces exactly the asymmetry seen in equities and *not* in currencies, where no such mechanism exists.

Crash risk. The equity skew dates from October 1987, as this section's case study shows. Before that day the market had not priced a 20% one-day decline; afterward it did. On this reading the skew is the price of a risk the lognormal model assigns essentially zero probability to. Rubinstein (1994), who documented the change, called it "crash-o-phobia."

Supply and demand. One-directional demand for low strikes and one-directional supply of high strikes (Section 19.8) will produce the observed shape whatever the distribution.

Case Study 19.1 The day the skew appeared

The four explanations above are hard to separate on cross-sectional data. History separates them rather sharply.

Studies of index option prices before October 1987 find implied volatilities close to *flat* across strikes, a smile so mild it was regarded as a minor pricing puzzle rather than a feature. The Black–Scholes–Merton model, on this evidence, described the market it was written for tolerably well for its first fourteen years.

After October 19, 1987, the skew appears, and it does not go away. Nearly four decades later the pattern of Figure 19.1 is a permanent fixture of every equity index option market in the world.

That timing is difficult to reconcile with the leverage explanation, which describes a mechanism that was equally true in 1986. It is difficult to reconcile with a change in the true distribution, since nothing about the economy changed that Monday. What changed was the market's belief about the probability of a very large decline, revised by a single observation and never revised back.

The supply-and-demand explanation gets support from the same event, which Section 40.2 tells in full: the failure of portfolio insurance destroyed the belief that crash protection could be manufactured by dynamic trading, which turned investors who had made their own puts into buyers of listed ones. Both readings point the same way, and both say the skew is priced crash risk rather than a measured distribution.

Because each predicts a similar shape, the best evidence is the next section's: a skew that merely described the distribution would be fairly priced on average, and it is not.

19.10 THE VOLATILITY RISK PREMIUM

Sections 10.12 and 12.4 asserted a result this chapter owes: implied volatility exceeds subsequently realized volatility, on average, across most markets and most periods. That average gap is the volatility risk premium that Section 15.7 named.

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The magnitude is typically two to four volatility points on equity indices (an implied volatility of 18% against a realized 15%, say), and the gap is larger for low strikes than at the money, so the skew is steeper than the realized distribution justifies.

Two readings are offered, and both are partly right.

It is compensation. Selling options means accepting a payoff that is worst precisely when markets fall and investors most need money. It is Chapter 18's short-gamma position, in economic terms. A risk that pays badly in bad states commands a premium, exactly as equities do relative to bills.

It is supply and demand. The buyers of protection described in Section 16.2 are price-insensitive relative to the sellers, who are constrained by capital. The premium is then a rent earned by whoever can bear the risk.

Section 12.4's straddle buyer pays this premium, so buying options is a bet that realized will exceed implied, not that the stock will move. Section 12.12's covered-call funds collect it, so their option income earns a genuine expected return, though part of each distribution is the investor's own capital handed back. And selling options earns a risk premium with a short-gamma position, profitable on average and, as Section 18.10's gap example showed, capable of losing more than the whole premium in a single move.

This also settles the question Section 15.7 raised. When an implied volatility of 29.85% meets a historical estimate of 20.4%, there are two readings: the market anticipates something the history does not contain, or buyers are paying the premium. Within a single implied volatility the two components cannot be told apart; averaged over enough observations, the forecast-error component tends to zero, and what remains is the premium.

19.11 HOW THE SMILE MOVES

For a desk, what matters more than the smile at one instant is what happens to it when the underlying moves, and Section 18.10 warned that the answer changes the Greeks.

Two reference conventions describe it, and a third completes Derman's (1999) taxonomy.

Sticky strike holds that the implied volatility of a *given strike* stays put as the spot moves. The smile is fixed in strike space, and Black–Scholes deltas are correct as computed.

Sticky delta (or sticky moneyness) holds that the implied volatility of a given *moneyness* stays put, so the whole smile translates with the spot. A fixed strike then acquires a different implied volatility as the market moves.

Sticky implied tree, named for the tree form of local volatility, holds that the local-volatility function of Section 19.4 stays put, so a rise in the spot *lowers* the implied volatility at a fixed strike and the delta falls below the Black–Scholes delta. Derman associated sticky strike with a range-bound index, sticky delta with a trending one, and the implied tree with a market that fears a sharp fall.

Figure 19.4 shows what each convention does to the one-year smile when the spot rises from 100 to 105.

Under sticky strike nothing happens to the curve, and the 100-strike option keeps its 20.00% volatility. Under sticky delta the curve slides to the right in proportion to the spot, and the 100-strike option, now 5% in the money rather than at the money, picks up the 21.45% that used to belong to the 95.2 strike. The same option at the same instant carries volatilities 1.45 volatility points apart under the two conventions, and Example 19.5 shows what that does to the hedge.

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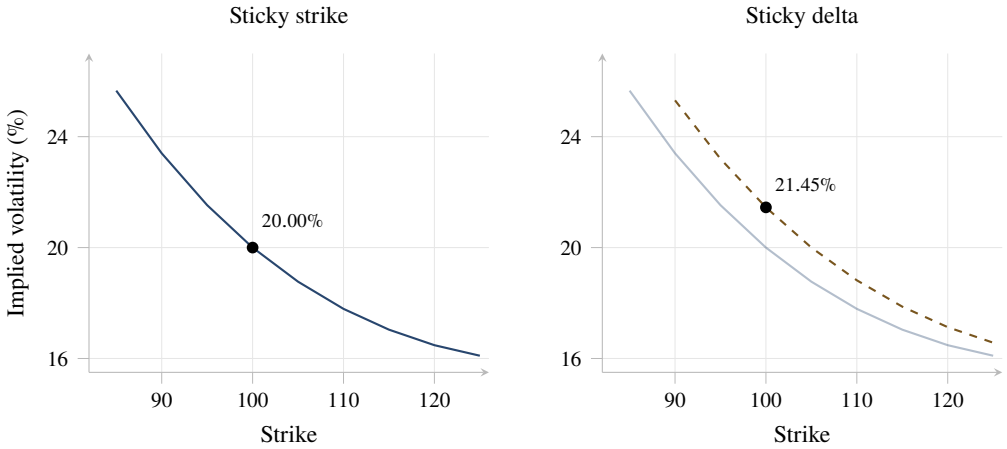


Figure 19.4: The one-year smile after the spot rises from 100 to 105. The marked point is the 100-strike option. Faint curve: the smile before the move.

Example 19.5 The delta correction

Under sticky delta the spot enters the price twice, directly and through the implied volatility the smile hands the option as it slides, so the total sensitivity to the spot is a total derivative:

$$\Delta_{\text{total}} = \frac{dc}{dS} = \frac{\partial c}{\partial S} + \frac{\partial c}{\partial \sigma}\frac{\partial \sigma}{\partial S} = \Delta_{\text{BS}} + \mathcal{V}\frac{\partial \sigma}{\partial S} \tag{19.4}$$

The first partial is the usual delta, computed with the smile held still; the second is vega times the rate at which the sliding smile assigns the option a new volatility. In Chapter 18's expansion, equation (18.1), the delta and vega terms were treated as independent; a sliding smile makes $\Delta\sigma$ move with ΔS, so part of the vega term is first order in the stock and belongs in the delta hedge.

For the one-year skew of Figure 19.1 with $S = 100$ and $r = 0$:

Strike	Type	σ	Δ_{BS}	Correction	Δ_{total}
90	put	0.2340	−0.2853	+0.1256	−0.1596
100	call	0.2000	+0.5398	+0.1092	+0.6490
110	call	0.1779	+0.3275	+0.0683	+0.3958

The corrections run from 0.07 to 0.13 in delta, a 20% error on the at-the-money hedge, or 10,920 shares on a position of 100,000 options. A desk hedging with Black–Scholes deltas while the market behaves in a sticky-delta way has every delta too low: too few shares against a call, too many against a put.

Under an equity skew, volatility falls as strike rises. Sticky delta means the smile moves with spot, so a rise in S lowers the option's moneyness K/S and therefore *raises* its implied volatility. Since vega is positive, the option's value rises by more than delta alone predicts, so every delta in the table moves up: a call's hedge grows and a put's shrinks.

Which convention holds is an empirical question, and for equity indices the measured answer lies outside the first two: when the index falls, at-the-money volatility rises by more than sticky strike implies, though by less than local volatility predicts (Bergomi 2009). The correction for an index therefore has the opposite sign to this table's, and the minimum-variance delta lies below the Black–Scholes delta, as Section 18.10 reported from Hull and White (2017). FX is closer to sticky delta, which is why Section 16.4 quotes in delta terms. Desks measure the relationship rather than assuming it, and the measured slope enters (19.4) directly.

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19.12 THE SURFACE IN PRACTICE

A desk maintains a volatility surface per underlying, and it is the single most important piece of market data an option business owns.

What one actually looks like. Table 19.3 is an S&P 500 surface in the form a desk reads it: implied volatility in percent, strikes across as a percentage of spot, maturities down the side. Its wings are milder than Figure 19.1's.

Table 19.3: Implied volatility (%) by strike, as a percentage of spot, and by maturity. A calm market; illustrative, not market data.

	Strike (% of spot)						
Maturity	**80**	**90**	**95**	**100**	**105**	**110**	**120**
1 month	30.0	22.4	19.6	17.5	15.8	14.9	15.6
3 months	27.5	21.9	19.9	18.5	17.1	16.0	14.8
6 months	26.0	21.7	20.3	19.3	18.2	17.2	15.6
1 year	25.0	21.9	20.8	20.0	19.3	18.5	17.2
2 years	24.2	22.0	21.3	20.8	20.2	19.7	18.7

Every row falls from left to right through the money, which is the skew of Section 19.1. A one-month put struck 20% below spot is marked at 30.0% against 17.5% at the money; the same strike two years out is 24.2% against 20.8%. Measured between the 90 and 110 strikes the slope runs 3.75 volatility points per 10% of spot at one month, 1.70 at one year and 1.15 at two, so two-thirds of the skew is gone by the two-year point. Measured in Section 19.1's standardized coordinate instead, the same rows give 1.89 volatility points per standard deviation at one month against 3.37 at two years: the skew *steepens*.

Read down the at-the-money column instead and the second dimension appears: 17.5% rising to 20.8%, the calm-market term structure of Section 19.6.

Two features are worth pointing out because they look like errors and are not. Total variance rises down every column: at the 110 strike $\sigma^2 T$ runs from 0.0019 at one month to 0.0776 at two years. That is Section 19.5's calendar condition holding, and a column in which it failed anywhere would be a calendar-spread arbitrage rather than a steep surface. And the top right corner turns back up (the one-month 120 strike at 15.6% against the 110's 14.9%), because no one sells a far out-of-the-money one-month call at the price a straight continuation of the skew would imply.

It is built from quotes, not from a model. Each expiration's forward is read first from put–call parity near the money (Section 11.6), and time runs on Section 15.8's trading-day clock, refined to Chapter 23's business time, which weights each day by its expected variance; both matter most at the short end, where Section 23.9 prices the same-day smile this way. Liquid strikes and maturities are then marked to the market; everything else is interpolated subject to the constraints of Section 19.5.

It prices the illiquid from the liquid. Section 15.8's practical rule was to use each option's own implied volatility. The surface is what makes that possible for an option that does not trade: locate it in strike–maturity space, interpolate, price. *This is what the Black–Scholes–Merton formula is actually for.* Example 19.6 carries it through on Table 19.3.

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Example 19.6 Pricing an option that does not trade

A client asks for a nine-month put struck at 4,600 on an index trading at 5,000. No line of Table 19.3 is that option: the strike is 92% of spot and the maturity falls between two quoted rows. It sits inside the rectangle formed by the 90 and 95 strikes and the six-month and one-year maturities, and those four corners are what price it.

Interpolate across the strikes first, linearly, at each of the two maturities:

$$\sigma(92, 0.5) = 21.7 + 0.4 \times (20.3 - 21.7) = 21.14\%$$
$$\sigma(92, 1) = 21.9 + 0.4 \times (20.8 - 21.9) = 21.46\%$$

Then move down the maturities, but not in volatility. Interpolate in total variance $\sigma^2 T$, the quantity Section 19.5 requires to be nondecreasing:

$$w(0.5) = 0.2114^2 \times 0.5 = 0.02234, \qquad w(1) = 0.2146^2 = 0.04605$$

At nine months $w = 0.03420$, so $\sigma = \sqrt{0.03420/0.75} = 21.35\%$. Interpolating the two volatilities directly would have given 21.30%. Five hundredths of a point is nothing here, but only the variance rule always respects the calendar bounds the two quoted maturities impose.

With $r = 4\%$ and a dividend yield of 1.5%, the Black–Scholes–Merton formula at 21.35% gives a put worth 157.9 index points.

Interpolated down the 100 column the same way, the nine-month at-the-money volatility is 19.77%; priced at that single volatility, the same put is worth 136.2. A desk doing so would have sold the option 13.8% below its market value, and the missing 21.7 index points are the skew: the option's own strike was never allowed to enter its price.

It generates the risk report. Chapter 18's vega is computed bucket by bucket against this surface, and the buckets are chosen where the surface actually moves independently.

Its failures are exotic risk. A vanilla option is priced by reading the surface. An exotic is not: Chapter 34 shows that barrier and path-dependent payoffs depend on how the surface *evolves*, which no static surface specifies. Two models that fit today's surface perfectly can price the same barrier option far apart. Section 21.11 measures the gap at nearly half the price for a reverse knock-out (a call knocked out by a barrier above its strike, so it dies where it is in the money), and at about one percent for a down-and-out call struck at the money (knocked out by a fall through a barrier below). That gap, not the vanilla price, is the model risk that Section 13.4's completeness discussion predicted, and its size depends on where the payoff sits relative to the region the surface leaves unconstrained.

SUMMARY

Implied volatility is not constant across strikes. Equities show a skew (low strikes expensive, high strikes cheap) that has persisted since October 1987; currencies show a roughly symmetric smile whose tilt changes sign with sentiment. The skew flattens with maturity, mostly because a fixed strike gap spans fewer standard deviations at longer horizons; persistent volatility keeps the long-dated skew from vanishing as independent shocks alone would let it. A skew number means nothing without the moneyness convention it was measured in.

By Breeden–Litzenberger, $g(K) = e^{rT}\partial^2 c/\partial K^2$, so a complete set of option prices contains the market's entire risk-neutral distribution. Applied to a skew it gives a fat left tail and a thin right one, with the market assigning a large decline several times the probability the lognormal does; run on a quoted chain rather than a fitted curve, butterflies give the density and call spreads give the tails. Beyond the lowest traded strike, extrapolation decides the tail's shape.

Because the density must be nonnegative, Section 11.3's convexity condition becomes a

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constraint on any surface, alongside monotonicity in strike and nondecreasing total variance in maturity. Fitting a smooth curve is not the same as fitting an arbitrage-free surface: a wing can steepen only so far across a range of strikes before the quotes are themselves an arbitrage.

A volatility index is an output of the surface: a $1/K^2$-weighted strip of out-of-the-money options prices variance, and the skew lifts it above at-the-money volatility. Models that fit the smile trade one virtue against another: local volatility fits today's surface exactly and keeps the market complete but mispredicts how it moves, while stochastic volatility and jumps capture the dynamics and short-dated tails at the cost of completeness.

Four explanations contribute: a distribution that is not lognormal, leverage, crash risk priced since 1987, and one-directional demand. Leverage and crash risk give equities a direction that currencies lack; the timing of 1987 and the index's steeper skew point to priced crash risk and one-directional demand.

The term structure moves on its own logic: at-the-money volatility slopes upward in calm markets and inverts in stressed ones, which is why vega is bucketed by maturity. Component and index volatilities together imply a correlation, and selling it against realized correlation is the dispersion trade. Implied volatility on equity indices also exceeds subsequently realized volatility by a few points on average, so buying a straddle is a bet that realized will exceed implied rather than a bet that the market will move.

How the smile moves determines the hedge. The total delta is $\Delta_{\mathrm{BS}} + \mathcal{V}\,\partial\sigma/\partial S$, a correction too large to ignore; sticky delta makes it positive, while on equity indices, where fixed-strike volatility falls as the index rises, it is negative, so desks measure it rather than assume it. The surface is built from quotes subject to those constraints, prices an option that does not trade consistently with ones that do, and generates Chapter 18's vega buckets. Its limitation is that it says nothing about its own evolution, which is where exotic model risk lives.

FURTHER READING

Breeden and Litzenberger (1978) is the original statement of (19.1) and is short. Derman (1999) sets out the sticky-strike, sticky-delta and sticky-implied-tree regimes of Section 19.11, and Bergomi (2009) measures how index smiles actually move against them. Daglish, Hull and Suo (2007) derive the no-arbitrage condition a moving surface must satisfy and test the sticky rules against it and against S&P 500 data; a skewed surface is consistent with neither rule in its basic form. Gatheral's *The Volatility Surface* (2006) is the standard reference for arbitrage-free parameterizations and the constraints of Section 19.5; Gatheral and Jacquier (2014) give SSVI and Lee (2004) the wing bound. On the volatility risk premium, Carr and Wu (2009) measure it from synthetic variance swaps across indices and single stocks, and Bollerslev, Tauchen and Zhou (2009) relate it to expected stock returns; Chapters 21 and 22 return to it when volatility becomes a traded asset.

PRACTICE QUESTIONS

19.1 Explain why equity options show a skew while currency options show a roughly symmetric smile, giving the structural reason in each case.

19.2 Using the three-month and six-month rows of Table 19.3, compute the slope between strikes 90 and 110 per 10% of spot and per standard deviation of standardized moneyness (use each row's at-the-money volatility), and explain why the two disagree about whether the skew flattens.

19.3 State the Breeden–Litzenberger result and explain, without algebra, why the second derivative of the call price with respect to strike is a probability density.

19.4 Using the table in Example 19.1, state by what factor the market's probability of a 30%

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decline exceeds the lognormal's, and explain what feature of the smile produces it.

19.5 Using Example 19.3, state the largest rise in implied volatility the 115/120 spread can carry, describe the arbitrage that appears beyond it, and explain why the permitted slope shrinks as strikes move further out of the money.

19.6 An at-the-money option has vega 40 and Black–Scholes delta 0.54. Under sticky delta with $\partial\sigma/\partial S = 0.0015$, compute the corrected delta and the error in shares on a 50,000-option position, and state which convention gives the larger hedge for a call under an equity skew.

19.7 Explain why the volatility risk premium implies that a straddle buyer is not simply betting the market will move.

19.8 In Example 19.2 the bucket probabilities and the two tail probabilities sum to exactly 100%. Explain why this had to happen and what it would mean if it did not; then replace the \$50 call's midquote of 1.475 with 1.050, and say which butterfly turns negative and what arbitrage it offers.

19.9 An index of 100 equally weighted stocks has component implied volatilities of 25% and an index implied volatility of 16%. Compute the implied correlation, and explain why index skew is steeper than single-stock skew.

19.10 Explain why the volatility term structure inverts in a crisis, and why this makes total vega a poor summary of an option book's exposure.

FURTHER QUESTIONS

19.11 Example 19.1 notes that its quadratic smile, extended below the observable range, implies a density that turns negative below a strike of about 6 and totals 1.06 when integrated upward from there. Explain why this is not a numerical error to be corrected, describe two different extrapolation choices and how each would distribute the probability below the lowest observable strike (a total the chapter shows is fixed), and state what a risk manager should conclude about tail estimates derived this way.

19.12 Section 19.9 gives four explanations for the skew. Design an empirical test that would distinguish the leverage explanation from the supply-and-demand explanation, stating what data you would need and what each hypothesis predicts.

19.13 Section 19.10 states that the volatility risk premium is larger at low strikes than at the money. Explain what this implies about the profitability of selling puts versus selling straddles, and why the strategy with the higher average return is not obviously the better one.

19.14 Show that under sticky delta the at-the-money-forward implied volatility does not change when the spot moves, even though the correction in (19.4) for the option struck there today does not vanish (Example 19.5), and explain why this makes the at-the-money-forward straddle of Section 11.4 a natural reference point for a desk.

19.15 Section 19.12 claims two models fitting today's surface perfectly can price the same barrier option far apart, and that the size of the gap depends on the payoff. Explain what information a barrier price requires that a vanilla price does not, connect this to Section 13.4's completeness argument, and explain why a reverse knock-out should be far more exposed to the modeling choice than a down-and-out struck at the money.

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Chapter 20
Numerical Procedures

Chapter 13 opened by saying that its tree was a teaching device rather than a computational method, and that how to build trees efficiently, how many steps they need, and how to make them converge belonged here. Chapter 15 said the same of the free boundary in an American option, Chapter 14 of the choice between two ways to discretize a process, and Chapter 17 of the simulation needed to value a relative-performance award; the smile-fitting models of Section 19.4, local volatility above all, have no formula and are computed the same way. This chapter answers those questions.

The subject is narrow and the stakes are not. Every price in Chapters 13 through 19 that does not come from a closed-form formula is produced by one of the three methods in this chapter, and the difference between a method that converges and one that merely appears to is the difference between a book of trades that is valued correctly and one that is not.

Section 20.1 says what numerical methods are for. Sections 20.2 to 20.5 are trees in practice: the control variate, trinomial trees, and dividends. Sections 20.6 to 20.10 are Monte Carlo: discretizing the process, variance reduction, several assets, and Greeks from a simulation. Section 20.11 is American Monte Carlo and Section 20.12 finite differences; Section 20.13 chooses among the three methods, and Section 20.14 records where they came from.

20.1 WHAT NUMERICAL METHODS ARE FOR

A closed-form price exists only when three things happen to be true at once: the payoff is simple, the process is simple, and there is no early exercise. Break any one and a number must be computed.

Table 20.1: Why a problem needs a numerical method, and which one suits it.

Feature	Example	Natural method
Early exercise	American put, Bermudan swaption	Tree or finite difference
Path dependence	Asian, lookback, barrier	Monte Carlo
Many state variables	Basket, quanto basket, CVA	Monte Carlo
Both at once	Callable structured note	American Monte Carlo
Greeks across a grid	A whole book, revalued	Finite difference

Table 20.1 is the chapter in one page, and the pattern in it has a reason. [TN15] showed that the pricing partial differential equation and the risk-neutral expectation are the same statement (the Feynman–Kac relation), so a price can be obtained either by solving the equation backward or by simulating the expectation forward. The two directions have opposite strengths, and everything in Table 20.1 follows from that:

Backward methods (trees, finite differences) sweep from expiration to today and therefore know, at every node, the value of continuing. That is exactly what an early-exercise test needs. They also produce the value at every state, so the Greeks come out of the same computation.

Their weakness is dimension: a grid with m points along each of d state variables has m^d nodes (a million at $m = 100$ and $d = 3$), which is fatal beyond about three.

Forward methods (Monte Carlo) generate whole paths and average. **Path dependence**, a payoff that depends on the route and not only the destination, therefore costs no more to evaluate than a terminal payoff does: the whole path has already been generated, and an Asian or a lookback payoff simply reads it. The cost also grows linearly rather than exponentially in the number of state variables. Their weakness is the mirror image: at any intermediate date a simulated path does not know its own continuation value, which is why early exercise is hard and why Section 20.11 exists.

Transform methods are a fourth route, and a narrow one. Where a model's characteristic function (the Fourier transform of its terminal distribution) is known in closed form (Heston, Bates and the jump models of Chapter 21), a European price is a single integral over it, which the fast Fourier transform of Carr and Madan (1999) or the cosine expansion of Fang and Oosterlee (2008) evaluates fast enough for Chapter 21 to calibrate Heston to a whole surface inside an optimizer. Having no early exercise and no path dependence, the route supplements the other three rather than replacing any.

Convergence is the whole subject. Every method here produces an answer for any settings at all; the question is always whether that answer is close, and how one would know. The chapter's recurring warning is that a number that looks stable can be wrong, and that the cheapest defense is to compute something whose answer is already known.

20.2 BINOMIAL TREES IN PRACTICE

Chapter 13 built the tree and Section 13.9 observed that its value settles to a limit at a rate of about $1/n$, oscillating between odd and even n. Here is that statement made precise, because both halves of it matter.

Take a European put with $S_0 = K = 50$, $r = 5\%$, $\sigma = 30\%$, and six months to expiration. The Black–Scholes–Merton value is 3.582934.

Table 20.2: Cox–Ross–Rubinstein tree against the Black–Scholes–Merton value.

n	Tree	Error	n	Tree	Error
5	3.789185	+0.206251	100	3.572449	−0.010485
10	3.479418	−0.103516	101	3.592980	+0.010046
11	3.675951	+0.093017	200	3.577688	−0.005246
20	3.530789	−0.052145	201	3.587979	+0.005045
21	3.631453	+0.048519	500	3.580835	−0.002099
50	3.561990	−0.020943	1000	3.581884	−0.001050

Two features of Table 20.2 deserve attention.

The error halves when n doubles. From 100 to 200 to 500 to 1000 the error runs −0.0105, −0.0052, −0.0021, −0.0011. This is first-order convergence, and it is bad news: to gain one decimal place you must take ten times as many steps, and the work at n steps is of order n^2 nodes, written $O(n^2)$, so a factor of ten in accuracy costs a factor of a hundred in time.

The sign alternates with the parity of n. Even n undershoots, odd n overshoots, and they do so by almost the same amount. The reason is geometric: with $S_0 = K$, an even number of steps places a terminal node exactly at the strike while an odd number straddles it, so the discretized distribution sits differently against the payoff's kink in the two cases.

The alternation is a property of this contract, not of trees. What sets the error is where the strike falls between two terminal nodes, and a node lands on it exactly when $\kappa + n$ is an even

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integer, for $\kappa = \ln(K/S_0)/(\sigma\sqrt{\Delta t})$. At the money $\kappa = 0$ and that reads "n even." Otherwise κ grows like $\sqrt{n}$, the strike drifts through the lattice as steps are added, and the signs stop alternating: at $K = 55$ the errors at $n = 26, 27, 28$ run −0.006, +0.026, +0.003. Section 20.3 turns this into a rule for choosing n at any strike.

How many steps. There is no universal answer, but the shape of Table 20.2 gives the discipline. Never report a tree value without computing it at two step counts. If 200 and 400 steps disagree in the third decimal, the third decimal is not real. Dealers' pricing systems typically run 200 to 1000 steps for a vanilla American option and use the correction of the next section to do better with fewer.

20.3 THE CONTROL VARIATE

This section treats one technique, the main repair for the slow, oscillating convergence Section 13.9 deferred: it reaches far beyond trees, and the rest of the chapter keeps returning to it.

The idea is general. Suppose you want A, which you can only estimate, and there is a related quantity E that you can estimate *by the same method* and whose true value E^{true} you already know. Then

$$\boxed{\hat{A}_{\text{corrected}} = \hat{A} + \left(E^{\text{true}} - \hat{E}\right)} \tag{20.1}$$

where $\hat{A}$ is the method's estimate of the target, $\hat{E}$ its estimate of the companion (the same method at the same settings, on the same tree or the same paths), E^{true} the companion's known exact value, and $\hat{A}_{\text{corrected}}$ the number reported. The correction is the method's own error on a problem where the error is visible.

For an American option the natural companion is the European option with the same strike and expiration: price both on the same tree, and correct the American value by the tree's error on the European, which the Black–Scholes–Merton formula gives exactly.

Example 20.1 The control variate on an American put

Take the same contract, now American: $S_0 = K = 50$, $r = 5\%$, $\sigma = 30\%$, six months. Its value is 3.69702, computed two ways that share no arithmetic (corrected trees at thousands of steps, and the finite-difference grid of Section 20.12), against the European 3.582934, an early-exercise premium of 0.114.

n	Plain tree	Error	Corrected	Error
50	3.68552	−0.01150	3.70646	+0.00944
51	3.71691	+0.01989	3.69700	−0.00002
100	3.69135	−0.00567	3.70183	+0.00481
101	3.70717	+0.01015	3.69712	+0.00010
200	3.69419	−0.00283	3.69944	+0.00242
201	3.70213	+0.00511	3.69709	+0.00007

On odd step counts the correction is extraordinary: at $n = 201$ the error falls from +0.00511 to +0.00007, a factor of about 75, for roughly twice the work. On even step counts it changes the sign of the error and shrinks it by less than a fifth.

The asymmetry says exactly what the control variate does. With $S_0 = K$ and odd n, no node sits at the strike, and the American and the European tree carry the same large error from straddling the payoff kink—a *shared* error, which (20.1) removes completely. With even n a node sits exactly on the strike and the two errors part company: the European's is as large as before, but the American's is only about half of it, because the American value also carries the error in locating

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the early-exercise boundary, which the European option does not have and therefore cannot correct.

A control variate removes what the two problems share and nothing else. The practical corollary is to choose a companion problem as close to the target as possible.

The step count is the second lever, and it works at any strike. Nothing in that explanation mentions parity: what decides the matter is whether a node sits on the strike, which by Section 20.2 happens when $\kappa + n$ is even and is furthest from happening when $\kappa + n$ is odd. At the money $\kappa = 0$ and that reads "take n odd," which is Section 20.2's alternation turned into an instruction. Away from the money it reads "choose n so that $\kappa + n$ is close to an odd integer," whatever the parity of n.

At $K = 55$, step counts that leave the strike near a midpoint are about sixteen times more accurate for the same work.

Notice also, in Figure 20.1, what the correction does to the shape of the sequence. The plain tree oscillates, so extrapolating across it is dangerous; the corrected sequence oscillates too, but its odd branch already sits on zero and its even branch decays smoothly toward it, so extrapolating along a branch is safe. Following the correction with **Richardson extrapolation** (taking $2V(2n) - V(n)$, which cancels an error of the form C/n provided both step counts carry the same C) is the standard recipe in practice, and it reaches four-decimal accuracy at a few hundred steps. The proviso is not automatic, because C depends on the strike's **offset**, its position between the two terminal nodes that bracket it: the two step counts must share an offset, which at the money means both even.

Other parameterizations remove the oscillation at its source. The equal-probability tree fixes $p = 1/2$ and puts the drift in u and d; Leisen and Reimer (1996) choose u, d and p by inverting the normal distribution so that the tree is centered on the strike, which makes convergence monotone and close to second order with no extrapolation. It is a common choice for vanilla American equity options.

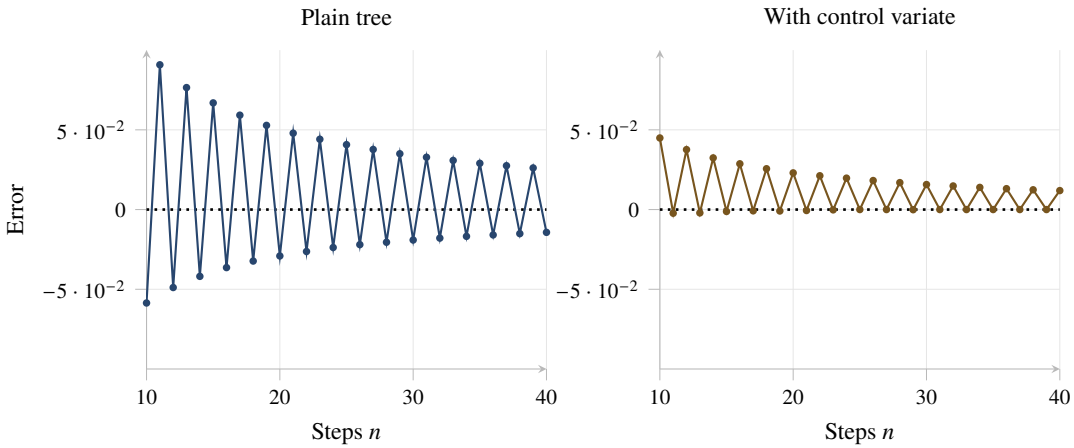


Figure 20.1: American put error against step count, same vertical scale.

20.4 TRINOMIAL TREES

Let the price move up, down, *or stay*. With the log-price $x = \ln S$ spaced Δx apart and drift $\nu = r - q - \sigma^2/2$,

$$p_u = \frac{1}{2}\left(\frac{\sigma^2\Delta t + \nu^2\Delta t^2}{\Delta x^2} + \frac{\nu\Delta t}{\Delta x}\right), \quad p_d = \frac{1}{2}\left(\frac{\sigma^2\Delta t + \nu^2\Delta t^2}{\Delta x^2} - \frac{\nu\Delta t}{\Delta x}\right), \quad p_m = 1 - p_u - p_d \tag{20.2}$$

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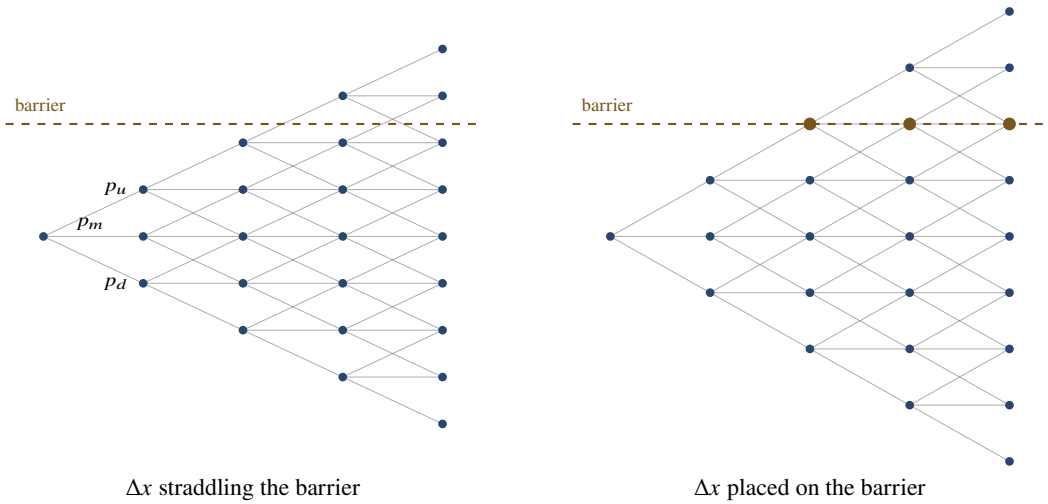
These are not a definition but a consequence. Over one step the change δx in the true log-price has

$$E[\delta x] = \nu\,\Delta t, \qquad E[\delta x^2] = \sigma^2\Delta t + \nu^2\Delta t^2$$

while the tree moves x by $+\Delta x$, 0 or $-\Delta x$ and so delivers $(p_u - p_d)\,\Delta x$ and $(p_u + p_d)\,\Delta x^2$. Setting each pair equal, together with $p_u + p_m + p_d = 1$, gives three equations in three unknowns, and (20.2) is their solution. The tree is built to reproduce the first two moments of the process it stands for, which is what Chapter 13's binomial tree does with two branches rather than three.

That leaves Δx free, and $\sigma\sqrt{3\Delta t}$ is the common choice: it puts $p_u + p_d$ at about $1/3$, so the middle branch carries roughly $2/3$ of the probability and the two outer ones stay safely positive.

Figure 20.2 shows the geometry, and the recombination is the property that matters: a step adds two nodes rather than tripling them, so the work grows with the square of the number of steps instead of with 3^n.



Δx straddling the barrier Δx placed on the barrier

Figure 20.2: Four steps of a trinomial lattice, drawn against the same barrier at two choices of Δx.

The extra branch buys two things and does *not* buy a third.

It does not converge faster. On the European put above, the trinomial errors at $n = 50, 100, 200$ are -0.0207, -0.0103, -0.0052, against binomial errors of -0.0209, -0.0105, -0.0052. Both are $O(1/n)$ and they are numerically almost identical. Anyone who reaches for a trinomial tree expecting accuracy for free will not get it.

It buys a free parameter. In (20.2), Δx and Δt are chosen independently, whereas the binomial tree ties them by $u = e^{\sigma\sqrt{\Delta t}}$. That freedom is what lets a node be placed exactly on a barrier, as the right panel of Figure 20.2 does and the left one fails to. Chapter 34 shows that a barrier option priced on a tree whose nodes straddle the barrier converges badly, because the barrier is effectively moved to the nearest node.

It buys room for a varying volatility. A local-volatility model in the sense of Section 19.4 needs a different σ at each node; in a binomial tree that changes u from node to node, and the paths stop meeting. The trinomial keeps Δx fixed and absorbs σ in the probabilities.

It is also a finite difference scheme. Section 20.12 shows that the explicit finite difference method *is* a trinomial tree, with the three probabilities appearing as the coefficients of the discretized derivatives. The two subjects are one subject.

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20.5 DIVIDENDS AND THE NON-RECOMBINING TREE

Section 13.10 flagged this as the first place the tree's convenience breaks. Two of the three ways a stock can pay a dividend cost the tree nothing. A *continuous yield* is handled in the drift: replace r by $r - q$ and the geometry is untouched. A *discrete proportional* dividend (a known fraction δ of whatever the price turns out to be at the ex-date t_d) is free for a different reason: every node at that date is multiplied by the same $1 - \delta$, so the lattice is rescaled rather than distorted and recombines exactly as before. A known cash dividend is not free. If the price drops by \$2 on the ex-date, an up-then-down path and a down-then-up path no longer meet, because \$2 is subtracted at different price levels: subtracting a constant moves nodes by different proportions, where multiplying by one does not. Every node standing at the ex-date then starts a tree of its own, so each dividend multiplies the node count by the number of nodes at its date, and with several ex-dates the count grows geometrically until the tree is unusable. Figure 20.3 shows the difficulty.

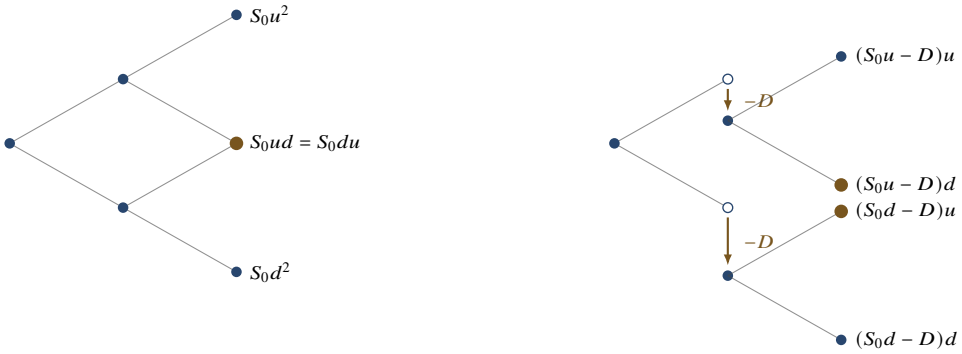


Figure 20.3: Two steps around an ex-date, without and with a cash dividend D.

The standard remedy is the **displaced tree**, which splits the stock price into a part that behaves like a lognormal and a part that does not.

Write $S = S^* + De^{-r(t_d - t)}$ for $t < t_d$, where D is the dividend and t_d its date. The second term is a known, riskless amount, the present value of a certain payment. Build the tree on S^*, which has no dividend and therefore recombines, then add the present value of the dividend back at every node when computing the exercise decision and the payoff. One caveat goes with it: the σ on the tree is the volatility of S^*, so a volatility quoted on S must be scaled up to roughly $\sigma S/S^*$. Example 20.2 takes its 30% as the volatility of S^*.

Figure 20.4 draws the construction rather than its result. The dashed lattice is S^*: it is the tree that is actually built, it recombines, and it runs straight through the ex-date without noticing it. The solid nodes above it are what the exercise test compares against, S^* with the dividend's present value added back, and past the ex-date that value is zero and the two are one lattice. The recombination lives in S^* and the economics in S, which is the trick.

Example 20.2 A cash dividend done three ways

Give the American put of Example 20.1 a \$2.00 dividend at $t = 0.25$.

The displaced spot is $S^* = 50 - 2e^{-0.05\times 0.25} = 48.0248$, and a tree built on it gives 4.6063, 4.6096 and 4.6100 at 100, 400 and 1,600 steps. That sequence looks settled and is not: the converged American put value is 4.6098, which the tree reaches only after tens of thousands of steps and the finite-difference method of Section 20.12 confirms. The corresponding European

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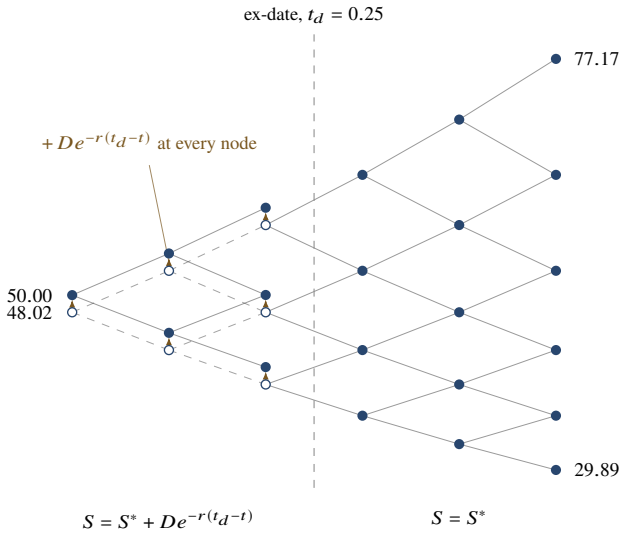


Figure 20.4: Five steps on the displaced tree of Example 20.2: dashed, S^*; solid, S^* plus the dividend's present value. $S_0 = 50$, $D = 2$ at $t_d = 0.25$, $\sigma = 30\%$, $r = 5\%$.

value is 4.4689 from the Black–Scholes–Merton formula on S^*.

Compare that with two shortcuts:

Treatment	Value	Error
Displaced tree (correct)	4.6098	—
Dividend ignored	3.6970	−19.8%
Spot reduced by the full \$2.00	4.6385	+0.6%

Ignoring the dividend is catastrophic and obviously so. Subtracting the undiscounted dividend instead of its present value is a small error here (\$0.03, or 0.6%), but it is the kind that scales: with a larger dividend, a longer time to the ex-date, or a higher rate, the discounting matters more, and the error is one-directional rather than random.

The early-exercise premium also moves. Without the dividend it was 0.114; with it, 4.6098 − 4.4689 = 0.141. The rise comes from the lower price, which leaves the put deeper in the money, and not from the dividend itself: on a non-paying stock at the same spot of 48.0248 the premium is 0.157. Before the ex-date a dividend makes early exercise of a *put* less attractive, not more: the put holder wants the stock price low, and the dividend does that for them without their having to exercise. That runs opposite to the call case of Section 11.10.

Holding the dividend *amount* fixed shows what its form and timing are worth. A 4% proportional dividend at t_d has a present value of exactly $\delta S_0 = \$2.00$, since $e^{-rt}S_t$ is a martingale, against \$1.9752 for the \$2.00 of cash, and the put is worth correspondingly more: 4.6223 against 4.6098. Spread the same 4% smoothly over the six months instead ($q = -\ln(0.96)/0.5 = 8.16\%$) and the terminal distribution is unchanged, $0.96S_0e^{(r-\sigma^2/2)T+\sigma W_T}$ either way, so the European value is 4.4810 under both; but the American falls to 4.4812: the early-exercise premium drops from 0.1414 to 0.0002, because a continuous yield lowers the drift to $r - q = -3.16\%$ everywhere, and against a stock expected to fall a put is worth holding rather than exercising. The form of the payment moves the value by 0.3%, its timing by more than 3%.

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20.6 MONTE CARLO SIMULATION

Simulate the risk-neutral process forward, compute the payoff on each path, average, and discount:

$$\hat{f} = e^{-rT}\,\frac{1}{M}\sum_{i=1}^{M}\text{payoff}\big(\text{path}_i\big) \tag{20.3}$$

This is the definition of a risk-neutral price from Section 13.2, evaluated by brute force. Nothing about it is subtle, and its two properties follow immediately from the law of large numbers and the central limit theorem.

It is unbiased. Each sampled payoff has the price as its discounted expectation, and their average converges to it regardless of the payoff's shape, which is why a path-dependent payoff needs no special handling.

Its error falls like $1/\sqrt{M}$. Write $f_i = e^{-rT}\,\text{payoff}(\text{path}_i)$ for the discounted payoff on path i. The standard error of (20.3) is

$$\text{s.e.}(\hat{f}) = \frac{s}{\sqrt{M}}, \qquad s^2 = \frac{1}{M-1}\sum_{i=1}^{M}\left(f_i - \hat{f}\right)^2 \tag{20.4}$$

where s is the sample standard deviation of the M discounted payoffs and $\hat{f}$ is their average, which is (20.3) itself. Both come out of one pass over the paths, so a simulation reports its own accuracy, which nothing else in this chapter does. To gain one decimal place you must run a hundred times as many paths, and Example 20.3 shows what that costs.

Example 20.3 The cost of a decimal place

Simulate the European put of Table 20.2, whose exact value is 3.582934:

Paths	Estimate	Standard error	Actual error
1,000	3.71346	0.16249	+0.13053
10,000	3.63958	0.05062	+0.05664
100,000	3.58181	0.01595	−0.00113
1,000,000	3.58916	0.00504	+0.00623

The standard error falls by $\sqrt{10} \approx 3.16$ for every tenfold increase in paths, which is the $1/\sqrt{M}$ rule holding to within 2%. The actual errors bounce around inside it (at a million paths the estimate happens to be worse than at a hundred thousand), which is the point of a standard error: it is a statement about the distribution of the answer, not about this answer.

A million paths bought a standard error of half a cent on an option worth \$3.58. A 200-step binomial tree, costing a small fraction of the time, was already inside a cent. For a vanilla European option, Monte Carlo is the wrong tool, and reporting a simulated value where a formula exists is a sign that something has gone wrong in the design rather than in the code.

The standard error also has a use beyond reporting: it is a diagnostic. A simulation whose standard error is not falling like $1/\sqrt{M}$ has a bug, a badly behaved payoff, or a random number generator with structure. Always compute it, and always compare it with the observed error on a case where the answer is known. Example 20.4 does both, on a calculation Chapter 9 needed.

Example 20.4 The exposure profile of Chapter 9

Chapter 9 valued counterparty risk on a five-year swap using an expected exposure profile it did not simulate. The calculation is a good one to meet first because the answer can be checked in closed form.

A dealer holds a five-year payer swap on \$100 million, struck at the market so that it is worth

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zero today. Take the swap rate to diffuse as a normal random walk, so that at a horizon t it stands at $S_t = S_0 + \sigma_r \sqrt{t}\, Z$ for a standard normal Z, with $\sigma_r = 1.456\%$ a year an absolute volatility rather than a proportional one. That is the normal-volatility convention of Section 19.1, which rates markets adopted because it permits a negative rate. It also makes the closed form below available: $V(t)$ is linear in the rate and so inherits its distribution. By equation (7.6) the swap is then worth

$$V(t) = L\,(S_t - K)\,A(t)$$

where L is the \$100 million notional, $K = S_0$ is the fixed rate, and $A(t)$ is the annuity of the $5 - t$ years still to run, which we take as $5 - t$ years of unit payments.

The simulation. Draw Z, form $V(t)$, keep $\max(V(t), 0)$, and average. Ten thousand paths at $t = 2.5$ give \$2.29 million, with a standard error of about \$0.03 million.

The closed form. What the simulation just estimated is the expected exposure $\mathrm{EE}(t) = E[\max(V(t), 0)]$: the average over states of what would be lost if the counterparty failed at t, with the states in which the dealer owes money contributing nothing. Because $V(t)$ is normal with mean zero, equation (9.3) evaluates that same expectation directly, with $\sigma_V(t)$ the standard deviation of the swap's value at that horizon:

$$\mathrm{EE}(t) = 0.3989\,\sigma_V(t) = 0.3989\,L\,\sigma_r\sqrt{t}\,A(t)$$

At mid-year points this gives, in millions, \$1.848, \$2.490, \$2.296, \$1.630 and \$0.616, the column Example 9.2 used; that example and Figure 9.1 explain the shape of the profile.

The closed form exists only because the swap's value is linear in a normal rate. Add an option, a barrier, or a second currency, and the expectation stops being available in closed form, and that is when the three-stage simulation of Section 9.1 stops being an illustration and becomes the only method.

20.7 DISCRETIZING THE PROCESS

Section 14.4 set out two ways to step a geometric Brownian motion forward and said the choice would return here.

The **Euler** scheme discretizes the stochastic differential equation directly:

$$S_{t+\Delta t} = S_t + \mu S_t \Delta t + \sigma S_t \varepsilon \sqrt{\Delta t} \tag{20.5}$$

The **exact** scheme uses the solution rather than the equation:

$$S_{t+\Delta t} = S_t \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\varepsilon\sqrt{\Delta t}\right] \tag{20.6}$$

They are not equivalent, and the difference matters in two ways.

Equation (20.6) has no discretization error at all. It is the exact distribution of $S_{t+\Delta t}$ given S_t, so for geometric Brownian motion a single step from 0 to T is as accurate as a thousand. Any intermediate steps are needed only because the *payoff* looks at them: an Asian option needs the dates its average is taken on, a barrier option needs a fine grid.

Equation (20.5) can go negative. A large negative ε drives S below zero, which is impossible for the process it claims to simulate. With $\sigma\sqrt{\Delta t}$ small this is rare; with a large step or a high volatility it is not, and a simulation that produces negative prices will produce nonsense payoffs without necessarily producing an error message.

So use (20.6) whenever it exists, which for lognormal processes it always does. The reason Euler schemes are ubiquitous is that for most other processes (stochastic volatility, mean-reverting rates, jump models) no exact solution is available, and then the discretization error is real and must be managed.

A **predictor–corrector** step addresses one of its two parts, the error in the drift. Where the drift depends on the state being simulated, (20.5) freezes that drift at the value it has when the interval opens, and the step is biased whenever the state moves appreciably before the interval

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closes. Write the process as

$$dX = a(X,t)\,dt + b(X,t)\,dW$$

Take the Euler step first, as a prediction,

$$\hat{X} = X_t + a(X_t,t)\,\Delta t + b(X_t,t)\,\varepsilon\sqrt{\Delta t} \tag{20.7}$$

then step again from X_t, averaging the drift at the two ends of the interval,

$$X_{t+\Delta t} = X_t + \frac{1}{2}\big[a(X_t,t) + a(\hat{X},t+\Delta t)\big]\Delta t + b(X_t,t)\,\varepsilon\sqrt{\Delta t} \tag{20.8}$$

The same ε stands in both, the draw of (20.7): drawing again would add a second independent source of noise where the intention was to remove a bias. The cost is one extra drift evaluation per step and no extra random numbers, and Example 20.5 measures what it buys.

Example 20.5 What a predictor–corrector corrects

Geometric Brownian motion lets the drift error be priced rather than simulated. Taking expectations through (20.5) multiplies S by $1 + \mu\Delta t$ over a step, the first two terms of $e^{\mu\Delta t}$, and through (20.8) by $1 + \mu\Delta t + \frac{1}{2}(\mu\Delta t)^2$, the first three. Each scheme therefore arrives at a different forward, and the Black–Scholes–Merton formula prices each of those forwards exactly.

Take $S_0 = K = 100$, $r = 10\%$, $\sigma = 20\%$ and $T = 1$. The exact forward is 110.5171 and the call is worth 13.2697; at twelve steps Euler reaches a forward of 110.4713 and the corrector 110.5170. The premiums those imply:

Steps	Euler	Error	Pred.–corr.	Error	Ratio
1	12.931946	0.337730	13.258455	0.011222	30
4	13.180623	0.089053	13.268935	0.000742	120
12	13.239625	0.030052	13.269593	0.000083	360
52	13.262708	0.006968	13.269672	0.0000045	1560

Halving the step halves the Euler error and quarters the corrector's: the drift is discretized to first order in Δt by (20.5) and to second order by (20.8).

One error is absent from the table. Pricing each scheme's forward under Black–Scholes isolates the drift, and an Euler simulation carries a second error on top: its terminal distribution is a product of normals rather than a lognormal, which is the same defect that lets (20.5) go negative. A predictor–corrector repairs the first and leaves the second alone.

The limit of the method is the distributional error just named. The diffusion term of (20.8) is the Euler one unchanged, so a quantity that depends on the fine structure of the path rather than on its expectation gains nothing from correcting the drift. Nor does the correction help a continuously monitored barrier, whose error comes from crossings missed between dates: that needs, on each step, the probability that the path crossed the barrier between the two simulated points, or Chapter 34's continuity correction.

That leaves a rule with two branches. Use the exact scheme where one exists. Use a predictor–corrector where the drift depends on the state and is not small beside the diffusion. Chapter 32's market model is where the predictor–corrector is normally applied, because there the drift of each rate is a function of all the others, and Example 32.4 shows it earning almost nothing, for the reason this section supplies: at a 3% rate the drift is about 1% of the diffusion, so correcting it cannot move the answer.

20.8 VARIANCE REDUCTION

Since the error falls like $1/\sqrt{M}$, buying accuracy with paths is expensive. Buying it with structure is cheap. Four techniques follow.

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Antithetic variates. For every path generated with shocks $\varepsilon_1, \ldots, \varepsilon_n$, generate its mirror with $-\varepsilon_1, \ldots, -\varepsilon_n$ and average the two payoffs. Each path has the right distribution, and for a payoff monotone in the shocks the two are negatively correlated, so the variance of the average is lower. On the European put, the standard error at 100,000 paths (50,000 antithetic pairs) falls from 0.0159 to 0.0112, a factor of 1.4. It costs almost nothing and is worth doing for monotone payoffs, but it is not transformative, and for a payoff symmetric in the shocks, such as a straddle at the money, it is useless.

Control variates. Equation (20.1) returns in a more powerful form. Estimate a quantity Y whose true value is known alongside the target X, and report

$$\hat{X}_{\text{corrected}} = \hat{X} - \beta\left(\hat{Y} - Y^{\text{true}}\right), \qquad \beta = \frac{\operatorname{Cov}(X, Y)}{\operatorname{Var}(Y)} \tag{20.9}$$

With β chosen as in (20.9), the variance is reduced by a factor $1/(1-\rho^2)$, where ρ is the correlation between X and Y. Everything depends on ρ, and the difference between a good control variate and a mediocre one is enormous.

Example 20.6 A control variate that is worth 58 million paths

Price an arithmetic-average Asian call: $S_0 = K = 50$, $r = 5\%$, $\sigma = 30\%$, one year, twelve monthly fixings. There is no formula.

There is a formula for the *geometric*-average version, because the geometric average of lognormals is lognormal. It gives 4.012352 exactly. And the geometric and arithmetic averages of the same twelve prices are nearly the same number, so their payoffs are nearly perfectly correlated.

At 100,000 paths:

	Estimate	Standard error
Plain Monte Carlo	4.23399	0.02020
Geometric control variate	4.23671	0.00084

The measured correlation is $\rho = 0.99914$ and $\beta = 1.0485$. The variance falls by a factor of 583: to reach the same standard error, plain Monte Carlo would need 58 million paths.

The pattern that produced it is the one that gave the tree its accuracy in Example 20.1: find a problem as close as possible to the target and solvable in closed form, let the method price both, and correct the target by the error on the companion.

Importance and stratified sampling. Both put the paths where the answer is decided. A deep out-of-the-money option pays on a few paths in a thousand; **importance sampling** shifts the drift of the simulation toward the payoff region and reweights each path by how much likelier it was under the true drift than under the shifted one, which keeps the estimate unbiased, and it is the standard tool for deep out-of-the-money prices and for the rare tail losses of credit portfolios. **Stratified sampling** draws a fixed number of paths from each of several equal-probability bins of the normal draw that drives the path, so the sample cannot overweight a region by chance; moment matching, which rescales the draws to a sample mean of exactly 0 and variance of exactly 1, is a cruder relative, at the cost of a small bias.

Quasi-Monte Carlo. Replace pseudo-random numbers with a low-discrepancy sequence (Sobol or Halton) designed to fill the space more evenly than randomness does. Convergence can approach $1/M$ rather than $1/\sqrt{M}$ when a few of the draws decide most of the answer. The cost is that the usual standard error is no longer valid, since the points are not random; estimating the error requires several randomized repetitions. It works best when a few of the

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random numbers drawn for each path decide most of its payoff, and paths are often built in a special order to make that so.

20.9 SIMULATING SEVERAL ASSETS

Monte Carlo's advantage over trees is dimension, and correlated assets are the reason to use it.

To simulate d assets with correlation matrix $\mathbf{R}$, factor $\mathbf{R} = \mathbf{L}\mathbf{L}^\top$ by Cholesky decomposition and set $\boldsymbol{\varepsilon} = \mathbf{L}\mathbf{z}$, where $\mathbf{z}$ is a vector of d independent standard normals. Here $\mathbf{L}$ is lower triangular (zero above the diagonal), and that shape is not incidental: it is what makes the construction sequential, with ε_1 built from z_1 alone, ε_2 from z_1 and z_2, and each shock in turn from the draws already used plus one new one. For two assets it reduces to

$$\varepsilon_1 = z_1, \qquad \varepsilon_2 = \rho z_1 + \sqrt{1-\rho^2}\, z_2$$

which is all that most problems need.

It is the correlation matrix that is factored, not the covariance matrix. Each asset's volatility enters separately, in its own price update, so the ε_i must come out with unit variance, which the two-asset form shows, since $\rho^2 + (1-\rho^2) = 1$. Factoring the covariance matrix instead returns shocks that already carry σ_i, and applying σ_i to them a second time is an error that produces plausible prices at the wrong volatility.

The factorization also needs an internally consistent matrix (technically positive semidefinite: one that gives no portfolio a negative variance), and one assembled pair by pair, from estimates over different windows or from traders' manual adjustments, often is not; Cholesky then fails partway through. Risk systems replace such a matrix with the nearest consistent one, by Higham's (2002) method.

Example 20.7 Valuing a relative-TSR performance award

Section 17.7 said that a performance award with a market condition is valued by simulation at grant and never trued up, and pointed here for the machinery.

The award: for each target unit, the executive receives shares at vesting in three years, in a quantity that scales linearly from 0% if the company's total shareholder return trails the index by 30% or more, through 100% if the two are equal, to a cap of 200% if the company beats the index by 30% or more. Settlement is in shares.

Take $S_0 = 50$, company volatility 35%, index volatility 20%, correlation 0.60, $r = 5\%$, three years. Simulating 400,000 correlated pairs gives:

Expected units earned, $E[u]$	0.8361
Expected share price, $E[S_T]$	58.1795
Naive product, $e^{-rT}E[u]\,E[S_T]$	41.8704
Correct value, $e^{-rT}E[u\,S_T]$	60.2468

Before reading anything off that table, apply the chapter's own rule to it. The exact $E[S_T]$ is $50e^{0.15} = 58.0917$, so the run came in 0.15% high, about one and a half standard errors, which is what 400,000 paths should deliver. The award value carries a standard error of roughly 0.13.

Two results follow.

The expected number of units is below one, at 0.836, even though the payout schedule is symmetric around a 100% payout. The company's higher volatility drags its median return below the index's (the $-\sigma^2/2$ term of Section 14.9), so the relative performance is more often slightly negative than slightly positive, while the 200% cap truncates the compensating upside.

And yet the award is worth 60.25 per unit, against a share price of 50. Expected units times expected share price gives 41.87, which understates the award by 30%; equivalently, the correct value is 44% higher. The gap is entirely the covariance: units are earned precisely on the paths where the share price is high, so $E[u\,S_T] \neq E[u]E[S_T]$.

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Section 17.7's asymmetry now has a number attached: that 60.25 is expensed in full whether or not the award ever pays. Simulation is the natural method because the payoff, a capped and floored function of two correlated prices multiplied by one of them, is trivial to write inside a loop and awkward to integrate in closed form.

20.10 GREEKS FROM A SIMULATION

A simulated price is one number; a desk needs its sensitivities, because they are the coefficients of Chapter 18's expansion, equation (18.1), whose terms its hedges are built to cancel. The obvious approach, to revalue with S_0 bumped by a small h, difference the two prices, and divide by h, has a trap in it.

The trap is independent draws. If the bumped and unbumped simulations use independent random numbers, the difference of two estimates each carrying a standard error of, say, 0.016 has a standard error of 0.023, while the delta being measured might move the price by 0.05. The noise swamps the signal, and no number of paths fixes it efficiently. With common random numbers (the same draws reused in both runs) the two runs differ only through the bump, the noise largely cancels, and the estimator is usable.

This is the same principle as the control variate: the two runs share an error, and sharing it lets it cancel.

None of this difficulty arises on a lattice or a grid, which produce the value at every state and hand over delta, gamma and theta as differences of numbers already computed; Section 20.12 gives the formulas. Simulation is the method that has to work for them.

Two refinements remove the bump instead of taming it. **Pathwise differentiation** differentiates the payoff along each path, holding the draw fixed, which for a call is one application of the chain rule:

$$\frac{\partial}{\partial S_0}\, e^{-rT}(S_T - K)^+ = e^{-rT}\,\mathbf{1}\{S_T > K\}\,\frac{\partial S_T}{\partial S_0} = e^{-rT}\,\mathbf{1}\{S_T > K\}\,\frac{S_T}{S_0}$$

The second step is that $S_T = S_0 \exp[(r - \sigma^2/2)T + \sigma\sqrt{T}\,\varepsilon]$ is proportional to S_0 once ε is fixed, so $\partial S_T/\partial S_0 = S_T/S_0$. Average the right-hand side over the paths and the delta is in hand, from the paths that already produced the price, with no second simulation. The average is exactly unbiased here, because $e^{-rT}\mathbb{E}[S_T\mathbf{1}\{S_T > K\}] = S_0 N(d_1)$ makes the estimator's mean $N(d_1)$, the Black–Scholes delta itself. At $S_0 = K = 100$, $r = 10\%$, $\sigma = 20\%$ and $T = 1$ that delta is 0.7257, and 400,000 paths give 0.7262.

What this buys is not lower variance. Against a bump run on common random numbers the two variances agree (the ratio is 0.976 at $h = 1$ and 1.000 at $h = 0.01$), because as h falls the difference quotient becomes the derivative it approximates. What it buys is that there is no h to choose: the $h = 1$ bump reports 0.7344 against the exact 0.7257, an error of 0.0086 where the standard error is 0.0008. That error is bias. For a continuous payoff a small bump, or one taken on both sides of S_0, removes it at no cost in noise; the choice of h bites for a discontinuous payoff, where the bumped difference is nonzero on a vanishing fraction of paths and its variance grows like $1/h$.

It does require a differentiable payoff, and the failure is worse than it sounds. For a cash-or-nothing digital the payoff *is* the indicator, its derivative is zero wherever it exists, and the estimator returns zero on every path against a true delta of 0.0167: not a noisy answer but a confidently wrong one.

The **likelihood-ratio method** differentiates the density instead, which leaves the payoff untouched. Writing the price as $e^{-rT}\int \Pi(s)\,p(s;S_0)\,ds$, with Π the payoff and p the density

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of S_T, and differentiating under the integral,

$$\frac{\partial}{\partial S_0}\, e^{-rT}\,\mathbb{E}[\Pi] = e^{-rT}\,\mathbb{E}\left[\Pi(S_T)\,\frac{\partial \ln p}{\partial S_0}\right] = e^{-rT}\,\mathbb{E}\left[\Pi(S_T)\,\frac{\varepsilon}{S_0\sigma\sqrt{T}}\right]$$

where the last step is a short calculation: $\ln S_T$ is normal with mean $\ln S_0 + (r - \sigma^2/2)T$, so $\partial \ln p/\partial S_0$ reduces to $(\ln S_T - \text{mean})/(\sigma^2 T S_0)$, and that is the draw ε divided by $S_0\sigma\sqrt{T}$. The estimator is therefore the discounted payoff times the draw, scaled: one multiplication on top of the price, for any payoff whatever. On the digital it returns 0.0166 against the exact 0.0167, with a standard error of 0.00004.

Nothing is free. Because the payoff is left whole and the derivative is pushed onto the draw, the estimator inherits the payoff's full variability: on the call it carries 8.6 times the variance of the pathwise estimator, which is 2.9 times the standard error and so about nine times the paths for the same precision. Dealers' pricing systems use pathwise where they can and likelihood ratio where they must. Applied to every input at once, pathwise differentiation becomes adjoint algorithmic differentiation (AAD). A simulated price is a chain of elementary steps (build the exponent, exponentiate, take the payoff, discount, average), and the chain rule can run along that chain in either direction. Run forward, it carries the derivative with respect to one input through every step, so n inputs cost n passes, no better than bumping. Run backward, it starts from the price and asks at each step how much the price moves per unit change in that step's result, a number called the step's **adjoint**; one backward pass then delivers the derivative with respect to every input at once. AAD software records the steps during the pricing run and replays them in reverse, so all the sensitivities together cost a fixed multiple of one price, typically three to five times, however many inputs there are; Section 32.13 shows what that does to the cost of a whole portfolio's sensitivities. The cost is memory, since every intermediate value must be kept for the backward pass, and the limit is pathwise differentiation's: a digital has no derivative to pass back. Example 20.8 runs it on one path.

Example 20.8 One path, run backward

Take the call above ($S_0 = K = 100$, $r = 10\%$, $\sigma = 20\%$, $T = 1$) and a single draw $\varepsilon = 0.5$. The forward pass computes and stores

$$x = \left(r - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}\,\varepsilon = 0.18, \qquad S_T = S_0 e^{x} = 119.72, \qquad V = e^{-rT}(S_T - K)^+ = 17.84$$

The backward pass, writing $\bar{y}$ for the adjoint of a quantity y, starts from $\bar{V} = 1$ and multiplies by one local derivative per step,

$$\bar{S}_T = e^{-rT}\,\mathbf{1}\{S_T > K\} = 0.9048, \qquad \bar{x} = \bar{S}_T\, S_T = 108.33$$

and each input reads its sensitivity off the step where it entered:

$$\frac{\partial V}{\partial S_0} = \bar{S}_T\,\frac{S_T}{S_0} = 1.083, \quad \frac{\partial V}{\partial \sigma} = \bar{x}\left(\sqrt{T}\,\varepsilon - \sigma T\right) = 32.50, \quad \frac{\partial V}{\partial r} = \bar{x}\,T - T\,V = 90.48, \quad \frac{\partial V}{\partial K} = -0.905$$

One backward pass has produced four sensitivities, each the pathwise estimate on this path; averaged over the paths they are delta, vega, rho and the sensitivity to the strike. Bump-and-revalue would have needed four more simulations.

20.11 AMERICAN MONTE CARLO

American Monte Carlo is the one genuinely difficult technique in the chapter. The problem, as Section 20.1 noted, is structural: a simulated path standing at an intermediate date does not know its own future, and an early-exercise decision requires exactly that.

The Longstaff–Schwartz method resolves it by *estimating* the continuation value with a

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regression across paths.

Carry two numbers per path. Let C_i be the cash flow path i receives under the rule built so far, and τ_i the date it is received; initialize C_i to the payoff at expiration and τ_i to T. Now work backward through the exercise dates. At each date t_k:

1. Select the paths that are in the money at t_k. The decision is trivial elsewhere, and confining the fit to the region where it is close is what keeps the regression useful.

2. For each selected path form $Y_i = C_i e^{-r(\tau_i - t_k)}$: the value at t_k of the cash flow that path is currently scheduled to receive. Regress Y on a few functions of the current price, the regression's **basis functions** (1, S, S^2 will do), across those paths only.

3. Take the fitted value at each path's own S_{t_k} as its continuation value, and exercise where the immediate payoff exceeds it. On those paths overwrite C_i with the immediate payoff and τ_i with t_k, discarding the later cash flow: a path can be exercised once.

4. Step back to the previous date.

There is no regression at time zero. The estimate is the average of the discounted cash flows, $\frac{1}{M}\sum_i C_i e^{-r\tau_i}$, compared with the immediate payoff if the option may be exercised at once.

The regression is doing something specific and limited: it is estimating a conditional expectation from the cross-section of paths, using the fact that many paths pass near the same state. Example 20.9 runs the algorithm by hand.

Example 20.9 Longstaff–Schwartz on ten paths

Take the American put of Example 20.1 ($S_0 = K = 50$, $r = 5\%$, $\sigma = 30\%$, six months), with three exercise dates a sixth of a year apart and ten paths. Ten is far too few to price anything; it is enough to run the algorithm by hand.

Path	$t = 1/6$	$t = 2/6$	$t = 3/6$
1	58.83	55.93	62.73
2	47.62	56.43	57.46
3	43.85	54.54	46.88
4	42.21	43.86	45.49
5	51.71	47.49	48.36
6	48.92	47.20	46.39
7	64.47	71.06	70.94
8	40.90	41.89	49.10
9	55.01	49.11	61.99
10	42.38	39.52	38.54

At expiration $C_i = \max(50 - S_T, 0)$, which is nonzero on paths 3–6, 8 and 10.

At $t = 2/6$ six paths are in the money. Each carries $Y_i = C_i e^{-0.05/6}$, and the fitted regression is $\hat{Y} = 306.71 - 12.925\,S + 0.13674\,S^2$:

Path	S	Y	Continue	Exercise	Decision
4	43.86	4.4726	2.8718	6.1400	exercise
5	47.49	1.6264	1.2975	2.5100	exercise
6	47.20	3.5800	1.2909	2.8000	exercise
8	41.89	0.8925	5.2346	8.1100	exercise
9	49.11	0.0000	1.7579	0.8900	continue
10	39.52	11.3649	9.4837	10.4800	exercise

Path 9 is the instructive row: its own future cash flow is zero, so $Y_9 = 0$, yet the regression

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assigns it a continuation value of 1.76 from what happened to the *other* paths near $S = 49$, and since that exceeds the 0.89 on offer, the rule continues. That is the idea: the continuation value is estimated from the cross-section, not read off the path's own future.

At $t = 1/6$ the cash flows just written are discounted back one more step, and the fit is $\hat{Y} = 375.53 - 15.604\,S + 0.16274\,S^2$:

Path	S	Y	Continue	Exercise	Decision
2	47.62	0.0000	1.5224	2.3800	exercise
3	43.85	3.0684	4.2281	6.1500	exercise
4	42.21	6.0890	6.8491	7.7900	exercise
6	48.92	2.7768	1.6620	1.0800	continue
8	40.90	8.0427	9.5716	9.1000	continue
10	42.38	10.3930	6.5367	7.6200	exercise

Path 10 shows what the method is and is not. It was scheduled to collect 10.48 at $t = 2/6$, worth 10.39 here, but the regression puts its continuation value at 6.54 and the immediate payoff is 7.62, so the rule exercises and the path collects 7.62. The rule is estimated, not clairvoyant, and on this path it gives up value that waiting would have earned.

Discounting each path's surviving cash flow from its own exercise date and averaging over all ten gives 3.6940. The converged value is 3.6970, and the agreement is luck: the European option priced on the same ten paths comes out at 2.46 against its true 3.5829, which is the honest measure of what ten paths are worth.

Three details decide whether an implementation works.

The regression needs far more paths than basis functions. With three basis functions and only three in-the-money paths the fit is exact, each fitted continuation value equals that path's own realized cash flow, and the exercise rule silently acquires perfect foresight, reporting a confident answer that is far too high. Keep the in-the-money count comfortably above the number of basis functions, or use fewer of them.

Scale the basis. At $S \approx 50$ the three columns hold values near 1, 50 and 2,500, and least squares on columns differing that much in scale loses precision. Replacing S by S/K throughout the basis makes the columns 1, S/K and $(S/K)^2$, all of order one near the strike, which removes the problem.

Two biases run in opposite directions. Fitting the regression and valuing the option on the same paths leaks a little foresight, which pushes the answer up. The suboptimal exercise rule pushes it down, and by more. Example 20.10 shows the net at path counts that matter. The low estimate also has a partner: Rogers (2002) and Haugh and Kogan (2004) showed independently that any exercise rule can also be turned into an *upper* bound on the value (the dual method), and Andersen and Broadie (2004) made that bound computable, so a least-squares price can be bracketed from both sides and the width of the bracket measures how good the rule is.

Example 20.10 Longstaff–Schwartz against the tree

The target is the American put of Example 20.1, worth 3.69702. With 25 or 50 exercise dates the method prices a **Bermudan** option, one exercisable only on a fixed set of dates, so that is its benchmark: on a tree that may exercise only on the same dates it is worth 3.6907 and 3.6938, below the American by 0.0063 and 0.0032. Using antithetic paths, with each pair counted once in the standard error, and a quadratic regression:

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Paths	Dates	Estimate	Std. error	Bermudan	Error
20,000	25	3.69681	0.0175	3.6907	+0.0061
100,000	25	3.66563	0.0079	3.6907	−0.0251
100,000	50	3.68495	0.0077	3.6938	−0.0089
400,000	50	3.68705	0.0038	3.6938	−0.0068

In one run the rows scatter about their Bermudans by a standard error or more, and the first lands above; what is structural is the direction of the error on average, not its size in any row, because the regression produces an imperfect exercise rule and any rule other than the optimal one earns less. Adding paths shrinks the standard error but not the bias; a better regression basis reduces the bias, and more exercise dates move the target closer to the American.

A 200-step binomial tree gave this value to within 0.003 in a fraction of the time, so for a one-dimensional American option Longstaff–Schwartz is strictly the wrong choice. Its value is that it is the only method that works when the tree cannot be built: a callable note on a basket of five assets, a Bermudan swaption in a multi-factor model, or the exposure profiles of Chapter 9, where the early-exercise decision sits inside a simulation that must run anyway.

20.12 FINITE DIFFERENCE METHODS †

This section develops the third route to a price: solving the Black–Scholes–Merton partial differential equation numerically.

Write $V(S,t)$ for the price of the derivative being valued. Chapter 15 called it f; V is the usual symbol in this literature and is used throughout the section. Section 15.9 derived the form that carries a dividend yield,

$$\frac{\partial V}{\partial t} + (r-q)S\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} = rV$$

Lay a grid over (S,t), replace each derivative by a difference quotient, and solve backward from the known terminal condition.

Three edges of the grid are filled in before the calculation can work backward from expiration, and they are what the scheme is solved from. The terminal column comes from the payoff. The two price boundaries come from what the option is worth when the underlying is worthless or very large: for an American put, $V(0,t) = K$, since it would be exercised at once, and $V(S_{\max},t) = 0$. A European put takes the discounted strike, $V(0,t) = Ke^{-r(T-t)}$, since it cannot be exercised before expiration. $S_{\max}$ has to sit far enough out that the option really is worthless there, and the scale to measure it against is $\sigma\sqrt{T}$, the standard deviation of $\ln S_T$: on the contract used below, an $S_{\max}$ three of these above $\ln S_0$ already makes the error from cutting the grid off less than 10^{-6}, and the examples' 200 (four times the strike) is more than six above it. Reaching far enough out is easy; the trap is that $S_{\max}$ also fixes $\Delta S = S_{\max}/M$, where M is now the number of price steps in the grid (not the path count of Section 20.6), and unless $S_0/\Delta S$ is a whole number the grid has no node at today's price at all. With $S_{\max} = 150$ and $M = 400$ the nearest node sits 0.125 below S_0, and the put comes back 0.053 too high, which is that 0.125 times the delta of −0.43 that Example 20.13 reports, and nothing to do with the scheme. The *coarser* $\Delta S = 0.5$, which does land on S_0, is out by 0.002. Choose M together with $S_{\max}$ so the grid has a node at S_0, or interpolate between the two that straddle it. Figure 20.5 shows the arrangement.

In the log variable $x = \ln S$ the coefficients stop depending on the state,

$$\frac{\partial V}{\partial t} + \nu\frac{\partial V}{\partial x} + \frac{1}{2}\sigma^2\frac{\partial^2 V}{\partial x^2} = rV, \qquad \nu = r - q - \frac{1}{2}\sigma^2$$

and that ν is the drift of (20.2). The examples below set $q = 0$, matching the trees they are

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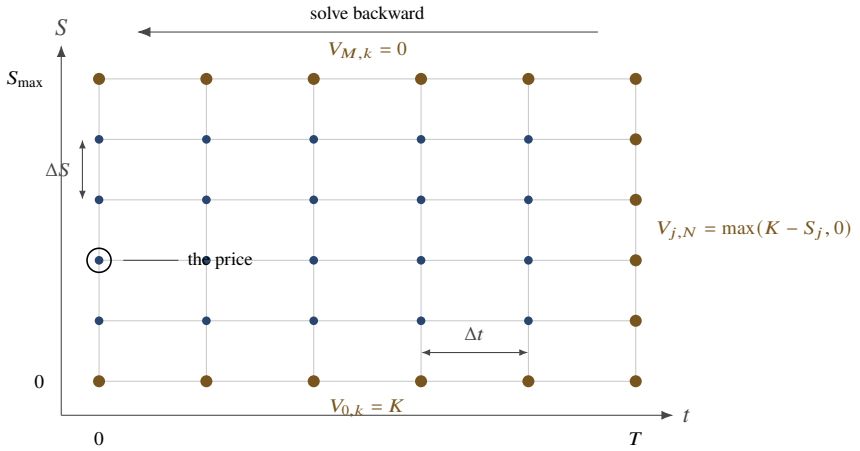


Figure 20.5: The grid for an American put. Larger (amber) nodes are boundary values; the rest are computed backward to $t = 0$.

measured against.

Three schemes differ in *when* the derivatives in S are evaluated. Write $V_{j,k}$ for the option's value at the grid point $S_j = j\,\Delta S$ and the date $t_k = k\,\Delta t$ (the first index counts price steps, the second counts dates), and $\mathbf{V}_k$ for the whole column of values at one date. The calculation runs backward, from the terminal column $\mathbf{V}_N$ to $\mathbf{V}_0$. Write $\mathcal{L}$ for the operator the equation puts against the time derivative,

$$\mathcal{L}V = rV - (r-q)S\frac{\partial V}{\partial S} - \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}, \qquad \frac{\partial V}{\partial t} = \mathcal{L}V$$

Each derivative is then replaced by a difference over neighboring grid points:

$$\frac{\partial V}{\partial S} \approx \frac{V_{j+1,k} - V_{j-1,k}}{2\,\Delta S}, \qquad \frac{\partial^2 V}{\partial S^2} \approx \frac{V_{j+1,k} - 2V_{j,k} + V_{j-1,k}}{\Delta S^2}, \qquad \frac{\partial V}{\partial t} \approx \frac{V_{j,k+1} - V_{j,k}}{\Delta t}$$

The two space differences are centered on node j and carry an error of order ΔS^2; the time difference is one-sided and carries an error of order Δt. Substituting them, and using $S_j = j\,\Delta S$, makes every ΔS cancel, leaving three coefficients that depend on the node index alone:

$$(\mathcal{L}\mathbf{V}_k)_j = A_j V_{j-1,k} + B_j V_{j,k} + C_j V_{j+1,k}$$

$$A_j = \frac{1}{2}\left[j(r-q) - \sigma^2 j^2\right], \qquad B_j = r + \sigma^2 j^2, \qquad C_j = -\frac{1}{2}\left[j(r-q) + \sigma^2 j^2\right]$$

That the grid spacing has vanished is why the stability condition below is a statement about how many nodes there are rather than about how wide they are. With θ a weight between 0 and 1, the share of the operator taken at the earlier date, the scheme is

$$\frac{\mathbf{V}_{k+1} - \mathbf{V}_k}{\Delta t} = \theta\,\mathcal{L}\mathbf{V}_k + (1-\theta)\,\mathcal{L}\mathbf{V}_{k+1}$$

and gathering the unknown column on the left turns it into one linear system per step:

$$(\mathbf{I} + \theta\,\Delta t\,\mathcal{L})\,\mathbf{V}_k = (\mathbf{I} - (1-\theta)\,\Delta t\,\mathcal{L})\,\mathbf{V}_{k+1}$$

That equation is the method. Its right-hand side is known, its left-hand side is not, and each step delivers the column one date earlier. The three schemes are three readings of it, and Figure 20.6 draws what each costs.

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Explicit, $\theta = 0$. The left side is $\mathbf{V}_k$ by itself, so nothing is solved and every interior node is read off directly:

$$V_{j,k} = a_j V_{j-1,k+1} + b_j V_{j,k+1} + c_j V_{j+1,k+1}, \qquad j = 1, \ldots, M-1$$

with $a_j = -\Delta t A_j$, $b_j = 1 - \Delta t B_j$ and $c_j = -\Delta t C_j$. Those three weights are the arrows in the figure's left panel, and on the log grid they approximate the probabilities of (20.2).

Implicit, $\theta = 1$. Now $(\mathbf{I} + \Delta t\, \mathcal{L})\mathbf{V}_k = \mathbf{V}_{k+1}$, which one row at a time reads

$$\Delta t A_j V_{j-1,k} + \left(1 + \Delta t B_j\right) V_{j,k} + \Delta t C_j V_{j+1,k} = V_{j,k+1}$$

for $j = 1, \ldots, M-1$. The first and last rows reach past that range, to $V_{0,k}$ and $V_{M,k}$, which are the boundary values of Figure 20.5 and therefore known, so they move across:

$$\begin{pmatrix} 1+\Delta t B_1 & \Delta t C_1 & & \\ \Delta t A_2 & 1+\Delta t B_2 & \ddots & \\ & \ddots & \ddots & \Delta t C_{M-2} \\ & & \Delta t A_{M-1} & 1+\Delta t B_{M-1} \end{pmatrix} \begin{pmatrix} V_{1,k} \\ V_{2,k} \\ \vdots \\ V_{M-1,k} \end{pmatrix} = \begin{pmatrix} V_{1,k+1} - \Delta t A_1 V_{0,k} \\ V_{2,k+1} \\ \vdots \\ V_{M-1,k+1} - \Delta t C_{M-1} V_{M,k} \end{pmatrix}$$

The matrix is tridiagonal, so it is never inverted. Write the row for $V_{j,k}$ as $\alpha_j V_{j-1,k} + \beta_j V_{j,k} + \gamma_j V_{j+1,k} = d_j$, with $\alpha_j = \Delta t A_j$, $\beta_j = 1 + \Delta t B_j$, $\gamma_j = \Delta t C_j$, and d_j the right-hand side above. The **Thomas algorithm** is Gaussian elimination that uses the three bands and touches nothing else: a forward pass clears the subdiagonal and a back substitution reads the column off. Starting from $\gamma'_0 = d'_0 = 0$, the forward pass runs for $j = 1, \ldots, M-1$:

$$\gamma'_j = \frac{\gamma_j}{\beta_j - \alpha_j \gamma'_{j-1}}, \qquad d'_j = \frac{d_j - \alpha_j d'_{j-1}}{\beta_j - \alpha_j \gamma'_{j-1}}$$

and the back substitution, starting from $V_{M-1,k} = d'_{M-1}$, runs back down for $j = M-2, \ldots, 1$:

$$V_{j,k} = d'_j - \gamma'_j V_{j+1,k}$$

Each pass is a single loop over M points, so a column costs $O(M)$ operations, the same order as an explicit step. And since the matrix is the same at every step, the γ'_j depend on nothing that changes: compute them once before the time loop begins, and each column costs only the d'_j recursion and the back substitution.

Crank–Nicolson, $\theta = 1/2$. The same system with half the operator on each side,

$$\left(\mathbf{I} + \frac{\Delta t}{2}\,\mathcal{L}\right)\mathbf{V}_k = \left(\mathbf{I} - \frac{\Delta t}{2}\,\mathcal{L}\right)\mathbf{V}_{k+1}$$

so a step costs the implicit scheme's solve plus one tridiagonal multiplication to form the right-hand side.

And, for an American option, one more line. Once a column has been computed, the early-exercise condition is imposed across it:

$$V_{j,k} \leftarrow \max\left(V_{j,k},\ K - S_j\right)$$

This is the free boundary Chapter 15 deferred, handled by projection. The boundary is never located; it is enforced wherever it binds, and where it does the node holds the payoff instead of the solved value. For the explicit scheme the projection is exact; after an implicit or Crank–Nicolson solve it carries an error of order Δt, which costs Crank–Nicolson its second

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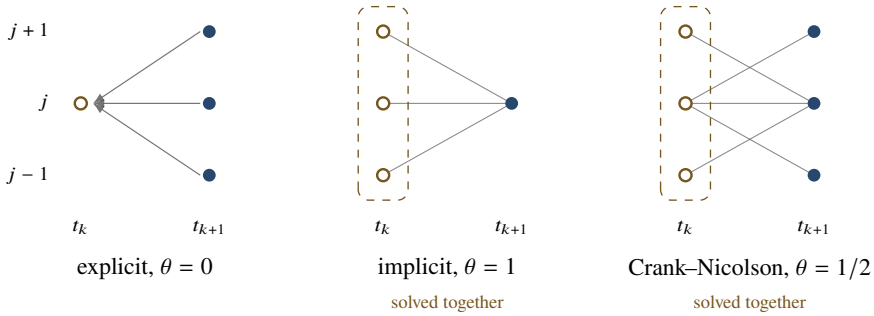


Figure 20.6: The nodes entering one equation under each weight. Filled nodes are known, hollow ones are being solved for, and t_k is the earlier date.

order. Imposing the condition inside the solve restores it, by Brennan and Schwartz's ordering of the elimination for a put or by projected iteration in general.

The weight is what sets the order in time. At $\theta = 1/2$ the right-hand side is the average of the operator at the two dates, which is its value at the midpoint between them to within $O(\Delta t^2)$, so the one-sided time difference above becomes a centered one about that midpoint. That is why Crank–Nicolson is second-order in time for a smooth problem where the other two are first-order, and Example 20.12 measures what survives of it on an American put.

The explicit scheme is a trinomial tree. At $\theta = 0$ each new value is an explicit weighted average of three neighbors, and on the log grid those weights are (20.2) to first order in Δt. The tree carries a $\nu^2 \Delta t^2$ term the scheme drops, and discounts by $e^{-r\Delta t}$ where the scheme multiplies by $1 - r\Delta t$. Both differences fall by a factor of ten for every factor of ten in Δt, so Section 20.4's remark holds in the limit: the two methods are the same computation described in two vocabularies.

And it is only conditionally stable. The weights are probabilities only if Δt is small relative to $1/(\sigma^2 M^2)$, where M is the number of space steps. Violate that and the scheme does not merely lose accuracy: it explodes.

Example 20.11 What instability looks like

Solve the American put of Example 20.1 explicitly on a grid running to $S_{\max} = 200$:

Space steps	Time steps	Value
100	1,000	3.6766
200	1,000	beyond 10^{40}
200	4,000	3.6918

Refining the space grid while leaving the time grid alone destroyed the calculation. The middle row is not slightly wrong; it is dozens of orders of magnitude wrong, because the scheme amplifies rounding error geometrically once the stability condition fails, and its exact size depends on nothing more than the order of the arithmetic.

Doubling the space resolution therefore requires four times as many time steps. That is the practical objection to the explicit scheme, and it is why the implicit and Crank–Nicolson schemes exist: both are unconditionally stable, at the cost of solving a tridiagonal system at each time step, which the Thomas algorithm does in $O(M)$ operations.

There is one consolation. An unstable explicit scheme fails loudly, and a number beyond 10^{40} cannot be mistaken for a price. The failures worth fearing are the quiet ones, and Crank–Nicolson

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has one: it damps error that flips sign from one node to the next only slowly, so the payoff's kink leaves oscillations near the strike that persist for many steps and show in gamma before the price. Rannacher's repair is to take the first few steps fully implicit.

Figure 20.7 draws the boundary that the three rows sit on either side of. It is a parabola, so the time steps a finer space grid needs rise with the square of its resolution: the middle row was not unlucky, only on the wrong side of the curve.

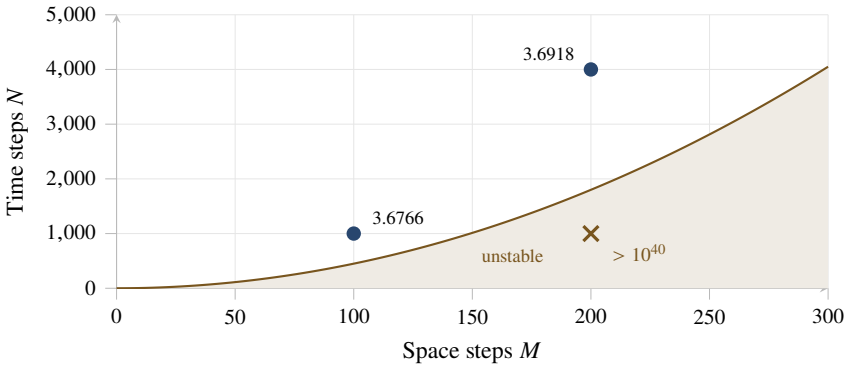


Figure 20.7: Stability of the explicit finite-difference scheme on the American put of Example 20.1. Below the curve $N = T\sigma^2 M^2$ the scheme diverges; the marks are the three grids of Example 20.11.

Example 20.12 Implicit against Crank–Nicolson

The same American put, benchmark 3.69702, gives:

Space	Time	Implicit	Error	Crank–Nicolson	Error
100	50	3.65776	−0.03926	3.67363	−0.02339
200	100	3.68196	−0.01506	3.69033	−0.00669
400	200	3.69056	−0.00646	3.69494	−0.00208
800	400	3.69403	−0.00299	3.69631	−0.00071

The implicit errors fall by roughly a factor of two per refinement, which is first order. The Crank–Nicolson errors fall by about three and a half, then three, then less: the symmetric time-averaging delivers second order on the smooth part of the problem, and the projection's first-order error shows through as the grid is refined. For nearly the same work Crank–Nicolson is more accurate at every grid size, and its lead grows with refinement, from under twice as accurate on the coarsest grid to four times on the finest, which is why it is the default.

Both converge from below, as the European put does on the same grids: the sign is the discretization's, not the exercise constraint's.

Why a desk uses finite differences at all. The method produces the value at every grid point, so the quantities a desk needs are already differences of numbers it holds. Writing j_0 for the node nearest the spot,

$$\Delta \approx \frac{V_{j_0+1,0} - V_{j_0-1,0}}{2\,\Delta S}, \qquad \Gamma \approx \frac{V_{j_0+1,0} - 2V_{j_0,0} + V_{j_0-1,0}}{\Delta S^2}, \qquad \Theta \approx \frac{V_{j_0,1} - V_{j_0,0}}{\Delta t}$$

which are the same three quotients the scheme is built from. A tree gives them as cheaply, from the nodes at steps one and two. For a book that must be revalued, and its risk recomputed, across a range of spot prices every evening, that is decisive.

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Vega and rho do not come free: volatility and the rate are parameters of the equation rather than state variables, so each costs a second solve at a bumped input, which is why a desk needing a full set of Greeks reruns twice rather than five times.

Example 20.13 Greeks from the tree and from the grid

Ask each method for the Greeks of the American put of Example 20.1 as well as its price, at no extra cost in either case:

	Value	Δ	Γ	Θ
Tree, 6,000 steps	3.6969	−0.4300	0.03953	−3.187
Crank–Nicolson, 800×400	3.6963	−0.4299	0.03953	−3.190

Two methods sharing no arithmetic agree to three significant figures on all four numbers: Section 20.13's second habit applied to the Greeks, and the cheapest check available on an implementation.

The European option's closed-form delta and gamma are −0.4114 and 0.03668, against the American values above. The early-exercise feature shows more strongly in the Greeks than in the price: the value differs by 3%, the delta by 5%, and the gamma by 8%. A desk hedging an American book off European Greeks is wrong by more than the pricing error would suggest.

With a second solve, raising σ by one volatility point moves the price by 0.1373, and raising r by a basis point moves it by −0.0009.

20.13 CHOOSING A METHOD

The chapter reduces to a short list of decisions.

One state variable and early exercise → tree or finite difference. Use a control variate against the European, then extrapolate between step counts that share an offset. A few hundred steps suffice.

Path dependence → Monte Carlo, with a control variate. Look hard for a closely related closed form before accepting plain simulation; Example 20.6 is what that search is worth.

More than three state variables → Monte Carlo, necessarily. Grid methods are out of reach.

Both early exercise and high dimension → Longstaff–Schwartz, accepting the downward bias and reporting it.

Greeks across a grid of spot prices → finite difference, which delivers them as a by-product.

Desks divide along the same lines. Market makers in listed American equity options price on trees, fast enough to update quotes across every strike and expiration continuously; rates desks run one-factor callable and Bermudan books on trees or grids (Chapter 31) and simulate with regression only when the model has several factors (Chapter 32); XVA desks simulate a netting set's many state variables and take their sensitivities by AAD. Stochastic-volatility models are calibrated by transform pricing, and a market-condition award is valued for the accounts by simulation, as in Example 20.7.

Two habits matter more than the choice.

Always price something you already know. Run the European option through the same machinery and compare with the Black–Scholes–Merton formula. It costs almost nothing, it catches most implementation errors, and by (20.1) the discrepancy is not merely a diagnostic but a correction.

Always compute the answer twice. Two step counts, two path counts, two grid resolutions. A number computed once is not a number; it is a hypothesis. Between them the two habits catch most failures in this chapter: the instability of Example 20.11 falls to the second immediately, while the bias of Example 20.10, which a second path count cannot reveal, shows only against an independent answer, the Bermudan tree there or the dual upper bound where no tree exists.

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20.14 WHERE THIS CAME FROM

Boyle (1977) introduced Monte Carlo to option pricing, but early exercise resisted simulation for two decades. Methods for it appeared through the 1990s, among them Broadie and Glasserman's (1997), and it was Longstaff and Schwartz's (2001) regression of the continuation value on the paths themselves that made American Monte Carlo routine, and that lets the exposure simulations of Chapter 9 price Bermudan options at all. The finite-difference route is older: Brennan and Schwartz (1977) priced the American put that way, and in 1978 showed the explicit scheme to be a lattice approximation of the diffusion, the identity Section 20.12 uses.

On lattices the pathologies came before the fixes. Boyle and Lau (1994) explained why a barrier priced on a tree converges badly and how node placement repairs it; Broadie, Glasserman and Kou (1997) gave the continuity correction for discretely monitored barriers that Chapter 34 checks against simulation. Barone-Adesi and Whaley (1987) supplied a fast analytic approximation to the American premium, which lattices and later methods now match in accuracy at little cost.

SUMMARY

A closed form requires a simple payoff, a simple process, and no early exercise; break one and a number must be computed. The pricing equation and the risk-neutral expectation are the same statement, so a price may be obtained backward by solving the equation or forward by simulating the expectation, and the two directions have opposite strengths: backward methods handle early exercise but fail in high dimension, forward methods handle path dependence and dimension but not early exercise.

Binomial trees converge at $O(1/n)$, with an error set by where the strike falls between two terminal nodes; at the money that is the parity of n. The control variate exploits it, correcting the target by the method's own error on a companion whose answer is known and cutting the American put's error by orders of magnitude; it removes what the two problems share and nothing else, so the companion should sit as close to the target as possible.

Trinomial trees do not converge faster. What they buy is an independent choice of Δx, which lets a node be placed on a barrier, and the explicit finite difference scheme is a trinomial tree in another notation. A known cash dividend destroys recombination (a proportional one, discrete or continuous, does not), and the remedy is to build the tree on the price net of the dividend's present value and add it back at each node.

Monte Carlo is unbiased with a standard error falling like $1/\sqrt{M}$, so a decimal place costs a hundredfold increase in paths, and structure beats brute force: a geometric-average control variate for an arithmetic Asian option removes so much variance that matching it with plain paths would take an impractical number of them. Lognormal paths are simulated exactly, and an Euler step, which can go negative, is a fallback for processes without a solution. Greeks from a simulation need common random numbers if bumped; the pathwise estimator needs a differentiable payoff and the likelihood-ratio estimator does not, and adjoint differentiation returns all sensitivities at once for the cost of a few prices. Correlated assets go through a Cholesky factor, and the reason to bother shows in a relative total shareholder return award, where multiplying expected units by expected price understates the value badly, because units are earned on precisely the paths where the share price is high. Longstaff–Schwartz regresses the continuation value on the cross-section of in-the-money paths and is biased downward, since a suboptimal exercise rule earns less; a dual calculation supplies the matching upper bound, and the method is the right choice only where a tree cannot be built.

Finite difference schemes differ in when the derivatives in S are evaluated. The explicit scheme is only conditionally stable, while implicit and Crank–Nicolson are unconditionally stable, and Crank–Nicolson is second-order in time because its weight centers the time

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difference, an order that early exercise imposed after the solve gives back. A grid returns delta, gamma and theta as differences of values it has already computed; vega and rho need a second solve. Two habits govern everything: price something whose answer is already known, and compute every answer twice.

FURTHER READING

Glasserman's *Monte Carlo Methods in Financial Engineering* is the standard reference, and the one that explains why variance reduction works rather than only how to implement it. Longstaff and Schwartz (2001) is the original least-squares Monte Carlo paper and is short; the subsequent literature on the method's bias and on choosing regression bases is large and worth sampling before implementing it. Wilmott, Howison and Dewynne treat the partial differential equation route with the numerical analysis intact, which is the place to go for stability conditions derived rather than quoted. Capriotti (2011) is the practitioner account of adjoint differentiation for Greeks. On trees, Broadie and Detemple (1996) smooth the terminal payoff by putting the Black–Scholes–Merton value at the last step, and Boyle and Lau (1994) show how node placement interacts with a barrier; between them they explain a family of convergence pathologies that Chapter 34 meets again. For low-discrepancy sequences, the practical literature diverges sharply from the theoretical bounds, and any implementation should be tested against a known answer at the dimensions actually used.

PRACTICE QUESTIONS

20.1 A binomial tree gives 3.5724 at 100 steps and 3.5777 at 200. Estimate the exact value by Richardson extrapolation, state what you must check before trusting it, and explain why the binomial error alternates in sign with the parity of n when $S_0 = K$.

20.2 A tree gives an American put value of 3.6914 and a European put value of 3.5724 at the same step count, and the Black–Scholes–Merton formula gives 3.5829 for the European. Compute the control-variate corrected American value, and explain why the correction helps enormously at odd step counts and only slightly at even ones.

20.3 State two things a trinomial tree buys over a binomial tree and one thing it does not, and name the finite difference scheme that is a trinomial tree in another notation.

20.4 Explain why a known cash dividend destroys the recombination of a binomial tree, and describe the standard remedy.

20.5 A Monte Carlo estimate has a standard error of 0.04 at 10,000 paths. How many paths are needed for a standard error of 0.005?

20.6 Explain why the exact lognormal simulation step is preferable to an Euler step, giving both reasons.

20.7 A control variate has correlation 0.95 with the target. By what factor is the variance reduced? Compare with the 0.99914 achieved by the geometric Asian control variate and explain why the improvement is so much larger than the correlations differ.

20.8 Explain why bumping the spot price and revaluing gives a poor delta if the two simulations use independent random numbers.

20.9 State the direction of the Longstaff–Schwartz bias, explain why it goes that way, and state when the method is nonetheless the right choice.

20.10 Explain why the explicit finite difference scheme becomes unstable when the space grid is refined, and state how the time grid must respond.

20.11 Explain why a relative total shareholder return award cannot be valued as expected units

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times expected share price, and state the direction and rough size of the error.

FURTHER QUESTIONS

20.12 Richardson extrapolation can return an answer worse than either of the values it is built from. Derive the condition on the error sequence under which extrapolation improves the estimate, explain how the offset of Section 20.3 decides whether two step counts satisfy it, and propose a diagnostic a pricing system could run automatically to detect a violation before reporting a price.

20.13 Example 20.1 argues that a control variate removes the error the two problems share and nothing else. Formalize this: write the tree's American and European errors as a shared component plus an idiosyncratic one, show what (20.1) does to each, and use the result to predict how the correction's effectiveness changes as the option moves deep into or out of the money.

20.14 The Asian control variate of Example 20.6 achieves a variance reduction of 583 through a correlation of 0.99914. Explain why the geometric and arithmetic averages are so highly correlated, determine how the correlation and hence the reduction would change if the number of fixings rose from 12 to 250, and discuss whether the closed form for the geometric case remains available.

20.15 Example 20.7 finds that the naive product of expected units and expected share price understates the award's value by 30%, so that the correct value is 44% higher. Decompose the correct value into the naive product plus a covariance term, explain the economic content of that term, and state what it implies about a company that grants relative total shareholder return awards and simultaneously reports their value using an expected-payout methodology.

20.16 Section 20.11 states that Longstaff–Schwartz is biased low and that adding paths does not remove the bias. Explain why the bias is not a small-sample effect, describe how one could construct an *upper*-biased estimate to bracket the true value, and discuss what a risk manager should report when only the lower estimate is available.

20.17 Example 20.11 shows an explicit scheme returning a number beyond 10^{40}. Derive the stability condition for the scheme, show that it is equivalent to requiring the three trinomial weights of (20.2) to be nonnegative, and explain why this equivalence means the instability is a statement about probabilities rather than about arithmetic.

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Part IV

Volatility as an Asset Class

Volatility stopped being a parameter and became something traded in its own right. This part covers the models that reproduce a surface, the instruments that give direct exposure to variance, and what follows in market structure when the hedging those instruments require is large enough to move the market. Chapter 21 carries the dagger; Chapters 22 and 23 do not, and describe episodes (the February 2018 unwind, the rise of same-day options) that a reader of the financial press has already met.

21 Volatility Modeling: Local, Stochastic, Jump, and Rough
the model families, and which failure each one accepts

22 Volatility Products: VIX, Variance Swaps, and Dispersion
trading variance directly

23 Short-Dated Options, Dealer Positioning, and Market Feedback
when hedging flows move the underlying

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Chapter 21

Volatility Modeling: Local, Stochastic, Jump, and Rough †

Section 19.4 named three families of model that resolve the contradiction between a formula with one volatility and a market that quotes many, and said what each one buys and what it costs. That is enough to tell them apart; it is not enough to choose one and defend the choice. This chapter develops how each is built, how it is calibrated, and what its particular failure costs, and adds a fourth, rough volatility, which is the most recent development and only became usable after about 2015. The mathematics is deferred to [TN21]; what stays here is what each model captures, where it fails, and how one would tell.

The organizing question is not "which model is right." No model here is right. The question is which model is wrong in a way that does not matter for the trade in front of you, and answering it requires knowing precisely how each one fails.

The order is logical rather than historical (Section 21.13 gives the history). Local volatility comes first, in Section 21.2, because an exact fit to today's surface forces it on you, and Section 21.3 shows what the fit costs in dynamics; Heston's stochastic volatility, Merton's jumps and their combination in Bates follow. Section 21.7 measures all three against one fact, the $T^{-0.4}$ power law the index skew follows, and none of them reproduces it. That fact produced the fourth family; the later sections turn to the hybrid models desks run, to calibration, and to what the disagreement between models costs.

One convention holds for every worked example in the chapter: $S_0 = 100$ and $r = q = 0$. The forward then equals the spot, "at the money" means a strike of 100 at every maturity, and the numbers measure the models rather than the carry. The equations keep r and q, because the models are used with them; the examples drop them, because they would only obscure the comparison.

21.1 WHAT A VOLATILITY MODEL MUST DO

Two requirements compete, and almost every model satisfies one at the expense of the other.

Fit today's surface. A model that does not reproduce the quoted vanilla prices is arbitrageable against the market by anyone who trades the vanillas. This is a hard constraint, not a preference.

Get the dynamics right. The model must also say how the surface will move, because the value of anything path-dependent depends on that. Section 19.11 showed the point in miniature: the delta of a plain call already depends on whether the smile is sticky in strike or in moneyness, and that is a question about dynamics, not about today's prices.

Nothing in the vanilla market pins down the dynamics. Today's surface is a snapshot of the risk-neutral distribution at each maturity; infinitely many processes produce the same set of terminal distributions and behave completely differently in between. That degree of freedom is the subject of this chapter, and Section 21.11 measures what it is worth in money.

Table 21.1 states the trade each of the four families makes, and it is worth reading before

the sections that develop them: no row buys both requirements, which is why the chapter ends in a choice rather than in a winner.

Table 21.1: What each family buys and what it costs.

Family	Buys	Costs
Local volatility	Fits any arbitrage-free surface exactly; stays complete	Dynamics wrong: the forward smile (quoted on a future date) flattens; the market's does not
Stochastic volatility	Realistic dynamics, stationary forward smile	Cannot fit the short end; market incomplete
Jumps	Short-dated skew a diffusion cannot produce	Skew dies too fast with maturity; unhedgeable by construction
Rough volatility	The observed skew term structure from one parameter	Not Markovian; simulation is expensive

21.2 LOCAL VOLATILITY

Let volatility be a deterministic function of price and time:

$$dS = (r - q)\,S\,dt + \sigma_{\text{loc}}(S, t)\,S\,dW \tag{21.1}$$

Dupire's result is that for any arbitrage-free surface of call prices $C(K, T)$ there is exactly one function σ_{loc} in (21.1) reproducing every one of them, and it can be read off directly, evaluated at $S = K$, $t = T$:

$$\sigma_{\text{loc}}^2(K, T) = \frac{\dfrac{\partial C}{\partial T} + (r - q)K\dfrac{\partial C}{\partial K} + qC}{\frac{1}{2}K^2\dfrac{\partial^2 C}{\partial K^2}} \tag{21.2}$$

Three things about (21.2) are easy to read past.

The denominator is the implied density. It is $\frac{1}{2}K^2e^{-rT}$ times the Breeden–Litzenberger density of (19.1). So local volatility is a ratio of two things the surface already contains, and it exists precisely when the density is positive and the numerator is not negative, which are Section 19.5's butterfly and calendar conditions again, now as an existence condition for a model.

The model is complete. There is one Brownian motion and one traded asset, so Section 13.4's replication argument survives intact and every option is hedgeable with the underlying. That is a real advantage: a local-volatility book has a delta that means something.

The fit is exact by construction, not by calibration. There are no parameters to search over. This is why local volatility remains the baseline for any payoff fixed by today's surface alone, such as an unquoted European.

The simplest local volatility is a single power of the price, the constant elasticity of variance model of Cox (1975), $\sigma_{\text{loc}}(S) = \sigma S^{\beta-1}$ with σ a scale constant: with $\beta < 1$ volatility rises as the price falls, a skew from one parameter, and $\beta = 1$ and $\beta = 0$ are the lognormal and normal cases. It fits no particular surface, but its exponent survives as SABR's β (Section 29.9).

Example 21.1 Local volatility is about twice as steep as implied

Take the surface Heston's model (Section 21.4) gives at the base case of Figure 21.2 (any arbitrage-free surface will do) and apply (21.2). Its call prices come from Heston's semi-analytic formula, and the strike derivatives are central differences in log-strike $y = \log K$ (with $C_y = \partial C/\partial y$, $K^2\,\partial^2 C/\partial K^2 = C_{yy} - C_y$), which keeps the wings numerically stable, and $\partial C/\partial T$ is a central

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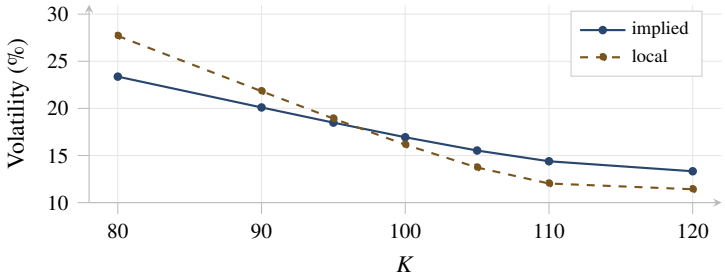


Figure 21.1: Local and implied volatility at one year, from the same Heston surface.

difference in maturity. These are formula prices, not quotes, so none of [TN21]'s differencing noise arises here. Both functions are read at the same argument K, but it means different things on the two sides: for σ_{impl} it is the strike of an option expiring in a year, and for σ_{loc} it is the price level at which that volatility is in force, a year from now. With the chapter's convention, $S_0 = 100$ and $r = q = 0$:

Strike	Implied vol	Local vol
80	23.37%	27.69%
90	20.10%	21.82%
95	18.49%	18.92%
100	16.94%	16.15%
105	15.53%	13.73%
110	14.39%	12.03%
120	13.33%	11.42%

Measured as slopes in $k = \log(K/S_0)$ at the money, across the ±5% pair (the convention (21.3) fixes), implied volatility falls at -0.296 per unit of k (about 3 volatility points from 95 to 105) and local volatility at -0.518, a ratio of **1.75**.

Figure 21.1 shows the two curves together. This is the "rule of two," and the reason for it is an averaging argument. An implied volatility is the single number that reproduces the price of an option struck at K, so it is an average of the local volatilities the price path visits on its way to K. Averaging flattens. To produce a given slope in the average, the underlying function must slope about twice as steeply.

So a steep local-volatility surface, however alarming its wings look, is not a calibration failure; it is arithmetic.

21.3 WHAT LOCAL VOLATILITY GETS WRONG

Section 19.4 said local volatility fits today's prices exactly and forecasts their evolution badly. Here is the mechanism and its size.

Because σ_{loc} is a fixed function of S and t, the model has no freedom left once it is calibrated. In particular it has a definite prediction about the **forward smile**: the smile that will be quoted at some future date, for options starting then. And the prediction is wrong in a specific direction.

One measurement convention is needed first, because from here on the chapter reports skews as numbers rather than shapes. Section 19.1 used "skew" for the shape: implied volatility falling as the strike rises. Measured, it is the slope of implied volatility in **log-moneyness**, taken at the money, and the tables below report the second derivative beside it:

$$\text{skew}(T) = \left.\frac{\partial \sigma_{\text{impl}}}{\partial k}\right|_{k=0}, \qquad \text{curvature}(T) = \left.\frac{\partial^2 \sigma_{\text{impl}}}{\partial k^2}\right|_{k=0} \tag{21.3}$$

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with $k = \log(K/F)$, where F is the forward price to the option's maturity, $F = S_0 e^{(r-q)T}$ for an asset paying a continuous dividend yield q. Under the chapter's convention $r = q = 0$, so $F = S_0$ throughout and "at the money" and "at the forward" name the same strike. **Curvature** is the smile's convexity in the same variable: zero for a straight line, and positive when both wings sit above the tangent at the money. Both are estimated in practice by finite differences across a pair of strikes either side of the forward (and the forward itself, for the curvature): placed symmetrically in k, at Fe^{-h} and Fe^{h}, where a table states a bump h, and at 95 and 105 (the "±5% strike pair") where it does not. At so narrow a spacing the two placements barely differ.

Three properties are worth fixing now, because every table in this chapter reports them. *Log-moneyness rather than strike* makes the number scale-free, so two underlyings trading at 40 and 4,000 report the same skew for the same shape. *The sign is negative on an equity surface*, and a steeper skew is a more negative number. *The value depends on the width of the strike pair*, because the smile's slope changes away from the money: the one-year base-case smile of Section 21.4 measures −0.296 across the ±5% strike pair and −0.282 at $h = 0.10$. Each table states its own bump, and skews are comparable down a column rather than across tables.

Example 21.2 The forward smile flattens

Take Example 21.1's surface and measure both skews ±5% around the forward, the local one at the date equal to each maturity:

Maturity	Implied skew	Local-vol skew
3 months	−0.479	−0.981
6 months	−0.403	−0.770
1 year	−0.296	−0.518
2 years	−0.196	−0.320
3 years	−0.148	−0.231

The local volatility function's own skew decays by a factor of more than four between three months and three years. A one-year option starting in three years is priced from the local volatilities between years three and four, whose skew starts at −0.231, where today's one-year option is priced from skews of −0.98 to −0.52; so the model predicts that the three-year-forward smile will be far flatter than today's spot smile.

The market does not agree. Forward-starting options (options whose strike is set at the money on a future date) and cliquets, chains of them that reset one after another, trade with a skew comparable to today's: the smile is roughly *stationary* in moneyness, not decaying. So a local-volatility model marks anything that gains when the future skew steepens too cheap (an out-of-the-money forward-starting put, a forward-starting risk reversal that is long the put and short the call) and its mirror image too expensive, and, worse, hedges both with the wrong vega profile.

Two products with identical vanilla hedges can have quite different forward-smile exposure, and local volatility, holding one flattened forward smile, cannot tell them apart.

There is a second, subtler consequence. Because local volatility ties σ to S, a fall in the spot mechanically raises volatility along the calibrated function. That makes the model's delta systematically different from the Black–Scholes delta, in a direction Section 19.11's vocabulary can name: a rise in the spot *raises* a fixed strike's implied volatility under sticky delta and *lowers* it under local volatility, so the local-volatility delta sits below even the sticky-strike (Black–Scholes) delta. The measured index behavior of Section 19.11 lies between the two: below Black–Scholes, but not as far below as local volatility puts it.

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21.4 STOCHASTIC VOLATILITY

Let volatility be a second random process, correlated with the price:

$$dS = (r-q)S\,dt + \sqrt{v}\,S\,dW_1, \qquad dv = \kappa(\theta - v)\,dt + \sigma_v\sqrt{v}\,dW_2, \qquad dW_1\,dW_2 = \rho\,dt \quad (21.4)$$

The variance v mean-reverts at rate κ toward a long-run level θ, with volatility-of-volatility σ_v, and it is correlated with the price at ρ. The square root in the diffusion keeps v nonnegative, though it can touch zero unless $2\kappa\theta \geq \sigma_v^2$, the Feller condition, which fitted equity parameters usually violate (the base case used throughout the chapter, $v_0 = \theta = 0.04$, $\kappa = 1.5$, $\sigma_v = 0.6$, $\rho = -0.7$, has $2\kappa\theta = 0.12$ against $\sigma_v^2 = 0.36$); that matters for simulation, where a discretized variance can go negative, not for the formula. The square root is also what makes the model tractable: the characteristic function $\varphi(u) = \mathbb{E}\left[e^{iu\log S_T}\right]$ (risk-neutral, $i = \sqrt{-1}$) is available in closed form, and a call price follows from it by Fourier inversion rather than by simulation,

$$C = S_0 e^{-qT} P_1 - K e^{-rT} P_2, \qquad P_j = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \mathrm{Re}\left[\frac{e^{-iu\log K}\varphi_j(u)}{iu}\right] du \quad (21.5)$$

where $\varphi_2 = \varphi$ and $\varphi_1(u) = \varphi(u - i)/\varphi(-i)$, which is the same function under the measure that takes the stock itself as numeraire; P_2 is the risk-neutral probability that the call finishes in the money and P_1 the same probability under that stock measure, Heston's $N(d_2)$ and $N(d_1)$. And φ is elementary. Writing $b = \kappa - \rho\sigma_v iu$ and

$$d = \sqrt{b^2 + \sigma_v^2\left(u^2 + iu\right)}, \qquad g = \frac{b-d}{b+d}$$

the characteristic function is

$$\varphi(u) = \exp\left\{iu\left(\log S_0 + (r-q)T\right) + A(T) + B(T)\,v_0\right\}$$

with

$$B(T) = \frac{b-d}{\sigma_v^2}\cdot\frac{1-e^{-dT}}{1-ge^{-dT}}, \qquad A(T) = \frac{\kappa\theta}{\sigma_v^2}\left[(b-d)T - 2\log\frac{1-ge^{-dT}}{1-g}\right]$$

That is the whole of it: a square root, an exponential and a logarithm in u, with no integral and no recursion anywhere. This is what "semi-closed form" means: no formula for the price, but none needed, because everything except a single one-dimensional integral is written above. It is also why a whole surface calibrates in seconds. One warning: d and the logarithm are complex, and the algebra pins only d^2, which either square root satisfies, and the two give algebraically equivalent formulas for A and B. Take the principal root, as written above, and the logarithm stays continuous; take the other and it jumps by $2\pi i$, destroying the pricing integral at long maturities. [TN21] derives all of it and calls that failure the "little Heston trap."

Heston is the standard model for three reasons that have little to do with realism. The semi-closed form is the first. It has five parameters (v_0, θ, κ, σ_v, ρ), which is enough to fit a smile and few enough to interpret. And every parameter maps onto a feature of the surface a trader already thinks in terms of.

Figure 21.2 shows the two parameters a trader adjusts most. Table 21.2 puts numbers on four of the five; the fifth, v_0, stays at 0.04 throughout. Its skew and curvature are first and second central differences of implied volatility in log-moneyness at $h = 0.10$.

Correlation produces the skew, and produces it for the right reason. At $\rho = 0$ the skew vanishes: the smile is symmetric to three decimals, which is Section 19.1's currency shape. Making ρ negative tilts it, and the economics is Section 19.9's leverage explanation written as a parameter: when volatility rises as the price falls, low strikes become expensive.

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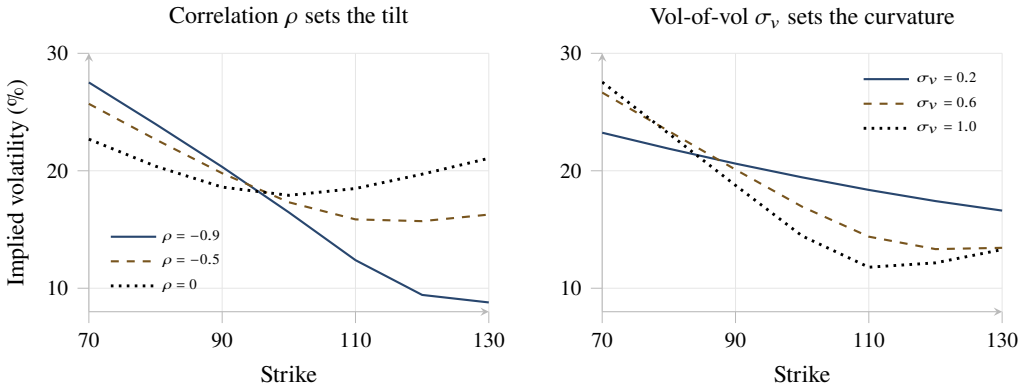


Figure 21.2: One-year Heston smiles. Base case $v_0 = \theta = 0.04$, $\kappa = 1.5$, $\sigma_v = 0.6$, $\rho = -0.7$; each panel varies one parameter.

Table 21.2: One-year at-the-money level, skew, and curvature as each parameter is varied from the base case.

	ATM (%)	Skew	Curvature
Base ($\rho = -0.7$, $\sigma_v = 0.6$, $\kappa = 1.5$, $\theta = 0.04$)	16.94	−0.282	0.36
$\rho = -0.9$	16.46	−0.398	−0.58
$\rho = 0$	17.91	0.000	1.27
$\sigma_v = 0.2$	19.44	−0.112	−0.01
$\sigma_v = 1.0$	14.45	−0.340	1.40
$\kappa = 5$	18.94	−0.155	0.03
$\theta = 0.09$	22.70	−0.227	0.08

Vol-of-vol produces the curvature. At $\sigma_v = 0.2$ the curvature is −0.01 and the smile is a straight line to two decimals; at $\sigma_v = 1.0$ it is 1.40 and both wings lift. This is the mechanism of Section 19.9: uncertainty about volatility fattens both tails of the terminal distribution.

The two interact, and that is the practical difficulty. Raising σ_v from 0.2 to 1.0 also steepens the skew from −0.112 to −0.340 and drops the at-the-money level from 19.44% to 14.45%. No parameter moves one feature alone, so calibration is a genuine optimization rather than a sequence of one-dimensional fits, and the fitted parameters are correlated enough that reporting them individually is misleading.

Mean reversion flattens everything. At $\kappa = 5$ the skew falls to −0.155 and the curvature to 0.03: fast mean reversion gives variance less time to wander, so the terminal distribution is closer to lognormal. This is the averaging that flattens the skew with maturity beyond Section 19.1's change of unit, and it is why κ is hard to identify from a single maturity's smile: its effect looks like a longer maturity.

Diffusive variance fails at the short end. Variance is a diffusion, so over a vanishingly short horizon it has not had time to move, and however the parameters are set the at-the-money skew approaches a finite limit as $T \to 0$ instead of steepening as the market's does. In the base case the one-month skew is 1.77 times the one-year skew (slopes at $h = 0.01$), where the market's power law gives about 2.7. Section 21.7 measures the gap; Sections 21.5 and 21.6 are the classical alternatives, jumps alone and jumps added to Heston.

Rate markets use SABR instead, built for refitting speed: its volatility is stochastic too, but an asymptotic expansion gives the implied volatility as a formula, so a whole grid of swaption smiles can be refitted many times a day (Section 29.9).

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21.5 JUMPS

Allow the price to move discontinuously. In Merton's model a diffusion is punctuated by jumps arriving at rate λ, each multiplying the price by a lognormal factor Y:

$$\frac{dS}{S} = (r - q - \lambda\bar{k})\,dt + \sigma\,dW + (Y-1)\,dN, \qquad \bar{k} = \mathbb{E}[Y] - 1 \tag{21.6}$$

where N is a Poisson process of intensity λ (a count of jumps that rises by one at random moments, λ times a year on average) and $\log Y$ is normal with mean $\mu_J = \log(1+\bar{k}) - \delta^2/2$ and variance δ^2. The $-\lambda\bar{k}$ in the drift is a compensator: it subtracts the jumps' expected contribution, so that jumps negative on average do not drag the price below the forward and $e^{-(r-q)t}S_t$ stays a martingale.

Conditional on exactly n jumps the terminal price is lognormal again, so the option is a Poisson-weighted sum of Black–Scholes–Merton values, which is what makes the model closed form:

$$C = \sum_{n=0}^{\infty} \frac{e^{-\lambda^* T}(\lambda^* T)^n}{n!}\, C_{\mathrm{BSM}}(\sigma_n, r_n)\,, \qquad \lambda^* = \lambda(1+\bar{k})$$

with $\sigma_n^2 = \sigma^2 + n\delta^2/T$ and $r_n = r - \lambda\bar{k} + n\log(1+\bar{k})/T$. Each term is the price on the paths that happen to take n jumps; the intensity is λ^* rather than λ because the sum discounts at r while each term discounts at its own r_n.

Merton's jumps are few and large, about one every two years at the $\lambda = 0.5$ used below. The variance-gamma model of Madan, Carr, and Chang (1998) and its generalization by Carr, Geman, Madan, and Yor (2002), known as CGMY, go to the other extreme: infinitely many small jumps and typically no diffusion. They fit short-dated skew well and price by the same Fourier inversion as Heston, but their skew flattens with maturity just as Merton's does, and they are used mainly to fit and interpolate vanilla smiles.

The reason to reach for jumps is entirely about short maturities.

Example 21.3 What a jump does that a diffusion cannot

Take Merton's model with $\sigma = 17\%$, jump intensity $\lambda = 0.5$ a year, mean jump $\bar{k} = -10\%$, jump volatility $\delta = 13\%$:

Strike	1 month	3 months	1 year
85	30.99%	23.82%	21.21%
90	25.07%	21.78%	20.81%
95	20.57%	20.35%	20.47%
100	18.77%	19.49%	20.18%
105	18.25%	18.99%	19.94%
110	18.42%	18.73%	19.74%

At one month the smile falls from 31.0% at 85 to a low of 18.2% at 105 and turns up again beyond it, a range of 12.7 volatility points. At one year it falls all the way across the grid, from 21.2% at 85 to 19.7% at 110, a range of 1.5 points.

The smile's range collapses by a factor of more than eight as maturity goes from one month to one year. A diffusion cannot do this, because over a short horizon a continuous path simply cannot travel far enough to make a 15%-out-of-the-money put worth anything; only a discontinuity can.

That is also the criticism. Measured as the at-the-money skew (slopes at $h = 0.01$), the jump model's ratio between one month and one year is **3.7**; the power law index skews follow (Section 21.7) gives about **2.7**. Jumps overshoot the short end by more than a third, where a diffusion undershoots, and a model fitted to one-month options will misprice one-year options unless the jump parameters are allowed to vary with maturity—which is to say, unless the model is

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abandoned.

Section 19.4 noted that jumps break completeness for a sharper reason than stochastic volatility does: a jump of unknown size cannot be hedged by any position established before it. Stochastic volatility leaves a risk no position in the stock alone can hedge, but a second option hedges it, canceling the vega term of Chapter 18's expansion, equation (18.1), as Example 18.7 did; jump risk is not hedged by the stock and any finite set of options, because no such set pays off correctly for every possible jump size. A jump is a move too large for that expansion to describe, so canceling its terms, however completely, does not reach it. Section 13.4's argument said an incomplete market has no unique price; with jumps, it has no perfect hedge either: options can reduce the exposure, but no finite set of them removes it.

Who is paid for jump risk. Because the risk cannot be hedged away, someone must hold it, and they are paid to: part of Section 19.10's volatility risk premium compensates for diffusive variance a delta hedger cannot capture perfectly, and part for a discontinuity no hedge covers at all. The second part sits in the wings and at short maturities, where Example 21.3 put the jump model's contribution, which is convenient for calibration and treacherous for inference: a steep short-dated put wing is consistent both with a high real-world chance of jumps and with a large premium for bearing them, and option prices, which are risk-neutral, cannot separate the two.

For the desk, the consequence is a rule about who should be short. A delta-hedged book can hold diffusive variance risk and earn the premium on it. Jump risk does not diversify across stocks, because jumps cluster across underlyings (Chapter 23 shows the same clustering intraday), so a book that has systematically sold short-dated out-of-the-money options holds a correlated tail, and Chapter 22's case study of February 5, 2018 shows a crowd of short-volatility positions meeting such a tail.

21.6 STOCHASTIC VOLATILITY WITH JUMPS

One mechanism undershoots the short end and the other overshoots it, so the obvious move is to run both at once. Bates (1996) does exactly that: Heston's variance process, with Merton's jumps added to the price:

$$\frac{dS}{S} = (r - q - \lambda\bar{k})\,dt + \sqrt{v}\,dW_1 + (Y-1)\,dN, \qquad dv = \kappa(\theta - v)\,dt + \sigma_v\sqrt{v}\,dW_2 \quad (21.7)$$

with $dW_1\,dW_2 = \rho\,dt$, and N, Y and $\bar{k}$ as in (21.6). The jumps are taken independent of both Brownian motions.

That independence is what makes the model usable: $\log S_T$ is a sum of independent pieces, so its characteristic function is a product:

$$\varphi_{\text{Bates}}(u) = \varphi_{\text{Heston}}(u) \cdot \exp\left\{\lambda T\left(e^{\,iu\mu_J - u^2\delta^2/2} - 1\right) - iu\lambda\bar{k}T\right\}, \qquad \mu_J = \log(1+\bar{k}) - \frac{\delta^2}{2}$$

with φ_{Heston} exactly as Section 21.4 writes it. The price follows from the same inversion (21.5), subject to the same warning about the branch of d. One integral still prices it, which is why Bates is widely used for short-dated skew; [TN21] derives the product and recovers Merton's series from it.

Three parameters are added to Heston's five, and they act where Heston cannot reach:

- λ, the arrival rate, sets how much of the smile is made by jumps rather than by the diffusion.
- $\bar{k}$, the mean jump, tilts the short-dated smile, and a negative value is what makes short-dated puts expensive.

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- δ, the jump volatility, spreads the short-dated wings.

Example 21.4 shows the division of labor in the smiles themselves.

Example 21.4 Where each mechanism acts

Take Heston's base case with its diffusive variance cut to $v_0 = \theta = 0.025$ and Example 21.3's jumps added ($\lambda = 0.5$, mean jump -10%, jump volatility 13%), so that the two mechanisms together carry the base case's variance rate of 0.04 a year:

Strike	1 month	3 months	1 year
85	31.57%	26.08%	21.67%
90	26.52%	23.58%	20.36%
95	21.67%	20.68%	19.00%
100	17.17%	17.33%	17.55%
105	13.74%	14.08%	16.03%
110	15.06%	13.21%	14.57%

At one month the smile falls from 31.6% at 85 to 13.7% at 105, a range of 17.8 volatility points, and turns up again beyond it. Over the same strikes Heston's own one-month smile spans 11.1 points and the jump model's 12.7: the combination is steeper than either component, because the jump wing now sits on a diffusion that is itself skewed. At one year the range is 7.1 points, against Heston's 7.3 and the jump model's 1.5.

At one month the shape of the smile (the put wing and the upturn beyond 105) is the jump model's; at one year the smile is Heston's, tilt and all. Each mechanism carries the maturities the other cannot reach.

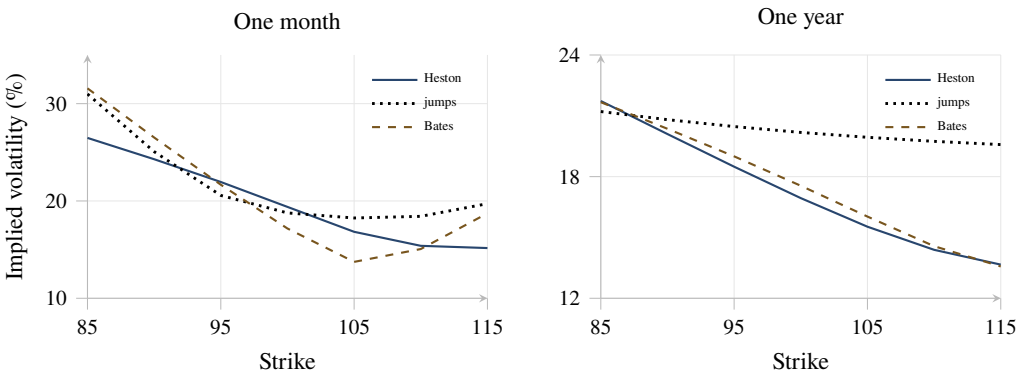


Figure 21.3: One-month and one-year smiles under Heston, Merton's jumps and Bates, with the parameters of Examples 21.3 and 21.4.

Figure 21.3 draws the comparison. In the left panel the Bates curve tracks the jump curve down the put wing and turns up with it past 105, sitting below it from the money upward because its diffusive variance is smaller, 0.025 against the jump model's $0.17^2 = 0.029$; in the right panel it lies on top of Heston's, and the jump curve is nearly flat. Table 21.3 puts numbers on what each of the three new parameters does to that picture. Every row resets $v_0 = \theta$ to hold the variance rate at the base case's; skew and curvature are central differences at $h = 0.10$, and the last column divides the one-month by the one-year slope at the money ($h = 0.01$, as in Figure 21.4), so it is not the quotient of the skew columns.

Read the one-year columns first. Down every row the skew stays between -0.24 and -0.36 and the curvature stays near zero, so a year out the three jump parameters are close to invisible and the smile is the one Table 21.2 described. At one month they are everything.

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Table 21.3: One-month and one-year at-the-money level, skew, and curvature as each jump parameter is varied from the Bates base case ($\lambda = 0.5$, $\bar{k} = -0.10$, $\delta = 0.13$).

		One month			One year			
	$v_0 = \theta$	ATM (%)	Skew	Curv.	ATM (%)	Skew	Curv.	**Ratio**
Base	0.0250	17.17	−0.531	7.11	17.55	−0.289	−0.44	2.92
$\lambda = 0.25$	0.0325	18.35	−0.495	3.42	17.23	−0.287	−0.02	2.23
$\lambda = 1.0$	0.0101	13.96	−0.584	17.02	18.19	−0.287	−1.28	6.20
$\bar{k} = -0.05$	0.0297	18.07	−0.380	6.34	17.35	−0.253	0.17	2.55
$\bar{k} = -0.15$	0.0169	15.19	−0.703	10.33	17.75	−0.346	−1.36	3.63
$\delta = 0.05$	0.0330	18.92	−0.538	1.29	17.49	−0.246	0.01	2.77
$\delta = 0.20$	0.0121	13.13	−0.400	19.15	16.64	−0.359	−0.13	3.18

The intensity sets how much of the smile the jumps make. Doubling λ to 1.0 leaves only a quarter of the variance in the diffusion, and the one-month at-the-money level falls from 17.2% to 14.0% while the curvature rises from 7.1 to 17.0: variance that used to sit under the whole distribution now sits in its tails. The one-year row barely notices.

The mean jump sets the tilt. From $\bar{k} = -0.05$ to -0.15 the one-month skew steepens from −0.380 to −0.703, and the one-month/one-year ratio goes from 2.55 to 3.63. This is the parameter that makes short-dated puts expensive, and it is the one a calibration to the put wing pins down first.

The jump volatility sets the wings. From $\delta = 0.05$ to 0.20 the one-month curvature goes from 1.3 to 19.2, and at the top of that range the skew flattens to −0.400, because a wide jump distribution fattens both tails rather than the left one alone.

What the combination buys is the last column. The base case's one-month slope at the money is **2.92** times its one-year slope, against the market's 2.70, where Heston alone reaches 1.77 and jumps alone 3.70; Section 21.7 draws all four against maturity. Mixing two errors of opposite sign cancels most of the short-end miss.

The long end is worse, not better. At three years the skew is 0.372 of its one-year value, against the market's 0.644, almost exactly the jump model's 0.367, and further off than Heston's own 0.494. Moving variance out of the diffusion and into jumps does not lift the term structure; it rotates it: jump skew is concentrated at short maturities and is negligible at long ones, so the short-end repair is paid for at the long end. Lowering the intensity trades one end against the other (at $\lambda = 0.25$ the three-year ratio improves to 0.42 while the one-month falls to 2.23, below the market); raising it misses both ends by more (6.20 and 0.317 at $\lambda = 1.0$).

Eight parameters are enough to put the fit where a particular book needs it, which is why many equity desks use some version of it for short-dated vanillas. They do not supply the power law the market's skew follows across maturities (Section 21.7): two unrelated mechanisms, each fitted to a different part of the surface, are a patch rather than an explanation.

21.7 THE SKEW TERM STRUCTURE, AND WHY IT BROKE ALL THREE MODELS

Sections 21.4, 21.5 and 21.6 each made a claim about how the skew behaves as maturity shortens. Put the three side by side against the market and the claims resolve into a single, sharp, measurable failure, the one the next section finally explains.

Write $\psi(T)$ for the *magnitude* of the at-the-money skew (21.3),

$$\psi(T) = -\left.\frac{\partial \sigma_{\text{impl}}}{\partial k}\right|_{k=0}$$

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which is positive on an equity surface, so that a steepening skew is a rising ψ. Ask how it behaves as $T \to 0$.

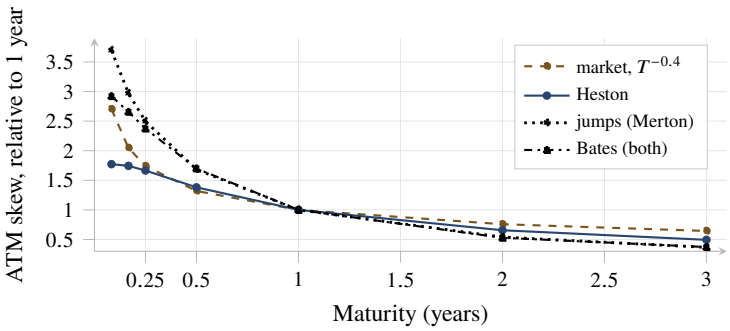


Figure 21.4: At-the-money skew against maturity, normalized to one year: Heston's base case, Example 21.3's jumps, Example 21.4's Bates, and the power law $T^{-0.4}$; slopes at the money, $h = 0.01$.

The empirical answer is a power law. Across equity indices, $\psi(T)$ is roughly proportional to $T^{-\alpha}$ with α around 0.4, at least from about a month to a few years: the skew keeps steepening as maturity shortens. Whether it goes on steepening below a month is disputed, and Section 21.8 returns to it.

A classical stochastic-volatility model cannot produce that. Section 21.4 said why: variance is a diffusion, so $\psi(T)$ approaches a finite constant as $T \to 0$ no matter how the parameters are set. In Figure 21.4 the Heston skew rises only 1.77 times from one year to one month, where the market rises 2.70 times.

At the long end the failure reverses. Heston's skew decays to 0.494 of its one-year value by three years, against the market's 0.644: too flat at the short end and decaying too fast at the long end simultaneously, which is not a parameter problem but a structural one. Fitting Heston to a whole surface therefore means fitting it badly everywhere rather than well somewhere.

Jumps fix the short end and overshoot it, as Example 21.3 showed: ratio 3.7 against the market's 2.7, the third curve in Figure 21.4. Bates, the fourth curve, splits the difference at the short end (2.92 against 2.70) and inherits the jump model's long end, 0.372 against 0.644, for the reason Section 21.6 gave.

None of the three produces the power law. In each of them the variance is either constant or a Markov diffusion, with or without jumps appended to the price, and the short-end skew of any such model either saturates or is set by hand, maturity by maturity, through the jump parameters. The market's curve is not a compromise between the other three; it is a different shape, and matching it needs either a different kind of process or several ordinary ones with widely separated speeds. That was the position for two decades: a short-maturity problem, patched and never explained.

21.8 ROUGH VOLATILITY

In a paper circulated in 2014 and published in 2018, Gatheral, Jaisson, and Rosenbaum estimated the smoothness of realized volatility directly from high-frequency data on equity indices and liquid futures, and found a pattern that had been in the data all along.

Log-volatility behaves like the fractional Brownian motion of Section 14.13, with Hurst parameter $H \approx 0.1$. Ordinary Brownian motion has $H = 0.5$; at 0.1 the path is far *rougher* (more jagged, with successive increments correlated at -0.43 rather than zero), and the rough path of Figure 14.5 is what such a series looks like. The estimate was strikingly stable: roughly

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the same H across twenty-one equity indices and for Bund, crude oil and gold futures, between 0.06 and 0.20, and nowhere near 0.5, when the sample was split in two, and, in later work, across close to two thousand individual US stocks.

Two things made this more than an empirical curiosity.

It produces the observed skew term structure with no extra machinery. A model whose volatility is driven by a fractional Brownian motion with Hurst parameter H generates

$$\psi(T) \sim T^{H-1/2} \quad \text{as } T \to 0 \tag{21.8}$$

where $\sim$ means proportional to, in that limit. Setting $H = 0.1$ gives $\psi(T) \sim T^{-0.4}$—the observed power law, from one parameter, with $H = 0.5$ recovering the flat short-end behavior of the classical models. Figure 21.4's "market" curve *is* equation (21.8) at $H = 0.1$.

And roughly the same H appears on both sides. The value estimated from realized volatility under the physical measure (the real-world probabilities that generated the data) roughly matches the value implied by option prices under the risk-neutral measure. A parameter estimated from time-series data has no obligation to match one estimated from cross-sectional prices, and when it does, that is evidence about the process rather than about the fit.

Case Study 21.1 How roughness became usable

Rough volatility was, at first, a model no one could use. Fractional Brownian motion is not Markovian: the expected future path of volatility depends on its entire history, so there is no state variable to put on a tree and no partial differential equation to solve. Every price required a simulation, and simulating the model's volatility exactly, jointly with the price, costs $O(n^2)$ per path in the number of time steps after a Cholesky factorization, against $O(n)$ for an ordinary diffusion.

Three developments changed that between roughly 2015 and 2021.

Fast simulation. Hybrid schemes reduced the cost of generating rough paths to something comparable with ordinary Monte Carlo, by computing exactly the heavily weighted most recent shocks and approximating the weight on older ones.

Markovian approximations. The rough weighting of past shocks can be approximated by a sum of a small number of ordinary mean-reverting factors with widely separated speeds. This restores a state vector, typically of four to eight dimensions, and with it ordinary Monte Carlo and, in low dimension, the partial differential equations of Chapter 20. It is an approximation, but a controlled one, and it is how most implementations work.

Neural network calibration. Calibrating a rough model by repeated simulation is prohibitive. Training a network offline to map parameters to implied volatility surfaces, then inverting it at run time, turns a calibration that took hours into one that takes milliseconds. The network forecasts nothing; it interpolates a known function that is merely expensive to evaluate, so it replaces repeated simulation without changing the model being calibrated.

Adoption has been uneven. Non-bank market makers and some hedge funds took the models up first, for short-dated index options and relative-value trades between SPX and VIX options; banks have been slower, and most exotics books remain on the local-stochastic volatility of Section 21.9.

The lesson is about what a modeling constraint really is. For twenty years the profession assumed volatility must be modeled as a Markov diffusion, and most failures at the short end were patched by adding jumps. The Markov assumption was never an economic claim; it was a computational one, and when the computation became affordable the assumption went. It is worth asking which of today's modeling conventions are economics and which are just arithmetic we have not yet made cheap enough.

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The standard implementation is the **rough Bergomi** model of Bayer, Friz, and Gatheral (2016), and it is worth writing out, because each of its three parameters has one main job:

$$\begin{aligned} \frac{dS}{S} &= (r-q)\,dt + \sqrt{v_t}\,dW_{1,t}, \qquad dW_1\,dW_2 = \rho\,dt \\ v_t &= \xi_0(t)\exp\left\{\eta\sqrt{2H}\int_0^t (t-s)^{H-1/2}\,dW_{2,s} - \frac{\eta^2}{2}\,t^{2H}\right\} \end{aligned} \tag{21.9}$$

Read it from the outside in. $\xi_0(t)$ is the **forward variance curve**, today's price of the variance to be realized at t, which follows from the variance swap rates Section 19.3 replicated from the vanilla surface and which Chapter 22 trades. It is an input rather than a parameter: the exponential has mean one, so $\mathbb{E}[v_t] = \xi_0(t)$ and the curve is reproduced exactly whatever the other three do. The integral is a version of Section 14.13's fractional Brownian motion (the Riemann–Liouville form: the same roughness H, but increments that are not stationary), written as a weighted sum of past shocks: each $dW_{2,s}$ enters with weight $(t-s)^{H-1/2}$, the **kernel** of Section 14.13 (Heston's is $e^{-\kappa(t-s)}$), and the $\sqrt{2H}$ normalizes the variance to t^{2H}, which is why $-\frac{1}{2}\eta^2 t^{2H}$ keeps the mean of the exponential at one. η scales the volatility of log variance and ρ the correlation, with the same jobs as Heston's σ_v and ρ.

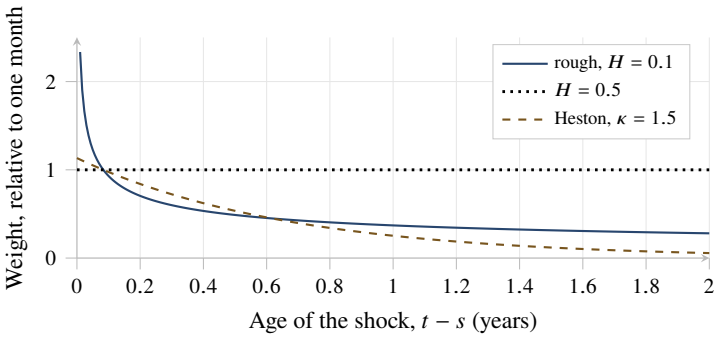


Figure 21.5: Weight on a past variance shock as a function of its age, normalized to one at one month.

The kernel is where the model differs from everything before it, and Figure 21.5 draws it against Heston's. For $H < 1/2$ the exponent is negative and the weight rises without limit as $s \to t$: the most recent shocks dominate, which is roughness. But the weight decays as a power rather than an exponential, so a shock from two years ago still carries 28% of the weight of one from a month ago, where Heston's $e^{-\kappa(t-s)}$ at $\kappa = 1.5$ leaves it 6%. Rough volatility is both more jagged and longer-remembering than a mean-reverting diffusion, and the combination is what a single exponential cannot produce: to move that much over short horizons, Heston needs a very large σ_v; to remember that long, a very small κ; and the two settings fight over the same one-year smile. Table 21.4 puts numbers on the three parameters in the format of Tables 21.2 and 21.3. Its figures are simulated (200,000 paths, hybrid scheme), so skews are reliable to about ±0.01 and the ratio to about ±0.05; skew and curvature are central differences at a half-width $h = 0.10$, and the ratio uses the ±5% strike pair, since Monte Carlo noise rules out Table 21.3's $h = 0.01$, so it is not the quotient of the skew columns either.

Two of the three read exactly as they did for Heston. *ρ sets the tilt*: at $\rho = 0$ the skew is zero at both maturities and the smile is symmetric. *η sets the curvature*, and drags the level with it as σ_v did: from $\eta = 1.0$ to 3.0 the one-month curvature goes from −0.1 to 13.3 and the at-the-money level from 19.3% to 15.2%.

The third has no counterpart in Heston. From $H = 0.10$ to 0.45 the one-month/one-year skew ratio falls from 2.6 to 1.3, where (21.8) predicts $12^{1/2-H}$: 2.70, 1.86 and 1.13, which is

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Table 21.4: One-month and one-year at-the-money level, skew, and curvature as each parameter is varied from the rough Bergomi base case ($H = 0.10$, $\eta = 1.9$, $\rho = -0.9$, flat $\xi_0 = 0.04$).

	One month			One year			
	ATM (%)	Skew	Curv.	ATM (%)	Skew	Curv.	**Ratio**
Base	17.8	−0.63	4.3	16.3	−0.31	0.0	2.6
$H = 0.25$	18.6	−0.73	0.1	15.1	−0.39	−0.3	2.0
$H = 0.45$	19.5	−0.47	−0.3	15.3	−0.36	−0.2	1.3
$\eta = 1.0$	19.3	−0.51	−0.1	18.6	−0.19	−0.1	2.7
$\eta = 3.0$	15.2	−0.47	13.3	12.8	−0.30	2.1	2.1
$\rho = -0.5$	18.3	−0.34	5.9	17.1	−0.17	0.9	2.4
$\rho = 0$	18.6	0.00	6.7	17.7	0.00	1.2	—

close, given that (21.8) describes the skew as maturity shrinks and is only approximate between a month and a year (Heston, a diffusive-variance model with $H = 1/2$, gives 1.77, not 1). The one-year skew, meanwhile, stays between −0.31 and −0.39 across the same three rows. H is a term-structure parameter: it decides how the skew is distributed across maturities, not how much of it there is at a year, and Heston has nothing that does that job. Two further points attach to the table. The parameters interact: the standard deviation of log variance over a horizon t is $\eta\, t^H$, so with the same η a rougher process swings more within a month, and the one-month level falls as H falls; over a year those swings largely average out, so the rough row's average variance is the least dispersed and its one-year level the highest. And on index surfaces three parameters, with the forward variance curve taken from the market, fit better than Heston's five (Bayer, Friz, and Gatheral, 2016), which is the strongest argument for the model.

Estimating H. The number did not come from option prices. Gatheral, Jaisson, and Rosenbaum built a daily series of realized volatility, σ_t on day t being the square root of that day's realized variance from five-minute returns, took its logarithm, and for each lag Δ from one to about thirty days computed the average p-th power of the change over that lag,

$$m(p, \Delta) = \frac{1}{n} \sum_{t} \left| \log \sigma_{t+\Delta} - \log \sigma_t \right|^p$$

over the n pairs of days in the sample, for powers p from about 0.5 to 3. Averaging over start days measures a single quantity because the processes they fitted, fractional Brownian motion and its slowly mean-reverting version, have stationary increments: a change over Δ days has the same distribution whichever day it starts on. Self-similarity then gives $m(p, \Delta) \propto \Delta^{pH}$, so a regression of $\log m$ on $\log \Delta$ has slope pH; the slopes came back proportional to p, as the model requires, with H of order 0.1, and nowhere near 0.5, for every index examined. Two cautions attach to the number. Realized variance is an estimate of the variance, not the variance itself, and measurement noise in it makes a series look rougher than it is; how much of the measured roughness is noise is still debated, as the end of this section explains. And the estimate is made under the physical measure; the H that prices options is read off the skew term structure under the risk-neutral measure, and the two roughly agreeing, as the start of this section noted, is the evidence, not an assumption.

Where the frontier is. Two developments since 2019 matter. Rough Heston replaces the exponential decay of a variance shock in Heston (Figure 21.5) with the power-law kernel. Because the model is still affine (the logarithm of its characteristic function is linear in today's forward variance curve, as Heston's is in v_0 above), its characteristic function survives: it

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solves a fractional Riccati equation numerically rather than in closed form, but one integral still prices a vanilla and the machinery of Section 21.4 carries over. And the joint calibration problem, one model fitting the index smile and the VIX smile of Chapter 22 at the same time, long defeated the classical models and was narrowed but not closed by rough volatility. Among the models that fit both, path-dependent volatility makes today's volatility a weighted function of the index's own past path, its recent returns and their recent squares, rather than of a separate factor. That is Section 19.9's leverage effect, seen from the modeling side.

Two caveats follow. Rough models fit vanilla surfaces very well and their advantage for pricing exotics is real but smaller than the vanilla fit suggests. And the choice of H is not innocuous for anything sensitive to the very short end, where the model's predictions are most distinctive and market data are thinnest.

What is still contested. Each leg of the case for roughness has been questioned since 2022. On S&P 500, Euro Stoxx 50 and DAX surfaces, El Amrani and Guyon (2023) find that the power law fits from about a month to three years but not below a month, where a skew rising toward a finite limit fits better. Cont and Das (2024) show that realized volatility estimated from a process that is not rough can still measure as rough, so the time-series estimate of H may be partly an artifact of estimation error. And the quintic Ornstein–Uhlenbeck model (a polynomial in one mean-reverting Gaussian factor) of Abi Jaber, Illand, and Li (2023) fits SPX and VIX smiles jointly with a single Markov factor and no rough paths, its steep short-end skew coming from fast mean reversion and a large vol-of-vol. Rough volatility remains the most economical account of the skew term structure; whether volatility is in fact rough is open.

21.9 LOCAL-STOCHASTIC VOLATILITY

Desks divide the work. Local volatility serves for payoffs the vanilla surface alone determines, and jump models mainly the short-dated skew; for path-dependent equity and foreign-exchange exotics, what desks run is neither the model of Section 21.2 nor that of Section 21.4. It is both:

$$dS = (r - q)S\,dt + L(S,t)\sqrt{v}\,S\,dW_1 \tag{21.10}$$

with v following a stochastic-volatility process as in (21.4). The function $L(S,t)$, the **leverage function**, is chosen so that the model reproduces the vanilla surface exactly.

The stochastic part supplies realistic dynamics; the leverage function absorbs whatever is left over so the fit is exact. What follows says what L is, what the one free choice in the construction does to a price, and who gets to make that choice.

What the leverage function is. A model reproduces the vanilla surface if and only if the average of its instantaneous variance, conditional on the spot at each date, equals the Dupire local variance at that spot, the result [TN21] derives from Gyöngy's theorem; local variance is therefore itself an average, of the instantaneous variance over the paths that pass through (S,t). For (21.10) the instantaneous variance is $L^2 v$, so

$$L^2(S,t) = \frac{\sigma^2_{\text{loc}}(S,t)}{\mathbb{E}\big[\, v_t \mid S_t = S \,\big]} \tag{21.11}$$

The numerator is the variance the surface demands at (S,t); the denominator is the variance the stochastic factor delivers there on average. The leverage function is the part of the smile the stochastic factor did not explain: above one where the stochastic model under-delivers, below one where it over-delivers, and exactly one where it already fits. The equation is circular, since the conditional expectation on the right depends on the paths and the paths depend on L. It is solved forward in time, one step at a time, by a forward partial differential equation or

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by a particle method that estimates the expectation from the simulated paths as they advance. [TN21] sets out the fixed point and both ways of solving it.

The mixing parameter. Scale the volatility of volatility, $\sigma_v \to \zeta\,\sigma_v$ with $\zeta \in [0, 1]$, and recalibrate L at each setting. At $\zeta = 0$ the variance is deterministic, the denominator of (21.11) is just $v(t)$, and $L = \sigma_{\text{loc}}/\sqrt{v}$: pure local volatility. At $\zeta = 1$ the stochastic factor explains as much of the smile as it can and L carries only the residual. Every setting reproduces the vanillas exactly, which is what makes ζ different from every other parameter in this chapter: it is not a *fit* parameter. It is a choice about dynamics that the vanilla market cannot make for you, and Figure 21.6 shows what it does to L. The vanilla surface there is the whole implied-volatility surface, every strike and maturity, that Heston produces at the base case of Figure 21.2, and the stochastic factor is that same base-case variance process, so at $\zeta = 1$ the stochastic model already fits and L equals one everywhere—the check that the construction is working. At $\zeta = 0$ the leverage function is the entire local-volatility shape, 1.51 at a spot of 75 falling to 0.57 at 115, and at $\zeta = 0.5$ it is flatter but not a midpoint: just below the money it dips below both, because the half-strength stochastic factor delivers more at-the-money variance than the surface wants.

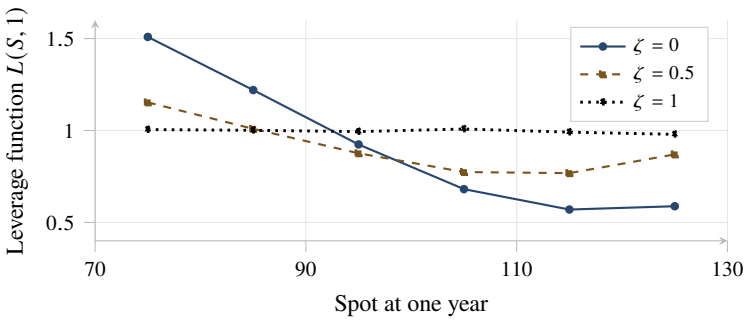


Figure 21.6: The leverage function at one year on Heston's own surface, at three settings of the mixing parameter.

Example 21.5 The mixing parameter as a dial

Heston's base case generates the vanilla surface; the local-stochastic model (21.10) is calibrated to it at five settings of ζ by the particle method, and prices a one-year up-and-out call, strike 100, barrier 110, monitored daily (extinguished when the spot first reaches 110); P(survive) is the share of paths that never touch the barrier. The particle method runs 80,000 Monte Carlo paths; the vanilla check is the one-year implied volatility the model returns at three strikes, against Heston's 23.37%, 16.94% and 13.33%.

	One-year vanilla fit				
Mixing ζ	$K = 80$	$K = 100$	$K = 120$	**Up-and-out 110**	**P(survive)**
0	23.35%	16.96%	13.30%	0.420	44.8%
0.25	23.34%	16.94%	13.29%	0.463	45.7%
0.5	23.27%	16.89%	13.28%	0.572	47.7%
0.75	23.21%	16.85%	13.23%	0.707	50.2%
1	23.24%	16.85%	13.23%	0.845	52.7%

The vanilla columns are flat. Every row matches the target to within 0.2 volatility points, which is the noise of the simulation and of averaging within spot ranges: the fit is exact at every setting, as the construction promises. *The barrier column doubles.* From $\zeta = 0$ to 1 the price of the same

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option, against the same vanilla surface, goes from 0.420 to 0.845, and it moves monotonically, so any value between the two can be produced by a setting of the dial. Both ends are legitimate calibrated models. Nothing the vanilla market quotes distinguishes them. The two ends are the two models of Example 21.7, whose independent simulation prices the same option at 0.4244 and 0.8402; the differences from 0.420 and 0.845 here are within the two runs' standard errors (about 0.005 and 0.008).

Section 21.11 returns to this option and explains the direction: a stochastic-volatility path that ends between the strike and the barrier is far more likely to have stayed below the barrier.

That is why the mixing parameter is not a nuisance. *It is the model risk, made into a dial*: a desk that prices a barrier at both ends of it has measured its own exposure to the modeling choice without building a second model, and the range (here a factor of two) is the number to take to a risk committee, not either end of it.

Who sets the dial, and how. The model is the standard on two kinds of desk, and they set ζ differently. In foreign exchange, where local-stochastic volatility has been the standard model for barrier and touch products since the mid-2000s, one-touch and double-no-touch options (which pay a fixed amount if the exchange rate touches a level, or if it never leaves a range, before expiration) trade actively enough to calibrate to: the desk picks the ζ that reproduces their prices, and the market has, in effect, priced the dial. In equities there is no such instrument. Barriers, autocallables and cliquets trade over the counter and are priced by the same desks that would need them as inputs, so ζ is set by judgment (from the historical volatility of implied volatility, from how the forward smile has behaved, or by house policy), and Section 21.11 will argue that the honest output is then the range rather than a point. The products most sensitive to the setting are the ones whose value depends on the forward smile: autocallables and the snowballs of Chapter 35, whose knock-in put has its barrier where the put is in the money, the reverse-barrier case on which Example 21.7 shows the models disagree.

Living with it. Two practical consequences follow from (21.11). Because L is a function of today's surface, it is not calibrated once and kept; it is re-derived whenever the surface moves, which in practice means daily, and the recalibration is the expensive part of running the model. And because the denominator is a conditional expectation estimated from finitely many paths, L is noisiest in the wings, where paths are few. A leverage function that looks jagged there is an estimator artifact, not a discovery, and practitioners smooth it before use. The delta and vega the model produces lie between local volatility's and Heston's, so the hedge, not only the price, moves with ζ.

21.10 CALIBRATION IN PRACTICE

Every model above has to be fitted before it can be used, and the fitting is where most of the practical damage is done. Three decisions matter more than the choice of model.

What to fit to. A surface contains hundreds of quotes of wildly different quality: the at-the-money one-month option trades continuously, the five-year 10-delta put trades twice a month. Fitting to all of them equally lets the least reliable quotes drive the parameters.

The standard remedies are to fit implied volatilities rather than prices (equivalently, to divide each price error by the option's vega) and to weight each quote by its liquidity, through bid–ask width or open interest. Dividing by vega is the most common choice, because it measures every error in volatility points, the unit in which options are quoted and bid–ask spreads are set, so a wing option's error counts as much as an at-the-money one. Weighting volatility errors *by*

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vega does the opposite, turning them back into price errors, which the at-the-money options dominate.

Fit quality against parameter stability. A model refitted every day to the whole surface will produce parameters that jump. Some of that is real; most of it is the optimizer trading one parameter against another along the flat directions of the objective function, which the overlapping effects in Table 21.2 create.

The consequence is not cosmetic. A book revalued with jumping parameters generates phantom profit and loss, and a desk cannot distinguish a genuine mark-to-market move from a calibration artifact. Standard practice is therefore to penalize movement away from the previous day's parameters (a regularization term), accepting a slightly worse fit in exchange for parameters that mean something over time.

Which risk the calibration is for. A model fitted to the whole surface is fitted to no part of it especially well. If the book's exposure is concentrated in one-year 25-delta puts, that is what the calibration should target, and the fit elsewhere matters little.

The objective should match the hedge: a model that reproduces every quote and mishedges the one position on the book has failed. Example 21.6 shows the objective alone moving the fit.

Example 21.6 Two fits, one surface

Fit Heston to the jump model's surface of Example 21.3, which no Heston parameters reproduce, at one month, three months and one year, strikes 80 to 120. One fit minimizes squared price errors; the other first divides each price error by the option's vega, which to first order makes it an implied-volatility error.

Price errors are dominated by at-the-money options, which have the largest vega, so a given volatility miss costs most there. The one-month 80 put is worth \$0.05, so the first fit leaves its implied volatility 7.08 points too low while missing the one-month at-the-money option by 0.12 points. The second cuts the wing miss to 4.01 points at a cost of 0.19 points at the money, and the average absolute one-month miss falls from 1.72 points to 1.11. Weighting volatility errors *by* vega instead nearly gives back the first fit (6.67 points in the wing).

Neither is right in general. The first is right for a desk whose book is at-the-money straddles; the second for a desk whose book is in the wings, like the structured-product issuers of Section 19.8. The objective encodes an assumption about the book; choose it rather than inherit a software default.

21.11 MODEL RISK MADE CONCRETE

Section 19.12 asserted that two models fitting today's surface perfectly can price the same barrier option far apart. That assertion is now checkable, because Sections 21.2 and 21.4 give two models that agree on every vanilla by construction.

Example 21.7 The same vanillas, two barrier prices

Take the Heston parameters of Figure 21.2, and derive the local-volatility surface from Heston's own vanilla prices by (21.2). The two models now agree on *every* European option: priced by simulation, the one-year 100-strike call comes to 6.766 under Heston and 6.767 under local volatility, against the semi-analytic 6.751, equal within Monte Carlo error.

Now price barrier options under each, one year, monitored daily, using the *same random numbers* for both models so that the difference is measured directly rather than as the difference of two noisy prices. This is Section 20.10's common-random-numbers technique.

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Barrier	Heston	Local vol	Difference (± s.e.)	Relative
Down-and-out call, strike 100				
80	6.5286	6.5821	+0.053 ± 0.015	+0.8%
85	6.2180	6.2962	+0.078 ± 0.015	+1.3%
90	5.5388	5.6138	+0.075 ± 0.015	+1.4%
Up-and-out call, strike 100				
130	5.8138	5.5787	−0.235 ± 0.015	−4.0%
120	4.0046	3.4193	−0.585 ± 0.013	−14.6%
115	2.4071	1.7138	−0.693 ± 0.010	−28.8%
110	0.8402	0.4244	−0.416 ± 0.005	**−49.5**%

On the down-and-out the two models agree to within about one percent, a difference a desk could ignore. On the up-and-out they disagree by nearly half the price.

So model risk is not a property of "barrier options." It is a property of where the payoff lives relative to where the models disagree. A down-and-out call struck at the money pays off in a region well above the barrier where the two models describe the distribution almost identically, because the vanillas pin it down. A reverse knock-out (the case Section 19.12 described, as this up-and-out call is) pays off only in the narrow band between the strike and the barrier, and that band is exactly where the surface constrains the models least: no vanilla option isolates the joint behavior of "finish in this band" and "never touch the top of it."

The direction is worth stating carefully, because a plausible-sounding argument gets it backward. One might reason that local volatility is low at high spot (Example 21.1 gives 12.0% at a spot of 110 a year out, against 16.2% at 100), so paths approach the upper barrier slowly, knock out less, and the option should be worth *more*. It is worth less, and the premise is false: local-volatility paths knock out more often (44.8% survive against 52.7% in Example 21.5). Both models put the same probability on every terminal level, so what matters is survival *given* where a path ends. Of the paths finishing between 108 and 110, 39% have never touched 110 under Heston and 14% under local volatility; of those finishing between 100 and 102, 77% against 57%. Under Heston some paths are quiet and some wild, and the quiet ones reach the band without touching the barrier; local volatility reproduces the same terminal distribution with one volatility at each spot and date, so it has no quiet paths to offer.

So the vanilla market does not price this product, no recalibration will make it, and the number to report is the range the mixing parameter of Example 21.5 produces, which matters only where it is wide; this table says which products those are, before the trade.

Case Study 21.2 When the model choice showed up in the profit and loss

Autocallables are the clearest case of this chapter's model choice paid for in cash.

The structure, described in Section 12.12 and Chapter 35, embeds a down-and-in put that the investor sells to the issuer and a series of discrete observation dates. Its value depends on three things the vanilla surface does not determine: how the smile evolves between observation dates, volatility near the barrier, and, for notes on several underlyings, how their correlation rises in a decline.

Issuers hedge these books dynamically, and the hedge is computed from a model. Through the 2010s many issuers priced them on local volatility, because it fitted the vanillas exactly and the fit was auditable. Example 21.2 identifies the flaw in that assumption. The issuer is long the knock-in put but short the coupon and early-redemption payments, which are digitals, and a steeper forward skew raises the value of both, so the sign of the net exposure depends on the barriers and the coupon. On many notes the digitals dominate, the issuer loses when forward skew steepens, and local volatility, which predicts

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a flattening forward smile, *understates* what the issuer owes; on any note it puts the vega profile in the wrong maturities.

The consequence stays hidden while the market is calm and appears when the index falls toward the knock-in region, when the forward smile matters most and the misspecified hedge has to be adjusted quickly. Section 18.12 described the resulting flow (issuers buying index futures into the decline, then selling, and withdrawing that bid, at the barrier); the model question is why the hedge needed adjusting then rather than earlier.

Desks moved to local-stochastic volatility for these books, and the mixing parameter of Section 21.9 is now a first-order risk number rather than a technical setting. They moved because the simpler model's error had a price, discovered in a drawdown.

The general lesson is that a model's failure mode determines what it may price: local volatility fits and mis-forecasts, stochastic volatility forecasts and mis-fits, and products that depend on both need the two combined, with the remaining uncertainty in the combination's free parameter.

21.12 CHOOSING A MODEL

The decision is not about realism. It is about which failure the trade is exposed to.

Marking a vanilla book → no model at all: interpolate the surface of Section 19.12.

Pricing a barrier or an autocallable → local-stochastic volatility, and report the range across the mixing parameter, since here the choice has a material cost.

Forward-starting products, cliquets → stochastic or rough volatility. Local volatility gets the forward smile structurally wrong (Example 21.2), and the forward smile is the product.

Very short-dated options → a smile fitted directly on a business-time clock (Section 23.9), with rough volatility or jumps as the check that some process could produce it. Classical stochastic volatility cannot make the skew, and Chapter 23's 0DTE market lives entirely in this region.

Swaptions and rate smiles → SABR, which Section 29.9 sets out and calibrates.

Two habits follow, parallel to Chapter 20's.

Calibrate to what you will hedge with. Fit to the instruments that will actually be in the hedge, and check the fit there rather than in aggregate.

Price it twice, with different models. The spread is not noise to be averaged away; it is the number the trade's model risk is measured in, and Example 21.7 shows how far two models can differ while agreeing on everything observable.

21.13 WHERE THIS CAME FROM

The Black–Scholes–Merton formula was accurate enough that its failures took more than a decade to become commercially serious, and the crash of 1987 made them so. The equity smile appeared and did not go away.

Three answers were soon available, two of them already in print before the crash, and all three survive. Merton (1976) had already added jumps, which produce a smile at short maturities and were the natural response to a market that had just gapped. Hull and White (1987), with Wiggins (1987) and Scott (1987) independently, made volatility itself stochastic (the standard reference, and still the paper to read for the structure of the problem). Stein and Stein (1991) added a tractable mean-reverting form, and Heston (1993) displaced them all—not because its dynamics are better but because its characteristic function is known in closed form and the price is therefore a single integral. Carr and Madan (1999) made that computational accident decisive by showing how to invert such transforms fast.

The other answer was to give up choosing the dynamics. Dupire (1994), with Derman

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and Kani (1994) and Rubinstein (1994) arriving from trees, showed that exactly one local volatility surface reproduces every quoted vanilla. This chapter is largely about the price of that exactness. Bates (1996) combined jumps with stochastic volatility, the form in which jump models are now mostly used. Hagan and co-authors (2002) then gave SABR, whose asymptotic formula made a stochastic-volatility smile quotable from a formula and became the market's language for rates. And Gatheral, Jaisson, and Rosenbaum (2018) reopened the question by measuring that volatility is rougher than any of these assume.

Section 21.11's method, reading the spread between models fitted to one vanilla surface as the model risk, is Hull and Suo's (2002). They applied it to the implied volatility function model (local volatility) and found what Section 21.11 finds: agreement on every quoted vanilla leaves some exotics nearly fixed and barrier options far apart. The contribution was to make model risk a measurement rather than a warning, and it recurs in Chapters 34 and 39.

SUMMARY

A volatility model must fit today's surface and predict how it moves. Nothing in the vanilla market pins down the second requirement, and that degree of freedom is what every model in this chapter spends differently.

Local volatility makes σ a deterministic function of price and time, read directly off the surface by Dupire's formula, so the fit is exact by construction and the market stays complete. The local volatility is about twice as steep in strike as the implied volatility, because an implied volatility is an average of the local volatilities a path visits, and averaging flattens. Its failure is dynamic: the local-volatility skew decays steeply with maturity, so the model predicts a nearly flat forward smile where the market's is roughly stationary, and forward-starting products are mispriced and hedged with the wrong vega.

Heston makes variance a mean-reverting square-root diffusion correlated with the price. Correlation produces the skew, vol-of-vol produces the curvature, and mean reversion flattens both. No parameter moves one feature alone, so calibration is a genuine optimization and individual fitted parameters are not separately meaningful. Jumps produce short-dated skew no diffusion can, but that skew collapses with maturity faster than the market's does.

The at-the-money skew follows a power law $\psi(T) \sim T^{-0.4}$ from about a month to a few years; below a month the evidence is disputed. Heston saturates instead, rising too little toward the short end, and decays too fast at the long end: missing both ends at once is a structural failure, not a parameter problem. Bates, which runs a variance diffusion and jumps together, brings the short end close to the market's and pays for it at the long end. Eight parameters place the fit; they do not produce the law.

Rough volatility is the most economical resolution, though each leg of its case is now contested. Log-volatility behaves like a fractional Brownian motion with $H \approx 0.1$, which yields $\psi(T) \sim T^{H-1/2} = T^{-0.4}$ from a single parameter, and roughly the same H appears in realized volatility and in option prices. Rough Bergomi, given the market's forward variance curve, needs three parameters and fits index surfaces better than Heston. What made it usable was fast simulation, Markovian approximations that restore a state vector, and neural-network calibration: the Markov assumption it displaced was a computational convention, never an economic claim.

Exotics desks run local-stochastic volatility, a stochastic variance process multiplied by a leverage function calibrated to make the vanilla fit exact, and the mixing parameter between the two is the model risk turned into a dial. Two models agreeing on every vanilla can still price the same barrier far apart, because they differ in how the price travels to each terminal distribution, so the output is a range, not a price. Calibration is a modeling choice too: the objective decides which quotes the fit honors and should match the book's hedge, and parameters that stay stable

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from day to day are bought with a worse fit.

FURTHER READING

Gatheral's *The Volatility Surface* treats local and stochastic volatility together and is the natural next step from this chapter. Dupire (1994) is short and worth reading in the original for the argument rather than the formula. On Heston, the original 1993 paper sets out the model, but the numerically important material (branch cuts in the characteristic function, and the "little trap" formulation that avoids them) is in Albrecher and co-authors and in Gatheral, and any implementation that skips it will produce silently wrong prices at long maturities. Hagan and co-authors (2002) is the SABR paper; the wing corrections that repair its negative densities are later and scattered. For rough volatility, Gatheral, Jaisson, and Rosenbaum (2018) is the empirical paper that started it and is explicit about what was measured; Bayer, Friz, and Gatheral (2016) introduce rough Bergomi, and El Euch and Rosenbaum (2019) give rough Heston its characteristic function, which is the paper to read for how the affine machinery survives roughness. Bates (1996) is the paper that adds jumps to stochastic volatility, and Duffie, Pan and Singleton (2000) the general affine framework that explains why its characteristic function factorizes. For the local-stochastic model, Guyon and Henry-Labordère's *Nonlinear Option Pricing* (2013) is the account of the particle method by the people who made it practical, and is clear that the difficulty is the conditional expectation rather than the calibration. Guyon and Lekeufack (2023) is the path-dependent volatility paper behind the frontier remark in Section 21.8. On calibration itself, Christoffersen and Jacobs (2004) is the paper to read before choosing an objective function: fitting the same model to dollar price errors and to implied-volatility errors produces different parameters and different out-of-sample performance, which is why Section 21.10's weighting question is a modeling decision rather than a numerical one. The literature on Markovian approximations and on neural-network calibration is recent and moving, and a reader should expect the state of the art to have advanced.

PRACTICE QUESTIONS

21.1 Explain why the denominator of Dupire's formula (21.2) is proportional to the implied density, and what that implies about when a local-volatility model exists.

21.2 Explain the "rule of two" using the averaging argument, and state whether a steep local-volatility surface is evidence of a calibration error.

21.3 Using Example 21.2, explain why a local-volatility model misprices a forward-starting option and in which direction.

21.4 A Heston fit returns $\rho = -0.05$. What shape of smile does this imply, and in which market would you expect to see it?

21.5 Explain why raising σ_v changes the at-the-money level as well as the curvature, and what this implies about interpreting a single fitted parameter.

21.6 State what jumps produce that a diffusion cannot, and state the corresponding defect in the term structure they generate.

21.7 Define the at-the-money skew $\psi(T)$, state the empirical power law it obeys, and explain why a classical stochastic-volatility model cannot reproduce a skew that keeps steepening as maturity shortens.

21.8 State the relationship between the Hurst parameter and the skew term structure, and give the value of H that reproduces the observed exponent.

21.9 Explain what the leverage function in a local-stochastic volatility model does, and what

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the mixing parameter measures.

21.10 Two models reproduce every quoted vanilla price and disagree on a barrier. Explain how this is possible without either committing an arbitrage, and state what a desk should report as a result.

21.11 Explain why a calibration objective is usually stated in implied volatility (price errors divided by vega) rather than in price, what a price-error objective would be blind to, and what a desk gains and gives up by penalizing day-to-day parameter movement.

FURTHER QUESTIONS

21.12 Example 21.1 finds a local-to-implied slope ratio of 1.75 rather than exactly 2. Derive the averaging argument that produces the factor of two, identify the approximation under which it is exact, and explain what features of the surface make the realized ratio depart from it.

21.13 Section 21.3 argues that local volatility gets the forward smile wrong while fitting today's surface exactly. Construct two processes that produce identical terminal distributions at every maturity but different forward smiles, and use them to explain precisely what information about the process the vanilla surface does not contain.

21.14 Table 21.2 shows that a rise in κ and a fall in σ_v both flatten the smile. Show that they do so through different mechanisms, explain why this makes them hard to identify jointly from one maturity, and describe what data would separate them.

21.15 Section 21.7 shows Heston too flat at the short end and decaying too fast at the long end simultaneously. Explain why adding parameters cannot fix both, and state what this implies about the practice of calibrating a single model to an entire surface.

21.16 Equation (21.8) gives $\psi(T) \sim T^{H-1/2}$. Explain why $H < 1/2$ makes the skew diverge as $T \to 0$, discuss whether an infinite short-maturity skew is economically sensible, and state what the model is really claiming about very short-dated options.

21.17 The case study in Section 21.8 argues that the Markov assumption in volatility modeling was a computational convention rather than an economic claim. Identify two other modeling conventions in this book that may be of the same kind, state what would have to become cheap for each to be abandoned, and explain how one would tell the difference between a convention and a claim.

21.18 Table 21.3 holds the variance rate fixed while the jump parameters change, and the one-year skew barely moves while the one-month skew moves a great deal. Using the fact that a jump component's contribution to skewness decays like $T^{-1/2}$ while a correlated diffusion's grows with maturity at first, explain why, and state what this implies about which options a Bates calibration should be run on to pin down the jump parameters.

21.19 Example 21.5 shows the up-and-out price rising monotonically with the mixing parameter while the vanilla fit is unchanged. Explain the mechanism, state why a foreign-exchange desk can calibrate the mixing parameter from market prices while an equity desk cannot, and describe what an equity desk should report instead of a single price.

21.20 Example 21.7 produces two barrier prices from models that agree on every vanilla. Explain why the local-volatility price is the lower of the two, construct a portfolio of vanillas that would partially hedge the difference, and explain why no portfolio of vanillas can hedge it completely.

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Chapter 22

Volatility Products: VIX, Variance Swaps, and Dispersion

Section 12.4 said that a trader who wants volatility exposure must delta-hedge an option and that the residual is a bet on realized against implied. Chapter 18 came within a step of saying what that residual is: a delta-hedged position is long gamma and short theta, and Section 18.4 found that it comes out level on a day the stock moves by one daily standard deviation at its implied volatility. Section 22.1 takes the step. Both descriptions make volatility something you get *left with* after removing what you did not want.

This chapter is about the instruments that deliver it directly. A variance swap pays realized variance against a fixed rate with no delta to hedge and no dependence on where the price moved. A VIX futures contract references an index computed from option prices. A dispersion trade isolates correlation. Each of these turns a by-product into a contract, and each is imperfect in a different and instructive way.

Why anyone wants such contracts shapes the whole market. Everyone who holds equities is short volatility whether they know it or not: drawdowns arrive when volatility rises, and the hedges that would soften them cost the most at exactly that moment. So there is a natural buyer of volatility, the holder of equity, and a natural seller, the intermediary or fund paid to bear what the holder wants to shed. Section 19.10 put the price of that transfer at two to four points of implied over realized volatility on equity indices and called it the volatility risk premium. The instruments here let that premium be bought and sold on its own, without the delta, the strike and the rebalancing that come with an option. That is their appeal. It is also why the sellers tend to be crowded, and why, with investors in aggregate structurally short volatility through their equity, the episodes in this chapter (2008, February 2018, March 2020) recur in the same shape.

The route follows the order in which the market built the instruments: the variance swap and its caps (Sections 22.2 to 22.5), VIX, its futures and its products (Sections 22.6 to 22.8), then the dealer's view (Sections 22.9 to 22.13).

22.1 THREE WAYS TO OWN VOLATILITY

Delta-hedged options. Buy an option and hedge the delta. Section 18.1's dealer, hedging a written call, stood on the other side of this trade. The hedge cancels the delta term of Chapter 18's expansion, equation (18.1), and what is left are its gamma and theta terms: the buyer earns $\frac{1}{2}\Gamma(\Delta S)^2$ on each day's move and pays theta, whose cost over the day, with the interest term dropped, is $\frac{1}{2}\Gamma S^2\sigma^2\,\Delta t$ at the implied volatility σ. With the day's return written $\Delta S/S$, the day's profit is

$$\frac{1}{2}\Gamma S^2\Big[\left(\Delta S/S\right)^2-\sigma^2\,\Delta t\Big]$$

realized less implied variance, weighted by $\frac{1}{2}\Gamma S^2$. Summed over the position's life this is Section 12.4's residual, but weighted by gamma: it depends on *where* the underlying moved,

not only how much, and two paths with the same realized volatility pay differently. The identity holds for every option (a put, a butterfly, the straddle traders use because it starts close to delta-neutral); only Γ changes. It also holds σ fixed: if implied volatility moves, the vega term of equation (18.1), Section 18.5's vega profit or loss, is added.

Variance swaps. Contract directly on realized variance. There is no delta, no gamma weighting and no rebalancing. The exposure is exactly what the name says.

VIX futures. Trade an index constructed from option prices. This is the most accessible and the least direct: a VIX futures contract is not a claim on volatility but a claim on the *level of a computed index at a future date*, and Section 22.7 shows how far apart those can be.

The three are not substitutes, and most of the trouble in this market comes from treating them as though they were.

22.2 THE VARIANCE SWAP

A variance swap pays, at maturity,

$$N_{\text{var}}\left(\sigma_R^2 - K_{\text{var}}^2\right) \tag{22.1}$$

where σ_R^2 is the annualized realized variance of log returns over the life of the contract, K_{var} is the strike, quoted as a volatility, and N_{var} is the variance notional, paid per variance point (volatilities in percentage points, so a realized variance of 0.04 enters as $20^2 = 400$).

The vega notional convention. Variance notionals are awkward to think in, so the market quotes **vega notional**, the slope of the payoff in realized volatility, taken at the strike, so that a one-point move in realized volatility earns N_{vega} to first order. Since the slope of (22.1) at $\sigma_R = K_{\text{var}}$ is $2K_{\text{var}}N_{\text{var}}$,

$$N_{\text{var}} = \frac{N_{\text{vega}}}{2K_{\text{var}}} \tag{22.2}$$

so that, by (22.2), the payoff (22.1) becomes $N_{\text{vega}}(\sigma_R^2 - K_{\text{var}}^2)/(2K_{\text{var}})$. A desk saying it is "short \$100,000 of vega at 20" means it loses roughly \$100,000 if realized volatility comes in at 21 rather than 20—*roughly*, because the payoff is not linear in volatility: the exact loss is $100{,}000 \times (21^2 - 20^2)/(2 \times 20) = \$102{,}500$, and the extra 2.5% is the convexity that is the subject of Section 22.4.

Realized variance is defined, not observed. The contract specifies the observation times (daily closes), the return convention (log returns), the annualization factor (252), and, importantly, whether the mean is subtracted:

$$\sigma_R^2 = \frac{252}{n}\sum_{i=1}^{n}\left(\ln\frac{S_i}{S_{i-1}}\right)^2 \tag{22.3}$$

with S_0 the close on the trade date and S_i the official close on the ith of the n scheduled observation dates. That sum is the contract. It is what the confirmation says and what settlement is computed from, and no integral appears in any document. No mean is subtracted in it, and that is standard practice: it is what lets the option strip of Section 22.3 replicate the payoff, and it makes the measure differ slightly from a sample variance. These are not details: two contracts with the same strike and different terms are different contracts, and the cap that Section 22.5 describes, standard on single-stock contracts and found on index contracts since 2008, is one such term.

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22.3 REPLICATING VARIANCE

The result that makes this market work is that variance can be replicated from vanilla options held *statically*: no model, no volatility forecast, and nothing traded as the price moves except a stock position of constant dollar size.

The route is Chapter 12's. Section 12.8 built any piecewise-linear payoff from a bond, a forward, and options at its kinks; a smooth payoff is the limit of piecewise-linear ones, so it is built the same way from a continuous strip of options. Apply that to the smooth payoff $-2\ln(S_T/F)$, the **log contract**, with T the contract's maturity in years and F today's forward price for that date. Two facts combine, and each takes a paragraph.

Delta-hedging a log contract earns exactly realized variance. Let the price follow

$$\frac{dS}{S} = \mu_t\,dt + \sigma_t\,dW$$

with μ_t any drift and σ_t any volatility process (constant, stochastic, path-dependent); nothing below depends on which. Itô's lemma applied to $\ln S$ gives

$$d(\ln S) = \Big(\mu_t - \frac{1}{2}\sigma_t^2\Big)dt + \sigma_t\,dW, \qquad \text{so} \qquad \frac{dS}{S} - d(\ln S) = \frac{1}{2}\sigma_t^2\,dt$$

The gap between the simple return and the log return accumulates half the variance, whatever the drift and whatever the path. Integrating over the life of the contract and multiplying by $2/T$,

$$\frac{1}{T}\int_0^T \sigma_t^2\,dt = \frac{2}{T}\left[\int_0^T \frac{dS}{S} - \ln\frac{S_T}{S_0}\right] \tag{22.4}$$

The left side is realized variance, in continuous time: the quantity the contract's sum (22.3) approximates with daily closes, and the limit that sum tends to as the observations become dense, by the $dW^2 = dt$ of Section 14.7. The right side is a trading strategy: the integral is the profit, before financing and dividends, of holding $1/S_t$ of the stock at every instant (a continuously rebalanced position with a constant dollar exposure), and the logarithm is the payoff of a contract on $\ln S_T$. Realized variance is therefore two things a desk can hold, and no probability has entered.

The gamma of the log contract does the same job one rebalance at a time. Its payoff is $-2\ln(S_T/F)$, and its value at time t, V_t, is that payoff's risk-neutral expectation discounted to t. Take σ constant for the moment and write $F_t = S_t e^{(r-q)(T-t)}$ for the forward at t, with r the risk-free rate and q the dividend yield. Under the risk-neutral measure $\mu_t = r - q$, and integrating the $d(\ln S)$ equation above from t to T gives $\ln S_T$ the mean $\ln F_t - \sigma^2(T-t)/2$, so

$$V_t = e^{-r(T-t)}\,E_t^{Q}\Big[-2\ln\frac{S_T}{F}\Big] = e^{-r(T-t)}\Big[-2\ln\frac{F_t}{F} + \sigma^2(T-t)\Big]$$

and its gamma is

$$\Gamma = \frac{\partial^2 V_t}{\partial S^2} = \frac{2e^{-r(T-t)}}{S^2}$$

which is the payoff's own second derivative, $2/S^2$, multiplied by the discount factor. The dividend yield sits inside F_t and does not reach the second derivative. The daily gamma gain of Section 18.3 is therefore

$$\frac{1}{2}\Gamma(\Delta S)^2 = \frac{1}{2}\cdot\frac{2e^{-r(T-t)}}{S^2}\,(\Delta S)^2 = e^{-r(T-t)}\Big(\frac{\Delta S}{S}\Big)^2$$

the present value of the day's squared return, whatever the level of S. That is precisely the weighting that makes the summed gamma gains independent of where the moves happened: an

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ordinary option's gamma is concentrated near its strike, and the same move earns it more there than elsewhere. Carried to expiration at the risk-free rate, each day's gain is worth exactly the squared return, and that is the e^{rT} in (22.5) below: the strip's option prices are present values, and the replication multiplies them back up.

The log contract's payoff is replicated by a strip of options weighted $1/K^2$. So the contract can be built from what trades. Section 12.8 built piecewise-linear payoffs by placing an option at each kink, in a quantity equal to the change in slope there. A smooth payoff $f(S_T)$ is the limit: a kink at every strike, with slope change $f''(K)\,dK$. Expanding around the forward, with puts below it and calls above so that every option used is out of the money,

$$f(S_T) = f(F) + f'(F)(S_T - F) + \int_0^F f''(K)\,(K - S_T)^+\,dK + \int_F^\infty f''(K)\,(S_T - K)^+\,dK$$

For $f(S_T) = -2\ln(S_T/F)$ the pieces are

$$f(F) = 0, \qquad f'(F) = -\frac{2}{F}, \qquad f''(K) = \frac{2}{K^2}$$

a short position in $2/F$ forward contracts, which costs nothing to enter, and $2\,dK/K^2$ of an option at every strike.

Write σ^2_{var} for the fair variance swap rate: the strike K^2_{var} at which (22.1) is worth zero today, which is the risk-neutral expectation of realized variance, $\sigma^2_{\text{var}} = E^Q[\sigma^2_R]$. Take that expectation of (22.4). Under the risk-neutral measure the stock's expected return is $r - q$, so the stock position's expected profit is

$$E^Q\left[\int_0^T \frac{dS}{S}\right] = (r - q)\,T = \ln\frac{F}{S_0}$$

and combining it with the log term gives

$$\sigma^2_{\text{var}}\,T = -2\,E^Q\left[\ln\frac{S_T}{F}\right]$$

so the fair total variance $\sigma^2_{\text{var}}T$ is the expected payoff of the log contract. Take that expectation term by term in the expansion above: the forward term has zero expectation, and each option's expected payoff is its price grown at the risk-free rate, $e^{rT}P(K)$ or $e^{rT}C(K)$, with $P(K)$ and $C(K)$ today's prices of the put and call struck at K and expiring at T. The result is

$$\sigma^2_{\text{var}} = \frac{2e^{rT}}{T}\left[\int_0^F \frac{P(K)}{K^2}\,dK + \int_F^\infty \frac{C(K)}{K^2}\,dK\right] \tag{22.5}$$

which is (19.2), now derived rather than asserted.

Figure 22.1 shows what the $1/K^2$ weighting looks like across the strip: the low strikes carry most of the weight (a strike at 50 four times the at-the-money strike's, a strike at 200 a quarter of it), which is why a variance swap is far more sensitive to the downside wing than a trader used to at-the-money vega would expect.

Example 22.1 Building the strip, and what truncation costs

Take a flat 20% surface, one year, $S_0 = 100$, $r = 2\%$, no dividends (forward 102.02); the answer must be exactly 20%.

Build the strip from options out of the money relative to the forward, at strikes every 5 points from 50 to 200, weight each by $\Delta K/K^2$, and apply (22.5):

replicated volatility = **19.96%**

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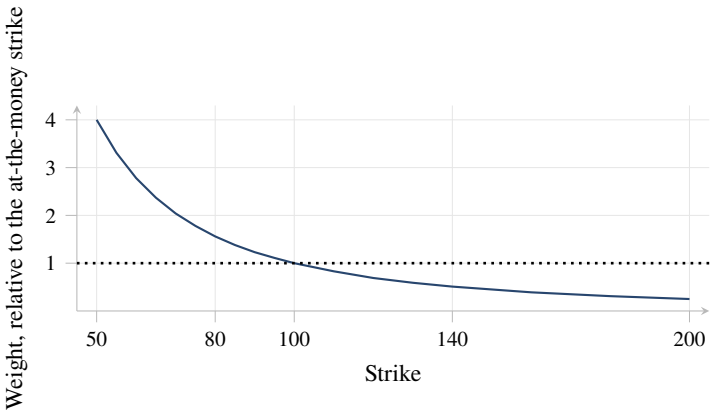


Figure 22.1: Weight on each option in the variance strip, relative to the at-the-money strike ($S_0 = 100$).

against a true 20.00%. The obvious suspect for the 4 basis points of error is the truncation of the integral at 50 and 200, and the table shows it is the wrong one.

The table widens the range and, in its last row, narrows the spacing:

Strike range and spacing	Replicated volatility
50–200, step 5	19.956%
30–300, step 5	19.956%
20–500, step 5	19.956%
10–800, step 2	20.015%

Two lessons follow. Widening the range beyond 50–200 buys nothing, because at a 20% volatility a one-year put struck at 50 is worth a fraction of a cent and contributes nothing despite its large weight. But the spacing matters, and not because the integrand is hard. The error depends on where the grid meets the forward, 102.02, at which the strip switches from puts to calls and the out-of-the-money price peaks in a kink: a strike sitting on that peak credits its whole cell with the peak value, so a strike exactly on the forward overshoots by +10 basis points, while this 5-point grid, straddling it, falls 4 short. The error shrinks with the square of the spacing, so the 2-point grid, whose strike at 102 is close to that grid's worst placement, is within 1.5 basis points.

The ranking of the two errors reverses when volatility is high. At 60% volatility the options beyond 50 and 200 are worth something real, the truncation error dominates, and Section 19.3's warning applies with force: the strip stops where the quoting rule stops, not where the integral does.

22.4 WHY VARIANCE AND NOT VOLATILITY

The obvious contract would pay realized *volatility*, which is what people think in. The market trades variance instead, and the reason is Section 22.3: variance is replicable and volatility is not. Volatility swaps exist, and trade mostly in foreign exchange, where they are often quoted in place of variance swaps, but they are priced off variance swaps with an adjustment, and Example 22.2 shows that the adjustment is large enough to matter.

Example 22.2 The variance–volatility spread

Suppose realized volatility over the period is, under the risk-neutral measure, lognormal with mean 20%. The variance swap strike is $\sqrt{E^Q[\sigma_R^2]}$ and the volatility swap strike is $E^Q[\sigma_R]$, and by Jensen's inequality the first exceeds the second.

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Vol-of-vol ν	Variance swap	Volatility swap	Spread
20%	20.40%	20.00%	0.40 pts
30%	20.92%	20.00%	0.92 pts
50%	22.66%	20.00%	2.66 pts
70%	25.55%	20.00%	5.55 pts

Since for a lognormal $E^Q[\sigma_R^2] = (E^Q[\sigma_R])^2 e^{\nu^2}$, the spread is $\bar{\sigma}\left(e^{\nu^2/2} - 1\right)$, where $\bar{\sigma}$ is the mean of realized volatility over the period and ν the standard deviation of its logarithm over the same period (a vol-of-vol measured over one period), so the spread grows quickly with ν. If the period's realized volatility has a log standard deviation of 40% to 60%, as is realistic for a one-year index swap, the variance swap strike sits one and a half to four volatility points above the volatility swap strike on the same underlying.

Two consequences follow. A variance swap strike is not a volatility forecast. Quoting a 22.66% variance swap and calling it "the market's expected volatility" overstates the forecast by 2.66 points. This is not a further overstatement on top of the excess over at-the-money volatility that Section 19.3 attributed to the skew: at-the-money implied volatility sits close to the volatility swap strike (a little below it when spot and volatility move inversely), so that excess is largely this same gap.

And the spread is itself a traded view. Buying a volatility swap against a variance swap is a position in vol-of-vol, one of the few ways to trade it without options on variance.

22.5 THE CONVEXITY IS NOT FREE

The same convexity that makes the strike high makes the payoff asymmetric, and Figure 22.2 shows it directly: the variance swap's payoff curves away from the straight line a volatility swap would pay, touching it at the strike and lying above it everywhere else. The gap grows with the square of the distance from the strike, equally on both sides, but realized volatility can rise much further above the strike than it can fall below it.

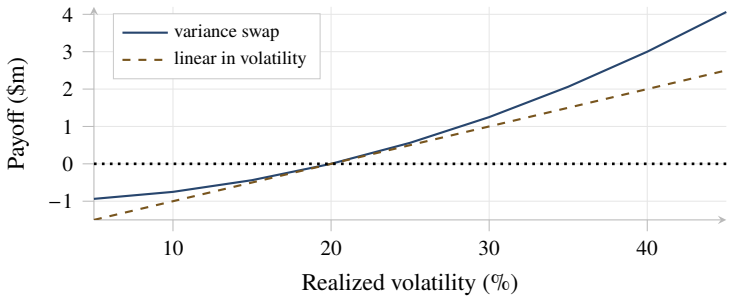


Figure 22.2: Payoff of a long variance swap struck at 20%, vega notional $100,000 per point.

Example 22.3 The buyer's convexity, and the seller's problem

A long variance swap struck at 20% with $100,000 of vega notional pays:

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Realized	Payoff	If linear in vol	Convexity gain
10%	−$750,000	−$1,000,000	+$250,000
15%	−$437,500	−$500,000	+$62,500
20%	$0	$0	$0
25%	+$562,500	+$500,000	+$62,500
30%	+$1,250,000	+$1,000,000	+$250,000
40%	+$3,000,000	+$2,000,000	+$1,000,000

The buyer gains from convexity in both directions, which is why the strike is above the volatility swap's. The seller pays for it in one direction only in practice, because realized volatility cannot fall below zero but can rise without limit: at 40% realized the seller loses $3 million on a position whose best realistic gain is the $750,000 at 10%. The seller's loss has no upper bound, while its gain can never exceed $1 million, the payoff at zero realized volatility.

In 2008 realized volatility on equity indices reached levels several times the strikes at which variance had been sold, and sellers, many of them dealers who treated the short variance as an offset to the options they were long, took losses far beyond anything the vega notional suggested.

The market's response was structural. A **cap** on realized volatility, conventionally 2.5 times the strike, had been standard on single-stock variance swaps from the start, because their gaps make the tail far worse; after 2008 that market nearly stopped trading even with its caps, and capped index contracts appeared too, though uncapped ones still trade. A capped variance swap is a variance swap less a call on realized variance struck at the cap, so it is no longer statically replicable by (22.5); the cap must be priced with a model of how forward variance moves, such as the rough-volatility models of Section 21.8.

So the instrument that was attractive because it needed no model acquired one, as a direct result of the payoff its replication made possible.

Case Study 22.1 March 2020, and what a cap is worth

The 2008 lesson of Example 22.3 was tested twelve years later, and where the repair had been written into the contract, it held.

Realized volatility on major equity indices ran above 80% annualized over the worst month of the pandemic sell-off, against variance swap strikes of 12% to 18% set in the weeks before it. On uncapped terms (the pre-2008 terms, on which much of the index market still traded), that is a loss no seller's capital was sized for.

Take a one-month swap struck at 15% just before the sell-off began, with $100,000 of vega notional, and suppose realized volatility over its life came in at 80%.

Uncapped, the seller loses

$$100{,}000 \times \frac{80^2 - 15^2}{2 \times 15} = 100{,}000 \times 205.83 = \$20.6 \text{ million}$$

on a position whose stated vega notional was one hundred thousand dollars. That factor of 206 is the whole problem with the instrument, and it involves no model error—it is what the payoff formula does when realized volatility comes in at more than five times the strike.

Capped at 2.5 times the strike, the loss stops at

$$100{,}000 \times \frac{37.5^2 - 15^2}{2 \times 15} = 100{,}000 \times 39.375 = \$3.9 \text{ million}$$

The cap reduced the loss by a factor of **5.2**, and, more to the point, it converted an unbounded exposure into a known one that could be capitalized against in advance.

Two observations follow.

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The repair was structural, not behavioral. No one forecast the pandemic. What the market did after 2008 was change the contract so that a failure to forecast would not be fatal, which is a different and more reliable kind of prudence.

The cost was paid in model risk. Example 22.3 noted that a capped variance swap is no longer statically replicable. The sellers of capped variance who came through March 2020 with bounded losses did so holding an instrument they had to price with Chapter 21's machinery rather than with a strip of options. They traded a known unbounded risk for a smaller risk they could only estimate, and on this occasion that was clearly the right trade.

22.6 VIX: AN INDEX, NOT A PRICE

VIX is the variance swap rate for a 30-day maturity on the S&P 500, expressed as a volatility. But the published number is not (22.5). It is a discrete sum over a strip that Cboe, the exchange that publishes the index, selects by a rule fixed in advance, and every peculiarity of the index is a property of that rule rather than of the theory: why it sits above at-the-money implied volatility, why it depends on how thickly the wings are quoted, why it can move on a day when no volatility changed. So the rule is worth writing out.

The forward comes from the options, not from the index. For each expiration, with T its time to expiration and r the matching risk-free rate, let K^* be the strike at which the call and the put are closest in price. The construction sets

$$F = K^* + e^{rT}\big(C(K^*) - P(K^*)\big) \tag{22.6}$$

which is put–call parity read backward. Parity holds at every strike, so any listed strike would return the same F from error-free quotes; Cboe uses K^* because there both options are near the money and liquid, and the quote error in $C - P$ is smallest. K^* is a choice of estimator, not a requirement of the identity. Neither the S&P 500's level nor a dividend forecast enters. Whatever the market believes about carry over the next month is already inside the option prices, and (22.6) extracts it rather than assuming it, so the forward VIX is built around is not the S&P 500's quoted level.

The strip is the out-of-the-money options around K_0. Let K_0 be the highest listed strike at or below F. The strip takes puts at strikes below K_0, calls above, and at K_0 itself the average of the two, so that every option used is out of the money except the pair at K_0, whose call is slightly in the money because $K_0 \le F$, and none is used twice. Each carries ΔK_i, half the distance between its neighbors. Walking outward from K_0 in each direction, the calculation stops after two consecutive strikes with no bid. Writing $Q(K_i)$ for the bid–ask midpoint of the option used at K_i (Cboe's notation, not the measure), the expiration's variance swap rate σ^2 (σ^2_{var} of Section 22.3) is

$$\sigma^2 = \frac{2}{T}\sum_i \frac{\Delta K_i}{K_i^2}\, e^{rT} Q(K_i) \;-\; \frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2 \tag{22.7}$$

The sum is (22.5) discretized. The subtracted term is not, and it is the one piece of the method that does not follow from Section 22.3.

Where that last term comes from. The replication splits the strip at F; the listed strip splits at K_0, which is a listed strike and so in general is not F. Expanding the logarithm about K_0 instead and taking risk-neutral expectations returns (22.5) with K_0 in place of F, plus the cost of having moved the split. With $x = F/K_0 - 1$, that cost is $2[\ln(1+x) - x] = -x^2 + O(x^3)$,

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the second-order remainder of a logarithm, and divided by T it is the subtracted term. Its size is bounded by $(\Delta K/K_0)^2/T$, which on the S&P's five-point grid near 4500 is about **0.005 VIX points** (at a VIX near 17, 24 days from expiration); Cboe carries it anyway because nothing guarantees a fine grid, and on a 25-point grid the bound is about 0.13 of a point. [TN22] derives it in full, states the exact correction the quadratic replaces, and measures it against the two larger errors that separate the published index from the variance swap rate.

Thirty days is a blend of two expirations. In general no option expires in exactly 30 days, so (22.7) is evaluated on the two nearest expirations that bracket it (the near term with more than 23 days to run, the next term with fewer than 37), and the results are interpolated *in total variance*, not in volatility:

$$\text{VIX} = 100\sqrt{\left[T_1\sigma_1^2\,\frac{N_2 - N_{30}}{N_2 - N_1} + T_2\sigma_2^2\,\frac{N_{30} - N_1}{N_2 - N_1}\right]\frac{N_{365}}{N_{30}}} \tag{22.8}$$

where σ_1^2 and σ_2^2 are (22.7) for the near and next expirations, N_1 and N_2 are the minutes from now to the near and next expirations, $N_{30} = 30 \times 1{,}440 = 43{,}200$ is the minutes in thirty days, $N_{365} = 525{,}600$ the minutes in a year, and $T_i = N_i/N_{365}$. The bracket is a linear interpolation of total variance to thirty days (variances over successive periods add; volatilities do not); the factor N_{365}/N_{30} annualizes it. The calculation counts minutes, not days, so the weights slide continuously through the session rather than stepping once a day. Weekly expirations have been eligible since 2014, which keeps the two brackets tight around 30 days; before that the window was wider and the interpolation did more work.

Example 22.4 Computing the index from a strip

Let the S&P 500 stand at 4500, with $r = 4.2\%$, options quoted on a five-point grid, and a downward-sloping skew: 17% at the forward, steepening to the downside. The near term expires in 24 days, the next in 31. The prices were generated with a 1.4% dividend yield, which the construction never sees.

Start with the forward. Equation (22.6) returns $F =$ **4508.29** for the near term, 8.3 points above the S&P's 4500 (the 24-day carry), so $K_0 = 4505$.

Then build the strip. The no-bid rule keeps 295 strikes, running from 3475 to 4945:

K	**Option**	ΔK	**Price**	$\Delta K e^{rT} Q/K^2$ ($\times 10^6$)
3500	put	5	0.060	0.024
4000	put	5	2.302	0.721
4300	put	5	20.540	5.570
4505	put/call avg.	5	78.347	19.355
4700	call	5	11.785	2.675
4900	call	5	0.207	0.043
Total over all 295 strikes				1020.478

Note how far the strip reaches on each side: 22.9% below the forward but only 9.7% above. The skew keeps deep puts bid long after the corresponding calls have lost their bids, so the strip is asymmetric before any truncation rule is applied.

Assembling (22.7):

$$\frac{2}{T}\sum \frac{\Delta K_i}{K_i^2}e^{rT}Q(K_i) = 0.0310395, \qquad \frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2 = 0.0000081$$

so $\sigma_1^2 = 0.0310314$ and $\sigma_1 =$ **17.62%**. The correction moved the answer by 0.0023 VIX points, consistent with the bound above, and a reminder that the term is a safeguard rather than a working

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part.

The 31-day expiration gives σ_2 = **17.82%**. The weights in (22.8) are ratios of the N, so their unit cancels; taken at a moment when the expirations are exactly 24 and 31 days away, $N_1 = 24 \times 1{,}440$ and $N_2 = 31 \times 1{,}440$ minutes, the weights are $(31 - 30)/(31 - 24) = 1/7$ and $6/7$, and

$$\text{VIX} = \mathbf{17.80}$$

closer to the 31-day figure than to the 24-day one, because 30 days is closer to 31.

Now thin the wings. Raise the price below which an option shows no bid, so the near-term strip, for example, runs from 3890 to 4835 instead of 3475 to 4945 (190 strikes rather than 295). VIX falls to **17.70**. No option changed its implied volatility; the index fell a tenth of a point because fewer options were quoted.

Three consequences follow from that experiment and from (22.7) and (22.8).

The $1/K^2$ weighting makes VIX exceed the at-the-money implied volatility, because low strikes carry more weight and the skew makes them expensive. Apply (22.5) to the one-year skew of Figure 19.1 (standing in for a 30-day skew), interpolating between its plotted strikes and holding the smile flat beyond them, with zero rates, over strikes from 50 to 200: it returns 24.38% against an at-the-money implied volatility of 20.00%, while a flat 20% smile returns exactly 20.00%. The 4.38-point gap is the skew premium, and it widens as the skew steepens. On the strikes that skew actually quotes, 70 to 130, the answer falls to 21.99% and the flat control to 19.56%, so truncation costs something even with no skew at all.

The truncation rule makes VIX depend on quoting depth. The calculation stops including strikes when it meets two consecutive strikes with no bid, so on a quiet day the wings are thin, the sum is truncated early, and the index is measured on a narrower strip than on a busy one. That is worth a tenth of a point in Example 22.4, and more when volatility is high and the wings matter most. On August 5, 2024, before the US open, VIX printed 65.73 while S&P 500 futures were down about 3 to 4%: the formula was reading the midpoints of very wide pre-open SPX quotes, with few trades behind them. It closed at 38.57. That is Example 22.4's effect at full scale.

And the 30-day constant maturity is an interpolation. Two listed expirations bracketing 30 days are blended in total variance, so VIX refers to a synthetic maturity that no traded option has. As one expiration rolls out of the window and another rolls in, the weights in (22.8) shift, and around the roll the index can move for reasons that have nothing to do with volatility.

Putting these together, VIX is a well-defined number that is not the price of anything that can be held. The strip that defines it today is not the strip that defines it tomorrow, so no position kept over time tracks it and it cannot be arbitraged against.

22.7 VIX FUTURES AND THE TERM STRUCTURE

A VIX futures contract is marked to market daily, like any futures contract, against a settlement price taken from trading at the close. Its final settlement, at expiration, is in cash against the **Special Opening Quotation** (SOQ): the VIX formula applied to the opening prices, set in a special opening auction on the settlement morning, of S&P 500 (SPX) options that expire exactly 30 days later, so no interpolation is needed; settlement falls on the Wednesday 30 days before that standard (third-Friday) SPX expiration. Like the index options of Section 10.3, the contract settles to a value that is computed rather than traded, and it can differ noticeably from both the previous close and the index a few minutes later. So the SOQ fixes the futures price on the final day, and the price converges to it. Before then the price is *not* tied to the index's current level, and the reason is worth being precise about.

A VIX futures contract is not a forward on VIX. Chapter 5's cost-of-carry argument requires

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that the underlying can be bought and stored, and VIX cannot be held. So there is no carry arbitrage linking the futures price to the spot index. A futures contract costs nothing to enter and is marked to market daily, so under the risk-neutral measure its price has zero expected growth (Section 13.10) and converges to the index at expiration: the futures price is the risk-neutral expectation of VIX_T, the index's value at the futures expiration T. Since VIX_T^2 is a 30-day variance rate and the square root is concave, Section 22.4's Jensen effect places the futures price below the square root of the forward variance swap rate, the fair variance rate for the 30 days that follow T (Section 22.12).

The practical consequence is the term structure. Because volatility mean-reverts (Section 19.6), the curve is usually upward sloping when spot volatility is low (which is most of the time) and inverts in stress: in Chapter 5's terms, contango in calm markets and backwardation in stressed ones. Mean reversion also makes the futures move less than the index: between the two curves of Figure 22.3 the index rises from 15.5 to 40 but the front contract only from 16.4 to 33, about two-thirds as far, and the five-month contract under a quarter as far. A hedge sized on spot VIX therefore under-hedges, and more so the further out the contract. Besides the issuers of Section 22.8's products, who hedge in the first two contracts, the curve's main traders are hedge funds positioning on its slope.

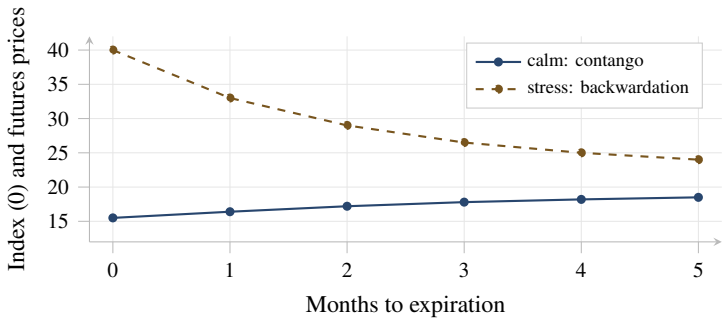


Figure 22.3: Representative VIX futures curves. Illustrative.

Example 22.5 What contango costs a long position

Using the calm curve of Figure 22.3: the front month is 16.4 and the second month 17.2.

A fund holding constant one-month exposure must roll: it buys the second month at 17.2, and one month later, if the curve has not moved, that contract has become the front month and is marked at 16.4.

$$\text{loss} = 17.2 - 16.4 = 0.8 \text{ points}, \qquad 0.8/17.2 = \mathbf{4.65\% \text{ per month}}$$

Compounded, $(1 - 0.0465)^{12} - 1 = -43.5\%$ **per year**, with the VIX index unchanged throughout.

This is Section 5.11's roll yield with no carry behind it, unusually large because the curve is unusually steep. It has two consequences, one for each side of the trade.

Long VIX futures products are structurally poor hedges to hold. An investor who buys volatility exposure as portfolio insurance and holds it for a year pays something like 40% of the position for the privilege, and the payoff must beat that before the hedge has done anything.

And short VIX futures products earn that roll as a carry, which is the volatility risk premium of Section 19.10 collected through a different instrument. Section 22.8 is about what happened to the people collecting it.

The premium and the roll are the same thing seen from two sides. Mean reversion alone would tilt the curve upward from a low spot level, but a futures price that merely forecast the index would cost a long holder nothing on average; the curve slopes up by *more* than that because sellers demand compensation for bearing the risk, and that excess is both the roll cost to the buyer and the

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compensation to the seller. Neither is a market inefficiency, and a strategy built on "harvesting contango" is not exploiting anything—it is being paid to take a risk that occasionally arrives.

22.8 VIX EXCHANGE-TRADED PRODUCTS

Section 18.12 counted February 2018 among the episodes in which hedging moved the market. Here is the mechanism from the product's side.

An **inverse VIX product** is an exchange-traded note (a bank's debt) or fund whose value moves opposite, day by day, to an index of a rolling position in the first two VIX futures contracts: it rises 1% for every 1% that index falls, and falls 1% for every 1% it rises. It is a short position in the front two VIX futures contracts, packaged so that a retail investor can collect the roll carry of Example 22.5 without a futures account, and by early 2018 the two largest, XIV and SVXY, held about $3 billion between them. Offering $-1\times$ exposure *daily* means the product must rebalance every day to restore that exposure, exactly as the leveraged funds of Section 14.9 do.

Example 22.6 Why an inverse product must buy into a rally

A fund with net asset value (NAV) A maintaining $-1\times$ exposure is short futures with notional A. Suppose futures return R over the day.

The NAV becomes $A(1-R)$, so the new target short is $A(1-R)$.

The existing short has grown to $A(1+R)$ of exposure.

So the fund must buy

$$A(1+R) - A(1-R) = \mathbf{2AR}$$

of futures at the close. It buys as the price rises. The rebalance is destabilizing, and it is Section 18.12's short-gamma feedback in a different form.

With roughly $3 billion in inverse VIX products, the rebalancing works out as follows:

Futures move	NAV after	Buying required
+10%	$2.70bn	$0.60bn
+25%	$2.25bn	$1.50bn
+50%	$1.50bn	$3.00bn
+96%	$0.12bn	$5.76bn

The demand grows as the fund shrinks. At a 96% move the products have lost 96% of their capital and must buy $5.76 billion of futures (nearly twice their original size) into a market that is already moving against them.

The buying was due in the fifteen minutes before the futures close, when the market normally trades a small part of its daily volume. The rebalance was not a large order in a deep market; it was a large order in a thin one, announced in advance.

Figure 22.4 puts the two quantities on one axis, and the crossing is the whole problem. The purchase the fund owes the market rises with the move while the fund itself shrinks, and on any day the futures rise by more than a third the buy exceeds everything the fund has left. February 2018 was the point where those lines had already crossed.

Case Study 22.2 February 5, 2018, from the volatility products

The S&P 500 fell about 4% that day—a bad day, not a historic one. VIX futures roughly doubled.

For a $-1\times$ product, a 100% move in the underlying futures is total loss: the NAV multiplier is $(1-R)$, and at $R = 1$ it is zero. The largest inverse product, XIV, had a term

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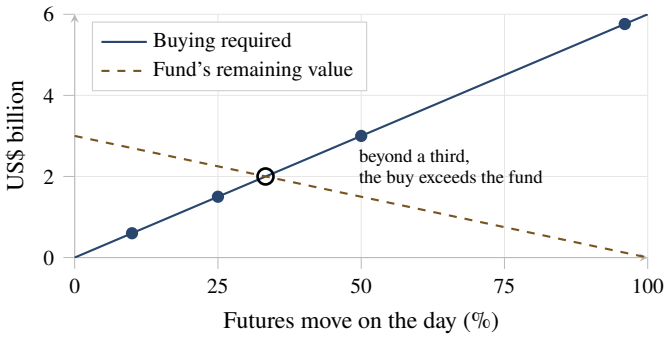


Figure 22.4: Daily rebalancing of a $-1\times$ inverse VIX product with \$3 billion of assets. Marks are the moves of Example 22.6.

in its prospectus allowing early termination if the value the issuer published during the day (its intraday indicative value) fell more than 80% from the prior close. It did. The note was redeemed early, and holders received \$5.99 a note, against more than \$100 a few days earlier. SVXY, a fund holding the futures directly rather than a bank's note and carrying no such clause, lost about 90% on the same day but survived, and its sponsor cut the exposure to $-0.5\times$ three weeks later, where it remains.

Three things about the episode deserve to be separated.

The product did what it said. A $-1\times$ daily inverse product loses everything on a 100% daily move, and the prospectus said so. This was not mis-selling in the narrow sense.

The rebalance amplified the move that destroyed it. Example 22.6 gives the size. The products had to buy into the close precisely because the price had risen, and their buying was a material fraction of the volume available. A product whose hedging requirement grows as it loses money is a product that helps arrange its own destruction.

And the underlying was already fragile. Two years of low realized volatility had built large short-volatility positions well beyond the exchange-traded products: in the option-selling strategies of Section 19.10, in dealers' books, and in funds that size equity by its volatility (risk-parity and volatility-target funds), whose allocations fall mechanically when realized volatility rises. All of these were positioned the same way.

The general lesson is the one Section 18.12 drew from five separate episodes and Section 35.7 meets again in the autocallable market. Crowding plus mechanical hedging is the mechanism, and the instrument is incidental. 1987's portfolio insurance, 2018's inverse exchange-traded products, and 2024's snowball knock-ins are the same structure with different wrappers, and the recurring feature is that the hedging demand is a known function of the price, in 2018 a published one, calculable in advance by anyone who cared to.

22.9 THE PREMIUM, SEEN THROUGH EACH INSTRUMENT

Section 19.10 reported that implied volatility exceeds subsequently realized volatility by two to four points on equity indices. In variance-swap form that premium is measured as VIX^2 against the next 30 days' realized variance, as Carr and Wu (2009) measured it, and it varies over time. The three instruments of Section 22.1 all collect that premium, and they do not collect the same amount. That is a useful check on whether one understands what each contract is, and Example 22.7 applies it.

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Example 22.7 One premium, three numbers

Suppose one-year implied volatility is 18% and realized comes in at 15%.

A short delta-hedged straddle earns, by Section 22.1, the difference between implied and realized *variance*, gamma-weighted along the path:

$$18^2 - 15^2 = 324 - 225 = 99 \text{ variance points (squared volatility points)}$$

but the realization depends on where the underlying spent its time. If the stock spends the year far from the strike, the position has little gamma when the moves occur and collects a fraction of the 99.

A short variance swap at 18% earns exactly 99 variance points. Per unit of vega notional, by (22.2) the short position's payoff is $(K_{\text{var}}^2 - \sigma_R^2)/(2K_{\text{var}})$, here

$$\frac{18^2 - 15^2}{2 \times 18} = \textbf{2.75 volatility points}$$

with no dependence on where the moves occurred. It is less than the 3-point gap in volatility because the payoff is convex in σ_R (Section 22.4): the short gains less when realized volatility falls short of the strike than it would lose if it overshot by the same amount. It is the straddle's quantity, collected cleanly, which is the reason the contract exists.

A short VIX futures position earns something else. It collects the roll of Example 22.5 (0.8 points a month on the illustrative curve), which is a claim about the *shape of the curve*, not about the gap between one-year implied and one-year realized volatility. The two are related, but they are different contracts on different quantities and can move apart.

Instrument	What is collected	What can go wrong
Delta-hedged straddle	Implied − realized variance, gamma-weighted	Path: the gamma may not be where the moves are
Variance swap	Implied − realized variance, exactly	Convexity: the loss is quadratic
Short VIX futures	The roll down the curve	The curve inverts, and does so before realized volatility rises

The third row contains the trap. The VIX curve inverts on the *expectation* of volatility, which is to say immediately, while realized volatility rises only as the moves actually occur. A short VIX futures position therefore takes its loss at once, while a short variance swap's settlement loss accrues only as the moves occur; its mark-to-market also loses as implied variance rises, but the payoff waits for the realized path. Anyone treating the two as interchangeable will find the correlation between them is high in calm markets and much lower in exactly the week it matters.

22.10 WHAT GOES WRONG IN A VARIANCE SWAP BOOK

The replication of Section 22.3 is exact under assumptions, and a dealer's business is the gap between those assumptions and the world. Four gaps matter.

Jumps. The log-contract argument assumes a continuous path. A gap of log size J adds J^2 to realized variance (before annualization), while the replicating portfolio collects $2(e^J - 1 - J)$, which differs by about $J^3/3$ and falls short on a downward gap. The replication under-hedges precisely on the days that matter, and this is the residual that Section 21.5 said cannot be hedged by any position taken beforehand.

Dividends. Realized variance is computed from price returns, and a price drops on the ex-dividend date. Unless the contract adjusts for it (index contracts usually do not; single-stock contracts often add back the dividend actually paid), a large special dividend registers as volatility that never happened.

Single stocks. Section 22.5 recorded that single-stock variance swaps, capped from the outset, nearly stopped trading after 2008. The reason is the convexity of Example 22.3 applied

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to an underlying that can gap 40% on an earnings miss or a takeover bid: the seller's loss, even capped, is quadratic up to the cap in a move whose distribution has a much fatter tail than an index's. Dispersion desks that would naturally use single-stock variance often use straddles or other options on the constituents instead, because the capped swaps trade thinly and the cap cuts off the payoff on the large single-stock moves the component leg is long. That substitution reintroduces exactly the path dependence the instrument was designed to remove.

The strip is not the contract. The dealer holds a fixed portfolio of options; the client holds a claim on realized variance. If the underlying moves far enough that the strip's strikes are all deep in or out of the money, the replication's gamma no longer matches the log contract's, and the hedge degrades exactly when realized variance, and so the payoff owed, is largest. This is the same truncation issue as Example 22.1, now as a running risk rather than a pricing error.

None of these is a flaw in the theory. Each is a place where a contract defined to be model-free requires a judgment, and a variance-swap desk is in the business of pricing those judgments.

22.11 DISPERSION

Section 16.2 named dispersion traders and Section 19.7 gave the arithmetic: index and single-stock implied volatilities together imply a correlation. The dispersion trade of Example 19.4, selling index volatility against single-stock volatility, is a position in that correlation, and with variance swaps it can be constructed cleanly.

Take the numbers of Example 19.4, its implied volatilities taken as one-year variance swap strikes: 50 equally weighted stocks at 30%, the index at 18%, implying a correlation of 34.7%; the 2.6-point gap Example 19.4 found at a realized 25% is 2.38 points as a variance payoff (first row below).

The weight on the component leg follows from Example 19.4's equal-weight form of (19.3). A parallel move in component variance σ_c^2 moves index variance σ_I^2, at fixed correlation, by the factor $\sigma_I^2/\sigma_c^2 = (0.18/0.30)^2 = 0.36$, so 0.36 units of variance notional spread equally over the 50 component swaps against 1 unit of index variance makes the position neutral to volatility and exposed only to correlation. In general the weight is $(K_I/K_c)^2$, the square of the index strike K_I over the component strike K_c.

Figure 22.5 plots the resulting profit against realized correlation, and the line is the point of the structure: it crosses zero at the implied 34.7% whatever the component volatility, and is linear in correlation with a slope set by component variance. Example 22.8 reads four outcomes off it and adds a calm one.

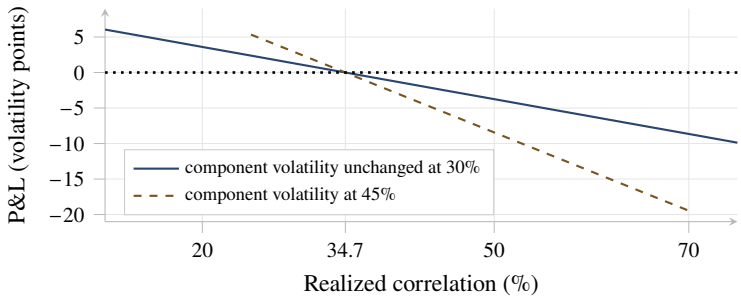


Figure 22.5: Profit on a variance-neutral dispersion trade against realized correlation.

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Example 22.8 What dispersion earns and what it loses

The position is short 1 unit of index variance at 18% and long 0.36 units of component variance at 30%.

Component vol	Realized corr	P&L	
30%	25.0%	+2.38	correlation below implied
30%	34.7%	0.00	correlation realizes at implied
30%	45.0%	−2.53	correlation rises, volatility unchanged
20%	20.0%	+1.60	calm: everything falls
45%	70.0%	**−19.46**	crash: volatility and correlation both jump

(P&L in volatility points per unit of index vega notional.)

The second row is the check. When realized correlation comes in exactly at the implied level, the trade makes precisely zero, whatever volatility does, as long as the components realize a common volatility. No parameter was tuned to produce it; it follows from the weighting, and it confirms that the position is a pure correlation bet.

The last row is the trade. The first row, correlation ten points below implied, earns 2.38 points; the crash loses 19.46. The ratio is one to eight, and the loss arrives in the same weeks as everything else in the portfolio.

Two causes combine. The correlation loss is scaled by realized component variance, and in a crash both rise together: at an unchanged 30% volatility, 70% correlation costs 8.65 points; at the 45% that accompanies it, 19.46. And correlation starting at 34.7% has nearly twice as far to rise as to fall, so the largest possible moves are the ones that run against the short index leg.

This is why the trade is described as "earning steadily and losing in blocks," and why it is run by desks that can size it against a capital base rather than by funds that must report monthly. Cboe publishes implied correlation as an index (COR1M, COR3M). In July 2024, with index returns concentrated in a few very large stocks, the three-month index fell to its lowest level since 2006 and the trade had become crowded; when volatility spiked on August 5 (Section 22.6), correlation rose with it, in the direction of the last row.

Correlation swaps. A correlation swap pays realized average pairwise correlation against a strike directly, which sounds cleaner and is used less. The reason is replication: dispersion is built from variance swaps that can themselves be replicated from options, whereas a correlation swap cannot be replicated, and the dealer who writes one hedges it only approximately, with a dispersion position whose payoff is a variance-weighted proxy for the same correlation. The market trades the replicable proxy rather than the clean instrument, which is the same preference that made variance rather than volatility the standard contract. The buyers are mostly structured-product dealers, short correlation from worst-of notes (Section 34.11); the writers are hedge funds and dispersion desks already holding that view through variance.

22.12 FORWARD VARIANCE AND OPTIONS ON VOLATILITY

Section 21.8 observed that rough-volatility models specify the forward variance curve, today's fair variance rate for each future window, directly. That curve is a traded object, and this is where it trades.

A **forward-starting variance swap** contracts on variance over a future window, and it is replicated by a spread of two spot-starting variance swaps: long the longer, short the shorter, weighted by their maturities. Since both legs are replicable from options, so is the forward: the whole forward variance curve is spanned by today's vanilla surface, and it needs no storable underlying, which Chapter 5's forward prices do. Dealers use it to hedge the forward-volatility exposure of the cliquets and autocallables they issue (Section 34.12, Chapter 35), and volatility funds to take term-structure positions. Example 22.9 works one through.

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Example 22.9 Reading the forward variance off two spot swaps

Six-month variance swaps are quoted at 18% and one-year at 20%. Total variance to each date is TK^2_{var}: $0.5 \times 0.0324 = 0.0162$ and $1 \times 0.0400 = 0.0400$. The variance over the second half-year is the difference, spread over the half-year it covers:

$$\sigma^2_{\text{fwd}} = \frac{1 \times 0.0400 - 0.5 \times 0.0324}{1 - 0.5} = \frac{0.0238}{0.5} = 0.0476, \qquad \sigma_{\text{fwd}} = \mathbf{21.82\%}$$

The forward sits above both spot strikes, as a forward rate sits above both spot rates on an upward-sloping curve in Chapter 4, and for the same reason: a two-point rise in the term structure over a horizon that is only half the maturity has to be made up entirely in the later half. The check is the telescoping identity: the two half-year variances average to the one-year, $(0.0324 + 0.0476)/2 = 0.0400$.

Long one unit of the one-year swap and short half a unit of the six-month swap, in variance notional, is long 0.5 units of forward variance; scaled to one unit, long 2 against short 1, which is what "weighted by their maturities" meant.

What is *not* spanned is anything nonlinear in future variance. VIX futures (a square root of variance), options on realized variance and on VIX, and the caps of Section 22.5 all require a model of how variance moves, not merely its expectation. This is the boundary Chapter 21 drew.

VIX options are the largest instance of this. They are European and settle in cash to the same SOQ as the futures, and their forward is the futures price for the same expiration, not spot VIX, so put–call parity runs against the futures. Pricing them requires a joint model of the S&P 500 and of the volatility surface from which VIX is computed. Their implied volatilities carry an upward skew, the reverse of the equity skew, because VIX spikes up and not down, and the market pays more for the right tail of a quantity that spikes. Tail hedgers buy the out-of-the-money calls as crash protection; volatility sellers and dealers write them. The VIX formula applied to a strip of VIX options gives VVIX, the implied volatility of VIX itself. Fitting the SPX smile and the VIX smile with one model is the joint calibration problem of Section 21.8, and the steep VIX skew is what makes it hard: it demands a volatility of volatility the SPX smile alone does not reveal.

22.13 WHO TRADES WHAT

Buyers of volatility are the protection buyers of Section 16.2 using a different instrument. A pension fund buying VIX calls or entering a long variance swap is buying convexity against its equity book. It pays the premium of Section 19.10 and the roll of Example 22.5, and it does so knowingly: the alternative is holding less equity.

Sellers of volatility are collecting that premium. They divide into those who can hold the position through a drawdown (dealers with capital, insurers, and endowments) and those who cannot, which is most of the rest. Allianz Global Investors' Structured Alpha funds show a seller less hedged than it said: they sold index options against crash puts their investors were told they held, but the managers had bought cheaper puts further out of the money, and in March 2020 funds that had run more than $11 billion lost over $7 billion. The firm's US arm pleaded guilty to securities fraud in May 2022 and paid about $6 billion.

Dealers intermediate and end up long or short depending on what clients trade, hedging with the vanilla strip of Section 22.3; their business is the *difference* between that strip and the contract (Section 22.10).

Dispersion desks trade correlation, as Section 22.11 described, and are the natural counterparty to index protection buyers: the index put demand that steepens the index skew is what makes implied correlation high enough to be worth selling. Worst-of note issuance is a

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second source of that flow: the investors are long correlation, so the issuing dealers are short it (Section 34.11) and buy it back by buying index volatility against single-stock volatility, which lifts implied correlation further.

Volatility-target and risk-parity funds do not trade volatility products at all, but they are among the largest volatility *exposures* in the market, because their equity allocation is a decreasing function of realized volatility. A fund targeting 10% volatility holds equity in proportion $10\%/\sigma_R$, so realized volatility doubling from 12% to 24% halves its equity weight from 83% to 42%; a hypothetical $100 billion of such funds then sells about $42 billion of equity. They are mechanically short volatility without holding a single option, and Section 22.8's snapshot showed what that means when realized volatility jumps.

Investors, taken together with their equity, are structurally short volatility, and the premium is the price of shedding that risk. What the episodes in this chapter show is that the capacity to bear it is smaller than the appetite to collect it, and the gap is discovered at inconvenient moments.

22.14 WHERE THIS CAME FROM

The instruments in this chapter rest on a result that needs no model at all. Breeden and Litzenberger (1978) showed that the option chain reveals the terminal distribution outright, and Carr and Madan later showed that any payoff written on that distribution can be replicated from options. Neuberger's log contract (1994), the payoff whose delta-hedged profit is exactly realized variance, is what variance swaps were built on, and Demeterfi, Derman, Kamal and Zou (1999) wrote the replication out in the form desks use, the source of Section 22.3's $1/K^2$ weighting. The CBOE rebuilt VIX on the same replication in 2003, which is why the index is a variance calculation rather than an average of implied volatilities, and Carr and Wu (2009) then measured the premium the trade earns, Section 19.10's premium in variance terms.

SUMMARY

Volatility can be owned three ways (delta-hedged options, variance swaps, VIX futures), and they are not substitutes. The first pays realized minus implied variance weighted by gamma, so it depends on where the underlying moved; the second pays realized variance exactly; the third is a claim on the level of a computed index.

A variance swap pays $N_{\text{vega}}(\sigma_R^2 - K_{\text{var}}^2)/(2K_{\text{var}})$ on a realized variance the contract defines. It can be replicated statically from out-of-the-money options weighted $1/K^2$, because delta-hedging a log contract earns realized variance on every path and the log contract's payoff is replicated by that strip. Variance rather than volatility is traded because variance is replicable; the variance swap strike exceeds the volatility swap strike by an amount that grows quickly with the volatility of volatility, so it is not a volatility forecast. The same convexity makes the seller's loss quadratic: after 2008 the single-stock market, capped from the start, shrank, and capped index contracts appeared too; a capped contract needs a model, and March 2020 showed what a cap is worth.

VIX is the 30-day variance swap rate on the S&P 500 as Cboe's rule computes it: above at-the-money implied volatility because the $1/K^2$ weights load on the low strikes the skew makes expensive, dependent on quoting depth because of the truncation rule, and an interpolation between two expirations. It is a well-defined number that is not the price of anything that can be held. VIX futures are not forwards, because VIX cannot be held; rolling a long position on a contango curve loses steadily, at a rate that compounds over a year to a large share of the position, and that cost to the buyer is the premium to the seller.

An inverse VIX product with net assets A must buy $2AR$ of futures when futures return R, so its hedging demand grows as it loses money; on February 5, 2018 that demand overwhelmed

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the liquidity at the close. Dispersion sells index variance against component variance weighted $(K_I/K_c)^2$, makes zero when correlation realizes at the implied level, and loses many times its normal gain in a crash, because correlation rises when everything falls and the loss on it is scaled by component variance, which rises with it. The forward variance curve is spanned by today's vanilla surface; anything nonlinear in variance is not, and needs the models of Chapter 21.

FURTHER READING

Demeterfi, Derman, Kamal and Zou's paper on variance swaps is the original practitioner derivation of (22.5); it works the replication through in a way this chapter compresses. Carr and Lee survey what can and cannot be replicated in volatility, and their survey is the place to go for the volatility-swap adjustment stated properly rather than approximated. Cboe's VIX methodology documents the construction including the truncation rule and the roll, and is worth reading precisely because the index's peculiarities are all in the mechanics rather than the theory. On February 2018, the Bank for International Settlements' analysis in its March 2018 Quarterly Review is the standard account; the mechanism of Example 22.6 is arithmetic, and the question it addresses is how much of the day's move the rebalance actually caused. Bergomi's book (2016) is the practitioner reference for forward-variance models and for options on variance and on VIX.

PRACTICE QUESTIONS

22.1 A variance swap is struck at 20% with \$100,000 of vega notional. Compute the variance notional and the payoff at 35% realized, compare it with what a linear-in-volatility contract would give, and explain what determines the gap between a variance and a volatility swap strike.

22.2 Explain why delta-hedging a log contract earns realized variance regardless of the path, what feature of the log contract's gamma is responsible, and why the replication strip weights options by $1/K^2$ rather than equally.

22.3 Using Example 22.1, state whether widening the strike range or narrowing the strike spacing is the more effective way to improve a replication at low volatility, and explain why the answer changes at high volatility.

22.4 Six-month variance swaps are quoted at 25% and one-year variance swaps at 22%. Compute the forward volatility for the second half-year, explain why it lies below both spot strikes, and state the positions in the two swaps, in variance notional, that make up one unit of forward variance.

22.5 Explain why variance swaps, single-stock ones above all, are now often written with caps, and what the cap does to the replication argument.

22.6 Explain why VIX exceeds the at-the-money implied volatility of 30-day S&P options, and why a VIX futures contract is not a forward contract on VIX in the sense of Chapter 5.

22.7 Using the curve of Figure 22.3, compute the monthly and annualized roll cost of a constant one-month long VIX futures position, and state who receives it.

22.8 A $-1\times$ inverse VIX product has \$2bn of NAV. Compute the rebalancing trade required if futures rise 30% in a day, state its direction, and explain why the requirement grows as the product loses money.

22.9 Explain why a dispersion trade weighted $(K_I/K_c)^2$ makes exactly zero when realized correlation equals implied correlation, whatever the components' common volatility, and

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why it loses so much more in a crash than it earns in a normal quarter.

22.10 Implied volatility is 22% and realized comes in at 18%. Compute what a short variance swap earns in volatility points, and explain why a short delta-hedged straddle need not earn the same and why it typically earns less.

22.11 A special dividend of 8% is paid on a stock during the life of a variance swap that does not adjust for dividends. Explain what happens to measured realized variance and who gains.

22.12 Explain why dispersion desks often use straddles rather than variance swaps on single stocks, and what they give up by doing so.

FURTHER QUESTIONS

22.13 Example 22.1 finds that widening the strike range beyond 50–200 buys nothing at 20% volatility, while changing the spacing moves the result. Derive the truncation error and the discretization error, show how each scales with volatility and maturity, and determine the volatility at which they are equal for a one-year contract on strikes from 50 to 200 spaced 5 apart, with the forward on a listed strike.

22.14 Section 22.5 argues that capping a variance swap destroys its static replication and therefore requires a model. Decompose a capped variance swap into a variance swap and a call on variance, explain what information about the process the call requires that the swap does not, and connect this to Section 21.11's measurement of model risk.

22.15 Example 22.5 finds a 43.5% annual roll cost and asserts that this is compensation rather than an inefficiency. Construct the argument that a systematic short VIX futures position is a risk premium, then construct the argument that it is a crowded trade earning a rent, and state what evidence would distinguish them.

22.16 Example 22.6 shows that an inverse product's rebalancing demand is $2AR$. Derive the corresponding expression for a $+2\times$ leveraged long product, explain why both signs of leverage are destabilizing, and discuss whether a product rebalancing at a random time of day rather than at the close would be safer or merely less predictable.

22.17 Section 22.13 claims that volatility-target funds are mechanically short volatility without holding options. Formalize this: write the fund's equity allocation as a function of realized volatility, derive its sensitivity to a change in realized volatility, and express that sensitivity in units comparable to an option's vega. Then discuss what this implies about measuring the market's aggregate short-volatility position.

22.18 Example 22.8 shows a one-to-eight ratio between the first row's gain and the crash loss. Derive the general condition on the implied correlation under which a variance-neutral dispersion trade has positive expected value, given a joint distribution for realized correlation and component volatility, and explain why a desk might rationally run the trade even when that condition fails.

22.19 Example 22.9 reads a forward variance off two spot variance swaps. Repeat it for a curve that is *inverted* (six-month at 30%, one-year at 24%), and show that the forward strike falls below both spot strikes. Then find the six-month strike at which the implied forward variance would be zero, and explain what a quote above that level would mean and how it would be arbitraged.

22.20 Section 22.12 states that the whole forward variance curve is spanned by today's vanilla surface with no storable underlying required. Explain why the variance case needs no storage, identify the assumption that makes it work, and state what fails if that assumption is relaxed.

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Chapter 23
Short-Dated Options, Dealer Positioning, and Market Feedback

Every chapter so far has treated the option market as a place where risk is transferred. This one is about what happens when the transfer becomes large enough, and fast enough, that the hedging it requires becomes a force in the underlying market.

The occasion is a market structure that did not exist when the previous generation of textbooks was written. Options expiring the same day they are traded, the 0DTE options of Section 10.12, went from a curiosity to the majority of S&P 500 option volume in under a decade. Section 10.12 recorded the growth and deferred the mechanism here; Section 18.12 supplied the machinery, a dealer whose hedging moves the market it is hedging in; and Section 21.12 observed that this market lives where classical stochastic volatility cannot make a skew.

Three demands meet in the contract: a hedger whose risk lasts one session (a data release, a policy decision) and who used to buy a month of protection for it, a retail buyer wanting a leveraged intraday bet, and a systematic seller collecting the volatility risk premium of Section 19.10 at the shortest horizon there is (Section 23.13). Whether their hedging stabilizes or destabilizes the index depends on how they net.

The chapter has an unusual shape, because the answer to its central question is that we do not yet know. It first calculates the mechanism by which 0DTE could destabilize the market (Sections 23.2 to 23.11), then weighs the evidence, which so far says mostly no, and why that is not reassuring (Section 23.12), and ends with what would have to go wrong (Section 23.14).

23.1 WHAT 0DTE IS, AND HOW IT GREW

An option expiring on the day it trades is not new—monthly and quarterly expirations have always had a final day. What is new is that there is one *every* day.

S&P 500 index options (SPX: European, settled in cash, $100 per index point, the weekly and daily series against the closing level) acquired weekly expirations progressively: Fridays first, then Wednesdays and Mondays in 2016, Tuesdays and Thursdays in 2022. Since then every trading day has been somebody's expiration, and a trader can put on a same-day position at any time.

The share of volume moved with it. 0DTE contracts were a single-digit percentage of SPX option volume in the mid-2010s, passed half at the end of 2024, and ran close to three-fifths through 2025. The most heavily traded S&P 500 option is now one that will not exist tomorrow.

Two things about that sentence are worth separating.

Volume is not risk transfer. A 0DTE contract has a fraction of the vega of a monthly option and lives for hours rather than weeks, so measuring the market in contracts overstates how much exposure has changed hands. It is the most common error in commentary on the subject.

But volume is hedging. Every contract traded is hedged by somebody, and the hedging of a 0DTE contract is concentrated into hours rather than spread over months. The flow is compressed even where the risk is not, and compression is what makes a market impact.

23.2 WHY SHORT-DATED OPTIONS BEHAVE DIFFERENTLY

Everything peculiar about this market follows from what happens to the Greeks as $T \to 0$. Take the S&P 500 at 5000 with a 15% implied volatility (a normal quiet market), with a 4% rate and a 1.5% dividend yield. Time is counted in trading hours throughout, 6.5 to the day and 252 days to the year, because that is the clock a 0DTE option lives on; every number in the chapter is priced this way, and "30 days" means thirty trading days, about six calendar weeks.

Table 23.1: At-the-money call as expiration approaches. The last two columns are the delta-hedged profit per unit of the index on a 1% rise.

Time left	Price	Delta	Gamma	Gamma term	Exact P&L
30 days	110.47	0.5323	0.001534	1.92	1.89
7 days	51.58	0.5158	0.003188	3.98	3.90
1 day	19.10	0.5060	0.008443	10.55	9.60
4 hours	14.94	0.5047	0.010763	13.45	11.65
1 hour	7.43	0.5024	0.021528	26.91	17.55
15 minutes	3.71	0.5012	0.043056	53.82	21.25

Gamma at one hour is fourteen times gamma at thirty days (Table 23.1), and it keeps rising: at-the-money gamma diverges like $1/\sqrt{T}$, which Section 18.3's formula makes obvious once one looks at it. That one number gives the scale of the problem.

But the more important fact is not how large the gamma is—it is how *narrow* it is (Figure 23.1). The table's last two columns show it. Both are the hedged profit on a 1% rise, once as the gamma term $\frac{1}{2}\Gamma\,(0.01\,S)^2$ of the Taylor expansion and once by repricing the option; at one hour a 1% move is 2.7 hourly standard deviations, and gamma falls away as the index leaves the strike, to 3% of its at-the-money value by the end of the move. The exact profit is earned on a gamma that falls as the index moves, while the gamma term keeps the starting value, so the profit comes out a third below the gamma term, and at fifteen minutes it is two-fifths of it.

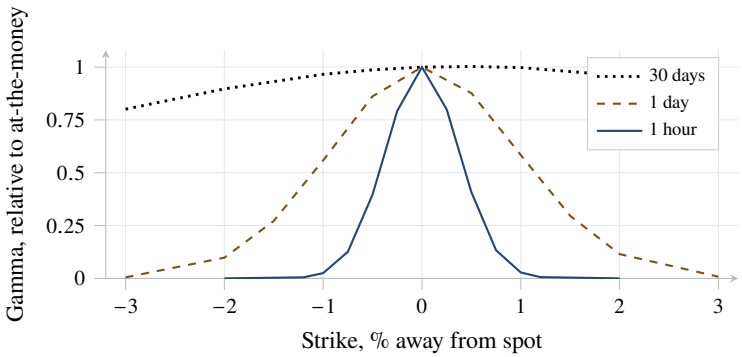


Figure 23.1: Gamma by strike, normalized to the at-the-money value.

Call the range of strikes over which gamma exceeds half its at-the-money value the **gamma band**. Its edges, which are symmetric in the logarithm of the strike about a point slightly above spot rather than about spot itself, close in as follows:

Time to expiration	Edges of the gamma band, above / below spot
30 days	+6.76%/−5.52%
1 day	+1.13%/−1.09%
1 hour	+0.44%/−0.43%

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This is the structural fact of the chapter. A month-dated book has gamma spread across a wide range of index levels, so a move of one percent changes the hedging requirement gradually. A 0DTE book has four-fifths of its gamma within half a percent of spot and nearly all of it within one percent—and therefore the hedging requirement is a step function of where the index happens to be. Move the index one percent and a dealer's entire gamma exposure can switch off, or switch on, or change sign.

23.3 THE BUYER'S ARITHMETIC

Before the market-structure question comes the individual trade. A 0DTE option is almost pure time value, and it has hours to live.

Figure 23.2 traces that life through one session: with the index unchanged, the option loses value steadily for most of the day and then falls away in the final hour.

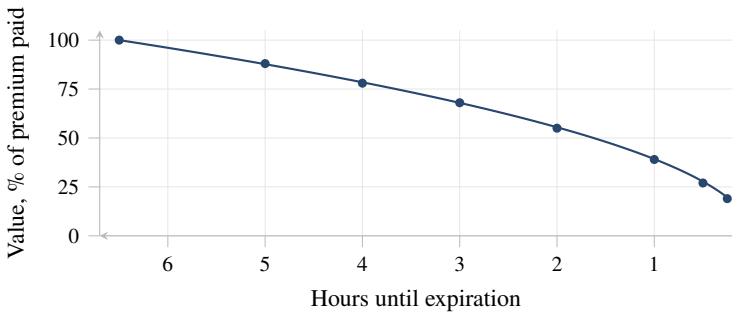


Figure 23.2: Value of an at-the-money 0DTE call through the session, with the index unchanged. Time runs left to right.

Example 23.1 What the buyer faces

An at-the-money 0DTE call on the index at 5000, bought at the open, costs 19.10, or **0.38% of spot**. A one-standard-deviation daily move is 47 points, or 0.94%.

The table gives the option's value through the day if the index does not move at all, and the rise the index needs for the buyer to break even on selling at each point:

Time left	Value, index unchanged	% of premium paid	Break-even rise
6.5 hours	19.10	100%	—
4 hours	14.94	78%	+0.15%
2 hours	10.53	55%	+0.28%
1 hour	7.43	39%	+0.35%
15 minutes	3.71	19%	+0.38%
Expiration	0	0%	+0.38%

An earlier exit needs a smaller rise, because the option is sold with time value still in it: with four hours left, 11.47 of the 19.10 recovered is time value.

Held to expiration, the buyer needs a 0.38% move in the right direction merely to get their money back, against a daily standard deviation of 0.94%. That is a hurdle of about 0.4 standard deviations before any profit. The option is not a bad bet in the way a lottery ticket is; the option is priced without obvious error, and its expected return is negative, set mainly by the volatility risk premium of Section 19.10.

What makes it feel different is the *shape*. Sixty-one percent of the premium is gone by the last hour if nothing happens, so the position is either right quickly or worthless. The instrument converts a modest edge against the holder into a rapid and highly visible sequence of total wins

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and total losses, which is a design that human beings find much more engaging than a slow loss. Section 23.13 returns to that.

Note also what this does to bid–ask cost. At 19.10 a one-tick spread of ten cents (SPX's minimum tick for an option priced at $3 or more) is 0.52% of the premium; on the same option with fifteen minutes left, priced at 3.71, the same dime is 2.7%. Trading the same contract later in the day costs several times as much in percentage terms, and the volume is heaviest late.

23.4 CHARM: THE GREEK THAT ONLY MATTERS HERE

Section 18.6 named charm, the rate at which delta changes with the passage of time, and moved on, because for a monthly option it is a rounding error. Writing f for the option's value,

$$\text{charm} = \frac{\partial \Delta}{\partial t} = \frac{\partial^2 f}{\partial S \, \partial t}$$

with t the time elapsed. It is the cross-derivative that pairs with gamma: gamma is how the hedge changes when the price moves, charm is how it changes when the clock moves and the price does not. It is a cross term in $\Delta S \, \Delta t$ that Chapter 18's expansion, equation (18.1), leaves out, because for a monthly option the delta barely drifts with the clock alone. For 0DTE it is the dominant flow.

The reason is visible in Figure 23.1. As expiration approaches, an out-of-the-money option's delta must go to zero and an in-the-money option's to one. The whole delta surface is collapsing into a step, and that collapse happens on the clock rather than on the price.

Figure 23.3 shows that collapse on a single contract: with the index unchanged from open to close, the delta of a call struck just above spot falls the whole way down, and every point of that fall is a hedge somebody has to trade.

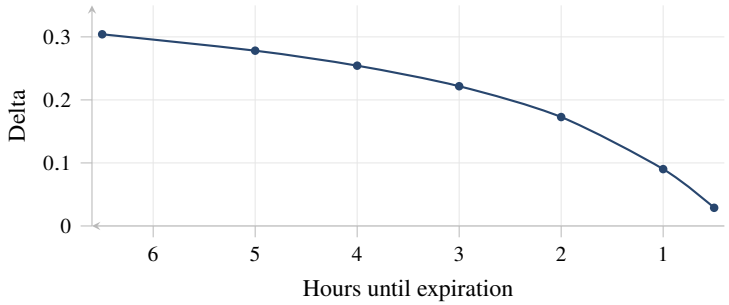


Figure 23.3: Delta of a call struck 0.5% above spot, with the index unchanged all day. Time runs left to right.

Example 23.2 A billion dollars of hedging, with the index unmoved

Take a call struck 0.5% above spot (about half a daily standard deviation, an ordinary strike for a 0DTE trade). With the index held at 5000 all day:

Time left	Delta	Charm, per hour
6.5 hours	0.3041	−0.0146
4 hours	0.2543	−0.0274
2 hours	0.1728	−0.0614
1 hour	0.0902	−0.1099
0.5 hours	0.0288	−0.1254

A dealer short 10,000 of these contracts (index multiplier 100) hedges by holding 304,095

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index units long at the open ($1.52 billion) and 90,164 units with one hour left, or $451 million.

The dealer must *sell* roughly $1.07 billion of index by the last hour, with the index unmoved.

Three properties distinguish this flow from the gamma hedging of Section 18.12.

It is one-directional. Gamma hedging buys on the way up and sells on the way down, netting out over a round trip. Charm hedging runs one way all day, set by the clock.

It is predictable. Anyone who knows the open interest by strike can compute it in advance, because the only input is the passage of time.

And it accelerates. The size of charm rises from 0.015 to 0.125 per hour, more than eightfold, so the flow is fastest in the final hour, which is also when 0DTE volume is heaviest and the day's other closing trades are competing for the same buyers and sellers.

Whether this stabilizes or destabilizes depends on the sign of dealer positioning, and that is Section 23.7. What is certain is that a large, predictable, clock-driven flow now exists in the final hour of the US equity session, and that ten years ago it was a fraction of its size.

23.5 THE OVERNIGHT GAP IS NOT IN THE CONTRACT

A quiet measurement problem sits underneath every comparison of 0DTE implied volatility with anything else, and it accounts for a discrepancy that is easily misread.

A 0DTE option bought at the open expires at the close. It covers the session and not the night, yet it is quoted with a full trading day as its time to expiration. If the overnight period (close to next open) carries a fraction ω of daily variance, then the session carries $1 - \omega$, and the implied volatility that reproduces the option's price is

$$\sigma_{\text{session}} = \sigma_{\text{daily}} \sqrt{1 - \omega}$$

where σ_{daily} is the annualized volatility of close-to-close returns, the figure a 30-day implied volatility measures.

Overnight share ω	Implied volatility shown	Against 15% close-to-close
0%	15.00%	—
10%	14.23%	−5.1%
20%	13.42%	−10.6%
30%	12.55%	−16.3%
40%	11.62%	−22.5%

For a US index, where most news arrives during the session, ω is typically in the 15–25% range, so on the trading-day clock used here 0DTE implied volatilities print about a tenth below the 30-day implied volatility for purely mechanical reasons. (A calendar-minute clock makes the opposite error, showing 0DTE volatility well above the daily figure.) Cboe's one-day volatility index, VIX1D, launched in April 2023, publishes this session-only volatility, and comparing it with the 30-day VIX needs the same adjustment.

This can be misread as evidence that short-dated options are cheap. They are not; the measurement window is shorter than the quoted maturity. Section 19.6's term structure is a real phenomenon, and this is not part of it; mixing the two produces a term structure with a spurious kink at the front.

The effect has a practical corollary that matters more than the misreading. A 0DTE option cannot hedge an overnight gap, because the gap happens after it has expired. An institution using daily options to hedge event risk must hold a contract expiring after the event, and for an announcement made after the close that means tomorrow's contract, not today's.

23.6 THE SMILE AT ZERO MATURITY

Section 21.12 said that this chapter's market lives entirely in the region where classical stochastic volatility cannot produce a skew, and Section 21.7 gave the empirical law: the at-the-money

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skew grows like $T^{-0.4}$ as maturity shortens, measured from about a month to a few years. Whether it continues below a month is disputed; rough volatility (Section 21.8) says it does. Extrapolated to this market, the law is dramatic.

Maturity	Skew, relative to 30 days	Volatility points per 1% of moneyness
30 days	1.00	2.5
7 days	1.79	4.5
1 day	3.90	9.7
1 hour	8.24	20.6

A one-day smile is roughly four times as steep as the 30-day smile, and an hour before expiration it is eight times as steep. The ratios are the law's; the 30-day level of 2.5 points is a stylized anchor, several times the one-month slope of Table 19.3's surface (about 0.4 points per 1%), and every slope below scales with it. The slope is negative, so it is the puts that carry it: a put one percent below spot can show an implied volatility twenty points above the at-the-money level. The same slope cannot be run the other way for a call at the same distance (twenty points below a 15% at-the-money volatility is not a volatility), so the call wing flattens and turns up instead, and the smile at this horizon is a steep put wing against a shallow, convex call wing. Figure 23.4 draws the four maturities on one axis: from a month to an hour the put wing goes from a gentle slope to a nearly vertical one, while the call wing moves by a few points.

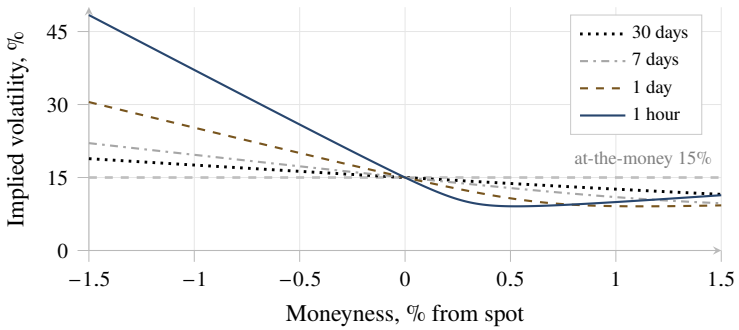


Figure 23.4: Implied volatility by moneyness at four maturities. Schematic: the at-the-money slope follows the $T^{-0.4}$ law of the table, extrapolated below one month, the at-the-money level is held at 15%, and one stylized wing shape is used throughout.

Three consequences follow, and the third is the one that matters for market structure.

The Black–Scholes implied volatility becomes a poor coordinate. Sections 15.7 and 19.12 described implied volatility as a language for quoting prices. At these maturities the language distorts: small differences in price map to enormous differences in implied volatility, and quoting in volatility terms conveys less information than quoting the price.

Only a model with jumps, roughness or very fast mean reversion can fit it. Example 21.3 showed jumps producing this shape (a smile whose range across strikes is 12.7 volatility points at one month and 1.5 at one year, almost all of it in the put wing), and Section 21.8 showed rough volatility producing the exponent directly. Whatever generates the 0DTE smile is not a classical diffusion with smoothly varying volatility, and this is the sharpest available evidence for that claim, because at these horizons such a path cannot travel far enough to give an out-of-the-money option any value; it takes jumps, very rough volatility, or a factor mean-reverting fast enough to act like one (Section 21.8).

And the steepness is where the retail flow is. The out-of-the-money puts that an intraday buyer of protection, or of a bet on a fall, naturally reaches for are precisely the strikes whose

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implied volatilities are most extreme. A buyer paying twenty points of skew is not making an obviously bad trade, since the steepness reflects a real probability of a downward jump, but they are buying the most expensive part of the surface, measured in the units the market itself uses. The call buyer at the same distance is on the cheap side of the same surface, paying less than the at-the-money volatility.

23.7 DEALER POSITIONING AND THE FEEDBACK LOOP

Section 18.12 established the mechanism, and the gamma term of equation (18.1) is its source: a move ΔS changes the dealer's delta by $\Gamma\,\Delta S$, and rehedging it trades with the move when dealers are *short* gamma, amplifying it, and against it when they are *long*, damping it.

Applying that to 0DTE requires knowing which side dealers are on, and here the market's structure matters more than any model.

Retail buys options. The retail flow that Section 10.12 described is predominantly premium-paying: buying calls and puts outright. Against that flow dealers are short options and short gamma, and the feedback is amplifying.

But institutions sell them. A large and growing share of 0DTE volume is systematic premium selling—put spreads, call spreads, and iron condors (Section 12.3: a put spread below spot and a call spread above it, sold together) sold by funds harvesting the premium of Section 19.10 at the shortest available horizon. Against that flow dealers are long options and long gamma, and the feedback is dampening.

The two roughly offset, and the net is small and changes sign. This is the single most important empirical fact about the market and the one most often ignored. Studies of 0DTE order flow (Dim, Eraker, and Vilkov among them; see Further Reading) generally find it close to **two-sided**: customers buy options from dealers and sell options to dealers in roughly equal amounts, so dealers' net position is the difference of two large gross flows, and net dealer gamma from 0DTE contracts comes out a small fraction of that from longer-dated positions.

Example 23.3 How large the hedging flow is

Desks state gamma as the notional of the index that must be traded to rehedge a 1% move. A 1% move in the index changes the delta of one option by $0.01\,S\,\Gamma$ index units, so a position of n contracts with multiplier m needs

$$\text{hedge notional per } 1\% = 0.01\,n\,m\,\Gamma S^2$$

of index traded to restore the hedge. This is a different number from the gamma term of Table 23.1, the profit or loss on the same move, $\frac{1}{2}\Gamma S^2 \times 0.01^2$ per unit; the two describe one gamma, one as a trade and one as a P&L.

Suppose the dealer community is net short \$500 million of gamma per 1% from 0DTE positions. From Table 23.1 that is the gamma of about 930 at-the-money contracts an hour before expiration ($0.01 \times 100 \times 0.021528 \times 5000^2$ = \$538,200 each, the 100 being the multiplier m), or 13,000 at thirty days. Being short gamma, the dealers gain delta as the index falls and lose it as the index rises, so they must sell futures into a fall and buy into a rise; to first order, holding gamma fixed, the amount is \$500 million for each 1%:

Index move	Index futures dealers must sell (fall) or buy (rise)
0.5%	\$250m
1.0%	\$500m
2.0%	\$1,000m
3.0%	\$1,500m

Each trade follows the move that caused it, which makes the flow amplifying. Gamma does not stay fixed at this horizon (Section 23.10): for the one-hour book the true figures are about \$190m

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on a 0.5% move and stop near $230m, the whole of the book's delta, beyond 1%; for the thirty-day book the table is close.

Against S&P futures volume of several hundred billion dollars notional a day, these are small numbers in aggregate. A billion dollars of futures is minutes of normal volume.

They are not small at 3:55 p.m. New York time. The flow is concentrated in the final hour, by both charm and gamma, and within that window it is all on one side. The close is the busiest part of the futures day, but a hedger who must sell now does not get to trade against the day's volume. They trade against what other participants are willing to buy at that moment at the best bid, which is a few hundred E-mini S&P 500 futures contracts (each $50 times the index, about $250,000 at 5000), or tens of millions to about a hundred million dollars. That is the right comparison, not daily volume, and against it a few hundred million dollars that all wants the same side is material.

23.8 JUMPS WITHIN THE DAY

Section 21.5 observed that jumps cluster across underlyings (the events that make one stock gap tend to make others gap) and left the intraday side to this chapter. For 0DTE the scheduled events behind that clustering are not a refinement—they are the dominant term.

Volatility within a session is not uniform. It concentrates at the open, at scheduled data releases, at policy announcements, and into the close. For a monthly option this averages away; for an option with six hours to live it does not.

A 0DTE implied volatility is therefore not a number but a *schedule*, and a public one.

Example 23.4 What a two o'clock announcement does to a morning option

Take Section 23.5's session: 15% close-to-close volatility, 20% of variance overnight, so an ordinary session's standard deviation is 0.845% and a 0DTE option shows 13.4% implied volatility.

Assume ordinary variance is spread evenly through the session. Now add a scheduled announcement whose move has standard deviation s_e. Variances add, so the session's standard deviation becomes $\sqrt{0.845\%^2 + s_e^2}$; the last column is the implied volatility as a multiple of the 13.4% with no event:

Event move (s.d.)	Session s.d.	Implied volatility shown	Multiple
none	0.845%	13.4%	—
0.50%	0.982%	15.6%	1.16
0.75%	1.130%	17.9%	1.34
1.00%	1.309%	20.8%	1.55
1.50%	1.722%	27.3%	2.04

A single scheduled event with a one-percent standard deviation raises the morning's 0DTE implied volatility by more than half. Nothing about the market's ordinary volatility has changed; one known future minute has been added to the measurement window.

The sharper effect is what happens when the minute passes. Compare the option at 1:59 p.m. and at 2:01 p.m., with two hours left, the index unchanged, and the 1.00% event of the table:

	Remaining s.d.	Annualized
Just before the announcement	1.104%	31.6%
Just after	0.469%	13.4%

The implied volatility falls by a factor of 2.36 in two minutes. The event variance has simply left the window.

Three points follow.

This is not a crash in volatility; it is the calendar. A trader who buys the option at 1:59 and sells it at 2:01 without a view on the announcement has bought the event and sold the aftermath, which is a coherent trade. A trader who buys at 1:59 hoping to profit from a rise in volatility has

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misunderstood the instrument.

And it makes the constant-volatility assumption untenable in a way that is not a modeling nicety. Section 15.1 listed constant volatility among the Black–Scholes assumptions and Chapter 21 relaxed it. At this horizon the failure is not that volatility is stochastic—it is that volatility is *scheduled*, and everyone has the schedule. Desks price 0DTE against a calendar of event weights rather than a single number, which is a different kind of model from anything in Chapter 21; Section 23.9 describes it.

Finally, the clustering that Section 21.5 attributed to jumps is visible here directly. The events that make one index gap are the same events (data, policy, geopolitics) that make others gap, so intraday jump risk is correlated across underlyings exactly when it is largest. A book short out-of-the-money 0DTE options on several indices is not diversified; it is one position.

23.9 PRICING A 0DTE OPTION

Every number so far has come from the Black–Scholes–Merton formula with a 15% volatility and a clock in trading hours, and for the Greeks that was the right tool: the shape of gamma, charm and decay does not depend on the details. A reader who has followed the chapter this far will have two questions the formula does not answer. Which options are actually being traded? And if constant volatility is untenable, what does a desk use instead?

Which strikes trade. Mostly they are not the at-the-money ones. Cboe's own tabulation of 2023 volume and the academic samples named in the Further Reading agree on the shape. The typical contract is near the money but out of it (roughly a quarter to one percent away, a delta between about 0.10 and 0.35), and puts and calls trade in roughly equal numbers, the puts reaching further from spot than the calls do. Far out-of-the-money strikes, one to two percent away on a quiet day, account for many contracts and very little premium, because they cost a few dollars at most, and cents by the afternoon; counted by premium rather than by contract, the center of the market moves closer to the money. The distribution also moves through the day: in the morning the traded strikes span about a percent either side of spot, and by the last hour nearly everything is within about half a percent, the one-hour band of Figure 23.1, because that is the only place where any time value is left. In-the-money options trade little. By put–call parity, an in-the-money call is the out-of-the-money put at the same strike plus a long forward position: its time value is the put's, and the rest is index exposure bought at an option's wider bid–ask spread. That time value drains away through the day: at the open a call half a percent in the money carries about 15 points of it, an hour before the close about 4, and with fifteen minutes left about 1, at the skewed volatility of Section 23.6. A trader who wants the exposure buys the futures contract, and one who wants the option buys the out-of-the-money side. Retail flow is mostly single-leg, an outright call or put opened and often closed within the session, while institutional flow is mostly spreads: a put spread half to one percent below spot, a call spread the same distance above, or both together as an iron condor, held to expiration. Section 23.13 says who these people are; the point here is that the market's pricing problem is concentrated in a narrow band of near-the-money strikes and a steep wing of puts just below it.

Why one volatility will not do. The Black–Scholes–Merton model makes three assumptions that are tolerable for a monthly option and fail together at this horizon. Volatility is constant, so variance accrues evenly with the clock; but Section 23.5 showed that a fifth of the day's variance falls in a window the contract does not cover, and Example 23.4 showed a scheduled minute carrying several times as much variance as the rest of the afternoon. The price path is continuous, so a strike several standard deviations away is nearly worthless; but at one hour a strike one percent below spot is 2.7 standard deviations away, and Section 23.6 showed

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the market pricing it as though the distance were far smaller, because a jump can cover it in an instant. And the hedge is continuous and free, so the price is the cost of replication; but Table 23.1 put gamma at fourteen times its monthly value and Example 23.1 put a dime of bid–ask at 2.7% of the premium, so the replication cost is neither small nor well defined. Example 23.5 shows how large the resulting error is.

Example 23.5 What a flat volatility gets wrong an hour before the close

Price a put below spot in the chapter's setting (index 5000, 4% rate, 1.5% dividend yield), first with a flat 15% volatility and then with the volatility given by the at-the-money slope of Section 23.6, extended linearly: 15% plus 2.5 points per percent of moneyness at thirty days, plus 20.6 points per percent at one hour.

Time left	Strike	Flat 15%	With the skew	Ratio
30 days	0.5% below spot	83.99	92.45 (at 16.2%)	1.1
30 days	1.0% below spot	73.30	89.90 (at 17.5%)	1.2
1 hour	0.5% below spot	0.75	3.72 (at 25.3%)	5.0
1 hour	1.0% below spot	0.02	2.75 (at 35.6%)	148

At thirty days the smile is a correction of ten or twenty percent to a price the flat model gets roughly right. At one hour it is the price. The flat model values the put one percent below spot at two cents, because under a continuous path with a 0.37% hourly standard deviation the index almost never gets there; the market values it at about 2.75, over a hundred times as much, because a jump can. A desk that priced this put with the at-the-money volatility would offer it at the minimum tick and hedge it as if it were worthless.

What desks use instead. There is no one replacement; practice is closer to Chapter 19's than to Chapter 21's: the surface is the model, with the pieces assembled around it.

The first piece is the clock. Time to expiration is measured not in minutes but in expected variance remaining: an intraday profile heavy at the open and into the close with a thin middle, plus the weights of the day's scheduled events. This is **business time**, and the Black–Scholes–Merton formula is used with $\sigma^2 T$ replaced by that variance budget. It is why the option of Example 23.4 lost more than half its implied volatility in two minutes with nothing else changing: the budget was spent, not the volatility revised.

The second is the forward, which comes from the E-mini futures contract, the instrument the position will be hedged with, or from put–call parity within the chain as the VIX construction of Section 22.6 does. The smile is quoted against that forward, not against the cash index.

The third is the smile itself, fitted expiration by expiration in the standardized moneyness of Section 19.1,

$$\frac{\ln(K/F)}{\sigma\sqrt{T}},$$

the strike's distance from the forward in standard deviations, with σ the at-the-money implied volatility and T the business time remaining. The form is low-parameter and arbitrage-free by construction (the SVI parameterization of Section 19.5 is common), refitted through the day. In raw moneyness the smile steepens sharply as expiration approaches; in standardized coordinates most of that steepening disappears, and what is left, a slope proportional to $T^{0.1}$ (the $T^{-0.4}$ times the $\sqrt{T}$ the standardization divides by), is the residual the fit has to carry. The wings are where a view on jumps enters: a desk that expects an event to be larger than the calendar weight implies lifts the put wing directly, rather than changing a parameter in a dynamic model. The models of Chapter 21 are used off the desk, to check that the fitted surface is one some process could have produced. Jump models give a short-end skew that overshoots

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the market's, as Section 21.7 found, and rough volatility gives the $T^{-0.4}$ the market shows, but they do not set the quote.

The last piece is inventory. The spread and the skew adjustment around the fitted midpoint are set by the dealer's own position and by customers' orders: quotes are shaded against the book's gamma and charm strike by strike, widen into the close, and move with the flow. What the model supplies is the hedge ratios that make that possible, and for an actively traded 0DTE strike that is its whole job. The price is set by the bids and offers already in the market, not by anyone's formula.

23.10 PINNING, EVERY DAY

Section 10.9 introduced **pinning** alongside pin risk at the monthly expiration: the underlying is drawn toward a strike with large open interest, because dealers hedging a long gamma position sell into rallies and buy into declines around it. With a daily expiration, that mechanism now runs every session.

Example 23.6 computes the size of the restoring force, because it is far larger than the monthly intuition suggests, and Figure 23.5 plots it.

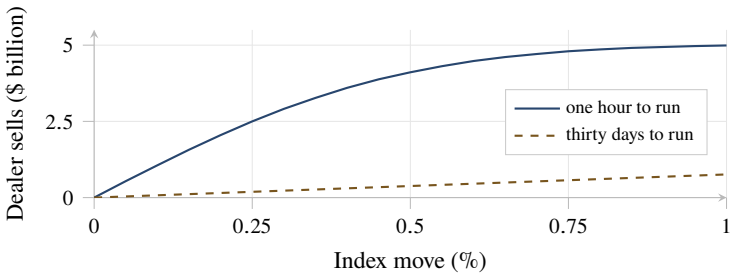


Figure 23.5: Dealer sale against index move for 20,000 calls held long at the money.

Example 23.6 The restoring force at one hour

Dealers are long 20,000 calls at one strike, the index sits at it, and one hour remains.

Index move	Delta goes from	Dealer sells
+0.10%	0.502 → 0.609	$1,063m
+0.25%	0.502 → 0.752	$2,499m
+0.50%	0.502 → 0.912	$4,114m
+1.00%	0.502 → 0.996	$4,990m

With thirty days to run the same book sells $382 million on a 0.5% move and $760 million on a 1% move, the straight line of Figure 23.5: doubling the move doubles the sale, which the one-hour curve plainly does not.

Two features of the one-hour figures matter and they point in opposite directions.

Small moves meet an enormous restoring force. A tenth of a percent generates a billion dollars of stabilizing supply. This is genuine pinning, and it is why the index often trades in a narrow range near a strike with large open interest.

But the force saturates. By a 1% move the delta has reached 0.996 and there is nothing left to sell: the last half-percent of the move generated only $876 million of additional hedging against $4.1 billion for the first half. The stabilizer is strongest for moves that did not need stabilizing and absent for the ones that did.

That asymmetry is the defensible form of the worry in Section 23.14. A market that suppresses

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small fluctuations and cannot suppress large ones does not merely fail to help in a crisis—it makes the crisis look more surprising, because the small moves that would ordinarily have signaled building stress were damped away.

23.11 MEASURING DEALER GAMMA, AND WHY THE NUMBER IS UNRELIABLE

Everything in Section 23.7 depends on the sign of net dealer gamma, and that quantity is not observable.

The standard construction, which Section 18.12's case study introduced, takes the open interest OI_i of each listed contract i (a call or a put at some strike), computes its gamma Γ_i, and assigns the dealers a side, $s_i = +1$ (long) or -1 (short). The usual rule assumes them long all call open interest and short all put open interest. The published "gamma exposure," in the form SqueezeMetrics (2017) first set out, is then $\sum_i s_i \,\mathrm{OI}_i \, m \, \Gamma_i \, S^2 \times 0.01$, with m the contract multiplier (100 for SPX): the hedge notional per 1% of Example 23.3 (Section 18.7's gamma in dollars) with the signs assumed, and the "flip level" is the index level at which it changes sign.

The trouble is that the sign assumption is asserted, not measured. Commercial versions infer the side from trade data rather than assume it, which narrows the problem without removing it; Example 23.7 shows how much of the answer the assumption decides.

Example 23.7 The assumption is the answer

Take 100,000 calls and 100,000 puts outstanding at the same strike, with identical gamma per contract. Assume dealers are on one side of every contract. Let w_c be the fraction of the calls dealers are long (they are short the rest) and w_p the fraction of the puts they are short (they are long the rest), so that the standard rule is $w_c = w_p = 100\%$. Net dealer gamma is then 200,000 $(w_c - w_p)$ contracts' worth.

Assumed split (calls long / puts short)	Net dealer gamma (thousands of contracts)
100% / 0%	+200
80% / 20%	+120
60% / 40%	+40
50% / 50%	0
40% / 60%	−40
100% / 100%	0

The sign flips between the 60/40 and the 40/60 assumption, and nothing in public data determines which is right. The last row is the standard rule itself: applied to equal call and put open interest it returns exactly zero, so here the published number would be zero not because dealers are flat but because the assumption is symmetric. Open interest records that a contract exists; it does not record who is on which side.

A published gamma-exposure level is therefore what dealer positioning would be if the assumed split were true, a weaker statement than it usually receives. Three points follow:

The assumption is worst exactly where it matters. It is roughly defensible for a market dominated by call overwriting (investors selling calls against stock they hold) and institutional put buying. Section 23.7 argued that 0DTE flow is close to two-sided, which is the case in which the naive split is furthest from the truth.

And the flip level inherits the error. A published "gamma flip" at some index level is a function of the assumed signs, so two analysts with different assumptions produce different flip levels from identical data.

None of this means the concept is useless. Dealer gamma is real, and Sections 18.12 and 23.10 depend on it; published estimates simply measure open interest precisely and positioning by

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assumption.

23.12 DOES 0DTE AMPLIFY OR DAMPEN?

This is the question the market cares about, and the answer is that the evidence is mixed and the sample is short.

The case that it amplifies. The mechanism of Section 23.7 is real. Gamma is enormous and concentrated in a narrow band, so hedging demand is a step function of the index level. Charm produces a large one-directional flow into the close. And the positions are concentrated and the hedging mechanical, which is the pattern that produced 1987 (Section 40.2), February 2018 (Section 22.8) and the snowball knock-ins of Section 35.7.

The case that it dampens. Short-dated option sellers, in aggregate, supply gamma to the market. A dealer long gamma sells into rallies and buys into declines, which is stabilizing. If systematic premium selling dominates, 0DTE makes the intraday market *less* volatile, not more.

What the data show. Realized intraday volatility on the S&P 500 has not risen in a way that tracks 0DTE growth. There has been no episode in which 0DTE hedging has been convincingly identified as the cause of a large move, though two stress days have tested the market since 0DTE became its largest segment: August 5, 2024 and the April 2025 tariff sell-off, when VIX closed above 50. Accounts of both trace the moves to news and to leveraged positions being unwound, and on August 5 to wide pre-open option quotes that inflated the VIX itself, not to 0DTE hedging. And the aggregate net dealer gamma attributable to 0DTE appears small relative to the longer-dated book. Against this, Brogaard, Han, and Won find a rise in volatility and attribute it to speculative retail trading rather than to dealer hedging.

So the fair summary is: the mechanism exists and it has not yet produced an incident. Four reasons stand out. The flow is two-sided, the main reason and the most fragile. Expiration is a natural stop-loss, since tomorrow the position does not exist. Margin is collected daily and the positions are cleared: the 2018 episode ran through an exchange-traded note, XIV, whose prospectus allowed early termination (Section 22.8), while a listed option position is margined by the clearing house of Chapter 2 against a position that self-liquidates. And the underlying is the deepest equity market in the world. The last three are structural rather than behavioral, which, by the argument of Section 22.5's case study, is the kind that holds.

That summary deserves caution. *The stabilizing property is a property of the flow, not of the instrument.* Nothing guarantees that 0DTE order flow stays two-sided. It is two-sided because two groups happen to want opposite things at the same horizon, and either could stop.

Positions do not accumulate, which is genuinely reassuring. A 0DTE book is gone by the close. Compare Section 22.8's inverse products, whose short position was carried from day to day for years, or the snowball books of Chapter 35, whose knock-in exposure grew as the market fell over months. Whatever risk 0DTE creates is created and destroyed within one session, which puts a hard bound on how large it can get.

But the bound is on size, not on speed. A structure that resets daily cannot build a slow-motion catastrophe; it can still produce a violent hour. And the events that would test this market (a large gap that arrives while dealers are short gamma near the close) are rare.

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Case Study 23.1 The same mechanism in a shallow market

The argument of Section 23.12 rests heavily on the depth of the S&P 500. What the identical mechanism does without that condition is well documented.

In January 2021 a heavily shorted small-capitalization US stock attracted very large retail call buying, much of it in short-dated, far out-of-the-money strikes. The dealers who sold those calls were short gamma, and Section 23.7's arithmetic applied unchanged: as the stock rose, their delta grew, and they had to buy stock to stay hedged.

Everything is the same as the index case except one number. The S&P 500 absorbs a billion dollars of hedging in minutes. A stock with a small free float, a large fraction of which was already lent out to short sellers, could not absorb the equivalent at all. The hedging demand was large relative to the shares actually available, so it moved the price, which increased the hedging demand, which moved the price again.

Three points need separating, since the episode is usually told as one about intent.

The dealers were not speculating. They were hedging a position they had sold, and a hedge that moves the thing it hedges is a structural feature of delta hedging in a market too small for it.

The short interest (shares sold short and not yet bought back) mattered as much as the options. Short sellers covering are buyers too, and the two flows reinforced one another. Attributing the whole move to option hedging overstates a mechanism that was real but not alone. That is the same caution Section 23.12 applies to 0DTE.

And the binding constraint was free float, not leverage. The options were not unusually leveraged; the underlying was unusually small, and that is the variable that decides whether the feedback of Section 23.7 is a curiosity or an event.

The lesson is therefore not that 0DTE is safe. The mechanism is settled, and the outcome depends on the ratio of hedging demand to available liquidity, which is small for the S&P 500 and was not for a single small-cap stock. To ask whether 0DTE is dangerous is to ask whether that ratio can change, and Section 23.14 lists the ways it could.

23.13 WHO TRADES 0DTE, AND WHY

Retail directional traders buy 0DTE options as a leveraged intraday bet. Example 23.1 gives the economics: a modest negative expected return, delivered as a rapid sequence of complete wins and complete losses.

Systematic premium sellers sell 0DTE spreads to harvest the volatility risk premium at the shortest horizon, where the annualized premium looks largest and, crucially, where a position cannot be carried into a multi-day drawdown. Selling a defined-risk spread that expires in hours is a different risk from selling a monthly straddle: the loss is bounded and the position cannot be held past its worst moment.

Institutional hedgers use 0DTE tactically around events within the session (a data release, a policy decision), and buy the next day's expiration for one announced after the close, where the risk is genuinely one-day and paying for a month of protection is waste. This is the least remarked and possibly the largest legitimate use, and it is a straightforward improvement in market completeness: a one-day risk can now be hedged with a one-day instrument.

Dealers and market makers intermediate, and their positioning is the transmission mechanism of Section 23.7.

The instrument itself is neutral: Section 12.8's replication argument, run across expiration dates instead of strikes, says a finer set of maturities strictly enlarges the space of achievable payoffs. What is new is the concentration of hedging it induces, which should be assessed separately.

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23.14 WHAT WOULD HAVE TO GO WRONG

The useful form of the question is not "is 0DTE dangerous" but "what configuration would make it so," because that is checkable in advance.

One-sided flow. If the systematic sellers withdrew (after a bad quarter, or a capital charge, or a change in fund mandates), the retail buying would leave dealers uniformly short gamma.

A gap into concentrated gamma. The dangerous configuration is a large move arriving in the last hour, when the gamma band of Figure 23.1 is at its narrowest and the charm flow of Example 23.2 is at its fastest. A 1% move then takes the index outside the band, and the hedging that would have absorbed it runs out partway through, as Example 23.6 showed.

Liquidity that is not there. All of the numbers in Example 23.3 are small against average volume and large against the depth available in a fast market. The mechanism that matters is the one Section 2.9 identified in margin procyclicality: in a fast market the participants who would provide liquidity are the same ones who suddenly need it. And short-dated hedging is reaching shallower markets: since January 2026 stocks with a market value above $700 billion also carry Monday and Wednesday expirations (Section 10.2), which is the setting of Section 23.12's case study rather than the index's.

None of this is a prediction. It is a specification of what to watch, and the watchable quantities (the open-interest distribution near spot, the share of volume that is premium-selling, and estimates of dealer gamma by strike, read with Section 23.11's caution) are published or computable. The risk is not hidden, and unlike the episodes of Sections 22.8, 35.7 and 40.2, the profession is watching this one while it is small.

SUMMARY

Same-day options went from a single-digit share of SPX option volume to more than half in under a decade. Volume overstates the risk transferred; it does not overstate the hedging, which is compressed into hours.

Everything peculiar follows from the Greeks as $T \to 0$. Gamma an hour from expiration is many times gamma at thirty days, and the gamma band collapses to a narrow range around spot, so hedging demand becomes a step function of the index level. Charm becomes the dominant flow: a dealer short calls struck just above spot unwinds a large hedge in a day the index does not move, one way, predictably, fastest in the final hour. The buyer of an at-the-money call pays well under one standard deviation of the day's move and has lost most of the premium by the last hour if nothing happens.

Three things are routinely misread. A 0DTE option covers the session but is priced with a full day, so it shows less implied volatility than the daily figure and cannot hedge an overnight gap. The skew grows like $T^{-0.4}$, steepening by nearly an order of magnitude from thirty days to the last hour before expiration. And volatility within the day is scheduled: an announcement lifts the morning's implied volatility, and the remaining option's falls sharply when it passes. One volatility therefore cannot price the market: a put just below spot an hour before the close is worth almost nothing at a flat volatility and many times as much with the skew—so desks price on a business-time clock against a smile fitted in standardized moneyness, and the model supplies hedge ratios, not the price.

Whether hedging amplifies or dampens depends on the sign of dealer gamma, which is not observable: retail buying and systematic selling roughly offset, and published estimates measure open interest precisely and positioning by assumption. Pinning happens daily and saturates: the hedging flow is heaviest on a small move and spent by a larger one. There has been no incident because the flow runs in both directions, positions expire before they accumulate, margin is daily, and the underlying is deep; only the first of these could change, and the configuration to watch is flow running one way meeting a large move in the final hour.

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FURTHER READING

This is the newest topic in the book and the literature is accordingly unsettled; a reader should expect the empirical picture to have moved. Cboe's own research notes on 0DTE volume composition (2023) are the source for what is actually traded and by whom, and they separate contract counts from risk transfer, which most commentary does not. On order flow and dealer positioning, Brogaard, Han, and Won (2023, revised 2026) find that 0DTE trading raises the volatility of the underlying and attribute the effect mainly to speculative retail trading rather than to dealer hedging, Dim, Eraker, and Vilkov (2024) find market makers' net 0DTE gamma small and, on average, long, with calmer rather than wilder trading after it, and Beckmeyer, Branger, and Gayda (2023) document the retail side (who buys, at what strikes, and with what returns); together they are the evidence Section 23.12 calls mixed, and they disagree less about dealer hedging than about whether 0DTE trading moves volatility at all. Bandi, Fusari, and Renò (2023) is the pricing paper, with time-varying skewness and kurtosis and jumps for the wings. On the feedback mechanism itself, the literature on option hedging and market impact predates 0DTE by decades and applies unchanged, and Grossman (1988) and the Brady Commission report are the place to start.

PRACTICE QUESTIONS

23.1 Explain why 0DTE contract volume overstates the amount of risk being transferred.

23.2 State how at-the-money gamma behaves as $T \to 0$, give the ratio between one-hour and thirty-day gamma from Table 23.1, and explain why the narrowing of the gamma band matters more than the growth in gamma itself.

23.3 An at-the-money 0DTE call costs 0.38% of spot. State the move required to break even at expiration, compare it with a one-standard-deviation daily move, and explain why the bid–ask spread is a much larger percentage cost late in the day than early.

23.4 Define charm, explain why it is negligible for a monthly option and dominant for a 0DTE option, and use Example 23.2 to state three properties that distinguish charm hedging from gamma hedging.

23.5 The overnight period carries 25% of daily variance. Compute the implied volatility a 0DTE option will show in a market whose close-to-close volatility is 16%, explain why this is not a term-structure effect, and state which contract would hedge an announcement made after the close.

23.6 Using the $T^{-0.4}$ law, compute how much steeper a four-hour smile is than a 30-day smile, and explain why quoting a 15-minute option in implied volatility terms conveys less information than quoting its price.

23.7 Explain why the pinning force of Example 23.6 saturates, and why a market that damps small moves but not large ones may be worse than one that damps neither.

23.8 A published gamma-exposure figure assumes dealers are long all calls and short all puts. State what is measured and what is assumed, and explain when the assumption is least defensible.

23.9 A 0DTE option shows 13.4% implied volatility with no event scheduled. A 2 p.m. announcement adds a move of 0.8% standard deviation. Compute the morning implied volatility and its value just after 2 p.m., with two hours left, and explain why the fall is not a fall in volatility.

23.10 Explain why retail flow makes dealers short gamma and systematic premium selling makes them long. A dealer book is net short \$800m of gamma per 1%. Compute the

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hedging trade a 1.5% move requires, state its direction, and explain why the same flow can be negligible against daily volume and material against the depth at 3:55 p.m.

23.11 State the four reasons Section 23.12 gives for the absence of an incident, identify which is behavioral rather than structural, and explain why daily expiration bounds the size of the risk but not its speed.

23.12 Explain what happens to dealer hedging demand when a move takes the index outside the gamma band, and why this is the dangerous configuration.

FURTHER QUESTIONS

23.13 Section 23.2 reports the gamma band's upper edge closing in from +6.76% to +0.44%. Derive the band's edges as a function of σ and T, show why the band is symmetric in log-strike about a point slightly above spot rather than about spot, verify the three pairs of figures, and determine how the band would change if implied volatility doubled. Discuss what this implies about whether 0DTE is more or less dangerous in a high-volatility regime.

23.14 Example 23.2 finds a $1.07bn hedge unwind with the index unmoved. Derive the charm flow between the open and the last half hour for a book with a given open-interest distribution across strikes, explain why over that window the flow is largest for strikes roughly half a daily standard deviation from spot rather than at the money, say where the at-the-money strike's flow goes instead, and discuss what this implies about where in the strike distribution to look for the largest predictable flow.

23.15 Reproduce the one-hour rows of Example 23.5, then price the same two puts in Merton's jump-diffusion of Section 21.5 with the diffusion volatility set to 15%, and find the jump intensity and mean jump size that reproduce the skewed prices. Comment on how many parameter pairs do so, and on what that says about calibrating a dynamic model to a single expiration.

23.16 Section 23.12 argues the market is benign mainly because the flow is two-sided, and that this is the fragile reason. Construct a model of the two groups (retail buyers and systematic sellers) in which the equilibrium is stable, then identify the parameter changes that would break it, and state what observable series would give warning.

23.17 Section 23.14 claims that hedging demand "saturates and then stops" when a move takes the index outside the gamma band. Formalize this: write dealer hedging demand as a function of the index level for a book concentrated at one strike, show where it is increasing and where it vanishes, and assess whether a mechanism that absorbs small moves and not large ones is worse than one that absorbs neither.

23.18 Section 23.13 argues that a daily expiration is a genuine improvement in market completeness. Make the argument precise using Section 12.8's replication result, then construct the counter-argument that finer maturities can reduce welfare by inducing hedging concentration, and state what would have to be measured to decide between them.

23.19 Compare 0DTE with portfolio insurance (Section 40.2), snowball autocallables (Section 35.7) and inverse volatility products (Section 22.8) along the dimensions of crowding, mechanical hedging, accumulation, and observability. Explain which features 0DTE shares and which it does not, and use the comparison to argue either that the profession's current level of concern is about right or that it is misdirected.

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Part V

Credit

Credit risk is priced twice over: once as the chance a borrower fails, and again as the cost of facing a counterparty who might. This part runs from default modeling through the CDS market and the securitization machinery that failed in 2007–2009, and ends with the counterparty and capital regime built in response. The crisis chapter is history with a purpose—most of what Chapters 8, 9, and 27 require exists because of something in it.

24 Credit Risk and Default Modeling
hazard rates, structural models, and the credit triangle

25 Credit Derivatives and the CDS Market
the contract, its valuation, and what the market is saying

26 Securitization, Structured Credit, and the 2007–2009 Crisis
tranching, correlation, and the failure

27 Counterparty Credit Risk and Capital
exposure, netting, SA-CCR, and the capital that follows

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Chapter 24
Credit Risk and Default Modeling

Chapter 9 computed a credit valuation adjustment and took two inputs as given: a default rate read off a CDS spread through the credit triangle, assumed flat, and a recovery rate assumed to be 40%. It pointed here for the machinery behind both. This chapter supplies it, and then asks what the inputs actually mean.

Credit risk is among the oldest financial risks and the largest one most banks carry. Bank loan books, the corporate and sovereign bond markets, the credit default swaps of Chapter 25, the securitizations of Chapter 26, and the counterparty exposures of Chapter 27 all price off the two quantities this chapter is about, and anyone who lends, holds a bond, or faces a counterparty is exposed to them whether or not they model them. Section 24.16 says who those holders are and how the risk has moved between them since the reforms.

The subject divides cleanly. Credit risk is two questions (whether a borrower defaults, and how much is recovered if it does), and almost every difficulty in this part of the book comes from the fact that market prices identify only their product. Sections 24.1 to 24.4 establish that, with the credit triangle and the hazard-curve bootstrap on the way. Sections 24.5 to 24.8 are the two families of model that try to say more: structural models, which explain default as an option finishing out of the money and fail at short maturities, and reduced-form models, which assume it and fit anything. Sections 24.9 to 24.12 are the gap between what prices imply and what happens: risk-neutral against historical default rates, ratings and migration, and the sovereign case in which default is a choice. Section 24.13 introduces the correlation machinery, Section 24.14 turns it into the credit VaR against which a loan book is capitalized, Section 24.15 reads a hazard rate from a bond price and sets it against the CDS, and Section 24.17 records where the models came from.

24.1 THE TWO QUESTIONS

A defaultable bond promises \$100 and delivers either \$100 or something less. Valuing it requires a probability and a payoff.

Default probability. The default probability is not a single number but a curve: the chance of default by each future date. Write τ for the random time of default and $Q(t) = \mathbb{Q}(\tau > t)$ for the **survival probability**, the chance that the borrower is still performing at t. The convention is to work not with Q itself but with the **hazard rate** $\lambda(t)$, the probability of defaulting in the next instant given survival to t, per unit of time:

$$\lambda(t) = \lim_{\Delta t \to 0} \frac{\mathbb{Q}(t < \tau \le t + \Delta t \mid \tau > t)}{\Delta t} = -\frac{Q'(t)}{Q(t)} = -\frac{d}{dt} \ln Q(t) \tag{24.1}$$

The second equality is the definition of a conditional probability: the probability of defaulting in $(t, t + \Delta t]$ is $Q(t) - Q(t + \Delta t)$, and dividing by the probability $Q(t)$ of being there to default gives the conditional rate. Integrating the last form gives the survival probability to time T,

$$Q(T) = \exp\left(- \int_0^T \lambda(u)\, du \right) \tag{24.2}$$

For a constant hazard this is $e^{-\lambda T}$, and the density of default at t is $\lambda e^{-\lambda t}$.

The hazard rate is not the default probability. A 3% hazard rate is an intensity of 3% per year: *given survival to the start of a year*, the chance of defaulting during it is $1 - e^{-0.03} = 2.96\%$, and over ten years the cumulative default probability is $1 - e^{-0.3} = 25.9\%$, not 30%. The distinction matters as soon as the horizon is long or the borrower is weak.

Under which probability? Two versions of every quantity above will appear in this chapter, and they are not close. The one that prices a contract is the risk-neutral probability of Chapters 13 and 28 (written $\mathbb{Q}$ there and here), and it is the one the market reveals: a CDS spread read through the next two sections gives a risk-neutral λ, because the spread is a price. The one that forecasts is the **real-world** probability ($\mathbb{P}$), estimated from default records. Everything up to Section 24.8 is risk-neutral, and should be read as pricing rather than prediction; Section 24.9 sets the two side by side and finds them apart by a factor of several.

Recovery. Recovery is what the claim is worth after default, conventionally quoted as a fraction R of face value, with $1 - R$ the **loss given default**. Forty percent is the market's standard assumption for senior unsecured corporate debt, and Section 24.4 explains why that number is used far beyond the evidence supporting it.

Everything else in this chapter is a way of estimating one of these two, or an admission that only their product is observable.

24.2 THE CREDIT TRIANGLE

A credit default swap pays $1 - R$ per unit of notional if the **reference entity** (the borrower the contract is written on) defaults before maturity T, in exchange for a spread s per year paid by the protection buyer until default or maturity. Its par spread is the s at which the two legs have the same value today, and both legs are integrals over the risk-neutral survival curve of Section 24.1. Write $P(0,t)$ for the risk-free discount factor and take the premium to be paid continuously for the moment. The premium leg collects $s\,dt$ over each instant the borrower is still alive, so its value is

$$\text{premium leg} = s \int_0^T P(0,t)\, Q(t)\, dt.$$

The protection leg pays $1 - R$ at the moment of default, and by (24.1) the probability of that moment falling in $(t, t + dt]$ is $\lambda(t)Q(t)\,dt$, the hazard rate times the probability of having survived to t, so

$$\text{protection leg} = (1 - R) \int_0^T P(0,t)\, \lambda(t)\, Q(t)\, dt.$$

If the hazard rate and the interest rate are both flat, $Q(t) = e^{-\lambda t}$ and $P(0,t) = e^{-rt}$, the integrand decays at $r + \lambda$, and

$$\text{protection leg} = (1 - R)\,\lambda \int_0^T e^{-(r+\lambda)t}\, dt = (1 - R)\,\lambda\, \frac{1 - e^{-(r+\lambda)T}}{r + \lambda}.$$

For a flat hazard rate on any discount curve, λ comes outside the integral, the same integral $\int_0^T P(0,t)Q(t)\,dt$ appears in both legs, and setting them equal cancels it:

$$\boxed{s = \lambda\,(1 - R)} \tag{24.3}$$

Neither the discount curve nor the level of the survival probability remains in the answer; the par spread is the default rate times the loss when default happens, which is what the price of insurance ought to be. The result is usually called the **credit triangle**, because any two of s, λ and $1 - R$ give the third. Reading a spread as a hazard rate therefore requires only a recovery assumption, not a discount curve, which is why traders can do the conversion mentally; and a

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change in the discount curve alone barely moves the par spread, which is why CDS spreads did not move mechanically when the discounting convention changed in the reforms of Chapter 4.

This is usually presented as an approximation valid for small λ, and the standard statement puts the error in the wrong place.

Example 24.1 The credit triangle is exact for a continuous premium, and the error people quote is the accrued premium

Price a five-year CDS with a flat hazard rate $\lambda = 1\%$, discounting at $r = 3\%$, recovery $R = 40\%$, three ways: with the premium paid continuously; quarterly, with the premium accrued since the last payment date paid on default, as real contracts do; and quarterly with no accrual. The protection leg is the same in all three; from the flat-rate formula above, with $r + \lambda = 0.04$, it is $0.6 \times 0.01 \times (1 - e^{-0.2})/0.04 = 0.027190$ per unit of notional. The premium legs are the spread s times an annuity, and the annuities differ: 4.5317, 4.5148 and 4.5091 ([TN25] works them out one quarter at a time). The par spread sets each premium leg equal to the protection leg:

Continuous premium	$4.5317\, s = 0.027190$	$\Rightarrow\ s = 60.00$ bp
Quarterly, with accrual	$4.5148\, s = 0.027190$	$\Rightarrow\ s = 60.23$ bp
Quarterly, no accrual	$4.5091\, s = 0.027190$	$\Rightarrow\ s = 60.30$ bp

The first is $\lambda(1 - R) = 0.01 \times 0.6$ exactly, as the derivation of (24.3) says it must be. The second is 0.376% higher, because a premium paid at the end of a quarter is discounted for half a quarter longer, on average, than the same money paid continuously through it; to first order the excess is $r\Delta/2$ for a payment interval $\Delta = 0.25$, and Chapter 25 carries the same effect as about a basis point on a 250 bp contract. The third drops the accrued premium, and that is the "approximation error" of the textbook statement: a tenth of a percent here. It is the value of a contract feature, not an error in the triangle, and unlike the 0.376% it grows with the hazard rate: at $\lambda = 20\%$ the three spreads are 1200.00, 1204.47 and 1235.17 bp, and the missing accrual is worth 2.5%.

24.3 BOOTSTRAPPING A HAZARD CURVE

Spreads are quoted at several maturities and they are not flat. Extracting the implied hazard curve is the credit analog of Chapter 4's zero-curve bootstrap: work outward, taking each quote as determining the hazard rate over the interval that quote is the first to cover.

Example 24.2 From a spread curve to a hazard curve

An issuer's CDS curve is quoted at the maturities below; recovery is assumed at 40% and rates at 3%:

Maturity	Spread (bp)	Bootstrapped λ	Credit-triangle guess
1y	80	1.333% (0–1y)	1.333%
3y	130	2.617% (1–3y)	2.167%
5y	170	3.998% (3–5y)	2.833%
7y	190	4.222% (5–7y)	3.167%
10y	200	3.885% (7–10y)	3.333%

The second row shows the step. The one-year quote fixes $\lambda_1 = 0.0080/0.6 = 1.333\%$ exactly by (24.3), because a single bucket is the constant-hazard case. The three-year quote must then be reproduced by a contract whose first year runs at λ_1 and whose next two years run at an unknown λ_2. With a continuously paid premium both legs are integrals of $e^{-rt}Q(t)$, and the integral splits at the knot: write $A_1 = \int_0^1 e^{-(r+\lambda_1)t}\,dt = 0.9786$ for the first year and $A_2 = \int_1^3 e^{-rt}Q(t)\,dt = Q(1)\,e^{-r}\int_0^2 e^{-(r+\lambda_2)u}\,du$ for the next two (on $[1,3]$, $Q(t) = Q(1)\,e^{-\lambda_2(t-1)}$; substitute $u = t - 1$), where $Q(1) = e^{-\lambda_1} = 0.9868$ is the probability of reaching the second bucket alive. The par

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condition is

$$s_3\,(A_1 + A_2) = (1 - R)\,(\lambda_1 A_1 + \lambda_2 A_2),$$

one equation in λ_2, since A_2 depends on it. At $\lambda_2 = 2.617\%$, $A_2 = 1.8115$; the premium leg is $0.0130 \times 2.7902 = 0.036272$ and the protection leg is $0.6 \times (0.013049 + 0.047405) = 0.036272$, so the equation balances. Each later bucket repeats the step with one more known piece and one unknown. The credit triangle applied to the three-year spread gives $0.0130/0.6 = 2.167\%$ instead, close to the time-weighted average of the two bucket rates, $(1.333\% + 2 \times 2.617\%)/3 = 2.19\%$, but not the rate for years one to three.

The last column applies (24.3) directly to each quoted spread. It is the same as the bootstrapped rate for the first bucket and different after that, most of all in the 3–5 year bucket, and the reason is the reason a swap rate is not a forward rate: a spread is an average over the whole life of the contract, and the hazard rate for a later bucket must exceed that average when the curve is upward sloping.

At the 3–5 year bucket the difference is large: 3.998% against 2.833%, a factor of 1.41. Anyone using the five-year spread to price something that only pays off in years three to five will understate the hazard rate for those years by almost 30%.

The implied cumulative default probabilities are

	1y	3y	5y	7y	10y
Cumulative default probability	1.32%	6.36%	13.55%	20.55%	29.29%

Note the shape of the hazard curve: rising through seven years and then falling. An upward-sloping spread curve is the norm for **investment-grade** issuers, those rated BBB (Baa) or higher; an issuer rated below that line is **speculative grade**. The upward slope reflects the same mean reversion that Section 19.6 found in volatility: a currently healthy firm is more likely to deteriorate than to improve. The fall after seven years is arithmetic: the ten-year spread is only 10 bp above the seven-year one, so the last bucket cannot carry a hazard rate as high as the one before it. Distressed issuers show the same effect in extreme form, with an inverted spread curve, for the same reason inverted volatility curves appear in stress: if the firm survives the next year, the danger has largely passed and the hazard falls.

Figure 24.1 draws the table's last two columns.

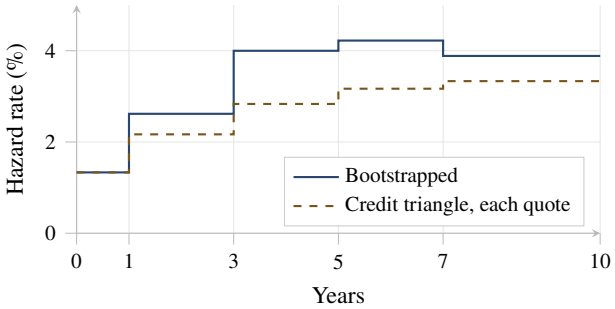


Figure 24.1: Hazard rates bootstrapped from the spread curve of Example 24.2, against the credit triangle applied to each quoted spread directly.

24.4 RECOVERY: THE OTHER UNKNOWN

The bootstrap above assumed $R = 40\%$. Where does that come from, and what happens if it is wrong? The answer to the first question is convention, supported by an average that conceals almost everything of interest.

Recovery varies by seniority. Table 24.1 gives Moody's long-run averages measured two ways: the market's bid for the defaulted debt shortly after default, close to what a CDS auction

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Table 24.1: Average recovery rates by priority of claim (%), issuer-weighted for trading prices. Moody's, March 2023 study.

	Trading price, 1983–2022	Ultimate, 1987–2022
First-lien bank loan	65.7	71.7
Second-lien bank loan	33.3	–
Senior unsecured bank loan	46.3	–
First-lien bond	55.0	61.2
Second-lien bond	44.2	–
Senior unsecured bond	37.7	47.1
Senior subordinated bond	31.0	–
Subordinated bond	32.0	27.8
Junior subordinated bond	22.3	–

sets, and the ultimate recovery when the default is resolved, usually a year or two later (North American nonfinancial issuers, for all term loans and all senior secured bonds, set against the first-lien rows). The ranking broadly follows priority, from about two-thirds on first-lien loans to about a fifth on junior subordinated bonds, though not strictly. Senior unsecured bonds, the class the 40% convention describes, recovered 37.7% at the trading price and 47.1% at resolution: the second figure is about ten points higher, though the two come from different samples.

And it is correlated with the default rate. Recoveries are lowest in the years when defaults are highest, because a firm defaulting in a recession is selling its assets into a market full of other distressed sellers.

Example 24.3 Hazard and recovery are not separately identified

A five-year CDS trades at 170 bp. Which (λ, R) pair produced it?

Recovery	Implied λ	5-year default probability
20%	2.125%	10.08%
40%	2.833%	13.21%
60%	4.250%	19.14%
80%	8.500%	34.62%

All four price the CDS identically. The market observes only the product $\lambda(1 - R)$, so the recovery assumption is not extracted from the spread—it is imposed on it.

Usually this does not matter, because the thing being priced also depends only on the product. It matters a great deal when it does not, and three cases show how much.

A digital default swap pays a fixed amount on default regardless of recovery, so its value depends on λ alone. The table's four rows price it roughly in proportion to the last column, a range of more than three to one; between the 40% and 60% rows alone a five-year digital is worth 45% more. The contract has only ever traded thinly, mostly between dealers separating a view on default from a view on recovery; its use here is as the sharpest test of the split.

A first-to-default basket pays $1 - R$ on the first default among a group of borrowers. To first order its value is the sum of the borrowers' $\lambda_i(1 - R)$, the product again, so most of the ambiguity cancels; what survives comes from the probability of at least one default being concave in the λ_i. On five independent borrowers each quoted at 170 bp, the basket prices about 14% lower at 60% recovery than at 40%, where the digital prices 45% higher.

And any calculation of expected loss for capital or provisioning needs the split, because regulation asks for probability of default and loss given default separately.

The practical convention is to fix R by seniority and sector and let λ absorb everything, which is defensible for pricing and indefensible for risk measurement, for the reasons the case study below

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sets out.

Case Study 24.1 Why the 40% convention survives evidence against it

The recovery assumption of Section 24.4 is used with a confidence the data do not support, and the gap follows a pattern that recurs throughout this part of the book.

Realized recoveries are enormously dispersed. The 40% figure is a long-run average across senior unsecured corporate bonds. Year by year the average moves a good deal: senior unsecured bonds recovered 48.7% in 2021 and 31.1% in 2022, when defaults by Russian issuers (recovering 15.5%) and Chinese property developers (25.3%) pulled it down.

The dispersion is systematic, not noise. Recovery depends on seniority, on whether the debt is secured and by what, on the industry's asset tangibility, on the jurisdiction's insolvency regime, and on where in the credit cycle the default occurs. An asset-heavy firm in a creditor-friendly jurisdiction defaulting in a boom is a different proposition from an asset-light firm in a debtor-friendly one defaulting in a recession, and the two have almost nothing in common but the letter on the rating.

And it moves with the thing it is multiplied by. A portfolio model that fixes recovery at 40% and lets only the default rate vary understates tail losses twice over: once because more borrowers default, and once because each one recovers less. In a bad year in which twice the average number of borrowers default and recovery falls from 40% to 25%, the loss is $2 \times 0.75/0.60 = 2.5$ times the average year's, where the fixed-recovery model says 2.

So why does the convention hold?

Because for pricing a CDS it does not matter. Example 24.3 showed that the spread identifies only $\lambda(1 - R)$, so a desk that fixes R and lets λ absorb the residual prices the contract correctly whatever R truly is. The convention is not a belief about recovery; it is a coordinate choice, and within its domain it is harmless.

The damage is done when the coordinate is exported. Multiplying the extracted λ back by the same 60% loss given default is harmless: it returns the spread, whatever the true recovery. The harm comes from pairing that λ with a recovery from somewhere else (a risk system that takes the λ extracted at 40% and applies its own loss-given-default estimate of 75% has used two recoveries for one borrower), or from feeding λ into a calculation that is nonlinear in the default probability, such as the worst-case default rate of Section 24.14, where the product no longer carries through. A capital calculation that requires probability of default and loss given default separately is asking for a decomposition the market never provided.

The general lesson is one this book has met before. Section 21.10 observed that a calibration encodes an assumption about what the model is for; Section 22.6 observed that VIX is well defined and is not the price of anything. A number can be exactly right for the purpose that produced it and badly wrong one step away, and credit is the area where that distance is shortest and least visible.

24.5 STRUCTURAL MODELS: MERTON

The two families of model differ in whether default is explained or assumed.

Merton's insight is that a firm's equity is a call option on its assets. If the firm's assets are worth V_T at the date its debt of face value D matures, shareholders receive $\max(V_T - D, 0)$: they keep the surplus and walk away from the shortfall. The strike is the face value of the debt.

Figure 24.2 shows the consequence for the other side. Debt holders receive $\min(V_T, D)$,

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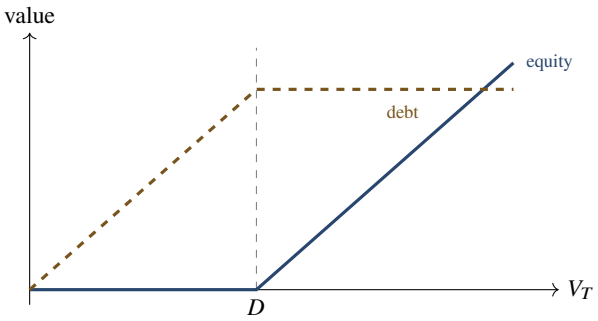


Figure 24.2: Equity and debt payoffs at the debt's maturity.

which is D less a put struck at D: lending is selling a put on the borrower's assets, and the credit spread is the premium on that put.

This is the sharpest available statement of the conflict Section 17.11 flagged. Equity is a call, so equity holders gain from volatility; debt is short a put, so debt holders lose from it. An executive paid in options is paid in a claim whose value rises with firm risk, and every increase in that risk, at a given firm value, is a transfer from creditors. Nothing in the payoff structure aligns the two, and no amount of governance language changes the derivative.

24.6 CALIBRATING MERTON FROM WHAT IS OBSERVABLE

Firm asset value and asset volatility are not observable. Equity value and equity volatility are, and two equations link them.

Write V for the firm's asset value today and σ_V for its volatility, with D and T the face value and maturity of the debt as in Section 24.5. The first equation is Chapter 15's formula applied to the call that equity is:

$$E = V\,N(d_1) - De^{-rT}N(d_2), \qquad d_1 = \frac{\ln(V/D) + \left(r + \tfrac{1}{2}\sigma_V^2\right)T}{\sigma_V\sqrt{T}}, \qquad d_2 = d_1 - \sigma_V\sqrt{T} \tag{24.4}$$

The second follows from applying Itô's lemma to E as a function of V, which is Chapter 14's calculation and produces the elasticity of the call to the firm, the percentage change in equity value for a one percent change in asset value:

$$\sigma_E E = N(d_1)\,\sigma_V V \tag{24.5}$$

Two equations, two unknowns, and everything on the left-hand side is quoted. There is no closed form for V and σ_V; the pair is solved numerically, which is all Example 24.4 does.

The output the model is actually used for is the argument of the second normal:

$$\mathrm{DD} = d_2 = \frac{\ln(V/D) + \left(r - \tfrac{1}{2}\sigma_V^2\right)T}{\sigma_V\sqrt{T}} \tag{24.6}$$

This is the **distance to default** (how many standard deviations of asset value separate the firm from its debt), and the risk-neutral default probability is

$$\mathbb{Q}(V_T < D) = 1 - N(d_2) = N(-\mathrm{DD}).$$

Example 24.4 Distance to default from a share price

A firm has equity worth 40 with 55% volatility, and debt of face value 80 maturing in five years. Rates are 5%.

Solving the two equations simultaneously:

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Implied asset value	95.87
Implied asset volatility	27.2%
Distance to default	0.41 standard deviations
Risk-neutral 5-year default probability	34.26%
Implied credit spread	218 bp

The last row follows from the first. Debt is the firm less its equity, $95.87 - 40 = 55.87$, against $80e^{-0.25} = 62.30$ if it were riskless, and the continuously compounded spread that discounts one to the other is

$$s = -\frac{1}{T}\ln\frac{55.87}{62.30} = 2.18\%.$$

The first two rows can be checked by substitution. At the solution, before rounding, $d_1 = 1.0131$ and $d_2 = 0.4054$, so (24.4) gives $95.87 \times 0.8445 - 62.30 \times 0.6574 = 80.96 - 40.96 = 40.00$, the observed equity value, and (24.5) gives $\sigma_E = 0.8445 \times 0.272 \times 95.87/40 \approx 55\%$, the observed volatility. The same $d_2 = 0.4054$ is the third row, and $N(-0.4054) = 34.26\%$ is the fourth.

The distance to default of (24.6) is the model's most durable output. What Merton's model establishes is a monotonic relationship between distance to default and default probability: a firm further from its **default point** (the asset value at which it defaults, the face value D in Merton's model), measured in standard deviations of its own asset value, defaults less often. That ordering is more robust than the default probability that follows from it, because it does not require the normality assumption to be right, only the monotonicity, which is the license the commercial mapping below takes.

Two observations about the numbers follow.

Equity is twice as volatile as the firm. Equity volatility is twice asset volatility (55% against 27.2%), and the ratio is the equity's elasticity to the firm, $N(d_1)V/E = 0.8445 \times 95.87/40 = 2.02$: the leverage $V/E = 2.40$ scaled down by the call's delta. It rises as the firm approaches default, which is why distressed equities are so volatile and why Section 19.9's leverage explanation of the skew is this model in miniature.

And the implied recovery is high. The expected loss is 10.3% against a 34% default probability, an implied recovery near 70%. Merton recovers too much because, on default, the bondholders take over the whole of the firm's assets at no cost. With 27% asset volatility those assets are still worth 55.88 on average at the debt's maturity when the firm defaults (a risk-neutral expectation), about 70% of the debt's face, and the model has no bankruptcy costs or fire-sale discounts to reduce that. Real defaults destroy value in ways the model has no room for.

What the commercial versions change. The framework became a product before it became a consensus, and the standard implementation is the one built by **KMV** (the firm Kealhofer, McQuown and Vasicek founded, named from their initials), now part of Moody's Analytics. It departs from Merton in three ways, each replacing one of the model's weakest assumptions with data rather than with more theory.

The default point is not the face value of debt. Firms do not fail the moment assets touch total liabilities, because long-dated debt need not be repaid yet. The default point is set at short-term debt plus half of long-term debt. That rule is not derived from the model; KMV chose it because, in its database of defaulted firms, asset value at the time of default was typically around that level.

The default probability does not come from the normal distribution. Instead of reading $N(-\text{DD})$, distance to default is mapped to an **expected default frequency** (EDF) through an empirical table built from a large default history, and the distance itself is computed with the firm's expected asset return in place of r, because the target is a real-world frequency rather than a price. This is the change that matters, because Section 24.7 is about to show that the normal tail is far too thin at short horizons. It is also why distance to default outlives the model that produced it: the mapping uses only the ordering, so (24.6) can be right about which firms

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are riskier while $N(-\text{DD})$ is wrong about how risky they are.

And the calibration is iterative. Rather than solving (24.4) and (24.5) once, the implied asset series is used to re-estimate σ_V and the system is re-solved to convergence, which is more stable when the equity volatility going in is itself an estimate.

The pattern recurs. A commercial version of an academic model is usually that model with its weakest assumption replaced by a fitted table, and the table holds both the value and the secrecy. Chapter 26 shows the same substitution made for default correlation, and made badly.

24.7 WHAT MERTON GETS WRONG

The model's characteristic failure is at the short end, and it can be measured precisely.

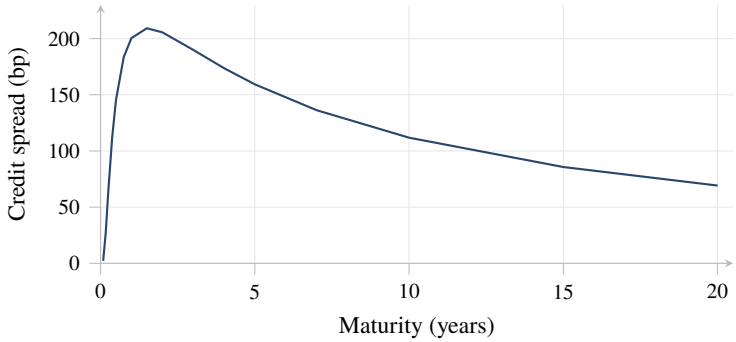


Figure 24.3: Credit spread by maturity in Merton's model. Assets 100, debt 80, asset volatility 25%, $r = 5\%$.

Figure 24.3 has the right shape in the middle and the wrong shape at the left edge.

Maturity	Spread	Default probability
1 week	0.0000 bp	$6 \times 10^{-9}\%$
1 month	2.2 bp	0.09%
3 months	64.6 bp	3.42%
1 year	200.5 bp	16.66%
2 years	205.6 bp	23.05%

The spread goes to zero as maturity shortens. A firm whose assets exceed its debt cannot default tomorrow if asset value follows a continuous path, because it would have to pass through every intermediate level first and there is no time for it to travel.

Real short-dated credit does not behave this way. One-year CDS on investment-grade issuers trades in the tens of basis points and does not collapse toward zero as maturity shortens; short-dated spreads on stressed issuers can be enormous.

This is the same defect, in the same place, for the same reason as Section 21.7's. There, a diffusive volatility model could not produce the observed short-dated skew because continuous paths cannot generate short-horizon tails. Here, a diffusive asset process cannot produce short-dated credit spreads for the same reason. Two literatures discovered the same limitation of continuous paths independently, and the repairs are related: jumps in both fields and, in credit, default made to arrive as a surprise rather than as the outcome of a diffusion.

That second repair is the reduced-form approach.

24.8 REDUCED-FORM MODELS

Rather than explain default, model it directly. Let the default time τ be the first arrival of a Poisson process whose intensity λ_t is itself a stochastic process: a **Cox process**, or doubly

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stochastic Poisson process. Conditional on the whole path of the intensity, survival to T has the probability (24.2) already gave; averaging over intensity paths,

$$Q(T) = \mathbb{E}\left[\exp\left(-\int_0^T \lambda_u \, du\right)\right] \tag{24.7}$$

which is (24.2) with the expectation restored. Default is a surprise: however healthy the firm looks, the probability that it defaults in the next instant is $\lambda_t \, dt$, not zero. That is what the diffusion of Section 24.7 could not deliver, and it is why this construction repairs the short end: with default arriving as a jump, a one-month contract carries a spread of $\lambda(1 - R)$ from its first day.

Now price a bond with the same expectation as (24.7), because that step is where the framework shows its value. A zero-coupon bond paying 1 at T and nothing if the issuer has defaulted is worth

$$\bar{P}(0,T) = \mathbb{E}\left[e^{-\int_0^T r_u \, du} \, \mathbf{1}_{\{\tau > T\}}\right] = \mathbb{E}\left[\exp\left(-\int_0^T (r_u + \lambda_u) \, du\right)\right] \tag{24.8}$$

By (24.8) the hazard rate enters exactly where the short rate does, and that is the whole reason the approach took over. A defaultable bond is a riskless bond discounted at $r + \lambda$, so everything built for short-rate models in Chapter 31 (the trees, the closed forms, the calibration machinery) transfers to credit unchanged, with the intensity playing the part of a second rate factor. Nothing else in credit modeling is this convenient, and convenience is why the market prices with it.

Recovery slots into the same place. Suppose that on default the claim is worth a fraction R of what it was worth just before, so the loss is $1 - R$ of its *market value*, not of its face. Follow the bond over a short interval dt, writing $\bar{P}_t$ for its value at t. With probability $1 - \lambda_t \, dt$ the borrower survives and the bond is worth $\bar{P}_{t+dt}$; with probability $\lambda_t \, dt$ it defaults and the bond is worth $R \, \bar{P}_{t+dt}$. Under the risk-neutral measure the bond must earn the riskless rate in expectation, so

$$\begin{aligned} \bar{P}_t &= e^{-r_t \, dt}\left[(1 - \lambda_t \, dt) \, \bar{P}_{t+dt} + \lambda_t \, dt \, R \, \bar{P}_{t+dt}\right] \\ &= e^{-r_t \, dt}\big(1 - \lambda_t (1 - R) \, dt\big)\bar{P}_{t+dt} \approx e^{-(r_t + \lambda_t (1-R)) \, dt} \, \bar{P}_{t+dt}. \end{aligned}$$

Figure 24.4 draws the step. Default has added $\lambda_t(1 - R)$ to the one-period discount rate: the intensity of the event times the fraction it destroys. Chaining these steps from 0 to T and taking the expectation over the paths of r and λ gives

$$\bar{P}(0,T) = \mathbb{E}\left[\exp\left(-\int_0^T \big(r_u + \lambda_u(1 - R)\big) \, du\right)\right] \tag{24.9}$$

so the credit spread is $\lambda(1 - R)$. That is (24.3), Section 24.2's credit triangle, arriving as the constant-intensity case of a general result rather than as a rule of thumb. It also makes Example 24.3's identification problem structural rather than accidental: λ and R enter the price only through their product, so no bond price and no CDS spread can ever separate them.

One convention changed along the way. Section 24.2 took recovery as a fraction of *face*, which is how a CDS settles and how the market quotes R; the derivation above takes it as a fraction of the bond's *market value* an instant before default, which is what makes the algebra close. For a bond trading near par the two conventions agree, because face and market value then coincide. They differ for a bond far from par (a bond at 50 that recovers 40 has recovered 40% of its face but 80% of its market value), and for a CDS on such a bond the face convention is the one that matters, since the contract pays $1 - R$ of face whatever the bond was worth. Chapter 25 keeps the face convention throughout for that reason; the market-value convention

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is used here because it is the one under which the hazard rate enters a bond price exactly as a spread.

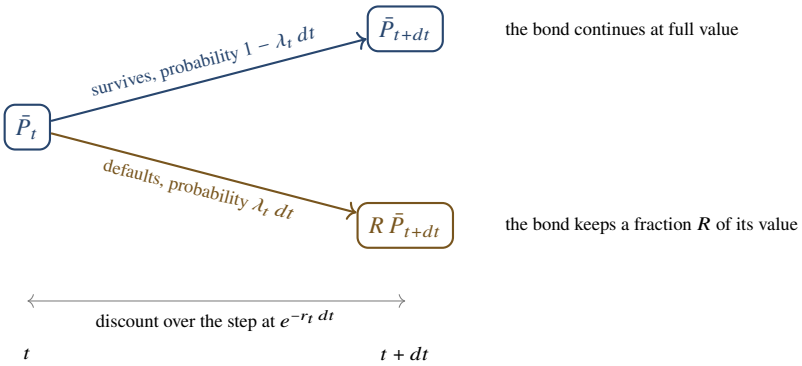


Figure 24.4: One step of the argument behind (24.9). The bond's value today is the discounted risk-neutral expectation of the two outcomes, which shortens to a discount rate of $r_t + \lambda_t(1 - R)$.

Each family has its strengths and its weaknesses.

Reduced-form models fit. Because λ is a free function, any observed spread curve can be reproduced exactly, which is Section 24.3's bootstrap, and it is the same property that made local volatility the standard choice in Chapter 21.

Structural models explain. They say why a spread is what it is, connect it to the equity market, and produce a hedge: if equity is a call on the firm, credit can be hedged with equity, and capital-structure arbitrage is that trade.

And they say different things about correlation, which turns out to be where it matters most. In a structural model two firms default together because their asset values are correlated, a mechanism with an economic story attached. In a reduced-form model correlation must be imposed on the hazard rates, and the imposition is arbitrary. Chapter 26 is largely about what happened when the market chose a convenient imposition and treated it as a fact.

The market uses reduced-form for pricing and structural for thinking. A CDS desk bootstraps hazard rates; a credit analyst asks how far the firm is from default. The two are not in conflict, because they answer different questions.

24.9 REAL-WORLD AGAINST RISK-NEUTRAL

Section 9.2 computed CVA from the counterparty's CDS spread rather than from its historical default rate, and said that the default rate a spread implies has run several times the historical one. Here is how much.

Example 24.5 Most of a credit spread is not expected loss

Five-year average cumulative default rates by rating for global corporate issuers, 1981–2024, from S&P Global Ratings' 2024 annual default study, against the risk-neutral probabilities implied by representative mid-cycle five-year spreads at 40% recovery. The spreads are illustrative; the historical rates are S&P's. AAA sits above AA only because few issuers were ever rated AAA, a small-sample artifact.

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Rating	Spread	Historical	Risk-neutral	Ratio
AAA	40 bp	0.34%	3.28%	9.6×
AA	55 bp	0.28%	4.48%	16.0×
A	80 bp	0.39%	6.45%	16.5×
BBB	130 bp	1.36%	10.27%	7.5×
BB	300 bp	5.75%	22.12%	3.8×
B	550 bp	15.60%	36.77%	2.4×

Decomposing the 130 bp BBB spread: expected loss at the historical default rate is the annual hazard $-\ln(1-0.0136)/5 = 0.274\%$ times $1-R$, or 16.4 bp, leaving 113.6 bp, so 87% of the spread is not expected loss.

And the ratio is largest for high-grade issuers. An AA or A issuer's risk-neutral default probability is about sixteen times its historical rate and a single-B issuer's under two and a half times; AAA is lower only because its small-sample historical rate exceeds AA's. Figure 24.5 draws the two columns on a log scale, where the vertical gap is the ratio, and the gap narrows steadily from A down to B. Whatever the excess compensates for, it does not scale with default risk.

Three things are in the residual, and separating them is a live question rather than a settled one. *A risk premium*, because defaults cluster in bad states and bearing them is unattractive for the reason Section 19.10 gave for volatility. *Illiquidity*, because corporate bonds trade rarely and the spread includes the cost of that. And *a premium for parameter uncertainty*, about the inputs themselves: the recovery of Example 24.3 is assumed, not known, and a seller of protection is being compensated for that too.

The practical consequence follows. Use risk-neutral probabilities to price and hedge, historical probabilities to forecast losses, and never mix them. An institution that provisions using risk-neutral probabilities will over-reserve by a factor of seven or more on investment-grade portfolios; one that prices using historical probabilities will systematically sell protection too cheaply and discover the difference when it tries to hedge.

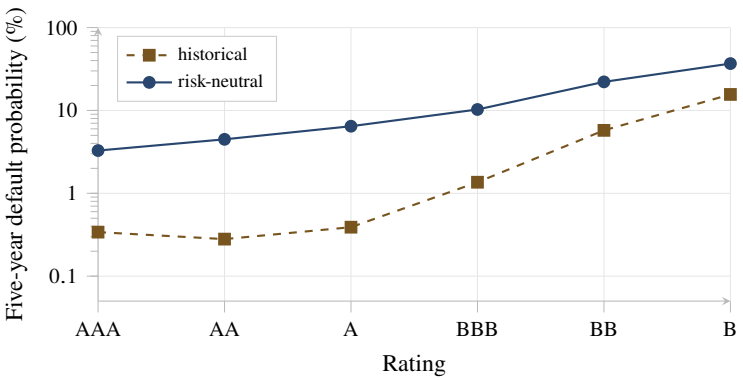


Figure 24.5: Five-year default probability by rating, historical and implied by spreads at 40% recovery, from Example 24.5. Log scale, so the vertical gap is the ratio.

24.10 RATINGS AND THEIR LIMITS

Ratings are an ordinal ranking of credit quality produced by agencies paid, for corporate and structured issues, by the issuer. Three properties determine how they should be used.

They rank well and calibrate poorly. The ordering has held up: over long samples, higher-rated issuers default less. But a corporate rating is defined as an ordinal opinion of

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relative credit risk, and neither agency attaches a fixed probability to the letter. The mapping users work with is assembled after the fact from the agencies' own default studies (the source of Example 24.5), and it is much less stable than the ordering, drifting with the credit cycle and differing across asset classes. The one place an explicit letter-to-probability table did exist is structured finance. A **collateralized debt obligation** (CDO) is a pool of bonds or loans whose losses are divided into **tranches**, layers that absorb those losses in turn, and Moody's idealized default rates and the default-rate assumptions in S&P's CDO model were used to size the tranches. Chapter 26 is about what happened to that table: the same letter applied to a corporate bond and to a tranche described very different risks, because the tranche's rating depended on a correlation assumption that the letter did not disclose.

They are through-the-cycle by design. Agencies deliberately avoid re-rating on every market move, which makes ratings stable and makes them lag. An issuer whose spread has tripled may hold its rating for months. This is a feature for a bond mandate and a defect for a risk system, and confusing the two produces a risk report that says nothing has changed while the market says otherwise.

And they are used in regulation and contracts, which creates mechanical effects that have nothing to do with information. A downgrade below investment grade forces sales by mandate-constrained holders, triggers collateral calls under CSAs of the kind Chapter 8 described, and can raise funding costs discontinuously. A rating is therefore not merely an opinion about default; it is an event with contractual consequences, and the consequences can cause the deterioration the rating anticipated.

24.11 MIGRATION: HOW DEFAULT ACTUALLY HAPPENS

Firms rarely jump from health to default. They deteriorate, and the deterioration is what a rating records. Modeling that path requires a **transition matrix**: the probability of moving from each rating to each other rating over a year. Default is absorbing: an issuer that reaches it stays there.

Table 24.2: Illustrative one-year rating transition probabilities (%); rows are the starting rating, D is default.

	AAA	AA	A	BBB	BB	B	CCC	D
AAA	90.09	9.01	0.60	0.10	0.10	0.10	0.00	0.00
AA	0.60	90.59	7.91	0.60	0.10	0.20	0.00	0.00
A	0.05	2.10	91.40	5.50	0.60	0.20	0.10	0.05
BBB	0.02	0.20	4.20	89.90	4.30	0.90	0.20	0.28
BB	0.02	0.06	0.30	5.60	83.00	8.50	1.00	1.52
B	0.00	0.05	0.20	0.40	5.80	82.00	5.10	6.45
CCC	0.00	0.00	0.30	0.60	1.50	12.00	55.00	30.60

The matrix is illustrative, and from A down it is harsher than the long-run averages of Example 24.5: compounded to five years it gives default probabilities of 2.89% for BBB, 12.0% for BB and 31.9% for B, against S&P's 1.36%, 5.75% and 15.60%, so its pattern, not its levels, is what to carry over, and the two should not be mixed. Raising the matrix to the nth power gives n-year transitions, and the default column gives cumulative default probabilities. Horizons that are not whole years need one more step, which practitioners take and this chapter only names: find the **generator matrix** Λ with $P = e^{\Lambda}$, whose off-diagonal entries are the intensities of moving from one rating to another, extending the hazard rate of Section 24.1 from one exit to seven, and then $P(t) = e^{t\Lambda}$ gives the transition matrix for any horizon t. Not every annual matrix has a valid generator; Lando's book in the Further Reading treats the fitting.

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Example 24.6 Downgrade is the route to default

Where a BBB issuer stands after five years, from Table 24.2:

	AAA	AA	A	BBB	BB	B	CCC	D
Probability (%)	0.10	1.37	14.64	62.17	12.90	5.06	0.87	2.89

So 78.3% is still investment grade, 18.8% has fallen to speculative grade (junk) without defaulting, and 2.89% has defaulted.

Now compare that 2.89% with the naive calculation. The one-year BBB default rate is 0.28%; compounding it as though each year were independent gives

$$1 - (1 - 0.0028)^5 = 1.39\%$$

against the matrix's 2.89%, a factor of 2.08.

The matrix gives more because default is reached through downgrade. A BBB issuer that is still alive in year four is disproportionately likely to be a BB or B issuer by then, facing a much higher hazard than the 0.28% it started with.

The route shows in the year-by-year numbers. Dividing each year's increase in the cumulative default probability by the probability of having survived to its start gives the **forward default probability** for that year, the discrete counterpart of the hazard rate of Section 24.1:

Year	1	2	3	4	5
Cumulative default probability (%)	0.28	0.72	1.31	2.03	2.89
Forward default probability (%)	0.28	0.44	0.59	0.74	0.87

The forward rate triples over five years although the starting rating has not changed, because the issuers still alive in year five are a different population from the ones that started: the ones that stayed BBB, plus the ones that fell to BB and B and have not yet defaulted.

This is the same structure as the hazard curve of Example 24.2, arrived at from a different direction: the forward hazard rate of an investment-grade issuer rises with horizon, and here one can see the mechanism rather than merely the fitted curve.

Migration matters in two further ways. It is a risk even without default, because a downgrade widens spreads and triggers the contractual consequences of Section 24.10, so a portfolio marked to market bears it whether or not anything defaults. And the matrix is the natural input to a credit portfolio model; Section 24.14 builds one from it and measures that migration risk.

One caution follows: the matrix is not stable. Transition probabilities are strongly cyclical (downgrades cluster in recessions), so a matrix estimated over a long sample understates the risk of a bad year and overstates it in a good one, in the way Section 24.13's correlation is designed to capture.

24.12 SOVEREIGN CREDIT IS DIFFERENT IN KIND

A **sovereign** is a national government borrowing in its own name, and its debt comes in two kinds that the market names differently and this book has so far kept apart. **Treasury debt** is issued in the government's own currency (the Treasuries of Chapters 4 to 6, and gilts; euro-area bonds such as Bunds are the borderline case, since the issuing government does not control the euro), and Part I treated it as the risk-free benchmark; it appeared later as the collateral everyone else posts. **External debt** is debt in a currency the government does not issue: dollars or euros borrowed by an emerging-market government from bondholders, banks, or official lenders. **Sovereign credit** is the risk that a government does not pay, and it lives mainly in the second kind, because a government can always print the currency it controls and cannot print anyone else's. CDS on emerging-market sovereigns reference that external debt; CDS on euro-area sovereigns, which do not control their currency, price the risk of redenomination as much as outright default. Everything above assumes a borrower that can be put through a

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bankruptcy process. A sovereign cannot, and the differences run deeper than the mechanics.

Ability and willingness are separate. A company defaults when it runs out of money. A country defaults when repaying costs more, politically, than not repaying—and it may have the money. Default is therefore a choice in a way that corporate default is not, which makes structural models of the kind in Section 24.5 inapplicable: there is no asset value crossing a debt threshold, because the relevant asset is the government's capacity to tax and its willingness to use it.

There is no court and no seniority. A defaulting sovereign cannot be liquidated, and the ranking of its obligations is whatever it chooses to respect. Restructurings are negotiated, protracted, and settled by a mixture of law and pressure. **Collective action clauses** (standard in New York-law sovereign bonds since Mexico's 2003 issue, and strengthened in 2014, after the Argentine holdout litigation, to a single vote aggregated across all of an issuer's series) bind dissenting holders to a majority agreement, and their presence or absence materially changes what a bond is worth in default.

The two kinds of debt are rated and priced apart. Default on local-currency debt is a policy choice with an inflationary alternative, so the market prices outright default and inflation together, as two routes to the same loss: a government that will not default outright may still repay in a currency worth less. Foreign-currency debt has no such escape, which is why the same issuer carries a local-currency and a foreign-currency rating, and why the second is usually the lower.

And the definition of default becomes contestable. A payment missed because the sovereign refuses is clearly a default; a payment missed because a third party blocked the transfer is less clear.

The pricing consequence is a wide and persistent basis between sovereign CDS and sovereign bond spreads, the difference Section 24.15 defines. The two should be close under the arbitrage that section describes, and they often are not, because the CDS references a definition of default while the bond references a cash flow, and because the natural sellers of sovereign protection are often institutions domiciled in the sovereign, which is Section 9.9's wrong-way risk in its purest form.

The point generalizes beyond sovereigns. Wherever default is a legal event rather than an economic one, the contract's definition matters as much as the borrower's finances, and Chapter 25 is largely about what happens when those two come apart.

24.13 DEFAULT CORRELATION

A portfolio of loans or bonds is not the sum of its parts, and the difference is entirely about correlation.

State the problem first, because the device that solves it looks arbitrary until the problem is clear. Sections 24.2 to 24.4 deliver, for each borrower separately, the probability that it defaults by T; a portfolio's loss distribution needs the probability of every *combination* of defaults, and with n borrowers there are 2^n of them. Nothing in the data comes close to pinning that down: joint defaults are rare events, and for any particular group of borrowers a century of history holds few or none. So the joint distribution has to be built, and it has to be built under three constraints. It must reproduce each borrower's own default probability exactly, because the market quoted that and a dealer who disagrees with it has a position, not a model. It must carry dependence with a small number of parameters, because there is no data for more. And it must be computable for a portfolio of a hundred or more.

Correlating the default events themselves does not work. A default indicator is a zero-or-one variable, and the correlation of two such variables with different default probabilities is capped well below one. For probabilities $p_1 < p_2$ the ceiling is $\sqrt{p_1(1-p_2)/\big(p_2(1-p_1)\big)}$, which is

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0.43 at 2% and 10%: the rarer event sets it, because the two can default together no more often than the rarer one defaults at all. Nor can such correlations be chosen freely for more than two borrowers at once, since many matrices of pairwise indicator correlations correspond to no joint distribution at all. Merton's model of Section 24.5 suggests the way around: default happens when a continuous quantity, the firm's asset value, crosses a threshold, and continuous quantities are easy to correlate. So give each borrower a continuous latent variable standing in for its asset value, correlate the latent variables through the common state of the economy, and set each threshold so that the probability of crossing it equals that borrower's own market-implied default probability. The result is a Merton model calibrated to CDS curves instead of balance sheets, and it is what the credit market calls a **copula**: a function that supplies the dependence between variables whose marginal distributions are taken as given. Sklar showed in 1959 that any joint distribution can be split that way into its marginals and a copula, so the separation is always available; the choice made here is to take the copula from a multivariate normal.

The standard device is therefore the **one-factor Gaussian copula**. Give each borrower i a latent variable

$$X_i = \sqrt{\rho}\, M + \sqrt{1-\rho}\, Z_i, \qquad M, Z_1, \ldots, Z_n \text{ independent } N(0,1) \tag{24.10}$$

so that every X_i is standard normal and any two are correlated at ρ. Write $\pi_i(T) = 1 - Q_i(T)$ for borrower i's own probability of *defaulting* by T: the complement of the survival probability (24.2), and the marginal that Section 24.3's bootstrap produced from that borrower's CDS curve. Borrower i defaults when its latent variable falls below the threshold that marginal implies:

$$\text{default by } T \iff X_i < N^{-1}\big(\pi_i(T)\big) \tag{24.11}$$

The threshold is what makes this a copula. Each $\pi_i(T)$ is whatever the market said it was; the construction does not touch it. All (24.10) supplies is the *dependence* between the borrowers. Marginals from the market, dependence from an assumption: that separation is the entire appeal of a copula, and it is also what let a single number be asked to carry the joint behavior of the 125 issuers in a standard CDS index of Chapter 25.

The consequence is the one the section needs: conditional on the common factor, the borrowers are independent. In a homogeneous portfolio, where every borrower has the same default probability π by T, so that $\pi_i(T) = \pi$ for every i, each defaults given M with probability

$$p(M) = N\left(\frac{N^{-1}(\pi) - \sqrt{\rho}\, M}{\sqrt{1-\rho}}\right) \tag{24.12}$$

so the number of defaults among n borrowers is a binomial averaged over the factor, weighted by the standard normal density ϕ of M:

$$\mathbb{P}(k \text{ defaults}) = \int_{-\infty}^{\infty} \binom{n}{k} p(M)^k \big(1 - p(M)\big)^{n-k}\, \phi(M)\, dM \tag{24.13}$$

One integral, and the whole portfolio is priced; Example 24.8 carries one case through it. That tractability, and not any evidence about how firms actually fail, is why this model became the market standard.

Two features of (24.12) recur in everything that follows. A bad common factor raises everyone's default probability at once: $p(M)$ rises as M falls, which is the clustering the model is built to produce. And the idiosyncratic channel diversifies away while the common one does not, so in a large portfolio the loss distribution is driven almost entirely by M.

The limit can be written down. Conditional on M the borrowers are independent with common default probability $p(M)$, so by the law of large numbers the fraction of a very large

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homogeneous portfolio that defaults is $p(M)$ itself:

$$L(M) = p(M), \tag{24.14}$$

a deterministic function of one standard normal draw, whose quantile Section 24.14 takes.

Figure 24.6 shows what correlation does to the distribution of defaults, and Example 24.7 measures it.

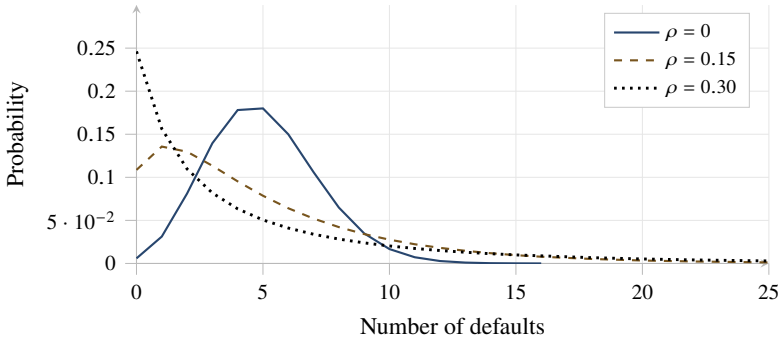


Figure 24.6: Distribution of defaults in a portfolio of 100 borrowers, each with 5% default probability. Expected defaults are 5 in every case.

Example 24.7 Correlation moves probability to both tails

One hundred borrowers, each with a 5% default probability over the horizon.

ρ	Expected	Std. dev.	P(no defaults)	P(20 or more)
0.00	5.00	2.18	0.59%	0.00%
0.05	5.00	3.22	3.13%	0.11%
0.15	5.00	4.89	10.86%	1.85%
0.30	5.00	7.12	24.62%	4.91%
0.60	5.00	11.56	53.02%	7.87%

The expected loss does not move at all. What correlation does is take probability out of the middle and put it at *both* ends, which is why the same expected loss can describe a safe portfolio or a dangerous one: at $\rho = 0.6$ there is a 53% chance that nothing at all defaults and an 8% chance that at least a fifth of the portfolio does.

That double-tailed behavior is what tranche pricing turns on. A senior tranche, one that suffers only if losses exceed some large threshold, is a bet on correlation and almost nothing else, because its expected loss is negligible at low correlation and material at high correlation while the portfolio's expected loss is unchanged. An equity tranche (the first-loss layer, which absorbs the portfolio's losses before any other tranche does) is the opposite: it benefits from correlation, because correlation raises the chance that nothing happens at all.

So a tranche is a correlation instrument that looks like a credit instrument, and pricing one requires a number, ρ, that no observable market quoted before tranches themselves began to trade. The circularity in that sentence is Chapter 26's subject.

Example 24.8 Applying the copula, one value of the common factor at a time

Here is how the $\rho = 0.15$ row of Example 24.7 is produced. With $\pi = 0.05$ the threshold in (24.11) is $N^{-1}(\pi) = -1.6449$, and conditional on the common factor M each of the 100 borrowers defaults

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independently with probability

$$p(M) = N\left(\frac{-1.6449 - \sqrt{0.15}\,M}{\sqrt{0.85}}\right),$$

which is (24.12). Given M, the number of defaults K is binomial with parameters 100 and $p(M)$, so every quantity of interest (the expected count $100\,p(M)$, the probability of no defaults $(1-p(M))^{100}$, the probability of twenty or more) is an elementary binomial calculation. The table evaluates them at six values of M, with the standard normal density $\phi(M)$ that weights each state.

M	$\phi(M)$	$p(M)$	$\mathbb{E}[K \mid M]$	$\mathbb{P}(K = 0 \mid M)$	$\mathbb{P}(K \geq 20 \mid M)$
−3	0.0044	30.0%	30.0	0.00%	99.1%
−2	0.0540	17.3%	17.3	0.00%	27.1%
−1	0.2420	8.6%	8.6	0.01%	0.03%
0	0.3989	3.7%	3.7	2.26%	0.00%
1	0.2420	1.4%	1.4	25.03%	0.00%
2	0.0540	0.4%	0.4	64.72%	0.00%

The unconditional numbers are each column integrated against $\phi(M)$, which is (24.13) applied to the event in question: expected defaults 5.000, no defaults 10.86%, twenty or more 1.85%. These are the $\rho = 0.15$ row. Two things are visible in the table that the row alone hides. The expected count comes out at five because $p(M)$ is raised above 5% in the bad states and lowered below it in the good ones by amounts that average out exactly. And the two tails live in different states: three-quarters of the probability of twenty or more defaults comes from $M \leq -2$, states with about 2% of the probability, while two-thirds of the probability of none at all comes from $M \geq 1$. Raising ρ steepens $p(M)$ in M, feeding both tails at once from the middle, which is the redistribution Figure 24.6 draws.

24.14 CREDIT VAR

Chapter 38 asks how much a trading book could lose over a few days. A loan book needs the same question asked differently, for three reasons. Its positions cannot be sold in a few days, so the horizon is the time it takes for losses to be realized and capital to be rebuilt, conventionally one year. Its loss distribution is nothing like a trading book's roughly symmetric profit and loss: in most years a few borrowers default and the loss is close to its average, and in rare years many default together, so the distribution is bounded on one side and long on the other. And its expected loss is not a risk in the sense that matters for capital. It is a cost, charged in the loan's spread and set aside in provisions: the expected credit loss of IFRS 9, and the current expected credit loss (CECL) standard in the United States. What capital has to absorb is the loss in excess of that cost.

Credit VaR is therefore defined as a high quantile of the one-year portfolio loss minus the expected loss, a difference called the **unexpected loss**. The confidence level is far higher than Chapter 38's: 99.9% for regulatory capital, and typically 99.95% or more for a bank's own **economic capital**, a level chosen so that the bank's probability of being wiped out in a year matches the default rate of the rating it wants to hold.

The large-portfolio limit of Section 24.13 delivers this quantity directly. In a large homogeneous portfolio whose borrowers share a one-year default probability π, the fraction that defaults within the year is $L(M) = p(M)$ of (24.14), a decreasing function of the common factor, so its quantile at confidence q is $p(M)$ evaluated at $M = -N^{-1}(q)$. That quantile is the **worst-case default rate** (WCDR), the default rate exceeded in only a fraction $1-q$ of years:

$$\mathrm{WCDR}(q) = N\left(\frac{N^{-1}(\pi) + \sqrt{\rho}\,N^{-1}(q)}{\sqrt{1-\rho}}\right) \tag{24.15}$$

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With exposure A and recovery R, the credit VaR is

$$\text{Credit VaR}(q) = A\,(1 - R)\,\left[\,\text{WCDR}(q) - \pi\,\right] \tag{24.16}$$

The feature that matters most is visible at $\rho = 0$, where (24.15) returns π at every confidence level. A large portfolio of independent loans has no credit VaR at all, because the law of large numbers pins its default rate to the mean. Every dollar of a granular portfolio's unexpected loss comes from correlation, which Example 24.9 puts in numbers.

Example 24.9 All of the credit VaR is correlation

A bank holds a \$100 million book of corporate loans, numerous and small enough to be granular, each with a one-year default probability of 1% and a loss given default of 40%, the Basel foundation value for a senior unsecured loan to a nonfinancial corporate. The expected loss is $100 \times 0.40 \times 0.01 = \$0.40$ million whatever the correlation. The worst-case default rate (24.15) and the credit VaR (24.16) are not:

ρ	WCDR, 99%	WCDR, 99.9%	Credit VaR, 99%	Credit VaR, 99.9%
0.00	1.00%	1.00%	\$0.00m	\$0.00m
0.05	3.19%	4.67%	\$0.88m	\$1.47m
0.10	4.68%	7.75%	\$1.47m	\$2.70m
0.20	7.53%	14.55%	\$2.61m	\$5.42m
0.30	10.43%	22.44%	\$3.77m	\$8.58m

Take the $\rho = 0.20$ row, close to the correlation the Basel formula assigns to a borrower with this default probability (Section 39.10). In the worst year in a thousand, 14.55% of the book defaults, against 1% on average, and the loss is $100 \times 0.40 \times 0.1455 = \$5.82$ million. Subtracting the \$0.40 million already priced leaves a credit VaR of \$5.42 million, 13.6 times the expected loss.

Two consequences follow. Correlation sets the ratio of credit VaR to expected loss, so two books with identical expected loss can need capital that differs by a factor of almost six (\$1.47 million at $\rho = 0.05$ against \$8.58 million at 0.30). And the confidence level matters as much as the model: moving from 99% to 99.9% raises the credit VaR to between 1.67 and 2.3 times its 99% value, more at higher correlation, because the tail is long and correlation is what lengthens it.

Two assumptions sit inside (24.15), and each understates the risk of a real book. The portfolio is taken to be infinitely granular. A book with a few large exposures keeps idiosyncratic risk that does not diversify, and its credit VaR is higher than the formula says, by an amount that grows with the concentration. And the dependence is the Gaussian copula, with the absence of tail dependence that Section 38.12 described and Chapter 26 shows to matter; a copula with fatter joint tails puts more defaults in the bad years and raises the quantile further.

Migration, not only default. The measure above counts a loss only when a borrower defaults, which is right for a loan held at amortized cost. A portfolio marked to market loses money whenever a borrower is downgraded and its spread widens, which is the migration risk of Section 24.11. **CreditMetrics**, published by J.P. Morgan in 1997, measures it. It revalues each position at the horizon in every rating state, discounting the remaining cash flows on that rating's spread curve, and weights the values by a row of the transition matrix. To correlate migrations across borrowers it reuses the latent variable of (24.10), cutting that variable into bands rather than at a single default threshold: a BBB borrower defaults if X_i falls below $N^{-1}(0.28\%)$, ends the year at CCC or worse if it falls below $N^{-1}(0.48\%)$, and so on up the row. The same ρ then correlates downgrades as well as defaults, and a portfolio's distribution is found by simulating M and the Z_i.

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Example 24.10 Downgrade, not default, sets the 99% VaR of a BBB bond

A five-year BBB bond pays an annual coupon of 5% on \$100 face. A year from now it will have paid one coupon and have four years left, and its value then depends on its rating. The risk-free rate is 3% for every maturity, compounded annually, each rating trades at the spread in the table, the spreads of Example 24.5 extended to CCC, and the bond recovers \$40 if it defaults during the year, without the coupon. The probabilities are the BBB row of Table 24.2.

Rating in one year	Probability	Spread	Value
AAA	0.02%	40 bp	110.89
AA	0.20%	55 bp	110.32
A	4.20%	80 bp	109.38
BBB	89.90%	130 bp	107.52
BB	4.30%	300 bp	101.53
B	0.90%	550 bp	93.54
CCC	0.20%	1,200 bp	76.45
Default	0.28%	—	40.00

Each value is the coupon just received plus the remaining flows discounted at 3% plus the spread; at BBB it is $5 + 5/1.043 + 5/1.043^2 + 5/1.043^3 + 105/1.043^4 = 107.52$. The probability-weighted mean is 106.97.

Reading up from the bottom of the table, the bond ends the year in default with probability 0.28%, at CCC or worse with 0.48%, and at B or worse with 1.38%. The first percentile therefore falls in the B state, and the 99% credit VaR is $106.97 - 93.54 = 13.44$ per \$100 of face. The 0.1% quantile falls in default, and the 99.9% credit VaR is $106.97 - 40.00 = 66.97$.

Now drop migration and keep only default, so the bond is worth 107.52 unless it defaults. Its mean is 107.33, and since default has a probability below 1% the first percentile is 107.52 itself, above the mean: the default-only model reports no 99% credit VaR at all, where the migration model reports 13.44. At 99.9% the two agree to within half a dollar, because both quantiles sit in default.

So the moderate quantile of a high-grade bond is a question about downgrade and the extreme quantile is a question about default. That is why the Basel formula of Section 39.10 carries an adjustment for maturity, which charges a long loan for the downgrade risk it carries before it can default. It is also why the trading book's default charge of Section 39.3 leaves migration to the spread sensitivities, which already measure what a downgrade costs.

Who uses it. A bank's credit portfolio management function runs a model of this kind across the whole loan book to set concentration limits and to allocate economic capital loan by loan, and the allocation enters pricing: under **risk-adjusted return on capital** (RAROC), a loan must earn its hurdle return on the capital it consumes, so a loan to a borrower highly correlated with the rest of the book carries a wider spread than one that diversifies it. The allocation is a contribution to the portfolio quantile, the same decomposition Section 38.15 applies to market risk. Where the model finds a concentration, the bank buys CDS protection on it (Chapter 25) or transfers a tranche of the book in a synthetic securitization (Chapter 26), and Section 27.11 shows how much regulatory capital each of those relieves. Rating agencies used the same machinery on the pools behind structured credit. And supervisors turned (24.15) into the formula that sets most banks' credit capital, with a prescribed correlation and a maturity adjustment. They could do so because, as Gordy (2003) showed, in a one-factor model of a granular portfolio the portfolio's quantile is the sum of each loan's own worst-case loss, so (24.15) can be applied one loan at a time with each loan's own π. Section 39.10 takes the formula up beside the output floor that now limits it.

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24.15 THE CDS–BOND BASIS

Two instruments reference the same issuer: a bond, whose spread over the risk-free curve compensates for default risk, and a CDS, whose spread does the same. They should agree, but they do not.

The bond's spread needs defining first, because there are several. The one that compares with a CDS is the **Z-spread**: the constant spread that, added to every rate on the risk-free curve, discounts the bond's cash flows to its market price. The curve is the SOFR swap curve, which is what a CDS dealer funds and discounts on; a spread over Treasuries mixes in the swap spread of Chapter 7. A close cousin is the par **asset-swap spread**, the spread over SOFR received when the bond's fixed coupons are swapped to floating, which is how a basis trade is packaged in practice and is close to the Z-spread for a bond near par. From the Z-spread, a hazard rate follows the same two ways it did for a CDS: roughly through the credit triangle, and exactly by pricing the bond.

Example 24.11 A hazard rate from a bond price

A five-year bond pays an annual coupon of 5% and trades at 98.55. The SOFR curve is flat at 3%, continuously compounded. Discounted on that curve alone the cash flows would be worth

$$5\,(e^{-0.03} + e^{-0.06} + e^{-0.09} + e^{-0.12}) + 105\,e^{-0.15} = 108.94,$$

and the bond's Z-spread, the z that makes the same sum at $3\% + z$ equal 98.55, is 220 bp.

The credit triangle reads the spread as $\lambda(1 - R)$, so at 40% recovery $\lambda = 0.022/0.6 = 3.67\%$.

Pricing the bond with a constant hazard rate, recovery of 40% of face paid at default, and coupons only while the issuer survives:

$$98.55 = \sum_{t=1}^{5} 5\,e^{-(0.03+\lambda)t} + 100\,e^{-(0.03+\lambda)5} + 40\,\lambda \int_0^5 e^{-(0.03+\lambda)t}\,dt$$

gives $\lambda = 3.63\%$, close to the triangle, as it should be for a bond near par. The gap of $108.94 - 98.55 = 10.39$ between the riskless and the actual price is the present value of the expected loss from default, and the equation above is that statement written out.

The same issuer's five-year CDS trades at 170 bp, which implies $\lambda = 2.83\%$. The bond and the CDS disagree about the issuer's hazard rate by about 0.8 percentage points, and Example 24.12 is about why that disagreement can persist.

The arbitrage argument appears complete. Buy the bond, buy protection on the issuer, and the combination has no credit risk: if the issuer defaults, the protection makes up the loss. So the bond's spread should equal the CDS spread, and the difference, the **basis**, should be zero.

$$\text{basis} = s_{\text{CDS}} - s_{\text{bond}} \tag{24.17}$$

Example 24.12 shows why a nonzero basis can persist.

Example 24.12 Why a negative basis is not an arbitrage

The bond of Example 24.11 trades at a Z-spread of 220 bp; the five-year CDS trades at 170 bp. The basis (24.17) is −50 bp.

The trade suggests itself: buy the bond, buy protection, and collect $220 - 170 = 50$ bp on a package with no credit exposure.

Except that the bond must be financed. Buying it requires cash, raised in repo against the bond itself. A corporate bond does not repo at the risk-free rate: with a haircut and a funding spread the all-in cost might be 60 bp over the curve the 220 bp was measured against, and $50 - 60 = -10$ bp. The trade loses money. And it consumes balance sheet, which under the frameworks of Chapter 27

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carries a capital charge of its own.

So the negative basis is not an arbitrage but a measure of what balance sheet costs, and it persists while funding costs more.

Several other factors move the basis, each in a known direction.

The cheapest-to-deliver option pushes the basis up. A CDS settles at the recovery price an auction sets for the cheapest qualifying obligations, so the protection buyer is in effect delivering the cheapest bond. That right raises the CDS spread by its value; it is the structure of the Treasury futures delivery option of Chapter 6.

Counterparty risk pushes it down. Protection is worth less than face if the seller might fail, and least when seller and reference entity are likely to fail together, which is Section 9.9's wrong-way risk.

And the contracts define default differently. A CDS pays on credit events that may include restructuring, a broader trigger than the missed payment that defaults a bond, so protection is worth more and the basis rises (Section 24.12).

At the trough of the 2008 crisis the basis on investment-grade issuers reached about −250 bp on average, and about −650 bp on speculative-grade issuers, an enormous riskless profit on the arbitrage logic. It was not taken, because those who would have taken it could not fund the bond at any price, the limit of arbitrage of Section 5.15.

24.16 WHO BEARS CREDIT RISK

Banks originate it and, since the capital reforms of Sections 27.11 and 26.13, increasingly distribute it; what they retain is capitalized under the internal-ratings formula of Section 39.10, the regulatory form of credit VaR, and the charge often decides whether a loan is made. **Bond investors** hold it deliberately, paid by the spread of Example 24.5, mostly not for expected loss. **Insurers and pension funds** are the natural long-horizon holders, because they can collect the spread's illiquidity premium without being forced to sell. **Protection sellers in the CDS market** take it synthetically, through Chapter 25's contract.

And private credit funds have taken a growing share of what banks once held. The shift has two sides: the risk sits largely with locked-up long-horizon capital, an improvement (though retail semi-liquid vehicles rationed redemptions in 2026), but it is valued infrequently and by the holder, so its price is far harder to observe. Chapter 26 records what happened the last time credit risk migrated to holders whose marks no one could check.

24.17 WHERE THIS CAME FROM

Merton (1974) began the subject: equity is a call on the firm's assets, and debt a short put. Black and Cox (1976) added the covenants that let default happen before maturity, and Leland (1994) made the capital structure itself endogenous. The structural approach is illuminating but inaccurate: it predicts short-maturity spreads near zero. The reduced-form answer modeled the arrival of default instead: Jarrow and Turnbull (1995) and Duffie and Singleton (1999) founded it, Lando (1998) supplied the Cox-process formulation, and Hull and White (2000) gave Chapter 25's CDS valuation.

Altman (1968) had earlier combined accounting ratios in a discriminant score to predict bankruptcy. On the portfolio side, KMV commercialized distance to default in the 1990s; J.P. Morgan published CreditMetrics in 1997; Li (2000) put a Gaussian copula on default times; and Vasicek's large-portfolio limit became Basel II's risk-weight formula.

SUMMARY

Credit risk is two questions, default probability and recovery, and market prices identify only their product. The hazard rate is the instantaneous conditional default rate, so survival is

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$\exp\left(-\int_0^T \lambda(u)\,du\right)$ and default probabilities compound rather than add.

The credit triangle $s = \lambda(1 - R)$ is exact for a continuously paid premium, because survival and discounting cancel between the legs. Hazard and recovery are not separately identified: the same CDS spread is reproduced by a low recovery with a low hazard or a high recovery with a high hazard, which is harmless for anything that depends only on the product and decisive for a digital default swap or a regulatory calculation that asks for the two separately. Recovery itself ranks with priority of claim and is lowest when defaults are highest, so a fixed recovery convention understates tail losses twice.

In Merton's model equity is a call on the firm's assets struck at the face value of debt, so lending is selling a put on those assets, and equity holders gain from the volatility that debt holders lose from. The model's spread goes to zero at short maturity, for the reason Section 21.7 found diffusive volatility models unable to make a short-dated skew. Reduced-form models repair it by making default a surprise with a free hazard function, which fits any curve and explains nothing.

Risk-neutral default probabilities exceed historical ones, and most of an investment-grade spread is not expected loss: price with risk-neutral probabilities, forecast losses with historical ones, and never mix them. Ratings rank well, calibrate poorly, and lag by design. Default is reached through downgrade, so a rating transition matrix says more than compounding the one-year rate, and for a high-grade bond the moderate quantile is a question about downgrade and the extreme one a question about default. A sovereign defaults by choice, so structural models do not apply and the contract's definition of default matters as much as the borrower's finances.

Correlation leaves a portfolio's expected loss unchanged and moves probability to both tails, which is why a senior tranche is a bet on correlation and why Chapter 26 is about a number no market quoted. The same mechanism sets credit VaR, the loan book's version of Chapter 38's question: a one-year horizon, a confidence level of 99.9% or more, and only the loss beyond the expected loss, which is priced and provisioned rather than held as capital. With independent defaults a large portfolio's worst-case default rate equals the average, so all of a granular book's credit VaR comes from correlation. The CDS–bond basis persists because closing it takes funded balance sheet, and it widens when that is scarce.

FURTHER READING

Merton (1974) is short and worth reading in the original: Merton states his assumptions plainly, and the model's limitations follow directly from them. Crosbie and Bohn (2003) document the KMV implementation of Section 24.6, including the default point and the mapping to expected default frequencies. Duffie and Singleton's *Credit Risk* (2003) is the standard treatment of reduced-form models and takes up the identification problem of Example 24.3; Lando's *Credit Risk Modeling* (2004) is the other standard, and the one to consult for rating-transition models and the generator matrix that Section 24.11 names. Vasicek's (2002) short note is the source of the large-portfolio limit (24.14). Gupton, Finger and Bhatia's *CreditMetrics—Technical Document* (1997) is the original statement of the migration approach of Section 24.14, and Gordy (2003) is the paper that shows why the large-portfolio quantile can be computed one loan at a time, the property the Basel formula rests on. On recovery, the empirical literature is larger than the 40% convention suggests, and less reassuring; the positive correlation between default rates and loss given default is the finding that recurs across samples. S&P Global Ratings' annual default study is the source of the historical figures in Example 24.5 and Moody's of Table 24.1; both are worth consulting directly, both for the data and for the methodological notes that explain why the numbers move between vintages. On the decomposition of credit spreads, the "credit spread puzzle" literature is large and unresolved. Li (2000) is the paper

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that put the copula of Section 24.13 into the market, and is worth reading beside Chapter 26's account of what followed.

PRACTICE QUESTIONS

24.1 A borrower has a constant hazard rate of 4%. Compute the probability of default within five years and within ten, and explain why a 3% hazard rate does not mean a 30% probability over ten years.

24.2 A CDS trades at 240 bp with assumed recovery of 40%. Estimate the hazard rate, and explain why a change in the discount curve alone barely moves a par CDS spread.

24.3 Explain why the credit triangle is exact rather than approximate, and what the usually quoted error actually measures.

24.4 Using Example 24.2, explain why applying the credit triangle to a five-year spread understates the hazard rate over years three to five.

24.5 A five-year CDS trades at 170 bp. State the implied hazard rate at 40% recovery and at 60%, and name one instrument the two assumptions would price differently.

24.6 Explain why recoveries are lowest in the years when defaults are highest, and what this does to portfolio losses.

24.7 Explain why equity is a call option on the firm and debt is short a put, what this implies about the shareholder–creditor conflict, and what the ratio of equity to asset volatility measures as the firm approaches default.

24.8 Explain why Merton's credit spread goes to zero as maturity shortens, and name the chapter in which the same defect appeared in a different market.

24.9 Historical five-year default rates for BBB issuers are about 1.36% and the spread is about 130 bp. Compute the expected loss, state what fraction of the spread it accounts for, and explain why the ratio of risk-neutral to historical default probability is much larger for investment-grade than for single-B issuers.

24.10 A portfolio of 100 borrowers has 5% default probability per borrower. Explain why raising correlation leaves expected defaults unchanged but raises both the probability of zero defaults and the probability of twenty or more, and why a senior tranche is a bet on correlation while an equity tranche benefits from it.

24.11 Using Table 24.2, state the probability that a BBB issuer is still investment grade after five years and the probability it has fallen to speculative grade without defaulting, then explain why compounding a one-year default rate understates the five-year probability and give the factor from Example 24.6.

24.12 State two reasons a structural model cannot be applied to a sovereign borrower, and explain why the natural sellers of sovereign protection create wrong-way risk.

24.13 Explain why fixing recovery at 40% is harmless for pricing a CDS and damaging for a capital calculation.

24.14 A large, granular loan book has a one-year default probability of 2%, a loss given default of 45% and a copula correlation of 0.15. Compute the worst-case default rate at 99.9% and the credit VaR per \$100 million, and explain why the credit VaR would be zero if the loans defaulted independently.

24.15 Using Example 24.10, explain why the 99% credit VaR of a BBB bond is set by downgrade and its 99.9% credit VaR by default, and state what a model that ignores migration reports

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at each level.

24.16 Explain why credit VaR subtracts the expected loss while Chapter 38's VaR is measured from zero, and why its horizon is a year rather than a few days.

24.17 A five-year bond with a 5% annual coupon trades at 97.00 against the flat 3% SOFR curve of Example 24.11. Find its Z-spread and the hazard rate it implies at 40% recovery by the credit triangle, and state the CDS–bond basis if the issuer's five-year CDS trades at 170 bp. Explain why a basis trader would not simply buy the bond and buy protection.

FURTHER QUESTIONS

24.18 Example 24.1 shows the credit triangle is exact for a continuously paid premium and 0.376% off for a quarterly contract with accrual. Derive the cancellation, identify precisely which contract features are required for exactness, and determine how the relation changes for a contract that pays the premium annually in arrears with no accrual on default.

24.19 Example 24.3 shows that a CDS spread does not identify λ and R separately. Construct a pair of instruments whose prices together *would* identify them, explain why such instruments are illiquid or absent in practice, and discuss what this implies for the regulatory requirement to report probability of default and loss given default separately.

24.20 Section 24.7 argues that Merton's short-maturity failure is the same defect Section 21.7 found in diffusive volatility models. Make the parallel precise: state the shared mathematical property, show why it forces both spreads and skews to vanish at short horizons, and evaluate whether the repairs adopted in the two literatures are the same repair.

24.21 Example 24.5 finds that 87% of a BBB spread is not expected loss and that the ratio falls steeply from the high-grade ratings to single-B. Propose a decomposition of the residual into risk premium, illiquidity, and parameter uncertainty; state what data would identify each component; and explain why the rating-dependence of the ratio is evidence about which component dominates.

24.22 Section 24.5 states that lending is selling a put on the borrower's assets. Use this to derive the effect on debt value of an increase in asset volatility, a dividend payment to shareholders, and a debt-financed acquisition. Then explain which covenants in a real loan agreement correspond to which of these, and why covenants are the creditor's substitute for a hedge they cannot execute.

24.23 Example 24.7 shows correlation moving probability to both tails. Derive the limiting loss distribution as $\rho \to 1$, explain what it says about the meaning of a AAA rating on a senior tranche, and use this to state precisely what a rating agency was asserting when it rated such a tranche.

24.24 Section 24.16 argues that the migration of credit from banks to private funds genuinely has two sides. Set out the argument that it has made the system safer and the argument that it has made risk unobservable, identify what evidence would distinguish them, and explain why the absence of a market price is not the same as the absence of a loss.

24.25 Section 24.14's worst-case default rate assigns each loan a capital charge that depends only on that loan. Show that in the one-factor large-portfolio model the portfolio's loss quantile is the sum of the loans' own worst-case losses, explain why a regulator writing a capital rule needs this property, and identify two features of a real book that break it (a concentration and a second systematic factor), stating the direction in which each moves the true quantile.

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Chapter 25
Credit Derivatives and the CDS Market

Chapter 24 treated the credit default swap as a source of information: a spread from which a hazard rate could be extracted. This chapter treats it as a contract, and the difference matters more here than anywhere else in the book.

A CDS does not pay on default. It pays on a **credit event**, which is a list of things written into a legal document. Most of the time the two coincide and the distinction is pedantic. When they come apart, someone loses a great deal of money for reasons that have nothing to do with whether the borrower could pay—in March 2023 roughly CHF 16 billion of Credit Suisse Additional Tier 1 (AT1) instruments were written down to zero while shareholders kept about CHF 3 billion, and no credit event occurred at all. Sections 25.10 to 25.12 are about the occasions on which the two have come apart.

The chapter runs from the contract outward. Sections 25.1 to 25.5 are the single-name contract itself: its legs, its credit events, the 2009 standardization that made the risky annuity the whole of valuation, how a position is marked, and how a defaulted contract is settled. Sections 25.6 to 25.8 build portfolios out of it (index CDS and options on it, index tranches, and basket default swaps) and show that the same 125 issuers can be five different instruments depending on one correlation parameter, which is the bridge to Chapter 26. Section 25.9 is the total return swap, the other contract that separates economics from ownership. Sections 25.10 to 25.12 are the cases in which the definition and the economics came apart, and Section 25.13 turns them into a taxonomy of what a CDS actually insures. Sections 25.14 to 25.17 are the neighboring instruments: convertible bonds, credit-linked notes, hybrid structures, and sovereign CDS paid in a foreign currency. Sections 25.18 and 25.19 close with what the post-crisis reforms did to the market and who trades what is left of it.

Section 9.9 pointed here to show that its wrong-way risk example (sovereign protection bought from a bank in that country) was not hypothetical, and Section 25.17 takes it up. Section 10.10 promised convertible bonds here, and Section 25.14 supplies them, because a convertible is the other major instrument in which credit and an option are fused.

25.1 THE CONTRACT

A credit default swap has two legs on a notional N referencing an entity and a class of its obligations.

The premium leg. The buyer of protection pays the premium quarterly until maturity or a credit event, whichever comes first. The market calls the fixed rate at which it is paid the **coupon**, a word borrowed from bonds for a contract that is not one, and this chapter follows the market. The ISDA definitions do not: there the rate is the **Fixed Rate** and the protection buyer is the Fixed Rate Payer, so a reader who opens the legal document will find no coupon in it, and a confirmation that says Fixed Rate 100 basis points and a trader who says 100 coupon are describing the same term. On a credit event the buyer also pays the accrual from the last

payment date, a feature Example 24.1 valued at about a tenth of a percent of the spread for an investment-grade issuer.

The protection leg. On a credit event, the seller pays $\mathcal{N}(1 - R)$, where R is determined by the auction of Section 25.5.

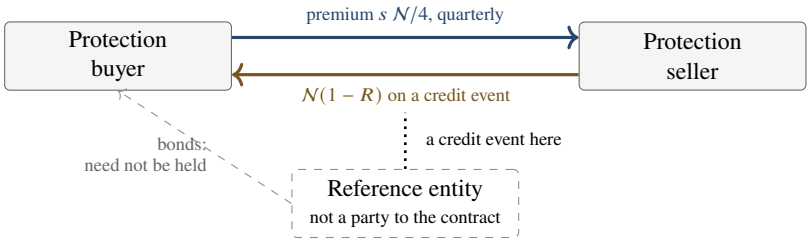


Figure 25.1: The premium and protection legs run between buyer and seller. The dashed links are the ones that may not exist.

Figure 25.1 is the contract, and the two dashed lines are all that is peculiar about it. The reference entity is named by the contract and is party to nothing. It is not consulted, not notified, and receives nothing; it may be unaware that protection on its debt trades at all. A borrower's cost of funds can therefore be quoted in a market it cannot participate in, which is the mechanism behind Section 25.13's question about what a CDS actually insures.

And the link from the reference entity to the buyer is dashed because it need not exist. A protection buyer who holds no bond is not hedging anything, and Section 25.6's index market and Section 25.5's settlement problem both follow from the fact that this link is optional.

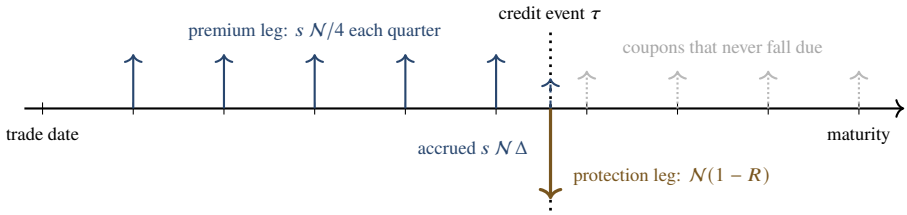


Figure 25.2: Quarterly premium against a single contingent payment. Schematic; the accrual stub is drawn at exaggerated length.

Figure 25.2 puts the same two legs on a calendar. Everything difficult about a CDS is a consequence of the asymmetry in that picture: the premium leg is a stream of small certain payments and the protection leg is one large uncertain one, so the contract's value is dominated by an event whose probability no one observes and whose payout depends on an auction that has not been held.

The buyer of protection is short the credit. They profit if the entity deteriorates, in the same way a put buyer profits if a stock falls, and they need not own the underlying bond—which is the feature that made the instrument useful and the feature that made it controversial.

It completes the market. A bank with a loan it cannot sell can hedge it. An investor who thinks a credit is mispriced can act on that view in either direction. Before CDS, credit was a one-sided market in which the only way to be short was not to own.

It decouples the size of the exposure from the size of the debt. More protection can be outstanding than there are bonds, which matters at settlement (Section 25.5) and which means the instrument's gross notional does not measure what is paid on a credit event, as Section 1.6 warned about derivatives generally.

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25.2 CREDIT EVENTS, AND WHY THE DEFINITION IS THE CONTRACT

The ISDA definitions specify which events trigger payment. For corporates the standard list is bankruptcy, failure to pay, and, depending on the region, restructuring. For financial reference entities the 2014 definitions add governmental intervention, a write-down or bail-in of the reference-level debt imposed by a resolution authority. For sovereigns it adds **repudiation or moratorium** (the government disowns its debt or declares a suspension of payments on it) and, for emerging-market sovereigns, **obligation acceleration**: a default of some other kind on one of its obligations that lets the holders demand repayment at once, before the debt falls due.

Each is defined in detail, and the detail is the contract.

Failure to pay requires a payment actually missed, after any grace period, above a threshold. A payment made late but within the grace period is not a credit event. A payment made by a guarantor is not a credit event.

Bankruptcy is a list of legal states, and different jurisdictions enter them differently. A firm may be economically insolvent for a long time without entering any of them.

Restructuring is the difficult one, and Section 25.10 takes it separately.

Whether a credit event has occurred is decided by a determinations committee. Since 2009 that question has gone to a committee of dealers and investors rather than to a court or the parties to the trade, and its decisions bind the market. This is efficient, because it prevents thousands of bilateral disputes over the same question. It also concentrates the outcome in a body whose members have positions, a governance problem the market long noticed. ISDA's 2025 reform added an independent governance committee and reasoned decisions, but the members of the determinations committees still hold positions.

The gap between economic distress and a contractual credit event is the chapter's recurring theme. A protection buyer who is right about the credit and wrong about the definition receives nothing.

25.3 STANDARDIZATION: THE BIG BANG

Until 2009 CDS traded at par: the coupon was set so the contract had zero value at inception, so every trade had a different coupon and no two positions could be netted.

The 2009 reforms, the "Big Bang," fixed the coupon at a small menu of standard values and moved the difference into an upfront payment:

$$\text{upfront} = \left(s_{\text{par}} - c_{\text{fixed}}\right) \times A \tag{25.1}$$

where A is the **risky annuity**, the present value of one basis point paid until default or maturity.

For North American corporate issuers the menu is 100 basis points for investment grade and 500 for high yield, and those two are by far the most traded; this chapter's examples use them. The menu is wider elsewhere because the upfront has to stay moderate:

- Europe adds 25 for the tightest credits and 300, 750, and 1,000 for wide ones, so that a distressed issuer is not quoted at half its notional upfront;
- sovereign CDS trades mostly on 25 or 100;
- loan CDS uses 100, 250, and 500;
- Japan uses 25 alongside 100 and 500.

A contract's coupon is therefore one of a small number of round numbers chosen to sit near the issuer's spread, and every quoted spread is converted to an upfront against whichever of them the contract carries.

The risky annuity carries the whole valuation, so it is worth building once.

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A CDS has two legs, and both are expectations over the survival curve Q of Section 24.1. Write $t_1 < \cdots < t_n$ for the premium dates, Δ_i for the accrual fraction of the period ending at t_i, and $D(t)$ for the risk-free discount factor. The accrual fraction is a day-count fraction, not a quarter of a year: the standard contract accrues Actual/360 between its quarterly payment dates (March 20, June 20, September 20 and December 20, which the CDS market calls IMM dates although they are not the third-Wednesday futures dates of Chapter 6), so Δ_i runs from about 0.2500 to 0.2556 with the calendar and averages about 0.2536, and Example 25.1 measures what setting it to 0.25 costs. The **premium leg** collects the coupon on every date the issuer survives to, plus the part-period accrued if it defaults in between:

$$A = \underbrace{\sum_{i=1}^{n} \Delta_i \, D(t_i) \, Q(t_i)}_{\text{scheduled}} + \underbrace{\frac{1}{2} \sum_{i=1}^{n} \Delta_i \, D(t_i) \big[Q(t_{i-1}) - Q(t_i) \big]}_{\text{accrued on default}} \tag{25.2}$$

per unit of spread. That is the risky annuity. The $1/2$ is the textbook simplification that default lands, on average, halfway through the period; Hull and White (2000) and O'Kane and Turnbull (2003) both use it. The market's own model does not: the ISDA CDS Standard Model, through which every quoted spread is converted to an upfront, integrates the accrued premium over each period against the hazard rate. At quarterly frequency the two agree to a tenth of a percent of the accrual term (0.0012857 against 0.0012866 per unit of spread for the first quarter of Example 25.1), because, conditional on default inside a period, the expected default time sits only about $\lambda\Delta/12$ of the period before its midpoint, and discounting from that date rather than the period end nearly offsets it. The simplification is kept here for the algebra it makes possible in (25.5). The **protection leg** pays $1 - R$ whenever default arrives:

$$V = (1 - R) \sum_{i=1}^{n} D(t_i) \big[Q(t_{i-1}) - Q(t_i) \big] \tag{25.3}$$

Setting the two equal defines the par spread:

$$s_{\text{par}} = \frac{V}{A} \tag{25.4}$$

A CDS spread is a ratio of two present values, not a yield, and the denominator is what makes it behave unlike one. Everything else in this chapter is a consequence of that denominator: the upfront of (25.1), the mark-to-market of Section 25.4, the index wedge of Section 25.6.

Why the credit triangle is nearly exact. Under a flat hazard rate, a flat discount curve, and a constant accrual fraction Δ the sums in (25.2) and (25.3) are geometric and cancel term by term, leaving

$$s_{\text{par}} = \frac{2(1 - R)}{\Delta} \tanh\!\left(\frac{\lambda \Delta}{2} \right) \tag{25.5}$$

Two things follow, and both answer questions the book has already raised. The discount rate has vanished entirely, which is why, with a flat hazard rate, a change in the risk-free curve alone does not move a par CDS spread: the survival probabilities and discount factors appear identically in both legs and divide out. With a sloping hazard curve the discount factors survive only as weights, and the spread barely moves. And as $\Delta \to 0$ the expression becomes $\lambda(1 - R)$, so Section 24.2's credit triangle is the continuous-premium limit of this contract rather than an approximation to it. At quarterly premiums and the 4.167% hazard of Example 25.1 the gap is two thousandths of a basis point. [TN25] does the integrals behind both legs without the half-period rule, derives the closed forms under flat curves, and sets the three payment conventions (at default, half-period, period end) side by side.

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Example 25.1 takes a single contract from a quoted spread to a cash amount; the table in Example 25.2 is the same calculation repeated across spreads.

Example 25.1 Valuing a CDS from first principles

$10 million of five-year protection on an issuer quoted at a par spread of 250 basis points, traded against the standard 100 basis point coupon. Recovery is 40%, the risk-free rate is 3% continuously compounded, and premiums are quarterly.

A hazard rate comes first. The credit triangle turns the quote into a default intensity,

$$\lambda = \frac{s}{1-R} = \frac{0.0250}{0.60} = 4.167\%$$

and the survival probability follows as $Q(t) = e^{-\lambda t}$:

Year	1	2	3	5
$Q(t)$	0.9592	0.9200	0.8825	0.8119
Cumulative default	4.08%	8.00%	11.75%	**18.81%**

The premium leg. The buyer pays the coupon quarterly, but only while the issuer survives, and pays the accrued fraction of a period if it defaults part-way through one. Per unit of spread,

$$A = \underbrace{\sum_i \Delta_i D(t_i) Q(t_i)}_{\text{scheduled}} + \underbrace{\sum_i \frac{\Delta_i}{2} D(t_i)\left[Q(t_{i-1}) - Q(t_i)\right]}_{\text{accrued on default}} = 4.1647 + 0.0218 = \mathbf{4.1865}$$

which is the risky annuity, and the 250 bp row of Example 25.2.

The protection leg. The seller pays $1 - R$ of notional when default arrives, discounted to today:

$$(1-R)\sum_i D(t_i)\left[Q(t_{i-1}) - Q(t_i)\right] = \mathbf{10.4662}\% \text{ of notional}$$

The two legs should reproduce the quoted spread. Dividing the protection leg by the annuity gives the 250 bp back:

$$s = \frac{0.104662}{4.1865} = 250.00 \text{ bp.}$$

The exactness is a property of the convention rather than a theorem: with the default payment and the accrued coupon both made at the quarter end the discount factors cancel, and the closed form (25.5) gives 249.998 bp, within two thousandths of a basis point. The accrual term is what makes it work: worth only 0.0218 of the 4.1865, half of one percent of the annuity, but leaving it out puts the par spread at 251.31 bp.

Pay the protection when default actually happens, which is what the contract says, and the leg is worth 10.5056% instead of 10.4662%, the par spread is 250.94 bp, and the upfront becomes 6.3191% ($631,908), some $3,941 more. So the credit triangle is exact for a continuously paid premium and about a basis point light for a quarterly contract.

None of what has been assumed is innocent. Default arrives with constant intensity, so the credit curve is flat. The risk-free rate is flat and independent of the default time, so no wrong-way risk appears. Recovery is known at 40% rather than uncertain and correlated with default. And $\Delta_i = t_i - t_{i-1}$ is taken as exactly 0.25: real contracts accrue Actual/360 between the March, June, September, and December payment dates, making Δ_i nearer 0.2536, which lifts the annuity by 1.44% and, with the quote held at 250 bp, raises the upfront by $8,154. The conventions are worth more than the third decimal place of the model.

Now the cash changes hands. Protection is worth more than the 100 bp coupon stream, and the buyer pays the difference at inception:

$$\text{upfront} = 10.4662\% - 100\,\text{bp} \times 4.1865 = 10.4662\% - 4.1865\% = \mathbf{6.2797}\%$$

or **$627,966** on $10 million. Equation (25.1)'s short form, $(250 - 100)\,\text{bp} \times 4.1865$, gives 6.2798%. The par spread's 0.002 bp shortfall from 250 bp, times the annuity, is the 0.0001% between them: $9 on $10 million, $627,975 against $627,966.

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Example 25.2 From a spread to an upfront

Five-year contracts on a 100 bp coupon, recovery 40%, rates 3%:

Par spread	Hazard rate	Risky annuity	Upfront
40 bp	0.667%	4.5513	−2.73%
100 bp	1.667%	4.4428	0.00%
150 bp	2.500%	4.3550	+2.18%
250 bp	4.167%	4.1865	+6.28%
500 bp	8.333%	3.8027	+15.21%
800 bp	13.333%	3.4044	+23.83%

The annuity falls from 4.55 to 3.40 as the spread widens, because a contract on a deteriorating credit is expected to terminate early. The risky annuity is the credit market's duration, and it shrinks as the credit worsens, because default ends the payments rather than only discounting them.

The purpose of the convention is fungibility. With a fixed coupon, standard payment dates, and a standard effective date for the protection, two contracts on the same issuer and maturity differ only in the upfront paid at inception, a cash flow already settled, so they net, they can be cleared, and a position can be closed by trading the offsetting contract. The reform was infrastructure, and it did more for the market's resilience than any pricing insight in this chapter.

25.4 MARKING A POSITION

Under the standard contract a protection position's value is the difference between the current par spread and the fixed coupon, multiplied by the current risky annuity,

$$\text{MTM} = \left(s_t - c\right) A_t \tag{25.6}$$

where c is the standard coupon the contract pays (100 or 500 bp for a North American corporate) and not the par spread at which it was traded. That spread has already been settled: the buyer paid or received the upfront of (25.1) on the trade date, and the cash is gone. What remains is a contract paying c, and (25.6) is what it is worth. Two things are easy to miss: that c is the coupon, not the traded spread, and that *both* objects on the right carry the subscript t. Example 25.3 shows what each mistake costs.

Example 25.3 The annuity moves too

A desk bought five-year protection when the par spread was 120 bp, on the standard 100 bp coupon. The five-year risky annuity at 120 bp was 4.4074, so by (25.1) it paid $(120 - 100)\text{ bp} \times 4.4074 = 0.88\%$ of notional upfront, \$88,148 on \$10 million. A year later the four-year par spread is 300 bp.

What the desk holds is a four-year contract paying 100 bp, and the four-year risky annuity *at the new spread* is 3.4103, so

$$\text{MTM} = (300 - 100)\text{bp} \times 3.4103 = 6.82\% \text{ of notional},$$

or \$682,067, what a dealer would pay to take the contract over. The gain since inception is that less the upfront paid, $6.82\% - 0.88\% = 5.94\%$, or \$593,919, before the year of 100 bp coupons the desk has paid to carry the position.

Two errors are common. Marking against the traded spread, $(300 - 120)\text{ bp} \times 3.4103 = 6.14\%$, mixes a cash flow that has already happened into the value of what remains: the 120 bp was settled in the upfront and does not enter the mark. And using the annuity at the old spread (the four-year annuity at 120 bp is 3.6118) gives $(300 - 100)\text{ bp} \times 3.6118 = 7.22\%$, an overstatement of 5.9%. The annuity must be computed on the current curve, because the premium stream actually received will be the one the market now expects.

The direction of the second error is systematic. As a credit deteriorates the annuity shortens, so protection gains less than a naive duration calculation suggests. The position is concave in the

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spread, and the concavity works against the protection buyer. A desk sizing a hedge on duration alone will be under-hedged exactly when the hedge is working.

The standard risk measure is the **CS01**, the change in value for a one-basis-point rise in the spread. It equals the risky annuity times one basis point only when the spread is at the coupon; here the annuity's fall as the spread widens makes the CS01 $3,195 per $10 million, 6% below the $3,410 that the annuity alone suggests, so it must be recomputed at the current spread.

25.5 SETTLEMENT AND THE AUCTION

When a credit event occurs, the contract must be settled, and the mechanism has a history.

Physical settlement was the original design: the protection buyer delivers a defaulted bond and receives par. It works only if there are enough bonds, and Section 25.1 noted that protection outstanding can exceed the debt outstanding. When it does, protection buyers bid for deliverable bonds and their price is driven *up*, which reduces the payout on exactly the contracts that are supposed to be paying out. This happened, and it is why the mechanism changed.

The auction replaced it. Run under separate protocols before 2009 and written into the standard contract by the 2009 reforms, a two-stage auction determines a single recovery rate for the whole market, and all contracts settle in cash against it. The first stage collects dealer markets and physical settlement requests to establish an initial price and a **net open interest** (the amount of bonds still wanted for purchase or offered for sale once offsetting physical requests are matched); the second runs a limit-order auction to clear that interest, and the clearing price is the final recovery.

The design solves the squeeze and creates a different problem. A single number now determines the payout on every contract, so the auction's outcome is worth a great deal to a great many people, and the incentive to influence it is proportionate. The mechanism's defenses are that it is open, that participation is broad, and that the two stages make manipulation expensive—not that it is impossible.

And the auction recovery is not the eventual recovery. It is the market's price for the defaulted claim a few weeks after the event, which is a forecast made under duress. Realized recoveries after a completed restructuring have differed from auction recoveries substantially in both directions, which matters for Chapter 24's Example 24.3: the R used to convert spreads into hazard rates is a convention, and even the observed R at settlement is a market price rather than a realized outcome.

The Lehman Brothers auction of October 2008 is the standard illustration. It set the recovery at 8.625 cents on the dollar, so sellers of protection owed 91.375% of face, yet the Depository Trust and Clearing Corporation (DTCC) reported about $72 billion of gross notional on Lehman and a net transfer of only about $5.2 billion, against rumors near $400 billion. That is netting, as Section 1.6 said, not a failure of the contracts.

25.6 INDEX CDS

Most CDS risk trades through indices (CDX in North America, iTraxx in Europe), each a standardized portfolio of equally weighted issuers (125 for the investment-grade indices, which the examples assume; 100 for CDX high yield and 75 for iTraxx Crossover), rolling to a new series every six months.

The index is built by rule, not by valuation. An administrator selects the constituents at each roll, weights them equally, and fixes the coupon (100 bp for the investment-grade indices, 500 for high yield) for the life of the series. For CDX Investment Grade the constituents are 125 North American issuers rated investment grade and ranked by dealers' liquidity polls; the high-yield and European indices apply the same procedure to a different universe. A new series, which becomes the *on-the-run* series, is launched every March and September; the old one

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continues to trade as an *off-the-run* series. A constituent that suffers a credit event between rolls is not replaced. Its $1/125$ share is settled through the auction of Section 25.5 (the protection buyer receives $(1 - R)/125$ of the index notional), and it drops out: the contract's notional is scaled by an **index factor**, $124/125 = 0.992$ after one event, the coupon is paid on the reduced notional from then on, and the series is relabeled a new version so that contracts traded before and after the event remain the same instrument. The survivors carry on unchanged to the series' own maturity; the next roll launches a new series and neither alters nor terminates the old one. High-yield indices, with 100 constituents and more frequent events, see the factor move most; investment-grade series often finish their life at a factor of one.

An index CDS is therefore a single contract paying protection on each constituent as it defaults, with the notional reduced accordingly. It is the credit market's principal liquidity: bid–ask spreads on the index are a fraction of those on constituents, and a desk hedging a portfolio will usually hedge with the index and accept the basis.

Example 25.4 Why the index does not trade at the average spread

Consider a 125-issuer index whose constituents average 80 bp: 115 issuers at 55 bp and 10 distressed issuers at 367.5 bp.

The arithmetic average is 80.0 bp. The spread that makes the index's protection and premium legs balance is **77.3 bp**.

The 2.7 bp difference is arithmetic, not a mispricing. Both legs are additive across constituents. The index's protection leg is the sum of the constituents' protection legs, $\sum_j V_j$, each on its $1/125$ share; and because every constituent pays the same coupon until its own default, the index's premium leg per unit of coupon is the sum of their risky annuities, $\sum_j A_j$. Applying (25.4) to the two sums gives

$$s_{\text{index}} = \frac{\sum_j V_j}{\sum_j A_j} = \frac{\sum_j A_j\, s_j}{\sum_j A_j} \tag{25.7}$$

so the index's fair spread is the annuity-weighted average of its constituents' par spreads, not the arithmetic one. This is what the constituents imply the index should be worth, its *intrinsic* level, rather than how the index is built or quoted: the traded index has its own price, and the standard model converts that quote to an upfront with a single flat hazard curve for the whole index rather than constituent by constituent. A wide-spread issuer has a shorter risky annuity (Example 25.2's effect), so it carries less weight in (25.7) than in a simple mean, and the index trades below it. The wider the dispersion, the larger the gap.

The difference between the level at which the index actually trades and the intrinsic level (25.7) computed from its constituents is the **index skew**. It is a real trading opportunity when it is large enough to cover the cost of trading 125 individual issuers against one index. It widens when the index is used heavily as a macro hedge, because that demand moves the index without moving the constituents. This is Section 19.7's index protection demand appearing in a different market.

Options on a CDS. A **payer option** gives the right to buy protection at a fixed spread K on the expiration date T, and a **receiver option** the right to sell it. With the forward risky annuity A as numeraire (the premium leg of (25.2) starting at T, valued today), a payer on a forward spread s_F with volatility σ is worth $A\,[\,s_F N(d_1) - K N(d_2)\,]$, where $d_{1,2} = [\ln(s_F/K) \pm \sigma^2 T/2]/(\sigma\sqrt{T})$; this is a payer knocked out by default before T, and one that is not adds the **front-end protection**. With $s_F = 250$ bp on five-year protection starting at $T = 1$, so that $A = e^{-(0.03+0.0417)} \times 4.1865 = 3.897$, $K = 300$ bp and an illustrative $\sigma = 60\%$, $d_1 = -0.004$, $d_2 = -0.604$ and the payer is worth $3.897 \times (250 \times 0.4985 - 300 \times 0.2730) \approx 166.5$ bp of notional. Macro funds buy payers on the investment-grade index to cap the cost of a widening at the premium; dealers carry the volatility.

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25.7 INDEX TRANCHES

Slice the index's loss into layers and sell each separately. A tranche $[a, b]$ absorbs portfolio losses between a and b percent of notional: a is its **attachment point**, the pool loss at which it begins to take losses, b its **detachment point**, the pool loss at which it is exhausted, and the width $b - a$ is its own notional as a fraction of the pool's. The 0–3% **equity** tranche takes the first losses; the 15–30% **senior** tranche only suffers after everything below it is exhausted; the layers in between are the **mezzanine** tranches.

Section 24.13 established the essential point: correlation leaves the portfolio's expected loss unchanged and redistributes it between the layers. The procedure comes before the numbers, because a tranche is valued the same way whatever the pool.

Valuing a tranche. Take a pool of n equally weighted issuers with a common default probability $\pi(t)$ by each date t, recovery R, and the one-factor copula of Section 24.13 with correlation ρ. Four steps take a tranche from that description to a par spread, and only the second of them is where the model lives.

Step 1: the tranche's loss as a function of the pool's. Write $K(t)$ for the number of defaults by t and $L(t) = (1 - R)\,K(t)/n$ for the pool's loss fraction. A tranche $[a, b]$ has lost, as a fraction of its own notional,

$$T_{[a,b]}(L) = \frac{\min\{\max(L - a,\ 0),\ b - a\}}{b - a}, \tag{25.8}$$

zero until the pool's loss reaches the attachment point a, then rising one for one until the detachment point b exhausts it. Its remaining notional at t is $1 - T_{[a,b]}(L(t))$.

Step 2: the pool's loss distribution at each date, conditional on the factor. Given the common factor M (the standard normal variable that stands for the state of the economy in Section 24.13's copula), each issuer has defaulted by t independently with probability

$$p_t(M) = N\!\left(\frac{N^{-1}\big(\pi(t)\big) - \sqrt{\rho}\,M}{\sqrt{1-\rho}}\right), \tag{25.9}$$

which is equation (24.12) with the threshold set at date t's default probability, so $K(t) \mid M$ is binomial with parameters n and $p_t(M)$. The tranche's conditional expected loss is therefore a finite sum,

$$\mathbb{E}\big[T_{[a,b]}(L(t)) \,\big|\, M\big] = \sum_{k=0}^{n} \binom{n}{k} p_t(M)^k \big(1 - p_t(M)\big)^{n-k}\, T_{[a,b]}\!\left(\frac{(1-R)\,k}{n}\right). \tag{25.10}$$

Step 3: integrate out the factor. The unconditional expected tranche loss at t is

$$E_{[a,b]}(t) = \int_{-\infty}^{\infty} \mathbb{E}\big[T_{[a,b]}(L(t)) \,\big|\, M\big]\,\phi(M)\,dM, \tag{25.11}$$

where ϕ is the standard normal density, which is equation (24.13) with the tranche loss as the integrand. In practice the integral is a Gauss–Hermite quadrature (a weighted sum of the integrand at a few dozen fixed values of M) repeated at every payment date.

Step 4: the two legs. On payment dates $t_1 < \cdots < t_m$ with accrual fractions Δ_j and discount factors $D(t_j)$, the protection leg pays the increase in the expected tranche loss over each period, and the premium leg pays the spread on the expected remaining notional, with the half-period accrual of (25.2) on the part lost during the period, which is the same as paying it on the

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expected remaining notional averaged over the period:

$$
\begin{aligned}
V_{[a,b]} &= \sum_{j=1}^{m} D(t_j)\big[E_{[a,b]}(t_j) - E_{[a,b]}(t_{j-1})\big], \\
A_{[a,b]} &= \sum_{j=1}^{m} \Delta_j\, D(t_j)\Big[1 - E_{[a,b]}(t_j) + \frac{1}{2}\big(E_{[a,b]}(t_j) - E_{[a,b]}(t_{j-1})\big)\Big] \\
&= \sum_{j=1}^{m} \Delta_j\, D(t_j)\Big[1 - \frac{1}{2}\big(E_{[a,b]}(t_{j-1}) + E_{[a,b]}(t_j)\big)\Big],
\end{aligned}
\tag{25.12}
$$

and the par spread is $s_{[a,b]} = V_{[a,b]}/A_{[a,b]}$, exactly as in (25.4) with the tranche's expected loss in the place the default probability occupied for a single issuer. Equity tranches are quoted, like the CDS of distressed issuers, as an upfront against a fixed running coupon of 500 basis points through (25.1); that was the equity convention from the first standardized index tranches, and since the 2009 move to fixed coupons the other tranches have been quoted the same way, as an upfront against 100 or 500 basis points.

Everything in Steps 1, 3 and 4 is bookkeeping that any model must do; Step 2 is where the Gaussian copula enters, and replacing it (with random recovery, a heavier-tailed factor, or a different copula) changes only the conditional probability (25.9). The same is true of the pool's description. Equal weights are exact for the standard indices, but a common $\pi(t)$ is a simplification: a dealer's model gives each constituent its own hazard curve, bootstrapped as in Section 24.3, and the binomial in (25.10) becomes a recursion that adds the constituents one at a time (Andersen, Sidenius, and Basu, 2003; Hull and White, 2004), with nothing else in the four steps moving. Example 25.5 runs the four steps on one pool and tabulates what comes out at the five-year date.

Example 25.5 The same portfolio, five different instruments

A 125-issuer index, each issuer with a 7% five-year default probability and 40% recovery. The portfolio's expected loss is $0.07 \times 0.60 = 4.200\%$ of notional at every correlation, because expectation is linear: the pool's loss is the average of 125 indicator variables times $1 - R$, the expectation of each indicator is its own default probability, and no joint probability (hence no ρ) enters the sum.

The tranche numbers come from the copula exactly as Example 24.8 applied it, with a loss given default and a tranche attached. Each default costs $1 - R = 0.6$ of its $1/125$ share, so the pool's loss fraction is $L = 0.6K/125$ for K defaults, and a tranche $[a, b]$ loses $T_{[a,b]}(L)$ of (25.8). This is Step 1. At $\rho = 0.30$ the threshold for a default probability of 7% is $N^{-1}(\pi) = -1.4758$; conditional on the common factor M, defaults are binomial with parameters 125 and $p(M)$, which is Step 2; the conditional pieces (the pool's expected loss and, from (25.10), the expected loss of the equity and senior tranches as fractions of their own notional) are:

M	$p(M)$	$\mathbb{E}[K \mid M]$	$\mathbb{E}[L \mid M]$	$\mathbb{E}[T_{[0,\,0.03]} \mid M]$	$\mathbb{E}[T_{[0.15,\,0.30]} \mid M]$
−3	57.9%	72.4	34.8%	100.0%	99.7%
−2	32.5%	40.6	19.5%	100.0%	30.1%
−1	13.4%	16.7	8.0%	100.0%	0.0%
0	3.9%	4.9	2.3%	71.8%	0.0%
1	0.8%	1.0	0.5%	15.6%	0.0%
2	0.1%	0.1	0.1%	2.1%	0.0%

Integrating each column against $\phi(M)$, which is Step 3 and (25.11), gives the pool 4.200%, the equity tranche 62.59% and the senior tranche 1.88% (the $\rho = 0.30$ row below). The columns show why the two tranches move in opposite directions. The equity tranche is wiped out in every state

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from $M = -1$ down, so it gains nothing further from the bad states that correlation makes more likely, while it is spared in the good states that correlation also makes more likely. The senior tranche loses only in states below about $M = -1.6$, which give it a 4.97% chance of any loss; at $\rho = 0$ the factor moves nothing and that chance is essentially zero. Raising ρ moves probability into those states, and that is the whole of the redistribution.

Expected loss as a percentage of each tranche's own notional, with the 30–100% layer that absorbs everything above the senior tranche:

ρ	0–3%	3–7%	7–10%	10–15%	15–30%	30–100%
0.00	95.76%	32.73%	0.59%	0.00%	0.00%	0.00%
0.10	81.81%	33.49%	9.44%	2.16%	0.10%	0.00%
0.20	71.40%	31.51%	13.50%	5.48%	0.78%	0.00%
0.30	62.59%	29.30%	15.21%	7.87%	1.88%	0.03%
0.50	47.36%	24.75%	15.81%	10.40%	4.27%	0.22%
0.80	26.59%	17.11%	13.35%	10.75%	6.96%	1.05%

Weight each column by its layer's width (0.03, 0.04, 0.03, 0.05, 0.15, 0.70) and every row adds to 4.200%: correlation redistributes the loss and cannot create or destroy any of it. The last layer is what makes the accounting close, carrying nothing at low correlation and, at $\rho = 0.80$, about a sixth of the pool's expected loss.

The equity tranche's expected loss falls from 95.8% to 26.6% as correlation rises; the senior tranche's rises from zero to 6.96%. *Selling equity protection is buying correlation; selling senior protection is selling it.*

The senior tranche is where the danger is, and the right panel of Figure 25.3 draws it on its own scale. A senior tranche valued on a correlation of 0.10 (expected loss 0.098%) and actually facing 0.50 (4.272%) has an expected loss 43 times larger. Nothing about the underlying credits has changed: the same 125 issuers, the same 7% default probabilities, the same 4.2% portfolio loss. Only a parameter that no market quoted has changed.

Step 4, equation (25.12), turns the expected losses into prices. Take the default probability to build up at a flat hazard rate fitted to the five-year figure, $\lambda = -\ln(0.93)/5 = 1.451\%$, so that $\pi(t) = 1 - e^{-\lambda t}$; quarterly payments; and 3% discounting. At $\rho = 0.30$ the five tranches' par spreads are

	0–3%	3–7%	7–10%	10–15%	15–30%
Five-year expected loss	62.59%	29.30%	15.21%	7.87%	1.88%
Risky annuity $A_{[a,b]}$	2.86	4.02	4.35	4.50	4.60
Par spread (bp)	2,062	671	318	158	37

The spread is not the expected loss divided by five. The equity tranche's annuity is only 2.86 because most of its notional is expected to be gone before maturity, so the running spread is correspondingly high. That is why the market quotes that tranche as an upfront instead: against the 500 bp coupon, $(0.2062 - 0.05) \times 2.86 = 44.7\%$ of notional at inception. Repeating the calculation for the senior tranche at $\rho = 0.10$ and 0.50 gives 1.9 and 85.5 bp against 37 at 0.30: the 43-fold move in expected loss is a move of about 45-fold in the spread, because the annuity barely changes for a tranche that is rarely touched.

The rating on such a tranche is a statement about ρ. Section 24.10 said the letter did not disclose the correlation assumption. The table says what that omission was worth: the difference between an instrument that almost never loses and one that loses 4.3% of its notional in expectation.

And the market's own correlation number is not a correlation. Quoting a tranche in terms of the ρ that reproduces its price (**implied correlation**) produces different values for different tranches on the same portfolio, which is logically impossible if ρ were a property of the portfolio. The market moved to **base correlation**: instead of each tranche's own ρ, quote the ρ that prices the **first-loss tranche** $[0, D]$ for each detachment point D (the equity tranche widened to detach at D), and price a mezzanine tranche $[a, b]$ as the difference of the two first-loss tranches $[0, b]$

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and $[0, a]$, each at its own base correlation. A first-loss tranche's value is monotone in ρ, so its implied correlation is unique and the curve can be interpolated between quoted detachment points; a mezzanine tranche's is not: Example 25.5's 3–7% tranche loses 32.73% at $\rho = 0$, 33.49% at 0.10 and 31.51% at 0.20, so one price can correspond to two correlations or to none. The device removes the inconsistency by construction without removing the fact that generated it. This is Section 19.1's implied volatility smile again: a quoting convention rather than a belief.

Chapter 26 is about what happened when it was treated as a belief.

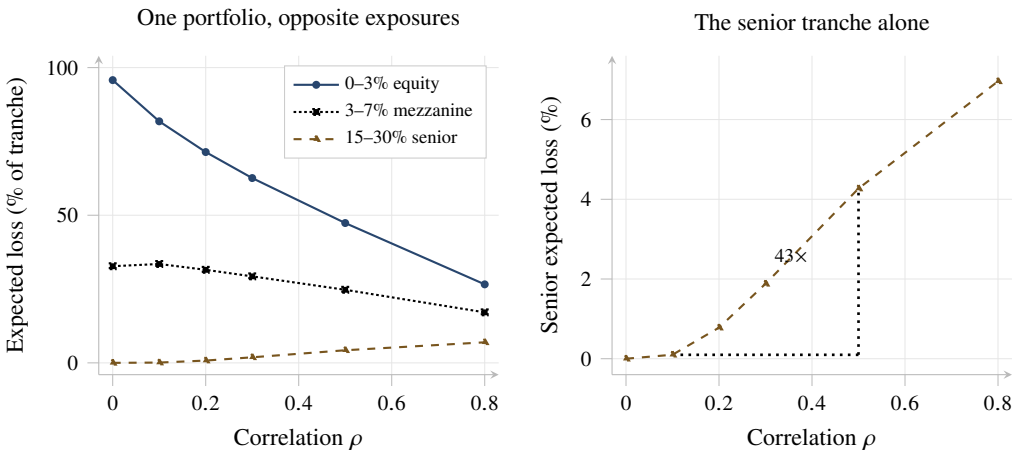


Figure 25.3: Example 25.5's 125-issuer portfolio. The pool's expected loss is 4.200% at every point on both panels.

Figure 25.3 is the picture to carry out of this chapter. The left panel's three curves are one portfolio, so nothing about the credits changes along the horizontal axis, yet the curves go in three different directions; the mezzanine tranche is nearly flat and so says least about ρ. The right panel puts the senior curve on its own scale, where what looked like zero is the forty-three-fold move the example measured.

25.8 BASKET DEFAULT SWAPS

A tranche slices a large portfolio by loss. A **basket default swap** slices a small one (five to ten issuers) by *count*. An nth-to-default basket pays protection on the nth issuer to default and then terminates; the first-to-default basket is the common case, and Section 24.4 has already used it as an instrument whose price depends on the hazard rate at first order.

Nothing new is needed to value one. Write $K(t)$ for the number of issuers that have defaulted by t, whose distribution is Section 24.13's equation (24.13). The basket pays exactly when K crosses n, so it is a credit default swap on a synthetic issuer whose survival probability is

$$Q_n(t) = \mathbb{P}\big(K(t) < n\big) \tag{25.13}$$

Substitute Q_n for Q in (25.2) and (25.3) and the par spread is (25.4) unchanged. The basket is a CDS whose reference entity is the nth order statistic of the portfolio's default times, and once that is seen there is nothing left to derive. Example 25.6 prices all five baskets on one small portfolio.

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Example 25.6 The same five issuers, five different instruments

Five issuers, each trading at 150 basis points with 40% recovery, discounted at 3%, five years, quarterly premiums. Par spreads in basis points:

ρ	1st	2nd	3rd	4th	5th
0.00	750	133	16	1	0
0.15	669	160	34	5	0
0.30	591	179	56	15	2
0.60	439	196	99	47	17
0.90	269	185	138	102	67

At zero correlation the first-to-default costs the sum of its parts. Five issuers at 150 bp give 750 bp, and the basket prices at 750: if defaults never coincide, protection on the first is protection on all five. Correlation is what makes a basket cheaper than the sum, because coincident defaults waste the protection: the second and later defaults arrive when the contract has already terminated.

One entry is worth working in full. Each issuer's hazard rate is $\lambda = 0.0150/0.6 = 2.5\%$ by the credit triangle, so its five-year survival probability is $e^{-5\times0.025} = 0.8825$. At $\rho = 0$ the first-to-default survives only if all five issuers do, $Q_1(5) = 0.8825^5 = 0.5353$. At $\rho = 0.30$, (25.13) is the copula integral $Q_1(5) = \int \big(1 - p_5(M)\big)^5 \phi(M)\, dM = 0.6149$. It is higher because correlation crowds the defaults into the same states and leaves more states with none. Substituting Q_1 for Q at each of the twenty quarterly dates gives a risky annuity of 3.6473 and a protection leg of 0.2157, so $s_1 = 0.2157/3.6473 = 591$ bp, against 3.4665, 0.2600 and 750 bp at $\rho = 0$. The annuity rose and the protection leg fell, and both moved for the same reason.

The opposite movement of the two ends is Section 25.7's, in a different instrument. As correlation rises the first-to-default falls from 750 to 269 bp while the fifth-to-default rises from nothing to 67. A first-to-default is the equity tranche of this portfolio and an nth-to-default with n large is its senior tranche, and the buyer of one and the buyer of the other want opposite things from the same number.

And the five instruments add up. Since $\sum_{n=1}^{5} \mathbf{1}\{K \geq n\} = K$, the five protection legs must sum to the five single-name protection legs at every correlation, and they do, to machine precision. Every default that occurs is paid for exactly once in either portfolio, at the moment it occurs (by the CDS on the issuer that defaulted, or by the basket whose turn it was), so the two sets of cash flows coincide on every path, not merely on average. Correlation redistributes the premium among the baskets and cannot create or destroy any of it, which is Section 24.13's statement about tranches, arrived at from the other direction.

25.9 TOTAL RETURN SWAPS

Section 7.12 introduced the total return swap and said Chapter 27 would examine what it made possible in 2021. The instrument belongs here because it is the principal way credit and equity exposure is taken without ownership.

One party pays the total return on a reference asset (coupons, dividends, and price change) and receives a financing rate. The receiver of the total return has the full economics of owning the asset without owning it, and without paying for it: the position is funded by the swap.

Three properties follow, and they are the same three that recur whenever an instrument separates economics from ownership.

Leverage is set by the margin, not by the asset. A cash buyer of a bond posts its full price. A total return receiver posts initial margin, which may be a small fraction. The economics are identical and the balance sheet is not.

The exposure is not visible in ownership disclosures. The swap counterparty holds the asset as a hedge and appears as the owner. Positions built this way do not show up where people look for them, which is Section 7.12's phrase about leverage accumulating quietly, and Chapter 27's

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case study is what that enabled.

And the risk is counterparty risk presented as market risk. The dealer paying total return is hedged in the market and exposed to the client; if the asset falls, the client owes variation margin, and if the client cannot pay, the dealer holds the asset. The hedge is perfect against the market and worthless against the client, the configuration Chapters 8 and 9 dwelt on.

25.10 RESTRUCTURING

Restructuring is the most contested credit event, and the one where the definition works hardest.

A restructuring (a maturity extension, a coupon reduction, a change in seniority, imposed on holders) is economically a loss but not a missed payment. Whether it should count as a credit event has been answered differently in different places and different decades.

The problem it creates is the cheapest-to-deliver problem in its sharpest form. After a bankruptcy, all senior unsecured obligations are worth roughly the same, so which one is delivered hardly matters. After a restructuring, the entity is still alive, and its obligations trade at very different prices depending on maturity: a thirty-year bond of a distressed but living issuer is worth far less than a one-year bond of the same issuer. A protection buyer could deliver the cheapest, collecting a payout far exceeding their actual loss.

The market's responses were successive restrictions on what could be delivered (limits on maturity, then buckets) and, in North America, largely dropping restructuring as a trigger for corporate issuers. Europe kept it, mainly because bank regulators require it: a hedge that does not pay on restructuring does not obviously transfer the risk of the loan it hedges, and Example 25.7 prices what the clause is worth.

Example 25.7 What the clause is worth

Take an issuer with a 3% annual intensity of bankruptcy or failure to pay, recovering 40%. Suppose including restructuring adds a further 15% to that intensity (0.45 percentage points) but restructuring events recover 70%.

No-restructuring contract	180.0 bp
With restructuring	193.5 bp
Value of the clause	13.5 bp (7% of the spread)

The point is not the number, which depends entirely on the assumed intensities, but that it is small and stable right up until it is neither. When a restructuring is actually negotiated, one contract pays and the other does not, and the gap between them goes from 13 bp to the whole protection payment. A risk system that treats them as substitutes because their historical basis is small has mismeasured a binary exposure as a small one.

25.11 WHEN THE DEFINITION AND THE ECONOMICS COME APART

Three episodes each established something about what a CDS actually insures.

Greece, 2012. The sovereign's debt was restructured through a package that was initially presented as voluntary, and a voluntary exchange is not a credit event. Collective action clauses were then applied retroactively to bind holdouts, and that application—compulsion—triggered the contracts. Protection paid, but only after a period in which it was genuinely unclear whether the largest sovereign restructuring in history would trigger the instrument designed to insure against exactly that.

Manufactured events. In the 2010s several transactions were arranged specifically to trigger or avoid triggering CDS: a distressed borrower persuaded to miss a small payment so that a protection holder would be paid, or refinanced in a way designed to leave protection worthless. ISDA amended the definitions in 2019 to address the most obvious form. The episodes are

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instructive because they show that when a contract references an event rather than a loss, the event can be purchased.

Russia, 2022. Payments on sovereign debt were made in rubles, or blocked by sanctions, or made but unable to reach holders. The question of whether a payment prevented by a third party is a failure to pay had no clean answer, and the determinations process had to resolve a case its drafters had not anticipated. A contract that insures against a borrower's unwillingness or inability is silent about a borrower who is willing, able, and prevented. The committee found a failure to pay in June 2022, on $1.9 million of accrued interest, and sanctions then delayed an auction that set the recovery near 56 cents.

The same gap runs through the corporate market. A liability-management exercise of the kind Section 26.15 describes can cut a lender's recovery while no payment is missed and, in the North American contract, no restructuring counts, so the protection pays nothing, or pays on an obligation whose price has fallen less than the loan's.

25.12 CREDIT SUISSE, MARCH 2023

The most instructive recent case triggered nothing at all, which is the point.

Case Study 25.1 The AT1 write-down and what subordination means

Additional Tier 1 instruments were created after the 2008 crisis to do something specific: absorb losses while a bank is still operating, so that it can be recapitalized without a resolution. They pay a high coupon, and they contain contractual triggers under which they are written down or converted to equity.

In March 2023, as part of the arrangements accompanying Credit Suisse's acquisition, roughly CHF 16 billion of AT1 instruments were written down to zero. Shareholders received approximately CHF 3 billion.

The capital structure was inverted. An instrument senior to equity was extinguished while equity retained value. That is the opposite of the ordering that the word "subordination" is supposed to guarantee.

Three observations follow, and the third is the one for this chapter.

It was contested in law, not only in markets. The Swiss AT1 terms allowed write-down on a viability event determined by the regulator, and holders had been told so. In October 2025, however, the Federal Administrative Court found that no such event had occurred and that the Swiss regulator's order lacked a legal basis; the appeal is pending, and the bonds remain written down. Whichever way it ends, almost no one had priced the possibility, because the instruments had been sold and analyzed as if the ordinary hierarchy applied.

The repricing was immediate and general. AT1 instruments across European banks fell sharply, and several regulators issued statements affirming that in *their* jurisdictions equity would absorb losses first. The market had been pricing a legal ordering that turned out to be jurisdiction-specific.

And no credit event occurred. Credit Suisse did not default; it was acquired. Its senior obligations performed throughout, and when the determinations committee was asked in May 2023 whether the write-down was a governmental intervention, it ruled that it was not: the AT1s ranked below every obligation the contracts reference, subordinated CDS included. So CDS on Credit Suisse did not pay, while holders of an instrument ranking above equity lost everything.

That last sentence is Section 25.2's thesis in its cleanest form. A CDS insures against a list of events, not against loss. A protection buyer who had correctly identified Credit

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Suisse as the weakest large European bank, and had bought protection to express it, would have been right about the institution and received nothing—while an AT1 holder with no view at all lost the entire position.

The general lesson extends past this instrument: wherever a payoff is defined by a legal event rather than by an economic outcome, the document is the instrument, and the residual risk is the risk that the document does not say what everyone assumed.

25.13 WHAT A CDS ACTUALLY INSURES

The episodes of Sections 25.11 and 25.12 are usually told as anecdotes. They are more useful as a taxonomy, because a hedger can check each item in advance.

A protection buyer who wants to be made whole on a credit loss faces four distinct residual risks, and only the last is normally measured.

Definitional risk. The loss occurs and no listed event does. This is the Credit Suisse case, and it is the largest of the four because it is binary and uncorrelated with anything a risk system tracks. The check is to read the credit event definitions against the specific way this borrower could plausibly impose a loss; for a bank, resolution, recapitalization, and default are three routes with three different answers.

Determination risk. An event occurs and the committee rules that it did not, or takes long enough that the position must be marked in the meantime. Greece and the 2022 sovereign cases are both instances. The check is whether the borrower's plausible distress route is one the committee has ruled on before. Novelty is the risk.

Settlement risk. The event is determined, the contract pays, and the auction recovery differs from the loss actually suffered on the hedged asset. A hedger holding a long-dated bond who receives an auction recovery struck on the cheapest deliverable has been made whole on a different instrument.

Basis risk. The protection references an entity and the exposure is to a different one: a subsidiary, a guaranteed affiliate, a proxy issuer entirely, as in the CVA hedging of Section 9.10. This is the one desks do measure, and it is the smallest.

The ordering is the reverse of the attention these risks receive: basis risk is quantified, reported, and hedged, while definitional risk, the one the Credit Suisse case exposed, is not quantified at all.

25.14 CONVERTIBLE BONDS

Section 10.10 deferred convertibles here because they are the other instrument that fuses credit with an option, and the fusion is what makes them interesting.

A convertible is a bond that the holder may exchange for a fixed number of shares, the **conversion ratio**. So it is a corporate bond plus a call on the issuer's own equity, and Section 17.4's dilution analysis applies to that call.

Example 25.8 Decomposing a convertible

Face 1000, 2% annual coupon, five years, conversion ratio 20. The stock is at 40, its volatility 35%, rates 4%, and the issuer's credit spread 300 bp.

How the split is made. The floor is the 2% bond discounted at 4% + 3% = 7%, continuously compounded; the option is a European call on 20 shares struck at the conversion price, valued by the Black–Scholes–Merton formula at the 4% risk-free rate with no dividends ($d_1 = 0.362$, $d_2 = -0.421$, 11.86 a share); and the convertible is their sum. The additive split is an approximation: it treats conversion as exercisable only at maturity, ignores any issuer call, and lets the option survive a default that would extinguish it. Example 25.9 measures each omission. Tsiveriotis and

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Fernandes (1998) is the widely used practitioner model, pricing the debt and equity components on one tree at different discount rates.

Bond floor (straight bond discounted at 7%)	786.14
Conversion value, or *parity* (20×40)	800.00
Conversion price (1000/20)	50.00
Conversion premium over the share price	25.0%
Embedded call on 20 shares struck at 50	237.13
Convertible value	**1023.28**
Equity delta	12.82 shares (64% of the 20)

The instrument's behavior across share prices shows what the issuer sold:

Share price	Convertible	Parity	Delta (shares)
20	827.87	400.00	6.00
30	908.55	600.00	9.95
40	1023.28	800.00	12.82
50	1162.13	1000.00	14.82
70	1484.96	1400.00	17.18

The convertible is bond-like when the stock is low and equity-like when it is high, with the transition governed by the embedded call's delta. That is its selling point to an issuer: a 2% coupon on debt that would otherwise cost 7%, for equity upside above 50: Section 17.9's bond-plus-warrant, with the components fused rather than separable.

Section 13.7 said why the additive split is not a valuation: conversion and call are node-by-node decisions that interact, so a bond and its embedded options cannot be valued separately and added; and credit belongs on the same tree, since a default ends the conversion right too. Hull's construction puts all three on one tree: each step of length Δt the stock moves up by u, down by $d = 1/u$, or the issuer defaults with probability $1 - e^{-\lambda \Delta t}$, leaving the stock worthless and the bond worth its recovery RF:

$$u = e^{\sqrt{(\sigma^2 - \lambda)\,\Delta t}}, \qquad p_u = \frac{e^{r\Delta t} - d\,e^{-\lambda \Delta t}}{u - d}, \qquad p_d = \frac{u\,e^{-\lambda \Delta t} - e^{r\Delta t}}{u - d}, \tag{25.14}$$

chosen so that the stock's expected growth, default paths included, is the risk-free rate and its total variance per unit time, jump included, is σ^2. The credit triangle gives $\lambda = s/(1 - R)$. With conversion ratio n and call price K_c, each node's rollback is

$$V = \max\left\{ nS,\ \ \min\left(e^{-r\Delta t}\left[p_u V_u + p_d V_d + (1 - e^{-\lambda \Delta t})\, RF \right],\ K_c \right) \right\} + \text{coupon due}, \tag{25.15}$$

read from the inside out: the bracket is the continuation value; the issuer calls when continuation exceeds K_c; and the holder converts whenever n shares are worth more than what is left. Without a call the min drops out; with conversion only at maturity the outer max acts only there.

Example 25.9 Pricing the convertible on a tree with default

Example 25.8's bond on a tree of five annual steps, so that the coupon dates are the nodes. From $s = 300$ bp and $R = 40\%$, $\lambda = 0.03/0.6 = 5\%$; then (25.14) gives $u = 1.3090$, $d = 0.7639$, $p_u = 0.5763$, $p_d = 0.3749$, default probability 0.0488 and discount factor $e^{-0.04} = 0.9608$ per step. Figure 25.4 is the whole tree for the callable case.

One node is worth working in full, through the rollback (25.15). At $t = 4$ with the stock back at 40, the surviving successors are $S = 52.36$, where the bond converts at maturity for 1047.19 plus the final coupon, 1067.19, and $S = 30.56$, where it is redeemed for 1020.00 with the coupon; default pays 400. Continuation is $0.9608 \times (0.5763 \times 1067.19 + 0.3749 \times 1020.00 + 0.0488 \times 400) = 977.09$, against $20 \times 40 = 800$ for converting now, so the holder holds, and the node is worth 977.09 plus

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the year-4 coupon, 997.09. The issuer's call at 1100 from year 2 shows up in two forms. At $t = 2$ with $S = 68.54$ continuation is 1426.81, so the issuer calls, and the holder, offered 1100, converts instead for 1370.77: the call forced conversion three years early and cost the holder 56. At $t = 3$ with $S = 52.36$ continuation is 1148.00, the issuer calls, and this time 20 shares are worth only 1047.19, so the holder takes the 1100 in cash and loses 48 against holding on.

	Five steps	**500 steps**
Bond floor (no conversion right)	793.50	795.10
Conversion at maturity only, or at any time	1046.38	1057.82
Conversion at any time, callable at 1100 from year 2	994.47	988.40
Additive split of Example 25.8, for comparison	1023.28	

Three readings deserve comment. The floor is 795 rather than 786 because recovery is modeled as a payment of 400 at default rather than folded into a 7% discount rate. Conversion at maturity only gives 1058 against the additive 1023: 263 rather than 237 for the option, the rest being the higher floor. The default branch makes the call dearer, not cheaper, because a stock that can drop to zero must grow at $r + \lambda$ conditional on surviving (Merton's 1976 jump-to-ruin result), which outweighs the lower diffusive volatility of 26.9%. The American right is worth nothing here, since without dividends the holder never converts early (Section 11.8). What changes the value is the issuer's call, −52 on five steps and −69 converged, because it forces conversion whenever continuing would be worth more than 1100; that is the interaction Section 13.7 said the additive method discards, and it is the largest term.

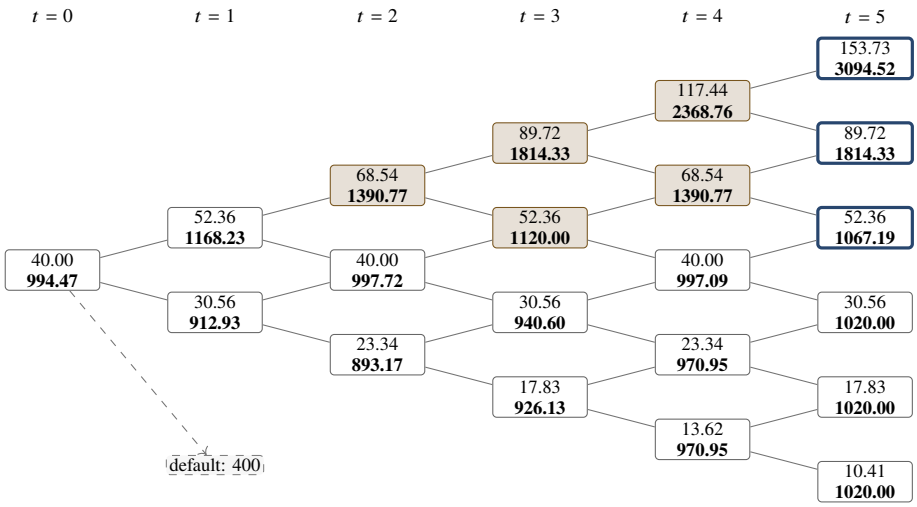


Figure 25.4: Example 25.9's callable convertible on the five-step tree: share price above, bond value (with the coupon paid that date) below. Shaded nodes are called, heavy-bordered nodes converted; the default branch worth 400 exists at every node and is drawn once.

Convertible arbitrage. The classic hedge fund trade buys the convertible and shorts the delta, leaving a position that is:

Long gamma and vega, because the embedded call is now unhedged in second order. This is Chapter 18's dealer position with the sign reversed.

Short credit, because the bond floor is a risky bond.

And long carry, from the coupon less the cost of the short.

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The trade's characteristic failure is that its two exposures go wrong together. A 100 bp widening in the issuer's spread, with the stock unchanged, costs 36.66 on the bond floor (3.58% of the convertible's value). And credit widens precisely when equity volatility spikes, so the position's volatility gain arrives at the same moment as its credit loss.

That would net out if both were realized on the same timetable. They are not. The credit loss is a mark-to-market that arrives immediately; the volatility gain is earned only by gamma-hedging through the move, which requires the fund to still be there. In 2008 many were not, and convertible arbitrage, a strategy that is genuinely hedged in the sense that its Greeks offset, produced one of the worst drawdowns of any hedge fund style. Being hedged in the model and being able to hold the position are different things, which is Section 5.15's limits-of-arbitrage argument in yet another market.

25.15 FUNDED AND UNFUNDED FORMATS

A CDS is **unfunded**: no principal changes hands at inception, and the protection seller's obligation is backed only by their creditworthiness and their Chapter 8 CSA's collateral.

A **credit-linked note** converts the same exposure into a funded one. The investor buys a note, paying cash upfront; the issuer invests the proceeds and pays a coupon; if a credit event occurs on the reference entity, principal is reduced accordingly. Economically the investor has sold protection and prepaid the collateral in full.

The difference is entirely about counterparty risk, and it runs both ways.

The protection buyer is fully secured. They hold the investor's cash, so there is no exposure to the seller at all. That is why credit-linked notes are the format of choice when the seller is an entity no one wants unsecured exposure to.

The investor acquires exposure to the issuer. Their principal now sits with the note's issuer and is invested in something. A credit-linked note is therefore two credit exposures, not one, and the second is easy to overlook because it is not what the note is named after. Several structures in 2008 delivered losses through the collateral rather than the reference entity.

The same funded-unfunded distinction organizes Chapter 26. A cash securitization is funded: investors buy notes and the proceeds buy the assets. A synthetic one is unfunded: investors sell protection on a reference portfolio that stays where it is. The synthetic version can be created without anyone originating a loan, which is why the outstanding amount of synthetic exposure was not limited by the amount of debt.

25.16 HYBRID CREDIT STRUCTURES

The previous section ended by observing that a credit-linked note is two credit exposures rather than one. The same construction generalizes, and it has become the part of the credit market where the modeling is hardest.

A **hybrid** structure makes a payment contingent on a credit event *and* on the level of something else: an interest rate, an exchange rate, an equity index. A note might pay an equity-linked coupon for as long as a reference issuer survives, redeem at a rate-linked amount unless a portfolio takes losses, or pay in a currency the reference entity does not borrow in. Chapter 35 treats the note wrapper and the distribution question; what belongs here is what the credit leg does to the valuation.

The temptation is to price the two legs separately and add, and it does not work. Write τ for the default time and X for the other factor. The payoff is $\mathbf{1}\{\tau > T\}\, f(X)$, and its value involves

$$\mathbb{E}[\mathbf{1}\{\tau > T\}\, f(X)] \;\neq\; \mathbb{P}(\tau > T)\, \mathbb{E}[f(X)]$$

unless default and the factor are independent. The two legs are individually observable (survival probabilities come from the CDS curve of Section 25.3, the distribution of X from its own

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option market), and the quantity that joins them is not observable from either. Example 25.10 puts a number on what that costs.

Example 25.10 The correlation a hybrid needs and no market quotes

A five-year note paying a single 8% coupon at maturity, and paying it only if the reference issuer has survived *and* an equity index finishes above its initial level. Risk-free rate 4%, dividend yield 2%, index volatility 30%, and a CDS curve implying a five-year survival probability of 0.90.

The index condition is met with probability $N(d_2)$, where

$$d_2 = \frac{(r - q - \sigma^2/2)T}{\sigma\sqrt{T}} = \frac{(0.04 - 0.02 - 0.045) \times 5}{0.30\sqrt{5}} = -0.1863$$

so $N(d_2) = 0.4261$.

Couple the two with a Gaussian copula, Section 24.13's device with two variables rather than a pool: default if the credit factor falls below the 10% point of the standard normal, -1.2816, and the index above strike if its factor exceeds $-d_2 = 0.1863$. The coupon is paid when both conditions hold, which is a bivariate normal probability.

ρ	P(coupon paid)	Expected coupon	Effect on the coupon
−0.4	0.3551	2.8408%	−22.7 bp
0.0	**0.3835**	**3.0678%**	**reference**
0.4	0.4091	3.2727%	+20.5 bp

At $\rho = 0$ the probability is exactly 0.90×0.4261, which is the control on the calculation. At a correlation of 0.4 the expected coupon moves by about **20 basis points**—on a trade whose two legs are each quoted to a fraction of a basis point.

The realistic sign is positive. Equity levels and credit quality move together, so the coupon tends to pay in the states where the issuer also survives, and the note is worth *more* than the independent calculation says. A desk that prices the credit leg off the CDS curve and the equity leg off the volatility surface and multiplies the two is therefore not making a small error in an unknown direction; it is understating the note.

Three things follow.

The correlation cannot be implied. Vanilla equity options imply the index's volatility and CDS implies the issuer's hazard rate, and no liquidly traded instrument implies the dependence between them. It is estimated from history, which is Section 26.6's position on default correlation, arrived at in a market that was supposed to have learned that lesson.

The bounds are wide but not useless. Any joint distribution with these marginals pays the coupon with probability between $\max(0.90 + 0.4261 - 1, 0) = 0.3261$ and $\min(0.90, 0.4261) = 0.4261$, so the note's value is confined to a band of $(0.4261 - 0.3261) \times 8\%$, or 80 basis points of coupon, whatever the dependence. That width is $1 - 0.90$ times the coupon and nothing else: for any index probability between 0.10 and 0.90 the two bounds differ by exactly the default probability, so the band is set by the credit leg alone. A reserve sized to the width of that band is defensible without agreeing on a number, which is the same response Section 34.7 offers to model risk in exotics.

And the hedge is cross-gamma. The sensitivity that matters is not delta to the index or spread sensitivity to the issuer but the cross term between them, and there is no instrument whose payoff is that cross term. It is hedged by rebalancing two positions, which means the hedging error is a function of realized rather than implied dependence.

Who trades them, and why. The buyer is an investor who wants a yield the single-factor version cannot produce: attaching a second condition to a coupon raises it, as the range accrual of Section 33.6 does with a rate, because the investor has sold something. The seller is one of the few dealers that can warehouse and partially hedge the cross-risk. Insurers and pension funds buy where the structure lets them reach a yield target within a ratings constraint, the same demand that Chapter 26's senior tranches met.

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What has changed. Section 9.9's wrong-way risk is the same mathematics as this section's correlation: an exposure that grows as the counterparty's credit deteriorates. What has changed is that the difficulty has migrated from the valuation adjustment into the products themselves. A hybrid note needs a model in which credit, rates and an equity index move together, calibrated to three separately quoted markets, and the number it produces is the price the client pays.

25.17 SOVEREIGN CDS AND THE CURRENCY OF THE PAYOUT

Section 24.12 established that sovereign credit is different in kind. The CDS market adds a feature that has no corporate analog and that connects directly to Chapter 16.

Sovereign CDS is usually denominated in a currency not the sovereign's own. Protection on an emerging market sovereign trades in dollars; protection on a euro-area sovereign has traded in dollars and euros. The reason is obvious: protection paying in a currency the defaulting sovereign controls is worth less than it appears. The gap is priced.

The two contracts differ by what Chapter 16 would call a quanto effect. A euro-denominated contract on a euro-area sovereign pays euros in a state of the world in which the euro may have depreciated sharply, precisely because that sovereign defaulted. The dollar contract pays the same notional in a currency that has not moved, so it is worth more, and the **quanto spread** between the two is the market's price for devaluation conditional on default.

It is a jump-at-default expectation. If the dollar contract trades meaningfully above the euro one, the market is pricing a substantial currency move conditional on the credit event, not a general currency view, since an unconditional move affects both contracts alike.

During the euro-area crisis it was a market-implied redenomination probability. The quanto spread on peripheral sovereigns widened when the possibility of leaving the currency union was being discussed and narrowed when it receded. A market with no instrument for pricing the breakup of a currency union priced it anyway, in the basis between two CDS contracts.

The pricing requires exactly the machinery of Section 16.7: a correlation between the credit event and the exchange rate, entering as a quanto adjustment. It is this book's plainest case of a credit instrument and a currency instrument that must be priced jointly (the credit-and-currency hybrids of Section 25.16 are the bespoke version).

The seller matters as much as the currency. Section 9.9's textbook case of wrong-way risk is protection on a sovereign bought from a bank in that country: the contract pays in exactly the states in which a bank holding its own sovereign's bonds, and funded in its currency, is least able to pay. In the euro-area crisis of 2010–2012 the case was live rather than hypothetical: Italian and Spanish banks held large amounts of their own governments' bonds, and their CDS spreads rose and fell with their sovereigns', so protection on Italy sold by an Italian bank was weakest exactly when it would have paid. Protection on a sovereign is therefore worth most when it is bought from a seller outside the jurisdiction and fully collateralized, and a desk that buys it from a domestic bank has to price that correlation into its CVA.

25.18 THE MARKET AFTER THE REFORMS

The instrument this chapter has described trades in a market a small fraction of the size of the one that made it famous.

Case Study 25.2 The market that shrank

Gross notional outstanding peaked at about $58 trillion at the end of 2007 and fell by roughly an order of magnitude over the following decade; at the end of 2025 it stood at $11 trillion, 70% of it cleared. Three forces did it, and they are worth separating because only one of them reduced risk.

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Compression removed offsetting trades. A dealer with a bought and a sold contract on the same issuer, maturity, and coupon holds two contracts and no risk. The Big Bang's fixed coupons and standard dates made those contracts fungible, and multilateral compression services then tore them up in bulk. This eliminated enormous gross notional and no net exposure at all.

Clearing concentrated and netted the rest. Index CDS moved to central clearing under the mandates of Chapter 2, which nets multilaterally rather than bilaterally and shrinks the reported figure further.

And single-name activity genuinely declined. This is the force that reduced risk, because fewer net positions are now taken, and it reduced the market's usefulness with it. Capital charges on dealer inventory rose, the instrument acquired a reputational cost after the crisis, and several large participants withdrew. Liquid single-name CDS now exists for a few hundred reference entities, which Section 9.10 already flagged as the reason most CVA is hedged by proxy.

Against: a bank hedging a loan to a mid-sized borrower usually cannot buy protection on it, so the risk-transfer function that justified the instrument works only for the largest issuers. The market that remains serves index macro hedging and the CVA desks of Chapter 9 better than it serves the original purpose.

For: the market that remains is standardized, cleared, compressed, and settled by auction, and none of those was true in 2007. The instrument that failed in the crisis and the instrument that exists now differ in every respect that mattered then.

The reforms traded breadth for robustness. That is probably the right trade, and it is a trade rather than a free improvement.

25.19 WHO TRADES CREDIT DERIVATIVES, AND WHY

Banks buy protection to hedge loan books and to reduce their capital, and this was the instrument's original purpose. Under the substitution rule of Section 27.11 the hedged loan takes the protection seller's risk weight, so the capital it saves can justify a hedge on its own (Example 27.6).

XVA desks buy protection on their counterparties, as Section 9.10 described. This is the largest structural demand for single-name protection, and it is why the market's liquidity is concentrated in the few hundred issuers that are also large derivatives counterparties.

Asset managers use index CDS to adjust credit exposure faster and more cheaply than by trading bonds. Selling index protection is a synthetic long credit position that requires no cash and no bond inventory.

Macro funds buy index protection as a hedge against a general deterioration, and this demand is what moves the index skew of Example 25.4.

Basis traders arbitrage the CDS–bond relationship of Section 24.15, subject to the funding constraint that section identified.

Correlation desks and hedge funds trade the equity and mezzanine tranches against the index or single names, as in the 2005 trade of Section 26.7; insurers, pension funds, and banks held the senior layers for a yield above their rating, the demand Chapter 26's senior tranches met. **Basket default swaps** let an investor earn more than any one issuer's spread for bearing the first default among a handful of names; the dealer buying the protection hedges with the constituents and keeps the correlation exposure. **Hedge funds** receive total return to hold financed positions they would otherwise fund and disclose (Section 7.12), and buy most new convertibles, alongside long-only convertible funds, for the arbitrage of Section 25.14; issuers are growth companies that would pay far more for straight debt.

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The market's structural position is that dealers are net sellers of protection to banks and buyers from asset managers, with the imbalance hedged in the index. A general widening therefore produces a demand for the same hedge from everyone at once. That is Chapter 22's crowding argument transplanted to credit, and Chapter 26 records the last time.

SUMMARY

A CDS pays on a credit event, not on default. When the two differ, the contract's definitions govern, and a protection buyer who is right about the credit and wrong about the definition receives nothing. Since 2009 a determinations committee of market participants, not a court, rules on whether an event occurred.

The 2009 standardization fixed the coupon at a few standard values (100 or 500 basis points for North American corporates) and moved the difference between the coupon and the issuer's spread into an upfront payment. The risky annuity falls as the spread widens, because a deteriorating credit's premium stream is expected to be cut short, so a hedge sized on duration is under-hedged exactly when it is working.

Physical settlement failed because protection outstanding can exceed deliverable debt, and the auction replaced it with a single market-wide recovery rate, a forecast made under duress, not the realized recovery. Slicing an index into tranches redistributes a fixed expected loss: as correlation rises, loss moves out of the equity tranche and into the senior one, so a senior tranche valued at a low correlation and facing a high one carries many times the expected loss it was priced for. Base correlation removes the inconsistency between the tranches' implied correlations without removing its cause. An index trades at the annuity-weighted, not the arithmetic, average of its constituents' spreads, and correlation lowers the price of a first-to-default basket while raising that of the last.

Restructuring is the contested credit event because it leaves the issuer alive with obligations at very different prices. Greece in 2012, the manufactured events of the 2010s, Russia in 2022, and Credit Suisse in March 2023, whose AT1 bonds were written to zero with no credit event, all show that where a payoff is defined by a legal event, the document is the instrument. The market shrank by an order of magnitude after the crisis, trading breadth for robustness.

A credit-linked note is a funded CDS with a second exposure to the note's issuer; a total return swap gives the economics of an asset without ownership; sovereign CDS pays in a currency the sovereign does not control, and the quanto spread is the price of devaluation conditional on default. A convertible is a corporate bond plus a call on the issuer's equity, but default, the conversion right, and the issuer's call interact and must be priced together on one tree. Convertible arbitrage is long gamma and vega, short credit, and long carry, with the credit loss marked immediately and the volatility gain earned only by hedging through the move. A hybrid coupon that needs both survival and an equity level cannot be priced as a product of marginals; the unquoted dependence moves it within a band set by the credit leg alone.

FURTHER READING

The ISDA credit derivatives definitions are the primary source, and Iwashita's OpenGamma note on single-name CDS conventions collects the coupon menus, day counts, and roll dates that the definitions leave to market practice; anyone who intends to rely on a CDS should read the credit event definitions rather than a summary of them. The determinations committee publishes its questions and rulings, and reading a contested one (Greece 2012 or the 2022 sovereign cases) conveys the character of the problem. On the auction mechanism, the academic work on its design and on whether the auction price is biased relative to eventual recovery is small, and the finding of Chernov, Gorbenko, and Makarov (2013) and Du and Zhu (2017) that biases exist and are related to the net open interest is worth knowing before relying on

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auction recoveries as data. On the Credit Suisse AT1 write-down, the Swiss regulator's own explanation and the subsequent statements from other supervisors are the documents to read, and the Federal Administrative Court's October 2025 judgment, now on appeal to the Federal Supreme Court, is the authoritative legal account so far. For the tranche and basket valuation of Sections 25.7 and 25.8, Hull and White (2004) and Andersen, Sidenius, and Basu (2003) are the sources of the four steps, and O'Kane and Turnbull (2003) is the practitioner treatment of the standard model. On convertibles, Tsiveriotis and Fernandes (1998) is the model behind most vendor systems, and the practitioner literature on convertible arbitrage sets out what happened in 2008.

PRACTICE QUESTIONS

25.1 Explain why a protection buyer need not own the reference obligation, giving one benefit and one problem this creates, and explain why gross CDS notional outstanding is not a measure of the risk transferred.

25.2 A five-year CDS has a par spread of 250 bp and trades on a 100 bp coupon with a risky annuity of 4.19. Compute the upfront payment, and explain why the risky annuity falls as the spread widens.

25.3 A desk bought five-year protection on the standard 100 bp coupon when the par spread was 150 bp, paying an upfront of 2.18% of notional. One year later the spread is 400 bp, and the risky annuity on the remaining four years at that level is 3.31. Compute the position's value and its gain since inception as percentages of notional, explain why marking against 150 bp or using the annuity at the original spread misstates them, and state in which direction the annuity error biases a hedge.

25.4 Explain why physical settlement failed, what the auction replaced it with, and why an auction recovery is not the same as the eventual realized recovery.

25.5 Explain why an index CDS trades below the arithmetic average of its constituents' spreads, what the index skew is, and why it widens when the index is used as a macro hedge.

25.6 Explain why restructuring creates a cheapest-to-deliver problem that bankruptcy does not, and why a risk system treating restructuring and no-restructuring contracts as substitutes has mismeasured the exposure.

25.7 State what the Credit Suisse AT1 write-down established about the meaning of subordination, and explain why CDS on Credit Suisse did not pay.

25.8 Explain why an equity tranche's expected loss falls as correlation rises while a senior tranche's rises, given that the portfolio's expected loss is unchanged, and use Example 25.5 to state the factor by which a 15–30% tranche's expected loss grows between $\rho = 0.10$ and $\rho = 0.50$.

25.9 Explain why implied correlation differing across tranches on the same portfolio is logically impossible, and what base correlation does about it.

25.10 A convertible has a conversion ratio of 25 and the stock trades at 30. Compute the conversion value and the conversion price on a face of 1000, and explain why the instrument behaves like a bond at low share prices and like equity at high ones.

25.11 State the four exposures of a delta-hedged convertible arbitrage position, and explain why two of them go wrong together and why the timing of their recognition differs.

25.12 Explain why a total return swap position does not appear in ownership disclosures, and who bears the risk if the receiver cannot meet a margin call.

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25.13 Distinguish the three causes of the CDS market's post-crisis shrinkage, and state which of them reduced risk rather than merely reducing gross notional.

25.14 Explain why a credit-linked note contains two credit exposures rather than one, and explain what the quanto spread between a dollar- and a euro-denominated CDS on a euro-area sovereign measures.

25.15 State the four residual risks a protection buyer faces even when right about the credit, and explain which of them is least measured and why it is the one most likely to produce the largest loss.

25.16 Five issuers each trade at 150 bp. Explain why the first-to-default basket costs the sum of the five spreads at $\rho = 0$ and falls as ρ rises while the fifth-to-default rises, and why the five baskets' protection legs add up to the five single-name legs (Example 25.6).

FURTHER QUESTIONS

25.17 Example 25.2 shows the risky annuity falling from 4.55 to 3.40 as the spread widens. Derive this dependence, show that it makes the upfront a concave function of the spread, and explain what this implies for a desk that hedges a portfolio of wide-spread issuers with a position in a tight-spread one.

25.18 Section 25.5 argues that the auction solved the squeeze and created a different problem. Set out the incentives of a participant with a large protection position, explain what the two-stage design does to constrain them, and propose a modification that would further reduce the incentive to influence the outcome, stating what it would cost.

25.19 Example 25.4 attributes the index-average gap to differences in risky annuity. Derive the exact relation between the index spread and the constituent spreads, show that the gap grows with dispersion, and determine the dispersion at which it becomes large enough to justify trading 125 issuers against the index.

25.20 Example 25.7 finds the restructuring clause worth 7% of the spread. Explain why this number is nearly useless as a risk measure, construct a measure that would capture the exposure correctly, and discuss why a bank hedging a loan book might rationally pay far more than 7% for the clause.

25.21 The Credit Suisse case study argues that where a payoff is defined by a legal event, the document is the instrument. Identify three other instruments in this book whose payoff depends on a legal or administrative determination rather than a market price, and for each state what the residual risk is and who is best placed to bear it.

25.22 Re-price Example 25.9's convertible with a 3% dividend yield on the stock and show that early conversion now has value; then add a holder's put at par at year 3 and state which way each feature moves the value and why.

25.23 Section 25.14 argues that convertible arbitrage was hedged in the model and failed in practice. Distinguish the model risk, the funding risk, and the crowding risk in that failure; explain which of the three the Greeks would have revealed; and relate your answer to Section 5.15 and to Chapter 22's account of 2018.

25.24 Using Example 25.10, show that the 80 bp band on the hybrid coupon does not depend on the index probability for any value between 0.10 and 0.90, and explain what that implies for how a desk should reserve against an unquoted correlation.

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Chapter 26
Securitization, Structured Credit, and the 2007–2009 Crisis

Section 24.13 established that correlation redistributes a portfolio's loss without changing it, and Section 25.7 put numbers on the redistribution: a senior tranche valued at a correlation of 0.10 and facing 0.50 has an expected loss forty-three times larger. This chapter is about what happened when an industry was built on a number no one could observe.

Every mechanism in this book that produced a loss elsewhere appears here at once: the model risk of Chapter 21, the crowding of Chapter 22, the funding fragility of Chapter 8, the rating-trigger effects of Section 24.10, and the gap between a contract and the risk it was meant to transfer from Section 25.13. It remains the best available case study because it is the only episode in which all of them fired together.

The chapter runs from the mechanism to the history and back to the present. Sections 26.1 and 26.2 are the structure: what pooling and ranking do, and why a tranche is a call spread on the pool's loss, which makes it convex in every input. Section 26.3 is the counter-example: the agency mortgage market, larger than everything that failed, where the option being sold was known and priced. Sections 26.4 to 26.6 are how the deals were built and modeled: the three sources of credit support and why they shrink when they are needed, re-securitization, and the Gaussian copula and what it was asked to do. Sections 26.7 to 26.12 are the history, ending with the order in which things broke. Sections 26.13 to 26.16 ask what was repaired, which structure survived, and where the same pattern operates now, and Section 26.17 records where the models came from.

26.1 WHAT SECURITIZATION DOES

Pool a set of loans, sell claims on the pool's cash flows, and rank those claims. That is the whole idea, and in its plain form it is useful.

It converts illiquid assets into tradeable ones. A mortgage cannot be sold easily; a security backed by ten thousand of them can.

It matches risk to appetite. A pension fund that cannot hold individual mortgages can hold a senior claim on a pool of them. Someone else takes the first-loss piece.

And it transfers risk out of banks, which is what the capital frameworks of Chapter 27 are designed to encourage when the transfer is genuine.

None of this is objectionable, and the market for plain mortgage-backed securities functioned through the crisis and functions now. What failed was not securitization; it was a specific set of things built on top of it, and keeping the two separate is necessary to draw any useful lesson.

26.2 THE WATERFALL

Cash from the pool is distributed by a **waterfall**: senior claims are paid first, then mezzanine, then equity. Losses arrive in the reverse order, so the equity tranche absorbs the first losses and the senior tranche only suffers once everything below it is exhausted.

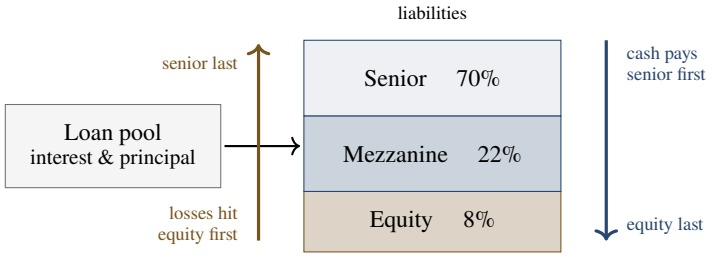


Figure 26.1: Cash descends the stack by seniority; losses climb it from the bottom. Tranche sizes illustrative.

Figure 26.1 is the entire structure, and the two arrows pointing in opposite directions are the reason a single pool can support securities with genuinely different risk. *Nothing has been created*: the pool's total cash and total loss are unchanged, and every currency unit the senior tranche is protected from is one the equity tranche absorbs.

A tranche is therefore an option on the pool's loss. Write L for the pool's loss as a fraction of its notional. The $[a, b]$ tranche absorbs

$$L_{[a,b]} = \frac{\min\big(\max(L-a,\,0),\;b-a\big)}{b-a} \tag{26.1}$$

as a fraction of its own notional: a call spread on L struck at a and b, normalized, which is Section 12.2's structure applied to a loss rather than a price. The attachment point a is the strike, the detachment point b the cap, and every property of a tranche in this chapter follows from that.

The family is named after its collateral, and the names are worth fixing because the rest of the chapter uses them. A tranched pool of mortgages is a mortgage-backed security; a tranched pool of bonds or loans is a collateralized debt obligation (CDO); when the collateral is leveraged corporate loans it is a **collateralized loan obligation** (CLO), the member of the family that survived, which Section 26.14 treats on its own terms. A CDO whose collateral is itself tranches of other deals is a **CDO-squared**, and Section 26.5 is about what that does. *Nothing in the mathematics distinguishes them*: (26.1) is the same function in every case, which is precisely why the differences that matter are in the collateral and in who manages it.

Two immediate consequences follow, and the rest of the chapter rests on them. *Adding up the tranches returns the pool*: $\sum(b_j - a_j)L_{[a_j,b_j]} = L$ when the layers partition $[0, 1]$, so securitization redistributes loss and creates none. And a tranche's expected loss is nonlinear in the pool's, because (26.1) is kinked at both ends, which is where Example 26.1's leverage comes from.

Identifying a tranche as a call spread does most of the analytical work in this chapter, because it means everything Chapter 12 said about spreads applies. A senior tranche is a deeply out-of-the-money option: worth almost nothing in expectation, enormously sensitive to the assumed distribution, and, critically, *convex in the parameters*.

Figure 26.2 shows both halves of the argument: the call-spread payoff on the left, and on the right what that payoff does to an error in the input.

Example 26.1 What a tranche does to a parameter error

A pool of 100 loans, each with a 5% default probability and 60% loss given default, correlation 0.15. The pool's expected loss is 3.00%, the 12–30% senior tranche's 0.266%.

Now suppose the true default probability was higher than assumed:

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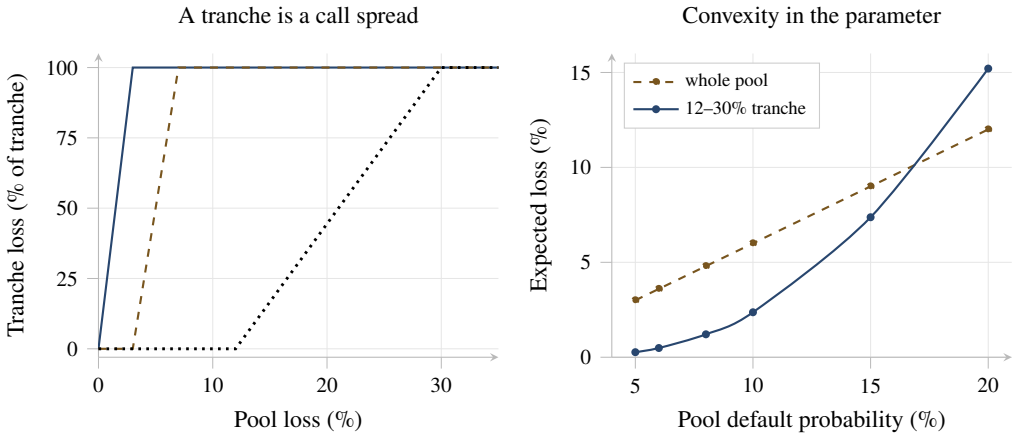


Figure 26.2: Left: equity (0–3%), mezzanine (3–7%) and senior (12–30%) losses against pool loss. Right: expected loss against default probability.

True default probability	Pool expected loss	Senior tranche expected loss
5%	3.00%	0.266%
6%	3.60%	0.486%
8%	4.80%	1.208%
10%	6.00%	2.369%
15%	9.00%	7.374%
20%	12.00%	15.205%

The numbers come from the copula exactly as Example 24.8 applied it. The tranche is untouched until 21 of the 100 loans default, so the pool's loss moves with the default probability and the tranche's with the probability of reaching that count.

Tripling the default probability triples the pool's loss and multiplies the senior tranche's by twenty-eight.

This is the chapter's central mechanism and it requires no story about fraud or stupidity. A tranche is an option, options are convex, and convexity amplifies parameter error.

The same convexity works in the parameter Section 25.7 identified. Both the default probability and the correlation enter a senior tranche's value with high leverage, and neither was observable. Two unobservable inputs, each amplified by a factor of tens, is not a model that should be used to rate an instrument AAA.

26.3 THE SECURITIZATION THAT WORKS

The counter-example comes first, because it is larger than the failures and it did not fail.

Agency mortgage-backed securities are issued or guaranteed by Ginnie Mae, Fannie Mae or Freddie Mac, and most are **pass-throughs**: the pool's monthly interest and principal, prepayments included, are passed straight through to investors in proportion to their holdings. They carry no meaningful credit risk: the guarantor absorbs default, and the investor receives principal and interest regardless. What the investor bears instead is prepayment risk, and it is a pure option problem with no correlation parameter in it.

A borrower may repay at any time without penalty. So the investor in a mortgage pool is short a call on the loan, exercised when refinancing is attractive, which is when rates have fallen.

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Example 26.2 Negative convexity, measured

A 30-year 5.5% pass-through on a pool of level-payment mortgages (a constant monthly installment over the loan's life), paying monthly, at a 5% yield compounded monthly. The speed is the market's conditional prepayment rate (CPR), the fraction of the outstanding balance repaid early in a year beyond scheduled amortization, applied monthly as $1-(1-\text{CPR})^{1/12}$ of the balance left after scheduled principal; it responds to the refinancing incentive: slow while market rates sit above the coupon, steep once they fall below it, leveling off once those who can refinance have. Prices are per unit of face at 5% + Δy; "fixed speed" keeps the 12% base rate; the comparison is a seven-year 5.5% semiannual bullet, a bond repaying all its principal at maturity, seven years being about the pool's average life at the base speed (6.6 years), so the comparison isolates the option rather than maturity.

Yield change	CPR	Pass-through price	Price at fixed 12% CPR	7y bullet
−2.0%	35%	1.0534	1.1399	1.1568
−1.0%	22%	1.0496	1.0797	1.0908
0	12%	1.0253	1.0253	1.0292
+1.0%	8%	0.9698	0.9759	0.9718
+2.0%	6%	0.9035	0.9307	0.9181

	Effective duration	Effective convexity
Pass-through, prepayments respond	3.89	**−304**
Same bond, fixed prepayment speed	5.07	+48.5
Seven-year bullet	5.78	+39.9

Effective duration and convexity are read off the ±1% prices: for the responding pass-through, $(1.0496-0.9698)/(2\times1.0253\times0.01) = 3.89$ and $(1.0496+0.9698-2\times1.0253)/(1.0253\times0.01^2) = -304$.

The convexity is negative, and strongly so. A two-percent rally lifts the bullet by 12.4% and the pass-through by 2.7%; a two-percent sell-off costs the bullet 10.8% and the pass-through 11.9%. The investor gets a fraction of the upside and all of the downside, because the borrowers refinance when rates fall and stay put when they rise.

The mechanism is exactly Chapter 18's. The investor is short an option, so they are short gamma, and Section 18.12's account of what short gamma does to a hedger applies directly: as rates fall the pool's duration shortens, so a mortgage portfolio hedged to a duration target requires buying duration as rates fall and selling it as they rise. That is mechanical buying into a rally and selling into a sell-off, pushing rates further in the direction they have already moved, by an investor base measured in trillions, and it is one of the largest identifiable feedback flows in any fixed-income market.

Two lessons follow for this chapter.

Risk that is genuinely understood can be borne at enormous scale. The agency mortgage-backed securities market is bigger than everything that failed in 2008 put together, its central risk is harder to model than credit default, and it worked. The difference is that the option being sold was known to be an option, priced as one, and hedged as one.

And the failure of Section 26.5 was not about complexity. Prepayment modeling is more technically demanding than a Gaussian copula. What distinguishes the two is not difficulty but whether the model's key input could be checked against anything, and prepayment speeds are observed monthly while default correlation was observed never.

26.4 HOW A DEAL IS ACTUALLY BUILT

A bond's **credit support** is the loss the pool absorbs before the bond takes any, and it comes from three sources. **Subordination** is Section 26.2's tranching: for a senior bond attaching at 20%, the 20% of bonds beneath it. **Overcollateralization** is the excess of assets over liabilities:

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\$980 million of bonds against \$1,000 million of loans gives 2% of cushion. **Excess spread** is what is left of the pool's coupon after the bonds' coupon and the servicing fees are paid; it is the first thing losses consume, before any tranche is touched, and if it is retained rather than released to the equity it accumulates, so the support grows every year the pool performs.

This is why the structures looked so robust, and the flaw is visible once stated. The support grows *provided the excess spread is actually earned*, and it is earned only while the loans keep paying. A pool that starts defaulting loses its spread at the moment it begins consuming the cushion it already has. This correlation between the buffer and the loss is structurally the wrong-way risk of Section 9.9 and the recovery–default correlation of Section 24.4: a protection that shrinks in the state where it is needed. Example 26.3 runs the two paths side by side.

Example 26.3 Where the protection comes from

A \$1,000 million pool of loans with an 8.5% weighted-average coupon backs \$980 million of bonds costing 5.5%; servicing and fees take 0.5%, and every percentage below is of the original \$1,000 million. The excess spread is $8.5 - 5.5 - 0.5 =$ **2.5%** a year on the paying balance, \$25 million in a default-free year, and the overcollateralization is 2%.

The promised path. A senior bond attaching at 20% starts with 20% + 2% = 22% of support. With the excess spread retained, each year adds 2.5%: 24.5% at the end of year 1, then 27.0%, 29.5%, 32.0% and 34.5%, the table's "promised" column.

The stressed path. Suppose 8% of the original balance defaults each year at a 40% recovery. The pool then loses $8\% \times 60\% = 4.8\%$ a year, but its performing balance falls by the full 8%: a defaulted loan pays no interest, and what is recovered on it comes back as principal, not as a loan still earning the coupon. Excess spread is earned on the average performing balance of the year, and losses are absorbed by the overcollateralization and the spread earned before they touch subordination:

	Promised	Realized under stress			
Year	Support	Spread earned	Loss	OC left	Support
1	24.5%	2.40%	4.80%	0.00%	19.60%
2	27.0%	2.20%	4.80%	0.00%	17.00%
3	29.5%	2.00%	4.80%	0.00%	14.20%
4	32.0%	1.80%	4.80%	0.00%	11.20%
5	**34.5%**	1.60%	4.80%	0.00%	**8.00%**

Years 1 and 2 are worth working in full. In year 1 the performing balance falls from 100% to 92%, averaging 96%, so the deal earns $2.5\% \times 0.96 = 2.40\%$ of spread against 4.80% of loss. The 2% of overcollateralization and the 2.40% together absorb 4.40%, so the overcollateralization is gone and the remaining 0.40% comes out of subordination: support ends the year at $22.0 + 2.40 - 4.80 = 19.60\%$. In year 2 the balance falls from 92% to 84%, averaging 88%, earning $2.5\% \times 0.88 = 2.20\%$ against another 4.80%; with the overcollateralization gone, the whole 2.60% shortfall comes out of subordination, and support falls to $19.60 - 2.60 = 17.00\%$.

The support does not rise to 34.5%; it falls to 8.00%, a swing of 26.5 points, and through two channels at once: the losses consume it, and the shrinking performing balance earns less of it each year, 1.60% in year five against 2.40% in year one. Cumulative losses of 24.0% have consumed the overcollateralization, all ten points of spread earned, and twelve of the twenty points of subordination; at more than three points a year, two and a half more years exhaust it. *The senior bond is on course not to survive.*

Figure 26.3 draws this table beside Example 26.2's: on the left the pass-through's price bending the wrong way, on the right the two support paths parting from the first year. Nothing in the example requires a copula, a correlation, or a view about house prices: it requires only that defaults arrive and that the spread be earned on what is left. The models that projected

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the accumulating column ran only scenarios in which it accumulated. The protection was a projection presented in a column that looked like a fact.

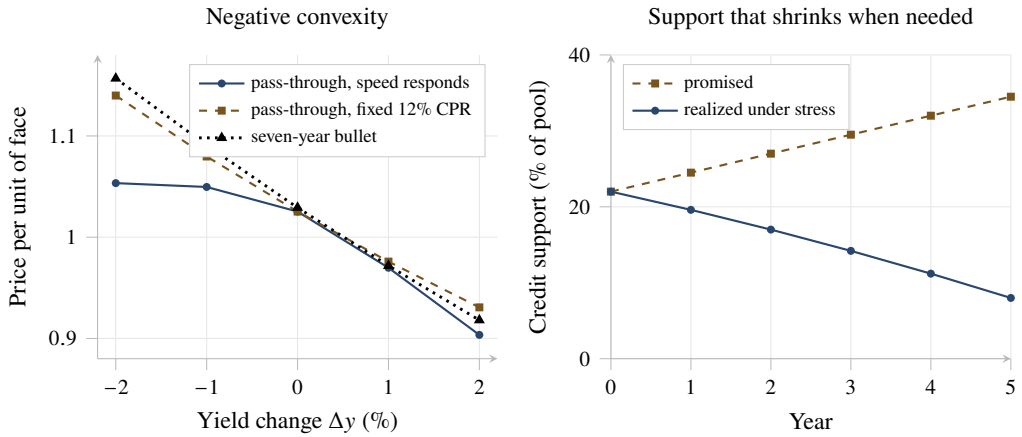


Figure 26.3: Left: the prices of Example 26.2. Right: the promised and stressed support paths of Example 26.3.

26.5 RE-SECURITIZATION

The step that turned a bad decade into a crisis was doing it twice.

Mezzanine tranches are hard to sell: too risky for senior buyers, not levered enough for equity buyers. The solution the market found was to pool them and tranche the pool.

Example 26.4 A CDO of mezzanine tranches

Take the pool of Example 26.1. At a correlation of 0.15 its tranches have expected losses:

Equity 0–3%	64.07%
Mezzanine 3–7%	20.19%
Mezzanine 7–12%	4.45%
Senior 12–30%	0.27%
Super senior 30–100%	0.00%

Now take the 3–7% mezzanine piece from one hundred *different* deals and pool those. Each has an expected loss of 20.19%. Tranche the new pool, and everything depends on the correlation *between* the mezzanine pieces. Model it as the copula applied twice, in Section 24.13's notation. Loan j of deal i has $X_{ij} = \sqrt{0.15}\,M_i + \sqrt{0.85}\,Z_{ij}$, so the loans in a deal share that deal's factor M_i at the correlation 0.15 of Example 26.1; and the hundred deal factors themselves load on one market factor M, $M_i = \sqrt{\rho}\,M + \sqrt{1-\rho}\,Z_i$, where ρ is now the inter-deal correlation. Conditional on M the hundred pieces are independent, each with the loss distribution its own tranche loss function gives it, so the new pool's loss distribution is built exactly as in Section 25.7's Step 2, with a piece's loss distribution in the place a single default occupied. The four layers of the re-securitization then lose:

Inter-deal ρ	"AAA" 30–100%	"AA" 12–30%	3–12%	Equity 0–3%
0.10	1.08%	44.13%	94.36%	99.96%
0.30	3.83%	40.89%	80.19%	97.71%
0.60	7.55%	36.64%	63.34%	86.84%
0.90	11.25%	32.18%	49.18%	69.72%

Weight the columns by width (0.70, 0.18, 0.09, 0.03) and every row adds to 20.19%, the pool's

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expected loss, as it must. The senior tranche of the re-securitization goes from losing 1.08% of its notional to 11.25% as the inter-deal correlation rises, a factor of ten, driven entirely by a parameter about which no market quoted anything.

And the parameter was not 0.10. The mezzanine tranches of one hundred subprime deals from the same vintage are exposed to one thing: national house prices. They are not a hundred independent bets on a hundred pools; they are one bet, made a hundred times, on whether house prices keep rising.

So the diversification that justified the structure did not exist. Pooling a hundred claims removes idiosyncratic risk only if the risk is idiosyncratic, and the claims being pooled had already had their idiosyncratic risk stripped out by the first securitization, which is precisely what a mezzanine tranche is. *Re-securitizing tranches pools the systematic residual, and the systematic residual does not diversify.*

26.6 THE COPULA AND WHAT IT WAS ASKED TO DO

Section 24.13's one-factor Gaussian copula was the industry standard, and it deserves a more careful verdict than it usually receives.

As a device it is unobjectionable. It is a way of writing down a joint distribution with a single correlation parameter, and it makes the loss distribution computable, which is a legitimate reason to choose a model. Section 25.7 noted that the market's own use of it, quoting tranches by the implied correlation that reproduces their prices, was self-evidently a quoting convention rather than a belief: a single-parameter model producing different parameters at different attachment points is announcing its own misspecification, exactly as the volatility smile does in Chapter 19.

Combining (26.1) with that factor model gives every tranche number in this chapter and the last, and the calculation is Steps 2 and 3 of Section 25.7: conditional on the common factor the defaults are binomial with the probability $p(M)$ of equation (24.12), and one integral over the factor and one binomial sum price the whole capital structure, which is why the model was adopted and could be run on every deal. As the pool grows, its loss fraction becomes $\ell\, p(M)$, a function of the factor alone, and the **large-pool** loss distribution is Chapter 24's worst-case default rate (24.15) read in reverse: with each issuer's default probability π, correlation ρ and loss given default $\ell = 1 - R$, the pool loses more than $\ell\,\mathrm{WCDR}(q)$ with probability exactly $1 - q$.

That is the formula underneath the Basel capital charge and, read backward, the reason a senior tranche is a statement about ρ. As $\rho \to 0$ the distribution collapses onto $\ell\pi$ and a senior tranche can never be touched; as $\rho \to 1$ it becomes a coin flip between nothing and everything. Rating such a tranche is choosing a point between those two, which is why the letter cannot be read without ρ.

Figure 24.6 drew the matching distribution of defaults for Example 26.1's pool. What is visible there is the middle draining into both tails as ρ rises, the mean pinned at five defaults, a loss of $\ell\pi = 3\%$; what the senior tranche cares about is the part the figure can barely show (the mass beyond the 12% attachment point, 21 defaults, at the right edge of its axis), which the large-pool approximation puts at 1.24% at $\rho = 0.15$, against the 1.50% that Example 26.1's exact count of a hundred loans gives. That mass is the senior tranche. Figure 26.4 draws the probability that the pool's loss exceeds x on the logarithmic scale the tail needs: left of the 12% line the loss is absorbed by the layers beneath; the height of each curve at the line is the probability the senior tranche is touched at all. The curves are nearly straight because this tail is close to exponential in x; what ρ sets is its slope.

The loss distribution is not yet a price. Section 25.7 gave the four steps from one to the other and Example 25.5 ran them on a 125-issuer index portfolio, so the calculation is not

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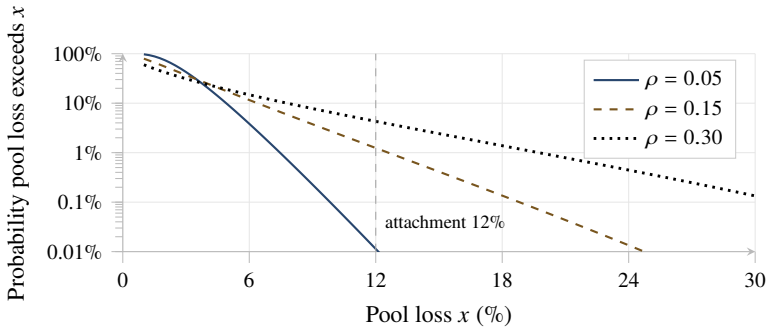


Figure 26.4: Probability that the pool's loss exceeds x, from the large-pool limit of equation (24.14), inverted as in equation (24.15), with $\pi = 5\%$, $\ell = 60\%$, evaluated every 0.25%; logarithmic scale.

repeated; only the principle is restated. The risky annuity is the present value of the notional that survives and the protection leg of the notional that does not, and where a CDS weighted by $Q(t_i)$, a tranche weights by $1 - E_{[a,b]}(t_i)$, the expected fraction still outstanding. Change the product and only the definition of "survives" changes (an issuer's default, a layer's erosion, a pool's amortization for Section 26.4's bonds), which is why the CLO of Section 26.14, whose cash flows are governed by tests rather than by a copula, is priced by projecting the same two quantities through its waterfall, not by a new theory.

Three things went wrong, and only one of them is a criticism of the model.

The Gaussian copula has no tail dependence. **Tail dependence** is the tendency of extreme outcomes to arrive together: the probability that one variable is extreme given that another is, as "extreme" is pushed ever further into the tail. Under the Gaussian copula, extreme events are asymptotically independent: conditional on one issuer defaulting catastrophically, the probability that another does converges to its unconditional value. *This is empirically false for credit*, where defaults cluster in exactly the states the model treats as independent, and it biases senior tranches, the ones that only matter in those states, in the direction of looking safe.

The correlation was calibrated to a period that contained no test. Whatever number was used came from data in which US house prices had not fallen nationally, so the estimate could not contain the event that mattered. This is not a modeling error at all; it is the sampling problem of Section 19.9's pre-1987 discussion, and no choice of copula would have repaired it.

And the output was treated as a measurement. A model that turns an assumption into a rating, run by an institution the issuer pays, produces a number whose precision is inherited from an input no one checked. This is the failure, and it is institutional, not mathematical.

The verdict is that the copula was a bad model used badly, and that the second clause carries most of the weight. A better model calibrated to the same data by the same institutions with the same incentives would have produced approximately the same ratings.

26.7 THE WARNING NO ONE READ: MAY 2005

Two years before the crisis, the correlation market produced a clean, self-contained failure that demonstrated the model's defect precisely. It is worth recounting because everything that mattered in 2007 is visible in it, at a scale small enough that no one was harmed much.

The standard trade was to be long the equity tranche of an index and short the mezzanine, in a ratio that made the package neutral to the index spread. Section 25.7's table explains the position: the equity tranche is long correlation and the mezzanine is much less so, so the package was a way to be long correlation while hedged against the credit market's direction. It carried positively, it had performed for years, and it was widely held.

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In May 2005 two large US automakers were downgraded to junk.

The shock was idiosyncratic, and the model had no room for idiosyncratic shocks. In the one-factor copula of Section 24.13, an issuer defaults because the common factor moves or because its own draw is bad, and the second channel is a residual that diversifies away in a 125-issuer index. A downgrade large enough to move two issuers' spreads violently without moving the other 123 is, in the model's terms, an event of negligible probability.

The effect on the two legs went the wrong way on both.

The equity tranche fell, because idiosyncratic widening in a few issuers is exactly what an equity tranche is exposed to. *The mezzanine did not widen to compensate*, because the hedge ratio had been computed on a model in which spreads move together: the short mezzanine leg was sized to gain from a widening that never came, and as implied correlation fell the mezzanine spread tightened instead.

The trade lost money on both legs simultaneously, which the model said could not happen. Implied correlation moved in ways no single parameter could reconcile, and the base-correlation apparatus of Section 25.7 was in part a response to the resulting inconsistencies.

Three lessons were available in 2005 and were drawn in 2008 instead.

The hedge ratio was model output, not market data. A position hedged by a model is hedged against the model's world, and Section 21.11's measurement of model risk is the same observation with a barrier option in place of a tranche.

A parameter that must be inferred from the prices it explains is not a parameter. The market was calibrating ρ to tranche prices and then using ρ to hedge tranches, which is circular in a way that produces sensible numbers right up until the relationship it assumes fails.

And crowding turned a model error into a market event. The trade was widely held in the same direction, so the unwind moved the prices being unwound—the mechanism of Chapters 22 and 23 in a market where no one had yet thought to look for it.

The episode was small, contained, and correctly diagnosed at the time. What did not happen is the transfer of that diagnosis to the much larger book of the same model being used, on much less observable portfolios, to rate instruments AAA.

26.8 RATINGS: THE SAME LETTER, DIFFERENT RISK

Section 24.10 said the rating on a structured tranche depended on a correlation assumption the letter did not disclose. Here is what that meant in practice.

A AAA corporate bond and a AAA senior tranche were notionally calibrated to the same idealized default probability; their loss distributions in bad states were entirely different.

A corporate bond's rating can be wrong by one notch. If the issuer deteriorates it becomes AA, then A, and a holder can sell into a market that reprices gradually.

A tranche's rating is closer to binary. Example 26.1 took a senior tranche from 0.27% to 15.2% expected loss on a default-probability move that raised the pool's own expected loss only fourfold, from 3% to 12%. There is no gradual path; the tranche is fine until the attachment point is threatened and then it is not.

Two structural features made this worse.

Rating triggers were embedded in contracts. Downgrades forced sales by mandate-constrained holders and triggered collateral calls, which is Section 24.10's mechanism operating on an instrument whose downgrades, when they came, came in blocks of many notches at once.

And the ratings were used as a substitute for analysis. An investor holding a AAA tranche was, in practice, holding an agency's correlation assumption. The rating was not a summary of an analysis the investor could have done themselves; it was the entire analysis, which is a different thing and was not disclosed as such.

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26.9 THE INCENTIVE CHAIN

The technical story above is sufficient to produce large losses. It is not sufficient to produce a crisis, and the additional ingredient was that no one in the chain retained the risk they created.

A mortgage was originated by a broker paid a fee per loan, sold to a bank that pooled it, tranched by an arranger paid a fee per deal, rated by an agency the arranger paid, and sold to an investor who trusted the rating. At every step the pay was for volume and the risk moved on.

Originate-to-distribute is not inherently wrong. It is the same separation of origination from risk-bearing that makes any capital market work, and Section 24.16's discussion of banks distributing credit describes the same thing functioning correctly.

What made it fail was that the distribution was not real. Banks retained **super-senior** tranches (the top layer of the capital structure, above even the AAA-rated senior notes) because they were unsellable and appeared riskless, held them in off-balance-sheet vehicles funded with short-term commercial paper, and provided liquidity backstops to those vehicles. *The risk did a full circle and came back*, and the capital framework of the time did not capture it because the exposure was contingent and the vehicle was not consolidated.

That is the mechanism Chapter 27 exists to address, and Section 26.13 records the response.

26.10 THE EUROPEAN ALTERNATIVE

Europe funded mortgages differently, and the contrast isolates the feature that did the damage.

A **covered bond** is a bank's bond secured on a pool of mortgages that *stays on its balance sheet*. The holder has **dual recourse**: a claim on the bank and, if it fails, a preferential claim on the ring-fenced pool. The pool is dynamic: the issuer must replace loans that deteriorate.

No risk is transferred at all, which is exactly the point.

Covered bonds have a very long history and a default record close to nil, through the crisis included. The reasons map one-to-one onto the failures of this chapter.

The originator keeps the credit risk, so Section 26.9's chain has no one to pass it to. An issuer that writes bad loans must replace them out of its own assets.

There is no tranching, so Example 26.1's convexity cannot arise.

And there is no re-securitization, because there are no tranches to re-pool.

The trade-off is equally clear and should be stated. *Covered bonds do not free up bank capital*, because the risk never leaves. They are a funding instrument, not a risk-transfer instrument, and a banking system financed entirely this way would lend less per unit of capital.

So the comparison is not that Europe was wiser. It is that two systems made different choices about whether risk transfer was the objective, and the system that did not attempt it did not suffer the failures that attempting it produced. Securitization's benefits and its failure modes come from the same feature, and a reader deciding what to think about the instrument should hold both halves of that sentence at once.

26.11 SYNTHETIC CDOS AND THE SUPPLY PROBLEM

Section 25.15 established the distinction. A cash CDO buys the assets; a synthetic one sells protection on a reference portfolio via the CDS of Chapter 25.

The synthetic format removed the constraint that there had to be a loan. Demand for AAA-rated yield exceeded the supply of mortgages, and a synthetic structure could reference the same pool as many times as there were counterparties willing to take the other side.

This has two consequences, and they should be separated because they are often conflated.

It multiplied the exposure without multiplying the risk of default. A hundred synthetic deals referencing the same mortgages do not make them more likely to default; they create a hundred times the notional of contracts whose value depends on whether they do.

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And it therefore multiplied the interconnection. The loss on the mortgages was bounded; the transfer of that loss through the system was not. A bounded shock propagated through an unbounded web of contracts is a different problem from a large shock, and it is why the crisis's transmission was through counterparty exposure rather than through mortgage losses alone.

A fair counter-argument runs as follows. Synthetic structures also let the risk be held by whoever wanted it, including investors who correctly thought subprime overvalued and used the format to be short. The instrument was symmetric; the understanding was not.

26.12 WHAT ACTUALLY FAILED

Case Study 26.1 The order in which things broke

The sequence matters, because the popular account compresses it and loses the mechanism.

2006–2007: the pool assumptions were wrong. Subprime delinquencies rose above anything in the calibration sample, for reasons (loosening underwriting, adjustable-rate resets, house prices flattening) visible in the data before they reached the ratings.

Mid-2007: tranches repriced nonlinearly. Example 26.1's convexity worked: bad pool losses became catastrophic tranche losses, mezzanine and re-securitized tranches first. In stylized round numbers, not the record of any one pool: if falling house prices double cumulative defaults from 15% to 30% of a pool's balance, and a foreclosed loan loses half its balance, the pool's loss doubles from 7.5% to 15%. A mezzanine tranche attaching at 8% and detaching at 20% loses nothing at 7.5% and $(15 - 8)/12 = 58\%$ of its principal at 15%.

Late 2007: the funding withdrew. The off-balance-sheet vehicles of Section 26.9 funded long-dated tranches with short-dated paper. When the paper could not be rolled, the sponsoring banks took the assets back onto their balance sheets: Chapter 8's funding-fragility problem, arriving by a route the capital framework had not anticipated.

2008: the counterparty channel opened. Institutions that had sold protection on senior tranches faced collateral calls as the marks fell. The largest case was an insurer that had written protection without capital against the mark-to-market exposure, reasoning that the tranches would never default; they did not need to, because the collateral calls on the marks sufficed.

And then valuation itself failed. With no trades, the marks came from models, and Chapter 21's model-risk warning arrived at the worst moment: institutions marking the same instrument differently, disputing collateral calls, unable to say who was solvent. Chapter 9's XVA machinery and Chapter 8's collateral standards exist because of it.

Three observations follow for a reader who was not there.

The mortgage losses alone were survivable. Realized US subprime losses were large but below the banking system's capital. The crisis was in the transmission, not the shock.

The instruments performed as written. Senior tranches largely did pay, eventually. Holders who could fund them to maturity mostly recovered. The losses were concentrated in those who had to sell, Section 5.15's limits of arbitrage at systemic scale.

And the risk had not been transferred. That is the finding that should be carried forward from this chapter, because it is the one that generalizes: a structure that appears to move risk out of a system, while the system retains the funding, the liquidity backstop, and the counterparty exposure, has not moved the risk.

26.13 WHAT WAS FIXED

More was fixed than is usually acknowledged, and a fair account matters for Section 26.16.

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Risk retention. Sponsors must retain an economic interest (typically 5%) in what they securitize. This addresses Section 26.9 directly, and its weakness is that 5% of a structure is a small fraction of the fee income from arranging it.

Consolidation and capital. Off-balance-sheet vehicles are consolidated; liquidity backstops carry capital; re-securitizations carry punitive risk weights that have made CDO-squared structures uneconomic rather than merely unfashionable.

Rating agency reform. Methodologies must be published, ratings on structured products are labeled distinctly, and many regulatory rules no longer rely on ratings. That is the most important change, since the mechanical use of ratings is what made downgrades destructive.

And the infrastructure of Chapters 2 and 8. Central clearing, initial margin on uncleared trades, and standardized collateral terms address the counterparty channel that transmitted the crisis.

The structure that failed cannot easily recur. A CDO-squared of subprime mezzanine, rated by an issuer-paid agency, held in an unconsolidated vehicle funded with commercial paper, is blocked at each link: the re-securitization by risk weights, the vehicle by consolidation, the paper funding by the capital its backstop carries, and the rating by rules that no longer use it.

What has not been fixed is the general pattern, and that is Section 26.16.

26.14 THE STRUCTURE THAT SURVIVED

A CLO is worth setting out on its own terms rather than only by contrast, because it is the member of the family a reader is most likely to meet.

The loans come first. A manager accumulates them in a **warehouse**, a credit line from the arranging bank with the future equity investors bearing the price risk; at closing the deal issues perhaps $400–500 million of notes from AAA down through BB with an unrated equity piece beneath, repays the warehouse, and completes the portfolio in a short ramp-up: typically 150–300 leveraged corporate borrowers, floating rate, senior secured. During a **reinvestment period** of about five years, principal received is reinvested rather than returned, subject to eligibility and diversity constraints written into the indenture. The manager trades within those constraints and usually holds part of the equity.

The feature that matters most is the one the 2007 structures lacked: a set of **coverage tests** that redirect cash automatically as the portfolio deteriorates. The senior **overcollateralization test** compares collateral par with the notes it supports,

$$\text{OC} = \frac{\text{collateral par}}{\text{par of this tranche and everything senior to it}} \tag{26.2}$$

and if it breaches its trigger, interest due to the equity amortizes the senior notes until it passes.

Example 26.5 The test that deleverages the deal automatically

A $400 million CLO with $256 million of AAA notes and a senior overcollateralization trigger of 120%. The opening ratio (26.2) is $400/256 = 156.25\%$.

The test fails once collateral par falls below $1.20 \times 256 = 307.2$, after $92.8 million of erosion, or 23.2% of the pool.

Suppose it does. At $300 million of par the ratio is 117.2%, a breach of 2.8 points of overcollateralization. Restoring it requires paying down the AAA by X where $300/(256-X) = 1.20$, so

$$X = 256 - \frac{300}{1.20} = 6.0$$

Six million dollars, one and a half percent of the deal, cures a breach that looked like 2.8 points. The diversion works on the *denominator*: paying down senior notes shrinks the base the ratio is measured against.

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At $280 million of par the ratio is 109.4% and the required pay-down is $22.7 million, so the cure gets rapidly more expensive as the collateral erodes.

Two consequences follow. The deal deleverages itself with no one deciding to, which Chapter 8's funding spirals show does not happen when deleveraging needs a decision under stress. And *the equity holder pays for it* (the diverted cash is the equity's distribution), so the party with the most information and the most incentive to gamble for resurrection is the one whose cash is taken first. Compare Example 26.1, where nothing stood between a parameter error and the senior tranche.

A CLO is structurally the same object as the instruments in Section 26.5 (a pool, a waterfall, a AAA senior piece), and CLOs are now the largest structured credit market. They performed well through 2008 and through 2020, and the reasons are specific enough to be diagnostic.

The collateral is not exposed to one factor. Loans to a hundred companies in many industries carry real idiosyncratic risk; the subprime mortgages of one vintage did not: Example 26.4's point. The diversification a CLO claims is real, and that is the largest difference.

There is no re-securitization. CLOs hold loans, not other deals' tranches; Section 26.5's compounding cannot start.

The liabilities are term-funded and match the assets. A CLO issues notes with maturities matched to the loan portfolio and cannot be forced to sell when spreads widen. Section 26.12's funding withdrawal has no analog—and this, rather than the credit quality, is probably the decisive difference. The instruments that failed in 2008 were largely fine on a hold-to-maturity basis; what killed their holders was having to sell.

And a manager has capital at risk. A CLO is managed within constraints, and the manager holds equity: Section 26.9's alignment problem addressed by construction, not regulation.

What is worth watching. Loan documentation has weakened over the cycle: **covenant-lite** loans, which carry no maintenance covenant (a financial ratio the borrower must keep meeting while the loan is outstanding), are now standard, so the early warnings a covenant breach used to provide arrive later. Recoveries on leveraged loans have been trending below historical averages, which Section 24.4's identification problem makes easy to underweight. And the investor base for the AAA tranches is concentrated.

26.15 THE LEVERAGED LOAN MARKET

The previous section described the structure and named its collateral in passing: leveraged corporate borrowers, floating rate, senior secured. That collateral is a market in its own right, and the two can no longer be understood separately: collateralized loan obligations are the largest single buyer of leveraged loans, and leveraged loans are very nearly the whole asset side of collateralized loan obligations.

The scale is the first thing to state. The US institutional leveraged loan market stood at about **$1.55 trillion** at the end of 2025 on the Morningstar LSTA index, more than three times the $465 billion it measured at the end of 2010, with Europe at more than €300 billion; collateralized loan obligations own about **two-thirds** of the US total. A market that was a bank funding channel in 2005 has therefore grown past the US high-yield bond market, which stood at about $1.48 trillion of par, and it did so while the credit derivatives market of Chapter 25 was shrinking by an order of magnitude. Any statement about where corporate credit risk is traded has to account for both movements.

Why it is a credit market and not a loan book. A leveraged loan is originated by a bank and then distributed. The arranging bank underwrites the whole facility, syndicates it to institutional investors, and retains little or none of it, so the instrument is priced by a market rather than held to maturity by its originator. It trades in a secondary market on standardized documentation, it

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is quoted daily, it settles through an assignment process, and index providers publish total-return indices on it. The economics are those of a floating-rate senior secured bond with a call at par, and the borrower is typically a company owned by a private equity sponsor, which is what supplies the leverage the name refers to.

Who is on each side, and why. The borrower is a sponsor-owned company financing an acquisition or a recapitalization, and it wants a floating-rate instrument it can prepay without penalty, because the sponsor intends to exit. The lender is a collateralized loan obligation buying assets for a portfolio it has already sold liabilities against, a loan mutual fund or exchange-traded fund offering retail investors a floating-rate credit exposure, or a separately managed account. The arranging bank wants the fee and not the asset. Each of those preferences is visible in the instrument: the prepayment option at par exists because the borrower wanted it, the floating rate exists because the largest buyer funds itself floating, and the covenant package is light because the arranger must clear the deal in a competitive market.

That last point is where the interesting arithmetic is. The previous section observed that recoveries on leveraged loans have been running below their historical averages, and the natural reading, that the borrowers have become weaker, is not the main explanation. Example 26.6 separates what changed from what did not.

Example 26.6 Why a senior claim recovers less without the borrower getting worse

Three capital structures on the same company. Enterprise value at origination $1,200 million and total debt $1,000 million in each; only the split between first-lien debt (the senior secured loan, with the first claim on the collateral) and junior debt differs.

Structure	First lien	Junior	Decline that first impairs the first lien
Traditional	400	600	66.7%
Intermediate	700	300	41.7%
Loan-only	**1,000**	**0**	**16.7%**

A first lien is impaired when enterprise value falls below its own par, so what protects it is everything beneath it, junior debt and equity alike, measured against its own size, and historically the senior claim was a small slice of the capital structure. With total debt fixed, shrinking the junior debt to zero makes the first lien the whole of the debt and moves the enterprise-value decline that first touches the first lien from **66.7% to 16.7%**.

Now apply one distress outcome to all three. Enterprise value falls to $700 million, a decline of 41.7%:

Structure	First-lien recovery	Junior recovery
Traditional	100.0%	50.0%
Intermediate	100.0%	0.0%
Loan-only	**70.0%**	—

The same enterprise outcome pays the first lien 100% or 70% according to how much debt sits beneath it, and nothing about the borrower differs between the three columns. An average first-lien recovery computed across a market that has shifted from the first row to the third will fall even if every borrower is identical and every enterprise outcome is identical.

Two further mechanisms push the loan-only figure down from there, and they compound rather than substitute.

Covenant-lite defers the event. A maintenance covenant forces a restructuring while enterprise value is still comparatively high; without one, the borrower continues until it cannot pay, and the recovery is measured on a smaller enterprise. A further 10% of erosion before the event takes the 70.0% to **63.0%**.

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A liability-management exercise moves the collateral. Loan documents that permit transfers to unrestricted subsidiaries (affiliates outside the lenders' collateral and covenants), or that allow new money to prime the existing lenders (rank ahead of them), let a distressed borrower and some of its lenders act against the rest. If 20% of enterprise value leaves the collateral package, the recovery falls to **50.4%**.

A claim can therefore be senior, secured, and first in the waterfall, and still recover half, without any of the three words having become false. Section 24.4's point about recovery rates was that they are the least reliable input in credit modeling; this is the mechanism behind that unreliability, and it is a documentation variable rather than a credit variable.

What this does to the structure above it. A collateralized loan obligation built on loan-only borrowers is holding collateral whose recovery distribution is worse and wider than the historical series its ratings were calibrated on, which is Section 26.8's problem arriving by a new route: the letter is the same and the loss behind it is not. The coverage tests of Example 26.5 still work (they are measured on par rather than on recovery, and they still divert cash automatically), but they trigger later relative to the eventual loss than they would have on a traditional capital structure. Nothing here undoes the four structural defenses the previous section set out; it sharpens one of the weaknesses that section named, the falling recoveries.

26.16 WHERE THE SAME STRUCTURE IS NOW

Two current markets share enough of the pattern to be worth examining, and the point of examining them is not to predict a crisis but to ask which of the specific failures each shares.

Significant risk transfer. Banks buy protection on the first-loss piece of their own loans, retaining the senior exposure, to reduce regulatory capital; the market has grown fast.

What is better. The bank keeps the loans and the senior tranche, so Section 26.9's incentive problem inverts: in 2007 the originator kept none of what it made, while here it keeps every loss beyond the first-loss piece it sold, and so most of the exposure to its own underwriting. Protection is usually funded, so Section 25.15's counterparty problem is largely absent, and the loans are the bank's own, which it underwrote and can monitor.

What is the same. The capital relief depends on a supervisory judgment that the transfer is genuine, and the history of this chapter is a history of transfers that appeared genuine and were not. And the buyers are concentrated: a few specialist funds, often themselves levered.

Private credit. Section 24.16 raised it and this is the natural place to finish the argument.

What is better, and it is substantial. The institutional holders are closed-end funds with locked-up capital that cannot be run on, so there is no maturity transformation of the kind that broke the vehicles in Section 26.12. The retail "semi-liquid" vehicles are the exception: they promise periodic redemptions, usually capped at 5% of net asset value a quarter, and several had to ration them in 2026. The loans are held whole, not tranched, so Example 26.1's convexity is absent.

What is worth watching. Valuations are infrequent and produced by the holder, so the marks that would signal deterioration may not exist. That is not the same as the risk being absent, and Section 26.12 records that the last time marks became unreliable was the moment the crisis became unmanageable. Leverage has entered at the fund level. And the banks that no longer make these loans often finance the funds that do, which rebuilds the exposure at one remove.

Neither is a repeat. Both differ from the 2007 structure in ways that address its failures; both make the risk's location harder to observe. The chapter's lesson is not that any structure is dangerous but that a system loses the ability to see a risk before it loses the ability to bear it.

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26.17 WHERE THIS CAME FROM

Portfolio credit went wrong step by step, each step defensible.

Vasicek's one-factor loan-portfolio model (published 2002, circulated far earlier) underlies Basel's capital formula and is sound within its assumptions. Li (2000) then applied a Gaussian copula to default times, and the tranche market was built on it. Andersen and Sidenius (2004) had shown how to relax the assumptions that mattered (random recovery and random factor loadings), and Hull and White (2004) had given a valuation without simulation. Coval, Jurek and Stafford (2009) stated this chapter's convexity argument formally, once the losses had arrived. Gorton and Metrick (2012) supplied the funding-run account Section 26.12 rests on.

SUMMARY

Securitization pools loans and ranks claims on them, making illiquid assets tradeable and matching risk to appetite. Plain mortgage-backed securities worked through the crisis and work now; what failed was what was built on top.

A tranche is a call spread on the pool's loss, so it is an option, and options are convex in their parameters: raising the pool's default probability raises the pool's expected loss in proportion and multiplies the senior tranche's many times over. Two unobservable inputs, each amplified that strongly, are not a basis for a AAA rating. Agency mortgage-backed securities carry pure prepayment risk, harder to model than a Gaussian copula, and they worked. The difference is not difficulty but whether the key input can be checked: prepayment speeds are observed monthly, default correlation never.

Re-securitization compounded it. Re-tranching mezzanine tranches produced a "AAA" piece whose expected loss depended almost entirely on the correlation between deals, which no one could measure—and the subprime mezzanines of one vintage are one bet on national house prices.

The Gaussian copula has no tail dependence, but the failure was institutional: a model calibrated to a sample with no national house-price decline, its output treated as a measurement by issuer-paid institutions. May 2005 had staged the rehearsal. The risk that was distributed came back: banks kept super-senior tranches in unconsolidated vehicles funded with paper they backstopped. Mortgage losses were survivable; the crisis was in the transmission.

Collateralized loan obligations are the same object and performed through 2008 and 2020: diversified collateral, whole loans, term funding and a manager holding equity remove four of the crisis's five mechanisms, and coverage tests deleverage the deal at the equity's expense. The loans beneath a CLO are a market of their own, and a senior secured claim there can recover far less than its history suggests without any of its three adjectives having become false, because recovery depends on how much debt sits beneath it, a documentation variable. Significant risk transfer and private credit share parts of it, and what they share is a risk whose location is harder to observe. A system loses the ability to see a risk before it loses the ability to bear it.

FURTHER READING

The Financial Crisis Inquiry Commission's report is the public account of record and its dissents are worth reading alongside it, since the disagreements are about causation, not facts. Gorton's work on the repo run treats Section 26.12's funding mechanism and corrects the impression that the crisis was mainly about mortgage losses. On the copula, Li's original paper is short and makes none of the claims its critics attribute to it. The May 2005 episode of Section 26.7 is set out as it happened in Belsham, Vause and Wells's Bank of England *Financial Stability Review* article on credit correlation, a diagnosis two years early. Coval, Jurek and Stafford's analysis of structured finance is the formal statement of Example 26.1's convexity argument. On what was fixed, the Basel documents on securitization capital are specific, and the right source for

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judging whether a current structure shares the defects. On private credit and significant risk transfer, the supervisory literature is where the structures and their risks are documented, and both markets are moving.

PRACTICE QUESTIONS

26.1 State three legitimate purposes of securitization, and explain why the failure of 2007–2009 does not condemn all of them.

26.2 Explain why a tranche is a call spread on the pool's loss, identify the strikes, and use Example 26.1 to state the factor by which the pool's and the senior tranche's expected losses each rise when the default probability triples.

26.3 Explain why an agency mortgage pass-through has negative convexity, identify the option and who holds it, and use Example 26.2 to give the effective convexity of the pass-through and of the same bond at a fixed prepayment speed.

26.4 Explain why a mortgage investor hedging to a duration target buys duration as rates fall, and relate this to Section 18.12.

26.5 Name the three sources of credit support in a securitization, explain which is consumed first, and explain why the accumulating-support calculation of Example 26.3 overstates protection in exactly the scenario where protection is needed.

26.6 Explain why mezzanine tranches were pooled and re-tranched rather than sold directly, why doing so did not diversify, and use Example 26.4 to state the range of the "AAA" tranche's expected loss across inter-deal correlations.

26.7 Explain what tail dependence means, why the Gaussian copula's lack of it matters specifically for senior tranches, and why calibrating the correlation to pre-2006 data could not have captured the event that mattered.

26.8 Explain why a AAA tranche's rating is closer to binary than a AAA corporate bond's, and why rating triggers made downgrades destructive rather than merely informative.

26.9 Explain why the May 2005 episode caused losses on both legs of a trade that was supposed to be hedged, and why inferring a correlation from tranche prices and then using it to hedge tranches is circular.

26.10 Describe the originate-to-distribute chain, identify at which point the risk was supposed to stop, and explain how the risk that banks appeared to have distributed returned to their balance sheets.

26.11 Explain why a synthetic CDO multiplies notional without multiplying the underlying default risk, and what this does to interconnection.

26.12 State the sequence in which the crisis unfolded, identify the step at which mortgage losses stopped being the binding problem, and explain why senior tranches largely paid in the end and yet produced enormous losses.

26.13 State four respects in which a CLO differs from a 2007 subprime CDO, say which of them is probably decisive, and use Example 26.6 to explain what is nonetheless worth watching in its collateral.

26.14 Explain what dual recourse means in a covered bond, why it removes the incentive problem, why covered bonds free up no bank capital, and what that implies about lending capacity.

26.15 Explain why the incentive problem in significant risk transfer is inverted relative to 2007,

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and state one respect in which private credit is safer than the pre-crisis structure and one that warrants monitoring.

FURTHER QUESTIONS

26.16 Example 26.1 shows a senior tranche's expected loss rising twenty-eight-fold when the pool's rises threefold. Derive the elasticity of a tranche's expected loss with respect to the pool's default probability, show how it depends on the attachment point, and determine where in the capital structure the elasticity is largest.

26.17 Example 26.4 argues that re-securitization pools the systematic residual. Formalize this: decompose a mezzanine tranche's loss into systematic and idiosyncratic components, show what pooling does to each, and derive the condition under which re-securitization adds diversification rather than concentration.

26.18 Section 26.6 argues that the copula was a bad model used badly, with the second clause carrying most of the weight. Construct the counter-argument that a model with tail dependence would have produced materially different ratings; identify what data would settle the question; and state what your answer implies about the value of model improvement relative to institutional reform.

26.19 Section 26.13 says risk retention addresses the incentive problem but that 5% is small relative to arranging fees. Model the arranger's decision, derive the retention level at which incentives are aligned, and discuss why regulators chose a number below it.

26.20 Section 26.11 states a fair counter-argument that synthetic structures let those who understood the risk be short. Evaluate whether the existence of informed short sellers made the market more or less efficient in this case, and reconcile your answer with the fact that prices did not adjust until the losses arrived.

26.21 Section 26.16 claims that a system loses the ability to see a risk before it loses the ability to bear it. Apply this to private credit: specify what would have to be observed to detect deterioration early, explain why infrequent holder-produced valuations obstruct it, and propose a disclosure regime that would restore observability without forcing mark-to-market on holders who need not sell.

26.22 Compare the 2007–2009 episode with the three episodes of Chapters 18, 22 and 23 along the dimensions of crowding, mechanical hedging, leverage, funding fragility, and observability. Identify which features are common to all four, which are unique to the crisis, and use the comparison to state what a supervisor should monitor that is not specific to any instrument.

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Chapter 27
Counterparty Credit Risk and Capital

Chapter 9 priced counterparty risk. This chapter capitalizes it, and the two are not the same activity.

CVA is a price: what the risk is worth, computed with risk-neutral probabilities and hedgeable in principle. Regulatory capital is a buffer: what must be held against the possibility that the price was wrong. *Capital in this chapter always means regulatory capital*: equity and other loss-absorbing instruments a bank is required to hold, set by the Basel framework of Section 9.7 and written into national rules by each jurisdiction's regulator, which is why the same standard binds differently in Europe, the United States, and China. Section 9.7 flagged that a firm can reduce one without reducing the other, and the reason is that they are computed by different rules for different purposes: one by the firm's own model against market prices, the other by a standardized formula against a supervisory scenario.

The chapter runs from the exposure to the framework that measures it, and then to what the framework misses. Sections 27.1 and 27.2 are the object itself and the legal clause that reduces it most. Sections 27.3 to 27.5 are the standardized approach and the capital it feeds, and Sections 27.6 to 27.9 the charges around it, including wrong-way risk. Sections 27.10 to 27.12 are the leverage backstop and a bank's whole requirement. Sections 27.13 to 27.17 are the limit: what a bilateral framework cannot see, and who bears it.

Section 7.10 said residual counterparty risk is capitalized under this framework and a 2021 episode showed how quickly it returns; Section 7.12 said total return swaps are the instrument through which leverage accumulates quietly and pointed here. Section 8.7 deferred wrong-way risk in general. Section 27.14 is where all three arrive together.

27.1 WHY IT NEEDS ITS OWN FRAMEWORK

A loan's exposure is known: it is the amount lent. A derivative's is not.

The exposure is an option on the market. A swap that is worth nothing today may be worth $50 million in two years, and the counterparty's failure matters only in the states where it is worth something to you. Section 9.1 made this point and Figure 9.1 drew the profile; the consequence for capital is that the amount at risk must be modeled rather than looked up.

Three features follow, and each requires machinery a loan book does not need.

It is bilateral. Either party may end up owing the other, so exposure can change sign, and both parties hold credit risk on the same contract at different times.

It is portfolio-dependent. The exposure to a counterparty is a property of the whole netting set, not of any trade in it. A new trade can reduce total exposure, which no loan ever does.

And it is mitigated by mechanisms that can themselves fail. Netting depends on enforceability, collateral on operational performance and on the value of what is posted, which Chapters 1 and 8 addressed and which the capital framework largely assumes will work.

Figure 27.1 draws that option over the life of a swap, as the expected exposure and the

95th-percentile potential future exposure (PFE) that Section 9.1 defined. Section 9.1 explained the hump: per unit of notional the PFE is about $(T - t) \times 1.645\,\sigma\sqrt{t}$ at a normal rate volatility σ, and the expected exposure is the same product with 0.40 in place of 1.645, so diffusion and the shrinking number of payments peak against each other about a third of the way through. The lowest curve is the same swap under daily variation margin with a ten-day margin period of risk. Margin settles the mark-to-market each day, so the only move that can accumulate is the one over the ten days it takes to close out a defaulted counterparty: the curve replaces $\sqrt{t}$ by $\sqrt{10/250}$, which is why it has no hump and simply declines with the payments left. It is also positive even at inception, when the swap is worth nothing and no margin is posted: a counterparty that fails on the first day still leaves ten days of moves uncovered. [TN27] derives all three curves. The gap between the margined curve and the others is what the maturity factor of Section 27.3 is about.

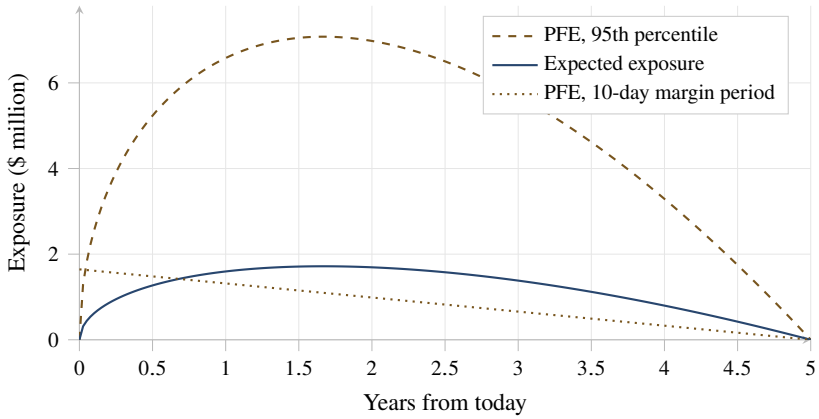


Figure 27.1: Exposure on a five-year par swap, $100 million notional, at a normal rate volatility of 100 basis points a year. Illustrative, not market data.

27.2 WHAT NETTING IS WORTH

Before any formula, Example 27.1 measures the single largest exposure reducer in the framework.

Example 27.1 The netting benefit

Five trades with one counterparty, current mark-to-market values in millions:

$$+40, \quad -25, \quad +15, \quad -30, \quad +10$$

Without an enforceable netting agreement, the exposure is the sum of the positive values: the counterparty's administrator would enforce the trades you owe on and pay only a recovery fraction on the trades owed to you. That is Chapter 1's cherry-picking, and it gives

$$40 + 15 + 10 = \$65 \text{ million}$$

With netting, the exposure is the net value:

$$\max(40 - 25 + 15 - 30 + 10,\ 0) = \$10 \text{ million}$$

An 85% reduction, from a legal clause rather than a hedge—and that is only today's exposure. The formula of Section 27.3 adds a second component for how much worse it could become before a default is closed out, and netting reduces that too, because offsetting trades partly cancel before it is computed.

This is why Section 1.5 called the enforceability of close-out netting a precondition for a derivatives market existing at all. The whole framework assumes netting is enforceable, and where

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it is not (no close-out netting statute, or a counterparty type carved out), exposure and capital revert to the gross figure. That is why a jurisdiction's **netting opinion** (the legal opinion, commissioned by the industry for each jurisdiction, that close-out netting will be upheld in a counterparty's insolvency there) is worth more to a dealer than any risk model.

27.3 SA-CCR, AND WHAT MARGINING BUYS

The **standardized approach for counterparty credit risk** (SA-CCR) computes exposure at default (EAD), the amount the capital rules treat as lent to the counterparty, for each netting set (Chapter 8), the trades with one counterparty that a single enforceable netting agreement covers:

$$\text{EAD} = \alpha \times (\text{RC} + \text{PFE}), \qquad \alpha = 1.4 \tag{27.1}$$

The two components are computed as follows; the scalar α is Section 27.5's subject.

Replacement cost. RC is what the counterparty's default would cost today. Write V for the current market value of the netting set to the bank, the sum of its trades' values, and C for the net collateral the bank holds against it after haircuts. Without a margin agreement,

$$\text{RC} = \max(V - C,\ 0) \tag{27.2}$$

With one, RC is the largest exposure that would *not* trigger a margin call:

$$\text{RC} = \max(V - C,\ \text{TH} + \text{MTA} - \text{NICA},\ 0) \tag{27.3}$$

where TH is the threshold below which no variation margin is called, MTA the minimum transfer amount, and NICA the net independent collateral amount (initial margin received less initial margin posted). The middle term is there because a margin agreement does not hold the exposure at zero: no call is made below the threshold, and then only above the minimum transfer, so the exposure can sit at TH + MTA with the counterparty in full compliance, and only initial margin in hand offsets that (Figure 27.2, left). With zero threshold and minimum transfer amount and no initial margin, the term vanishes and RC is the uncollateralized mark-to-market as before.

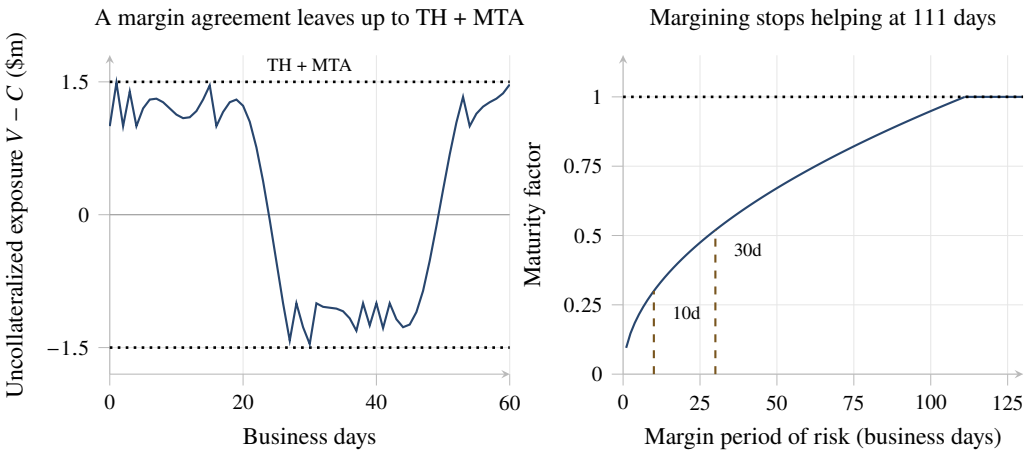


Figure 27.2: Left: uncollateralized exposure under a margin agreement (TH = \$1m, MTA = \$0.5m), illustrative path; each drop is a margin call. Right: equation (27.7), capped at one.

Potential future exposure. PFE is a supervisory estimate of how much worse the exposure

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could become before a default is closed out:

$$\text{PFE} = \text{multiplier} \times \sum_{\text{asset classes}} \text{AddOn} \tag{27.4}$$

The **add-on** for an asset class stands in for how far that class's positions can move over the horizon that matters. Within interest rates, the class Example 27.2 uses, trades are grouped into **hedging sets**, one per currency, and within a hedging set into three maturity buckets: under one year, one to five years, over five years. Each trade i contributes an **effective notional**

$$D_i = \delta_i \times d_i \times \text{MF}_i \times N_i \tag{27.5}$$

where N_i is the notional; δ_i is the **supervisory delta**, $+1$ for a position that gains when rates rise (paying fixed), -1 for one that loses, or a Black–Scholes delta for an option; d_i is the **supervisory duration**, the present value at a fixed 5% of one unit a year from the trade's start date t_i to its maturity T_i,

$$d_i = \frac{e^{-0.05 t_i} - e^{-0.05 T_i}}{0.05} \tag{27.6}$$

and MF_i is the **maturity factor**, which scales the position to the horizon over which it is exposed:

$$\text{MF} = \sqrt{\min(M, 1)} \ \text{ unmargined}, \qquad \text{MF} = 1.5\sqrt{\text{MPOR}/250} \ \text{ margined} \tag{27.7}$$

with M the remaining maturity in years and MPOR the margin period of risk in business days, the time assumed to pass between the last margin exchange and close-out: ten days for a daily bilateral agreement, five for client-cleared trades, twenty above 5,000 trades or with illiquid collateral, and double after repeated margin disputes. The 250 is the framework's business days in a year, so MPOR/250 is that window in years.

The product is not a notional in any economic sense. The first three factors are the trade's sensitivity to the rate ($\delta_i d_i N_i$ is its signed annuity, so Example 27.2's $D = 442.40$ million says the swap gains or loses \$44,240 per basis point), and MF_i carries the square root of the horizon over which that sensitivity is exposed. The volatility that turns the two into money sits in the supervisory factor below. So D_i is the amount of a one-year reference trade carrying the same exposure, which is what *effective* means in the name.

Effective notionals for the three maturity buckets $k = 1, 2, 3$ are summed within each, $D_k = \sum_{i \in k} D_i$, so offsetting trades cancel, and the buckets then combine with partial offset into the hedging set's effective notional EN:

$$\text{EN} = \sqrt{D_1^2 + D_2^2 + D_3^2 + 2 \times 0.7\, D_1 D_2 + 2 \times 0.7\, D_2 D_3 + 2 \times 0.3\, D_1 D_3} \tag{27.8}$$

Equation (27.8) is the standard deviation of a sum of correlated normal variables, with correlations of 0.7 between adjacent buckets and 0.3 between the outer two, and those correlations are what *partial* means. Take a payer of $D_2 = +100$ in the one-to-five-year bucket against a receiver of $D_3 = -100$ in the over-five-year bucket. Then (27.8) returns $\sqrt{100^2 + 100^2 - 2 \times 0.7 \times 100 \times 100} = 77$, not the 0 of a perfect hedge or the 141 of none: the position counts as mostly, not wholly, offset. Within a bucket the offset is complete, because there the D_i are added before anything is squared.

The hedging set's add-on is then

$$\text{AddOn}_{\text{hedging set}} = \text{SF} \times \text{EN} \tag{27.9}$$

with the **supervisory factor** SF = 0.50% for interest rates, and the asset class's add-on is the sum over its hedging sets. The other classes follow the same pattern with their own factors and hedging sets (4% per currency pair for foreign exchange, 32% for a single stock with partial

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offset across issuers for equity), and the Basel standard has the details. The approach covers derivatives of every kind, cleared as well as bilateral (which is what lets Example 27.3 put three regimes in one table, since clearing changes the risk weight and not the measure), but not securities financing.

The **multiplier** in (27.4) recognizes collateral in excess of the current exposure, or a netting set that is out of the money, both of which cushion future moves:

$$\text{multiplier} = \min\left\{1,\ 0.05 + 0.95\,\exp\left(\frac{V - C}{2 \times 0.95 \times \sum \text{AddOn}}\right)\right\} \tag{27.10}$$

It equals one whenever $V \geq C$, so for a netting set at or in the money the PFE is simply the sum of the add-ons, and it falls toward a floor of 5% as $C - V$ grows relative to the add-on.

None of this is arbitrary. The formula is a closed-form approximation of the internal-model quantity of Section 27.5: underneath it the netting set's value is normal with no drift, as in the model behind Figure 27.1, and the target is the effective expected positive exposure over one year (an average of the expected exposure over that year, defined precisely in Section 27.5), which α scales as it scales a bank's own model. Each piece corresponds:

- the supervisory factor is the one-year expected positive exposure of a unit at-the-money position at a stressed *normal* volatility, a move in rate rather than a percentage return (0.50% for rates corresponds to about 190 basis points a year);
- the maturity factor is the average of $\sqrt{t}$ over the horizon, discussed below;
- the bucket correlations behind (27.8) are a supervisory judgment that maturities move together but not identically;
- the multiplier is a floored exponential fitted to the exact normal expectation for a position that starts out of the money.

[TN27] works each correspondence through.

Example 27.2 A single swap, and what margin does to it

A five-year interest rate swap, \$100 million notional, in which the bank pays fixed. It is currently at zero mark-to-market, with no collateral and no margin agreement.

Replacement cost. $V = 0$ and $C = 0$, so by (27.2) RC = 0.

Effective notional. Paying fixed gains when rates rise, so $\delta = +1$. The trade starts now and runs five years, so $t = 0$ and $T = 5$, and by (27.6)

$$d = \frac{1 - e^{-0.25}}{0.05} = \frac{1 - 0.7788}{0.05} = 4.4240$$

Unmargined with five years remaining, MF $= \sqrt{\min(5, 1)} = 1$, so $D = 1 \times 4.4240 \times 1 \times 100 =$ \$442.40 million. It is the only trade in its hedging set and sits in the one-to-five-year bucket, so EN = 442.40.

Add-on, PFE, and EAD. AddOn $= 0.50\% \times 442.40 =$ \$2.212 million. With $V - C = 0$ the multiplier is one, so PFE = 2.212 and

$$\text{EAD} = 1.4 \times (0 + 2.212) = \$3.10 \text{ million}$$

In the money. If the swap were instead worth \$5 million to the bank, RC = 5, the multiplier stays at one, and EAD $= 1.4 \times (5 + 2.212) =$ \$10.10 million.

Out of the money. If it were worth −\$5 million, RC = 0 and the multiplier does its work:

$$\text{multiplier} = 0.05 + 0.95\,e^{-5/(2\times0.95\times2.212)} = 0.05 + 0.95 \times 0.3043 = 0.339$$

so PFE $= 0.339 \times 2.212 =$ \$0.75 million and EAD $= 1.4 \times 0.75 =$ \$1.05 million. A swap the

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bank owes money on still carries exposure, because rates can move more than $5 million in its favor, but a third of the at-the-money amount.

Now add a margin agreement with daily exchange, zero threshold and minimum transfer amount, and a ten-day margin period of risk. At zero mark-to-market (27.3) still gives RC = 0, and the maturity factor becomes MF = $1.5\sqrt{10/250} = 0.30$, so $D = 0.30 \times 442.40 = 132.72$, the add-on falls to $0.50\% \times 132.72 = \$0.664$ million, and

$$\text{EAD} = 1.4 \times 0.664 = \$0.93 \text{ million}$$

A 70% reduction in the add-on, purely from margining. This is the framework's central incentive and it is deliberate: the margin period of risk is the only window in which an unhedged move can hurt, so shortening it shortens the exposure. Chapters 2 and 8 described clearing and margin as regulatory mandates; this example is the economic reason a dealer would want them anyway. Capital, not virtue, is what moved the market.

Example 27.2's 70% reduction came from the maturity factor alone, and (27.7) contains a limit that is not obvious. The margined factor is capped at the unmargined value of one, and setting $1.5\sqrt{\text{MPOR}/250} = 1$ gives $\text{MPOR} = 250/1.5^2 = 111$ business days: beyond about five months of close-out risk, margining provides no add-on relief at all. The formula's logic is that exposure grows with the square root of time, so a shorter close-out window means a smaller possible move. But past 111 days the window is as long as the unmargined horizon and there is nothing left to shorten.

The 1.5 is a ratio of two averages. The supervisory factors are calibrated to an unmargined exposure averaged over one year, and an exposure growing as $\sqrt{t}$ averages to two-thirds of its one-year value, since $\int_0^1 \sqrt{t}\, dt = 2/3$; a margined exposure is the same ten-day move at every date, so it averages to the full $\sqrt{\text{MPOR}/250}$. Dividing by $2/3$ puts the two on one scale, and $1/(2/3) = 1.5$.

The practical content is in the middle of the curve. A position that genuinely takes thirty days to unwind (a concentrated equity book, an illiquid credit portfolio) keeps 52% of the unmargined add-on even under a daily margin agreement. The right panel of Figure 27.2 shows where the relief comes from and how quickly it runs out.

And that is the assumption Section 27.14 breaks. The regulatory ten days is a claim about liquidity, not the calendar; a dealer that applies it to a position worth many days of trading volume has used an input it did not verify, which is the failure mode of Chapter 21 arriving in a capital formula rather than a pricing model.

27.4 CAPITAL UNDER THREE REGIMES

EAD is not capital. Multiplied by the counterparty's risk weight RW it gives **risk-weighted assets** (RWA), and those times the required capital ratio CR give the capital the trade consumes:

$$\text{RWA} = \text{EAD} \times \text{RW}, \qquad \text{capital} = \text{RWA} \times \text{CR} = \text{EAD} \times \text{RW} \times \text{CR} \tag{27.11}$$

Neither factor is the bank's to choose. The supervisory risk weights are prescribed by counterparty type:

- 0% for a high-grade sovereign;
- 2% for a trade against a **qualifying CCP** (one licensed and supervised under rules consistent with the international standards for clearing houses);
- 20% for a highly rated bank;
- 100% for a typical unrated corporate;

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- more for a speculative-grade one.

CR is the minimum ratio Section 1.12 described, 8% plus the buffers a jurisdiction imposes. A bank with model approval estimates some risk weights itself, within supervisory floors, which is the internal-ratings route of Section 39.10.

Put the steps together and the framework's effect on market structure becomes a single table.

Example 27.3 Why everything moved to clearing

Example 27.2's swap at zero mark-to-market, against a corporate at a 100% risk weight, with a 10.5% capital requirement (the 8% minimum plus the 2.5% capital conservation buffer, a layer a bank may draw down under stress only at the price of limits on its dividends). The EAD is the $3.10 million computed there, carried to a third decimal so that the capital column can be checked.

	EAD	Risk wt.	RWA	Capital
Uncleared, unmargined	$3.097m	100%	$3.097m	$0.3252m
Uncleared, margined (10d)	$0.929m	100%	$0.929m	$0.0975m
Cleared at a qualifying CCP (5d)	$0.657m	**2%**	$0.013m	$0.00138m

The ratio is 236 : 71 : 1: clearing costs 0.4% of the capital of leaving the trade uncleared and unmargined.

Two changes drive it, and they compound. Clearing shortens the margin period of risk to five days, so MF = $1.5\sqrt{5/250} = 0.21$ and EAD falls 79% to $0.657m; the 2% weight for trade exposures against a qualifying CCP is then a further factor of fifty.

Turning capital into a running charge at a 10% hurdle rate (Section 9.7's KVA calculation) gives the number a client actually sees:

	bp per year of notional
Uncleared, unmargined	3.25
Uncleared, margined	0.98
Cleared	0.01

Three basis points a year sounds small, and on a swap that trades at a fraction of a basis point of bid–ask it is decisive. A dealer quoting the same swap to a cleared and an uncleared client is quoting from different cost bases, and the gap is larger than the spread.

This is the answer to a question Chapter 2 raised and did not settle. The clearing mandate covered standardized products; it did not force the rest of the market to clear voluntarily, and much of it did anyway. *The table is why.* Capital rules did more to move the market than the mandate did.

The table omits one qualification. A clearing member also holds capital against its default fund contribution, which the 2% line above excludes. The charge is set for the member, not the trade: it is the member's share, in proportion to its contribution, of the clearing house's own hypothetical capital requirement, computed as though the clearing house were a bank, and it can be no lower than that of a 2% risk weight on the contribution. For an illustrative $100 million contribution the floor alone is $2\% \times 100 \times 10.5\% = \$0.21$ million; spread over 10,000 cleared swaps like this one it adds $21 to each, against the $96,000 that clearing saves on each margined uncleared swap. The risk-sensitive charge is larger than its floor, but it would have to be thousands of times larger to change the ordering. It is the price of the mutualization of Chapter 2, a charge against a loss caused by somebody else's counterparty.

27.5 THE MEANING OF $\alpha = 1.4$

The multiplier in (27.1) is the most misread number in the framework. It looks like a measurement—exposures are 40% larger than they appear—and it is not one.

Under the internal model method, a bank computes an exposure profile by simulation, reduces it to **effective expected positive exposure** (the running maximum of expected exposure,

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averaged over the first year) and treats the result as a loan-equivalent amount. But a loan equivalent computed this way understates the capital a full portfolio credit model would require, because it ignores the correlation between exposures and defaults across counterparties and the granularity of the portfolio: a book concentrated in a few large counterparties is less diversified than the loan formula assumes.

α is the supervisory bridge over that gap. Its 1.4 is a calibration from studies of representative portfolios, with a floor of 1.2 for banks permitted to estimate their own.

So the 40% uplift is a buffer for model error, not a property of any trade; reading it as a real 40% higher exposure leads a desk to try to hedge something that does not exist.

A bank with supervisory approval keeps the internal model method because it simulates the actual netting set, margin terms and collateral, where SA-CCR applies prescribed factors. The benefit has a limit. Under the output floor, a bank on the internal model method computes counterparty exposure under SA-CCR as well, for the standardized base of the floor, so its total risk-weighted assets cannot fall below 72.5% of what that base gives.

27.6 THE CAPITAL CHARGE FOR CVA

A separate charge covers the risk that CVA itself moves: that a counterparty's credit spread widens, marking down the value of the derivative book, without anyone defaulting. The Basel Committee's 2011 announcement of the charge put roughly two-thirds of the crisis's counterparty credit losses in these mark-to-market movements, which the earlier framework had not capitalized. Section 9.7 explained why the charge exists and why an action that lowers the CVA price need not lower it.

The framework offers two approaches, and a simpler option for small banks. The **basic approach** (BA-CVA) is the default: a formula on exposures and on supervisory risk weights set by the counterparty's sector and rating. A bank whose non-centrally-cleared derivatives fall below a notional of €100 billion may instead set the charge equal to its counterparty credit capital. The **standardized approach** for CVA (SA-CVA) is a sensitivity-based calculation, delta and vega against credit spreads and the market factors driving the exposure, and is closer to how a desk actually hedges; it requires supervisory approval. It is a different standardized approach from Section 27.3's, which measures the exposure this charge sits on.

27.7 HEDGING CVA, AND THE EXEMPTIONS

The standardized CVA charge is sensitivity-based, which means eligible hedges reduce it. Which hedges are eligible, and by how much, turns out to determine what the desk of Section 9.10 actually trades.

A single-name CDS on the counterparty is recognized almost in full under the standardized approach, because the charge is computed on that counterparty's spread and the CDS is written on exactly it: buy the protection and the charge falls by what the protection removes. The basic approach caps the relief: however complete the hedge, it removes at most three-quarters of the charge.

An index CDS is recognized in part. The framework imposes a correlation between the index and the issuer, so some of the hedge's benefit is disallowed however well it works. A proxy (a CDS on a different issuer in the same sector and region) is recognized less still. This matters because for most counterparties there is no single-name CDS at all. Chapter 25's case study of the market that shrank established that liquid single-name CDS now exists for a few hundred issuers, while a dealer has thousands of counterparties.

So the instrument a desk can actually buy is the one the framework recognizes least. The consequence is a genuine dilemma rather than a rule to memorize: an index hedge removes most of the economic risk, earns partial capital relief, and introduces a basis (the index-to-single-name

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gap of Section 25.6) that the CVA charge still capitalizes, because the index is recognized only in part, and that shows up in earnings every quarter. Hedging the risk and hedging the capital charge are different trades, and the desk that does both holds a position it wanted neither of.

Then there are the exemptions, and they are large. The European framework exempts transactions with non-financial counterparties, sovereigns, and pension funds from the CVA capital charge, on the argument that these clients cannot post collateral and would be priced out of hedging. The United States and Canada did not adopt it. There the relief comes only from the framework's own carve-out, under which a trade cleared through a qualifying CCP is outside the CVA charge altogether, so what matters is how a trade is cleared rather than who is on the other side of it.

Whatever one thinks of the policy, the mechanical consequence should be stated. *The exposure is unchanged; only the capital is.* A European dealer's CVA on its corporate book is a real, volatile, unhedged number producing earnings swings and no capital requirement—and a dealer that *does* hedge it takes on market risk that is capitalized, while the CVA capital charge it is hedging against was already zero and cannot fall further.

The rule therefore discourages hedging the very exposures it chose not to charge for. That is an unusually clean example of the problem this chapter keeps returning to: a capital framework does not merely measure behavior, it selects it, and leniency in one place reappears as distortion in another.

27.8 DVA, AND WHY CAPITAL REFUSES IT

Section 9.3 established that a firm's own default risk produces a debit valuation adjustment (a gain when the firm's own credit deteriorates), that accounting standards require it to be recognized, and why regulators refuse it: a gain that can be monetized only by defaulting is not a buffer against default. The rule is that DVA gains and losses are filtered out of **common equity tier 1**, the highest-quality layer of capital (ordinary shares and retained earnings), so a bank whose spread widens may book a large accounting profit of which none counts as capital. Fair value and capital are answering different questions about the same liability, and the same reasoning is why Section 27.6's capital charge is computed on CVA gross of DVA while the accounting number is net.

27.9 WRONG-WAY RISK

Section 9.9 defined wrong-way risk (exposure that grows precisely where the counterparty is least able to pay) and showed that it makes the exposure conditional on default, not the average, the amount CVA must insure. Section 8.7 deferred the general treatment to here, and Example 27.4 puts a size on that substitution for CVA.

Example 27.4 What the correlation costs

A counterparty with a 3% annual default probability, 60% loss given default, and an average exposure of \$10 million has, under independence, CVA $= 0.60 \times 0.03 \times 10 =$ \$0.18 million. Putting $E[\text{exposure} \mid \text{default}]$ in place of the \$10 million multiplies that by the ratio of the two, so a correlation strong enough to double the conditional exposure doubles the charge, to \$0.36 million. The multiplier is linear in the conditional exposure ratio, and it is not small.

The calculation contains no correlation parameter. The conditional exposure ratio is the only thing that matters, and it is one number rather than a joint distribution, which is convenient, and also the reason the quantity is so hard to estimate. It is a conditional expectation on an event that has never occurred for this counterparty.

Wrong-way risk divides in two, and the distinction determines what can be done about it.

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Specific wrong-way risk is structural, like Section 9.9's sovereign case: taking a counterparty's own shares as collateral, or entering a commodity swap with a producer whose solvency depends on that commodity's price. These are identifiable in advance from the trade's economics. Where the counterparty is legally connected to the underlying or reference entity, as it is to its own shares, the regulatory response is to require the exposure to be computed as if that entity had already defaulted: a jump to default, with no correlation estimated. It is a rule rather than a model, because the model would need a correlation no one can estimate.

General wrong-way risk is milder: the counterparty's credit is correlated with the market factors driving the exposure. It is handled through the α of Section 27.5, whose calibration allows for that correlation, and by calibrating the exposure model to a period of stress.

The recurring lesson from three chapters generalizes. Section 24.4's recovery-default correlation, Section 26.4's excess spread that vanishes with losses, and this section's exposure-default correlation are the same failure: a protection whose value is negatively correlated with the need for it. Once named, it is checkable: for any mitigant, ask what it is worth in the state that makes it necessary.

27.10 THE LEVERAGE RATIO AS A BACKSTOP

Everything so far is risk-based: exposure is scaled by a risk weight, so a safer counterparty costs less capital. Alongside it sits a deliberately risk-*insensitive* constraint. The **leverage ratio** is tier 1 capital (common equity tier 1 plus the perpetual instruments that rank just behind it) divided by total exposure (assets, derivative exposure, securities financing and off-balance-sheet items, none risk-weighted), and it must clear a minimum LR, so every trade costs capital whatever its quality.

Two constraints on the same trade means one of them binds, and which one is arithmetic. Both sides are capital held against the trade. The leverage rule asks for $\text{EAD} \times \text{LR}$, since the trade adds its EAD to total exposure whatever its quality; the risk-based rule asks for the $\text{EAD} \times \text{RW} \times \text{CR}$ of (27.11). A bank must satisfy both ratios at once, and the same capital counts toward each, so a trade costs the larger of the two and never their sum. The first is larger when

$$\text{EAD} \times \text{LR} \;>\; \text{EAD} \times \text{RW} \times \text{CR} \qquad \Longleftrightarrow \qquad \text{RW} \;<\; \text{LR}/\text{CR}$$

so with a 10.5% capital ratio:

Leverage requirement	Leverage binds below a risk weight of
3% (baseline)	29%
5% (largest US banks, before April 2026)	48%
6%	57%

Sovereigns at 0%, qualifying CCPs at 2%, highly rated banks at 20%: all of the highest-quality exposures fall below the crossover. For exactly those businesses, capital is set by a measure that does not care that the risk is low.

The crossover is only a first approximation, and both refinements move it the leverage ratio's way. The leverage exposure measure uses its own version of EAD, recognizing collateral more grudgingly than (27.10) does, so a well-collateralized book looks larger to it than to the risk-based rule. And the two ratios are measured against different capital: tier 1 for leverage, total capital for the 10.5%, and since tier 1 is the narrower and costlier layer, a point of leverage requirement weighs more than a point of the risk-based one.

This is intended. A backstop that varied with risk weights would inherit whatever error is in them, and Chapter 26's pre-crisis experience is a story about risk weights being wrong. The leverage ratio bounds what the risk-based framework can be talked into.

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But the cost is specific and has been observed. The businesses it makes expensive (Treasury market-making, repo intermediation, client clearing) are low-margin, high-volume, and low-risk. A constraint that is insensitive to risk pushes a dealer out of its safest activities first, because those are where the constraint bites hardest relative to the revenue.

In March 2020 this became visible. Treasury market functioning deteriorated under a surge in volume, dealer balance sheets were constrained, and the supervisory response included temporarily excluding Treasuries and central bank reserves from the leverage exposure measure. The exemption expired in March 2021 and was not renewed; the recalibration of the US enhanced leverage ratio, finalized in November 2025 and effective April 1, 2026, lowered the requirement on all exposures instead, replacing the largest banks' fixed 2% buffer with half of each bank's global systemically important bank surcharge. The underlying question, whether a backstop should be relaxed at exactly the moment the activity it constrains becomes systemically important, has no comfortable answer.

Chapters 6 and 8 both rest on dealers intermediating at low margin, so a rule that taxes intermediation by volume rather than risk determines market liquidity and should be read as doing so.

27.11 HOW A BANK'S CAPITAL REQUIREMENT IS BUILT

Everything so far has priced one trade's capital through (27.11). A bank's requirement is assembled from many such pieces, and seeing the whole explains both what a derivative's capital competes with and which lever a desk can actually pull. The assembly has a numerator, the capital the bank holds, and a denominator, the risk-weighted assets it holds it against.

The numerator. Capital comes in layers, ranked by how readily they absorb loss, and each layer must be at least a set share of RWA. These are the **Pillar 1** minimums:

- *Common equity tier 1* (CET1), the ordinary shares and retained earnings of Section 27.8: at least 4.5%.
- *Tier 1*, which adds perpetual instruments that can be written down or converted: at least 6%.
- *Total capital*, which adds subordinated debt: at least 8%.

On top sit buffers, all of them met with CET1:

- the **capital conservation buffer**, 2.5%;
- the **countercyclical buffer**, 0 to 2.5%, which each jurisdiction sets according to its credit cycle;
- for a **global systemically important bank** (G-SIB), a surcharge of 1% to 3.5%, set by the bank's size, interconnection and complexity.

A bank that dips into the buffers is not in breach, but its dividends and bonuses are restricted until it rebuilds them. In the United States the conservation buffer of the largest banks is replaced by the stress capital buffer of Section 39.12, which is at least as large. Two further layers complete the framework:

- **Pillar 2**, the supervisor's bank-specific add-on for risks the formulas miss, such as concentration and interest rate risk in the banking book;
- **Pillar 3**, the public disclosure that lets the market check the arithmetic.

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The denominator. Total RWA is the sum of one term per risk type, each computed by its own rule:

$$\text{RWA} = \text{RWA}_{\text{credit}} + \text{RWA}_{\text{CCR}} + \text{RWA}_{\text{CVA}} + \text{RWA}_{\text{market}} + \text{RWA}_{\text{op}} + \text{RWA}_{\text{sec}} \tag{27.12}$$

- *Credit*: loans and bonds in the banking book, weighted by Table 27.1 below or by the internal-ratings formula of Section 39.10. It is usually by far the largest term.
- *Counterparty credit*: this chapter's SA-CCR exposure times the counterparty's risk weight.
- *CVA*: the charge of Section 27.6, times 12.5.
- *Market*: the trading-book charge of Chapter 39, times 12.5.
- *Operational*: losses from failed processes, people and systems, fraud and litigation among them. The charge is a standardized formula on a **business indicator** built from the bank's income, multiplied by coefficients of 12% to 18% that rise with size and, where the supervisor allows, scaled by the bank's own loss history; times 12.5.
- *Securitization*: a framework of its own, sketched below.

The factor 12.5 is the reciprocal of 8%: it turns a capital charge into the RWA that would require it. Under the output floor of Section 39.10, total RWA from internal models may not fall below 72.5% of what the standardized rules give for every term together.

Example 27.5 From risk-weighted assets to a capital requirement

A G-SIB in the 1.5% surcharge bucket, in a jurisdiction with a 0.5% countercyclical buffer, reports RWA in billions of dollars:

Risk type	RWA ($bn)
Credit (banking book)	600
Counterparty credit (SA-CCR)	40
CVA	30
Market (Chapter 39)	60
Operational	100
Securitization	20
Total	**850**

The buffers add 2.5% + 1.5% + 0.5% = 4.5% of CET1 to every layer, so the requirements are

	Ratio	Capital ($bn)
CET1	4.5% + 4.5% = 9.0%	76.50
Tier 1	6.0% + 4.5% = 10.5%	89.25
Total capital	8.0% + 4.5% = 12.5%	106.25

before any Pillar 2 add-on. The bank's leverage exposure is $2,500 billion, and a G-SIB's leverage requirement is the 3% minimum plus half its surcharge, 3% + 0.75% = 3.75%, or $93.75 billion of tier 1. That exceeds the risk-based tier 1 figure of $89.25 billion, so the leverage ratio binds. The reason is Section 27.10's crossover: the bank's average risk weight is 850/2,500 = 34%, below the 3.75%/10.5% = 35.7% at which the two rules cost the same.

Read the table for what it says about derivatives. Counterparty and CVA risk together are 70 of 850, about 8%, and the market term another 7%: the capital a derivatives business consumes is a minority of the bank's, which is why its marginal cost depends on what the rest of the balance sheet is doing. Here, with leverage binding, a desk that lowered its risk weights would save nothing, and one that shrank its exposure would.

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Standardized credit risk weights. A bank without model approval, and every bank for the output floor, weights its banking-book exposures by Table 27.1. The weight depends on the type of borrower and, in the Basel text, on its external rating.

Table 27.1: Standardized risk weights for credit exposures in the Basel framework (%), by external rating. Jurisdictions that do not use ratings, the United States among them, substitute their own schedules.

	AAA to AA	A	BBB	BB	B	Below B	Unrated
Sovereign	0	20	50	100	100	150	100
Bank	20	30	50	100	100	150	40–150
Corporate	20	50	75	100	150	150	100

Regulatory retail 75; residential mortgage 20 to 70 by loan-to-value; equity 250; subordinated debt 150

An off-balance-sheet item enters through a **credit conversion factor**, the share of it treated as drawn:

- 100% for a guarantee that substitutes for a loan;
- 40% for an undrawn commitment;
- 10% for a commitment the bank can cancel without notice;
- 20% for a short-term trade letter of credit.

A \$200 million undrawn revolving facility to a BBB corporate therefore carries $200 \times 0.40 \times 0.75 =$ \$60 million of RWA before a cent is lent.

Credit risk mitigation. The weights apply to the exposure the bank actually bears, and three things reduce it:

- *Collateral* reduces the exposure itself. Under the comprehensive approach the exposure after mitigation is $E^* = \max\{0,\ E(1+H_e) - C(1 - H_c - H_{fx})\}$, where the supervisory haircuts H grow with the collateral's volatility (zero for cash, a few percent for a short government bond, 20% or more for equities), with a further 8% if the collateral is in another currency.
- *Guarantees and credit derivatives* work by substitution: the protected part of the exposure takes the risk weight of the protection seller instead of the borrower's.
- *Netting*, the subject of Section 27.2, reduces exposure for derivatives and securities financing.

Substitution is how a bank's purchase of CDS protection (one of the uses Chapter 25 lists) turns into capital relief (Example 27.6), and it comes with conditions:

- the protection must be direct, irrevocable and unconditional;
- if the CDS does not count restructuring as a credit event, only 60% of it is recognized, because the loss a restructuring imposes falls on the lender unprotected;
- if the protection runs out before the loan, it is recognized only in proportion, as $P_a = P\,(t - 0.25)/(T - 0.25)$ for protection P with t years left against a loan with T, and not at all if it has three months or less to run.

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Example 27.6 What a CDS hedge does to a loan's capital

A bank has a $100 million five-year loan to an unrated corporate, weighted 100%, at the 10.5% requirement of Example 27.3. It buys five-year protection from a bank rated AA, weighted 20%.

	RWA ($m)	Capital ($m)
Unhedged	100.00	10.50
Hedged, CDS covers restructuring	20.00	2.10
Hedged, CDS without restructuring	52.00	5.46
Hedged with a three-year CDS	53.68	5.64

The full hedge moves the whole loan to the protection seller's 20%. Without restructuring, 60% of it is recognized: $60\times0.20+40\times1.00 = 52$. With a three-year contract against a five-year loan, the recognized amount is $100\times(3-0.25)/(5-0.25) = 57.89$, and $57.89\times0.20+42.11\times1.00 = 53.68$.

The full hedge saves $8.40 million of capital. At the 10% hurdle rate of Example 27.3 that is worth $0.84 million a year, 84 bp of the loan. So for a borrower whose CDS trades below 84 bp the capital saved covers the hedge's cost on its own, and for one above it the hedge is bought for the risk it removes, with the capital relief covering only part of its cost. The comparison also leaves out two costs the hedge adds back: the CDS is itself a derivative, with a counterparty exposure under SA-CCR and a CVA charge, both small for a single bank-sold contract. A $100 million loan secured by $60 million of cash, by contrast, needs no counterparty at all: $E^* = 100 - 60 = 40$, and RWA fall to $40 million.

Securitization. A securitization position has its own risk-weight function, because a tranche's risk depends on where it sits in the waterfall of Section 26.2 as much as on the pool. The framework computes the capital the pool would need if held whole, K, and charges each tranche according to how much of that it absorbs:

- a tranche below K is weighted 1,250%, the reciprocal of 8%, so it costs capital dollar for dollar;
- a tranche above it is charged by a formula that falls with the attachment point, down to a floor of 15% for a senior tranche (10% for one meeting the simple, transparent and comparable criteria).

Those two numbers explain the significant risk transfer trades of Chapter 26. A bank that sells the first-loss and mezzanine tranches of a loan book, and keeps the senior one at or near the floor, replaces the loans' risk weights with a fraction of them. The regulator's test for whether the transfer is real is exactly how much risk has left the bank.

The pieces fit one way. A desk's capital is a share of a requirement whose binding constraint (the risk-based ratios, the leverage ratio, the output floor, or a stress test) is set by the whole bank. Which one binds decides which lever moves capital: risk weights matter only while a risk-weighted rule binds, exposure only while the leverage ratio does, and neither while the floor does.

27.12 CLIENT CLEARING AND THE COST OF BEING AN INTERMEDIARY

Example 27.3 showed the capital benefit of clearing to the party whose trade is cleared. Somebody has to provide the access, and that role's economics are worse than the table shows.

A client does not face the clearing house directly. *A clearing member stands between them*, facing the CCP on one side and the client on the other, and the two legs do not cancel: the member owes the CCP whatever the client fails to pay, and holds the client's margin against it.

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Three capital consequences follow, and together they explain a surprising market structure.

The member's exposure to the client is a derivative exposure and capitalized as one, even though the member has no market risk on it: each leg's market moves offset the other's, but the two face different counterparties and so cannot be netted.

Client margin is segregated, so the member cannot use it to fund. Chapter 8's segregation requirement, which exists so that the client's collateral survives the member's failure, means the member funds its own contribution to the CCP without the benefit of what it holds. That is Section 9.5's MVA arriving as an operating cost rather than as a charge on a trade.

And the leverage ratio counts the exposure. Section 27.10's crossover applies with force here: the risk weight against a qualifying CCP is 2%, so the risk-based charge is negligible and the leverage requirement binds. Whether segregated client initial margin may be recognized as reducing the leverage exposure measure has been changed more than once and remains among the most consequential technical questions in the framework, because it determines whether client clearing is a viable business.

The observable result is concentration. Client clearing in the major currencies is provided by a small number of banks, and several large institutions have exited. A framework designed to reduce concentration of counterparty risk has concentrated access to the mechanism that reduces it—a reminder that capital rules allocate activity as well as loss-absorbency.

And it interacts with Chapter 2's central concern. Porting a defaulted member's client positions requires another member able to take them, so the fewer members there are, the less credible porting becomes—and porting is what the whole client-protection story rests on.

27.13 CONCENTRATION AND THE LIMITS OF BILATERAL MEASUREMENT

Everything above measures a dealer's exposure to a counterparty. Nothing above measures a counterparty's exposure to the market, and that gap is the chapter's most important weakness.

The framework is bilateral by construction: each firm computes its own exposure to each of its counterparties, using data it has. A client's positions at other dealers are not data it has.

So a counterparty that is prudently margined at every dealer individually can be catastrophically levered in aggregate, and no participant in the system is in a position to observe it. The only parties who could are the client, who has no obligation to disclose, and the regulators, whose reporting captures cleared and reported trades but not, in useful time, a consolidated view of one client's leverage across brokers.

In March 2021 that gap cost the banks that had measured their own exposures correctly more than $10 billion.

27.14 ARCHEGOS

Case Study 27.1 What total return swaps made possible

In March 2021 a family office defaulted on positions built through total return swaps at several prime brokers. Losses to the banks totaled over $10 billion, with a single institution losing more than $5 billion—one of the largest single-counterparty losses in the derivatives market's history, and it involved no exotic instrument and no pricing error.

The arithmetic is simple enough to be alarming.

Take a fund with $10 billion of capital taking five times leverage through total return swaps, split across six prime brokers:

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Total gross exposure	\$50bn
What each broker sees	\$8.3bn
Initial margin held by each, at 15%	\$1.25bn

The 15% is a prime brokerage number for a diversified equity book. Now the underlying positions, concentrated in a small number of issuers, fall:

Move	Loss on gross	
−10%	\$5.0bn	survivable
−20%	\$10.0bn	client's capital exhausted
−30%	\$15.0bn	client insolvent; loss beyond the margin falls on the brokers

The client's capital is exhausted at a 20% move, which for an equity portfolio concentrated in a few stocks is an ordinary week. At 30%, each broker holds \$1.25 billion of margin against a \$2.5 billion loss—*a shortfall of \$1.25 billion each*. This assumes a broker can look only to the margin it holds, since the client's other \$2.5 billion of capital is not posted. Were all of it collected, the total shortfall would be \$5 billion rather than \$7.5 billion.

Four failures stand out, and they are worth separating because only one is about the instrument.

Invisibility. Section 7.12 said total return swaps are how leverage accumulates quietly, and this is what that meant. Each broker's exposure was individually unremarkable. The swap format also kept the positions out of ownership disclosures, so the concentration was not visible to the market either.

Margin that was not risk-based. Initial margin was set at levels reflecting competition for the client's business rather than the concentration and liquidity of the underlying. The margin period of risk assumption of Example 27.2 (that a portfolio can be closed out in ten days) was false for a position worth many days of trading volume in each issuer.

Concentration measured against the wrong thing. An \$8.3 billion exposure is modest against a large bank's balance sheet and enormous against the free float of a mid-cap stock. The relevant denominator is liquidity, not capital, which is the same lesson Section 23.12's single-stock snapshot drew from the other direction.

And a liquidation that moved the price. When the brokers sold, they sold the same stocks into each other, which is Chapter 22's crowding and Chapter 23's feedback in the least sophisticated form: six firms with the same collateral trying to sell it in the same week. The brokers that sold first lost least, which converted a credit problem into a race.

What the framework did and did not catch. SA-CCR would have registered each broker's exposure and required capital against it; no bilateral framework registers that five other institutions hold the same exposure to the same client. The measurement was correct and the aggregate invisible, which is the chapter's conclusion.

Supervisory responses have focused on margin practices for concentrated positions, prime brokerage risk governance, and disclosure of large synthetic equity positions. None fully solves the aggregation problem, which needs either a consolidated view no institution has or a disclosure obligation on the client.

27.15 INITIAL MARGIN AS THE REAL ANSWER

Chapter 8 described the uncleared margin rules as documentation and operations. What they buy is capital, and the two margin types buy different parts of it.

Variation margin removes current exposure: it is settled daily and brings the mark-to-market back to zero, which is the RC of (27.1). *Initial margin removes the gap risk*: the move between

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the last margin call and the close-out, which is precisely the margin period of risk that sets the maturity factor in Example 27.2.

The two differ in kind. Variation margin is a transfer of value that has already happened; initial margin is a buffer against value that has not, which is why it must be segregated and cannot be rehypothecated.

Its cost is Section 9.5's MVA, and its benefit is the EAD reduction of Example 27.2. A dealer's decision about how much initial margin to demand is therefore a trade between a funding cost it pays and a capital cost it avoids, and Archegos is what happens when competition pushes that trade toward the client.

The structural point is that initial margin is the only mitigant that works without information. Netting requires legal certainty; wrong-way risk requires identifying the correlation; concentration limits require knowing the aggregate. Initial margin protects against a move of a given size whatever its cause, which is what makes it right when what you cannot see is what hurts you.

27.16 WHAT THE FRAMEWORK DOES NOT CAPTURE

The framework leaves four things unmeasured, and each has cost someone money.

Aggregate leverage. Section 27.13's gap, which nothing bilateral sees.

Liquidity of the underlying. SA-CCR's supervisory factors are by asset class, not by the liquidity of the specific position. A concentrated position in an illiquid issuer and a diversified one of the same notional carry the same add-on.

Correlated liquidation. The framework asks what a firm loses if one counterparty fails. It does not ask what it loses if that failure forces five other firms to sell the same assets.

And the model risk in the exposure itself. The exposure profile is a simulation output, and Chapter 21 established that model choice moves such outputs materially; SA-CCR's standardization is partly a response, at the cost of being wrong in a known direction outside its calibration.

None of this is an argument against the framework. The framework substantially improved on what preceded it, and the CVA charge in particular addressed a loss that had gone entirely uncapitalized. The argument is that a capital framework measures what a firm can see, and the losses that matter most have repeatedly come from what the whole system could not.

Capital is a buffer, not a control. It fixes what must be held once a position exists; it does not decide whether to take the position. Every episode in this chapter turned on a judgment made inside a firm (how much initial margin to demand of a client, which limit to override, how fast a concentrated holding could be sold), and no calibration reaches those. Regulation can make imprudence expensive, which is most of what the post-crisis framework achieved; declining to be imprudent is left to the institution, and Section 27.14 is what it costs when a firm does not.

27.17 WHO BEARS WHAT

Dealers bear counterparty risk, capitalize it under this chapter, price it under Chapter 9, and hedge it in the credit market of Chapter 25: three activities with three different numbers, which Section 9.11 warned would not reconcile.

Clearing houses bear it mutualized, under the default waterfall of Chapter 2. Moving a trade to a CCP converts a bilateral exposure into a contribution to a shared pool, which is a real reduction in system-wide exposure and a real concentration of it in one place.

Clients bear the cost, through the charges Chapter 9 enumerated, seen or not.

And the system bears the residual, which is the part this chapter says cannot be measured bilaterally. *Whoever bears that is decided after the fact*, and in 2008 and 2021 it favored whoever sold first.

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SUMMARY

CVA is a price and regulatory capital is a buffer, computed by different rules for different purposes. A derivative's exposure is an option on the market, bilateral, belonging to the whole netting set, and dependent on mitigants that can fail. Netting is the largest reducer of it: a portfolio whose trades largely offset can carry a net exposure that is a small fraction of its gross, a reduction obtained from a legal clause, so a jurisdiction's netting opinion is worth more to a dealer than any risk model.

SA-CCR sets EAD = $1.4 \times$ (RC + PFE), with RC the collateralized mark-to-market (or the exposure a margin agreement's threshold leaves uncalled) and PFE built from add-ons by asset class. Daily margin cuts a swap's exposure sharply through the maturity factor alone, which scales the add-on to the margin period of risk rather than to the trade's maturity, yet a position that takes weeks to unwind keeps much of its add-on. The capital a swap consumes falls by a large multiple from uncleared to margined and again from margined to cleared, which is why capital rules moved more of the market to clearing than the mandate did. The α of 1.4 is a buffer for model error, not a property of any trade.

A separate CVA charge covers spreads widening without default, two-thirds of the crisis's counterparty losses and uncapitalized before it; hedge recognition is uneven, and Europe's exemptions discourage hedging exactly the exposures the charge was built for. Capital filters DVA out of common equity tier 1 because a gain realizable only in default is not a buffer. Wrong-way risk, recovery–default correlation, and vanishing excess spread are one failure mode: protection whose value falls as the need for it rises.

The leverage ratio binds wherever the risk weight is below LR/CR, so it makes the safest, thinnest businesses expensive first: Treasury market-making, repo, and client clearing. A bank's whole requirement stacks minimums of 4.5%, 6% and 8% of risk-weighted assets for common equity, tier 1 and total capital, then conservation, countercyclical and systemic buffers. Standardized credit weights rise as the rating falls; collateral reduces the exposure, and a guarantee or CDS substitutes the protection seller's weight, recognized only in part if the contract omits restructuring or ends before the loan. Which rule binds for the whole bank (risk-based, leverage, floor or stress test) decides whether a desk saves capital by cutting risk weights, by cutting exposure, or not at all. And the framework is bilateral: in 2021 a family office that looked prudently margined at each of its prime brokers, and was heavily levered across them through total return swaps, cost them over $10 billion with no exotic instrument and no pricing error. The measurement was correct and the aggregate was invisible. Initial margin is the one mitigant that works without information, and a capital framework measures only what a firm can see.

FURTHER READING

The consolidated Basel Framework, published online by the Bank for International Settlements, is the primary source for Section 27.11: its credit risk chapters hold the standardized weights, the conversion factors and the mitigation rules, and the Basel Committee's *Basel III: Finalising Post-Crisis Reforms* (2017) is the reform that set them. The Basel standards on SA-CCR and on CVA risk are the primary sources and are specific enough to compute from; anyone implementing either should read them rather than a summary, since the hedging-set definitions carry most of the practical content. Gregory's book on counterparty credit risk and XVA is the standard practitioner treatment and covers the pricing of Chapter 9 and the capital of this chapter in one place. On Archegos, the independent report commissioned by the largest loss-making bank sets out how risk limits were set and overridden; the supervisory reviews that followed say more about what changed than about what happened. On wrong-way risk, the literature is small, and the regulatory rule (compute as if default had occurred) remains more

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widely used than most of the models. On the aggregation gap of Section 27.13, the work on position transparency and on consolidated leverage reporting is active and unresolved.

PRACTICE QUESTIONS

27.1 Explain the difference between CVA as a price and CVA capital as a buffer, and why reducing one need not reduce the other.

27.2 Four trades with a counterparty mark at +30, −12, +8, −20. Compute the exposure with and without netting, and explain why a jurisdiction's netting opinion is described as worth more than any risk model.

27.3 Compute the SA-CCR exposure at default for a portfolio with a replacement cost of \$8 million and a PFE add-on of \$3 million. Then compute the maturity factor for a ten-day margin period of risk, state what it does to the add-on, and state what assumption the margin period embodies.

27.4 Compute the margin period of risk at which the maturity factor reaches one, and state what this means for an illiquid portfolio.

27.5 A bank pays fixed on a three-year \$100 million swap, unmargined, with no collateral. Compute its supervisory duration, effective notional and add-on. It then adds a seven-year receive-fixed swap of \$47.2 million, chosen so that the two have equal and opposite dollar duration. Compute the hedging set's effective notional and add-on, and explain why a position with no net rate sensitivity still carries most of the add-on, and what would change if the second trade had a four-year maturity instead.

27.6 Using Example 27.3, state the ratio of capital across the three regimes and the two changes that produce it, then convert a capital requirement of \$0.20 million on \$100 million of notional into a running charge in basis points at a 12% hurdle rate.

27.7 Explain why the clearing mandate is described as less important than the capital rules in moving the market to clearing.

27.8 Explain what $\alpha = 1.4$ is calibrated to bridge, and why it is not a statement about any individual trade.

27.9 Explain what the CVA capital charge covers that the default charge does not, why this loss went uncapitalized before the charge existed, and why the hedge a desk can actually buy is the one the framework recognizes least.

27.10 Explain why regulatory capital filters DVA out of common equity tier 1 while accounting requires it to be recognized.

27.11 A counterparty has a 4% default probability, 60% loss given default, and an average exposure of \$20 million. Compute CVA under independence, and again if the exposure conditional on default is twice the average. Distinguish specific from general wrong-way risk and state how each is treated.

27.12 Give three examples from this book of a protection whose value is negatively correlated with the need for it.

27.13 Derive the risk weight below which the leverage ratio binds at a 10.5% capital requirement and a 5% leverage requirement, and explain why a risk-insensitive constraint makes a dealer's safest businesses expensive first.

27.14 Explain why a clearing member's capital requirement is not eliminated by the fact that its client-facing and CCP-facing legs offset, and how the resulting concentration weakens the porting mechanism of Chapter 2.

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27.15 In the Archegos arithmetic, state the move at which the client's capital is exhausted and the shortfall per broker at a 30% move, taking margin as all a broker can collect, and explain why the relevant denominator for concentration is liquidity rather than the dealer's capital.

27.16 Explain why initial margin is described as the only mitigant that works without information, and state four things the capital framework does not capture.

27.17 In Example 27.5, the countercyclical buffer rises from 0.5% to 2.0%. Compute the new CET1, tier 1 and total capital requirements, and state whether the leverage ratio still binds and why the buffer change does not affect the leverage requirement.

27.18 A bank has a $50 million loan with four years to run to a BBB-rated corporate and hedges it with a two-year CDS, covering restructuring, bought from a bank rated A. Using Table 27.1, compute the loan's RWA before and after the hedge, and explain why protection shorter than the loan is recognized only in part.

27.19 Explain why a securitization tranche below the pool's own capital requirement is weighted 1,250%, and why the floor on the senior tranche's weight is what makes a significant risk transfer trade worth doing.

FURTHER QUESTIONS

27.20 Example 27.2 shows margining cutting the add-on by 70%. Derive the maturity factor's dependence on the margin period of risk, determine the period at which margining ceases to reduce the add-on, and discuss what this implies for a portfolio whose true close-out period is thirty days.

27.21 Section 27.9 argues that recovery-default correlation, disappearing excess spread, and exposure-default correlation are one failure mode. Formalize the common structure, propose a single diagnostic that would detect all three, and apply it to two mitigants used elsewhere in this book.

27.22 Section 27.13 identifies the aggregation gap. Design a reporting regime that would have revealed the Archegos position: specify what would be reported, by whom, to whom, and at what frequency; then state the costs (competitive, operational, and in terms of what the disclosure itself might cause).

27.23 The Archegos snapshot notes that brokers who sold first lost least. Model this as a game between the brokers, identify the equilibrium, and explain why a coordinated liquidation would have produced a better aggregate outcome and why it was not achievable. Relate your answer to Chapter 2's default waterfall, which is designed to solve exactly this problem for cleared trades.

27.24 Section 27.15 argues initial margin works without information. Construct the counter-argument that margin levels themselves require information about volatility, liquidity, and concentration, and reconcile the two positions by distinguishing what must be known to set the level from what must be known for the protection to function.

27.25 Section 27.16 says a capital framework measures what a firm can see. Compare this with Chapter 26's conclusion that a system loses the ability to see a risk before it loses the ability to bear it, and with Chapter 23's observation that 0DTE hedging demand is computable in advance. Use the three to argue either that observability is the binding constraint on financial stability or that it is not.

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Part VI

Interest Rate Derivatives (Advanced)

The book's quantitative core. Chapters 28, 30 and 32 carry the dagger: they are aimed at readers who will implement or validate a rate model. Chapters 29 and 31 do not, and are needed by anyone who prices a rate option at all. Chapter 33 returns to market practice and stands on its own: a course that skips the rest of this part can take it directly after the swap material in Chapter 7. One idea runs through all six: choose the numeraire that makes the awkward part of a payoff disappear.

28 Martingales and Measures
the change-of-numeraire machinery, and why the choice is the valuation

29 Standard Market Models for Rate Derivatives
Black's formulas for caps and swaptions, and where they hold

30 Convexity, Timing, and Quanto Adjustments
three corrections, and how large each one actually is

31 Short-Rate Models
calibration, trees, and the Bermudan swaption

32 HJM, LMM, and Multi-Curve Term Structure Models
the drift condition, factor reduction, and calibration

33 Swaps Revisited
the non-standard book: amortizing, compounding, in-arrears, and inflation

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Chapter 28
Martingales and Measures †

Placed here deliberately: everything in Part VI takes it for granted.

Chapter 29 prices a swaption by treating the annuity as a numeraire, Chapter 30 changes measure three times to derive three adjustments, Chapter 32 derives drift conditions from no-arbitrage, and Chapter 33 needs all of it at once. Reading it first means none of those chapters has to borrow machinery it has not been given.

A reader who wants only the working rules can take Section 28.1 and Section 28.6 and move on. The rest is the justification, and it is here because the vocabulary in Section 28.1 is used by every practitioner in the field and is almost never written down precisely. Anyone who has traded a derivative has heard "risk-neutral probability" and "equivalent martingale measure"; comparatively few could state what either one is, and the terms are not hard—they are merely never defined.

The chapter runs from the vocabulary to what it buys. Sections 28.1 to 28.3 fix the terms: what a measure is, what the market price of risk is the price of, and what changes when uncertainty has more than one source. Sections 28.4 to 28.6 are the machinery: martingales, the change of measure written out explicitly, and the numeraire that names each measure. Sections 28.7 and 28.8 put it to work on why a forward price is an expectation and on the annuity measure from which Chapter 30's CMS adjustment comes. Sections 28.9 to 28.11 restate the familiar results in this language, including what $N(d_1)$ is; Sections 28.12 and 28.13 are where a numeraire fails and what the theory is for.

28.1 THE VOCABULARY, STATED PRECISELY

Seven terms carry the whole subject. Each is stated here in the form a practitioner needs, which is less than a probabilist would want and more than the phrases usually get.

A probability measure P assigns a probability to each event, that is, each set of possible paths the market might follow. Two measures can disagree about probabilities while agreeing about what is possible, and that agreement is what the next definition captures.

Equivalent measures. Two measures P and Q are **equivalent**, written $P \sim Q$, if they agree on which events have probability zero:

$$\text{for every event } A: \quad P(A) = 0 \iff Q(A) = 0. \tag{28.1}$$

They may disagree about how likely anything is, but not about what can happen. This is the precise sense in which a change of measure reweights the world without inventing or destroying scenarios, and it is exactly what a no-arbitrage argument needs, because an arbitrage is a payoff that is impossible to lose on, and impossibility must survive the change.

A self-financing portfolio is one whose value changes *only* because the prices of its holdings change: no cash is added and none is withdrawn after it is set up. Rebalancing is permitted, provided every purchase is paid for by a sale. The condition is what makes a replication argument an argument about prices, rather than about someone's willingness to keep funding a position, and it is why a payoff can be replicated at a cost that is known today. It is also the condition hidden in the next definition: an asset qualifies as a numeraire only if its quoted price is the value of a self-financing position, which is why Section 28.12 has to disqualify a stock that pays dividends.

A numeraire is any traded asset with a strictly positive price, used as the unit in which other prices are quoted. Which one is chosen is not innocuous, and the freedom to choose is the technique that Part VI lives on: pick the numeraire that makes the payoff you are pricing simple, and the hard part of the calculation disappears into the choice.

A martingale is a process with zero drift: a process X is a martingale under Q if

$$E^{Q}[X(T) \mid \mathcal{F}_t] = X(t) \qquad \text{for all } t \le T, \tag{28.2}$$

where $\mathcal{F}_t$ is everything observable by time t, the **information** available then. The best forecast of a martingale's future value is its present value. A martingale is not a process that cannot move; it is one whose movements are unforecastable.

An equivalent martingale measure for a numeraire N is a measure $Q^N \sim P$ under which *every* traded asset price, divided by N, is a martingale:

$$\boxed{\frac{V(t)}{N(t)} = E^{N}\left[\frac{V(T)}{N(T)}\,\middle|\,\mathcal{F}_t\right]} \qquad \text{for every traded } V. \tag{28.3}$$

In a market free of arbitrage every numeraire has one, which is what makes the choice above a choice of measure. The word "equivalent" carries a condition and is usually skipped. It is what forbids the measure from assigning zero probability to a state the real world admits—without it, one could price away a genuine risk by declaring it impossible.

The risk-neutral measure is the equivalent martingale measure associated with one specific numeraire: the money market account $B(t)$, which grows at the instantaneous short rate. Under it,

$$V(0) = E^{Q}\left[e^{-\int_0^T r_s\,ds}\,V(T)\right],$$

and every asset has expected return r. The name misleads, so state what it is not. It is not a claim that investors are indifferent to risk, and the probabilities in it are not anyone's forecast. It is a pricing tool rather than a description of the world: a set of weights under which discounted prices have no drift, so that expectation becomes a valid pricing operation.

One sentence connects all seven. A price is a numeraire times an expectation, and the measure is whatever makes that true. Everything after this section proves that statement, draws a consequence from it, or chooses the numeraire well.

28.2 THE MARKET PRICE OF RISK

Start with a quantity that unifies everything after it.

Suppose two assets depend on the same single source of uncertainty dW:

$$\frac{dg}{g} = \mu_g\,dt + \sigma_g\,dW, \qquad \frac{dh}{h} = \mu_h\,dt + \sigma_h\,dW$$

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No-arbitrage forces the two to offer the same compensation per unit of risk:

$$\frac{\mu_g - r}{\sigma_g} = \frac{\mu_h - r}{\sigma_h} \tag{28.4}$$

The common value is the **market price of risk** λ carried by the source dW, and the proof that the two must agree is a portfolio.

Example 28.1 What happens if they differ

Let $r = 3\%$, and take

$$\text{A} : \mu = 8\%,\ \sigma = 20\% \Rightarrow \lambda_A = 0.25, \qquad \text{B} : \mu = 6\%,\ \sigma = 15\% \Rightarrow \lambda_B = 0.20$$

Hold one unit of A and $-\sigma_A/\sigma_B = -1.3333$ units of B. The dW terms cancel exactly, so the portfolio is riskless. Its return is

$$1.0000 \times 8\% - 1.3333 \times 6\% = 0.00\%$$

and its cost is

$$1.0000 - 1.3333 = -0.3333$$

A portfolio with a negative cost (a net borrowing of 0.3333) and a zero return. Financing it at 3% costs $0.03 \times (-0.3333) = -0.0100$, so the position earns **1.00% of the notional of A, riskless, per year**.

Scale it and the profit is unbounded, which is what an arbitrage means. The requirement $\lambda_A = \lambda_B$ then pins B's expected return: at $\lambda = 0.25$ and $\sigma_B = 15\%$, μ_B must be $3\% + 0.25 \times 15\% = \mathbf{6.75}\%$.

This is the whole of asset pricing theory in one calculation, and everything in this chapter is a restatement of it. Chapter 13's risk-neutral valuation is the case $\lambda = 0$; the measure changes of Sections 28.6 and 28.9 are the general case.

The market price of risk is a property of the risk, not of the asset. Two assets exposed to the same dW must carry the same λ whatever else differs about them: their maturity, their issuer, their payoff structure. In Chapter 31's one-factor models every bond carries the market price of interest rate risk, which is why a single λ suffices there and why the whole curve is determined by one process.

28.3 SEVERAL SOURCES OF UNCERTAINTY

Let $W_1, \ldots, W_n$ be independent Wiener processes, and let the state of the economy be carried by n variables $\theta_1, \ldots, \theta_n$ (**factors**, in the asset pricing literature) following

$$\frac{d\theta_i}{\theta_i} = m_i\, dt + s_i\, dW_i, \qquad i = 1, \ldots, n$$

where m_i and s_i may depend on the factors and on time. The factors need not be traded, and usually are not: a short rate, an inflation index, a volatility, a default intensity. The force of the assumption is that nothing else moves: every price in this economy is a function of t and of those n numbers.

So let f be the price of a traded asset, $f(t) = F(t, \theta_1(t), \ldots, \theta_n(t))$. Itô's lemma in n variables (Section 14.11 states the two-variable case) then delivers the dynamics rather than assuming them,

$$\frac{df}{f} = \mu\, dt + \sum_{i=1}^{n} \sigma_i\, dW_i, \qquad \sigma_i = \frac{\theta_i}{F}\frac{\partial F}{\partial \theta_i}\, s_i$$

where μ collects the time and second-order terms, and no cross terms survive because the W_i are independent. Read σ_i as the asset's exposure to factor i: the elasticity of its price in that

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factor, times that factor's own volatility. Such an asset must satisfy

$$\mu - r = \sum_{i=1}^{n} \lambda_i \sigma_i \tag{28.5}$$

where λ_i is the market price of risk for dW_i. The claim being made is that the λ_i are the same numbers for every asset. Applied to one asset in isolation the equation restricts nothing; it merely names n quantities. Applied to all of them at once it is the whole of the theory.

The proof is Example 28.1 with more bookkeeping. There, one asset hedged another and the riskless residual was forced to earn r. Here n assets hedge one.

Begin with n traded assets that are **non-redundant** (no one of them is a portfolio of the others) and whose exposures **span** the risks,

$$\frac{df_j}{f_j} = \mu_j \, dt + \sum_{i=1}^{n} \sigma_{ji} \, dW_i, \qquad j = 1, \ldots, n$$

spanning meaning that the matrix of loadings $[\sigma_{ji}]$ has rank n, hence is invertible: no risk is unhedgeable, and no asset duplicates the others. Define $\lambda_1, \ldots, \lambda_n$ as the solution of

$$\mu_j - r = \sum_{i=1}^{n} \lambda_i \, \sigma_{ji}, \qquad j = 1, \ldots, n$$

n linear equations in n unknowns, solvable and uniquely so because the loadings are invertible. Nothing has been assumed at this point. The λ_i are simply being read off the market: they record what these n assets currently pay per unit of each risk.

Now take *any* other asset in the economy, with drift μ and exposures $\sigma_1, \ldots, \sigma_n$, and hedge it. Hold one dollar of it, and $-w_j$ dollars of each f_j, with the w_j chosen so that every exposure is matched:

$$\sum_{j=1}^{n} w_j \, \sigma_{ji} = \sigma_i, \qquad i = 1, \ldots, n$$

solvable, and uniquely, for the same reason as before. Every dW_i term now cancels and the portfolio is riskless. It costs $1 - \sum_j w_j$ to establish and drifts at $\mu - \sum_j w_j \mu_j$, so no-arbitrage requires the drift to be exactly r times the cost:

$$\mu - \sum_j w_j \, \mu_j = r \left(1 - \sum_j w_j \right)$$

Collecting the interest terms leaves a statement about excess returns alone,

$$\mu - r = \sum_j w_j \, (\mu_j - r)$$

and the remainder is substitution. Replace each $\mu_j - r$ by the defining relation above, exchange the order of summation, and the inner sum is the hedge condition:

$$\mu - r = \sum_j w_j \sum_i \lambda_i \, \sigma_{ji} = \sum_i \lambda_i \sum_j w_j \, \sigma_{ji} = \sum_i \lambda_i \, \sigma_i$$

which is (28.5).

Notice what the argument never did. The λ_i were pinned down by the n spanning assets and then never revisited, so they cannot depend on which asset was subsequently tested against them. The one-factor conclusion, that λ belongs to the risk and not to the asset, survives intact, and it survives for the same reason.

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Suppose (28.5) fails. If some asset's drift exceeds $r + \sum_i \lambda_i \sigma_i$ by ε, the portfolio just built is riskless, costs nothing once the interest leg is financed, and earns ε per dollar per year. Reverse it if the drift falls short. Scale it and the profit is unbounded, so no asset can sit off the hyperplane (28.5) traces out.

Now count the assets against the risks. The argument used n non-redundant traded assets against n independent Wiener processes; with only $k < n$ of them non-redundant, the defining relations become k equations in n unknowns. Solutions still exist (no-arbitrage puts excess returns in the span of the loadings whatever the market's structure), but they form an $(n - k)$-dimensional family, every member of which prices the traded assets correctly, so no observed price singles one out. The rule is that *the market prices of risk are unique exactly when the traded assets are at least as numerous as the sources of uncertainty they must hedge*, counting only the non-redundant ones. Below that count, one member of that family is chosen rather than read off the market; at or above it, every further asset must price consistently with the λ_i already pinned down, or be an arbitrage. That is the incomplete-market case. A market is complete (Section 13.4) when every payoff can be replicated by trading the assets it has, and incomplete when some cannot, and the arbitrariness that the **second fundamental theorem** of asset pricing describes (an arbitrage-free market is complete exactly when its equivalent martingale measure is unique) enters here, at the market price of risk. [TN28] states both theorems and proves them.

Two of the model families in this book sit below that count. Chapter 21's stochastic volatility models add a second Wiener process driving the variance, and variance is not traded: two risks, one traded underlying, so the price of variance risk cannot be read off the market. It is chosen. Heston takes it proportional to the variance v, which keeps the variance in the same square-root class under Q, with Chapter 21's mean-reversion speed κ and long-run variance θ becoming $\kappa + \lambda$ and $\kappa\theta/(\kappa + \lambda)$. This is a modeling decision made for tractability, not a defect. It costs little: calibrating to option prices delivers the risk-neutral parameters directly, so the split between them and λ never has to be made, which is why the model is written under Q from the start. The premium itself does not disappear (it is Section 19.10's volatility risk premium), and once an option trades, its price supplies what the count could not. One traded option completes the market.

Chapter 31's short-rate models are the same case, and the same kind of choice is made there. The short rate drives everything and is not itself traded, so the price of rate risk is specified rather than observed: constant in Vasicek, proportional to $\sqrt{r}$ in Cox–Ingersoll–Ross, each form picked so that the process keeps its class under Q, Gaussian staying Gaussian and square root staying square root. Writing the model under Q from the start, as Chapter 31 does, is that choice already made, and calibrating to today's curve means the real-world parameters and λ never have to be separated.

The economic content of (28.5) is that risks price additively. An asset's excess return is the sum of its exposures times the prices of those exposures, which is the structure of every factor model in asset pricing, arrived at here from no-arbitrage rather than from equilibrium.

And it identifies what a model must supply. Chapter 32's K-factor Heath–Jarrow–Morton (HJM) model needs K market prices of risk to move from the real world to the risk-neutral one. Section 32.2's drift condition is exactly the statement that this move has been made, which is why the drift is not free.

28.4 MARTINGALES

Section 28.1 defined a martingale through its conditional expectation. In the continuous-time notation of Chapter 14 the same condition reads

$$d\theta = \sigma\, dW$$

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a process with no dt term. Zero drift and $E[\theta_T] = \theta_0$ say the same thing under an integrability condition that holds throughout this book, and whose failure [TN28] describes.

The reason martingales matter is that they turn valuation into an expectation. If some ratio V/N is a martingale, then

$$\frac{V_0}{N_0} = E\left[\frac{V_T}{N_T}\right] \quad \Longrightarrow \quad V_0 = N_0\, E\left[\frac{V_T}{N_T}\right]$$

and the pricing problem is solved provided the expectation can be computed. The ratio is a martingale only when V and N are both self-financing, a condition Section 28.12 returns to.

Which ratios are martingales, and under which probability measure, is the entire subject of this chapter.

28.5 THE CHANGE OF MEASURE, MADE EXPLICIT

Before the general theory, it is worth seeing a measure change as an actual list of numbers. A two-period binomial tree is enough: in Figure 28.1 the nodes are fixed, the middle terminal node is reached two ways, and only the branch probabilities change with the measure.

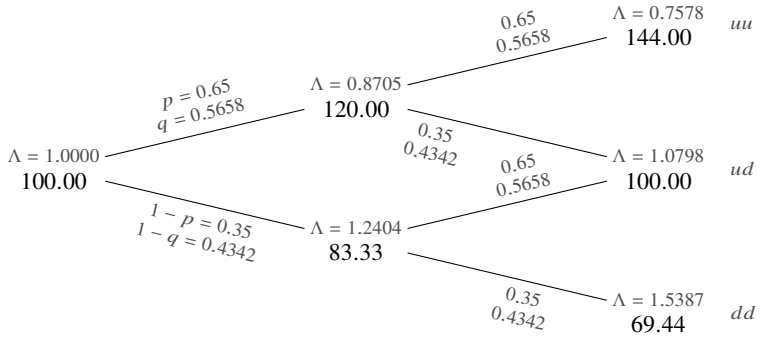


Figure 28.1: Example 28.2's tree: the real-world and risk-neutral probability on each branch, and Λ above the price at each node.

Example 28.2 The Radon–Nikodym derivative in four numbers

$S_0 = 100$, $u = 1.2$, $d = 1/1.2 = 0.8333$, $r = 4\%$, annual steps. The real-world probability of an up move is $p = 0.65$, an assumption about the world and one no one can observe directly.

The risk-neutral probability is not an assumption. It is forced:

$$q = \frac{e^{r\Delta t} - d}{u - d} = 0.565848$$

Both measures assign probabilities to the same three terminal nodes. The ratio of the two is the Radon–Nikodym derivative $\Lambda = dQ/dP$:

Path	$S(2)$	P	Q	Λ
dd	69.4444	0.122500	0.188488	1.538680
ud	100.0000	0.455000	0.491328	1.079842
uu	144.0000	0.422500	0.320183	0.757831
		1.000000	1.000000	$E^P[\Lambda] = 1.000000$

Three facts in that table are the whole of the theory.

$E^P[\Lambda] = 1$ exactly, and every value is positive: no path P allows is impossible under Q, which is what equivalence means. Together these are what make Λ a legitimate change of probability rather than an arbitrary reweighting.

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The reweighting is monotone in the outcome. Good states are weighted *down* (0.7578) and bad states *up* (1.5387). The direction is not automatic: it holds because $p > q$, which is this tree's way of saying the stock carries a risk premium, $e^{-2r}E^P[S(2)] = 106.02$ against $S_0 = 100$. That is what the market price of risk means, expressed as probabilities: a dollar delivered in a bad state is worth more to an investor than a dollar delivered in a good one, so a valuation that discounts every state at r has to make up the difference by treating bad states as more likely than they are. Were λ zero the two measures would coincide, and a negative premium would tilt dQ/dP the other way.

And the expectation comes out right: $e^{-2r}E^Q[S(2)] = 100.0000 = S_0$, so the discounted stock price is a Q-martingale.

Now price a two-year call struck at 100:

$e^{-2r}\,E^P[\text{payoff}]$	17.160733
$e^{-2r}\,E^Q[\text{payoff}]$	13.004930
Error from using the wrong measure	**+32.0**%

Same nodes, same payoffs—only the probabilities differ, and the price is wrong by a third.

In general the object has a name, and a theorem behind it. Equivalence says P and Q agree on which events are impossible, so neither puts weight where the other puts none; the **Radon–Nikodym theorem** says that this is enough for Q to be P reweighted by a single random variable. That variable is the **Radon–Nikodym derivative** $\Lambda = dQ/dP$, also called the **density** of Q with respect to P, positive, with

$$Q(A) = E^P[\Lambda\,\mathbf{1}_A] \quad \text{for every event } A, \qquad E^P[\Lambda] = 1. \tag{28.6}$$

The unit mean is forced by taking A to be everything, and expectations follow the same rule:

$$E^Q[X] = E^P[\Lambda X] \quad \text{for any payoff } X. \tag{28.7}$$

Changing measure is multiplying by Λ before averaging, never a change in which outcomes are possible. On a tree Λ is a list, one number per path (the list above); in continuous time it is a process, $\Lambda_t = E^P[\Lambda \mid \mathcal{F}_t]$, and Section 28.9 identifies which one. That conditional expectation is concrete. Λ is known only at the end, once the path is complete; Λ_t is its P-average over the paths still open at t. On the tree above, it is the probability-weighted average of the Λ values of the paths through the current node.

At $t = 0$ nothing is ruled out, so $\Lambda_0 = E^P[\Lambda] = 1$; at T one path is left and $\Lambda_T = \Lambda$. At the upper $t = 1$ node the open paths are uu and ud, so $\Lambda_1 = 0.65 \times 0.757831 + 0.35 \times 1.079842 = 0.870535$, which is q/p; the lower node gives $(1-q)/(1-p) = 1.240436$. On a tree the density is the running product of one-step ratios. It is a P-martingale by the **law of iterated expectations** (also called the tower property), which says that averaging a later conditional expectation over what is known earlier gives the earlier one, $E[E[X \mid \mathcal{F}_s] \mid \mathcal{F}_t] = E[X \mid \mathcal{F}_t]$ for $t \le s$ (here $0.65 \times 0.870535 + 0.35 \times 1.240436 = 1$ again), and it reweights any event settled by t: $Q(A) = E^P[\Lambda_t\,\mathbf{1}_A]$ for A in $\mathcal{F}_t$.

28.6 NUMERAIRES AND THE MEASURES THAT GO WITH THEM

Prices measured in units of the numeraire are what we make martingales, and the central result is that, in a market free of arbitrage, the measure to do it in always exists: for any numeraire N there is an equivalent measure Q^N under which V/N is a martingale for every traded asset V, which is (28.3). What matters is not that the measures exist but how to move between them, and that is one line.

The change-of-numeraire theorem. Let N and M be two numeraires with associated measures Q^N and Q^M. Their density, restricted to the information $\mathcal{F}_t$

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available at time t (the bar is read that way), is

$$\left.\frac{dQ^M}{dQ^N}\right|_{\mathcal{F}_t} = \frac{M(t)\,/\,M(0)}{N(t)\,/\,N(0)}. \tag{28.8}$$

The left side is the density process Λ_t of Section 28.5, written for this pair of measures. The proof is three lines and worth seeing: it shows the theorem restates the definition rather than adding a fact. For any traded V, the defining property of Q^M gives $V(0)/M(0) = E^M[V(t)/M(t)]$. Rewrite that expectation under Q^N by (28.7), with (28.8) as the density:

$$E^M\left[\frac{V(t)}{M(t)}\right] = E^N\left[\frac{V(t)}{M(t)}\cdot\frac{M(t)/M(0)}{N(t)/N(0)}\right] = \frac{N(0)}{M(0)}\,E^N\left[\frac{V(t)}{N(t)}\right] = \frac{N(0)}{M(0)}\cdot\frac{V(0)}{N(0)},$$

which is $V(0)/M(0)$, as required. The ratio in (28.8) is positive, both numeraires being strictly positive, and it is a Q^N-martingale with unit expectation, so it is a valid density. ■

One specialization generates most of Part VI. Taking $N = P(\cdot, T_2)$ and $M = P(\cdot, T)$, two zero-coupon bonds maturing in either order,

$$\left.\frac{dQ^T}{dQ^{T_2}}\right|_{\mathcal{F}_t} = \frac{P(t,T)\,/\,P(0,T)}{P(t,T_2)\,/\,P(0,T_2)}, \tag{28.9}$$

and because this density is a function of the interest rate path, (28.7) gives, for any X measurable at t,

$$E^T[X] = E^{T_2}\left[X\cdot\frac{dQ^T}{dQ^{T_2}}\right] = E^{T_2}[X] + \mathrm{Cov}^{T_2}\left(X, \frac{dQ^T}{dQ^{T_2}}\right), \tag{28.10}$$

since the density has unit mean. **Every adjustment in Chapter 30 is an application of** (28.10), with the relevant pair of numeraires in place of the two bonds. And the condition for no adjustment falls straight out: if X is uncorrelated with rates, the covariance vanishes and the two expectations agree. That is why a convexity, timing, or quanto correction is always a covariance, and why it is always zero when the variable is independent of the numeraire ratio: the discount rate for a convexity or timing correction, the exchange rate for a quanto.

Four numeraires account for essentially all of practice, and it is worth tabulating what each is for.

Numeraire	Measure	What becomes a martingale	Used for
Money market	Risk-neutral	V/B, $B(t) = \exp\left(\int_0^t r_s\,ds\right)$	Simulation
Zero-coupon bond $P(t,T)$	T-forward	Any T-maturity forward	Single payoffs
Swap annuity $A(t)$	Swap	The forward swap rate	Swaptions, CMS
Foreign money market	Foreign risk-neutral	Foreign assets	Quantos, FX

Two remarks are easy to miss.

The choice of numeraire does not change any price. That is the content of the theorem, and Section 31.7 verifies it numerically, because the invariance is worth seeing rather than accepting.

But it changes the difficulty enormously, and that is the entire practical point. A quantity that requires a simulation under one numeraire may have a closed form under another.

28.7 WHY THE FORWARD PRICE IS AN EXPECTATION

Chapter 5 derived the forward price from arbitrage, in the form $F(t,T) = S_t/P(t,T)$ that holds whatever rates do; Chapter 30 will open by asserting that a forward price is an expectation under the right measure. This section makes "the right measure" precise, and shows what goes wrong under the wrong one.

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Under the T-forward measure (numeraire $P(t,T)$), the forward price, which equals the asset divided by $P(t,T)$, is a martingale. For a stock,

$$\frac{S_0}{P(0,T)} = E^T\left[\frac{S_T}{P(T,T)}\right] = E^T[S_T]$$

since $P(T,T) = 1$. So $F(0,T) = E^T[S_T]$, exactly, which is the result Chapter 30 uses throughout.

While the forward price is a martingale under the T-forward measure, the futures price is a martingale under the risk-neutral measure, where the money market account is the numeraire. This can be shown as follows. The money market account $B(t)$ follows

$$dB = rB\,dt$$

Write $H(t)$ for the futures price for delivery at T. A futures contract costs nothing to enter and pays dH in cash as it accrues, so consider holding $B(t)$ contracts at every t and keeping the proceeds in the account. The strategy starts from nothing, and its value $V(t)$ is the balance in the account, earning interest r. Since a futures position is itself worth zero once it has been marked to market, the change in V can be written as

$$dV = \underbrace{rV\,dt}_{\text{interest on the balance}} + \underbrace{B\,dH}_{\text{settlement on } B(t) \text{ contracts}}$$

Those are the only cash flows there are, which is what makes the strategy self-financing. Dividing through by the account,

$$d\left(\frac{V}{B}\right) = dH, \qquad \text{so} \qquad \frac{V(t)}{B(t)} = H(t) - H(0)$$

Section 28.6 makes V/B a Q-martingale, because V is the value of a self-financing portfolio of traded assets; so the futures price H is a Q-martingale too. At delivery $H(T) = S_T$, which gives the pair

$$\text{futures price} = E^Q[S_T], \qquad \text{forward price} = E^T[S_T] \tag{28.11}$$

Two prices for the same delivery, two measures, and the difference between them is the futures–forward convexity adjustment of Section 30.3. Section 5.12 reaches it without the machinery, by asking when the variation margin arrives.

28.8 THE ANNUITY MEASURE, AND WHERE CMS COMES FROM

The third numeraire in the table is the one that is least intuitive and does the most work.

A swap that starts at T_0 pays its fixed leg on dates $T_1, \ldots, T_N$, and the **annuity**

$$A(t) = \sum_{i=1}^{N} \tau_i\, P(t,T_i)$$

is the present value of the fixed leg per unit of fixed rate on unit notional, τ_i being the year fraction of the ith period. It is a portfolio of zero-coupon bonds, hence a traded asset, hence an admissible numeraire.

The forward swap rate is

$$S(t) = \frac{P(t,T_0) - P(t,T_N)}{A(t)}$$

a traded asset divided by the annuity, so it is a martingale under the annuity measure Q^A. That is the entire justification for Section 29.6.

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Now notice what a swaption's payoff looks like. It is $A(T_0)\max(S(T_0)-K,0)$: the annuity appears in the payoff. Valuing it with A as numeraire,

$$V_0 = A(0)\,E^A[\max(S(T_0)-K,0)]$$

and the annuity cancels. With S a martingale and lognormal, Black's formula follows exactly, which is why the swaption convention works as well as it does.

That cancellation is this chapter's thesis in its cleanest form. Nothing was approximated and no model was chosen—a numeraire was picked so that the awkward part of the payoff divided out. Most of Part VI consists of cases where the cancellation is imperfect, and what fails to cancel is precisely what has to be computed.

The nearest such case differs from a swaption only in how the rate is paid, and it recurs throughout Part VI. In a **constant-maturity swap (CMS)**, one leg pays a long swap rate (the ten-year rate, say) observed afresh on each reset date, against a fixed rate or a short floating rate on the other leg. The floating leg of an ordinary swap references a rate whose own life matches the period it accrues over: the three-month rate set in January covers January to April, and is paid at the end of it. A CMS leg breaks that alignment deliberately. The ten-year rate set in January is paid three months later, on a quarterly accrual, and the ten-year swap it is the rate *of* never happens.

Investors buy the leg for the exposure it isolates: a coupon that follows long rates without the maturity, the duration, or the balance-sheet consequences of holding a ten-year bond, and dealers hedge the resulting book in swaptions. It is also the building block of steepener notes, which pay a multiple of a CMS spread, the difference between two CMS rates of different tenors, such as the ten-year rate less the two-year.

The coupon pays $S(T_0)$ on a *single* date T_p rather than over the annuity, so the numeraire it calls for is the zero-coupon bond $P(\cdot,T_p)$ rather than A. Under that measure S is no longer a martingale, the annuity no longer cancels, and what is left behind is the covariance between S and the ratio of the two numeraires.

That covariance is the CMS convexity adjustment, which Section 30.5 computes, sizes, and draws. What belongs here is where it comes from: it is not a correction for a modeling deficiency and not an empirical fudge, but the price of quoting a rate that is a martingale under one measure in a contract that settles under another.

Even its sign follows from the cancellation that failed. A swap rate that rises raises the coupon linearly but lowers the value of the annuity the coupon would have been paid over, and the CMS contract keeps the first effect while discarding the second—so the adjustment is positive. The holder of a CMS coupon has been handed that convexity, and the adjustment is what it costs.

28.9 GIRSANOV

Changing the measure changes the drift and leaves the volatility alone.

Formally: let dW^N be a Wiener process under Q^N with correlation ρ to the numeraire ratio M/N, whose volatility is $\sigma_{M/N}$. Moving to Q^M,

$$dW^M = dW^N - \rho\,\sigma_{M/N}\,dt$$

is a Wiener process under Q^M. It is a new process and not the old one relabeled (the two differ by that drift), and only the part of dW^N that moves with the numeraire is shifted, which is where ρ comes from. Every process therefore picks up a drift equal to the covariance between its own logarithm and the log of the numeraire ratio, and a process uncorrelated with that ratio keeps the drift it had. That single sentence generates Chapter 30's three adjustments and Chapter 32's two drift conditions, each derived where it is used.

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The simplest case shows the whole mechanism. Take two assets following geometric Brownian motions under the risk-neutral measure, with correlation ρ between their Wiener processes:

$$\frac{dS_1}{S_1} = r\,dt + \sigma_1\,dW_1, \qquad \frac{dS_2}{S_2} = r\,dt + \sigma_2\,dW_2$$

Now elect S_2 as the numeraire. The numeraire ratio is S_2/B, whose volatility is σ_2, so the rule above gives S_1 a drift equal to its covariance with $\ln(S_2/B)$, which is $\rho\sigma_1\sigma_2$, and gives S_2 its own variance σ_2^2:

$$\frac{dS_1}{S_1} = (r + \rho\sigma_1\sigma_2)\,dt + \sigma_1\,dW_1^{S_2}, \qquad \frac{dS_2}{S_2} = \left(r + \sigma_2^2\right)dt + \sigma_2\,dW_2^{S_2}$$

The two Wiener processes are shifted by different amounts, $dW_1^{S_2} = dW_1 - \rho\sigma_2\,dt$ and $dW_2^{S_2} = dW_2 - \sigma_2\,dt$, because S_2 moves with itself perfectly and with S_1 only through ρ.

Both drifts moved and neither volatility did. σ_1 and σ_2 multiply the new Wiener processes unchanged; what changed is the rate at which each asset is expected to grow, and by exactly the covariance the rule names.

The check is that the ratio is now a martingale. Itô's lemma applied to $X = S_1/S_2$ gives it a drift of

$$(r + \rho\sigma_1\sigma_2) - \left(r + \sigma_2^2\right) + \sigma_2^2 - \rho\sigma_1\sigma_2 = 0$$

as it must have, since X is a traded asset divided by the numeraire. Its volatility is $\sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$, so the change of numeraire has already produced the volatility that Section 28.11 prices an exchange option on, before any option has been mentioned.

Why volatility is invariant is the part worth understanding rather than memorizing. A change of measure reweights the probability of paths. It can make an upward path more likely and a downward path less likely, which moves the mean. But it cannot change how far apart the paths are, because the paths themselves are unchanged—only their probabilities move.

Two consequences follow directly and are used constantly.

Volatility is the same in every measure, so a volatility estimated from real-world data can be used in a risk-neutral pricing model. Without this, no calibration in this book would work, since historical volatility and implied volatility would not even be comparable quantities.

Expected returns are not. A traded asset drifts at μ under P and at r under Q, and $\mu - r = \lambda\sigma$, the λ of (28.4), reached here by a change of measure rather than by Example 28.1's hedge.

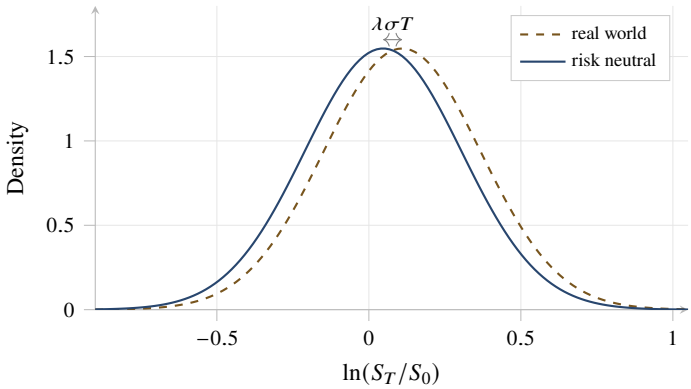


Figure 28.2: Example 28.2's tree in continuous time, at its two-year horizon: one density displaced by $\lambda\sigma T$.

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Figure 28.2 draws the theorem on Example 28.2's tree: one density twice, displaced by $\lambda\sigma T = 0.0626$ and not otherwise altered. What is invariant is the volatility, the standard deviation $\sigma\sqrt{T}$ of the log return, and that is a continuous-time fact, not a general one. Reweighting can change a variance, and in the two-step tree it does, mildly; what rules it out here is that volatility is pathwise, the quadratic variation of $\ln S$, and equivalent measures agree path by path. In the level of S_T the two give lognormals of visibly different shape, which is why the figure is drawn in logs.

28.10 WHAT $N(d_1)$ IS

Every numeraire so far has been the money market account, a bond, or a portfolio of bonds. Nothing in Section 28.6 required that, and dropping the restriction explains a formula the reader has known since Chapter 15.

Split the Black–Scholes call into the two pieces it obviously has:

$$c = e^{-rT}E^Q\left[S_T\mathbf{1}_{\{S_T>K\}}\right] \;-\; Ke^{-rT}\,Q(S_T > K) \tag{28.12}$$

The second piece is the easy one. $Q(S_T > K)$ is a probability under a measure in which $\ln S_T$ is normal with mean $\ln S_0 + (r - \sigma^2/2)T$, and it evaluates to $N(d_2)$. So $N(d_2)$ is the risk-neutral probability that the option finishes in the money, which is the one part of the formula most readers can already interpret.

The first piece is the awkward one, because it is the expectation of a *product*: a terminal price multiplied by an indicator of an event that depends on that same price. Done directly it means integrating S_T against a lognormal density over a half-line and completing a square.

Choose the numeraire to make the product disappear. The payoff $S_T\mathbf{1}_{\{S_T>K\}}$ is the promise to deliver one share if the share finishes above K. Value it using the *stock itself* as the numeraire, which is admissible since a non-dividend-paying stock is traded and strictly positive:

$$\frac{V_0}{S_0} = E^S\left[\frac{S_T\mathbf{1}_{\{S_T>K\}}}{S_T}\right] = E^S\left[\mathbf{1}_{\{S_T>K\}}\right] = Q^S(S_T > K)$$

Dividing by S_T is exactly what cancels the S_T in the payoff, and an expectation of a product has become a probability. So $V_0 = S_0\,Q^S(S_T > K)$.

What remains is that probability, and Girsanov supplies it. Moving from the money-market numeraire to the stock shifts every drift by its covariance with $\ln S$; for the stock itself that covariance is σ^2, so its log-drift under Q^S is $r + \sigma^2/2$ rather than the $r - \sigma^2/2$ it has under Q. *The variance is untouched*, and the same calculation that produced $N(d_2)$ now produces

$$Q^S(S_T > K) = N(d_1), \qquad d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

Substituting both pieces into (28.12) gives $c = S_0N(d_1) - Ke^{-rT}N(d_2)$.

This says something about the Black–Scholes–Merton formula that the usual derivation hides. $N(d_1)$ and $N(d_2)$ are the probability of *the same event*, the option finishing in the money, computed under *two different measures*. The formula is one event priced in two units of account: shares for the first term, cash for the second. Calling $N(d_1)$ "the delta" is true and unilluminating; it is a probability, and it differs from $N(d_2)$ only because it is measured in a different numeraire. Example 28.3 puts numbers on the gap.

Example 28.3 The same event, two probabilities

A one-and-a-half-year call, $S_0 = 100$, $K = 110$, $r = 4\%$, $\sigma = 28\%$. Then $d_1 = 0.068498$ and $d_2 = -0.274431$, so

$$N(d_1) = 0.527305, \qquad N(d_2) = 0.391877$$

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and the call is worth $100 \times 0.527305 - 110e^{-0.06} \times 0.391877 =$ **12.1344**.

The option is out of the money and the two probabilities disagree by 13.5 percentage points. Under the risk-neutral measure it finishes in the money **39.19%** of the time; under the stock measure, **52.73%**—more likely than not.

Neither number is a forecast, and the gap is not an error: the two $N(\cdot)$ terms are one event measured in two units of account.

The stock measure over-weights the paths that finish high, because it measures value in shares and shares are worth more precisely on those paths. That reweighting is the whole content of d_1 versus d_2, and the gap $\sigma\sqrt{T} = 0.3429$ between them is nothing but the volatility of the numeraire ratio.

28.11 EXCHANGING ONE ASSET FOR ANOTHER

The technique is at its most persuasive on a payoff that looks genuinely harder. An **exchange option** confers the right to give up asset 2 and receive asset 1:

$$\max(S_1(T) - S_2(T),\, 0)$$

Two assets, two volatilities, a correlation, and *a strike that moves*. Attacked directly this is a two-dimensional problem: a partial differential equation in two state variables, or a double integral over a bivariate lognormal.

Use asset 2 as the numeraire and the second dimension disappears. Asset 2 is traded and strictly positive, so it is admissible, and

$$\frac{V_0}{S_2(0)} = E^{S_2}\left[\frac{\max(S_1(T) - S_2(T),\, 0)}{S_2(T)}\right] = E^{S_2}[\max(X_T - 1,\, 0)]\,, \qquad X = \frac{S_1}{S_2}$$

The moving strike has become the number 1, which is what electing an asset as the unit of account buys. And X is a traded asset divided by the numeraire, so it is a martingale under Q^{S_2}, with no drift to estimate.

The volatility of X follows from Itô's lemma applied to a ratio:

$$\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\,\sigma_1\sigma_2} \tag{28.13}$$

A driftless lognormal struck at 1 is Black's formula, and unwinding the numeraire gives **Margrabe's formula**:

$$V_0 = S_1(0)\,N(d_1) - S_2(0)\,N(d_2) \tag{28.14}$$

where

$$d_1 = \frac{\ln(S_1(0)/S_2(0)) + \sigma^2 T/2}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}$$

Four features of the result are worth stopping on.

The interest rate is absent. In units of asset 2 there is nothing to finance, so no rate can appear. A trader who asks which curve to discount an exchange option on has misunderstood the instrument.

The Black–Scholes–Merton formula is the special case. Let asset 2 be K zero-coupon bonds maturing at T, so $S_2(0) = Ke^{-rT}$ and $\sigma_2 = 0$. Then (28.13) collapses to σ_1 and (28.14) reads

$$V_0 = S_1(0)\,N(d_1) - Ke^{-rT}\,N(d_2)$$

An ordinary call is an option to exchange cash for stock, and the two formulas agree to fifteen decimal places because they are the same formula.

And the reading of $N(d_1)$ and $N(d_2)$ from Section 28.10 survives the generalization.

$$N(d_2) = Q^{S_2}(S_1(T) > S_2(T))\,, \qquad N(d_1) = Q^{S_1}(S_1(T) > S_2(T))$$

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One event, priced first in units of asset 2 and then in units of asset 1. Which asset is the numeraire is a choice, and the formula is symmetric in a way the Black–Scholes special case conceals.

And the correlation is what the value turns on. Example 34.8 runs it from −0.5 to +0.99 and the value falls sevenfold, while no interest rate enters anywhere—the input a desk can observe does not affect the price, and the input it cannot observe determines it.

Exchange options are rarely labeled as such. An asset manager paid on performance against a benchmark holds one, since the fee is a call on the fund's value with the index as the strike. A stock-for-stock merger is one: a shareholder offered a fixed ratio of the acquirer's shares holds the right to exchange the target for the acquirer, which is why deal terms are quoted with collars and why the arbitrage community prices correlation. A gas-fired power station is a strip of them (Section 36.6's spark spread option is the option to exchange fuel for electricity, exercised hour by hour), one of the largest applications of (28.14) by notional. Chapter 34 develops the multi-asset family and Chapter 36 the energy applications; all of them are one line of change of numeraire.

28.12 WHEN A NUMERAIRE FAILS

Every change of numeraire in this chapter has assumed two things about the asset in the denominator: that it is **traded**, meaning its price is the value of a self-financing portfolio, and that its price is **strictly positive**. Both assumptions fail in practice. The four cases below run in increasing order of what they cost. Each of the first three violates one of the two conditions; the fourth violates neither, and is the most consequential of all.

1. The price is not a self-financing portfolio

A stock paying dividends is not a numeraire, because the holder receives cash along the way, so its price is not the value of a self-financing position. The repair is complete and costs nothing: use the *reinvested* stock instead. Chapter 5's dividend adjustments are that same repair.

2. The price can reach zero

Dividing by a numeraire that can vanish is not permitted, and defaultable assets vanish. An option on a credit default swap takes as its numeraire the forward risky annuity of Section 25.3 (the present value of one basis point paid until default or maturity), under which the forward CDS spread is a martingale. But that annuity is worth nothing in default, so the measure it defines gives probability zero to an event that Q does not, and the two are therefore not equivalent in the sense (28.1) requires.

Single-name convention steps around this by making the option knock out at default. Index options cannot, because they are written not to cancel: the protection buyer is owed the losses accumulated before expiration, which is what **front-end protection** is, and the correction matters most in precisely the state that destroys the numeraire. Morini and Brigo (2011) construct the conditional measure that repairs it. Negative rates are not an instance of this failure, despite the association (the cash annuity of the next subsection *rises* as rates fall), and what broke in the euro and yen swaption markets after 2014 was the lognormal quoting convention, repaired by Section 29.8's shifted and normal conventions.

3. The numeraire is not an asset at all

This is the quiet failure, and it is the one that costs money every day. The cash-settled swaption convention values the contract with $G(S) = \sum_{i=1}^{mn} m^{-1}(1 + S/m)^{-i}$ for an n-year swap paying m times a year, a formula applied to the swap rate rather than a portfolio of bonds, so it is not a traded asset, and it does not cancel against the numeraire the way Section 28.8 required. The

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cash annuity falls as rates rise: on a ten-year annual-pay swap, from 8.98 at a 2% swap rate to 7.36 at 6%, so it is smallest exactly where a payer swaption pays most, and that covariance survives. On an at-the-money five-year option on a ten-year swap, at a 3% forward rate and 25% volatility, it is worth about 9.9 basis points of a 66-basis-point premium (both quoted as a rate, per unit of annuity), or 15%, widening to 28% at a volatility of 40% because it is a convexity term.

The contract pays $G(S_T)(S_T - K)^+$ in cash at expiration, so with the true annuity as numeraire its value is

$$V_0 = A_0 \, E^A\left[\frac{G(S_T)}{A_T}\,(S_T - K)^+\right] \tag{28.15}$$

and the ratio inside is the whole difficulty: A_T is a portfolio of bonds depending on the entire curve at expiration, $G(S_T)$ a formula in one rate, and nothing cancels. As written, the expectation runs over the *joint* law of the swap rate and the curve around it, which no swaption smile supplies.

One step removes the joint law, and it approximates nothing. The payoff depends on S_T alone, so conditioning on S_T and applying the law of iterated expectations gathers every unknown into a single function of the swap rate:

$$V_0 = A_0 \, E^A\left[\alpha(S_T)\, G(S_T)(S_T - K)^+\right], \qquad \alpha(s) \equiv E^A\left[\frac{1}{A_T}\;\middle|\; S_T = s\right]. \tag{28.16}$$

This is an identity, not a model. What remains is a one-dimensional integral against the *marginal* law of S_T, which a smile does supply, with everything unknown pushed into α, the **annuity mapping function**. Nor is α arbitrary: a unit payoff at expiration must be worth the discount bond, forcing $A_0 E^A[\alpha(S_T)] = P(0,T)$. Freezing the annuity at its forward level makes α the constant $1/G(F)$, which satisfies it exactly on a flat curve and yields the figure quoted above; desks run linear terminal-swap-rate models instead, taking α affine in s, and the difference shows in the wings.

A physically settled swaption has a price once the volatility is given. A cash-settled one has a price once the volatility *and* a model of the annuity are given—and a convention that reads as a settlement detail has moved the contract from the first category into the second.

4. Both conditions hold and the numeraire is still wrong

The risk-free bank account is traded and strictly positive, and it is not the numeraire for a collateralized trade. Nothing in the theorem is violated; the wrong admissible asset has been chosen, and that is why this case is the expensive one. It closes a loop opened in Part I.

Chapter 7 argued that a collateral agreement determines the discount curve, and Chapter 8 that the terms of the CSA change the value of a trade. In this chapter's language, those are statements about which numeraire is correct. A trade collateralized in cash earning the overnight rate is replicated by a position financed at that rate, so the correct numeraire is the *collateral account*, not a hypothetical risk-free bank account.

Three chapters' worth of apparatus follows from that one identification. *Multi-curve discounting* is the observation that the numeraire changed and the forecast curve did not. *ColVA* of Section 9.6 is the value of the optionality in *which* numeraire applies, since a CSA permitting several collateral currencies gives the poster a choice, and a choice of numeraire that one party controls has value. *And FVA* of Section 9.4 is what appears when the trade is *not* collateralized, so that no traded numeraire replicates the financing, which is why the FVA debate was, in part, a disagreement about whether a firm's own funding cost can define a numeraire, and why it could not be settled by better data.

The question "what is the numeraire" is not abstract; it is the question of what portfolio actually replicates the trade, and every time the market's answer to that changed (to the collateral

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account once CSAs set the discount rate after 2008, to SOFR for dollar cleared trades when the clearing houses switched discounting in October 2020 (Chapter 7), to the firm's own funding where no collateral was posted), a valuation adjustment appeared.

28.13 WHAT THE THEORY IS FOR

A fair question, at the start of a part that will use this material on every page, is what it buys.

It supplies the drift conditions. Chapter 32's HJM condition and LIBOR market model (LMM) drifts are Girsanov applied twice. Without the theory they are unmotivated formulas, and a desk cannot validate an unmotivated formula.

It makes Black's formulas exact. Section 29.3's caplet and Section 29.6's swaption are not approximations under their own measures—they are exact consequences of the martingale property, and knowing which measure makes each exact is what tells you when to trust them.

It generates Chapter 30's adjustments. Convexity, timing, and quanto adjustments are all the same computation: a rate is a martingale under one measure and the payoff lives under another, so a covariance term appears.

It explains where models are needed and where they are not. A market is complete (every payoff replicable) if and only if its martingale measure is unique, so the second fundamental theorem says in advance which prices will depend on a modeling choice, which is a genuinely predictive statement rather than a retrospective one. [TN28] proves both theorems and works out the range of prices that incompleteness leaves open.

And it disciplines the use of probabilities. The measure that prices correctly is not the one that forecasts correctly—Section 28.1's distinction, missed often enough to matter.

What it does not do is tell you which model to use. The theory says a measure exists; it does not say what the volatility structure is or how many factors the curve has. Chapters 31 and 32 answer those questions with data, and the theory could not.

SUMMARY

The market price of risk $\lambda = (\mu - r)/\sigma$ must be the same for any two assets driven by the same uncertainty; if it is not, a portfolio of the two is riskless and earns more than r. With several sources, $\mu - r = \sum_i \lambda_i \sigma_i$: risks price additively, derived from no-arbitrage rather than equilibrium. Risk-neutral valuation is the case $\lambda = 0$.

A change of measure reweights paths without changing which paths are possible. It moves drifts and leaves volatilities alone, which is why a volatility estimated from history can be used in a risk-neutral model at all, and the error from pricing under the wrong measure is not small: even on a two-period tree, real-world probabilities misprice a call substantially.

Valuation is the search for a ratio that is a martingale. For any numeraire (any traded, strictly positive, self-financing asset) there is a measure under which every price divided by it is driftless, so $V_0 = N_0 E[V_T/N_T]$. The choice changes no price and changes the difficulty enormously.

Four choices cover most of practice. A zero-coupon bond makes the forward rate a martingale, so Black's caplet formula is exact and the forward price is an expectation under that measure, which in general it is not under the risk-neutral one. The annuity does the same for the swap rate, and because a swaption's payoff contains the annuity it cancels; a CMS coupon pays on a single date, the cancellation fails, and what survives is the convexity adjustment. The stock itself turns $E^Q[S_T \mathbf{1}_{\{S_T>K\}}]$ into a probability, which is what $N(d_1)$ is, and a second risky asset prices the right to exchange one asset for another, in which no interest rate appears at all. Two results that look unrelated, the gap between the forward and $E^Q[S_T]$ and the futures–forward adjustment of Section 30.3, are one covariance term under two names, and Chapter 31 shows a third: the error from discounting a caplet at today's zero rate.

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Both conditions on a numeraire fail in practice, and two further failures cost more: the cash-settled swaption annuity is not a traded asset at all, so nothing cancels, and the risk-free bank account is not the numeraire for a collateralized trade, which is what Chapter 7's discount curve, ColVA, and FVA are all statements about.

A market is arbitrage-free if and only if an equivalent martingale measure exists, and complete if and only if that measure is unique; [TN28] proves both and says what has to change in continuous time. The second theorem is where every difficulty in this book lives: when a price depends on a modeling choice, it is because the market is incomplete in a way that can be named.

FURTHER READING

Harrison and Kreps, and Harrison and Pliska, are the papers that established the two fundamental theorems; they are demanding and are the right place to go for anyone who wants the results stated precisely rather than usefully. Geman, El Karoui, and Rochet's paper on change of numeraire is the source for Section 28.6. Baxter and Rennie's short book gives the ideas without measure-theoretic machinery, and Shreve's two volumes are the standard graduate treatment for someone who wants the machinery. Björk's book sits between them and is the one most often used for a first serious course. On the real-world versus risk-neutral distinction, the actuarial literature treats it operationally, because the profession has had to make the distinction workable for regulatory reporting; the papers on economic scenario generators are the useful entry point.

PRACTICE QUESTIONS

28.1 Using Example 28.1, construct the riskless portfolio and compute its excess return, then explain why the market price of risk must be the same for two assets driven by the same source of uncertainty.

28.2 Using Example 28.2, state the three values of $\Lambda = dQ/dP$, explain why they average to one under P rather than under Q, and explain why the reweighting is monotone in the terminal price.

28.3 Using Example 28.2, state the size of the pricing error from using real-world probabilities and explain why it is positive for a call.

28.4 Explain what it means for a ratio to be a martingale, and why finding one is the whole of the valuation procedure.

28.5 Name the four numeraires that cover practice, state the measure each induces, and explain why the choice changes no price but changes the difficulty enormously.

28.6 Explain why Black's swaption formula is exact rather than approximate, being specific about what cancels.

28.7 Explain why a CMS coupon breaks the cancellation that makes Black's swaption formula exact, identify the numeraire its payoff calls for, and say what quantity the resulting adjustment consists of.

28.8 Explain why Girsanov changes drifts and leaves volatilities alone, and what this makes possible throughout the book.

28.9 Explain why $N(d_1)$ and $N(d_2)$ are probabilities of the same event, name the measure each is computed under, and say why they differ.

28.10 Derive Margrabe's formula by taking asset 2 as the numeraire, and explain why no interest rate appears in the result.

28.11 State the two conditions a numeraire must satisfy, and give an instance in which each has

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failed in a real market.

28.12 Explain why the discount curve of Chapter 7, ColVA, and FVA are all statements about the choice of numeraire, and why that made the FVA debate unresolvable by data.

28.13 Using [TN28], state both fundamental theorems, and explain why stochastic volatility, jumps, and credit each produce incompleteness for a different specific reason.

FURTHER QUESTIONS

28.14 Derive the market-price-of-risk condition from the requirement that a riskless portfolio earn r, then show that risk-neutral valuation is the special case $\lambda = 0$ and explain in what sense the risk-neutral measure is a computational device rather than a belief.

28.15 Prove that volatility is invariant under a change of measure, and use the result to explain why a stochastic volatility model calibrated to option prices can be tested against realized volatility at all; then state precisely what such a test can and cannot detect.

28.16 The second fundamental theorem says incompleteness produces a range of arbitrage-free prices. For a stochastic volatility model, characterize that range, explain what determines where inside it the market trades, and connect your answer to Section 19.10's volatility risk premium.

28.17 This chapter is placed first in Part VI, on the argument that four of the part's six chapters would otherwise borrow machinery they have not been given. Construct the opposing case for teaching it last, on the argument that the theory is easier to motivate after seeing what it explains; identify which chapters would become harder and which easier, and give your own judgment with reasons.

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Chapter 29
Standard Market Models for Rate Derivatives

The interest rate options that trade in the largest volume (caps, floors, and swaptions) are priced by the market using a small family of formulas that are, in essence, Black's model applied three times over. All three trade over the counter, dealer to client under ISDA documentation, in the market Section 10.10 described; none is exchange-listed in any size. This chapter sets out those formulas, the conventions attached to them, and the two respects in which the standard apparatus has had to change: once because rates went negative, and once because the reference rate is no longer known at the start of the period it applies to.

It is worth being clear at the outset about what these models are and are not. They are not descriptions of how interest rates behave. No one believes a single forward rate is lognormal with constant volatility, and Chapters 31 and 32 develop models that take term structure dynamics seriously. What the standard market models provide is a *quotation convention*: a one-to-one map between a price and a number called a volatility, agreed by everyone, so that traders can talk about relative value without arguing about models.

That distinction matters more than it might appear. When a trader says three-month expiration swaptions are trading at 90 basis points, they are not asserting that rates are normally distributed. They are quoting a price in units everyone can invert. The model is a language, and its defects matter only where they make the language ambiguous—which, as Section 29.8 explains, is precisely what happened when rates fell below zero.

The chapter runs from the formula to the surface the market quotes on. Sections 29.1 to 29.4 are Black's model and the instruments it prices: bond options, caps and floors, and the backward-looking caplets the transition created. Sections 29.5 to 29.7 are the conventions attached to them: flat against spot volatilities, European swaptions, and the correlation that makes a cap and a swaption different trades. Sections 29.8 and 29.9 are what negative rates did to the quoting convention and the cube that results; Section 29.10 is how the books are hedged.

29.1 BLACK'S MODEL

Chapter 16 introduced Black's model for options on futures. Its natural home turns out to be interest rate derivatives, for a reason Section 28.6 made precise: each of the products in this chapter has a *natural numeraire* under which the relevant forward rate is a martingale, and under that measure the forward rate can be modeled directly without reference to how it got there.

In its general form, consider a payoff at time T equal to

$$\max(V_T - K, 0)$$

where V_T is the value at T of some variable whose forward value today is F_0. At T the forward and the spot coincide, $F_T = V_T$, so the payoff is equally $\max(F_T - K, 0)$, which is what allows the forward, a martingale under the measure that goes with the product's own numeraire, to be

modeled in place of the spot. If V_T is lognormal with volatility σ, the value today is

$$\boxed{c = P(0,T)\,[F_0\,N(d_1) - K\,N(d_2)]} \tag{29.1}$$

with

$$d_1 = \frac{\ln(F_0/K) + \sigma^2 T/2}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T} \tag{29.2}$$

and the corresponding put

$$p = P(0,T)\,[K\,N(-d_2) - F_0\,N(-d_1)] \tag{29.3}$$

Three features make (29.1) the workhorse of this chapter. The discount factor $P(0,T)$ appears *outside* the expectation, so discounting and expectation separate cleanly. The drift of the underlying never appears—only its forward value, which is observable. And σ is the volatility of the *forward* value, not of the spot.

The entire content of the sections that follow is the identification, for each product, of what plays the role of F_0, what plays the role of the discount factor, and over what horizon σ is measured.

29.2 BOND OPTIONS

A bond option confers the right to buy or sell a bond at a specified price on a specified date. Standalone bond options trade, but the economically important ones are embedded: the issuer's right to call a bond before maturity, or the holder's right to put it back.

To apply (29.1), take V_T to be the bond price at option maturity T and F_0 the forward bond price:

$$F_0 = \frac{B_0 - I}{P(0,T)} \tag{29.4}$$

where B_0 is the bond's cash price today and I the present value of coupons paid before T. The volatility σ is that of the forward bond price. Example 29.1 works one through.

Yield volatilities. Bond price volatilities are inconvenient to think about, because a bond's price volatility falls mechanically as it approaches maturity. Traders therefore quote *yield* volatilities, converted using duration. From Section 4.11, $\Delta B \approx -BD^*\Delta y$. Both volatilities are proportional, so put the relation in proportional form:

$$\frac{\Delta B}{B} \;\approx\; -D^* y\,\frac{\Delta y}{y}$$

The price moves by D^*y times the yield's proportional move, and taking standard deviations,

$$\sigma \;\approx\; D^*\,y_0\,\sigma_y \tag{29.5}$$

where σ_y is the forward yield's proportional volatility and D^* the modified duration of the forward bond. A 5% yield with 15% yield volatility on a bond of modified duration 4 implies a price volatility of $4 \times 0.05 \times 0.15 = 3\%$. Example 29.2 shows what skipping the conversion costs.

Example 29.1 A European bond option

A one-year European call struck at 100 is written on a bond whose cash price today is 102.50. The bond pays coupons of 3 at three months and nine months. The relevant discount factors are $P(0, 0.25) = 0.9901$, $P(0, 0.75) = 0.9704$, and $P(0, 1) = 0.9608$. The forward bond price volatility is 4%.

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The present value of the coupons paid before option maturity is

$$I = 3 \times 0.9901 + 3 \times 0.9704 = 5.8815$$

so from (29.4) the forward bond price is

$$F_0 = \frac{102.50 - 5.8815}{0.9608} = 100.5605$$

Then

$$d_1 = \frac{\ln(100.5605/100) + \frac{1}{2} \times 0.04^2}{0.04} = \frac{0.0055893 + 0.0008}{0.04} = 0.15973$$

$$d_2 = 0.15973 - 0.04 = 0.11973$$

giving $N(d_1) = 0.56345$ and $N(d_2) = 0.54765$, so from (29.1)

$$c = 0.9608\,[100.5605 \times 0.56345 - 100 \times 0.54765] = 0.9608 \times 1.89596 = 1.822$$

or \$1.82 per 100 of face value. The coupons are subtracted because the option holder does not receive them.

Example 29.2 A quoted yield volatility, converted

Consider a two-year European call on a bond, struck at 98.00, when the forward bond price is 98.50 and $P(0,2) = 0.9124$. The screen quotes a *yield* volatility of 16%; the forward yield is 4.20% and the modified duration of the forward bond is 6.4.

Convert first. From (29.5),

$$\sigma \approx D^* y_0 \sigma_y = 6.4 \times 0.0420 \times 0.16 = 4.30\%$$

so the total volatility to expiration is $0.0430\sqrt{2} = 0.06082$. Then

$$d_1 = \frac{\ln(98.50/98.00) + \frac{1}{2} \times 0.06082^2}{0.06082} = 0.11408$$

$$d_2 = 0.11408 - 0.06082 = 0.05326$$

giving $N(d_1) = 0.54541$ and $N(d_2) = 0.52124$, so from (29.1)

$$c = 0.9124\,[98.50 \times 0.54541 - 98.00 \times 0.52124] = 0.9124 \times 2.64197 = 2.41$$

Now price the same option with 16% substituted directly into Black's formula, as though a yield volatility were a price volatility: the answer is **8.31**, three and a half times too large.

The conversion is not a refinement. A bond's price volatility and its yield volatility differ by the factor $D^* y_0$, which here is 0.269, and that factor is what the duration is doing in (29.5).

Callable and puttable bonds. A callable bond is an ordinary bond less a call option held by the issuer; a puttable bond is an ordinary bond plus a put held by the investor. Both can be valued by pricing the straight bond and the embedded option separately. The complication is that the embedded options are typically *Bermudan* (exercisable on any of several dates), which puts them beyond Black's model and into the tree methods of Chapter 31 and the simulation methods of Chapter 32. Black's model remains useful for a first approximation and for quoting.

Option-adjusted spread. Comparing callable bonds by yield is meaningless, because a high yield may reflect a wide credit spread or a valuable call the investor has written. The market's solution is the **option-adjusted spread** (OAS): the constant spread which, added to the entire benchmark discount curve, makes a model's price for the callable bond equal its observed market price.

The OAS is what remains of the bond's spread after removing the value of the embedded

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option, and it is the quantity across which callable bonds can legitimately be compared. Two cautions attach to it. First, it is thoroughly model-dependent: the option value depends on an assumed volatility, so a higher volatility assumption produces a larger option value and hence a smaller OAS for the same market price. An OAS quoted without its volatility assumption is not a well-defined number. Second, it is only as good as the exercise rule the model assumes: for a callable bond that rule is optimal exercise by the issuer, while for mortgages, whose prepayment option is the largest embedded option in the market, it has to be a behavioral prepayment model, because borrowers demonstrably do not exercise optimally.

29.3 INTEREST RATE CAPS AND FLOORS

An **interest rate cap** protects a floating-rate borrower against rates rising above a level K. It is written on a notional L over a sequence of accrual periods, and pays, for the period $[T_i, T_{i+1}]$ with accrual factor τ_i, at the period's end T_{i+1},

$$L\,\tau_i \max(R_i - K, 0) \tag{29.6}$$

where R_i is the reference rate for that period. A **floor** is the mirror image, paying $L\tau_i \max(K - R_i, 0)$.

The option covering one period is a **caplet**, or for a floor a floorlet, and the essential observation is that each one settles on its own period's rate, whatever the others do. A cap is a *portfolio* of caplets, not a compound option, so it can be valued one period at a time and summed.

Applying Black's model to a single caplet, with F_i the forward rate for $[T_i, T_{i+1}]$:

$$\text{caplet}_i = L\,\tau_i\,P(0,T_{i+1})\,[F_i\,N(d_1) - K\,N(d_2)] \tag{29.7}$$

with d_1, d_2 from (29.2) using the volatility σ_i of that forward rate over $[0, T_i]$: the rate fixes at T_i, so that is when the option's uncertainty is resolved, even though the payment does not arrive until T_{i+1}. Discounting is done with $P(0, T_{i+1})$, the bond maturing on the payment date, which, as Section 28.6 established, is exactly the numeraire under which F_i is a martingale, so no adjustment is required.

Caplets and bond options. A caplet is also a bond option, and the equivalence is an identity rather than an approximation. The reference rate is read off the bond price prevailing at T_i,

$$R_i = \frac{1}{\tau_i}\left(\frac{1}{P(T_i, T_{i+1})} - 1\right)$$

and the payoff (29.6) is already known at T_i, so discounting it back over the accrual period multiplies it by $P(T_i, T_{i+1})$ and the bond prices cancel:

$$L\,\tau_i\,P(T_i,T_{i+1})\max(R_i - K, 0) = L\,(1 + \tau_i K)\max\left(\frac{1}{1+\tau_i K} - P(T_i,T_{i+1}),\ 0\right)$$

A caplet is therefore $L(1 + \tau_i K)$ European *puts* on the T_{i+1} zero-coupon bond, expiring at T_i and struck at $1/(1 + \tau_i K)$. Puts, because the bond price falls when the rate rises. Nothing has been assumed about how the rate is distributed (the identity is between contracts, not between models), and it is what allows a model of the short rate to price a cap in closed form, since such a model delivers the distribution of $P(T_i, T_{i+1})$ directly. Section 31.7 does exactly that.

Example 29.3 Valuing a caplet

A caplet covers the three-month period beginning in one year, on a notional of \$10 million. The forward rate for that period is $F = 4.00\%$, the strike is $K = 4.25\%$, the volatility is $\sigma = 20\%$, the

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accrual factor is $\tau = 0.25$, and $P(0, 1.25) = 0.9520$.

Taking the horizon as $T = 1$ (the rate is fixed at the start of the period),

$$d_1 = \frac{\ln(0.04/0.0425) + \frac{1}{2} \times 0.20^2 \times 1}{0.20\sqrt{1}} = \frac{-0.060625 + 0.020}{0.20} = -0.2031$$

$$d_2 = -0.2031 - 0.20 = -0.4031$$

so $N(d_1) = 0.41953$ and $N(d_2) = 0.34344$. From (29.7),

$$\text{caplet} = 10{,}000{,}000 \times 0.25 \times 0.9520\,[0.04 \times 0.41953 - 0.0425 \times 0.34344]$$

$$= 2{,}380{,}000 \times (0.016781 - 0.014596) = 2{,}380{,}000 \times 0.002185 = \$5{,}200$$

Cap–floor parity. A useful identity follows from the payoffs: since $\max(R-K, 0) - \max(K-R, 0) = R - K$,

$$\text{cap} - \text{floor} = \text{payer swap struck at } K \tag{29.8}$$

A long cap and short floor at the same strike replicate a swap paying fixed at K and receiving floating. This is a model-free relationship, and it provides an immediate consistency check: cap and floor quotes that violate it are arbitrageable regardless of anyone's view on volatility.

Which caplets are included. Under LIBOR conventions the first caplet of a **spot-starting** cap (one whose first accrual period begins today) was conventionally excluded, because the first period's rate was already fixed and the caplet's payoff therefore known. That convention no longer makes sense, for the reason developed next.

Who buys a cap. The natural buyer is a borrower with floating-rate debt: a company or property owner whose loan resets quarterly buys a cap to hold its funding cost at the strike while keeping the benefit of lower rates, and lenders often require it.

Listed rate options. Caps and floors are over-the-counter. The exchange-traded counterparts are options on SOFR futures (including mid-curve options, which expire before the underlying futures contract) and on Treasury futures. They are priced with Black's model on the futures price, usually quoted in normal volatility, and need no convexity adjustment, because the futures price is already a martingale under the risk-neutral measure (Section 28.7). Asset managers and money-market funds use them to hold rate optionality without a bilateral contract.

29.4 BACKWARD-LOOKING CAPLETS

Section 29.3 quietly assumed the reference rate is fixed at the start of its accrual period, so that a caplet is an option expiring at T_i on a quantity known at T_i. Chapter 4 established that this is no longer how the reference rate works. A compounded risk-free rate accrues across $[T_i, T_{i+1}]$ and is not known until the period ends.

An option on such a rate is a **backward-looking** caplet, and it differs from its forward-looking predecessor in a way that is small in magnitude but conceptually important.

Nothing is exercised by decision here. A caplet settles automatically against the realized rate, so its "expiration" is the date the payoff becomes *known*, not a date on which somebody chooses: T_i when the rate is fixed in advance, T_{i+1} when it is compounded across the period. That is why the backward-looking version is an option at all despite the rate being undetermined at T_i—there is nothing to decide at T_i.

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How much uncertainty is there?

Consider the rate for the period $[T_1, T_2]$, with $\tau = T_2 - T_1$, and write σ for its volatility, in the sense of Section 29.1: the volatility of a forward value, taken as constant. Before T_1 no fixing has entered it yet, so it accumulates uncertainty exactly as a forward-looking rate does: variance $\sigma^2 t$ at time t. Its forward is the same F_0, as the pricing below shows. The two rates part company only once compounding starts.

Once compounding begins at T_1, something new happens. Each day's overnight fixing that has already been observed is locked in. As the period elapses, a growing fraction of the final compounded rate is already determined, so the remaining sensitivity to rate movements declines, linearly to a good approximation, reaching zero at T_2. This is the same decaying-delta effect that Section 30.3 notes for averaging futures.

Modeling that decay as linear, the rate's volatility at a time u inside the period is $\sigma(T_2 - u)/\tau$, falling from σ at T_1 to zero at T_2, and the variance accumulated between T_1 and T_2 is

$$\int_{T_1}^{T_2} \sigma^2 \left(\frac{T_2 - u}{\tau} \right)^2 du \;=\; \frac{\sigma^2 \tau}{3} \tag{29.9}$$

Adding the variance accumulated before T_1 gives the total:

$$\text{Var} = \sigma^2 \left(T_1 + \frac{\tau}{3} \right) \tag{29.10}$$

Figure 29.1 shows the two paths, drawn with $\tau = T_1$ so the effect is visible. The forward-looking rate stops accumulating uncertainty at T_1; the backward-looking rate continues past it, though at a decreasing rate as more of the average is locked in, adding a further $\tau/3$ years' worth of variance.

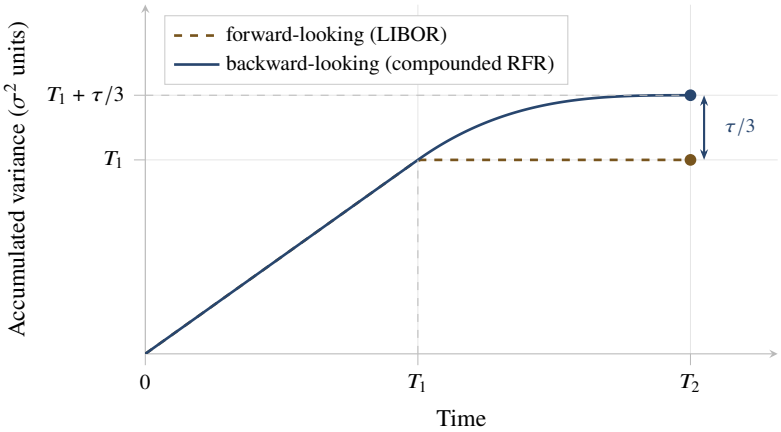


Figure 29.1: Uncertainty accumulated by a forward-looking and a backward-looking rate, drawn with $\tau = T_1$.

Pricing

The measure does not change: a caplet pays at T_2 either way, so $P(0, T_2)$ is the numeraire in both cases. What changes is the process modeled under it. Section 29.3 used the forward rate $F(t)$ for $[T_1, T_2]$, which stops moving at T_1, and a rate still accumulating at T_2 needs an extension of it.

The extension comes from a trading strategy: buy the T_1 bond and, at T_1, roll the proceeds at the overnight rate through to T_2. Its value $C(t)$ is a traded portfolio throughout, so $C(t)/P(t, T_2)$

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is a martingale under the T_2-forward measure, and therefore so is

$$R(t) = \frac{1}{\tau}\left(\frac{C(t)}{P(t,T_2)} - 1\right) \tag{29.11}$$

Before T_1, $C(t) = P(t,T_1)$ and $R(t)$ is the old forward rate unchanged; at T_2 it is the compounded rate the caplet settles against; and $R(0) = F_0$, so both caplets are struck against the same forward. Only the total variance changes: the martingale now runs to T_2 instead of freezing at T_1, and (29.10) is what it accumulates. So price a backward-looking caplet with Black's formula (29.7), but replace $\sigma^2 T_1$ with $\sigma^2(T_1 + \tau/3)$ wherever it appears. Equivalently, use an effective volatility

$$\sigma_{\text{eff}} = \sigma\sqrt{\frac{T_1 + \tau/3}{T_1}} \tag{29.12}$$

over the horizon T_1. Example 29.4 applies it to the caplet of Example 29.3.

Example 29.4 The same caplet, backward-looking

Take Example 29.3 and treat the rate as compounded in arrears rather than fixed in advance. Here $T_1 = 1$ and $\tau = 0.25$, so

$$\sigma^2\left(T_1 + \frac{\tau}{3}\right) = 0.04 \times 1.08333 = 0.043333, \qquad \sqrt{0.043333} = 0.20817$$

Repeating the calculation with $\sigma\sqrt{T}$ replaced by 0.20817:

$$d_1 = \frac{-0.060625 + \frac{1}{2} \times 0.043333}{0.20817} = -0.1871, \qquad d_2 = -0.3953$$

giving $N(d_1) = 0.42577$, $N(d_2) = 0.34631$, and

$$\text{caplet} = 2{,}380{,}000\,[0.04 \times 0.42577 - 0.0425 \times 0.34631] = \$5{,}505$$

against \$5,200 for the forward-looking version—about **5.9% more valuable**.

Two observations follow.

Backward-looking caplets are worth more. Since $T_1 + \tau/3 > T_1$ always, the option on a compounded rate is unambiguously the more valuable. It is exposed to rate movements for longer.

The first caplet is no longer trivial. Under LIBOR the first caplet of a spot-starting cap had $T_1 = 0$ and hence zero variance: the rate was already fixed, and the caplet was a known cash flow, conventionally excluded from the quote. Under a compounded rate, $T_1 = 0$ gives variance $\sigma^2\tau/3$, which is not zero. The first caplet of a spot-starting cap now carries genuine optionality, and the old convention of dropping it silently understates the price.

The effect scales with the accrual length. For quarterly periods $\tau/3$ is a month, negligible against a five-year expiration but material against a three-month one. For annual periods, or for the front of the curve, it is worth having right.

29.5 FLAT AND SPOT VOLATILITIES

A cap is a portfolio of caplets, each with its own volatility. The market, however, quotes a single number per cap. This creates two distinct volatility concepts that are easily confused and must be kept apart.

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- A **flat volatility** is the single volatility which, applied to *every* caplet in a given cap, reproduces the cap's market price. It is a quotation device, not a property of any individual caplet.
- A **spot volatility** (also called a caplet or forward volatility) is the volatility of one specific caplet.

A flat volatility is thus a kind of average of the spot volatilities of the caplets making up that cap. The two coincide only for the shortest cap, the one containing a single caplet. That averaging is visible in Figure 29.2: because a flat volatility summarizes every caplet up to its maturity, its curve peaks later, peaks lower, and decays more slowly than the spot curve underneath it.

Extracting spot volatilities from quoted flat volatilities is a bootstrap, exactly parallel to the zero curve bootstrap of Chapter 4. The price of a two-year cap less the price of a one-year cap is the value of the caplets falling in the second year; those caplets' spot volatilities are whatever reproduces that difference, given the spot volatilities already extracted for year one. Example 29.5 carries out one step.

Example 29.5 Bootstrapping spot volatilities

Annual-reset caps struck at 4.00% are quoted at flat volatilities of 20.0% for the two-year and 21.0% for the three-year. The relevant forward rates are 4.00% for $[1, 2]$ and 4.30% for $[2, 3]$; discount factors are $P(0, 2) = 0.9246$ and $P(0, 3) = 0.8890$. Work per unit of notional.

Two-year cap. It contains a single caplet, covering $[1, 2]$ (the first period is excluded, as under the LIBOR convention of Section 29.3), so its spot volatility is the quoted flat volatility, $\sigma_1 = 20.0\%$. At the money with $T = 1$, $d_1 = 0.1$ and $d_2 = -0.1$, so

$$\text{caplet}_{[1,2]} = 0.9246 \times 0.04\,(0.53983 - 0.46017) = 0.0029460$$

Three-year cap. Valuing *both* caplets at the quoted flat volatility of 21% gives 0.0030928 for $[1, 2]$ and 0.0058161 for $[2, 3]$, a total cap price of

$$0.0030928 + 0.0058161 = 0.0089088$$

The bootstrap step. The $[2, 3]$ caplet must account for the remainder:

$$0.0089088 - 0.0029460 = 0.0059628$$

Solving Black's formula numerically for the volatility that reproduces this value gives

$$\sigma_2 = 21.7\%$$

So a flat volatility curve of 20.0%, 21.0% corresponds to spot volatilities of 20.0%, 21.7%. The second caplet's own volatility exceeds the three-year cap's quote, because that quote is diluted by the lower-volatility first caplet inside it. Using 21% to value the $[2, 3]$ caplet in isolation would understate it by about 2.5%.

The characteristic shape in Figure 29.2, rising to a hump at two to three years and then declining, is remarkably persistent across currencies and decades. The usual explanation is that very short rates are anchored by the central bank's current policy setting and very long rates by expectations about the long run, while the intermediate maturities carry most of the uncertainty about the path of policy in between.

A flat volatility quoted for a five-year cap and a flat volatility quoted for a ten-year cap describe overlapping portfolios; the difference between them is not the volatility of anything. Comparing the two directly, or interpolating between them, is a common and expensive error. Convert to spot volatilities before doing anything analytical.

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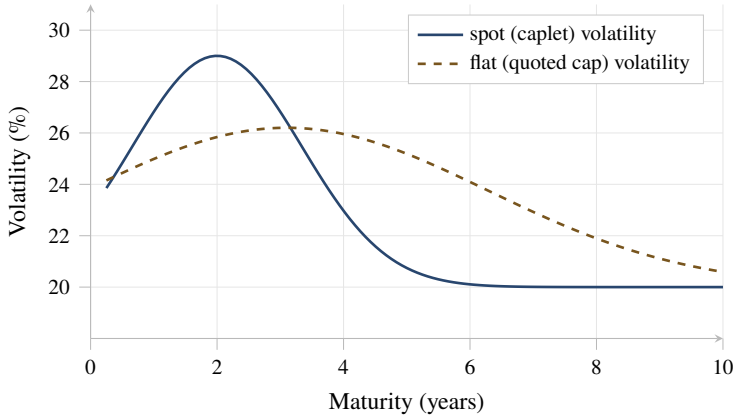


Figure 29.2: Representative shapes of flat and spot volatility curves. Illustrative rather than market data.

29.6 EUROPEAN SWAPTIONS

A **swaption** is an option to enter an interest rate swap. A *payer* swaption confers the right to pay fixed and receive floating; a *receiver* swaption the right to receive fixed. They are described by two maturities (the option expiration and the underlying swap's tenor), so a "5×10" is a five-year option on a ten-year swap.

The key to valuing them is to recognize what the option is really on. At expiration T, the holder of a payer swaption struck at K exercises if the prevailing swap rate S_T exceeds K, gaining $\tau(S_T - K)$ on each of the n payment dates. The total value at exercise is

$$L \sum_{i=1}^{n} \tau_i P(T, t_i) \max(S_T - K, 0) = L\, A(T) \max(S_T - K, 0)$$

where $T < t_1 < \cdots < t_n$ are the payment dates of the fixed leg of the swap the option delivers, τ_i the accrual factor for each, and $A(T)$ the annuity of Section 28.8, the present value at T of one unit paid on each of those dates. So a swaption is an option on the swap rate, with a payoff multiplied by the annuity. In the major currencies that fixed leg now pays annually (Section 7.1), so τ_i is about one and n is the swap's tenor in years, which is what Example 29.6 assumes. Legacy dollar swaps paid fixed semiannually against quarterly floating, and some currencies still do.

That structure points to the right numeraire. Taking the annuity as numeraire (Section 28.8), the swap rate is a martingale, and Black's model applies with the annuity replacing the discount factor:

$$\text{payer} = L\, A(0)\, [S_0\, N(d_1) - K\, N(d_2)] \tag{29.13}$$

$$\text{receiver} = L\, A(0)\, [K\, N(-d_2) - S_0\, N(-d_1)] \tag{29.14}$$

with d_1, d_2 from (29.2) using the forward swap rate S_0 and its volatility, and

$$A(0) = \sum_{i=1}^{n} \tau_i\, P(0, t_i) \tag{29.15}$$

The annuity is not merely a discount factor: it is the numeraire that makes the swap rate driftless, which is what licenses the use of Black's model in the first place. Chapter 30 exploits this directly: the CMS convexity adjustment arises precisely because a CMS coupon is paid at a single date rather than across the annuity.

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Example 29.6 Valuing an at-the-money swaption

Consider a five-year option on a five-year annual-pay swap, notional \$100 million. The forward swap rate is $S_0 = 4.00\%$, the strike is $K = 4.00\%$, and the volatility is $\sigma = 20\%$.

With a flat 4% curve, the discount factors for the five payment dates (years 6 through 10) are 0.790315, 0.759918, 0.730690, 0.702587, and 0.675564, so from (29.15)

$$A(0) = 3.659074$$

Because the option is at the money, $\ln(S_0/K) = 0$ and

$$d_1 = \frac{1}{2}\sigma\sqrt{T} = \frac{1}{2} \times 0.20\sqrt{5} = 0.22361, \qquad d_2 = -0.22361$$

giving $N(d_1) = 0.58847$ and $N(d_2) = 0.41153$. From (29.13),

$$\text{payer} = 100{,}000{,}000 \times 3.659074 \times 0.04 \times (0.58847 - 0.41153) = \$2{,}589{,}700$$

or about 2.59% of notional. At the money the payer and receiver have equal value, as put–call parity requires: their difference is a forward swap struck at the forward rate, which is worth zero.

Swaptions and bond options. A payer swaption is equivalent to a put option on a coupon bond: exercising the right to pay fixed at K is exercising the right to sell, at par, a bond paying coupon K. Receiver swaptions correspond to call options on the same bond, but the swap-rate formulation is the market standard because the swap rate, unlike a bond price, does not drift mechanically toward par as expiration approaches.

Who uses swaptions. A borrower planning to issue fixed-rate debt buys a payer swaption to cap the rate it will pay. Mortgage servicers and insurers with minimum-rate guarantees buy receiver swaptions, because falling rates bring prepayments or deepen the guarantees. Asset managers and pension funds sell swaptions to earn the premium, which is why the market's volatility is not set by hedgers alone.

29.7 CAPS VERSUS SWAPTIONS: THE ROLE OF CORRELATION

Caps and swaptions written on the same schedule and strike look superficially like the same trade. They are not, and the difference between them is one of the more instructive facts in interest rate derivatives.

A cap pays $\sum_i \tau_i \max(F_i - K, 0)$: it is a **portfolio of options**, one on each forward rate, exercised independently. A payer swaption pays $A \max(S - K, 0)$, and since the swap rate is approximately a weighted average of the same forward rates, it is an **option on a portfolio**.

The distinction has a definite direction. For any weights w_i summing to one,

$$\sum_i w_i \max(F_i - K, 0) \;\geq\; \max\left(\sum_i w_i F_i - K,\; 0\right) \tag{29.16}$$

because a sum of positive parts is at least the positive part of the sum. Hence, on comparable terms, **a cap is always worth at least as much as the corresponding swaption**.

The intuition is diversification, running backward. The cap holder collects on whichever rates happen to exceed the strike, and is not penalized for those that do not. The swaption holder owns one option on the average, so a rate that finishes far above the strike is partly canceled by another that finishes below it.

That immediately tells us what governs the size of the gap: the **correlation** between forward rates. If all forward rates started level and moved in perfect lockstep, they would always finish on the same side of the strike, no cancellation could occur, and (29.16) would hold with equality.

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The less correlated they are, the more the averaging suppresses the swap rate's variance, and the cheaper the swaption becomes relative to the cap.

This can be made quantitative. Suppose n forward rates each have volatility σ, enter the swap rate with equal weight, and have uniform pairwise correlation ρ. The variance of the average is then $\sigma^2\,[1/n + (1 - 1/n)\,\rho]$, so the swap rate's volatility is

$$\sigma_{\text{swap}} = \sigma\sqrt{\rho + \frac{1-\rho}{n}} \tag{29.17}$$

At $\rho = 1$ this returns σ, and the two products carry the same volatility. At $\rho = 0$ it falls to $\sigma/\sqrt{n}$. For a ten-period swap with $\rho = 0.7$, the swaption's volatility is $\sigma\sqrt{0.73} = 0.854\,\sigma$—some 15% below the caplet volatility, purely from imperfect correlation.

The practical significance is that this runs in reverse. Correlations between forward rates are not directly observable, but cap and swaption prices are. The ratio between them is therefore the market's main source of information about forward-rate correlation, and it is the primary calibration input for the multi-factor term structure models of Chapter 32. A model calibrated to caps alone will price swaptions badly, because caps contain almost no information about correlation: each caplet depends on one rate at a time.

29.8 NORMAL AND SHIFTED LOGNORMAL MODELS

Every formula so far assumes the underlying rate is lognormal. That assumption carries an implication which went unexamined for thirty years: a lognormal variable cannot be negative or zero. When European rates fell below zero in the middle 2010s, Black's model did not merely become inaccurate—it became *undefined*. A negative forward rate cannot be substituted into $\ln(F_0/K)$.

Case Study 29.1 When Black's model broke

The European Central Bank moved its deposit facility rate below zero in June 2014. Within a year three-month Euribor had fixed below zero for the first time, in April 2015, and short euro swap rates had followed; by 2016 substantial portions of the euro and Swiss curves traded at negative yields. The Bank of Japan followed in January 2016.

For the swaption market this was not an academic inconvenience. Brokers quote volatility, not price, and the standard quotation convention had suddenly stopped being able to represent the market. Screens showed blanks. Risk systems that inverted Black's formula to imply a volatility returned errors on live positions.

The market responded in two ways within a matter of months. Some desks adopted a **shifted lognormal** convention, adding a fixed displacement (typically 1% to 3%) to both rate and strike so that the shifted quantities stayed positive and Black's formula could be applied unchanged. Others moved to quoting **normal** volatilities under the Bachelier model, which places no restriction on sign. Normal volatility eventually became the euro and yen market standard and has since been widely adopted for dollar swaptions as well.

The episode is a useful reminder that a modeling assumption can survive for decades not because it is right but because it has never been tested in the region where it fails.

The normal (Bachelier) model

Assume instead that the rate is normally distributed, with $dF = \sigma_N\,dW$ for a constant **normal volatility** σ_N quoted in basis points per annum rather than as a percentage. The value of a

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payer-type option is

$$c = P(0,T)\left[(F_0 - K)\,N(d) + \sigma_N\sqrt{T}\,\varphi(d)\right], \qquad d = \frac{F_0 - K}{\sigma_N\sqrt{T}} \tag{29.18}$$

where φ is the standard normal density. The formula is defined for any real F_0 and K, positive or negative.

At the money the expression collapses, since $d = 0$, $N(0) = 1/2$, and $\varphi(0) = 1/\sqrt{2\pi}$:

$$c_{\text{ATM}} = P(0,T)\,\sigma_N\sqrt{\frac{T}{2\pi}} \tag{29.19}$$

An at-the-money option's value is proportional to the normal volatility and to the square root of time. Traders use this constantly for order-of-magnitude checks.

Converting between conventions

Because both conventions are one-to-one maps between price and volatility, a price quoted one way can always be restated the other way. Near the money the relationship is approximately

$$\sigma_N \;\approx\; \sigma_{\text{LN}} \times F_0 \tag{29.20}$$

where σ_{LN} is the lognormal (Black) volatility. This is just the statement that a proportional volatility applied to a level gives an absolute volatility. Example 29.7 tests it on the swaption of Example 29.6.

Example 29.7 Lognormal and normal volatility for the same option

The swaption of Example 29.6 has an at-the-money value of 2.59% of notional, with $A(0) = 3.659074$. Dividing out the annuity, the option value per unit of annuity is

$$0.04 \times (0.58847 - 0.41153) = 0.0070776$$

Setting this equal to the at-the-money normal value $\sigma_N\sqrt{T/2\pi}$ from (29.19),

$$\sigma_N = \frac{0.0070776\sqrt{2\pi}}{\sqrt{5}} = \frac{0.017741}{2.23607} = 0.007934$$

or **79.3 basis points**. The rule of thumb (29.20) predicts $0.20 \times 4.00\% = 80$ basis points—within a basis point, which is why the approximation is used so freely.

Which to use

The two models make genuinely different statements about how rates move. Black's σ is a proportional volatility ($dF = \sigma F\,dW$), so the standard deviation of the rate's change, measured in basis points, scales with the level: at a quoted 20%, a 5% rate has an annual standard deviation of 100 basis points against 50 for a 2.5% rate. Bachelier's σ_N is that standard deviation quoted directly, and is therefore the same number of basis points at any level.

The empirical evidence is regime-dependent, and not in one direction: rate moves look close to absolute, independent of the level, while rates sit in a middle band of roughly 2% to 6%, and closer to proportional both below that band and above it, the pattern Deguillaume, Rebonato, and Pogudin (2013) found across currencies and decades. Practice has settled on normal volatilities for quotation, with the understanding that the number will need to be re-marked as the level of rates changes.

Figure 29.3 shows the difference where it matters most, for a forward of 0.5%. The lognormal density is confined to positive values and cannot represent a negative outcome at all; the normal density assigns genuine probability to the shaded region below zero. When

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forwards themselves turned negative, Black's formula ceased to be merely inaccurate and became undefined.

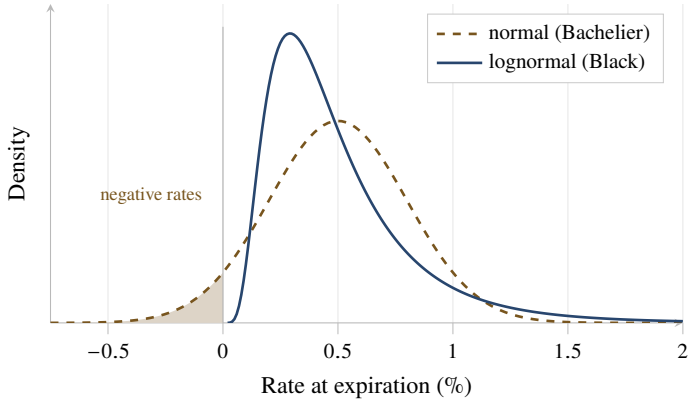


Figure 29.3: Distributions of a rate with a forward of 0.5% under the two conventions.

Shifted lognormal

The shifted lognormal convention is a compromise. Choose a displacement α large enough that $F_0 + \alpha$ and $K + \alpha$ are both comfortably positive, and apply Black's formula to the shifted quantities. It preserves the lognormal machinery, permits negative rates down to $-\alpha$, and interpolates between the two extremes: large α approaches normal behavior, $\alpha = 0$ recovers Black.

Its drawback is that α is an extra convention that must be agreed. A shifted lognormal volatility is meaningless without the shift that goes with it, and quotes carrying different shifts are not comparable. This is the same trap as flat versus spot volatilities, and the same discipline applies: establish the convention before comparing the numbers.

29.9 THE VOLATILITY CUBE

A single volatility number is enough to quote one option. Describing the whole swaption market requires three dimensions:

- **option expiration**, from one month to thirty years;
- **swap tenor**, the length of the underlying swap;
- **strike**, usually expressed as an offset from the at-the-money forward.

The resulting object is the **volatility cube**. Fixing the strike at the money gives the *volatility surface* over expiration and tenor; fixing expiration and tenor and varying strike gives a *smile*, of the kind Chapter 19 discussed for equities.

Two regularities are worth carrying away. Along the expiration dimension, volatility displays the hump of Figure 29.2. Along the tenor dimension, volatility generally *declines*: a thirty-year swap rate is an average of forward rates over thirty years, and averaging suppresses variance (exactly the mechanism of Section 29.7), so long-tenor swap rates move less than short-tenor ones. Figure 29.4 shows the at-the-money slice of the cube with both effects visible at once: the ridge running along two to three years' expiration, where uncertainty about the policy path is greatest, and the steady decline as swap tenor lengthens. Varying strike as well would add the third dimension.

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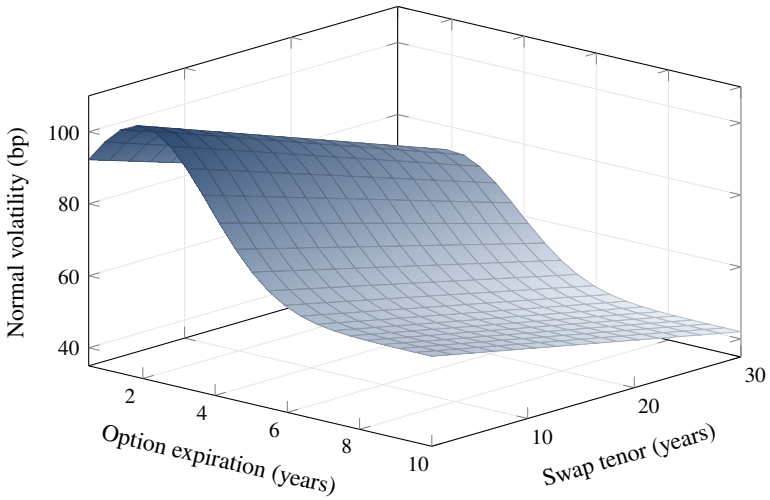


Figure 29.4: Normal volatility against option expiration and swap tenor. Representative rather than market data.

Fitting the cube: SABR

Interpolation within the cube is constrained: an arbitrary interpolation can produce a surface admitting arbitrage: calendar spreads with negative value, or butterfly spreads implying negative probability density. Practitioners therefore fit a parametric model, and for the strike dimension the standard choice is **SABR**, proposed by Hagan and co-authors in 2002. It is a stochastic-volatility model in the sense of Section 21.4, but built around a different requirement from Heston's: it must be fast enough to refit every swaption smile on the grid, many times a day. It posits a forward rate whose own volatility is stochastic:

$$dF = \alpha F^{\beta}\, dW_1, \qquad d\alpha = \nu\, \alpha\, dW_2, \qquad dW_1\, dW_2 = \rho\, dt \tag{29.21}$$

The model has four parameters, each with a distinct job:

- α sets the overall level of volatility, and is what the at-the-money quote pins down.
- β governs the **backbone**, how the at-the-money volatility moves when the forward rate itself moves. At $\beta = 1$ the model is lognormal and at-the-money volatility is unchanged by a shift in F; at $\beta = 0$ it is normal, and lognormal volatility moves inversely with F. This is the same choice as Section 29.8, now made continuously, and at $\beta = 0$ it lets SABR represent negative rates, where a lognormal model has nothing to say.
- ρ produces the skew. A negative correlation between rate and volatility tilts the smile downward to the right, which is the shape rate markets normally display.
- ν, the volatility of volatility, produces the curvature, how much the wings lift relative to a straight line through the middle.

That division is the same one Heston makes in Section 21.4. What made SABR the market standard is not the dynamics, which are unremarkable, but the closed-form approximation Hagan derived for the Black (lognormal) implied volatility $\sigma_B(K)$ they produce at strike K and expiration T, with no integral and no simulation:

$$\sigma_B(K) = \frac{\alpha}{(FK)^{(1-\beta)/2}\, D(K)} \cdot \frac{z}{x(z)} \cdot (1 + C\, T) \tag{29.22}$$

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where, writing $L = \ln(F/K)$,

$$D(K) = 1 + \frac{(1-\beta)^2}{24}L^2 + \frac{(1-\beta)^4}{1920}L^4, \qquad z = \frac{\nu}{\alpha}(FK)^{(1-\beta)/2}L$$

$$x(z) = \ln\left[\frac{\sqrt{1-2\rho z+z^2}+z-\rho}{1-\rho}\right],$$

$$C = \frac{(1-\beta)^2}{24}\frac{\alpha^2}{(FK)^{1-\beta}} + \frac{1}{4}\frac{\rho\beta\nu\alpha}{(FK)^{(1-\beta)/2}} + \frac{2-3\rho^2}{24}\nu^2$$

The formula has three factors, and each does one job: the first is the level, the second bends the smile away from it, and the third is the maturity correction. At $K = F$ the middle factor is 1 and the whole thing collapses to $\alpha F^{\beta-1}(1 + CT)$. A desk can fit a smile in microseconds and quote off it.

β is usually fixed rather than fitted. It is nearly redundant with ρ on a single smile, so fitting both produces unstable parameters that jump between days without the market having moved. [TN29] makes the redundancy exact: the skew depends on β and ρ only through one combination of them, and the curvature is where they separate again. Desks pick β from the observed backbone over weeks and then calibrate the other three daily.

The calibration recipe. Four parameters cannot be identified from a small number of quotes without structure, so desks proceed in a fixed order:

1. **Fix β** by judgment or by regressing historical at-the-money volatility on the level of the forward, never by fitting it to a single day's smile, for the reason just given. Rate desks commonly use $\beta = 0.5$.
2. **Solve α from the at-the-money quote**, given trial values of ρ and ν. This is a cubic in α, and forces the fit through the most liquid point exactly.
3. **Fit ρ and ν to the wings**, re-solving for α at each step so the at-the-money quote is never violated.

Example 29.8 Fitting SABR to a swaption smile

A five-year option on a five-year swap has a forward swap rate of $F = 4.00\%$. Three strikes are quoted, in lognormal volatility:

$$\sigma_B(3.00\%) = 23.6\%, \qquad \sigma_B(4.00\%) = 20.0\%, \qquad \sigma_B(5.00\%) = 18.3\%$$

Fix $\beta = 0.5$. Three quotes then determine the three remaining parameters, and solving (29.22) gives

$$\alpha = 0.0387, \qquad \rho = -0.30, \qquad \nu = 0.35$$

Checking the at-the-money point. With $F^{1-\beta} = \sqrt{0.04} = 0.2$, the leading term is $\alpha/0.2 = 0.1935$, and

$$C = \frac{0.25}{24}\frac{\alpha^2}{0.04} + \frac{1}{4}\frac{(-0.30)\times 0.5\times 0.35\alpha}{0.2} + \frac{2-3\times 0.09}{24}\times 0.35^2 = 0.006681$$

the three terms being 0.000390, −0.002540, and 0.008830. Hence $\sigma_B(F) = 0.1935\,(1+0.006681\times 5) = 0.1935\times 1.03340 = 20.00\%$, matching the quote.

Reproducing the wings. Substituting $K = 3\%$ gives $z = 0.4842$, $x(z) = 0.4403$, hence $z/x(z) = 1.0998$, and (29.22) returns 23.59%. At $K = 5\%$, $z = -0.4268$ and $z/x(z) = 0.9655$, giving 18.26%. Both agree with the quotes to the one decimal place at which they are quoted.

Using the fit. The value of the calibration is what it says about strikes no one quoted:

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Strike	2.5%	3.0%	3.5%	4.0%	4.5%	5.0%	5.5%
σ_B	26.25%	23.59%	21.54%	20.00%	18.92%	18.26%	17.91%

The smile is steeply downward-sloping near the money and flattens in the high-strike wing—the characteristic shape of a rate market, and in its downward slope much like the equity skew of Chapter 19.

Figure 29.5 plots the result. The at-the-money point is matched exactly by construction, since α is re-solved from it at every step of the fit; the wings are what the two remaining parameters must reproduce.



Figure 29.5: The SABR fit of Example 29.8, with $\beta = 0.5$, $\alpha = 0.0387$, $\rho = -0.30$, $\nu = 0.35$.

Where the approximation breaks. Equation (29.22) is an asymptotic expansion, accurate when $\nu^2 T$ is small. For long expirations or high volatility of volatility it degrades, and, more seriously, it can imply a *negative probability density* at low strikes, which is Section 19.5's constraint violated by an approximation rather than by the market. Since rates fell toward and below zero, that region stopped being a remote wing and became where options actually trade.

The responses parallel Section 29.8: a *shifted* SABR displaces F and K so the model operates away from zero, while *no-arbitrage* SABR replaces the expansion with a numerical solution guaranteed to produce a valid density. Both are in use at dealers, and which a desk employs is a material assumption rather than a technical detail.

The general point stands for the whole cube: it is fitted rather than interpolated, and the fitting model is itself a source of model risk. Two desks marking the same quoted points can carry materially different values for everything in between.

Case Study 29.2 The 2022 volatility regime change

For most of the decade following the financial crisis, interest rate volatility was low and falling. Policy rates sat near zero, central banks guided expectations explicitly, and the MOVE index, which tracks implied volatility on Treasury options, spent long stretches well below its historical average.

That ended abruptly in 2022. As inflation forced central banks into the fastest tightening cycle in forty years, rate volatility rose sharply, and the MOVE index climbed through 2022 to a peak during the March 2023 banking stress at its highest level since

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2008. Normal volatilities on short-dated dollar swaptions doubled or tripled from their pre-2022 levels, and longer-dated ones rose by less.

Two lessons follow for this chapter. First, positions that looked like carry trades in the low-volatility regime (selling swaptions, or the convexity trades of Section 30.3) turned out to be short volatility at prices set by a decade of calm. Second, the vol-of-vol embedded in the smile matters: a cube fitted in a quiet regime understates how much its own shape can move.

29.10 HEDGING INTEREST RATE DERIVATIVES

The Greeks of Chapter 18 carry over, with rate-specific names.

Delta for a rate derivative is its sensitivity to a shift in the curve. Because a curve is not a single number, delta is not either: desks compute a **DV01** (or PV01) against a parallel one basis point shift, and a vector of **bucketed deltas** against a one basis point shift in each curve segment separately. The bucketed vector is what actually gets hedged, since a portfolio can be flat to a parallel shift while carrying large offsetting exposures to steepening.

Vega is sensitivity to volatility, and inherits the dimensionality of the cube: a swaption book has vega to each expiration–tenor bucket, and hedging it requires trading options at comparable points.

Gamma measures how fast delta changes as the rate moves, and so how often a hedge must be rebalanced. Backward-looking products add a second source of change. As Section 29.4 showed, a compounded-rate option's sensitivity to the rate decays through the accrual period as fixings are locked in, so its delta changes with the passage of time even when rates do not move. A hedge that is correct at the start of the period is progressively too large, so these products require rebalancing through the accrual period rather than at reset dates alone.

The general caution from Chapter 18 applies with extra force here. These Greeks are computed within a model that is known to be a quotation convention rather than a description of reality. A delta computed under Black and a delta computed under Bachelier differ, and the difference is not small when the smile is steep. Which is right depends on how volatility actually moves when the rate moves—a question the quoting model does not answer.

SUMMARY

Caps, floors, and swaptions are valued with Black's model applied to the appropriate forward quantity under the appropriate numeraire. For a caplet the forward quantity is the forward rate and the numeraire is the zero-coupon bond maturing on the payment date; for a swaption it is the forward swap rate and the annuity. These are quotation conventions rather than descriptions of rate behavior: their function is to provide an agreed, invertible map between prices and volatilities.

A cap is a portfolio of separately settled caplets, so it is valued one period at a time. Cap–floor parity states that a cap less a floor at the same strike equals a payer swap, a model-free relationship useful as a consistency check. Market quotes use flat volatilities (a single number that reprices a whole cap), while analysis requires spot volatilities specific to each caplet, extracted by a bootstrap. The two must never be compared directly.

The move to compounded risk-free rates changed the caplet. A backward-looking rate keeps accumulating uncertainty across its accrual period, with sensitivity decaying linearly as fixings are locked in, giving total variance $\sigma^2(T_1 + \tau/3)$ rather than $\sigma^2 T_1$. Backward-looking caplets are therefore always worth more than their forward-looking equivalents, and the first caplet of a spot-starting cap, formerly a known cash flow, now carries real optionality.

Caps and swaptions on the same schedule are not equivalent: a swaption is a single option

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on a weighted average of the rates a cap options one by one. Because a sum of positive parts is at least the positive part of the sum, the cap is worth at least as much, and the gap is governed by the correlation among forward rates, closing to zero at perfect correlation. Those correlations are not observable while cap and swaption prices are, so the ratio between them is the market's principal source of correlation information and the main calibration input for multi-factor term structure models. Bond options are quoted in yield volatility and converted to price volatility by duration, and an option-adjusted spread means nothing without its volatility assumption.

Negative rates made the lognormal assumption not merely inaccurate but undefined, since $\ln(F_0/K)$ requires a positive forward. The market moved to normal (Bachelier) volatilities, under which an at-the-money option is worth $\sigma_N\sqrt{T/2\pi}$ per unit of numeraire, and to shifted lognormal conventions. Near the money, $\sigma_N \approx \sigma_{\mathrm{LN}}F_0$.

The swaption market is described by a volatility cube over expiration, tenor, and strike. Volatility humps along the expiration dimension and declines along the tenor dimension, since a longer swap rate averages more forwards. The cube is fitted with a parametric model, usually SABR, rather than interpolated, because arbitrary interpolation admits arbitrage; Hagan's expansion can itself imply a negative density at low strikes, which the shifted and no-arbitrage variants repair. Hedging inherits the cube's dimensionality: bucketed delta, vega by expiration and tenor, and a delta that decays through the accrual period of a backward-looking product.

FURTHER READING

Black's "The Pricing of Commodity Contracts" (1976) is the origin of the formula this chapter applies to caps and swaptions. Bachelier's 1900 thesis is the normal model, written five years before Einstein's treatment of Brownian motion and worth reading now that rates have made the normal convention standard again. Hagan, Kumar, Lesniewski, and Woodward's "Managing Smile Risk" (2002) introduced SABR and is the source of the expansion in (29.22). Andersen and Piterbarg's three-volume *Interest Rate Modeling* is the reference to reach for when the market conventions here are not enough, and Lyashenko and Mercurio (2019) is where the $T_1 + \tau/3$ result for backward-looking rates comes from.

PRACTICE QUESTIONS

29.1 Explain why Black's model can be applied to a caplet without any adjustment for the drift of the forward rate. Which discount factor must be used, and why that one?

29.2 A caplet covers a three-month period beginning in two years, on a notional of \$20 million. The forward rate is 3.60%, the strike is 3.75%, the volatility is 18%, and the discount factor to the payment date is 0.9280. Value the caplet, treating the rate as fixed in advance.

29.3 Repeat Question 29.2 treating the rate as compounded in arrears. By what percentage does the value increase?

29.4 State cap–floor parity and prove it from the payoffs. Explain how you would use it to check a set of broker quotes.

29.5 Explain the difference between a flat volatility and a spot volatility. Why is it wrong to compare the flat volatilities quoted for a three-year and a seven-year cap?

29.6 Explain why the first caplet of a spot-starting cap was conventionally excluded under LIBOR, and why that convention is no longer appropriate.

29.7 A five-year option on a three-year annual-pay swap has a forward swap rate of 3.80%, a strike of 3.80%, and a volatility of 22%. The annuity is 2.6540 per unit of notional. Value the payer swaption on a notional of \$50 million, and state the value of the corresponding receiver.

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29.8 Show that at the money the Bachelier formula reduces to $\sigma_N \sqrt{T/2\pi}$ per unit of numeraire.

29.9 An at-the-money swaption is quoted at a lognormal volatility of 24% with a forward swap rate of 3.20%. Estimate the equivalent normal volatility, and explain why the rule of thumb works.

29.10 Explain why the volatility of a swap rate generally declines as the tenor of the underlying swap increases.

29.11 Explain, without algebra, why a cap must be worth at least as much as a swaption written on the same schedule and strike. Under what condition would the two be equal?

29.12 Eight forward rates each have a volatility of 24% and enter a swap rate with equal weight. Using (29.17), find the implied swap rate volatility when the uniform pairwise correlation is 1.0, 0.6, and 0.2. Comment on what the results imply about calibrating a model to caps alone.

FURTHER QUESTIONS

29.13 Derive the result $\text{Var} = \sigma^2(T_1 + \tau/3)$ for a backward-looking rate, stating clearly the assumption made about how sensitivity decays through the accrual period. How would the answer change if the decay were assumed to be quadratic rather than linear?

29.14 A cap consists of four quarterly caplets beginning immediately, with $\tau = 0.25$. Using a flat volatility of 20%, a strike of 4.00%, and a forward rate of 4.00% for every period, and ignoring discounting, compare the total value of the cap when the caplets are treated as forward-looking against the value when they are treated as backward-looking. Comment on where the difference is concentrated.

29.15 A trader observes that three-month expiration swaptions are quoted at 85 basis points of normal volatility while one-year expiration swaptions on the same tenor are quoted at 78. Explain what this implies about the expected path of rate uncertainty, and identify at least two reasons this need not represent an arbitrage.

29.16 Explain why a delta computed under Black's model differs from one computed under Bachelier for the same option at the same price. Under what circumstances is the difference largest, and what additional information would you need to decide which is the better hedge ratio?

29.17 In 2022 the level of rates and the volatility of rates rose together. Explain, using the relationship between normal and lognormal volatility, why a desk quoting in normal volatilities and a desk quoting in lognormal volatilities would have reported very different percentage increases in "volatility" over that year, even though both observed the same market.

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Chapter 30
Convexity, Timing, and Quanto Adjustments †

Almost everything in the preceding chapters rests on the same approach: when the expected future value of a rate or a price is required, today's forward value is used in its place. Forward contracts were valued that way, swaps were valued that way, and Chapter 29 priced caps and swaptions by assuming the relevant forward rate was the mean of a lognormal distribution.

That approach is valid far more widely than one might expect, but it is not universal. It is the consequence of a specific and rather delicate alignment: the forward value of a variable equals its expected future value *only under one particular probability measure*, and which measure that is depends on when the payoff is made and in what currency.

When a contract respects that alignment (a rate observed at the start of a period and paid at the end of it, in its own currency), the forward rate is exactly the right expectation and no adjustment is needed. When a contract breaks the alignment, the forward rate is the wrong expectation, and using it introduces a pricing error that can run from a fraction of a basis point to tens of basis points.

This chapter is about the three ways the alignment breaks and the corrections each one requires:

- a **convexity adjustment**, when the instrument's settlement mechanics change the measure, most importantly when daily margining makes a futures rate behave differently from a forward rate;
- a **timing adjustment**, when a payoff is made at a date other than the one naturally associated with the variable;
- a **quanto adjustment**, when a payoff denominated in one currency is based on a variable belonging to another.

All three are the same phenomenon in a different form, and Section 30.1 explains why. The unifying idea is worth the effort: once you see that each adjustment is a change of measure, you can derive corrections for contracts this chapter never mentions, rather than memorizing a catalog of formulas.

These adjustments are not academic refinements. The convexity adjustment on a ten-year interest rate futures contract can exceed 50 basis points—larger than the bid–ask spread by two orders of magnitude, and large enough that a curve built without it will misprice every long-dated swap on the book. Desks employ people to get these numbers right.

The chapter runs from the principle to the three corrections. Section 30.1 states when a forward value is an expectation and under which measure. Sections 30.2 to 30.5 take the breaks in turn: a payment made at the wrong time, the daily settlement that separates a futures contract from a forward, and the swap rate paid outside its own annuity, which is the largest of the adjustments that come from payment timing. Section 30.6 is the currency break, and

Section 30.7 shows the three as one calculation done three times. Sections 30.2 through 30.6 can be read in any order, and each adjustment is derived where it is stated; those derivations rest on Chapter 28 and can be skipped without losing the results.

30.1 FORWARD VALUES ARE EXPECTATIONS—UNDER THE RIGHT MEASURE

The general principle comes first, because the three adjustments that follow are special cases of it.

The chapter requires a single concept, Chapter 28's numeraire: any asset with a strictly positive price in which all other prices are measured. What follows is the working statement of it, and nothing here requires more. If $N(t)$ is the price of a traded asset that pays no income, then there exists a probability measure (call it Q^N, the measure *associated with* the numeraire N) under which the price of every other traded asset, divided by N, is a martingale: for any traded asset with price $V(t)$,

$$\frac{V(t)}{N(t)} = E^N\left[\frac{V(T)}{N(T)}\,\middle|\,\mathcal{F}_t\right], \qquad t \le T$$

where E^N denotes expectation under Q^N and $\mathcal{F}_t$ the information available at t. This is the fundamental result of arbitrage pricing, and everything in this chapter follows from it. Only the case $t = 0$ is used below; the martingale property is the stronger statement that the equality holds at every intermediate date as well, which is what makes a change of numeraire legitimate at any point in a contract's life.

Two choices of numeraire dominate practice.

The money market account. Take $N(t) = B(t)$, the value of a bank account earning the instantaneous short rate. The associated measure is the familiar risk-neutral measure Q, in which expected returns equal the risk-free rate.

The zero-coupon bond. Take $N(t) = P(t,T)$, the price at time t of a bond paying \$1 at time T. The associated measure is called the **T-forward measure**, written Q^T. It has a property that makes it enormously convenient: since $P(T,T) = 1$,

$$V(0) = P(0,T)\,E^T[V(T)] \tag{30.1}$$

The value today of a payoff received at time T is the discount factor times the expected payoff under the T-forward measure: no stochastic discount factor appears inside the expectation, and discounting and expectation separate cleanly. This is why the T-forward measure is the natural home for any contract with a single payment date T.

Which rates are martingales, and under which measure

Now consider the forward rate. Let $F(t;T_1,T_2)$ be the forward rate observed at time t for the period $[T_1,T_2]$, with accrual factor $\tau = T_2 - T_1$, defined in the usual way from bond prices:

$$F(t;T_1,T_2) = \frac{1}{\tau}\left[\frac{P(t,T_1)}{P(t,T_2)} - 1\right] = \frac{P(t,T_1) - P(t,T_2)}{\tau\,P(t,T_2)} \tag{30.2}$$

Look at the right-hand expression. The numerator, $P(t,T_1) - P(t,T_2)$, is the price of a traded portfolio: long one T_1-bond, short one T_2-bond. The denominator is τ times the price of the T_2-bond. So F is the price of a traded asset divided by τ times a numeraire, and therefore F is a **martingale under** Q^{T_2}:

$$E^{T_2}[F(T_1;T_1,T_2)] = F(0;T_1,T_2) \tag{30.3}$$

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At $t = T_1$ the period begins immediately, so $F(T_1; T_1, T_2)$ is the spot rate for $[T_1, T_2]$, which later sections write $R(T_1)$. Equation (30.3) is therefore the precise statement of the approach relied on throughout: the forward rate is the expected future spot rate, but only under the T_2-forward measure, where T_2 is the date the rate is *naturally* paid.

The two results together give the chapter its subject. Equation (30.1) requires an expectation under Q^T to value a payoff received at time T; equation (30.3) supplies an unbiased expectation under Q^{T_2}. If the payment date T and the natural date T_2 coincide, the two align and the forward rate is used directly. If they do not, because the payment comes early, or late, or in a different currency, or because margining has changed the effective numeraire to the money market account, then the required expectation is under a different measure (Q^T if the payment moves, Q if margining makes the money market account the numeraire), while only $E^{T_2}[F]$ is known, and the difference is the adjustment. Call the measure under which the variable's forward value is an unbiased expectation its **natural measure** (here Q^{T_2}) and the measure the contract's payment calls for the **required measure**; each section below names both.

Why an adjustment is a covariance

Changing measure is done with a Radon–Nikodym derivative, and for forward measures it takes a particularly interpretable form. Moving from Q^{T_2} to Q^T requires reweighting each scenario by the ratio of the two numeraires' growth:

$$\left.\frac{dQ^T}{dQ^{T_2}}\right|_{\mathcal{F}_t} = \frac{P(t,T) \,/\, P(0,T)}{P(t,T_2) \,/\, P(0,T_2)} \tag{30.4}$$

Write Λ_t for this ratio (Chapter 28's density process, with Q^{T_2} in place of P and Q^T in place of Q) and Λ for its value at the date the variable is observed. It is not constant: the two time-zero prices are fixed, so it moves with $P(t,T)/P(t,T_2)$, which is the discount factor between the two dates, not with either bond price on its own, since a rise in rates lowers both. When the required date is the later one, $T > T_2$, that factor falls as rates rise, so high-rate scenarios are downweighted and low-rate scenarios upweighted; when $T < T_2$ the weighting runs the other way. The reweighting is monotone in rates either way, and that is what matters here. Therefore:

> **If the variable being adjusted is correlated with interest rates, reweighting the scenarios changes its mean. If it is uncorrelated, the mean is unchanged and no adjustment is needed.**

Every adjustment in this chapter is a covariance with that density, and at leading order a covariance between the variable and whatever drives it: an interest rate, or an exchange rate. The sign of that covariance gives the sign of the adjustment, which is enough to check any formula in this chapter without re-deriving it.

Section 28.6 derives (30.4) and one consequence that every adjustment below is an instance of. Because the density has unit mean, for any X known at t,

$$E^T[X] = E^{T_2}[X] + \mathrm{Cov}^{T_2}(X, \Lambda_t) \tag{30.5}$$

Every derivation in this chapter is an evaluation of (30.5) for a particular X, and the work each time is finding what Λ_t is a function of.

30.2 TIMING ADJUSTMENTS

Start with the simplest break in the alignment: a payoff based on a variable observed at time T^*, but paid at some later time T rather than at T^*.

Let V be the variable, with known forward value V_0, meaning $E^{T^*}[V(T^*)] = V_0$. What is required is $E^T[V(T^*)]$.

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Write $D = P(T^*, T)$ for the discount factor over the deferral period and $D_0 = P(0,T)/P(0,T^*)$ for its forward value today. Evaluating (30.4) at $t = T^*$ and using $P(T^*, T^*) = 1$, the density is simply the discount factor rescaled,

$$\Lambda \;=\; \frac{P(T^*,T)\,P(0,T^*)}{P(0,T)} \;=\; \frac{D}{D_0},$$

and $E^{T^*}[D] = D_0$ because $P(t,T)/P(t,T^*)$ is a traded asset over the numeraire. So (30.5) reads

$$E^T[V(T^*)] \;=\; V_0 \;+\; \frac{\mathrm{Cov}^{T^*}(V,D)}{D_0} \tag{30.6}$$

exactly. The timing adjustment is the covariance of the payoff with the discount factor that defers it, and nothing has been approximated yet.

What has to be approximated is the translation into quoted terms, because the market quotes the volatility of a rate rather than of a discount factor. Writing $\tau = T - T^*$ and y for the simply-compounded rate covering $[T^*, T]$, the two bond prices are defined by that rate,

$$D = \frac{1}{1+y\tau}, \qquad D_0 = \frac{1}{1+y_0\tau},$$

so $\Lambda = (1+y_0\tau)/(1+y\tau)$, with $\Lambda(y_0) = 1$ and $\Lambda'(y_0) = -\tau/(1+y_0\tau)$. Linearizing Λ about y_0 turns (30.6) into a covariance with the rate,

$$E^T[V] \;\approx\; V_0 - \frac{\tau}{1+y_0\tau}\,\mathrm{Cov}^{T^*}(V,y),$$

and for jointly lognormal V and y that covariance is $V_0 y_0 \rho \sigma_V \sigma_y T^*$, giving:

$$E^T[V(T^*)] \;\approx\; V_0\left[1 - \frac{\rho\,\sigma_V\,\sigma_y\,y_0\,\tau\,T^*}{1+y_0\tau}\right] \tag{30.7}$$

or, in the exponential form the market usually quotes, which agrees with (30.7) to first order and is exact when V and D are jointly lognormal ([TN30] shows why, and how much the linearization costs),

$$E^T[V(T^*)] \;\approx\; V_0 \exp\!\left(-\frac{\rho\,\sigma_V\,\sigma_y\,y_0\,\tau\,T^*}{1+y_0\tau}\right) \tag{30.8}$$

where

- y_0 is today's forward rate for the deferral period $[T^*, T]$, simply compounded over τ;
- σ_y is the volatility of that forward rate (proportional, that is, lognormal);
- σ_V is the volatility of the forward value of V;
- ρ is the correlation between V and y.

Reading the formula. The adjustment is negative when $\rho > 0$. The intuition is exactly the covariance argument of Section 30.1: deferring a payment means discounting it over $[T^*, T]$ at a rate that is not known today. If V tends to be large precisely when that rate is high, then the large payoffs are the ones that get discounted most heavily, and the deferred claim is worth less than a naive calculation suggests. If V and rates are uncorrelated, $\rho = 0$ and no adjustment is needed no matter how long the deferral.

Note also that the adjustment scales with τT^*, the product of the *deferral length* and the *time until observation*. A long deferral of a payoff observed tomorrow needs almost no adjustment, because there is almost no time for V and y to become correlated in their realized movements.

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Example 30.1 Payment deferred by one quarter

A structured note pays the five-year swap rate observed at the end of year 5. Payment, however, is made at the end of year 5.25 rather than immediately on observation.

The forward five-year swap rate is $V_0 = 4.00\%$ with volatility $\sigma_V = 20\%$. The forward rate covering the deferral quarter is $y_0 = 4.00\%$ with volatility $\sigma_y = 20\%$. The correlation between the two is $\rho = 0.7$. Here $T^* = 5$ and $\tau = 0.25$.

The exponent in (30.8) is

$$-\frac{0.7 \times 0.20 \times 0.20 \times 0.04 \times 0.25 \times 5}{1 + 0.04 \times 0.25} = -\frac{0.0014}{1.01} = -0.001386$$

so the adjustment factor is $e^{-0.001386} = 0.998615$ and

$$E^T[V] = 4.00\% \times 0.998615 = 3.9945\%$$

The timing adjustment is -0.55 basis points.

The half basis point of Example 30.1 may look like a rounding error, and on a single trade it is. On a \$5 billion structured note program it is \$275,000 per year of exposure, and it is the kind of error that is systematically in one direction, so it never averages out across the book. This is the general character of timing adjustments: individually small, collectively worth getting right.

30.3 CONVEXITY ADJUSTMENTS I: FUTURES VERSUS FORWARDS

The largest and most economically important convexity adjustment in fixed income arises from a difference in settlement mechanics rather than payment dates. It is the reason a futures rate is not a forward rate.

Why margining changes the measure

Recall Chapter 2. A forward contract settles once, at maturity. A futures contract is marked to market daily: gains and losses are paid in cash as they accrue.

That difference determines the numeraire. Because the cash flows on a futures contract occur continuously and are immediately available for investment (or must be immediately funded), the correct numeraire is the **money market account**, and Section 28.7 showed that the futures price is then a martingale under the risk-neutral measure Q. The forward price is a martingale under Q^T. So for an asset settling at T,

$$\text{Futures price} = E^Q[S_T], \qquad \text{Forward price} = E^T[S_T] \tag{30.9}$$

Two measures, two expectations of the same terminal price, and (30.5) compares them. The density taking Q to Q^T is the T-bond over the money market account, normalized,

$$\Lambda = \frac{1}{B(T)\,P(0,T)}$$

so

$$\text{Forward price} - \text{Futures price} = \text{Cov}^Q(S_T,\, \Lambda) \tag{30.10}$$

and Λ is large exactly when $B(T)$ is small, that is, when rates have been low. An asset positively correlated with interest rates therefore has a futures price above its forward price, and one negatively correlated has a futures price below it. The sign follows from the covariance alone; no model of rates has been assumed.

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Which is larger, and why

Section 5.12 settled the direction from the margin flows alone, before any of the measure language was available:

$$\text{Futures price} \;>\; \text{Forward price} \qquad \text{(asset positively correlated with rates)}$$

with a negative correlation reversing it. Equation (30.10) verifies that argument: the covariance it exhibits carries exactly the sign Chapter 5 reasoned to from the cash flows, and, being an expectation, it can also be evaluated.

An interest rate futures contract is the extreme case, which is why the adjustment is largest here: its underlying is the rate itself, so the correlation is perfect rather than incidental. Writing the futures rate as the rate the quoted price implies (100 less the quotation, for the contracts of Section 6.2), it is an affine function (a constant plus a fixed multiple) of a martingale and a martingale itself, so (30.9) becomes

$$\text{Futures rate} = E^{Q}[R(T_1)], \qquad \text{Forward rate} = E^{T_2}[R(T_1)] \tag{30.11}$$

the second by (30.3). The futures price falls as the rate rises, so the inequality above inverts when read in rates: the futures rate exceeds the forward rate by the convexity adjustment.

The size of the adjustment

Under the Ho–Lee model (the simplest no-arbitrage short-rate model, developed with its mean-reverting extension in Sections 31.5 and 31.6), the short rate follows

$$dr = \theta(t)\,dt + \sigma\,dW$$

with $\theta(t)$ a drift chosen so that the model reproduces today's term structure and σ the standard deviation of the *absolute* (not proportional) change in the short rate per year, and the adjustment takes a memorable form:

$$\text{Forward rate} \;=\; \text{Futures rate} \;-\; \frac{1}{2}\,\sigma^2\,T_1\,T_2 \tag{30.12}$$

where T_1 is the time until the futures contract's rate is determined and T_2 is the time until the end of the period the rate covers.

Strictly, (30.12) and (30.13) adjust the continuously compounded rate; for the simply compounded quarterly rate on which the contract settles, add about $\frac{1}{2}\sigma^2\tau T_1$, which is 0.1 basis points at one year and under 3% of the adjustment at ten. The structure of (30.12) carries the lesson. The adjustment grows with the square of maturity, so it is negligible for near-dated contracts and dominant for far-dated ones. Chapter 6 stated the same formula and worked it at a normal volatility of 100 basis points a year (Section 6.2 and Example 6.12): 0.63 basis points on the one-year contract, 33 at eight years and about 51 at ten, where treating futures rates as forward rates puts an error into the curve two orders of magnitude larger than the bid–ask spread.

Ho–Lee overstates the long end, because it has no mean reversion and mean reversion damps the volatility of long-dated rates. Adding it gives Hull–White,

$$dr = (\theta(t) - ar)\,dt + \sigma\,dW$$

where a is the speed at which the short rate is pulled back toward its mean, and the adjustment becomes

$$\text{Futures} - \text{Forward} = \frac{1}{\tau}\,B(T_1,T_2)\left[B(T_1,T_2)\left(1 - e^{-2aT_1}\right) + 2a\,B(0,T_1)^2\right]\frac{\sigma^2}{4a} \tag{30.13}$$

where

$$B(t_1,t_2) = \frac{1 - e^{-a(t_2-t_1)}}{a}$$

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As $a \to 0$ this must reduce to (30.12), and it does: $B(T_1, T_2) \to \tau$, $B(0, T_1) \to T_1$, and $1 - e^{-2aT_1} \to 2aT_1$, so the bracket becomes $2aT_1T_2$ and the whole expression collapses to $\frac{1}{2}\sigma^2 T_1 T_2$. The check matters because the mean-reverting form is easy to mistype, and a wrong a is invisible at the short end and large at the long one. Equation (30.13) is what desks use.

One caution applies to both formulas. The σ in each is a normal (absolute) volatility. Substituting a lognormal volatility of, say, 20% into (30.12) produces a number four hundred times too large. Volatility quoting conventions are the most common source of error here.

Figure 30.1 plots both versions across the strip. Two features repay attention. The first is the shape: the adjustment is almost invisible for the first two years and then accelerates, because it grows with the square of maturity. A desk can reasonably ignore convexity on the front contracts and cannot possibly ignore it on the back ones. The second is the gap between the curves. Mean reversion pulls long-dated rate volatility down, so the Hull–White adjustment falls progressively below Ho–Lee as maturity extends, by roughly a quarter at ten years in this calibration. Using Ho–Lee at the long end does not merely approximate the adjustment; it systematically overstates it.

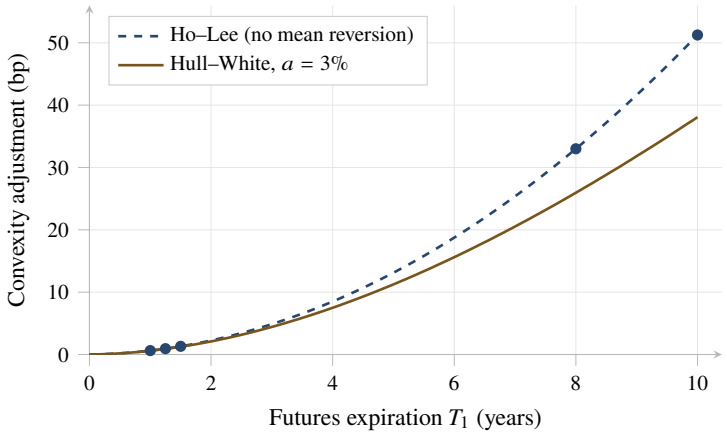


Figure 30.1: The futures–forward convexity adjustment against contract expiration, for the continuously compounded rate, three-month contracts and a short-rate volatility of 100 basis points per year. Markers show the values quoted in Chapter 6.

The modern complication: averaging futures

Equation (30.12) was developed for Eurodollar futures, which settled against a *term* rate: three-month LIBOR, fixed on a single date. The contracts that replaced them work differently, and the difference is worth understanding.

Three-month SOFR futures settle against the **compounded daily average** of the overnight rate over the reference quarter. One-month SOFR futures settle against the **arithmetic average** of the overnight rate over the calendar month. In neither case is there a single fixing date; the settlement rate accrues day by day across the contract's reference period.

This changes the risk profile in a way that matters for hedging more than for pricing:

- **The magnitude of the convexity adjustment is broadly similar.** At the leading order captured by Ho–Lee, an averaging contract and a term contract of the same expiration carry adjustments within a basis point of each other for typical quarterly periods. Equation (30.12) remains a reasonable first approximation.

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- **The delta decays during the reference period.** As each day's overnight rate is observed, that portion of the settlement rate is fixed and its sensitivity disappears. A three-month SOFR futures contract's rate exposure declines roughly linearly to zero across its reference quarter. A Eurodollar futures contract's exposure, by contrast, persisted at full size until the fixing date and then vanished at once.
- **Realized volatility of the settlement rate is lower**, because averaging suppresses the variance of the daily fixings, a point that matters more for options on the averaged rate than for the futures themselves. Chapter 29 computes that variance: the average carries only a third of the accrual period's variance, so it is below that of the period's last fixing, though still above that of a rate fixed when the period begins.

The practical consequence is that convexity *hedging* under SOFR requires continuous rebalancing through the reference period rather than a single roll, but curve construction still requires the same order-of-magnitude correction it always did.

Case Study 30.1 Convexity as a trade

The gap between futures and forward rates is not merely a modeling nuisance; it has long been a trade in its own right. A **convexity trade** buys futures and pays fixed on a swap covering the same period, locking in the higher futures rate on one side and the lower forward rate on the other and collecting the spread between them.

The position is not a riskless profit. The trader collects the convexity adjustment but takes on the exposure it compensates for: the trade loses when realized rate volatility exceeds the volatility implied in the adjustment. It is, in substance, a short volatility position in fixed-income form, with the risk profile that phrase implies. Convexity trades performed poorly through 2022, when realized rate volatility ran far above the levels embedded in convexity adjustments computed from the preceding decade's calm.

The general lesson recurs throughout the book: a pricing adjustment that compensates for a risk can always be harvested as a return, and the return is never free.

30.4 CONVEXITY ADJUSTMENTS II: RATES PAID AT THE WRONG TIME

The second family of convexity adjustments arises when a rate is paid at a date other than the end of the period it covers. Historically this was the domain of the *in-arrears swap*; under risk-free reference rates it has changed character in an instructive way.

The classical case: rates paid in arrears

Under LIBOR conventions, a rate for $[T_1, T_2]$ was *set* at T_1 and *paid* at T_2, the natural alignment. An **in-arrears swap** broke it by paying the rate at T_1, immediately on observation.

What is now required is $E^{T_1}[R(T_1)]$ rather than $E^{T_2}[R(T_1)] = F$. The natural measure is Q^{T_2} and the required measure is Q^{T_1}, so the frame is the same as before; what is different is the density. Since $P(T_1, T_2) = 1/(1 + R(T_1)\tau)$, evaluating (30.4) at $t = T_1$ with $T = T_1$ gives

$$\Lambda = \frac{1}{P(T_1, T_2)} \cdot \frac{P(0, T_2)}{P(0, T_1)} = \frac{1 + R(T_1)\tau}{1 + F\tau}$$

which has unit mean under Q^{T_2} precisely because $E^{T_2}[R] = F$. This density is *affine* in R, and that is what makes the case exact: where Section 30.2 had to linearize a density that was not, here (30.5) can be evaluated as it stands,

$$\mathrm{Cov}^{T_2}(R, \Lambda) = \frac{\tau}{1 + F\tau}\,\mathrm{Cov}^{T_2}(R, R) = \frac{\tau\,\mathrm{Var}^{T_2}(R)}{1 + F\tau}$$

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so that

$$E^{T_1}[R(T_1)] \;=\; F \;+\; \frac{\tau\,\mathrm{Var}^{T_2}(R)}{1+F\tau} \tag{30.14}$$

exactly, with no approximation beyond the model for the variance. If R is lognormal with volatility σ, then $\mathrm{Var}^{T_2}(R) = F^2\left(e^{\sigma^2 T_1}-1\right)$, giving

$$E^{T_1}[R(T_1)] \;=\; F + \frac{F^2\left(e^{\sigma^2 T_1}-1\right)\tau}{1+F\tau} \;\approx\; F + \frac{F^2\sigma^2 T_1\tau}{1+F\tau} \tag{30.15}$$

The adjustment is positive, unlike the timing adjustment of Section 30.2, and the reason is simple: here the payment is being *accelerated*, not deferred. Receiving a rate early is worth more when rates are high, so the scenarios that get upweighted are the favorable ones.

Example 30.2 In-arrears convexity

A swap pays the three-month rate for the period beginning in five years, but pays it at the start of that period rather than the end. The forward rate is $F = 4.00\%$ with volatility $\sigma = 25\%$; $T_1 = 5$ and $\tau = 0.25$.

The exact lognormal variance is

$$\mathrm{Var} = F^2\left(e^{\sigma^2 T_1}-1\right) = 0.0016\times\left(e^{0.3125}-1\right) = 0.0016\times 0.36684 = 5.8694\times 10^{-4}$$

so from (30.14) the adjustment is

$$\frac{0.25\times 5.8694\times 10^{-4}}{1.01} = 1.453\times 10^{-4} = 1.45\text{ bp}$$

The rate to use is 4.0145%. The first-order approximation $F^2\sigma^2 T_1$ in (30.15) gives 1.24 bp—an error of 15% in the adjustment itself. At long maturities and high volatilities the exact form is worth using.

What survives under risk-free rates

A compounded RFR is determined at the end of its accrual period and paid then, so the standard product no longer breaks the alignment and the classical adjustment of Example 30.2 largely dissolves; Section 33.4 prices what is left, the timing adjustment for the few days between the end of observation and payment, and finds it a few hundredths of a basis point even at ten years. The exception is the "last reset" convention of Chapter 4, which pays a rate determined at the end of the *previous* period one period late: that is a deferral, and it carries a timing adjustment in the opposite direction to (30.14), priced with Section 30.2's machinery.

30.5 CONVEXITY ADJUSTMENTS III: CMS AND THE ANNUITY MEASURE

The third convexity adjustment is the largest of the payment-timing family and, unlike Section 30.4's, was unaffected by the LIBOR transition. It concerns constant-maturity swaps.

The product and the problem

Recall from Section 28.8 that a CMS coupon pays a long *swap rate* (the ten-year rate, say) observed on each reset date. Unlike an ordinary floating leg, it is not matched to the period it is paid for.

A related and equally common structure is the **CMS spread**, paying the difference between two swap rates of different tenors (the 10-year less the 2-year, for instance). These are the building blocks of steepener notes, which pay attractive coupons unless the yield curve flattens.

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The measure problem here is structural. A swap rate S is the fixed rate that makes a swap worth zero; it equals the value of the floating leg divided by the annuity of Section 28.8 (the present value of the fixed leg per unit of fixed rate, so that one basis point on the fixed leg is worth 0.0001 A):

$$S(t) = \frac{\text{floating leg}(t)}{A(t)}, \qquad A(t) = \sum_{i=1}^{n} \tau_i \, P(t, t_i)$$

The annuity $A(t)$ is the price of a traded portfolio of bonds, so it is a legitimate numeraire, and S, being a traded asset divided by A, is a martingale under the associated swap measure Q^A. This is what justifies Black's formula for swaptions in Chapter 29, where the annuity appears naturally as the discounting term.

But a CMS coupon is a *single* cash flow at a *single* date T_2. Valuing it requires $E^{T_2}[S]$, and the annuity measure supplies $E^A[S] = S_0$. The numeraires do not match, so an adjustment is required.

In the frame of Section 30.1, the natural measure is Q^A, the required measure is Q^{T_2}, and the density between them is the T_2-bond over the annuity, normalized,

$$\Lambda_t = \frac{P(t, T_2) \,/\, P(0, T_2)}{A(t) \,/\, A(0)}$$

so the adjustment is $\mathrm{Cov}^A(S, \Lambda)$ by (30.5), exactly as in the two sections before this one. Evaluating that covariance needs Λ written as a function of the swap rate, and supplying it is what the next subsection's G does.

The adjustment

The correction depends on how the annuity varies with the swap rate, which is a question about the convexity of a bond price–yield relationship. Define $G(y)$ as the price, per dollar of principal, of a bond paying coupon S_0 with the swap's payment frequency and maturity, valued at flat yield y. Because the coupon equals the forward swap rate, $G(S_0) = 1$.

The trick is to adjust the bond price rather than the rate, because the bond price is the martingale. Its forward price satisfies $E[G(S(T_1))] = G(S_0)$, so expanding G to second order about S_0,

$$G(S_0) = E[G(S(T_1))] \approx G(S_0) + G'(S_0)\,(E[S] - S_0) + \frac{1}{2} G''(S_0)\,\mathrm{Var}(S)$$

the leading terms cancel and solving for $E[S]$ gives the adjustment directly. Then

$$E^{T_2}[S(T_1)] \;\approx\; S_0 \;-\; \frac{1}{2}\, S_0^2\, \sigma^2\, T_1\, \frac{G''(S_0)}{G'(S_0)} \tag{30.16}$$

where σ is the volatility of the forward swap rate.

Since bond prices fall as yields rise, $G' < 0$; since bond prices are convex in yield, $G'' > 0$. The ratio G''/G' is therefore negative and the whole adjustment is positive. The CMS rate to use always exceeds the forward swap rate.

The intuition is the familiar one from Chapter 4's treatment of duration and convexity. The holder of a CMS coupon has exposure to a long-maturity rate but does not hold the corresponding bond, so they capture the rate's movements without the offsetting convexity of a bond position. That asymmetry is worth something, and (30.16) prices it.

Figure 30.2 shows where the adjustment comes from geometrically (at the parameters of Example 30.3 the gap is a fraction of a percent of the price, far smaller than drawn), and the same picture explains every convexity adjustment in this chapter. What is pinned down is the *bond price*: G is a martingale, so $E[G(S)] = G(S_0)$. Now suppose the swap rate

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were distributed symmetrically about S_0. Because G is convex, the average of G over the two outcomes lies *above* G evaluated at the average: the chord sits above the curve, which is Jensen's inequality. That would violate the martingale condition. The only way to restore it is for the distribution of S to sit at a higher level, since G is decreasing. The distance the rate must shift is precisely the convexity adjustment.

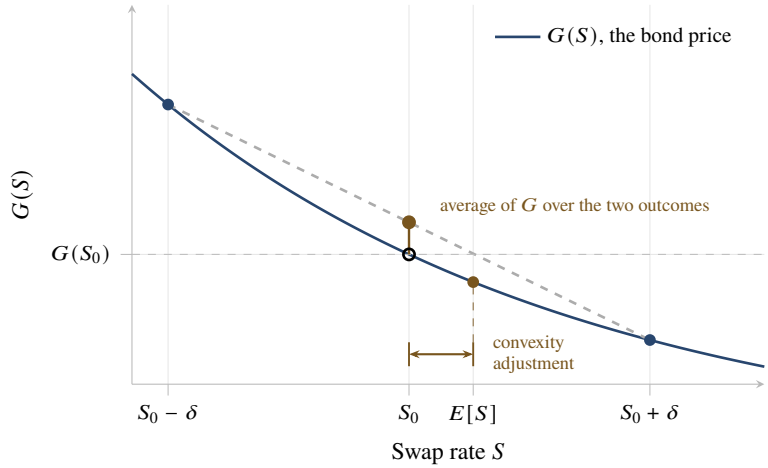


Figure 30.2: Why the expected swap rate exceeds the forward swap rate. Curvature exaggerated.

Example 30.3 CMS convexity adjustment

Consider a coupon paying the five-year swap rate observed in five years' time. The forward five-year swap rate is $S_0 = 4.00\%$, its volatility is $\sigma = 20\%$, payments are annual, and $T_1 = 5$.

$G(y)$ is the price of a five-year annual bond with a 4% coupon. Since the coupon equals S_0, the bond prices at par when $y = 4\%$. Differentiating,

$$G'(0.04) = -4.4518, \qquad G''(0.04) = 25.0125$$

so $G''/G' = -5.6185$ and

$$E^{T_2}[S] = 0.04 - \frac{1}{2} \times 0.04^2 \times 0.20^2 \times 5 \times (-5.6185) = 0.04 + 0.000899$$

The CMS rate is **4.090%** against a forward swap rate of 4.000%—a convexity adjustment of **9.0 basis points**.

Nine basis points on a rate of four percent is a 2.2% relative correction, and it grows with the square of volatility and linearly with time to reset. On a twenty-year CMS structure the adjustment can exceed 50 basis points. Unlike the timing adjustments of Section 30.2, this is not a refinement that can be safely ignored.

CMS spreads

The structures that trade in volume are usually not single CMS coupons but *spreads* between two swap rates. Since the payoff is a difference, it is tempting to assume the adjustments on the two legs largely offset. They do not: the ratio G''/G' that drives the adjustment depends strongly on the underlying swap's maturity. A ten-year bond has far more convexity relative to its duration than a two-year bond, so the long leg carries much the larger correction.

One arithmetic simplification makes these calculations tractable by hand. Because the bond in G is constructed with coupon equal to the forward swap rate, it prices at par, and for a par

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bond the first derivative is exactly the negative of the annuity:

$$G'(S_0) = -A(S_0), \qquad A(S_0) = \sum_{i=1}^{n} \frac{1}{(1+S_0)^i} = \frac{1-(1+S_0)^{-n}}{S_0} \tag{30.17}$$

The 5-year figure $G'(0.04) = -4.4518$ used in Example 30.3 is exactly the five-year annuity at 4%, and can be read off (30.17) without differentiating anything.

Example 30.4 A CMS spread coupon

A note pays the ten-year swap rate less the two-year swap rate, both observed in five years. The forward rates and their volatilities are

$$S_0^{10} = 4.40\%, \ \ \sigma_{10} = 18\%; \qquad S_0^{2} = 3.90\%, \ \ \sigma_{2} = 24\%$$

with annual payments and $T_1 = 5$. The forward spread is $4.40\% - 3.90\% = 50.0$ bp.

Each leg must be adjusted separately, using (30.16) with its own maturity, level, and volatility. Applying (30.17) for G' and the series of Technical Note 30 for G'':

	2-year leg	10-year leg
Forward swap rate S_0	3.90%	4.40%
Volatility σ	24%	18%
Annuity $A = -G'$	1.8888	7.9518
G''	5.4189	78.3883
$-G''/G'$	2.8690	9.8580
Convexity adjustment	6.28 bp	15.46 bp
Adjusted rate	3.9628%	4.5546%

The adjusted spread is therefore

$$4.5546\% - 3.9628\% = 59.18 \text{ basis points}$$

against a raw forward spread of 50.0 basis points—a net convexity effect of **+9.2 basis points**, which is 18% of the spread being valued.

The two-year leg is assumed *more* volatile than the ten-year, as short rates typically are, and this works against the effect. It is nowhere near enough to overcome it: the ratio $-G''/G'$ is 3.4 times larger for the ten-year leg, and that maturity mismatch dominates everything else. Had both legs referenced swaps of the same maturity, the adjustments would have largely canceled, leaving only a small net correction.

Two warnings attach to Example 30.4. First, it values the *linear* spread. Real steepener notes pay something like $\max(\text{spread}, 0)$, or a leveraged multiple of it, and an option on a difference depends on the **correlation** between the two swap rates, a quantity no convexity adjustment captures. Valuing the floored payoff requires a two-factor model or a replication argument, and the correlation involved is neither observable nor stable.

Second, the sign of the effect is systematic. Because the long leg carries the larger adjustment unless the short leg's volatility is far higher than in this example, convexity *widens* the fair spread on a steepener. A dealer who omitted the adjustment would sell every such note too cheaply, and would do so consistently rather than randomly.

Beyond the simple formula

Equation (30.16) assumes a single volatility for the swap rate, which is equivalent to assuming Black's model. Since the adjustment is fundamentally a statement about the *convexity* of the payoff, it is sensitive to the whole shape of the volatility smile, not just the at-the-money level.

Desks therefore price CMS by **static replication**: a CMS coupon decomposes into a

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portfolio of swaptions across all strikes, weighted by the payoff's second derivative. Write $H(s) = s\,\alpha_2(s)$, where $\alpha_2(s) = E^A[P(T_1,T_2)/A(T_1) \mid S(T_1) = s]$ is the annuity mapping function of Section 28.12 for payment at T_2. Then

$$V_0 = A(0)\,H(S_0) + \int_0^{S_0} H''(K)\,\mathrm{Rec}(K)\,dK + \int_{S_0}^{\infty} H''(K)\,\mathrm{Pay}(K)\,dK \tag{30.18}$$

where $\mathrm{Rec}(K)$ and $\mathrm{Pay}(K)$ are today's prices of receiver and payer swaptions struck at K: receivers below the forward swap rate, payers above, each weighted by $H''(K)$. This approach captures the whole volatility surface and is the market standard. Section 12.8 builds an arbitrary payoff from a strip of options and Section 22.3 applies the construction to variance; the CMS case is its most commercially important application in rates. Equation (30.16) remains valuable as a quick estimate and a sanity check on the replication machinery.

Case Study 30.2 Steepeners and the flat curve

CMS spread structures were sold widely to retail and mid-market institutional investors during the 2000s, typically as notes paying a high fixed coupon for a year or two, then switching to a multiple of the 10-year-less-2-year swap spread.

Their appeal rested on the curve remaining upward-sloping. As policy rates rose (the Federal Reserve's until mid-2006, the European Central Bank's into 2008), the curve flattened and in places inverted, and the floating coupons on many of these notes fell to zero. They came back only when the rate cuts of 2008 steepened the curve again. Investors who had understood the note as a high-yielding bond discovered they had bought a leveraged bet on the shape of the yield curve.

The episode shows, obliquely, why this chapter's arithmetic matters. The convexity adjustment is part of the price of the curve exposure embedded in these structures, and the floor on the spread is an option whose value turns on the correlation the adjustment does not capture. An investor who could compute neither could not tell what they had been paid for the risk—and, as it turned out, most had not been paid nearly enough.

30.6 QUANTO ADJUSTMENTS

The final adjustment concerns currency rather than timing. A quanto (Section 16.7) is a derivative whose payoff depends on a variable denominated in one currency but is settled in another, at a *fixed* exchange rate specified in the contract.

Examples are common. A dollar-settled option on the Nikkei 225 that pays \$1 per index point removes the yen exposure entirely: the investor gets the Japanese equity return without the currency return. A euro-settled swap referencing dollar interest rates works the same way.

Why the FX rate matters even though it is fixed

The apparent paradox of a quanto is that the exchange rate is contractually fixed, yet exchange rate volatility affects the price. The resolution lies in hedging.

A dealer short a quanto on a foreign asset hedges by holding that asset, so the hedge's gains and losses arrive in foreign currency and are worth, at home, whatever the exchange rate is when they arrive. The quanto's payoff ignores the exchange rate. If the asset tends to rise when the foreign currency is strong, the hedge's gains are converted at a high rate and its losses at a low one, so in domestic currency the hedge systematically outruns the payoff it replicates. If the asset tends to rise when the foreign currency weakens, it lags. Either way the effect recurs, a benefit or a cost according to the correlation's sign, and it must appear in the price.

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The adjustment

Adopt the convention that S is the exchange rate expressed as **units of domestic currency per unit of foreign currency** (for a US investor holding a Japanese asset: USD per JPY). The convention matters: reversing it reverses the sign of the answer. Let V be the foreign-currency price of the underlying asset, σ_V and σ_S their volatilities, and ρ the correlation between them.

The correction is the same change of numeraire as the rest of the chapter. A foreign asset's natural measure is the *foreign* risk-neutral measure Q^f, the one that goes with the foreign money market account B_f; a quanto settles in domestic currency, so the required measure is the domestic Q^d, whose numeraire is B_d. The density taking one to the other is the domestic account over the domestic value of the foreign one, normalized,

$$\Lambda_t = \frac{B_d(t)}{S(t)\,B_f(t)} \cdot S(0)$$

which moves *against* the exchange rate, so (30.5) charges minus the covariance between the asset and S, the quanto effect in the same form as every other adjustment here, and with the sign that makes a positively correlated asset drift below r_f. [TN30] carries the derivation out in full. Under Q^d the asset's drift is therefore not its foreign risk-free rate r_f but

$$\mu_V = r_f - \rho\,\sigma_V\,\sigma_S \tag{30.19}$$

This can also be checked without the density, in two steps. Under the domestic risk-neutral measure, an investor holding foreign cash owns a traded domestic asset worth $S(t)e^{r_f t}$, whose drift must be the domestic risk-free rate r_d, so $\mu_S = r_d - r_f$, which is covered interest parity. The foreign asset's domestic value, VS, is also a traded domestic asset, and Itô's product rule gives

$$\frac{d(VS)}{VS} = (\mu_V + \mu_S + \rho\,\sigma_V\sigma_S)\,dt + \sigma_V\,dW_V + \sigma_S\,dW_S$$

whose drift must also be r_d; substituting $\mu_S = r_d - r_f$ gives (30.19). The cross-variation term $\rho\sigma_V\sigma_S$ is the entire source of the quanto effect, and it is the covariance the density charges.

The term $-\rho\sigma_V\sigma_S$ is the **quanto drift adjustment**. The measure change is the machinery, the covariance is the economics, and the hedging argument above is what the covariance means on a desk. Hence the quanto forward relates to the ordinary foreign-currency forward F_0^f by

$$\boxed{F_0^{\text{quanto}} \;=\; F_0^f\,e^{-\rho\,\sigma_V\,\sigma_S\,T}} \tag{30.20}$$

Once (30.19) is in hand, quanto derivatives are priced with entirely standard machinery: use the adjusted drift, and otherwise apply the Black–Scholes–Merton model or Black's model as usual. A quanto call, for instance, is valued with the ordinary Black formula using F_0^{quanto} in place of the forward and discounting at the *domestic* rate.

Section 16.7 priced a quanto forward and call at a positive correlation, where the adjustment lowers the forward. For the dollar-settled Nikkei position above it raises it: Japanese equities have tended to rise as the yen weakens ($\rho < 0$ in dollars per yen), so the dealer's hedge earns its index gains in a weakened yen, and at $\rho = -0.3$, $\sigma_V = 20\%$ and $\sigma_S = 10\%$ the one-year quanto forward sits about 0.6% above the yen forward to pay for that shortfall.

Two observations follow. First, the adjustment vanishes when $\rho = 0$: an asset uncorrelated with its own exchange rate needs no quanto correction, however volatile either may be. Second, ρ here is a correlation between an asset and an exchange rate over the life of the contract, which is notoriously unstable: the Nikkei–yen correlation has changed sign across regimes. Quanto positions carry a **correlation risk** that cannot be hedged with liquid instruments, and desks reserve against it instead.

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30.7 A UNIFIED VIEW

Every adjustment here has been the same calculation: name the measure under which the variable's forward value is unbiased, name the measure the contract's payoff requires, write the density between them, and take the covariance. Only the density changes, as the table shows, where each arrow runs from the first measure to the second.

Adjustment	Measure change	Covariance that drives it
Futures/forward convexity (30.12)	$Q^{T_2} \longrightarrow Q$ (money market)	Margin flows with the financing rate
Timing (30.8)	$Q^{T^*} \longrightarrow Q^T$	Variable with the deferral rate
In-arrears (30.14)	$Q^{T_2} \longrightarrow Q^{T_1}$	Rate with itself
CMS (30.16)	Q^A (annuity) $\longrightarrow Q^{T_2}$	Swap rate with the annuity
Quanto (30.20)	Q^f (foreign) $\longrightarrow Q^d$ (domestic)	Asset with the exchange rate

Reading across the table, the recipe for any adjustment is the same:

1. Identify the measure under which the variable's forward value is an unbiased expectation, its **natural measure**, determined by the numeraire that appears in its definition.
2. Identify the measure required by the contract's actual payoff date and currency.
3. Compute the covariance of the variable with whatever drives the numeraire ratio.
4. The adjustment is that covariance, to first order.

Two practical warnings close the chapter.

Adjustments are model-dependent, and the dependence is not small. Every formula here required a distributional assumption: Ho–Lee in (30.12), lognormality in (30.8), (30.15), and (30.16). Each depends on a volatility, and (30.8) and (30.20) additionally depend on a correlation. These are estimates, not observables, and the adjustment inherits their uncertainty. The 9.0 bp CMS adjustment of Example 30.3, computed at 20% volatility, becomes a 20 bp adjustment at 30% volatility—an increase of more than double. Reporting a convexity adjustment to three decimal places implies a precision that the inputs do not support.

Adjustments are systematic, not random. Unlike bid–ask spread or execution slippage, an omitted adjustment is a bias in a known direction that accumulates across every trade of its type on the book. This is why a desk that tolerates a basis point of noise will not tolerate half a basis point of bias, and why these corrections are computed and monitored rather than ignored.

SUMMARY

Forward values equal expected future values only under the measure associated with the contract's natural numeraire. When a contract's settlement mechanics, payment date, or settlement currency depart from that natural arrangement, the forward value is a biased estimate and an adjustment is required. Every such adjustment is exactly a covariance between the variable being valued and the density that changes the numeraire; it can be evaluated exactly where that density is affine in the variable, and to leading order otherwise.

Because futures contracts are settled daily, futures rates are expectations under the risk-neutral measure while forward rates are expectations under the forward measure. The resulting convexity adjustment makes futures rates exceed forward rates, by an amount growing with the square of maturity, negligible at the front of the strip and material at long maturities.

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Averaging-style RFR futures carry adjustments of similar magnitude to the term-rate contracts they replaced, but with delta that decays through the reference period.

Timing adjustments apply when a payoff is deferred or accelerated relative to its natural payment date. They are negative for deferral when the variable is positively correlated with rates, and are typically small. The transition to risk-free rates largely eliminated the classical in-arrears convexity adjustment, since a compounded RFR is naturally determined and paid at the end of its period, but left smaller residual adjustments, from payment delays and from last-reset conventions that pay a stale rate one period late.

CMS convexity is the largest adjustment of the payment-timing family, arising because a swap rate is a martingale under the annuity measure while a CMS coupon is paid at a single date. It is always positive, large enough to matter in pricing, and is priced in practice by static replication with swaptions so that the full volatility smile is captured.

Quanto adjustments arise when a payoff based on a foreign variable is settled in domestic currency at a fixed exchange rate. Exchange rate volatility affects the price despite the fixed rate because it affects the dealer's hedging cost. The adjustment multiplies the foreign forward by $e^{-\rho\sigma_V\sigma_S T}$ and vanishes when the asset and exchange rate are uncorrelated.

All these adjustments depend on volatilities and correlations that are estimated rather than observed. Their importance lies less in their size than in their systematic direction: an omitted adjustment biases every affected trade the same way.

FURTHER READING

Geman, El Karoui, and Rochet (1995) is the foundational reference for the change-of-numeraire machinery this chapter runs on, and the source of the change-of-numeraire formulas used here. Brigo and Mercurio's *Interest Rate Models—Theory and Practice* treats convexity adjustments and CMS pricing in full in its Chapter 13. For the two adjustments computed here in closed form, Kirikos and Novak (1997) derive the Hull–White futures convexity adjustment and Hagan's "Convexity Conundrums" (2003) is the standard practitioner treatment of static replication for CMS.

PRACTICE QUESTIONS

30.1 Explain why a forward rate is a martingale under the T_2-forward measure but not under the risk-neutral measure. Which measure is relevant for valuing a payoff made at T_2?

30.2 Without performing any calculation, state the sign of the required adjustment in each case, and justify your answer in one sentence:

(a) a variable uncorrelated with interest rates, paid three months after its natural date;

(b) a ten-year swap rate paid as a single cash flow in five years;

(c) a quanto forward on a foreign equity index, where the index and the foreign currency are positively correlated.

30.3 A three-month interest rate futures contract has $T_1 = 3$ years and $T_2 = 3.25$ years. The short-rate volatility is 90 basis points per year. The futures rate is 3.85%. Calculate the forward rate using the Ho–Lee adjustment.

30.4 Repeat Question 30.3 with $T_1 = 8$ and $T_2 = 8.25$. By what factor does the adjustment increase, and does that match what equation (30.12) predicts?

30.5 Explain in words why a trader who is long the rate through a futures position benefits from daily settlement, and why this requires the futures rate to exceed the forward rate.

30.6 A payoff equal to the two-year swap rate is observed in three years and paid six months later. The forward two-year swap rate is 3.50% with 22% volatility; the forward rate for

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the deferral period is 3.20% with 25% volatility; the correlation is 0.8. Calculate the timing adjustment in basis points.

30.7 A rate for a six-month period beginning in four years is paid at the start of the period rather than the end. The forward rate is 3.75% with volatility 28%. Calculate the in-arrears convexity adjustment using both the exact form (30.14) and the approximation in (30.15), and comment on the difference.

30.8 Explain why the transition from LIBOR to compounded risk-free rates eliminated most of the in-arrears convexity adjustment. Identify two situations in which a convexity adjustment is still required under RFR conventions.

30.9 A quanto forward on a foreign stock index is settled in domestic currency. The index has 25% volatility, the exchange rate (domestic per foreign) has 12% volatility, and the correlation is 0.4. The one-year foreign-currency forward level is 5,000. Calculate the quanto forward level.

30.10 Continuing Question 30.9, the domestic interest rate is 3%. Value a one-year quanto call on the index, struck at the quanto forward level and paying one unit of domestic currency per index point, and compare it with the same call if the correlation were zero. Why does the correlation matter when the contract's exchange rate is fixed?

30.11 Without using algebra, explain why the convexity adjustments on the two legs of a CMS spread do not cancel. Under what condition would they cancel?

FURTHER QUESTIONS

30.12 A CMS coupon pays the ten-year swap rate observed in three years. The forward ten-year swap rate is 4.20%, its volatility is 18%, and payments are annual. Compute G' and G'' for a ten-year annual bond with a 4.20% coupon at a flat yield of 4.20%, and hence find the convexity adjustment. Compare the size of your answer with Example 30.3 and explain the difference.

30.13 Using the result of Question 30.12, recompute the adjustment at volatilities of 12% and 24%. Comment on what your answers imply about the precision with which convexity adjustments can be quoted.

30.14 A dealer sells a quanto call on a foreign index. Describe the dealer's hedge, explain how the hedge must be rebalanced as the index moves, and use this to give an economic derivation of the sign of the quanto adjustment in equation (30.20). Why can this position not be fully hedged in practice?

30.15 A structured note pays $3 \times$ (10-year swap rate $-$ 2-year swap rate) annually for ten years, subject to a floor of zero. Identify every adjustment from this chapter that its valuation requires, and explain which of them dominates.

30.16 Show that a rate futures contract's convexity adjustment can be interpreted as the value of an option-like exposure to realized rate volatility. Using this interpretation, explain why a convexity trade that appeared profitable on 2019 volatility estimates would have lost money through 2022.

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Chapter 31
Short-Rate Models

Chapter 29 priced a cap and a swaption with Black's model and Chapter 30 corrected the forward when the payoff arrives at the wrong time or in the wrong currency. Both took a quoted volatility and produced one number.

Neither built a model of the term structure, and there is a large class of instruments for which that is fatal. A **Bermudan swaption** can be exercised on any one of a fixed set of dates; its holder decides at year three whether to exercise, and that decision depends on the entire curve at year three—a curve Black's model never constructs. A callable bond's issuer faces the same decision annually for a decade. A structured note pays a coupon that depends on where rates are, conditional on where they have been.

For these, the model has to produce a consistent evolution of the whole curve, not a distribution for one rate at one date. A desk running a book needs more than that: one model that prices every deal on it consistently, so that a swaption, a callable bond and a Bermudan are valued off the same curve dynamics and their risks aggregate. The oldest and still most widely used way to do that is to model the short rate and derive everything else from it.

Section 31.12 is the one that changes how the rest of Part VI reads: it prices the defect that motivates Chapter 32, and it does so with a number rather than an assertion. A one-factor model prices a CMS spread option at 56.2 basis points when, at a realistic correlation between the two rates, it is worth 245.0.

The chapter runs from the models to the defect that motivates the next one. Sections 31.1 to 31.4 are the equilibrium models, Vasicek and Cox–Ingersoll–Ross (CIR), and why a model that cannot reprice today's curve is not used to price. Sections 31.5 to 31.8 are the no-arbitrage answer: Ho–Lee, Hull–White, and the tree that implements it. Sections 31.9 to 31.11 are what the tree is built for: Bermudan swaptions, what calibration cannot pin down, and callable bonds through option-adjusted spread. Sections 31.12 to 31.15 are the repairs and their limits: the one-factor failure priced in basis points, two factors, negative rates, and the transition; Sections 31.16 to 31.18 say where these models are still the right tool and who uses them.

31.1 WHAT A SHORT-RATE MODEL IS

Specify a stochastic process for the instantaneous short rate $r(t)$ (the rate for borrowing over the next instant, the limit of the zero rate as maturity shrinks to zero) under the risk-neutral measure. Every zero-coupon bond price then follows from

$$P(t,T) = \hat{E}_t\left[\exp\left(-\int_t^T r(s)\,ds\right)\right] \tag{31.1}$$

which is Chapter 13's risk-neutral valuation applied to a payoff of \$1.

This is a strong modeling choice. The entire term structure, at every future date, is generated by a single scalar state variable. That is the source of the approach's tractability and of the defect in Section 31.12, and the two cannot be separated.

Two properties make a model usable.

Analytic bond prices. If (31.1) evaluates in closed form, the model is fast enough to calibrate. The widest class for which it does is the **affine** models, in which $\ln P(t,T)$ is affine in r:

$$P(t,T) = A(t,T)\, e^{-B(t,T)\, r(t)} \tag{31.2}$$

Written in rates rather than prices, with $R(t,T)$ the continuously compounded zero rate of Section 4.8,

$$R(t,T) = -\frac{\ln P(t,T)}{T-t} = \frac{B(t,T)\, r(t) - \ln A(t,T)}{T-t}$$

so every rate on the curve is affine in the one state variable, and Example 31.1's curves follow from it.

And the ability to fit today's curve. A model whose output curve differs from the observed one misprices the underlying swaps and bonds themselves, before any option is considered. This requirement, not realism, is what separated the two generations of models, and Section 31.4 makes the separation concrete.

31.2 VASICEK AND THE EQUILIBRIUM APPROACH

The first generation asked what process for r is economically reasonable, and derived the curve from it. Vasicek's model is

$$dr = a(b - r)\, dt + \sigma\, dW \tag{31.3}$$

The drift pulls r toward b at speed a: **mean reversion**, and the property that distinguishes rates from equities. A stock price has no level it returns to; a short rate does, because a central bank and an economy both act on it. Figure 31.1 shows what reversion does to the whole distribution, not just to the expected rate. It is slow at plausible speeds: the half-life, the time over which the expected gap between r and b halves, is $\ln 2/a$, about seven years at $a = 0.10$, so over the life of a ten-year deal the pull toward b is partial rather than complete. Set $a = 0$ and the rate is a driftless random walk whose spread grows as $\sigma\sqrt{t}$ without limit; restore a and the spread converges to $\sigma/\sqrt{2a}$. That second half is what lets the model produce a finite long rate at all, and Section 31.5 returns to it, because the $a = 0$ case is a model in its own right. Note which rate reverts to b: the expected short rate does, as t grows, while the zero curve's long end sits at the lower level R_∞ marked on the figure, whatever the short rate happens to be.

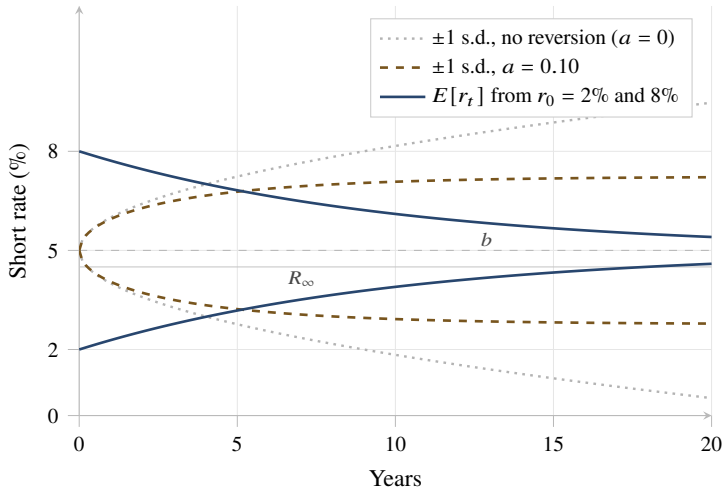


Figure 31.1: Mean reversion at $a = 0.10$, $b = 5\%$, $\sigma = 1\%$, against the same model with $a = 0$; R_∞ is the limiting zero rate of Example 31.1.

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Equation (31.3) is affine, with

$$B(t,T) = \frac{1 - e^{-a(T-t)}}{a}$$

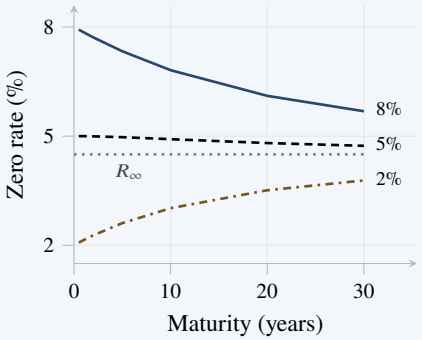
$$\ln A(t,T) = \frac{(B(t,T) - (T-t))\left(a^2 b - \sigma^2/2\right)}{a^2} - \frac{\sigma^2 B(t,T)^2}{4a}$$

Bond prices, and therefore the whole curve, follow from $r(t)$ and three constants. The two terms in σ^2 are the convexity correction. The $-\sigma^2/2$ inside the first is what pulls the long end of the curve below b in Example 31.1; the last term's contribution to the zero rate dies away as maturity grows.

Example 31.1 Three starting rates, one curve

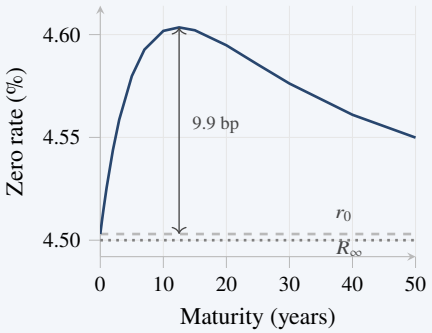
Take $a = 0.10$, $b = 5\%$, $\sigma = 1\%$, and compute the zero curve from three different starting short rates.

Maturity	$r_0 = 2\%$	$r_0 = 5\%$	$r_0 = 8\%$
0.5y	2.073%	5.000%	7.926%
1y	2.144%	4.999%	7.853%
2y	2.275%	4.994%	7.713%
5y	2.610%	4.971%	7.332%
10y	3.020%	4.916%	6.812%
20y	3.513%	4.810%	6.107%
30y	3.783%	4.734%	5.684%



The model produces upward-sloping, flat, and downward-sloping curves from the same parameters: the shape is determined by where r_0 sits relative to b.

A fourth shape is available, but barely. The curve is humped for $b - 3\sigma^2/4a^2 < r_0 < b$, a band 75 basis points wide at these parameters, and the largest hump anywhere in that band is 9.9 basis points, at $r_0 = 4.50\%$ peaking at twelve and a half years. Real curves hump by five to ten times that. *The shape is in the model; the magnitude is not.*



$a = 0.10$, $b = 5\%$, $\sigma = 1\%$, $r_0 = 4.503\%$

All three converge toward the same long rate, and it is not b:

$$R_\infty = b - \frac{\sigma^2}{2a^2} = 5\% - \frac{0.0001}{0.02} = \mathbf{4.50}\%$$

The gap is a convexity effect, and it is the same effect as Chapter 30's: a long bond's price is convex in the rate, so uncertainty raises its expected price and lowers its yield. Here the model produces the adjustment automatically instead of requiring it to be added.

But now count the degrees of freedom. Three parameters and one state variable generate the entire curve. The curve has hundreds of observable points and the model has four numbers, so it cannot match the market. That is not a calibration failure; it is arithmetic.

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31.3 COX–INGERSOLL–ROSS

Vasicek's r is normally distributed and can go negative. The best-known repair is

$$dr = a(b - r)\,dt + \sigma\sqrt{r}\,dW \tag{31.4}$$

The $\sqrt{r}$ has two consequences. Volatility falls as the rate falls, consistent with the pre-2008 data, and the diffusion coefficient vanishes at zero, so the rate remains nonnegative. Strict positivity requires more, namely the **Feller condition** $2ab > \sigma^2$; when it fails, zero is attained and acts as a reflecting boundary.

The model remains affine, so bond prices retain a closed form, and its coefficients are more involved than Vasicek's without being harder to evaluate. With $\tau = T - t$ and $\gamma = \sqrt{a^2 + 2\sigma^2}$,

$$B(t,T) = \frac{2\,(e^{\gamma\tau} - 1)}{(\gamma + a)\,(e^{\gamma\tau} - 1) + 2\gamma}$$

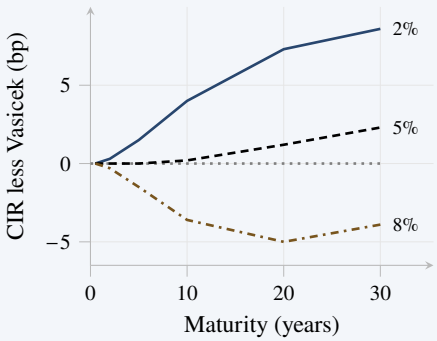
$$A(t,T) = \left[\frac{2\gamma\, e^{(a+\gamma)\tau/2}}{(\gamma + a)\,(e^{\gamma\tau} - 1) + 2\gamma}\right]^{2ab/\sigma^2}$$

The limiting zero rate is $R_\infty = 2ab/(\gamma + a)$, again below b and again by a convexity term, but one that now depends on the level of rates, because the volatility does. Example 31.2 measures what the change of diffusion does to the curves.

Example 31.2 What positivity costs

The two models are comparable only if their volatilities are. Vasicek's diffusion is σ and CIR's is $\sigma\sqrt{r}$, so the two agree at $r = b$ when CIR's σ is set to $0.01/\sqrt{0.05} = 4.47\%$. Take Example 31.1's $a = 0.10$ and $b = 5\%$ with that value, so both models carry a 1% instantaneous volatility at $r = 5\%$. The Feller condition holds comfortably, $2ab = 0.0100$ against $\sigma^2 = 0.0020$.

Maturity	$r_0 = 2\%$	$r_0 = 5\%$	$r_0 = 8\%$
0.5y	2.074%	5.000%	7.926%
1y	2.144%	4.998%	7.852%
2y	2.278%	4.994%	7.710%
5y	2.626%	4.971%	7.317%
10y	3.059%	4.918%	6.777%
20y	3.586%	4.821%	6.057%
30y	3.869%	4.757%	5.645%



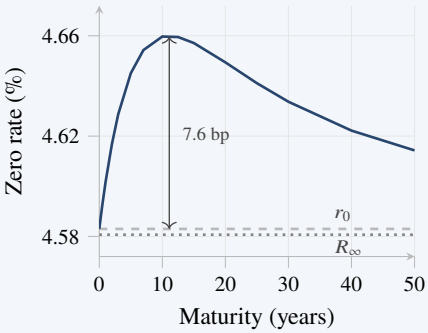
Left: CIR's zero curves. Right: those curves less Example 31.1's, in basis points.

The two sets of curves are nearly the same. The largest gap across thirty years is 8.6 basis points, and inside five years it is under two. Positivity cost almost nothing in the shape of the curve, which is why CIR held the field for two decades.

The gap that does appear has an instructive sign. From $r_0 = 2\%$ the CIR curve sits above Vasicek's, and from 8% below. A convexity correction grows with variance, and in CIR the variance moves with the rate: from a low start there is less of it to depress the long end, from a high start more, while Vasicek applies the same correction either way. The limiting zero rate records the same thing, $R_\infty = 4.58\%$ against Vasicek's 4.50%, a drag below b of 0.42 percentage points rather than 0.50.

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CIR humps too, and over a narrower band: from 4.40% to b, 60 basis points against Vasicek's 75. The largest hump in it is 7.6 basis points, at $r_0 = 4.58\%$ peaking at eleven years. *Both models put their best hump at R_∞ and neither can make it large.* Producing a realistic hump is not something a one-factor model does, whichever diffusion it carries.



$a = 0.10$, $b = 5\%$, $\sigma = 4.47\%$, $r_0 = 4.583\%$

The CIR model was for two decades the standard answer to Vasicek's negative rates, and what happened to that argument should be stated plainly. *Rates went negative*, in several major currencies, for years. A model that cannot produce a negative rate cannot be calibrated to a market that has one—not approximately, but at all, because there is no parameter value under which a process that cannot fall below zero assigns any probability to the observed state.

So the property that made CIR attractive is the property that retired it. Section 31.14 takes this up as a general point, because it recurs.

31.4 WHY EQUILIBRIUM MODELS ARE NOT USED TO PRICE

Example 31.1 contains the whole argument, and it deserves to be stated as a principle rather than a limitation of one model.

An **equilibrium model** treats today's term structure as an output. It asks what curve the assumed process implies, and the answer is a curve with three or four parameters' worth of freedom.

A trading desk needs today's term structure to be an input. If the model says the ten-year zero is 4.92% and the market says 4.10%, then every instrument priced on the model is wrong by an amount that has nothing to do with the option in it. A swaption priced on a curve that is eighty basis points off is not a swaption price.

The distinction is worth being precise about, because it is often stated as though equilibrium models were merely less accurate.

They are answering a different question, and it is a legitimate one. An economist asking what the curve *should* look like, or a risk manager generating scenarios that are internally consistent over decades, wants a model whose curve is derived rather than fitted. Real-world scenario generation for pension and insurance liabilities still uses models of this kind, for exactly that reason.

But a desk quoting a price cannot use one, and the second generation of models exists to remove this problem and nothing else.

31.5 HO–LEE AND THE NO-ARBITRAGE IDEA

Make the drift time-dependent and choose it so the model reproduces the observed curve. A model built this way, taking today's curve as an input rather than producing it, is a **no-arbitrage model**.

Start from what is observable. The input is today's zero curve, the discount factors $P(0,t)$ that Chapter 7 builds from quoted instruments. What a fitted model needs from them is not the zero rate $R(0,t)$ of Section 31.1 but the **initial forward curve**, the instantaneous rate for

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borrowing at t that today's prices imply:

$$f(0,t) = -\frac{\partial \ln P(0,t)}{\partial t} \qquad \text{equivalently} \qquad P(0,t) = \exp\left(-\int_0^t f(0,u)\,du\right)$$

The two descriptions carry the same information, and $R(0,t)$ is the average of $f(0,u)$ over $[0,t]$. The forward curve is the useful one here because a drift is a local quantity, so it has to be matched against a local description of the curve rather than against an average.

Ho–Lee is the simplest case:

$$dr = \theta(t)\,dt + \sigma\,dW \tag{31.5}$$

$\theta(t)$ is not a parameter to estimate; it is solved for:

$$\theta(t) = \frac{\partial f(0,t)}{\partial t} + \sigma^2 t$$

and the model fits the curve exactly by construction.

This is the whole idea of the no-arbitrage generation, and the surprising part is how cheap it is. One function of time, chosen mechanically, converts a model that cannot price into one that can. The cost is that $\theta(t)$ carries no information (it is whatever it has to be), so a model in this class explains nothing about why the curve has the shape it has. It is an interpolation device with a stochastic component, and that is the correct way to understand every model in the rest of this chapter.

Ho–Lee has one defect that rules it out in practice: *no mean reversion.* The volatility of the T-maturity forward rate is σ regardless of T, so the model says a thirty-year forward rate is as volatile as a one-year forward rate. Section 31.6 shows what the data say.

31.6 HULL–WHITE

Add mean reversion to Ho–Lee, or a fitted drift to Vasicek (the same model either way):

$$dr = \big[\theta(t) - ar\big]dt + \sigma\,dW \tag{31.6}$$

As in Ho–Lee, θ is solved for rather than estimated, and in terms of the same initial forward curve $f(0,t)$,

$$\theta(t) = \frac{\partial f(0,t)}{\partial t} + a\,f(0,t) + \frac{\sigma^2}{2a}\left(1 - e^{-2at}\right)$$

which returns Section 31.5's θ as $a \to 0$, the last term tending to $\sigma^2 t$. [TN31] derives this and explains why a single function of time can match every maturity at once. In practice it is never evaluated, because bond prices can be written with today's curve in them directly:

$$\ln A(t,T) = \ln\frac{P(0,T)}{P(0,t)} + B(t,T)\,f(0,t) - \frac{\sigma^2}{4a}\left(1 - e^{-2at}\right)B(t,T)^2$$

so θ disappears from everything that is computed. It was solved for the curve, and the curve is what remains.

This is the workhorse. It fits the initial curve exactly, has closed-form bond and European option prices, and builds on a recombining tree. Almost every desk that uses a short-rate model uses this one or a close relative.

The two constants do different jobs, and confusing them is the most common error in using the model.

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Example 31.3 What a and σ separately control

The instantaneous forward rate follows by differentiating (31.2):

$$f(t,T) = -\frac{\partial \ln P(t,T)}{\partial T} = -\frac{\partial \ln A(t,T)}{\partial T} + e^{-a(T-t)}\, r(t)$$

since $\partial B/\partial T = e^{-a(T-t)}$. Only r is stochastic, so the forward rate inherits its diffusion scaled by that factor, and its volatility seen from today is

$$\sigma\, e^{-aT} \tag{31.7}$$

T	$a = 0.01$	$a = 0.05$	$a = 0.10$	$a = 0.20$
1y	99.0	95.1	90.5	81.9
2y	98.0	90.5	81.9	67.0
5y	95.1	77.9	60.7	36.8
10y	90.5	60.7	36.8	13.5
20y	81.9	36.8	13.5	1.8
30y	74.1	22.3	5.0	0.2

Forward-rate volatility in basis points, at $\sigma = 100$ bp.

σ sets the level of the volatility term structure and a sets its shape. At $a = 0.01$ the curve is nearly flat. That is Ho–Lee, and it says a thirty-year forward is three-quarters as volatile as a one-year forward. At $a = 0.20$ the thirty-year forward has essentially no volatility at all, which is equally wrong in the other direction.

Because the two act on different features, they are separately identified by the data. Section 29.9's volatility cube contains caps and swaptions of many expirations, and the term structure of their volatilities determines a while their level determines σ.

That is the encouraging version. Section 31.10 shows that for the instruments that matter most, it is not true.

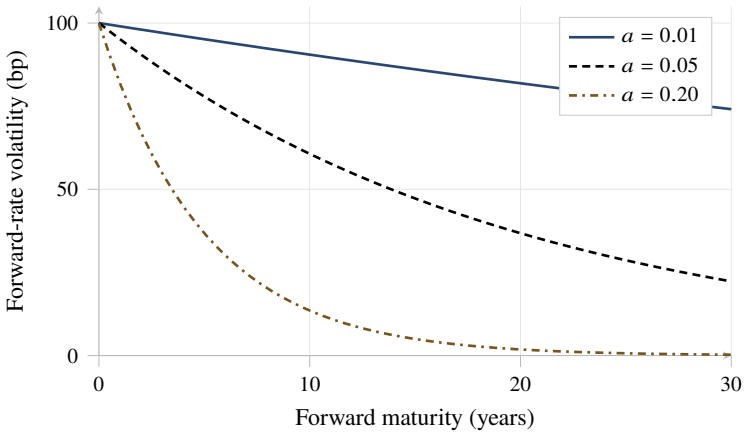


Figure 31.2: Equation (31.7) at $\sigma = 100$ bp.

Figure 31.2 is the parameter that makes Hull–White usable: a alone decides whether a thirty-year forward is nearly as volatile as a one-year forward or barely moves at all.

31.7 EUROPEAN OPTIONS IN CLOSED FORM

The model is usable before any lattice is built: every European rate option has a closed form, and they all come from one of them. Example 31.4 carries these formulas through a cap and a swaption.

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An option on a zero-coupon bond. Under the T-forward measure the forward bond price $P(t,S)/P(t,T)$ is a martingale, and since (31.2) makes $\ln P$ affine in a Gaussian r, it is a lognormal one. Its instantaneous volatility $\sigma e^{-a(T-t)}(1-e^{-a(S-T)})/a$ is deterministic, so the variance accumulates to a number rather than a random variable, and its square root over the option's life is

$$\sigma_P(T,S) = \frac{\sigma}{a}\left(1-e^{-a(S-T)}\right)\sqrt{\frac{1-e^{-2aT}}{2a}} \tag{31.8}$$

Lognormality with a deterministic volatility is the assumption on which Black's formula rests, so for a strike X and an expiration T on the bond maturing at S, the zero-bond call (ZBC) and put (ZBP) are, with Φ the standard normal distribution function,

$$\text{ZBC} = P(0,S)\,\Phi(h) - X\,P(0,T)\,\Phi(h-\sigma_P) \tag{31.9}$$

$$\text{ZBP} = X\,P(0,T)\,\Phi(\sigma_P - h) - P(0,S)\,\Phi(-h) \tag{31.10}$$

$$h = \frac{1}{\sigma_P}\ln\frac{P(0,S)}{X\,P(0,T)} + \frac{\sigma_P}{2}$$

the put (31.10) following from the call (31.9) by parity. Only today's curve and the two parameters appear. Both factors of (31.8) saturate (at σ/a in the bond's remaining life, at $\sigma/\sqrt{2a}$ in the option's), which is (31.7) seen from the other side. [TN31] derives (31.8) and works out both ceilings.

A cap. Section 29.3 showed that a caplet on notional L, struck at K on the rate for the accrual period $[T_i, T_{i+1}]$ of length τ_i, is $L(1+\tau_i K)$ puts on the T_{i+1} bond struck at $1/(1+\tau_i K)$. Substituting (31.10) prices a cap one period at a time, with no approximation anywhere.

A swap needs no formula at all. It is linear in bond prices,

$$V = L\left[P(0,T_0) - P(0,T_n) - K\sum_{i=1}^{n}\tau_i P(0,T_i)\right]$$

and neither a nor σ appears. A swap's value comes entirely from the curve. It could not be otherwise, since the model is fitted to that curve and reproduces it by construction. This locates where a model starts doing work. Its parameters do work only where there is optionality, and a model that misprices a swap has made a fitting error rather than a modeling one. That is Section 31.4's argument in one line.

A European swaption. Exercising a payer swaption is exercising the right to sell, at par, a bond paying $c_i = K\tau_i$ at each t_i, with the principal added to the last payment, $c_n = 1 + K\tau_n$; that is the equivalence of Section 29.6. What makes it tractable here is the single factor. Each $P(T,t_i)$ is a decreasing function of $r(T)$, so the coupon bond is too, and there is a unique r^* at which it is worth par. Above r^* *every* component bond is simultaneously below its value $X_i = P(T,t_i;\,r^*)$ there, and below r^* every one is above it. The exercise regions coincide, so

$$\text{payer swaption} = L\sum_{i=1}^{n} c_i\,\text{ZBP}(T,t_i,X_i) \tag{31.11}$$

This is **Jamshidian's decomposition**, and it is an identity rather than an approximation. It rests on monotonicity in a single state variable, so it survives neither a second factor (where the exercise region is a curve in the plane rather than a threshold) nor a Bermudan schedule, where the decision at one date turns on the value of the rights that remain. Both go to the tree of Section 31.8.

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Example 31.4 A cap and a swaption

Take a flat continuously compounded curve at 4%, so $P(0,T) = e^{-0.04T}$, with $a = 0.08$ and $\sigma = 1\%$ on a notional of 100. Left: a two-year semiannual cap struck at 4%, every period's forward rate being 4.040268%. Right: a one-into-three annual payer swaption struck at the par rate, whose annuity is 2.662181 and par rate 4.081077%, giving $r^* = 3.987895\%$ from (31.11).

Caplet	σ_P	Value
[0.5, 1.0]	33.98	0.142743
[1.0, 1.5]	47.12	0.190218
[1.5, 2.0]	56.60	0.222055
Cap		**0.555017**

Maturity	c_i	X_i	Value
2y	0.040811	0.960860	0.014025
3y	0.040811	0.923177	0.025775
4y	1.040811	0.886915	0.906913
Swaption			**0.946713**

σ_P in basis points.

The caplets rise in value with their expiration although the forward rate is identical in each, because σ_P grows with the option's life, and that growth is bounded, which is mean reversion at work. In the swaption almost all the value sits in the final strip, which carries the principal.

Both figures were also computed a second way, by integrating the payoff directly over the distribution of $r(T)$ with no decomposition and no bond-option formula, and the two routes agree to nine decimal places. That check is only correct under the T-forward measure, where

$$r(T) \sim \mathcal{N}\left(f(0,T),\ \frac{\sigma^2}{2a}\left(1 - e^{-2aT}\right)\right)$$

The mean is the forward rate exactly, while under the risk-neutral measure it carries a further $(\sigma^2/2a^2)(1 - e^{-aT})^2$. Integrating the swaption against that instead returns 0.952366 rather than 0.946713, an error of 0.6% from nothing but the wrong measure in the last step, which is the subject of the two examples that follow.

Chapter 28 proved that the choice of numeraire cannot change a price, and that the right choice can change the work enormously. With a model in hand, both claims can be put to a number.

Example 31.5 Invariance, and what it buys

A caplet on the simple rate over [2.0, 2.5], paid at 2.5, struck at 4%, notional 100. The economy is (31.6) with $a = 0.08$ and $\sigma = 1\%$, on a market curve carrying $P(0,2.0) = 0.932913$ and $P(0,2.5) = 0.914898$. The forward rate follows:

$$F_0 = \frac{1}{\tau}\left(\frac{P(0,T_1)}{P(0,T_2)} - 1\right) = 2.0 \times \left(\frac{0.932913}{0.914898} - 1\right) = 3.9381\%$$

(a) Under the 2.5-year forward measure. The forward rate is a martingale (no drift to model), and by the identity of Section 29.3, with $1 + \tau K = 1 + 0.5 \times 4\% = 1.02$, the caplet is $L(1 + \tau K) = 102$ puts on the 2.5-year bond struck at $1/(1 + \tau K) = 0.980392$. That bond is lognormal here, because (31.2) makes $\ln P$ affine in r and Hull–White's r is Gaussian; its forward price carries a deterministic volatility:

$$\sigma\, B(T_1,T_2)\sqrt{\frac{1 - e^{-2aT_1}}{2a}} = 64.1 \text{ bp}$$

As with the zero-bond options of (31.10), that makes Black's formula exact here rather than an approximation:

$$\text{caplet} = L \cdot P(0,T_2)\left[(1 + \tau F_0)\Phi(d_1) - (1 + \tau K)\Phi(d_2)\right] = \mathbf{0.22478353}$$

A closed form, evaluated in microseconds.

(b) Under the risk-neutral measure. Now the numeraire is the money market account, so the payoff must be discounted along each simulated path of r. Two million antithetic paths, two

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hundred steps:

0.22498411 (standard error 0.00023576)

The two agree to 0.85 standard errors. This is the theorem, confirmed: the numeraire is a choice of units and prices do not depend on it.

(c) The error the measure change exists to prevent. Suppose one computes the expected payoff under the risk-neutral measure and discounts it at today's zero rate, treating the payoff and the discount factor as independent:

0.23044665 an error of **+2.52**%

That 2.52% is exactly the covariance between the payoff and the discount factor. A caplet pays when rates are high, and when rates are high the discount factor is small. Ignoring that correlation overstates the value.

This is what the forward measure does. Using $P(t,T_2)$ as numeraire folds the discounting into the numeraire, so no covariance term survives. The reason Black's formula for a caplet is exact rather than approximate is that the convention already made this choice.

A measure change is not a formality; it is the difference between an exact formula and a simulation, and between a correct answer and one wrong by the covariance you forgot.

The same economy settles a second question. A forward price is an expectation under the T-forward measure, as Section 28.7 shows, and is not one under the risk-neutral measure; the difference is a covariance.

Example 31.6 And it is false under the risk-neutral measure

$S_0 = 100$, equity volatility 25%, $T = 5$, with the curve of Example 31.5 so that $P(0,T) = 0.824639$. The forward price is

$$F(0,T) = \frac{100}{0.824639} = 121.2652$$

and this is a no-arbitrage fact: it does not depend on any correlation.

Now compute $E^{Q}[S_T]$ under the risk-neutral measure, in the Hull–White economy of Example 31.5, for a range of correlations between the stock and the short rate. That expectation has a name. A futures price is a martingale under Q, as Section 28.7 established, so $E^{Q}[S_T]$ *is* the futures price for delivery at T, and the ratio below is the futures–forward basis of Section 30.3, computed inside a model rather than read off a screen:

$\rho(S,r)$	$E^{Q}[S_T]/F(0,T)$	**Gap**
−0.50	0.989442	−1.056%
−0.25	0.996260	−0.374%
0.00	1.003125	+0.313%
+0.25	1.010037	+1.004%
+0.50	1.016997	+1.700%

The forward price is fixed and $E^{Q}[S_T]$ moves with ρ, so with stochastic rates the two agree only at one particular correlation, by coincidence; in general they coincide only when rates are deterministic. Zero correlation is not enough: the $\rho = 0$ row still runs 0.31% high.

The gap has two parts. The part that moves with ρ is the covariance between the asset and the discount factor, the same quantity that produced the 2.52% error in Example 31.5. A caplet mispriced by ignoring that covariance and a futures price standing above a forward are one term under two names. The part that survives at $\rho = 0$ is the discount factor's own convexity:

$$E^{Q}\left[e^{\int_0^T r\,ds}\right] = \frac{e^{\mathrm{Var}\left(\int_0^T r\,ds\right)}}{P(0,T)}$$

This exceeds $1/P(0,T)$ whenever the short rate is stochastic.

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A practical warning follows. A Monte Carlo engine that simulates under the risk-neutral measure and reports "the expected terminal price" is not reporting the forward; it is reporting the futures price. At $\rho = 0.5$ over five years the difference is 1.7%, which is larger than the bid–ask on the forward it is being compared with.

31.8 BUILDING THE TREE

The closed forms of Section 31.7 cover the European cases. *Everything else needs a lattice*, and Hull–White's is built in two stages—which is the construction's central idea, not a computational detail.

Step one: the geometry and the probabilities. Stage one builds a tree for the process with no drift level,

$$dx = -ax\,dt + \sigma\,dW, \qquad x(0) = 0 \tag{31.12}$$

Nodes sit at $x = j\Delta x$. Over one step from level j the move has mean and variance

$$M = -a\,j\Delta x\,\Delta t, \qquad V = \sigma^2\Delta t$$

and the three branching probabilities are fixed by reproducing both, together with the requirement that they sum to one. Writing $\eta = M/\Delta x = -aj\Delta t$ and branching to $j+1$, j, $j-1$,

$$p_u + p_m + p_d = 1, \qquad p_u - p_d = \eta, \qquad p_u + p_d = \frac{V}{\Delta x^2} + \eta^2$$

The spacing is chosen to make this tidy. Setting $\Delta x = \sigma\sqrt{3\Delta t}$ turns $V/\Delta x^2$ into $1/3$, and the three equations solve to

$$p_u = \frac{1}{6} + \frac{\eta^2 + \eta}{2}, \qquad p_m = \frac{2}{3} - \eta^2, \qquad p_d = \frac{1}{6} + \frac{\eta^2 - \eta}{2} \tag{31.13}$$

At the center $\eta = 0$ and these are $1/6$, $2/3$, $1/6$. Nothing about the curve has entered yet, which is why stage one can be built once and reused.

Where symmetric branching breaks down. Write $u = aj\Delta t$, so $\eta = -u$. Both p_u and p_d in (31.13) stay positive for every u (each is a quadratic in u with negative discriminant), but the middle one does not:

$$p_m = \frac{2}{3} - u^2 < 0 \qquad \text{once} \qquad u > \sqrt{2/3} = 0.8165$$

In the node index that reads $j > 0.8165/(a\Delta t)$: far enough above the center, the drift is too strong for symmetric branching to match.

The remedy is to branch downward from the top node, to j, $j-1$, $j-2$. The same three equations then give

$$p_u = \frac{7}{6} + \frac{u^2 - 3u}{2}, \qquad p_m = -\frac{1}{3} - u^2 + 2u, \qquad p_d = \frac{1}{6} + \frac{u^2 - u}{2} \tag{31.14}$$

and again the middle probability is the binding one. Solving $-\frac{1}{3} - u^2 + 2u \geq 0$ gives

$$1 - \sqrt{2/3} \;\leq\; u \;\leq\; 1 + \sqrt{2/3}, \qquad \text{that is} \qquad 0.1835 \leq u \leq 1.8165$$

So each pattern is valid on an interval, and the intervals overlap on $0.1835 \leq u \leq 0.8165$. The switch has to happen somewhere inside that window, and taking it as early as possible keeps the tree narrowest. So $j_{\max}$ is the smallest integer satisfying

$$j_{\max} \;\geq\; \frac{1 - \sqrt{2/3}}{a\,\Delta t} \;=\; \frac{0.1835}{a\,\Delta t} \tag{31.15}$$

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that is, the first level at which the switched branching is legal. *The bottom of the tree mirrors the top:* at $-j_{\max}$ the branching runs up-up-middle, with (31.14) read at $-u$ and p_u and p_d exchanged.

Step two: displacing the tree onto the curve. Stage two shifts every level by an $\alpha(i)$, so that $r = x + \alpha(t)$, with x the process of (31.12), reprices today's zero curve. The curve is the one used throughout this chapter,

$$R(T) = 4.2\% - 1.2\%\, e^{-0.25T}, \qquad P(0,T) = e^{-R(T)\,T}$$

continuously compounded; the discount factors of Examples 31.5 and 31.6, and of every tree example that follows, come from it. The displacements are computed forward, one time step at a time, from the Arrow–Debreu prices. Write $Q(i,j)$ for the price, today, of a security that pays \$1 if the tree reaches node (i,j) and nothing otherwise (Section 13.3's state price, under the name the rates literature uses). The displacements are then *solved*, not fitted, and Example 31.7 carries that out with every number shown.

One feature of the geometry is worth a figure, because it is where the construction stops being obvious.

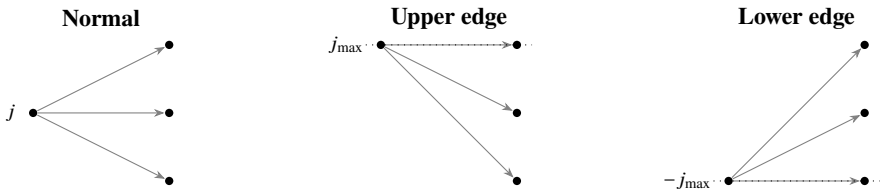


Figure 31.3: Branching switches at $\pm j_{\max}$. Schematic; spacing exaggerated.

Figure 31.3 is (31.15) drawn: below $j_{\max}$ the branching is symmetric, at the top node it runs down-down-middle, and at the bottom node up-up-middle.

The tree therefore stops widening. That is not an approximation—it is what mean reversion means, expressed in the geometry.

Example 31.7 Building the tree, step by step

Take this chapter's curve with $a = 0.08$ and $\sigma = 1\%$, but $\Delta t = 1$ year, so that every number fits on the page. The production tree described after this example runs the identical construction at $\Delta t = 0.25$.

Stage one fixes the geometry and never changes. $\Delta x = \sigma\sqrt{3\Delta t} = 0.017321$ and $j_{\max}$ is the smallest integer at or above $0.1835/(a\Delta t) = 2.294$, so $j_{\max} = 3$. The branching probabilities depend on j alone (not on the curve, not on the step), because stage one knows nothing about either:

j	p_u	p_m	p_d	**branches to**
+3	0.835467	0.089067	0.075467	$j,\ j-1,\ j-2$
+2	0.099467	0.641067	0.259467	$j+1,\ j,\ j-1$
+1	0.129867	0.660267	0.209867	$j+1,\ j,\ j-1$
0	0.166667	0.666667	0.166667	$j+1,\ j,\ j-1$
−1	0.209867	0.660267	0.129867	$j+1,\ j,\ j-1$
−2	0.259467	0.641067	0.099467	$j+1,\ j,\ j-1$
−3	0.075467	0.089067	0.835467	$j+2,\ j+1,\ j$

The drift shows in the asymmetry: above the center the down probability exceeds the up, and the imbalance grows with $|j|$. That is mean reversion, discretized. The two end rows come

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from (31.14), and their probabilities attach to different nodes, which is why p_u is the largest of the three at $j = +3$, where it is the probability of *staying* at j rather than of rising above it. Four steps is enough to reach them: the tree touches $j = \pm 3$ at level 3, and the switched rows are what carry it from there to level 4.

Stage two solves one equation per step. Begin with $Q(0,0) = 1$. At level i the displacement that makes the tree reproduce the next discount factor is

$$\alpha(i) = \frac{1}{\Delta t}\left[\ln \sum_j Q(i,j)\, e^{-j\Delta x\,\Delta t} \;-\; \ln P\big(0,(i+1)\Delta t\big)\right] \tag{31.16}$$

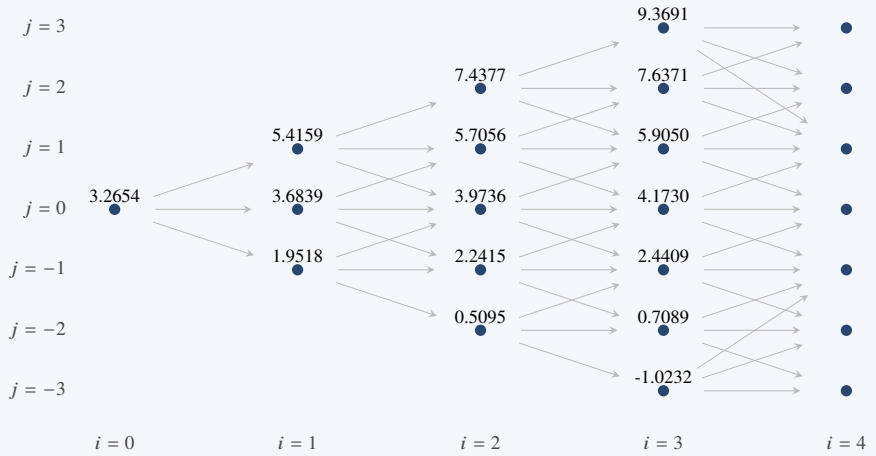
after which the Arrow–Debreu prices roll forward,

$$Q(i+1,k) = \sum_j Q(i,j)\; p(j \to k)\; e^{-(\alpha(i)+j\Delta x)\Delta t}$$

in which $p(j \to k)$ is whichever of p_u, p_m, p_d carries level j to level k, and zero if none of them does, so the *branches to* column above is what decides which. Four steps of that give

	$i=0$	$i=1$	$i=2$	$i=3$	$i=4$
$Q(i,+3)$	—	—	—	0.0018	0.0048
$Q(i,+2)$	—	—	0.0198	0.0369	0.0477
$Q(i,+1)$	—	0.1613	0.2045	0.2092	0.2017
$Q(i,\ 0)$	1.0000	0.6452	0.4799	0.3907	0.3356
$Q(i,-1)$	—	0.1613	0.2081	0.2165	0.2123
$Q(i,-2)$	—	—	0.0205	0.0395	0.0528
$Q(i,-3)$	—	—	—	0.0020	0.0056
$\alpha(i)$	3.2654%	3.6839%	3.9736%	4.1730%	—
$P(0,(i+1)\Delta t)$	0.9679	0.9329	0.8967	0.8604	—

The last column carries no α: level 4 is where this tree stops, and $\alpha(4)$ would be fixed by $P(0,5)$.



Short rates $r(i,j) = \alpha(i) + j\Delta x$, in percent; level 4 is terminal and carries none. *Levels 3 and 4 have the same seven nodes.* That is the tree refusing to widen once it reaches $\pm j_{\max}$.

At each level i the sum $\sum_j Q(i,j)\, e^{-r(i,j)\Delta t}$ returns $P(0,(i+1)\Delta t)$, the bottom row of the table, exactly, to the last bit.

Three features stand out. The Q's are not symmetric (0.0198 against 0.0205 at $i = 2$) because the discounting differs across nodes even though the probabilities at j and $-j$ mirror each other. α rises along the curve, from 3.27% to 4.17%, because it is absorbing the upward slope of the forward rates. And the bottom node at level 3 carries -1.0232%: a Gaussian short rate goes negative, which Section 31.14 takes up as a property rather than a defect.

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Nothing here is iterated. Equation (31.16) is one equation in one unknown with everything on the right already computed, which is why the curve comes back to machine precision rather than to a tolerance.

The production tree of this chapter runs the same construction at $\Delta t = 0.25$, over eleven years: 44 steps, $\Delta x = 0.008660$, $j_{\max} = 10$. Summing the Arrow–Debreu prices at each of those 44 levels returns the input discount factors with errors at or below one unit in the last place: machine precision, not convergence, because $\alpha(i)$ solves one equation exactly rather than being fitted.

That is worth remembering, because it separates two kinds of error. The tree's discretization error affects option values, which is Section 31.9's subject. It does not touch the underlying curve at all, so a mispriced swaption on this tree is never a symptom of a badly fitted curve.

31.9 BERMUDAN SWAPTIONS

Here is what the tree is for.

Example 31.8 One option with five dates, not five options

A receiver swaption on a swap paying 4% fixed and maturing at $t = 6$, notional 100, on the production tree of Section 31.8.

First check the machinery. A European receiver expiring in one year has a closed-form value by Jamshidian's decomposition (a European option on a coupon bond is a portfolio of options on the zeros), and that gives **1.1465**. The tree should converge to it:

Steps/year	Tree	Error
4	1.1637	+0.0172
8	1.1696	+0.0231
12	1.1667	+0.0202
24	1.1586	+0.0121
48	1.1502	+0.0037

The error shrinks as the grid is refined but not monotonically: eight steps a year is worse than four. Section 20.2 established why: a lattice's error depends on where the nodes fall relative to the payoff's kink, and refining the grid moves them. A desk that extrapolates two tree values without checking for this will get a worse answer than either.

Value the Bermudan exercisable annually at $t = 1, 2, 3, 4, 5$, at 24 steps per year, alongside each European:

Exercise right	Value
European, 1y only	1.1586
European, 2y only	1.1953
European, 3y only	1.0559
European, 4y only	0.7974
European, 5y only	0.4412
Sum of all five Europeans	4.6484
Bermudan, any one of the five	**1.6995**

The Bermudan is worth 42.2% more than the best single European and 37% of their sum.

It exceeds any European because the right to choose has value. The holder does not have to guess in advance which date will be best.

It falls far short of the sum because the rights are substitutes, not additions. Exercising at year two extinguishes the option at years three, four, and five. A portfolio of five Europeans is five options; a Bermudan is one option with five dates, and the gap between 1.70 and 4.65 is the entire

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economic content of that sentence.

And Black's model cannot produce the 1.6995. It prices each row of the top block, because each is a European option on a single rate. It has nothing to say about the choice, because making the choice requires knowing the curve at year three, and Black's model never builds one.

31.10 CALIBRATION, AND WHAT IT CANNOT DETERMINE

What calibration does here is narrower than the word suggests. Section 31.6 has already fitted the curve: $\theta(t)$ is solved so that today's $P(0,t)$ comes back exactly, and nothing about it is open to choice. What is left unfixed is the pair (a,σ), and those two govern volatility rather than level. *So calibration in a fitted short-rate model means choosing two constants from option prices* (the caps and swaptions of Section 29.9's volatility cube), and it never means matching the curve, which is automatic.

The procedure is the ordinary one. Take a set of quoted instruments, price each on the model at a trial (a,σ), and minimize a weighted sum of squared differences,

$$\min_{a,\,\sigma} \sum_i w_i \left[V_i^{\text{model}}(a,\sigma) - V_i^{\text{market}}\right]^2$$

with the weights w_i chosen as Section 21.10 describes. The closed forms of Section 31.7 are what make this fast enough to run every morning, which is the practical reason for deriving them.

Which instruments go into that sum decides what the calibration can see. Example 31.3 said a and σ are separately identified. The following example is the most important in this chapter, and it says that where it matters most, they are not.

Example 31.9 Every European matched, the Bermudan unmatched

Hold the one-year-into-five-year European receiver fixed at its tree value of 1.1586. For each mean reversion a, solve for the σ that reproduces it exactly. Then reprice the Bermudan of Example 31.8.

a	σ	**European**	**Bermudan**	**vs.** $a = 0.08$
0.01	0.00823	1.1586	1.6474	−3.1%
0.03	0.00871	1.1586	1.6622	−2.2%
0.08	0.01000	1.1586	1.6995	—
0.15	0.01202	1.1586	1.7541	+3.2%
0.30	0.01724	1.1586	1.8705	+10.1%

Every row prices the calibration instrument identically, to four decimals, and the Bermudan spans 13.1% of its own value.

The mechanism is the reason two-factor models exist. A Bermudan's value depends on the *relationship* between rates at different exercise dates: how much the curve at year three tells you about the curve at year four. That relationship is governed by a, and a single European swaption contains no information about it, because it observes the curve once.

There are three responses, and a desk should know which it is using.

Calibrate to a co-terminal strip. Use the swaptions that expire on each of the Bermudan's exercise dates and mature on its final date. This is the standard practice, and it constrains a far better than a single option because the strip observes several dates.

Fix a from history and calibrate only σ. This is cruder, and defensible when the strip is not liquid. Section 29.9 reaches the same conclusion about SABR's β, which desks fix rather than fit: when the data cannot support a parameter, choose it outside the calibration rather than letting the fit invent it.

Or quote a range. Section 21.11 argued that model risk should be reported rather than hidden, and here the range is computable from a five-row calculation any desk can run.

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What is not defensible is calibrating to one European and reporting one Bermudan price to four decimals, which the table shows to be four decimals of a number determined by an unobserved parameter.

31.11 CALLABLE BONDS AND OPTION-ADJUSTED SPREAD

The largest application of short-rate trees is not swaptions. It is the callable bond market, and the number the model produces there has a name that overstates what it is.

Example 31.10 What the Z-spread hides, and what the OAS assumes

A ten-year 5% annual bond, callable at par each year from year three, priced on the production tree of Section 31.8.

Straight bond, zero spread	106.9013
Callable bond, zero spread	101.6743
Embedded call option	**5.2270**

Now suppose the bond trades at 99.00. Two spreads can be quoted.

The Z-spread is the constant addition to the curve that reproduces the price while ignoring the call: **94.6 basis points**.

The option-adjusted spread (OAS) is the constant addition that reproduces the price with the call modeled: **52.1 basis points**.

The 42.5 basis point difference is the call, expressed as a running yield. A portfolio manager comparing this bond with a non-callable one on Z-spread would see 94.6 and conclude it offered generous credit compensation. Roughly half of that is payment for an option the investor has written, and the OAS exists precisely to strip it out.

Now vary the model's parameters:

a	σ	**OAS**
0.03	0.010	39.5 bp
0.08	0.010	52.1 bp
0.20	0.010	71.5 bp
0.08	0.015	20.9 bp

The OAS ranges from 20.9 to 71.5 basis points across parameter choices no market quotes. The Z-spread, for all its faults, is a fact about the price; *the OAS is a fact about the price and a model*, and the word "adjusted" does not convey how much of it is the second.

The practical rule follows immediately. OAS comparisons are meaningful *within* a book run on one model with one calibration, and close to meaningless across two houses that have not agreed on a and σ. Comparing a dealer's OAS with your own is comparing two models, not two bonds.

31.12 THE ONE-FACTOR DEFECT

Every model so far has one source of uncertainty, and it has a consequence that no amount of calibration can remove.

At any future date, every rate on the curve is a deterministic increasing function of $r(t)$. A two-year rate and a ten-year rate cannot move in opposite directions, cannot move by unrelated amounts, and are perfectly correlated. The model does not merely approximate a correlation of one; it asserts it.

For a swaption or a callable bond this is tolerable, because the payoff depends on one rate or on a monotone function of the state. For anything whose payoff depends on the curve's *shape*, it is fatal, and it is straightforward to price how fatal.

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Example 31.11 What perfect correlation costs

A five-year option on the spread between the ten-year and two-year swap rates is a **CMS spread option**, one of the most widely sold structured coupons. Forward rates 3.40% and 2.80%, normal volatilities 90 and 75 basis points a year, annuity 4.20. The forward spread is 60 basis points, and the option is struck at the money.

The spread's volatility is $\sqrt{\sigma_{10}^2 + \sigma_2^2 - 2\rho\,\sigma_{10}\sigma_2}$:

ρ	Spread vol	Option value	Ratio
1.00	15.0 bp	56.2 bp	1.00
0.95	30.0 bp	112.4 bp	2.00
0.90	39.7 bp	148.7 bp	2.65
0.80	54.1 bp	202.6 bp	3.61
0.70	65.4 bp	245.0 bp	4.36
0.50	83.5 bp	312.9 bp	5.57

A one-factor model is forced to the first row. It returns 56.2 basis points—and the 15.0 is not even a correlation assumption, it is just $|\sigma_{10} - \sigma_2|$, the residual from two rates moving in lockstep at slightly different speeds.

At a market-plausible correlation of 0.70 the option is worth 245.0 basis points, 4.4 times as much.

Three things follow.

No calibration repairs this. One could force a one-factor model to reproduce the 245 by inflating σ, but then every swaption in the book would be mispriced. The error is in the model's structure, not its parameters, which is exactly the distinction Section 21.11 drew for equity models.

And the error has a sign. A one-factor model *always* undervalues a spread option, because $\rho = 1$ minimizes the spread's volatility. A desk short these options on a one-factor model is systematically short at a price it computed to be too low.

The magnitude is the argument for Chapter 32. A factor of 4.4 is not a refinement.

31.13 TWO-FACTOR MODELS

The repair is a second source of uncertainty. The standard construction adds a stochastic mean-reversion level:

$$dr = \big[\theta(t) + u - ar\big]dt + \sigma_1\,dW_1, \qquad du = -b\,u\,dt + \sigma_2\,dW_2 \tag{31.17}$$

with $dW_1\,dW_2 = \rho\,dt$, where b is now the speed at which u reverts to zero rather than a level. Equivalently, r is the sum of two zero-mean mean-reverting factors and a deterministic shift chosen to fit the curve. This is the **G2++** form, whose name is a description of the construction: Gaussian, two factors, and "++" for the shift. That is how the model is usually implemented, because two zero-mean factors discretize more cleanly than a stochastic mean-reversion level does.

Now the short rate and the long rate are driven by different combinations of two shocks, so rates at different maturities are imperfectly correlated and the curve can steepen, flatten, and twist. The model remains affine, so bond prices stay closed-form.

The costs are real and should be weighed.

Five parameters instead of two, and the calibration is correspondingly less stable, a familiar problem from Section 21.4's account of Heston, where no parameter moved one feature alone.

And the lattice becomes two-dimensional. A two-factor tree is a 3×3 branching structure whose node count grows as the square, which is why two-factor implementations usually price by Monte Carlo or by finite differences on a two-dimensional grid rather than on a tree. There is a further obstruction on the tree: the natural correlation correction to the nine branching probabilities is admissible only for $|\rho| \le 1/3$. [TN31] gives that obstruction and the change of

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variable that removes it, together with the two-factor bond and option formulas and the reason Jamshidian's decomposition does not survive the second factor.

Section 32.17 sets out the rule for choosing among all these models; for the number of factors it comes down to whether the payoff depends on the level of rates or on their shape, and Example 31.11's question (what is the payoff worth at $\rho = 1$?) tells which.

31.14 NEGATIVE RATES AND THE REPAIRS

Hull–White's r is normally distributed, so it goes negative with positive probability, and Example 31.12 shows the probability is not small.

Example 31.12 How often the model says rates are below zero

A flat forward curve at 3%, and the probability the model assigns to a negative short rate, with $r(T)$ normal around the forward rate and the standard deviation the model implies:

	1y	2y	5y	10y
$a = 0.05$, $\sigma = 1.0\%$	0.1%	1.3%	6.5%	11.6%
$a = 0.05$, $\sigma = 1.5\%$	2.0%	6.9%	15.7%	21.3%
$a = 0.10$, $\sigma = 1.0\%$	0.1%	1.0%	4.6%	7.5%

At ten years the middle row assigns better than one chance in five to a negative short rate. Before rates went negative (in Denmark in July 2012, in the euro area in June 2014), this was cited as the model's principal flaw and as the reason to prefer CIR or a lognormal short rate.

Then rates went negative and stayed there for years, and the property became the reason Hull–White survived while its competitors did not. A Gaussian model priced the market that existed; a positive model could not be calibrated to it at all.

The general lesson is the one Section 19.9 drew about 1987 and Section 26.6 drew about house prices. A constraint imposed because the unconstrained case "cannot happen" is a statement about the sample, and the sample is always shorter than the future. A model's implausible region is where its next test comes from.

Where positivity or a floor is genuinely wanted, the standard repairs are a **shifted** model (apply a positive model such as CIR or a lognormal short rate to $r + s$ for a constant shift s, which moves the lower bound from zero to $-s$ rather than removing it) or the displaced and normal-versus-lognormal machinery of Section 29.8, which is the same idea applied to the quoting convention rather than to the process.

A lognormal short rate such as Black–Karasinski, in which $\ln r$ mean-reverts, keeps positivity but gives up the closed forms. It has no closed-form bond price, so Jamshidian's decomposition is unavailable, and the tree's displacement $\alpha(i)$ must be found by Newton iteration at each step instead of from (31.16). That is the one genuine difference from Hull–White on the tree.

31.15 AFTER THE LIBOR TRANSITION

Chapter 7 rebuilt the curve framework around risk-free rates and Chapter 29 rewrote the caplet. What did that do to this chapter?

Less than one would expect, and for an instructive reason. A short-rate model models the instantaneous risk-free rate, which is closer to what SOFR and €STR actually are than to what LIBOR ever was. The transition removed an inconsistency rather than creating one: pre-2022, the model's r was a risk-free rate while the products referenced a credit-sensitive rate, and the gap was absorbed by convention.

Two things did change.

Discounting and forecasting are now the same curve for cleared trades in the overnight rates, so the multi-curve complication of Section 7.5 does not reach into the short-rate model at

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all. Where a spread is needed (a credit-sensitive benchmark, or a currency basis), it is added as a deterministic spread rather than modeled, which is a modeling choice worth being aware of and is usually adequate.

And compounded-in-arrears fixings introduce a short averaging window that Section 29.4 handled by adjusting the caplet's variance. On a short-rate tree it needs no adjustment at all: the tree simulates the path of r, so an average over the accrual period is computed directly. That is a small but genuine advantage of this class of model over a forward-rate model, and it is the reverse of the usual ordering.

31.16 WHERE THESE MODELS ARE STILL THE RIGHT TOOL

Chapter 32 introduces models that are more realistic in every respect and harder to use in most. The case for short-rate models is not nostalgia, and it rests on four properties.

They price American and Bermudan exercise naturally. A tree gives the continuation value at every node, so an early-exercise decision costs nothing extra. A forward-rate model does not, and needs the regression machinery of Section 20.11. For the callable and Bermudan books, which are among the largest users of interest-rate models by notional, this is decisive.

They are fast. A 44-step tree prices a Bermudan in milliseconds, which matters when a desk revalues thousands of positions and computes their sensitivities every night.

They have few parameters, which is a defect when the market is rich and a virtue when it is not. Example 31.9 showed that two parameters were already one too many for the available data.

And they handle path dependence in the short rate directly, which Section 31.15 illustrated with compounded fixings.

The case against is Section 31.12 and nothing else, and Section 32.17 turns it into the rule for when to reach for something else.

31.17 WHO USES THEM, AND FOR WHAT

Callable-bond desks and the investors who buy their paper. The OAS of Example 31.10 is produced by a short-rate model at essentially every institution that quotes one, and Section 26.3's mortgage prepayment models sit on top of the same machinery: a prepayment model needs a rate path to respond to, and this is what generates it.

Bermudan swaption desks, where Example 31.9's calibration question is a daily decision rather than a textbook point.

Insurers and pension funds valuing liabilities with embedded guarantees: a minimum crediting rate is an interest rate floor with a fifty-year tenor, and it must be valued on a model that can generate a coherent curve that far out.

Structured-note issuers, for the level-dependent coupons. *Not for the spread-dependent ones*, which is the division Section 31.13 drew and the reason a single desk often runs both a one-factor and a two-factor model and routes trades between them.

And risk managers, for scenario generation, where, unusually, the equilibrium models of Section 31.2 come back. A real-world scenario generator wants a curve that is derived from an economically coherent process, not one fitted to today's quotes, because the question being asked is what could happen rather than what a hedge costs.

31.18 WHERE THIS CAME FROM

The apparatus Chapter 15 built for equity options does not transfer to bonds, because a bond's volatility must vanish at maturity and because the whole curve, not one price, is the state variable. Sorting that out took twenty years.

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Vasicek (1977) is the first serious answer: a mean-reverting short rate from which the entire curve follows in closed form. *Its defect is fundamental and instructive*—being an equilibrium model it produces a curve, and that curve is not today's. Cox, Ingersoll and Ross (1985) gave the approach its general-equilibrium foundation and a square-root diffusion that keeps rates positive, and the same defect remained.

Ho and Lee (1986) inverted the problem: take today's curve as given and fit the model to it. *That is the no-arbitrage turn*, and everything since sits on the far side of it. Hull and White (1990) combined the two, adding mean reversion to a fitted model, which is the formulation this chapter uses; Black, Derman and Toy (1990) and Black and Karasinski (1991) did the same in lognormal form. Jamshidian (1989) showed that an option on a bond portfolio decomposes into a portfolio of options in a one-factor model, which is why European swaptions had closed forms in these models from the start; Bermudans waited for the lattice.

SUMMARY

Black's model prices one option on one rate, and cannot price a Bermudan swaption or a callable bond, which need the whole curve at each future decision date. A short-rate model generates that curve from a single state variable, the source of both its tractability and its central defect. A handful of parameters cannot match a curve with hundreds of points, so the no-arbitrage generation makes the drift a solved function of time: $\theta(t)$ carries no information, and every model in this class is an interpolation device with a stochastic component. CIR's $\sqrt{r}$ keeps rates positive, which is why it could not be calibrated once rates went negative.

In Hull–White, σ sets the level of the volatility term structure and a sets its shape, forward-rate volatility decaying as σe^{-aT}. The tree reprices the curve to machine precision, because the displacement $\alpha(t)$ is solved rather than fitted, and converges at roughly first order but not monotonically.

A Bermudan exercisable at several dates is worth clearly more than the best single European and far less than the sum of all of them: a portfolio of Europeans is several options, a Bermudan is one option with several dates. Holding the calibrating European fixed and varying a leaves every European identical and still moves the Bermudan materially, because a single European observes the curve once and contains no information about a. Calibrating to one European and quoting the Bermudan to four decimals is not defensible.

On a callable bond the Z-spread exceeds the OAS by enough that much of the apparent credit compensation pays for a written option, but the OAS itself moves widely across ordinary parameter choices. The Z-spread is a fact about the price; the OAS is a fact about the price and a model, so comparing OAS across houses compares models rather than bonds.

In one factor every rate is an increasing function of one state variable, so all rates are perfectly correlated—asserted, not approximated. A CMS spread option is forced to $\rho = 1$ and is worth a fraction of its value at a realistic correlation; no calibration repairs it, and the error always has the same sign. Use one factor for level products and two for shape products, and test a payoff by what it is worth at $\rho = 1$.

Gaussian rates go negative, with a probability far from negligible at long horizons. Cited as the model's principal flaw before rates went negative, that became the reason it survived: a model's implausible region is where its next test comes from. Short-rate models remain right for early exercise, for speed, and when the data cannot support more parameters—and wrong whenever the payoff depends on the curve's shape.

FURTHER READING

Vasicek's and Cox, Ingersoll, and Ross's original papers are short and state what an equilibrium model is trying to do; reading them after this chapter is more useful than before it, because the

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thing they are trying to do is the thing Section 31.4 says a desk cannot use. Hull and White's papers on the extended model and on the trinomial construction are the primary sources for Sections 31.6 and 31.8, and the tree paper describes the two-stage procedure precisely enough to code from. Brigo and Mercurio's book is the standard reference for the whole family, including the G2++ form of Section 31.13 and the shifted variants of Section 31.14, and it is where to go for the algebra this chapter states without deriving. Andersen and Piterbarg's three volumes are the practitioner's reference and treat the calibration question of Example 31.9 at much greater length, reaching the same conclusion. On OAS, the investor literature is where the parameter dependence of Example 31.10 is set out from the holder's side (what a quoted spread assumes about volatility and the exercise policy), while the dealer literature documents the models the quotes come from; read the first to interrogate a quoted OAS, the second to reproduce one.

PRACTICE QUESTIONS

31.1 Explain why Black's model cannot price a Bermudan swaption, and be specific about what it fails to produce.

31.2 In Vasicek's model with $a = 0.10$, $b = 5\%$, $\sigma = 1\%$, compute the limiting long rate and explain why it is below b. Then explain why an equilibrium model cannot match an observed term structure, and why this is arithmetic rather than calibration.

31.3 Explain what $\theta(t)$ does in Ho–Lee, why it carries no information, and state the defect that rules the model out in practice together with the quantity that exposes it.

31.4 Using σe^{-aT}, compute the ten-year forward-rate volatility at $\sigma = 100$ bp for $a = 0.05$ and $a = 0.20$, and explain which feature of the volatility term structure is set by a and which by σ.

31.5 Describe the two stages of the Hull–White tree construction, state which quantities are solved rather than fitted, and explain why the tree reproduces the initial curve to machine precision while still mispricing options.

31.6 Explain why the branching pattern must change at $\pm j_{\max}$, and what would go wrong if it did not.

31.7 In Example 31.8, explain why the Bermudan is worth more than any single European and much less than their sum, and why the tree's convergence to the Jamshidian value is not monotone.

31.8 Explain why a single European swaption contains no information about the mean-reversion rate, use Example 31.9 to state the range of Bermudan values consistent with an exactly matched European, and describe three defensible responses and one that is not.

31.9 Explain the difference between a Z-spread and an option-adjusted spread, and what the difference between them measures.

31.10 Using Example 31.10, state the OAS at $a = 0.08$ with $\sigma = 1.0\%$ and with $\sigma = 1.5\%$, and explain what this implies about comparing OAS across dealers.

31.11 Explain why a one-factor model forces perfect correlation between all rates, and why this is an assertion rather than an approximation.

31.12 Using Example 31.11, state the CMS spread option's value at $\rho = 1$ and at $\rho = 0.70$, and explain why the one-factor error always has the same sign and why that sign is the dangerous one.

31.13 State the decision rule for choosing between a one-factor and a two-factor model, and give one product on each side of it.

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31.14 Explain why Gaussian short rates were regarded as defective before rates went negative and as necessary afterward, and name four properties that keep short-rate models in use together with the single property that rules them out.

31.15 Using Example 31.5, state the caplet's value under the forward measure and under the risk-neutral measure and comment on their agreement, then explain what the 2.52% error in part (c) consists of and its sign.

31.16 Using Example 31.6, state the range of $E^{Q}[S_T]/F(0,T)$ across the correlations shown, explain what quantity produces it, and name the two other results in this book that are the same quantity.

FURTHER QUESTIONS

31.17 Example 31.1 shows three curves converging to $R_\infty = b - \sigma^2/2a^2$. Derive this limit, explain the convexity argument that produces the second term, and determine the parameter values at which the correction becomes large enough to matter for a thirty-year bond.

31.18 Example 31.9 shows the Bermudan spanning 13.1% of its value with the European held fixed. Extend the calculation to a co-terminal strip: determine how many co-terminal swaptions must be matched before the Bermudan's range falls below 1%, and explain what this implies for a currency in which only two of them are liquid.

31.19 Example 31.10 shows the OAS moving from 20.9 to 71.5 basis points. Propose a reporting convention that would make OAS comparable across institutions, state what each party would have to disclose, and explain why the market has not adopted one.

31.20 Example 31.11 prices the one-factor defect at a factor of 4.4. Derive the spread's volatility as a function of ρ, show that $\rho = 1$ minimizes it for any $\sigma_{10} \neq \sigma_2$, and determine the conditions under which a one-factor model could be made to price a single spread option correctly, and then explain what it would then get wrong.

31.21 Section 31.13 says a two-factor model costs five parameters and a two-dimensional lattice. Compare this trade-off with Section 21.4's account of Heston, where additional parameters also failed to move features independently. State the general principle governing when an extra factor is worth its calibration cost, and apply it to both cases.

31.22 Section 31.14 argues that a model's implausible region is where its next test comes from, citing 1987 and 2007. Identify a constraint currently imposed on rate models because the alternative "cannot happen," argue both sides, and propose how a desk should hedge the possibility that it can.

31.23 Section 31.16 claims short-rate models remain right for Bermudan exercise while Chapter 32's models are more realistic. Construct the strongest case that a forward-rate model with American Monte Carlo should replace them for the callable book, state what would have to be true about accuracy and computation for that case to win, and give your own judgment with reasons.

31.24 Example 31.5 finds a 2.52% error from ignoring the payoff-discount covariance. Derive that covariance term analytically for a caplet under Hull–White, determine its dependence on rate volatility and option maturity, and identify the configuration in which it would exceed 10%.

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Chapter 32

HJM, LMM, and Multi-Curve Term Structure Models †

Section 31.12 put a number on the one-factor defect: a one-factor short-rate model prices a CMS spread option at 56.2 basis points when it is worth 245.0. This chapter is about the class of models built to fix that, and about the price of the fix.

There is a second and less often stated motivation. A short-rate model models a quantity no one trades. The instantaneous rate $r(t)$ is not observable, not quoted, and not hedgeable; every market instrument is a function of it, and the model's parameters must be inferred backward through those functions. The models in this chapter model the rates that are actually quoted, and that turns out to matter more for calibration than for realism.

The chapter runs from the general framework to the model that is actually implemented. Sections 32.1 to 32.3 are HJM: the whole forward curve at once, the drift no-arbitrage forces on it, and why no one implements it in that form. Sections 32.4 to 32.8 are the LIBOR market model: the measure and the drift that follows, how many factors are needed, and the shapes instantaneous volatility and correlation have to take. Sections 32.9 to 32.13 are practice: calibration, simulation, an incompatibility with the swap market model whose size depends on the trade, early exercise, and why sensitivities cost more than prices. Sections 32.14 to 32.16 are what requires this machinery, the smile, and multi-curve after the transition; Sections 32.17 to 32.19 choose among the models and say who runs them.

32.1 HJM: MODEL THE WHOLE FORWARD CURVE

The object being modeled is a family of curves. Section 31.5's $f(0,t)$ was today's forward curve; HJM needs $f(t,T)$, the curve as it stands at any future date t. It is defined as a limit of quoted forward rates. The simply compounded forward rate for $[T, T+\tau]$ seen from t is

$$F(t;T,T+\tau) = \frac{1}{\tau}\left(\frac{P(t,T)}{P(t,T+\tau)} - 1\right)$$

and letting the accrual shrink to nothing gives the instantaneous forward rate,

$$f(t,T) = \lim_{\tau \downarrow 0} F(t;T,T+\tau) = -\frac{\partial \ln P(t,T)}{\partial T} \tag{32.1}$$

Two relations follow, and the chapter uses both throughout. Integrating (32.1) returns the bond price,

$$P(t,T) = \exp\left(-\int_t^T f(t,u)\,du\right) \tag{32.2}$$

so the forward curve and the discount curve carry the same information: either determines the other, and a model of one is a model of both. Evaluating the forward curve at its lower endpoint returns the short rate,

$$r(t) = f(t,t) \tag{32.3}$$

That identity carries the chapter's argument. A short-rate model specifies the single point $f(t,t)$ and derives the rest of the curve from it, which is why the shapes it can produce are constrained. HJM specifies $f(t,T)$ for every T simultaneously, and derives nothing.

Heath, Jarrow, and Morton proceeded directly. Instead of a process for one rate, specify a process for every instantaneous forward rate:

$$df(t,T) = m(t,T)\,dt + \sum_{k=1}^{K} s_k(t,T)\,dW_k \tag{32.4}$$

The HJM processes are written under the risk-neutral measure Q, and each W_k is a Wiener process under it; the market model later changes numeraire, and Section 32.5 says what that does to the drift. No market price of risk appears anywhere in what follows and none is needed: a price is an expectation under Q, and the calibration of Section 32.9 fits Q-parameters to quoted prices directly. That is also what makes the next section's result possible.

The state is now the entire curve $\{f(t,T) : T \geq t\}$, driven by K independent shocks. Today's curve enters as the initial condition, so the fitting problem of Section 31.4 does not arise: it is satisfied by construction rather than solved.

The number of factors is now a modeling choice rather than a consequence. Section 32.6 makes that choice with data.

32.2 THE DRIFT IS NOT A CHOICE

Once the volatility functions $s_k(t,T)$ are specified, the drift $m(t,T)$ is determined. No-arbitrage forces

$$m(t,T) = \sum_{k=1}^{K} s_k(t,T) \int_t^T s_k(t,\tau)\,d\tau \tag{32.5}$$

which is the **HJM drift condition**.

It follows from expressing the bond price in terms of the forward curve and identifying its drift. Under Q a zero-coupon bond earns the short rate, so the bond's drift is determined before any modeling choice is made, and the drift of the forward curve must be the one consistent with it.

Take one factor for the derivation and drop the subscript; the K-factor result follows at the end. Take logs in (32.2) and name the result,

$$Y(t,T) \;\equiv\; \ln P(t,T) \;=\; -\int_t^T f(t,\tau)\,d\tau$$

Differentiation produces two terms. The integrand evolves, contributing $-\int_t^T df(t,\tau)\,d\tau$; and the lower limit advances, evaluating the integrand at that limit and contributing $+f(t,t)\,dt$. That second term is where the short rate enters, by (32.3):

$$dY = r(t)\,dt - \int_t^T \big[m(t,\tau)\,dt + s(t,\tau)\,dW\big]d\tau = \big[r(t) - M(t,T)\big]dt - S(t,T)\,dW$$

where

$$M(t,T) = \int_t^T m(t,\tau)\,d\tau, \qquad S(t,T) = \int_t^T s(t,\tau)\,d\tau$$

are the drift and the volatility of the forward curve accumulated across the bond's remaining life. That accumulation is the essential feature: a bond responds to the entire strip of forward rates it spans, not to any single one of them.

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Apply Itô's lemma to $P = e^Y$ to get

$$\frac{dP}{P} = \left[r(t) - M(t,T) + \frac{1}{2} S(t,T)^2 \right] dt - S(t,T)\, dW$$

A zero-coupon bond is a traded asset and (32.4) is written under Q, so the bond's expected return is the short rate, which leads to

$$M(t,T) = \frac{1}{2} S(t,T)^2$$

That equation is the entire content of the drift condition: the accumulated drift is not a free input, but is determined by the accumulated volatility.

M and S are both integrals over $[t,T]$, so differentiating in T gives $\partial M/\partial T = m(t,T)$ and $\partial S/\partial T = s(t,T)$, with the chain rule on the right:

$$m(t,T) = S(t,T)\, s(t,T) = s(t,T) \int_t^T s(t,\tau)\, d\tau$$

which is (32.5) with one factor. With K shocks, S becomes a vector and the variance term becomes $\frac{1}{2} \sum_k S_k^2$; the shocks are independent, so no cross terms appear, each factor contributes the one-factor result, and the contributions add.

The derivation used only that bond prices are traded and that $f(t,t) = r(t)$. No equilibrium argument and no utility function appear, which is why the condition binds every arbitrage-free term-structure model rather than a family of them.

Three consequences follow, and the third is the one that shapes practice.

The model has exactly one free input. Choose the volatility structure and everything else follows. That is a remarkably clean statement about a term-structure model, and it is the analog of Chapter 13's result that a derivative's price does not depend on the underlying's expected return: drift is not an independent degree of freedom once you insist on no arbitrage.

Every arbitrage-free term-structure model is an HJM model. Chapter 31's two Gaussian models are (32.4) under particular choices of $s(t,T)$:

- **Ho–Lee**: $s(t,T) = \sigma$, the same volatility at every maturity, so a shock shifts the whole curve in parallel;
- **Hull–White**: $s(t,T) = \sigma e^{-a(T-t)}$, a volatility decaying with the forward rate's distance, so a shock moves the short end more than the long.

Chapter 31 was therefore working inside HJM, with a volatility structure chosen for tractability rather than for fit. The two chapters are not competing frameworks but one framework and a family of choices within it.

And the drift depends on an integral of the volatility, which is why the framework is hard to implement. Section 32.3 takes that up.

32.3 WHY HJM IS NOT IMPLEMENTED DIRECTLY

Two obstacles stand in the way, and they are not of the same severity. The first removes every numerical method of Chapter 31; the second only requires the model to be restated in other variables.

The process is generally non-Markov. The drift (32.5) depends on the volatility structure along the whole curve, so the future evolution depends on the path, not just on the current state. A non-Markov process cannot be put on a recombining tree, which removes every technique of Chapter 31 and leaves only Monte Carlo—and Monte Carlo, by Section 20.11, is awkward for early exercise.

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There is an important exception, and it explains Chapter 31. If $s(t,T) = \sigma e^{-a(T-t)}$ (volatility decaying exponentially in time to maturity), the model *is* Markov in a single state variable, and that state variable is the short rate. Hull–White is exactly the HJM model that happens to be Markov, which is why it can be put on a tree while its generalizations cannot. The tractability that made Chapter 31 useful was never an accident of the short-rate formulation; it was a restriction on the volatility structure.

The exponential case is worth following through, because it is the one that collapses. With $s(t,T) = \sigma e^{-a(T-t)}$ the integral gives $S(t,T) = \sigma B(t,T)$, Hull–White's own function, and integrating the forward-rate dynamics leaves

$$f(t,T) = f(0,T) + \phi(t,T) + e^{-a(T-t)}x(t), \qquad x(t) = \int_0^t \sigma e^{-a(t-u)}\,dW(u)$$

with ϕ deterministic. The entire curve is a deterministic function of the single scalar $x(t)$, an **Ornstein–Uhlenbeck process** (a Gaussian process pulled back at rate a toward a fixed level, here zero, of the kind Vasicek's model makes of the short rate), and setting $T = t$ shows x is the short rate up to a deterministic shift.

The step that collapses it is a factorization: $e^{-a(T-u)} = e^{-a(T-t)}e^{-a(t-u)}$ separates the volatility's dependence on calendar time from its dependence on maturity. That is the precise sense in which Hull–White is the simplest Markov HJM model; a second exponential buys a second state variable, and the pattern continues.

The exception is a family, not one model. An HJM volatility that separates as $s(t,T) = g(t)h(T)$ is Markov in a small state vector, and the Cheyette and linear Gaussian Markov (LGM) models are of this kind. They are used when a Bermudan needs more than one factor but not a full LMM.

The second obstacle concerns calibration, and it is not that instantaneous forward rates are untraded. The short rate is not traded either, and Chapter 31 put it on a trinomial tree and priced Bermudan swaptions there; an unobservable state variable is the normal case in this subject, not an objection to it. The difficulty is one of dimension. HJM's free input $s_k(t,T)$ is a function of two variables (a volatility for every pair of calendar date and maturity), while the market quotes a finite set of volatilities, on rates that accrue over three or six months rather than instantaneously. No market instrument pays off on a single $f(t,T)$, so there is no quote to read any one of those volatilities from. A model written in terms of $f(t,T)$ must therefore be discretized along the maturity axis before it can touch a quote at all, and that discretization is a modeling choice the framework does not supply, which is to say the modeling happens in the discretization rather than in (32.4).

Both obstacles point the same way: work with rates that are quoted, over the accrual periods they are quoted over. That is the market model.

32.4 THE LIBOR MARKET MODEL

Fix the tenor structure $T_0 = 0 < T_1 < \cdots < T_N$ of an actual instrument (the reset dates of a cap or a swap) and model the forward rates $F_k(t)$ for the periods $[T_k, T_{k+1}]$ directly, each fixing at the start of its own period, T_k. Taking $T_0 = 0$ has one consequence used throughout: F_0 fixes today, so it is known rather than modeled, carries no caplet, and still supplies the discount factor for the first period. The caplets of a cap on this structure are therefore on F_1 through F_{N-1}, and the one on F_k has k whole periods of life. The convention costs no generality, because the grid and the instrument are separate things: a structure that starts later (a cap beginning in five years, or the swap underlying a swaption) keeps this same grid from today and simply carries no caplets on the forwards before it starts. Those forwards are still modeled, since the discount

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factors to the later payment dates run through them. Assume each forward is lognormal:

$$\frac{dF_k(t)}{F_k(t)} = \mu_k(t)\,dt + \sigma_k(t)\,dW_k \tag{32.6}$$

with $dW_i\,dW_j = \rho_{ij}\,dt$.

The point of the assumption is that it makes the model consistent with the market's own quoting convention. Section 29.3 priced a caplet with Black's model, which assumes exactly this lognormality. So the LMM reproduces the market's caplet prices by construction rather than by calibration: the model is built on the convention instead of being fitted to it.

That is the historical significance of the model and it is easy to understate. Before it, every model priced caps approximately and the market quoted them in a formula the models did not contain. Afterward, the quoting formula was a special case of the model. The name is now an anachronism (the rates are SOFR, €STR and EURIBOR forwards, not LIBOR), but the construction is unchanged, and Section 32.16 says why.

32.5 WHICH MEASURE, AND THE DRIFT THAT FOLLOWS

Under the T_{k+1}-forward measure $Q^{T_{k+1}}$ (numeraire $P(t,T_{k+1})$), the forward F_k is a martingale and $\mu_k = 0$. But different forwards are martingales under different measures, and a simulation needs all of them under one.

Pick a single numeraire and the drifts follow. Under the terminal measure (numeraire $P(t,T_N)$),

$$\mu_k(t) = -\sigma_k(t) \sum_{j=k+1}^{N-1} \frac{\rho_{kj}\,\tau_j\,F_j(t)\,\sigma_j(t)}{1+\tau_j F_j(t)} \tag{32.7}$$

The derivation is a change of numeraire performed once per accrual period, and one object does all of the work. For each j write

$$\Lambda_j(t) \;=\; \frac{P(t,T_{j+1})}{P(t,T_j)}$$

for the ratio of two consecutive numeraires. By the definition of F_j that ratio is a function of a single forward rate and of nothing else,

$$\Lambda_j(t) \;=\; \frac{1}{1+\tau_j F_j(t)}$$

It carries the letter this book uses for a Radon–Nikodym derivative because it is one up to a constant:

$$\left.\frac{dQ^{T_{j+1}}}{dQ^{T_j}}\right|_{\mathcal{F}_t} \;=\; \frac{P(0,T_j)}{P(0,T_{j+1})}\,\Lambda_j(t)$$

and a constant multiple leaves a volatility unchanged.

Since F_j has volatility σ_j, applying Itô's lemma to $\Lambda_j = 1/(1+\tau_j F_j)$ gives

$$\frac{d\Lambda_j(t)}{\Lambda_j(t)} \;=\; \mu_{\Lambda_j}(t)\,dt \;-\; \lambda_j(t)\,dW_j, \qquad \lambda_j(t) \;=\; \frac{\tau_j F_j(t)\,\sigma_j(t)}{1+\tau_j F_j(t)}$$

The drift μ_{Λ_j} is never used and is not evaluated. The volatility of Λ_j is λ_j: F_j's own volatility, damped by the factor $\tau_j F_j/(1+\tau_j F_j)$, the damping that makes the drift small, as the end of this section shows. The minus sign in the display is no part of that volatility, since $-W_j$ is a Wiener process as much as W_j is; it records that Λ_j falls when F_j rises, and that is where the minus sign on the covariance below comes from. Which measure dW_j belongs to does not matter here, since the rule stated next says a change of numeraire leaves volatilities alone.

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Section 28.9 derived this as Girsanov's theorem: a change of numeraire adds to a process's drift its covariance with the numeraire ratio, whatever that drift already is, and leaves the volatility unchanged. For consecutive bond numeraires the ratio is Λ_j, and because Λ_j moves against F_j, a forward F_k of volatility σ_k and correlation ρ_{kj} with F_j has covariance $-\rho_{kj}\,\sigma_k\,\lambda_j$ with it. Stepping from Q^{T_j} to $Q^{T_{j+1}}$ therefore adds exactly that to μ_k.

The walk goes one numeraire at a time, from the forward's own measure to the terminal one. Under $Q^{T_{k+1}}$ the forward F_k is a martingale and $\mu_k = 0$; this is the measure the caplet on F_k is priced in. Take one step, to $Q^{T_{k+2}}$. The numeraire ratio for that step is Λ_{k+1}, and with $dW_k\,dW_{k+1} = \rho_{k,k+1}\,dt$ from (32.6) the rule gives

$$\mu_k(t) \;=\; -\,\sigma_k(t)\,\rho_{k,k+1}\,\lambda_{k+1}(t)$$

The correlation enters for one reason: the covariance the rule asks for is between the diffusion of F_k and that of Λ_{k+1}, and F_{k+1} is the only forward rate Λ_{k+1} contains.

Take the next step, to $Q^{T_{k+3}}$. The ratio is now Λ_{k+2}, the identical calculation runs with $j = k+2$, and a second term $-\sigma_k\rho_{k,k+2}\lambda_{k+2}$ is added to the drift F_k already carries. In general the step from Q^{T_j} to $Q^{T_{j+1}}$ uses Λ_j and contributes $-\sigma_k\rho_{kj}\lambda_j$.

The contributions add rather than interact, because consecutive ratios multiply:

$$\frac{P(t,T_N)}{P(t,T_{k+1})} \;=\; \prod_{j=k+1}^{N-1}\Lambda_j(t)$$

The logarithm of a product is the sum of the logarithms, so the diffusion terms add (not the volatilities, since the factors are driven by different Wiener processes), and the whole change from $Q^{T_{k+1}}$ to Q^{T_N} carries $-\sum_{j=k+1}^{N-1}\lambda_j\,dW_j$. Walking all $N-k-1$ steps and collecting,

$$\mu_k(t) \;=\; -\,\sigma_k(t)\sum_{j=k+1}^{N-1}\rho_{kj}\,\lambda_j(t)$$

which is (32.7). Only forward rates longer than F_k appear, because the walk crosses exactly Λ_{k+1} through Λ_{N-1} and no others.

Simulating under the **spot measure** instead (whose numeraire is a rolling investment, rolled at each reset date into the bond maturing at the next) gives sums over $j \le k$, with the opposite sign. Neither choice is more correct, and implementations pick whichever puts the zero drift where the payoff is most sensitive.

Three observations matter for implementation.

The drift is state-dependent, since F_j appears on the right. That is what makes the simulation of Section 32.10 non-trivial and is the source of its discretization error.

The drift is a no-arbitrage consequence, not a parameter. It is the same statement as (32.5), in the discretized language.

It is also small. At a 3% rate with annual accrual, $\tau F/(1+\tau F) \approx 0.029$, so each term in the sum is roughly three percent of a product of volatilities. That smallness has a consequence worth stating in advance of Section 32.10: the predictor–corrector scheme exists because a plain Euler step freezes this state-dependent drift across the interval, and switching to it moves the answer by 0.15 basis points, while raising the step count from four to two hundred moves it by about 6. The discretization error is not where the state dependence would put it.

32.6 HOW MANY FACTORS

The whole motivation was Section 31.12's factor count, so it should be settled with data rather than convention.

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Example 32.1 The curve's factor structure

Model the correlation between forward rates as $\rho_{ij} = e^{-\beta|T_i - T_j|}$, take thirty annual forwards, and compute the eigenvalues of the correlation matrix. The cumulative share of variance explained is:

β	1 factor	2	3	4	5
0.02	82.8%	92.5%	95.3%	96.6%	97.4%
0.05	64.8%	82.2%	88.6%	91.7%	93.5%
0.10	46.6%	68.1%	78.3%	83.9%	87.3%
0.20	28.9%	49.0%	61.9%	70.2%	75.9%

This is **principal component analysis**: the eigenvectors are the factors, the eigenvalues their shares of variance. They carry the reading Litterman and Scheinkman found in bond returns and every major market has reproduced since: a roughly parallel shift (**level**), one sign change (**slope**), two (**curvature**). Further Reading has the reference.

Empirical estimates of β for major currencies fall around 0.05 to 0.10, which puts the first factor at 47–65% and the first three at 78–89%.

The form is an assumption, not a finding. What recommends it is that it is a valid correlation matrix for every $\beta \geq 0$, so no admissibility condition has to be checked (not true of every parameterization one might write down), and that it decays monotonically in tenor distance, the robust qualitative feature of measured correlation. Only β is estimated; the shape is imposed.

Neither the volatility structure nor the drift can move these percentages, since the matrix is a correlation matrix. The number of forwards can, by more than one would guess: at $\beta = 0.05$ the first factor explains 85.5% of ten annual forwards, 64.8% of thirty, 57.5% of forty. A factor count describes a strip, not a curve.

The form also sends ρ_{ij} to zero far out, where measured correlation flattens to a floor; the standard repair is $\rho_{ij} = \rho_\infty + (1 - \rho_\infty)e^{-\beta|T_i - T_j|}$. A floor is itself a common factor, so it lifts every entry: at $\rho_\infty = 0.3$ one factor explains 75.2% and three 92.0%. The table is conservative, and 92% is still not 99%.

The same table supports two readings, and both are correct.

Against one factor the case is decisive: it leaves a third to a half of the variance unexplained, and Example 31.11 priced what that omission does to a spread option.

Against the often-repeated "three factors explain 99% of the curve," it is a corrective: that figure comes from principal components of *yield levels*, dominated by a common trend. On the correlation structure a spread option depends on, three factors explain closer to 80%—and the residual is exactly the relative movement the option is written on.

The working answer is three factors for most books and two for many. The marginal factor is cheap here: it adds a column to the Cholesky factor and one more Wiener increment per step, so the cost of a factor grows linearly under simulation, where on a lattice it would grow exponentially with the dimension.

32.7 INSTANTANEOUS VOLATILITY AND ITS SHAPE

Section 29.5 distinguished flat volatility (one number per cap) from spot volatility, the piecewise-constant caplet volatilities that reproduce it. *The LMM needs a third object*: the *instantaneous* volatility $\sigma_k(t)$, a function of calendar time for each forward.

They are related by the requirement that total variance match:

$$\sigma_k^{\text{Black},2}\, T_k = \int_0^{T_k} \sigma_k(t)^2\, dt \tag{32.8}$$

The market determines the integral and not the integrand, so the shape of $\sigma_k(t)$ is a modeling choice with no caplet price to discipline it.

The standard choice is Rebonato's parametrization, in which volatility depends only on the time left to maturity, $T_k - t$, and not on where the calendar happens to stand. A specification

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with that property is **time-homogeneous**, and the term recurs below:

$$\sigma_k(t) = \big[a_R + b_R(T_k - t)\big]e^{-c_R(T_k - t)} + d_R \tag{32.9}$$

Four parameters produce a humped shape, which is what the data show: a forward rate two years out is more volatile than one thirty years out and also more volatile than one settling next week.

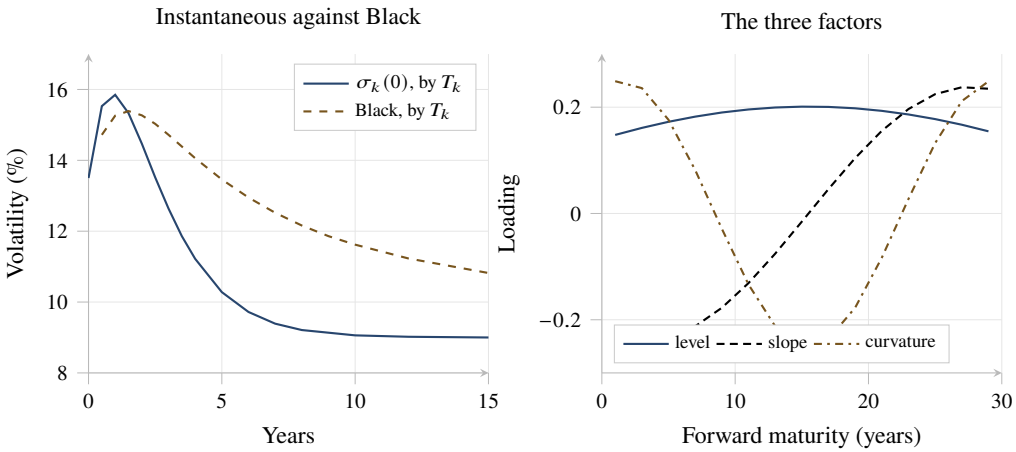


Figure 32.1: Left: equation (32.9) at $a_R = 0.045$, $b_R = 0.10$, $c_R = 0.75$, $d_R = 0.09$, and the Black volatilities it implies through (32.8). Right: the first three eigenvectors of Example 32.1 at $\beta = 0.05$. Illustrative parameters, not a calibration.

Figure 32.1's left panel shows why the distinction matters. Both curves are today's, and both are indexed by the caplet's own maturity: at T_k the solid curve is that forward's instantaneous volatility now, the dashed one its Black volatility. That the two can share an axis at all is owed to time homogeneity. As t runs from 0 to T_k the argument $T_k - t$ sweeps $[0, T_k]$, which is exactly the interval the solid curve covers, so the instantaneous volatilities a caplet meets over its life are the values the solid curve takes to the left of its own maturity, and the dashed curve is the running root-mean-square of the solid one. Were $\sigma_k(t)$ to depend on calendar time separately, the two would not be two views of one object, and the panel could not be drawn.

The instantaneous volatility peaks at 15.87% about ten and a half months out (at $1/c_R - a_R/b_R$) and falls to $d_R = 9\%$ in the tail; the Black volatilities it implies peak at 15.39% and are still at 10.8% fifteen years out, where the panel ends. Both curves tend to d_R in the end (a running root-mean-square of a function approaching 9% approaches 9%), but the dashed one takes far longer to get there, reaching 9.95% only by thirty years. Being a running average, the dashed curve is smoother, flatter, and shifted. A desk that reads a quoted Black volatility as though it were the instantaneous volatility of that forward has the wrong number by more than three volatility points around five years, and still by nearly two at fifteen.

The right panel shows the eigenvectors of Example 32.1: the first is positive throughout, the second changes sign once near fifteen years, the third twice.

A second specification is in wide use, and it is worth knowing because it makes the freedom in (32.8) concrete. Instead of a smooth function, take $\sigma_k(t)$ **piecewise constant** (one level per accrual period) and time-homogeneous. Write $\hat{\sigma}_i$ for the level in the i-th period counting backward from the reset, so that $\hat{\sigma}_1$ is the level over the last period before the rate fixes.

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Then (32.8) is a sum rather than an integral,

$$\sigma_k^{\text{Black},2}\,T_k \;=\; \sum_{m=1}^{k} \hat{\sigma}_{k-m+1}^2\,\tau_m \tag{32.10}$$

counting forward: period m runs for τ_m and carries the level with $k-m+1$ periods still to run. When the accruals are all of one length τ, the right-hand side is $\tau\sum_{i=1}^{k}\hat{\sigma}_i^2$, and differencing consecutive caplets inverts it:

$$\hat{\sigma}_k^2\,\tau \;=\; \sigma_k^{\text{Black},2}\,T_k \;-\; \sigma_{k-1}^{\text{Black},2}\,T_{k-1} \tag{32.11}$$

Each caplet in turn supplies one more level. No optimizer is involved, and the fit to the caplet strip is exact by construction. That is the attraction: the instantaneous volatilities come off the quoted strip with a subtraction, and a reader can do it by hand.

Only equal accruals reduce to that subtraction, and the condition is not a technicality: on a strip alternating one-year and six-month periods, using (32.11) anyway gets $\hat{\sigma}_2$ wrong by a third and $\hat{\sigma}_6$ by more than half. The general case costs one more line rather than an optimizer. The level with k periods to run appears in (32.10) for caplet k exactly once, in the $m=1$ term, so each caplet still brings exactly one new unknown and the bootstrap still runs one caplet at a time; what gets subtracted is the remainder of the sum rather than the previous caplet's total variance:

$$\hat{\sigma}_k^2\,\tau_1 \;=\; \sigma_k^{\text{Black},2}\,T_k \;-\; \sum_{m=2}^{k} \hat{\sigma}_{k-m+1}^2\,\tau_m$$

The cost is the mirror image. There are as many free levels as caplets, so nothing smooths the result, and nothing prevents (32.11) from returning a negative $\hat{\sigma}_k^2$, which it does whenever the quoted strip rises steeply enough that a later caplet's total variance falls short of an earlier one's. The parametric form cannot fail that way, and cannot fit exactly either.

The two specifications look very different as functions even when they agree on every caplet. On the strip implied by (32.9) at the parameters above, the bootstrap gives

Periods to reset	1	2	3	4	5	6	7	8	9	10
$\hat{\sigma}_i$, instantaneous	15.26	15.28	13.54	11.89	10.72	9.98	9.54	9.29	9.16	9.08
Caplet Black vol	15.26	15.27	14.72	14.06	13.46	12.95	12.52	12.16	11.86	11.62

in percent. The instantaneous levels fall to 9.1% where the caplet volatilities they reproduce fall only to 11.6%, because each caplet volatility averages the instantaneous one over the whole life. Figure 32.1's left panel is the same relationship for the smooth form.

The reason to prefer a time-homogeneous form is not fit but stability: the volatility structure such a model implies for one year from now looks like today's. A model that is not time-homogeneous predicts that the market's volatility surface will change shape by tomorrow—not because anything happened, but because the calendar advanced—and a desk hedging vega on such a model is hedging an artifact. Section 21.3 made the same criticism of local volatility's forward smile, and it is the same defect.

What the choice between the two forms is worth in money is a question with an answer, and Example 32.2 computes it on three products built to see the difference.

Example 32.2 What the shape is worth: three nonstandard caps

Two specifications that price every caplet identically must differ somewhere, and the products that find the difference are the ones whose payoff depends on the path of the forward-rate vector rather than on one rate at one date. Three standard ones, each a cap whose strike or exercise rule is not fixed in advance:

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- a **ratchet cap** resets each strike to the rate just observed plus a spread, $K_{k+1} = R_k + s$;
- a **sticky cap** resets it to the previous *capped* rate plus a spread, $K_{k+1} = \min(R_k, K_k) + s$, so the strike follows the rate down but rises by at most the spread each period;
- a **flexi cap** is an ordinary cap on which only the first N caplets that finish in the money pay.

Take annual periods on the grid of Section 32.4, so that the cap's ten caplets are the ones on F_1 through F_{10} and the terminal measure is $Q^{T_{11}}$. Put the forwards flat at 5%, take the correlation of Example 32.1 at $\beta = 0.05$, a spread of 25 basis points, and a flexi cap limited to five exercises. Calibrate both specifications to the same caplet strip (the parametric form to (32.9)'s parameters, the piecewise-constant form by the bootstrap (32.11) from the volatilities that form implies), so that every vanilla caplet price agrees to eight decimals. Then simulate both on the same draws:

	Parametric	Piecewise	Difference	Std. err.
Ratchet cap	1.8169	1.8065	−0.57%	0.04%
Sticky cap	3.5472	3.5369	−0.29%	0.03%
Flexi cap	2.5339	2.5333	−0.02%	0.07%
Vanilla cap	3.6729	3.6733	0.01%	0.02%

Per 100 of notional, from 12,000 paired paths at sixteen steps a year with common random numbers.

The vanilla cap is a control rather than a result. Both specifications reproduce its price by construction, so any difference there is numerical, and 0.01% against a standard error of 0.02% is what says the experiment is sound. The ratchet's −0.57% is therefore shape and not noise, at more than a dozen standard errors; the sticky cap's −0.29% likewise; the flexi cap's −0.02% is inside its own error: the flexi cap does not see the shape at all.

The control is not decoration. At four steps a year instead of sixteen, the vanilla cap reads +0.14% against the same 0.02% standard error (discretization bias, which is larger for a step volatility than for a smooth one, and now large enough to be detected), and the ratchet reads −1.60%. That is nearly three times the true figure, and nothing inside the ratchet's own number reveals it. This is Example 32.4's lesson met from the other side: a quantity computed as a difference between two models inherits both models' discretization error, and the only defense is an instrument whose answer you already know.

The size carries a lesson of its own. A ratchet at 0.57% of premium is inside most desks' bid–ask on a ten-year structure but not far inside it, and the exposure is one-directional: the smooth form is dearer on both path-dependent products. The reason is not where the volatility sits: each bootstrapped level is the root-mean-square of the smooth curve over its own period, so the two forms put the same variance into every period. It is how neighboring forwards move together within a period. The step function moves them in lockstep, so it correlates successive fixings slightly more, and a ratchet or sticky cap, whose strike is set by the previous fixing, is an option on the change from one fixing to the next. A desk that calibrates one way and hedges against a counterparty who calibrates the other keeps that difference as an unexplained mark.

32.8 CORRELATION: INSTANTANEOUS AND TERMINAL

Two distinct quantities carry the same name and are routinely confused.

Instantaneous correlation ρ_{ij} describes how the two forwards move together over an instant. *Terminal correlation* describes how their values at some horizon are related, and that is what a spread option pays on.

Example 32.3 Do the two differ enough to matter?

With the Rebonato form (32.9) at $a_R = 0.045$, $b_R = 0.10$, $c_R = 0.75$, $d_R = 0.09$, terminal

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correlation over a horizon H is

$$\frac{\rho_{ij}\int_0^H \sigma_i\sigma_j\,dt}{\sqrt{\int_0^H \sigma_i^2\,dt\int_0^H \sigma_j^2\,dt}}$$

Set the instantaneous correlation to 1 and see what the volatility shapes alone produce:

T_i	T_j	**Horizon**	**Terminal corr. at $\rho = 1$**
2	3	2y	0.9968
2	5	2y	0.9970
2	10	2y	0.9993
5	10	5y	0.9936
10	20	5y	0.9994

At most 0.6% of decorrelation, which invites being ignored. It should not be. A spread's variance is $V_i + V_j - 2\rho\sqrt{V_iV_j}$ with $V_i = \int_0^H \sigma_i^2\,dt$, and the subtraction nearly cancels at high ρ, so the surviving error is amplified: treating the instantaneous ρ as the terminal one understates the 5y/10y spread volatility by 1.27% at $\rho = 0.85$ and 4.18% at $\rho = 0.99$. An at-the-money premium moves with its volatility, so that is at or above the bid–ask—and worst where spread options are actually struck.

So in the LMM the correlation you calibrate is close to the correlation you get, which makes the correlation matrix the object to spend calibration effort on and the model's spread-option prices interpretable. It does not make the shape correction droppable: a multiplier of 0.994 is not a multiplier of one once the cancellation levers it.

The contrast with Chapter 31 sharpens too. A one-factor short-rate model has $\rho_{ij} = 1$ by construction and no offsetting shape effect either, so its terminal correlations are essentially 1 as well. The defect of Section 31.12 cannot be repaired with clever volatility structures; it takes more factors.

32.9 CALIBRATION IN PRACTICE

Caplet volatilities come first and are exact. Equation (32.8) determines the total variance of each forward from the market's caplet quotes, and the LMM reproduces them by construction. *No optimization is involved*, which is the model's chief practical attraction.

The volatility shape is then fitted subject to that constraint. The four parameters of (32.9) are chosen to fit swaption volatilities (which, unlike caplets, depend on the shape and on correlation), with the total variances held at their caplet values.

Correlation is fitted last and is the hard part. It can be estimated historically, parametrized (the $e^{-\beta|T_i-T_j|}$ form of Example 32.1 is standard), or implied from spread options where those are liquid. Historical and implied correlations differ systematically, and by more than the volatility analogs of Section 19.10. That should be expected. A **correlation risk premium** (a persistent gap between the correlation priced into options and the correlation the rates go on to show) arises for the same reason as a volatility risk premium, but far less trading can close it: volatility is bought and sold directly, through variance swaps and a deep listed options market, whereas correlation trades only through the dispersion trade of Chapter 22 (which ties up capital and carries other exposures) or through correlation swaps that barely trade. A gap between historical and implied correlation can therefore persist with no one able to arbitrage it away.

Then the correlation matrix is reduced to K factors, which changes it. A rank-K approximation to a full correlation matrix does not reproduce the original entries, and Example 32.1's table is exactly the statement of how much is lost.

LMM calibration is therefore one exact step, one well-posed fit, and one genuinely underdetermined estimate. A desk that reports the three separately is doing better than one that

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reports a single fitting error.

32.10 SIMULATION

The LMM has no lattice, so everything is Monte Carlo, and the drift (32.7) is state-dependent.

That state dependence rules out the exact scheme of Section 20.7, which is exact only because a geometric Brownian motion's drift is deterministic. What is used instead is an Euler step taken on the logarithm. Itô's lemma turns (32.6) into

$$d\ln F_k = \left(\mu_k(t) - \frac{1}{2}\sigma_k(t)^2\right) dt + \sigma_k(t)\, dW_k$$

and stepping that over $[t, t+\Delta t]$ gives

$$F_k(t+\Delta t) \;\approx\; F_k(t)\exp\left[\left(\mu_k(t) - \frac{1}{2}\sigma_k(t)^2\right)\Delta t + \sigma_k(t)\,\varepsilon\sqrt{\Delta t}\right]$$

It is an approximation in both coefficients: each is held at the value it has when the interval opens, and neither is constant across it, σ_k least of all, since Section 32.7 gives it a hump.

The two are not on the same footing, though. Because σ_k is deterministic, its approximation is removable: use the variance the model actually accumulates over the step, $\int_t^{t+\Delta t} \sigma_k(u)^2\, du$, in place of $\sigma_k(t)^2\Delta t$, and the diffusion becomes exact. The drift admits no such treatment, because μ_k depends on the forwards themselves and is not known until the step has been taken. One approximation can be integrated away and the other has to be corrected for, which is why a predictor–corrector exists for the drift and nothing analogous is needed for the diffusion.

Stepping the logarithm rather than F_k itself keeps the forward positive, since it is an exponential, so the failure of a direct Euler step never arises. With the variance integrated exactly, the frozen drift would be the only error left anywhere, and it is what the predictor–corrector removes. The comparison below keeps the step as displayed, with both coefficients frozen, so that it can measure the two errors side by side: the gap between the two schemes is the drift's error, and the movement as the step count rises is mostly the volatility's.

The standard concern is that freezing the drift introduces bias, and that the predictor–corrector of Section 20.7 removes it.

That concern deserves to be measured rather than repeated, and measuring it needs a product the model cannot price in closed form. A cap will not do: the LMM reproduces every caplet exactly by construction, so simulating one tests the simulation against an answer already in hand. A swaption is the simplest instrument whose value depends on the forwards jointly (on their correlations as well as their volatilities), so the model has no formula for it and it has to be simulated. It is also the trade Section 32.11 needs, which is why the same one appears there.

Example 32.4 Where the simulation error actually is

A one-year option on a five-year swap, five annual forwards flat at 3%, instantaneous correlation 0.85, Rebonato volatility as in Example 32.3. Four hundred thousand antithetic pairs were used.

Each simulated price is inverted through Black's swaption formula of Section 29.6 to give the Black volatility the third column reports. That assumes nothing about the swap rate: Black's value is monotone in volatility, so the inversion is a change of units, and it states the error in basis points of volatility rather than in currency. The last column then measures each figure against Rebonato's approximation, which Section 32.11 sets out; the approximation serves only as a fixed yardstick here, so what the table is about is the movement down the column rather than the distance across to it.

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Steps	Scheme	Implied vol	vs. approximation
4	Euler (frozen drift)	12.52%	−6.2 bp
4	Predictor–corrector	12.52%	−6.3 bp
12	Euler	12.53%	−5.2 bp
12	Predictor–corrector	12.53%	−5.3 bp
50	Euler	12.58%	−0.2 bp
50	Predictor–corrector	12.58%	−0.2 bp
200	Euler	12.58%	−0.1 bp
200	Predictor–corrector	12.58%	−0.1 bp

The Monte Carlo standard error is 2.2 basis points of volatility on each level, and rows at different step counts do not share draws, so a difference between two of them carries about 3.1. The difference between the two schemes at one step count is another matter: it shares its draws, and is measured to a thousandth of a basis point.

Two findings follow, and the second contradicts the usual advice.

Discretization bias is real and it goes away quickly. Four steps a year leaves the answer about 6 basis points low. That is only two standard errors of the difference, but the shortfall is not noise: it is what holding σ_k at its value at the start of each step does across the hump, and computing that effect directly, with no simulation, gives 7.7 basis points at four steps and 0.6 at fifty. Fifty steps closes the gap to inside the Monte Carlo error, and two hundred does not improve on fifty. A scheme that needs fifty steps a year is not an obstacle.

The predictor–corrector makes no measurable difference here. On shared draws the two schemes differ by 0.15 basis points at four steps a year and by 0.003 at two hundred—smaller than the bias it is supposed to remove, at every step count.

The reason is Section 32.5's third observation. Every term of (32.7) carries a product $\sigma_k\sigma_j$, so the drift is second order in volatility where the diffusion is first order, and the factor $\tau F/(1+\tau F) \approx 0.029$ damps it further. Over this option's one-year life the largest drift moves $\ln F$ by 0.0019, against a diffusion standard deviation of 0.153. Correcting a term that small cannot move the answer, and the residual bias is coming from the frozen volatility (the error that integrating the variance removes) rather than from the frozen drift.

The drift correction earns its cost when the drift is large (high rates, and many forwards between the simulated rate and the numeraire) and not otherwise. The accrual convention matters less than it looks: halving τ halves each term of (32.7) but doubles the number of terms over the same stretch of curve. A desk running a five-forward book at 3% that pays for a predictor–corrector has bought nothing, while one running a thirty-forward book in a high-rate currency needs it. This is Section 20.7's rule: a bias-reduction technique should be validated on the problem at hand, not adopted on the strength of the case that motivated it.

32.11 THE SWAP MARKET MODEL, AND WHAT THE INCOMPATIBILITY COSTS

Section 29.6 priced swaptions with Black's model applied to the forward swap rate, which assumes the *swap rate* is lognormal. Section 32.4 assumes the *forward rates* are lognormal.

Behind the first assumption is a model built exactly as the LMM is. Section 28.8 showed the forward swap rate $S(t)$ to be a traded asset divided by the annuity, and therefore a martingale under the annuity measure Q^A. Give it a deterministic volatility and

$$\frac{dS(t)}{S(t)} = \sigma_S(t)\, dW^A$$

which is the **swap market model**. A swaption is then priced by Black's formula exactly, in the same sense and for the same reason that the LMM prices a caplet exactly: each model takes the rate its own market quotes, makes that rate lognormal under the measure in which it is already a martingale, and so reproduces the quoting convention by construction rather than by calibration.

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That is the one-rate case, and it is all the argument below needs. The swap market model in full carries a family of *co-terminal* swap rates (swaps starting on different dates and all ending on the same one) under a single annuity measure, with drift conditions derived exactly as Section 32.2's were and with swap rates in place of forwards; it is Jamshidian's, and this chapter states it without developing it, because the derivation would repeat Section 32.2 with the letters changed while the LMM is the model more commonly implemented. Its natural application is the Bermudan swaption, whose exercise dates are precisely a co-terminal strip, so there the model's state variables are the exercise decisions themselves, which is the other route to the problem Section 32.12 takes in LMM terms.

Both cannot be exactly true. A swap rate is a weighted sum of forwards with stochastic weights, and a weighted sum of lognormals is not lognormal. This is a theorem, and it means the market's two standard quoting conventions are mutually inconsistent.

That statement usually ends the discussion. The useful question is how large the inconsistency is, and it has a number.

Example 32.5 Measuring the inconsistency

Rebonato's approximation expresses the swaption's Black volatility as a weighted average of forward volatilities, with weights

$$w_k = \frac{\partial S}{\partial F_k} \cdot \frac{F_k}{S}$$

For the swap of Example 32.4 these are 0.2120, 0.2058, 0.1998, 0.1940, 0.1884, *summing to 1.0000*, which is the check that the approximation is a genuine weighted average and not merely a fitted formula.

The approximation gives a Black volatility of **12.5818%**. Simulating at 800 steps gives **12.58%**, which is under a basis point away against a Monte Carlo error of about two.

So for this trade the incompatibility has no measurable price, and the effort spent on it in the literature looks out of proportion to its size.

Now change the trade, since a short option on a short swap is where an approximation has the least to do. A five-year option on a ten-year swap, ten annual forwards flat at 6%, instantaneous correlation 0.70:

	1y into 5y at 3%	**5y into 10y at 6%**
Rebonato approximation	12.5818%	9.0048%
Simulation, 800 steps	12.58%	9.00%
Error	under 1 bp	under 1 bp

Within the noise, the longer trade is no worse than the shorter one. Both errors are under a basis point of volatility, both are well inside the simulation's own error of about two, and neither is visible against a bid–ask measured in half-points. Four separate estimates were taken of each (two step counts under each of two numeraires), and they bracket the approximation rather than sitting to one side of it.

That the weights sum to one is an identity, not a normalization. The forward swap rate is homogeneous of degree one in the forwards (scaling every F_k by λ scales S by λ), so Euler's theorem gives $\sum_k F_k \, \partial S/\partial F_k = S$, hence $\sum_k w_k = 1$. This is why the check in Example 32.5 is informative: a weight vector that misses one signals an error in the discount-factor recursion rather than a modeling choice.

The approximation looks cruder than it is. It freezes the weights w_k at today's values, and the weights do move: more with more forwards, with longer expiration, and with larger τF, all three of which grow between the two columns. But the swap rate is a weighted average, so a weight that drifts up is offset by others drifting down, and the error enters at second order in those displacements rather than at first. Freezing a set of numbers that must keep summing to one is a much smaller sin than freezing a single one of them would be.

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So the swap/forward incompatibility is a theorem with no measurable price, at either end of the range tested. A desk may calibrate to caps and price swaptions with Rebonato's formula and not think about it again—which is what desks in fact do, and this is the calculation that says they are entitled to.

The general lesson is one this book has drawn twice before, and this is the first time it comes out the other way. Section 21.3 found local volatility's forward-smile error structural and large; Section 31.12 found the one-factor correlation error structural and a factor of four. Here the structural error is real as mathematics and negligible as a price—which is the most comfortable of the three answers and the rarest.

32.12 EARLY EXERCISE

A Bermudan swaption needs a continuation value at every exercise date, and Monte Carlo runs forward while the exercise decision needs backward induction. Section 20.11's Longstaff–Schwartz regression is the standard answer: regress realized continuation values on state variables at each exercise date, and use the fitted function as the exercise rule.

Three practical points follow, since this is where implementations go wrong.

The regression basis must span the state. With a high-dimensional forward-rate vector, the natural choices are the swap rate and the annuity, sometimes with a second swap rate to capture the curve's shape. Choosing too small a basis produces a suboptimal exercise rule and therefore a downward-biased price, which looks like conservatism and is not.

The estimate is biased in both directions and the biases are separable. Using the same paths to fit the rule and to price gives an upward foresight bias; using a suboptimal rule gives a downward bias. The standard discipline is to fit on one set of paths and price on another, which removes the first and leaves the second, so the reported price is a genuine lower bound. Upper bounds via the dual formulation (which values exercise with perfect foresight, less a martingale chosen to charge for that foresight) exist and are expensive.

And the whole apparatus is slower than a tree by orders of magnitude. Example 31.8's Bermudan takes milliseconds on a trinomial tree; the same product in a three-factor LMM with regression takes seconds to minutes. For a book of thousands of Bermudans revalued nightly with sensitivities, that difference decides the model, which is the reason short-rate models still run the callable book.

32.13 SENSITIVITIES, AND WHY THEY COST MORE THAN PRICES

A price is one number; a book needs its derivatives with respect to every calibration input, and in an LMM there are many.

The obvious method is **bump-and-revalue**: move one input by a small amount, revalue, and take the difference against the unperturbed price. It scales badly. With M curve points and V volatility points, a full sensitivity run is $M + V + 1$ simulations (the base, plus one for each input). At fifty curve points and thirty volatility points, that is eighty-one Monte Carlo runs for one product. That is why an LMM book's overnight risk run, not its pricing, is the binding computational constraint.

Common random numbers are not optional here. If the bumped and base simulations use different draws, the difference of two noisy estimates carries the noise of both, which can exceed the bump's whole effect, so the delta comes out smaller than its own standard error. Section 20.10 made the point for equity Greeks, where sharing the draws improves an estimate; in an LMM risk run it decides whether there is an estimate at all, and Example 32.6 measures the gap.

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Example 32.6 Why the draws must be shared

A swaption prices at 100.0000 with a Monte Carlo standard error of 0.0500 at 100,000 paths, and a one basis point bump to the curve moves it by 0.0480.

With independent draws the difference of the two runs has standard error $0.0500\sqrt{2} = 0.0707$:

$$\frac{\text{signal}}{\text{noise}} = \frac{0.0480}{0.0707} = 0.68$$

The delta is smaller than the uncertainty in the delta—the run has produced no information at all, and a risk report built this way is reporting sampling noise with a sign.

With common random numbers the two runs differ only through the bump, most of the sampling error cancels, and a standard error near 0.0004 is routine:

$$\frac{\text{signal}}{\text{noise}} = \frac{0.0480}{0.0004} = 120$$

The cost of not sharing the draws is not a factor of two; it is a factor of 31,250 in paths. Matching that precision with independent draws requires

$$\left(\frac{0.0707}{0.0004}\right)^2 \times 100{,}000 \approx \textbf{3.1 billion paths}$$

per bump, and there are eighty bumps. One line of implementation—seed the bumped run identically—buys what no amount of hardware can.

The same arithmetic explains why the input count is what kills the method rather than the path count. Adding a currency to the book does not make the simulation slower; it makes the number of simulations larger:

Curve points	Vol points	Bump-and-revalue	AAD
50	30	81 runs	~4 runs
80	45	126 runs	~4 runs
120	70	191 runs	~4 runs

At twelve minutes per pricing run, eighty-one runs is **16.2 hours** and does not fit in an overnight window; four is **48 minutes**. *The right column does not move*, which is the whole of the argument for adjoint methods.

Where the payoff permits it, Section 20.10's pathwise differentiation beats bumping: differentiating the simulated payoff with respect to the input along each path gives an unbiased estimator out of a single simulation, with no bump size to choose. *It fails for discontinuous payoffs* (a digital, or a barrier), where the payoff has no derivative to take, and the repairs are to smooth the payoff or to fall back on the likelihood-ratio method.

And adjoint algorithmic differentiation changed the economics of the whole exercise. AAD (Section 20.10) computes all $M + V$ sensitivities in a fixed multiple (typically three to five times) of the cost of one price, independently of how many inputs there are. *Eighty-one simulations become four*, which is a factor of twenty and is the reason models of this complexity are viable on a trading desk at all.

This deserves a place in a chapter on models rather than a footnote in one on computation. Section 9.1 noted that cost, not method, is the binding constraint on an XVA desk's exposure calculation; the same logic operates here. A model that cannot produce its own risk overnight will not be used, however correct it is, and AAD is what moved several models across that line after 2010. It is also why the LMM's cost relative to a tree, which Section 32.12 put at orders of magnitude, is smaller for sensitivities than for a single price.

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32.14 PRODUCTS THAT REQUIRE IT

The abstract case for more factors is Section 31.12. Here is the concrete list, because the model exists for these and for very little else.

CMS spread options and the steepener notes built on them. A coupon paying $\max(S_{10} - S_2 - K, 0)$, sold to investors who want yield and will be long the curve's slope to get it: the coupon pays only while the ten-year rate exceeds the two-year by more than K, which a merely flat curve already fails, let alone an inverted one. Example 31.11 measured what one factor costs here. These are the largest single use of the model.

Range accruals on two rates. A coupon accruing only on days when a rate (or the spread between two) sits inside a band. The payoff depends on the joint distribution at many dates, not on one date's marginal.

Ratchets and target redemption notes. A ratchet's coupon is capped relative to the previous coupon, which makes it path-dependent in the forward-rate vector; a **target redemption note (TARN)** terminates once cumulative coupons reach a threshold, which makes the termination date itself a functional of the path. Neither is a function of a single state variable at a single date, which is precisely the condition under which Chapter 31's tree stops being applicable.

Callable structured notes, which combine the above with the exercise problem of Section 32.12, and are the hardest products in the rate market for exactly that reason.

Who is on each side is worth stating, because it explains the market's shape. The buyers are yield-seeking investors: private banks, insurers, regional banks. In a callable note or a range accrual they are short optionality and are paid a coupon for it, but the steepener's buyer is long a CMS spread option, and the steepener is what sets the correlation market. The dealer is therefore short spread optionality and must hedge it, which means trading swaptions and CMS in the correlations of Section 32.8. A spread option is worth more the less the two rates are correlated, so the large and persistent one-way demand that structured-note issuance creates bids spread volatility up, and that is why dealers' implied correlations sit systematically below historical ones. It is the same asymmetry Section 19.10 identified in equity volatility, arising the same way.

32.15 SMILE

The LMM as stated is lognormal, so it produces no smile, and Section 29.8 established that rate markets have a pronounced one. Two repairs are in use.

The shifted or displaced LMM models $F_k + s$ as lognormal, which produces a skew and, by the same mechanism as Section 31.14's shifted model, accommodates negative rates. It is cheap and it is what most books use for the skew alone.

SABR-LMM gives each forward its own stochastic volatility with the SABR dynamics of Section 29.9, correlated across forwards. This is the state of the art and it is expensive: the parameter count multiplies, and the correlation structure now has to cover volatility-volatility and rate-volatility correlations as well as rate-rate.

The market's actual practice is worth recording because it is inconsistent and works. Swaptions are quoted through SABR per expiration-tenor cell (the volatility cube of Section 29.9), and the LMM is calibrated to those quotes. So SABR does the smile interpolation and the LMM does the term-structure dynamics, and the two are not a single consistent model. Section 21.12's judgment applies: a model used for interpolation and a model used for dynamics are being asked different questions, and requiring one object to do both has costs that are usually larger than the inconsistency.

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32.16 MULTI-CURVE, AFTER THE TRANSITION

Chapter 7 rebuilt the framework around a discount curve and one or more forecast curves. What does that do to a model whose state variable is a forward rate?

Less than it did to the short-rate models of Chapter 31, and for the opposite reason. A short-rate model models a risk-free rate and had to have the spread added to it. An LMM models whatever forward rates you tell it to, so a multi-curve LMM simply carries more of them (the forwards of each index's forecast curve, plus the discount curve), with the drift condition applying to each set under the chosen numeraire.

Two genuine changes follow.

The numeraire is now the discount curve's bond, so the drift (32.7) is taken with respect to the discounting curve while the forwards being modeled come from forecast curves. That separation is precisely what (32.2) ruled out: a forward curve and a discount curve carrying the same information, either determining the other. The identity still holds within each curve, but it no longer runs between them, so the two must be carried and calibrated separately. The correction is a covariance term between the two curves, and most implementations set it to zero by treating the basis between those two curves as deterministic. That is reasonable while the basis is small and stable, which the move to overnight rates largely made it (Section 33.9 traces the tenor basis shrinking to term premia and compounding mechanics once the credit component went), and it is wrong for anything whose payoff depends on the basis itself, such as the currency option inside a two-currency CSA that Section 9.6 prices as ColVA.

And backward-looking rates change what a forward is. Section 29.4's compounded-in-arrears rate is not known at the start of its period, so the "forward rate" for a SOFR period does not fix at the start of its accrual period: it keeps moving, as a martingale, until the period ends. The clean formulations of this (the forward-market model with backward-looking rates) were developed after 2018, and they show that the received LMM machinery survives with the accrual convention changed. The transition altered what is compounded, not how the model is built, which is a fair summary of its effect on this whole chapter.

32.17 CHOOSING A MODEL

The reader who skips the rest of the chapter should take this section.

Use a one-factor short-rate model when the payoff depends on the level of rates and the product has early exercise: callable bonds, Bermudan swaptions, mortgage books. Chapter 31's speed and natural treatment of exercise decide it, and Example 31.9's calibration warning is the price.

Use a two-factor short-rate model when the payoff has mild shape dependence and still needs exercise. It keeps the lattice, at the cost of a two-dimensional one.

Use an LMM when the payoff depends on several rates at once and on their joint distribution: CMS spread options, ratchets, range accruals on more than one rate, anything where Section 32.8's correlation matrix is the answer to the pricing question. Accept that early exercise will cost you regression and time.

Use SABR alone when the question is interpolation across strikes at one expiration, which is most of what a swaption desk does most days.

And the general principle, which outlasts all four: a model should be chosen by which of its errors you can afford, not by which is most realistic. Chapter 31's one-factor model is unrealistic in a way that costs a factor of four on a spread option and nothing on a callable bond. The LMM is realistic in a way that costs seconds instead of milliseconds. Neither fact is a defect; both are the specification.

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32.18 WHO USES THESE

Exotic rate desks at the large dealers, which is where the LMM lives. The products it exists for (CMS spread options, structured coupons on more than one rate) are dealer-intermediated and issued into structured notes, so the model's user and the product's ultimate holder are several steps apart.

Model validation groups, disproportionately. An LMM implementation has more places to be wrong than any other model in this book (drift, measure, factor reduction, regression basis, discretization), and the validation effort is correspondingly large. Example 32.4 is the kind of test that belongs in it.

Insurers with variable annuity books (investment-linked savings policies that carry a guaranteed minimum benefit), where the guarantee depends jointly on equity and rate paths and a consistent multi-asset simulation is unavoidable.

And academics and vendors, more than trading desks. *This should be said plainly*: the LMM is the most-written-about interest rate model and not the most-used one. Chapter 31's models price more notional every day. The LMM's importance is that it defines what the simpler models are approximating, and Example 32.5 is an instance—you cannot know that a structural inconsistency is worth under a basis point without a model in which it is absent.

32.19 WHERE THIS CAME FROM

Heath, Jarrow and Morton (1992) generalized Chapter 31's models by dropping the short rate altogether: model the whole forward curve, and derive the condition its drift must satisfy for no arbitrage. *HJM is a framework rather than a model*, and its practical difficulty is the one Section 32.3 identifies: no instrument is quoted on an instantaneous forward rate, so the object being modeled is not the object being quoted.

Brace, Gatarek and Musiela (1997), Jamshidian (1997), and Miltersen, Sandmann and Sondermann (1997) fixed that within a year of each other by modeling the rates the market actually quotes. The result was not only a usable model but a justification: it showed that the Black formula the market had been applying to caps and swaptions for years was consistent with an arbitrage-free term structure, which until then no one had established.

SUMMARY

HJM models every instantaneous forward rate at once, with today's curve as the initial condition, so Chapter 31's fitting problem is built in rather than solved. Once the volatility functions are chosen, no-arbitrage determines the drift, and every arbitrage-free term-structure model is an HJM model. The framework is not implemented directly because the process is generally non-Markov and so has no recombining tree; Hull–White is the volatility structure that happens to be Markov, which is why Chapter 31 could use a lattice at all.

The LIBOR market model fixes a tenor structure and makes each forward lognormal, so Black's caplet formula is reproduced by construction rather than by calibration. The number of factors it needs depends on the correlation structure a product is written on: one factor leaves much of that variance unexplained, and the often-quoted 99% comes from principal components of yield levels, not of the correlations a spread option depends on.

The market fixes each forward's total variance and not the shape of its instantaneous volatility, so Rebonato's humped time-homogeneous form is chosen for stability rather than fit. Terminal and instantaneous correlation differ by little, so the correlation you calibrate is essentially the correlation you get, and Chapter 31's one-factor defect needs more factors rather than cleverer volatility.

In simulation, the predictor–corrector is worth its cost only when rates are high. Forwards and swap rates cannot both be exactly lognormal, so the market's two quoting conventions are

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inconsistent, but the inconsistency is smaller than the simulation's own error: it is a theorem without a price. Smile enters through a displaced or SABR-based LMM, and in market practice SABR interpolates across strikes while the LMM supplies the dynamics. The move to overnight rates changed what is compounded, not how the model is built.

Sensitivities, not prices, are the binding constraint: bump-and-revalue costs a simulation per input, while adjoint algorithmic differentiation delivers all of them at a small multiple of the cost of one price. The products that require the model (CMS spread options, two-rate range accruals, ratchets, and TARNs) depend on several rates at several dates, and dealers' one-way issuance flow is why their implied correlations sit below historical ones. Early exercise costs the LMM its main advantage, which is why short-rate models still price the callable book. Choose a model by which of its errors you can afford, not by which is most realistic.

FURTHER READING

Heath, Jarrow, and Morton's paper is the origin of Section 32.2 and the original statement of why the drift is not free; it is harder going than the market-model papers and worth the effort once. Brace, Gatarek, and Musiela, and independently Jamshidian and Miltersen, Sandmann, and Sondermann, established the market model and its consistency with Black's formula, and the original papers are the place to see the historical point of Section 32.4 as it was first made. Litterman and Scheinkman is the source of the principal-component reading of Example 32.1, and is where level, slope and curvature were first named as the factors of a curve; it is short. Rebonato's books are the practitioner's reference on everything in Sections 32.7 through 32.9, he treats the volatility-correlation trade-off in calibration at length, and the parametric correlation forms of Example 32.1, including the long-end floor, are his. Hull and White take the other route through the same material: the piecewise-constant instantaneous volatility of Section 32.7, bootstrapped period by period off the caplet Black volatilities, is theirs, and the same paper relates forward-rate to swap-rate volatilities, which is the approximation Section 32.11 measures the simulation against. Brigo and Mercurio cover the algebra of the drift conditions and the SABR-LMM of Section 32.15. Andersen and Piterbarg are the reference for simulation and for the early-exercise machinery of Section 32.12, including the dual upper bounds this chapter only mentions. On backward-looking rates and the post-transition formulations of Section 32.16, the literature is recent and still settling; the paper of Lyashenko and Mercurio is the standard entry point.

PRACTICE QUESTIONS

32.1 Explain what the state variable is in an HJM model, why today's curve poses no fitting problem, and state the HJM drift condition in words together with what it says about the model's degrees of freedom.

32.2 Explain in what sense Ho–Lee and Hull–White are HJM models, why a general HJM model cannot be put on a recombining tree, and identify the volatility structure that makes one Markov.

32.3 Explain why the LIBOR market model reproduces Black's caplet formula by construction rather than by calibration, and why the forward rates cannot all be martingales under one measure.

32.4 Compute $\tau F/(1+\tau F)$ at $F = 3\%$ and at $F = 8\%$ with annual accrual, and explain the relevance to the drift.

32.5 Using Example 32.1, state the share of variance explained by one and by three factors at $\beta = 0.05$, and explain why the often-quoted figure of 99% for three factors is misleading in this context.

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32.6 Distinguish flat, spot, and instantaneous volatility, state which the market quotes, and explain why the market determines the integral of instantaneous variance but not its shape.

32.7 Explain why a time-homogeneous volatility parametrization is preferred, and relate this to Section 21.3.

32.8 Distinguish instantaneous from terminal correlation, state which one a spread option pays on, use Example 32.3 to give the maximum decorrelation produced by volatility shape alone, and explain why that result strengthens rather than weakens Chapter 31's criticism.

32.9 Describe the three stages of LMM calibration, state which is exact and which is underdetermined, and explain what is lost when a full correlation matrix is reduced to K factors.

32.10 Using Example 32.4, state the change in implied volatility between 4 and 200 time steps, explain why the predictor–corrector makes no difference there, and state the conditions under which it would.

32.11 Explain why forward rates and swap rates cannot both be exactly lognormal, explain the significance of Rebonato's weights summing to one, and use Example 32.5 to state the size of the inconsistency in each of the two cases and compare it with a typical bid–ask.

32.12 Compute the number of simulations a bump-and-revalue risk run requires for a book with forty curve points and twenty-five volatility points, and explain why common random numbers are essential rather than merely helpful.

32.13 State the condition under which pathwise differentiation fails and name one repair, then explain what adjoint algorithmic differentiation delivers and why its cost does not grow with the number of inputs.

32.14 Name four products that require a multi-factor model, state for each what makes a single state variable insufficient, and explain why dealers' implied correlations sit systematically below historical ones.

32.15 Explain why using the same Monte Carlo paths to fit and to apply an exercise rule biases the price upward, and why a suboptimal rule produces a lower bound rather than an error of unknown sign.

32.16 State which model you would use for a callable bond, for a CMS spread option, and for quoting a single swaption, with reasons.

FURTHER QUESTIONS

32.17 Derive the HJM drift condition for a one-factor model by imposing that discounted bond prices are martingales. Then verify that $s(t,T) = \sigma e^{-a(T-t)}$ reduces the resulting dynamics to a Markov process in a single state variable, and identify that variable.

32.18 Example 32.1 shows three factors explaining 78–89% of variance on a correlation structure, against the often-quoted 99% on yield levels. Reconcile the two figures precisely, determine which is relevant for a CMS spread option and which for a duration hedge, and state the general principle for deciding which decomposition a given problem needs.

32.19 Example 32.3 finds the volatility-shape channel contributes at most 0.6% of decorrelation under Rebonato's parametrization. Determine whether this is a property of the parametrization or a general feature: construct a volatility structure that maximizes the gap between instantaneous and terminal correlation, state how large it can be made, and judge whether such a structure is economically plausible.

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32.20 Example 32.4 shows discretization bias dominating drift-correction error. Identify the source of the residual bias, propose a scheme that would remove it, and estimate the number of time steps at which the remaining bias falls below the Monte Carlo standard error for a book of one million paths.

32.21 Example 32.5 finds the swap/forward incompatibility worth under a basis point of volatility on both a short and a long trade, although the weights w_k that Rebonato's approximation freezes demonstrably move over the option's life. Derive the approximation's error term explicitly, show that it is second order rather than first order in the weights' displacements, and explain what the constraint $\sum_k w_k = 1$ has to do with it.

32.22 Section 32.15 notes that the market lets SABR do smile interpolation and the LMM do term-structure dynamics, which is inconsistent. Construct the argument that this should be replaced by a single consistent model, construct the argument that it should not, and state what evidence would settle it.

32.23 Section 32.17 argues a model should be chosen by which of its errors you can afford. Apply this to a specific book (a portfolio of Bermudan swaptions and CMS spread options in one currency) and design the model configuration you would run, stating explicitly which errors you are accepting on which trades and how you would monitor them.

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Chapter 33
Swaps Revisited

Chapter 7 priced the plain vanilla swap and built the curves it needs. It also deferred almost everything else, because the non-standard swaps require the adjustments of Chapter 30 and the measure machinery of Chapter 28, and those had not been developed.

They have now, and this chapter is where the deferred instruments are priced. The organizing observation is that a non-standard swap is almost always a standard swap plus a convexity adjustment, and that the adjustment is computable, has a sign that can be reasoned about, and is large enough to matter.

There is a second reason to return to swaps here. The transition to risk-free rates changed some of these instruments profoundly and others not at all, and the pattern of what changed is instructive. Section 33.4 contains the sharpest case: a convexity adjustment the market spent twenty years computing became roughly sixty times smaller overnight, not because anyone solved it but because the underlying convention changed.

The chapter runs from the variations that need no new theory to the ones that need all of it. Sections 33.1 to 33.3 change the notional or the payment dates and price by discounting alone. Sections 33.4 to 33.6 are where convexity enters: in-arrears swaps before and after the transition, CMS swaps, and range accruals, which are a strip of digitals in disguise. Sections 33.7 to 33.10 are the swaps whose payoff references something else: an embedded option, a commodity, a basis, an inflation index. Sections 33.11 to 33.13 are the equity and total return swaps whose funding leg Chapter 27 made expensive, how a position is unwound, and who uses these.

33.1 AMORTIZING, ACCRETING, AND ROLLER-COASTER SWAPS

The simplest variation is a notional that changes over time. *An amortizing swap* steps the notional down, typically to match a loan being repaid; *an accreting swap* steps it up, matching a construction drawdown; *a roller-coaster swap* does both.

Pricing needs nothing new. The schedule is known in advance, so each leg is discounted as it stands.

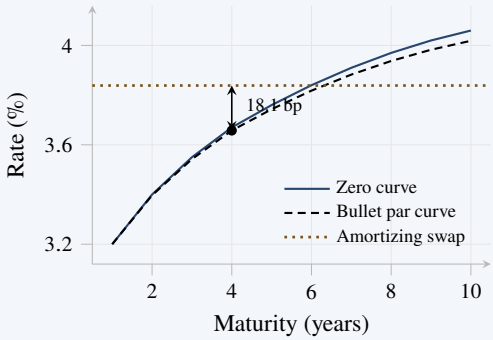
Example 33.1 Why the average-life approximation fails

Take a ten-year annual swap amortizing linearly from 100 to zero (notional 100 in the first year, 90 in the second, down to 10 in the tenth). The zero curve is upward-sloping and flattening, at 3.20, 3.40, 3.55, 3.67, 3.76, 3.84, 3.91, 3.97, 4.02 and 4.06 percent for one through ten years, annually compounded. The par rate is

$$\frac{\sum_k N_k \left[P(0,t_k) - P(0,t_{k+1})\right]}{\sum_k N_k \, P(0,t_{k+1})} = \mathbf{3.8386}\%$$

For comparison, against **bullet swaps**, ordinary swaps whose notional stays at 100 until maturity:

Bullet 10-year par rate	4.0187%
Bullet 5-year par rate	3.7432%
Bullet 4-year par rate	3.6576%
Amortizing par rate	**3.8386%**



The tempting shortcut is to price the amortizing swap as a bullet swap at its weighted average life. Weighting each payment date by the notional outstanding over the period ending on it, that life is exactly 4.0 years here, and the bullet par rate at four years is 3.6576%, an error of −18.1 **basis points**.

Which definition is used matters. Weight instead by principal repaid, which is what *weighted average life* means in the bond market, and the answer is 5.5 years, where the bullet rate is 3.7775% and the error is 6.1 basis points. Two standard summaries of one schedule differ by eighteen months, so the shortcut's error is not even a well-defined number until the convention is fixed.

Weighted average life summarizes the schedule by its first moment, and the par rate is a nonlinear function of the discount factors. Replacing a schedule by its mean is exact only when the function being applied is linear, and the par rate is not. The error grows with the curvature of the curve, which is why the shortcut is at its worst precisely when the curve is most interesting.

A desk that uses it is making the same class of error as Section 20.9's naive expectation: replacing a distribution by its mean inside a nonlinear function.

33.2 FORWARD-STARTING AND DELAYED SWAPS

A forward-starting swap (also called a delayed or deferred-start swap) begins at a future date. It requires no new machinery: the forward swap rate is read off the curve as in Section 7.3, and the value is zero at the forward par rate by construction.

What it requires is care about which curve. A ten-year swap starting in five years is priced from the five- and fifteen-year points, and an error of one basis point in the fifteen-year rate moves its rate by about *one and a half*, because the forward rate is a difference of two long rates divided by a shorter span, and differences amplify errors.

This is why forward-starting swaps are the instrument that exposes curve-construction disagreements between desks, and why the interpolation choices of Section 7.6, which look like housekeeping, are argued about.

33.3 COMPOUNDING SWAPS

In a compounding swap, the floating payments are not paid as they accrue; they are reinvested at a specified rate and paid as a single amount at maturity.

The reinvestment rate is the whole of the pricing question. Three conventions exist and they price differently.

Flat compounding reinvests at the floating rate itself. The leg's value is then the same as an ordinary floating leg's: the payments are simply rolled up at the rate that would have been earned by receiving and reinvesting them, so no adjustment arises. This is the common convention and the reason compounding swaps are usually treated as an operational rather than a valuation matter.

Compounding at the floating rate plus a spread is the standard convention where the leg

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carries a credit spread. Here the leg's value is *not* the same, because the spread is being compounded at a rate that would not have been earned. The correction is small but real, and it is the reason ISDA's compounding definitions run to several pages.

Compounding at a fixed rate makes the leg a portfolio of forward rate agreements with deterministic weights: straightforward, and rare.

The general point is the one Chapter 28 supplies. A payment moved from its natural date to another date is a change of measure, and whether an adjustment arises depends on whether the rate remains a martingale under the measure that goes with the new payment date. Flat compounding happens to preserve it. The other conventions do not.

33.4 IN-ARREARS SWAPS, BEFORE AND AFTER

This is the instrument the transition changed most.

The classic LIBOR-in-arrears swap pays the rate for $[T_k, T_{k+1}]$ at T_k, at the *start* of the period it applies to, rather than the end. Since LIBOR was set in advance, this was operationally possible and was done.

But the rate is a martingale under the T_{k+1}-forward measure, and the payment is under the T_k one. Section 30.4 derived the resulting adjustment, and Example 33.2 puts numbers on it:

$$E^{T_k}[F_k] \approx F_k + \frac{F_k^2\,\sigma^2\,\tau\,T_k}{1 + F_k\tau} \tag{33.1}$$

where F_k is the forward rate for the period, τ its length in years, and σ the rate's volatility.

Example 33.2 A convexity that the transition mostly removed

$F = 3.5\%$, $\tau = 0.5$, $\sigma = 25\%$:

T_k	Adjustment	Adjusted rate
0.5y	0.19 bp	3.5019%
1y	0.38 bp	3.5038%
2y	0.75 bp	3.5075%
5y	1.88 bp	3.5188%
10y	3.76 bp	3.5376%

The transition removed the mismatch rather than the volatility. A compounded-in-arrears SOFR rate is determined at the end of its accrual period and paid then, so the payment is no longer early and (33.1) has nothing to correct. What is left is the short delay between the end of the observation period and the payment date, typically two business days:

T_k	Adjustment at a 2-day lag
1y	0.0061 bp
5y	0.0304 bp
10y	0.0607 bp

At ten years the adjustment falls from 3.76 basis points to 0.06, about sixty times smaller.

That changes how the product should be classified. The rate replacing LIBOR is the overnight rate compounded in arrears, which is what the floating leg of Section 7.1's overnight indexed swap pays, so the instrument this section files as non-standard is now the standard one. Paying in arrears is the convention, and it is a rate fixed in advance that needs a special name and a separate construction, which is what Chapter 4's Term SOFR is. The convention traded places with its own exception.

It also means the old product is not extinct, only currency-specific. Paying a period's rate at that period's start requires knowing it at the start, so the structure needs a forward-looking term rate, and one survives wherever the transition left it standing. The Euro Interbank Offered

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Rate (EURIBOR) was never retired, so a EURIBOR-in-arrears swap carries the classic adjustment unchanged: 3.76 basis points at ten years, not 0.06. Term SOFR would serve in dollars, but the ARRC's recommendation limits it to business loans and the derivatives hedging them, not the swap market at large.

A convexity adjustment is a property of a convention rather than of a market: nothing about interest rate volatility changed, the timing convention did, and the adjustment followed it—into the currencies that kept a term rate. Twenty years of literature on in-arrears convexity was solving a problem the market later removed by changing the contract.

The new convention has a problem of its own, in a different place. Section 29.4's backward-looking caplet needs an adjustment the forward-looking version did not, because the rate is unknown until the period ends. The complexity moved rather than vanished, which is true of most convention changes in this book.

33.5 CMS SWAPS

A constant-maturity swap exchanges a fixed rate for a long swap rate (the ten-year rate, say) observed and paid periodically. It is the largest non-standard swap market and the one where the adjustments of Chapter 30 are unavoidable.

Example 33.3 What ignoring the adjustment costs

Pay the ten-year swap rate annually for ten years against a fixed rate. Forward ten-year rate flat at 3.40%, swap-rate volatility 25%, discounting at 3.2%.

Section 30.5's adjustment applies to each payment, growing linearly in its date:

Payment	Adjustment	Adjusted rate
1	2.72 bp	3.4272%
3	8.17 bp	3.4817%
5	13.62 bp	3.5362%
7	19.06 bp	3.5906%
10	27.23 bp	3.6723%

The fair fixed rate is the annuity-weighted average of the adjusted rates:

Fair fixed rate, with adjustments	3.5426%
Fair fixed rate, ignoring them	3.4000%
Difference	**14.26 bp**

The difference is fourteen basis points on the fixed leg, every year for ten years. A ten-year swap trades at a bid–ask of well under a basis point, so this is not a refinement—a desk quoting a CMS swap without the adjustment is not slightly wrong, it is not in the market.

And the adjustment depends on a volatility, which is the structural point. An ordinary swap is priced off the curve alone; a CMS swap requires a swaption volatility, so a linear-looking product carries a vega. That has three consequences.

The CMS desk is a volatility desk. Its hedges are swaptions, not swaps, and its risk report has a vega column.

The adjustment uses the whole smile, not one volatility. The replication of Section 30.5 integrates over strikes (the same $1/K^2$-flavored argument as Section 22.3), so the number depends on the smile's wings, which Section 29.9 noted are the least reliable part of the volatility cube.

And the sign is always the same. The adjustment is positive, so a payer of CMS always pays more than the naive forward. An investor buying a CMS-linked note is being sold that convexity, and whether they are being sold it at a fair price is not something the note's headline coupon reveals.

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33.6 RANGE ACCRUALS

A range accrual coupon accrues only on days when a reference rate sits inside a band. It is a strip of digital options, one per observation day, and it is the most widely sold rate-linked structured coupon after CMS.

Example 33.4 The enhancement is the short vega

A coupon accruing only when the three-month rate is inside $[2.50\%, 4.50\%]$. Forward rate 3.50%, volatility 25%, observation at two years.

Under the forward measure the rate is lognormal, so each digital is a Black-style probability. For a strike K,

$$P(R < K) = N(-d_2), \qquad d_2 = \frac{\ln(F/K) - \sigma^2 T/2}{\sigma\sqrt{T}}$$

$P(\text{rate} < 2.50\%)$	0.2192
$P(\text{rate} < 4.50\%)$	0.8126
P(**inside**)**—the expected accrual fraction**	**0.5934**

So a coupon of $3.50\%/0.5934 = 5.8980\%$, paid only on in-range days, has the same value as a vanilla 3.50% coupon. The headline enhancement is **240 basis points, 69% above the vanilla coupon**, which is what sells the note.

Now see what the investor gave up:

Realized volatility	Accrual fraction	Value of the 5.8980% coupon
15%	0.8322	4.9082%
20%	0.7012	4.1357%
25%	0.5934	**3.5000%**
35%	0.4422	2.6079%
50%	0.3095	1.8254%

At the pricing volatility the coupon is worth exactly the vanilla 3.50%, by construction. Above it the investor receives less, below it more.

The enhancement is the short vega, priced. A range accrual investor is selling volatility, and the 240 basis points is the premium for that sale spread over the coupon rather than paid up front. Nothing is being given away.

Two structural points follow.

The dealer is long volatility and hedges with a strip of digitals or, since digitals are hedged as tight call spreads, with a great many options near the barriers. Section 12.8's building-an-arbitrary-payoff argument is the machinery, and the practical consequence is that the dealer's hedge is concentrated exactly at the range boundaries, which is where the note's value is most sensitive and where the smile is least reliable.

And the investor's loss is asymmetric in a way the table understates. Volatility rising is correlated with the rate moving away from its forward, so the accrual fraction and the reinvestment opportunity deteriorate together. The coupon fails in the states where its holder most needs income, which is Section 27.9's wrong-way structure arriving in a retail product.

Figure 33.1's left panel is the picture a range accrual prospectus does not contain. The coupon is a decreasing function of the very quantity the investor is unlikely to be forecasting, and the curve is steepest near the pricing volatility.

33.7 CANCELLABLE AND EXTENDABLE SWAPS

A cancellable swap may be terminated early by one party; an extendable swap may be prolonged. Both are a vanilla swap plus a Bermudan swaption, and Chapter 31 priced exactly that option.

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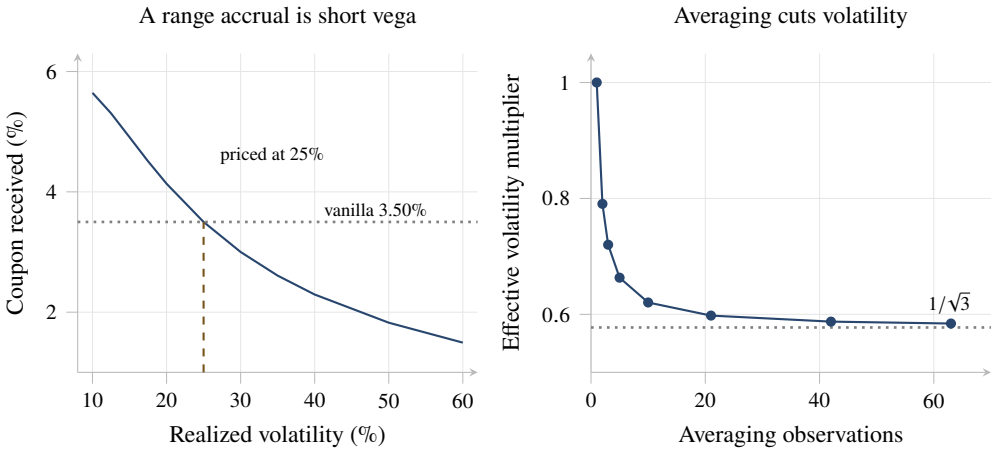


Figure 33.1: Left: Example 33.4's coupon against realized volatility. Right: the multiplier $\sqrt{(n+1)(2n+1)/6n^2}$ of Example 33.6.

Example 33.5 carries its value onto the swap by dividing it by the swap's annuity, which turns a one-off option premium into a running spread on the fixed leg.

Example 33.5 The Bermudan, annuitized

Take Example 31.8's six-year swap with annual cancellation rights from year one, whose Bermudan value was **1.6995** per 100 of notional.

Annuity of the six-year swap	5.272406
Bermudan option value	1.6995
Spread on the fixed leg	**32.23 bp**
Vanilla six-year par rate	3.9862%
Cancellable swap fixed rate	**4.3085%**

To first order: Example 31.8's option is struck at 4%, not at the cancellable rate of 4.3085% that this calculation produces, so the spread slightly overstates the option's value, and a full solution would iterate the strike to a fixed point.

A borrower whose debt may be prepaid or refinanced is the natural buyer, and it pays 32 basis points a year for the right to walk away—the whole economics of the instrument.

The reason to put this here rather than in Chapter 31 is what it does to the calibration problem. Example 31.9 showed the Bermudan's value spanning 13.1% across mean-reversion assumptions that all reproduce the calibrating European exactly. Annuitized, that becomes

$$0.131 \times 32.23 = \mathbf{4.2} \text{ basis points}$$

of ambiguity in a quoted swap rate, to first order.

Four basis points is well outside the bid–ask on a six-year swap, which is a fraction of a basis point. So the model ambiguity of Chapter 31 is not a theoretical concern that gets absorbed in spreads—it is several times the spread, and two dealers quoting the same cancellable swap from the same European volatilities can differ by more than the market they are quoting into.

Model risk is therefore a pricing problem and not only a risk-management one.

33.8 COMMODITY SWAPS AND THE COST OF AVERAGING

A commodity swap exchanges a fixed price for a floating one, and the floating one is almost always an average of daily prices over the delivery month rather than a single fixing, the one

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published price observed on a stated date. That convention exists because the underlying physical exposure is itself an average (a refiner buys crude every day), and it has a large valuation consequence.

Example 33.6 What averaging does to volatility

The average of a lognormal price over n equally spaced dates $T/n, 2T/n, \ldots, T$ has, in logs, variance

$$\frac{(n+1)(2n+1)}{6n^2}\,\sigma^2 T$$

which tends to $\sigma^2 T/3$ as n grows. This is exact for the geometric average and a close approximation for the arithmetic average a contract actually settles on. The effective volatility multiplier is therefore

n	Variance ratio	Vol multiplier
1	1.0000	1.0000
5	0.4400	0.6633
10	0.3850	0.6205
21	0.3575	**0.5979**
63	0.3413	0.5842
∞	0.3333	$1/\sqrt{3} = 0.5774$

Twenty-one fixings, at one every two and a half weeks over a year, already take the multiplier to within four percent of its continuous limit, so a modest number of fixings does essentially all the work continuous averaging would. What the multiplier does assume is that the averaging spans the option's whole life; an average over only the last month of a longer option cuts far less.

The pricing consequence is large. An at-the-money one-year option on an \$80 commodity with a 35% volatility, settled either on one fixing or on 21 fixings spread across the year:

	Effective volatility	Premium
On a single fixing at expiration	35.00%	11.1136
On the 21-fixing average	20.93%	6.6669

The averaging alone removes 40% of the option premium, with no change to the underlying's volatility.

Two implications follow, and the second is a trap.

A commodity option quoted on an average is not comparable with one quoted on a fixing, and comparing their implied volatilities without adjusting is a common and expensive error.

And the swap itself is not affected while the option is. A commodity *swap* settles against the average, but its fair fixed price is the average of the forwards, a linear function, so averaging changes nothing. It is only when optionality is attached, as in a commodity cap or a swaption on the average, that the multiplier bites. The distinction between what averaging does to a linear payoff and to a convex one is the same distinction Example 33.1 drew about weighted average life.

33.9 BASIS SWAPS, AND WHAT THEY PRICE

Section 7.7 introduced tenor basis swaps (three-month against six-month) and Section 7.8 the cross-currency basis. Here is what a basis swap is, in this Part's language.

A tenor basis swap prices the difference between two forecast curves. Before 2022 that difference was largely credit and liquidity: three-month LIBOR embedded less bank credit risk per unit time than six-month. After the transition to overnight rates compounded over the relevant period, the credit component is gone, and what remains is a much smaller basis reflecting term premia and the mechanics of compounding.

The tenor basis market therefore shrank rather than transformed, which is a rare case of a market being genuinely simplified.

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The cross-currency basis did not. It prices the relative scarcity of funding in two currencies, which is not a convention and does not go away. Section 7.8's account stands, and Section 9.6's ColVA is its valuation-side counterpart: a CSA permitting collateral in either currency contains an option on the basis, and that option is worth something.

The resettable notional

One feature of the cross-currency market deserves naming because it solves a problem the rest of this chapter would otherwise leave open.

A fixed-notional cross-currency swap exchanges principal at inception and at maturity at the same rate. If the exchange rate moves, the present value of the final exchange moves one-for-one with it, so the swap's mark-to-market grows with the move. Variation margin does cover that mark, but a mark that large means large and growing collateral flows, a large exposure over the margin period of risk, and a large SA-CCR add-on, all on a swap whose principal is paid once at the end.

A mark-to-market or resettable cross-currency swap fixes this by resetting one leg's notional at each payment date to the current spot equivalent, with the difference paid in cash. The mark is then reset to zero periodically rather than accumulating, and the swap's exposure profile of Section 9.1 is dramatically flatter.

Two consequences follow.

The resettable version prices differently, and the difference is a convexity adjustment. Resetting the notional means the interest paid on it depends on the exchange rate at the reset date, so the payment is a product of a rate and an FX level, which is the quanto configuration of Section 30.6. The correction is the covariance between the exchange rate and the interest rate, and it is why resettable and fixed-notional swaps quote at different bases.

And the choice is driven by Chapter 27, not by pricing. The resettable structure is now the market standard for interbank cross-currency swaps precisely because it cuts exposure at default and therefore capital. A convention adopted for capital reasons carried a valuation consequence with it, which is the same pattern as Section 33.4's in-arrears adjustment, running in the opposite direction: there a convention change removed a convexity; here one introduced it.

Example 33.7 puts both halves of that trade-off in numbers, because the exposure saving is large and obvious while the valuation cost is small and easy to omit.

Example 33.7 What the reset saves and what it costs

A five-year $100 million cross-currency swap paying semiannually, spot 1.10, a 3% euro forward rate with 25% rate volatility and 10% FX volatility. The table shows three of the ten payment dates; the average is over all ten.

The saving in margin and exposure is the reason it exists. Under a fixed notional, a move in spot changes the swap's mark by the move times the notional. Variation margin covers the mark, but the amount to be margined, and the exposure over the margin period of risk, grow with it. The table shows the size of the mark:

FX move	Mark, fixed notional	Mark after reset
+10%	$10.0m	≈ $0
+20%	$20.0m	≈ $0
+30%	$30.0m	≈ $0

Thirty percent of notional is a very large mark to margin daily, and Section 27.3's add-on prices it accordingly.

The valuation cost runs the other way and is not zero. Interest accrues on a notional that is

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itself a function of spot, so each payment is a product of a rate and an FX level and the forward rate must be adjusted by $-F\rho\,\sigma_S\,\sigma_F\,T_i$, where ρ is the correlation between the exchange rate and the euro rate, σ_S and σ_F are their volatilities, and T_i is the payment date:

Payment date	$\rho = 0$	$\rho = 0.3$	$\rho = 0.6$
1 year	0.00 bp	−2.25 bp	−4.50 bp
3 years	0.00 bp	−6.75 bp	−13.50 bp
5 years	0.00 bp	−11.25 bp	−22.50 bp
Average	**0.00 bp**	**−6.19 bp**	**−12.38 bp**

At a correlation of 0.3 the reset feature is worth about 6 basis points on the floating spread, the same order as the year-on-year inflation adjustment of Section 33.10 and two orders of magnitude larger than the post-transition in-arrears correction. It grows linearly in maturity and linearly in the correlation, and it vanishes only if rates and the exchange rate are independent, which for a currency pair whose basis is a funding variable is not a natural assumption.

So the two effects are not comparable in kind and both are real: tens of millions of mark to margin against single-digit basis points of value. The market chose the exposure saving, correctly, and then had to learn to price the residue—which is the general shape of every convention change in this chapter.

33.10 INFLATION SWAPS

Inflation swaps are the largest swap market this book has not yet priced, and one where the convexity structure is unusually clean.

A zero-coupon inflation swap exchanges a fixed compounded rate for the realized index ratio $I_T/I_0 - 1$, settled once at T. One payment, one observation ratio, no timing mismatch—no convexity adjustment. The quoted rate is the **breakeven inflation rate** directly: the fixed inflation rate at which the swap is worth zero, which is the market's price for inflation over the term.

A year-on-year swap pays $I_k/I_{k-1} - 1$ each year. Each payment is a ratio of index levels observed at two different dates and paid at the later one, and that is exactly the configuration Chapter 28 says produces an adjustment. Example 33.8 sizes it.

Example 33.8 Why the two inflation markets quote separately

Breakeven inflation flat at 2.50%, inflation volatility 1.2% in normal terms.

Year	Year-on-year convexity adjustment
1	0.00 bp
2	1.44 bp
5	5.76 bp
10	12.96 bp

Averaged over ten years: 6.48 basis points.

So a ten-year year-on-year swap does not strike at the ten-year zero-coupon breakeven, and the gap is not a quoting error. It is the accumulated convexity of ten ratio payments.

This is why the two inflation markets are quoted independently rather than one being derived from the other. Whenever a market quotes two instruments that look like transformations of each other, the difference is usually a convexity adjustment and is usually worth locating.

Two features of inflation swaps have no analog in rate swaps.

Indexation lag. The index used for a payment is typically the one published two or three months earlier, because the statistics office needs time. The swap therefore references a rate that is already partly known, which reduces its volatility near each reset and complicates the interpolation

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between monthly index points.

Seasonality. Consumer price indices have a pronounced and stable annual pattern. A year-on-year swap spanning exactly twelve months is nearly seasonality-neutral; anything else is not, and a curve built without a seasonal adjustment will misprice every off-anniversary payment in a predictable direction.

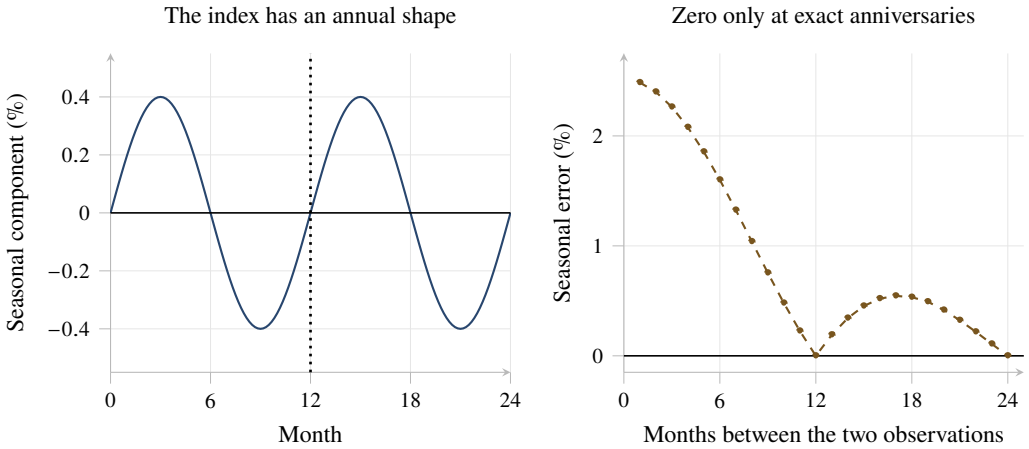


Figure 33.2: Illustrative seasonal amplitude of 0.4%, not market data. Right panel: worst case over the ending month, annualized.

Figure 33.2 is why an inflation curve is quoted by calendar month rather than by tenor. The left panel's pattern cancels exactly when the two observations are twelve months apart, which is what makes a standard year-on-year swap clean. The right panel is what happens when they are not: a swap observing three months apart carries a seasonal error of over two percentage points, annualized, and it is a *predictable* error rather than a risk: the same months produce the same sign every year. An unadjusted curve therefore does not misprice off-anniversary payments at random; it misprices them in a direction a counterparty can look up.

The users are worth naming, because the flow is one-directional and it shows in the price. Pension funds and insurers with inflation-linked liabilities are structural receivers of inflation, and there is no natural payer of comparable size; inflation-linked bond issuance by governments is the main one, and it is smaller. So breakeven inflation from swaps has tended to sit above breakeven from bonds (the gap between a nominal government bond's yield and an inflation-linked bond's real yield) in markets such as the sterling market, where pension demand is concentrated, and the gap is a demand imbalance rather than a forecast disagreement. This is the same structure as Section 7.9's pension demand compressing swap spreads, in a different market. In sterling, the retail prices index (RPI) is also to be aligned with CPIH in February 2030, as the government and the UK Statistics Authority announced in November 2020; the change reprices long-dated RPI swaps and index-linked bonds alike, and it falls on the pension schemes that hold them.

33.11 EQUITY AND TOTAL RETURN SWAPS

Section 7.12 introduced these and Section 27.14 showed what they made possible. The pricing is the easy part and the funding is not.

An equity swap exchanges the total return on an index or single stock for a floating rate plus a spread. Its fair spread is not a forecast of the equity's return (the equity leg's expected return under the pricing measure is the financing rate, by Chapter 13) but a function of the dealer's

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cost of holding the hedge:

$$\text{spread} \approx \text{funding} + \text{borrow cost} + \text{dividend leakage} + \text{balance sheet charge}$$

The borrow cost is the fee for borrowing stock to hedge a client's short; **dividend leakage** is the part of the dividends the dealer loses to withholding tax and so cannot pass on; the balance sheet charge is the cost of the capital the hedge position consumes. Not every term applies to every trade. A dealer hedging a client's long buys the stock, so it pays funding and loses the leakage but borrows nothing; one hedging a client's short borrows the stock and sells it, so it receives cash rather than needing it and holds no dividends to lose, and only the borrow cost and the balance sheet charge remain.

Each term is observable to the dealer and none is to the client, which is why equity swap spreads vary far more across dealers than swap rates do, and Example 33.9 shows how little a client learns from the quote.

Example 33.9 Four costs behind one number

A one-year total return swap on a $50 million basket yielding 2.0%. The dealer's costs:

Component	bp	Determined by
Funding above the risk-free rate	25	the dealer's own credit
Dividend leakage: 15% withholding on 2.0%	30	the dealer's tax residence
Balance sheet under the leverage ratio	15	Section 27.10's constraint
Quoted spread, synthetic long	**70**	
Stock borrow, if the client is short	40	securities lending supply
Balance sheet under the leverage ratio	15	Section 27.10's constraint
Quoted spread, synthetic short	**55**	

The short-side spread comes off the floating rate the client receives, and it is not the long-side spread plus the borrow: of the long side's three costs only the balance sheet charge survives.

The client sees one number and cannot decompose it. The 25 bp between dealers quoting 70 and 95 basis points can be assembled in more than one way (the cheaper dealer funding 10 bp below the other and a tax treaty sparing it 15 bp of dividend leakage, say, or the treaty plus a leverage ratio that binds it less), and the client cannot tell which, because none of the three is disclosed and all three move independently of the equity market.

The practical consequences run in both directions. The dividend term is a tax fact presented as a market price, so a client that could hold the shares directly and reclaim withholding is paying 30 bp for a service it does not need. And the balance sheet term is the one that moves without warning: it is a function of the dealer's leverage ratio headroom rather than of anything about the trade, so a spread can widen at a quarter-end for reasons the client will never observe. That is Section 27.10's constraint reaching a customer who has never heard of it.

Three uses account for most of the market, and they explain the spread's components.

Synthetic long exposure with leverage, which is the Archegos configuration of Section 27.14. Here the balance-sheet charge is the component that grows with the client's book: on a single trade it is the smallest of the three, as in Example 33.9, but once one client's exposure is large enough to move the dealer's leverage ratio it becomes the cost that limits how much the dealer will extend.

Synthetic short exposure, where the borrow cost dominates and Section 5.2's hard-to-borrow stocks carry spreads of hundreds of basis points.

And dividend and tax positioning, where the swap's treatment of dividends differs from direct ownership. This is the least discussed and historically the most controversial use, and the regulatory attention it has received in several jurisdictions is the reason many dealers now

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restrict it.

33.12 UNWINDS, NOVATIONS, AND COMPRESSION

A swap can be ended in three ways, and they are not equivalent.

Unwind: the two parties agree on a termination payment equal to the mark-to-market. Simple, and it requires the original counterparty's cooperation and pricing.

Novation: a third party steps into one side. This preserves the trade and changes the counterparty, and it requires consent because the remaining party is acquiring a new credit exposure, which is Chapter 27's subject appearing as a documentation clause.

Offsetting: enter an opposite trade. The market risk is flat and the counterparty risk is doubled, which is exactly the wrong trade under Chapter 27's framework and is done anyway when an unwind cannot be agreed.

Compression exists because of the third. Multilateral compression services tear up offsetting trades across many participants while preserving each one's net market risk. Section 25.18 recorded compression, alongside clearing and a genuine decline in activity, as one of the three forces that cut the CDS market's gross notional by roughly an order of magnitude; it has done the same in rates, and the notional statistics that headline the derivatives market are largely a story about compression rather than about activity.

The point for this chapter is that a swap's life does not end when its risk does, and the gap between the two is a capital cost under Chapter 27 and an operational cost always.

33.13 WHO USES NON-STANDARD SWAPS

Corporate borrowers take amortizing and accreting swaps because their debt amortizes: the non-standard swap exists to make the hedge match the exposure, and a bullet swap against an amortizing loan leaves a residual that grows over time.

Insurers and pension funds are the structural receivers in inflation swaps and the largest users of long-dated forward-starting swaps, for the reason Section 7.9 gave: their liabilities are long and they hedge duration they cannot buy in bond markets deep enough.

Structured note investors are the ultimate holders of the CMS convexity of Example 33.3, usually without seeing it separated from the coupon.

Hedge funds use equity and total return swaps for leverage, shorting, and, as Section 27.14 documented, for the position invisibility that the format provides.

And dealers sit in the middle of all of it, running the vega that CMS creates, the basis that cross-currency creates, and the balance sheet that equity swaps consume. The recurring pattern of this chapter is that a non-standard swap moves a risk from a client who does not want it to a dealer who prices it, and the price is a convexity adjustment the client usually cannot compute.

SUMMARY

A non-standard swap is almost always a standard swap plus a convexity adjustment, and the adjustment is computable, has a reasoned sign, and is large enough to matter. It arises wherever a payoff is nonlinear in the rate it depends on, or a rate is paid at a time other than its natural one. An amortizing swap priced at its weighted average life shows the first case, because the par rate is not linear in the schedule.

The classic LIBOR-in-arrears adjustment grows with maturity, while a compounded-in-arrears SOFR rate is determined and paid at the end of its period, so the adjustment all but vanishes: a convexity adjustment is a property of a convention, not of a market. The complexity moved rather than vanished, since Section 29.4's backward-looking caplet needs an adjustment its forward-looking predecessor did not. On a CMS swap the adjustment is many times the

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bid–ask spread, and because it depends on a volatility, a linear-looking product carries vega hedged off the whole smile.

Other products carry options inside them. A range accrual coupon is a strip of digitals whose enhancement is short vega, and it fails in the states where its holder most needs income. A cancellable swap is a vanilla plus a Bermudan, and Example 31.9's calibration ambiguity, quoted as a rate, makes model risk a pricing problem rather than only a risk-management one. A commodity swap settles against an average, which removes much of an option's premium and is neutral only inside a linear payoff.

Basis and inflation swaps show what a spread prices. Tenor basis priced bank credit before 2022 and prices much less now, while the cross-currency basis prices funding scarcity and did not shrink. Zero-coupon and year-on-year inflation swaps differ by a convexity adjustment, which is why the two markets quote independently, and pension and insurance demand can put swap breakevens above bond breakevens. An equity swap's spread is built from funding, dividends, borrow and balance sheet, every term observable to the dealer and none to the client.

Unwind, novation, and offsetting are not equivalent: offsetting flattens market risk and doubles counterparty risk, which is why compression exists and why headline notional describes compression more than activity. The recurring pattern is a non-standard swap moving a risk from a client who does not want it to a dealer who prices it, at a convexity adjustment the client cannot compute.

FURTHER READING

The ISDA definitions are the primary source for every convention in this chapter; the compounding and rate-cut-off provisions of Section 33.3 in particular repay direct reading, because the summaries in textbooks (including this one) elide distinctions that matter operationally. Hull's chapter on swaps is a short treatment of the non-standard varieties. On CMS, Hagan's papers on convexity conversion are the practitioner standard and are where Example 33.3's replication argument is developed properly. On inflation, Deacon, Derry, and Mirfendereski is the standard book and covers the seasonality and indexation-lag questions of Section 33.10 in the detail they require. On compression and the notional statistics of Section 33.12, the BIS semiannual survey's methodology notes explain what the headline notional counts. For the post-transition conventions, the market-practice recommendations published by the currency-specific working groups are the record of what was adopted, and they differ across currencies in ways the academic literature tends to ignore.

PRACTICE QUESTIONS

33.1 Using Example 33.1, state the amortizing par rate and the error from the weighted-average-life shortcut, and explain why the error grows with the curve's curvature.

33.2 Explain why a forward-starting swap amplifies curve errors, and why compounding at a spread produces an adjustment while flat compounding does not.

33.3 Using Example 33.2, state the classic in-arrears adjustment at ten years and the post-transition adjustment at ten years, and explain why the ratio between them says nothing about rate volatility.

33.4 Using Example 33.3, state the fair fixed rate with and without convexity adjustments and the difference, and explain why a product with a linear-looking payoff carries a vega.

33.5 Explain why a CMS convexity adjustment always has the same sign, and what that implies for the investor in a CMS-linked note.

33.6 Using Example 33.4, state the accrual fraction, the enhanced coupon, and the coupon's

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value if realized volatility is 50%, then explain what the enhancement actually is.

33.7 Explain why a range accrual's coupon fails in the states where its holder most needs income, and name the earlier section whose structure this repeats.

33.8 Using Example 33.5, state the ambiguity in the quoted cancellable swap rate implied by Example 31.9 and compare it with a typical bid–ask, then explain why this makes model risk a pricing problem.

33.9 Derive the variance ratio for an average of n observations, evaluate it at $n = 21$, and state its limit. Explain why averaging changes an option's value and not the swap's.

33.10 Explain why an implied volatility quoted on an average is not comparable with one quoted on a single fixing.

33.11 Explain why the mark-to-market of a fixed-notional cross-currency swap grows with the exchange rate, what that does to collateral flows and exposure, and what the resettable structure does about it.

33.12 Explain why a resettable cross-currency swap prices differently from a fixed-notional one, and identify the configuration from Chapter 28 that produces the correction.

33.13 Using Example 33.8, state the year-on-year adjustment at ten years and its average, and explain why the zero-coupon market needs no adjustment.

33.14 Explain why, where pension demand for inflation protection is concentrated, inflation swap breakevens can sit above bond breakevens, and why this is a demand imbalance rather than a forecast disagreement.

33.15 Distinguish unwind, novation, and offsetting, state what each does to market risk and to counterparty risk, and explain why headline notional statistics describe compression more than activity.

FURTHER QUESTIONS

33.16 Example 33.1 finds an 18.1 basis point error from the average-life shortcut. Derive the error's dependence on the curve's shape, determine the curve under which the shortcut is exact, and state how large the error can become on a realistically steep curve.

33.17 Example 33.2 shows the in-arrears adjustment falling by a factor of about sixty. Derive both adjustments from the change-of-numeraire argument, verify the factor, and determine the payment lag at which the post-transition adjustment would again become material.

33.18 Example 33.3 finds 14.26 basis points on a ten-year CMS swap. Determine how this scales with the CMS tenor, the swap's maturity, and the volatility level; then identify the configuration in which the adjustment exceeds 100 basis points and judge whether such a product would be sellable.

33.19 Section 33.5 argues that a CMS-linked note investor is being sold convexity without seeing its price. Design a disclosure that would make the convexity component visible, state what the issuer would have to reveal, and discuss why such disclosure is not standard.

33.20 Example 33.8 attributes the zero-coupon/year-on-year gap to convexity. Construct the arbitrage that would exist if the two markets quoted at the same breakeven, determine what would have to be traded to capture it, and explain what practical frictions prevent the two from being kept in line.

33.21 Section 33.10 says pension demand makes the inflation swap market one-directional. Model the resulting price impact, compare it with Section 7.9's swap-spread compression

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and Section 26.3's mortgage convexity flows, and state the general conditions under which a structural demand imbalance persists rather than attracting offsetting supply.

33.22 Example 33.4 shows the range accrual coupon falling from 4.91% to 1.83% as volatility rises from 15% to 50%. Derive the coupon's vega, determine the volatility at which it is largest, and explain why that volatility is close to the one the note is priced at rather than far from it.

33.23 Example 33.5 converts a 13.1% model ambiguity into 4.2 basis points on a quoted rate. Work out what a dealer should do about an ambiguity several times its bid–ask: consider reserving, widening, hedging the parameter, and declining the trade, and state the conditions under which each is the right response.

33.24 Section 33.9 says the resettable cross-currency structure was adopted for capital reasons and carried a pricing consequence. Derive the covariance correction between the resettable and fixed-notional versions, estimate its size for a major currency pair, and judge whether it is large enough that the capital saving might be partly illusory.

33.25 This chapter argues that a non-standard swap moves risk from a client who cannot price it to a dealer who can. Assess this as a description of the market: identify where the client is compensated for the information asymmetry, where they are not, and what a regulator concerned with the second could usefully require.

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Part VII

Exotics and Structured Products

Two chapters on payoffs that are designed rather than listed. The first covers the exotic families and the replication arguments that price them; the second follows those payoffs into the retail market, where the same structure is sold as a note with a headline coupon. Read together they make one point: the exotic is priced by the dealer and bought by someone reading a different document.

34 Exotic Options
barriers, lookbacks, Asians, baskets, and the arguments that price them

35 Structured Products and Autocallables
the same payoffs wrapped as notes, and what the wrapper hides

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Chapter 34
Exotic Options

Five chapters have deferred material to this one. Everything in it trades over the counter, in the market Section 10.10 described; an exotic option is a dealer's product, and that fact shapes how it is priced, hedged, and sold.

Section 15.2 said every payoff whose value depends on S and t alone satisfies the Black–Scholes–Merton equation without modification, and that a path-dependent one adds a state variable. Section 18.6 said vanna and volga matter here. Section 18.13 said that where gamma diverges, dynamic hedging becomes an impossibility problem and a static hedge is the only answer. Section 19.12 said barrier prices depend on how the surface *evolves*, which no static surface specifies. Section 20.4 promised that a barrier priced on a tree whose nodes straddle the barrier converges appallingly. Section 28.11 said the multi-asset family of options is developed here.

The unifying question is which exotics are genuinely harder than vanillas and which merely look it. The answer is not about payoff complexity. It is about whether the payoff can be replicated by instruments the market quotes, and Sections 34.5 and 34.13 give the test.

34.1 WHAT MAKES AN OPTION EXOTIC

Three features separate an exotic from a vanilla, with very different consequences. Two turn on the path rather than the endpoint, so the notation comes first. Throughout, S_t is the underlying and K the strike, and the path statistics are

$$M_T = \max_{0\le t\le T} S_t, \qquad m_T = \min_{0\le t\le T} S_t, \qquad A_T = \frac{1}{T}\int_0^T S_t\,dt$$

the running maximum, the running minimum, and the average.

Path dependence. The payoff depends on the route, not only the destination. A **barrier option** is a vanilla multiplied by an indicator of whether the path has reached a barrier level H:

$$\text{down-and-out call} = (S_T-K)^+\mathbf{1}_{\{m_T>H\}}, \qquad \text{down-and-in call} = (S_T-K)^+\mathbf{1}_{\{m_T\le H\}} \tag{34.1}$$

with the up-variants replacing m_T by M_T. Written this way, in–out parity (knock-in plus knock-out equals the vanilla, equation (34.10) in Section 34.3) stops being a result and becomes an observation: the two indicators sum to one, so the two options sum to the vanilla.

A **lookback option** replaces the strike or the terminal price by an extreme of the path:

$$\begin{aligned} \text{floating call} &= S_T - m_T, & \text{floating put} &= M_T - S_T,\\ \text{fixed call} &= (M_T - K)^+, & \text{fixed put} &= (K - m_T)^+ \end{aligned} \tag{34.2}$$

The floating pair can never finish out of the money, which is most of the reason they cost what they cost. An **Asian option** replaces the terminal price by the average A_T in one of two places:

$$\text{average price} = (A_T - K)^+, \qquad \text{average strike} = (S_T - A_T)^+ \tag{34.3}$$

and a **cliquet** sums period returns $r_i = S_{t_i}/S_{t_{i-1}} - 1$ over reset dates $0 = t_0 < t_1 < \cdots < t_n = T$, each floored at F and capped at C:

$$\text{cliquet} = \sum_{i=1}^{n} \min\big(\max(r_i,\ F),\ C\big) \tag{34.4}$$

The cap is inside the sum rather than outside it, and Section 34.12 is about what that does to the sign of the volatility exposure.

This is where the model risk lives, because a model calibrated to terminal distributions does not pin down paths. Five sections of this chapter go to the barrier alone for that reason (Sections 34.3 through 34.7), and Section 34.9 takes the lookback and the Asian together.

Multiple underlyings. **Basket options** pay on a weighted sum of several assets, **rainbow options** on the best or the worst of them, and **exchange options** on one asset against another; four payoffs matter:

$$\begin{aligned} \text{basket} &= \Big(\sum_i w_i S_i(T) - K\Big)^+, & \text{exchange} &= \big(S_1(T) - S_2(T)\big)^+, \\ \text{worst-of put} &= \big(K - \min_i S_i(T)\big)^+, & \text{best-of call} &= \big(\max_i S_i(T) - K\big)^+ \end{aligned} \tag{34.5}$$

Here $S_i(T)$ is the price of the ith asset at expiration and w_i the number of units of it the basket holds; the basket of Example 34.8 is $(S_1 + S_2)/2$, so $w_1 = w_2 = 1/2$. The new input is correlation, which Section 19.7 established is not observable in the way volatility is. It enters every one of these payoffs and is quoted in none of them, which is Section 34.10's subject and the reason the worst-of put in Section 34.11 behaves as it does.

Non-standard exercise or payoff. Bermudans, compounds, choosers, and the digital pair (options paying either a fixed cash amount Q or the asset itself if they finish in the money), the last of these carrying the identity that ties it to Black–Scholes:

$$\begin{aligned} \text{cash-or-nothing} &= Q\,\mathbf{1}_{\{S_T>K\}}, & \text{asset-or-nothing} &= S_T\,\mathbf{1}_{\{S_T>K\}}, \\ (S_T - K)^+ &= S_T\,\mathbf{1}_{\{S_T>K\}} - K\,\mathbf{1}_{\{S_T>K\}} \end{aligned} \tag{34.6}$$

A vanilla call is one asset-or-nothing less K cash-or-nothings, which is Section 28.10's reading of $N(d_1)$ and $N(d_2)$ written as a payoff rather than as a price. These are usually the easy ones, because a single-asset payoff at a single date is a function of the terminal distribution, which the vanilla market already reveals.

The taxonomy sorts the chapter: the third group is priced almost model-independently, the first two are not. A desk that treats "exotic" as one category prices some trades far too carefully and others not carefully enough.

34.2 OPTIONS THAT ARE BARELY EXOTIC

Several instruments carry exotic names and almost no model risk.

Packages are combinations of vanillas (Chapter 12's spreads, straddles, and collars) sold as one contract and priced by adding up the pieces.

Non-standard American options, exercisable only in windows or with a strike that steps, need Chapter 20's lattice but no new theory. A **gap option** pays $S_T - K_1$ if $S_T > K_2$ and nothing otherwise, which is a vanilla struck at K_2 plus $K_2 - K_1$ in a digital, so it is priced exactly and inherits the digital's hedging problem of Section 34.8. A **shout option** lets the holder lock in the intrinsic value at one chosen date, and pays at expiration the greater of that locked value and the ordinary payoff: a cliquet whose single reset the holder chooses.

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Compound options, an option on an option, have closed forms under Black–Scholes and are genuinely sensitive to volatility in a way vanillas are not: a compound option is a bet on volatility twice over, since both the outer and inner values depend on it. Their vega is correspondingly large relative to their price. Geske's formula for a call on a call (the right, on payment of K_1 at t_1, to acquire a call struck at K_2 expiring at T) is

$$c_{\text{call}} = S_0 e^{-qT} M(a_1, b_1; \rho) - K_2 e^{-rT} M(a_2, b_2; \rho) - K_1 e^{-rt_1} N(a_2), \qquad \rho = \sqrt{t_1/T} \quad (34.7)$$

where $M(\cdot, \cdot; \rho)$ is the bivariate standard normal distribution function,

$$a_1 = \frac{\ln(S_0/S^*) + (r - q + \sigma^2/2)t_1}{\sigma\sqrt{t_1}}, \qquad b_1 = \frac{\ln(S_0/K_2) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}},$$

$a_2 = a_1 - \sigma\sqrt{t_1}$, $b_2 = b_1 - \sigma\sqrt{T}$, and S^* is the asset price at t_1 at which the inner call is worth exactly K_1. That critical price is the only part of the formula requiring numerical work, and it is a one-dimensional root search on Black–Scholes. The correlation entering M is $\sqrt{t_1/T}$ because one Brownian motion carries the price through both dates, which is the whole reason a compound option is a two-dimensional integral rather than a one-dimensional one. The other three differ only in signs. Writing p_{call} for a put on a call, and c_{put} and p_{put} for a call and a put on a put,

$$\begin{aligned}
p_{\text{call}} &= K_2 e^{-rT} M(-a_2, b_2; -\rho) - S_0 e^{-qT} M(-a_1, b_1; -\rho) + K_1 e^{-rt_1} N(-a_2) \\
c_{\text{put}} &= K_2 e^{-rT} M(-a_2, -b_2; \rho) - S_0 e^{-qT} M(-a_1, -b_1; \rho) - K_1 e^{-rt_1} N(-a_2) \\
p_{\text{put}} &= S_0 e^{-qT} M(a_1, -b_1; -\rho) - K_2 e^{-rT} M(a_2, -b_2; -\rho) + K_1 e^{-rt_1} N(a_2)
\end{aligned}$$

Two things change with the inner option rather than the outer one. In the last two, S^* is the price at which the inner put is worth K_1, not the inner call, so it is a different root search; and a put loses value as the asset rises, so the outer option is exercised below S^* for a call on a put and above it for a put on a put.

Only two of the four are independent. Holding a call on a call and writing a put on the same call delivers that call at t_1 against K_1 paid, whatever the asset does in between, so

$$c_{\text{call}} - p_{\text{call}} = c(S_0, K_2, T) - K_1 e^{-rt_1}, \qquad c_{\text{put}} - p_{\text{put}} = p(S_0, K_2, T) - K_1 e^{-rt_1}$$

and those two identities are the cheapest check available on an implementation of any of the four.

Figure 34.1 puts the volatility claim in numbers. At a 10% volatility the compound is worth 3.02 against the vanilla's 6.81, and its value moves **2.2 times** as much for a given relative move in volatility; by 50% that ratio is down to 1.2. The ratio is one of volatility elasticities, $\sigma\,(\partial V/\partial\sigma)/V$, the percentage change in value for a one percent change in volatility. The compound's barely moves: it sits near 1.1 throughout, so its value is close to proportional to volatility, while the vanilla's climbs from 0.50 to 0.86. Paying twice for volatility does not buy a larger vega; it buys a vega that stays large relative to a much smaller price. A compound option suits a firm bidding for a foreign-currency contract it may not win: it fixes the cost of the hedge if the bid succeeds and costs only the first premium if it fails.

Adding the second premium at its present value, $K_1 e^{-rt_1} = 4.88$, gives what ending up with the same call costs in total: 7.90 against the vanilla's 6.81 at a 10% volatility, 22.59 against 21.79 at 50%. The total exceeds the vanilla at every volatility (committing to exercise buys the identical instrument, so it cannot be cheaper), and the gap is what the right to walk away at t_1 is worth: **1.10** at 10%, narrowing to **0.80** at 50%, because the higher the volatility the less likely the holder is to decline.

Chooser options let the holder decide at t_1 whether the option is a call or a put, both struck at K and both expiring at T. A simple chooser, one whose call and put share a strike and

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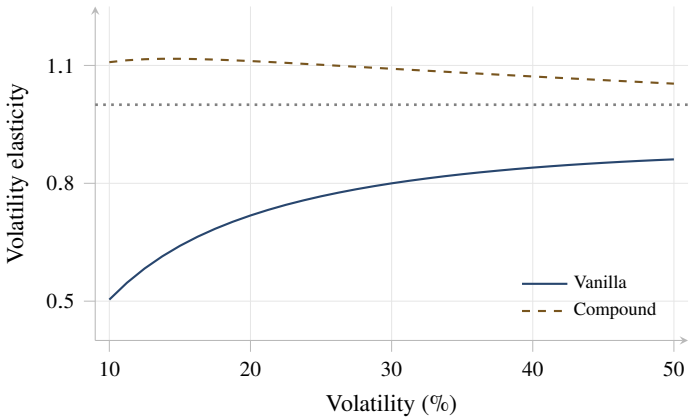


Figure 34.1: $\sigma\,(\partial V/\partial\sigma)/V$ at $S_0 = K_2 = 100$, outer premium $K_1 = 5$ paid at $t_1 = 0.5$, $T = 1$, $r = 5\%$, no dividends.

an expiration as here, decomposes exactly into vanillas. Write $c(S, K, T)$ and $p(S, K, T)$ for Black–Scholes–Merton values. At the choice date the holder takes whichever is worth more, so the chooser is then worth

$$\max\bigl\{c(S_{t_1}, K, T - t_1),\ p(S_{t_1}, K, T - t_1)\bigr\}$$

Put–call parity at t_1 gives $p - c = Ke^{-r(T-t_1)} - S_{t_1}e^{-q(T-t_1)}$, the value of a short forward, which carries no optionality at all. Substituting it,

$$\begin{aligned}\max\{c,\ p\} &= c + \max\{0,\ p - c\}\\ &= c(S_{t_1}, K, T - t_1) + e^{-q(T-t_1)}\max\Bigl(Ke^{-(r-q)(T-t_1)} - S_{t_1},\ 0\Bigr)\end{aligned}$$

where $e^{-q(T-t_1)}$ has been taken outside the maximum. Both terms are now instruments the vanilla market prices. The first is a call struck at K expiring at T, carried through the choice date untouched, and so worth $c(S_0, K, T)$ today. The second is $e^{-q(T-t_1)}$ puts struck at $Ke^{-(r-q)(T-t_1)}$, and they expire at t_1 rather than at T, because the maximum is taken on the price at the choice date. Hence

$$V_0 = c(S_0, K, T) + e^{-q(T-t_1)}\,p\Bigl(S_0,\ Ke^{-(r-q)(T-t_1)},\ t_1\Bigr) \tag{34.8}$$

Choosers are rare outright; their buyers are those who must commit to a volatility position before they know the direction.

Forward-start options are struck at the money on a future date. Under Black–Scholes their value scales with the spot, so an option struck at αS_{t_1} and expiring at T is worth

$$V_0 = e^{-qt_1}\,c(S_0,\ \alpha S_0,\ T - t_1) \tag{34.9}$$

a vanilla's price times e^{-qt_1}, with no dependence on anything observed between now and t_1. Under any realistic model the value is not this simple, because it depends on the *forward* smile—and Section 21.3 measured how badly local volatility gets that wrong. A cliquet is a strip of forward-starts, so it is the purest instrument in this chapter for trading forward volatility, and the hardest to price for exactly the same reason.

34.3 BARRIER OPTIONS

Barriers are the largest exotic market, and they take up most of this chapter.

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A barrier option is a vanilla that is extinguished (**knock-out**) or created (**knock-in**) if the underlying touches a level H. With up and down, in and out, and calls and puts, there are eight basic types, and the in-out parity

$$\text{knock-in} + \text{knock-out} = \text{vanilla} \tag{34.10}$$

means only four need to be priced.

They exist because they are cheaper. A down-and-out call costs less than the vanilla because it can be extinguished, and a buyer who believes the barrier will not be touched is paying for the part of the distribution they think matters.

All eight have closed forms under Black–Scholes with continuous monitoring, and the whole family is assembled from four expressions. Write $\phi = +1$ for a call and -1 for a put, $\eta = +1$ for a down-barrier and -1 for an up-barrier, and set

$$\lambda = \frac{r - q + \sigma^2/2}{\sigma^2}, \qquad x_1 = \frac{\ln(S_0/K)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}, \qquad x_2 = \frac{\ln(S_0/H)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T},$$
$$y_1 = \frac{\ln\big(H^2/(S_0K)\big)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}, \qquad y_2 = \frac{\ln(H/S_0)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}.$$

Then

$$\begin{aligned}
A &= \phi S_0 e^{-qT} N(\phi x_1) - \phi K e^{-rT} N\big(\phi(x_1 - \sigma\sqrt{T})\big),\\
B &= \phi S_0 e^{-qT} N(\phi x_2) - \phi K e^{-rT} N\big(\phi(x_2 - \sigma\sqrt{T})\big),\\
C &= \phi S_0 e^{-qT} \left(\frac{H}{S_0}\right)^{2\lambda} N(\eta y_1) - \phi K e^{-rT} \left(\frac{H}{S_0}\right)^{2\lambda-2} N\big(\eta(y_1 - \sigma\sqrt{T})\big),\\
D &= \phi S_0 e^{-qT} \left(\frac{H}{S_0}\right)^{2\lambda} N(\eta y_2) - \phi K e^{-rT} \left(\frac{H}{S_0}\right)^{2\lambda-2} N\big(\eta(y_2 - \sigma\sqrt{T})\big),
\end{aligned} \tag{34.11}$$

and the four knock-in values are read off the table below. Knock-outs follow from the parity (34.10).

Knock-in option	**Value, by where the barrier sits relative to the strike**			
Down-and-in call	$H \le K$:	C	$H > K$:	$A - B + D$
Up-and-in put	$H \ge K$:	C	$H < K$:	$A - B + D$
Up-and-in call	$H \ge K$:	$B - C + D$	$H < K$:	A
Down-and-in put	$H \le K$:	$B - C + D$	$H > K$:	A

Three things can be read off (34.11) directly.

A is the vanilla, since $x_1 = d_1$; the barrier enters only through B, C, and D. B is A with the barrier replacing the strike inside the normal arguments: the strike still sets the payoff, but the barrier now sets the boundary. C and D are the reflections of A and B in the barrier, weighted by the powers of H/S_0 that the reflection principle produces. That principle pairs each path that touches the barrier with its mirror image in the barrier from the moment of touching, and when the price has a drift the mirror image is not equally likely, which is what the powers correct for, and why Section 34.5's static hedge is exact only when the drift is zero.

The two entries equal to A are the vanilla itself, because a barrier lying between the spot and the strike is one the option cannot finish in the money without having crossed: an up-and-in call with $H < K$ knocks in on every path that pays it anything.

And the formula contains the static hedge. When $r = q$ the exponent λ collapses to $1/2$, the two weights become H/S_0 and S_0/H, and the down-and-in call's C reads term for term as K/H puts struck at H^2/K. Section 34.5's reflection result is visible in the formula, which is a good reason to write the formula out rather than call a library.

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The formulas assume no **rebate**, a fixed amount R paid to the holder when a knock-out option is extinguished. A rebate is valued separately, and for a down barrier both versions have closed forms in the same λ as (34.11). If it is paid at expiration it is a digital on the running minimum, paying if $m_T \le H$, and is worth Re^{-rT} times the risk-neutral probability that the barrier is touched,

$$P(m_T \le H) = N\left(\frac{\ln(H/S_0)}{\sigma\sqrt{T}} - (\lambda-1)\sigma\sqrt{T}\right) + \left(\frac{H}{S_0}\right)^{2\lambda-2} N\left(\frac{\ln(H/S_0)}{\sigma\sqrt{T}} + (\lambda-1)\sigma\sqrt{T}\right)$$

whose weight $(H/S_0)^{2\lambda-2}$ is the one that already appears in C and D. If it is paid on the touch it is a first-passage claim, discounted from a random date, and is worth

$$R\left[\left(\frac{H}{S_0}\right)^{\lambda-1+\psi} N(z) + \left(\frac{H}{S_0}\right)^{\lambda-1-\psi} N\big(z - 2\psi\sigma\sqrt{T}\big)\right]$$

where

$$\psi = \sqrt{(\lambda-1)^2 + \frac{2r}{\sigma^2}}, \qquad z = \frac{\ln(H/S_0)}{\sigma\sqrt{T}} + \psi\sigma\sqrt{T}$$

An up barrier reverses the signs of the arguments of N in both. For the down-and-out call of Example 34.1 below, $\lambda = 1.3$, and with no dividends $\psi = \lambda$ exactly; the barrier is touched with probability 0.652, so a rebate of 1 paid at expiration is worth $e^{-0.05} \times 0.652 = 0.620$, and the same rebate paid on the touch, which arrives earlier on average, is worth 0.643. Either is added to the no-rebate value to price the contract with it.

Example 34.1 What the barrier removes

$S_0 = 100$, $K = 100$, $H = 90$, $r = 5\%$, $\sigma = 25\%$, $T = 1$, continuously monitored.

Vanilla call	12.335999
Down-and-out call	9.111221
Down-and-in call (the difference)	3.224778

The down-and-in value is the first row of the knock-in table, the single expression C with $\lambda = 1.3$: $(H/S_0)^{2\lambda} = 0.760380$ and $(H/S_0)^{2\lambda-2} = 0.938740$, $y_1 = -0.517884$, and

$$C = 100 \times 0.760380 N(-0.517884) - 100 e^{-0.05} \times 0.938740 N(-0.767884) = 3.224778$$

The barrier removes 26.1% of the premium—a discount large enough to be the entire reason the instrument is bought.

Two observations shape everything after this.

The knock-out is not simply a cheaper call. Approaching the barrier makes the option *less* valuable while approaching the strike makes it more valuable, and between those two pulls a knock-out's delta can exceed one in magnitude and its gamma can change sign. Figure 34.4 measures both. *A reverse knock-out* (barrier above the strike, so the option is knocked out when it is deep in the money) has a discontinuous payoff at the barrier and is the most difficult vanilla-like instrument in the market.

And the price depends on the whole path distribution. The vanilla depends only on the terminal distribution, which Section 19.2 showed the option chain reveals. The barrier depends on the probability of touching, which it does not.

34.4 WHY BARRIERS BREAK TREES

Section 20.4 promised this, and the failure is instructive rather than merely inconvenient.

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Example 34.2 More steps, worse answer

Price Example 34.1's down-and-out call ($S_0 = K = 100$, $H = 90$, $r = 5\%$, $\sigma = 25\%$, $T = 1$, no dividends, European exercise) on a Cox–Ross–Rubinstein (CRR) tree, knocking out at any node at or below H. The tree can knock out only at a time step, while the analytic answer is the continuously monitored one: **9.111221**.

Steps	Tree	Error	Effective barrier
50	9.142046	+0.031	89.94
51	10.494157	**+1.383**	86.93
100	9.944477	+0.833	88.25
200	9.144117	+0.033	89.94
400	9.438414	+0.327	89.36
1600	9.153506	+0.042	89.92

Fifty-one steps is forty-five times worse than fifty. And 1600 steps is worse than 200.

The mechanism is entirely in the last column. The tree can only knock out at a node, so the barrier it actually enforces is the highest node level at or below H. As n changes, that level jumps around: at $n = 50$ it is 89.94 and at $n = 51$ it is 86.93. The tree is pricing a different contract at every step count, and the error is dominated by which contract rather than by discretization.

The error is also one-signed. The effective barrier is always at or below the true one, so the option is always over-priced, which is worse than a random error, because averaging two grids does not help (the binomial error of a vanilla alternates in sign with the parity of n, which is why averaging neighboring grids works there; here it never changes sign) and Richardson extrapolation, which Section 20.3 already warned about, makes it worse.

Now fix it. Choose n so that a layer of nodes falls exactly on the barrier. In a CRR tree the node levels are $S_0 u^{\pm j}$ with $u = e^{\sigma\sqrt{T/n}}$, so this requires

$$\frac{\ln(S_0/H)}{\sigma\sqrt{T/n}} = m \quad \text{for an integer } m \qquad \Longrightarrow \qquad n = \left\lfloor \frac{m^2\sigma^2 T}{\ln^2(S_0/H)} \right\rfloor$$

m	Steps	Tree	Error
4	90	9.111965	+0.00074
6	202	9.119059	+0.00784
8	360	9.113142	+0.00192
9	456	9.111153	**−0.00007**
12	810	9.113370	+0.00215
16	1441	9.112680	+0.00146

At 90 steps with the layer on the barrier, the error is 0.0007. At 100 steps without it, the error is 0.833—more than a thousand times larger, with more steps.

The lesson generalizes well beyond barriers. A numerical method that resolves the payoff's discontinuities converges; one that does not, does not—and refining a grid that misses the discontinuity refines the wrong problem. Section 20.3's control variate worked for the same underlying reason, and Section 20.4's trinomial tree exists to give the free parameter that makes this placement possible for any n.

Never let a user choose the step count on a barrier product. The pricing library should choose it from the barrier.

In Figure 34.2's left panel, the sawtooth is not noise: each descending run is the effective barrier drifting up toward H as n increases, and each jump is a node crossing the barrier and resetting it.

Or choose a method whose grid is free. The binomial tree's trouble is that it has no spare parameter: $u = e^{\sigma\sqrt{\Delta t}}$ is fixed once σ and Δt are, so the node levels are determined and the

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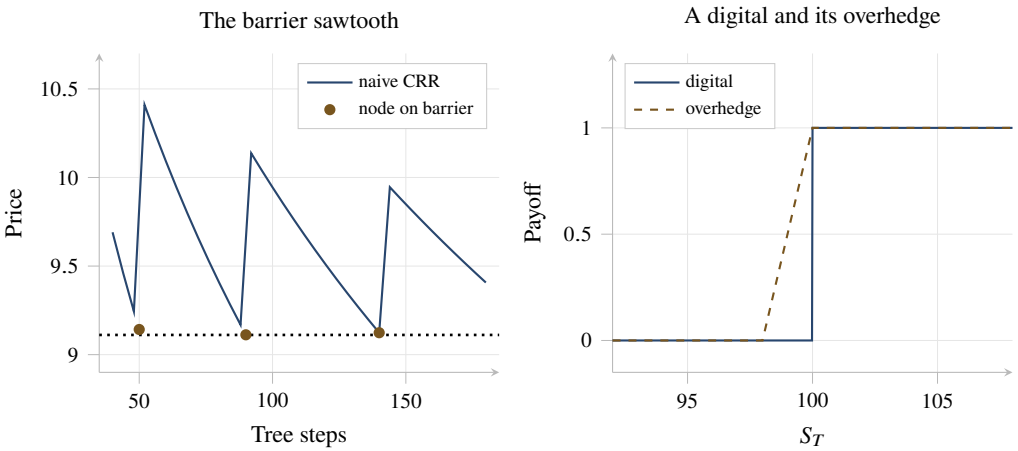


Figure 34.2: Left: Example 34.2; the dotted line is the analytic value. Right: a one-sided call spread dominating a digital.

barrier falls where it falls. A trinomial tree carries a space step that can be set independently of the time step, so choosing $\Delta x = \ln(S_0/H)/m$ for an integer m puts a layer of nodes on the barrier at *every* step count rather than at lucky ones. In $\ln S$ that tree *is* the explicit finite-difference scheme of Section 20.12, and an explicit scheme pays for the freedom with a stability condition, $\Delta x \geq \sigma\sqrt{\Delta t}$: at 50 time steps it allows only two space steps between the spot and the barrier, so hitting the barrier costs a coarse grid. An implicit scheme has no such condition. There the barrier is made the lower boundary of the grid, where the knock-out is the exact condition $V = 0$ rather than a test applied at interior nodes, and the space step is refined as far as wanted independently of the time step. That is the sense in which these are the natural methods for barriers: the discontinuity is put on the mesh by construction instead of being hunted for. On Example 34.2's contract the largest error over 50, 51, 100, 200 and 400 steps falls from **1.383** to **0.017**, and the jump from 50 steps to 51 that cost 1.35 becomes 0.02. This is why barrier books are priced on grids rather than on binomial trees.

34.5 STATIC REPLICATION

Section 18.13 argued that where gamma diverges, a static hedge is the only workable answer. Barriers are the main place that argument pays off.

Example 34.3 A barrier hedged with vanillas and never touched again

When the drift is zero ($r = q$), a **reflection** argument gives an exact static replication. A down-and-in call with strike K and barrier $H < K$ has the same value as K/H puts struck at H^2/K.

The argument has two cases. If the barrier is never touched, the call never comes alive, and the puts expire worthless too: they are struck at H^2/K, below H, and the price has stayed above H. If the barrier is touched, then at that moment K/H puts struck at H^2/K are worth exactly one call struck at K (a symmetry of the lognormal distribution that holds when $r = q$), so the puts are sold and the call bought at no cost, and the hedge has become the option.

Nothing has to be computed to confirm it, because (34.11) contains it. A down-and-in call with $H \leq K$ is the single block C, and at $r = q$ the exponent is $\lambda = 1/2$, so the two weights collapse to H/S_0 and S_0/H:

$$C = e^{-rT}\left[H\,N(y_1) - K\frac{S_0}{H}\,N\big(y_1 - \sigma\sqrt{T}\big)\right], \qquad y_1 = \frac{\ln\big(H^2/(S_0K)\big)}{\sigma\sqrt{T}} + \frac{\sigma\sqrt{T}}{2}$$

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Now price the puts. One struck at H^2/K has $d_1 = \ln\big(S_0K/H^2\big)/(\sigma\sqrt{T}) + \sigma\sqrt{T}/2$, so $-d_1 = y_1 - \sigma\sqrt{T}$ and $-d_2 = y_1$, and K/H of them are worth

$$\frac{K}{H}\left[\frac{H^2}{K}e^{-rT}N(y_1) - S_0e^{-rT}N\big(y_1 - \sigma\sqrt{T}\big)\right] = e^{-rT}\left[H\,N(y_1) - K\frac{S_0}{H}\,N\big(y_1 - \sigma\sqrt{T}\big)\right]$$

term for term the same expression. The replication is therefore exact rather than close, at any common level of $r = q$: the bracket contains no r, so raising the carry only discounts the same number. The condition is zero *drift*, not zero rates.

Now break the condition. At $r = 5\%$, $q = 0$:

Down-and-in call	3.224778
Replicating puts	1.861532
Static hedge error	**−42.3%**

The replication is not robust to drift, and 42% is not a rounding error. Refinements exist (replicating with a strip of vanillas at several maturities rather than one, which trades exactness for robustness), and they are what desks actually use.

But the reason to want a static hedge survives the error. A static hedge is a portfolio of vanillas set up once and traded again at most once, when the barrier is touched. It requires *no rebalancing*, so it has no running transaction costs and no gamma at the barrier; and it does not depend on the surface dynamics that Section 19.12 said barrier prices depend on. A hedge that is wrong by a known amount and never has to be adjusted can be better than one that is right in a model and must be adjusted precisely where the model is least reliable.

34.6 DISCRETE MONITORING

Every formula so far assumes the barrier is monitored continuously. *No contract is*: barriers are observed at a daily close, sometimes weekly or monthly. The gap between the two is larger than it looks and has a clean correction.

Example 34.4 What a daily close is worth

Broadie, Glasserman, and Kou showed that a barrier monitored m times behaves, to first order, like a *continuous* barrier shifted *away* from the spot, and pricing with the shifted barrier is the **continuity correction**:

$$H_{\text{eff}} = H\,e^{\pm\beta\sigma\sqrt{T/m}}, \qquad \beta = -\frac{\zeta(1/2)}{\sqrt{2\pi}} = 0.5826 \tag{34.12}$$

where ζ is the Riemann zeta function, $\zeta(1/2) = -1.4604$, and the sign in the exponent is chosen so the barrier moves further from the current spot: down for a down-barrier, up for an up-barrier.

Applied to Example 34.1's down-and-out call, whose continuous value is 9.111221:

Monitoring	m	**Effective** H	**Price**	**vs. continuous**
Daily	252	89.18	9.527218	+4.57%
Weekly	52	88.20	9.968231	+9.41%
Monthly	12	86.29	10.675121	+17.16%
Quarterly	4	83.68	11.365166	+24.74%

Even daily monitoring is worth 4.6% of the option, and monthly monitoring 17%. A knock-out priced as continuous is undervalued, because a path that dips below the barrier between closes is not caught.

Verify it independently. Simulating the daily-monitored contract directly with 400,000 antithetic paths:

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	Value	Against Monte Carlo
Monte Carlo (daily)	9.523044	(s.e. 0.028437)
Corrected formula (34.12)	9.527218	+0.15 standard errors
Uncorrected (continuous)	9.111221	**−14.5 standard errors**

The correction lands within a fifth of a standard error and the uncorrected formula is fourteen and a half away. The correction looks like a fudge and is not: β comes from the expected overshoot of a random walk past a level, and the same constant appears wherever a discretely observed process is compared with a continuous one.

Three consequences for practice follow.

The correction is not optional at low monitoring frequencies. Quarterly observation moves the price by a quarter, which no bid–ask absorbs.

It interacts with Section 34.4. A tree with a node layer on the *stated* barrier prices the continuous contract; to price the discrete one, the layer goes on H_{eff}. A misaligned layer and the correction both move the barrier away from spot, so an unaligned tree that skips the correction can land near the discrete price by accident: on this contract it is 0.42 high at 100 steps and 0.39 low at 50. Neither adjustment substitutes for the other, and a tree that is right for the wrong reason at one step count is wrong at the next.

And it matters most for the products of Chapter 35. An autocallable (a note that redeems early if the underlying is above a call barrier on an observation date, and exposes the investor to the underlying's loss if a lower knock-in barrier is breached) observes its knock-in barrier daily and its call barrier monthly, so a single instrument carries two different monitoring frequencies and needs two different corrections.

34.7 BARRIERS AND THE VOLATILITY SURFACE

A barrier price depends on the volatility surface in a way a vanilla price does not, and Section 21.11 measured by how much. Two models, local volatility and a stochastic-volatility model, were calibrated to the *same* vanilla surface and verified to reproduce it. They then priced a one-year up-and-out call, spot and strike at 100, barrier at 110, monitored daily: 0.8402 under the stochastic-volatility model against 0.4244 under local volatility, a difference of **49.5%** on an option worth less than a point.

The design of that experiment is Hull and Suo's (2002), set out in Section 21.11. What the spread measures is not an error in either model—both are right about every quoted vanilla—but the size of the question the market has left unanswered.

The reason is Section 19.12's. A vanilla depends on the terminal distribution and a barrier depends on the path, and two models can agree on every terminal distribution while disagreeing about how the underlying gets there. Calibration to vanillas is therefore necessary and not sufficient, and a desk that reports a barrier price to four decimals from a calibrated model is reporting the model.

Three practical responses follow, in ascending order of what they give up.

Price with the model, reserve against the range. Compute under two or three models and hold the spread as a reserve. Section 21.11's argument for reporting model risk applies exactly.

Hedge statically where possible, which converts the model question into a residual and a known error, as Section 34.5 described.

Or refuse the trade. A reverse knock-out at a barrier where the two models differ by half the premium is a trade whose price the desk does not know, and there are market conditions in which the correct response is to say so.

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34.8 DIGITALS AND THE OVERHEDGE

This section treats digitals at more length than their own market would justify, because a digital is the simplest payoff with a jump in it. Everything that makes a discontinuous payoff hard to hedge dynamically appears here with nothing else in the way: a delta that grows without bound as expiration approaches, a hedge that has to be overhedged rather than replicated, and a bid–ask spread set by what the overhedge costs. A barrier option carries the same jump at its barrier, which is why Example 34.6 returns to the argument for the reverse knock-out.

Section 12.8 showed a digital has no finite replicating portfolio. Here is what that costs.

The closed forms are the two halves of Black–Scholes taken apart. A **cash-or-nothing** call pays Q if $S_T > K$; an **asset-or-nothing** call pays S_T in the same event. Their values are

$$V_{\text{cash}} = Qe^{-rT}N(d_2), \qquad V_{\text{asset}} = S_0e^{-qT}N(d_1) \tag{34.13}$$

with d_1 and d_2 the usual Black–Scholes arguments, and the put versions replacing $N(\cdot)$ with $N(-\cdot)$. With $Q = 1$ the vanilla call is $V_{\text{asset}} - K\,V_{\text{cash}}$, which is Section 15.4's reading of $N(d_1)$ and $N(d_2)$ restated as two instruments rather than two integrals.

The formula is also the limit of the call spreads the example below prices, writing $c(K)$ for the value of a call struck at K:

$$\frac{c(K-\varepsilon/2) - c(K+\varepsilon/2)}{\varepsilon} \xrightarrow[\varepsilon\to 0]{} -\frac{\partial c}{\partial K} = e^{-rT}N(d_2) \tag{34.14}$$

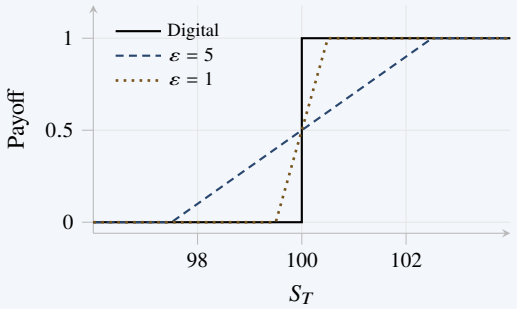
A digital is the strike-derivative of the vanilla, one differentiation short of Section 19.2's implied density, which is the second. The price is therefore a derivative of quoted prices, and a derivative of a quoted price is not itself quoted.

Example 34.5 What certainty costs

A cash-or-nothing call paying \$1 if $S_T > 100$, with $S_0 = 100$, $r = 5\%$, $\sigma = 25\%$, $T = 1$ and no dividends: worth **0.504049**.

Replicate with a call spread of width ε straddling the strike (long $K - \varepsilon/2$, short $K + \varepsilon/2$), held in quantity $1/\varepsilon$. Each spread pays at most ε, so $1/\varepsilon$ of them pay nothing below $K - \varepsilon/2$, the digital's full dollar above $K + \varepsilon/2$, and rise linearly between them. That is (34.14) evaluated at finite ε:

ε	Spreads	Value
5.00	0.2	0.504159
1.00	1.0	0.504054
0.10	10.0	0.504050
0.01	100.0	0.504049



The value converges and the position diverges. The plot shows why the value converges (the payoff approaches the step) and also why this spread is not yet a hedge: it crosses the digital, paying more below the strike and less above it.

What the dealer has sold has no finite greeks. With τ the time remaining to expiration, differentiating $e^{-r\tau}N(d_2)$ once and twice, and collecting $d_1 = d_2 + \sigma\sqrt{\tau}$,

$$\Delta = \frac{e^{-r\tau}N'(d_2)}{S\sigma\sqrt{\tau}}, \qquad \Gamma = -\frac{e^{-r\tau}N'(d_2)\,d_1}{S^2\sigma^2\tau}$$

At the money $d_1 = 1.3\,\sigma\sqrt{\tau}$, since $(r - q + \sigma^2/2)/\sigma^2 = 1.3$ here, so both greeks grow like $1/\sqrt{\tau}$: each doubles whenever the time remaining is quartered. Away from the money it is the d_1

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in the numerator that carries the gamma's sign, so the gamma changes sign at the delta's peak, $Ke^{-(r-q+\sigma^2/2)\tau}$. That point closes on the strike as expiration approaches (three cents below it with a day to run, under a hundredth of a cent with a minute), while $|\Gamma|$ five cents to either side of it reaches 32. The delta is 0.25 with a day to run, 0.65 with an hour, 5.0 with a minute and 38.8 with a second.

Holding $1/\varepsilon$ spreads caps the delta at $1/\varepsilon$, which is the whole reason for holding spreads rather than the digital; and even at that cap, with $\varepsilon = 0.10$, ten shares of stock have to be traded for every dollar of payoff while the price sits inside the window in the last moments before expiration, with no time to trade them. That is what Section 18.13's "gamma diverges" means when someone has to do it.

So dealers overhedge. To **overhedge** is to hedge with a position whose payoff is at least the sold option's everywhere, paying a known premium in place of an unhedgeable jump. Sell the digital and buy $1/\varepsilon$ call spreads struck at $K - \varepsilon$ and K, so that the upper strike stays at the digital's strike and the whole window lies *below* it. That position already pays the full dollar at K and something below it, so it dominates the digital's payoff everywhere. The table widens the window and compares each cost with the digital's 0.504049:

ε	Strikes	Cost	Premium over the digital
1	[99, 100]	0.511635	+1.5%
2	[98, 100]	0.519249	+3.0%
5	[95, 100]	0.542210	+7.6%

That premium is what certainty costs. A dealer who will not carry the residual cannot sell the digital below the dominating spread, and buying it cannot pay above the mirror image: a spread struck entirely *above* the strike, which the digital dominates. At a one-point window those bounds are 0.4965 and 0.5116 around a digital worth 0.5040, so no two-way price narrower than **3.0%** is hedged on both sides. What a desk actually quotes is wider: under a skew the digital's own value moves (Example 19.3 differentiates along the smile), a jump through the strike can leave the overhedge short, every one of the $1/\varepsilon$ spreads is bought at the offer, and the overhedged book ties up capital. None of these is in the model here; this is the floor the payoff alone puts under the quote. Figure 34.2's right panel shows the dominance.

The width is a business decision, not a modeling one. A tighter spread is a better price, a worse hedge, and a higher cap on the position the desk may have to trade: one share at a point wide, ten at a tenth, a hundred at a hundredth. The dealer chooses where on that trade-off to sit based on how much of the position they expect to hold to expiration.

And this generalizes to every discontinuous payoff in the chapter. A reverse knock-out is a digital in disguise. It pays a positive amount right up to the barrier and nothing beyond it, which is why Section 34.3 called it the most difficult vanilla-like instrument in the market. The overhedge for it is a barrier struck slightly further away, and the premium is much larger, as Example 34.6 shows.

Who trades digitals. Very little of the market trades a digital on its own. Most digital risk sits inside other products as a coupon trigger: each observation date on an autocallable pays its coupon only if the underlying is above a level, and a range accrual pays a day's interest only if a rate stays inside a band, so both are strips of digitals, which Chapter 35's investors are short and the issuing dealers long. The one active standalone market is in foreign exchange, where European digitals and their path-dependent relatives (the **one-touch**, which pays if the rate trades at a level at any time before expiration, and the **no-touch**, which pays if it never does) are standard dealer products. Macro hedge funds buy them to take a view on a level for a fraction of a vanilla's premium.

Professionals have little need for a listed digital, because a tight call spread already is one. Cboe listed binary options, as digitals are also called, on the S&P 500 and the VIX in 2008, and they drew almost no volume. Binaries found their buyers among retail investors instead,

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sold by offshore platforms that were the counterparty to every trade, usually at maturities of minutes or hours. The European Securities and Markets Authority restricted their sale to retail clients from July 2018 (a measure renewed in three-month steps and later carried into national rules), and Israel closed its binary-options industry outright.

The growth since 2024 has come from **event contracts**: cash-or-nothing contracts on an inflation release, a central-bank decision or an election, listed on exchanges regulated by the Commodity Futures Trading Commission and offered on prediction markets, a venue structure that is still changing as of this writing. They are digitals in the strict sense of this section, but on outcomes that no vanilla option spans. There is no call spread to overhedge them with, so Example 34.5's argument does not reach them: the price is a probability set by the order book, and the market maker carries the jump at the event rather than hedging it.

Example 34.6 The instrument that is short its own underlying

An up-and-out call struck at 100 with the barrier at 120, one year, 25% volatility, and $r = q = 0$: zero carry, so the reflection results of Section 34.5 apply exactly. Its value is the vanilla A less the up-and-in call $B - C + D$ from the third row of Section 34.3's table, by in–out parity. The vanilla is worth 9.9476; the knock-out is worth **0.6709**, *6.7% of it*. Almost the whole premium has been given away, because the option is extinguished in precisely the states where a call is valuable.

Its delta is the striking part:

Spot	Value	Delta
100	0.6709	−0.0149
110	0.4081	−0.0359
115	0.2132	−0.0414
118	0.0863	−0.0430

The delta is negative from a spot of 94.3 upward. This is a call option whose holder loses money when the underlying rises, because approaching 120 removes more value through the knock-out than it adds through moneyness. Its holder hedges by buying stock, as though short a call—the reverse of every intuition built up in Chapter 18—and the sign flips back the moment the barrier is removed.

The overhedge is where the difficulty becomes a price. Dominating the payoff means pricing and hedging the contract as a knock-out with the barrier further away, and the cost rises steeply:

Barrier used	Value	Cost over the contract
122	0.8947	+33.4%
124	1.1513	+71.6%
126	1.4377	+114.3%

Moving the barrier out by two points, 1.7% of its level, costs a third of the option's value, against 7.6% for the digital's much wider five-point window in Example 34.5. That ratio is the whole of why reverse knock-outs quote in percentage points while down-and-outs quote in basis points, and it is a property of the payoff rather than of any model: the cliff is 20 points high on an option worth 0.67.

Figure 34.3 plots it. The value is a hump whose peak sits at 94.3, six points *below* the strike, because above that level a rise in spot removes more value by raising the probability of a knock-out than it adds by moving the call further into the money, and the delta in the right panel is simply the slope of the left one, crossing zero exactly at the peak. A call option with an interior maximum is not a call option in any sense Chapter 18 would recognize, and the hedging difficulty is that the maximum is inside the trading range rather than at the edge of it.

Figure 34.4 takes the same option with a few weeks to run rather than a year, and adds the gamma. The delta climbs to 0.67 by a spot of 104, turns down, crosses −1 at 115.7, and

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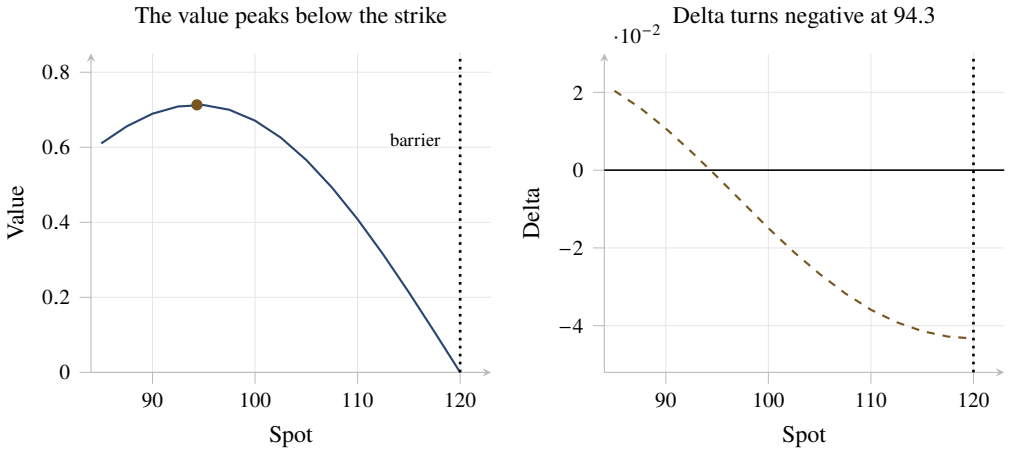


Figure 34.3: Example 34.6's up-and-out call: $K = 100$, $H = 120$, $\sigma = 25\%$, $T = 1$, $r = q = 0$.

reaches -1.46 at the barrier, where a vanilla's delta is $+1$; with a year to run it never left $[-0.043, 0.027]$. The gamma is negative above 104.4, where a vanilla's is positive everywhere, and at -0.23 near a spot of 113 its magnitude is three times the largest gamma the vanilla ever has.

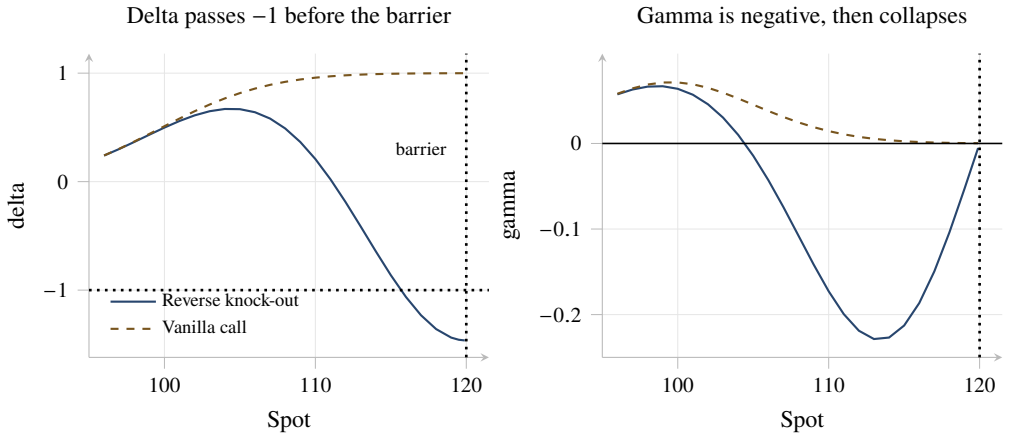


Figure 34.4: Up-and-out call against a vanilla call: $K = 100$, $H = 120$, $\sigma = 25\%$, $T = 0.05$, $r = q = 0$. The dotted vertical line is the barrier, the dotted horizontal one $\Delta = -1$.

The hedging problem is in the last few points of the right-hand panel. Approaching the barrier the value becomes nearly linear in the spot (0.146 at a spot of 119.9, against 0.146 for the straight line through the barrier), so the gamma collapses to -0.005 while the delta does not: it is still -1.46. A hedge of 1.46 shares against an option worth 0.15 means a tenth of a point of spot moves the hedge by the option's entire value, and at the barrier the option ceases to exist, so that delta has to come off at once. Gamma, which normally warns a hedger that rebalancing is about to become expensive, is closest to zero exactly where the position is most dangerous. In the terms of Chapter 18's expansion, equation (18.1), the gamma term is what announces that the delta term's coefficient is about to change, and here it is silent because the change is not gradual: at the barrier the delta jumps from -1.46 to zero, which no second-order

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term of an expansion around a smooth value can represent.

34.9 LOOKBACKS AND ASIANS

This section compares two path-dependent payoffs, one cheaper than the vanilla and one dearer. Both have closed forms, which come first.

Two averages are in use. The traded contract takes the arithmetic average of its fixings, $\frac{1}{n}\sum_{i=1}^{n} S_{t_i}$, the discrete form of the A_T in (34.3). The **geometric average** multiplies the fixings and takes the nth root,

$$G_T = \Big(\prod_{i=1}^{n} S_{t_i}\Big)^{1/n} = \exp\Big(\frac{1}{n}\sum_{i=1}^{n} \ln S_{t_i}\Big),$$

which becomes $G_T = \exp\left(\frac{1}{T}\int_0^T \ln S_t\, dt\right)$ when the averaging is continuous, and which never exceeds the arithmetic average of the same prices. Its logarithm is an average of normal variables and is therefore normal, so the geometric-average Asian call, paying $(G_T - K)^+$, is exactly lognormal: it is Black–Scholes with two substitutions. Averaging continuously over $[0, T]$,

$$c_G = S_0 e^{(b_G - r)T} N(d_1) - K e^{-rT} N(d_2), \qquad \sigma_G = \frac{\sigma}{\sqrt{3}}, \qquad b_G = \frac{1}{2}\left(r - q - \frac{\sigma^2}{6}\right) \tag{34.15}$$

with $d_1 = \left[\ln(S_0/K) + (b_G + \sigma_G^2/2)T\right]/(\sigma_G\sqrt{T})$ and $d_2 = d_1 - \sigma_G\sqrt{T}$. The $1/\sqrt{3}$ is Section 33.8's variance ratio; b_G is the drift the geometric average inherits, which is below $r - q$ because a geometric mean sits below an arithmetic one.

The arithmetic-average Asian, the one actually traded, has no closed form, because a sum of lognormals is not lognormal. Two devices cover the gap. Matching the average's first two moments to a lognormal and pricing with Black's formula gives 6.883 for the contract in Example 34.7, against 6.85 by simulation: about half a percent high, which is the size of error the whole approximation literature is arguing about, though it grows with volatility, to 1.4% at 50%. The arithmetic version is worth 5% more than the geometric one priced in that example, the gap between an arithmetic and a geometric mean, and that gap rises roughly in proportion to volatility, from 2% at a 10% volatility to 11% at 50%. And (34.15) serves as a control variate for simulation: the two payoffs correlate at 0.9994, and subtracting the geometric payoff one for one cuts the variance by a factor of 330, or about 850 with the regression-optimal coefficient. The instrument that is traded is the one without a formula, and it barely matters—which is the reverse of the situation for every barrier in this chapter.

The floating-strike lookback call paying $S_T - S_{\min}$, where $S_{\min}$ is the running minimum observed so far, is worth

$$c_{\text{lb}} = S_0 e^{-qT} N(a_1) - S_0 e^{-qT}\,\kappa\, N(-a_1) - S_{\min} e^{-rT}\left[N(a_2) - \kappa\, e^{Y_1} N(-a_3)\right], \qquad \kappa = \frac{\sigma^2}{2(r-q)} \tag{34.16}$$

where

$$a_1 = \frac{\ln(S_0/S_{\min}) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}}, \qquad a_3 = \frac{\ln(S_0/S_{\min}) + (-r + q + \sigma^2/2)T}{\sigma\sqrt{T}},$$

$a_2 = a_1 - \sigma\sqrt{T}$, and $Y_1 = -2(r - q - \sigma^2/2)\ln(S_0/S_{\min})/\sigma^2$. The floating-strike put, paying $S_{\max} - S_T$ with $S_{\max}$ the running maximum, is the mirror image: the maximum takes the place of the minimum and the drift terms change sign. The formula is singular at $r = q$ (κ diverges), though the value is not, the limit being finite; an implementation that does not special-case zero drift will return nonsense on precisely the parameter set Section 34.5 cares about.

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Example 34.7 Averaging destroys optionality; looking back manufactures it

All at $S_0 = K = 100$, $\sigma = 25\%$, $r = 5\%$, $T = 1$:

	Value	Relative to vanilla
Vanilla at-the-money call	12.3360	1.00
Floating-strike lookback call	20.5522	**1.67**
Geometric-average Asian call	6.5298	**0.53**

Same underlying, same volatility, and a factor of three between the two exotics.

The Asian is cheap because an average has less variance than an endpoint. Section 33.8 computed the ratio: $(n+1)(2n+1)/6n^2$, tending to $1/3$, so the effective volatility is about $1/\sqrt{3}$ of the true one. An Asian option is a vanilla on a less volatile quantity, and that explains most of the discount: at $25\%/\sqrt{3} = 14.4\%$ an at-the-money call is worth 8.38, about two-thirds of the vanilla. The rest comes from the average's lower forward, the drift b_G of (34.15), which takes the ratio to 0.53.

The lookback is expensive because it removes the timing decision. A floating-strike lookback call pays $S_T - \min S$, so the holder buys at the best price achieved. Nothing is left to get wrong, which is worth 67% more than the vanilla.

Figure 34.5 repeats the comparison across volatilities, and it separates what is optionality from what is not. At zero volatility the vanilla and the lookback are worth the same 4.88: the path then rises steadily at the 5% rate, its minimum is the starting price, and both pay only the drift. The Asian starts lower, at 2.41, because the average of a rising path sits below its end, and that gap is carry rather than optionality. Between 25% and 50%, each volatility point adds about 0.61 to the lookback, 0.38 to the vanilla and 0.18 to the Asian.

Three practical notes follow.

Asians dominate commercial usage and lookbacks barely exist. A corporate hedging a year of purchases has an average exposure and wants an Asian; no one has an economic exposure to the minimum of a price path. Lookbacks are taught more than they are traded.

The Asian's model risk is low and the lookback's is high. An average smooths the path, so an Asian is less sensitive to the path distribution than the vanilla is to the terminal one; a lookback is a bet on the extremes and is very sensitive to it.

And discrete monitoring matters in both, in opposite directions. A monthly-averaged Asian is worth more than a continuously-averaged one (fewer observations, more variance), while a daily-monitored lookback is worth less than a continuous one, because a discretely observed minimum is higher. Section 34.6's continuity correction is the same adjustment, and the sign follows from which way discretization moves the observed extreme.

34.10 BASKETS, RAINBOWS, AND EXCHANGE OPTIONS

Multiple underlyings are the second source of genuine difficulty, and their new input is correlation.

Example 34.8 What correlation is worth

Two assets at 100 with volatilities 25% and 30%, $T = 1$, $r = 5\%$, no dividends.

A Margrabe exchange option, the right to give up asset 2 and receive asset 1, has the closed form derived in Section 28.11 as equation (28.14):

$$V_0 = S_1 e^{-q_1 T} N(d_1) - S_2 e^{-q_2 T} N(d_2), \qquad \hat{\sigma} = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2} \tag{34.17}$$

with $d_1 = \left[\ln\left(S_1 e^{-q_1 T}/S_2 e^{-q_2 T}\right) + \hat{\sigma}^2 T/2\right]/(\hat{\sigma}\sqrt{T})$ and $d_2 = d_1 - \hat{\sigma}\sqrt{T}$. Correlation enters only through $\hat{\sigma}$, so the table below is a table of Black–Scholes–Merton prices at five volatilities:

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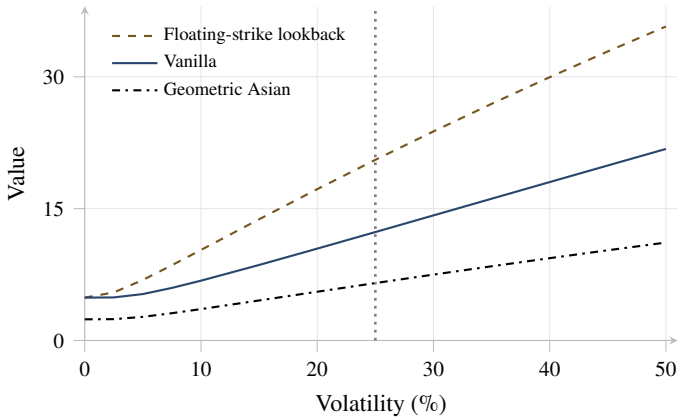


Figure 34.5: At-the-money calls against volatility: $S_0 = K = 100$, $r = 5\%$, $T = 1$, no dividends. The dotted line marks Example 34.7's 25%.

ρ	Effective volatility	Value
−0.50	47.70%	18.8495
0.00	39.05%	15.4808
+0.50	27.84%	11.0703
+0.90	13.23%	5.2737
+0.99	6.32%	2.5227

The value falls by a factor of seven across the correlation range. No interest rate appears, for the reason Section 28.11 gives: in units of asset 2 there is nothing to finance.

Now compare a *basket call* on $(S_1 + S_2)/2$ struck at 100 with a *basket of calls*, $(c_1 + c_2)/2$, half a 100-strike call on each asset:

ρ	Basket call	Basket of calls	Ratio
−0.50	8.3481	13.2836	0.628
0.00	10.3260	13.2836	0.777
+0.50	11.9088	13.2836	0.896
+0.90	13.0178	13.2836	0.980
+1.00	13.2792	13.2836	1.000

An option on a basket is always cheaper than a basket of options, with equality only at perfect correlation and equal volatilities, which is why the last row still shows a gap of 0.0044. The gap is the diversification the basket option does not let the holder keep: an option on each asset pays on whichever rises, while an option on the average pays only if the average rises.

And unlike the exchange option, the basket has no closed form, for the reason the arithmetic Asian does not: a weighted sum of lognormals is not lognormal. The workhorse approximation matches the first two moments of the basket $B = \sum_i w_i S_i$, with w_i units of asset i, to a lognormal,

$$M_1 = \sum_i w_i F_i, \qquad M_2 = \sum_i \sum_j w_i w_j F_i F_j \, e^{\rho_{ij}\sigma_i\sigma_j T}, \qquad \sigma_B = \sqrt{\frac{1}{T}\ln\frac{M_2}{M_1^2}} \tag{34.18}$$

where $F_i = S_i e^{(r-q_i)T}$, and then prices with Black's formula at forward M_1 and volatility σ_B. The approximation degrades exactly where the option is most interesting: against the exact values in the table it overstates by 0.06% at $\rho = 0.9$ and by 0.75% at $\rho = -0.5$, because a negatively correlated basket is the least lognormal. The geometric basket $\prod_i S_i^{w_i}$, by contrast, is exactly lognormal and has a closed form, which makes it the natural control variate here just as (34.15) is for the Asian.

That gap is what Section 22.11's dispersion trade, short the index option and long the

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components, trades: it earns the difference between implied and realized correlation, and the ratio column shows how much is at stake.

The modeling difficulty is that ρ has to come from somewhere. With two assets and a liquid option on each plus a liquid basket option, it can be implied. With fifty assets it cannot, because one basket price cannot determine 1,225 pairwise correlations, and the standard device (assume a single correlation for all pairs) is a one-factor approximation whose error is exactly Section 32.9's factor-reduction problem in a different market.

34.11 WORST-OF OPTIONS

One multi-asset payoff is the single most common exotic sold to retail investors, and its correlation exposure runs the opposite way to a basket's.

With two assets it has a closed form and the derivation is one identity. Since $\min(S_1, S_2) = S_1 - \max(S_1 - S_2, 0)$, a forward on the minimum is a long position in asset 1 against a short exchange option (34.17), and the strike-K put on the minimum follows from

$$p_{\min}(K) = Ke^{-rT} - \underbrace{\left[S_1 e^{-q_1 T} - V_0^{\text{exch}}\right]}_{\text{PV of receiving } \min(S_1,S_2)} + c_{\min}(K) \tag{34.19}$$

Stulz's formula supplies the last term in bivariate normal probabilities:

$$\begin{aligned} c_{\min}(K) = {} & S_1 e^{-q_1 T} M(y_1,\, -d_1;\, -\rho_1) + S_2 e^{-q_2 T} M(y_2,\, d_2;\, -\rho_2) \\ & - Ke^{-rT} M\left(y_1 - \sigma_1\sqrt{T},\; y_2 - \sigma_2\sqrt{T};\; \rho\right) \end{aligned} \tag{34.20}$$

where $\hat{\sigma}$, d_1, and d_2 are the exchange option's (34.17), and

$$y_i = \frac{\ln\left(S_i e^{-q_i T}/Ke^{-rT}\right) + \sigma_i^2 T/2}{\sigma_i\sqrt{T}}, \qquad \rho_1 = \frac{\sigma_1 - \rho\sigma_2}{\hat{\sigma}}, \qquad \rho_2 = \frac{\sigma_2 - \rho\sigma_1}{\hat{\sigma}}$$

The exchange option appears twice, once as an instrument in (34.19) and once as the correlation structure inside M, which is the sense in which every two-asset payoff in this chapter is built from Margrabe.

Three assets is where it stops. The same construction extends to n assets (Johnson's formula), but the bivariate M becomes an n-variate normal distribution function, and quadrature for those is already slow in three dimensions and unreliable above four. The three-asset worst-of that dominates the retail market is therefore simulated, which is why Example 34.9 reports a table rather than a formula, and why the note desk's risk numbers are Monte Carlo numbers with all the noise that implies.

Example 34.9 Why the coupon is high

Three assets, each at 100 with 28% volatility, $T = 1$, $r = 5\%$. A **worst-of put** struck at 100 pays on the *minimum* of the three. The benchmark is a vanilla put on any one of them, also struck at 100, which is worth 8.5955. The worst-of values are simulated, with 4 million antithetic pairs at each correlation; every standard error is below 0.005.

ρ	$E[\min S_T]$	Worst-of put	Relative to single-asset put
0.20	83.87	17.64	**2.05**
0.40	86.59	16.26	1.89
0.60	89.86	14.71	1.71
0.80	94.23	12.81	1.49
0.95	99.62	10.64	1.24

The worst-of put is worth more as correlation falls, exactly the opposite of the basket in

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Example 34.8.

A basket averages, so low correlation *reduces* dispersion and cheapens the option. A worst-of takes a minimum, so low correlation *increases* the chance that at least one asset falls a long way. At a correlation of 0.20 the expected minimum at one year is 83.87 against a starting level of 100—the drift is upward and the minimum still falls sixteen percent, purely from picking the loser three times.

A **worst-of note** is a bond whose principal is repaid in full only if the worst of the three assets finishes above the strike; otherwise the investor bears that asset's loss. The investor has therefore sold the issuer a worst-of put, and the issuer pays for it as a coupon above its own funding rate. At $\rho = 0.20$ the put is worth 17.64 per 100 of notional, roughly 18 percentage points of coupon on a one-year note, and twice what a note on any single one of the assets could pay.

So the investor in such a note is long correlation, losing as it falls: the lower the correlation, the more the put is worth and the more the investor has given away. In a crash correlation rises, which on its own would lower the put's value, but it rises because everything is falling together, and with all three assets below the strike the put is deep in the money whatever the correlation. The investor therefore carries two losses for one coupon: the one that low correlation pays for, a single asset collapsing while the others hold up, and the market-wide fall, against which the rise in correlation offers no protection.

Adding assets to a worst-of always raises the coupon and always raises the risk, and a note advertised as "linked to three blue-chip stocks" is riskier than one linked to any of them individually. Chapter 35 returns to this as the central example of the gap between how these products are described and what they are.

Figure 34.6 puts this example and Example 34.8 on one axis. Both ratios come to one at perfect correlation, where the assets move as one, and they leave it in opposite directions as correlation falls: the basket call loses value against its components, while the worst-of put gains it. A worst-of note's investor is short the upper curve, which is why the coupon is highest exactly where the assets are least alike.

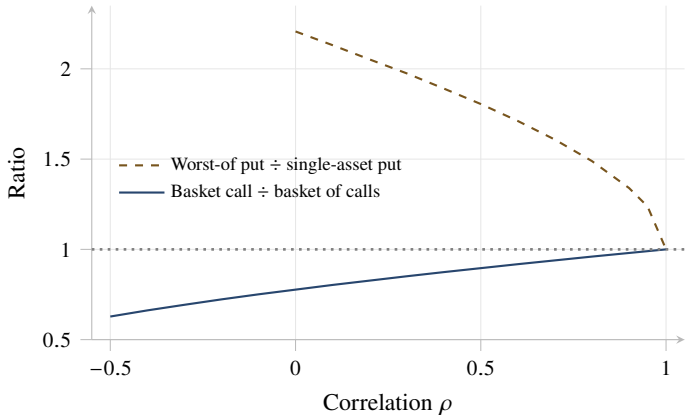


Figure 34.6: Examples 34.8 and 34.9 across correlation. The worst-of values are simulated.

34.12 CLIQUETS, AND THE CAP THAT REVERSES THE SIGN

Section 34.2 called a cliquet the purest instrument for trading forward volatility. The version actually sold is not, and the reason is instructive.

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Example 34.10 What the cap does

A three-year annual-reset cliquet paying $\sum_i \max(r_i, 0)$, where r_i is the year's return, each period capped at 8%. Under Black–Scholes each period is a forward-start at-the-money call spread, so its value depends only on that year's forward volatility. By the scaling in (34.9), the period payoff $\min\big(\max(r_i, 0),\ C\big)$ is worth, at its own reset date,

$$V = c(1,\ 1,\ \tau) - c(1,\ 1 + C,\ \tau) \tag{34.21}$$

per unit of notional, where $c(S, K, T)$ is the Black–Scholes–Merton call of Section 34.2, evaluated here at a spot of 1 and the period's forward volatility, $C = 8\%$ is the cap, and $\tau = t_i - t_{i-1}$ is the length of a period, one year here. That is a number, not a function of the level, so the whole contract is $\sum_i e^{-rt_{i-1}} V$. The table reports the undiscounted sum $3V$, which isolates the volatility comparison; discounting the second and third periods scales the total by about 0.95.

Forward volatility	Per period	Three-year cliquet
15%	0.03800	0.11401
20%	0.03670	0.11009
25%	0.03561	0.10682
30%	0.03463	0.10388
35%	0.03372	0.10115

The value falls as volatility rises. An uncapped cliquet would be long volatility; the cap makes this one short it, because the 8% cap sits close to the one-year forward and higher volatility helps the short leg of the spread more than the long one. That reversal is what lets it be sold at an attractive headline rate: it is marketed on upside participation and is, in volatility terms, a *short*-volatility position.

And the model risk is severe for a reason Section 21.3 supplies. The value sits entirely in the forward smile: the second and third periods are forward-starting, so what matters is the smile prevailing a year and two years from now. Section 21.3 measured what local volatility predicts. The skew of its local-volatility function (the slope of volatility against log-moneyness at the money) decays from -0.981 at three months to -0.231 at three years, so the model expects the smile a few years out to be far flatter than today's, while the market's forward smile is roughly stationary. A local-volatility model prices the later periods off a smile it has flattened, and a cliquet's value is concentrated exactly there.

This is the clearest case in the chapter where calibrating to today's surface is not merely insufficient but actively misleading, because the model reproduces every vanilla and gets the one quantity the product depends on structurally wrong.

Figure 34.7 shows the reversal against the contract the cap was applied to. The two slope in opposite directions, and an investor told they are buying participation in rising markets is holding the line that falls when volatility arrives. Both lines are close to straight because both options involved, the at-the-money call and the call at the cap, are struck near the forward, where an option's vega barely changes with volatility.

34.13 WHAT THEY HAVE IN COMMON

The chapter's material sorts on one question, put as a test rather than a list.

Can the payoff be replicated by traded instruments?

If it can be replicated statically (a package, a chooser, a digital as a limit of call spreads), then the vanilla market determines the price, model risk is small, and the residual is a hedging cost rather than a valuation uncertainty.

If it can only be replicated dynamically, then the price depends on the assumed dynamics, and calibration to vanillas constrains but does not determine it. Barriers, cliquets, lookbacks, and everything multi-asset live here.

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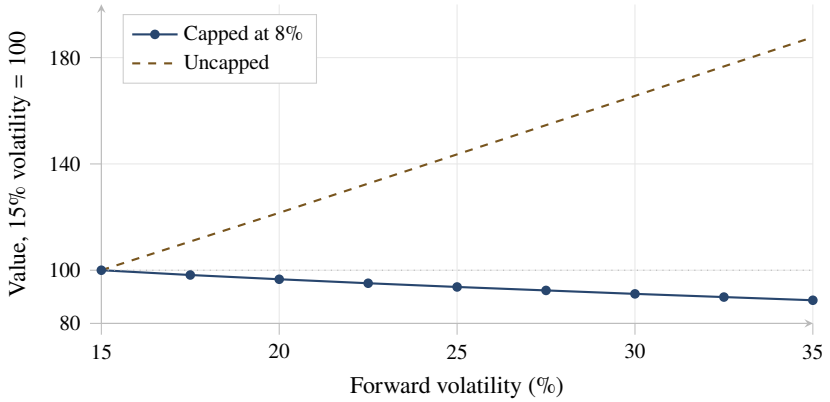


Figure 34.7: Three-year annual-reset cliquet of Example 34.10, each indexed to its own value at 15% volatility. In levels the capped contract falls from 0.11401 to 0.10115.

The test is more useful than the taxonomy of Section 34.1 because it predicts the right thing. A compound option sounds harder than a barrier and is easier; a forward-start option sounds trivial and carries as much model risk as anything in the book.

And it is a different question from whether a formula exists, which is worth putting side by side because the two are so easily confused.

Instrument	Closed form under Black–Scholes	Model risk
Package, chooser	Exact, (34.8)	None
Compound	(34.7), plus one root search	Low
Digital	(34.13)	Low; the cost is hedging
Barrier, continuous	(34.11), all eight	**Severe**
Barrier, discrete	(34.11) with (34.12)	**Severe**
Lookback	(34.16)	High
Asian, geometric	(34.15)	Low
Asian, arithmetic	**None**; moment match or simulate	Low
Exchange	(34.17)	Correlation only
Two-asset max/min	(34.20)	Correlation
Basket, worst-of ($n \geq 3$)	**None**; simulate	Correlation, unhedgeable
Forward-start, cliquet	(34.9), (34.21)	**Severe**

The two columns are almost uncorrelated. Barriers, lookbacks, and cliquets all have closed forms and carry the worst model risk in the chapter; the arithmetic Asian has no formula at all and carries almost none. A closed form is a statement about Black–Scholes, not about the market: it returns the same number every time it is called, which is precision, not accuracy. The formulas are worth having because they are fast, exact within the model, and show where the model enters, not as evidence that the price is known.

And it explains a market structure fact that otherwise looks arbitrary. The exotics with static hedges trade at narrower spreads than those without, and the ordering follows the test rather than the complexity of the payoff description.

Case Study 34.1 The structure that sold a discount and bought a multiple

The **accumulator** was sold across private banks in Hong Kong and Singapore through 2007 and into 2008, and it is this chapter's mechanism assembled into a single retail contract.

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The terms sound like a discount. The client agrees to buy a fixed quantity of a stock on every trading day for a year, at a strike set a few percent below the spot price at inception. Nothing is paid at the outset. Two clauses complete it: the contract *knocks out* if the stock trades a few percent above the initial spot, and the daily quantity *doubles* if the stock is below the strike.

Decomposed, it is three of this chapter's instruments. A strip of daily forwards at a discount; short a strip of up-and-in calls, which is what the knock-out clause amounts to, since it surrenders the forwards' gains once the barrier trades; and short a leveraged strip of puts, which is what the doubling clause is. The discount is the premium of the options the client has sold, and the client is short volatility, short the downside at twice the size, and capped on the upside at a few percent.

What happened is what the structure specifies. Hong Kong and mainland equities peaked in October 2007 and fell through 2008. Because prices fell, the knock-out never triggered, so the contracts ran their full term rather than terminating early as they did in the case the salesman described. The doubling clause then required the client to buy twice the daily quantity, every day, at a strike far above the market, with the obligation growing precisely as the losses grew. The instrument acquired a nickname that was a pun on its own name: *I kill you later*.

The failure was not mathematical. Nothing here needed a model the seller did not have. The knock-out was disclosed, the doubling was disclosed, and both behaved exactly as written. What was not conveyed is the asymmetry the two clauses create together—a payoff described by its best state and sized by its worst.

Three lessons connect it to the rest of the book. A barrier that removes the upside is worth more than it looks when it is one of two clauses rather than the only one. A zero-premium structure has a premium; Chapter 35's treatment of a note's estimated initial value gives the machinery for finding it, and here it is the discount itself. And leverage written into a contract's quantity is harder to see than leverage written into its notional, because nothing in the term sheet is labeled with a multiple.

34.14 WHO TRADES EXOTICS, AND WHY

Corporate hedgers are the economically motivated end of the market. An Asian option matches an averaged exposure and a barrier matches a hedge that is only wanted in some states: an importer who needs FX protection but not if the currency has already collapsed past the point where the business is unviable. These are exotics bought because the vanilla is the wrong shape.

Structured-note issuers are the volume. Chapter 35 is about them, and almost every payoff in this chapter appears inside a retail note: barriers as knock-in protection, digitals as coupon triggers, baskets as "worst-of" underlyings, cliquets as annual ratchets.

Dealers intermediate, and the pattern is one-directional. The client is almost always short the exotic optionality and the dealer long it, because the exotic feature is what cheapens the note or the hedge. So dealers run a structural long-exotic, short-vanilla book, which they hedge in the vanilla market, and the marks Section 19.8 traces on the surface and Section 22.11's implied correlation both reflect that flow.

Hedge funds take the other side selectively, and the trades named in this book (dispersion in Section 22.11, forward-volatility positions against cliquets) are mostly bets that the dealer's hedging flow has pushed a price away from fair.

The recurring structure is worth naming once. An exotic option transfers a risk the client does not want to price to a dealer who does, and the dealer's compensation is the difference

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between what the exotic can be hedged for and what it was sold for. Sections 34.5 and 34.13 determine how large that difference has to be, which is the practical reason the test in this chapter matters.

SUMMARY

Exotics divide by path dependence, multiple underlyings, and non-standard exercise, but the useful division asks whether the payoff can be replicated by traded instruments. Statically replicable payoffs are priced by the vanilla market with small model risk; dynamically replicable ones depend on assumed dynamics, which calibration to vanillas constrains without determining.

A down-and-out call is worth well below the vanilla, and a reverse knock-out loses value as the underlying rises toward its barrier. Barriers are hard to compute as well as to hedge: a binomial tree's value jumps erratically with the number of steps unless a node layer sits on the barrier, and correcting for daily rather than continuous monitoring moves the price materially. The reflection hedge is exact only when $r = q$ and errs badly when they differ, and two models calibrated to the same vanilla surface priced one barrier far apart.

A digital is the limit of call spreads whose value converges while the position diverges, so dealers overhedge with a spread that dominates the payoff, at a modest premium over the digital. Every discontinuous payoff in the chapter is priced this way.

A lookback is worth substantially more than the vanilla and a geometric Asian substantially less: an Asian is a vanilla on a less volatile quantity, and a lookback removes the timing decision.

Correlation enters every multi-asset payoff and is rarely quoted, so an option on a basket is cheaper than a basket of options. A worst-of put on several assets is worth far more than a single-asset put, because a minimum rewards dispersion where a basket's average punishes it, and that premium is what pays a worst-of note's coupon. A capped cliquet is short forward volatility, losing value as volatility rises, and its value sits in the forward smile, which today's surface does not pin down.

A closed form is a statement about Black–Scholes, not about the market. Barriers, lookbacks and cliquets have exact formulas and the worst model risk in the chapter; the arithmetic Asian has no formula and almost none.

Corporates buy exotics because the vanilla is the wrong shape, and structured-note issuers supply the volume. Clients are almost always short the exotic optionality, so dealers run a long-exotic, short-vanilla book hedged in the vanilla market.

FURTHER READING

Zhang's *Exotic Options* and Haug's *Complete Guide to Option Pricing Formulas* are the reference works for closed forms; Haug states each formula with a worked number, which makes it the place to check an implementation against, and both should be read knowing that a closed form exists only under assumptions the market violates. Derman, Ergener and Kani on static replication of barriers is the paper behind Section 34.5 and develops the strip-of-maturities refinement this chapter only mentions. Carr and Chou extend the reflection argument considerably further. Boyle and Lau (1994) is the source of Example 34.2's node-placement rule. Broadie, Glasserman and Kou on discrete barrier monitoring supplies the continuity correction of Section 34.6. On the multi-asset material, Margrabe's original paper is short and is the source of the exchange option; Stulz (1982) is the source of (34.20), and states put prices through the identity (34.19); and Johnson (1987) extends the result to n assets. The literature on basket approximations is large, and its methods differ mainly in accuracy, so decide what accuracy is needed before sampling it. For the model-risk question of Section 34.7, the practitioner literature on barrier reserves is thin and mostly internal; the academic treatment

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under uncertain volatility is the public substitute.

PRACTICE QUESTIONS

34.1 State the replication test that organizes this chapter, and use it to explain why a compound option carries less model risk than a forward-start.

34.2 Using Example 34.1, state the fraction of the vanilla premium the barrier removes, and explain why knock-out deltas can exceed one.

34.3 Using Example 34.2, state the tree prices at 50 and 51 steps, explain the difference, and state the general lesson in a form that does not mention barriers.

34.4 Explain why grid-averaging and Richardson extrapolation both make the barrier tree error worse rather than better.

34.5 Derive the step count that places a node layer on the barrier, and state the error it achieves against an unaligned grid with more steps.

34.6 Using Example 34.3, state the static hedge error when $r = 5\%$ and $q = 0$, and explain why the condition for exactness is zero drift rather than zero rates.

34.7 Explain why a static hedge that is wrong by a known amount can be preferable to a dynamic hedge that is right in a model.

34.8 Explain why two models calibrated to the same vanilla surface can price a barrier 49.5% apart, and state the three defensible responses.

34.9 Using Example 34.5, state the cost of the $[95, 100]$ overhedge relative to the digital, and explain why the replicating call spread's notional diverges while its value converges.

34.10 Explain why an Asian option is worth about half a vanilla and a lookback about 1.67 times it, and why Asians dominate commercial usage while lookbacks do not.

34.11 Using Example 34.4, state the value of daily and of monthly monitoring relative to continuous, and explain why an unaligned tree that omits the continuity correction can land near the right price for the wrong reason.

34.12 Using Example 34.9, state the worst-of put's value relative to a single-asset put at $\rho = 0.20$, and explain why the correlation dependence runs opposite to a basket's.

34.13 Using Example 34.8, state the basket call's value relative to the basket of calls at $\rho = -0.5$ and at $\rho = 1$, explain what the gap represents, and explain why fifty assets cannot be handled with a single correlation.

34.14 Explain why a capped cliquet is short volatility, why its value sits entirely in the forward smile, and why that makes calibration to today's surface actively misleading rather than merely insufficient.

34.15 Explain why dealers run a structural long-exotic, short-vanilla book, and why exotics with static hedges trade at narrower spreads than others.

34.16 Using (34.11), price the down-and-in call of Example 34.1 from the single expression C, and explain why the up-and-in call with $H < K$ is worth the vanilla.

34.17 Name three instruments in this chapter with closed forms and severe model risk, and one with no closed form and very little. Explain why the two properties are nearly unrelated.

FURTHER QUESTIONS

34.18 Set $r = q$ in (34.11) and show that the down-and-in call's C is exactly K/H puts struck at H^2/K. Then show where the argument fails when $r \neq q$, and relate the failure to the

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exponents 2λ and $2\lambda - 2$.

34.19 Example 34.2 shows a thousandfold error reduction from node placement. Derive the effective barrier as a function of n, show that the resulting price error is first-order in the barrier displacement, and use that to explain the sawtooth's shape between jumps.

34.20 Derive the reflection result of Example 34.3 from the symmetry of the lognormal density under $r = q$. Then determine how the static hedge error scales with $(r - q)T$, and design a multi-maturity strip that reduces it, stating what it costs.

34.21 Section 34.7 reports a 49.5% model spread on a barrier. Propose a reserving policy: specify which models define the range, how the reserve is released over the trade's life, and how you would prevent the policy from being gamed by choosing convenient models.

34.22 Example 34.5 shows the overhedge premium rising with the spread's width. Derive the dealer's optimal width as a trade-off between the premium given up and the expected hedging loss, and state how it depends on the expected holding period.

34.23 Example 34.8 shows a basket option below a basket of options at all $\rho < 1$. Prove the inequality in general, determine the conditions for equality, and connect the result to the convexity argument of Section 11.3.

34.24 This chapter argues that spreads follow replicability rather than payoff complexity. Test the claim against the exotics in this chapter: rank them by replicability, rank them by the spreads you would expect, and identify any instrument where the two orderings disagree. Then explain the disagreement.

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Chapter 35
Structured Products and Autocallables

This is where most people meet derivatives. Far more households own a structured note than have ever traded an option, and the note is sold by a bank branch or a private banker rather than by a broker.

Chapter 12 said structured products are packaged strategies and took two apart with their option prices assumed. Chapter 34 priced the components. This chapter puts them together and asks the question the previous two could not: what does the buyer actually own, and is the price fair? Section 35.10 answers with a number: a $1,000 note worth $979 the day it is sold, the $21 sitting inside the coupon rather than beside it.

The chapter is deliberately blunt about a conflict. Every instrument here is legitimate and most serve a real purpose. But the products are sold on a headline number (a participation rate, a coupon), and the risk sits in terms that do not appear in the headline. That gap is the subject.

Section 35.1 takes the common notes apart into a bond and options, and Sections 35.2 and 35.3 price the two that dominate, the reverse convertible and the autocallable, and show what the autocallable's return distribution looks like. Sections 35.4 and 35.5 cover the secondary market and the remaining families. Sections 35.6 to 35.8 turn to the issuer's side: how the embedded option is hedged, what happens when many notes share a barrier, and why most notes reference the worst of several underlyings. Sections 35.9 to 35.11 are about what the buyer pays for (the issuer's credit and the fee) and give a procedure for evaluating a note; Section 35.12 asks what disclosure can and cannot fix, and Section 35.13 who buys these products and why.

35.1 TAKING A NOTE APART

Every structured note decomposes into a bond and a derivative. Doing the decomposition is the single most useful skill in this chapter, because it converts a marketing document into a portfolio.

The decompositions are short, and the rest of the chapter refers back to them. Write F for the face amount, S_0 and S_T for the underlying at issue and at maturity, and $N = F/S_0$ for the number of option units a note of face F carries. In every case the redemption is a bond plus a position in options, and the only question is which options and on which side.

Principal-protected note. The buyer is promised the full face back at maturity whatever the underlying does, plus a share p, the participation rate, of any rise in the underlying over the note's life. A note with 90% participation on an index that rises 20% repays 118 per 100 of face; if the index falls, it repays 100. At maturity the redemption is

$$V_T = F + p\,N\,(S_T - S_0)^+ \tag{35.1}$$

a zero-coupon bond plus pN calls struck at the initial level. The bond secures the guarantee, and whatever the bond leaves over buys the calls, which is why the participation rate is a residual

rather than a promise, and why Example 35.1 finds it governed by the issuer's funding cost.

Capped participation. The same promise, except that the share of the rise is paid only up to a ceiling: above a level C the redemption stops growing. In exchange the participation below the cap is higher, often above 100%. Capping the upside at C turns the call into a call spread:

$$V_T = F + p\,N\left[(S_T - S_0)^+ - (S_T - C)^+\right] \tag{35.2}$$

The sold call is what pays for the higher headline participation. A note advertising a participation above 100% has almost always sold something, and this is usually what.

Reverse convertible. The buyer receives a fixed coupon c well above deposit rates. At maturity the face is repaid in cash if the underlying stock or index ends at or above a barrier H; if it ends below, the buyer receives shares, or their cash value, instead and so takes the whole fall from the initial level:

$$V_T = F(1 + c) - N\,(S_0 - S_T)^+\,\mathbf{1}_{\{S_T < H\}} \tag{35.3}$$

a bond and a coupon, less a put the investor has written: struck at the initial level, but paying only when the underlying ends below H. Setting $H = S_0$ deletes the indicator and leaves a plain short put, which is the un-barriered version of the product. *The indicator is also the discontinuity*: the note pays $F(1 + c)$ just above H and $F(1 + c) - N(S_0 - H)$ just below it, and Section 35.6 is about what that does to the hedge.

Shark fin. Principal is protected and the buyer shares in the rise, usually at a high participation rate, but only as long as the underlying never trades above an upper level H during the note's life. If it ever does, the participation is lost and the buyer receives a small fixed rebate R instead. The name comes from the shape of the payoff, drawn in the bottom-right panel of Figure 35.1: flat at par below the initial level, rising steeply along the participation line up to H, then dropping back to par plus the rebate, the outline of a shark's dorsal fin. At maturity

$$V_T = F + p\,N\,(S_T - S_0)^+\mathbf{1}_{\{\max_{t\le T} S_t < H\}} \;+\; R\,\mathbf{1}_{\{\max_{t\le T} S_t \ge H\}} \tag{35.4}$$

a bond, an up-and-out call, and the rebate. The knock-out is cheaper than the vanilla call of (35.1), so the same budget buys more of it: that, and not a better view, is where a shark fin's headline participation above 100% comes from. Shark fins come in two forms. Some observe the barrier only at maturity, so the payoff depends on S_T alone and Figure 35.1 draws it exactly; others observe it daily or continuously, which is what (35.4) writes and Example 35.4 prices, and then a path that touches H and falls back is knocked out too.

Autocallable. The buyer receives a high coupon, but the note can end early. On each of a series of observation dates, if the underlying is at or above a trigger level, the note is *called*: it repays the face plus all the coupons accrued so far, and stops. If it is never called, it repays the face with all the coupons at maturity unless the underlying ends below a barrier, in which case the buyer is repaid in shares, takes the loss as in a reverse convertible, and forfeits the coupons. In symbols, take observation dates $t_1 < \cdots < t_n = T$, an autocall trigger K_a, a coupon c per annum with each t_i measured in years, and a barrier H, observed here only at maturity for simplicity; in practice it is usually observed daily, as in Example 35.2. Let τ be the first date on which $S_{t_i} \ge K_a$, and $\tau = \infty$ if that never happens. Then

$$V = \sum_{i=1}^{n} F(1 + c\,t_i)\,\mathbf{1}_{\{\tau = i\}} \;+\; \Big[F(1 + cT)\,\mathbf{1}_{\{S_T \ge H\}} + N S_T\,\mathbf{1}_{\{S_T < H\}}\Big]\mathbf{1}_{\{\tau = \infty\}} \tag{35.5}$$

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a strip of digital payments, each conditional on not having been called already, plus, if the note is never called, the written put of (35.3): below the barrier the buyer receives shares worth NS_T and forfeits the coupons. Two things follow that the **term sheet** (the document setting out the note's terms for the buyer) does not say. *The investor is short volatility twice*: once in the digitals, which want the underlying quietly above K_a, and once in the put. And a reverse convertible is the case $n = 1$ with no trigger, except that it keeps its coupon below the barrier. The two share their downside, a written put that pays only below the barrier, and are sold to the same buyer on the same high coupon, but their risk profiles differ: early redemption cuts the autocallable's expected life well below its stated term, so the buyer collects the coupon for less time while carrying the same tail, and the outcome depends on the path as well as on where the underlying ends.

Dual-currency deposit. The buyer places a deposit in one currency and earns a rate above the ordinary deposit rate. At maturity the bank chooses the currency of repayment (the original one, or a second currency at a conversion rate fixed at the outset), and it will choose whichever is worth less to the depositor. Write X_T for the exchange rate at maturity, in units of the second currency per unit of the deposit currency, K for the conversion rate, and c for the interest, paid in the deposit currency whatever happens. Measured in the deposit currency, the redemption is

$$V_T = F(1 + c) - \frac{F}{X_T}\,(X_T - K)^+ \tag{35.6}$$

a deposit less a written call on the deposit currency struck at the conversion rate. When the deposit currency strengthens past K, the principal comes back converted at K and is worth only FK/X_T. Example 35.5 carries the arithmetic, and the enhancement is the option premium, as it is in every structure above.

One identity organizes all of them. At issue,

$$\text{issue price} = \underbrace{\text{PV(principal)} + \text{PV(coupons)}}_{\text{the bond}} + \underbrace{\text{options bought} - \text{options sold}}_{\text{the derivative}} + \text{fee} \tag{35.7}$$

Figure 35.1 draws the first four on one scale, and the shapes carry the argument. In each panel the dashed line is what the same 100 would return left on deposit, at 4% for a year, so the gap between it and the solid line is what the embedded options add or take away against simply earning interest. The principal-protected note repays 100 where the deposit pays 104, which is the interest given up to buy the participation; the reverse convertible pays 110 above its barrier, the extra 6 being the premium for the put the buyer has written, and falls far below the deposit once the barrier is breached. Where each solid line crosses the dashed one marks the range of the underlying in which the buyer comes out ahead:

- above 104.4 for the principal-protected note;
- above 104 for the capped note;
- anywhere at or above the barrier of 70 for the reverse convertible;
- only between 103.6 and the knock-out at 130 for the shark fin.

Each note is a bet that the underlying ends in its own range, and the reverse convertible's range is the widest because its losses, when they come, are the largest. Every term but the fee is quotable, and that is what makes the decomposition worth doing: price the bond at the issuer's own funding curve, price the options in the market, and the residual is what the note costs to own.

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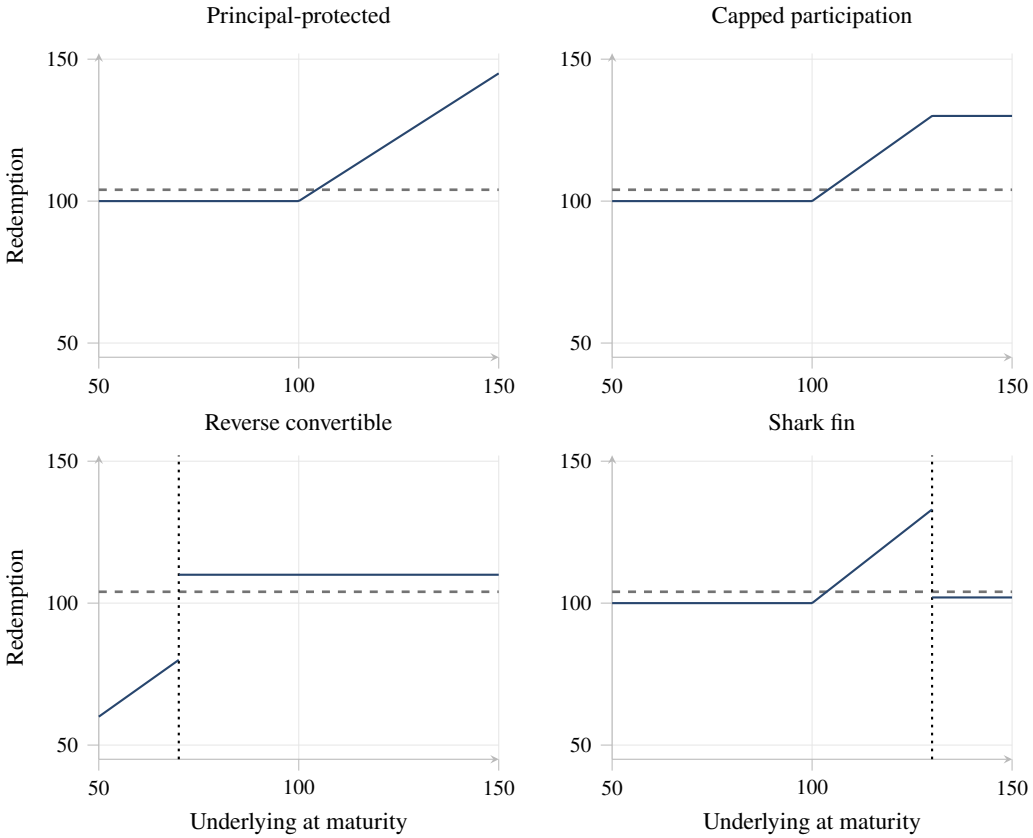


Figure 35.1: Redemption per 100 of face, initial level 100. Participation 90%, cap 130, barrier 70 with a 10% coupon, knock-out 130 with a 2% rebate. Dashed: 100 left on deposit at 4% for a year. Illustrative.

Example 35.1 Where the participation rate comes from

Example 12.7 took this note apart: five years, a zero-coupon bond worth \$79.85 per \$100 of face at 4.5% (\$798.52 on the \$1,000 face used here), the option budget it leaves, and what a 2% fee takes out of that budget. It held the call's price fixed. Here the index yields 2.0% with 20% volatility, and the issuer's own funding rate discounts the at-the-money call as well as the bond (the call costs \$209.60 at 4.5%), so both legs move with funding:

Issuer funding rate	Participation (no fee)	(2% fee)
2.0%	59.4%	46.9%
3.0%	77.7%	66.6%
4.5%	96.1%	86.6%
6.0%	107.3%	99.0%

The participation rate is mostly a statement about the issuer's funding cost, not about the index. It nearly doubles as funding goes from 2% to 6%, with the index unchanged.

So across issuers at one moment, a higher participation rate signals a weaker credit. The issuer paying more to fund can buy more optionality, the best-looking note in a comparison is systematically the one with the most credit risk, and Section 35.9 says what that has cost.

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35.2 REVERSE CONVERTIBLES AND AUTOCALLABLES

The reverse convertible comes first, because an autocallable is a reverse convertible with early redemption added. The investor in a reverse convertible is short the put of (35.3), and the coupon is the premium, as Example 12.8 showed: a note paying 10% when deposits pay 4% is not offering 6% of extra yield; it is paying 6% for a put the investor has written. The mis-selling risk is structural rather than incidental. The product is described in the vocabulary of fixed income (coupon, maturity, principal), and its risk is entirely equity, so a buyer who compares it with a bond is comparing the wrong thing.

The autocallable is the dominant structured product globally, and the rest of this chapter is built on it.

The mechanics are simple to state and hard to price. The note pays a coupon and terminates early ("autocalls") if the underlying is above a level on an observation date. If it never autocalls, principal is returned at maturity together with the accrued coupons *unless* the underlying has breached a lower knock-in barrier, in which case the investor takes the loss and forfeits the coupons.

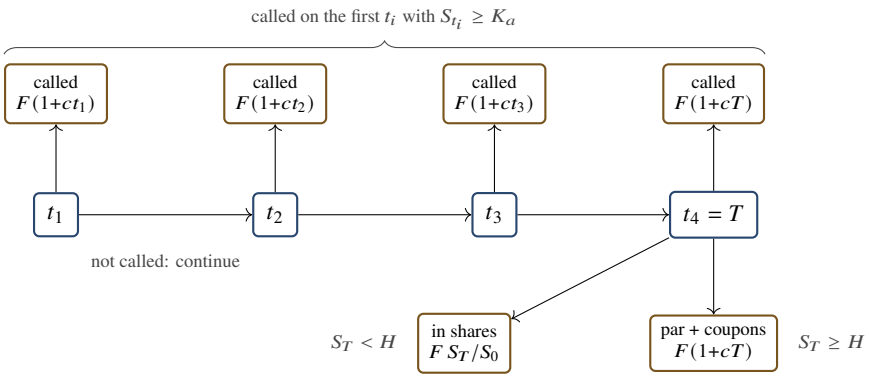


Figure 35.2: Each observation either redeems the note with the coupons accumulated to that point or passes it on. Only the final date can end below the barrier.

Figure 35.2 is (35.5) drawn. The note is a sequence of chances to be redeemed early, each paying the coupons banked so far, and only the last one can end in shares.

Example 35.2 Pricing one, and what the buyer owns

A two-year note on an index at 100. Autocall at 100% observed monthly after a two-month **lockout period**, during which the note cannot be called, so the first observation is at month three; knock-in at 75% observed daily. The coupon accrues at an annual rate c: a note called at the end of month m repays $1 + c\,m/12$, and one that is never called and never knocked in repays $1 + 2c$ at maturity. If knocked in and never autocalled, the investor receives $\min(S_T/100, 1)$ per unit of face and no coupon. Volatility 25%, rate 2.5%; 80 million antithetic paths (200 seeds of 400,000), which put the fair coupon's standard error at 0.005 of a percentage point. The coupon c is the unknown: it is set so that the note is worth exactly its issue price of 1.00, which makes it the *fair coupon*.

P(autocalls at some point)	0.8311
P(knocks in at some point)	0.4132
P(**knocked in and never autocalled**)	**0.1645**
Fair coupon	**18.42% per annum**

An 18.4% coupon, and the note prices at par. Nothing is being given away: the coupon is

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exactly what the embedded short put is worth.

The probability structure is what sells it and what should worry the buyer. The note produces its advertised return about 83% of the time. That is why these products are easy to sell and why holders report satisfaction for years at a stretch.

The remaining 17% is where all the loss is, and it is not a small loss—a knocked-in note pays $\min(S_T/100, 1)$, so the investor absorbs the full downside from par.

This is a negatively skewed payoff (frequent small gains, rare large losses) sold on its mean and its hit rate, which is the same structure as Section 22.8's inverse volatility products and Section 33.6's range accrual, and it recurs because it is what sells.

Now consider the exposure the buyer does not see:

Realized volatility	**Note value at** $c = 18.42\%$
15%	1.0520
20%	1.0204
25%	**1.0000**
30%	0.9847
40%	0.9574

The note is short volatility. It was priced at 25%; realized volatility above that makes it worth less than par, and the investor never agreed to a volatility view.

And it is short volatility in the worst way, because volatility rises when markets fall, which is exactly when the knock-in is approached. The two risks are not independent—this is Section 19.9's leverage effect and Section 27.9's wrong-way structure appearing in a retail product.

35.3 WHAT THE RETURN DISTRIBUTION LOOKS LIKE

A fair price says the mean is right. It says nothing about the shape, and the shape is what the buyer is actually choosing.

Example 35.3 The same expected return, a different distribution

Simulate Example 35.2's note at its fair 18.42% coupon on the same 80 million paths, and record the investor's total return, then do the same for simply holding the index for two years.

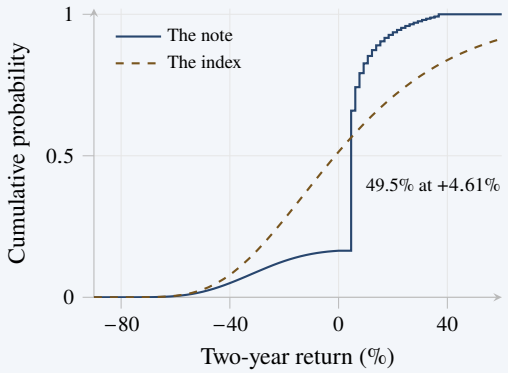
	The note	**The index**
Mean total return	+1.60%	+5.13%
Standard deviation	17.25%	38.36%
Skewness	**−1.59**	**+1.14**
Excess kurtosis	2.83	—
P(positive)	**83.55%**	48.59%
P(loss > 25%)	11.02%	—
5th percentile	−40.03%	−44.79%

Everything here is under the pricing measure, so both instruments earn the risk-free rate by construction. The note's lower mean is not a worse deal: it terminates early, after 0.70 years on average, so it earns that rate over a shorter life. $2.5\% \times 0.70 = 1.75\%$, against the simulated 1.60%, which falls a little short because the paths that run longest, and so carry the largest growth factor, are the knocked-in ones that lose; that is the internal check that the pricing is consistent.

With the mean neutralized, the comparison is purely about shape, and the shape is the product.

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Median return	+4.61%
E[return | positive]	+8.22%
E[return | negative]	**−32.05**%
Average of the worst 1%	−60.23%



Wins average 8% and losses average 32%—a four-to-one asymmetry, at a five-to-one win rate. Those two ratios are the product, and they multiply out to a fair price.

Three observations follow.

Skewness measures a distribution's asymmetry: it is negative when the long tail lies on the loss side, so that small gains are frequent and large losses rare, and positive when the long tail lies on the gain side. (Excess kurtosis, the row below it, measures how much more of the distribution sits in the tails than a normal distribution's, for which it is zero.) The note's skewness is negative and the index's is positive. A buyer who moves from the index into the note has not reduced risk; they have exchanged a positively skewed exposure for a negatively skewed one at the same expected return, which is the opposite of what "capital-efficient" and "defensive" suggest.

The 25th and 50th percentiles are identical, at +4.61%. Nearly half the distribution, 49.5%, is a single point (the note autocalled at the first opportunity), so every percentile from the 17th to the 65th is +4.61%, which is why holders experience these products as reliable. The plot beside the second table draws both distributions cumulatively: the note's single point is the 49.5-point vertical step at +4.61%. Below zero the note's curve lies under the index's, because a loss is far less likely (16.45% against 51.41%), but the note can never pay more than +36.84%, the level by which the index has reached only its 82nd percentile.

And the standard deviation is genuinely lower than the index's, 17.2% against 38.4%. A risk measure that sees only the second moment will report this product as safer, which it is on that measure and is not on any measure that looks at the tail. Chapter 38's warning that a single-number risk measure can miss what sits in the tail is not abstract; *this is the product it describes*.

35.4 THE SECONDARY MARKET

A structured note is quoted daily and is not liquid, and the distinction matters more than it sounds.

The issuer is usually the only bid. There is no exchange, no competing market maker, and no obligation to quote. A buyer who wants out sells back to the party who sold it to them, at a price that party computes.

Three consequences follow.

The bid embeds a second spread. Having paid the issue premium of Section 35.10, an early seller pays again on the way out. Round-tripping a note in the first year can cost several percent of principal before any market move.

The quoted price is a model output. These are Level 2 or Level 3 valuations in the accounting sense (prices computed from a model, with observable or unobservable inputs respectively, because no active market quotes the instrument), and Section 26.12's observation about the crisis applies in miniature: when the only price comes from the party on the other side, the holder cannot distinguish a mark from a quote.

And illiquidity interacts with the payoff. An investor most wants to exit after a sharp fall,

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when the knock-in is close and the outlook has changed, and that is exactly when the bid is lowest and the embedded option's value has moved most against them. The exit is worst when it is most wanted, which is this book's recurring wrong-way structure appearing one last time.

Case Study 35.1 A market that learned this from its losses

South Korea developed one of the world's largest retail markets for autocallable notes, sold as **equity-linked securities** through bank branches to ordinary savers. A large share referenced the Hang Seng China Enterprises Index (HSCEI), chosen because its high volatility supported high coupons.

The structure worked as designed for years. Notes autocalled, holders were repaid early with the advertised coupon, and the proceeds were rolled into new notes: the 83% outcome of Example 35.3, repeated often enough to look like a rule rather than a probability.

Then the index fell a long way and stayed there. Notes issued in 2021 carried knock-in barriers set against 2021 levels; by the time those notes reached maturity in 2024, the index had not recovered, and the loss that Example 35.2 gives a 17% probability became the realized outcome for a large cohort at once.

Four features of the episode are general and worth separating from the specifics.

The losses were synchronized because the issuance was. Notes sold during one period share a vintage, a barrier level, and a maturity. A product with a 17% loss probability sold to a hundred thousand people in the same quarter does not produce 17% of them losing independently; it produces all or none.

The distribution channel mattered as much as the product. The notes were sold through bank branches to savers, alongside deposits, by staff whose training and incentives were not designed for a negatively skewed derivative.

The regulatory response focused on sales practice rather than on the instrument: **suitability** assessment (a check, before the sale, that a product fits the buyer's circumstances, knowledge and objectives), recording requirements, limits on where such products may be sold. That is the right diagnosis if one accepts that the instrument is legitimate and the mismatch was between the payoff and the buyer.

And the mechanism was entirely visible in advance. Everything in Sections 35.2 through 35.3 could have been computed at issue from the term sheet. Nothing about this episode required hindsight, which is what distinguishes it from most of the case studies in this book.

35.5 OTHER FAMILIES

Four more structures account for most of the rest of the market, and each is a recombination of what has already been priced.

Shark-fin notes pay an enhanced participation up to a barrier and a small fixed rebate if the barrier is touched: an up-and-out call plus a rebate. *The buyer is selling the tail of the upside*, and the appeal is that giving up the potential of returns above 30% feels costless to someone who expects 10%. Example 35.4 prices what the feeling costs.

Example 35.4 Where the headline participation comes from

A one-year \$1,000 note at a 3% funding rate, on an index at 100 with 25% volatility. The zero-coupon bond costs 970.45, leaving **29.55** for the option and its rebate.

An at-the-money call is worth 11.3485 per 100 of index, so that budget buys **26.0%** participation, an unattractive number to print on a term sheet. Adding a knock-out at 130 drops the option

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to 2.1766, but a shark fin also pays a rebate, here 2% of face at maturity if the barrier is touched. The touch probability, from Section 34.3's formula, is 0.292, so the rebate costs $1{,}000 \times 2\% \times e^{-0.03} \times 0.292 = 5.68$, and the remaining 23.88 buys

$$\frac{23.88}{21.766} = \textbf{109.7\% participation}$$

The participation rate has more than quadrupled, and what has been given up is every path that touches 130 at any time before maturity, not only those that end above it.

The question the term sheet does not answer is how often those outcomes occur. That touch probability, **29.2%** under the pricing measure, is the answer: in nearly a third of scenarios the 109.7% participation is extinguished and the note pays its small rebate—and the knock-out happens in the states where the index rallied hardest. The buyer has not sold a remote tail; they have sold roughly the three strongest rallies in ten.

Two features make this harder to see than the autocallable's. When the barrier is monitored continuously, as here, a single intraday touch during a rally destroys the participation for the rest of the year even if the index closes the year at 128. And the rebate is quoted as a percentage of face rather than as what it replaces: a 2% rebate against a forgone 109.7% participation on a 30% move is 2% against 32.9%. The instrument is Example 34.6's reverse knock-out sold retail, with the same discontinuity and none of the warning.

Dual-currency deposits pay an above-market rate in one currency, with the issuer holding the right to repay in another at a preset rate. *It is a deposit plus a short FX option*, and it is among the most widely sold structured products in Asian private banking. The presentation as a "deposit" is the problem: the principal is at risk in currency terms and the vocabulary says otherwise. Example 35.5 shows that the entire enhancement is the option premium, to the basis point.

Example 35.5 The deposit that is not one

A three-month dollar deposit of $100,000 with a conversion rate of 153 against a USD/JPY spot of 150. Dollar deposits pay 4.5%, yen 0.5%, and three-month volatility is 10%. The depositor has written a dollar call struck at 153: if the dollar strengthens past it, the bank repays in yen at 153. Interest is paid in dollars whatever happens; only the principal can be converted.

Chapter 16's Black–Scholes–Merton formula, with the yen rate for discounting and the dollar rate as the yield q (the form known as Garman–Kohlhagen), values that option at **1.2776** yen per dollar, which is 0.852% of the dollar notional over the quarter, or **3.41% annualized**. So:

$$4.5\%\ \text{(deposit)} \ + \ 3.41\%\ \text{(written option)} \ = \ \textbf{7.91\%}$$

The advertised 7.9% rate is the deposit rate plus the premium, and nothing else. A depositor who understood the decomposition would ask whether 3.41% a year is a fair price for the option—which is the question the product's presentation prevents, because a deposit rate is not something one negotiates.

The outcomes are asymmetric in the familiar direction:

USD/JPY at maturity	Repaid in	Quarter's return
140	dollars	+1.98%
150	dollars	+1.98%
153	dollars	+1.98%
160	yen at 153	−2.40%
165	yen at 153	−5.30%
175	yen at 153	−10.59%

The gain is capped at one quarter's coupon and the loss is not capped at all, which is Section 35.2's shape in a three-month product and a different asset class. And the conversion

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arrives at the worst moment: a dollar-based investor is repaid in yen precisely when the yen is weak.

Credit-linked notes embed the short CDS position of Section 25.15. The whole of Part V's credit machinery applies, and the retail versions are the ones where the reference entity's identity is least understood by the buyer.

And range accruals appear as notes as well as swaps, with the coupon arithmetic and the short-vega exposure of Section 33.6 unchanged.

Hybrids attach a second condition from another asset class to the credit one (a coupon paid only while an issuer survives *and* an index is above its strike), and they are the exception to this section's arithmetic. The two legs cannot be priced separately and added, because the value depends on a correlation between default and the other factor that neither market quotes; Section 25.16 works the example, and the practical consequence for a buyer is that the usual decomposition is unavailable to them as well as to the dealer.

The pattern across the first four is the one this chapter began with. The enhanced headline term is paid for by an option the buyer has written, and the option is described by its trigger rather than by its risk.

The exchange-traded buffer fund is the contrast case. A defined-outcome fund holds a deep in-the-money call, a put spread that supplies the buffer, and a sold call that finances the cap, and it resets each year. Its cap and buffer are quoted, its terms are rules rather than a negotiated term sheet, no issuer's credit is involved, and its cost shows in the expense ratio, so it supplies the comparison for Sections 35.9 and 35.10.

35.6 ISSUER HEDGING, AND THE DISCONTINUITY

The investor's position implies the issuer's, and the issuer's hedging is where these products stop being a private matter between two parties.

The investor is short a down-and-in put; the issuer is long it. A long put has negative delta, so the issuer hedges by holding the index, and the size of that holding is what matters.

Example 35.6 Why the hedge stabilizes, until it does not

The embedded option: a down-and-in put struck at 100 with a barrier at 75, one year, with Example 35.2's 25% volatility and 2.5% rate and no dividends. Valued exactly with the down-and-in put row of Section 34.3's barrier formulas. The note's knock-in is observed daily, which Section 34.6's correction handles by moving the barrier away from the spot, to 74.32; that lowers the value at 100 from 6.160 to 5.956 and leaves the shape of the delta, and its jump at the barrier, essentially unchanged, so the continuous barrier is used below.

Index	Option value	Delta
130	0.706	−0.055
110	3.128	−0.220
100	6.160	−0.399
90	11.365	−0.652
80	19.262	−0.921
76	23.124	−1.006
75.33	23.802	**−1.017**
74.67	24.411	**−0.827**
74	24.968	−0.836
70	28.414	−0.885

Above the barrier, delta steepens monotonically from −0.055 to −1.017 as the market falls, and it *exceeds one in magnitude*, a barrier feature Section 34.3 flagged.

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The issuer, long this option, must therefore hold a growing long position in the index as the market falls. From 130 down to the barrier the hedge grows from 5.5% to 101.7% of notional. The issuer is buying into a falling market, continuously, and that flow is stabilizing. It is the gamma term of Chapter 18's expansion, equation (18.1), with the sign Section 18.12 showed dampens a move: above the barrier the issuer is long gamma, so keeping its delta term canceled means buying as the market falls.

Then the barrier is touched and the delta jumps from −1.017 to −0.827: a change of +0.19.

Two things happen at once and the second is the larger. The issuer must *sell* about 19% of notional immediately. And *the buying stops*: below the barrier the option is an ordinary put whose delta steepens far more slowly, so the persistent bid that had been supporting the market on the way down simply disappears.

The popular account of these episodes emphasizes the forced selling. The arithmetic says the withdrawal of the bid is the bigger effect, and stating it that way changes what a regulator should monitor: not the size of the sell order at the barrier, but the size of the stabilizing flow that is about to vanish.

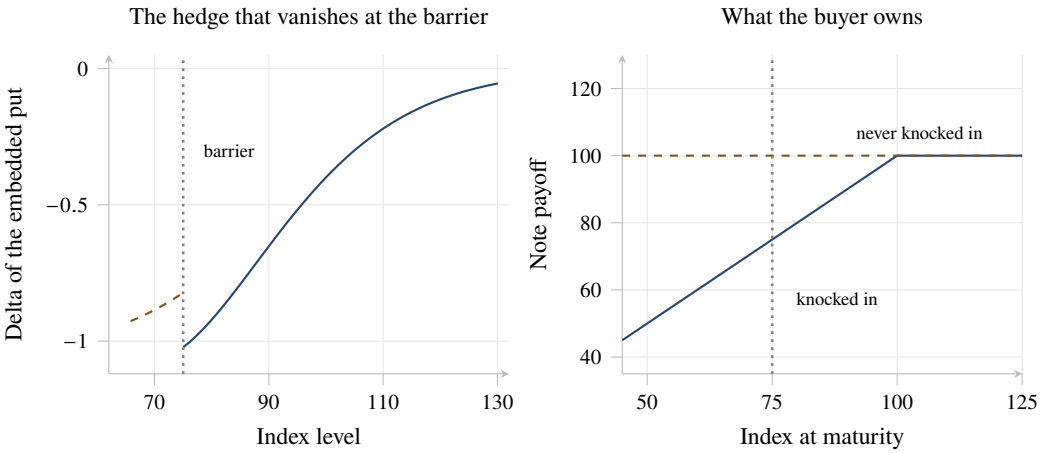


Figure 35.3: Left: Example 35.6's delta. Right: maturity payoff, coupon excluded.

Figure 35.3's left panel is the mechanism in one picture. The gap at the barrier is the discontinuity, and the steep approach to it on the left is the stabilizing flow that ends there. Its right panel is the maturity payoff, coupon excluded: flat at par if the barrier was never touched, and the index level itself, up to par, if it was.

35.7 WHEN MANY NOTES SHARE A BARRIER

One note's hedging is irrelevant to a market. The problem is that barriers cluster.

Notes are issued continuously against a rolling reference level, so a sustained decline of a given size brings a large cohort of barriers into range at once. The clustering is not coincidence but arithmetic: barriers are set at round fractions (75%, 80%) of the level prevailing at issue, so notes issued during any period of stable prices share nearly the same absolute barrier.

Three consequences follow, in order of increasing seriousness.

The stabilizing flow is larger than any one issuer knows. Each dealer sees its own book, which is Section 27.13's aggregation problem in a different market.

The withdrawal is synchronized. When the index passes the cluster, many issuers' bids disappear within the same few sessions.

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And the feedback is self-reinforcing while it lasts. The buying on the way down slows the decline, which keeps the index above the barriers longer, which sustains the buying—until it does not. A stabilizer that works until it stops is not a stabilizer, and Section 23.10's saturating pin risk made the identical point about a different product.

Case Study 35.2 A cluster passed through: China, January–February 2024

China's structured-product market is built around the **snowball**, the domestic form of the autocallable, and its standard terms track Example 35.2 closely: a CSI 500 or CSI 1000 underlying (the China Securities Index Company's mid-cap and small-cap indices), twelve to twenty-four months, monthly knock-out (autocall) observation after a lockout of about three months, a knock-in observed daily at 70–80%, and coupons of 15–20% a year. The underlying is chosen for its volatility, which is what pays the coupon (the same reason Korea's notes referenced an offshore China index), and that coupon range is the same order as the 18.42% of Example 35.2, because it is the same option at the same volatility.

One feature of the market differs from Korea's in the way that matters most. Snowballs are sold by securities firms acting as counterparty, wrapped in asset-management products with a minimum subscription around RMB 1 million, so the buyer is a qualified investor rather than a depositor. The second of the Korean episode's four general features, that the distribution channel matters as much as the payoff, had been addressed institutionally, and in advance.

What had not been addressed was the synchronization. Barriers were set at 75% or 80% of the index level prevailing at issue, so notes written through a comparatively stable 2023 shared nearly the same absolute knock-in level. The decline of January and February 2024 went straight through it.

Both of Example 35.6's effects are identifiable in what followed, and the popular account has their sizes the wrong way round. Dealers hedging the long down-and-in put had been buying index futures as the market fell, and that flow was supporting it. At the barrier the immediate sale is about 19% of notional, while the larger effect is that the bid supporting the market on the way down disappeared. So snowball hedging did not cause the fall; it was a mechanism that first slowed the decline and then stopped slowing it—and in a price series, "stopped slowing" looks a great deal like "caused."

Attribution is harder still because a second leveraged chain, quantitative funds' equity positions financed through broker swaps, was unwinding in the same week, and separating the two needs dealer-level position data that is not public.

The regulatory response is the informative part, because most of it ran ahead of the event. Suitability requirements for snowball sales, an explicit instruction that the product is not principal-protected, and later limits on how much of a private fund could be put into snowballs (reported from 2022) and on how much a dealer could issue against its capital were all in place before 2024. *None of them addressed the clustering*, because suitability governs who may buy, and the cluster is a fact about how many investors bought notes with nearly the same barrier level in the same quarter. A rule aimed at individual prudence does not solve a problem about an aggregate, which is Section 27.13's conclusion, reached again in a different market. The measures after the cluster confirmed the diagnosis. Press reports in April 2024 said regulators had told some large brokers to halt any rise in their net exposure to over-the-counter derivatives linked to A shares, snowballs included, and on October 25, 2024 the Securities Association of China issued measures on income certificates under which a floating-income certificate's loss on its underlying may not exceed its fixed-income amount.

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35.8 WORST-OF UNDERLYINGS

Section 34.11 priced the mechanism; here is what it does commercially.

Most autocallables reference the *worst* of two to four underlyings. *This roughly doubles the coupon*, and Example 34.9 showed why: at a correlation of 0.20 a worst-of put on three assets is worth 2.05 times a single-asset put.

The disclosure problem is acute and simple. A note "linked to three major indices" sounds diversified. It is the opposite: the investor is exposed to whichever of the three performs worst, so adding an underlying can only increase the risk and can only increase the coupon.

And the coupon's size is the tell. A coupon far above what a single-underlying note pays is not a better deal; it is a measure of how much extra risk was added, and it is the one number in the term sheet that reliably signals the risk it was designed to obscure.

35.9 THE ISSUER'S CREDIT

A structured note is an *unsecured obligation of the issuer*. The derivative is embedded in a bond, and if the bond defaults the derivative goes with it.

Principal protection means protection from the market, not from the issuer. That distinction was academic until September 2008, when retail holders of notes issued by a failed investment bank discovered that "100% principal protected" ranked alongside every other senior unsecured claim.

The pricing consequence is Example 35.1's uncomfortable corollary. The issuer's funding spread *is* the source of the note's attractive terms, so, other terms equal, a better-looking note is a worse credit. The investor is being paid for taking issuer credit risk, and in almost no case is that presented as the reason the terms are good.

Two developments have improved this and neither removes it. Disclosure rules now require the issuer's credit to be stated, and some jurisdictions require an estimated initial value (the note's fair value net of fees) to be printed on the term sheet. The estimated-value disclosure is the single most effective reform in this chapter's subject, because it makes Example 35.7's fee arithmetic visible without requiring the buyer to do it.

35.10 WHERE THE FEES ARE

The economics of the business are worth stating plainly, because they explain the product design.

A structured note carries no explicit management fee. The compensation is the difference between the note's issue price and its fair value, typically 1.5% to 3% of face for a simple structure and more for a complex one, earned at issue.

This creates a specific incentive: complexity is profitable because it is hard to price. A buyer can check a participation rate against a competitor's; the fair value of a worst-of autocallable on three underlyings with a monthly autocall and a daily knock-in is not something a buyer can check at all. The margin is largest where the price is hardest to verify, which is the same structure Section 33.13 identified in non-standard swaps and Section 34.13 in exotics.

The fee is not an estimate once the fair coupon is known, and Example 35.7 extracts it from the numbers already computed.

Example 35.7 Reading the fee out of the coupon

Example 35.2's note prices at par on a coupon of **18.42%**. Suppose the same structure is issued at 1,000 paying **12.65%**.

The factor that converts a coupon into value is not the 0.70-year average life of Section 35.3. A note that knocks in and is never called runs the full two years and pays no coupon at all, and those

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paths carry nearly half of that average; what remains is then discounted. Simulating the note gives the factor directly: each point of annual coupon is worth **0.366** points of face. So the value given up is

$$(18.42\% - 12.65\%) \times 0.366 = 2.11\% \text{ of face}$$

and the note's fair value at issue is about **979** against an issue price of 1,000. The $21 is the fee, and it sits inside the coupon rather than beside it.

Three things follow that a term sheet does not say. The fee is paid at issue and cannot be recovered by holding to maturity: it is not an annual charge of 2.11% but a single 2.11% taken on day one, which on a note with a 0.70-year expected life is an annualized **3.0%**. *The buyer who sells early pays it twice*, once at issue and again in Section 35.4's exit spread. And *the fee is invisible in every scenario table*, because those tables are computed from the stated coupon and the stated coupon is already net of it.

This is precisely what the estimated-initial-value disclosure repairs. Printing "estimated initial value: $979 per $1,000" converts an argument about fairness into a number a buyer can compare across issuers, without requiring anyone to understand a single thing about barrier options. It does not attempt to explain the product; it prices it.

Illustrative magnitudes follow, scaled to complexity:

	Fair value	Gap to par
Principal-protected note, one index	0.9800	2.00%
Reverse convertible, one index	0.9750	2.50%
Autocallable, one index	0.9700	3.00%
Autocallable, worst-of three	0.9550	**4.50%**

The gap widens exactly as the buyer's ability to check it narrows.

The defense, which deserves stating, is that the alternative is worse for the buyer. These structures cannot be assembled retail: the components are OTC, the sizes are institutional, and the hedging is continuous. A margin for manufacturing a payoff that could not otherwise be bought is a legitimate charge. The question is its size and whether it is visible, not whether it should exist.

35.11 HOW TO EVALUATE A NOTE

The chapter's material assembles into a procedure. It is short, it requires no model the reader does not now have, and it answers the questions a term sheet is designed not to raise. It has five steps:

1. *Decompose it.* Every note is a bond plus one or more options. Identify the bond, identify each option, and, critically, identify which side of each option the buyer is on. In Example 35.1 the buyer is long a call; in Section 35.2's reverse convertible and Example 35.2 the buyer is short a put. If the headline term is attractive, the buyer is short something.

2. *Price the bond leg at the issuer's own curve.* Example 35.1 showed that an attractive participation rate comes mostly from the issuer's funding spread: an issuer that pays more to borrow spends less on the zero-coupon bond and has more left to buy options. A note from an AA issuer and one from a BBB issuer are not comparable on their headline terms, and the difference between them is credit compensation, not investment skill.

3. *Value the option leg and compare with par.* The gap is the fee. Where an estimated initial value is disclosed, this step is done for the reader; where it is not, the table in Section 35.10 gives the order of magnitude, and the rule that it widens with complexity is reliable.

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4. *Simulate the distribution, not the scenarios.* A term sheet presents three or four illustrative outcomes. Example 35.3 shows what the term sheet omits: the skewness, the conditional loss given a loss, and the fact that the median and the 25th percentile can be the same number. A buyer who sees only $E[\text{return}]$ and $P(\text{positive})$ has seen the two statistics the product was engineered to optimize.

5. *Ask what the exit is worth.* Section 35.4: the issuer is the only bid, the spread applies again, and the bid is lowest when the need to sell is greatest. A note should be evaluated as a hold-to-maturity instrument, because that is what it is.

Applied to Example 35.2's note, the five steps produce a description no marketing document contains: a two-year unsecured loan to the issuer, bundled with a written down-and-in put on an index, sold, at Example 35.7's 12.65% coupon, about 2.1% above fair value, returning a positive amount with 83% probability (most often 3.2%, on average 5.6%) and on average −32% with 17% probability, and not sellable except back to the issuer.

That description is neither hostile nor favorable. It is the instrument, and a buyer who wants it after reading it is making an informed choice—which is the entire objective.

35.12 SUITABILITY, AND WHAT DISCLOSURE CANNOT FIX

Disclosure is necessary and it is not sufficient, and this chapter's material shows precisely where it stops working.

What disclosure fixes well. Fees and fair value: the estimated-initial-value requirement converts an invisible cost into a printed number, and it works because the quantity is a single figure the issuer already computes. Section 35.10's gap is exactly the kind of thing disclosure is good at.

What it fixes poorly. A distribution cannot be disclosed into intuition. A buyer told that the product has a skewness of −1.59 has been given a true fact and no understanding, and a buyer shown three illustrative scenarios has been given an understanding that is wrong. The gap between Example 35.3's two columns is not a disclosure failure; it is a comprehension problem, and jurisdictions that have tried to solve it with more disclosure have generally produced longer documents and unchanged outcomes.

What it cannot fix at all is the correlation between the product's failure and everything else the buyer owns. An autocallable knocks in when equity markets fall, which is when the buyer's other assets have fallen, their employer is under pressure, and their appetite for risk has collapsed. No disclosure of the product in isolation reveals that, because the fact is about the portfolio and not about the note.

That points at the reform this chapter would actually endorse: not more disclosure, but a suitability test framed around the payoff's shape rather than the buyer's stated risk tolerance, because a buyer who says they want "income with capital preservation" has described a payoff the product does not have, and no amount of the buyer confirming that they understand the risks changes that mismatch.

Two limits on that recommendation need stating, so that the argument is fair. Suitability regimes are paternalistic and impose real costs, including on buyers who understand the product perfectly well. And the demand is genuine: the yield problem these products solve is real, and a regime that removes them without addressing that demand pushes it somewhere less regulated. This is a trade-off rather than a solved problem, which is why Section 35.13 names the buyers rather than dismissing them.

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35.13 WHO BUYS THESE, AND WHY

Retail and private-bank clients are the volume, and the demand is genuine rather than manufactured. The dominant motive is yield in a world where deposits pay less than the buyer needs, and the second is a preference for defined outcomes over open-ended ones. Both are real preferences and neither is irrational.

Insurers and pension funds buy principal-protected structures for the reason Section 31.17 gave: their liabilities contain guarantees, and a note with a floor matches them better than an unhedged equity holding.

Corporates occasionally buy them as treasury investments, and the suitability question is sharper there than in retail, because the sums are larger and the governance is often thinner.

Issuers sell them for funding and for the margin. The funding motive is underrated: a structured note is term funding raised from a source that does not otherwise lend to banks, and it is stickier than wholesale funding because the buyer cannot easily exit.

And dealers, frequently a different institution from the issuer, take the hedging risk, which is where Section 35.6's flows originate.

The recurring conclusion of the preceding three chapters applies here in its sharpest form. The client is short the optionality and the dealer is long it, and here the client is a household. That does not make the trade illegitimate, but it does mean the disclosure question is not a formality—it is the whole of the consumer-protection issue, and Section 35.9's estimated-value requirement is the reform that addresses it most directly.

SUMMARY

Every structured note decomposes into a bond and a derivative, and doing the decomposition converts a marketing document into a portfolio.

A principal-protected note spends most of its face on the zero-coupon bond and buys participation with the rest, and the distributor's fee cuts that participation noticeably. Participation rises steeply with the issuer's funding rate while the index is unchanged, so the best-looking note on offer is systematically the weakest credit. A reverse convertible is a bond plus a short put that pays only below its barrier: its coupon above deposit rates is payment for a written put, described in fixed-income vocabulary while its risk is entirely equity.

An autocallable with a daily knock-in prices at par on a coupon far above deposit rates. It pays its advertised return most of the time and loses in the rest, with small frequent gains set against rare large losses, so its returns are negatively skewed where the index's are positively skewed. A buyer moving from the index into the note exchanges a positively skewed exposure for a negatively skewed one at the same expected return, and a variance-based risk measure reports the note as the safer of the two. It is also short volatility at the worst time, since volatility rises as the knock-in is approached.

Korea's retail market showed what this does at scale: the usual outcome repeated until it looked like a rule, and then a single vintage's barriers, maturities, and losses arrived together. A loss probability sold to a hundred thousand people in one quarter does not produce a matching fraction of them losing independently; it produces all or none.

The issuer is long the embedded down-and-in put, whose delta steepens sharply just above the barrier and jumps at it. The withdrawal of the issuer's bid matters more than the forced sale, and because barriers cluster at round fractions of the level at issue, the withdrawal is synchronized across issuers who each see only their own book. Worst-of underlyings raise the coupon substantially, and a note "linked to three major indices" is the opposite of diversified.

A note is unsecured issuer debt, so principal protection is protection from the market, not from the issuer. The fee is largest where the price is hardest to verify; disclosure fixes fees well and distributions poorly. The secondary market is the issuer alone, so the exit is worst when it

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is most wanted. The reform this chapter would endorse is a suitability test framed around the payoff's shape rather than the buyer's stated risk tolerance.

FURTHER READING

The regulatory literature is where this subject is treated, because supervisors have had to define what a fair disclosure looks like: the estimated-value rules and the key information document of the European Union's PRIIPs Regulation (No. 1286/2014) are worth reading as primary sources, and their differences across jurisdictions are informative about what each regulator thought the problem was. On the academic side, the work measuring issue prices against model fair values (Henderson and Pearson; Célérier and Vallée) is consistent enough across markets and decades to be treated as settled: notes are issued at a premium to fair value, and the premium rises with complexity. Bergstresser and others on the retail structured-products market document the demand side. On autocallable hedging flows the public literature is thin and the useful material is in central-bank financial stability reports, which have covered the topic increasingly since 2022. For the pricing machinery, the references of Chapter 34 apply unchanged, since an autocallable is a barrier product with an early-termination feature and nothing in it is new.

PRACTICE QUESTIONS

35.1 Using Example 35.1, compute the participation rate before and after a 2% fee, and explain why the same note's participation would be 59.4% at 2% issuer funding and 107.3% at 6%.

35.2 Explain why the best-looking note in a comparison across issuers at one moment is systematically the weakest credit.

35.3 Decompose a reverse convertible paying 10% when deposits pay 4%, and state what the 6% actually buys.

35.4 Using Example 35.2, state the fair coupon and the three probabilities, and explain why a payoff sold on its hit rate is negatively skewed.

35.5 Explain why an autocallable is short volatility in the worst possible way, being specific about when volatility rises.

35.6 State the note's and the index's skewness from the simulated distribution, and explain what a buyer switching between them is exchanging when both earn the risk-free rate.

35.7 State the average win, the average loss, and the worst one percent, and reconcile them with the 83.55% probability of a positive return.

35.8 Explain why the 25th and 50th percentiles of the note's return coincide, and why that makes the product unusually persuasive.

35.9 Explain why a variance-based risk measure reports this note as safer than the index, and what that says about variance as a risk measure.

35.10 Explain why the exit from a structured note is worst precisely when it is most wanted.

35.11 Explain why a product with a 17% loss probability sold to a hundred thousand people does not produce 17% of them losing independently.

35.12 Using Example 35.6, state the delta just above and just below the barrier and the jump, then explain why the withdrawal of the issuer's bid matters more than the forced sale at the barrier.

35.13 Explain why barriers cluster across issuers, why no issuer can observe the clustering, and

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what that implies about what should be monitored.

35.14 Explain why a note linked to the worst of three indices pays roughly double the coupon, and why that is a measure of risk rather than of value.

35.15 List the five evaluation steps of Section 35.11, state what each is designed to reveal, and apply them to Example 35.2's note in one sentence.

FURTHER QUESTIONS

35.16 Example 35.1 shows participation nearly doubling with the issuer's funding rate. Derive the participation rate as a function of funding, maturity, and volatility; determine which of the three a buyer can most easily verify; and design a comparison metric that would rank notes across issuers on a like-for-like basis.

35.17 Example 35.2 finds a fair coupon of 18.42% with a 16.45% probability of loss. Compute the distribution of the investor's return rather than its mean, state its skewness, and use it to argue either that the product is or is not suitable for a buyer whose stated objective is income with capital preservation.

35.18 Example 35.6 shows the hedge delta jumping by 0.19 at the barrier. Aggregate the flow: assume a market-wide notional and a barrier distribution, derive the total hedging demand as a function of the index level, and identify the level at which the withdrawal of the bid is largest.

35.19 Section 35.7 argues that a stabilizer which works until it stops is not a stabilizer. Assess the claim rigorously: construct the case that autocallable hedging reduces realized volatility on average, construct the case that it redistributes volatility into the tail, and state what data would settle which effect dominates.

35.20 Section 35.10 argues the margin is largest where the price is hardest to verify. Test this across the products in Chapters 33–35: rank them by verifiability and by typical margin, and state whether the relationship holds and what explains any exception.

35.21 Section 35.4 argues the exit is worst when most wanted. Quantify it: model the issuer's bid as fair value less a spread that widens with volatility, and compute the loss an investor takes by exiting after a 20% fall relative to holding to maturity.

35.22 Section 35.12 argues that a suitability test should be framed around payoff shape rather than stated risk tolerance. Design such a test: specify the statistics it would require, the thresholds it would apply, and how it would classify each product in this chapter. Then identify the product it misclassifies and say why.

35.23 The estimated-initial-value disclosure tells a buyer what the note is worth on the day it is issued. Determine what it does *not* tell them, propose one further disclosure that would close the largest remaining gap, and explain why an issuer would resist it.

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Part VIII

Commodity, Environmental, and Digital Markets

Markets where the underlying is not a financial asset, and the standard assumptions bend. Storage costs, delivery constraints, weather, emissions allowances, and a perpetual contract with no expiration each break something the earlier chapters took for granted. Both chapters can be read on their own; where they draw on Parts VI and VII, the point is restated where it is used.

36 Energy, Commodity, and Environmental Derivatives
carry with storage, spreads, weather, and carbon

37 Crypto Derivatives
perpetuals, funding rates, and derivatives without a settlement asset

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Chapter 36
Energy, Commodity, and Environmental Derivatives

Section 5.8 priced a commodity forward with storage costs and a convenience yield, and Section 5.14 left to this chapter the markets where the term structure carries the most information. What makes it carry information is a single asymmetry. In equity and rates, the forward curve is derived from the spot by arbitrage; a violation is a trade. In commodities the arbitrage runs one way only—you can buy and store, but you cannot short and un-store—so the curve carries information about physical conditions that no arbitrage forces into line.

That asymmetry generates everything in this chapter. The curve's shape means something, seasonality survives, volatility depends on maturity, and one commodity, electricity, breaks the framework entirely because it cannot be stored at all.

The chapter then follows the asymmetry to its limit. Emissions allowances are storable at no cost, so the carry relation binds tighter than for any physical commodity and its residual means something else; allowance options have a bimodal density and a smile that is not a smile; and weather and catastrophe contracts have an underlying that cannot be held, so the writer cannot hedge and the price is an expectation rather than a replication cost. Those last sections show how much of Parts II and III depended on a hedge existing.

Sections 36.1 to 36.3 read the curve: what its shape says about inventories, why seasonality survives, and why volatility depends on maturity. Sections 36.4 to 36.8 take electricity and the contracts written on it, the spreads that turn one commodity into another, storage as an option, and the models these markets need; Sections 36.9 to 36.12 cover basis, the stack-and-roll hedge, financial participation, and the remaining markets. Sections 36.13 to 36.16 follow the asymmetry to its limit through emissions, weather, and catastrophe risk, and Section 36.17 sorts what is priced by arbitrage from what is priced by expectation. Section 36.18 says who trades these markets and why, and Section 36.19 how the chapter's results were obtained.

36.1 THE CURVE IS THE INFORMATION

Section 5.8's relation is

$$F(t,T) = S(t)\, e^{(r+u-y)(T-t)} \tag{36.1}$$

with u the storage cost and y the convenience yield, the benefit of holding the physical commodity rather than a claim on it.

The convenience yield is not observed: it is the residual that makes the equation hold, and treating it as a parameter with an independent existence is the most common error in commodity finance. But the residual is informative. A high convenience yield (a market in **backwardation**) says physical inventory is scarce and someone will pay to hold it now. A low one (**contango**) says inventory is plentiful and the curve is compensating storage.

Two consequences follow that have no analog in financial markets.

The upper bound on the curve is enforced and the lower bound is not. If F rises far above (36.1), anyone can buy spot, store, and sell forward. If F falls below it, the arbitrage requires

borrowing the physical commodity, and for most commodities that market is thin or nonexistent. So contango is capped by storage economics and backwardation is not capped at all, which is why the deepest backwardations occur in genuine shortages and can be extreme.

And storage capacity is finite. When it fills, the cap disappears: the forward can trade above full-carry because there is nowhere to put the commodity. The most-quoted case is the April 2020 crude oil episode, when a front-month contract settled below zero because holders faced physical delivery with no storage available. That was not a market failure but the arbitrage bound behaving as it must when the constraint binds, and the clearest modern demonstration that a commodity futures contract is a claim on a physical thing.

Example 36.1 Reading inventory off the curve

Crude spot \$78.00, financing 4.5%, storage 6% of value per year. Inverting (36.1),

$$y = r + u - \frac{1}{T}\ln\frac{F}{S}$$

	Backwardated		Contango	
Months	F	**Implied** y	F	**Implied** y
1	77.40	19.77%	78.35	5.13%
3	76.30	19.31%	79.10	4.90%
6	75.00	18.34%	80.20	4.94%
12	73.20	16.85%	82.00	5.50%
24	71.50	14.85%	84.50	6.50%

In backwardation the implied yield far exceeds $r + u$, which is the market paying nearly 20% a year for the privilege of holding a physical barrel now.

The term structure of y is itself a forecast: it declines with maturity in the backwardated case, which says the shortage is expected to resolve, where a flat implied-yield curve would say the tightness is expected to persist.

A negative implied convenience yield is the diagnostic for full storage. It means the curve is steeper than full carry, which cannot be arbitraged away because there is nowhere to put the commodity. That steepness is the precondition for the April 2020 episode, but not the episode itself: (36.1) makes F a positive multiple of S, so no convenience yield, however negative, can take a futures price below zero. What went negative was the spot value of a barrel at the delivery point. Holders of the expiring contract had to take delivery with no tank to put it in, so owning a barrel had become a disposal cost, and on April 20, 2020 the May contract settled at −\$37.63 while June, a month later, settled at +\$20.43.

36.2 SEASONALITY

Financial forward curves are smooth; commodity curves are not. Natural gas, power, and agricultural curves have a stable annual shape driven by consumption:

Month	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
Gas forward (\$/MMBtu)	2.85	2.88	2.95	3.05	3.08	3.02	3.10	3.45	3.80	3.95	3.85	3.30

The winter premium is not a forecast that prices will rise. It is the cost of the storage cycle: gas injected in summer and withdrawn in winter must earn the storage cost, the financing, and a return on the facility.

Two practical consequences follow.

A "roll yield" computed across a seasonal boundary is meaningless. Rolling from December into January in gas is not a term-structure trade; it is a seasonal step. Section 5.11's roll analysis applies within a season and not across one.

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And seasonality must be removed before a volatility is estimated: a series that rises every winter has a partly deterministic variance, and treating it as risk overstates the volatility. This is Section 33.10's point about inflation, with a larger amplitude.

36.3 THE SAMUELSON MATURITY EFFECT

The **Samuelson maturity effect**, named for Paul Samuelson, who predicted it in 1965, is the tendency of a futures contract's volatility to rise as its expiration approaches. Seen across the curve on any one day, it means near contracts are more volatile than far ones. The reason is mean reversion in the spot.

Example 36.2 Volatility falls with maturity

If the spot mean-reverts at rate a, a shock decays before a distant contract expires, so the futures volatility is

$$\sigma_F(T) = \sigma_S\, e^{-a(T-t)}, \tag{36.2}$$

with σ_S the volatility of the spot price and $T - t$ the contract's time to expiration. [TN36] derives it from a mean-reverting log spot price. The decay is the one Section 31.6 found for Hull–White forward rates, and for the same reason: mean reversion damps a shock before the distant date arrives. The quantities differ, though. There the short rate mean-reverts and the result is a basis-point volatility of a forward rate; here the logarithm of the spot price mean-reverts and the result is a percentage volatility of a futures price. At a 60% spot volatility:

Months out	$a = 0.5$	$a = 1.5$	$a = 3.0$
1	57.6%	52.9%	46.7%
3	52.9%	41.2%	28.3%
6	46.7%	28.3%	13.4%
12	36.4%	**13.4%**	3.0%
24	22.1%	3.0%	0.1%

At $a = 1.5$ the one-month contract carries 52.9% volatility and the twelve-month 13.4%, a factor of 3.96.

Three things follow.

Options on different contract months are not comparable at the same strike. A 40% implied volatility on the front month and on the one-year are describing very different markets, and a volatility surface for a commodity needs a maturity dimension that means something different from an equity surface's.

Hedge ratios across maturities are not one-for-one. Hedging a two-year exposure with front-month futures over-hedges the volatility by the ratio in the table, which is the commodity version of the duration mismatch that Section 6.7's duration-based hedge corrects.

And the effect competes with a second one. Storage links adjacent contracts, which *raises* correlation between them; mean reversion *lowers* the far contract's volatility. A commodity with cheap, ample storage has a flat volatility term structure and a highly correlated curve; one with costly or full storage has a steep one and a curve whose near and far ends can move independently. The volatility term structure is therefore a direct read on storage economics.

Figure 36.1 shows both effects: futures volatility falling with maturity, and a gas curve whose shape is set by the calendar rather than by the price level.

36.4 ELECTRICITY

Power cannot be stored, and that single fact removes most of this book's machinery.

(36.1) does not hold, because there is no cash-and-carry: you cannot buy power today and deliver it tomorrow. So the forward is not linked to the spot by arbitrage at all: it is an

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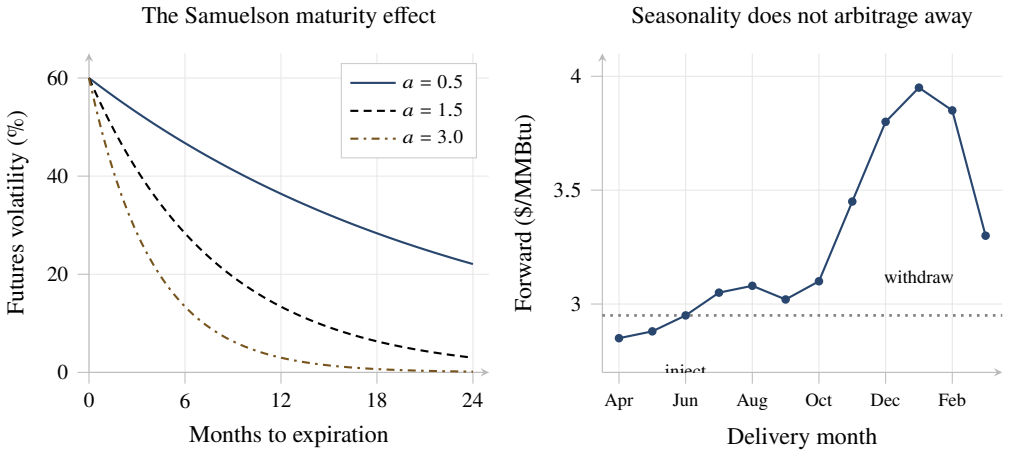


Figure 36.1: Left: equation (36.2) at $\sigma_S = 60\%$. Right: Section 36.2's gas curve.

expectation plus a risk premium, and nothing enforces it.

The price process is correspondingly unusual. Supply is a stack of generators ordered by marginal cost, and the price is set by the last unit needed. When demand approaches capacity the stack becomes near-vertical, so a small demand increase produces an enormous price move—and then reverts within hours when the weather changes.

A simple model captures this. On an ordinary day the price is lognormal; on a small fraction p of days demand reaches the steep end of the stack, and the day's price is multiplied by a random spike factor J:

$$P = m\, e^{\sigma Z} J^{B}, \qquad Z \sim N(0,1), \quad B \sim \text{Bernoulli}(p), \quad \ln J \sim N\big(\mu_J, s_J^2\big), \tag{36.3}$$

with Z, B and J independent. Here m is the median ordinary-day price (half of the days without a spike settle below it, and their mean is the slightly higher $m\, e^{\sigma^2/2}$), σ is the standard deviation of its logarithm, and $E[J] = e^{\mu_J + s_J^2/2}$ the average size of a spike. Call this the **spike model**. Because J is lognormal, the price on a spike day is lognormal too, so the price is a mixture of two lognormals: with probability $1 - p$ its logarithm is normal with mean $\ln m$ and variance σ^2, and with probability p normal with mean $\ln m + \mu_J$ and variance $\sigma^2 + s_J^2$. Its mean is

$$E[P] = m\, e^{\sigma^2/2}\big(1 - p + p\, E[J]\big),$$

and its distribution function is the probability-weighted sum of the two lognormals' distribution functions. The q-th percentile x_q therefore solves

$$(1-p)\, N\!\left(\frac{\ln(x_q/m)}{\sigma}\right) + p\, N\!\left(\frac{\ln(x_q/m) - \mu_J}{\sqrt{\sigma^2 + s_J^2}}\right) = q,$$

which has no closed-form solution but has exactly one root, because the left side rises steadily from 0 to 1 as x_q increases, so a few steps of bisection find it.

A contract on the day's price is valued the same way, by conditioning on whether the day spikes. A daily cap struck at K, paying $\max(P - K, 0)$, is worth

$$c(K) = (1-p)\, c^{\mathrm{B}}\big(F_0, K, \sigma\big) + p\, c^{\mathrm{B}}\Big(E[J]\, F_0,\ K,\ \sqrt{\sigma^2 + s_J^2}\Big), \qquad F_0 = m\, e^{\sigma^2/2}, \tag{36.4}$$

before discounting to the delivery date, where $c^{\mathrm{B}}(F, K, v) = F\, N(d_1) - K\, N(d_1 - v)$ with $d_1 = [\ln(F/K) + v^2/2]/v$ is Black's formula from Chapter 16 written with the total volatility

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v in place of $\sigma\sqrt{T}$. Each term is one regime: an ordinary day has mean F_0 and log standard deviation σ; a spike day has a mean $E[J]$ times larger, and the spike factor's own uncertainty adds to the variance. A floor is valued the same way with Black's put, and a forward on the day's price is worth $E[P]$. So every figure the model produces is exact.

The model has no time dimension. It describes the distribution of one day's price, with each day drawn independently and a spike lasting exactly one day: the "reverts within hours" above, taken to its limit. That is all a daily cap needs, because its payoff depends on one day's price and a strip of daily caps is worth the sum of its days. A payoff that depends on several days (a cap on a monthly average, a swing contract, a hedge carried through a spike) needs the path as well, and there the standard choice is a mean-reverting log price with jumps that decay within days, so that spikes arrive in clustered episodes rather than as independent days.

Example 36.3 Why a lognormal model misprices a price cap

Take the spike model (36.3) with an ordinary-day median $m = \$43$ and $\sigma = 30\%$. Spikes occur on 2.4% of delivery days ($p = 2.4\%$), about nine days in a 365-day year, with a spike factor of mean $E[J] = 5$ and $s_J = 60\%$. Against it, fit a single lognormal with the *same mean ($49.30) and the same standard deviation ($40.42)* and compare.

	Spike model	Matched lognormal
Mean	\$49.30	\$49.30
Median	\$43.39	\$38.12
99th percentile	\$206.81	\$202.15
99.9th percentile	\$573.80	\$349.60

Now price daily caps with (36.4). At a \$100 strike, $F_0 = 43\,e^{0.045} = \$44.98$, so an ordinary day is worth $c^{\mathrm{B}}(44.98, 100, 0.30) = 0.0238$; a spike day, with mean \$224.90 and total volatility $\sqrt{0.09 + 0.36} = 67.1\%$, is worth $c^{\mathrm{B}}(224.90, 100, 0.671) = 130.24$. Weighting by the probabilities,

$$c(100) = 0.976 \times 0.0238 + 0.024 \times 130.24 = 0.0232 + 3.1257 = 3.1490,$$

so 99% of the cap's value comes from the nine or so spike days a year. The matched lognormal is a single application of Black's formula, at the forward \$49.30 and the 71.7% volatility that reproduces the standard deviation of \$40.42. Across strikes:

Strike	Spike model	Lognormal	Ratio
\$ 60	5.3300	10.4989	0.51
\$100	3.1490	4.1344	0.76
\$200	1.6578	0.6518	**2.54**
\$400	0.5056	0.0483	**10.48**
\$800	0.0739	0.0016	46.70

Matched on mean and on volatility, the two models disagree by a factor of ten at a \$400 strike—and they disagree in opposite directions at the two ends.

The lognormal spends its variance on everyday fluctuation and the real distribution spends it on nine or so days a year. So the lognormal over-prices near-the-money caps, because it makes moderate moves too likely, and drastically under-prices far ones.

A utility buying a price cap is buying precisely that difference, and it is the single largest model risk in energy trading. No amount of calibrating a lognormal to at-the-money options fixes it. The lognormal already over-prices the cap near the money, so fitting that price lowers its volatility, and a lower volatility cheapens the far caps further.

Two further features of electricity have no financial-market analog.

Prices can be negative, and routinely are in systems with large inflexible or subsidized generation, because a generator may prefer to pay to keep running rather than shut down and

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Table 36.1: Electricity contracts, their users, and how each is priced.

Contract	What it pays	Main users	Priced by
Block futures and forwards	A fixed price for power over a month, quarter or year, baseload or peak hours; usually cash-settled on the average hourly spot price	Generators sell; retailers and large consumers buy	The market: the curve is an input
Swaps	The same, over the counter: fixed against the average spot price at a hub or zone	Utilities, industrial consumers	As forwards
Options on block futures	Calls and puts on a block forward	Generators and retailers shaping hedges	Black's formula with the block's own volatility
Daily and hourly options; caps and floors	One option per delivery day or hour on that period's spot price	Retailers capping cost; owners of peaking plants	A spot model with spikes, Example 36.3
Spark and dark spread options; tolling	Power less fuel; a tolling agreement rents a plant's output	Plant owners and buyers of capacity	Exchange option, Example 36.5; dynamic programming for operating constraints
Swing and load-following contracts	A fixed price with volume at the holder's choice, within daily and total limits	Retail suppliers whose load follows the weather	Dynamic programming or least-squares Monte Carlo
Financial transmission rights	The price difference between two nodes, the congestion rent	Generators and load-serving utilities hedging location	Auction prices and network simulation; no replication
Power purchase agreements	A fixed price for a plant's output as produced; a virtual agreement settles the difference in cash	Corporate buyers of renewable power; wind and solar developers	A forward at the output-weighted price, plus shape, volume and basis risk

restart. Any model requiring positivity, and any option formula written in log-space, fails outright, which is why power desks work in normal rather than lognormal terms, exactly as Section 29.8's rate markets did for the same reason. Neither model in Example 36.3 can produce one: both multiply positive factors, so the spike model repairs the upper tail and is silent about the lower. Its cap prices are unaffected, because a cap pays only in the upper tail. Where negative prices matter (a floor, or a generator with an obligation to keep running), the ordinary-day price has to be modeled additively, as a normal variable or a lognormal shifted below zero, with the spike added on top.

And the commodity is defined by location and by hour. Power at one node and at another is a different good, connected by transmission that can bind. The basis between locations is itself a traded market, and its behavior is closer to a congestion option than to a spread.

36.5 ELECTRICITY CONTRACTS AND HOW THEY ARE PRICED

Because nothing links the forward to the spot, an electricity market is a family of contracts, each written on a different slice of the same price (a delivery period, a day, an hour, a location), and each priced by its own method. Table 36.1 sets them out.

Three rules organize the last column.

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The forward curve is an input. Power cannot be carried, so (36.1) produces no forward price. A block forward is the expected average spot price over its delivery period under the pricing measure, and nothing arbitrages it toward today's spot. Desks therefore take the curve from the market and calibrate every other model to it, which reverses Chapter 5's order: there the spot and the cost of carry gave the forward, and here the forwards are the data.

The settlement period picks the model. A contract settled on an average over a month depends on how the whole delivery period moves and is priced with a model of the forward curve. A contract settled on one day or one hour depends on spikes and is priced with a model of the spot. A block forward averages hundreds of hourly prices, so its volatility is a fraction of the spot's, and the Samuelson maturity effect lowers it further the farther away delivery is. An option on a block is therefore cheap next to a strip of daily caps at the same strike. The next example shows by how much, and why the answer turns on something Example 36.3 could leave out.

Flexibility is a chain of linked options. The clearest case is the **swing contract**, a supply agreement at a fixed price in which the buyer chooses each day how much to take, between a daily minimum and maximum, subject to a minimum and maximum over the whole contract and a penalty for taking less than the total minimum. A gas retailer holding one takes the maximum on cold days, when spot gas costs more than the contract price, and the minimum on mild ones; what links the days is the total limit, because volume taken in December is not available in February. Tolling agreements and storage work the same way: the holder chooses how much, or when, on many dates, and each choice uses up capacity that a later one would have needed. Their value therefore depends on the whole price path rather than on one day's price, and it is found by dynamic programming over the price and the remaining volume, the method of Section 36.7, or by Chapter 20's least-squares Monte Carlo when the state is too large for a lattice.

Example 36.4 A daily cap and a monthly cap are different instruments

The example shows that two caps on the same price at the same strike can differ in value by a factor of eight, and that the gap depends on how spikes are spread through time, which the spike model of Example 36.3 does not specify. Take a 30-day month, a strike of \$60, and 1 MWh a day, with each day's price drawn from that example's spike model. A strip of daily caps pays $\sum_d \max(P_d - 60, 0)$; a cap on the monthly average pays $30\max(\bar{P} - 60, 0)$, with $\bar{P}$ the month's average price. Values are undiscounted.

The strip is worth $30 \times 5.3300 = \$159.90$, whatever the relation between days, because each of its terms depends on one day only. The average cap depends on the joint behavior of the days. The table varies that behavior while holding each day's distribution fixed: spikes arrive either on independent days or in episodes of three or five consecutive days, with the probability that a given day spikes kept at 2.4%. The average cap has no formula like (36.4), because its payoff depends on a sum of 30 prices, so it is valued by simulation: draw each day's ordinary price, whether the day spikes and its spike factor, average the month, and take the mean payoff over 400,000 simulated months for each row:

Spikes arrive	S.d. of $\bar{P}$	Monthly-average cap	Daily strip
On single days	\$ 7.41	\$19.3	\$159.90
In three-day episodes	\$10.11	\$44.3	\$159.90
In five-day episodes	\$12.15	\$65.9	\$159.90

Averaging removes most of a spike. A spike day that multiplies the price by five raises a 30-day average by only about \$6, so with independent days the monthly cap is worth an eighth of the strip. That is why options on block futures trade at volatilities far below those of daily options in the same market: the block's payoff is an average, and an average does not see a single day.

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Clustering raises the monthly cap and leaves the strip alone. Spikes in five-day episodes (a heat wave rather than a hot afternoon) more than triple the monthly cap, from \$19.3 to \$65.9. The dependence across days that Example 36.3 could ignore is the whole input to a monthly contract. A model calibrated to daily options says nothing about it, and a desk that prices monthly options as if days were independent will value them at less than half their worth if spikes in fact come in three-day episodes, and at less than a third if they come in five-day ones.

36.6 SPARK, DARK, AND CRACK SPREADS

The most important commodity derivatives are not on commodities but on the *margin between them*, because that is what an industrial owner actually earns.

Example 36.5 A power plant is an exchange option

A gas-fired generator earns $\max(P_e - \text{HR} \cdot P_g, 0)$ per MWh, where P_e is the power price per MWh, P_g the gas price per million British thermal units (MMBtu), and HR the **heat rate**, the MMBtu of gas the plant burns to produce one MWh. The margin $P_e - \text{HR} \cdot P_g$ is the **spark spread**. The plant runs when it is positive and shuts when it is not, so the plant *is* a strip of call options on the spark spread—and by Section 34.10 each is an **exchange option**: power against HR units of gas.

The example asks what one of those options is worth, and shows that the answer turns almost entirely on the correlation between power and gas. Take a peaking plant, one that runs only in the hours when power prices are high: power \$42.00/MWh, gas \$3.20/MMBtu, heat rate 12.5, six months, power volatility 55%, gas volatility 45%. Fuel cost is \$40.00/MWh, so the intrinsic spread is \$2.00. The option values below are forward values, undiscounted.

ρ	Effective vol	Option value	Time value
0.9	24.4%	\$3.929	\$1.929
0.7	39.8%	\$5.658	\$3.658
0.5	50.7%	\$6.891	\$4.891
0.3	59.7%	\$7.900	\$5.900
0.0	71.1%	\$9.171	\$7.171

Modeling power and gas as independent values the option on each megawatt-hour at \$9.171 instead of \$3.929, 133% too high.

Power and gas are strongly correlated because gas frequently sets the marginal price of power, so the spread is far less volatile than either leg. *High correlation makes a spread option cheap*, which is the opposite of the intuition that two volatile inputs must make a volatile output.

This is the single most consequential correlation assumption in energy finance, because it values the entire generation fleet: a merchant generator's enterprise value is the value of a portfolio of spark-spread options, and Section 34.10's warning that correlation must come from somewhere applies here with a whole plant's value at stake.

The same structure recurs. A **dark spread** is the coal-fired version. A **crack spread** is the refiner's:

Crude	\$78.00/bbl
Gasoline \$2.45/gal	\$102.90/bbl
Heating oil \$2.60/gal	\$109.20/bbl
3-2-1 crack $= (2 \times 102.90 + 109.20 - 3 \times 78.00)/3$	**\$27.00/bbl**

A refiner is long this and hedges by selling it: buying crude futures and selling product futures in the 3:2:1 ratio. The ratio is a technology statement, not a market convention: it approximates the yield of a typical US refinery, and a refiner with a different configuration hedging at 3-2-1 has a residual exposure to its own yield slate.

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36.7 STORAGE IS AN OPTION

A storage facility is the right, not the obligation, to buy at one date and sell at another. Its value therefore splits the way an option's does.

Example 36.6 Intrinsic and extrinsic

Using Section 36.2's curve, a facility can lock in a spread today: inject in the five cheapest months (April through September, skipping August, averaging \$2.95) and withdraw in the five most expensive (November through March, averaging \$3.67).

The intrinsic value is \$0.72 per MMBtu per cycle, and it can be hedged immediately with calendar spreads.

The extrinsic value is the right to change one's mind: to withdraw early if winter arrives early and cold, to re-inject if the spread widens. It is the time value of a strip of calendar-spread options, and it depends on volatility and on how correlated adjacent months are:

Monthly vol	$\rho = 0.90$	$\rho = 0.75$	$\rho = 0.50$
30%	0.2%	2.2%	7.1%
45%	1.7%	8.3%	17.4%
60%	5.1%	15.7%	27.0%
80%	11.0%	25.0%	**37.4%**

The entries are extrinsic value as a share of the total, with the strip valued as one exchange option (Margrabe's formula) on the \$3.67 withdrawal average against the \$2.95 injection average over six months, both at the volatility shown and correlation ρ between them.

At plausible parameters the extrinsic share is 10–30%, and it is the part a spreadsheet does not capture.

This explains a market structure fact. Storage capacity is often leased by trading firms rather than operated by the utility that built it, and the reason is not that traders are better at running compressors. A utility that books the intrinsic spread has sold the extrinsic value to whoever leases the capacity, because capturing it requires continuous re-optimization, which is an option-hedging operation, not a physical one.

And the same decomposition prices every physical asset with flexibility: a pipeline is a locational spread option, a swing contract on gas is a strip of daily options linked by its volume limits, and a liquefied natural gas (LNG) cargo with destination flexibility is an option on the basis between two markets.

36.8 MODELING

Four features have to be accommodated and no single model does all of them well.

Mean reversion generates Section 36.3's volatility term structure and is the minimum requirement.

Seasonality is handled deterministically (a seasonal function added to the log price) rather than stochastically, because it is predictable and modeling it as risk overstates volatility.

Use two factors, at least, for anything spread-related. A one-factor model makes all contract months perfectly correlated, which, once the model is fitted to a realistic curve, prices a calendar spread option at nearly zero. This is Section 31.12's defect in another market, and the fix is the same: a short-term mean-reverting factor plus a long-term factor with a random walk, which is the standard two-factor commodity model and produces both a decaying volatility term structure and imperfect correlation along the curve. Example 36.7 shows how much the second factor is worth.

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Example 36.7 Why a calendar spread needs two factors

The example shows that a one-factor model fitted to the same market as a two-factor model can value a calendar spread option at about a twelfth of the two-factor price. Let the market follow the two-factor model

$$\ln S = \chi + \xi, \qquad d\chi = -\kappa\chi\, dt + \sigma_\chi\, dW_\chi, \qquad d\xi = \mu\, dt + \sigma_\xi\, dW_\xi,$$

with independent shocks, a short-term factor at $\sigma_\chi = 40\%$ and $\kappa = 2$, and a long-term factor at $\sigma_\xi = 20\%$. The argument of [TN36], applied to each factor, gives the futures volatility at time to delivery τ as $\sigma_F(\tau) = \sqrt{\sigma_\chi^2 e^{-2\kappa\tau} + \sigma_\xi^2}$. Fit a one-factor model, $\sigma e^{-a\tau}$, to the one-year and two-year contracts, which is what a desk pricing a spread between them would do. Matching their 20.7% and 20.0% needs $a = 0.035$ and $\sigma = 21.5\%$:

Months to delivery	Two-factor (market)	One-factor fit
1	39.3%	21.4%
6	24.8%	21.1%
12	20.7%	20.7%
24	20.0%	20.0%

Now price an option, expiring in six months, to receive the two-year contract and deliver the one-year, both at \$80: Example 36.5's exchange option, with both legs on the same curve. Its value is Margrabe's formula with the variance of $\ln(F_2/F_1)$ accumulated over the option's life. In the two-factor model the long-term factor moves both contracts by the same amount and drops out of the ratio, so only the short-term factor contributes:

$$v^2 = \sigma_\chi^2 \left(e^{-\kappa T_1} - e^{-\kappa T_2}\right)^2 \frac{e^{2\kappa T_0} - 1}{2\kappa},$$

with $T_0 = 0.5$, $T_1 = 1$ and $T_2 = 2$. That gives $v = 5.92\%$ and, since the two forwards are equal, a value of $80\,[2N(v/2) - 1] = \$1.89$. The one-factor model has the same formula with σ and a in place of σ_χ and κ, and gives $v = 0.50\%$ and \$0.16. Both values are forward values, undiscounted.

The one-factor model misses because of how it had to be fitted. To carry a 20% volatility out to two years it must mean-revert slowly, so the two legs have almost the same volatility; being perfectly correlated as well, they move almost in step, and the spread barely moves. The two-factor model carries the long end with ξ and leaves χ free to revert fast, and it is the fast factor that moves one contract against the other. The two legs' correlation in the two-factor model is still 0.974, and the small gap from one carries all of the option's value.

And the misfit shows in the table. A desk that fitted it to front-month options instead would get the opposite error at the back. No single σ and a fit both ends of the curve, and that is the practical sign that a second factor is needed.

And add jumps, for power and increasingly for gas. Example 36.3 showed what omitting them costs.

Contracts on the forward curve are modeled on the curve itself. A model that gives each delivery period's forward its own volatility and its correlation with the others (Chapter 32's HJM approach, applied to commodity forwards) prices block forwards and the options on them consistently with the quoted curve. Spot models with spikes and forward-curve models are therefore complements rather than rivals: the first prices what settles on a day, the second what settles on a period, and Example 36.4 shows that a spot model calibrated to daily options leaves the value of a monthly contract open by a factor of three.

Commodity modeling is less standardized than rates or equities, because the physical constraints differ by commodity and a model that fits crude does not fit power. The practical discipline is to name the physical feature each model term is representing (this factor is inventory, this jump is a capacity constraint), because a commodity model with terms that correspond to

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nothing physical will be calibrated to fit and will fail out of sample.

36.9 BASIS: GRADE, LOCATION, AND TIMING

Section 3.3 defined basis risk generally. In commodities it is not a residual but the main event, because a hedger essentially never owns the contract grade at the contract location on the contract date.

Example 36.8 What the hedge leaves behind

A producer's realized price against a light sweet crude contract:

Contract (light sweet, Cushing, December)	\$78.00/bbl
Grade adjustment (heavier, higher sulfur)	−3.20
Location adjustment (wellhead to Cushing)	−2.40
Timing adjustment (November production)	−0.35
Realized price	**\$72.05/bbl**
Total basis	−\$5.95 **(7.6%)**

Call the futures price itself the **flat price**, to distinguish it from the basis. At a 32% volatility of the flat price and a 28% volatility *of the basis*:

Unhedged standard deviation of the realized price	\$24.96/bbl
Hedged standard deviation (basis only)	\$1.67/bbl
Risk reduction	**93.3%**

The hedge works: it removes 93% of the risk, which is why producers hedge.

But the residual is real and it is different in kind. Flat price is liquid, continuously quoted, and hedgeable; *basis is none of those*. Grade differentials move with refinery configuration and with regulation on sulfur content; location differentials move with pipeline capacity and can jump when a line is constrained; timing basis moves with local storage.

Three consequences follow.

Basis is hedgeable in some markets and not others. Where basis swaps trade (major US gas hubs, some crude grades), the residual can be reduced further. Where they do not, a producer's remaining exposure is to a variable with no market and no forecast.

Basis risk is worst exactly when flat price risk is worst. A pipeline constraint that sharply widens a location differential typically coincides with the demand event that moved the flat price. This is the wrong-way structure of Section 27.9 in a physical market, and it means the 93% figure overstates the protection in precisely the episodes the hedge was bought for.

And a hedger who reports the position as "hedged" has renamed the exposure rather than removed it: flat price is hedged and basis is not, and the two should be reported separately, for the reason Section 9.11 kept three valuation numbers apart.

36.10 STACK AND ROLL

The most instructive hedging failure in commodities is not a model error but a hedge that was economically correct and cash-flow wrong, and nothing in the pricing gives it away.

Example 36.9 A hedge that is right and still fails

A supplier sells ten years of fixed-price delivery, one million barrels a year. The far-dated futures are illiquid, so it hedges by **stacking** the whole ten-year exposure in the front month and rolling monthly.

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Total commitment	10,000,000 bbl
Stacked in the front month	10,000,000 bbl
Exposure arising in year one	1,000,000 bbl
Stack ratio	**10×**

Three problems follow, and each is separately sufficient.

One: the roll costs money, on the stacked notional. Section 5.11's roll yield applies ten times over:

Monthly contango	Annualized	Cost per year on a $780m stack
0.5%	6.2%	$48.1m
1.0%	12.7%	$98.9m
2.0%	26.8%	$209.2m

Two: the margin timing. A price fall makes the futures lose cash immediately while the offsetting gain on the supply contract accrues over a decade. The present value discounts each year's gain from mid-year at 5%, continuously compounded:

Price fall	Futures loss (cash, now)	PV of contract gain
−10%	$ 78.0m	$ 61.4m
−20%	$156.0m	$122.7m
−30%	$234.0m	$184.1m
−40%	$312.0m	$245.5m

A hedge that is economically right and cash-flow wrong can end a firm, which is Section 8.9's margin-as-liquidity-risk in its purest form and the mechanism behind more than one famous episode.

Three: the stack ratio is wrong even in principle. Two separate reasons apply, and they compound.

Discounting. At −10% the PV of the gain is $61.4m against a $78.0m loss, or **79%**. A one-for-one stack over-hedges in present-value terms by construction, because ten years of future gains are worth less than an immediate loss of the same nominal size.

The Samuelson maturity effect adds to it. Example 36.2 showed the front contract is roughly four times as volatile as the twelve-month at $a = 1.5$. Hedging a ten-year exposure with front-month futures therefore over-hedges the volatility as well.

The remedy is a maturity-weighted stack (fewer front contracts, weighted by e^{-aT} and by the discount factor), and the correct hedge ratio requires the mean-reversion parameter, which cannot be estimated reliably: Example 36.7 shows that no single a fits both ends of the curve, so the value obtained depends on which end it is fitted to. So the choice is between a hedge that is knowably wrong and one that is wrong by an unknown amount, and the first is preferable only if its error is reported.

The same mismatch returned in 2022, one generation after Metallgesellschaft (Section 3.10): European generators that had sold their output forward owed variation margin in cash as gas and power prices rose several-fold, while the offsetting gain on their plants would arrive only on delivery. Example 8.9 works the arithmetic and Section 40.12 tells the episode.

36.11 FINANCIAL PARTICIPATION AND THE CURVE

Commodity index investment (long-only, front-month, rolled mechanically) grew large enough to change the market it was invested in, by Section 22.8's mechanism. A published roll schedule creates a predictable flow: index funds sell the expiring contract and buy the next over a known window every month. A flow that is a published function of the calendar is a flow that can be anticipated, and the roll's cost to investors is the counterpart of someone else's return.

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And the aggregate position changed the curve's shape. Persistent long-only demand in the front months contributed to periods of contango in markets that had historically been backwardated, which raised the roll cost and, in turn, made the index products perform worse than the spot returns they were sold on.

The general lesson has now appeared four times in this book: portfolio insurance in Chapter 18, inverse volatility products in Section 22.8, snowball hedging in Chapter 35, and commodity index rolls here. A strategy whose trades are a published function of an observable is a strategy whose trades other people can price, and the resulting cost shows up as underperformance rather than as a loss anyone can point to.

36.12 FREIGHT, METALS, AND OTHER MARKETS

Three markets round out the picture, and each illustrates a point the chapter has already made.

Freight derivatives settle against published route indices: forward freight agreements on the cost of moving a cargo between two ports. Freight is the extreme case of non-storability: a voyage not made today cannot be banked, so the spot rate is set entirely by the balance of available tonnage and cargoes, and the market behaves more like power than like oil. Its economic role is to complete the locational spread of Section 36.7: an LNG cargo's destination option is only monetizable if the shipping leg can be priced, and freight is what prices it.

Base metals trade on a structure inherited from a physical warehousing system: delivery on individual days, the *prompt dates*, rather than in monthly contracts, and delivery against warrants, the title documents to metal in approved warehouses. The warehouse queue is a real economic variable: metal in a warehouse with a long load-out queue is not immediately available, so it supports the convenience yield of Section 36.1 even though the inventory statistics count it.

And a short-squeeze in a metal is the one place where the missing arbitrage of Section 36.1 becomes acute: a party short a physical contract cannot manufacture the metal, so the backwardation has no ceiling. The 2022 nickel episode is treated in Section 2.9 as a clearing and market-structure case; the price behavior that produced it is this chapter's asymmetry taken to its limit.

Agricultural markets add the feature that supply is annual and known only after harvest, so the curve breaks at the crop year. A calendar spread across a crop-year boundary is a different instrument from one within a year. It is the same warning as Section 36.2's, with a sharper discontinuity, because old-crop and new-crop are physically distinct goods.

The common thread is that in each market the financial contract is anchored to a physical constraint (a ship, a warehouse queue, a harvest), and the constraint, not the contract, sets the behavior the model has to reproduce.

36.13 EMISSIONS ALLOWANCES

A regulator caps emissions, issues allowances equal to the cap, and requires one surrendered per metric ton emitted; allowances trade, so *the price is the marginal cost of abatement*, the cost of the last ton of emissions cut to meet the cap. Three properties are all that matter here: they are **bankable**, so an allowance not surrendered this year can be kept for a later one; holding one is a registry entry; and borrowing one is hard, since the lending market is thin and holders with a compliance need will not lend.

So $u = 0$ and, absent doubt about future validity, $y = 0$, and (36.1) collapses to $F = Se^{r(T-t)}$—*the limiting case of a storable commodity*, in which the carry relation binds tighter than for anything physical in this chapter.

Where allowances trade. Allowances are created by governments, so the primary market is an auction run by the regulator or on its behalf, and the secondary market is mostly in futures

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rather than in the allowances themselves. A futures contract delivers allowances into the buyer's registry account at expiration, and most volume sits in each year's December contract, which lines up with the annual surrender deadline. The main schemes:

Scheme	Unit	Where it trades	Main users
EU Emissions Trading System (from 2005)	EU allowance (EUA)	Auctions on the European Energy Exchange (EEX); futures and options on ICE and EEX	Power generators; steel, cement, chemicals and refining; airlines on European routes; shipping from 2024
UK Emissions Trading Scheme (from 2021)	UK allowance (UKA)	Auctions and futures on ICE	Power generators, heavy industry, airlines on domestic routes
California and Québec cap-and-trade	California carbon allowance (CCA)	Joint quarterly auctions; futures on ICE and Nodal Exchange	Power generators and importers, large industry, suppliers of transport fuel and natural gas
Regional Greenhouse Gas Initiative (US Northeast)	RGGI allowance	Quarterly auctions; futures on ICE and Nodal Exchange	Fossil-fuel power generators only
China national emissions trading scheme (from 2021)	Chinese emission allowance (CEA)	Spot only, on the Shanghai Environment and Energy Exchange	Power generators; steel, cement and aluminum added from 2025

The EU market is the deepest, and the euro prices in this section's examples are of the order of an EU allowance. The last column lists the compliance entities, which hold allowances because they must surrender them. In every scheme they are joined by dealers who finance compliance purchases and funds that hold allowances as a view on the cap. From 2026 importers of steel, cement, aluminum, fertilizer and electricity into the EU must buy border-adjustment certificates priced off the allowance, a new class of buyer with a compliance horizon rather than a financial one. Section 36.18 returns to what each party wants.

Example 36.10 A storable asset with no storage cost

Spot €72.00, financing 3.5%. With $u = y = 0$, (36.1) gives

Years	Theoretical forward
0.5	€73.27
1	€74.56
2	€77.22
3	€79.97

The one-sided bound of Section 36.1 survives, and it points the same way. A curve *steeper* than carry is arbitraged (buy spot, bank, sell forward), and here the arbitrage is unusually clean, because banking is costless and unlimited, so the upper bound binds tightly. A curve flatter than carry cannot be arbitraged, since that requires borrowing an allowance.

The residual is therefore the whole content. Suppose the forward trades 3% below carry. Inverting (36.1) as in Example 36.1:

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Horizon	Implied y
0.5y	6.09%
1y	3.05%
2y	1.52%
3y	1.02%

A positive implied convenience yield here cannot be physical: there is no warehouse and no queue. It prices a probability that a banked allowance will not be usable as expected, and the $1/T$ decay in the table is the signature of a one-off event risk rather than a running yield: a constant y would not decay, while a single expected loss of value spread over a longer horizon does.

The same algebra that measured warehouse tightness in Example 36.1 is measuring something else entirely, and only the decay pattern distinguishes them.

Two things contaminate that residual before it can be used. Dealers finance compliance inventory (an installation buys allowances forward from a dealer holding the spot), so the curve prices a financing trade as much as an environmental scarcity and the residual is event risk *net of* a financing spread the market does not publish. And where banking stops at a phase boundary, the end of one of a scheme's multi-year trading periods, the allowance is a claim that expires, so the carry relation holds within a phase and not across one: Section 36.12's crop-year discontinuity with a legal calendar in place of a biological one, and a spread straddling the boundary is not a calendar spread.

Example 36.11 A convex supply curve meeting a vertical one

Abatement has a merit order, cheap measures first:

Cumulative abatement (Mt)	Marginal cost (€/t)
50	8
120	25
200	45
260	70
300	110
320	190

Supply is fixed, so a demand shift moves along this curve instead of being absorbed. Equal 20 Mt increases in required abatement:

Required abatement	Clearing price	Change
190 → 210 Mt	€42.50 → €49.17	+15.7%
250 → 270 Mt	€65.83 → €80.00	+21.5%
270 → 290 Mt	€80.00 → €100.00	+25.0%
290 → 310 Mt	€100.00 → €150.00	**+50.0%**

The same 20 Mt shift moves the price 15.7% in one place and 50.0% in another, with nothing changed but the position on the curve.

Two consequences follow for anyone carrying the risk. *Volatility rises as the cap tightens*, because the curve steepens, so a maturing allowance market becomes more volatile rather than less, which is the reverse of what happens as a financial market deepens. And a value at risk, the loss that should be exceeded only with a small stated probability, estimated on a historical window understates the risk systematically, because the elasticity that generated the history is not the elasticity now in force. The correct state variable is the position on the abatement curve, not the price level, and a risk model that conditions on the second is fitting the wrong thing.

Two adjacent instruments are routinely priced off allowances, and each fails for a reason

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this book has already named. A *voluntary credit* claims a reduction against an unobservable counterfactual, so credits are not fungible and a standardized futures contract on them prices the worst deliverable. That is Chapter 6's cheapest-to-deliver problem with integrity in place of duration, and it is why the market stays bilateral. A *renewable certificate* carries an obligation proportional to sales rather than a cap, so supply grows with generation and Example 36.11's vertical supply does not apply; and where the obligation can be discharged at a fixed **buy-out** price, that price is a hard cap and the certificate is a capped payoff by construction. An option on it inherits the cap, which is Section 34.12's capped cliquet in another market: past a threshold the vega changes sign with no change in the contract.

36.14 OPTIONS ON AN UNDERLYING WITH A BIMODAL DENSITY

Allowance options are quoted in Black's model like every other commodity option, and the quotes need reading differently. The reason is distributional rather than institutional, so the treatment here applies to any underlying whose uncertainty is dominated by a scheduled binary event (an election, a court ruling, a trial readout), and carbon is simply the case where the structure is permanent.

Example 36.12 What a bimodal density does to the smile

Model the one-year allowance price as a mixture: with probability $p = 0.35$ a tightening takes the forward to €110, otherwise it stays at €45, and within each regime the price is lognormal with 25% volatility. The mixture forward is

$$F = 0.35 \times 110 + 0.65 \times 45 = €67.75$$

and a call is priced by conditioning:

$$c(K) = e^{-rT}\Big[\, p\, c^{\mathrm{B}}(F_h, K, \sigma_r) + (1-p)\, c^{\mathrm{B}}(F_l, K, \sigma_r)\,\Big] \tag{36.5}$$

with c^{B} Black's undiscounted call, $F_h = 110$ and $F_l = 45$ the two regimes' forwards, $\sigma_r = 25\%$ the volatility within each regime, and $r = 3.5\%$, Example 36.10's financing rate. Inverting each price for a single Black volatility gives

Strike	Call	Implied vol	Delta error
€40	28.167	45.5%	−0.036
€50	21.934	51.1%	−0.069
€60	17.418	55.0%	−0.059
€67.75	14.535	**56.4%**	−0.026
€80	10.552	56.3%	+0.034
€90	7.769	54.7%	+0.067
€100	5.474	52.6%	+0.078
€120	2.407	48.4%	+0.060

Delta error is the mixture's true delta minus the Black delta computed at the fitted implied volatility.

Three results follow.

The implied volatility humps instead of smiling. It peaks at the money at 56.4% and falls to 45.5% and 48.4% in the wings—*the opposite shape to an equity index*. The density has two modes and a valley between them, so an at-the-money option straddles the valley and is expensive, while the wings have less mass than a lognormal of the same variance and are cheap. A desk that fits a standard skew parametrization to these quotes will fit the wrong sign and will extrapolate in the wrong direction.

The at-the-money quote understates the total uncertainty. The mixture's standard deviation is €36.32, which a lognormal matched on variance would report as **50.3%**—but that is not the

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at-the-money implied volatility, which is 56.4%, and neither number is the volatility of anything. Decomposing the variance: the regime uncertainty contributes 73% and the within-regime diffusion 27%, so nearly three-quarters of what a model calls volatility is not diffusion at all.

And the fitted volatility gives the wrong hedge. The delta error reaches −0.069 at €50 and **+0.078** at €100, eight delta points on a position that the trader believes is hedged. The sign flips across the strikes, so errors at different strikes offset only by accident: the book's net error depends on its strike mix, and it accumulates a residual that looks like gamma P&L and is actually model error. The remedy is to hedge with the mixture's delta and use the implied volatility only as a quoting convention.

The right diagnostic is the density, not the volatility. Section 19.2's Breeden–Litzenberger construction recovers Figure 36.2's left panel directly from the option prices and displays the bimodality that any single volatility number averages away.

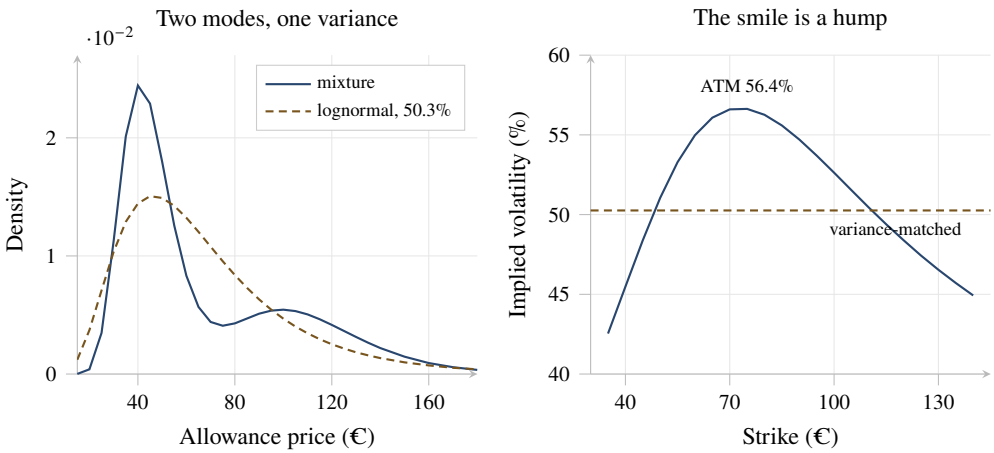


Figure 36.2: Example 36.12: mixture of lognormals, $p = 0.35$ on €110 against €45, 25% within each regime.

Figure 36.2 puts the two results side by side. The densities in the left panel have identical means and identical variances, and the lognormal's mode nearly coincides with the mixture's lower peak while standing only 61% as tall. From there it decays monotonically, *filling the valley the mixture leaves empty* (0.0095 against 0.0041 at €75) *and missing the upper mode entirely*. The right panel is what that misfit looks like through the only lens the market quotes in, and the hump is the diagnostic: an implied volatility surface that peaks at the money and falls in both wings is reporting a bimodal density, not a volatility.

36.15 WEATHER DERIVATIVES

Everything so far in this book has an underlying that can be held. *A weather index cannot be*, and the consequences for pricing are complete rather than partial.

A weather derivative settles against a measured index, most commonly **heating degree days** (HDD), HDD $= \max(65°\text{F} - T_{\text{avg}}, 0)$ summed over a period, with T_{avg} the day's average temperature, or its cooling counterpart, cooling degree days (CDD), which sum $\max(T_{\text{avg}} - 65°\text{F}, 0)$ instead. *The exposure it hedges is volume, not price*: a gas utility sells less gas in a warm winter, and no commodity hedge addresses that, because the utility's problem is that it has nothing to sell rather than that the price is wrong.

The index is a sum of many bounded daily quantities, so it is close to normal rather than lognormal, and the Bachelier form is the appropriate one. With index mean μ, standard

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deviation σ, strike K, tick value τ (the payment per index point) and $d = (\mu - K)/\sigma$,

$$P = \tau\big[\sigma\,\varphi(d) - (\mu - K)\,\Phi(-d)\big], \qquad C = \tau\big[\sigma\,\varphi(d) + (\mu - K)\,\Phi(d)\big] \tag{36.6}$$

with φ and Φ the standard normal density and distribution. There is no discounting of a forward and no volatility input in the usual sense: σ is the standard deviation of an index, estimated from history.

Example 36.13 Pricing without a hedge

November–March HDD at a reference station: mean 1200, standard deviation 140. A utility buys an HDD put struck at 1100 with a \$5,000 tick. Here $d = 100/140 = 0.7143$, and (36.6) gives

$P(\text{HDD} < 1100) = \Phi(-d)$	23.8%
$E[\text{shortfall} \mid \text{below}]$	82.2 HDD
Value of the put	**\$97,618**
Value of the matching call	\$597,618

Parity checks the pair: $C - P = \tau(\mu - K) = \$500{,}000$ exactly, which is the Bachelier analog of forward–strike parity and holds with no discounting because the index is not an asset.

That \$97,618 is an actuarial expectation, not a replication cost. No one can buy a degree day, so the writer cannot delta-hedge and the price is the expected payout plus a risk load set by capital and competition. Weather derivatives are priced like insurance and traded like derivatives, and every disagreement about value in this market is a disagreement about the distribution rather than about the model.

That makes the estimate of σ, and of μ, the entire pricing problem. Two conventions compete. *Burn analysis* takes the empirical distribution of the last 20 to 40 years of the index and averages the payoff; *index modeling* fits a daily temperature process and simulates. The first is transparent and assumes stationarity; the second imposes structure and can be wrong in ways burn analysis cannot. *And a warming trend makes stationarity false*, so detrending is a pricing input rather than a refinement: a 30-year window with a 1 HDD per year trend misstates μ by roughly 15 HDD. At this tick that is \$75,000 on a contract linear in the index, and it moves this put's value by about \$19,000, a fifth of its premium.

The geographic basis is the other half of the problem. The index is measured at one station; the utility's load is spread over a service area. Hedging load exposure L with the station index S, the minimum-variance ratio is $h^* = \rho\,\sigma_L/\sigma_S$ and the residual standard deviation is $\sigma_L\sqrt{1-\rho^2}$:

Station–load ρ	Variance reduction	Residual s.d.
0.95	90%	31%
0.90	81%	44%
0.85	**72%**	**53%**
0.70	49%	71%
0.50	25%	87%

At $\rho = 0.85$ the hedge removes 72% of the variance and leaves 53% of the standard deviation, and, unlike Section 36.9's commodity basis, *there is no instrument to hedge the remainder*, because a contract on the utility's own service area would have no counterparty willing to price it.

That asymmetry sets the market's structure. Weather risk is diversifiable across regions and perils but not hedgeable within one, so the natural writer is a party holding many uncorrelated exposures (a reinsurer or a diversified fund), and the market's participants look accordingly more like insurance than like a derivatives desk.

36.16 CATASTROPHE BONDS

The same economics can be funded. A catastrophe bond pays a coupon and returns principal unless a defined event occurs, in which case principal is written down and paid to the sponsor,

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the insurer or reinsurer that issued the bond to transfer the risk. Structurally it is Section 25.15's funded credit protection with a physical trigger, and Part V's machinery applies to the note's mechanics unchanged.

Pricing is an expected loss plus a load, and the load is most of it. The market quotes the ratio of spread to modeled expected loss as the **multiple**:

Expected loss	Spread	Multiple	Risk load
1.00%	4.50%	4.50×	3.50%
2.00%	6.50%	3.25×	4.50%
4.00%	10.00%	2.50×	6.00%
8.00%	16.00%	2.00×	8.00%

The multiple falls as the expected loss rises, which is the same pattern as credit: remote risks carry proportionally more compensation because the investor is being paid for the uncertainty in the loss estimate as well as for the loss. And the loss estimate comes from a vendor catastrophe model rather than from a market, so the multiple is partly a discount for model risk, which is why it widens after an event that the models did not anticipate, with no change in the physical hazard.

The trigger is the design decision and the trade-off is unavoidable. An **indemnity** trigger pays on the sponsor's actual losses, which is perfect for the sponsor and gives the investor exposure to the sponsor's underwriting quality. That is Section 26.9's incentive problem exactly. A **parametric** trigger pays on a measured physical quantity, wind speed or ground acceleration at a location, which removes the incentive problem and introduces basis risk of the kind Example 36.13 quantified. A trigger that eliminates basis risk transfers the moral hazard and one that eliminates the moral hazard transfers the basis risk, and no third option exists.

The diversification claim is unusually solid. Catastrophe risk is close to uncorrelated with financial markets because the mechanism is physical rather than statistical, which distinguishes it from almost every other diversification claim in this book, where the correlation is estimated from a sample and fails in the state where it is needed.

36.17 WHAT IS PRICED BY ARBITRAGE AND WHAT IS PRICED BY EXPECTATION

The instruments of the last four sections sort by how much of the book applies, and the sort matters because the formulas look identical in all three cases.

Allowance forwards: the machinery applies. Banking enforces the carry relation in one direction, and Example 36.10's residual is interpretable with the tools of Section 36.1.

Allowance options: the mechanics apply and the assumptions do not. (36.5) prices them correctly and Black's formula reports the answer in units of a volatility that is not one. The hedge ratio, not the price, is where the misreading costs money: Example 36.12's 0.078.

Weather and catastrophe: almost none of it applies. No replication, no arbitrage bound, no hedge for the writer. These are priced actuarially, and the derivatives contribution is contract design (the index, the trigger, the settlement) rather than the valuation.

The valuation machinery of Parts II and III derives a price from the cost of a hedge, and it returns a meaningful number only when the hedge exists. Where it does not, the same formula still returns a number, and that number is an expectation under an assumed distribution with nothing enforcing it. Both are legitimate quantities; confusing them is the error, and it is easier to make here than anywhere else in the book precisely because the arithmetic is unchanged.

One practical marker separates them. If the model's output changes when you change the hedging instrument, the price is a replication cost; if it changes only when you change the

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distribution, it is an expectation. A desk that cannot say which of the two it is quoting does not know what its risk report means.

36.18 WHO TRADES THESE MARKETS, AND WHY

Producers hedge to protect a capital-intensive investment made years before the revenue arrives. A producer's hedge is usually a term structure position, not a spot one, because the exposure is a production profile.

Consumers (airlines, utilities, food processors) hedge input costs, and their exposure is an average, which is why Section 33.8's Asian settlement is the norm.

Merchants and physical traders are the market's distinctive participant and have no analog in financial markets. They own the optionality of Section 36.7: storage, shipping, blending, and the right to deliver in one place rather than another. Their profit is the extrinsic value that the physical owner did not price, and it is earned by continuous re-optimization.

Utilities and generators are long the spreads of Section 36.6, whether or not they hedge them, and a generator that does not model its plant as a portfolio of options holds one anyway. In a market with an allowance price they are long the *clean* spread (power less fuel less carbon), so Section 36.6's exchange option becomes a three-asset spread requiring a correlation matrix rather than a single ρ, and the carbon leg is the one whose correlation is least stable.

Compliance buyers are the structural demand in allowances, and their hedging horizon is set by a surrender calendar rather than by a business plan, which makes their demand inelastic on known dates—an unusual and exploitable feature, and the reason well-designed schemes stagger the deadlines.

Utilities again, and reinsurers, are the two sides of weather. The absence of a natural opposite side within a region is why this market stayed small, and the point generalizes: a derivatives market needs not a risk but two parties with opposite exposures to it, which is what Section 33.10 established about inflation in a different setting.

Financial investors seek diversification and inflation sensitivity, and Section 36.11 describes what their participation did to the curve. In catastrophe risk their diversification claim is unusually well founded; in allowances their position is a directional bet on scarcity described as a commodity trade.

The recurring theme is that these markets are unusually close to something outside finance. The convenience yield is a fact about warehouses, the Samuelson maturity effect is a fact about how fast a shortage resolves, a negative power price is a fact about the cost of shutting down a turbine, and a carbon implied volatility is a fact about a cost curve. A model that loses sight of that will fit the data and misprice the constraint.

36.19 WHERE THIS CAME FROM

This subject is older than the rest of the book and grew from the study of storage rather than of options. Keynes (1930) argued that the futures price sits below the expected spot because hedgers pay speculators to bear risk, the normal backwardation of Section 5.13. Working (1949) and Brennan (1958) explained the shape of the curve instead by the cost and the benefit of holding the physical, which is where the convenience yield comes from.

The modern treatments make that yield stochastic, as Gibson and Schwartz (1990) did, and then ask how many factors a given commodity actually needs. *Schwartz (1997) is the standard comparison* and remains the natural starting point for choosing between one-, two- and three-factor specifications, because it compares them on the same data rather than advocating one.

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SUMMARY

One asymmetry generates the chapter. The cash-and-carry arbitrage runs one way (buy and store, never short and un-store), so contango is capped by storage economics and backwardation is not, and the cap disappears when storage fills. The convenience yield is the residual that makes the carry relation hold, and read off the curve it measures how scarce physical inventory is now and how long the scarcity is expected to last.

Storage also sets the volatility structure. Mean reversion makes near contracts more volatile than far ones, so options on different delivery months describe different markets and cannot be compared at one volatility. Electricity, which cannot be stored at all, is the extreme case: its forward is an expectation rather than an arbitrage price, its spot price spikes, and a lognormal matched on mean and variance misprices caps in both directions, overpricing those near the money and badly underpricing those far out. Which model a power contract needs depends on what it settles on. A daily cap depends only on each day's price; a contract on a monthly average depends on how spikes cluster in time, which no model of a single day can see.

Physical assets hide options. A power plant is a strip of exchange options whose value turns on the correlation between power and gas, and high correlation makes it cheap. Storage facilities, swing contracts and tolling agreements add flexibility, valued by dynamic programming because each choice uses up capacity that a later choice would have needed. Spreads between contract months need at least two factors, since a one-factor model fitted to a realistic curve moves the months in step and prices the spread close to zero. Hedging leaves a basis residual that is worst exactly when the hedge is needed, and a short-dated hedge of a long-dated commitment can be economically right and still fail for want of cash.

An emissions allowance is the limiting storable commodity. Banking is free, so the carry relation binds tightly, and whatever residual remains measures doubt about the scheme rather than physical scarcity. The price is set where a fixed supply meets a convex abatement curve, so volatility rises as the cap tightens. Policy events make the density of a future allowance price bimodal, so implied volatility humps instead of smiling, and the Black delta errs in a direction that flips across strikes.

At the far end the underlying cannot be held. Weather contracts and catastrophe bonds give the writer no hedge, so they are priced as an expectation plus a load for bearing the risk, and the buyer's residual exposure to its own location or portfolio cannot be removed. The distinction the chapter exists to make follows: if the output changes when you change the hedging instrument it is a replication cost, and if it changes only when you change the distribution it is an expectation.

FURTHER READING

Geman's *Commodities and Commodity Derivatives* is the standard reference, and the one that works through the physical constraints that generate the financial behavior. Clewlow and Strickland on energy derivatives is the practitioner's introduction to the modeling of Section 36.8 and is short. Schwartz and Smith's two-factor model is the paper behind that section's recommendation and is worth reading for the way it separates short-term and long-term uncertainty. On electricity specifically, Huisman and Mahieu's regime-switching model is the one behind (36.3), which is its one-day marginal; Cartea and Figueroa, and Geman and Roncoroni, add the mean reversion and jumps that Example 36.4 shows a monthly contract needs; and Weron's book surveys both families and the evidence that their fit varies from one market to the next. Eydeland and Wolyniec on energy and power risk management treats at length the spread and storage valuation of Sections 36.6 and 36.7. On the financialization question of Section 36.11, the empirical literature is contested and worth sampling from both sides; the disagreement is mostly about magnitude rather than direction. For Section 36.14, the mixture-of-lognormals treatment of event risk is developed in the literature on option prices

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around scheduled announcements. The carbon application is a special case of it, and reading the announcement papers first makes the hump in Figure 36.2 look inevitable rather than exotic. Weather derivatives have a small and stable literature of which the pricing material is mostly actuarial; the useful question, how to detrend, is treated mainly in the insurance journals rather than the finance ones. On catastrophe bonds the market's own offering documents set out the trigger trade-off of Section 36.16 and are worth reading in preference to summaries.

PRACTICE QUESTIONS

36.1 State the Samuelson relation and compute the twelve-month volatility at $\sigma_S = 50\%$ and $a = 1.0$. Using Example 36.2, state the one-month to twelve-month ratio at $a = 1.5$ and explain what a cross-maturity hedge ratio of one would cost.

36.2 Using Example 36.3, state the ratio of spike-model to lognormal cap values at \$200 and at \$400, and explain why calibrating the lognormal to at-the-money options does not close the gap.

36.3 Using Example 36.5, state the value of the option on one megawatt-hour at $\rho = 0.9$ and at $\rho = 0$, and explain why high power–gas correlation makes the spread option cheap rather than expensive.

36.4 Compute the 3-2-1 crack spread at crude \$80/bbl, gasoline \$2.50/gal, heating oil \$2.55/gal, and state what changes if the refinery's actual yield is 2-1-1.

36.5 Using Example 36.6, state the extrinsic share at 60% volatility and $\rho = 0.75$, and explain why the facility is usually leased by a trader rather than operated by the utility that built it.

36.6 Invert the carry relation to give the implied convenience yield, and compute it for $S = 80$, $F = 77$ at six months, $r = 4\%$, $u = 5\%$. State what a negative value would indicate and why it cannot be arbitraged away.

36.7 Using Example 36.8, state the risk reduction achieved by hedging the flat price, and explain why that figure overstates the protection in exactly the episodes the hedge was bought for.

36.8 Using Example 36.9, state the annual roll cost at 1% monthly contango on a \$780m stack, then state the cash loss and the present value of the offsetting gain at a 10% price fall and explain why the difference can end a firm.

36.9 Give the two independent reasons a one-for-one stack is the wrong size, state the remedy, and explain what the remedy requires that cannot be reliably estimated.

36.10 Compute the two-year allowance forward at a spot of €65 and a 4% rate, and state which side of that value is enforced by arbitrage.

36.11 Explain why a positive implied convenience yield in allowances cannot be physical, what its decay with maturity indicates, and how dealer financing of compliance inventory contaminates the reading.

36.12 Using Example 36.11, state the price change from two different 20 Mt shifts in required abatement, and explain what that implies for a value-at-risk estimated on a historical window.

36.13 Explain how a buy-out price caps a renewable certificate and what that does to the vega of an option written on it.

36.14 Using Example 36.12, state the at-the-money implied volatility and the two wing values, explain the shape, and state what a standard skew parametrization fitted to these quotes would get wrong.

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36.15 Explain why the variance-matched lognormal volatility of 50.3% differs from the 56.4% at-the-money implied volatility, and what each measures.

36.16 Using Example 36.12, state the largest delta error, explain why a book spread across strikes does not reliably net it out, and state what the residual will be mistaken for.

36.17 Explain why the Bachelier form rather than Black's is appropriate for a degree-day index, and use (36.6) to verify the put–call parity relation in Example 36.13.

36.18 Explain the difference between burn analysis and index modeling, state what each assumes, and quantify the effect of a 1 HDD-per-year trend on Example 36.13's contract.

36.19 Using Example 36.13, state the minimum-variance hedge ratio and the residual standard deviation at $\rho = 0.85$, and explain why the remainder cannot be hedged.

36.20 Explain why a catastrophe bond multiple falls as the expected loss rises, and why multiples widen after an unanticipated event with no change in the physical hazard.

FURTHER QUESTIONS

36.21 Example 36.2 gives the volatility term structure under mean reversion. Derive it from a mean-reverting spot process, then determine what the term structure looks like when storage is abundant, and use the result to explain why the observed term structure steepens when inventories fall.

36.22 Example 36.5 shows a 133% valuation error from ignoring correlation. Derive the spread option's sensitivity to ρ, determine where in the heat-rate spectrum the error is largest, and explain why a fleet of plants with different heat rates cannot be valued with a single correlation assumption.

36.23 Example 36.6 decomposes storage into intrinsic and extrinsic value. Formulate the storage operator's problem as a dynamic program with injection and withdrawal rate constraints, state how the constraints reduce the extrinsic value, and explain why a fast-cycling facility is worth more per unit of capacity than a slow one.

36.24 Example 36.8 finds a 93.3% risk reduction with a residual that is worst when the hedge is most needed. Model the basis as correlated with the flat price under stress, recompute the risk reduction conditional on a large flat-price move, and state how a producer should size the hedge given that conditional figure.

36.25 Example 36.10 reads the deviation from carry as a price of event risk and observes that it decays as $1/T$. Formalize this: model the allowance as paying off only if the scheme survives to T, derive the implied survival probability from the curve, state what else could produce the same discount, and propose a test on the term structure that would distinguish the explanations.

36.26 Example 36.12 produces a delta error that changes sign across strikes. Derive the mixture's delta in closed form from (36.5), determine the strike at which the error is largest as a function of p, and construct the portfolio of vanilla options that would hedge the regime exposure directly rather than through delta.

36.27 Example 36.13 finds 53% residual standard deviation at a 0.85 correlation. Determine how many independent station contracts would be needed to bring the residual below 20%, state the correlation structure your answer assumes, and explain why the market has not developed such multi-station products.

36.28 Example 36.4 holds each day's price distribution fixed and varies only how spikes cluster. Prove that the strip of daily caps cannot depend on the clustering, derive the variance of

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the monthly average as a function of the episode length, and explain why options on block futures trade at implied volatilities far below those of daily options in the same market.

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Chapter 37
Crypto Derivatives

Chapter 36 worked through markets where the valuation machinery of Parts II and III fails, and in each case it failed because of the underlying: electricity cannot be stored, an emissions allowance has a supply a legislature sets, a degree-day index cannot be held at all.

This chapter's underlying breaks none of those rules. A bitcoin is storable at essentially no cost, perfectly fungible, divisible to eight decimals, pays no dividend, has no convenience yield, no seasonality, and no delivery constraint. By the standards of Chapter 36 it is the best-behaved underlying in the part—better behaved than crude oil, which has to be put somewhere, and far better behaved than an allowance whose validity is a political question.

So Chapter 5's carry relation applies without modification. With no income and no storage cost, the arbitrage-free futures price is $F = Se^{rT}$, the cash-and-carry arbitrage can be run in both directions, and the basis trade of Chapter 6 transfers across with the algebra unchanged. When market prices depart from that relation, as Section 37.1 shows they do by several percentage points a year, the reason is not the underlying.

What is missing is everything around the contract. There is no central counterparty, no segregation of client collateral, no daily settlement cycle, because the offshore market never closes and so has no overnight pause in which to collect margin or manage a default, no enforceable netting opinion of the kind Chapter 27 called worth more than any risk model, and no lender of last resort. The machinery of Parts II and III assumes an infrastructure it never names, and this is the market where that assumption is visible because it is absent.

The distinction between the contract and the infrastructure around it organizes the chapter, and it is not a technicality. The largest custody failure in the market's history was not caused by a mispricing, a model failure, or a move in the underlying, and Section 37.10 argues that describing it any other way misplaces the lesson.

Sections 37.1 to 37.5 take the contracts in turn: listed futures and their basis, the effect of a staking yield, perpetual futures, options, and the inverse contract with its choice of numeraire. Section 37.6 looks at the producer and Section 37.7 at the spot ETFs that narrowed the gap to the regulated market; Sections 37.8 and 37.9 are the decentralized venue and the settlement asset. Section 37.10 is the custody failure, and Section 37.11 says who trades these derivatives and why.

37.1 LISTED FUTURES AND THE BASIS

Regulated futures on bitcoin have traded since 2017, and the offshore venues list futures on hundreds of tokens. The pricing is Chapter 5's, unchanged.

Example 37.1 The basis trade, transplanted

Spot at \$60,000 and the three-month futures price at \$61,800. Crypto markets quote the basis the other way round from Section 3.3's spot minus futures, and as a percentage: the futures premium over spot, $(F - S)/S$. Here that is $1{,}800/60{,}000 = 3.00\%$ over the quarter, or **12.55% annualized** with quarterly compounding $(1.03^4 - 1)$, against a dollar financing cost of 5% a year on the same

compounding basis. In Section 5.9's terms that 5% is the whole cost of carry, since a coin pays no income and costs nothing to store, while the futures contract is priced as if the cost of carry were 12.55%. Buying spot, selling the futures contract and waiting therefore earns the gap: **7.55%** a year over the cost of carrying the position.

Traders rarely put up the full position. Leverage here is the position's notional divided by the capital posted against it (for one unit of a futures contract, the futures price divided by the margin on it), so 20% initial margin is 5× leverage, and the return on that capital is the 7.55% multiplied by the leverage:

Initial margin	Leverage	Return on capital
50%	2.0×	15.1%
35%	2.9×	21.6%
20%	5.0×	37.8%

This is Section 6.5's Treasury basis trade with the numbers moved up by two orders of magnitude, and the reason the spread is 12.55% rather than a few basis points is not that the arbitrage is harder. It is that three things Chapter 6 took for granted are absent: *there is no CCP*, so the position is an unsecured exposure to a venue; margin is set at the venue's discretion rather than by a published risk model, and can be raised without notice; and *the collateral is held by the counterparty*, so the trade's return is conditional on the counterparty surviving the quarter.

The basis is therefore not an inefficiency but a price. This is Section 5.15's observation that arbitrage requires a balance sheet, in a market where it also requires trusting the venue with the balance sheet.

The return in Example 37.1 does not show the trade's main risk, which is a shortage of cash rather than a loss of value.

Example 37.2 A hedged trade that a rally can close

The basis trade has no net exposure to the price: the spot leg and the short futures position move together, so the profit is locked in at entry. *The cash flows are not flat.* The futures position is margined daily; the spot is not. So a rally calls cash on one leg against an unrealized gain on the other.

At $60,000 spot the futures price is $61,800, and 20% initial margin on the contract is $12,360 per unit. The cash called is the variation margin the short futures position pays as the futures price rises; with the basis held at 3%, a rally of x raises the futures price by $x \times \$61{,}800$:

Rally	Cash called	Share of margin	Value of the position
+10%	$6,180	50%	unchanged
+20%	$12,360	100%	unchanged
+33%	$20,394	165%	unchanged
+50%	$30,900	250%	unchanged

A 20% rally, the margin rate itself, exhausts the margin on a position that has not lost a cent. The trade is liquidated for want of cash while the thing it is hedged against is working exactly as intended, and the liquidation forfeits the 7.55%, which was locked in only for a position held to expiration. The futures loss is realized in cash at once, while the offsetting gain on the coin exists only as a mark until the coin is sold; sold alongside, it recovers nearly all of that loss but none of the carry, and kept, it is an unhedged long.

This is Section 8.9's margin call as a liquidity risk, and the reader has met it twice already: it is the problem that brought down Metallgesellschaft's oil hedge, and it is why Chapter 8 insisted that a hedge which is correct in value and wrong in timing is not a hedge. What this market adds is that the venue sets the margin, so the schedule in the table is not a parameter of the trade but a decision someone else may revise mid-position.

Hence the leverage in Example 37.1 is not free. At 5× the headline return is 37.8%, and the

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rally that ends the trade is 20%. At the 70% volatility this market commonly prices (Example 37.7 below), a coin with no expected drift in its price rises 20% at some point within a quarter with a probability of about 55%. The trade is more likely than not to be liquidated before the futures contract expires and the 7.55% is earned.

The direction of the basis is informative in the same way Chapter 36's curve was. A persistent premium says leveraged long demand exceeds the capital willing to finance it, and the premium collapses, occasionally inverting, when that demand unwinds. It shows how traders are positioned, measured by a market price rather than a survey, and it is easier to read than Chapter 36's convenience yield. That yield is never quoted: it has to be backed out of the curve as the residual left after interest and storage costs. Here the premium itself is quoted, and because a coin pays no income and costs nothing to store, subtracting the dollar interest rate leaves the excess over carry of Example 37.1 with nothing else to estimate.

37.2 WHEN THE UNDERLYING PAYS A YIELD

This chapter opened by saying the underlying pays nothing. That is true of bitcoin and false of the second-largest asset in the market, and the correction is worth making because it moves the carry relation from Chapter 5's case of an asset that pays nothing to its case of an asset with a known yield.

A **proof-of-stake** asset pays its holder for committing coin to validating the network. That is a continuous yield on the asset itself, so the forward price is Chapter 5's known-yield case:

$$F = Se^{(r-q)T}. \tag{37.1}$$

The yield q is not set by anyone's discretion. On Ethereum the protocol determines it, and it has two parts. The larger is newly issued coin paid to validators, at a rate that falls as more coin is staked: approximately $166.3/\sqrt{N}$ a year when N ether in total is staked, or about 2.6% at the roughly 41 million staked in 2026. The more coin is committed, the thinner the reward is spread, so the yield regulates how much of the supply is staked. The smaller part is the transaction fees and the payments for ordering transactions that a validator collects when it proposes a block; these vary with network activity and typically add a few tenths of a percent. A staking provider keeps a commission, commonly around 10%, out of both.

Two consequences follow for (37.1). First, q is not known in advance, since both the amount staked and the network's activity move; a forward embeds the yield expected over its life, not a contractual rate. Second, the issuance part is paid in new coin, so it is partly a transfer from holders who do not stake, whose share of the supply shrinks. Unlike a dividend, it is not paid out of anyone's earnings, and a holder who does not stake is diluted rather than merely forgoing income.

Example 37.3 A dividend the asset did not previously have

Spot \$3,000, a 5% dollar rate and the staking yield q, both continuously compounded, and three months, so the forward is $3{,}000\,e^{(0.05-q)\times 0.25}$:

Staking yield q	Forward	Quarterly basis	Annualized
0.0%	\$3,037.74	+1.258%	+5.13%
2.5%	\$3,018.81	+0.627%	+2.53%
3.5%	\$3,011.27	+0.376%	+1.51%
5.0%	\$3,000.00	0.000%	0.00%
7.0%	\$2,985.04	−0.499%	−1.98%

The basis is computed as in Example 37.1: the premium $(F-S)/S$ over the quarter, annualized

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by compounding four quarters. At $q = r$ the futures price equals spot, and above it the curve inverts. The inversion says nothing about scarcity. The asset now pays a yield, and a persistent backwardation that would signal scarcity in Chapter 36 signals here only that the staking yield exceeds the funding rate.

So Example 37.1's basis is not a clean positioning indicator for a staked asset. The observed premium mixes leveraged demand with the yield, and a desk reading it as sentiment will misread it whenever q moves. The two are separable only if the staking yield is quoted independently, which it is, so the correction is available and is routinely not made.

The first US spot ether ETFs of Section 37.7 were launched without staking, and that has a cost. A staked fund followed in March 2026: BlackRock's trust stakes 70 to 95% of its ether and keeps the rest unstaked to meet redemptions. A fund holding coin without staking it forgoes the yield: on a $10 billion fund at 3%, that is $300 million a year, and because most of the yield is new issuance, the fund's holders are diluted by it rather than merely missing it. What the fund buys in exchange is liquidity.

Staked coin is not immediately deliverable. Exiting a validating position takes a queue (usually short, but longer when many exit at once, while the entry queue can run to weeks), so the deliverable supply is smaller than the outstanding supply. A fund that must meet redemptions daily cannot hold all its assets behind any queue, which is why a staked fund keeps an unstaked sleeve, and the cash-and-carry arbitrage of Section 37.1 needs unstaked coin to close.

This qualifies the chapter's opening claim in a specific way. The underlying is storable at no cost *and* it now offers a yield for making it temporarily undeliverable, so a holder faces a genuine choice between earning q and retaining delivery capacity. That is Chapter 36's convenience yield reappearing—not as an inference from a curve, but as a posted rate a holder gives up to keep the asset liquid.

37.3 PERPETUAL FUTURES

A **perpetual futures** contract (a **perpetual** for short) has no maturity. It never delivers and never expires, so nothing forces convergence to spot. *What replaces convergence is a payment*: at fixed intervals, usually every eight hours (some venues hourly), longs pay shorts a **funding rate** proportional to the gap between the perpetual's price and a spot index, and when the perpetual trades below the index the payment reverses.

The mechanism deserves a moment because it inverts the usual construction. A dated futures contract is held to spot by the *prospect* of delivery, and the cost of carry is embedded in the forward price. A perpetual is held to spot by an explicit cash flow instead, and that cash flow takes the place of the premium in a dated contract. A long in a dated contract pays the carry through the price: it buys at a premium over spot, and the premium shrinks to nothing by delivery. A perpetual's price never converges, so there is no premium to shrink, and without a payment a long perpetual would be leveraged exposure to the coin with nothing paid for the financing. The funding rate is that payment. It is not a cost of carry in Chapter 5's sense, a quantity fixed by interest rates and storage, but a price set by the demand for leverage. It has to cover at least what an arbitrageur who sells the perpetual and holds the coin pays to hold it, and on top of that it carries whatever premium leveraged longs pay, just as the dated basis of Example 37.1 does.

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Example 37.4 Funding is the basis, quoted differently

Funding is paid every eight hours, so there are $3 \times 365 = 1{,}095$ periods a year:

Funding per 8 hours	Annualized	Equivalent 3-month premium
+0.0100%	+10.95%	+2.74%
+0.0300%	+32.85%	+8.21%
+0.0500%	+54.75%	+13.69%
−0.0200%	−21.90%	−5.47%

A quoted funding rate of 0.01% per eight hours and a three-month futures contract at a 2.74% premium are the same statement about the price of a levered long, and a desk running both should see them agree. When they do not, the difference is the cost of the two venues' credit and margin terms rather than anything about bitcoin.

In one respect this market reports more than the regulated one does. Chapter 5 had to *infer* a convenience yield as the residual that made an equation hold, and Chapter 36 spent a section extracting one from a curve. The corresponding quantity here, the price of holding a levered long, *is printed every eight hours* rather than backed out of a curve.

The perpetual exists because the infrastructure for delivery does not. A dated contract requires a settlement process, a delivery window, and a mechanism for failures; a contract that never delivers requires none of them. The instrument is an adaptation to missing infrastructure, which is the chapter's theme in a single product.

Liquidation replaces the margin call. There is no cure period, and no one at a broker deciding whether to give the client more time. When margin falls below a threshold the venue closes the position at once. In traditional markets the same trigger produces a margin call instead (from the broker on exchange-traded futures, from the counterparty under a credit support annex on OTC derivatives, from the broker on a securities margin account), and a margin call starts a clock: the client has about a day to raise cash, and can see whether the move reverses before committing it. The difference lies with the venue, not the coin, since bitcoin futures on CME carry margin calls like any other futures contract. On an offshore venue the trigger ends the position. The client can add collateral at any moment up to that point, but must do so before knowing whether it will be needed, so protection means collateral posted early and left idle, or a position watched around the clock—and a fast market can cross the threshold before anyone can act. Section 8.9's collateral spiral runs here in minutes rather than days, and because the liquidation levels of leveraged positions cluster at round prices, the forced flow is a known function of the price, which is Section 35.7's barrier clustering in another market, with one difference that Example 37.5 draws out.

Example 37.5 A cascade with published barriers

Take \$12 billion of long-biased perpetual open interest, distributed across the leverage the venues actually offer. A position at ℓ times leverage has margin worth $1/\ell$ of its notional, so it liquidates on a fall of roughly $1/\ell$ (a little less in practice, since the venue closes it before the margin is entirely gone):

Leverage	Liquidates on	Notional in tier	Cumulative forced selling
25×	−4%	\$1.2bn	\$1.2bn
20×	−5%	\$1.8bn	\$3.0bn
10×	−10%	\$4.2bn	\$7.2bn
5×	−20%	\$3.6bn	\$10.8bn

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The table assumes that every position's leverage is measured from the current price and that no one adds margin as the price falls.

A 10% decline mechanically forces $7.2 billion of selling, which is 60% of the open interest, and the selling arrives into a market already falling. *The second-round effect is the point*: that flow pushes the price further, which triggers the next tier. It is the same feedback as the issuer hedging of Section 35.6, except that no issuer decides when to trade: the venue's engine does.

But the disclosure is the opposite of Section 35.7's. Autocallable barriers cluster because issuers independently choose round fractions, and *no issuer can see the aggregate*; that was the whole difficulty. Here open interest and leverage tiers are *published by the venues in real time*, so the liquidation profile is computable by anyone who wants it.

That does not make the cascade smaller. It changes who gains from the cascade: the flow is visible in advance to the traders best placed to trade ahead of it, and the resulting anticipation is itself part of why the moves are abrupt. Transparency relocated the problem rather than removing it, which is worth setting beside Chapter 2's argument that disclosure is what makes a clearing system safe.

The largest recorded case is October 10, 2025, when, amid new US tariff announcements aimed at China, trackers counted about $19 billion of liquidations in roughly a day, and at least one large venue closed profitable positions by auto-deleveraging.

37.4 OPTIONS

Options trade on the offshore venues and, since 2024, on the regulated ETFs of Section 37.7. The pricing is Section 16.5's. The offshore contracts are European and cash-settled, and the venues price and hedge each one against the futures contract of the same expiration, so the model is Black's, with the futures price in place of spot. That choice matters more here than anywhere in Part III. Black's formula needs only the forward, so it takes in the basis of Section 37.1 and the staking yield of Section 37.2 without anyone having to estimate them, which is Section 16.8's rule of getting the forward right. A value computed from spot and the dollar rate assumes a forward of Se^{rT} and misprices the option by the whole basis, as Example 37.6 shows. Time is counted in calendar days over 365, since the market never closes.

Example 37.6 Pricing off the forward the market trades

Spot $60,000, the three-month futures price of $61,800 from Example 37.1, a strike of $60,000, 70% volatility and a 5% dollar rate:

Forward used	Call	Put
Futures price, $61,800 (Black)	$9,273	$7,496
Se^{rT}, $60,755	$8,661	$7,916

Pricing from spot and the dollar rate undervalues the call by $612, or 7%, and overvalues the put by $420, or 5%. The two errors are the same mistake: put–call parity, $c - p = e^{-rT}(F - K)$, ties both prices to the forward, and the second row uses the wrong one. The hedge is in the same contract. The call's delta against the futures contract is $e^{-rT}N(d_1) = 0.595$, so a dealer who sells the call buys 0.595 futures contracts per coin.

The implied volatility surface of these options differs from an equity index's in three ways.

The level is high, and the term structure inverts readily. At-the-money implied volatilities of 50% to 80% are ordinary, and short-dated volatility jumps above long-dated in every sharp move, a shape equity indices show mainly in a crisis (Section 19.6).

The skew changes sign. Equity index skew has pointed one way since 1987; *crypto skew reverses with the market*, with calls in demand in a rally and puts in demand in a decline. This is the pattern Section 19.1 found in currencies, whose tilt also changes sign as sentiment moves,

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rather than the one it found in equity indices, and for a similar structural reason: there is no leverage effect (the tendency of a stock's volatility to rise as its price falls, because a falling share price raises the firm's debt relative to its equity) and no natural buyer of protection, so the skew reflects whichever side is currently buying options.

And the volatility risk premium is large and unstable. Section 19.10's two to four volatility points on equity indices is often ten points or more here, which is why so much of the retail product in this market is a covered call in some form. What that level of volatility costs is worth seeing in currency rather than in percentage points.

Example 37.7 What a 70% volatility costs

A thirty-day at-the-money straddle on a $60,000 spot:

Volatility	Straddle	Share of spot	Break-even move
15%	$2,059	3.43%	±3.43%
30%	$4,116	6.86%	±6.86%
50%	$6,857	11.43%	±11.43%
70%	$9,591	**15.99%**	±15.99%
90%	$12,318	20.53%	±20.53%

An equity index at 15% costs 3.43% of spot to straddle for a month; this market at 70% costs 15.99%. The buyer needs a 16% move in thirty days merely to return the premium, and the seller collects 16% of spot for a month of risk, which is why the yield products of Section 37.11 can advertise returns that look impossible and are not.

The premium is real and it is not a riskless profit. To first order, selling variance earns $(\sigma_{\text{imp}}^2 - \sigma_{\text{real}}^2)/2$ a year:

Implied	Realized	Variance premium
70%	60%	+6.5% a year
70%	55%	+9.4% a year
70%	75%	−3.6% a year

Ten volatility points of edge is worth 6.5% a year and five points of adverse surprise costs 3.6%, so the strategy's sign depends on a quantity no one observes in advance, which is Section 22.4's point about variance swaps, in a market where the numbers are several times larger and the estimates far less stable.

37.5 INVERSE CONTRACTS AND THE CHOICE OF NUMERAIRE

A large part of this market trades in contracts that settle in the coin rather than in dollars, and the convention is not cosmetic. An **inverse** contract has a notional fixed in dollars, margin posted in the coin, and settlement paid in the coin: a long position of notional Q entered at F_0 pays

$$\Pi_{\text{coin}} = Q\left(\frac{1}{F_0} - \frac{1}{S_T}\right). \tag{37.2}$$

This is Section 28.6's numeraire question posed by a contract specification rather than by a model. Convert (37.2) at the terminal price and the dollar payoff is $Q(S_T/F_0 - 1)$—*exactly linear*. Leave it in coin and it is not.

Example 37.8 Linear in one currency, curved in the other

A long inverse position, $100,000 notional, entered at $60,000:

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Price	Move	P&L in coin	P&L in dollars
\$30,000	−50%	−1.6667	−\$50,000
\$42,000	−30%	−0.7143	−\$30,000
\$48,000	−20%	−0.4167	−\$20,000
\$72,000	+20%	+0.2778	+\$20,000
\$90,000	+50%	+0.5556	+\$50,000
\$120,000	+100%	+0.8333	+\$100,000

The dollar column is a straight line and the coin column is not. A 50% fall costs **1.6667 coin** while a 50% rise earns only 0.5556, and the coin gain is capped at $Q/F_0 = 1.6667$ however far the price rises while the coin loss is unbounded. A long inverse position is short gamma measured in coin, with $\partial^2 \Pi_{\text{coin}}/\partial S^2 = -2Q/S^3$.

The practical consequence is that margin is computed in the unit where the position is curved. Start with 0.5 coin of margin:

Price	Equity (coin)	Equity (dollars)
\$54,000	+0.3148	\$17,000
\$48,000	+0.0833	\$4,000
\$42,000	−0.2143	−\$9,000

Equity in coin falls faster than the price does, because the loss is denominated in an asset that is itself losing value, so liquidation arrives earlier than a dollar-margined trader would compute from the same leverage. The collateral and the exposure are the same asset, which is Chapter 27's wrong-way risk in its purest form, written into the contract rather than arising from a correlation.

The convention has a practical origin. An inverse contract needs no dollar leg, so a venue can run it without touching the banking system: the whole contract lives in the asset it references. That mattered when venues could not get bank accounts, and it is why the oldest and deepest contracts in this market are inverse ones while the newer dollar-margined contracts dominate flow from institutions that have banks.

And the pricing follows Chapter 28 mechanically once the numeraire is named. Valuing (37.2) in dollars means valuing a coin-denominated payoff, so the coin is the numeraire, and the change of measure that goes with it is the one Section 30.6 performs for a quanto. The convexity correction that follows is the same object as a quanto adjustment, and a desk that prices inverse contracts by discounting the dollar payoff is dropping it.

37.6 THE PRODUCER: MINING AS A SPREAD POSITION

This market has a producer, as Chapter 36's power market does. A miner spends electricity and capital to receive coins, which makes it *long the coin and short power*, which is Section 36.6's spark spread with a different output. The comparison holds closely enough to be useful, and the hedging problem lies where it fails.

Example 37.9 A miner's break-even, and the hedge the protocol supplies

A miner produces 500 coins a year against \$14 million of electricity and \$4 million of other costs. *Break-even is \$36,000 a coin*, and because the costs are fixed while revenue moves with the price, the margin moves much more than the price does:

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Coin price	Revenue	Costs	Margin	Margin/revenue
\$30,000	\$15.0m	\$18m	−\$3.0m	−20.0%
\$45,000	\$22.5m	\$18m	+\$4.5m	+20.0%
\$60,000	\$30.0m	\$18m	+\$12.0m	+40.0%
\$90,000	\$45.0m	\$18m	+\$27.0m	+60.0%

The output is not a fixed quantity, and that is what separates this from the spark spread. The network issues new coins at a fixed rate, a batch with each block of transactions, and a miner wins blocks in proportion to its share of the **hash rate**, the computing power all miners together apply to the puzzle each block requires. The protocol sets how hard that puzzle is, and its **difficulty adjustment** resets it at intervals so that blocks keep arriving on schedule however much hardware is running. If a price fall drives competitors off, the difficulty falls and the survivor's share rises.

Suppose the price falls from \$60,000 to \$42,000 and 20% of the network's hash rate leaves:

	Revenue	Change
Before	\$30.0m	—
After, at constant share	\$21.0m	−30%
After the difficulty adjustment	\$26.3m	**−12.5%**

The protocol converts a 30% revenue shock into a 12.5% one. The miner is long the price, and it is also long the exit of competitors, which happens when the price falls; the two offset, so the exposure is to price × share, and share moves against price.

Three qualifications keep this from being a free hedge:

- the difficulty adjustment comes at intervals of roughly two weeks, so the offset arrives after the cash-flow problem it would have solved;
- the protection comes from the miners who leave, and those are the high-cost ones, so a high-cost miner is near the exit itself when it needs protection most, while the low-cost miner, which needs it least, gains the most; and
- the hedge works through quantity, so a forward sale of a fixed amount fits neither direction: after a fall the miner produces more than it sold and the surplus is unhedged, and after a rise it produces less and must buy coin at the high price to deliver.

This explains how miners hedge. They sell a fraction of expected output forward rather than all of it, and they buy puts more often than they sell forwards, because a forward that must be delivered into is a short position in a market where the quantity is itself uncertain. *This is volume risk*. The contract built for it is the swing contract of Section 36.5, which lets its holder choose how much to take or deliver within limits; a miner would want the delivering side of one, and no such contract trades on coin output. So miners approximate it with a strip of listed puts and accept the basis.

Power contracting is the other half of the position and it is where the interesting structures are. A miner with an interruptible power contract has sold an option to the grid: it agrees to shut down on demand in exchange for a lower price. Interruptible contracts are old (aluminum smelters have signed them for decades), but a miner is an unusually good counterparty for one, because what it gives up by shutting down is coin, which can be valued at any moment: the shutdown cost is observable at every instant as the coin price times the foregone production, so the option's exercise value is marked to a screen. Several grids now contract with miners this way.

37.7 SPOT ETFS AND THE NARROWING OF THE GAP

The most consequential development in this market was not a derivative. United States spot bitcoin exchange-traded funds began trading in January 2024, and options on those funds followed in November 2024.

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What that changed is precisely the thing this chapter says is missing. An ETF share sits in ordinary custody, its options clear through the Options Clearing Corporation, the positions net under an enforceable framework, and the collateral is segregated. *None of it altered bitcoin*, which behaved identically before and after. It altered the infrastructure wrapped around it.

The gap between the two venues is therefore measurable, and it is the price of that infrastructure. An option on the ETF and an option on the offshore venue reference nearly the same underlying and differ in counterparty, segregation, netting, and margin treatment. The spread between their implied volatilities is not a modeling puzzle: it is Chapter 27's capital and Chapter 8's collateral, expressed in volatility points, and it is the cleanest available estimate of what the missing infrastructure was worth.

37.8 AUTOMATED MARKET MAKERS

A decentralized exchange has no order book. It holds an inventory of two assets and quotes by a rule: the **constant-product** rule keeps $x\,y = L^2$, where x is the inventory of a token and y the inventory of a dollar stablecoin (a token its issuer promises to redeem for one dollar, backed by a reserve of dollar assets, and the settlement asset of Section 37.9), counted here as dollars, and anyone may trade against the pool at the price that rule implies. That price is the ratio of the inventories, $p = y/x$, because along the curve $xy = L^2$ a small trade exchanges dx tokens for $dy = -(y/x)\,dx$ dollars. The constant is written as a square so that $L = \sqrt{xy}$, the geometric mean of the two inventories, measures the pool's size; it is called the pool's **liquidity**, and a deeper pool moves its price less for a trade of a given size. Neither inventory is fixed; only their product is. A buyer of the token pays stablecoin in and takes tokens out, so y rises, x falls and the price y/x goes up, while L changes only when providers deposit or withdraw.

Providers of that inventory are told they are earning fees. They are, and Example 37.10 shows what they are paying for them.

Example 37.10 Providing liquidity is selling gamma

At a price p the two conditions $xy = L^2$ and $y/x = p$ pin the inventory to $x = L/\sqrt{p}$ and $y = L\sqrt{p}$, so the position is worth

$$V(p) \;=\; x\,p + y \;=\; 2L\sqrt{p}. \tag{37.3}$$

Two derivatives tell the whole story. The delta is $V'(p) = L/\sqrt{p} = x$, *exactly the token inventory*, so the pool is automatically delta-hedged to its own holdings. And

$$V''(p) \;=\; -\frac{L}{2\,p^{3/2}} \;<\; 0,$$

so the position is concave in the price everywhere. *A liquidity provider is short gamma*, and Chapter 18 established what a short gamma position earns and what it loses.

Against simply holding the initial basket, the shortfall (the market calls it *impermanent loss*, a name suggesting the loss will reverse, which it does only if the price returns to where it started) depends only on the price ratio k:

$$\mathrm{IL}(k) \;=\; \frac{2\sqrt{k}}{1+k} - 1$$

Price ratio	0.25×	0.5×	0.8×	1.25×	2×	4×
Shortfall	−20.00%	−5.72%	−0.62%	−0.62%	−5.72%	−20.00%

It is symmetric in k and $1/k$, which is the signature of a short straddle: a payoff concave in the underlying is a short position in options.

The cost of being short gamma, the rate at which the pool falls behind the basket and which the market calls its bleed, has a simple closed form. Chapter 18's gamma cost per unit time is

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$\frac{1}{2}|V''|\sigma^2 p^2 = L\sigma^2\sqrt{p}/4$, and dividing by the position's value $2L\sqrt{p}$:

$$\text{cost per year} = \frac{\sigma^2}{8} \quad \text{of the position, whatever the price and whatever the pool size.} \tag{37.4}$$

Volatility	40%	60%	80%	100%	120%
Fee yield needed to break even	2.00%	4.50%	**8.00%**	12.50%	18.00%

The exact expected shortfall also has a closed form, $1 - e^{-\sigma^2 T/8}$, and it confirms (37.4) and shows where it stops applying: at 80% volatility over three months the exact loss is −1.980% against a predicted −2.000%, and over a year −7.688% against −8.000%. *The bleed is the small-$\sigma^2 T$ expansion of that exponential*, so it always overstates the loss: by 4% of the exact figure at the one-year horizon, but at 120% volatility over three years, where $\sigma^2 T = 4.32$, its 54.0% overstates the exact 41.7% by 29%.

So the question a liquidity provider should ask is the one an option seller asks: is the premium above or below the volatility I am short? At 80% volatility the pool must return 8% a year in fees before the provider is even, and a pool advertising a 6% fee yield on a pair that volatile is advertising a loss.

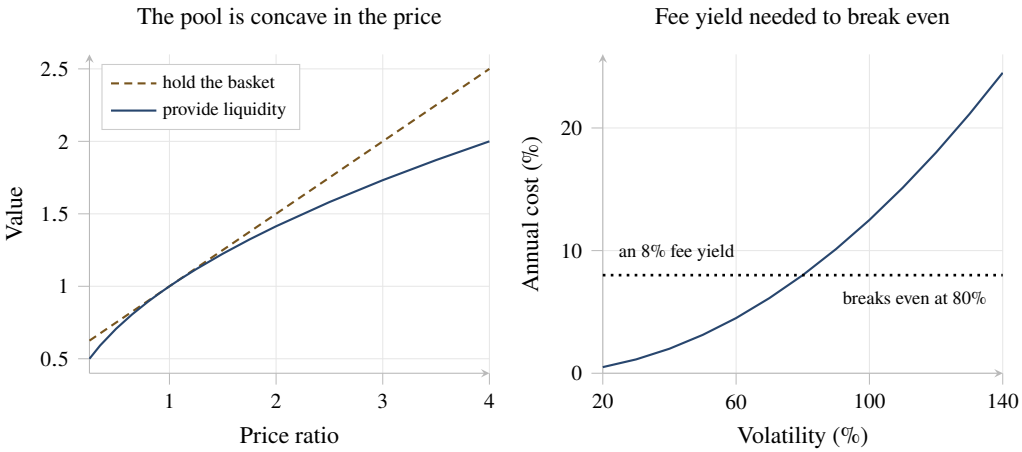


Figure 37.1: Left: pool value against holding the same initial basket, normalized to 1. Right: (37.4).

Figure 37.1 is the reason this section is in a derivatives textbook rather than a technology one. The left panel is a written straddle drawn in unfamiliar coordinates, and the concavity is the product, not a defect of it. A provider who understood the position as a short option would ask about the premium; one who understands it as "earning yield" will not.

The concentrated-liquidity variant makes the option clearer. Providers may confine their inventory to a price range, which concentrates the gamma and raises both the fee income and the bleed—and outside the range the position is entirely one asset and the delta is constant. *That is a short straddle with knock-outs*, which is Chapter 34's territory, and it is sold to retail users with no options vocabulary at all.

37.9 THE SETTLEMENT ASSET

Almost every trade on the offshore venues settles in a **stablecoin** rather than in dollars, and that changes what the quoted prices mean. A perpetual quoted at 60,000 is quoted in a token that its issuer promises to redeem at a dollar, so the futures price carries the issuer's credit whether or

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not anyone prices it that way.

The peg is not a law of the market; it is an arbitrage with a cost. If the token trades below a dollar, someone buys it and redeems it at par for the profit. That trade works only while redemption works, which requires the reserve to be liquid, the issuer to be operating, and the redemption window to be open. Remove any of the three and the token stops being a dollar and becomes a claim on a portfolio, valued as Part V values any other claim.

Example 37.11 What the discount was pricing

In March 2023 a large stablecoin disclosed that $3.3 billion of its roughly $40 billion reserve, about 8%, sat at a bank that had just failed. Take the rest of the reserve as sound and price the token as a claim:

Recovery on the exposed reserve	Fair value	Discount
100%	1.0000	0.00%
50%	0.9600	4.00%
0%	0.9200	8.00%

The token traded to about 0.8774, a 12.3% discount, against an arithmetic worst case of 8.00%. *No recovery assumption produces that price*, so the extra 4.3 points were not pricing the disclosed exposure at all.

They were pricing the redemption mechanism itself. The arbitrage that holds the peg needs an open window; a holder who fears the window closes before their turn will sell at any price above what they expect a queue to yield. So the discount decomposes into a credit loss on a known exposure and an option-like charge for access to the redemption facility, and only the first is in the disclosure.

The link to the rest of the book is direct. The failure of Silicon Valley Bank (Section 40.14) reached the derivatives market of this chapter through the settlement asset—not through any position, model, or counterparty in it. A trader perfectly hedged in coin terms lost 12.3% on the units the hedge was denominated in, which is Section 7.8's currency basis argument with the funding currency issued by a private firm.

Two consequences follow for anyone marking a book in this market. First, a coin-denominated profit and loss is not a dollar profit and loss, and the two diverge exactly when the market is stressed—which is when the distinction matters. Second, margin posted in a stablecoin is collateral with credit risk and correlation to the very events that trigger margin calls, so Chapter 27's wrong-way risk applies to the collateral rather than to the exposure. A haircut on the settlement asset is the obvious remedy and is almost never applied, because applying it would concede that the unit of account is not a dollar. Regulation is now supplying part of what the haircut did not. The US GENIUS Act of 2025 requires payment-stablecoin reserves of high-quality liquid assets and redemption at par, and in February 2026 the CFTC allowed futures commission merchants to accept stablecoins issued by national trust banks as margin. Whether a clearing member's margin schedule applies a haircut to them is the question to ask.

37.10 THE VENUE IS THE COUNTERPARTY

The failure of FTX in November 2022 is usually told as a story about crypto. It is more useful as a story about Chapter 2, because every layer that chapter described was missing and the absences can be listed one by one.

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Case Study 37.1 A default waterfall with no layers

Chapter 2 set out the sequence in which a clearing house absorbs a member default. Set that sequence beside what a large crypto venue had in place of each layer:

Chapter 2's waterfall	The venue
Defaulter's initial margin	Held by the venue, unsegregated
Defaulter's default fund contribution	None
The CCP's own capital, ahead of members	No distinction between the venue and its operator
Surviving members' mutualized fund	An "insurance fund" of undisclosed size and composition
Assessments and recovery tools	Losses socialized by unilateral decision

Bitcoin settled every block throughout. No contract mispriced, no model failed, and the underlying did what it always does. *Client assets were not segregated*, so a customer's exposure was never to the market: it was to the venue, and the venue was the same legal person as the trading operation on the other side.

Section 27.2 called a jurisdiction's netting opinion (a lawyer's opinion that close-out netting will be enforced if a counterparty fails) worth more to a dealer than any risk model. This market shows what that means by its absence. There was no netting opinion, no bankruptcy-remote structure, and no supervisor to ask for either, so the entire apparatus of Part V (exposure at default, close-out netting, the margin period of risk) had nothing to attach to.

The lesson is not that the instruments were exotic. They were the plainest instruments in this book: futures, perpetuals, and a spot balance. What was exotic was the absence of the infrastructure every earlier chapter assumed.

Where a clearing house would draw on its default fund, a crypto venue uses **auto-deleveraging**, which has no analog in Part I. When a liquidation cannot be filled at a price that leaves the venue whole, the shortfall is met by closing profitable positions on the other side, ranked by their profit and leverage. A trader who is right about the market and fully margined can have the position taken away to fund somebody else's loss, which is Chapter 2's assessment powers without the mutualization, the disclosure, or the governance.

37.11 WHO TRADES CRYPTO DERIVATIVES, AND WHY

Leveraged directional traders are the dominant flow and the reason perpetuals exist: a contract with no expiration, no delivery, and high leverage is the cheapest way to express a view. Their liquidation levels are the forced flow of Section 37.3, which makes their aggregate position a market-structure variable rather than a private one.

Basis traders and market-neutral funds harvest the basis of Example 37.1, and they are the capital that keeps the futures anchored to spot. *They are also the first to leave*, since their return is a spread and their risk is the venue, so the basis widens exactly when the venue is in question.

Miners are the natural commercial hedgers, with a revenue stream in coin and costs in fiat and electricity, *the configuration of Chapter 36's generator*, and they hedge it by selling forward and buying puts.

Companies and funds holding coin sell covered calls for yield, giving up the price rise above the strike for a premium. It is the same trade, with the same asymmetry, as the yield products of Chapter 35.

Liquidity providers supply the inventory of Section 37.8, mostly without pricing the option

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they have written.

The market's participant mix is unusual in one respect. In every other chapter of this book the natural hedger came first and the speculator followed. Here the speculative flow built the market and the commercial hedgers arrived afterward, which is why the instruments are optimized for leverage rather than for delivery, and why the infrastructure lagged the volume by a decade.

SUMMARY

The underlying is the easy part and the infrastructure is the hard part. A bitcoin is storable at no cost, fungible, divisible, and pays nothing, so Chapter 5's $F = Se^{rT}$ is the arbitrage-free price. What is absent is the CCP, the segregation, the daily settlement cycle, the netting opinion, and the lender of last resort: the infrastructure Parts II and III assume without naming it.

That explains the prices. A futures basis far wider than the cost of financing is the venue's credit being priced, not an inefficiency. On a staked asset the basis also mixes in the yield, so it no longer reads as positioning alone, and options are priced off the forward the market trades rather than off spot. The perpetual is the only genuinely new contract: with nothing forcing convergence, a funding rate paid every eight hours does delivery's work and prints the price of a levered long that Chapter 5 had to infer. Liquidation replaces the margin call, and because leverage tiers are published the forced flow is computable in advance, which relocates Section 35.7's barrier clustering rather than removing it.

Two conventions carry more risk than they appear to. An inverse contract is linear in dollars and curved in the coin, so liquidation arrives earlier than dollar leverage suggests. And almost every offshore trade settles in a stablecoin, so the quoted price carries an issuer's credit and the value of access to redemption. In both cases collateral and exposure are the same risk: Chapter 27's wrong-way risk sitting in the contract and in the unit of account.

The producer and the liquidity provider carry options they do not price as such. A miner holds Section 36.6's spark spread, long the coin and short power; the difficulty adjustment is an automatic hedge that arrives late and works through quantity, which is why miners buy puts rather than sell forward, and a miner on an interruptible power contract has sold the grid an option to shut it down. Providing liquidity to a constant-product pool is a written straddle that bleeds $\sigma^2/8$ a year whatever the price, so only substantial fees let the provider break even at this market's volatilities.

FTX is a story about Chapter 2. Every waterfall layer was missing, yet bitcoin settled every block throughout: no contract mispriced and no model failed. Client assets were not segregated, so the exposure was to the venue rather than to the market. The regulated route of the 2024 ETFs supplies the missing infrastructure. The instruments were never the problem.

FURTHER READING

The primary sources in this market divide sharply: venue documentation of funding-rate and liquidation mechanics is precise and worth reading directly, while almost everything written about valuation restates Chapters 5 and 15 with new vocabulary. On the perpetual contract, the mechanism papers are short and the specification documents are shorter, and both are the place to start, before the secondary literature. On automated market makers, Milionis, Moallemi, Roughgarden and Zhang on loss-versus-rebalancing derive the $\sigma^2/8$ rate of Section 37.8 from the Black–Scholes setting, and the connection to Chapter 18's gamma cost is immediate. The FTX bankruptcy filings are the account of what Section 37.10 describes, and they enumerate what was and was not held. On the regulated products, the fund prospectuses set out the custody and creation mechanics that Section 37.7 argues are the whole of the change.

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PRACTICE QUESTIONS

37.1 Using Example 37.1, state the annualized basis and the return on capital at 20% initial margin, and give three reasons the spread is wider than Chapter 6's Treasury basis.

37.2 Using Example 37.2, state the rally that exhausts the posted margin on a fully hedged basis trade, and explain why a position that has lost nothing can be closed.

37.3 Explain why the leverage in Example 37.1 and the rally in Example 37.2 must be quoted together, and which earlier chapter's case has the same shape.

37.4 Using Example 37.3, state the forward price and basis at a 3.5% staking yield, and the yield at which the future trades at spot.

37.5 Explain why a persistent backwardation in a staked asset does not mean what the same shape means in Chapter 36, and what must be quoted to separate the two.

37.6 Explain what holds a perpetual futures contract to spot in the absence of delivery, and why the mechanism is an adaptation to missing infrastructure.

37.7 Using Example 37.4, convert a funding rate of 0.03% per eight hours into an annualized rate and into the equivalent three-month futures premium.

37.8 Explain why liquidation on a crypto venue is not a margin call, and what the client gives up when it replaces one.

37.9 Using Example 37.5, state the forced selling triggered by a 10% decline, and explain why publishing the liquidation profile does not make the cascade smaller.

37.10 Explain why crypto skew changes sign with the market while equity index skew does not, and name the chapter whose pattern it resembles.

37.11 Using Example 37.8, state the coin profit and loss of a long inverse position at a 50% fall and at a 50% rise, and explain why the two are not symmetric.

37.12 Explain why an inverse contract's valuation is a change of numeraire, and which adjustment a desk drops by discounting the dollar payoff instead.

37.13 Using Example 37.9, state the break-even coin price, and the revenue change from a 30% price fall before and after the difficulty adjustment.

37.14 Explain why the difficulty adjustment is only a partial hedge, giving the three qualifications, and why miners prefer puts to forwards.

37.15 Using Example 37.11, compute the fair value at 50% recovery on the exposed reserve, and explain what the observed 12.3% discount priced that the disclosure did not.

37.16 Explain why margin posted in a stablecoin is wrong-way risk in the collateral, and what remedy Section 37.9 says is almost never applied.

37.17 Using Example 37.10, state the shortfall at a price ratio of 2 and of 4, and explain why the function is symmetric in k and $1/k$.

37.18 Derive the $\sigma^2/8$ bleed of (37.4) from Chapter 18's gamma cost, and explain why it does not depend on the pool size.

37.19 Explain why the FTX failure is described as a custody failure rather than a market failure, and what the underlying did throughout.

37.20 Explain what auto-deleveraging is, why it has no analog in Part I, and why a correctly positioned and fully margined trader can still lose the position.

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FURTHER QUESTIONS

37.21 Example 37.4 treats the funding rate and the futures basis as two quotations of one price. Construct the arbitrage that enforces the relation, determine what frictions would let the two persist at different levels, and estimate how large a gap those frictions could sustain.

37.22 Example 37.5 treats the liquidation profile as static. Model it dynamically, so that the selling from one tier moves the price into the next, determine the condition on the leverage distribution under which the cascade is self-limiting rather than self-sustaining, and state what a venue could change to enforce that condition.

37.23 Example 37.9 shows that the difficulty adjustment offsets part of a price fall with a lag. Formulate the miner's revenue as a product of price and share, model the share as a delayed response to price, and determine the forward-sale fraction that minimizes the variance of revenue.

37.24 Example 37.11 decomposes a discount into a credit loss and a charge for redemption access. Build a model of the second component as an option on the window's remaining capacity, calibrate it to the observed 12.3%, and state what the calibrated parameter would have to mean.

37.25 Example 37.10 values a pool under geometric Brownian motion. Redo the analysis under a jump-diffusion, determine whether the $\sigma^2/8$ result survives, and state what a provider should charge for the jump component that the diffusion formula omits.

37.26 Concentrated liquidity confines the inventory to a price range. Formulate the resulting position as a combination of Chapter 34's barrier options, derive the fee yield needed to break even as a function of the range width, and determine the width that maximizes expected profit at a given volatility.

37.27 Section 37.7 argues the ETF-to-offshore volatility spread prices market infrastructure. Design the measurement: state which contracts to compare, what must be controlled for, and how you would separate the infrastructure premium from a pure liquidity difference.

37.28 The snapshot in Section 37.10 maps a crypto venue onto Chapter 2's waterfall. Design a waterfall that would work for a venue with no members in Chapter 2's sense, state who funds each layer, and identify which layer is hardest to fund and why.

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Part IX

Risk, Capital, and Regulation

The last part turns from pricing a trade to holding a book of them. Value at risk and expected shortfall measure the exposure, the capital rules set what it costs, and the mishaps chapter shows where losses actually came from: mostly from what measures and rules cannot see (cash, size, information, the venue) rather than from the instruments. Machine learning, which changes what is computable rather than what it costs, is treated in the online technical note [TN39]. If only one chapter from this part can be assigned, it is Chapter 40.

38 **Value at Risk and Expected Shortfall**
the measures, their backtests, and where each one misleads

39 **Market Risk Capital and Model Risk**
FRTB, the P&L attribution test, and non-modellable factors

40 **Derivatives Mishaps and What We Can Learn**
the cases, and the control that was missing in each

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Chapter 38
Value at Risk and Expected Shortfall

Earlier chapters took one instrument or one market at a time: how it works, who uses it, and what it does to the risks of the people who hold it. This one asks what a whole portfolio of positions can lose, which is a different question and has a worse answer.

The difficulty is not computational. A trading book's loss distribution can be simulated to any precision the machine allows. The difficulty is that a distribution is not a number, and a risk report, a capital rule, a limit framework, and a margin call each need one. Everything in this chapter is a way of choosing which number, and the choices are not equivalent—one of them, the one the industry used for twenty years, can penalize diversification.

Two things make this chapter necessary rather than optional. Chapter 39 sets capital from these measures, so an error here is an error in what a bank must hold. And expected shortfall is a standard objective in deep hedging, the learned hedging strategies of the online note [TN39], so a reader who does not have it defined cannot read what that work optimizes.

Sections 38.1 to 38.4 define the two measures and the property that separates them. Sections 38.5 to 38.9 compute them by historical simulation, by a model, for books with options, under fat tails, and for losses not yet observed; Sections 38.10 to 38.14 cover the inputs: volatility, correlation, copulas, factor reduction, and scaling across horizons. Sections 38.15 to 38.18 put the measures to use: attribution to positions, backtesting, stress testing, and stressed calibration. Section 38.19 says what the measures cannot see, Section 38.20 who uses them and why, and Section 38.21 where the ideas came from.

38.1 WHAT A RISK MEASURE IS FOR

A **risk measure** maps a loss distribution to a single number. Three uses pull in different directions and it is worth separating them before choosing.

A limit is a control. It must be stable enough that a trader can plan against it and responsive enough to bind when it should, and it is compared across desks, which requires it to add up sensibly.

Capital is a buffer. It answers "how much loss must be survivable," so it cares about the size of bad outcomes, not their frequency.

A margin requirement is a forecast over a specific horizon, the margin period of risk of Section 8.4, which is set by how long a close-out takes rather than by anyone's risk appetite.

These are three different questions and the market answers them with variants of one idea. A measure that appears to behave strangely is often being asked a question it was not built for.

38.2 VALUE AT RISK

Value at risk (VaR) at confidence c over horizon h is the loss that is exceeded with probability $1 - c$:

$$\boxed{\mathbb{P}\big(L > \mathrm{VaR}_c\big) = 1 - c}, \tag{38.1}$$

where L is the portfolio's loss over the horizon h, positive when value is lost.

It is a quantile, and the whole of its behavior follows from that. It says where the tail begins and nothing whatever about what is in it.

Where the loss distribution jumps, as a bond portfolio's does at default, or is a finite sample, (38.1) can have no solution or many. VaR is then the smallest loss l with $\mathbb{P}(L \le l) \ge c$, the lower quantile, which is how Examples 38.2 and 38.3 read it.

Its appeal was never statistical. A single currency amount, attached to a stated probability, can be put in front of a board. Section 38.20 argues that this is the reason it displaced everything that preceded it, not any property in (38.1).

38.3 EXPECTED SHORTFALL

Expected shortfall (ES) at confidence c is the average loss over the worst $1 - c$ of outcomes. Each level p beyond c has its own VaR, and expected shortfall is their average:

$$\boxed{\mathrm{ES}_c = \frac{1}{1-c}\int_c^1 \mathrm{VaR}_p \, dp} \,. \tag{38.2}$$

Because it averages *all* the VaRs beyond c, it reads the whole tail rather than one point on it.

When the loss distribution is continuous, (38.2) is the mean loss given that the VaR is exceeded, $\mathbb{E}[L \mid L > \mathrm{VaR}_c]$, a quantity called the conditional tail expectation. Where the distribution jumps at the VaR, as in Example 38.2, the two differ, and (38.2) is the definition, because it is the form that keeps the measure coherent. The same quantity goes by other names: conditional VaR (CVaR) in the optimization literature and in deep hedging, and tail VaR in insurance. Deep hedging often minimizes (38.2) directly, and [TN38] derives the representation that makes it optimizable. Example 38.1 computes the two measures side by side on a normal book.

Example 38.1 The two measures on the same book

Consider a \$100 million position with a 2.0% daily volatility (a daily standard deviation of \$2.0 million) and normally distributed returns. Under normality both measures are multiples of the standard deviation σ: $\mathrm{VaR}_c = z\sigma$ and $\mathrm{ES}_c = \sigma N'(z)/(1-c)$, where $z = N^{-1}(c)$ is the standard normal quantile at confidence c and N' is the standard normal density:

Confidence	z	VaR	ES	ES/VaR
95.0%	1.6449	\$3.29m	\$4.13m	1.254
97.5%	1.9600	\$3.92m	\$4.68m	1.193
99.0%	2.3263	\$4.65m	\$5.33m	1.146
99.9%	3.0902	\$6.18m	\$6.73m	1.090

The ratio falls as the confidence rises, because the normal tail thins so fast that conditioning on being past a distant point adds little. That ratio is a property of the normal distribution and not of the portfolio, which is exactly why it cannot be trusted: on a fatter tail it widens, and Example 38.6 shows by how much.

The regulatory switch reads directly off this table. The Basel framework of Chapter 39 replaced 99% VaR with 97.5% ES, and under normality the two nearly coincide:

$$\mathrm{VaR}_{99\%} = 2.3263\,\sigma, \qquad \mathrm{ES}_{97.5\%} = 2.3378\,\sigma, \qquad \text{a ratio of } 1.0049.$$

The confidence level was chosen to make the change roughly capital-neutral for a normal book, so that what moved was not the level of the requirement but its sensitivity to the shape of the tail. Taken on its own, the swap of measure charges a bank with a genuinely fat-tailed book more and a bank with a normal book the same; the framework changed other things at the same time, which Chapter 39 sets out. Nor does the switch buy precision. Estimated from 500 days, a 97.5% ES

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averages only about a dozen losses, each of them widely dispersed, and for the fat-tailed book of Example 38.3 its standard error, from the large-sample variance of a tail average, is about 14% of its value, the same relative precision Section 38.5 derives for the 99% VaR it replaced.

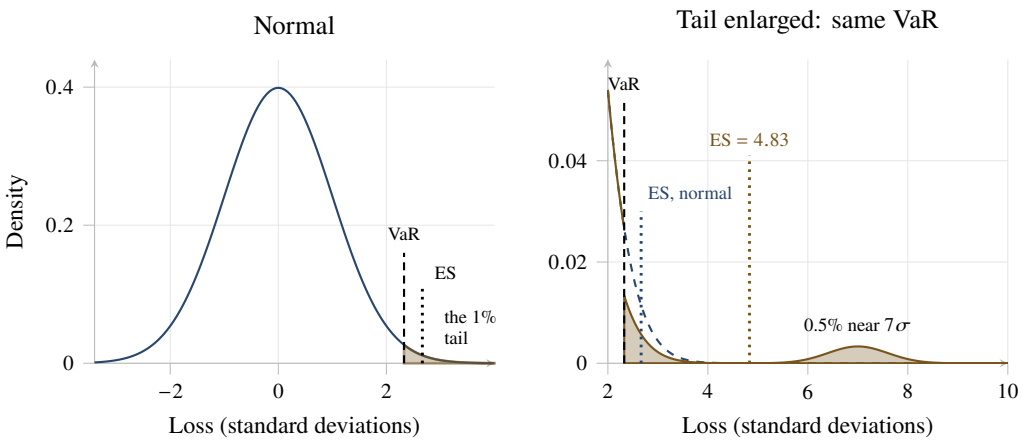


Figure 38.1: Losses at 99%, in standard deviations. Left: the standard normal. Right, the tail enlarged: the same distribution below VaR, with half its 1% tail moved to a cluster near 7σ. Illustrative.

Figure 38.1 carries the whole distinction. VaR is the dashed line: it locates the shaded region's left edge and is indifferent to everything inside it. Two portfolios with identical VaR can have any expected shortfall whatever, because the shading can be redistributed arbitrarily without moving the edge. The right panel does exactly that: the two distributions agree up to the VaR, so the VaR is the same 2.3263σ, but half of the 1% tail has been moved to a cluster of losses near 7σ (the profile of a book that has sold deep out-of-the-money protection), and the expected shortfall nearly doubles, from 2.67σ to 4.83σ. A trader whose limit is set in VaR can exploit that, and Section 38.4 shows the exploitation is not even subtle.

38.4 COHERENCE, AND WHERE VAR FAILS

A risk measure ρ is called **coherent** if it has four properties, each a piece of common sense about risk. It is monotonic: a position that loses more than another in every outcome carries at least as much risk. It is positively homogeneous: doubling a position doubles its risk. It is translation-invariant: adding an amount of cash to a position reduces its risk by exactly that amount. And it is **subadditive**, the property that matters here:

$$\rho(A + B) \;\le\; \rho(A) + \rho(B). \tag{38.3}$$

Subadditivity is the formal statement that diversification cannot hurt. It is also what makes a measure usable as a limit: if (38.3) fails, desk limits do not aggregate, and a firm can be within every limit and over its own total.

Expected shortfall satisfies (38.3) always; [TN38] proves this and the other three properties. VaR does not, and the counterexample needs no exotic structure at all: Example 38.2 builds it from two bonds.

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Example 38.2 Two bonds that break the measure

A book holds two independent corporate bonds, \$10 million each, with a 0.8% default probability and zero recovery. Measure at 99%.

Each bond alone. A loss occurs with probability 0.008, which is *less* than the 1% tail being measured. The no-default branch carries 99.2% of the probability, so the quantile is zero up to that level and jumps to the full loss beyond it:

$$\mathrm{VaR}_c(\text{one bond}) = \begin{cases} \$0, & c \le 99.2\%, \\ \$10\text{m}, & c > 99.2\%. \end{cases}$$

At 99% the bond sits just below the jump, so $\mathrm{VaR}_{99\%}(\text{one bond}) = \$0$.

The two together. The loss distribution is:

Loss	Probability	Cumulative
\$0m	0.984064	0.984064
\$10m	0.015872	0.999936
\$20m	0.000064	1.000000

Since $0.984064 < 0.99$, the 99th percentile falls in the \$10m outcome:

$$\mathrm{VaR}_{99\%}(\text{both}) = \$10\text{m} \;>\; \$0 + \$0 = \mathrm{VaR}_{99\%}(A) + \mathrm{VaR}_{99\%}(B).$$

Diversifying made the measured risk go from nothing to \$10 million. Nothing pathological was constructed: two ordinary bonds, independent, with a default probability smaller than the tail being measured.

Expected shortfall behaves. It averages the worst 1% rather than reading its boundary. For one bond, $\mathrm{ES}_{99\%} = \$8\text{m}$, against a VaR of zero on the same position. For the pair:

$$\mathrm{ES}_{99\%}(\text{both}) = \frac{(1 - 0.999936) \times 20 + (0.999936 - 0.99) \times 10}{0.01} = \$10.06\text{m} \;\le\; \$16\text{m}.$$

The mechanism is worth naming because it generalizes. VaR fails most readily where the loss distribution has a *jump* near the chosen quantile, which describes every credit portfolio, every book with digital or barrier payoffs, and every position with a default or knock-out. It can fail on continuous distributions too, if they are skewed or fat-tailed enough, and it is guaranteed to be subadditive at any confidence level above one half when losses are **elliptical**: jointly distributed so that every portfolio's loss has the same shape, differing only in location and scale, as jointly normal and jointly Student's t losses do. So it is well behaved on smooth, roughly elliptical distributions and unreliable on precisely the books that need measuring.

And the incentive follows directly. A desk measured on VaR can sell far-out-of-the-money options, collect premium, and report no additional risk, because the loss lives beyond the quantile. Expected shortfall prices that trade at what it is, which is a practical argument for the switch, and Section 38.8 gives a sharper one.

38.5 HISTORICAL SIMULATION

The dominant method is also the simplest. Take the last N trading days (500, about two years, is common, and the Basel rules require at least one year) and apply each day's market moves to *today's* portfolio, producing N hypothetical profits and losses. The measure is read off that sample.

What it assumes is one thing and it is a large one: that the last N days are a fair sample of tomorrow. It makes no distributional assumption at all, which is its great advantage: fat tails, skew, and the actual joint behavior of markets are captured without being modeled.

What it cannot do is produce a loss that has not happened. A 500-day window at 99% is estimating from five observations, and it cannot represent a move larger than the largest in the sample. A market that has been calm for two years reports a small number regardless of what

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is in the book, which is Section 38.18's problem and the reason the Basel capital framework does not rely on this window alone.

Weighting the observations helps and does not solve it. Exponentially weighting recent days makes the measure responsive, at the cost of shrinking the effective sample exactly when the tail is being estimated. Example 38.3 reads both measures off a 500-day window and then weights the same window twice, first with the bad days old, then with them recent.

Example 38.3 Five losses out of five hundred

The book is $10 million in an equity index, scored against the last 500 trading days. The 500 daily returns are simulated for illustration, not market data: Student's t with four degrees of freedom, scaled to a 1.25% daily standard deviation, the fat-tailed shape Section 38.8 examines. Each day's return applied to today's $10 million gives one scenario loss. The six worst losses are:

		As recorded		Calendar reversed	
Rank	Loss ($000)	Days ago	Weight	Days ago	Weight
1	581	428	0.064%	73	0.379%
2	437	458	0.055%	43	0.441%
3	424	446	0.059%	55	0.415%
4	408	249	0.157%	252	0.155%
5	396	442	0.060%	59	0.407%
6	357	383	0.080%	118	0.303%

With equal weights each scenario carries 0.2%, so the five worst losses are exactly the 1% tail. The 99% VaR is the loss with exactly five scenarios beyond it, the sixth worst, which is how (38.1) reads on a sample of 500: $\text{VaR}_{99\%} = \$357{,}000$. Expected shortfall is the mean of the five losses beyond it:

$$\text{ES}_{99\%} = \frac{581 + 437 + 424 + 408 + 396}{5} = \$449{,}000.$$

Some systems report the fifth-worst loss, $396,000, instead. That the convention alone moves the number by 11% is a measure of how little five observations are.

Now weight by recency, giving the scenario i days ago the weight

$$\pi_i = \frac{\lambda^i}{\sum_{j=1}^{500} \lambda^j}, \qquad \lambda = 0.995,$$

so that yesterday counts 0.544% and the oldest day 0.045%. The VaR is then the first loss, counting down from the worst, at which the running weight passes 1%. Four of the five worst losses are a cluster 428 to 458 days old, and the five together carry only 0.394%. The running weight does not pass 1% until the twelfth-worst loss, and the weighted VaR is $287,000, 20% below the equal-weighted figure, because the window's bad episode is old. Reverse the calendar, so that the same cluster is 43 to 73 days old, and the three worst losses alone carry 1.24%: the weighted VaR is the third-worst loss, $424,000, 19% above.

The weights moved the VaR between $287,000 and $424,000 without adding a scenario. They can only choose among losses already in the window, so whatever the weights, neither measure can exceed $581,000, the largest loss in the sample. A window whose worst day was mild reports a mild number however its days are weighted.

The weighting in the example is the scheme of Boudoukh, Richardson and Whitelaw (1998). A second refinement rescales the scenarios to today's volatility instead of reweighting them, and Section 38.10 takes it up once the volatility estimators it needs are in place.

Five observations are also a small sample in a precise sense. An estimated quantile has a

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sampling standard error of approximately

$$\text{s.e.}\big(\widehat{\text{VaR}}_c\big) \approx \frac{1}{f(\text{VaR}_c)}\sqrt{\frac{c\,(1-c)}{N}}.$$

The square root is the standard error of a proportion: the fraction of the N days that land beyond the estimate is uncertain by that much. Dividing by $f(\text{VaR}_c)$, the density of the loss at the quantile, converts that uncertainty in probability into uncertainty in dollars, and the tail is where the density is lowest, so a small error in probability becomes a large error in the loss.

For the book of Example 38.3, the daily loss on the \$10 million position has the example's 1.25% standard deviation, \$125 thousand, and is a standard t_4 variable rescaled to it. A standard t_4 has variance 2, so the scale, in thousands of dollars, is $125 \times \sqrt{1/2} = 88.39$. The t_4's 99% quantile is 3.7469, and its density is $f_4(t) = \frac{3}{8}(1 + t^2/4)^{-5/2}$. The loss's own density is that density divided by 88.39: stretching a variable by a factor spreads the same probability over a range that much wider, so the probability per thousand dollars is that much smaller. With $c = 0.99$ and $N = 500$, the calculation runs

$$\begin{aligned}
\sqrt{c\,(1-c)/N} &= \sqrt{0.99 \times 0.01/500} = 0.004450,\\
\text{VaR}_{99\%} &= 88.39 \times 3.7469 = \$331.2 \text{ thousand},\\
f(\text{VaR}_{99\%}) &= \frac{f_4(3.7469)}{88.39} = \frac{0.008682}{88.39} = 0.0000982 \text{ per thousand dollars},\\
\text{s.e.} &= \frac{0.004450}{0.0000982} = \$45.3 \text{ thousand},
\end{aligned}$$

14% of the true VaR. The example's \$357,000 is a little over half a standard error from the truth, well within what a 500-day window can resolve.

In practice the density is not known. The **bootstrap** estimates the standard error without it: draw many new samples of 500 from the window itself, with replacement, recompute the VaR on each, and take the standard deviation of the results. For a quantile this far out the answer is rough, because every resample is built from the same few tail losses and none can contain a loss larger than the largest of them.

38.6 THE MODEL-BUILDING APPROACH

The alternative assumes a distribution and computes. For a portfolio linear in n risk factors with position vector $\mathbf{w}$ and factor covariance Σ,

$$\text{VaR}_c = z\sqrt{\mathbf{w}^\top \Sigma\, \mathbf{w}}, \tag{38.4}$$

where z is the standard normal quantile at confidence c.

Equation (38.4) is fast, transparent, and differentiable, which is why it survives despite its assumptions: Section 38.15 needs that derivative and historical simulation does not supply one cleanly. Example 38.4 applies it to two stocks.

Example 38.4 Two stocks

A book holds \$10 million of stock 1, with a 2.0% daily volatility, and \$5 million of stock 2, with 1.0%; the correlation between them is 0.3. In dollars the daily standard deviations are \$0.200m and \$0.050m, and the square root in (38.4) is the portfolio's daily standard deviation:

$$\sigma_P = \sqrt{0.200^2 + 0.050^2 + 2 \times 0.3 \times 0.200 \times 0.050} = \sqrt{0.0485} = \$0.2202\text{m}.$$

At 99%, $z = 2.3263$, so the one-day VaR is $2.3263 \times 0.2202 = \$0.512$m. Over ten days the square-root-of-time rule of Section 38.14 multiplies by $\sqrt{10}$, giving \$1.620m. It is a rule, not a result: it holds only if daily returns are independent, and that section shows what happens when

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they are not.

Standalone, the two positions have VaRs of $2.3263 \times 0.200 = \$0.465$m and $2.3263 \times 0.050 = \$0.116$m, which sum to \$0.582m. The portfolio's \$0.512m is \$0.069m less, a diversification benefit of 11.9% that exists only because the correlation is below one.

Equation (38.4) makes two assumptions, and both are about shape. It assumes normality, which Example 38.6 prices; and it assumes the book is linear in its factors, which fails for any book with options in it. The usual repair for the second is a **delta–gamma** expansion, which adds a quadratic term so that convexity is at least approximated. Example 38.5 applies it.

Example 38.5 Sold puts on the same stock

Keep the book of Example 38.4, but hold the position in stock 1 through sold put options instead, with the same dollar delta of \$10 million and a dollar gamma of −\$100 million. A sold put gains as the stock rises, so its delta is positive, and like any sold option it has negative gamma. (The dollar delta $\Delta_\$$ and dollar gamma $\Gamma_\$$ are the delta times the stock price and the gamma times its square.) A return r on the stock then changes the position's value by approximately

$$\Delta V \approx \Delta_\$ \, r + \frac{1}{2} \, \Gamma_\$ \, r^2 = 10\, r - 50\, r^2 \qquad (\$ \text{ millions}).$$

This is Chapter 18's expansion, equation (18.1), restricted to its stock terms and written per unit return: with $\Delta S = S\, r$, its delta term $\Delta\, \Delta S$ becomes $\Delta_\$ \, r$ and its gamma term becomes $\frac{1}{2}\Gamma_\$ \, r^2$. Taken alone, the stock's 99% fall is $r = -2.3263 \times 2.0\% = -4.653\%$. The linear term loses \$0.4653m, the standalone figure above; the quadratic term adds $50\, r^2 = \$0.1082$m, for a loss of \$0.5735m, 23% more than the linear term alone. The loss grows as the stock falls, and a rise produces a loss only beyond 20%, ten standard deviations away, so the loss at the 99% fall is the position's 99% VaR under the expansion. Equation (38.4) would still report \$0.465m, because it sees the dollar delta and nothing else.

Two terms suffice here because the sold puts' gamma, large as it is, changes little over a 4.65% move, so the parabola built from today's delta and gamma stays close to the position's true value. The approximation weakens when gamma itself changes quickly (an option near expiration with the stock close to its strike) and fails outright at a barrier. Had the puts carried a knock-out barrier 3% below the spot, the position would change form as the stock crossed it, and an expansion built from today's delta and gamma could not see a change it had not yet reached.

A portfolio of barriers and autocallables from Chapters 34 and 35 is therefore not well approximated by two terms, and for those books full revaluation under Chapter 20's Monte Carlo is the only reliable route. Section 38.7 sets out that route, and a shortcut for the smooth option books in between.

38.7 NONLINEAR BOOKS: CORNISH–FISHER AND MONTE CARLO †

Between the linear book of Example 38.4 and a book of barriers lies most of what a derivatives desk holds: options whose value is curved but smooth. Two methods serve them, one analytical and one by simulation, and Table 38.1 sets them beside the two already met.

Where the quadratic approximation is good, its loss distribution can be summarized without simulation. With normally distributed factors, a delta–gamma P&L has known moments. Its **skewness** $\gamma_1 = \mathbb{E}[(L-\mu)^3]/\sigma^3$ measures asymmetry: zero for a symmetric distribution, positive when the long tail lies on the loss side. Its **excess kurtosis** $\gamma_2 = \mathbb{E}[(L-\mu)^4]/\sigma^4 - 3$ measures how much heavier its tails are than a normal distribution's, for which it is zero. The **Cornish–Fisher expansion** turns these moments into a quantile by correcting the normal one:

$$z_{\text{CF}} = z + \frac{z^2-1}{6}\,\gamma_1 + \frac{z^3-3z}{24}\,\gamma_2 - \frac{2z^3-5z}{36}\,\gamma_1^2, \qquad \text{VaR} \approx \mu + z_{\text{CF}}\,\sigma,$$

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where $z = N^{-1}(c)$ is the standard normal quantile at the confidence level, 2.3263 at 99%, and μ and σ are the mean and standard deviation of the loss. For the sold puts of Example 38.5, the loss $-10r + 50r^2$ with r normal at 2.0% has mean \$0.020m, standard deviation \$0.2020m, skewness 0.590 and excess kurtosis 0.466. A normal quantile on those two moments gives \$0.490m; the skewness term lifts it to \$0.578m and the fourth-order terms, kurtosis and squared skewness, bring it to \$0.573m, against an exact \$0.5735m. (The exact figure is the example's, because over any plausible range the loss rises monotonically as the stock falls, so its 99% quantile is the loss at the stock's 99% fall.) The expansion is reliable for moderate skewness like this and can misbehave for strongly nonlinear books, which is where simulation takes over.

Monte Carlo VaR draws the risk factors from their joint distribution many times, values the portfolio in each draw, and reads the quantile and the tail mean off the simulated losses. In **full revaluation** each draw reprices every position with its own pricing model; in **partial simulation** each draw applies the delta–gamma approximation, which is faster and inherits that approximation's limits.

Either way the result carries sampling error, and the standard-error formula of Section 38.5 measures it, with the number of draws M in place of the number of days. For the sold puts, the loss density at the VaR is the stock return's density there divided by the rate at which the loss grows with the return, which is $|-10 + 100\,r| = 14.65$ at $r = -4.653\%$; that gives $1.3326/14.653 = 0.0909$ per million dollars. With $M = 10{,}000$ draws the standard error is $\sqrt{0.99 \times 0.01/10{,}000}/0.0909 = \$0.011$m, 1.9% of the \$0.5735m VaR, and with 100,000 draws it is 0.6%. Unlike a historical window, a simulation can be lengthened as far as the machine allows, so its error is a cost of computing rather than a shortage of data.

Method	Distribution	Options	Typical users
Historical simulation	the window's, as observed	repriced in each day's scenario	bank trading desks; clearing houses, volatility-scaled
Variance–covariance	normal	delta only	limit and allocation systems; fund VaR limits
Delta–gamma with Cornish–Fisher	normal factors	quadratic approximation	quick checks on smooth option books
Monte Carlo, full revaluation	any that can be simulated	repriced in each draw	exotic and structured books; insurers' internal models

Table 38.1: Four ways to compute VaR or expected shortfall.

The table sets out the trade. Each step down handles curvature more faithfully and costs more computing; historical simulation alone takes its distribution from data rather than from an assumption, and Monte Carlo alone can be made as precise as the budget allows. Section 38.20 says why each user ends up where the last column puts it.

38.8 FAT TAILS, AND A COUNTERINTUITIVE RESULT

The normal distribution is a poor description of daily returns: large moves occur far more often than it allows. The standard alternative is **Student's t distribution** with ν degrees of freedom, a bell curve whose density falls off as a power of the loss, $|x|^{-(\nu+1)}$, rather than exponentially in its square. The fewer the degrees of freedom, the fatter the tails, and as ν grows the t approaches the normal. Its variance is $\nu/(\nu-2)$ for $\nu > 2$, so a t variable multiplied by $\sqrt{(\nu-2)/\nu}$ has unit variance and can then be scaled to any chosen standard deviation; for $\nu = 4$, a common fit to daily equity returns, the factor is $\sqrt{1/2}$, which is where Section 38.5's 88.39 came from. Example 38.6 uses it to change a book's shape while holding its volatility fixed.

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Example 38.6 The same volatility, a different shape

Take the $100 million book of Example 38.1 and change nothing except the distribution: Student's t with four degrees of freedom, rescaled to the *identical* 2.0% daily standard deviation. The mean and volatility are the same; only the tail is fatter.

	VaR		ES		
Conf.	**Normal**	**Fat-tailed**	**Normal**	**Fat-tailed**	**ES error**
95.0%	$3.29m	$3.01m	$4.13m	$4.53m	+10%
99.0%	$4.65m	$5.30m	$5.33m	$7.38m	+39%
99.9%	$6.18m	$10.14m	$6.73m	$13.70m	+103%

Figure 38.2 draws the two densities. The t is taller in the middle and thinner in the shoulders, and on the logarithmic scale of the right panel its tail lies orders of magnitude above the normal one; the vertical lines mark the two 99.9% VaRs, $6.18m and $10.14m.

Read the 95% row again. The fat-tailed VaR is *lower* than the normal one: $3.01m against $3.29m, some 8% *less* risk by that measure. Fat tails do not raise VaR at every confidence level. A distribution with the same variance and a heavier tail must be thinner somewhere, and at 95% that is where the quantile sits.

So a bank that fits a fat-tailed model and measures at 95% reports a risk reduction. As a quantile the number is right; it answers a question that ignores the tail, and nothing in the report says so.

Expected shortfall does this far less. It rises at every level in the table (10%, 39%, 103%) because it integrates the tail rather than sampling one point of it. It too falls below the normal at a low enough level, but its crossover sits near 89% against VaR's near 97.5%, so at the levels risk is actually measured at it moves the right way. This is a sharper argument for expected shortfall than subadditivity, because it does not require a constructed example: it happens on the most ordinary alternative distribution anyone would fit.

And it shows what the normal assumption costs where it matters: at 99.9%, the level at which credit and default-risk capital is set, the normal model understates the true expected shortfall by a factor of two.

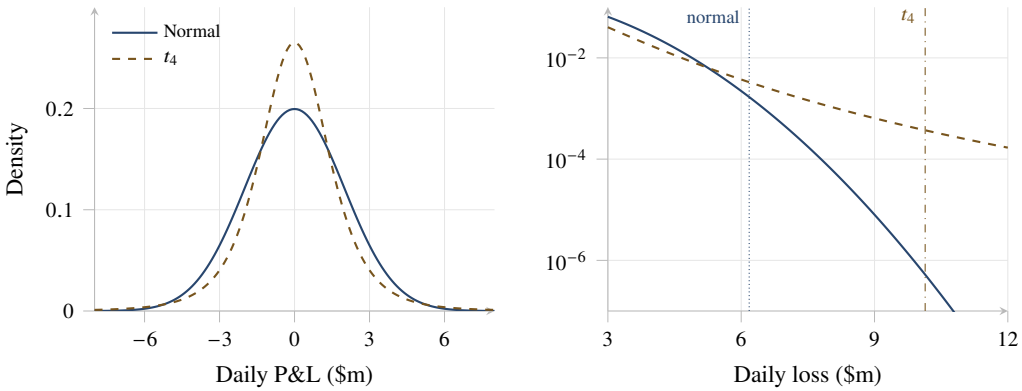


Figure 38.2: Daily P&L of the $100m book under a normal and a t_4 with the same 2.0% standard deviation. Left: the densities. Right: the loss tail on a log scale, with each model's 99.9% VaR marked.

38.9 ESTIMATING A LOSS THAT HAS NOT HAPPENED †

Section 38.5 left a problem unanswered. Historical simulation cannot report a loss larger than the largest in its window, and a 99.9% measure of a daily loss asks about a percentile where a

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500-day sample has half an observation.

Extreme value theory is the standard answer, and it is worth seeing measured rather than described. The result that the excesses over a high threshold of almost any distribution follow a **generalized Pareto** law lets the tail be *fitted* from the observations that exist and then *extrapolated* past them.

Fix a high threshold u and look only at the losses beyond it. As u rises, the excess $Y = L - u$ of a loss over the threshold has, for almost any underlying distribution, approximately the distribution function

$$\mathbb{P}(Y \le y) = 1 - \left(1 + \frac{\xi\, y}{\psi}\right)^{-1/\xi},$$

with a scale ψ and a shape ξ that carries the tail's thickness:

- $\xi > 0$ is a power-law tail (a Student's t with ν degrees of freedom has $\xi = 1/\nu$, so the book of Example 38.6 has $\xi = 0.25$);
- $\xi = 0$ is an exponential-type tail, the limit for the normal, the lognormal and most other unbounded distributions whose tails die off faster than any power;
- $\xi < 0$ is a tail with an end point.

Fitting ψ and ξ to the losses above u and reading the quantile off the fitted law is the **peaks-over-threshold** method, and Example 38.7 sets it beside the two simpler estimates. [TN38] derives the results it rests on, from the threshold limit to the formulas the example uses.

Example 38.7 Three ways to estimate a quantile you cannot observe

Take the fat-tailed book of Example 38.6: a daily loss in millions of dollars, distributed as a Student's t with four degrees of freedom and scaled to a standard deviation of 2.0, so that the true 99.9% VaR is \$10.14m and the expected shortfall \$13.70m. A risk manager sees only 500 days of losses drawn from it. One simulated window of 500 has mean loss $\bar{L} = -\$0.0011$m and standard deviation $s = \$1.7953$m, and its 26 largest losses, ranked from the largest down, are:

Ranks 1–5	6–10	11–15	16–20	21–26
7.2509	5.4179	3.7229	3.1511	3.0356
6.8962	4.6913	3.7217	3.1395	3.0329
6.1266	4.4446	3.5340	3.0841	3.0045
5.9353	4.1928	3.4639	3.0812	2.9119
5.7588	4.1022	3.3718	3.0789	2.8497
				2.8481

Those figures are all the data the three estimates below use.

Normal fit. With $z = N^{-1}(0.999) = 3.0902$ from Example 38.1,

$$\text{VaR} = \bar{L} + z\, s = -0.0011 + 3.0902 \times 1.7953 = \$5.55\text{m}.$$

Empirical quantile. Placing the kth smallest of the 500 losses at percentile $(k-1)/499$ puts the 99.9th percentile at $k = 1 + 0.999 \times 499 = 499.501$, 50.1% of the way from the second-largest loss to the largest:

$$6.8962 + 0.501 \times (7.2509 - 6.8962) = \$7.07\text{m}.$$

Peaks over threshold. Set the threshold at the 26th-largest loss, $u = \$2.8481$m, so that 25 losses, the worst 5%, exceed it. The generalized Pareto law is fitted to their excesses y_i over u by **maximum likelihood**: choose the ψ and ξ under which the observed excesses are most probable,

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which means maximizing the logarithm of their joint density,

$$\ln \mathcal{L}(\psi, \xi) = -25 \ln \psi - \left(1 + \frac{1}{\xi}\right) \sum_{i=1}^{25} \ln \left(1 + \frac{\xi\, y_i}{\psi}\right),$$

over $\psi > 0$ and $\xi \geq -1/2$ (below $-1/2$ the maximum-likelihood estimates lose their usual large-sample behavior). That gives $\hat{\psi} = 1.1511$ and $\hat{\xi} = 0.0983$. Since 25 of the 500 days exceed u, a probability of $1 - c = 0.001$ is a fraction $(500/25) \times 0.001 = 0.02$ of the excesses, and solving the fitted law for that fraction gives

$$\text{VaR} = u + \frac{\psi}{\xi}\left[\left(\frac{N}{N_u}(1-c)\right)^{-\xi} - 1\right] = 2.8481 + \frac{1.1511}{0.0983}\left(0.02^{-0.0983} - 1\right) = \$8.34\text{m},$$

with $N = 500$ days, $N_u = 25$ exceedances and $c = 0.999$. Averaging the fitted tail beyond that point gives the expected shortfall:

$$\text{ES} = \frac{\text{VaR} + \psi - \xi u}{1 - \xi} = \frac{8.3389 + 1.1511 - 0.0983 \times 2.8481}{1 - 0.0983} = \$10.21\text{m}.$$

All three estimates fall short of the true \$10.14m. The window's largest loss, \$7.25m, is below it, so the empirical quantile cannot reach it, and the 25 excesses suggest a thinner tail than the true one ($\hat{\xi} = 0.10$ against $\xi = 0.25$, or 0.19, the shape of the generalized Pareto law that best approximates the true t_4 excesses at this threshold, which [TN38] derives), so the extrapolation stops short too. Figure 38.3 draws the three against the data.

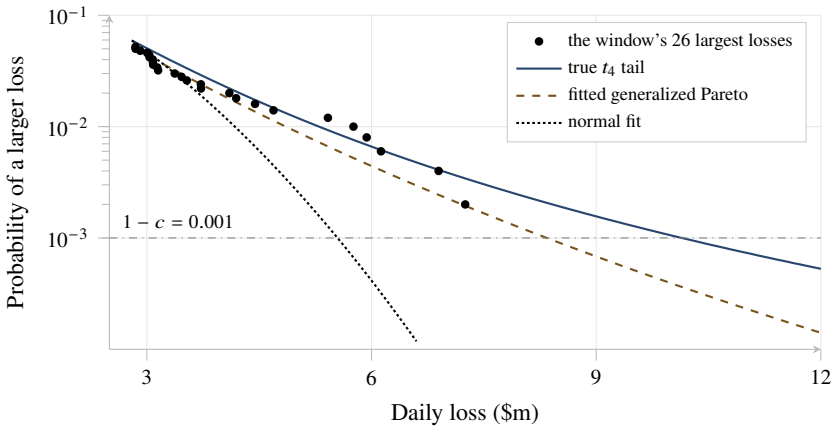


Figure 38.3: The tail of the window in Example 38.7, with the true distribution, the fitted generalized Pareto law and the normal fit. Each curve crosses the 0.001 line at its 99.9% VaR.

The figure shows what each method does where the data run out. The points stop at the largest loss, so the empirical quantile has nowhere further to go. The normal fit's tail falls away from the points almost at once and crosses the 0.001 line at \$5.55m. The generalized Pareto law follows the points and continues past them, reaching \$8.34m, but falls away faster than the true tail, which does not cross until \$10.14m.

One window shows how each calculation works, not how well each method performs. The table repeats the three calculations on 400 independent 500-day windows drawn from the same book, of which the example's is the first. The **mean VaR estimate** is the average of a method's 400 answers, the **bias** is that average divided by the true \$10.14m, less one, and the **spread** is the standard deviation of the 400 answers.

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Method	Mean VaR estimate	Bias	Spread across windows
Empirical quantile	\$9.22m	−9.1%	\$2.85m
Normal fit	\$6.19m	**−39.0%**	**\$0.70m**
Peaks over threshold	\$10.11m	−0.3%	\$3.32m

These are simulation results: the standard error of each mean is its spread divided by $\sqrt{400}$, under 2% of the true VaR for all three methods, and the chapter's notebook regenerates the 400 windows from a fixed seed, together with the further statistics quoted below.

The normal fit is by far the most precise method and it is wrong by 39%. Precision and accuracy point in opposite directions here, and a risk report showing a stable number month after month is displaying the property that a normal assumption is good at producing. Peaks over threshold removes the bias, its −0.3% being well inside its standard error, and pays for it with the widest spread of the three. Its expected shortfall shows the cost: because the formula divides by $1 - \hat{\xi}$, the few windows whose fitted shape comes out high report very large values, so the median window's estimate, \$11.48m, falls below the true \$13.70m while the mean, \$15.19m, is pulled above it.

The ceiling on historical simulation is not hypothetical either. A window's largest loss stays below the true 99.9% VaR with probability $0.999^{500} = 61\%$, so in most windows no confidence level gets the empirical quantile to the right answer, because the right answer is off the end of the sample.

Two cautions keep this from being a solution. The threshold is a choice, and the estimate moves with it: a parameter the analyst picks, in a calculation the analyst is being judged by. And the method extrapolates the tail's *shape* from data in the tail's shoulder, which assumes the shape continues. It cannot see a mechanism that switches on only in the extreme, which is exactly what Section 38.11's correlations and Section 38.12's tail dependence describe.

So extreme value theory converts an impossible estimate into a difficult one. That is a real gain (a bias within sampling error of zero, against 39%), and it is not the same as knowing the answer.

38.10 ESTIMATING VOLATILITY

The model-building methods above need a volatility, and historical simulation can use one to rescale its scenarios. Two estimators dominate and they differ in one structural respect.

Exponentially weighted moving average (EWMA) weights past squared returns geometrically:

$$\sigma_n^2 = \lambda \sigma_{n-1}^2 + (1 - \lambda)\, u_{n-1}^2, \tag{38.5}$$

where σ_n^2 is the variance estimate for day n, made at the end of day $n-1$, and u_{n-1} is the return on day $n-1$. The usual $\lambda = 0.94$ is the value J.P. Morgan's RiskMetrics (1996) chose because it minimized one-day forecast error across a broad set of daily series. The same recursion updates a covariance, with the product of two assets' returns in place of the squared return, which is how the correlations of Section 38.11 are usually kept current.

The **generalized autoregressive conditional heteroskedastic** (GARCH) model, in its GARCH(1,1) form, adds a constant, and the constant changes everything:

$$\sigma_n^2 = \omega + \alpha\, u_{n-1}^2 + \beta\, \sigma_{n-1}^2, \qquad V_L = \frac{\omega}{1 - \alpha - \beta}, \tag{38.6}$$

where V_L is the long-run variance to which the forecast reverts.

The three parameters are estimated by maximum likelihood on the return history, as the tail parameters were in Example 38.7: the chosen ω, α and β are those under which the observed returns are most probable given the variances the model assigns them. Setting $\omega = 0$ and

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$\alpha + \beta = 1$ in (38.6) recovers (38.5), so EWMA is GARCH with no long-run level to return to. That single difference decides the entire shape of the forecast, as Example 38.8 shows.

Example 38.8 One bad day, two forecasts

A market with a 1.00% daily volatility takes a −3% day.

Both estimators react almost identically. EWMA at $\lambda = 0.94$ moves to 1.2166%; GARCH with $\omega = 0.000002$, $\alpha = 0.06$, $\beta = 0.93$ moves to 1.2207%. A single day raises the estimate by about 22%.

They disagree completely about what happens next. GARCH has $\alpha + \beta = 0.99$ and therefore a long-run daily volatility of $\sqrt{0.000002/0.01} = 1.4142\%$, or 22.4% annualized; a daily volatility is annualized by multiplying it by $\sqrt{252}$, the square root of the number of trading days in a year. Writing σ^2_{n+1} for the variance just estimated, the GARCH forecast of the variance k days later is

$$\mathbb{E}\left[\sigma^2_{n+1+k}\right] = V_L + (\alpha + \beta)^k \left(\sigma^2_{n+1} - V_L\right),$$

so each day closes 1% of the remaining gap to the long-run level, and a deviation from it halves in about 69 trading days. Averaging that forecast over the days to a horizon and annualizing gives 19.4% for one day, 19.5% for ten days, 20.2% for three months and 21.4% for a year. EWMA's forecast is 19.3% at every horizon: its best guess for volatility in a year is today's estimate.

The practical consequence is that the two are fit for different jobs. A one-day margin or limit calculation cannot tell them apart, and EWMA's simplicity wins. Anything with a term structure needs the mean reversion: the volatility term structures of Chapter 19 slope the way they do because volatility reverts, and a model that denies reversion produces a flat term structure that the option market visibly contradicts.

Note also which way the current estimate sits. At 19.4% it is still below the market's 22.4% long-run level, and the GARCH forecast is *rising*. A risk system built on EWMA would report no such tendency, which matters most at exactly the moment it is least welcome.

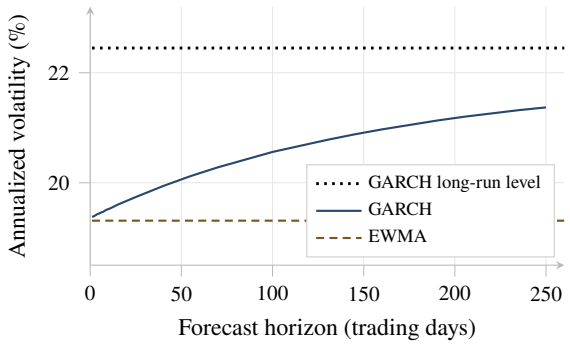


Figure 38.4: Average volatility to the horizon after the shock of Example 38.8.

Figure 38.4 is the argument in one picture. The gap between the two lines is not an estimation disagreement: both estimators saw the same data and agree about today. It is a disagreement about whether volatility has a level, and only one of them can produce a term structure.

Either estimator also repairs historical simulation, which Section 38.5 left reading every past day at the volatility it happened to have. **Volatility-scaled historical simulation**, as Hull and White (1998) proposed it, multiplies each day's return by $\sigma_{\text{today}}/\sigma_t$, the ratio of today's volatility estimate to the estimate on the day it occurred. A calm-period day is scaled up when markets are volatile and a crisis day scaled down when they are calm, so the scenarios keep the historical shape and take today's scale. Unlike any weighting, rescaling can produce a loss larger than the largest in the window. **Filtered historical simulation** (Barone-Adesi,

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Giannopoulos and Vosper 1999) goes a step further: it divides each historical return by that day's GARCH volatility and draws the resulting standardized shocks at random, passing them through the recursion (38.6) to build scenarios over several days. Clearing houses set margin with volatility-scaled scenarios (Section 38.20).

38.11 CORRELATION, AND WHY IT IS THE WEAK INPUT

Example 38.8's estimators supply the diagonal of the covariance matrix in (38.4). The rest of it is harder. Volatilities are estimated from many observations of a single series. Correlations are estimated from the joint behavior of series that spend most of their time telling you nothing about the tail, and they are the least stable input in (38.4). Example 38.9 shows how far the measure moves with one.

Example 38.9 One number, revalued

Position A is \$60m at 1.0% daily volatility; position B is \$40m at 2.5%. At a correlation of 0.30 the 99% one-day VaR is \$3.051m. Change only the correlation:

Correlation	VaR	Change
−0.50	\$2.028m	−34%
0.00	\$2.713m	−11%
0.30	\$3.051m	—
0.60	\$3.355m	+10%
0.90	\$3.634m	+19%
1.00	\$3.722m	+22%

A move from 0.30 to 0.90 raises the reported risk by 19% with no change in any position and no change in any volatility.

And correlations move most in exactly the episodes the measure exists to describe. Assets that are unrelated in normal conditions become one asset in a liquidation, because the common factor in a crisis is not economic—it is that the same leveraged holders are selling all of it. So the input is estimated in calm periods and applied in violent ones, and for a book that is long both assets, as this one is, the error has a consistent sign: measured risk is too low.

This is not a call for conservative correlations. It is a reason the framework does not let a bank choose them freely, and Chapter 39's constrained diversification rules exist because of this table.

38.12 COPULAS AND TAIL DEPENDENCE

Correlation summarizes dependence in one number, and one number is not enough. Two markets can have identical correlation and behave completely differently in the tail, which is what a **copula** separates out: the marginal distributions on one side, the dependence structure on the other. Example 38.10 holds the margins and the correlation fixed and changes only the copula.

Example 38.10 Same margins, same correlation, different joint extremes

A book holds two exposures, each normally distributed, with correlation 0.50. The only thing that differs is the copula: Gaussian against Student's t with four degrees of freedom.

Both beyond	Gaussian copula	t_4 copula	Ratio
95th percentile	0.01219	0.01694	1.39×
99th percentile	0.001294	0.002877	2.22×
99.5th percentile	0.0004963	0.001385	2.79×
99.9th percentile	0.00005426	0.0002635	**4.86**×

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The discrepancy grows as the event gets worse, which is the opposite of what a modeler wants from an approximation.

The limiting behavior is the cleanest statement. The coefficient of upper tail dependence (the limiting probability that one exposure is extreme given the other is) is exactly zero for a Gaussian copula at every correlation below one, and 0.2532 for this t copula. Under a Gaussian copula, the chance that one exposure is in catastrophe given that the other is falls to zero far enough out, however correlated the assets are; under this t copula it never falls below a quarter.

Section 26.6 made this point about CDO tranches, where a Gaussian copula priced the senior tranches as if joint defaults could not cluster in the extreme. Here it decides something more basic: whether a portfolio risk measure can see simultaneous losses at all. A firm that models dependence with correlations alone has assumed the answer to the question it is asking.

Figure 38.5 shows where the difference lives. It draws 1,500 joint outcomes from each copula, and in the middle the two clouds are hard to tell apart. In the corners they are not: in this draw the t_4 copula puts 26 points beyond both 95th percentiles against the Gaussian's 16, and 7 beyond both 99th percentiles against 1. The two copulas agree in the body and diverge exactly where a risk measure is supposed to be doing its work.

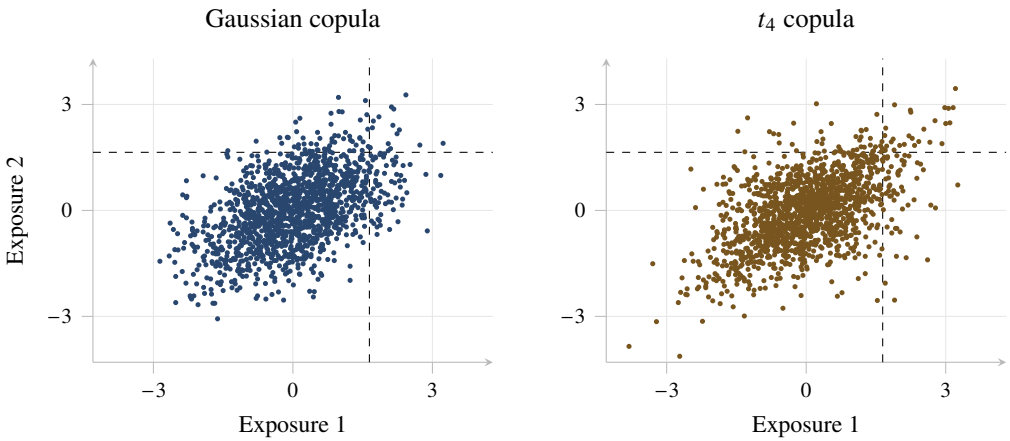


Figure 38.5: Joint outcomes, 1,500 under each copula, in standard deviations, with standard normal margins and correlation 0.50. Dashed lines mark each exposure's 95th percentile.

38.13 FACTOR REDUCTION †

A swap book has sensitivities to twenty or more curve points, and (38.4) then needs a 20×20 covariance matrix estimated from data that barely supports it. The standard remedy is to replace the buckets with a few factors: Chapter 32's factor reduction, applied to a risk measure instead of a simulation.

The factors are the principal components of the curve: the eigenvectors of Σ, the directions of movement that account, in order, for the largest shares of its variance. On a yield curve they are famously interpretable. Example 38.11 shows why interpretable is not the same as sufficient.

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Example 38.11 Three factors, two portfolios, two verdicts

The curve has nine tenors from three months to thirty years. Their daily rate-change volatilities, and a directional book's sensitivities to each, are:

Tenor (years)	0.25	0.5	1	2	3	5	7	10	30
Volatility (bp a day)	3.5	4.4	5.3	6.0	6.2	6.0	5.7	5.4	4.8
Directional book ($000 per bp)	10	5	−8	15	12	−10	20	18	6

The correlation between the rate changes at tenors T_i and T_j is

$$\rho_{ij} = 0.7 + 0.3\, e^{-0.3\,|\ln T_i - \ln T_j|},$$

so neighboring tenors move almost together and the three-month and thirty-year rates have a correlation of 0.77. The model is illustrative rather than fitted; its short-end correlations are higher than daily Treasury data would give, which flatters the level factor. The covariance matrix is $\Sigma_{ij} = \sigma_i \sigma_j \rho_{ij}$, and its leading factors are:

Factor	**Variance explained**	**Cumulative**	**Shape**
Level	90.98%	90.98%	all tenors move together
Slope	3.94%	94.92%	short end against long end
Curvature	1.96%	96.88%	wings against the belly
Fourth	1.10%	97.97%	—

Three factors explain 96.88% of the variance of the curve, which is the number usually quoted and the reason three-factor models are standard.

Now measure two books against it. The first is the directional book above, with sensitivities of both signs scattered across the curve. For a vector $\mathbf{w}$ of sensitivities the full-covariance 99% one-day VaR is $2.3263\sqrt{\mathbf{w}^{\top}\Sigma\,\mathbf{w}}$, as in (38.4), which for this book is $800,607. The second is a **butterfly**, which on a yield curve means a position of one sign at two tenors, the wings, against the opposite sign at a tenor between them, the body. The name is borrowed from the option butterfly of Section 12.3, whose wings and body sit at strikes instead. Sized so that the sensitivities sum to zero, it is flat under a parallel shift and is a position on the curve's curvature, not its level. This one is long $20,000 per basis point at two years and five years and short $40,000 at three, with a full VaR of $117,925. Keeping k factors replaces $\mathbf{w}^{\top}\Sigma\,\mathbf{w}$ by $\sum_{j\le k} \ell_j (\mathbf{v}_j^{\top}\mathbf{w})^2$, where ℓ_j and $\mathbf{v}_j$ are the eigenvalues and eigenvectors of Σ, and the table reports each book's VaR together with the share of its own variance that survives. (The VaRs use the unrounded quantile $N^{-1}(0.99) = 2.326348$; with 2.3263 the last digits move by up to $16.)

	Directional book		**Butterfly**	
Factors kept	**VaR**	**Variance kept**	**VaR**	**Variance kept**
1	$795,377	98.7%	$20,390	3.0%
2	$797,262	99.2%	$23,225	3.9%
3	$797,364	99.2%	$38,194	10.5%
5	$797,380	99.2%	$62,434	28.0%
9	$800,607	100%	$117,925	100%

The same three factors that keep 99% of the first book's variance keep 10% of the second's. In VaR terms a three-factor system would report the butterfly at $38,194 against an actual $117,925, about a third of its risk, and it would do so while displaying the 96.88% figure as justification.

The reason is that the two books sit on different parts of the decomposition, and Figure 38.6 shows it. The leading factors are smooth across the curve; the butterfly is three adjacent tenors with alternating signs, a shape none of them resembles, so it projects onto them weakly. The 96.88% is the share of the *curve's* variance in three factors. What a portfolio needs is the share of *its own* variance, and a book constructed to be neutral to level and slope has deliberately projected itself onto the directions that were discarded. Factor adequacy is a property of the portfolio, not of the

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curve—and the portfolios most likely to fail the test are the ones a relative-value desk exists to hold.

That gives the practical rule. A risk system may reduce factors, but it must report the residual variance *for each book*, not for the market.

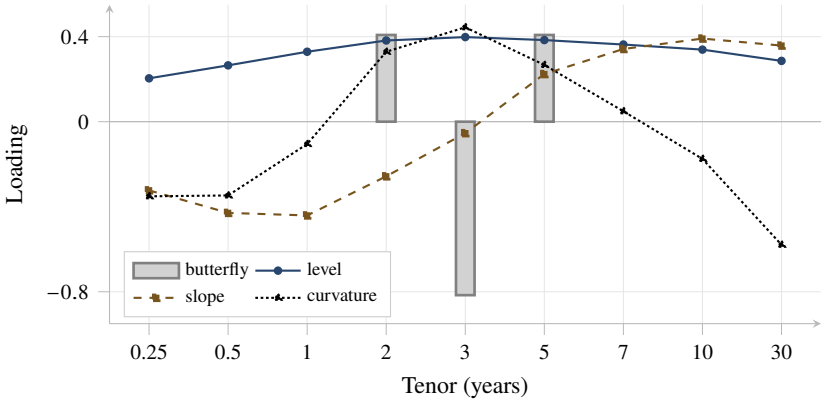


Figure 38.6: Loadings of the three leading factors of Example 38.11 by tenor, and the butterfly's sensitivities scaled to unit length.

38.14 SCALING ACROSS HORIZONS

Every figure so far has been a one-day measure, but capital and limits are often set over longer horizons, and the conversion is usually the square root of time:

$$\mathrm{VaR}_{h\text{-day}} \approx \sqrt{h}\,\mathrm{VaR}_{1\text{-day}}. \tag{38.7}$$

Equation (38.7) requires daily returns that are independent and identically distributed with mean zero, and for the quantile to scale like the standard deviation it also needs them normal; a fat-tailed one-day quantile does not scale by $\sqrt{h}$ even when the days are independent. The first condition is the one that fails most visibly, because returns are not quite independent. Suppose the correlation between returns k days apart is ρ^k, the pattern of a first-order autoregressive process with daily autocorrelation ρ. The variance of an h-day return is then the daily variance times $h + 2\sum_{k=1}^{h-1}(h-k)\rho^k$, one term for each pair of days, so under normality

$$\mathrm{VaR}_{h\text{-day}} = \mathrm{VaR}_{1\text{-day}}\sqrt{h + 2\sum_{k=1}^{h-1}(h-k)\,\rho^k}, \tag{38.8}$$

which reduces to (38.7) when $\rho = 0$. The table applies (38.8) to the one-day 99% VaR of \$4.65m from Example 38.1, scaled to ten days:

Daily autocorrelation	10-day VaR	Error in (38.7)
0.00	\$14.71m	—
0.10	\$16.10m	−9%
0.25	\$18.48m	−20%

The error column is the square-root figure's shortfall as a share of the true ten-day VaR, $(14.71 - \text{true})/\text{true}$: an autocorrelation of 0.10 leaves (38.7) about 9% too low and one of 0.25 a fifth too low, and positive autocorrelation is typical of the forced, one-directional selling of stressed markets, where the number is used. A mean-reverting market, with negative autocorrelation, breaks it the other way, so the rule is not conservative—it is merely wrong in a direction that depends on the market.

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The deeper problem is that the horizon should not be a constant at all. Section 8.4's margin period of risk is set by how long a position takes to close, and that is a property of the position: a liquid futures hedge and a bespoke structured trade cannot share a horizon. Chapter 39's liquidity horizons exist to fix precisely this, replacing one number with several by risk-factor class.

38.15 WHERE THE RISK ACTUALLY SITS

A total is not a management tool. The useful question is which position is producing the risk, and the derivative of (38.4) answers it.

Marginal VaR is $\partial \mathrm{VaR}/\partial w_i$, the rate at which the VaR grows as position i grows. Differentiating (38.4),

$$\frac{\partial \mathrm{VaR}}{\partial w_i} = z\,\frac{(\Sigma \mathbf{w})_i}{\sigma_P}, \qquad \sigma_P = \sqrt{\mathbf{w}^\top \Sigma\, \mathbf{w}}, \tag{38.9}$$

where σ_P is the portfolio's standard deviation and $(\Sigma \mathbf{w})_i$ is the covariance of factor i's return with the portfolio's P&L. **Component VaR** is $w_i\,\partial \mathrm{VaR}/\partial w_i$, and because (38.4) is homogeneous of degree one in the positions, the components sum exactly to the total, $\sum_i w_i z(\Sigma \mathbf{w})_i/\sigma_P = z\sigma_P$, an Euler allocation (Euler's theorem for functions homogeneous of degree one), not an approximation. Example 38.12 applies it to the book of Example 38.9.

Example 38.12 Sixty percent of the money, thirty percent of the risk

The book of Example 38.9: A is \$60m at 1.0% volatility, B is \$40m at 2.5%, correlation 0.30.

	Position	Standalone VaR	Component VaR	Share of risk
A	\$60m (60%)	\$1.396m	\$0.958m	31.4%
B	\$40m (40%)	\$2.326m	\$2.093m	**68.6%**
Total	\$100m	\$3.722m	\$3.051m	100%

B is 40% of the money and 69% of the risk, and the standalone numbers do not show this: they sum to \$3.722m against a portfolio VaR of \$3.051m, an 18.0% diversification benefit that has to be allocated somehow. Component VaR allocates it in the only way that both adds up and agrees with the marginal figures.

The marginal numbers are what a trader should be shown. In return units $\Sigma \mathbf{w} = (0.0090,\ 0.0295)$ and $\sigma_P = \$1.311\text{m}$, so (38.9) gives $2.3263 \times 0.0090/1.311 = 0.0160$ for A and $2.3263 \times 0.0295/1.311 = 0.0523$ for B. At the margin, a dollar moved into B adds 0.0523 of VaR against A's 0.0160, so shifting \$10m from A to B should add about $10\times(0.0523-0.0160) = \$0.36\text{m}$, or 11.9%; revalued exactly, the total rises by 12.8% while the book's size is unchanged. A limit on total VaR permits that trade as long as there is room under it, and records only that the total rose; component VaR shows that the whole rise came from B.

One caution applies, and it is the same one as everywhere in this chapter. Component VaR sums exactly only because VaR is homogeneous; it does not require VaR to be subadditive, and Example 38.2 shows it is not. So the allocation can be exact and the total still wrong. Expected shortfall admits the same Euler decomposition and is coherent; under simulation its components are simply each position's average loss on the scenarios in the tail, $\mathbb{E}[L_i \mid L > \mathrm{VaR}_c]$, which is one reason it is increasingly used for allocation as well.

Incremental VaR answers a third question: how much the VaR changes if a position is removed altogether. In the example, removing B leaves A's standalone \$1.396m, so B's incremental VaR is \$3.051m − \$1.396m = \$1.655m; removing A leaves \$2.326m, so A's is \$0.725m. The two sum to \$2.380m, not to the \$3.051m total, because each credits the whole \$0.671m diversification benefit to the position being removed, so the two together deduct it

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twice, once more than the total does: \$3.722m − 2 × \$0.671m = \$2.380m. Incremental figures answer whether to keep or close a whole position; component figures are the ones that add up, and marginal figures the ones that guide a small trade.

38.16 BACKTESTING

A risk number is a testable prediction: at 99%, losses should exceed the VaR on 1% of days. Counting exceptions is the only genuine validation any of this receives, and Example 38.13 shows how little a year of it can establish.

Example 38.13 Two hundred and fifty days is not much evidence

At 99% over 250 days, the expected number of exceptions is 2.5. With x exceptions in n days, a target exception rate $p = 1\%$ and the observed rate $\hat{p} = x/n$, the Kupiec likelihood-ratio (LR) statistic is

$$\mathrm{LR} = -2\ln\frac{(1-p)^{n-x}\,p^{x}}{(1-\hat{p})^{n-x}\,\hat{p}^{x}},$$

which under a correct model is approximately chi-squared with one degree of freedom, so the test rejects when it exceeds 3.841, the 95th percentile of that distribution, which gives it a nominal 5% significance level (the exact binomial probabilities in the table are the alternative when, as here, few exceptions are expected):

Exceptions	Probability	Kupiec LR	Reject?	Basel zone
0	0.0811	5.03	**yes**	green
1	0.2047	1.18	no	green
2	0.2574	0.11	no	green
3	0.2149	0.09	no	green
4	0.1341	0.77	no	green
5	0.0666	1.96	no	yellow
7	0.0097	5.50	**yes**	yellow
10	0.0002	12.96	**yes**	red

Anything from one to six exceptions is statistically indistinguishable from a correct model. Those counts are exception rates from 0.4% to 2.4%, so a model whose year looks like 97.6% coverage and one whose year looks like 99.6% both pass as 99% models.

Zero exceptions is a rejection too, and that is the row people forget: a model that never breaches is not conservative but miscalibrated, in the direction that wastes capital. The test is two-sided because a risk measure is a forecast, not a promise. The evidence is thin, though. A correct model shows no exceptions in 8.1% of years, and the 3.841 cutoff, a large-sample approximation that is coarse when only 2.5 exceptions are expected, rejects a correct model 9.5% of the time (zero exceptions, or seven or more) rather than the nominal 5%.

Note where the Basel zones, set in 1996 and kept by the current framework for its bank-wide test, and the statistics disagree. Five exceptions moves a bank into the yellow zone, where the 1996 rules raised the multiplier applied to its VaR from 3 to between 3.40 and 3.85 (4 in the red zone); the current framework keeps the zones and applies them to a multiplier of 1.5 to 2.0, set out in Chapter 39. Yet at five exceptions the Kupiec statistic is 1.96 and does not reject, and at seven to nine the test rejects while the zone is still yellow. The zones answer a different question from a two-sided test. A correct model lands in yellow or worse in 10.8% of years, a rate the supervisor accepted in exchange for power: a model whose true coverage is only 97% is flagged in 87% of years, against 63% for a Kupiec rejection.

Figure 38.7 puts the two verdicts on one axis. The shaded band is the range the Kupiec test tolerates and the dashed rules are the Basel boundaries, and they do not line up: a bank with five exceptions has left the green zone and pays a higher multiplier while the statistic sees

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nothing worth reporting. Note also how much probability mass sits under a correct model at counts a supervisor would treat as a warning.

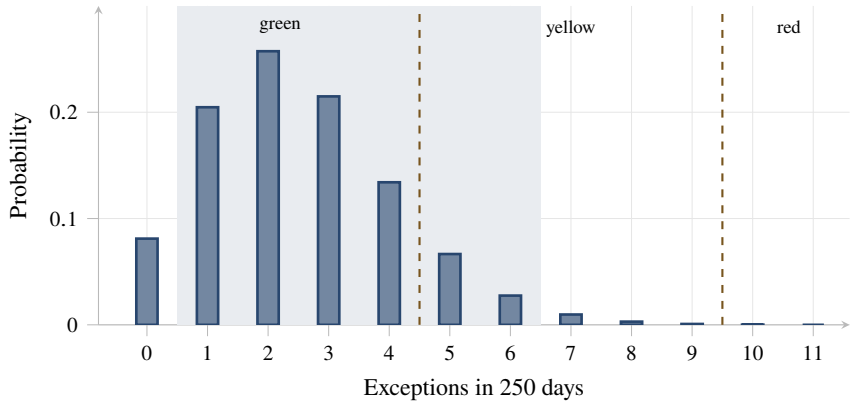


Figure 38.7: Distribution of 99% VaR exceptions in 250 days, Binomial(250, 0.01). Shading marks the range the Kupiec test does not reject; the dashed rules are the Basel zone boundaries.

And counting is only half of the test. Kupiec asks whether the number of exceptions is right and cannot ask when they arrived, so a model that breaches on five consecutive days in one stressed week passes it exactly as a model that breaches five times spread through the year. Those two models are not equally good, and the second one is the one the bank wants: clustered exceptions mean the model's variance is stale, which is precisely the failure that matters, because it arrives when the losses do.

Christoffersen's conditional-coverage test adds the missing half. Alongside the coverage statistic it tests **independence** (whether an exception yesterday raises the chance of one today) by comparing the observed transition frequencies against the hypothesis that they do not, and the two statistics add, each with one degree of freedom, so the combined statistic is compared with a chi-squared distribution with two degrees of freedom (5.991 at 5%). A model can pass on coverage and fail on independence.

The practical reading is that a clean exception count is weaker evidence than it looks. Chapter 39's supervisory framework counts exceptions and does not test their independence, so a desk that wants to know whether its model works has to run that test itself.

And expected shortfall cannot be backtested this way at all. There is no event to count: ES is a conditional mean, so validating it requires estimating a mean from the handful of observations in the tail. Tests built for it exist: Acerbi and Székely (2014) compare each tail loss with the forecast ES on the days VaR was exceeded. So does a separate obstacle to ranking competing ES forecasts: ES alone is not **elicitable** while the pair of VaR and ES is; [TN38] sets out both. The measure the framework moved to is the one that is harder to check, which is a real cost of the switch and the reason Chapter 39 backtests the VaR while capitalizing the ES.

38.17 STRESS TESTING

Every measure so far answers a question about probability: how large a loss is reached once in a hundred days, and how large the losses are beyond it. A **stress test** answers a different question. It revalues the portfolio under a specified scenario (a set of moves in prices, rates, volatilities, correlations and liquidity) and reports the loss, with no probability attached.

That is its weakness and its use. A number with no likelihood cannot be compared across books the way a VaR can, and it cannot be added to anything. But it does not depend on the

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distribution estimated from the window, so it can describe events the window never contained and joint moves that the estimated correlation matrix all but rules out. It is the tool aimed at the gaps Section 38.19 lists.

Scenarios come from two sources. **Historical scenarios** replay the moves of an actual episode on today's positions (October 19, 1987, the autumn of 2008, March 2020, the September 2022 gilt crisis of Section 8.9), and their merit is that nobody can call them implausible. **Hypothetical scenarios** are written by the firm for the risks it actually holds, and a good one moves correlations and liquidity as well as prices, because Section 38.11 showed that the joint behavior is what changes in a crisis. **Sensitivity analysis**, moving one risk factor at a time, is easier to run and to explain, and it misses exactly those joint moves. Example 38.14 sets a hypothetical scenario beside a sensitivity report on the same book.

Example 38.14 Two assets falling together

Take the book of Example 38.9: \$60m in asset A at a 1.0% daily volatility and \$40m in asset B at 2.5%, correlation 0.30, with a 99% one-day VaR of \$3.051m. The scenario is a liquidation day on which both assets fall by four of their own daily standard deviations, A by 4% and B by 10%.

The loss is $60 \times 4\% + 40 \times 10\% = \$2.4\text{m} + \$4.0\text{m} = \$6.4\text{m}$, 2.10 times the VaR. A sensitivity report that moves each asset alone by the same four standard deviations shows \$2.4m and \$4.0m, and its worst case, \$4.0m, falls 37.5% short of the joint scenario.

Under the model's own assumptions (normal returns, correlation 0.30), the portfolio's daily standard deviation is \$1.311m, so a \$6.4m loss lies 4.88 standard deviations out. Its probability is about 5 in 10 million a day, once in some 7,500 years of trading days. The stress test does not dispute that figure. It reports the loss and leaves its likelihood to judgment, because the figure is computed from the very correlation the scenario assumes has broken.

Reverse stress testing inverts the question. Instead of asking what a scenario costs, it fixes a loss (the one that would breach a limit, exhaust capital, or end the business) and asks which scenario produces it. It is the exercise most likely to find what the model misses, because it starts from the outcome rather than from the model's risk factors.

For a linear book with normally distributed factors the question has a closed form. Fix the loss at l. Many moves $\mathbf{r}$ of the factors lose exactly l on positions $\mathbf{w}$, and no single one has a probability, but they differ in how plausible they are: the joint density is highest close to the center of the distribution. Distance from the center is measured in standard deviations along whatever direction the move takes, by the **Mahalanobis distance** $\sqrt{\mathbf{r}^\top \Sigma^{-1} \mathbf{r}}$, and the most plausible move is the one that minimizes it subject to $\mathbf{w}^\top \mathbf{r} = -l$. The answer is

$$\mathbf{r}^* = -\, l \, \frac{\Sigma \mathbf{w}}{\mathbf{w}^\top \Sigma \, \mathbf{w}}, \tag{38.10}$$

and it lies l/σ_P standard deviations from the center, where $\sigma_P = \sqrt{\mathbf{w}^\top \Sigma \, \mathbf{w}}$ is the portfolio's standard deviation. Each factor moves in proportion to $(\Sigma \mathbf{w})_i$, its covariance with the portfolio, which is the same vector that gives marginal VaR in (38.9): the scenario loads most heavily on the positions with the highest marginal risk. For a nonlinear book there is no formula, and the scenario has to be found by searching over moves with full revaluation. Example 38.15 applies (38.10) to the book of Example 38.14.

Example 38.15 The most plausible way to lose \$6.4 million

Keep the book of Example 38.14 and fix the loss at the scenario's \$6.4m. In return units,

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$\Sigma\mathbf{w} = (0.0090,\ 0.0295)$ and $\mathbf{w}^{\top}\Sigma\,\mathbf{w} = 1.72$, so (38.10) gives

$$r_A^* = -6.4 \times \frac{0.0090}{1.72} = -3.35\%, \qquad r_B^* = -6.4 \times \frac{0.0295}{1.72} = -10.98\%.$$

The loss is $60 \times 3.35\% + 40 \times 10.98\% = \$6.4\text{m}$, as required, and the move lies $6.4/1.311 = 4.88$ standard deviations out.

The hand-written scenario of Example 38.14 lies 4.96 standard deviations out, so the reverse stress test has found a route to the same loss at which the joint density is about one and a half times as high. It also says where the risk is: on the most plausible path B falls 4.39 of its standard deviations and A only 3.35 of its own, as B's 69% share of the VaR in Example 38.12 would lead one to expect. The scenario a risk manager would write by symmetry, both assets falling equally hard, is not the one that matters most.

Figure 38.8 draws the geometry. The ellipse is the set of moves as plausible as the reverse-stress scenario, the dashed line every move that loses exactly \$6.4m, and the scenario is where the two touch; any other point on the line, including the hand-written one, lies outside the ellipse, where the density is lower.

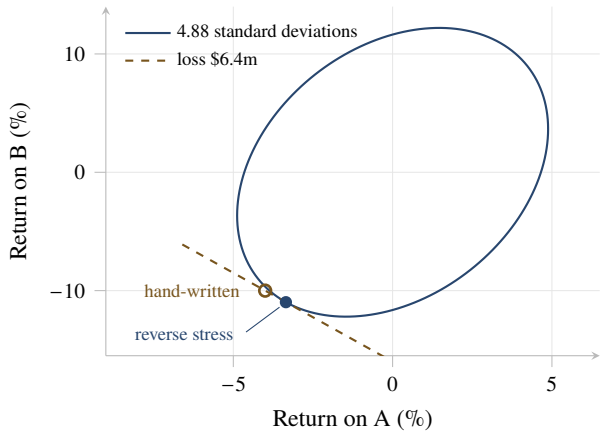


Figure 38.8: The most plausible move that loses \$6.4m on the book of Example 38.14: the point where the iso-loss line touches a contour of the return distribution.

Stress tests and probability-based measures are used side by side rather than combined. A typical limit framework sets a VaR or ES limit and, separately, a ceiling on the loss under each of a handful of named scenarios; a breach of either requires action. Some firms attach rough probabilities to their scenarios to fold them into a single number, but that reintroduces the estimate the stress test was built to avoid. The next section turns to the regulatory use of a crisis window, which keeps the probability machinery and changes only its data.

38.18 STRESSED MEASURES AND PROCYCLICALITY

Every estimator in this chapter is fitted to a window, and a window shrinks the measure as markets calm.

The same \$100m position at 99%, one day: at a calm 1.0% daily volatility the VaR is \$2.33m; estimated on a stressed window at 3.2%, it is \$7.44m. The same book comes out a factor of 3.2 apart, and the difference lies entirely in the sample.

The consequence is perverse and it is structural. Risk-based capital falls through a calm period, permitting more risk to be taken, and rises after volatility arrives, forcing positions to be

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cut into a falling market. The measure amplifies the cycle it is meant to guard against—which the reader has already met as Section 8.9's collateral spiral, appearing here as a capital spiral.

The repair is to add a measure that does not depend on the current window. A **stressed VaR**, added by Basel 2.5 in 2009, runs the same portfolio through a fixed twelve-month period of significant stress, so a bank's requirement cannot fall simply because the recent past was quiet. It is a crude device and it works, and Chapter 39 gives the current form.

What it does not fix is the composition problem. A stressed window contains the crisis that happened, so it protects against the last crisis rather than the next one. And books are optimized against the measure, which means the positions that look cheap under it are precisely the ones whose risk it does not see.

38.19 WHAT THESE MEASURES DO NOT SEE

A risk measure describes a distribution, and everything that matters about a book is not in its distribution.

Liquidity is absent. The measure assumes the position can be exited at the marked price; Section 8.9's episodes are all cases where it could not.

Concentration is invisible past a point. Two books with the same distribution have the same measure, whether the position is 1% of the market or 40% of it—and Amaranth, Archegos and the nickel short in Chapter 40 are all of the second kind.

Funding is a separate risk. A position can be correct, solvent, and still fail because variation margin arrives before the gain does. Example 37.2's basis trade loses nothing and can still be closed for want of cash, and Chapter 40's Metallgesellschaft and LDI cases are the same mechanism at scale.

And the model itself is a risk. Every number here is conditional on the mapping from market factors to portfolio value being right, the mapping that Chapter 21's calibration and, increasingly, trained networks produce. Chapter 39 treats that as its own subject, because it is the one exposure the risk system cannot measure with a risk system.

Case Study 38.1 Twenty-five sigma, several days running

In August 2007 a number of quantitative equity funds lost heavily over the same few days. The chief financial officer of Goldman Sachs, asked to explain, said the firm's funds had been seeing twenty-five standard deviation moves, several days in a row.

Take the statement literally and it is not a description of an event. Under a normal distribution a twenty-five sigma move has a probability of roughly 10^{-138}. There have been about 10^4 trading days in a working lifetime and something like 10^{17} seconds since the universe began. Several such days in one week is not a rare event; it is a refutation of the distribution used to measure it.

So the sentence has two readings and only one survives. Either something astronomically improbable happened repeatedly, or the model assigning the probabilities was wrong. Section 38.8 has already priced how wrong a normal assumption can be in the tail, and this is that error stated by someone who had just paid for it.

What the measure could not see was crowding. The positions were not unusual individually. They were held, in similar amounts and the same direction, by many funds running similar signals, so the deleveraging of one marked the book of the next. The exposure was to other people's positions, and a measure calibrated on a firm's own history contains no variable for that, which is why concentration belongs in this section rather than among the estimators.

And backtesting could not have raised the alarm in time. Example 38.13 showed

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that a year of exceptions cannot reject a 99% model whose realized coverage looks like 97.6%, and that even a model whose true coverage is 97% stays in the green zone one year in eight. A book can run on a badly shaped distribution for a long while and produce an exception count no supervisor would question.

The practical rule is the one the episode states out loud. Backtesting checks the prediction, but the test that matters is not whether the count of breaches looks reasonable but whether the shape is right. A model that reports twenty-five sigma has already told you it is the wrong model; no further evidence is required, and none should be waited for.

38.20 WHO USES THESE NUMBERS, AND WHY

Regulators use them to set capital, which is the use that decides the definitions: Chapter 39's rules are the reason the industry computes expected shortfall at 97.5% rather than at the 99% at which it had computed VaR.

Boards and senior management use VaR because it is one number with a probability attached, and that is genuinely the reason it won: the measure that can be explained in a sentence displaced measures that could not, and its statistical defects were formalized only after it had been adopted.

Risk managers use component and marginal figures rather than totals, because the total is not actionable and Example 38.12 is.

Clearing houses and prime brokers use short-horizon measures to set margin, where the horizon is a close-out estimate and the confidence level is a business decision about how often to be wrong. The large swap clearing houses compute initial margin by volatility-scaled historical simulation, which the industry often calls filtered historical simulation, taking an expected shortfall over a lookback of several years that includes a period of stress, so that a quiet year does not shrink the margin. LCH's SwapClear, the largest of them, uses expected shortfall at 99.7% on volatility-scaled scenarios from a ten-year look-back, with a five-day close-out for house accounts.

Insurers use the same idea at a different confidence and horizon. Solvency II, the European Union's insurance regime, sets the solvency capital requirement at the 99.5% VaR of an insurer's basic own funds, its assets less its liabilities, over one year, and an insurer with an approved internal model computes it by simulation.

Asset managers meet VaR as a limit on their funds. A European fund authorized under the rules for Undertakings for Collective Investment in Transferable Securities (UCITS) that measures its exposure by VaR must keep its 99%, twenty-day VaR below 20% of net asset value, or below twice that of a reference portfolio. In the United States, the SEC's Rule 18f-4, with which funds have had to comply since August 2022, imposes the same pair of limits on an open-end fund whose derivatives exposure exceeds 10% of its net assets.

Nonfinancial companies rarely report VaR, because their exposure is to operating cash flow rather than to the market value of a book. The variant they use is **cash-flow-at-risk**: the shortfall of a quarter's or a year's operating cash flow below its forecast that is exceeded with a stated probability, which is what a treasurer hedging commodity inputs or foreign revenue is trying to keep small.

Traders use them mainly as constraints, and a constraint is a thing to optimize against—which is why a measure's blind spots become positions. The London Whale of Section 40.10 is the plainest case: the measure changed and the book grew into it.

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38.21 WHERE THIS CAME FROM

Value at risk was a practitioner construction that regulation adopted, and the theory arrived afterward to tell it what was wrong.

Artzner, Delbaen, Eber and Heath (1999) is that paper. It sets out the axioms a risk measure should satisfy, shows that VaR violates subadditivity, and thereby names a defect desks had already met without being able to explain. Expected shortfall in its integral form satisfies them, as Acerbi and Tasche (2002) and Rockafellar and Uryasev (2002) showed for distributions with atoms, and the Basel framework moved to it seventeen years later, in 2016, an unusually clean case of a theoretical result changing a rule.

Kupiec (1995) is the standard backtesting reference and the source of the likelihood-ratio test of Section 38.16. Embrechts, Klüppelberg and Mikosch (1997) is the extreme-value treatment behind Section 38.9's tail estimation, and McNeil, Frey and Embrechts (2015) is the modern synthesis.

SUMMARY

A risk measure turns a distribution into a number, and the choice of number is the whole subject. VaR is the quantile of (38.1): it says where the tail starts and nothing about what is in it, while expected shortfall averages the tail beyond that point. The Basel move from 99% VaR to 97.5% ES was calibrated to be capital-neutral under normality, so what changed was not the level but the sensitivity to tail shape.

That sensitivity is the case for ES. VaR is not subadditive: two ordinary bonds, each with a VaR of zero, can have a large VaR together, wherever losses jump near the quantile. ES is coherent. The sharper argument is shape: a fat-tailed fit with the same volatility can lower VaR while raising ES, so the normal assumption understates exactly the part of the tail that ES measures.

Judgment enters through the inputs. EWMA is GARCH without a long-run level, so only GARCH produces a volatility term structure. Correlation is the weak input, and it moves most in the episodes the measure exists to describe; nor is correlation dependence, since a t copula makes joint extremes far more likely than a Gaussian one with the same correlation. Factor adequacy is a property of the portfolio, not of the curve, and allocation needs three figures, component, marginal and incremental VaR, because it asks three questions.

Estimation is where most of the error lives. Historical simulation is only as good as its window, and no reweighting adds information about a tail the window never saw; extreme value theory reaches beyond the data at a price in variance, and option books need a nonlinear treatment. Stress testing prices a named scenario with no probability attached, and reverse stress testing runs backward from a loss to the most plausible scenario that produces it.

Validation is weaker than anyone would like. At 99% over 250 days a wide range of exception counts is consistent with a correct model, and expected shortfall cannot be backtested by counting at all, which is why Chapter 39 backtests the VaR and capitalizes the ES. Because every estimator is fitted to a window, the requirement is smallest just before it is needed, which stressed measures exist to correct. And liquidity, concentration, funding and the model's own correctness lie outside any risk measure; Chapter 39 prices some of these and Chapter 40 shows the rest.

FURTHER READING

The coherence axioms and the subadditivity failure of Example 38.2 come from Artzner, Delbaen, Eber and Heath (1999), which is short and states the counterexample plainly; Acerbi and Tasche (2002) settles why the integral form of expected shortfall, and not the conditional mean, is the coherent one. On estimation, J.P. Morgan's RiskMetrics technical document

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(1996), which made EWMA the industry's default, states why $\lambda = 0.94$ and what it costs, and the GARCH literature, which begins with Bollerslev (1986), is best approached through its forecasting results rather than its estimation theory; Example 38.8's term structure is the part that matters for derivatives. On copulas, read Section 26.6's sources first: the tail-dependence result of Example 38.10 was known well before it was needed and ignored anyway, so read them for the history rather than the mathematics. On backtesting, Kupiec (1995) and the Basel Committee's 1996 backtesting framework are worth reading side by side precisely because they disagree, and the supervisory document sets out how its zones trade false alarms against the power to catch a bad model. Christoffersen (1998) is the paper for the other half of the question: it is short, and it is where the independence test of Section 38.16 comes from, together with the observation that coverage and independence are separable. Hull and White (1998) is the short paper behind volatility scaling, and Barone-Adesi, Giannopoulos and Vosper (1999) the one behind filtered historical simulation. On backtesting expected shortfall, Acerbi and Székely (2014) is the place to start, and Gneiting (2011) and Fissler and Ziegel (2016) supply the theory of why ES cannot be scored alone and how the pair with VaR can be. Chapter 39 gives the sources for the current capital framework, and [TN38] derives the representation that makes expected shortfall optimizable.

PRACTICE QUESTIONS

38.1 Using Example 38.1, state the 99% one-day VaR and ES for a \$100m book at 2% daily volatility, and explain why the ratio between them falls as confidence rises.

38.2 Show that 99% VaR and 97.5% ES nearly coincide for a normal loss, and explain what the regulator was trying to achieve by choosing those two levels.

38.3 State the subadditivity condition and explain why a measure that violates it cannot be used for desk limits.

38.4 Using Example 38.2, compute the 99% VaR of each bond alone and of the two together, and explain what feature of the loss distribution produces the failure.

38.5 Compute the 99% expected shortfall for the same two bonds, and verify that it satisfies subadditivity.

38.6 Explain why a desk measured on VaR has an incentive to sell far-out-of-the-money options, and what expected shortfall does to that incentive.

38.7 Explain what historical simulation assumes, and state the largest loss it is capable of reporting.

38.8 Using Example 38.6, explain why the fat-tailed VaR at 95% is *lower* than the normal VaR, and why the expected shortfall is not.

38.9 Compute, from Example 38.6, the ratio of the fat-tailed to the normal expected shortfall at 99.9% and the same ratio for VaR, and explain why the expected shortfall error matters more than the VaR error.

38.10 Using the simulation that follows Example 38.7, compare the bias and the variability of the three estimators, and explain why the most precise one is the most wrong.

38.11 Compute the probability that a 500-day window's largest loss falls below the true 99.9% VaR, and explain what this means for historical simulation at that confidence.

38.12 Give the two reasons extreme value theory converts an impossible estimate into a difficult one rather than into a known answer.

38.13 Using Example 38.8, compute the long-run volatility implied by $\omega = 0.000002$, $\alpha = 0.06$,

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$\beta = 0.93$, and state the one-year forecast under each estimator.

38.14 Explain why EWMA is adequate for a one-day margin calculation and inadequate for a volatility term structure.

38.15 Using (38.4), recompute the VaR of Example 38.9's book at correlations of 0.30 and 0.90, and explain why the estimation error has a consistent sign.

38.16 Using Example 38.10, compute the probability that the second exposure is beyond its 99.9th percentile given that the first is, under each copula, and compare the two with the coefficient of tail dependence.

38.17 Using Example 38.11, explain why three factors keep 99% of the directional book's variance and 10% of the butterfly's, and state what a risk system must report as a result.

38.18 State the square-root-of-time rule, the assumptions it requires, and the direction of its error in a trending market.

38.19 Using Example 38.12, state each position's share of the money and share of the risk, and explain what a total-VaR limit fails to notice.

38.20 Explain how an allocation can be exact while the total it allocates is unreliable.

38.21 Using Example 38.13, state the range of exception counts consistent with a correct model at 99% over 250 days, and explain why zero exceptions is also a rejection.

38.22 Explain where the Basel zones and the Kupiec test disagree, and what the supervisor gains by setting the zones where they are.

38.23 Explain why expected shortfall cannot be backtested by counting exceptions, and how the capital framework works around this.

38.24 Using Examples 38.14 and 38.15, explain what a stress test reports that VaR does not, why a one-factor-at-a-time sensitivity report understated the scenario loss, and what the reverse stress test found about where the book's risk lies.

38.25 A book's daily loss is normal with a standard deviation of \$2 million. Using the standard-error formula of Section 38.5, compute the standard error of a 99% VaR estimated from 250, 500 and 1,000 days, each as a share of the true VaR, and state how much longer a window must be to halve it.

38.26 Repeat Example 38.5 with a dollar gamma of −\$200 million. Compute the 99% one-day loss under the delta–gamma expansion, and state by how much it exceeds the delta-only figure.

38.27 Using the GARCH parameters of Example 38.8, show that a deviation of the variance from its long-run level halves in about 69 trading days, and compute the annualized volatility forecast for the single day 100 trading days after the variance just estimated.

38.28 Using (38.8), compute the ten-day 99% VaR of the one-day \$4.65m book of Section 38.14 at a daily autocorrelation of −0.10, and state the size and direction of the square-root rule's error.

38.29 Redo Example 38.12 at a correlation of 0.60: compute the portfolio VaR, each position's marginal, component and incremental VaR, and state which of the three sets sums to the total.

38.30 For the two-stock book of Example 38.4, find the most plausible pair of returns that loses \$1.0 million, state how many standard deviations out it lies, and compare it with the scenario in which both stocks fall by four of their own standard deviations.

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FURTHER QUESTIONS

38.31 Example 38.2 constructs a subadditivity failure with two bonds. Determine the conditions on default probability, recovery, and confidence level under which the failure occurs, state how the region changes as the number of bonds grows, and explain what this implies for a real credit portfolio measured at 99%.

38.32 Example 38.6 shows the fat-tailed VaR falling below the normal VaR at 95%. Find the confidence level at which the two cross for a t distribution with ν degrees of freedom scaled to a fixed variance, determine how that crossing point moves with ν, and state what it implies for a firm that measures at 95% and fits fat tails.

38.33 Example 38.8 compares forecast term structures. Fit both estimators to a real equity index series, compare their forecasts against subsequently realized volatility at horizons from one day to one year, and determine the horizon beyond which the mean reversion earns its extra parameter.

38.34 Section 38.12 shows that a Gaussian copula has zero tail dependence. Construct a portfolio of credit exposures whose 99.9% expected shortfall is materially different under Gaussian and t copulas calibrated to the same pairwise correlations, and determine what data would be needed to choose between them.

38.35 Section 38.14 treats the horizon as a property of the position rather than of the firm. Design a liquidity-horizon scheme for a book containing listed futures, cleared swaps, and bespoke structured trades; state how you would estimate each horizon; and determine how much the aggregate measure changes against a uniform ten-day assumption.

38.36 Section 38.16 notes that expected shortfall resists backtesting. Survey the proposed tests, implement one, and determine how many observations it needs to distinguish a correct model from one miscalibrated by 20%. Then state whether that number is available in practice.

38.37 Section 38.17 solves reverse stress testing in closed form for a linear book. Combine the sold puts of Example 38.5 with the \$5 million in stock 2 of Example 38.4, find numerically the most plausible pair of returns that loses \$1.0 million under the delta–gamma expansion, compare it with the linear answer, and explain why the gamma term moves the scenario.

38.38 Section 38.18 argues that books are optimized against whatever is measured. Take a measure from this chapter, construct the portfolio that maximizes true risk subject to a fixed reported value of it, and state what supervisory rule would prevent the construction.

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Chapter 39
Market Risk Capital and Model Risk

Chapter 38 built measures of what a book can lose. This chapter is about what a regulator does with them, and about the exposure that no measure in Chapter 38 can see: the risk that the model producing the numbers is wrong.

The two subjects belong together for a reason that becomes obvious once stated. A capital rule is a contract with a bank about a number, and the number comes from a model the bank builds. So the rule must either specify the model or test it, and the modern framework does both—which is why so much of it is machinery for deciding whether a bank's own model may be used at all.

One caution comes before any of the detail. The implementation dates in this area have moved repeatedly and in every major jurisdiction. As of October 2026, they stood as follows:

- the Basel standard's own implementation date was January 1, 2023;
- China had applied it since January 1, 2024;
- the European Union was due to apply its market risk rules from January 1, 2027, after two postponements, with a temporary adjustment to the capital effect proposed for three years;
- the United Kingdom was due to apply them from the same date, with its internal model approach for market risk deferred to January 1, 2028 (the PRA consulted on adjustments to it in June 2026);
- the United States had re-proposed its version in March 2026 with no final rule.

Check the calendar against the date you read this. The structure below is stable, and the calendar is not.

Sections 39.1 and 39.2 explain why the trading book has its own capital rule and where its boundary sits. Sections 39.3 to 39.8 are the rule itself: the standardized approach, internal models, non-modellable risk factors, the P&L attribution test, desk-level backtesting, and the approval cliff those tests create. Sections 39.9 to 39.12 assemble the total and cover the three channels that sit above or beside it: the output floor, prudent valuation, and stress testing. Section 39.13 asks what the framework changed and who bears the cost, and Sections 39.14 and 39.15 turn to model risk, including the validation of models that cannot be verified.

39.1 WHY THE TRADING BOOK NEEDS ITS OWN RULE

Banking book credit exposures are held to maturity and capitalized against default. Trading positions are held to be sold, so the loss that matters is a price move over a short horizon, not a default over years.

The framework that preceded the current one was built on Chapter 38's VaR and failed in 2008 in three identifiable ways. It used a ten-day 99% VaR calibrated on recent data, so it was procyclical exactly as Section 38.18 describes. It capitalized the trading book far more lightly than the banking book for the same economic exposure, which produced a decade of moving assets across the boundary. And it treated a position's liquidity as irrelevant, assuming everything could be exited in ten days.

The current framework is the Basel Committee's **Fundamental Review of the Trading Book**, known universally by its initials, **FRTB**, which is the name to search for and the one every vendor document uses; the US implementation was folded into the package called the **Basel III endgame**, whose scope has been contested and revised since it was first proposed. It answers each of the three failures:

- *procyclicality*, by calibrating the internal-model measure to a period of stress rather than to recent data, so that it cannot fall because markets are calm (Section 39.4);
- *the light trading book*, by fixing which book a position belongs in with a prescriptive list, and by denying any capital benefit to moving it (Section 39.2);
- *liquidity*, by giving each risk factor a horizon of 10 to 120 days, so that exposure to a risk factor that takes longer to exit is charged for it (Section 39.4).

It is worth reading that way: every complication below exists because something specific broke. The framework offers two ways to compute the charge, the **standardized approach**, which every bank computes, and the **internal models approach**, which a bank may use desk by desk once its models are approved.

Case Study 39.1 The patch, and why it was not enough

The response to 2008 came in two stages, and the gap between them is instructive.

The immediate patch arrived in July 2009 and is known as **Basel 2.5**. It added a **stressed VaR** computed on a crisis window, and an **incremental risk charge** for default and migration in the trading book. Both were added to the existing measure rather than replacing it, and supervisory estimates at the time put the combined effect at roughly a doubling to tripling of trading-book capital.

That the patch worked is not in dispute; that it was a patch is the point. Stressed VaR addressed procyclicality by adding a second calculation rather than fixing the first, so banks ran two measures on the same book and held the sum. The incremental risk charge covered default in the trading book because the boundary had let credit exposures migrate there, treating the symptom rather than the boundary; it did use liquidity horizons, with a three-month floor, but only for default and migration. Nothing in the patch addressed liquidity in the VaR measures themselves: a position was still assumed exitable in ten days.

Of the three failures of Section 39.1, the patch had met only procyclicality, and only by adding a calculation. The boundary and the horizon problems survived it, and so did a defect in the measure itself; those three were left to the current framework. The boundary was still intent-based, until Section 39.2's prescriptive list. The horizon was still a constant ten days, until Example 39.3's schedule of liquidity horizons. And the measure was still VaR, with Section 38.4's subadditivity failure and Section 38.8's finding that a fatter tail can lower VaR intact. That is the direct reason the current framework is written around expected shortfall.

The general lesson is about how frameworks fail. The 2009 patch was fast, quantitatively effective, and left every structural defect in place. Its successor took a decade and

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is still being implemented, which is the trade a supervisor faces whenever the diagnosis is architectural and the pressure is immediate.

39.2 THE BOUNDARY BETWEEN THE TRADING BOOK AND THE BANKING BOOK

The first fix concerns the line between the two books of Section 39.1: the trading book, capitalized against price moves under this chapter's rules, and the banking book, capitalized against default under the credit rules of Section 27.11 and Section 39.10. It is the least mathematical fix and was the most consequential. If capital differs between the two books, the boundary between them is a valuable option, and banks exercised it. A bank holding $500 million of corporate bonds that attracted 3% capital in the trading book and 8% in the banking book had $25 million of capital riding on a classification, and under the old rule the classification turned on the bank's stated intent.

The current rule makes the boundary prescriptive rather than intent-based. A presumptive list says what sits where:

- instruments held for short-term resale, positions in the correlation trading portfolio (securitization tranches and nth-to-default credit derivatives), net short credit or equity positions, and anything arising from underwriting or market making are presumed to be in the trading book;
- unlisted equities, real estate, retail and small-business credit, and equity investments in funds whose holdings the bank cannot identify and value daily are in the banking book.

A bank may depart from the presumption only with supervisory approval and documented reasons.

And, critically, reassignment between the books requires supervisory approval, is permitted only in extraordinary circumstances, and may not produce a capital benefit: if moving a position lowers its charge, the bank holds a disclosed surcharge equal to the difference. The option was not tightened; it was removed, which is the right response to an incentive rather than a modeling problem.

39.3 THE STANDARDIZED APPROACH

Every bank must compute the standardized charge, whether or not it has model approval, and the charge has three parts.

The sensitivities-based method is the bulk of it. A desk reports delta, vega, and curvature sensitivities to prescribed risk factors; the regulator supplies risk weights and correlations. The charge then combines them by a prescribed formula: each sensitivity is multiplied by its risk weight, and the products are combined the way a portfolio's standard deviation is built from the variance of equation (38.4) in Chapter 38's model-building approach: as the square root of their squares plus their correlated cross-products. Example 39.1 works it through. The three sensitivities are the coefficients of Chapter 18's expansion, equation (18.1), read as a capital rule: delta charges its first-order term, vega its volatility term, and curvature its second-order term. The correlations are given, not estimated. Section 38.11 showed that correlation is the least stable input to a risk measure: estimated in calm periods and applied in violent ones, so that a long book's measured risk comes out too low. The framework settles the question by fiat: the supervisor would rather be wrong in a known direction than depend on a bank's estimate.

The default risk charge covers **jump-to-default**, the loss if an issuer defaults overnight with no warning in its spread, on traded bonds, credit derivatives and equities. **The residual risk**

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add-on is a flat charge on payoffs the sensitivities do not describe. The next three subsections take the parts in turn.

39.3.1 Delta and vega

Sensitivities are grouped into **buckets** (one currency's interest rate curve, one equity sector) and aggregated in two stages, within each bucket and then across buckets. The delta charge is worth working through, because its structure contains a device that has no analog in Chapter 38 and that determines which books the framework treats harshly.

Example 39.1 Three correlation scenarios, and which one binds

Consider one currency and three points on its interest rate curve, at 2, 5 and 10 years. For each point k the bank reports its net sensitivity s_k: the change in value of all its positions on this curve when the rate at that point alone moves by one basis point, in dollars per basis point, the DV01 of Chapter 29, measured point by point rather than for a parallel shift, so that receiving and paying fixed at the same point carry opposite signs. The regulator supplies a risk weight RW_k for each point, here 130, 110 and 110 basis points—not the percentage of an exposure that Section 27.11's credit weights are, but the size of the rate move it prescribes as a shock—and the correlations

$$\rho_{kl} = \max\left(e^{-0.03\,|T_k - T_l|/\min(T_k, T_l)},\ 0.40\right),$$

which give 0.9560 between 2 and 5 years, 0.9704 between 5 and 10 years, and 0.8869 between 2 and 10 years, where T_k is the tenor of point k in years. Both the risk weights and this correlation are the Basel standard's own, not choices made for the example (for the major currencies a bank may divide the weights by $\sqrt{2}$). The correlation belongs to the family Chapter 32 used for forward rates in Example 32.1, decaying with the distance between tenors toward a floor, with two differences: the distance is measured relative to the shorter tenor, so the short end decorrelates faster (2 and 5 years at 0.956, against 0.991 for the same three-year gap between 10 and 13 years), and the floor of 0.40 is a cutoff rather than a blend. The **weighted sensitivity** $WS_k = s_k \times RW_k$ is then the dollar loss from that prescribed move at point k, and the bucket charge is

$$K_b = \sqrt{\max\left(0,\ \sum_k WS_k^2 + \sum_{k \neq l} \rho_{kl}\, WS_k\, WS_l\right)},$$

where the floor at zero matters only when the correlations are moved, as they are next.

The charge is computed three times, once with the prescribed correlations and once with each of

$$\rho_{kl}^{\text{high}} = \min\left(1.25\,\rho_{kl},\ 1\right), \qquad \rho_{kl}^{\text{low}} = \max\left(2\rho_{kl} - 1,\ 0.75\,\rho_{kl}\right).$$

The capital number is the largest of the three charges, not the charge at the smallest correlations: which set of correlations binds depends on the book, as the two books below show. (Basel takes this maximum after the buckets have been aggregated; with a single bucket, as here, that is the same thing.)

Take two books. The first receives fixed at all three points, with sensitivities of \$2,000, \$3,000 and \$4,000 per basis point. The second is a butterfly, the curve trade of Example 38.11, with wings at 2 and 10 years and the body at 5: it receives fixed at 2 and 10 years (\$2,200 and \$2,000 per basis point) and pays fixed at 5 years (\$4,600), sized so that its weighted sensitivities sum to zero: the prescribed shocks, 130 basis points at 2 years and 110 at 5 and 10, leave it unchanged, although a parallel one-basis-point move changes its value by \$400.

For the directional book the weighted sensitivities are \$260,000, \$330,000 and \$440,000, so under the prescribed correlations

$$\begin{aligned} K_b^2 &= 260^2 + 330^2 + 440^2 + 2 \times (0.9560 \times 260 \times 330 \\ &\qquad + 0.8869 \times 260 \times 440 + 0.9704 \times 330 \times 440) = 1{,}018{,}894, \end{aligned}$$

in thousands of dollars squared and with the correlations unrounded, so K_b = \$1,009,403. The three scenarios give:

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Scenario	Directional book	Butterfly
Low correlation	\$988,376	**\$100,854**
Medium (as prescribed)	\$1,009,403	\$71,315
High correlation	**\$1,030,000**	\$0
Charge (worst)	**\$1,030,000**	**\$100,854**
Excess over medium	+2.0%	+41.4%

The two books are punished by opposite scenarios. The directional book gets worse as the curve moves together, so *high* correlation binds, and the penalty is mild, 2.0%, because a directional book is nearly as risky under any correlation. The butterfly is the reverse: under high correlation its charge falls to *exactly zero*, since a perfectly correlated curve makes the hedge perfect, while under low correlation the legs stop offsetting and the charge is 41.4% above the prescribed case.

So the hedged book pays the larger proportional penalty, and the reason is the design working as intended: the rule refuses to let a bank choose the correlation that flatters its book. A framework that used a single prescribed correlation could be arbitraged by constructing positions that are riskless under it: a book hedged only at the prescribed number is unhedged at any other. Taking the worst of three removes that possibility.

The cost is borne by genuine relative-value business. A butterfly is a real hedge whose risk really is small when the curve moves together, and the framework charges it as though the correlations it relies on might not hold. Section 38.11 showed that correlations fail in exactly the episodes capital exists for, so the charge is defensible even though it is expensive.

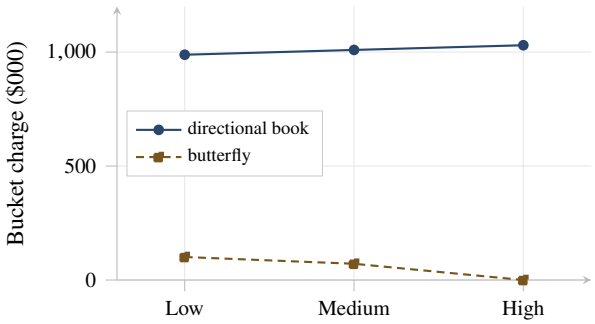


Figure 39.1: Bucket charge under the three correlation scenarios, for the two books of Example 39.1.

Figure 39.1 shows why one rule serves both books. The two lines slope in opposite directions, so no single correlation is conservative for both—and any rule that picked one would leave the other exploitable. Evaluating three scenarios and taking the worst is a simple way to be conservative against a quantity no one can estimate reliably.

Vega is aggregated the same way, with one difference that matters. Its risk weight is not a fixed number but $\min\left(55\% \times \sqrt{LH/10},\ 100\%\right)$, where LH is the liquidity horizon the framework assigns to the volatility, in days: 20 days for the volatility of a large-cap stock, giving 77.8%, and 40 days or more for every other volatility, giving 100%.

39.3.2 Curvature

Curvature is charged separately because delta and vega miss the convexity that Chapters 34 and 35 showed dominates structured books. The charge shocks the factor up and down by its delta risk weight, revalues in full, and asks how much worse the result is than the delta

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prediction:

$$CVR = \max\Big(0,\ -\min_{\text{up, down}}\big[V(\text{shocked}) - V(0) - \delta \times \text{shock}\big]\Big).$$

Here V is the position's value, δ its dollar delta, and the shock the delta risk weight, applied up and down. It revalues rather than multiplying a reported gamma by $\frac{1}{2}(\Delta S)^2$ because the shocks are large, and Section 18.10 showed that the expansion's gamma term misstates a large move. The charge has an asymmetry that surprises people, and the floor at zero is what produces it. (Basel applies that floor when it aggregates the risk factors; for a single factor, as here, the effect is the same.)

Take a long \$10 million notional at-the-money six-month call on a large-cap emerging-market stock, at 25% volatility with interest rates at zero. The option is worth \$704,000 and its delta is 0.5352, so its dollar delta is \$5.352m. The stock's delta risk weight, and so its curvature shock, is 55%, the Basel weight for a large-cap stock in an emerging market (advanced-economy large caps carry 30% to 50%). Up 55%, the option is worth \$5.505m against a delta prediction of \$0.704m + 0.55 × \$5.352m = \$3.648m; down 55%, it is worth almost nothing against a prediction of −\$2.239m. Revaluation beats the delta prediction by \$1.86m on the upside and \$2.24m on the downside, both favorable, because the holder is long gamma. The bracketed term is positive in both directions, so its minimum is positive, and the floor sets the charge to zero. Sell the same option and both figures change sign, giving a charge of **\$2.24m**. Figure 39.2 draws the two gaps.

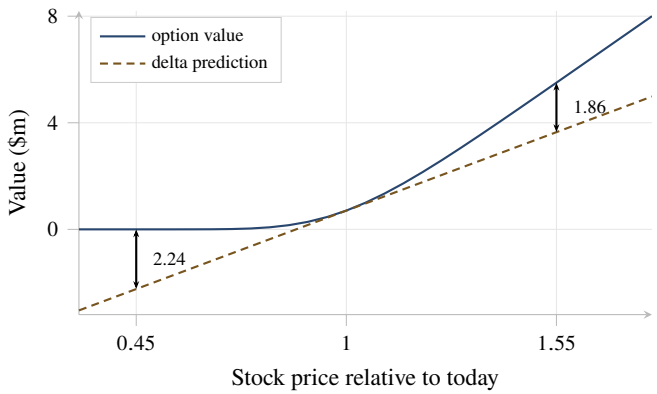


Figure 39.2: A long call's value and its delta prediction, with the gaps at the ±55% curvature shocks. Illustrative.

Both gaps lie on the same side of the line because the value curve is convex, and that is the whole of the asymmetry. Long optionality attracts no curvature charge and short optionality attracts all of it, which is the correct treatment and worth stating plainly because it is the first place in this book where a capital rule agrees exactly with Chapter 18's economics. The vega charge completes the picture, and the three components together answer a question a trading desk will ask immediately: does hedging reduce the charge?

Example 39.2 A delta-hedged short option position

Sell the \$10 million notional six-month at-the-money call above, on the same large-cap emerging-market stock at 25% volatility, and delta-hedge it.

Vega is reported as the sensitivity to a *relative* move in implied volatility, the desk's vega multiplied by the volatility itself, so at \$28,100 per volatility point the reported sensitivity is \$702,488. The volatility of a large-cap stock has a 20-day liquidity horizon, so its vega risk weight

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is $55\% \times \sqrt{2} = 77.8\%$ and the vega charge is \$546,407.

Component	Charge	Why
Delta	≈ \$0	removed by the hedge
Vega	\$546,407	the hedge does not touch it
Curvature	\$2,239,369	short gamma, both directions
Total	**\$2,785,776**	**27.9% of notional**

Delta-hedging removed the delta charge and none of the rest. A book that is flat by the measure a trader watches minute to minute attracts capital equal to more than a quarter of its notional, and 80% of that comes from the curvature term alone.

This is the framework doing deliberately what Chapter 18 says is true. Delta neutrality is a statement about the first derivative at a point; the risk in a short option is the second derivative and the volatility, and a capital rule that let delta-hedging extinguish the charge would be paying for a hedge that does not hedge the exposure.

It also explains a pricing fact from Chapter 35. An issuer selling optionality to retail carries this charge for the life of the note, on a delta-hedged book, so the capital cost is a running cost rather than a one-off, and Section 35.10's fee arithmetic has to cover it before anything is left over.

39.3.3 The default risk charge and the residual risk add-on

The **default risk charge** treats default as an event the sensitivities cannot see: a bond's spread sensitivity says what a widening costs, not what a default costs. Each position's jump-to-default loss is its loss given default on the notional, adjusted for the price already paid. Longs and shorts in the same issuer offset, offsets between different issuers in the same bucket are recognized only in part, and the net amount is multiplied by a risk weight set by credit quality. A desk long a bond and holding bought protection on the same issuer pays little; a desk long one issuer and short an equal amount of another in the same bucket still pays half the long position's charge, because the framework recognizes only part of an offset between two different issuers' defaults. The charge is calibrated as the banking book's credit capital is, to a one-year horizon at 99.9%, Section 24.14's worst-case default rate, so that a bond's default capital no longer depends on which book holds it; a bank on internal models computes it by simulating defaults over that year with at least two systematic factors. Migration is left out on purpose, because the spread sensitivities already measure what a downgrade costs.

The **residual risk add-on** is the framework's admission that some payoffs are beyond its sensitivities. It is 1% of the notional of instruments with an exotic underlying—weather, longevity, future realized volatility (a variance swap's underlying)—and 0.1% of the notional of other instruments bearing residual risks, such as gap risk, correlation risk or behavioral risk (a borrower's option to prepay, for instance); a desk with \$50 million of the first and \$200 million of the second carries \$0.7 million. A barrier or a correlation product attracts the add-on because the framework declines to model it, which is more honest than a false precision would be.

39.4 THE INTERNAL MODELS APPROACH

A bank with approval computes expected shortfall at 97.5%, Chapter 38's measure at the confidence level Example 38.1 explained, calibrated to a period of stress rather than to recent data. The resulting charge, the **internally modeled capital charge** (IMCC), weights equally the expected shortfall of the whole book, with diversification across risk classes, and the sum of the expected shortfalls of each risk class computed alone:

$$IMCC = \frac{1}{2}\,\mathrm{ES}_{\mathrm{all}} + \frac{1}{2}\sum_{c}\mathrm{ES}_{c}.$$

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The bank's own correlations thus enter with half weight.

Two design choices follow from Chapter 38's failures. Stress calibration answers Section 38.18's procyclicality directly: the window cannot shrink because markets are calm. And the horizon is no longer a constant.

Example 39.3 The horizon schedule is a capital multiplier

Risk factors are assigned **liquidity horizons** of 10, 20, 40, 60, or 120 days by class:

Horizon	Examples
10 days	major-currency interest rates, large-cap equity prices, major exchange rates
20 days	other interest rates, investment-grade sovereign spreads, small-cap equity prices, other exchange rates, large-cap equity volatility, energy and metal prices
40 days	investment-grade corporate spreads, other sovereign spreads, exchange rate volatility
60 days	high-yield corporate spreads, interest rate volatility, other equity volatility, energy and metal volatility, other commodity prices
120 days	other credit spreads, credit spread volatility, other commodity volatility

The ten-day expected shortfall is computed five times: ES_1 shocks every factor, and ES_j shocks only the factors whose horizon is at least LH_j, the jth horizon in the schedule. The results are aggregated as

$$\mathrm{ES} = \sqrt{\mathrm{ES}_1^2 + \sum_{j \geq 2} \left[\mathrm{ES}_j \sqrt{\frac{LH_j - LH_{j-1}}{10}} \right]^2},$$

with weights $\sqrt{(LH_j - LH_{j-1})/10}$ of 1.0000, 1.0000, 1.4142, 1.4142 and 2.4495. A factor with a 120-day horizon appears in all five terms, so its contribution is scaled by $\sqrt{1+1+2+2+6} = 3.46$ in total.

Take a desk whose ES values, for the factors surviving in each bucket, run 100, 90, 70, 45, 20. The aggregate is **185.34** against a flat ten-day figure of 100. The desk's capital is 1.85 times what a single-horizon measure would give, with no change in any position, volatility, or correlation.

The weights are the square-root-of-time rule of Section 38.14, applied to the stretch of horizon each bucket adds. Section 38.14 showed that rule failing when returns are autocorrelated, and the framework uses it anyway, because it needs a formula a supervisor can prescribe; the liquidity horizons fix the horizon problem and inherit the scaling problem. The horizon belongs to the position, not to the rule, so the framework replaces one horizon with five, and the price of that correctness is that a bank's capital now depends on a regulatory classification of its risk factors rather than on anything it measures. *The judgment moved rather than disappeared*, which is the recurring shape of this chapter.

39.5 NON-MODELLABLE RISK FACTORS

A factor may only be modeled if there is enough evidence of real prices for it. The **risk factor eligibility test** asks for at least 24 real price observations over the past twelve months, with no 90-day period containing fewer than four, or at least 100 observations over the twelve months. Anything failing that test is a **non-modellable risk factor** (NMRF) and is capitalized by a **stressed expected shortfall** (SES): a stress scenario calibrated factor by factor, no less severe than the expected shortfall's own calibration.

The penalty is not only that the stress is severe. It is that the non-modellable capital is added to the modeled charge rather than diversified with it: the modellable part gets the benefit of Chapter 38's covariance structure, and the non-modellable part does not. Within the stress add-on itself, idiosyncratic credit and equity factors are combined as if independent and the rest with a correlation of 0.6, so some diversification survives there, but none survives between the two parts.

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Example 39.4 Two penalties for one illiquid factor

A desk's risk splits into a modellable part with a stand-alone expected shortfall of 80 and an illiquid factor with a stand-alone expected shortfall of 30, correlated at 0.25. If every factor were modellable, the desk would hold

$$\sqrt{80^2 + 30^2 + 2 \times 0.25 \times 80 \times 30} = 92.2.$$

The illiquid factor fails the eligibility test. Its stress scenario, calibrated over its own longer horizon, is assumed here to be 1.6 times its expected shortfall, so its add-on is $1.6 \times 30 = 48$, and the desk holds $80 + 48 = 128$, 38.8% more for identical positions.

The 35.8 difference has two sources of almost equal size. Adding rather than diversifying costs $80 + 30 - 92.2 = 17.8$, and the harsher stress calibration costs $48 - 30 = 18.0$. Only the first is a property of the rule; the 1.6 is this example's assumption.

So identical risk costs more when it sits in an illiquid factor, and that is deliberate: it prices illiquidity, which Section 38.19 put first among what the measure cannot see. The framework cannot model liquidity, so it taxes illiquidity instead—a substitution worth noticing, because the tax falls on the bespoke structured business of Chapter 35.

39.6 THE P&L ATTRIBUTION TEST

This is the mechanism that decides whether a desk keeps its model, and it is the most quantitatively interesting thing in the framework. Approval is granted desk by desk, and a **trading desk** here is a unit the bank designates and the supervisor accepts: a group of traders or accounts running a well-defined business strategy, with its own risk management and a single head.

Each desk produces two daily numbers. **Hypothetical P&L** is the day's profit on yesterday's positions, computed by full front-office revaluation. **Risk-theoretical P&L** is the same thing computed using only the risk factors the risk model contains. If the risk model captures the desk, the two series agree.

The test compares them over 250 days with two statistics. The Spearman rank correlation asks whether the two series rise and fall on the same days. The **Kolmogorov–Smirnov** (KS) statistic is the largest gap between the two series' empirical distribution functions, and asks whether they have the same distribution. A desk is green if the rank correlation is above 0.80 and KS below 0.09, red if the rank correlation is below 0.70 or KS above 0.12, and amber otherwise.

Example 39.5 A third of the P&L may be unexplained

Let a fraction f of the desk's P&L variance come from factors the risk model does not contain, independent of the rest. The risk-theoretical series is then the hypothetical series with that part removed: it has variance $1 - f$ of the hypothetical one, and the two correlate at $\rho = \sqrt{1 - f}$. For jointly normal series the rank correlation is

$$\rho_S = \frac{6}{\pi} \arcsin\left(\frac{\rho}{2}\right).$$

KS has no closed form here, so the table gives its mean over 2,000 simulated 250-day samples:

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Unexplained variance	ρ	Spearman	KS (mean)	Zone
5%	0.9747	0.9722	0.043	green
20%	0.8944	0.8855	0.065	green
30%	0.8367	0.8243	0.081	green
33.8%	0.8135	0.8000	0.088	**amber begins**
40%	0.7746	0.7595	0.099	amber
48.6%	0.7167	0.7000	0.117	**red begins**
50%	0.7071	0.6902	0.120	red

A desk can leave a third of its profit-and-loss variance unexplained by its own risk model and remain in the green zone. It takes half before the model is disqualified.

The two statistics leave green within a couple of percentage points of each other (the rank correlation at 33.8%, the mean KS at about 35%) and reach red together near half, because a missing factor does two things at once. It weakens the day-by-day correlation, which the rank correlation sees, and it shrinks the risk-theoretical series' variance, which KS sees. A model error that did only one of the two would be caught by only one statistic.

The KS column is a mean, and a single desk's KS is noisy. At 30% unexplained variance, where the mean is 0.081, one 250-day sample in four exceeds 0.09, so an otherwise green desk can be pushed into amber by the sample alone.

The practical reading is not that the test is useless. It catches a desk whose risk model has missed a factor *entirely*, which is what it was built for and which was common. What it does not catch is a model that is systematically wrong by a moderate amount: one that leaves a quarter of the variance unexplained, or one that tracks the hypothetical P&L closely while sitting slightly off it every day.

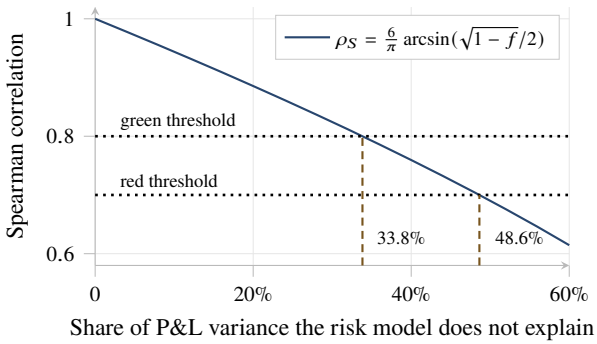


Figure 39.3: Rank correlation against unexplained P&L variance, with the zone thresholds.

Figure 39.3 is worth looking at before designing a risk model. The curve falls steadily, but the green threshold sits low on it: the rank correlation does not reach 0.80 until a third of the variance is unexplained. The test is permissive because of where its threshold is set, and the choice of statistic adds a second weakness: rank correlation was chosen because it is robust to outliers, and robustness to outliers is robustness to the days that matter.

39.7 BACKTESTING AT DESK LEVEL

The attribution test is not the only hurdle, and the two interact. Each desk also backtests: Chapter 38's exception counting, run at both 97.5% and 99% on the desk's own one-day VaR. A desk with more than 12 exceptions at 99%, or more than 30 at 97.5%, over the past 250 days loses its model approval and moves to the standardized approach.

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VaR is backtested even though capital is held against expected shortfall, for the reason Section 38.16 gave: exceptions can be counted, and expected shortfall offers no event to count. The framework checks the quantile and trusts the tail to come from the same model.

Separately, the bank's model as a whole is backtested at 99% across all its approved desks, and that count sets the multiplier of Section 39.9. The desk-level test decides whether a desk may use its model; the bank-level test decides how much the modeled number is scaled up.

The two desk tests fail for different reasons, which is the point of having both. Backtesting asks whether the measure is *calibrated*: are losses exceeding it at the stated rate? Attribution asks whether the model *contains the right factors*. A desk can pass either and fail the other.

A desk with a correctly sized but structurally wrong model passes backtesting and fails attribution: its VaR is the right magnitude because the missing factor's variance was absorbed into the calibration, while the daily series do not track.

A desk with the right factors and a stale calibration does the reverse: hypothetical and risk-theoretical P&L move together, so attribution passes, while exceptions pile up because the level is wrong.

The combination is more powerful than either, and it is the closest the framework comes to actually validating a model. It is still weak in the sense Section 38.16 established (250 days, and Example 39.5's permissiveness), but the two tests fail on different errors, so a model has to be wrong in a specific and rather narrow way to survive both.

39.8 DESK-LEVEL APPROVAL, AND THE CLIFF

Approval is granted and withdrawn *desk by desk*, which is the framework's most important structural choice. Figure 39.4 sets out the sequence of tests a desk passes through.

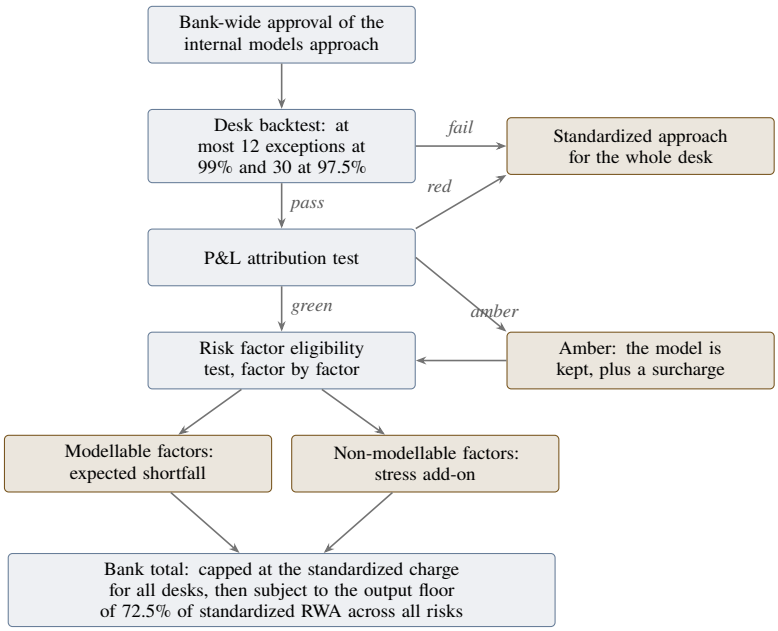


Figure 39.4: The tests a trading desk passes through, and where each outcome leads.

A failing desk does not disqualify the bank, so the consequence is proportionate. But a red desk, or one that fails its backtest, moves to the standardized charge, which for a book with genuine diversification can run well above the model number. An amber desk keeps its model

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but pays a surcharge: half the excess of the standardized over the modeled charge for the green and amber desks together, scaled by the amber desks' share of the standardized charge. The overall internal-model total is in turn capped at what the standardized approach would require for every desk.

The move to red is a cliff, not a slope, and it is triggered by a statistical test on 250 days, which Section 38.16 already showed to be a short sample. So a desk's capital can double on evidence that would not settle a scientific question, and the rational response is to run risk models that are conservative in the specific sense of containing every factor the front office prices with, whether or not they matter. The amber surcharge softens the first step down without removing the cliff at the bottom.

That incentive is the framework working as intended. It cannot verify a model, so it makes the cost of divergence high enough that banks align the two themselves.

39.9 ASSEMBLING THE TOTAL

The pieces above combine in a way worth seeing once, because the arithmetic explains where a trading book's capital actually comes from.

The modeled part is

$$\max\Big(IMCC_{t-1} + SES_{t-1},\ \ m_c \times \overline{IMCC} + \overline{SES}\Big),$$

where $IMCC$ is the internally modeled capital charge of Section 39.4, SES the non-modellable stress add-on, $t-1$ marks the previous day's figure, the bars denote sixty-day averages, and $m_c \geq 1.5$ is a multiplier that rises with the bank-wide backtesting exceptions of Section 39.7.

Example 39.6 Where the capital comes from

Take a bank with $IMCC_{t-1} = 420$, $\overline{IMCC} = 380$, $SES_{t-1} = 95$ and $\overline{SES} = 88$:

$$\max(420 + 95,\ 1.5 \times 380 + 88) = \max(515,\ 658) = 658.$$

The averaged, multiplied term binds, which is normal: the multiplier is calibrated so that it usually does, making the requirement depend on a sixty-day average rather than on yesterday. Only a sharp one-day jump reverses that: had yesterday's charge been 650, the first term would be 745 and would bind.

The multiplier is where backtesting enters. The bank-wide exception count over 250 days sets m_c, with the zone boundaries of Section 38.16's traffic light, whose yellow zone the current framework calls amber:

Multiplier	Modeled charge	Increase
1.50 (green)	658	—
1.70 (amber, five exceptions)	734	+11.6%
2.00 (red)	848	+28.9%

Adding the default risk charge on the approved desks and the standardized charge on the desks with no model approval gives the total. The residual risk add-on belongs to the standardized charge, so it sits inside the standardized-approach line:

Component	Amount	Share
Modeled measure (ES and NMRF)	658	60.2%
Default risk charge, approved desks	140	12.8%
Desks on the standardized approach	295	27.0%
of which residual risk add-on	35	3.2%
Total	**1,093**	100%

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More than a quarter of the requirement comes from desks whose models were not approved, and that share is the framework's real lever. Hedging moves this number at the margin; passing the attribution test on more desks moves it wholesale—which is why the test of Section 39.6, and not the measure of Section 39.4, is what trading-book capital management is actually about.

39.10 THE OUTPUT FLOOR

One provision sits above everything in this chapter and can nullify all of it.

The **output floor** caps the benefit of internal models across *all* risk types together: a bank's total risk-weighted assets (RWA) may not fall below 72.5% of what the standardized approaches would produce, once the floor is fully phased in, by 2028 under the Basel timetable, by 2030 in the European Union and the United Kingdom, and in China from 2024, where it replaced the earlier parallel-run floor. It is not a market risk rule, which is exactly why it matters here: the trading book's modeled capital is one input to a constraint that binds at the level of the whole bank.

Example 39.7 The same desk failure, three banks, three costs

Each bank has standardized RWA of 100 for trading and 900 for everything else, so its floor is $0.725 \times 1{,}000 = 725$. Its trading book is a single desk, whose RWA are 60 on models and 100 on the standardized approach.

	Modeled RWA	Floor	Held
Bank A — modeled non-trading RWA 700			
Desk keeps approval	760	725	760
Desk loses approval	800	725	800
Cost of failing			**40**
Bank B — modeled non-trading RWA 600			
Desk keeps approval	660	725	**725**
Desk loses approval	700	725	**725**
Cost of failing			**0**
Bank C — modeled non-trading RWA 640			
Desk keeps approval	700	725	**725**
Desk loses approval	740	725	740
Cost of failing			**15**

The identical desk, failing the identical test, costs one bank 40, another 15 and the third nothing. For Bank B the floor already binds whatever the desk does, so the trading book's modeled number is slack and improving it changes no requirement at all. Bank C sits between: the floor absorbs 25 of the desk's 40, and only the part that lifts the bank above the floor is a cost.

This inverts the incentive the rest of the chapter carefully constructs. Sections 39.6 and 39.8 argued that the attribution test disciplines banks because failing it is expensive. For a floored bank it is free, and so is every other refinement: a better model, a narrower NMRF population, a tighter liquidity classification. The tests still run, and they no longer change the requirement.

This explains a pattern that otherwise looks irrational. Section 39.13 notes that banks have chosen standardized capital over model approval on some desks. For a bank near the floor that is simply correct: model approval carries real operational cost and, below the floor, delivers no capital benefit whatever. The framework's preferred approach is unattractive not because the modeling is hard but because a different rule has already decided the answer.

Whether this is a defect depends on what the floor is for. It exists because supervisors do not believe model outputs are comparable across banks: the same distrust that produced every test in this chapter, applied at the top rather than desk by desk. Having both is redundant by

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construction, and the redundancy is deliberate: the tests police the model, and the floor limits the damage if the tests fail.

39.10.1 Where the modeled credit number comes from

Example 39.7 took each bank's modeled non-trading RWA as given. Most of it is credit risk in the banking book, and for a bank approved to use the **internal ratings-based approach** (IRB) it comes from a formula that is Section 24.14's worst-case default rate with the supervisor's parameters in it. The bank supplies each exposure's probability of default (PD), its loss given default (LGD) and its effective maturity M; under the foundation version it supplies only the PD, and the LGD of a senior unsecured claim on a nonfinancial corporate is set at 40%. The capital per dollar of exposure is

$$K = \text{LGD}\left[N\left(\frac{N^{-1}(\text{PD}) + \sqrt{R}\,N^{-1}(0.999)}{\sqrt{1-R}}\right) - \text{PD}\right]\frac{1 + (M-2.5)\,b}{1-1.5\,b} \tag{39.1}$$

and the risk weight is $12.5K$, the reciprocal of the 8% minimum capital ratio.

Three choices in (39.1) are the supervisor's rather than the bank's. The confidence is 99.9% over one year. The correlation is prescribed rather than estimated, falling from 0.24 for the safest corporate borrowers to 0.12 for the riskiest,

$$R = 0.12\,w + 0.24\,(1-w), \qquad w = \frac{1-e^{-50\,\text{PD}}}{1-e^{-50}}$$

on the view that a strong firm defaults mostly when the economy does and a weak one mostly for reasons of its own. And the last factor is a **maturity adjustment**, with $b = (0.11852 - 0.05478\ln\text{PD})^2$, which equals one at a one-year maturity and charges a longer loan for the downgrade risk that Example 24.10 measured. The PD itself is floored at 0.05%.

The formula can be applied one loan at a time because, in a one-factor model of an infinitely granular portfolio, the portfolio's quantile is the sum of the loans' own worst-case losses. That property is what lets a supervisor write a capital rule at all, and it is also the formula's blind spot: a book concentrated on a few large borrowers or exposed to a second systematic factor, a region or an industry, holds more risk than the sum says. Neither is charged in the formula; both are left to the supervisor's own review of each bank, outside the minimum requirement.

Example 39.8 Where the floor bites on a loan book

Take a corporate loan at each rating, with the one-year default probability from the last column of Table 24.2, an LGD of 40% and a maturity of 2.5 years, and compare its IRB risk weight with the standardized risk weight for a rated corporate in Table 27.1.

Rating	PD	R	WCDR	Maturity adj.	IRB RW	Std. RW	Ratio
A	0.05%	0.237	2.04%	1.752	17.5%	50%	0.35
BBB	0.28%	0.224	6.89%	1.411	46.7%	75%	0.62
BB	1.52%	0.176	16.95%	1.222	94.2%	100%	0.94
B	6.45%	0.125	32.42%	1.121	145.6%	150%	0.97
CCC	30.60%	0.120	72.59%	1.053	221.1%	150%	1.47

The BBB row runs as follows. The prescribed correlation is 0.2243, so the worst-case default rate is $N\big((-2.7703 + \sqrt{0.2243}\times 3.0902)/\sqrt{0.7757}\big) = N(-1.4837) = 6.89\%$. With $b = 0.1941$ the maturity adjustment is $1/(1-1.5\times 0.1941) = 1.411$, so $K = 0.40\times(0.0689-0.0028)\times 1.411 = 3.73\%$ and the risk weight is $12.5\times 3.73\% = 46.7\%$.

The ratio column is the one that matters for the floor. An investment-grade loan is modeled at well under 72.5% of its standardized weight (35% of it at A, 62% at BBB), so a bank whose book is mostly investment-grade lending, the typical book of a large bank lending to large companies, sits

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on the floor whatever its models say. Speculative-grade loans are modeled at close to or above their standardized weight, well above 72.5% of it, and leave the floor slack. Bank B of Example 39.7 is therefore not a curiosity. It is the bank with the highest-quality loan book, and for it the trading desk's model approval is worth nothing because its lending has already put it on the floor.

The comparison depends on the standardized side too. The United States does not use credit ratings in its current capital rules, and its standardized approach puts most corporate exposures at 100%, against which even the BBB loan is modeled at under half. The European Union went the other way and, until 2032, lets a bank's floor calculation weight an unrated corporate exposure at 65% where its own PD estimate is no more than 0.5%, precisely to soften the floor on investment-grade lending.

Figure 39.5 draws the whole curve. The IRB curve runs below 72.5% of the standardized weight everywhere in investment grade on the Basel table, so that is where a loan book pushes a bank onto the floor.

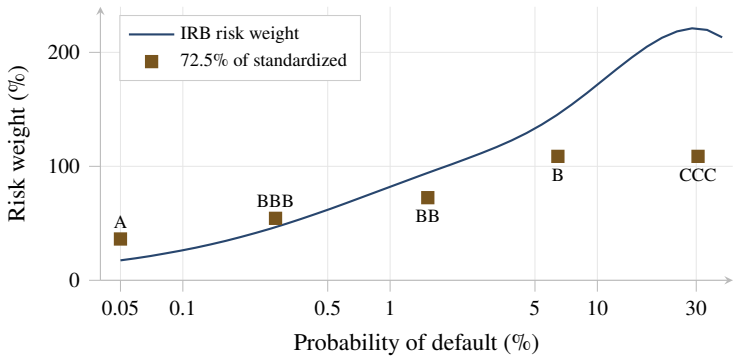


Figure 39.5: IRB risk weight for a corporate loan (LGD 40%, maturity 2.5 years) against 72.5% of the standardized weight at each rating's default probability.

39.11 PRUDENT VALUATION

There is a second capital channel that has nothing to do with the measures above, and for an exotic book it can matter more. In the European Union and the United Kingdom it is a rule; elsewhere supervisors expect valuation reserves of the same kind without prescribing them.

Fair value is an exit price, typically marked at mid within the bid–ask spread or at a model price. Prudent value is a price the position could actually be realized at with 90% confidence, and the difference, the **additional valuation adjustment** (AVA), is deducted from common equity directly rather than risk-weighted.

Example 39.9 The deduction on a $100 million book

An exotic book with a fair value of $100 million carries the following adjustments. Three of them, market price uncertainty, close-out cost and model risk, overlap in what they measure, so the standard (core) method of the European rules, as against the simplified method for small books, adds them and then counts half of the sum.

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Component	Basis points	Amount
Market price uncertainty	40.0	$400,000
Close-out cost (bid–ask)	35.0	$350,000
Model risk	60.0	$600,000
Sum, counted at 50%	67.5	$675,000
Concentrated position	25.0	$250,000
Unearned credit spread	10.0	$100,000
Investing and funding costs	8.0	$80,000
Future administrative costs	6.0	$60,000
Early termination	5.0	$50,000
Operational risk	4.0	$40,000
Total	**125.5**	**$1,255,000**

Model risk is the largest single component in this book, and it is the one a desk cannot reduce by trading better: a tighter market narrows the close-out cost and does nothing to the model uncertainty of a payoff nothing quotes. In a book of liquid instruments market price uncertainty usually dominates instead.

The capital arithmetic makes the size clear. A $1.255m deduction from common equity, at an assumed total requirement of 12% of RWA, removes the capital that would support $10.5m of risk-weighted assets, equity the bank can no longer use for anything else, whatever the position's own risk weight.

This is where Section 21.11's model spread becomes a number on a balance sheet. Two models agreeing on every vanilla and disagreeing on an exotic produce a valuation uncertainty; that uncertainty is not a modeling curiosity but the input to a deduction, and it is why dealers maintain model reserves against exotic books rather than marking to a single model.

The measurement that makes this possible is Hull and Suo's (2002), and the regulatory framing arrived afterward. Their procedure (calibrate competing models to a common surface, verify each reproduces it, then price what the surface does not determine) turns model risk from a caveat into a quantity, and a quantity is the only thing a capital rule can act on. The prudent valuation regime is that idea written as a requirement: it does not ask whether a model is right, which no rule could adjudicate, but how far defensible alternatives disagree.

39.12 STRESS TESTING AND WHAT SITS OUTSIDE THE FORMULA

Everything above is a formula, and supervisors do not rely on formulas alone.

Supervisory stress tests apply a common scenario across banks and compare. Their value is the comparison rather than the scenario: a bank whose losses are unusual under a shared shock is showing a position, and the exercise finds it without anyone specifying what to look for. In the United States the exercise also sets capital: the largest banks' trading books take an instantaneous global market shock, and the losses feed the **stress capital buffer**, a requirement on top of the minimum that rises with the bank's projected stress losses. For those banks a supervisory scenario, not the formula above, often decides how much the trading book costs.

Reverse stress testing inverts the question—not "what do we lose in this scenario" but "what scenario makes us fail"—and it is the only exercise in the framework that begins from the outcome. It is also the one most likely to find Section 38.19's blind spots, because it is not constrained to the risk factors the model contains. Section 38.17 works one through for a two-asset book, and finds that the most plausible route to a given loss runs through the position carrying the largest share of the VaR.

And the supervisory review sets capital above the formula where the formula is known to be inadequate; concentration, which Section 38.19 said no distribution-based measure sees, is

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the standard case.

The pattern across all three is the same. The formula handles what can be specified in advance; the discretionary machinery handles the rest, and the rest is where Chapter 40's losses live.

39.13 WHAT THE FRAMEWORK CHANGED, AND WHO BEARS THE COST

Three things changed, and only the first was the stated objective.

Capital went up for trading, and unevenly. The increase falls hardest on exotic and illiquid books (through the residual risk add-on, the NMRF treatment, and the long liquidity horizons) and lightly on high-volume trading in liquid instruments.

The internal models approach became less attractive than it was designed to be. Between the operational cost of the attribution test, the NMRF data requirements, and the cliff risk, many banks concluded that standardized capital on some desks was cheaper than model approval, which is the opposite of the intended direction, and Section 39.10 showed why the floor makes that conclusion correct for some of them. A framework that makes its own preferred approach unattractive has a design problem.

And the business consolidated. Fixed costs favor scale: the data infrastructure for modellability, the daily attribution machinery, and the desk-level governance are nearly independent of volume. Chapter 35's structured issuance and Chapter 27's client clearing both concentrated for the same reason, and this is a third instance of the same mechanism.

The cost lands in four places.

Dealers bear it directly and pass what they can into spreads, which is why the pricing of Chapter 35's structured issuance and Chapter 34's exotics carries a capital component that the models in those chapters do not contain.

Clients bear it as a wider bid–ask on precisely the trades the framework taxes (bespoke, illiquid, long-dated), and the effect is to push demand toward standardized products, which is intended.

Smaller dealers bear it as a fixed cost and exit, so the framework's second-order effect is concentration, and Section 27.12's concern about concentration is partly a consequence of the rules in this chapter.

And supervisors bear the residual: a framework that cannot verify models must set incentives instead, so the quality of the outcome depends on the calibration of penalties rather than on the correctness of any measurement.

39.14 MODEL RISK

Everything above assumes the model maps market factors to portfolio value correctly, and that assumption is itself an exposure.

Model risk is not a subset of market risk, which is why it needs its own treatment. A market risk measure is conditional on the model: the model that marks the position is also the one that measures its risk, so an error in it misstates both at once, and nothing inside the system disagrees with it. The measure reports no error because the only thing that could reveal one is the model itself.

Section 21.11 made this concrete. Two models calibrated to identical vanilla prices give different values for the same exotic, and the difference is not measurement error—it is the part of the payoff the calibration instruments do not span. No amount of data resolves it, because the disagreement is about what happens where nothing trades.

The governance response is well established and modest in its claims. Models are inventoried; each has an owner and a stated purpose; validation is performed independently

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of development; limitations are documented and the documented limitation is the basis for a reserve or a capital add-on. In the United States the supervisory guidance is SR 26-2 (2026), which replaced the 2011 guidance known as SR 11-7 and scales the effort to the risk a model creates; in the United Kingdom it is the Prudential Regulation Authority's SS1/23, in force since May 2024. Nothing in either list verifies a model, and neither claims to—they make the assumptions explicit and assign someone to own them.

39.15 VALIDATING WHAT CANNOT BE VERIFIED †

A model is *verified* when its output can be checked against something independent: an analytic formula, a converged numerical method, a market price. Validation is the weaker activity of testing a model that cannot be checked that way, and two kinds of model force the distinction.

The first is the calibrated model of Section 21.11. Local and stochastic volatility models fitted to the same vanilla surface agree on every vanilla and disagree on a barrier, and nothing that trades can say which is right. Validation can only measure the disagreement, which is what Hull and Suo's procedure and Section 39.11's model risk adjustment do, and cannot remove it.

The second is the learned model. A **learned pricer** is a neural network trained to reproduce a pricing model's outputs, so that a slow calculation can be evaluated in microseconds; a **learned hedging policy** is a network trained on simulated price paths to choose positions that minimize a risk measure such as Chapter 38's expected shortfall. The online note [TN39] treats both, and three of its findings bear on validation.

The training set is part of the model. A learned pricer degrades silently outside the range of inputs it was trained on, returning a plausible number rather than an error ([TN39], Section 3). A model inventory that records the network and omits the training range has recorded the less important half, and retraining on new data is a model change requiring revalidation, not a refresh.

Reproducibility is not the same as determinism. In Section 12 of [TN39], three training runs of one hedging policy differed only in the random number seed, agreed on the expected shortfall they achieved to within 0.8%, and held positions that differed by up to 1.7 percentage points of delta. A validation standard that asks whether results reproduce will pass all three, although the positions they would actually hold differ.

And validation reduces to out-of-sample testing in the region someone happened to sample. A Black–Scholes implementation can be checked against an analytic formula; a learned pricer can only be checked against the pricer it learned from, and the case study in Section 4 of [TN39] is the case where that pricer was wrong and the network smoothed the evidence.

The existing framework is adequate in structure and strained in practice. Inventory, ownership, independent validation, and documented limitations are the right categories, and SR 26-2 itself leaves generative AI outside its scope. What is missing is a validation technique with the power to distinguish a good model of either kind from a plausible one, and until there is one, the sensible response is the conservative one: use such models where a classical answer exists to check them against.

SUMMARY

A capital rule is a contract about a number the bank itself produces, so the framework both specifies the model and tests it. Each complication answers a specific failure: the prescriptive boundary removes an arbitrage, stress calibration answers procyclicality, and liquidity horizons make the horizon a property of the position. A desk whose horizons are spread across the schedule pays materially more than under a flat ten-day measure with no change in any position.

The standardized charge refuses to trust a correlation. It takes the worst of the prescribed correlations, a higher set and a lower set, and which scenario binds is a property of the book: a

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directional position is worst under high correlation and a butterfly under low, by a far wider margin. Curvature follows Chapter 18's economics: long optionality attracts no charge, and the same option sold attracts all of it. Non-modellable factors are capitalized by a stress scenario added without diversification, so a book with a sizable non-modellable share pays markedly more capital for identical risk.

The P&L attribution test is more permissive than its reputation. A desk may leave a sizable share of its P&L variance unexplained and stay green, because both statistics leave the green zone only once a missing factor carries roughly a third of the variance, so the test catches a missing factor but not a model that is moderately wrong. Desk-level backtesting checks the model's calibration and attribution its content, and a model has to be wrong in a narrow way to pass both. Approval is withdrawn desk by desk against a 250-day sample, so capital moves as a cliff, softened only by the amber surcharge.

The output floor caps risk-weighted assets at 72.5% of the standardized result when fully phased in, so for a bank where the floor already binds, failing the attribution test costs nothing, which is why declining model approval can be correct. Most of the modeled number the floor compares is banking-book credit, set by the IRB formula, which is the worst-case default rate at 99.9% with a prescribed correlation and a maturity adjustment; it models investment-grade lending at well under 72.5% of the standardized weight, so the floor binds hardest on banks with the best loan books. Prudent valuation deducts from equity directly, and for an exotic book model risk can be its largest component. Supervisory stress tests and review set capital where the formula cannot, and in the United States a stress scenario often decides the trading book's cost.

Model risk is not a subset of market risk: the model that marks a position also measures its risk, so an error in it misstates both at once, and nothing inside the system can reveal it. Governance makes assumptions explicit without verifying anything, and models that cannot be verified (calibrated models that disagree where nothing trades, and learned models) strain it most, so the conservative use is the defensible one.

FURTHER READING

The Basel Committee's *An Explanatory Note on the Basel II IRB Risk Weight Functions* (2005) derives the formula of Section 39.10 and explains each of its parameter choices, and Gordy (2003) proves the portfolio invariance that lets it be applied one loan at a time. The Basel Committee's *Minimum Capital Requirements for Market Risk* (2019) is the primary source for everything from Section 39.2 to Section 39.9; the sections motivating the boundary and the liquidity horizons are the ones to read first, and the industry quantitative impact studies are where the capital effects of Section 39.13 are actually measured. The output floor is in the Committee's *Basel III: Finalising Post-Crisis Reforms* (2017), which is the document to read for how the floor was meant to interact with internal models. On prudent valuation, the European Banking Authority's regulatory technical standards, adopted as Commission Delegated Regulation (EU) 2016/101, set out the nine adjustments and their aggregation. Jurisdictional implementation differs in ways that matter and in ways that do not; the divergences worth tracking are the treatment of internal models and the boundary, and both are documented in supervisory consultation papers rather than in the standard. On model risk, the US interagency guidance SR 26-2 (2026) and the Prudential Regulation Authority's SS1/23 (2023) are the current frameworks, and the earlier SR 11-7 (2011) is worth reading for the vocabulary both inherited and because it makes no claim to verify anything. Hull and Suo (2002) is the paper behind Section 39.11's measurement of model risk. On the questions raised in Section 39.15, SR 26-2's footnote on generative and agentic AI models is where US supervisors currently draw the line.

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PRACTICE QUESTIONS

39.1 Explain why the trading-book boundary was made prescriptive, compute the capital at stake in Section 39.2's bond portfolio, and explain why removing the capital benefit of reassignment matters more than tightening the definition.

39.2 Explain why the standardized approach supplies correlations rather than estimating them, and relate this to Section 38.11.

39.3 Using the weighted sensitivities and correlations of Example 39.1, compute the directional book's bucket charge under the high-correlation scenario, and explain why it equals the sum of the weighted sensitivities.

39.4 Using Example 39.1, explain why the directional book is worst under high correlation and the butterfly under low, why the butterfly's charge falls to zero under high correlation, and why the hedged book nonetheless pays the larger proportional penalty.

39.5 Explain why a long option attracts no curvature charge while the same option sold attracts the whole of it, using Figure 39.2.

39.6 Compute the curvature charge on a short \$10 million at-the-money six-month put on the stock of Section 39.3 at 25% volatility and zero interest rates, and explain why it equals the short call's.

39.7 Recompute Example 39.2 for a small-cap stock, whose volatility has a 60-day liquidity horizon, keeping the curvature shock at 55% so that only the vega term changes, and state the total charge as a share of notional and the curvature term's share of it.

39.8 A desk holds \$50 million notional of variance swaps and \$200 million of products carrying gap risk. Compute its residual risk add-on, and explain why the framework charges a flat percentage rather than modeling these payoffs.

39.9 In Example 39.3, a factor class is reclassified from a 40-day to a 60-day horizon, raising ES_4 from 45 to 60. Compute the new aggregate, and explain why the desk's capital changed although nothing it holds did.

39.10 State the risk factor eligibility test, and redo Example 39.4 with a correlation of 0.50 between the two parts, separating the cost of lost diversification from the cost of the stress calibration.

39.11 Using the closed form of Example 39.5, compute the share of unexplained variance at which the rank correlation equals 0.75, and state the zone the desk is in.

39.12 Explain why the rank correlation and the Kolmogorov–Smirnov statistic leave the green zone at nearly the same share of unexplained variance, and construct a model error that only one of them would detect.

39.13 Explain what kind of model failure the attribution test catches and what kind it does not, and relate the second to a learned pricing or hedging model.

39.14 A desk records 9 exceptions at 99% and 34 at 97.5% over 250 days. State the consequence, and explain why VaR is backtested when capital is held against expected shortfall.

39.15 Explain what backtesting tests that attribution does not, and construct a desk that passes each while failing the other.

39.16 Explain why desk-level approval is a cliff rather than a slope, what the amber surcharge changes, and what behavior the cliff induces.

39.17 Using Example 39.6, recompute the modeled charge when yesterday's internally modeled charge is 650, state which term binds, and explain why the averaged term usually does.

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39.18 State how the bank-wide backtesting exception count enters the capital formula, and compute the increase from a green to a red multiplier.

39.19 Using Example 39.6, state the share of the total requirement contributed by desks without model approval, and explain why this makes the attribution test the focus of capital management.

39.20 Using Example 39.7, find the modeled non-trading RWA at which failing the desk's test costs exactly 20, and explain what the floor does to the incentives of Section 39.6.

39.21 Explain why a bank near the output floor may rationally decline model approval, and why this is not evidence that its modeling is inadequate.

39.22 Using Example 39.9, compute the deduction if the three overlapping adjustments were added in full rather than at 50%, explain why model risk cannot be reduced by trading better, and compute the business the actual deduction displaces at a 12% capital requirement.

39.23 Explain how a supervisory stress test becomes trading-book capital in the United States, and why reverse stress testing is the exercise most likely to find what the formula misses.

39.24 Explain why model risk is not a subset of market risk, and why the error and the measure move together.

39.25 Using Section 39.15, explain why reproducibility is an inadequate validation standard for a learned hedging policy, and what should be required instead.

39.26 A senior unsecured loan to a nonfinancial corporate has a one-year PD of 1% and a maturity of 2.5 years, and the bank uses the foundation IRB approach. Compute the prescribed correlation, the worst-case default rate, the maturity adjustment, the capital requirement and the risk weight, and state whether the output floor binds on a book of such loans if the standardized weight is 100%.

39.27 Using Example 39.8, explain why the output floor binds on a bank that lends mainly to investment-grade companies and not on one that lends mainly to speculative-grade companies, and what this implies for the value of trading-book model approval at each.

FURTHER QUESTIONS

39.28 Example 39.5 assumes the unexplained component is independent noise. Redo the analysis with an unexplained component that is systematically related to the explained one (a missing factor correlated with the ones present) and determine whether the test becomes more or less powerful.

39.29 Example 39.5 shows the two statistics reacting to a missing factor at nearly the same point. Construct a desk whose model error makes the Kolmogorov–Smirnov statistic bind well before the rank correlation, and state whether that error is more or less serious than the one the correlation catches.

39.30 Section 39.8 argues the cliff induces banks to include factors that do not matter. Model the bank's optimization explicitly (capital cost against the probability of failing the test) and determine the conditions under which adding an immaterial factor is rational.

39.31 Example 39.3 treats liquidity horizons as a fixed schedule. Estimate empirical close-out horizons for three instrument classes, compare them with the regulatory assignment, and determine the capital effect of the difference.

39.32 Section 39.13 argues that fixed costs concentrate the market. Build a model of dealer entry and exit under a fixed compliance cost, calibrate it to the observed consolidation, and determine what level of fixed cost the observed market structure implies.

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39.33 Section 39.15 argues that no current validation technique separates a good learned model from a plausible one. Propose one, state what it would need to observe, and determine whether the required data exists for a pricing surrogate and for a hedging policy respectively.

39.34 The framework taxes illiquidity because it cannot model it. Design a capital treatment that charges for liquidity directly, state what it would require banks to report, and determine whether the reporting burden is smaller or larger than the current non-modellable-factor machinery.

39.35 Section 39.10 shows a floor-bound bank indifferent to its desk tests. Model a bank's choice, desk by desk, between internal models and the standardized approach as a function of its distance from the floor, and determine the distance at which the internal models approach stops paying for its fixed cost.

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Chapter 40
Derivatives Mishaps and What We Can Learn

This chapter is not a collection of cautionary tales. Every case in it is a controlled experiment that the profession did not intend to run, and each one tested a specific assumption made somewhere earlier in this book.

One finding organizes the chapter and it is not the expected one. In almost none of these losses did a derivative behave in a way its model did not predict. The instruments performed. What failed was around them—the cash, the exit, the information, the venue, the buyer's ability to price what it bought, or the decision not to hedge at all.

The last of those deserves saying at the start: the most recent case, Section 40.14, is a bank that failed because it did not hedge, and any chapter that presents derivatives as the source of danger has the sign wrong.

Chapters 38 and 39 built the measures and the rules; most of these losses fell in the gaps Section 38.19 lists, and Chapter 39's controls are read here against the cases behind them. Sections 40.1 to 40.14 take the cases one at a time; Section 40.15 draws out what repeats, Section 40.16 what changed, and Section 40.17 what institutions and regulators should take away.

40.1 METALLGESELLSCHAFT: CORRECT IN VALUE, WRONG IN TIME

In 1993 a German industrial group's US subsidiary had sold long-dated fixed-price oil supply contracts and hedged them with a **stack** of near-dated futures and short-dated swaps, rolled forward each month.

The hedge was sound in the only sense a valuation model recognizes. Whether the parent was right to liquidate it rather than fund it is still argued, and Chapter 3's case study names both sides of that debate; that the strategy never arranged the funding is not in dispute.

Example 40.1 The cash flow and the value point in opposite directions

Take the historical position, 160 million barrels of forward commitment, and treat it as hedged one-for-one with front-month futures; the real stack mixed exchange-traded futures with short-dated over-the-counter swaps. The price falls and the roll spread below are illustrative:

Price fall	Futures margin (cash out)	Supply contract (unrealized)
\$2.00/bbl	−\$0.32bn	+\$0.32bn
\$5.00/bbl	−\$0.80bn	+\$0.80bn
\$8.00/bbl	−\$1.28bn	+\$1.28bn

Every row is flat in undiscounted value and catastrophic in cash. The futures loss is paid in variation margin now; the supply contract's offsetting gain arrives over ten years and cannot be pledged. Chapter 3's case study records the historical outcome, margin calls of roughly \$0.9 billion by the end of 1993 and losses of about \$1.3 billion.

And the roll was a second exposure no one had chosen. A stack-and-roll pays or earns the calendar spread every month on the whole remaining stack, which converts a flat-price hedge into a calendar spread position; Section 36.10 works that arithmetic. On this position, with the stack averaging about 153 million barrels over the first year, a contango of $0.35 a barrel per roll costs about $0.64 billion in that year while the flat-price hedge was working perfectly.

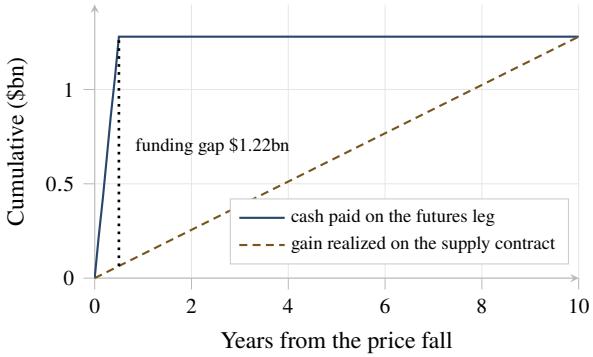


Figure 40.1: An $8/bbl fall spread over six months, on Example 40.1's position; illustrative.

Figure 40.1 is the mechanism of this chapter in one picture. Both curves reach $1.28bn, so the hedge is exact in undiscounted terms—and the vertical distance between them is a loan the strategy never arranged. The gap peaks at $1.22 billion at six months, once the margin has all been paid, and it closes only at the ten-year horizon.

Every case in Section 40.15's first category is that gap. The liability-driven investment (LDI) schemes, the margin calls at American International Group (AIG), and Example 37.2's basis trade all draw the same two curves with different labels. A valuation model computes where the curves end; the failure happens in between.

The lesson is Section 8.9's: margin calls are a liquidity risk, and Section 8.10's stress test measures them because it counts gross cash outflow, not net loss. On Example 40.1's position it reports $0.16 billion of variation margin for every dollar the oil price falls, where a value measure reports nothing. The funding line should be committed at that scale before the trade, and the roll budgeted as a calendar spread.

40.2 PORTFOLIO INSURANCE, 1987: WHEN THE HEDGE IS THE FLOW

This is the clearest case in the chapter of a hedging technique moving the market it was hedging against (the LDI selling of Section 40.8 is another), and it therefore deserves treatment as a mechanism rather than as an antique. Section 16.2 introduced the technique; this is the episode and its arithmetic.

Portfolio insurance replicated a protective put synthetically: rather than buy a put, sell index futures as the market falls and buy them back as it rises, in the proportion Chapter 18's delta prescribes (Sections 18.11 and 18.12). The replication is sound in theory, and the programs followed it as closely as the market allowed (on October 19, 1987 they fell behind schedule because the futures market could not absorb the selling), which makes it instructive.

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Example 40.2 The selling schedule was published by the mathematics

Take \$70 billion insured with a floor 5% below spot over three months at 20% volatility, with the interest rate set to zero for simplicity. The programs must be short $-\Delta$ of the index on a portfolio whose own value falls with the index, so the futures short is $|\Delta| \times \$70\text{bn} \times S/100$ at index level S:

Index	Put delta	Futures short	Incremental selling
100	−0.287	\$20.07bn	—
95	−0.480	\$31.92bn	+\$11.85bn
90	−0.688	\$43.35bn	+\$11.43bn
80	−0.952	\$53.33bn	+ \$9.98bn

A 10% decline obliges the programs to sell \$23.3 billion, and a 20% decline \$33.3 billion, with no discretion at any point.

The selling per point of the index is driven by gamma, so it is fastest near the floor and slows once the delta approaches −1; the last row covers ten points and the two above it five each. Near the floor the delta changes by about 0.042 per index point, which, net of the portfolio's own shrinking value, is about \$2.45 billion of futures per point. Figure 40.2 draws the whole schedule.

And the critical feature is what a synthetic put changes. A purchased put has a *seller*. Left unhedged, that seller's position calls for no trading at all; a dealer who delta-hedges the short put is short gamma and *sells* into the decline as well, as Section 18.12 shows. Listed options therefore spread the selling across writers, not all of whom hedge; they do not abolish it. Replication puts the whole of it in the holders' own programs, on a schedule anyone can compute in advance. The protection was real for each individual holder and, in aggregate, was manufactured out of the very selling that made it necessary.

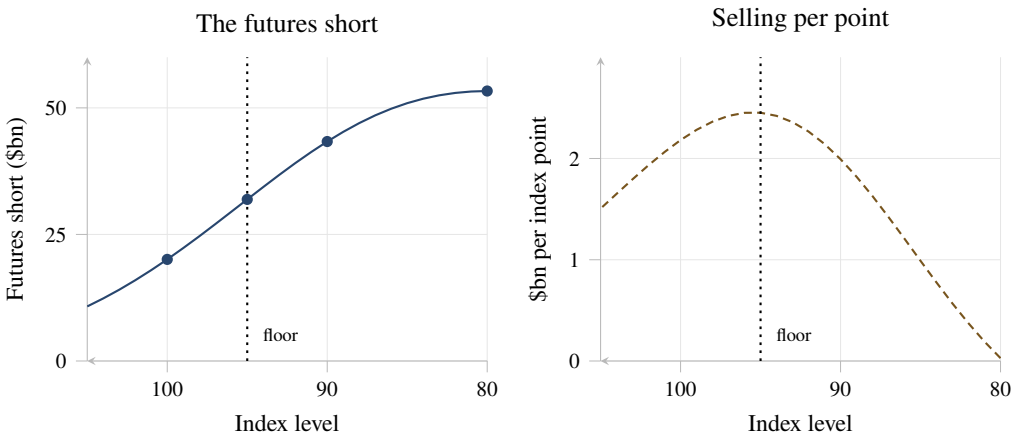


Figure 40.2: The portfolio insurers' selling schedule in Example 40.2, as the index falls from 105 to 80.

The mechanism has appeared three more times in this book. Section 35.6's issuer hedging of autocallables sells into a decline near a barrier; Section 37.3's liquidation cascade forces sales as a published function of price; and Section 23.7's dealer gamma positioning is the same arithmetic run daily on a much shorter horizon. In each, hedging demand is a mechanical function of the price, so the hedge and the move reinforce each other. That is the procyclicality of Section 38.18 with a hedge ratio in place of a risk measure: there a capital requirement that rises after volatility arrives forces positions to be cut into a falling market, and here a delta that grows as the index falls does the same. In both, the feedback needs the rule to be shared. One program selling on its own delta moves nothing; \$70 billion of insured portfolios following the

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same rule at once moved the index, and the move set the next round of selling.

What changed afterward is worth noting because it was not a ban. Circuit breakers were introduced (a mechanism that interrupts the feedback rather than the strategy), and the listed index option market grew. The fix was to make the protection purchasable rather than manufacturable, and that is the correct response to a mechanism rather than to an instrument.

Section 18.12's gamma arithmetic publishes a replicating hedge's flow in advance: the change in the futures short per index point (gamma on the insured notional, less the delta on the portfolio's own fall) peaks in Example 40.2 at about $2.45 billion at the floor (Figure 40.2). A program should set that number, summed over everyone running the same rule, against what the futures market can absorb, and where the market cannot absorb it, buy a listed put instead.

40.3 BARINGS AND SOCIÉTÉ GÉNÉRALE: A FAILURE OF CONTROLS, NOT OF DERIVATIVES

Two cases, thirteen years apart, are failures of the risk control system rather than of the instruments, and neither has any derivatives pricing content. In both, the checks that should have caught an unauthorized position (the **segregation of duties** that keeps trading apart from its settlement, independent reconciliation, and verification of what the trader reported) were in the hands of the person they were meant to check.

In both, positions reported as low-risk arbitrage were outright directional. At Barings one person controlled trading and the settlement of that trading, and performed the reconciliation that would have exposed the position. At Société Générale the trader had come from the middle office and knew its checks well enough to hide the positions behind fictitious offsetting trades.

	Position	Firm equity	Ratio
Barings, 1995	$27bn	$0.6bn	**45**×
Société Générale, 2008	EUR 50bn	EUR 30bn	1.7×

The same failure destroyed one firm and did not destroy the other, and what decided the outcome was the ratio. Barings held about forty-five times its equity in directional futures and option positions and lost £827 million on them; the French bank held a larger notional against a balance sheet that could absorb the EUR 4.9 billion it cost.

The instruments were exchange-traded: index futures in both, and at Barings also Nikkei options and Japanese government bond and Euroyen futures, all cleared, margined daily, and priced as in Chapters 5 and 15. No model failed and no instrument surprised anyone.

Separating trading, settlement, and reconciliation is necessary and will still sometimes fail, so a control failure's severity is set by a number the controls do not watch: the largest possible position against equity. Section 2.3's daily settlement lets a firm cap that number, because every exchange-traded loss is paid in margin the next morning, and capping the margin funding a desk can draw caps the position. At Barings that funding was the visible signal: the head office kept remitting margin to Singapore for a position it believed was hedged.

40.4 LTCM: LEVERAGE PRICES THE PATH, NOT THE VIEW

The 1998 failure of Long-Term Capital Management (LTCM) is usually told as a story about overconfidence in models: a fund whose partners included two Nobel laureates trusted spreads and correlations estimated in calm markets, and the Russian default of August 1998 broke them. That story is not wrong, but the arithmetic tells a narrower and more useful one.

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Example 40.3 A correct view at the wrong leverage

Equity of $4.7bn against a $125bn balance sheet is **26.6**× leverage, before derivatives notionals far larger than either.

Adverse move on assets	Loss	Share of equity
0.5%	$0.63bn	13%
1.0%	$1.25bn	27%
2.0%	$2.50bn	53%
4.0%	$5.00bn	**106%**

The wipeout move is 3.76%.

So the strategy did not require the convergence trades to be wrong. It required them to converge without the balance sheet first losing 3.8% of its value. And convergence trades widen before they converge, which is why they are available. At that leverage the fund was not expressing a view about value; it was expressing a view about the path, that spreads would not widen that far before converging, and it had not priced that path.

Leverage converts the path into a survival condition that a VaR at a horizon does not see, so size the trade by the path: Section 38.17's reverse stress test finds the adverse move the position must survive on the way to converging, 3.76% of the balance sheet in Example 40.3, and leverage should keep that move well beyond the widening such trades have shown. A position others also hold makes the move larger, because when the fund must sell, so must they, which is Section 40.2's shared rule in another form; LTCM's trades had been copied across the dealer community, and in September 1998 they were unwound together.

40.5 AMARANTH: EXIT COST IS A FUNCTION OF YOUR OWN SIZE

Amaranth's natural gas loss of about $6.6 billion in September 2006 shows that concentration, which Section 38.19 says a distribution-based measure cannot see, is not an academic gap.

The fund held positions approaching 40% of open interest in the contracts that mattered. At that share, the exit is not a market price—it is a schedule. Taking daily volume, for illustration, as equal to the open interest of those contracts:

Maximum participation in daily volume	Days to exit
10%	4.0 days
20%	2.0 days
30%	1.3 days

And every one of those days is a day the market knows a forced seller is present. VaR is read from the distribution of price changes, which does not depend on how much of the market the holder owns; only the exit does.

Set the horizon of the measure by the holder's own exit, not by the risk factor's class. At a 10% participation rate the table's exit takes four days, so Section 38.14's square-root-of-time rule scales the one-day VaR to the four-day exit by a factor of two if the position is held whole until sold—and it still assumes prices unmoved by the seller, which at 40% of open interest they are not. A risk manager should therefore limit the share of open interest directly and stress the exit with an impact cost, since Section 39.4's liquidity horizons, assigned by class, will not.

40.6 ARCHEGOS: THE SAME RISK AT MANY BROKERS

The 2021 losses fell on the dealers as well as the fund: the fund's capital was wiped out, and the dealers lost more than $10 billion. It was a failure of measurement, not of modeling.

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Example 40.4 Adequate margin against the wrong exposure

Take a family office holding concentrated single-stock exposure through total return swaps at six prime brokers. Each broker saw its own slice and margined it at a level it judged adequate for that slice. Section 27.14 works the per-broker arithmetic, which leaves $7.5 billion of margin across the system against a $50 billion gross position.

But the liquidation was not six independent liquidations. All six sold the same stocks at once, and with the client's capital gone only the margin stood against the loss:

Impact of simultaneous exit	Loss	Against $7.5bn margin
10%	$5.0bn	covered
25%	$12.5bn	**short by $5.0bn**
40%	$20.0bn	**short by $12.5bn**

The margin was set for the exposure each broker measured and was wrong for the exposure that existed. No broker's own book showed the difference, and Section 27.14 adds that the margin levels themselves were set by competition for the client's business rather than by the concentration of the positions.

The instrument choice was not incidental. A total return swap gives economic exposure without an ownership disclosure, so the aggregate was invisible by construction: the same property that made the product attractive made the risk unmeasurable. Section 27.13 argued that bilateral measurement has a limit, and this is the limit.

Margin a total return swap (Section 25.9) on the client's whole position and its exit, not on one broker's slice: Section 27.15's initial margin is meant to cover the close-out loss, and in Example 40.4 a joint exit at 25% impact loses $12.5 billion against $7.5 billion collected. A prime broker should ask for the positions held elsewhere, treat a refusal to disclose them as a risk in itself, and set the rate by Section 40.5's exit arithmetic, not by competition.

40.7 LME NICKEL: WHEN THE VENUE IS THE RISK

In March 2022 the nickel price, which had closed near $29,000 a metric ton the previous Friday, more than tripled in two trading days to above $100,000, against a very large short position held by a producer as a hedge. Take an illustrative 100,000-metric-ton short; the producer's actual position was larger, and much of it was held over the counter, out of the exchange's view.

Price	Loss on a 100,000-metric-ton short from $29,000
$40,000	$1.10bn
$50,000	$2.10bn
$80,000	$5.10bn
$100,000	$7.10bn

The exchange then canceled several hours of trades.

That cancellation is the case's real content, and it tests the assumption Section 2.10 flagged as a judgment about institutional resilience rather than a fact. Every calculation in Parts I and V treats the clearing house as the counterparty that does not fail, which is why central clearing improves on bilateral trading at all. Here the CCP faced the prospect of member defaults it judged could exceed the resources standing behind it, and its resolution was to void executed trades. No member defaulted, and the default waterfall was never drawn on.

So the risk that materialized was not the price. A trader who was long nickel, correct, and fully margined had the profit taken away, which the reader met in a different market as Section 37.10's auto-deleveraging, and which Chapter 2's account of this episode records as trade finality yielding to the clearing house's survival.

A CCP's guarantee is bounded by its resources and its governance, and when they are

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threatened it can become a negotiation; Section 2.8's waterfall has a bottom, and this case showed that a clearing house may act before it gets there. A member or end user should measure each layer of that waterfall against a move like March 2022's and read the venue's rules on canceling trades with it, which is Section 2.10's question of who absorbs the loss; the venue should know the full size of a large position, over-the-counter holdings included.

40.8 UK LDI: THE HEDGE THAT CONSUMED ITS OWN COLLATERAL

In September 2022 a rapid rise in gilt yields forced UK pension schemes into a self-reinforcing sale of the assets backing their hedges.

What makes the case valuable is that the schemes' hedges did their job.

Example 40.5 Better off and unable to pay

Take a system with roughly £1.5tn of liabilities at a duration near twenty, hedged with leveraged gilt and swap positions requiring variation margin. The schemes' reported hedge ratios were higher than the 40% used here, but part of each hedge was gilts held outright, which call for no collateral. Take the margined part as covering 40% of the liabilities' rate sensitivity, so each margin call is 40% of the fall in the liabilities' value:

Yield rise	Liability falls by	Margin call
50 bp	10%	£60bn
100 bp	20%	£120bn
150 bp	30%	£180bn
250 bp	50%	£300bn

The funding position improved at every row. Because the hedge covered only part of the liabilities' rate sensitivity, the liabilities fell faster than the assets, so the schemes became *better funded* as the crisis progressed—while unable to meet the margin calls.

Selling gilts to raise the collateral pushed yields higher, which raised the next margin call. The hedge's own success generated the demand that forced its unwinding.

This is Section 8.9 in its purest form—solvency and liquidity are different quantities, and only one of them forces a sale—and the lesson is Section 40.1's. The margined hedge in Example 40.5 is a DV01 of about £1.2 billion per basis point, so Example 8.10's arithmetic makes a 100 basis point move a £120 billion margin call, and Section 38.17's reverse stress test, run on the buffer a scheme holds, finds the move that exhausts it—the number to report beside the funding ratio. Plan for a larger move than any one scheme's stress, because forced gilt sales raise yields again: Example 8.11's loop, which only a supervisor can sum.

40.9 AIG: A COLLATERAL TRIGGER ON A POSITION THAT NEVER DEFAULTED

By the end of 2007 AIG's financial products unit had sold protection on about $527 billion of super-senior tranches, on a view that they attached above any plausible loss. Most of it was written for European banks seeking regulatory capital relief; the margin calls came mainly from about $78 billion written on multi-sector CDOs backed largely by subprime mortgages.

The view was defensible on realized defaults for much of the book, and it was irrelevant, because the contracts contained two features the view did not price: mark-to-market collateral against spread widening, and additional collateral on a downgrade of the seller.

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Spread widening	Collateral demanded on $78bn (duration held at 5)
100 bp	$3.9bn
200 bp	$7.8bn
400 bp	$15.6bn
800 bp	$31.2bn

No row requires a default. The position was a view on realized losses; the contract was a claim on marks, and Section 25.4's mark moves with the spread, default or not.

The downgrade trigger has no analog in a hedge. It converts a deterioration in the *seller's own* condition into an immediate cash demand, so the contract becomes most expensive exactly when the seller is least able to pay. That is Section 27.9's wrong-way risk written into the contract rather than arising from a correlation: the construction of Section 37.5's inverse contract, where collateral and exposure were the same asset.

Price the collateral terms, not only the view on defaults. Section 25.4's mark makes the collateral a duration calculation—$78 billion at a duration of 5 posts $39 million for every basis point of widening, with no default—so the seller's liquidity should be sized by Section 8.10's stress test applied to spreads, not by expected loss. The downgrade trigger belongs in the same test, assumed to arrive with the widening that produces it.

40.10 THE LONDON WHALE: THE MEASURE CHANGED AND THE RISK DID NOT

In 2012 a credit index book inside a bank's treasury function lost about $6.2 billion. The instructive part happened months earlier and cost nothing at the time.

Reported VaR under the previous model	$132m
Reported VaR under the new model	$66m
Reduction	**50%**
Positions changed on the day	**none**

A model change, hastily approved and containing a spreadsheet error, halved the desk's reported risk without anyone trading. Over the following months the book's notional roughly tripled under the looser limit. The eventual loss was 47 times the old figure and 94 times the new one—a one-day figure set against a cumulative loss, but the scale is the point.

This is Chapter 39's model risk in its purest administrative form. Section 39.14 argued that an error in the model misstates the mark and the measure together; here the measure moved alone, and the limit framework built on it loosened accordingly. A limit expressed in units of a model is a limit the model can relax.

It is also the case that most directly justifies Section 39.6's attribution test. The desk's risk model stopped describing its book, and the mechanism that now exists to detect that (comparing the risk model's own P&L against full revaluation, daily) is a direct descendant. Whether it would have caught this one is unclear, given Example 39.5's permissiveness.

Validate a VaR model change as Section 39.14 validates any model, before the new number reaches a limit: run old and new side by side, require the fall from $132 million to $66 million on an unchanged book to be explained, and ask for Section 38.16's backtest on both. A limit the model cannot relax, on gross notional, should sit beside the VaR limit, so that a book tripling in size is visible whatever the measure says.

40.11 COMPLEXITY SOLD TO A BUYER WHO COULD NOT PRICE IT

A fifth mechanism runs through several cases and is different in kind from the four that Section 40.15 draws from the cases so far, because the loss is not caused by cash, size, information, or a venue. It is caused by an asymmetry in who can value the trade.

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Example 40.6 Two leverages multiplied

Orange County, California, ran a $7.5 billion municipal investment pool, levered it to about $20.5 billion through repo (**2.73**×) and invested a large share of it in **inverse floaters**, whose coupon falls as rates rise.

The instrument carries its own leverage. An inverse floater with coupon leverage L has roughly $(1 + L)$ times the duration of its underlying, taken here as 4 years, so the two multiply. Were the whole book in such notes:

Coupon leverage	Instrument duration	Portfolio effective duration
1	8.0 yr	21.9 yr
2	12.0 yr	**32.8 yr**
3	16.0 yr	43.7 yr

The actual book mixed inverse floaters with other structured notes and plain agency paper, so its duration was lower than any row above. Its 1994 loss of about $1.64 billion, on a rise of roughly 300 basis points in short and intermediate yields, corresponds to an asset duration of about 2.67 years, an effective duration of about 7.3 years on the depositors' $7.5 billion:

Rate rise	Loss	Share of the $7.5bn
100 bp	$0.55bn	7.3%
200 bp	$1.10bn	14.6%
300 bp	$1.64bn	**21.9%**

Nothing was concealed. The coupon formula was in the term sheet and the repo was on the balance sheet. The pool reported its yield advantage rather than its duration, warnings during 1994 that further rate rises threatened it were dismissed, and the product was described, accurately in a sense, as an income enhancement. The county filed for bankruptcy in December 1994.

The same asymmetry produced the structured-swap disputes of the mid-1990s, in which corporate treasurers bought leveraged rate positions embedded in what were presented as financing trades, and later argued in court that they had not understood what they held. The legal outcomes varied; the mechanism did not.

And it is not a historical category. Chapter 35's retail structured products carry the same asymmetry in a regulated wrapper, Section 35.10's estimated initial value exists precisely to quantify it, and Section 37.8's liquidity providers are selling a straddle described to them as yield. The mechanism is permanent because the asymmetry is structural: the party that constructs a payoff can always price it better than the party that buys it.

The response to this category is disclosure and suitability, because no risk system helps a buyer who does not know which risk to measure. The measure Orange County needed was Section 4.11's duration taken through the whole structure: about 2.67 years on the assets times repo leverage of 2.73 is Example 40.6's 7.3 years on the depositors' money, and 21.9% lost on a 300 basis point rise. A buyer should not hold a payoff whose duration it cannot compute, and Sections 35.11 and 35.12 set out how a note is evaluated and what disclosure cannot fix.

40.12 EUROPEAN ENERGY 2022: THE SAME QUARTER, A DIFFERENT INDUSTRY

The LDI episode was not the only funding crisis of that quarter, and the second one matters because it suggests the mechanism is structural rather than a feature of pension funds.

A generator that has sold power forward holds a short position that is margined daily. The plant that will deliver the power is not. Example 8.9 works the arithmetic for one producer; at the scale of a large generator, 50 TWh of forward sales and a move from 50 to 400 EUR/MWh demand EUR 17.5 billion of cash against a position whose entire purpose was to remove

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risk. The generator's economics improved throughout (it owns a plant selling into a far higher market), and several such firms required state liquidity support to avoid failing. The support went to the firms that had hedged: those that had not faced no margin calls and took windfall profits as prices rose, which is what a system produces when it prices risk correctly and liquidity not at all.

This case bears on a question the earlier cases leave open. Metallgesellschaft could be dismissed as a strategy error and the LDI schemes as a leverage error. Two simultaneous instances in unrelated markets suggest the exposure is created by the market structure: it arises wherever a hedge is margined and the hedged item is not. Neither 2022 case was a product of mandatory clearing: most of the utilities hedged on exchanges by choice, and the pension schemes, exempt from the clearing mandate, met margin calls on repo and bilaterally margined swaps. But the post-2008 push toward margining, voluntary and mandatory, that Chapters 2 and 8 describe widened the footprint of that shape.

The lesson is Section 40.1's again, at the level of an industry, and Example 8.9 computed it for a single 10 TWh producer. At the scale of the large generator above, a forward sale of 50 TWh posts EUR 50 million of variation margin for every euro the power price rises, so the liquidity a generator commits before hedging should be sized by Section 8.10's stress test on a move like 2022's, not a typical month. A policymaker who pushes hedging into margined form should ask where that liquidity comes from in a stress.

40.13 MARK-TO-MODEL, AND POSITIONS WITH NO MARKET

A further mechanism applies to positions with no market price. Section 39.11's prudent valuation exists because the value carried for them is produced by the holder; this section adds the cases.

The failure mode is not fraud, though fraud is available. It is that a valuation that cannot be checked is a valuation that responds to pressure, and the pressure is asymmetric: no one objects to a mark that produces a profit.

The structural cases are Enron's long-dated energy contracts, booked at their full modeled value on the day of signing, which converts a decade of anticipated margin into current-period earnings, and illiquid structured credit carried at model prices while the market for it disappeared. In both, the position and the price came from the same place.

The safeguards are Chapter 39's. Independent price verification exists because the desk cannot be the source of its own marks. The model risk component of Example 39.9's adjustment, the largest single item, is a deduction for precisely this, and the reserve is taken whether or not anyone suspects a problem.

The connection to machine learning is direct. A learned pricer's silent extrapolation outside the training box ([TN39]) is mark-to-model with the model's confidence removed from view, which is why Section 39.15 insists the training set be recorded as part of the model.

40.14 SILICON VALLEY BANK: THE LOSS FROM NOT HEDGING

This is the most important case in the chapter.

A bank funded by concentrated, uninsured, rate-sensitive deposits held a large securities portfolio at meaningful duration and left the interest rate risk unhedged, having unwound during 2022 most of the pay-fixed swaps it had held against part of the book.

Example 40.7 What a swap would have cost, and what its absence did

Take a $120bn portfolio at duration 5.6:

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Yield rise	Economic loss	Share of a $16bn equity base
100 bp	$6.7bn	42%
200 bp	$13.4bn	84%
300 bp	$20.2bn	**126%**
400 bp	$26.9bn	**168%**

A pay-fixed swap on about half the duration would have cost negative carry (in 2021, paying a fixed rate near 1% against an overnight rate near zero, on the order of 100 basis points a year on the swapped notional) and would have removed about half of that column. The instrument was standard, liquid, cleared, and priced by Chapter 7's arithmetic. Nothing about the hedge was difficult.

Accounting completed the failure. Securities classified as held-to-maturity are not marked through capital, so the reported number stayed intact while the economic number did not. The unrealized loss, about $15 billion at the end of 2022, was visible in the footnotes but not in capital, until a loss-making sale and a capital raise in March 2023 put it in front of depositors at once.

So the sequence is: an unhedged exposure, an accounting treatment that concealed it, and a funding base that could leave in a day. Each is necessary; none alone is sufficient.

A bank failed because it did not hedge its rate risk, and the derivative it did without was the plainest instrument in this book. Section 3.2's case for hedging applies in full, and it is worth teaching against the reflex that derivatives create danger.

Banking-book rate risk sits outside Chapter 39's trading-book rule. The Basel Committee's 2016 standard treats it in Pillar 2, with a supervisory outlier test: a change in the economic value of equity, under prescribed rate shocks, above 15% of Tier 1 capital. The test looks through the reported number to the economics.

Hedge the economic duration with the tools of Sections 4.11 and 6.7: $120 billion at a duration of 5.6 is a DV01 of about $67 million against $16 billion of equity, and a pay-fixed swap valued by Section 7.3's arithmetic and sized to half of it would have halved every row of Example 40.7's table. Measure that DV01 with held-to-maturity securities included, because, as Section 3.8 shows, the swap would have been marked through earnings while the held-to-maturity securities were not.

40.15 WHAT REPEATS

Across the cases of this chapter, four mechanisms recur and the instruments do not.

Cash and value are different quantities. Metallgesellschaft, the LDI schemes, the 2022 European generators, AIG, and Example 37.2's basis trade are one mechanism: a position correct in value can be closed for want of cash, and no valuation model contains the closing.

Size and leverage change the exit. LTCM, Amaranth, Archegos, and the nickel case all failed at the point of liquidation rather than at the point of valuation. The measure describes the position's distribution; the loss came from the position's share of the market, or from leverage that made the path decisive, which are different variables.

Someone must see the whole. Barings, Société Générale, and Archegos are cases where the aggregate existed and no one charged with controlling it saw it (inside a firm in the first two, across firms in the third). The failure is informational, and in Archegos the instrument enabled it.

And the guarantee has a bottom. The nickel case and Section 37.10's crypto venue both show what happens when the mechanism that is supposed to make a market safe runs out of resources. The 2022 venue failure belongs in this list and is treated in Section 37.10, because its lesson is about custody, not any instrument: client assets were not segregated, the underlying settled normally throughout, and the exposure was to the venue, not the market.

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A fifth mechanism sits apart. Section 40.11's asymmetry is not a failure of measurement but of *who is equipped to measure*, and the only one here that recurs in regulated retail form. Section 40.13 adds a sixth: a mark that only the holder can produce.

Case Study 40.1 The information was already inside the firm

The most consistent finding across the official inquiries is also the least comfortable, because it rules out the easiest explanation.

In almost every case the facts needed to prevent the loss existed inside the institution before it happened. Barings' reconciliation discrepancies were reported and unresolved for months. The exchange-traded part of the nickel short was visible to the exchange, though much of the position was held over the counter. The credit index desk's own valuations were disputed internally before the loss was recognized. The unhedged securities portfolio of Section 40.14 had been the subject of supervisory findings. The Archegos post-mortem commissioned by Credit Suisse is explicit: the bank's risk function had escalated the concentration, and the escalation did not change the position.

So the failure is rarely detection and usually escalation. Someone saw it; the observation did not reach a person with both the authority and the incentive to act, or reached them in a form that competed with a revenue argument and lost.

That is why so little of the reform in Section 40.16 is quantitative. Segregating trading from settlement, requiring mandatory leave, making risk report outside the business line, and giving a desk-level test a capital consequence are all attempts to change what happens to an observation, not to generate a better one.

The corollary for a reader of this book is uncomfortable. Improving a measure has no effect if the existing one was already overridden, and Chapter 39's machinery tries to make overriding expensive rather than measurement better. That is the right design.

40.16 WHAT ACTUALLY CHANGED

The reforms these cases produced follow an unflattering pattern.

The control failures produced durable change. Segregation of trading from settlement, independent reconciliation, and mandatory leave became standard after Barings, and such losses are rarer, though Société Générale in 2008 and UBS in 2011 showed that controls are not self-enforcing. This, the profession's clearest success, required no modeling.

The clearing reforms after 2008 were substantial and are examined in Chapters 2 and 27. They also widened the exposure of Sections 40.8 and 40.12: the push to margin more of the market, cleared and bilateral, converts credit risk into liquidity risk, and the 2022 LDI and energy episodes were the bill for that conversion arriving.

The measurement failures produced the least. Amaranth's lesson about concentration was available in 2006, and Archegos repeated it in 2021 with a different instrument. Chapter 39's framework prices illiquidity and still does not measure concentration.

The accounting problem behind Section 40.14 is unresolved. The treatment that concealed the loss remains available, and the arguments for it are not frivolous: marking a held-to-maturity book through capital would import volatility into the accounts of institutions that genuinely intend to hold. The gap is real and the fix is not obvious, so the case will recur.

40.17 WHAT INSTITUTIONS AND REGULATORS SHOULD TAKE AWAY

Above the cases sits one rule: separate the instrument from the failure. In every case but nickel, where the venue voided the contract, the derivative did what its contract said, and a risk function that blames the instrument will look for the next problem in the wrong place. The

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failures were in cash, size, information, valuation and governance, and the advice follows them. For a financial institution:

- *Measure liquidity beside value.* Report, for every hedged position, the margin it can call under stress against the liquidity committed to meet it, gross and never netted against the hedged item (Metallgesellschaft, LDI, AIG, the 2022 generators).
- *Size positions by their exit.* Limit the share of open interest and the client's whole position across brokers, not only the notional or the VaR, and stress the exit with a price-impact cost (Amaranth, Archegos, nickel).
- *Cap what a control failure can cost.* Keep the segregation of duties, and cap the margin funding any one desk can draw, so that the largest position that could be hidden is small against equity (Barings, Société Générale).
- *Separate who marks from who profits.* Verify prices independently, reserve for model risk, and validate a risk-model change before it sets a limit (the London Whale, Section 40.13).
- *Start from the failure.* Run a reverse stress test for the move that exhausts equity or liquidity, and assume others hold the same position (LTCM, portfolio insurance, LDI).
- *Treat removing a hedge as a decision.* Leaving a risk unhedged is a position, and taking a hedge off deserves the review that putting it on received (SVB).

For a regulator:

- *Look at the aggregate.* Margin can be adequate in every bilateral relationship and inadequate in total; position reporting across prime brokers and supervisory review of concentration cover what no single firm sees (Archegos, Amaranth).
- *Supervise liquidity as closely as capital.* Margining converts credit risk into liquidity risk, so buffers should be sized to stressed margin calls, including at non-bank users such as pension schemes and utilities (LDI, the 2022 generators).
- *Interrupt feedback rather than ban strategies.* Where many holders follow the same rule, circuit breakers, margin that does not fall in calm markets, and backstops prepared in advance cut the loop (portfolio insurance, LDI).
- *Make a venue's guarantee, and its limits, known.* Test default resources against plausible extreme moves, and publish the rules for canceling trades before they are needed (nickel).
- *Protect the buyer who cannot price.* Require disclosure of duration, leverage and a payoff scenario table, and a suitability test on the payoff's shape (Orange County).
- *Look through the accounting.* Supervisory measures should capture economic duration and unrealized losses, whatever the accounting classification (SVB).

For governance and culture, at institutions and supervisors alike:

- *Question an unexplained profit as hard as an unexplained loss.* A book reported as low-risk arbitrage that earns like a directional one is directional, and no one objects to a mark that produces a profit (Barings, Société Générale, Section 40.13).
- *Fix escalation, not only detection.* Give the risk function an independent line to the chief executive and the board, and do not let the desk settle a risk manager's objection, because the concern had usually been raised and lost (Archegos, the London Whale).

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- *Pay for risk as it is realized.* Defer and claw back pay, and judge a desk on profit net of the risk and capital it uses, so that neither a decade of modeled margin booked at signing nor a halved VaR is rewarded (Enron, the London Whale).
- *Approve a position again as it grows.* A book approved at one size is a different book at three times that size, so growth past a set threshold should send it back to whoever approved it (the London Whale, Archegos).
- *Name an owner for each risk, and protect whoever raises it.* A discrepancy reported and left unresolved for months needs a named senior manager answerable for closing it, and the person who reported it needs protection from the revenue argument it contradicts (Barings, the case study in Section 40.15).

SUMMARY

In almost none of these losses did a derivative behave in a way its model did not predict. The instruments performed; what failed was around them: the cash, the exit, the information, the venue, the buyer's ability to price, or the decision not to hedge at all. Across the cases four mechanisms recur, and the instruments do not.

Cash differs from value. Metallgesellschaft's stack hedge was flat in value at every price and catastrophic in cash, because margin is paid now and the supply profit arrives over years. The UK LDI schemes and the European generators met the same shape in the same quarter of 2022, a hedge margined daily against a hedged item that is not, which suggests the exposure is created by market structure. AIG met it in another form, with collateral called on spread widening and no defaults at all.

Size and leverage change the exit, and someone must see the whole. LTCM, Amaranth and Archegos failed at the point of liquidation rather than of valuation. Archegos's prime brokers each margined their own slice of a position no one saw whole; at Barings and Société Générale a trader defeated the controls, and the ratio of the hidden position to equity decided which firm survived. Portfolio insurance in 1987 was the market-wide version: a hedge rule shared widely enough becomes the flow and deepens the decline it insures against.

The guarantee is bounded, and measures can bend. When nickel more than tripled in two days the exchange voided executed trades, so a central counterparty's guarantee depends on its resources and its governance. An approved model change halved the London Whale's VaR with no position changed. A fifth mechanism sits apart: Orange County could not price what it bought, though nothing was concealed.

The most recent case reverses the usual story. Silicon Valley Bank failed because it left its rate risk unhedged, so a reading that treats derivatives as the source of danger has the sign wrong. The lesson is to separate the instrument from the failure. In most cases the facts needed to prevent the loss already existed inside the firm, and what failed was escalation. The control failures produced durable reform; the measurement failures produced the least.

FURTHER READING

Most of these events produced an official inquiry or a commissioned review, and those reports are the primary sources. On Barings, the place to start is the Board of Banking Supervision's 1995 inquiry; on 1987, the Brady Commission's report of 1988, which Section 16.2 also draws on. On the London Whale, JPMorgan's own Management Task Force report and the US Senate Permanent Subcommittee on Investigations' case history, both of 2013, are worth reading together, since one is the bank's account and the other a legislative investigation of it. On Archegos, the Paul, Weiss report prepared for Credit Suisse (2021) is the primary source, particularly on what the bank's own risk function reported while the position was building. For

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the 2022 episodes, Oliver Wyman's independent review of the nickel suspension, prepared for the LME (2023), and the Bank of England's *Financial Stability Report* of December 2022 on the LDI intervention both state what the authorities did not know at the time, which is the part worth reading. On Silicon Valley Bank, the Federal Reserve's April 2023 review of its own supervision is the source, and it is explicit about examination failures. On Metallgesellschaft, Culp and Miller (1995) and Edwards and Canter (1995) are worth reading for the disagreement rather than the conclusion: whether the hedge was sound depends entirely on whether one counts the funding requirement as part of the strategy, and the two sides of that argument are the two halves of Section 40.1.

PRACTICE QUESTIONS

40.1 Using Example 40.1, explain why the stack hedge is flat in value and catastrophic in cash, and identify which leg produces the cash demand.

40.2 Using Figure 40.1, state the peak funding gap and when it occurs, and explain why the strategy can fail between the two points even though both curves end at the same total.

40.3 Explain how a stack-and-roll converts a flat-price hedge into a calendar spread, and state the first-year cost in Example 40.1's contango case.

40.4 Explain what Barings and Société Générale have in common, and why the same failure destroyed one firm and not the other.

40.5 Using Example 40.3, compute the move that exhausts LTCM's equity, and explain why this makes the strategy a view about the path rather than about value.

40.6 Using the Amaranth figures, compute the days required to exit at a 10% participation rate, and explain why both of the measure's assumptions fail simultaneously.

40.7 Using Example 40.4, explain why every prime broker was adequately margined and the system was not.

40.8 Explain what assumption of Chapter 2 the nickel case tested, and what the risk that materialized actually was.

40.9 Using Example 40.5, explain how a scheme's funding position could improve at every step while it was unable to meet collateral calls.

40.10 Explain why AIG's collateral demands required no defaults at all, and what distinction of Chapter 25 the case turns on.

40.11 Explain what changed and what did not change when the credit index desk adopted a new VaR model, and state the loss as a multiple of each reported figure.

40.12 Using Example 40.6, explain how portfolio leverage and instrument leverage multiply, and compute the loss on a 300 bp rise at the pool's effective duration of 7.3 years.

40.13 Explain why nothing was concealed in the Orange County case and the buyer still lost, and what this implies about the limits of disclosure.

40.14 Explain why a European generator that had sold power forward faced margin calls while its economics improved, and compute the demand on 50 TWh for a move from 50 to 400 EUR/MWh.

40.15 Explain why a valuation that cannot be independently checked responds to pressure, and why the pressure is asymmetric.

40.16 Explain the case study's finding that the failure is rarely detection, and why so little of the resulting reform is quantitative.

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40.17 Using Example 40.7, compute the economic loss on a $120bn portfolio at duration 5.6 for a 300 bp rise, and state it as a share of a $16bn equity base.

40.18 Explain the role of held-to-maturity accounting in the same case, and why a loss disclosed in the footnotes did not reach capital until a sale and a capital raise put it in front of depositors.

40.19 Name the four mechanisms that recur across these cases, and give one case for each.

40.20 Explain which category of failure produced durable reform and which produced the least, with an example of each.

40.21 Using Example 40.2 and Figure 40.2, compute the futures the programs must sell as the index falls from 100 to 90, and explain why the selling per index point peaks near the floor and slows once the delta approaches -1.

FURTHER QUESTIONS

40.22 Example 40.1 separates the value and cash profiles of a hedge. Construct a general measure of a hedge's funding requirement (the distribution of cumulative cash demanded before the offsetting gain is realized) and apply it to the LDI position of Example 40.5.

40.23 Example 40.4 shows margin adequate bilaterally and inadequate in aggregate. Design a reporting mechanism that would have revealed the total, state what each party would have to disclose and to whom, and determine whether the disclosure would itself have moved the market.

40.24 Section 40.5 treats exit time as a function of position share. Build a liquidation-cost model with price impact proportional to participation, determine the optimal unwind schedule for a position at 40% of open interest, and compare the resulting cost with the position's reported VaR.

40.25 The nickel case ended in trade cancellation. Design a CCP recovery arrangement that avoids voiding executed trades, state who funds it, and determine whether any participant would rationally join a clearing house on those terms.

40.26 Section 40.14 argues the accounting treatment concealed the exposure. Model a bank's optimal hedging decision under held-to-maturity accounting versus full fair value, and determine the conditions under which the accounting choice alone flips the hedging decision.

40.27 Section 40.16 argues that measurement failures produced the least reform. Take the concentration gap specifically, propose a supervisory measure that would capture it, and determine what data would be required and from whom.

40.28 Every case in this chapter was reconstructed after the fact. Choose one and identify the earliest point at which the information available inside the firm would have been sufficient to act, then state what report would have had to exist for someone to have seen it.

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Appendix A
Selected Derivations

The technical notes are a companion track to this book, not a part of it. They hold the derivations and the technical detail stripped out of the main text so that the chapters can stay intuitive, they are maintained and published separately, and the chapters point into them with markers such as [TN14].

This appendix reproduces the two the main argument leans on hardest, so that a reader working from the book alone is not stopped by them: the quadratic-variation and Itô arguments behind Chapter 14, and the integral behind Chapter 15's formula. Each keeps its own equation prefix, so a reference to (TN15.4) means the same thing here as it does in the companion track.

A.1 ITÔ'S LEMMA: DERIVATIONS

This note supplies the arguments Chapter 14 states without proof: why $dW^2 = dt$, how Itô's lemma follows, and why a change of measure alters the drift of a diffusion but not its volatility. The level is that of a careful heuristic rather than a measure-theoretic treatment; readers wanting the latter should consult the references at the end.

A.1.1 The Quadratic Variation of a Wiener Process

Everything rests on one fact: over a fixed interval, the sum of *squared* increments of a Wiener process converges to the length of the interval, and does so with vanishing randomness.

Partition $[0, T]$ into n intervals of length $\Delta t = T/n$, and write ΔW_i for the increment over the ith. From the definition, $\Delta W_i = \varepsilon_i \sqrt{\Delta t}$ with ε_i independent standard normals. Consider

$$Q_n = \sum_{i=1}^{n} \Delta W_i^2 = \Delta t \sum_{i=1}^{n} \varepsilon_i^2 \tag{TN14.1}$$

The mean. Since $\mathbb{E}[\varepsilon_i^2] = 1$,

$$\mathbb{E}[Q_n] = \Delta t \cdot n = T \tag{TN14.2}$$

independently of n.

The variance. For a standard normal, $\mathbb{E}[\varepsilon^4] = 3$, so $\mathrm{Var}(\varepsilon_i^2) = 3 - 1 = 2$. The ε_i are independent, so

$$\mathrm{Var}(Q_n) = \Delta t^2 \cdot n \cdot 2 = 2\,\frac{T^2}{n^2} \cdot n = \frac{2T^2}{n} \tag{TN14.3}$$

As $n \to \infty$ the variance vanishes while the mean stays at T. Therefore

$$\sum_{i=1}^{n} \Delta W_i^2 \;\longrightarrow\; T \quad \text{in mean square} \tag{TN14.4}$$

The limit is not a random variable. This is the content of the informal statement $dW^2 = dt$: squared increments accumulate deterministically, at rate one per unit time.

It is worth contrasting this with the increments themselves. By the same partition, $\sum_i \Delta W_i = W(T) - W(0) \sim N(0, T)$, which remains random however fine the partition. Squaring destroys the randomness; adding does not.

Why ordinary calculus fails. Consider instead a smooth function $x(t)$ with bounded derivative. Then $\Delta x_i = O(\Delta t)$, so $\sum_i \Delta x_i^2 = O(n\,\Delta t^2) = O(\Delta t) \to 0$. For smooth functions the quadratic variation is *zero*, which is precisely why second-order terms are discarded in ordinary Taylor expansions. A Wiener path has $\Delta W_i = O(\sqrt{\Delta t})$, so its quadratic variation is $O(n\,\Delta t) = O(T)$—finite and nonzero. *The whole difference between the two calculi is the difference between $O(\Delta t)$ and $O(\sqrt{\Delta t})$ increments.*

Two corollaries follow by the same argument and are used below:

$$\sum_i \Delta W_i\,\Delta t \to 0, \qquad \sum_i \Delta t^2 \to 0 \tag{TN14.5}$$

so in the shorthand of Chapter 14, $dW\,dt = 0$ and $dt^2 = 0$, while $dW^2 = dt$ alone survives.

A.1.2 Itô's Lemma

Let S follow the Itô process

$$dS = a(S,t)\,dt + b(S,t)\,dW \tag{TN14.6}$$

and let $f(S,t)$ be twice differentiable in S and once in t. Expand f to second order:

$$\Delta f = \frac{\partial f}{\partial S}\Delta S + \frac{\partial f}{\partial t}\Delta t + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\Delta S^2 + \frac{\partial^2 f}{\partial S\,\partial t}\Delta S\,\Delta t + \frac{1}{2}\frac{\partial^2 f}{\partial t^2}\Delta t^2 + \cdots \tag{TN14.7}$$

Now substitute the discrete form of (TN14.6), namely $\Delta S = a\,\Delta t + b\,\varepsilon\sqrt{\Delta t}$, and count orders.

The squared term is

$$\Delta S^2 = a^2\Delta t^2 + 2ab\,\varepsilon\,\Delta t^{3/2} + b^2\varepsilon^2\,\Delta t \tag{TN14.8}$$

Of these three, the first two are $o(\Delta t)$ and vanish when summed, by (TN14.5). The third is $O(\Delta t)$ and, by Section A.1.1, its sum converges to $\int b^2\,dt$ with no residual randomness. Hence

$$\Delta S^2 \;\longrightarrow\; b^2\,dt \tag{TN14.9}$$

The remaining second-order terms in (TN14.7) contain $\Delta S\,\Delta t = O(\Delta t^{3/2})$ and Δt^2, both of which vanish. Substituting (TN14.9) and (TN14.6) into (TN14.7) and passing to the limit gives **Itô's lemma**:

$$\boxed{df = \left(\frac{\partial f}{\partial S}a + \frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}b^2\right)dt + \frac{\partial f}{\partial S}\,b\,dW} \tag{TN14.10}$$

Setting $a = \mu S$ and $b = \sigma S$ recovers equation (14.8) of the main text.

The two-variable case. For $f(x,y,t)$ with $dx = a_1 dt + b_1 dW_1$, $dy = a_2 dt + b_2 dW_2$ and $dW_1 dW_2 = \rho\,dt$, the same expansion carries an additional cross term $f_{xy}\Delta x\Delta y$. By the argument of Section A.1.1 applied to the product,

$$\sum_i \Delta x_i\,\Delta y_i \;\longrightarrow\; \int b_1 b_2 \rho\,dt \tag{TN14.11}$$

which produces the extra drift term $f_{xy}b_1b_2\rho$ in equation (14.18).

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A.1.3 The Lognormal Solution

Chapter 14 asserts that geometric Brownian motion has the explicit solution used for simulation. Apply (TN14.10) to $G = \ln S$ with $a = \mu S$ and $b = \sigma S$:

$$\frac{\partial G}{\partial S} = \frac{1}{S}, \qquad \frac{\partial^2 G}{\partial S^2} = -\frac{1}{S^2}, \qquad \frac{\partial G}{\partial t} = 0$$

so

$$d \ln S = \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma \, dW \tag{TN14.12}$$

The coefficients are constant, so this integrates directly:

$$\ln S_T = \ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right) T + \sigma \, (W_T - W_0) \tag{TN14.13}$$

and since $W_T - W_0 \sim N(0, T)$,

$$\ln S_T \sim N\left(\ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right) T, \quad \sigma^2 T\right) \tag{TN14.14}$$

Exponentiating (TN14.13) gives the exact simulation scheme of equation (14.6).

Moments. If $\ln X \sim N(m, s^2)$ then, completing the square in the defining integral,

$$\mathbb{E}\left[X^k\right] = \mathbb{E}\left[e^{k \ln X}\right] = e^{km + k^2 s^2/2} \tag{TN14.15}$$

Setting $k = 1$ gives $\mathbb{E}[X] = e^{m + s^2/2}$; setting $k = 2$ and subtracting the square of the mean gives the variance quoted in the main text. Applying (TN14.15) with $k = 1$ to (TN14.14),

$$\mathbb{E}[S_T] = \exp\left(\ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right) T + \frac{\sigma^2 T}{2}\right) = S_0 e^{\mu T}$$

confirming that the Itô correction is exactly what makes the expected price grow at μ.

A.1.4 Why the Measure Change Preserves Volatility

Chapter 14 states that moving to the risk-neutral measure changes a diffusion's drift but leaves its volatility untouched. The reason is visible from Section A.1.1 without any formal apparatus.

A change of probability measure re-weights how likely each path is. It does not alter the paths themselves, and volatility is a property of a path rather than of its probability: by (TN14.4), the quadratic variation of S along any given path is $\int \sigma^2 S^2 \, dt$, computed from the path alone. Since two equivalent measures agree on which paths are possible—they differ only in the weights—they must agree on quadratic variation, and therefore on σ.

The drift enjoys no such protection. It is defined by an *expectation*, and expectations are exactly what re-weighting changes.

The formal statement. Girsanov's theorem makes this precise. If W is a Wiener process under P and λ is a suitably regular process, then

$$W_t^Q = W_t + \int_0^t \lambda_u \, du \tag{TN14.16}$$

is a Wiener process under a measure Q equivalent to P, with the change of measure given by the density

$$\frac{dQ}{dP} = \exp\left(-\int_0^T \lambda_u \, dW_u - \frac{1}{2}\int_0^T \lambda_u^2 \, du\right) \tag{TN14.17}$$

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Substituting (TN14.16) into $dS = \mu S\,dt + \sigma S\,dW$ gives

$$dS = (\mu - \lambda\sigma)S\,dt + \sigma S\,dW^{Q}$$

so choosing $\lambda = (\mu - r)/\sigma$—the **market price of risk**—produces the risk-neutral dynamics $dS = rS\,dt + \sigma S\,dW^{Q}$. The volatility is unchanged, as claimed, and only the drift has moved.

The quantity λ deserves a name because it recurs: it is the excess return per unit of volatility, and Chapter 28 shows that changing numeraire amounts to changing λ, which is why the convexity, timing, and quanto adjustments of Chapter 30 all take the form of a drift correction and never a volatility correction.

Further Reading

Neftci's *An Introduction to the Mathematics of Financial Derivatives* works through the arguments above at greater length and with more attention to what is being glossed over. Baxter and Rennie's *Financial Calculus* gives a short account of Girsanov's theorem and the martingale representation theorem, and is the place to start on both. Readers wanting the measure-theoretic treatment should go to Karatzas and Shreve, or to Shreve's second volume, which is written for exactly this audience.

A.2 THE BLACK–SCHOLES–MERTON FORMULA: DERIVATIONS

Chapter 15 states the formula and explains what its terms mean. This note carries out the integral that produces it, shows the equivalence with solving the differential equation, and derives the dividend and American-approximation results the chapter uses.

A.2.1 The Risk-Neutral Integral

Under the risk-neutral measure, [TN14] established that

$$\ln S_T \sim N\!\left(m,\, s^2\right), \qquad m = \ln S_0 + \left(r - \frac{\sigma^2}{2}\right)T, \qquad s = \sigma\sqrt{T} \tag{TN15.1}$$

and the call is the discounted expected payoff,

$$c = e^{-rT}\,\mathbb{E}^{Q}[\max(S_T - K,\,0)] = e^{-rT}\,\mathbb{E}^{Q}[(S_T - K)\mathbf{1}\{S_T > K\}] \tag{TN15.2}$$

Split it into two pieces:

$$c = e^{-rT}\underbrace{\mathbb{E}^{Q}[S_T\mathbf{1}\{S_T > K\}]}_{A} - e^{-rT}K\underbrace{Q(S_T > K)}_{B} \tag{TN15.3}$$

The second piece: B

Write $\ln S_T = m + sZ$ with Z standard normal. Then $S_T > K$ if and only if

$$Z > \frac{\ln K - m}{s}$$

so, using the symmetry $1 - N(x) = N(-x)$,

$$B = N\!\left(\frac{m - \ln K}{s}\right) = N\!\left(\frac{\ln(S_0/K) + \left(r - \sigma^2/2\right)T}{\sigma\sqrt{T}}\right) = N(d_2) \tag{TN15.4}$$

which identifies d_2 and confirms Chapter 15's statement that $N(d_2)$ is the risk-neutral probability of exercise.

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The first piece: A

Here the useful step is to complete the square. Writing $S_T = e^{m+sZ}$ and ϕ for the standard normal density,

$$A = \int_{z^*}^{\infty} e^{m+sz}\,\phi(z)\,dz = \frac{1}{\sqrt{2\pi}}\int_{z^*}^{\infty} \exp\!\left(m + sz - \frac{z^2}{2}\right) dz, \qquad z^* = \frac{\ln K - m}{s}$$

The exponent is

$$m + sz - \frac{z^2}{2} = m + \frac{s^2}{2} - \frac{(z-s)^2}{2}$$

so, substituting $y = z - s$,

$$A = e^{m+s^2/2}\,\frac{1}{\sqrt{2\pi}}\int_{z^*-s}^{\infty} e^{-y^2/2}\,dy = e^{m+s^2/2}\,N(s - z^*) \tag{TN15.5}$$

Two substitutions finish it. First, $e^{m+s^2/2} = S_0 e^{rT}$, the lognormal moment result of [TN14] and the same cancellation of $\sigma^2/2$ noted there. Second,

$$s - z^* = \frac{m - \ln K}{s} + s = d_2 + \sigma\sqrt{T} = d_1$$

so $A = S_0 e^{rT} N(d_1)$.

Assembling

Substituting A and B into (TN15.3):

$$\boxed{c = S_0 N(d_1) - K e^{-rT} N(d_2)} \tag{TN15.6}$$

The put follows from put–call parity rather than a second integral, since parity holds identically.

$N(d_2)$ arose as a probability; $N(d_1)$ arose from completing the square, that is, from the shift $z \mapsto z - s$ induced by weighting each state by S_T. That shift is the content of Chapter 15's remark that $N(d_1)$ is not a risk-neutral probability: it is the probability of exercise computed under the measure in which the stock, rather than cash, is the numeraire. Chapter 28 develops that idea in general.

A.2.2 Equivalence with the Differential Equation

Chapter 15 offers two routes to (TN15.6). They agree, and here is why.

Define $V(S,t) = e^{-r(T-t)}\mathbb{E}_t^{Q}[h(S_T)]$ for any payoff h. By the law of iterated expectations, the discounted value $e^{-rt}V$ is a Q-martingale. Applying Itô's lemma to $e^{-rt}V(S,t)$ under the risk-neutral dynamics $dS = rS\,dt + \sigma S\,dW^{Q}$:

$$d\big(e^{-rt}V\big) = e^{-rt}\left(V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV\right)dt + e^{-rt}\sigma S V_S\,dW^{Q}$$

A martingale has zero drift. Setting the dt coefficient to zero gives

$$V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} = rV \tag{TN15.7}$$

which is equation (15.1). *The differential equation and the risk-neutral expectation are the same statement*: one says the discounted price has no drift, the other computes the expectation that follows. This equivalence is the Feynman–Kac relation, and it is why Chapter 20 can solve pricing problems either by simulating expectations or by solving partial differential equations.

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A.2.3 Dividends

A continuous yield. With a yield q, the risk-neutral drift becomes $r - q$: the asset's total return must be r, and q of it arrives as income, so the price grows at $r - q$. Repeating Section A.2.1 with $m = \ln S_0 + (r - q - \sigma^2/2)T$ changes (TN15.5) to $A = S_0 e^{(r-q)T} N(d_1)$ and leaves (TN15.4) in the same form, giving

$$c = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2), \qquad d_1 = \frac{\ln(S_0/K) + \left(r - q + \sigma^2/2\right) T}{\sigma\sqrt{T}} \tag{TN15.8}$$

Known cash dividends, and the volatility subtlety. Chapter 15 subtracts the present value of dividends and applies the formula to $S^* = S - \mathrm{PV}(D)$. The justification is that S^* is the risky component: S equals a certain part, $\mathrm{PV}(D)$, plus an uncertain part which is assumed lognormal.

The subtlety the chapter flags can be made precise. If S^* has volatility σ^* and $S = S^* + \mathrm{PV}(D)$, then the volatilities of the two are related by

$$\sigma^* \approx \sigma \frac{S}{S^*} = \sigma \left(1 - \frac{\mathrm{PV}(D)}{S}\right)^{-1} \tag{TN15.9}$$

because the same absolute risk is carried by a smaller base. For a stock yielding 6% with a two-year option, $\mathrm{PV}(D)/S \approx 0.113$, so σ^* exceeds the observed σ by roughly 13%, which is not negligible. Using the unadjusted volatility understates the option.

A.2.4 Black's Approximation

Chapter 15 states the rule; here is why it takes that form.

Section 11.10 established that an American call on a stock paying a single dividend at t_1 can be optimally exercised only at two moments: immediately before t_1, or at expiration T. Exercise elsewhere is dominated, because the interest saved by deferring the strike payment exceeds any benefit until the dividend is imminent.

The value of an American call is the maximum over admissible exercise policies. Restricting attention to those two:

$$C \approx \max\left(\underbrace{c(S_0 - \mathrm{PV}(D),\, K,\, T)}_{\text{hold to } T},\ \underbrace{c(S_0,\, K,\, t_1)}_{\text{exercise at } t_1} \right) \tag{TN15.10}$$

The first term treats the option as European to T on the dividend-adjusted price. The second treats it as European to t_1 with no adjustment, since a holder exercising then captures the dividend and the adjustment does not apply.

The approximation *understates*, and the reason is now visible: it evaluates a fixed choice between two policies made today, whereas the true American holder decides at t_1 with knowledge of S_{t_1}. Being able to choose after observing is worth something, so

$$C \geq \max(\cdot, \cdot)$$

with equality only in the degenerate cases where one policy dominates in every state. The error is largest when the two terms are close, which is exactly when the exercise decision is genuinely uncertain.

Further Reading

Black and Scholes (1973) derive the formula through a hedged portfolio and the heat equation; Merton (1973) uses a self-financing replicating portfolio and also handles dividends and the American call. Cox and Ross (1976) is where the risk-neutral integral of Section A.2.1 first

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appears as a general method. For the Feynman–Kac equivalence of Section A.2.2, Baxter and Rennie is the short route and Shreve's second volume the measure-theoretic one.

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Appendix B
Major Derivatives Exchanges

Chapter 2 described what an exchange does; this appendix says who they are. The list covers venues that matter either because of their volume or because they own a benchmark that the rest of the market prices against, and for each it names the group that owns it, where it sits, and what actually trades there.

Two warnings belong before the tables.

Ownership and product lists move. Exchanges are acquired, renamed, and delisted; contracts are launched and retired. What follows reflects the position in early 2026, and any specific claim here should be checked against the venue's own contract specifications before it is relied on for a trade. *The structure is more durable than the details*, which is the reason the appendix closes with Section B.8 rather than with a table.

And there is no single ranking. Section B.1 explains why the two ways of measuring an exchange's size disagree by orders of magnitude, and why almost every published league table is answering a question the reader did not ask.

B.1 HOW TO READ THE TABLES

The two standard volume measures do not rank the same way, and the gap between them is not small.

Contracts traded counts tickets. It is the figure the Futures Industry Association publishes and the one exchanges quote in press releases. *It is also almost meaningless across venues*, because contract sizes differ by factors of thousands: an Indian index option on a lot worth a few thousand dollars and a Eurex Bund futures contract on €100,000 of notional each count as one.

Notional traded counts economic exposure. It ranks quite differently, and it is the figure that matters for anything in Chapters 27 and 39, where exposure and capital are the subject.

The consequence is plain. By contracts, the largest derivatives exchange in the world is in India; by notional it is in Chicago, and the two are not close in either direction. A reader who sees "largest exchange" in a headline should establish which measure is meant before drawing any inference about where risk actually sits.

Two further cautions apply to any of these figures.

Open interest is the better measure of what is held. Volume measures turnover, and a venue dominated by same-day expirations (Section 23.1's 0DTE flow, or the Indian weekly options market) generates enormous volume against small open interest. *Turnover and exposure are different questions.*

And exchange-traded volume is not the whole market. The largest interest rate and credit markets in this book are over-the-counter and clear through the houses of Section 2.4 rather than trading on any exchange listed here. *A directory of exchanges is a map of the listed market, not of derivatives.*

B.2 NORTH AMERICA

Exchange	Group	Principal derivatives
CME	CME Group	SOFR and Fed funds futures, E-mini S&P 500 and Nasdaq-100, FX futures, live cattle
CBOT	CME Group	US Treasury futures across the curve, corn, soybeans, wheat
NYMEX	CME Group	WTI crude, Henry Hub natural gas, refined products
COMEX	CME Group	Gold, silver, copper
Cboe Options	Cboe Global Markets	SPX options, VIX options, listed equity options
Cboe Futures (CFE)	Cboe Global Markets	VIX futures
ICE Futures U.S.	ICE	Sugar, coffee, cocoa, cotton, Russell 2000, USD index
Nasdaq PHLX, ISE, GEMX, MRX	Nasdaq	Listed equity and index options
MIAX Options, Pearl, Emerald, Sapphire	Miami International Holdings	Listed equity options
MGEX	Miami International Holdings	Hard red spring wheat
Montréal Exchange	TMX Group	Three-month CORRA futures, Canadian government bond futures, S&P/TSX 60 futures

The concentration is the point. Four of the eleven venues above belong to CME Group, and between them they carry the US benchmark contract in interest rates, equity indices, energy, and metals. *No other jurisdiction has anything comparable*, and Section B.8 returns to what that means for the clearing houses behind them.

One entry records a transition this book has already described. Montréal's BAX contract referenced CDOR and was retired with it; the three-month CORRA futures contract is its replacement, and Section 4.3's account of the benchmark transition is what that line in the table is about.

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B.3 EUROPE

Exchange	Group	Principal derivatives
Eurex	Deutsche Börse	Bund, Bobl, Schatz and Buxl futures, Euro Stoxx 50 futures and options, single-stock options, €STR and Euribor futures
ICE Futures Europe	ICE	Brent crude, gasoil, SONIA and Euribor futures, FTSE 100
ICE Endex	ICE	Dutch TTF natural gas, the European gas benchmark
London Metal Exchange	HKEX	Copper, aluminum, zinc, nickel, lead, tin, on the prompt-date structure
Euronext	Euronext	CAC 40, AEX, FTSE MIB, milling wheat, rapeseed, corn
EEX	Deutsche Börse	German and French power, EU emission allowances, gas
Nasdaq Nordic	Nasdaq	OMXS30 futures and options, Nordic equity derivatives
Borsa İstanbul (VIOP)	Borsa İstanbul	Equity index and single-stock futures, currency
Moscow Exchange	MOEX	Domestic index, FX, and commodity contracts; largely closed to international participants since 2022

Europe's map is drawn differently from America's, and the difference is the subject of Section B.8. Eurex clears its own contracts; most European over-the-counter interest rate risk clears at LCH, which is owned by neither an exchange group nor its members in the way a vertical silo is.

Two entries carry episodes from Chapter 40. The London Metal Exchange's nickel contract is where the March 2022 suspension and cancellation of trades happened, and the Dutch TTF gas contract is where the 2022 margin spiral of Section 40.11 ran. *Both are cases where the venue itself became the risk*, which is why they are named here rather than left to a footnote.

B.4 ASIA-PACIFIC

Exchange	Group	Principal derivatives
National Stock Exchange	NSE India	Nifty 50 and Bank Nifty index options and futures; the largest venue in the world by contracts traded
BSE	BSE India	Sensex and Bankex index options
MCX	MCX India	Gold, silver, crude oil, natural gas
Osaka Exchange	JPX	Nikkei 225 futures and options, TOPIX, JGB futures
TOCOM	JPX	Energy, rubber, precious metals
Korea Exchange	KRX	KOSPI 200 options and futures, three-year KTB futures
HKEX	HKEX	Hang Seng and HSCEI futures and options, MSCI contracts, USD/CNH futures
Singapore Exchange	SGX	FTSE China A50, Nikkei 225, iron ore, rubber, FX futures
TAIFEX	TAIFEX	TAIEX futures and options
ASX	ASX	SPI 200, three- and ten-year Treasury bond futures, 90-day bank bill
Bursa Malaysia Derivatives	Bursa Malaysia	Crude palm oil (FCPO), the global benchmark

The Indian entry deserves the qualification the table cannot carry. NSE has led the world

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by contract count for several years, on a market dominated by weekly index options with very small lot sizes, which is Section B.1's warning in its purest form. *Regulatory measures from late 2024* (larger lot sizes, and a limit of one weekly expiration per exchange) were introduced precisely because the authorities concluded that retail turnover of that shape was not serving the participants generating it. Volumes fell sharply and the ranking survived, which tells you how large the gap to second place was.

And one entry records a contract that moved countries. SGX's Nifty futures, long the offshore venue for Indian index exposure, transitioned in 2023 to NSE's international arm at GIFT City. *An offshore benchmark returning onshore is unusual*, and it is the mirror image of the Chinese pattern described next.

B.5 MAINLAND CHINA

China's exchanges are treated separately because the market is structured differently: six futures exchanges, each with a defined product remit, under a single regulator, with foreign access granted contract by contract rather than venue by venue.

Exchange	Location	Principal derivatives
Shanghai Futures Exchange (SHFE)	Shanghai	Copper, aluminum, zinc, nickel, rebar, hot-rolled coil, gold, silver, rubber
Shanghai International Energy Exchange (INE)	Shanghai	RMB-denominated crude oil, TSR20 rubber, low-sulfur fuel oil; open to overseas investors
Dalian Commodity Exchange (DCE)	Dalian	Iron ore, soybean meal and oil, palm olein, corn, coking coal, coke, LLDPE, LPG
Zhengzhou Commodity Exchange (ZCE)	Zhengzhou	PTA, methanol, cotton, sugar, rapeseed meal and oil, urea, soda ash, glass
China Financial Futures Exchange (CFFEX)	Shanghai	CSI 300, CSI 500, SSE 50 and CSI 1000 index futures and options; 2-, 5-, 10- and 30-year government bond futures
Guangzhou Futures Exchange (GFEX)	Guangzhou	Industrial silicon, lithium carbonate
Shanghai and Shenzhen Stock Exchanges	Shanghai, Shenzhen	ETF options on broad-based indices

The division of labor is administrative rather than historical, which is why the product split is so clean: ferrous and energy in Dalian, chemicals and soft commodities in Zhengzhou, non-ferrous metals in Shanghai, all financial contracts at CFFEX, and the new-energy complex at Guangzhou, which was created in 2021 for exactly that purpose. *No Western exchange group was assembled that tidily*, because none of them was assembled at all—they merged into their present shape.

Two features matter for anyone pricing these contracts. The exchanges impose daily price limits and position limits far more actively than their Western counterparts; and the government bond futures at CFFEX use a 3% notional coupon against the 6% of the CME's Treasury contracts, which changes which bond is cheapest to deliver.

Access is by contract, not by venue. INE's crude contract was built from the outset for overseas participation, and specific DCE and ZCE contracts have been opened since; most of the rest remain domestic. *Hong Kong is the offshore complement*, which is where the USD/CNH futures in Section B.4's table fit.

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B.6 LATIN AMERICA, THE MIDDLE EAST, AND AFRICA

Exchange	Location	Principal derivatives
B3	São Paulo	DI interest rate futures, Ibovespa futures and options, USD/BRL futures, single-stock options
MexDer	Mexico City	TIIE interest rate futures, IPC index, USD/MXN
Gulf Mercantile Exchange	Dubai	Oman crude, the principal East-of-Suez sour benchmark
ICE Futures Abu Dhabi	Abu Dhabi	Murban crude
Saudi Exchange	Riyadh	Index and single-stock futures
Johannesburg Stock Exchange	Johannesburg	Equity index and single-stock futures, currency, white and yellow maize, wheat

B3 is larger than its profile suggests. Its DI contract, a futures contract on the domestic interbank rate, trades in volumes that place it among the world's most active interest rate contracts by count, and Brazilian hedgers use it in place of the swap market that dominates elsewhere. *Where a listed contract is liquid enough, it displaces the over-the-counter equivalent*, which is the observation Chapter 2 makes in the abstract and Brazil demonstrates.

The two Gulf entries exist because of a gap in the crude benchmark map. Brent and WTI are light and sweet; much of what actually moves from the Gulf to Asia is neither, and both Oman and Murban were listed to give that barrel a transparent price. Section 36.9's discussion of benchmark grades is what these contracts are for.

B.7 CRYPTO VENUES

Chapter 37 treats these instruments; what belongs here is the distinction between the venues that trade them, because it is sharper than in any other asset class.

Regulated futures exchanges list bitcoin and ether futures and options under the same rules as any other contract. CME is much the largest of these, and its contracts are the ones institutional participants can hold without a separate custody and counterparty framework.

Offshore crypto-native venues carry most of the volume. They list the perpetual futures of Section 37.3 (a contract with no delivery date, held in line with spot by the funding-rate mechanism), and they clear, custody, and match on the same balance sheet, which is the vertical integration Section 37.10 identifies as the structural risk. *The largest of them trade more notional than the regulated venues by a wide margin*, and Deribit has long been the dominant venue for crypto options specifically.

The distinction is not about size but about what fails when something fails. A CME position is protected by the clearing arrangements of Chapter 2 and the segregation rules of Chapter 8. *On a venue that is simultaneously exchange, clearing house, and custodian, those three protections are one balance sheet*, and Section 37.10's account of what that has cost is the reason the appendix separates them.

B.8 WHAT THE MAP SHOWS

Four structural facts survive the details, and they are the reason to read a directory of this kind at all.

The listed market is concentrated in a small number of groups. CME Group, ICE, Cboe, Nasdaq, Deutsche Börse, HKEX, JPX, NSE, and B3 account for the large majority of exchange-traded derivatives volume worldwide. *Competition happens between groups, not between the venues inside them*, which is why a group's several exchanges rarely list the same contract against each other.

Liquidity does not migrate. A benchmark contract is a network good: it is liquid because

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everyone uses it, and everyone uses it because it is liquid. *Competing exchanges have repeatedly listed identical contracts against an incumbent and repeatedly failed*, which is why the Bund is at Eurex and Treasuries are at CBOT decades after either could in principle have moved.

The vertical silo is the dominant model outside Europe. An exchange that owns its clearing house captures the margin as well as the fee and, more consequentially, makes positions non-fungible with any competitor's. Europe's separation of LCH from the trading venues is the main exception, and Section 2.10's discussion of clearing concentration is where the consequences are worked out.

And the exchange is not where most derivatives risk sits. Interest rate swaps, credit default swaps, and the structured products of Chapter 35 are over-the-counter instruments; they clear centrally, but they do not trade on a venue in this appendix. *The exchanges hold the price discovery and the retail flow; the dealers and the clearing houses hold the exposure.* Anyone reading this list as a map of where the risk is has read it as the opposite of what it is.

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Appendix C
Glossary

Definitions are working ones: what the term means in this book and where it is developed, not what a dictionary would say. *A chapter number in parentheses is where to go*, and where a term is defined precisely in one place and used loosely elsewhere, the pointer is to the precise one.

A

Accreting swap A swap whose notional steps up over time, matching a drawdown (33).

Accrued interest The part of a bond's next coupon earned since the last coupon date, measured on the bond's day count and paid by the buyer to the seller on top of the quoted price (6).

Additional valuation adjustment (AVA) The gap between a position's fair value and a prudent value it could be realized at with 90% confidence, deducted directly from common equity under the EU and UK prudent valuation rules. Nine categories, among them market price uncertainty, close-out cost and model risk (39).

Add-on In SA-CCR, the supervisory stand-in for how far an asset class's positions can move over the horizon that matters: for each hedging set, the supervisory factor times the effective notional, summed over the class's hedging sets. Scaled by the multiplier, the add-ons make up potential future exposure (27).

Adjoint algorithmic differentiation (AAD) Computing every sensitivity of a simulated price by running the chain rule backward through the stored pricing calculation, from the price to each input, in a fixed multiple (typically three to five times) of the cost of the price itself, whatever the number of inputs. What makes a model with many calibration inputs viable on a desk (9, 20, 32).

Admissible strategy A trading strategy whose value never falls below a fixed floor. The restriction that makes the first fundamental theorem true in continuous time, by ruling out doubling strategies (TN28).

Affine model A short-rate model in which $\ln P(t,T)$ is affine in the short rate (linear plus a constant), so bond prices have a closed form and every yield on the curve is an affine function of one state variable. Vasicek, CIR and Hull–White are all in this class (31).

American Monte Carlo Simulation that handles early exercise, or the future revaluation of a portfolio, by estimating each path's continuation value with a least-squares regression across paths: the Longstaff–Schwartz method. Exposure systems borrow the regression to avoid revaluing every trade exactly on every path (9, 20).

American option An option exercisable at any time up to expiration (10, 20).

Amortizing swap A swap whose notional steps down, matching a loan being repaid. Pricing it at its weighted average life is a trap (33).

Annuity factor The present value of one unit paid on each fixed-leg payment date, weighted by the accrual factors, so one basis point of fixed rate on unit notional is worth 0.0001 times it; the natural numeraire for swaptions (7, 29, 30).

Annuity mapping function In CMS pricing, the conditional expectation $\alpha(s)$ of the reciprocal

of the annuity given that the swap rate finishes at s. It gathers every unknown about the curve's joint law into one function, so a CMS payoff becomes an integral against the swap rate's marginal law; freezing the annuity makes it constant, and a linear terminal swap rate model makes it affine (28).

Antithetic variates Pairing every simulated path with its mirror image, built from the same random shocks with their signs reversed, and averaging the two payoffs. The pair is negatively correlated, so the variance of the estimate falls at almost no cost (20).

Arbitrage A trading strategy that costs nothing to set up, can never lose money, and makes money in at least one state of the world. Its absence is the assumption that prices everything in this book (5, 11).

Arrow–Debreu price The value today of one unit paid at one node of a lattice and nothing anywhere else; the state price of Chapter 13 written on a tree. Summing them across a level returns that maturity's discount factor, which is what the Hull–White construction solves against (13, 31).

Asian option An option on the average price over a period. Cheaper than a vanilla because an average has less variance than an endpoint (34).

Assessment rights A CCP's right, near the bottom of its default waterfall, to demand further contributions from surviving clearing members once the default fund is exhausted. It allocates a loss rather than creating resources, and it transmits stress to firms already struggling (2).

Asset-swap spread The spread over SOFR received when a fixed-rate bond is bought at par and its coupons are swapped to floating; the measure of a bond's credit spread that a CDS–bond basis trade earns (24).

At-the-money Strike equal to the current spot or forward; the two conventions differ and the difference matters for quoting (19).

At-the-money forward A strike equal to the forward price, at which a European call and a European put have equal value by put–call parity. The usual reference point for quoting options, and the strike of the straddle used to trade volatility without a directional view (11, 19).

Attachment point The loss level at which a tranche begins to absorb losses (25, 26).

Autocallable A structured note that redeems early, repaying its face plus the coupons accrued, if the underlying is at or above a trigger level on an observation date. If it is never called, the buyer is short a put that knocks in below a barrier (35).

Automated market maker A venue that quotes from a formula over a pooled inventory rather than from an order book (37).

B

Backtesting / exception Checking a risk measure against realized losses by counting exceptions, the days on which the loss exceeds the VaR; at 99% over 250 days a correct model expects 2.5. Counting is weak evidence: anything from one to six exceptions is consistent with a correct model, zero is a rejection too, and the count cannot see whether exceptions cluster. The count sets the capital multiplier under internal models, and it is VaR that is backtested, because expected shortfall cannot be tested by counting (38, 40).

Backward induction Valuing a derivative on a tree by working back from expiration one step at a time, each node's value being the discounted risk-neutral expectation of the values at the nodes that follow it. For an American option each node also compares that value with immediate exercise (13).

Backward-looking caplet A caplet on a rate compounded in arrears over its accrual period, so its payoff is known only at the period's end. The rate keeps accumulating variance through

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the period, so Black's formula applies with $\sigma^2 T_1$ replaced by $\sigma^2(T_1 + \tau/3)$, and the caplet is worth more than its forward-looking counterpart (29).

Backwardation A forward curve sloping downward. Distinct from Keynes's *normal* backwardation, which compares the futures price with the expected spot (3, 5).

Banking Keeping an emissions allowance not surrendered this year for use in a later one. Because banking is costless and unlimited, the forward cannot exceed spot carried at the financing rate, and that upper bound binds tightly; because an allowance cannot be borrowed, the forward can sit below carry, and the residual measures doubt about the scheme. The limiting case of a storable commodity (36).

Barrier option An option extinguished or created when the underlying touches a level. The largest exotic market, and among the instruments with the most model risk, with cliquets and forward-start options (34).

Base correlation A quoting convention that removes the inconsistency of implied correlation across tranches without removing what caused it (25).

Basel framework The international bank capital standards agreed by the Basel Committee on Banking Supervision and implemented, with national variations, by each jurisdiction's regulator. It sets what a derivatives position costs in capital, and therefore what it costs to hold (9, 27, 40).

Basis The spot price minus the futures price, which converges to zero at delivery. What a hedger is left with when the hedge is lifted before delivery (2, 3). In credit, the CDS spread minus the bond's spread, which the arbitrage of buying the bond and buying protection says should be zero and which persists (24).

Basis swap A swap exchanging two floating rates, pricing the spread between them (7, 33).

Basket default swap Credit protection on a small basket of issuers, typically five to ten, that pays on the nth default and then terminates. The first-to-default basket is the common case; to first order its value depends on the issuers' hazard rates rather than on recovery, and correlation makes it cheaper than the sum of its parts, because coincident defaults waste the protection (24, 25).

Basket / rainbow option A basket option pays on a weighted sum of several assets; a rainbow option pays on the best or the worst of them, as a worst-of put does. A basket is worth more as correlation rises and a worst-of put more as it falls, which is what pays the high coupon on a worst-of note (34, 35).

Bermudan option An option exercisable on a fixed set of dates. The standard embedded option in callable bonds and cancellable swaps (31, 33).

Beta A portfolio's sensitivity to an index, and the hedge ratio for hedging it with index futures: the number of contracts is beta times the portfolio's value divided by the value of one futures contract (3).

Binomial tree A discrete lattice for the underlying, and the most direct route to risk-neutral valuation (13).

Black's approximation Valuing an American call on a dividend-paying stock as the greater of two European calls: one expiring at the option's expiration on the dividend-adjusted price, and one expiring just before the ex-dividend date on the unadjusted price (15).

Black's model The Black–Scholes–Merton model applied to a forward rather than a spot; the market standard for caps, swaptions, and futures options (16, 29).

Black–Scholes–Merton differential equation The partial differential equation satisfied by every derivative whose value depends only on the stock price and time. The stock's expected return does not appear in it; the instrument enters only through its boundary condition (15, 20).

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Black–Scholes–Merton formula The closed-form value of a European call or put on a lognormal stock, $c = S_0N(d_1) - Ke^{-rT}N(d_2)$. In practice it is used mostly in reverse, to convert a market price into an implied volatility (15).

Bootstrap (statistics) Estimating the sampling error of a statistic by recomputing it on many samples drawn with replacement from the data itself. For a far quantile it is rough, because no resample can contain a loss larger than the largest observed (38).

Bootstrapping Building a curve by solving instrument by instrument from the short end outward (4, 7).

Borrow fee The fee a short seller pays to borrow the asset: a few basis points for a liquid stock, tens of percent for one that is hard to borrow. It does not by itself widen the forward-pricing relation, because holders can defend it by selling; where they will not or cannot sell, it turns that relation and put–call parity into bands, and puts trade rich to calls (5, 10, 11).

Boundary condition The condition that, together with the Black–Scholes–Merton differential equation, determines a particular derivative's value: $\max(S - K, 0)$ at expiration for a call, $S - K$ for a forward. The equation is common to all of them; the instrument is the boundary condition (15).

Box spread A synthetic long position at one strike combined with a synthetic short at a higher one, paying the difference between the strikes at expiration whatever the underlying does. A zero-coupon bond assembled from four options, used to borrow and lend at the rate its price implies (11).

Bump-and-revalue Computing a sensitivity by moving one input a small amount and repricing; costs one valuation per input, which is what makes it impractical for a model with many (32).

Business time A clock that runs on expected variance rather than minutes (an intraday volatility profile plus the weights of scheduled events), so that a short-dated option can be priced with the Black–Scholes–Merton formula by replacing $\sigma^2 T$ with the variance remaining (23).

Butterfly spread Long one option at each of two outer strikes and short two at the strike midway between them, paying nothing outside the outer strikes and most at the middle one. Because its payoff is never negative, call prices must be convex in the strike; it is also how the curvature of the smile is traded. On a yield curve the name means the same shape across tenors: one sign at two tenors against the opposite sign at a tenor between them, flat under a parallel shift and so a position on the curve's curvature (11, 12, 19, 38, 39).

Buy-out price A fixed price at which an obligation to surrender renewable certificates can be discharged in cash instead. It caps the certificate's price, and an option on the certificate inherits the cap (36).

C

Calendar spread In futures, a position long one delivery month and short another; a bet on the shape of the curve, not its level. In options, a near-dated option sold against a longer-dated one at the same strike, which profits from the near option's faster time decay (5, 12).

Callable bond A bond the issuer may redeem early at a fixed price. The investor is short an option, which is why its spread over the curve overstates the credit compensation; see option-adjusted spread (31).

Cap A strip of interest rate call options, one per period, protecting a borrower against rates above a strike (29).

Capital conservation buffer A layer of common equity of 2.5% of risk-weighted assets above the minimums. A bank that draws on it is not in breach, but its dividends and bonuses are restricted until it is rebuilt (27).

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Capital valuation adjustment (KVA) The cost of the regulatory capital a trade consumes over its life: the equity a bank must hold against the position because the rules require it, not because it chooses to. On an uncollateralized trade it can exceed all the other adjustments combined (9).

Caplet One period of a cap (29).

Capped variance swap A variance swap whose realized volatility is capped, conventionally at 2.5 times the strike for single stocks. It is a variance swap less a call on variance, so unlike the uncapped contract it must be priced with a model (22).

Carry trade Borrowing in a low-rate currency to invest in a high-rate one. Hedged, it earns nothing, by covered interest parity; unhedged, it earns the interest differential and bears the currency risk, and whether that pays is an empirical question rather than an arbitrage (5, 16).

Cash-and-carry Borrowing cash, buying the asset, and carrying it to delivery against a short forward; the arbitrage that caps the forward price. The reverse cash-and-carry sells the asset and buys the forward, and needs either a stock loan or a holder willing to sell (5).

Cash-flow-at-risk The shortfall of a company's operating cash flow over a quarter or a year below its forecast that is exceeded with a stated probability: the variant of VaR a nonfinancial company uses, because its exposure is to cash flow, not to a book's market value (38).

Cash-or-nothing / asset-or-nothing Digital options that pay, if the underlying finishes past the strike, a fixed amount or the asset itself. Their values are the two terms of the Black–Scholes–Merton formula, and a vanilla call is an asset-or-nothing call less K cash-or-nothing calls (34).

CDS spread The running premium rate on a credit default swap: the rate at which the premium leg, paid until a credit event or maturity, has the same present value as the protection leg. CVA uses it as a price, not as an estimate of the probability of default (9, 24, 25).

Central counterparty (CCP) A clearing house that becomes buyer to every seller and seller to every buyer, replacing bilateral exposure with exposure to itself (2).

Certainty-equivalent value The sure amount of cash an employee would accept in place of an option grant. For an undiversified holder it is well below the grant's fair value, a real cost of paying in options that appears in no account (17).

Charm The rate at which an option's delta changes with the passage of time, $\partial^2 f / \partial S \, \partial t$. Negligible for a monthly option; the dominant, clock-driven hedging flow for a same-day option (18, 23).

Cheapest-to-deliver The bond a short futures position will actually deliver, given the conversion factors (6).

Clean price A bond's quoted price, which excludes accrued interest (6).

Clearing member A well-capitalized firm with direct access to a CCP. Everyone else clears through a member as a client, so if the member fails, what a client recovers turns on how its positions and collateral were held; see segregation and porting (2).

Cliquet A strip of forward-starting options. The purest instrument for trading forward volatility, and a capped one is short volatility (34).

Close-out netting The contractual right to terminate all trades with a defaulted counterparty and settle a single net amount. The whole capital framework assumes it is enforceable (1, 7, 27).

CMS spread A coupon paying the difference between two swap rates of different tenors, the building block of steepener notes; its value turns on the correlation between the two rates (30, 31, 32).

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Coherent risk measure A risk measure that is monotonic, positively homogeneous, translation-invariant and subadditive. Expected shortfall is coherent; VaR is not, because it can fail subadditivity (38).

Collar A put bought and a call sold against a holding, the call's premium paying for the put; it fixes a band of outcomes. A zero-cost collar on a currency is a range forward (12, 16).

Collateral Assets posted to secure an exposure. What converts credit risk into liquidity risk (8).

Collateral valuation adjustment (ColVA) The value of the ways a real CSA departs from daily cash collateral that exactly tracks the mark-to-market: thresholds and minimum transfer amounts, non-cash collateral, the option to choose the collateral currency, and the margin period of risk (9).

Collateralized debt obligation (CDO) A tranched claim on a pool of bonds or loans. Named for its collateral: a CLO holds leveraged loans, a CDO-squared holds tranches of other deals (24, 26).

Collateralized loan obligation (CLO) A CDO whose collateral is leveraged corporate loans, actively managed within indenture constraints and protected by coverage tests that divert cash from equity to the senior notes (26).

Common equity tier 1 The highest-quality layer of bank capital: ordinary shares and retained earnings. DVA gains and losses are filtered out of it, and with the perpetual instruments that rank just behind it, it forms the tier 1 capital of the leverage ratio (27).

Complete market A market in which every payoff can be replicated by trading the available assets. An arbitrage-free market is complete exactly when its equivalent martingale measure is unique (the second fundamental theorem), and jumps and stochastic volatility make real markets incomplete (13, 28).

Composite option (compo) An option on a foreign asset whose foreign-currency payoff is converted at the exchange rate prevailing at expiration, so the holder bears both the asset and the currency risk: in effect an option on the asset measured in domestic currency, with volatility $\sqrt{\sigma_S^2 + \sigma_X^2 + 2\rho\sigma_S\sigma_X}$. Contrast the quanto, which converts at a fixed rate (16).

Compound option An option on an option, such as the right to pay a fixed amount at one date for a call expiring at a later one. It has a closed form under Black–Scholes, and its value depends on volatility twice over, through both the outer and the inner option (34).

Conditional value at risk (CVaR) Another name for expected shortfall, used in the optimization literature and as the training objective of deep hedging (38, TN39).

Constant-maturity swap (CMS) A swap one of whose legs pays a long swap rate (the ten-year rate, say), observed afresh at each reset and paid on a single date, rather than across the annuity the rate belongs to. That mismatch is what the convexity adjustment prices (28, 30, 33).

Contango A forward curve sloping upward (3, 5).

Continuity correction Pricing a barrier monitored m times as a continuously monitored one with the barrier shifted away from spot by the factor $e^{\pm 0.5826\,\sigma\sqrt{T/m}}$, the Broadie–Glasserman–Kou result (20, 34).

Continuous compounding The limit of compounding m times a year as m grows without bound, under which an amount A grows to Ae^{Rn} in n years. A mathematical convenience used throughout the option-pricing chapters, not a market convention (4).

Control variate Estimating a quantity together with a related one whose true value is known, by the same method, and correcting the first estimate by the error measured on the second. It works for trees and for Monte Carlo alike (20, 34).

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Convenience yield The benefit of holding a physical commodity rather than a claim on it; what makes a commodity forward curve invert (5, 36).

Conversion factor The exchange-published multiplier making bonds of different coupons roughly interchangeable for futures delivery (6).

Convertible bond A bond the holder may exchange for a fixed number of the issuer's shares, the conversion ratio. A corporate bond plus a call on the issuer's own equity, so the dilution analysis of warrants applies to the call (10, 25).

Convexity Curvature of value in the underlying variable. A source of both hedging error and systematic price adjustment (4, 30).

Convexity adjustment The correction applied when a rate is paid at a date, or settled in a way, other than the one under which it is a martingale: the gap daily margining opens between a futures rate and a forward rate, or a swap rate paid on a single date as in a CMS (6, 30).

Convexity trade Buying interest rate futures and paying fixed on a swap over the same period, collecting the convexity adjustment between the futures rate and the forward rate. It is not riskless: it loses when realized rate volatility exceeds the volatility the adjustment implies (30).

Copula A function that imposes a dependence structure on given marginals: marginals from the market, dependence from an assumption. It is first met in wrong-way risk, built for default correlation as the one-factor Gaussian copula, and used to price tranches (9, 24, 26).

Cornish–Fisher expansion A correction of a normal quantile for skewness and excess kurtosis. It turns the known moments of a delta–gamma P&L into a VaR without simulation, and is reliable when the skew is moderate (38).

Correlation reserve Part of a book's model value held back against the chance that a correlation that cannot be hedged has been misestimated; the standard treatment for quanto positions (16, 30).

Correlation risk premium A persistent gap between the correlation priced into options and the correlation subsequently realized. It arises for the same reason as the volatility risk premium, but far less trading is available to close it (32).

Correlation swap A contract paying realized average pairwise correlation against a strike; unlike a dispersion trade it cannot be replicated from options (22).

Cost of carry The cost of holding an asset: financing, plus storage, minus income. For an investment asset the forward price is the spot price grown at the cost of carry; for a consumption asset the convenience yield is subtracted (5).

Countercyclical buffer A buffer of 0 to 2.5% of risk-weighted assets in common equity that each jurisdiction raises when credit grows fast and releases in a downturn. It does not apply to the leverage ratio (27).

Counterparty credit risk The risk that a derivatives counterparty defaults while owing you money. Distinct from lending risk because the exposure is itself uncertain (27).

Covenant-lite Of a loan: carrying no maintenance covenant, a financial ratio the borrower must keep meeting while the loan is outstanding. Now standard in leveraged loans, so the early warning a covenant breach used to give arrives later (26).

Coverage test A test written into a CLO's indenture that redirects cash automatically as the portfolio deteriorates. The overcollateralization test compares collateral par with the par of a tranche and everything senior to it; on a breach, the equity's interest pays down the senior notes (26).

Covered bond A bank's bond secured on a pool of mortgages that stays on its balance sheet. The holder has dual recourse, to the bank and then to the ring-fenced pool, and no risk is

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transferred (26).

Covered interest parity The relation that sets a forward exchange rate equal to the spot rate adjusted for the two currencies' interest rates, with no reference to any view of where the currency is heading. Violated persistently since 2008; the deviation is the cross-currency basis (5, 7).

Cox process A Poisson process whose intensity is itself random, also called a doubly stochastic Poisson process. The reduced-form model of default: survival is the expectation of $\exp\left(-\int \lambda\right)$ over intensity paths, and because default arrives as a surprise, short-dated spreads do not vanish (24).

Crack spread The refiner's margin: the value of refined products less the crude they are made from, quoted as a 3-2-1 combination of gasoline and heating oil against crude. A refiner is long it and hedges by selling it in futures (36).

Credit conversion factor The share of an off-balance-sheet item, such as an undrawn commitment or a guarantee, that the standardized approach treats as drawn before applying the risk weight (27).

Credit default swap (CDS) A contract paying on a credit event in exchange for a periodic premium (25).

Credit event The contractually defined trigger for a CDS payment. Not the same thing as default, and the gap is Chapter 25's recurring theme.

Credit-linked note A funded CDS: the investor pays principal upfront and loses it on a credit event, so the protection buyer has no counterparty exposure and the investor carries a second exposure to the note's issuer (25).

Credit support The loss a securitization's pool absorbs before a given bond takes any. It comes from subordination (the tranches beneath the bond), overcollateralization (assets in excess of liabilities) and excess spread, the pool interest left after bond interest and fees (26).

Credit support annex (CSA) The schedule to the ISDA master agreement under which the parties post collateral against their exposure to each other. It sets eligible collateral, thresholds, haircuts and the rate paid on collateral, and through that rate it fixes the curve on which the trade is discounted (1, 7, 8).

Credit triangle The relation $s = \lambda(1 - R)$ between spread, hazard rate, and recovery; exact for a continuously paid premium (24).

Credit valuation adjustment (CVA) The market value of counterparty credit risk on a derivatives portfolio (9).

Credit VaR A high quantile of a credit portfolio's one-year loss minus its expected loss, which is priced and provisioned rather than held as capital. In a large portfolio it is set almost entirely by default correlation (24, 39).

Critical stock price The stock price at each date below which immediate exercise of an American put is optimal; it rises toward the strike as expiration approaches. Traced through time it is the early-exercise boundary, the free boundary of the pricing equation (11, 13, 15).

Cross-currency basis The spread on a currency swap that covered interest parity says should be zero and is not; in practice it prices the relative scarcity of funding in the two currencies (7, 33).

Cross-currency swap A swap exchanging payments in two currencies. Unlike a single-currency swap, the notional principals are exchanged, at the start and again at maturity; the spread on its floating legs is the cross-currency basis (7, 33).

Cross-gamma The part of a position's response that comes from two variables moving together.

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For CVA, exposure depends on rates and currencies while the charge depends on credit, so hedging the credit sensitivity alone leaves the interaction exposed (9).

Cross hedge A hedge using futures on an asset other than the one being hedged. A one-for-one position is then generally not optimal, and the minimum-variance hedge ratio applies (3).

CVA capital charge Regulatory capital held against the risk that CVA itself moves as credit spreads change, which was where most counterparty losses in the 2008 crisis came from. Computed by the basic approach (BA-CVA) or the sensitivity-based standardized approach (SA-CVA), and part of KVA (9, 27).

D

Dark spread The coal-fired counterpart of the spark spread: the power price less the cost of the coal burned to produce it (36).

Debit valuation adjustment (DVA) The mirror image of CVA: the value to a dealer of its own option to default, which rises as its own credit deteriorates. Recognized in accounting, filtered out of regulatory capital (9, 27).

Default A counterparty's failure to meet its obligations on any of its contracts, or its bankruptcy or insolvency. It belongs to the counterparty, not to a trade, and reaches all its liabilities at once through cross-default, acceleration and bankruptcy; creditors differ only in what they recover. For derivatives, an event of default under the ISDA Master Agreement. A legal event, not a market price: a downgrade is not a default (9).

Default fund A pool to which every clearing member contributes, held to cover losses larger than a defaulter's margin can absorb. In the default waterfall the defaulter's own contribution is used first, then the CCP's own capital, then the surviving members' contributions (2).

Default point The asset value at which a firm defaults: the face value of its debt in Merton's model. Distance to default is measured from it (24).

Default risk charge The part of the trading-book capital charge for jump-to-default on traded bonds, credit derivatives and equities, which spread sensitivities cannot see. Offsets between different issuers are recognized only in part (39).

Degree day The number of degrees by which a day's average temperature falls below (heating) or rises above (cooling) 65°F, summed over a month or season. The index on which most weather derivatives settle (1, 36).

Deliverable basket The set of bonds a short Treasury futures position may deliver, defined by a maturity range on the delivery date, with conversion factors putting them on a comparable footing. The short's choice within it is what makes one bond cheapest to deliver (6).

Delta The rate of change of an option's value with the underlying; the hedge ratio (18).

Delta hedging Holding delta units of the underlying against an option so the position is locally insensitive to price. The strategy the Black–Scholes–Merton model assumes (15, 18).

Detachment point The loss level at which a tranche is exhausted (25, 26).

Difficulty adjustment The protocol rule that resets how hard a block is to mine, at intervals, so that blocks arrive on schedule however much computing power is running (37).

Digital option An option paying a fixed amount or nothing. Has no finite replicating portfolio, so it is quoted at the cost of an overhedge (12, 34).

Dilution The fraction of a company passing to option or warrant holders on exercise. Measured by share count, by the grant's value against market capitalization, or, in the accounts, by the incremental shares of the treasury stock method; the three differ (17).

Dirty price The amount actually paid for a bond: the clean price plus accrued interest. Also called the cash price (6).

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Discount curve The curve used to present-value a trade's cash flows. Determined by the collateral agreement rather than by the rate the trade references (7, 8, 28).

Dispersion trade Short index volatility against long single-stock volatility; a position in implied correlation (22).

Displaced tree A tree for a stock paying a known cash dividend, built on the stock price less the dividend's present value, which recombines, with that present value added back at every node before the dividend date (20).

Distance to default How many standard deviations of asset value separate a firm from its debt; the durable output of a structural model, and the input a commercial implementation maps to a default frequency (24).

Dividend equivalent Cash or extra units matching the dividends paid on the shares while restricted stock units are unvested. Without them, an RSU's grant-date value is the share price less the present value of the dividends forgone (17).

Duration The sensitivity of a bond's price to a parallel yield shift, to first order (4).

DV01 The change in a position's value for a one basis point move in rates; the same measure as PV01, named for the currency it is reported in (8, 29).

Dynamic hedging Holding a position and trading repeatedly to restore its hedge as the Greeks drift. It becomes impossible where gamma diverges (near a digital's strike at expiration, or at a barrier option's barrier), and a static hedge, options bought once and held, replaces it (18).

E

Economic capital The capital a bank's own model says it needs to survive its worst year at a chosen confidence, typically 99.95% or more for credit, and allocated to loans and businesses to price them (24).

Effective expected positive exposure In the internal model method for counterparty capital, the running maximum of expected exposure averaged over the first year, treated as a loan-equivalent amount (27).

Effective federal funds rate (EFFR) The rate at which US banks lend reserves to one another overnight, published by the New York Fed. Not the benchmark, because volume collapsed after 2008, but it is the policy rate, and the fed funds–SOFR basis between it and SOFR trades through swaps (4).

Effective notional In SA-CCR, a trade's notional scaled by its supervisory delta (its direction, or an option's delta), its supervisory duration (an annuity at a fixed 5%) and its maturity factor. Within a hedging set the three maturity buckets are aggregated with correlations of 0.7 between adjacent buckets and 0.3 between the outer two, so offsets across buckets are only partial; multiplied by the supervisory factor, the result is the add-on (27).

Elasticity (of an option) The percentage change in an option's value for a 1% change in the price of the underlying, $\Omega = S\Delta/f$: the option's leverage. Largest for out-of-the-money options and negative for puts; for equity viewed as a call on the firm it is $N(d_1)V/E$ (18, 24).

Elicitability The property of a risk measure that some scoring function is minimized, in expectation, by its true value, so that competing forecasts can be ranked by average score. VaR has it; expected shortfall does not on its own, but VaR and ES are elicitable jointly (38).

Elliptical distribution A joint distribution under which every portfolio's loss has the same shape, differing only in location and scale; jointly normal and jointly Student's t losses are the standard cases. On such losses VaR is subadditive at any confidence level above one half (38).

Emissions allowance A permit to emit one metric ton of carbon dioxide under a cap-and-trade scheme. Bankable, storable at no cost and impossible to borrow, so its forward price is capped

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at spot carried at the financing rate but can sit below it, and its price is the marginal cost of abatement (1, 36).

Equilibrium model / no-arbitrage model An equilibrium short-rate model produces today's term structure as an output of its assumed process; a no-arbitrage model makes the drift time-dependent so that today's curve is an input, reproduced exactly, as a trading desk requires (31).

Equity / mezzanine / senior tranche The equity tranche takes a pool's first losses, the senior tranche suffers only after everything below it is exhausted, and the mezzanine tranches lie between. The equity tranche gains from correlation, and the senior tranche is short it, losing as correlation rises. A first-loss tranche is the equity tranche widened to detach at a chosen point (24, 25, 26).

Equivalent martingale measure For a chosen numeraire, a measure equivalent to the real-world one (agreeing on which events are impossible) under which every traded asset's price divided by the numeraire is a martingale. The risk-neutral measure is the one for the money market account (28).

Estimated initial value A structured note's fair value net of fees, which some jurisdictions require the issuer to print on the term sheet. It turns an argument about fairness into a number a buyer can compare across issuers (35).

Euler scheme Stepping a stochastic differential equation by its increments rather than by its solution. Exact only when drift and volatility are constant, as for arithmetic Brownian motion; available for any process; and it can drive a positive process below zero (20).

European Market Infrastructure Regulation (EMIR) The EU regulation that mandates central clearing of standardized OTC derivatives, margin on uncleared trades and trade reporting. Its 2024 revision requires EU firms above the clearing threshold in euro and zloty interest rate derivatives and euro short-term interest rate derivatives to hold an active account at an EU CCP and clear a minimum number of trades through it (2).

Event contract A contract paying a fixed amount if a stated event occurs and nothing otherwise: an inflation release, a central-bank decision, an election. A digital on an outcome no option spans, so it is priced as a probability rather than by replication (34).

Event of default Under the ISDA Master Agreement, a failure to pay or deliver not cured within the grace period, bankruptcy or insolvency, or an agreed cross-default, among others. It entitles the other party to terminate and close out every trade under the agreement (7, 9).

Excess kurtosis The fourth standardized moment less three, $\mathbb{E}[(X-\mu)^4]/\sigma^4 - 3$: zero for a normal distribution and positive when the tails are heavier. With skewness it is one of the two corrections in the Cornish–Fisher expansion (38).

Ex-dividend date The first day on which a buyer of a stock is no longer entitled to the coming dividend. The price falls by about the dividend, which can make exercising an American call the day before rational (10, 11).

Exchange option An option to give up one asset and receive another, valued by Margrabe's formula with the asset given up as numeraire. No interest rate enters, and the correlation between the two assets is what the value turns on (28, 34, 36).

Exercise Taking up an option's right. The decision, not the contract (10, 11).

Exotic option Any option that is not a standard call or put. The useful test is not complexity but whether the payoff can be replicated by traded instruments (34).

Expected default frequency (EDF) A default probability read from an empirical table indexed by distance to default, rather than from the normal tail the structural model implies (24).

Expected exposure (EE) / expected positive exposure (EPE) Expected exposure is the av-

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erage exposure across simulated scenarios at each future date, the average of $\max(V(t), 0)$; expected positive exposure is its time average, a single number: over the life of the trade as a summary of the profile, and over the first year in the capital rules, whose effective version averages the running maximum. CVA is priced on expected exposure (9, 27).

Expected shortfall (ES) The average loss conditional on exceeding the VaR level. Coherent where VaR is not (38).

Explicit / implicit scheme The two basic finite difference schemes. The explicit scheme reads each node directly from the next time step and is a trinomial tree in disguise; the implicit scheme solves a tridiagonal system at each step. Crank–Nicolson averages the two and converges faster (20).

Exponentially weighted moving average (EWMA) A volatility estimate that weights past squared returns geometrically, $\sigma_n^2 = \lambda\sigma_{n-1}^2 + (1-\lambda)u_{n-1}^2$, with $\lambda = 0.94$ the RiskMetrics convention. It is GARCH(1,1) with no long-run level to revert to, so its forecast is flat at every horizon (38).

Exposure What a party would lose if its counterparty failed now, before recovery: the positive part of the portfolio's value, net of collateral. Not the notional, and not the value, which can be negative (9, 27).

Exposure at default (EAD) The regulatory measure of what is at risk when a counterparty fails (27).

Extreme value theory The theory of a distribution's extremes. Its central result, that losses beyond a high threshold follow a generalized Pareto distribution for almost any underlying law, lets a tail be fitted from a sample's worst observations (peaks over threshold) and extrapolated to quantiles the sample cannot reach. The fit assumes the tail's shape continues beyond the data, and the threshold is the analyst's choice (38).

F

Fair variance swap rate The strike at which a variance swap is worth zero today, the risk-neutral expectation of realized variance. Realized variance is replicated by a static portfolio of out-of-the-money options weighted by $1/K^2$ together with a dynamic position in the underlying, and the fair rate is the value of that strip grown at the risk-free rate; the construction behind the VIX (19, 22).

Feller condition The condition $2ab > \sigma^2$ under which the Cox–Ingersoll–Ross short rate stays strictly positive; when it fails, zero is reached and acts as a reflecting boundary (31).

Filtered historical simulation Historical simulation that divides each past return by that day's GARCH volatility and draws the standardized shocks at random through the GARCH recursion to build scenarios, due to Barone-Adesi, Giannopoulos and Vosper. It goes beyond volatility-scaled historical simulation by modeling how volatility evolves over the horizon (38).

Finite difference method Solving a derivative's pricing equation on a grid of prices and times, replacing its derivatives by differences and stepping back from expiration. The natural method for early exercise and for revaluing a whole book across a grid (20).

Flat price The futures price itself, as distinct from the basis. A hedge removes the flat-price risk and leaves the basis risk, and the two should be reported separately (36).

Flat / spot volatility A flat volatility is the single volatility which, applied to every caplet in a cap, reproduces the cap's price (a quotation device); a spot volatility, also called a caplet or forward volatility, is the volatility of one caplet. Spot volatilities are bootstrapped from flat ones, as zero rates are from par rates (29).

FLEX option An exchange-listed option whose strike, expiration and exercise style the parties

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choose, combining over-the-counter customization with a clearing house (10).

Flexi cap A cap on which only a limited number of caplets pay, the first that finish in the money. Path-dependent through the count, so it needs a term structure model (32).

Floor A strip of interest rate put options, protecting a lender against rates below a strike (29).

Forecast curve The curve from which forward rates are projected, distinct since 2008 from the curve used to discount them (7, 32).

Forward contract An agreement to trade an asset at a set price on a future date. Settled once, at maturity (1, 5).

Forward curve The settlement prices of successive delivery months plotted against those months, showing what the market will pay for the asset at every date it trades. Used by firms that never trade a contract. Its shape reflects the cost of carry and, for consumption commodities, a convenience yield (2, 5).

Forward freight agreement A forward fixing the cost of carrying a cargo on a given shipping route, settled against a published route index. Ship owners and charterers have the opposite exposures that let the market work (1, 36).

Forward points The difference between the forward and the spot exchange rate, in units of the last decimal place of the spot quote, which the client adds to spot to get the outright rate. Quoted in place of outright rates because they move far more slowly (5).

Forward price The delivery price that makes a forward contract worth zero at inception; it is not a forecast of the spot price, but the spot price carried to the delivery date at the cost of financing and holding the asset (1, 5).

Forward rate agreement (FRA) A forward contract on an interest rate (4, 6).

Forward-start option An option whose strike is set at a future date. Trivial under Black–Scholes and heavily model-dependent in practice, because it prices the forward smile (34).

Forward-starting variance swap A variance swap on a future window, replicated by a long position in a longer spot-starting variance swap and a short position in a shorter one, weighted by maturity. So the whole forward variance curve is spanned by today's vanilla surface (22).

Forward variance curve Today's price of the variance to be realized at each future time (forward, not spot, variance), read off the term structure of variance swap rates, the forward variance over an interval coming from two spot-starting swaps. What a stochastic volatility model has to reproduce, and the input the rough volatility models specify directly (21, 22).

Fractional Brownian motion A continuous Gaussian process whose variance over a horizon T is T^{2H}; increments are independent only at $H = 1/2$, where it is the Wiener process. Rougher than a Wiener path below one-half, and the model for volatility in rough volatility (14, 21).

Front-end protection In an index credit default swaption, the value of losses accumulated between trade date and expiration, owed to the protection buyer because the option does not cancel on default (28).

Full revaluation / partial simulation The two ways of computing Monte Carlo VaR: full revaluation reprices every position with its own model in each simulated scenario; partial simulation applies a delta–gamma approximation instead, which is faster and inherits that approximation's limits (38).

Fundamental Review of the Trading Book (FRTB) The Basel Committee's market risk capital standard, final in 2019: a prescriptive trading-book boundary, a sensitivities-based standardized approach, and internal models based on 97.5% expected shortfall with liquidity horizons, approved desk by desk (39).

Funding rate The periodic payment holding a perpetual futures contract in line with spot (37).

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Funding valuation adjustment (FVA) The cost of funding an uncollateralized derivatives position (9).

Futures contract An exchange-traded forward, standardized and settled daily (2).

Futures price The price at which a futures contract trades. It equals the forward price when rates are deterministic, and differs by a covariance between the asset and the financing rate when they are not (2, 5).

Futures rate The interest rate implied by an interest rate futures price, 100 less the quotation. Daily settlement makes it a martingale under the risk-neutral measure, so it exceeds the corresponding forward rate by the convexity adjustment (6, 30).

Futures-style margining Margining an option like a futures contract, with the premium posted and marked daily rather than paid up front. Black's model then loses its discount factor (16).

G

G2++ model A two-factor short-rate model in which the rate is the sum of two zero-mean, mean-reverting Gaussian factors and a deterministic shift fitted to today's curve. Rates at different maturities are then imperfectly correlated, so the curve can steepen and flatten (31).

Gamma The rate of change of delta with the underlying; the curvature a delta hedge cannot capture (18).

Gamma band The range of strikes over which an option's gamma exceeds half its at-the-money value. It narrows from several percent of spot a month before expiration to under half a percent in the final hour, which concentrates a same-day book's gamma at the current price (23).

General collateral / special A general collateral repo is one in which the cash lender accepts any security from an agreed class. A security in particular demand trades special, at a repo rate below general collateral and occasionally near zero, because the cash lender is giving up interest to borrow that security (4).

Generalized autoregressive conditional heteroskedastic (GARCH) model A volatility model; GARCH(1,1) updates the variance as $\sigma_n^2 = \omega + \alpha u_{n-1}^2 + \beta \sigma_{n-1}^2$. The constant ω gives a long-run variance $V_L = \omega/(1 - \alpha - \beta)$ toward which the forecast reverts, which is what produces a volatility term structure; EWMA is the special case $\omega = 0$, $\alpha + \beta = 1$ (38).

Generalized Pareto distribution The law that the excesses over a high threshold of almost any loss distribution approach, with a scale and a shape parameter; the shape is positive for a power-law tail, zero for an exponential one, negative for a tail with an end point (38).

Generalized Wiener process A process $dx = a\,dt + b\,dW$ with constant drift a and a constant scale b on the Wiener noise; the $b\,dW$ term is the diffusion. Over a horizon T the change is normal with mean aT and variance b^2T (14).

Generator matrix The matrix Λ of rating-transition intensities with $P = e^{\Lambda}$, from which the transition matrix for any horizon is $e^{t\Lambda}$; the hazard rate extended from one exit to many (24).

Geometric average The nth root of the product of n fixings. The geometric average of lognormal prices is lognormal, so an option on it has a closed form, which makes it the control variate for the arithmetic Asian option (20, 34).

Geometric Brownian motion The process $dS = \mu S\,dt + \sigma S\,dW$, in which the proportional return has a constant expected rate and a constant volatility. It makes the stock price lognormal and is the model underneath the Black–Scholes–Merton formula (14, 15).

Girsanov's theorem The result that a change of measure changes a Wiener process's drift and leaves its volatility alone. Moving to a new numeraire, every process picks up a drift equal to the covariance between its logarithm and the log of the numeraire ratio, which generates the convexity, timing and quanto adjustments (28, 30).

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Global systemically important bank (G-SIB) A bank designated by size, interconnection and complexity, which holds an extra 1% to 3.5% of common equity against risk-weighted assets and half that again on its leverage requirement (27).

Gold lease rate What a holder earns by lending gold to a bullion bank, quoted as the financing rate less the forward premium on the metal. Its small size is what an essentially zero convenience yield looks like in a real market (5).

Greeks The sensitivities of an option's value to its inputs (18).

Gross basis / net basis For a deliverable bond, the gross basis is its quoted price less the futures price times its conversion factor, and the bond that minimizes it is the cheapest-to-deliver. The net basis subtracts carry, coupon income less repo financing, and measures the cost of the short's delivery options (6).

H

Haircut The discount applied to collateral's market value to allow for its own price risk (8).

Hang Seng China Enterprises Index (HSCEI) An index of mainland Chinese companies listed in Hong Kong. Its high volatility supported high coupons, and a large share of Korean retail autocallables referenced it (35).

Hash rate The computing power all miners together apply to producing blocks; a miner's share of it is its share of new coin (37).

Hazard rate The probability of defaulting in the next instant, given survival to it, per unit of time. A rate rather than a probability: the proportional rate at which the survival probability declines (24).

Heat rate The gas, in MMBtu, a power plant burns to produce one MWh. The spark spread is the power price less the heat rate times the gas price (36).

Heath–Jarrow–Morton (HJM) drift condition The no-arbitrage requirement that fixes the drift of every forward rate once its volatility functions are chosen: for each factor, the volatility times its integral out to the forward's maturity, summed over the factors. Short-rate models are special cases of the framework it defines (32).

Hedge accounting Accounting that lets a firm which documents a hedging relationship in advance and shows it to be effective defer the derivative's gains and losses (for a cash flow hedge of a forecast transaction, in other comprehensive income) until the hedged item affects profit, so the two sides meet in the same period (3).

Hedge ratio The size of the futures position relative to the exposure. The minimum-variance hedge ratio, $h^* = \rho\,\sigma_S/\sigma_F$, minimizes the variance of the hedged position; it is not one in general, and least of all in a cross hedge (3).

Hedging set In SA-CCR, a group of trades within an asset class whose effective notionals may offset: for interest rates, one per currency, divided into maturity buckets of under one year, one to five years and over five years (27).

Historical / hypothetical scenario The two sources of stress scenarios: the moves of an actual episode replayed on today's positions, or moves written by the firm for the risks it holds, which should change correlations and liquidity as well as prices (38).

Historical simulation Estimating VaR or expected shortfall by applying each of the last N days' market moves (500 is standard) to today's portfolio and reading the measure off the resulting profits and losses. It assumes no distribution, so fat tails and joint behavior come for free, but it cannot produce a loss larger than any in its window and reports a small number after a calm period (38).

Hurst parameter The exponent H that sets how the variance of fractional Brownian motion

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grows with the horizon, T^{2H}; one-half for the Wiener process, and about 0.1 for realized volatility across equity indices and liquid futures markets (14, 21).

Hypothetical / risk-theoretical P&L A desk's daily profit on yesterday's positions, computed by full front-office revaluation (hypothetical) and using only the risk factors in the risk model (risk-theoretical). The P&L attribution test compares the two (39).

I

Implied correlation The single correlation that, with the constituents' implied volatilities, reproduces an index's implied volatility; for a tranche, the correlation that reproduces its price. Demand for index protection keeps it above realized correlation, which is what a dispersion trade sells (19, 22, 25).

Implied repo rate The annualized return a cash-and-carry trade earns: buy the asset, finance it, and deliver it into the futures contract. Comparing it with the rate at which the position can actually be financed is how practitioners test a futures price (5, 6).

Implied volatility The volatility that makes the Black–Scholes–Merton formula reproduce a market price. It reflects the market's view of future volatility together with a risk premium, and across strikes and maturities it serves as a quoting convention rather than as a model of the stock (15, 19).

In-arrears swap A swap paying a rate at the start of the period it applies to; the classic convexity adjustment, and one the benchmark transition largely removed (33).

Incentive stock option (ISO) A US employee option carrying favorable tax treatment: no ordinary income at exercise, but the spread is an alternative minimum tax preference, and holding periods apply to keep the treatment. The AMT is what creates the exercise-and-hold trap: a tax bill on a paper gain in shares that cannot yet be sold (17).

Incremental VaR The change in a portfolio's VaR when one position is removed entirely. It answers whether to keep a whole position; unlike component VaR, the incremental figures do not sum to the total (38).

Index CDS A standardized CDS on a fixed portfolio of reference entities (25).

Index factor The fraction of an index CDS's original notional that remains after constituents have suffered credit events and dropped out: $124/125$ after one event in an index of 125 issuers. The coupon is paid on the reduced notional (25).

Index skew The difference between the level at which an index CDS trades and the intrinsic level computed from its constituents. It widens when the index is used heavily as a macro hedge (25).

Initial forward curve The instantaneous forward rates implied by today's discount factors, $f(0,t) = -\partial \ln P(0,t)/\partial t$; what a no-arbitrage short-rate model fits its time-dependent drift to (31).

Initial margin Collateral posted at the outset against potential future exposure, as distinct from variation margin against current exposure (2, 8).

Internal models approach (IMA) Computing trading-book capital from a bank's own expected shortfall model, permitted desk by desk once the desk passes backtesting and the P&L attribution test (39).

Internal ratings-based approach (IRB) Computing banking-book credit capital from the bank's own estimates of probability of default and, in the advanced version, loss given default, put through a Basel formula that is the worst-case default rate at 99.9% with a prescribed correlation and a maturity adjustment (39).

International Money Market (IMM) dates The third Wednesdays of March, June, Septem-

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ber and December, between which three-month SOFR futures settle. Swaps traded on these dates can be hedged with futures without date mismatches (6, 7).

Intrinsic value What an option would be worth if exercised now (10).

Inverse VIX product An exchange-traded note or fund delivering minus one times the daily return of a VIX futures index: a packaged short position in the front VIX futures that collects the roll carry and must buy futures into a rally to rebalance (22).

Investment asset / consumption asset An investment asset is held by enough investors purely for investment that they are indifferent between it and a claim on it: they will sell it, or lend it, when the forward is out of line, so arbitrage holds the forward price from both sides. A consumption asset is held to be used, so its holders will not part with it, only the upper bound survives, and the gap is filled by a convenience yield (5).

Investment grade / speculative grade Ratings of BBB (Baa) and above are investment grade; below that, speculative grade. A downgrade across the line forces sales by mandate-constrained holders and can trigger collateral calls (24).

Iron condor A put spread below spot and a call spread above it, sold together, which collects premium if the underlying stays within a range. Systematic selling of such structures leaves dealers long gamma in same-day options (12, 23).

ISDA master agreement The document under which OTC derivatives between two parties form a single legal contract. The basis for close-out netting (1).

Itô integral The integral $\int b\,dW$ against a Wiener process, defined as the limit of sums with the integrand evaluated at the start of each interval, before the increment is revealed. That choice makes it a martingale (14).

Itô process A process $dx = a(x,t)\,dt + b(x,t)\,dW$ whose drift and volatility may depend on its current value and on time; the form taken by the book's diffusion models, which the jump and rough volatility models of Chapter 21 go beyond (14).

Itô's lemma The chain rule for functions of a stochastic process (14).

J

Jamshidian's decomposition Pricing a European swaption in a one-factor short-rate model as a portfolio of options on zero-coupon bonds, each struck at the bond's value at the rate that makes the swap worth par. An identity, not an approximation, but it fails with a second factor or a Bermudan schedule (31).

K

KMV The standard commercial implementation of Merton's structural model, built by the firm Kealhofer, McQuown and Vasicek founded and now part of Moody's Analytics. It sets the default point at short-term debt plus half of long-term debt, and maps distance to default to an expected default frequency through an empirical table rather than the normal tail (24).

Knock-in / knock-out A barrier option created, or extinguished, on touching the barrier. Related by in-out parity (34).

L

Large-pool loss distribution The distribution of a large homogeneous portfolio's loss rate under the one-factor Gaussian copula. It underlies the Basel capital charge and shows why a senior tranche is a statement about correlation (26).

Law of iterated expectations Averaging a later conditional expectation over what is known earlier gives the earlier one: $E[E[X \mid \mathcal{F}_s] \mid \mathcal{F}_t] = E[X \mid \mathcal{F}_t]$ for $t \le s$. Also called the tower property (28).

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Legging risk The risk that the market moves between the fills of a multi-leg option strategy executed one leg at a time, which can exceed the value of the spread itself. Exchanges run complex order books that match the legs as a single package (10, 12).

Leptokurtosis More probability mass in the center and in the tails than a normal distribution carries. The equity smile is what it looks like when measured from option prices rather than assumed (19).

Leverage A position's notional divided by the capital posted against it; for one unit of a futures contract, the futures price divided by the margin on it, or one over the margin rate (1, 37).

Leverage function In a local-stochastic volatility model, the function of spot and time that multiplies the stochastic variance so that the vanilla surface is reproduced exactly; the square root of local variance divided by the conditional expectation of the stochastic variance (21).

Leverage ratio Tier 1 capital divided by total exposure (assets, derivative exposure, securities financing and off-balance-sheet items), with no risk weighting. A deliberately risk-insensitive constraint, so every trade consumes capital whatever its quality (27).

Liability-driven investment (LDI) A mandate that hedges a pension scheme's liabilities rather than tracking an asset benchmark, typically with leveraged rate exposure. The strategy at the center of the 2022 UK gilt episode (8).

LIBOR The interbank offered rate that anchored derivatives markets until its replacement by risk-free rates (4).

Likelihood-ratio method Estimating a sensitivity by differentiating the probability density rather than the payoff; handles a discontinuous payoff, at the cost of higher variance (20, 32).

Liquidation preference Preferred shareholders' right to be paid back first when a company is sold or wound up. Because employees' common stock sits behind it, a private company's 409A price is typically well below the price of its latest preferred round (17).

Liquidity horizon The time a regulator assumes it takes to exit or hedge a risk factor, from 10 to 120 days by class. Expected shortfall is scaled across the horizons, so a desk's capital depends on how its factors are classified (38, 39).

Local volatility A model in which volatility is a deterministic function of spot and time; fits any surface exactly and gets the forward smile wrong (21).

Lockout period The initial period of an autocallable during which the note cannot be called (35).

Log contract The payoff $-2\log(S_T/F)$; delta-hedging it earns realized variance on every path, and it is replicated by a strip of options weighted $1/K^2$, which is why variance swaps are replicable (22).

Lognormal distribution The distribution of a variable whose logarithm is normal. Geometric Brownian motion makes the terminal stock price lognormal, and a binomial tree of small independent multiplicative moves converges to it (13, 14).

Long / short gamma A book long gamma gains from a move in either direction and hedges by selling rallies and buying dips, damping the move; a book short gamma, having sold options, hedges by trading with the move and amplifies it (18, 23).

Long-term equity anticipation securities (LEAPS) Listed options with expirations up to about three years on single stocks and five on the S&P 500 (10).

Lookback option An option on the maximum or minimum of the path. Taught more than it is traded (34).

Loss given default The fraction of a claim a creditor loses when the borrower defaults, $1 - R$. Conventionally 60% for senior unsecured debt, and not separable from the default probability

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in what a CDS spread observes (9, 24).

M

Mahalanobis distance The distance of a joint move $\mathbf{r}$ from the center of a distribution with covariance Σ, $\sqrt{\mathbf{r}^\top \Sigma^{-1} \mathbf{r}}$, counted in standard deviations along whatever direction the move takes. Under normality the density falls as it grows, so the most plausible scenario producing a given loss is the one that minimizes it (38).

Maintenance margin The balance below which a futures account triggers a margin call, set below the initial margin; the trader must then restore the account to the initial level (2).

Margin call A demand for additional collateral: on a futures account, when the balance falls below the maintenance level; under a CSA, when one party's valuation of the portfolio says the exposure has grown (2, 8).

Margin period of risk The assumed time between a counterparty's last margin payment and the close-out of its positions, conventionally five days for cleared trades and ten for bilateral ones (8, 27).

Margin valuation adjustment (MVA) The cost of funding initial margin over the life of a trade. Initial margin is posted gross, segregated and cannot be reused, so it must be funded from inception to maturity (8, 9).

Marginal / component VaR Marginal VaR is the derivative of VaR with respect to a position, $\partial \mathrm{VaR}/\partial w_i$; component VaR is the position times its marginal VaR. Because VaR is homogeneous of degree one in the positions, the components sum exactly to the total (an Euler allocation), though that does not make VaR subadditive (38).

Margrabe formula The closed form for an option to exchange one asset for another. Contains no interest rate, because the second asset is the numeraire (28, 34).

Market agreed coupon (MAC) A standardized swap whose fixed coupon is a round number in 25 basis point increments, the difference from the par rate being paid upfront. One product among many, not the general form of a swap: it buys fungibility and therefore cheaper compression and clearing, at the cost of cash at inception (7).

Market price of risk The excess return per unit of volatility, $(\mu - r)/\sigma$, that no-arbitrage forces to be the same for every asset driven by one source of uncertainty. It is unique only when the traded assets are at least as numerous as the sources of uncertainty they must hedge (28).

Marking to market Revaluing a position at current prices, and for futures, settling the change in cash daily (2).

Markov process A process in which only the present value matters for predicting the future; the path that led there carries no information. Assuming stock prices are Markov is a statement of weak-form market efficiency (14).

Martingale A process with no drift under a given measure. Choosing the measure that makes the right quantity a martingale is the technique of Chapter 28.

Maturity adjustment The factor in the IRB formula that raises the capital on a longer loan for the downgrade risk it carries before it can default. It equals one at a one-year maturity (39).

Maturity factor In SA-CCR, the scaling of a trade's effective notional to the horizon over which it is exposed: the square root of its remaining maturity, capped at one year, when unmargined, and a function of the margin period of risk when margined (27).

Maximum likelihood Estimating a model's parameters as those under which the observed data are most probable, by maximizing the logarithm of their joint density. It fits the generalized Pareto tail and the GARCH parameters in Chapter 38 (38).

Mean reversion The tendency of a rate to be pulled toward a long-run level, which distinguishes

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rates from equities. Most short-rate models have it; Ho–Lee does not, which is what rules it out in practice (31).

Minimum transfer amount The smallest collateral call a CSA permits; below it no transfer is made. With the threshold, it is what leaves an exposure uncollateralized between calls (8, 27).

Mixing parameter The scale on the volatility of volatility in a local-stochastic model, between zero (pure local volatility) and one; every setting fits the vanillas, so it measures model risk rather than fit (21).

Model risk The risk that the model mapping market factors to a position's value is wrong. It is not a subset of market risk, because an error in the model moves the measure and the position together (21, 39).

Modified duration Duration divided by $1 + y/m$: the sensitivity of a bond's price to its yield when the yield is quoted with compounding frequency m rather than continuously (4, 29).

Moneyness How far in or out of the money an option is; several conventions exist and they are not interchangeable (19).

Monte Carlo VaR VaR and expected shortfall read off losses simulated from the joint distribution of the risk factors. The only reliable route for strongly nonlinear or path-dependent books, at the cost of sampling error (38).

Multi-curve framework Discounting on one curve and projecting forwards on another, as practice has required since 2008 (7).

Multiple For a catastrophe bond, the ratio of its spread to its modeled expected loss. It falls as the expected loss rises, so remote risks carry proportionally more compensation (36).

Multiplier In SA-CCR, a factor between 0.05 and one that reduces potential future exposure when collateral exceeds the current exposure or the netting set is out of the money; it equals one otherwise (27).

N

Naked option An option written without a position in the underlying that covers it, such as a call sold without owning the shares. A naked call writer's exposure is unlimited and a naked put writer's is bounded by the strike; both must be collateralized with margin (10, 18).

Natural measure / required measure A variable's natural measure is the one under which its forward value is an unbiased expectation, set by the numeraire in its definition; the required measure is the one the contract's payment date and currency call for. Every convexity, timing and quanto adjustment is the covariance that moving from the first to the second introduces (30).

Negative convexity The property of a security whose price gains less in a rally than it loses in a sell-off, as a mortgage pass-through's does because borrowers prepay when rates fall. Hedging it means trading in the direction the market is already moving (6, 26).

Net open interest In a CDS auction, the amount of bonds still wanted for purchase or offered for sale once offsetting physical settlement requests are matched. The auction's second stage clears it, and the clearing price is the final recovery rate (25).

Netting opinion A legal opinion, commissioned by the industry for each jurisdiction, that close-out netting will be upheld in a counterparty's insolvency there. Where netting is not enforceable, exposure and capital revert to the gross figure (27).

Netting set The trades with one counterparty that a single enforceable netting agreement covers. The unit at which exposure is measured (8, 27).

No free lunch with vanishing risk (NFLVR) The strengthening of "no arbitrage" that is equivalent, in continuous time, to the existence of an equivalent local martingale measure (28).

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Non-modellable risk factor (NMRF) A risk factor that fails the risk factor eligibility test for lack of real price observations, and is capitalized by a stress scenario added to the modeled charge rather than diversified with it (39).

Non-qualified option (NSO) The ordinary US employee option, taxed as ordinary income on the spread at exercise and withheld through payroll. No alternative minimum tax complication and no holding period, so the choice against an incentive stock option is a genuine trade-off rather than one dominating the other (17).

Normal (Bachelier) volatility The standard deviation of a rate's change measured in basis points, as in Bachelier's model, rather than of its percentage change. The standard quote for rate options since policy rates went negative; roughly the lognormal volatility times the rate level (4, 19, 29).

Notional The reference amount on which payments are computed. Not an amount anyone owes or can lose, but not meaningless either: it scales operational load and is the base for several regulatory measures (1).

Novation The replacement of a contract between two parties by two contracts, each with a central counterparty. It is what allows a futures position to be closed out against anyone and makes contracts fungible whatever the identity of the seller (2). Bilaterally, a third party stepping into one side of an existing trade, which keeps the trade but changes the counterparty and so requires the remaining party's consent (33).

Numeraire The asset in whose units prices are quoted. Changing it is the central technique of Chapter 28.

O

One-factor Gaussian copula The standard copula for default correlation: each borrower's latent variable is $\sqrt{\rho}\,M + \sqrt{1-\rho}\,Z_i$, with M a common factor, and the borrower defaults when it falls below the threshold set by its own default probability. Conditional on M the defaults are independent, which is what makes the large-pool loss distribution tractable (24, 26).

One-touch / no-touch option A one-touch pays a fixed amount if the underlying trades at a level at any time before expiration; a no-touch pays it if the level is never reached. Standard dealer products in foreign exchange (34).

Open interest The number of contracts outstanding. Measures what is held, where volume measures turnover (2).

Option-adjusted spread (OAS) The spread that reconciles a callable bond's model price with its market price (29, 31).

Option chain The grid of a listed underlying's options, strikes down and expirations across, showing bid, ask, volume and open interest for each call and put (10).

Ornstein–Uhlenbeck process A Gaussian process pulled back toward a fixed level at a constant rate, the dynamics Vasicek's model gives the short rate. In the Hull–White form of HJM the whole forward curve is a function of one such process (31, 32).

Output floor The rule that a bank's total risk-weighted assets from internal models may not fall below 72.5% of what the standardized approaches give, across all risk types. A bank on the floor gains nothing from a better model (39).

Overhedge Selling a discontinuous payoff and hedging with a portfolio whose payoff is at least as large everywhere, paying a known premium in place of an unhedgeable jump. That premium sets the floor under the dealer's quote, which in practice is wider (34).

Overnight indexed swap (OIS) A swap exchanging a fixed rate for the compounded overnight rate. The instrument from which the discount curve is built (4, 7).

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P

Par rate The fixed rate making a swap worth zero at inception (4, 7).

Par yield The coupon rate at which a bond is priced at its face value (4).

Participation rate The share of an underlying's rise that a structured note pays, which may exceed 100%. A residual rather than a promise: it is set by how many calls the money left after buying the zero-coupon bond will buy, and so mostly by the issuer's funding cost (35).

Pass-through security A mortgage-backed security that passes the pool's monthly interest and principal, prepayments included, straight to investors in proportion to their holdings. An agency pass-through carries little credit risk; its risk is prepayment, which is an option problem (26).

Path dependence A payoff that depends on the route, not only the destination. Where the model risk in exotics lives (34).

Pathwise differentiation Estimating a sensitivity by differentiating the payoff along each simulated path, with the draws held fixed, and averaging. It uses the paths that produced the price and leaves no bump size to choose, though its variance is no lower than a bump on common random numbers; it requires a differentiable payoff, so it works for a call and fails for a digital (20, 32).

Payment for order flow Payment by a wholesale market maker to a retail broker for routing customer orders to it, which has funded zero-commission trading in the US since 2019. The cost moves from an explicit commission into the execution price (10).

Payoff What a contract delivers at expiration or settlement, before any cost of acquiring it: $\max(S_T - K, 0)$ for a call, a note's redemption, the amount a swap pays. Distinct from profit (1, 10).

Peaks over threshold Estimating an extreme quantile or expected shortfall by fitting the generalized Pareto distribution to the losses above a high threshold and extrapolating the fitted tail beyond the data (38).

Perpetual futures A futures contract with no delivery date, held near spot by a funding-rate mechanism (37).

Physical measure The real-world probabilities that generate observed data, as distinct from the risk-neutral measure under which prices are expectations (21).

Pillar 1 / 2 / 3 The three parts of the Basel framework: the formula minimums on risk-weighted assets, the supervisor's bank-specific add-on for risks the formulas miss, and disclosure (27).

Pin risk The uncertainty, for the writer of an option struck near the closing price at expiration, about whether it will be assigned. The writer learns the outcome only after the market has closed and is left with an unhedged position of unknown size and direction, which is why such positions are closed before expiration (10, 23).

Pinning The tendency of an underlying to be drawn toward a strike with large open interest as expiration approaches, because dealers long gamma there sell rallies and buy declines around it. With daily expirations it now operates every session (10, 18, 23).

Pip One unit in the last decimal place of an exchange-rate quote: 0.01 yen in a dollar–yen quote (5).

P&L attribution test The test of whether a desk's risk model captures its P&L: hypothetical and risk-theoretical P&L are compared over 250 days by Spearman rank correlation and the Kolmogorov–Smirnov statistic, placing the desk in a green, amber or red zone (39).

Plain vanilla swap The standard interest rate swap: fixed payments exchanged for floating payments on one notional principal in one currency, the notional itself never changing hands

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(7).

Poisson process A count of events that rises by one at random moments, at a given average rate. It drives the jumps in Merton's model and the arrival of default in reduced-form credit models (21, 24).

Portfolio insurance Replicating a protective put on an equity portfolio by selling index futures as the market falls and buying them back as it rises. Its mechanical selling into the decline of October 19, 1987 is the standard example of a dynamic hedge amplifying the move it hedges (16, 18).

Portfolio margining Setting margin on the net risk of a whole account, by stressing it across a grid of underlying prices and volatilities and charging for the worst outcome, rather than leg by leg. It substantially reduces requirements for hedged positions, and so permits larger ones (2, 10, 12).

Porting Transferring a failed clearing member's client positions, with their collateral, to a surviving member rather than closing them out. It preserves the client's hedge, and it needs margin computed gross, client by client, so that each client's collateral can move with it (2).

Potential future exposure (PFE) An estimate of how large an exposure could become over a horizon (9, 27).

Predictor–corrector An Euler step taken twice: once to reach a predicted point, then again with the drift averaged between the start and that point. The remedy for a state-dependent drift, at one extra drift evaluation per step (20, 32).

Premium-adjusted delta In foreign exchange, an option's delta less its premium expressed in units of the foreign currency, $\Delta - c/S_0$: the delta seen by a holder who pays the premium in that currency (16).

Prepayment risk The risk that mortgage borrowers refinance early, shortening a pass-through's life exactly when that hurts (26).

Price discovery The process by which trading assembles many participants' scattered information into a publicly visible price, a public good produced as a by-product of speculation (2).

Price limit A limit that halts trading in a futures contract if its price moves more than a set amount from the previous settlement. It does not remove the information driving the move, and a market locked at its limit is one in which no one can trade (2).

Prime broker A bank that finances and executes trades for hedge funds and similar clients, often through total return swaps. Archegos built positions that no one of its prime brokers could see in full (1, 27, 40).

Principal component analysis Decomposing the correlation matrix of curve moves into eigenvectors, the factors, and eigenvalues, their shares of variance. For interest rates the first three read as level, slope and curvature (32).

Principal-protected note A note that returns its full face at maturity plus a participation rate times any rise in the underlying: a zero-coupon bond plus a call, paid for with the interest the buyer gives up (12, 35).

Procyclicality The tendency of a requirement to move with the market cycle and so reinforce it. Margin rises when volatility rises, which is in a falling market, so every member must find cash at once and the natural source is selling into the decline (2, 8, 40).

Profit The payoff net of what the position cost. A profit diagram and a payoff diagram cross zero at different points: the breakeven and the strike respectively (1, 10, 12).

Prudent valuation Valuing a position at a price it could be realized at with 90% confidence

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rather than at fair value, and deducting the difference from equity. A rule in the EU and the UK (39).

Put–call parity The arbitrage relation linking a call, a put, the underlying, and a bond. Model-free (11).

Q

Quadratic variation The limit of the sum of a process's squared increments. For a Wiener process over $[0, T]$ it is exactly T, and Itô's lemma can be derived from it (14).

Qualified investor An investor meeting asset or subscription thresholds that permit the sale of complex products, such as China's snowballs, which are sold with minimum subscriptions of about RMB 1 million (35).

Qualifying CCP A central counterparty licensed and supervised under rules consistent with the international standards for clearing houses. Trades against one carry a 2% risk weight (27).

Quanto A payoff computed from an underlying denominated in one currency and paid in another at a fixed exchange rate agreed in advance, so the holder carries no currency exposure. Requires a correlation-driven adjustment to the underlying's drift (16, 30).

Quanto spread The difference between dollar- and euro-denominated CDS on a euro-area sovereign: the market's price for the currency's depreciation conditional on that sovereign's default (25).

Quasi-Monte Carlo Simulation with low-discrepancy sequences, such as Sobol or Halton, that fill the space more evenly than random numbers. Convergence can approach $1/M$ rather than $1/\sqrt{M}$, but the usual standard error no longer applies (20).

R

Radon–Nikodym derivative The random variable $\Lambda = dQ/dP$ that reweights one measure into an equivalent one, positive with $E^P[\Lambda] = 1$, so that $E^Q[X] = E^P[\Lambda X]$. Changing measure is multiplying by it before averaging, never a change in the possible outcomes (28).

Range accrual A coupon accruing only on days a reference rate is inside a band; a strip of digitals, and short volatility (33).

Range forward A zero-cost collar on a currency. An importer buys a call and sells a put, with strikes set so the premiums offset; an exporter does the reverse. Either locks in a band of exchange rates without paying a premium (16).

Ratchet cap A cap whose strike resets each period to the rate just observed plus a spread. Path-dependent in the forward-rate vector, and sensitive to the shape of instantaneous volatility that caplet prices leave undetermined (32).

Rebate A fixed amount paid to the holder of a knock-out option when it is extinguished, or at expiration if the barrier has been touched; valued separately from the option (34, 35).

Recovery rate The fraction of face value a defaulted claim realizes; in a CDS, set by auction (9, 24, 25).

Reference entity The borrower a CDS is written on. Not a party to the contract (24, 25).

Reference rate A published benchmark that contracts point to rather than negotiate individually. It is useful only if widely accepted and hard to manipulate; LIBOR satisfied the first condition and not the second (4).

Reinvestment period The period, typically about five years, during which a CLO reinvests the principal it receives rather than returning it to noteholders, within the indenture's eligibility and diversity constraints (26).

Replacement cost What it would cost to replace a contract on current terms if the counterparty

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defaulted: its value to the surviving party, net of collateral, when positive. Far smaller than a swap's notional, and the first component of exposure at default under SA-CCR (7, 27).

Repo A sale with an agreement to repurchase; economically a secured loan, and the mechanism behind much collateral funding (8).

Repudiation / moratorium Credit events for sovereign CDS: the government disowns its debt, or declares a suspension of payments on it. Contracts on emerging-market sovereigns also include obligation acceleration, a default on one obligation that lets its holders demand immediate repayment (25).

Residual risk add-on A flat charge in the standardized approach on payoffs its sensitivities do not describe: 1% of notional for exotic underlyings, 0.1% for other residual risks such as gap or correlation risk (39).

Restructuring The most contested credit event, because the entity survives and its obligations trade at very different prices (25).

Reverse convertible A note paying a high fixed coupon whose face is repaid in shares, taking the full fall, if the underlying ends below a barrier. A bond plus a written put; the coupon is the risk-free rate plus the put's premium, less the issuer's fee (12, 35).

Reverse knock-out A knock-out option whose barrier lies on the side where the option is in the money, such as an up-and-out call. It pays only in the band between strike and barrier, where the volatility surface constrains models least, and behaves like a digital at the barrier (21, 34).

Reverse stress testing Fixing a loss (one that would breach a limit or exhaust capital) and finding the scenario that produces it. For a linear book with normal factors the most probable such scenario moves each factor in proportion to its covariance with the portfolio (38, 40).

Rho The sensitivity of an option's value to the interest rate. On short-dated equity options it is of the same order as vega and still ignored, because rates barely move; it is first-order for long-dated and currency options (18).

Richardson extrapolation Combining values at n and $2n$ steps as $2V(2n) - V(n)$ to cancel an error proportional to $1/n$. Reliable on a tree only along one branch of its oscillation, once a control variate has left that branch decaying smoothly, and only when both step counts place the strike at the same offset between nodes (20).

Right-way risk Exposure that falls as the counterparty's credit deteriorates, the reverse of wrong-way risk; it reduces CVA (9).

Risk-adjusted return on capital (RAROC) A loan's or a business's return divided by the economic capital allocated to it, compared with a hurdle rate. It is how a portfolio's credit VaR enters the pricing of each loan (24).

Risk factor eligibility test The test of whether a risk factor may be modeled: at least 24 real price observations over twelve months with no 90-day period containing fewer than four, or at least 100 observations (39).

Risk-free rate (RFR) An overnight rate built from actual borrowing transactions, such as SOFR or €STR; the replacement for LIBOR. Contracts use it compounded in arrears over each interest period (4).

Risk measure A rule mapping a loss distribution to a single number, used as a limit, a capital requirement or a hedging objective. VaR and expected shortfall are the two in regulatory use (38, TN39).

Risk-neutral probability The probability under which every asset is expected to earn the risk-free rate. Not the real-world probability or anyone's forecast; it is what makes a derivative's value a discounted expectation (13).

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Risk-neutral valuation Pricing as a discounted expectation under a measure in which all assets drift at the risk-free rate. Not a claim about probabilities (13, 15).

Risk reversal An out-of-the-money call against the corresponding put. The FX market quotes it as the difference between their implied volatilities, the 25-delta call's less the 25-delta put's, which is how it quotes skew; as a position, it is the structure that trades skew (12, 16, 19).

Risk-weighted assets (RWA) Exposure multiplied by a supervisory risk weight set by counterparty type. Multiplied by the required capital ratio, it gives the capital a position consumes (27, 40).

Risky annuity The present value of one basis point paid until default or maturity, whichever comes first. Under the standard fixed-coupon CDS it carries the whole valuation, and it is the credit market's duration, shrinking as the credit worsens because default cuts the premium stream short (25).

Roll yield The gain or loss from rolling a futures position along the curve (5).

Rough volatility The finding that the path of an asset's volatility is far more jagged than a Wiener path, with a Hurst parameter near 0.1, and the models built on that finding, which reproduce the steep short-dated skew (14, 21).

S

SABR model A stochastic volatility model for a forward rate, $dF = \alpha F^{\beta}\, dW_1$ and $d\alpha = \nu\alpha\, dW_2$, fitted strike by strike to each swaption smile in the volatility cube: α sets the level, β the backbone, ρ the skew and ν the curvature. The market standard because of Hagan's closed-form approximation for the implied volatility it produces (29).

SA-CCR The standardized regulatory approach to counterparty exposure (27).

Samuelson maturity effect The tendency of a futures contract's volatility to rise as its expiration approaches, so that near contracts are more volatile than far ones. It follows from mean reversion in the spot price (36).

Scenario analysis Revaluing a whole book in full over a grid of moves in the underlying and in volatility and reading the worst cell, because the Greeks are local approximations that fail for large moves (18).

Second fundamental theorem of asset pricing An arbitrage-free market is complete exactly when its equivalent martingale measure is unique. In an incomplete market a model chooses a price rather than reading it from the market (28).

Segregation Keeping client collateral separate from a clearing member's own. Under individual segregation each client's collateral sits in its own account; under omnibus segregation clients are pooled, which is cheaper and offers less protection, since one client's shortfall can reach the others (2).

Segregation of duties The control that keeps the people who take positions apart from those who settle, reconcile and value them, so that no one checks their own work. Its absence, not any instrument, produced the Barings and Société Générale losses (40).

Self-financing portfolio A portfolio whose value changes only because the prices of its holdings change: rebalancing is allowed, but every purchase is paid for by a sale and no cash is added or withdrawn. The condition that makes a replication argument an argument about prices (28).

Sensitivities-based method The core of the standardized approach: prescribed risk weights and correlations applied to delta, vega and curvature sensitivities, computed under three correlation scenarios with the worst taken (39).

Sensitivity analysis Moving one risk factor at a time and reporting the change in value. Easy

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to run and explain, it misses the joint moves that make a crisis costly (38).

Shifted lognormal Black's model applied to a rate plus a constant displacement, which restores a usable quoting convention when rates are negative (29, 32).

Short hedge / long hedge A short hedge sells futures to protect an asset owned, or to be owned, against a fall in price; a long hedge buys futures to protect a planned purchase against a rise (3).

Short-rate model A model specifying the dynamics of the instantaneous rate, from which the whole curve follows (31).

SIMM ISDA's Standard Initial Margin Model, the industry method for computing initial margin on uncleared derivatives from weighted sensitivities. A common model as much to prevent disputes as to measure risk (8).

Skew Implied volatility falling as the strike rises, the shape of equity index and single-stock option prices since October 1987. Low-strike puts are expensive because the market assigns more probability to large falls than the lognormal allows; traded with the risk reversal (15, 19, 21).

Skewness A measure of a distribution's asymmetry, $\mathbb{E}[(X-\mu)^3]/\sigma^3$. For returns it is negative when the long tail lies on the loss side; for a distribution of losses the same tail makes it positive. An autocallable's return distribution is negatively skewed even when its underlying's is positively skewed (35, 38).

Smile The pattern of implied volatility across strikes, and evidence that the Black–Scholes assumption is wrong in a stable way (19).

Snowball The domestic Chinese form of the autocallable, path-dependent through its monthly knock-out and knock-in observations, and sold through asset-management products with a minimum subscription around RMB 1 million, so its buyers are qualified investors rather than depositors (35).

SOFR Index An index, published by the New York Fed, that accumulates daily compounded SOFR from a fixed start date, so the compounded rate between any two dates follows from two index values (4).

Sovereign credit The risk that a national government does not pay, chiefly on debt in a currency it does not issue, since it can always print its own. Default is a choice rather than an insolvency, there is no bankruptcy court, and local-currency debt can be inflated away, so structural models do not apply (24).

Spark spread The power price less the heat rate times the gas price: the margin of a gas-fired plant, which is therefore a strip of exchange options on it. Its value turns almost entirely on the correlation between power and gas (28, 36).

Special purpose acquisition company (SPAC) A listed shell company that raises cash from public investors to merge with a private operating company. Its units of a share and a fraction of a warrant made SPAC warrants the highest-volume recent example of dilutive warrants (17).

Spike model A model of power prices as ordinary lognormal days punctuated, with some probability, by spike days on which the price is multiplied by a lognormal jump. The price is a mixture of two lognormals, with a far heavier upper tail than a lognormal of the same mean and standard deviation (36).

Spot measure The measure whose numeraire is an investment rolled at each reset date into the bond maturing at the next one; an alternative to the terminal measure for simulating the LIBOR market model (32).

Spot-starting cap A cap whose first accrual period begins today. Under LIBOR conventions its first caplet, already fixed, was excluded; with backward-looking rates it is no longer trivial (29).

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Spring-loading Granting an option or other award just before good news is released, so that the grant price is set below where the news will take the stock. The successor to backdating: SAB 120, issued in 2021, requires the award's fair value to reflect the pending news, and Item 402(x), adopted in 2022, requires US companies to disclose grants made around such releases (17).

Stablecoin A token its issuer promises to redeem for one unit of a currency, usually a dollar, backed by a reserve of assets in that currency; the settlement asset of most crypto trading, and a claim on the issuer's credit (37).

Stack and roll Hedging a long-dated exposure by placing all of it in the nearest futures contract and rolling it into the next as each expires. Each roll is taken at whatever spread then prevails, which in contango costs money on the whole stack; the futures losses are cash today while the offsetting gains arrive over years; and a one-for-one stack is the wrong ratio, overhedging once discounting and the Samuelson maturity effect are counted (3, 6, 36).

Standardized approach (SA) The trading-book capital charge every bank computes from prescribed formulas: the sensitivities-based method, the default risk charge and the residual risk add-on (39).

State price The price today of a security that pays one unit if a particular state occurs and nothing otherwise, an Arrow–Debreu security. A market admits no arbitrage if and only if a set of positive state prices exists (13).

Static replication Hedging with a portfolio of vanillas set up once and traded again at most once (for a barrier, when it is touched), with no rebalancing in between. Where it is possible, model risk is small (34).

Sticky cap A cap whose strike resets to the previous *capped* rate plus a spread, so the strike follows the rate down but rises by at most the spread each period (32).

Sticky strike / sticky delta Competing assumptions about how the smile moves when spot moves. They imply different deltas for the same option (19).

Stochastic volatility A model in which volatility is itself random, producing a smile through the correlation between volatility and the underlying and through the volatility of volatility. Heston's version fits a surface badly at both ends: its skew is too flat at short maturities and decays faster than the market's at long ones (21).

Stock appreciation right (SAR) An award paying the spread between the share price and a strike, in cash, without the holder finding the strike. The same payoff as an option and none of the dilution, but cash settlement makes it a liability, remeasured to fair value every reporting date rather than fixed at grant (17).

Straddle A call and a put at the same strike. At the money it has almost no directional exposure, but only while the underlying stays near the strike; once it moves the position becomes directional, so a pure volatility position must be delta-hedged (12).

Straight-line / graded recognition Two ways to spread an equity award's grant-date value over the vesting period: straight-line treats the grant as one award expensed equally each year; graded treats each vesting tranche as a separate award, so the early years carry a share of every tranche (17).

Stress capital buffer A US capital requirement on top of the minimum, set by the losses a large bank projects in the supervisory stress test, including a global market shock to its trading book (39).

Stress test Revaluing a portfolio under a specified scenario of moves in prices, rates, volatilities, correlations and liquidity, and reporting the loss with no probability attached. It complements VaR and ES by describing events the estimation window never contained (38, 40).

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Stressed expected shortfall (SES) The stress-scenario capital for non-modellable risk factors, calibrated factor by factor and added to the modeled charge (39).

Stressed VaR VaR computed on a fixed historical period of significant stress rather than on the recent window, so that capital cannot fall merely because markets have been calm. Added by Basel 2.5 in 2009 alongside the existing VaR; the current framework keeps the idea by calibrating expected shortfall to a period of stress (38, 40).

Strike price The price at which an option's holder may buy (a call) or sell (a put) the underlying (1, 10).

Strip A set of contracts of the same type spanning consecutive dates, or a run of strikes, traded and priced as a single position (6).

Structured note A bond whose coupon or redemption is linked to a derivative payoff (35).

Student's t distribution A bell-shaped distribution with ν degrees of freedom whose density falls off as a power, $|x|^{-(\nu+1)}$, rather than exponentially; the fewer the degrees of freedom, the fatter the tails, and it approaches the normal as ν grows. Its variance is $\nu/(\nu-2)$, so it is rescaled to a chosen standard deviation before use. With $\nu = 4$ it is the book's standard fat-tailed alternative to the normal (38).

Subadditivity The property $\rho(A+B) \le \rho(A) + \rho(B)$: diversification cannot raise measured risk. It is what lets desk limits add up to a firm limit; expected shortfall always has it and VaR does not (38, 40).

Super-senior tranche The top layer of a securitization's capital structure, above even the AAA-rated senior notes. Banks retained it as unsellable and apparently riskless, which is how the risk came back to them in 2007–08 (26).

Survival probability The probability $Q(t)$ that a borrower is still performing at t, obtained from the hazard rate as $Q(T) = \exp\bigl(-\int_0^T \lambda(u)\,du\bigr)$. A curve rather than a number; both legs of a CDS are integrals over its risk-neutral version (24, 25).

Swap An agreement to exchange cash flows on a schedule (7).

Swap market model A model that makes the forward swap rate lognormal under the annuity measure, in which it is a martingale, so that Black's formula prices a swaption exactly, as the LIBOR market model prices a caplet (32).

Swap measure The measure whose numeraire is a swap's annuity, also called the annuity measure. The forward swap rate is a martingale under it, which justifies Black's formula for swaptions; valuing a CMS payment requires a change away from it (28, 30).

Swap spread The swap rate minus the yield on a government bond of the same maturity. Long-dated dollar swap spreads have frequently been negative since 2008, which prices the balance sheet a dealer needs to hold the bond, not a view that banks are better credits than the sovereign (7).

Swaption An option to enter a swap (29).

Swing contract A fixed-price supply agreement in which the buyer chooses each day how much to take, within daily and total volume limits; the total limit links the days, so it is valued by dynamic programming over price and remaining volume (36).

Synthetic short A bought put and a written call at the same strike and expiration, which gains and loses exactly as a short stock position does; its price reflects the cost of borrowing the stock (10, 11). Also short exposure taken through a total return swap, where the dealer borrows the stock to hedge and the borrow cost dominates the spread (33).

Systematic risk Risk that moves with the market as a whole and cannot be diversified away. An asset with none has a futures price equal to the expected spot price; most assets have some,

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so the futures price sits below it (5).

T

Tail dependence The tendency of extreme outcomes to arrive together, measured as the limiting probability that one variable is extreme given that another is. The Gaussian copula has none, which is empirically false for credit (26, 38).

Tailing the hedge Reducing a futures hedge by the discount factor over the remaining hedge life, because daily settlement lets gains earn interest until the hedge is lifted. Small for short horizons and growing with maturity and the level of rates (3).

Target redemption note (TARN) A structured note that terminates once its cumulative coupons reach a stated threshold, so the redemption date itself is a function of the path and not of any single date's state (32).

Tenor The length of the period a rate applies to (overnight, one month, three months), as distinct from the maturity of the instrument that pays it. A ten-year swap whose floating rate resets every three months has a maturity of ten years and a tenor of three months (4, 7). For a swaption, the length of the underlying swap, so a 5×10 is a five-year option on a ten-year swap (29).

Term rate A forward-looking rate for a period, known at its start, which was LIBOR's defining feature. Term SOFR is constructed from SOFR futures prices rather than from bank submissions (4).

Term sheet The document setting out a structured note's terms for the buyer. It states the payoff but not that the investor is short volatility, or how often the bad outcomes occur (35).

Term SOFR A forward-looking SOFR term rate, published by CME for one, three, six, and twelve months and derived from SOFR futures prices rather than from bank submissions. Recommended by the ARRC for business loans and the derivatives hedging them, but not for the general derivatives market (4, 7, 30).

Term structure The dependence of a rate or a volatility on maturity (4, 19).

Theta The rate at which an option loses value with the passage of time (18).

Thomas algorithm Gaussian elimination specialized to a tridiagonal system, a forward sweep and a back substitution that touch only the three bands; how the implicit and Crank–Nicolson schemes are solved (20).

Tick size The smallest permitted change in a futures contract's price (2).

Time-homogeneous Of a volatility specification: depending only on the time left to maturity and not on calendar time, so the structure it implies for a year from now matches today's. Preferred for stability rather than for fit (32).

Time value An option's value in excess of its intrinsic value (10).

Title transfer / security interest The two legal forms of collateral. Under title transfer, collateral becomes the recipient's property, to be returned in equivalent form and free to be rehypothecated; under a security interest the poster keeps ownership and the recipient holds a charge it can enforce only on default (8).

Total return swap A swap exchanging an asset's total return for a financing rate; the instrument through which leverage accumulates without appearing in ownership disclosures (7, 27, 33).

Total shareholder return (TSR) Share price appreciation plus dividends. A relative TSR target is a market condition: it is built into an award's grant-date fair value, usually by simulation, and expensed even if the award pays nothing (17, 20).

Trading book / banking book The two capital regimes for a bank's positions: the trading book for positions held for sale or market making, capitalized against price moves, and the

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banking book for positions held to maturity, capitalized against default. The boundary between them is now prescriptive (39).

Tranche A layer of a securitization's loss distribution. Economically a call spread on the pool's loss (24, 25, 26).

Transition matrix The probabilities of moving from each rating to each other rating, default included, over a fixed period, usually a year. Its generator matrix gives the matrix for any other horizon (24).

Trinomial tree A tree in which the price can move up, down or stay level at each step. It absorbs state-dependent volatility into its probabilities, and the explicit finite difference scheme is one in disguise (20, 31).

V

Undertakings for Collective Investment in Transferable Securities (UCITS) The European Union's framework for funds sold to the public. A UCITS fund that measures its derivatives exposure by VaR must keep its 99%, twenty-day VaR below 20% of net asset value, or below twice that of a reference portfolio (38).

Value at risk (VaR) A loss threshold exceeded with a stated probability over a stated horizon. Not coherent, and not a worst case (38).

Vanna, volga Second-order sensitivities: vanna to spot and volatility together, how delta moves with volatility; volga, also called vomma, to volatility twice, how vega moves with volatility. What a delta-vega hedge leaves behind (18).

Variable annuity An investment-linked savings policy carrying a guaranteed minimum benefit. The guarantee depends jointly on equity and interest rate paths (32).

Variance swap A contract paying realized variance against a fixed rate; replicable from a strip of options (22).

Variation margin Collateral covering the current mark-to-market, settled as it changes (2, 8).

Variation margin gains haircutting A CCP recovery tool that withholds part of the variation margin gains owed to members whose positions have profited, taking money from the winners to pay for the losers once the default waterfall is exhausted (2).

Vega The sensitivity of an option's value to volatility (18).

Vega notional The way a variance swap is quoted: the slope of its payoff in realized volatility at the strike, so that a one-point move earns the notional to first order; variance notional is vega notional over twice the strike (22).

Vertical spread A spread whose legs share an expiration and differ only in strike, such as a bull call spread or a bear put spread. With the outright option and the risk reversal, one of the few strategies whose dominant exposure is direction (12).

VIX The 30-day variance swap rate on the S&P 500, expressed as a volatility and computed by Cboe from a discrete strip of listed options selected by a fixed rule. A well-defined number that is not the price of anything: no portfolio replicates it (19, 22).

Volatility drag The gap of $\sigma^2/2$ between the growth rate of a lognormal stock's expected value and the growth rate of its median, the return realized on a typical path (14).

Volatility cube Swaption implied volatility as a function of option expiration, underlying swap tenor and strike. Fixing the strike at the money gives a volatility surface; fixing expiration and tenor gives a smile, to which SABR is fitted (29).

Volatility risk premium The amount by which implied volatility tends to exceed the volatility later realized: typically two to four volatility points on equity indices, and more at low strikes. Partly compensation for selling protection that pays out exactly when markets fall (15, 19, 22).

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Volatility-scaled historical simulation Historical simulation in which each past return is rescaled by the ratio of today's volatility estimate to the estimate on the day it occurred, so the scenarios keep their historical shape and take today's scale; due to Hull and White. Unlike reweighting, it can produce a loss larger than any in the window (38).

Volatility surface Implied volatility as a function of strike and maturity. A quoting device that constrains, but does not determine, exotic prices (19).

Volatility swap A contract paying realized volatility, rather than variance, against a strike. Not statically replicable; priced off the variance swap less a convexity adjustment that grows with the volatility of volatility (22).

W

Warehouse A credit line from the arranging bank in which a CLO manager accumulates loans before the deal closes, with the future equity investors bearing the price risk (26).

Warrant An option issued by a company on its own stock. Exercise creates new shares, so the payoff is diluted (10, 17).

Waterfall The order in which a securitization's cash is paid out and its losses absorbed, the two running in opposite directions (26). Also the default waterfall, the sequence in which a CCP uses its default resources (2).

Wiener process A process whose increments over non-overlapping intervals are independent and normal, with variance equal to the length of the interval; also called Brownian motion, and written W in this book (14).

Worst-case default rate The default rate of a large portfolio that is exceeded in only a fraction $1 - q$ of years, the confidence-q quantile of the one-factor model's large-portfolio limit (24, 39).

Worst-of note A structured note whose principal is exposed to the worst performer among several underlyings. The investor has sold a worst-of put, and the put's premium is what pays the high coupon (34, 35).

Wrong-way risk Exposure that grows precisely as the counterparty's credit deteriorates (9, 27).

X

XVA The family of valuation adjustments (CVA, DVA, FVA, and others) applied to a derivatives book (9).

Y-Z

Yield curve The relation between yield and maturity for a class of instrument (4).

Z-spread The constant spread over the curve that reprices a bond while ignoring any embedded call (24, 31).

Zero-days-to-expiration (0DTE) option An option traded on its expiration day. Since S&P 500 options began expiring every trading day, they have grown to more than half of that market's volume; with hours to live they have almost no time value and very large gamma (10, 23).

Zero rate The rate earned on an investment made today that pays nothing until a single maturity. It can be quoted at any compounding frequency; continuously compounded, $R(t,T) = -\ln P(t,T)/(T-t)$, it is the curve every model in Part VI is fitted to (4, 31).

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Appendix D
Bibliography

This is a record of where the subject came from, not only of what the chapters cite. *A paper is listed if it mattered*—if it opened a line of work, or settled a question, or was the standard treatment for a decade—whether or not the main text discusses it and whether or not its results have since been superseded. Hull and White's 1987 stochastic-volatility paper is here for that reason, as are the short-rate models the market has moved past: a reader who does not know what was tried before cannot judge what is used now.

The number after each entry is the chapter the work belongs to, so the list doubles as an index of what each subject rests on. Official-sector documents are cited by issuing body, since their authorship is institutional and their editions change.

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